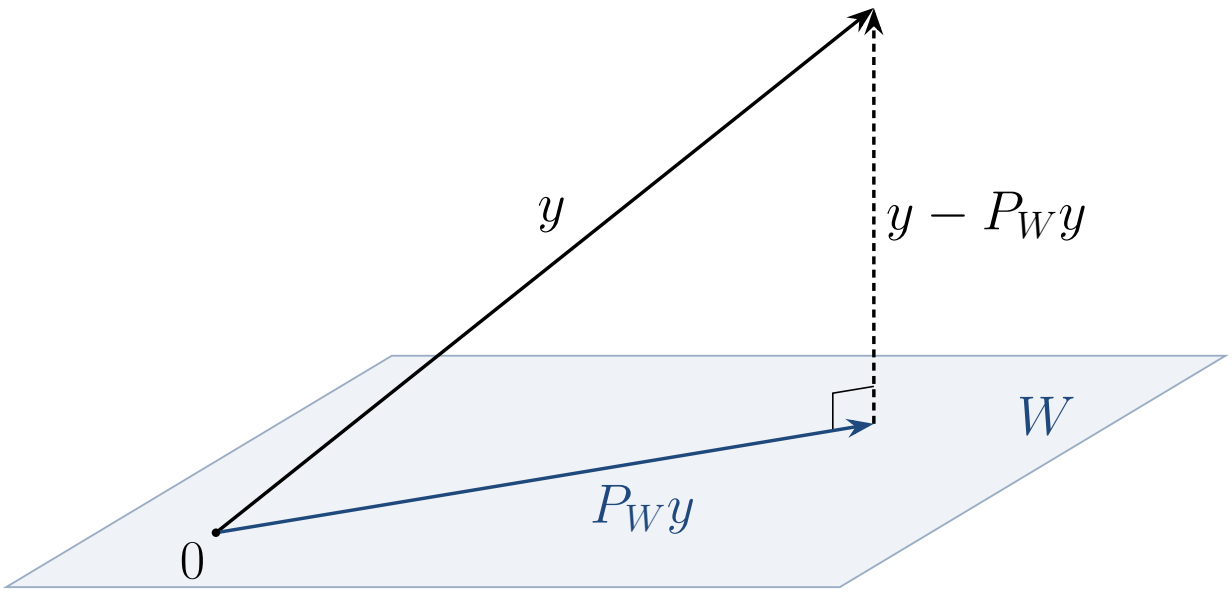
Linear Algebra

And Its Statistical Applications



Chapter 1. Matrices

Matrices and products

Definition 1.1 ([1, §1.3], [2, Ch. 2], [3, §1.1])**.** Let m, n be positive integers. An $m \times n$ **matrix** $A = (a_{ij})$ is a rectangular array of real numbers, its **entries**; a_{ij} stands in row i and column j ($1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$). The set of all $m \times n$ matrices is $\mathbb{R}^{m\times n}$.

- $A = B$ means that A and B have the same size and $a_{ij} = b_{ij}$ for all i, j.
- $\mathbb{R}^n = \mathbb{R}^{n\times 1}$ is the set of **column vectors** $x = (x_i)$, and $\mathbb{R}^{1\times n}$ the set of **row vectors**. A 1×1 matrix is identified with its entry.
- $\mathrm{col}_j(A) \in \mathbb{R}^m$ is the jth column and $\mathrm{row}_i(A) = (a_{i1}, \ldots, a_{in}) \in \mathbb{R}^{1\times n}$ the ith row of A. We write $a_j = \mathrm{col}_j(A)$, so $A = [a_1 \ \cdots \ a_n]$.
- Sums and scalar multiples are taken entrywise: $(A + B)_{ij} = a_{ij} + b_{ij}$, $(cA)_{ij} = ca_{ij}$ ($c \in \mathbb{R}$); $-A = (-1)A$ and $A - B = A + (-B)$. They obey the laws of numbers, such as $A + B = B + A$ and $c(A + B) = cA + cB$, because these hold entry by entry.
- The **zero matrix** 0 has all entries 0. The **identity matrix** is $I_n = (\delta_{ij}) \in \mathbb{R}^{n\times n}$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$; its columns $e_1, \ldots, e_n$ are the **unit vectors** of $\mathbb{R}^n$.

In statistics, measurements of k variables on n units are stored in an $n \times k$ **data matrix** (x_{ij}): row i holds the measurements on unit i, column j holds the n values of variable j, and x_{ij} is the value of variable j on unit i.

Definition 1.2 ([1, §1.5], [4, Ch. II, §3])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$. The **product** $AB \in \mathbb{R}^{m\times p}$ is defined by

$$(AB)_{ij} = \sum_{r=1}^{n} a_{ir}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \qquad (1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant p)$$

It is defined only when A has as many columns as B has rows.

Theorem 1.3 ([1, §1.5, Theorem 8], [4, Ch. II, Theorem 3.1])**.** Whenever the sizes allow the operations,

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC, \quad A(cB) = c(AB) = (cA)B$$

$$I_mA = A = AI_n, \qquad 0A = 0, \qquad A0 = 0 \qquad (A \in \mathbb{R}^{m\times n})$$

Proof. Let $A \in \mathbb{R}^{m\times n}$. In each rule both sides have the same size, so by Definition 1.1 it suffices to show that they have the same (i, j) entry. Every entry below is computed from Definition 1.2 and from the entrywise definition of sums and scalar multiples.

- $A(BC) = (AB)C$ for $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times q}$. Since $(BC)_{rj} = \sum_{s=1}^{p} b_{rs}c_{sj}$,

$$[A(BC)]_{ij} = \sum_{r=1}^{n} a_{ir}(BC)_{rj} = \sum_{r=1}^{n} a_{ir}\Big(\sum_{s=1}^{p} b_{rs}c_{sj}\Big) = \sum_{r=1}^{n}\sum_{s=1}^{p} a_{ir}b_{rs}c_{sj}$$

  Since $(AB)_{is} = \sum_{r=1}^{n} a_{ir}b_{rs}$,

$$[(AB)C]_{ij} = \sum_{s=1}^{p}(AB)_{is}c_{sj} = \sum_{s=1}^{p}\Big(\sum_{r=1}^{n} a_{ir}b_{rs}\Big)c_{sj} = \sum_{s=1}^{p}\sum_{r=1}^{n} a_{ir}b_{rs}c_{sj}$$

  Both double sums add the same np numbers $a_{ir}b_{rs}c_{sj}$ ($1 \leqslant r \leqslant n$, $1 \leqslant s \leqslant p$), only in a different order, so they are equal.

- $A(B+C) = AB + AC$ for $B, C \in \mathbb{R}^{n\times p}$:

$$[A(B+C)]_{ij} = \sum_{r=1}^{n} a_{ir}(b_{rj} + c_{rj}) = \sum_{r=1}^{n} a_{ir}b_{rj} + \sum_{r=1}^{n} a_{ir}c_{rj} = (AB)_{ij} + (AC)_{ij} = (AB + AC)_{ij}$$

- $(A+B)C = AC + BC$ for $B \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{n\times p}$:

$$[(A+B)C]_{ij} = \sum_{r=1}^{n}(a_{ir} + b_{ir})c_{rj} = \sum_{r=1}^{n} a_{ir}c_{rj} + \sum_{r=1}^{n} b_{ir}c_{rj} = (AC + BC)_{ij}$$

- $A(cB) = c(AB) = (cA)B$ for $B \in \mathbb{R}^{n\times p}$, $c \in \mathbb{R}$:

$$[A(cB)]_{ij} = \sum_{r=1}^{n} a_{ir}(cb_{rj}) = c\sum_{r=1}^{n} a_{ir}b_{rj} = [c(AB)]_{ij}$$
$$[(cA)B]_{ij} = \sum_{r=1}^{n}(ca_{ir})b_{rj} = c\sum_{r=1}^{n} a_{ir}b_{rj} = [c(AB)]_{ij}$$

- $I_m A = A = AI_n$. In $(I_m A)_{ij} = \sum_{r=1}^{m} \delta_{ir}a_{rj}$ every term with $r \neq i$ is 0, because $\delta_{ir} = 0$, and the term with $r = i$ is $1 \cdot a_{ij}$; so $(I_m A)_{ij} = a_{ij}$. In the same way $(AI_n)_{ij} = \sum_{r=1}^{n} a_{ir}\delta_{rj} = a_{ij}$.
- $0A = 0$ and $A0 = 0$: every term of $(0A)_{ij} = \sum_r 0 \cdot a_{rj}$ and of $(A0)_{ij} = \sum_r a_{ir} \cdot 0$ is 0. □

By the first rule, a product of several matrices needs no parentheses. A 1×1 factor in a product acts as a number: for $M \in \mathbb{R}^{m\times 1}$, $N \in \mathbb{R}^{1\times p}$ and a 1×1 matrix c, identified with its entry (Definition 1.1), Definition 1.2 gives $(Mc)_{i1} = m_{i1}c$ and $(cN)_{1j} = cn_{1j}$, so the products Mc and cN are the scalar multiples cM and cN, and hence $McN = M(cN) = c(MN)$ by the rules above. For $A \in \mathbb{R}^{n\times n}$ we put $A^0 = I_n$ and $A^{k+1} = A^k A$ $(k \geqslant 0)$.

Theorem 1.4 ([5, (2.2.7)–(2.2.11)], [1, §1.5], [3, §1.1])**.** Let $A \in \mathbb{R}^{m\times n}$ have columns $a_1, \ldots, a_n$ and $B \in \mathbb{R}^{n\times p}$ columns $b_1, \ldots, b_p$.

(a) $AB = [Ab_1 \ \cdots \ Ab_p]$, that is, $\mathrm{col}_j(AB) = Ab_j$.

(b) For $x \in \mathbb{R}^n$, $Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n$. In particular $Ae_j = a_j$.

(c) $\mathrm{row}_i(AB) = \mathrm{row}_i(A)B = a_{i1}\,\mathrm{row}_1(B) + \cdots + a_{in}\,\mathrm{row}_n(B)$.

(d) $(AB)_{ij} = \mathrm{row}_i(A)\,b_j$, and $AB = a_1\,\mathrm{row}_1(B) + \cdots + a_n\,\mathrm{row}_n(B)$, a sum of n products of a column and a row (the **outer product expansion**).

Proof. Write $B = (b_{rj})$. The column a_r of A has entries $a_{1r}, \ldots, a_{mr}$, the column b_j of B has entries $b_{1j}, \ldots, b_{nj}$, and $\mathrm{row}_r(B) = (b_{r1}, \ldots, b_{rp})$. By Definition 1.2,

$$(AB)_{ij} = \sum_{r=1}^{n} a_{ir}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Each part compares entries with this sum (Definition 1.1).

- (a) Fix j. The product Ab_j of $A \in \mathbb{R}^{m\times n}$ and $b_j \in \mathbb{R}^{n\times 1}$ lies in $\mathbb{R}^m$, and by Definition 1.2 its ith entry is

$$(Ab_j)_i = \sum_{r=1}^{n} a_{ir}(b_j)_r = \sum_{r=1}^{n} a_{ir}b_{rj} = (AB)_{ij} \qquad (1 \leqslant i \leqslant m)$$

  So the vector Ab_j and $\mathrm{col}_j(AB)$, whose entries are $(AB)_{1j}, \ldots, (AB)_{mj}$, have the same entries:

$$\mathrm{col}_j(AB) = \begin{pmatrix} a_{11}b_{1j} + \cdots + a_{1n}b_{nj} \\ \vdots \\ a_{m1}b_{1j} + \cdots + a_{mn}b_{nj} \end{pmatrix} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{1j} \\ \vdots \\ b_{nj} \end{pmatrix} = Ab_j$$

- (b) By Definition 1.2 (with $p = 1$), the ith entry of Ax is $a_{i1}x_1 + \cdots + a_{in}x_n$. By the entrywise definition of scalar multiples and sums (Definition 1.1), the ith entry of $x_1a_1 + \cdots + x_na_n$ is $x_1a_{i1} + \cdots + x_na_{in}$, the same number. Hence

$$Ax = \begin{pmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{pmatrix} = x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} = x_1a_1 + \cdots + x_na_n$$

  For $x = e_j$ we have $x_j = 1$ and $x_r = 0$ for $r \neq j$, so $Ae_j = 1 \cdot a_j = a_j$.

- (c) Fix i. The product of $\operatorname{row}_i(A) = (a_{i1}, \ldots, a_{in}) \in \mathbb{R}^{1\times n}$ and $B \in \mathbb{R}^{n\times p}$ lies in $\mathbb{R}^{1\times p}$, and by Definition 1.2 its jth entry is $\sum_{r=1}^{n} a_{ir}b_{rj} = (AB)_{ij}$, the jth entry of $\operatorname{row}_i(AB)$. So $\operatorname{row}_i(AB) = \operatorname{row}_i(A)B$. The jth entry of the row vector $a_{i1}\operatorname{row}_1(B) + \cdots + a_{in}\operatorname{row}_n(B)$ is $a_{i1}b_{1j} + \cdots + a_{in}b_{nj}$, the same sum. Hence

$$\operatorname{row}_i(AB) = \Big(\sum_{r=1}^{n} a_{ir}b_{r1},\ \ldots,\ \sum_{r=1}^{n} a_{ir}b_{rp} \Big) = \sum_{r=1}^{n} a_{ir}\,(b_{r1}, \ldots, b_{rp}) = \sum_{r=1}^{n} a_{ir}\operatorname{row}_r(B)$$

- (d) The product of $\operatorname{row}_i(A) \in \mathbb{R}^{1\times n}$ and $b_j \in \mathbb{R}^{n\times 1}$ is the 1×1 matrix with entry $\sum_{r=1}^{n} a_{ir}b_{rj} = (AB)_{ij}$. Next fix r. The product of $a_r \in \mathbb{R}^{m\times 1}$ and $\operatorname{row}_r(B) \in \mathbb{R}^{1\times p}$ is the $m \times p$ matrix whose (i, j) entry is the one-term sum $a_{ir}b_{rj}$ (Definition 1.2 with inner size 1):

$$a_r \operatorname{row}_r(B) = \begin{pmatrix} a_{1r} \\ \vdots \\ a_{mr} \end{pmatrix} (b_{r1},\ \ldots,\ b_{rp}) = \begin{pmatrix} a_{1r}b_{r1} & \cdots & a_{1r}b_{rp} \\ \vdots & & \vdots \\ a_{mr}b_{r1} & \cdots & a_{mr}b_{rp} \end{pmatrix}$$

  Adding these n matrices entry by entry, the (i, j) entry of $a_1\operatorname{row}_1(B) + \cdots + a_n\operatorname{row}_n(B)$ is $a_{i1}b_{1j} + \cdots + a_{in}b_{nj} = (AB)_{ij}$. □

Example 1.5. Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & 1 \\ 2 & 0 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 4}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 2 & -1 \end{pmatrix} \in \mathbb{R}^{4\times 2}$$

By Definition 1.2, each entry of $AB \in \mathbb{R}^{3\times 2}$ is a sum of four products:

$$\begin{aligned} (AB)_{11} &= 1\cdot 1 + 2\cdot 0 + 0\cdot 1 + 1\cdot 2 = 3, & (AB)_{12} &= 1\cdot 0 + 2\cdot 1 + 0\cdot 1 + 1\cdot(-1) = 1 \\ (AB)_{21} &= 0\cdot 1 + 1\cdot 0 + 3\cdot 1 + 1\cdot 2 = 5, & (AB)_{22} &= 0\cdot 0 + 1\cdot 1 + 3\cdot 1 + 1\cdot(-1) = 3 \\ (AB)_{31} &= 2\cdot 1 + 0\cdot 0 + 1\cdot 1 + 1\cdot 2 = 5, & (AB)_{32} &= 2\cdot 0 + 0\cdot 1 + 1\cdot 1 + 1\cdot(-1) = 0 \end{aligned}$$

so

$$AB = \begin{pmatrix} 3 & 1 \\ 5 & 3 \\ 5 & 0 \end{pmatrix}$$

The three views of Theorem 1.4 produce the same matrix.

- Columns, by (a) and (b): column j of AB is the combination of the four columns of A with coefficients from column j of B,

$$Ab_1 = 1\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + 0\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 5 \end{pmatrix}$$

$$Ab_2 = 0\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} - 1\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$

- Rows, by (c): row i of AB is the combination of the four rows of B with coefficients from row i of A,

$$\begin{aligned}\text{row}_1(AB) &= 1\,(1,0) + 2\,(0,1) + 0\,(1,1) + 1\,(2,-1) = (3,\ 1)\\ \text{row}_2(AB) &= 0\,(1,0) + 1\,(0,1) + 3\,(1,1) + 1\,(2,-1) = (5,\ 3)\\ \text{row}_3(AB) &= 2\,(1,0) + 0\,(0,1) + 1\,(1,1) + 1\,(2,-1) = (5,\ 0)\end{aligned}$$

- Outer products, by (d): AB is the sum of four 3×2 matrices $a_r\,\text{row}_r(B)$,

$$\begin{aligned}AB &= \begin{pmatrix}1\\0\\2\end{pmatrix}(1,\ 0) + \begin{pmatrix}2\\1\\0\end{pmatrix}(0,\ 1) + \begin{pmatrix}0\\3\\1\end{pmatrix}(1,\ 1) + \begin{pmatrix}1\\1\\1\end{pmatrix}(2,\ -1)\\ &= \begin{pmatrix}1&0\\0&0\\2&0\end{pmatrix} + \begin{pmatrix}0&2\\0&1\\0&0\end{pmatrix} + \begin{pmatrix}0&0\\3&3\\1&1\end{pmatrix} + \begin{pmatrix}2&-1\\2&-1\\2&-1\end{pmatrix} = \begin{pmatrix}3&1\\5&3\\5&0\end{pmatrix}\end{aligned}$$

Example 1.6 ([1, §1.5, Example 10])**.** (a) By Definition 1.2,

$$\begin{pmatrix}1&0&0&0\\-3&1&0&0\\2&0&1&0\\-1&0&0&1\end{pmatrix}\begin{pmatrix}1&-1&2&1\\3&4&8&5\\-2&3&-1&0\\1&1&3&3\end{pmatrix} = \begin{pmatrix}1&-1&2&1\\0&7&2&2\\0&1&3&2\\0&2&1&2\end{pmatrix}$$

By Theorem 1.4(c), its second row is $-3\,(1,-1,2,1) + 1\,(3,4,8,5) = (0,7,2,2)$, its third row is $2\,(1,-1,2,1) + 1\,(-2,3,-1,0) = (0,1,3,2)$, and its fourth row is $-1\,(1,-1,2,1) + 1\,(1,1,3,3) = (0,2,1,2)$.

(b) By Theorem 1.4(b),

$$\begin{pmatrix}2&1&0\\5&4&1\\1&3&2\\0&1&3\end{pmatrix}\begin{pmatrix}2\\-1\\1\end{pmatrix} = 2\begin{pmatrix}2\\5\\1\\0\end{pmatrix} - 1\begin{pmatrix}1\\4\\3\\1\end{pmatrix} + 1\begin{pmatrix}0\\1\\2\\3\end{pmatrix} = \begin{pmatrix}3\\7\\1\\2\end{pmatrix}$$

(c) For $N = \begin{pmatrix}0&1&0\\0&0&0\\0&0&0\end{pmatrix}$ and $M = \begin{pmatrix}1&-5&2\\2&3&4\\9&-1&3\end{pmatrix}$,

$$NM = \begin{pmatrix}2&3&4\\0&0&0\\0&0&0\end{pmatrix}, \qquad MN = \begin{pmatrix}0&1&0\\0&2&0\\0&9&0\end{pmatrix}$$

Row 1 of NM is row 2 of M, and column 2 of MN is column 1 of M. So $NM \neq MN$, although both are 3×3.

Remark 1.7 ([1, §1.5, Example 10], [2, §4.9])**.** Matrix products differ from products of numbers. In general $AB \neq BA$ (Example 1.6(c)); square matrices with $AB = BA$ are said to **commute**. A product can be 0 although no factor is 0: for

$$X = \begin{pmatrix}1&1&3\\2&2&6\\-1&-1&-3\end{pmatrix}, \qquad X^2 = 0$$

To see this, let $u = \text{col}_1(X)$, the column with entries $u_1 = 1$, $u_2 = 2$, $u_3 = -1$, and let $w_1 = 1$, $w_2 = 1$, $w_3 = 3$ be the entries of $\text{row}_1(X) = (1,1,3)$. Rows $1, 2, 3$ of X are $1, 2, -1$ times $\text{row}_1(X)$,

that is, $\mathrm{row}_i(X) = u_i\,\mathrm{row}_1(X)$; columns $1,2,3$ of X are $1,1,3$ times u, that is, $\mathrm{col}_j(X) = w_j u$. By Theorem 1.4(d) and the rule $A(cB) = c(AB) = (cA)B$ of Theorem 1.3,

$$(X^2)_{ij} = \mathrm{row}_i(X)\,\mathrm{col}_j(X) = \Big(u_i\,\mathrm{row}_1(X)\Big)(w_j u) = u_i w_j\,\Big(\mathrm{row}_1(X)\,u\Big)$$
$$\mathrm{row}_1(X)\,u = 1\cdot 1 + 1\cdot 2 + 3\cdot(-1) = 0$$

so every entry of X^2 is 0. Hence $AB = 0$ does not force $A = 0$ or $B = 0$ (take $A = B = X$), and $AB = AC$ does not force $B = C$ (take $A = B = X$ and $C = 0$).

Transpose and trace

Definition 1.8. The **transpose** of $A \in \mathbb{R}^{m\times n}$ is $A' \in \mathbb{R}^{n\times m}$ with $(A')_{ij} = a_{ji}$: the rows of A' are the transposed columns of A. We write column vectors as $x = (x_1, \ldots, x_n)'$. A square A is **symmetric** if $A' = A$.

Theorem 1.9 ([4, Ch. II, Theorem 3.3], [2, Example 5.6])**.** For matrices of sizes for which the operations are defined and $c \in \mathbb{R}$,

$$(A')' = A, \quad (A+B)' = A' + B', \quad (cA)' = cA', \quad (AB)' = B'A', \quad (ABC)' = C'B'A'$$

For $x, y \in \mathbb{R}^n$, $x'y = y'x = \sum_{i=1}^n x_i y_i$; in particular $x'x = \sum_i x_i^2$. The identity matrix is symmetric, $I_n' = I_n$, and for every $B \in \mathbb{R}^{m\times n}$, $B'B$ and BB' are symmetric.

Proof. In each identity both sides have the same size; we compare their entries, using $(A')_{ij} = a_{ji}$ (Definition 1.8).

- For $A, B \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}$, Definitions 1.8 and 1.1 give
$$((A')')_{ij} = (A')_{ji} = a_{ij}, \qquad ((cA)')_{ij} = (cA)_{ji} = ca_{ji} = (cA')_{ij}$$
$$((A+B)')_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A' + B')_{ij}$$
- Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$. Then $(AB)'$ and $B'A'$ are both $p \times m$. For $1 \leqslant k \leqslant p$ and $1 \leqslant i \leqslant m$, Definition 1.2 gives
$$(B'A')_{ki} = \sum_{j=1}^n (B')_{kj}(A')_{ji} = \sum_{j=1}^n b_{jk}a_{ij} = \sum_{j=1}^n a_{ij}b_{jk} = (AB)_{ik} = ((AB)')_{ki}$$
- For a third matrix C with ABC defined, the previous bullet applied first to the pair AB, C and then to the pair A, B gives
$$(ABC)' = ((AB)C)' = C'(AB)' = C'B'A'$$
- For $x, y \in \mathbb{R}^n = \mathbb{R}^{n\times 1}$, $x' \in \mathbb{R}^{1\times n}$ has entries $(x')_{1i} = x_i$, so $x'y$ is 1×1 with entry $\sum_{i=1}^n x_i y_i$. In the same way $y'x = \sum_{i=1}^n y_i x_i$, the same number since $y_i x_i = x_i y_i$. With $y = x$, $x'x = \sum_i x_i^2$.
- $(I_n')_{ij} = \delta_{ji} = \delta_{ij}$ for all i, j (Definition 1.1), so $I_n' = I_n$.
- For $B \in \mathbb{R}^{m\times n}$, the product rule and $(B')' = B$ give $(B'B)' = B'(B')' = B'B$ and $(BB')' = (B')'B' = BB'$. □

Definition 1.10 ([5, §5.1])**.** The **trace** of $A \in \mathbb{R}^{n\times n}$ is $\mathrm{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$.

Theorem 1.11 ([5, (5.1.3)–(5.1.5), Lemma 5.2.1, (5.2.9), (5.2.5), Lemma 5.3.1])**.** (a) $\mathrm{tr}(A+B) = \mathrm{tr}(A) + \mathrm{tr}(B)$, $\mathrm{tr}(cA) = c\,\mathrm{tr}(A)$ and $\mathrm{tr}(A') = \mathrm{tr}(A)$ for $A, B \in \mathbb{R}^{n\times n}$.

(b) $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ for $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$.

(c) $\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}(BCA)$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times m}$.

(d) $\operatorname{tr}(A'A) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2$ for $A \in \mathbb{R}^{m\times n}$; hence $\operatorname{tr}(A'A) = 0$ only if $A = 0$.

Proof. • (a) The ith diagonal entries of $A + B$, cA and A' are $a_{ii} + b_{ii}$, ca_{ii} and $(A')_{ii} = a_{ii}$ (Definitions 1.1 and 1.8). Hence, by Definition 1.10,

$$\operatorname{tr}(A+B) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii}, \qquad \operatorname{tr}(cA) = \sum_{i=1}^{n} ca_{ii} = c\sum_{i=1}^{n} a_{ii}, \qquad \operatorname{tr}(A') = \sum_{i=1}^{n} a_{ii}$$

- (b) Both $AB \in \mathbb{R}^{m\times m}$ and $BA \in \mathbb{R}^{n\times n}$ are square. By Definitions 1.10 and 1.2,

$$\operatorname{tr}(AB) = \sum_{i=1}^{m}(AB)_{ii} = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji}, \qquad \operatorname{tr}(BA) = \sum_{j=1}^{n}(BA)_{jj} = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij}$$

  Both double sums add the same mn numbers $a_{ij}b_{ji}$ ($1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$), so they are equal.

- (c) Since $AB \in \mathbb{R}^{m\times p}$ and $C \in \mathbb{R}^{p\times m}$, (b) applied to the pair AB, C gives $\operatorname{tr}((AB)C) = \operatorname{tr}(C(AB))$, that is, $\operatorname{tr}(ABC) = \operatorname{tr}(CAB)$. Since $A \in \mathbb{R}^{m\times n}$ and $BC \in \mathbb{R}^{n\times m}$, (b) applied to the pair A, BC gives $\operatorname{tr}(A(BC)) = \operatorname{tr}((BC)A)$, that is, $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$. Parentheses are dropped by Theorem 1.3.

- (d) The matrix $A'A$ is $n \times n$, and its jth diagonal entry is $(A'A)_{jj} = \sum_{i=1}^{m}(A')_{ji}a_{ij} = \sum_{i=1}^{m} a_{ij}^2$. Summing over $j = 1, \ldots, n$ gives $\operatorname{tr}(A'A) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2$. Every term a_{ij}^2 is $\geqslant 0$. If some $a_{kl} \neq 0$, then $\operatorname{tr}(A'A) \geqslant a_{kl}^2 > 0$. So $\operatorname{tr}(A'A) = 0$ forces every $a_{ij} = 0$, that is, $A = 0$. □

Special matrices; means and sums of squares

Definition 1.12 ([2, §4.7.8], [5, §1.3g])**.** For $d_1, \ldots, d_n \in \mathbb{R}$, $\operatorname{diag}(d_1, \ldots, d_n)$ is the $n \times n$ matrix with diagonal entries $d_1, \ldots, d_n$ and all other entries 0; a matrix of this form is **diagonal**. A square matrix A is **upper triangular** if $a_{ij} = 0$ for $i > j$, and **lower triangular** if $a_{ij} = 0$ for $i < j$.

Proposition 1.13 ([2, §4.7.8], [5, Lemma 1.3.1])**.** Let $D = \operatorname{diag}(d_1, \ldots, d_n)$ and $F = \operatorname{diag}(f_1, \ldots, f_n)$.

(a) Left multiplication by a diagonal matrix scales the rows, right multiplication scales the columns: for a matrix A with n rows, $(DA)_{ij} = d_i a_{ij}$ and $\operatorname{row}_i(DA) = d_i \operatorname{row}_i(A)$; for a matrix A with n columns, $(AD)_{ij} = a_{ij}d_j$ and $\operatorname{col}_j(AD) = d_j \operatorname{col}_j(A)$.

(b) $\operatorname{diag}(d_1, \ldots, d_n)\operatorname{diag}(f_1, \ldots, f_n) = \operatorname{diag}(d_1f_1, \ldots, d_nf_n)$.

(c) If $A, B \in \mathbb{R}^{n\times n}$ are upper (lower) triangular, then AB is upper (lower) triangular, with diagonal entries $a_{11}b_{11}, \ldots, a_{nn}b_{nn}$.

Proof. (a) The (i, r) entry of D is d_i if $r = i$ and 0 if $r \neq i$, so in each sum of Definition 1.2 only one term survives:

$$(DA)_{ij} = \sum_{r=1}^{n} D_{ir}a_{rj} = d_i a_{ij}, \qquad (AD)_{ij} = \sum_{r=1}^{n} a_{ir}D_{rj} = a_{ij}d_j$$

So row i of DA is d_i times row i of A, and column j of AD is d_j times column j of A.

(b) By (a), $(DF)_{ij} = d_iF_{ij}$ is d_if_i for $i = j$ and 0 for $i \neq j$, that is, $DF = \operatorname{diag}(d_1f_1, \ldots, d_nf_n)$.

(c) For upper triangular $A, B \in \mathbb{R}^{n\times n}$, the term $a_{ik}b_{kj}$ is 0 if $k < i$ (then $a_{ik} = 0$) and if $k > j$ (then $b_{kj} = 0$); so only the terms with $i \leqslant k \leqslant j$ remain:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} = \sum_{k=i}^{j} a_{ik}b_{kj}$$

For $i > j$ no k satisfies $i \leqslant k \leqslant j$, so $(AB)_{ij} = 0$; for $i = j$ only $k = i$ remains, so $(AB)_{ii} = a_{ii}b_{ii}$. For lower triangular A, B the term $a_{ik}b_{kj}$ is 0 if $k > i$ or $k < j$, so only the terms with $j \leqslant k \leqslant i$ remain, and $(AB)_{ij} = 0$ for $i < j$, $(AB)_{ii} = a_{ii}b_{ii}$. □

Definition 1.14 ([2, §§5.2, 5.3])**.** The **summing vector** $\mathbf{1}_n \in \mathbb{R}^n$ has all entries 1. Put

$$J_n = \mathbf{1}_n \mathbf{1}_n' \in \mathbb{R}^{n\times n}, \qquad \bar{J}_n = \tfrac{1}{n} J_n, \qquad C_n = I_n - \bar{J}_n$$

Every entry of J_n is 1, and C_n is the **centering matrix**. Subscripts are dropped when the size is clear. A square matrix A with $A^2 = A$ is **idempotent**.

Proposition 1.15 ([2, §5.2, (5.13)–(5.18)])**.** $\mathbf{1}'\mathbf{1} = n$, $J_n^2 = nJ_n$, and $\bar{J}_n$ and $C = C_n$ are symmetric and idempotent:

$$\bar{J}_n' = \bar{J}_n = \bar{J}_n^2, \qquad C = C' = C^2, \qquad C\mathbf{1} = 0$$

Proof. Write $\mathbf{1} = \mathbf{1}_n$, $J = J_n$, $\bar{J} = \bar{J}_n$, $I = I_n$.

- By Theorem 1.9, $\mathbf{1}'\mathbf{1} = \sum_{i=1}^n 1 \cdot 1 = n$.
- By Theorem 1.3 (associativity) and the paragraph after Theorem 1.3 (the 1×1 factor $\mathbf{1}'\mathbf{1} = n$ acts as a number),

$$J^2 = (\mathbf{1}\mathbf{1}')(\mathbf{1}\mathbf{1}') = \mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{1}\, n\, \mathbf{1}' = n\, \mathbf{1}\mathbf{1}' = nJ, \qquad \bar{J}^2 = \tfrac{1}{n^2} J^2 = \tfrac{1}{n^2}\, nJ = \tfrac{1}{n} J = \bar{J}$$

- By Theorem 1.9, $J' = (\mathbf{1}\mathbf{1}')' = (\mathbf{1}')'\mathbf{1}' = \mathbf{1}\mathbf{1}' = J$, so $\bar{J}' = \frac{1}{n} J' = \bar{J}$; and $I' = I$, by the same theorem. Hence $C' = I' - \bar{J}' = I - \bar{J} = C$.
- By Theorem 1.3 and $\bar{J}^2 = \bar{J}$,

$$C^2 = (I - \bar{J})(I - \bar{J}) = I \cdot I - I\bar{J} - \bar{J}I + \bar{J}^2 = I - 2\bar{J} + \bar{J} = I - \bar{J} = C$$

- By Theorem 1.3, the paragraph after it and $\mathbf{1}'\mathbf{1} = n$, $C\mathbf{1} = \mathbf{1} - \frac{1}{n}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1} - \frac{1}{n}\, n\, \mathbf{1} = 0$. □

Example 1.16 ([2, Example 5.6])**.** For data $x = (x_1, \ldots, x_n)'$, $\mathbf{1}'x = x_1 + \cdots + x_n$ (Theorem 1.9), so the mean is $\bar{x} = \frac{1}{n}\mathbf{1}'x$, and $\mathbf{1}'x = n\bar{x}$. By Theorem 1.3,

$$Cx = (I - \tfrac{1}{n}\mathbf{1}\mathbf{1}')x = x - \tfrac{1}{n}\mathbf{1}(\mathbf{1}'x) = x - \bar{x}\mathbf{1} = (x_1 - \bar{x},\ \ldots,\ x_n - \bar{x})'$$

the vector of **deviations** from the mean. Moreover $x'\mathbf{1} = \mathbf{1}'x = n\bar{x}$ (Theorem 1.9), so

$$x'Cx = x'x - \tfrac{1}{n}(x'\mathbf{1})(\mathbf{1}'x) = \sum_{i=1}^n x_i^2 - \tfrac{1}{n}(n\bar{x})(n\bar{x}) = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

Expanding each square and using $\sum_i x_i = n\bar{x}$,

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i^2 - 2\bar{x}x_i + \bar{x}^2) = \sum_{i=1}^n x_i^2 - 2\bar{x}\sum_{i=1}^n x_i + n\bar{x}^2 = \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

Hence the sum of squares about the mean is

$$\sum_{i=1}^n (x_i - \bar{x})^2 = x'x - n\bar{x}^2 = x'Cx$$

Example 1.17 (cf. [2, Example 11.1])**.** Four units ($n = 4$) are measured on two variables ($k = 2$); x_{ij} is the value of variable j on unit i:

$$X_0 = (x_{ij}) = \begin{pmatrix} 2 & 3 \\ 0 & 3 \\ 2 & 1 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 2}, \qquad \mathbf{1}'X_0 = (4,\ 8), \qquad \bar{x} = \tfrac{1}{4} X_0'\mathbf{1} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

The vector $\bar{x} = (\bar{x}_1, \bar{x}_2)'$ of column means $\bar{x}_j = \frac{1}{n}\sum_i x_{ij}$ is the **mean vector**. By Theorem 1.4(a), $\operatorname{col}_j(CX_0) = C\operatorname{col}_j(X_0)$, which by Example 1.16 is the vector of deviations of variable j from its mean $\bar{x}_j$. In matrix form, by Theorem 1.3 and $\mathbf{1}'X_0 = (X_0'\mathbf{1})' = n\bar{x}'$ (Theorem 1.9),

$$CX_0 = X_0 - \tfrac{1}{n}\mathbf{1}(\mathbf{1}'X_0) = X_0 - \mathbf{1}\bar{x}'$$

By Proposition 1.15, $C = C^2 = C'C$, so $X_0'CX_0 = X_0'C'CX_0 = (CX_0)'(CX_0)$ (Theorem 1.9). The centered data matrix and the matrix of **corrected sums of squares and products** are therefore

$$CX_0 = \begin{pmatrix} 2-1 & 3-2 \\ 0-1 & 3-2 \\ 2-1 & 1-2 \\ 0-1 & 1-2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \\ 1 & -1 \\ -1 & -1 \end{pmatrix}, \qquad X_0'CX_0 = (CX_0)'(CX_0) = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$

Here the (j,l) entry of $(CX_0)'(CX_0)$ is $\operatorname{col}_j(CX_0)'\operatorname{col}_l(CX_0)$ (Theorem 1.4(d)): $1+1+1+1 = 4$ for $(1,1)$ and $(2,2)$, and $1-1-1+1 = 0$ for $(1,2)$ and $(2,1)$. The same matrix comes from the uncorrected sums: in general, by Theorem 1.3 and $X_0'\mathbf{1} = n\bar{x}$,

$$X_0'CX_0 = X_0'(I - \tfrac{1}{n}\mathbf{1}\mathbf{1}')X_0 = X_0'X_0 - \tfrac{1}{n}(X_0'\mathbf{1})(\mathbf{1}'X_0) = X_0'X_0 - \tfrac{1}{n}(n\bar{x})(n\bar{x}') = X_0'X_0 - n\bar{x}\bar{x}'$$

and here

$$X_0'X_0 - 4\bar{x}\bar{x}' = \begin{pmatrix} 8 & 8 \\ 8 & 20 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ 8 & 16 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$

Unit vectors and the matrices E_{ij}

Proposition 1.18 ([2, §5.1.5], [5, (2.4.1)–(2.4.5), Lemma 2.3.2])**.** Let $A \in \mathbb{R}^{m\times n}$. Write e_i for the unit vectors of $\mathbb{R}^m$ or of $\mathbb{R}^n$, whichever the product requires, and put $E_{ij} = e_ie_j' \in \mathbb{R}^{m\times n}$ $(1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant n)$.

(a) $e_i'e_k = \delta_{ik}$, $Ae_j = \operatorname{col}_j(A)$, $e_i'A = \operatorname{row}_i(A)$ and $e_i'Ae_j = a_{ij}$.

(b) E_{ij} has (i,j) entry 1 and all other entries 0, and

$$A = \sum_{i=1}^{m} e_i \operatorname{row}_i(A) = \sum_{j=1}^{n} \operatorname{col}_j(A)\, e_j' = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}E_{ij}$$

In particular $I_m = e_1e_1' + \cdots + e_me_m'$.

(c) $E_{ij}E_{kl} = \delta_{jk}E_{il}$, where $E_{kl} = e_ke_l' \in \mathbb{R}^{n\times p}$ and $E_{il} = e_ie_l' \in \mathbb{R}^{m\times p}$.

(d) For $B \in \mathbb{R}^{m\times n}$, $A = B$ if and only if $Ax = Bx$ for every $x \in \mathbb{R}^n$.

Proof. (a)

- $e_i'e_k = \sum_r (e_i)_r(e_k)_r$ (Theorem 1.9), and the term $(e_i)_r(e_k)_r$ is 1 if $r = i = k$ and 0 otherwise. So $e_i'e_k = \delta_{ik}$.
- $Ae_j = \operatorname{col}_j(A)$ is Theorem 1.4(b). By Theorem 1.4(c), $e_i'A = (e_i)_1\operatorname{row}_1(A) + \cdots + (e_i)_m\operatorname{row}_m(A) = \operatorname{row}_i(A)$.
- Applied to the $m \times 1$ matrix $\operatorname{col}_j(A)$, the last statement gives $e_i'Ae_j = e_i'\operatorname{col}_j(A) = a_{ij}$.

(b)

- The (k,l) entry of e_ie_j' is $(e_i)_k(e_j)_l$ (Definition 1.2 with one term), which is 1 for $k = i$, $l = j$ and 0 otherwise.

- The outer product expansion (Theorem 1.4(d)) of $A = I_m A$, whose first factor has the columns $e_1, \dots, e_m$, gives $A = \sum_i e_i \operatorname{row}_i(A)$. As $I_n' = I_n$ (Theorem 1.9), the rows of I_n are $e_1', \dots, e_n'$, and the expansion of $A = AI_n$ gives $A = \sum_j \operatorname{col}_j(A) e_j'$.
- By Theorem 1.4(b), $\operatorname{col}_j(A) = I_m \operatorname{col}_j(A) = \sum_i a_{ij} e_i$. Substituting and using Theorem 1.3, $A = \sum_j \sum_i a_{ij} e_i e_j'$. For $A = I_m$, $\operatorname{row}_i(I_m) = e_i'$, and the first expansion reads $I_m = \sum_i e_i e_i'$.

(c) $E_{ij}E_{kl} = e_i(e_j'e_k)e_l' = \delta_{jk} e_i e_l'$ by Theorem 1.3 and (a), the 1×1 factor $e_j'e_k$ being a number.

(d) If $A = B$, then $Ax = Bx$ for every x. Conversely, $x = e_j$ gives $\operatorname{col}_j(A) = Ae_j = Be_j = \operatorname{col}_j(B)$ for every j, by (a); so $A = B$. □

For a data matrix $X_0 \in \mathbb{R}^{n \times k}$, $e_i' X_0$ is the row of measurements on unit i, $X_0 e_j$ the column of values of variable j, and $x_{ij} = e_i' X_0 e_j$. Several unit vectors select several variables or units: for instance, if $k \geqslant 3$ and $n \geqslant 4$, then by Theorem 1.4(a), $X_0[e_3\ e_1] = [\operatorname{col}_3(X_0)\ \operatorname{col}_1(X_0)]$ keeps variables 3 and 1, in this order; by Theorem 1.4(c), the rows of $[e_2\ e_4]' X_0$ are $\operatorname{row}_2(X_0)$ and $\operatorname{row}_4(X_0)$.

Powers and polynomials of a square matrix

Proposition 1.19 ([4, Ch. II, §3], [6, Exercises 2.33, 2.34], [7, Ch. II, §3, Exercise 30], [5, Lemma 1.3.1])**.** Let $A \in \mathbb{R}^{n \times n}$, and let $r, s \geqslant 0$ be integers.

(a) $A^r A^s = A^{r+s} = A^s A^r$.

(b) $(A^r)^s = A^{rs}$.

(c) $(A')^r = (A^r)'$. In particular, every power of a symmetric matrix is symmetric.

(d) $\operatorname{diag}(d_1, \dots, d_n)^r = \operatorname{diag}(d_1^r, \dots, d_n^r)$ for all $d_1, \dots, d_n \in \mathbb{R}$; here $t^0 = 1$ for every $t \in \mathbb{R}$, also for $t = 0$.

(e) If A is upper (lower) triangular, then A^r is upper (lower) triangular, with diagonal entries $a_{11}^r, \dots, a_{nn}^r$ (with $t^0 = 1$ for every $t \in \mathbb{R}$, as in (d)).

Proof. (a) Induction on s, with r fixed.

- $s = 0$: $A^r A^0 = A^r I_n = A^r$ (Theorem 1.3).
- If $A^r A^s = A^{r+s}$, then $A^{r+s+1} = A^{r+s} A = (A^r A^s) A = A^r (A^s A) = A^r A^{s+1}$ (Theorem 1.3).
- Exchanging the roles of r and s, $A^s A^r = A^{s+r} = A^{r+s}$.

(b) Induction on s: $(A^r)^0 = I_n = A^0$, and if $(A^r)^s = A^{rs}$, then $(A^r)^{s+1} = (A^r)^s A^r = A^{rs} A^r = A^{rs+r}$ by (a).

(c)

- Induction on r: $(A')^0 = I_n = I_n' = (A^0)'$ (Theorem 1.9). If $(A')^r = (A^r)'$, then by Theorem 1.9 and (a), $(A^{r+1})' = (A^r A)' = A'(A^r)' = A'(A')^r = (A')^{r+1}$.
- If $A' = A$, this reads $(A^r)' = A^r$.

(d) Induction on r: $\operatorname{diag}(d_1, \dots, d_n)^0 = I_n = \operatorname{diag}(1, \dots, 1) = \operatorname{diag}(d_1^0, \dots, d_n^0)$, and if the formula holds for r, then $\operatorname{diag}(d_i)^{r+1} = \operatorname{diag}(d_i^r) \operatorname{diag}(d_i) = \operatorname{diag}(d_i^{r+1})$ by Proposition 1.13(b).

(e) Induction on r: $A^0 = I_n$ is upper triangular with diagonal entries $1 = a_{ii}^0$. If A^r is upper triangular with diagonal entries a_{ii}^r, then $A^{r+1} = A^r A$ is a product of two upper triangular matrices, so it is upper triangular with diagonal entries $a_{ii}^r a_{ii} = a_{ii}^{r+1}$ (Proposition 1.13(c)). The lower triangular case is the same. □

Definition 1.20 ([4, Ch. II, §3, Exercise 39], [2, §5.3], [5, Ch. 8, Exercise 2])**.** A matrix $A \in \mathbb{R}^{n \times n}$ is **nilpotent** if $A^k = 0$ for some integer $k \geqslant 1$, and **involutory** if $A^2 = I_n$.

For instance, the matrix X of Remark 1.7 is nilpotent ($X^2 = 0$), and I_n, $-I_n$ and $\operatorname{diag}(1,-1,1,-1)$ are involutory.

Proposition 1.21 ([8, Ch. 1, Exercise 8])**.** Let $A \in \mathbb{R}^{n\times n}$ be **strictly upper triangular**, that is, $a_{ij} = 0$ for $i \geqslant j$, and let $p \geqslant 1$ be an integer.

(a) The (i,j) entry of A^p is 0 for $1 \leqslant i \leqslant n$ and $1 \leqslant j \leqslant \min(n, i+p-1)$.

(b) For $i \geqslant n-p+1$, the ith row of A^p is zero.

(c) $A^p = 0$ for $p \geqslant n$; so A is nilpotent.

The same holds for a **strictly lower triangular** A ($a_{ij} = 0$ for $i \leqslant j$), with the (j,i) entry in (a) and the ith column in (b).

Proof. Write b_{ik} for the (i,k) entry of A^p.

(a) Induction on p.

- $p = 1$: the range is $j \leqslant \min(n,i) = i$, and $a_{ij} = 0$ for $j \leqslant i$.
- Suppose (a) holds for p, and let $j \leqslant \min(n, i+p)$. Since $A^{p+1} = A^p A$,

$$(A^{p+1})_{ij} = \sum_{k=1}^{n} b_{ik}a_{kj} = \sum_{k=1}^{\min(n,i+p-1)} 0 \cdot a_{kj} + \sum_{k=i+p}^{n} b_{ik}a_{kj} = 0$$

  The first sum vanishes by the hypothesis on A^p. The second is empty if $i+p > n$; in it $k \geqslant i+p \geqslant j$, so $a_{kj} = 0$. This is (a) for $p+1$.

(b) For $i \geqslant n-p+1$, $i+p-1 \geqslant n$, so $\min(n, i+p-1) = n$, and by (a) all n entries of row i of A^p are 0.

(c) For $p \geqslant n$, $n-p+1 \leqslant 1$, so by (b) all rows of A^p are zero.

If A is strictly lower triangular, then A' is strictly upper triangular and $A^p = \left((A')^p\right)'$ (Proposition 1.19(c)); transposing (a)–(c) for A' gives the statements for A. □

Example 1.22 ([4, Ch. II, §3, Exercise 31])**.** (a) $N = \begin{pmatrix} 0 & 1 & 6 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix}$ is strictly upper triangular. By Theorem 1.4(c), $\operatorname{row}_1(N^2) = 1\ \operatorname{row}_2(N) + 6\ \operatorname{row}_3(N) = (0,0,4)$, $\operatorname{row}_2(N^2) = 4\ \operatorname{row}_3(N) = 0$ and $\operatorname{row}_3(N^2) = 0$; then $\operatorname{row}_1(N^3) = 4\ \operatorname{row}_3(N) = 0$. So

$$N^2 = \begin{pmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = N^2 N = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

as Proposition 1.21 says for $n = 3$. Here $N^2 \neq 0$, so the exponent n in Proposition 1.21(c) cannot be lowered in general.

(b) For the upper triangular

$$U = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}, \qquad U^2 = \begin{pmatrix} 1 & 3 & 1 & 3 \\ 0 & 4 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 4 \end{pmatrix}, \qquad U^3 = U^2U = \begin{pmatrix} 1 & 7 & 2 & 8 \\ 0 & 8 & 3 & 3 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

for instance $\operatorname{row}_1(U^2) = 1\ \operatorname{row}_1(U) + 1\ \operatorname{row}_2(U) + 1\ \operatorname{row}_4(U) = (1,3,1,3)$ and $\operatorname{row}_1(U^3) = 1\ \operatorname{row}_1(U) + 3\ \operatorname{row}_2(U) + 1\ \operatorname{row}_3(U) + 3\ \operatorname{row}_4(U) = (1,7,2,8)$. Both powers are upper triangular, with the diagonal entries $1^2, 2^2, (-1)^2, 2^2$ and $1^3, 2^3, (-1)^3, 2^3$ (Proposition 1.19(e)); and $(U')^3 = (U^3)'$ is lower triangular (Proposition 1.19(c)).

Definition 1.23 ([1, §4.2])**.** A (real) **polynomial** $f(t) = c_0 + c_1 t + \cdots + c_m t^m$ is given by its coefficients $c_0, \ldots, c_m \in \mathbb{R}$. For $A \in \mathbb{R}^{n\times n}$,

$$f(A) = c_0 I_n + c_1 A + \cdots + c_m A^m$$

Polynomials are added and multiplied as usual: for $g(t) = d_0 + d_1 t + \cdots + d_l t^l$, the product $(fg)(t) = \sum_{i=0}^{m}\sum_{j=0}^{l} c_i d_j t^{i+j}$ has the coefficient $\sum_{i+j=k} c_i d_j$ at t^k; in particular $fg = gf$.

Proposition 1.24 ([1, §4.2, Theorem 2])**.** Let f and g be polynomials, $c \in \mathbb{R}$ and $A \in \mathbb{R}^{n\times n}$.

(a) $(cf + g)(A) = cf(A) + g(A)$.

(b) $(fg)(A) = f(A)g(A)$. Hence $f(A)g(A) = g(A)f(A)$: polynomials in the same matrix commute.

(c) $f(A)' = f(A')$; in particular $f(A)$ is symmetric if A is.

Proof. Write $f(t) = \sum_i c_i t^i$ and $g(t) = \sum_j d_j t^j$, with coefficients 0 beyond the degrees.
(a) $cf + g$ has the coefficients $cc_i + d_i$, so $(cf + g)(A) = \sum_i (cc_i + d_i)A^i = c\sum_i c_i A^i + \sum_i d_i A^i$ by the laws of Definition 1.1.
(b)

- By Definition 1.23 and (a), $(fg)(A) = \sum_i \sum_j c_i d_j A^{i+j}$.
- $c_i d_j A^{i+j} = (c_i A^i)(d_j A^j)$ by Proposition 1.19(a) and Theorem 1.3.
- By the distributive laws (Theorem 1.3), $\sum_i \sum_j (c_i A^i)(d_j A^j) = \left(\sum_i c_i A^i\right)\left(\sum_j d_j A^j\right) = f(A)g(A)$.
- As $fg = gf$, $f(A)g(A) = (fg)(A) = (gf)(A) = g(A)f(A)$.

(c) By Theorem 1.9 and Proposition 1.19(c), $f(A)' = \sum_i c_i (A^i)' = \sum_i c_i (A')^i = f(A')$. □

Example 1.25. Matrices $aI_n + bJ_n$, with all diagonal entries $a+b$ and all other entries b, are frequent in statistics; $\bar{J}_n$ and $C_n = I_n - \bar{J}_n$ are of this form. They are polynomials of degree at most 1 in J_n, so any two of them commute (Proposition 1.24(b)). Since $J_n^2 = nJ_n$ (Proposition 1.15),

$$(aI_n + bJ_n)(cI_n + dJ_n) = acI_n + (ad + bc)J_n + bdJ_n^2 = acI_n + (ad + bc + nbd)J_n$$

is again of this form. For instance, $C_n = I_n - \frac{1}{n}J_n$ and J_n ($a = 1$, $b = -\frac{1}{n}$, $c = 0$, $d = 1$) give $C_n J_n = \left(1 + 0 - n \cdot \frac{1}{n}\right)J_n = 0$, and $J_n C_n = C_n J_n = 0$. For $n = 4$ and $a = b = c = d = 1$,

$$A = I_4 + J_4 = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}, \qquad A^2 = I_4 + 6J_4 = \begin{pmatrix} 7 & 6 & 6 & 6 \\ 6 & 7 & 6 & 6 \\ 6 & 6 & 7 & 6 \\ 6 & 6 & 6 & 7 \end{pmatrix}$$

For $f(t) = t^2 - 6t + 5 = (t-1)(t-5)$, Proposition 1.24(b) gives

$$f(A) = (A - I_4)(A - 5I_4) = J_4(J_4 - 4I_4) = J_4^2 - 4J_4 = 0$$

So $A^2 = 6A - 5I_4$, and every power of A reduces to a combination of I_4 and A, by induction on k: $A^1 = A$, and if $A^k = \alpha A + \beta I_4$, then $A^{k+1} = A^k A = \alpha A^2 + \beta A = (6\alpha + \beta)A - 5\alpha I_4$ (Theorem 1.3). For instance $A^3 = AA^2 = 6A^2 - 5A = 6(6A - 5I_4) - 5A = 31A - 30I_4$ (Proposition 1.19(a)), that is,

$$A^3 = \begin{pmatrix} 32 & 31 & 31 & 31 \\ 31 & 32 & 31 & 31 \\ 31 & 31 & 32 & 31 \\ 31 & 31 & 31 & 32 \end{pmatrix}$$

Symmetric, skew-symmetric and idempotent matrices

Definition 1.26 ([2, §5.1.6], [6, Ch. 2])**.** A matrix $A \in \mathbb{R}^{n\times n}$ is **skew-symmetric** if $A' = -A$, that is, $a_{ji} = -a_{ij}$ for all i, j.

Proposition 1.27 ([6, Exercises 2.22, 2.23])**.** Let $A \in \mathbb{R}^{n\times n}$.

(a) $A + A'$ is symmetric and $A - A'$ is skew-symmetric.

(b) $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$, the sum of a symmetric and a skew-symmetric matrix.

(c) If A is skew-symmetric, then $a_{ii} = 0$ for every i.

(d) If $B \in \mathbb{R}^{m\times m}$ is symmetric and $P \in \mathbb{R}^{m\times n}$, then $P'BP \in \mathbb{R}^{n\times n}$ is symmetric.

Proof. (a) By Theorem 1.9, $(A + A')' = A' + (A')' = A' + A$ and $(A - A')' = A' - A = -(A - A')$.
(b) By (a) and $(cM)' = cM'$ (Theorem 1.9), $\frac{1}{2}(A + A')$ is symmetric and $\frac{1}{2}(A - A')$ skew-symmetric; their sum is A, since the terms $\pm\frac{1}{2}A'$ cancel.
(c) A' has the same diagonal entries as A, so $A' = -A$ gives $a_{ii} = -a_{ii}$, that is, $a_{ii} = 0$.
(d) $(P'BP)' = P'B'(P')' = P'BP$ (Theorem 1.9). □

For instance, the matrix $X_0'C_nX_0$ of corrected sums of squares and products (Example 1.17) is symmetric by (d), since C_n is symmetric (Proposition 1.15).

Example 1.28. For

$$A = \begin{pmatrix} 1 & 4 & -1 & 2 \\ 0 & 2 & 3 & -1 \\ 3 & -1 & 5 & 3 \\ 0 & 1 & 1 & 3 \end{pmatrix}, \qquad A' = \begin{pmatrix} 1 & 0 & 3 & 0 \\ 4 & 2 & -1 & 1 \\ -1 & 3 & 5 & 1 \\ 2 & -1 & 3 & 3 \end{pmatrix}$$

Proposition 1.27(b) gives $A = S + K$ with

$$S = \tfrac{1}{2}(A + A') = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 2 & 1 & 0 \\ 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 \end{pmatrix}, \qquad K = \tfrac{1}{2}(A - A') = \begin{pmatrix} 0 & 2 & -2 & 1 \\ -2 & 0 & 2 & -1 \\ 2 & -2 & 0 & 1 \\ -1 & 1 & -1 & 0 \end{pmatrix}$$

S is symmetric, K is skew-symmetric with zero diagonal, and $S + K = A$ entry by entry.

Proposition 1.29 ([5, Lemma 10.1.2], [2, §5.3, Exercise 5.10(b)], [8, Ch. 10, Exercise 1(b)])**.** Let $A, B \in \mathbb{R}^{n\times n}$.

(a) A is idempotent if and only if A' is idempotent.

(b) A is idempotent if and only if $I_n - A$ is idempotent; and then $A(I_n - A) = (I_n - A)A = 0$.

(c) If A is idempotent, then $A^k = A$ for every integer $k \geqslant 1$.

(d) If A and B are idempotent and $AB = BA$, then AB is idempotent.

Proof. (a)

- If $A^2 = A$, then $(A')^2 = (AA)' = A'$ (Theorem 1.9).
- Conversely, if A' is idempotent, so is $(A')' = A$, by the first step.

(b)

- If $A^2 = A$, then $(I - A)^2 = I - 2A + A^2 = I - A$ (Theorem 1.3).

- Conversely, if $I - A$ is idempotent, so is $I - (I - A) = A$, by the first step.
- Then $A(I - A) = A - A^2 = 0$ and $(I - A)A = A - A^2 = 0$.

(c) Induction on k: $A^1 = A$, and if $A^k = A$, then $A^{k+1} = A^k A = A^2 = A$.
(d) $(AB)(AB) = A(BA)B = A(AB)B = A^2B^2 = AB$ (Theorem 1.3). □

Traces of products; a cancellation rule

Proposition 1.30 ([5, (5.1.1), (5.2.4), (5.2.6)], [6, Exercise 2.26(b)]). (a) $\operatorname{tr}(I_n) = n$, $\operatorname{tr}(J_n) = n$, $\operatorname{tr}(\bar{J}_n) = 1$ and $\operatorname{tr}(C_n) = n - 1$.

(b) For $A, B \in \mathbb{R}^{m\times n}$ with columns $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\operatorname{tr}(A'B) = \operatorname{tr}(AB') = \sum_{j=1}^{n} a_j' b_j = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ij}$$

In particular $\operatorname{tr}(ab') = a'b$ for $a, b \in \mathbb{R}^m$.

(c) $x'Ax = \operatorname{tr}(Axx')$ for $A \in \mathbb{R}^{n\times n}$ and $x \in \mathbb{R}^n$.

Proof. (a) The diagonal entries of I_n and of J_n are all 1. By Theorem 1.11(a), $\operatorname{tr}(\bar{J}_n) = \frac{1}{n}\operatorname{tr}(J_n) = 1$ and $\operatorname{tr}(C_n) = \operatorname{tr}(I_n) - \operatorname{tr}(\bar{J}_n) = n - 1$.
(b)

- The jth diagonal entry of $A'B$ is $\operatorname{row}_j(A')\, b_j = a_j' b_j = \sum_i a_{ij} b_{ij}$ (Theorems 1.4(d) and 1.9); summing over j gives the two sums.
- $\operatorname{tr}(AB') = \operatorname{tr}(B'A)$ (Theorem 1.11(b)), and the diagonal entries of $B'A$ are $b_j' a_j = a_j' b_j$, so $\operatorname{tr}(AB')$ is the same sum.
- For $n = 1$, $A = a$ and $B = b$, this reads $a'b = \operatorname{tr}(ab')$.

(c) By Theorem 1.11(b), applied to the $1 \times n$ matrix x' and the $n \times 1$ matrix Ax, $\operatorname{tr}\big(x'(Ax)\big) = \operatorname{tr}\big((Ax)x'\big)$; the trace of the 1×1 matrix $x'Ax$ is this number. □

Corollary 1.31 ([5, Corollaries 5.3.2, 5.3.3]). Let $A \in \mathbb{R}^{m\times n}$.

(a) $A'A = 0$ if and only if $A = 0$.

(b) For $B_1, B_2 \in \mathbb{R}^{n\times p}$, $A'AB_1 = A'AB_2$ if and only if $AB_1 = AB_2$.

(c) For $B_1, B_2 \in \mathbb{R}^{p\times n}$, $B_1A'A = B_2A'A$ if and only if $B_1A' = B_2A'$.

Proof. (a) If $A = 0$, then $A'A = 0$. If $A'A = 0$, then $\operatorname{tr}(A'A) = 0$, so $A = 0$ by Theorem 1.11(d).
(b)

- If $AB_1 = AB_2$, multiplying by A' on the left gives $A'AB_1 = A'AB_2$.
- Conversely, let $A'AB_1 = A'AB_2$. By Theorems 1.3 and 1.9, $(AB_1 - AB_2)' = (B_1' - B_2')A'$, so

$$(AB_1 - AB_2)'(AB_1 - AB_2) = (B_1' - B_2')(A'AB_1 - A'AB_2) = 0$$

- By (a), applied to the matrix $AB_1 - AB_2$, $AB_1 - AB_2 = 0$.

(c) By Theorem 1.9, $B_1A'A = B_2A'A$ is equivalent to $A'AB_1' = A'AB_2'$, and $B_1A' = B_2A'$ to $AB_1' = AB_2'$; these two are equivalent by (b). □

Example 1.32 ([2, Example 5.2])**.** Let

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{pmatrix}, \qquad A'A = \begin{pmatrix} 10 & 2 & 2 \\ 2 & 4 & -2 \\ 2 & -2 & 2 \end{pmatrix}$$

(the (j,l) entry of $A'A$ is $\mathrm{col}_j(A)'\mathrm{col}_l(A)$, Theorems 1.4(d) and 1.9; for instance the $(1,1)$ entry $1\cdot 1 + 3\cdot 3 = 10$ and the $(2,3)$ entry $2\cdot(-1) + 0\cdot 1 = -2$), and let $b_1 = (1,0,0)'$ and $b_2 = (0,2,3)'$. By Theorem 1.4(b),

$$A'Ab_1 = \mathrm{col}_1(A'A) = \begin{pmatrix} 10 \\ 2 \\ 2 \end{pmatrix}, \qquad A'Ab_2 = 2\,\mathrm{col}_2(A'A) + 3\,\mathrm{col}_3(A'A) = \begin{pmatrix} 4+6 \\ 8-6 \\ -4+6 \end{pmatrix} = \begin{pmatrix} 10 \\ 2 \\ 2 \end{pmatrix}$$

So $A'Ab_1 = A'Ab_2$, although $b_1 \neq b_2$: the factor $A'A$ cannot be cancelled. The factor A' can: by Corollary 1.31(b), with $B_1 = b_1$ and $B_2 = b_2$ ($p = 1$), $Ab_1 = Ab_2$. Indeed, by Theorem 1.4(b),

$$Ab_1 = \mathrm{col}_1(A) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad Ab_2 = 2\,\mathrm{col}_2(A) + 3\,\mathrm{col}_3(A) = \begin{pmatrix} 4-3 \\ 0+3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Sums of squares and products; linear combinations of variables

Proposition 1.33 ([2, §5.1.4, §10.3], [6, Exercise 2.17(a)])**.** Let $X_0 \in \mathbb{R}^{n\times k}$ be a data matrix with the rows $x_1', \ldots, x_n'$ ($x_i \in \mathbb{R}^k$ holds the measurements on unit i), let $y \in \mathbb{R}^n$, $\mathbf{1} = \mathbf{1}_n$, $C = C_n$, and let $\bar{x} = \frac{1}{n}X_0'\mathbf{1}$ be the mean vector of X_0 (Example 1.17).

(a) $X_0'X_0 = \sum_{i=1}^n x_i x_i'$ and $X_0'y = \sum_{i=1}^n x_i y_i$; the (j,l) entry of $X_0'X_0$ is $\sum_i x_{ij}x_{il}$.

(b) $CX_0 = X_0 - \mathbf{1}\bar{x}'$ has the rows $(x_1 - \bar{x})', \ldots, (x_n - \bar{x})'$, and

$$X_0'CX_0 = (CX_0)'(CX_0) = \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})' = X_0'X_0 - n\bar{x}\bar{x}'$$

Its (j,l) entry is $\sum_i (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l)$.

Proof. (a)

- The columns of X_0' are $x_1, \ldots, x_n$ and the rows of X_0 are $x_1', \ldots, x_n'$, so the outer product expansion (Theorem 1.4(d)) gives $X_0'X_0 = \sum_i x_i x_i'$.
- By Theorem 1.4(b), $X_0'y = y_1x_1 + \cdots + y_nx_n$.
- The (j,l) entry of $X_0'X_0$ is $\mathrm{col}_j(X_0)'\mathrm{col}_l(X_0) = \sum_i x_{ij}x_{il}$ (Theorems 1.4(d) and 1.9).

(b)

- $\mathbf{1}'X_0 = n\bar{x}'$, so $CX_0 = X_0 - \frac{1}{n}\mathbf{1}(\mathbf{1}'X_0) = X_0 - \mathbf{1}\bar{x}'$. The ith row of $\mathbf{1}\bar{x}'$ is $\bar{x}'$ (Theorem 1.4(c)), so the ith row of CX_0 is $(x_i - \bar{x})'$.
- $C' = C = C^2$ (Proposition 1.15), so $X_0'CX_0 = X_0'C'CX_0 = (CX_0)'(CX_0)$.
- (a), applied to the centered data matrix CX_0, gives the sum of outer products and the entries.
- As $X_0'\mathbf{1} = n\bar{x}$, $X_0'CX_0 = X_0'X_0 - \frac{1}{n}(X_0'\mathbf{1})(\mathbf{1}'X_0) = X_0'X_0 - n\bar{x}\bar{x}'$.

□

Proposition 1.34 (cf. [3, §10.1], [9, Definition 5.19])**.** Let $X_0 \in \mathbb{R}^{n\times k}$ be a data matrix with mean vector $\bar{x}$, let $C = C_n$, and let $A \in \mathbb{R}^{k\times q}$. The jth column of $Z = X_0A \in \mathbb{R}^{n\times q}$ is $X_0\,\mathrm{col}_j(A) = a_{1j}\,\mathrm{col}_1(X_0) + \cdots + a_{kj}\,\mathrm{col}_k(X_0)$ (Theorem 1.4(a),(b)): it holds the values, on the n units, of the linear combination of the k variables with the coefficients $a_{1j}, \ldots, a_{kj}$.

(a) The mean vector of Z is $A'\bar{x}$, and $CZ = (CX_0)A$.

(b) The matrix of corrected sums of squares and products of Z is $Z'CZ = A'(X_0'CX_0)A$.

(c) For $a, b \in \mathbb{R}^k$, the combinations $z = X_0a$ and $w = X_0b$ have the means $\bar{z} = a'\bar{x}$ and $\bar{w} = b'\bar{x}$, and

$$\sum_{i=1}^{n}(z_i - \bar{z})(w_i - \bar{w}) = a'(X_0'CX_0)b, \qquad \sum_{i=1}^{n}(z_i - \bar{z})^2 = a'(X_0'CX_0)a$$

Proof. (a) $\frac{1}{n}Z'\mathbf{1} = \frac{1}{n}A'X_0'\mathbf{1} = A'\bar{x}$ (Theorem 1.9), and $CZ = C(X_0A) = (CX_0)A$ (Theorem 1.3).
(b) $Z'CZ = (X_0A)'C(X_0A) = A'(X_0'CX_0)A$ (Theorems 1.9 and 1.3); by Proposition 1.33(b), applied to Z, it is the matrix of corrected sums of squares and products of the columns of Z.
(c)

- Take $A = [a\ b] \in \mathbb{R}^{k\times 2}$, so that $Z = [z\ w]$. By (a), the means are $a'\bar{x}$ and $b'\bar{x}$.
- By (b) and Proposition 1.33(b), the $(1,2)$ and $(1,1)$ entries of $A'(X_0'CX_0)A$ are the two sums; by Theorem 1.4(d) they are $a'(X_0'CX_0)b$ and $a'(X_0'CX_0)a$.

□

Example 1.35. Four students take three tests; x_{ij} is the score of student i on test j:

$$X_0 = \begin{pmatrix} 5 & 4 & 6 \\ 3 & 4 & 2 \\ 5 & 3 & 6 \\ 3 & 1 & 6 \end{pmatrix}, \qquad \bar{x} = \tfrac{1}{4}X_0'\mathbf{1}_4 = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}, \qquad C_4X_0 = X_0 - \mathbf{1}_4\bar{x}' = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & -3 \\ 1 & 0 & 1 \\ -1 & -2 & 1 \end{pmatrix}$$

By Proposition 1.33(b), $M = X_0'C_4X_0$ is the sum of the outer products of the rows of C_4X_0:

$$M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & -1 & 3 \\ -1 & 1 & -3 \\ 3 & -3 & 9 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 & 4 \\ 2 & 6 & -4 \\ 4 & -4 & 12 \end{pmatrix}$$

Its trace, $\operatorname{tr}(M) = 4 + 6 + 12 = 22$, is the sum of the squares of all entries of C_4X_0, since $M = (C_4X_0)'(C_4X_0)$ (Theorem 1.11(d)); row by row, $3 + 11 + 2 + 6 = 22$.

The total score $z = X_0\mathbf{1}_3 = (15, 9, 14, 10)'$ and the difference of the first two tests $w = X_0b = (1, -1, 2, 2)'$, $b = (1, -1, 0)'$, are linear combinations of the tests. By Proposition 1.34(c), $\bar{z} = \mathbf{1}_3'\bar{x} = 12$, $\bar{w} = b'\bar{x} = 1$, and

$$\sum_i (z_i - \bar{z})^2 = \mathbf{1}_3'M\mathbf{1}_3 = 26, \qquad \sum_i (w_i - \bar{w})^2 = b'Mb = 6, \qquad \sum_i (z_i - \bar{z})(w_i - \bar{w}) = \mathbf{1}_3'Mb = 6$$

($\mathbf{1}_3'M = (10, 4, 12)$, the column sums of M, and $Mb = (2, -4, 8)'$). Directly, $z - \bar{z}\mathbf{1}_4 = (3, -3, 2, -2)'$ and $w - \bar{w}\mathbf{1}_4 = (0, -2, 1, 1)'$ give $9 + 9 + 4 + 4 = 26$, $0 + 4 + 1 + 1 = 6$ and $0 + 6 + 2 - 2 = 6$. The coefficients of w add up to 0 ($\mathbf{1}_3'b = 0$): w is a **contrast** of the tests, and it does not change if the same number t_i is added to the three scores of each student i, since $(X_0 + t\mathbf{1}_3')b = X_0b + t(\mathbf{1}_3'b) = X_0b$ for every $t \in \mathbb{R}^4$.

Exercises

Exercise 1.1. Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 2 & 0 \\ -1 & 1 & 0 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{pmatrix}$$

(a) Find the $(2,3)$ entry of AB from $\mathrm{row}_2(A)$ and $\mathrm{col}_3(B)$.

(b) Find $\mathrm{col}_2(AB)$ as a combination of the columns of A.

(c) Find $\mathrm{row}_3(AB)$ as a combination of the rows of B.

(d) Compute AB and BA.

Solution. (a) By Theorem 1.4(d), $(AB)_{23} = \mathrm{row}_2(A)\,\mathrm{col}_3(B) = 0 \cdot 2 + 1 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 = -1$.
(b) By Theorem 1.4(a),(b), $\mathrm{col}_2(AB) = A\,\mathrm{col}_2(B) = 0\,a_1 + 1\,a_2 + 3\,a_3 + 0\,a_4$, with $a_j = \mathrm{col}_j(A)$:

$$\mathrm{col}_2(AB) = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 3 \\ 1 \end{pmatrix}$$

(c) By Theorem 1.4(c), $\mathrm{row}_3(AB) = 2\ \mathrm{row}_1(B) + 0\ \mathrm{row}_2(B) + 1\ \mathrm{row}_3(B) + 0\ \mathrm{row}_4(B) = (2,0,4,0) + (0,3,1,0) = (2,3,5,0)$.
(d) Row by row, $\mathrm{row}_1(AB) = 1\,(1,0,2,0) + 2\,(-1,1,0,1) = (-1,2,2,2)$, $\mathrm{row}_2(AB) = 1\,(-1,1,0,1) - 1\,(0,3,1,0) + 1\,(1,0,0,2) = (0,-2,-1,3)$ and $\mathrm{row}_4(AB) = 1\,(-1,1,0,1) + 1\,(1,0,0,2) = (0,1,0,3)$; with (c),

$$AB = \begin{pmatrix} -1 & 2 & 2 & 2 \\ 0 & -2 & -1 & 3 \\ 2 & 3 & 5 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix}$$

in agreement with (a) and (b). In the same way $\mathrm{row}_1(BA) = 1\,(1,2,0,0) + 2\,(2,0,1,0) = (5,2,2,0)$, $\mathrm{row}_2(BA) = -1\,(1,2,0,0) + 1\,(0,1,-1,1) + 1\,(0,1,0,1) = (-1,0,-1,2)$, $\mathrm{row}_3(BA) = 3\,(0,1,-1,1) + 1\,(2,0,1,0) = (2,3,-2,3)$ and $\mathrm{row}_4(BA) = 1\,(1,2,0,0) + 2\,(0,1,0,1) = (1,4,0,2)$, so

$$BA = \begin{pmatrix} 5 & 2 & 2 & 0 \\ -1 & 0 & -1 & 2 \\ 2 & 3 & -2 & 3 \\ 1 & 4 & 0 & 2 \end{pmatrix} \neq AB$$

Exercise 1.2 (cf. [6, Exercise 2.31])**.** Let

$$N = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Compute N^2, N^3 and N^4.

(b) Show that the matrices $B \in \mathbb{R}^{4\times 4}$ with $NB = BN$ are exactly

$$B = \alpha I_4 + \beta N + \gamma N^2 + \delta N^3 = \begin{pmatrix} \alpha & \beta & \gamma & \delta \\ 0 & \alpha & \beta & \gamma \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \alpha \end{pmatrix} \qquad (\alpha, \beta, \gamma, \delta \in \mathbb{R})$$

Solution. (a) By Theorem 1.4(a),(b), $Ne_1 = 0$ and $Ne_j = e_{j-1}$ for $j = 2, 3, 4$. Hence $N^2 e_j = e_{j-2}$ for $j = 3, 4$ and $N^2 e_j = 0$ for $j = 1, 2$; $N^3 e_4 = e_1$ and $N^3 e_j = 0$ for $j \leqslant 3$; and $N^4 e_j = 0$ for all j. Since $\operatorname{col}_j(N^k) = N^k e_j$,

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^4 = 0$$

This gives the matrix form of $\alpha I_4 + \beta N + \gamma N^2 + \delta N^3$ in (b).

(b) Let $B = (b_{ij})$. Since $\operatorname{row}_i(N) = e'_{i+1}$ for $i \leqslant 3$ and $\operatorname{row}_4(N) = 0$, Theorem 1.4(c) gives $\operatorname{row}_i(NB) = \operatorname{row}_{i+1}(B)$ for $i \leqslant 3$ and $\operatorname{row}_4(NB) = 0$. By (a), $\operatorname{col}_j(BN) = BNe_j = Be_{j-1} = \operatorname{col}_{j-1}(B)$ for $j \geqslant 2$ and $\operatorname{col}_1(BN) = 0$. So

$$(NB)_{ij} = \begin{cases} b_{i+1,j} & i \leqslant 3 \\ 0 & i = 4 \end{cases}, \qquad (BN)_{ij} = \begin{cases} b_{i,j-1} & j \geqslant 2 \\ 0 & j = 1 \end{cases}$$

Suppose $NB = BN$. Comparing the entries:

- row 4, $j \geqslant 2$: $b_{4,j-1} = 0$, so $b_{41} = b_{42} = b_{43} = 0$;
- column 1, $i \leqslant 3$: $b_{i+1,1} = 0$, so $b_{21} = b_{31} = b_{41} = 0$;
- $i \leqslant 3$, $j \geqslant 2$: $b_{i+1,j} = b_{i,j-1}$, so every entry equals the entry diagonally above and to its left.

The last condition gives $b_{32} = b_{21} = 0$; so all entries below the diagonal vanish. It also gives $b_{11} = b_{22} = b_{33} = b_{44}$, $b_{12} = b_{23} = b_{34}$ and $b_{13} = b_{24}$. With $\alpha = b_{11}$, $\beta = b_{12}$, $\gamma = b_{13}$, $\delta = b_{14}$, B has the stated form. Conversely, by Theorem 1.3 and $NN^k = N^{k+1} = N^k N$ (Proposition 1.19(a)),

$$N(\alpha I + \beta N + \gamma N^2 + \delta N^3) = \alpha N + \beta N^2 + \gamma N^3 + \delta N^4 = (\alpha I + \beta N + \gamma N^2 + \delta N^3)N$$

for all $\alpha, \beta, \gamma, \delta$.

Exercise 1.3. Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

(a) Compute AB and BA.

(b) Write AB as the sum of the four products $\operatorname{col}_j(A) \operatorname{row}_j(B)$ of Theorem 1.4(d).

(c) Compute $B'A'$, $\operatorname{tr}(AB)$, $\operatorname{tr}(BA)$ and $\operatorname{tr}(A'A)$, and show that $(BA)^2 = BA$.

Solution. (a) By Theorem 1.4(c), $\operatorname{row}_i(AB) = \operatorname{row}_i(A)B$ is a combination of the rows $(1, -1, 1)$, $(0, 1, -1)$, $(0, 0, 1)$, $(0, 0, 0)$ of B:

$$\begin{aligned} \operatorname{row}_1(AB) &= (1, -1, 1) + (0, 1, -1) = (1,\ 0,\ 0) \\ \operatorname{row}_2(AB) &= (0, 1, -1) + (0, 0, 1) = (0,\ 1,\ 0) \\ \operatorname{row}_3(AB) &= (0, 0, 1) + (0, 0, 0) = (0,\ 0,\ 1) \end{aligned}$$

Likewise each row of $BA \in \mathbb{R}^{4\times 4}$ is a combination of the rows of A:

$$\begin{aligned} \operatorname{row}_1(BA) &= (1, 1, 0, 0) - (0, 1, 1, 0) + (0, 0, 1, 1) = (1,\ 0,\ 0,\ 1) \\ \operatorname{row}_2(BA) &= (0, 1, 1, 0) - (0, 0, 1, 1) = (0,\ 1,\ 0,\ -1) \\ \operatorname{row}_3(BA) &= (0, 0, 1, 1), \qquad \operatorname{row}_4(BA) = (0, 0, 0, 0) \end{aligned}$$

Hence

$$AB = I_3, \qquad BA = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \neq I_4$$

(b) $\mathrm{col}_4(A)\,\mathrm{row}_4(B) = 0$, and

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}(1,\ -1,\ 1) + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}(0,\ 1,\ -1) + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}(0,\ 0,\ 1) = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = I_3$$

(c) $B'A' = (AB)' = I_3' = I_3$, by the rules $(AB)' = B'A'$ and $I_3' = I_3$ of Theorem 1.9. Next $\mathrm{tr}(AB) = 3$ and $\mathrm{tr}(BA) = 1{+}1{+}1{+}0 = 3$, equal by Theorem 1.11(b); and by Theorem 1.11(d), $\mathrm{tr}(A'A)$ is the sum of the squares of the entries of A, which is 6. Finally, by Theorem 1.3, $(BA)^2 = B(AB)A = BI_3A = BA$.

Exercise 1.4 (cf. [6, Exercise 2.5(a)])**.** Let $A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 2 & 1 & 0 & 1 \end{pmatrix}$, and let B be the matrix obtained from A by interchanging its first two rows. Compute AB', BA', $A'B$ and $B'A$, and compare their traces.

Solution. Write $r_1' = (1,2,1,0)$, $r_2' = (0,1,2,1)$, $r_3' = (2,1,0,1)$ for the rows of A ($r_1, r_2, r_3 \in \mathbb{R}^4$). The rows of B are r_2', r_1', r_3', so $B = \left(\begin{smallmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{smallmatrix}\right)$. Since $\mathrm{col}_j(B') = \mathrm{row}_j(B)'$, the (i,j) entry of AB' is $\mathrm{row}_i(A)\,\mathrm{row}_j(B)'$ (Theorem 1.4(d)). The six numbers $r_i'r_k$ needed are

$$\begin{aligned} r_1'r_1 &= 1+4+1+0 = 6, \quad & r_2'r_2 &= 0+1+4+1 = 6, \quad & r_3'r_3 &= 4+1+0+1 = 6 \\ r_1'r_2 &= 0+2+2+0 = 4, \quad & r_1'r_3 &= 2+2+0+0 = 4, \quad & r_2'r_3 &= 0+1+0+1 = 2 \end{aligned}$$

and $r_k'r_i = r_i'r_k$ (Theorem 1.9). Hence

$$AB' = \begin{pmatrix} r_1'r_2 & r_1'r_1 & r_1'r_3 \\ r_2'r_2 & r_2'r_1 & r_2'r_3 \\ r_3'r_2 & r_3'r_1 & r_3'r_3 \end{pmatrix} = \begin{pmatrix} 4 & 6 & 4 \\ 6 & 4 & 2 \\ 2 & 4 & 6 \end{pmatrix}, \qquad BA' = (AB')' = \begin{pmatrix} 4 & 6 & 2 \\ 6 & 4 & 4 \\ 4 & 2 & 6 \end{pmatrix}$$

by Theorem 1.9. Since $\mathrm{col}_k(A') = \mathrm{row}_k(A)' = r_k$, the outer product expansion (Theorem 1.4(d)) gives

$$A'B = \mathrm{col}_1(A')\,\mathrm{row}_1(B) + \mathrm{col}_2(A')\,\mathrm{row}_2(B) + \mathrm{col}_3(A')\,\mathrm{row}_3(B) = r_1r_2' + r_2r_1' + r_3r_3'$$

This sum is symmetric, since $(r_1r_2')' = r_2r_1'$ and $(r_3r_3')' = r_3r_3'$ (Theorem 1.9). Its (i,j) entry is $\mathrm{col}_i(A)'\,\mathrm{col}_j(B)$; for instance $(A'B)_{23} = 2\cdot 2 + 1\cdot 1 + 1\cdot 0 = 5$. Computing all sixteen entries in this way,

$$A'B = \begin{pmatrix} 4 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 \\ 2 & 5 & 4 & 1 \\ 3 & 3 & 1 & 1 \end{pmatrix} = (A'B)' = B'A$$

So $A'B$ is symmetric while AB' is not. The reason: $B = SA$ with the symmetric matrix $S = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, since $\mathrm{row}_1(SA) = \mathrm{row}_2(A)$, $\mathrm{row}_2(SA) = \mathrm{row}_1(A)$ and $\mathrm{row}_3(SA) = \mathrm{row}_3(A)$ (Theorem 1.4(c)). Then $A'B = A'SA$ and $(A'SA)' = A'S'A = A'SA$, whereas $AB' = AA'S$ need not be symmetric.

The four products have the same trace 14 ($4+4+6$ and $4+5+4+1$); this is no coincidence: $\mathrm{tr}(AB') = \mathrm{tr}(B'A)$ and $\mathrm{tr}(BA') = \mathrm{tr}(A'B)$ by Theorem 1.11(b), and $\mathrm{tr}(BA') = \mathrm{tr}\big((AB')'\big) = \mathrm{tr}(AB')$ by Theorems 1.9 and 1.11(a).

Exercise 1.5. Let $u = (1,2,3,4)'$, $v = (1,1,1,-1)'$, $w = (1,1,-1,0)'$, and put $A = uv'$ and $B = uw'$.

(a) Write out A and B, and compute $v'u$ and $w'u$.

(b) Show that $A^2 = 2A$, $A^3 = 4A$, $B^2 = 0$, $AB = 2B$ and $BA = 0$.

(c) Compute $\operatorname{tr}(A)$ and $\operatorname{tr}(B)$.

Solution. (a) Row i of uv' is $u_i v'$ (Theorem 1.4(c)), so

$$A = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 2 & 2 & 2 & -2 \\ 3 & 3 & 3 & -3 \\ 4 & 4 & 4 & -4 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 2 & 2 & -2 & 0 \\ 3 & 3 & -3 & 0 \\ 4 & 4 & -4 & 0 \end{pmatrix}$$

and $v'u = 1+2+3-4 = 2$, $w'u = 1+2-3+0 = 0$.

(b) By Theorem 1.3, a product of two such matrices collapses to a number times a column times a row: for instance $A^2 = (uv')(uv') = u(v'u)v' = (v'u)\,uv'$, because $v'u$ is a number. In this way

$$A^2 = u(v'u)v' = 2A, \qquad A^3 = A^2A = 2A^2 = 4A, \qquad B^2 = u(w'u)w' = 0$$
$$AB = u(v'u)w' = 2B, \qquad BA = u(w'u)v' = 0$$

So A and B do not commute, and $B \neq 0$ has $B^2 = 0$.

(c) $\operatorname{tr}(A) = 1+2+3-4 = 2$ and $\operatorname{tr}(B) = 1+2-3+0 = 0$. In general $\operatorname{tr}(uv') = \operatorname{tr}(v'u) = v'u$ by Theorem 1.11(b).

Exercise 1.6. Let $L \in \mathbb{R}^{4\times 4}$ be the lower triangular matrix with all entries on and below the diagonal equal to 1, and $B = I_4 - Z$, where Z has entries $z_{21} = z_{32} = z_{43} = 1$ and all others 0:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

(a) For $x \in \mathbb{R}^4$, compute Lx and Bx. Deduce that $BL = I_4 = LB$.

(b) Show that $LL' = (\min(i,j))$ and $L'L = (5 - \max(i,j))$, and write out both.

(c) Compute $B'B$ and show that $(LL')(B'B) = I_4$.

Solution. (a) $\operatorname{row}_i(L)x = x_1 + \cdots + x_i$, so $Lx = (x_1,\ x_1+x_2,\ x_1+x_2+x_3,\ x_1+x_2+x_3+x_4)'$ is the vector of partial sums, and $Bx = x - Zx = (x_1,\ x_2-x_1,\ x_3-x_2,\ x_4-x_3)'$ is the vector of differences. The differences of the partial sums of x are the entries of x, and the partial sums of the differences of x are again the entries of x: $B(Lx) = x = L(Bx)$ for every x. With $x = e_j$, $\operatorname{col}_j(BL) = BLe_j = e_j$ and $\operatorname{col}_j(LB) = e_j$ (Theorem 1.4(a),(b)); so $BL = I_4 = LB$.

(b) $(LL')_{ij} = \operatorname{row}_i(L)\operatorname{row}_j(L)'$ counts the indices k with $k \leqslant i$ and $k \leqslant j$, which is $\min(i,j)$ (Theorems 1.4(d) and 1.9). $(L'L)_{ij} = \operatorname{col}_i(L)'\operatorname{col}_j(L)$ counts the indices $k \leqslant 4$ with $k \geqslant i$ and $k \geqslant j$, which is $5 - \max(i,j)$. So

$$LL' = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \qquad L'L = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 3 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

both with trace 10.

(c) The columns of B are $b_1 = e_1 - e_2$, $b_2 = e_2 - e_3$, $b_3 = e_3 - e_4$, $b_4 = e_4$, and $(B'B)_{ij} = b_i'b_j$ (Theorems 1.4(d) and 1.9). Here $b_1'b_1 = b_2'b_2 = b_3'b_3 = 1+1 = 2$, $b_4'b_4 = 1$, $b_1'b_2 = b_2'b_3 = b_3'b_4 = -1$, and $b_i'b_j = 0$ for $|i-j| \geqslant 2$, since b_i and b_j then have no nonzero entry in the same position:

$$B'B = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

By Theorems 1.3 and 1.9 and (a), $(LL')(B'B) = L(L'B')B = L(BL)'B = LI_4B = LB = I_4$.

Exercise 1.7. Four units are measured on two variables ($n = 4$, $k = 2$), with data matrix

$$X_0 = \begin{pmatrix} 1 & 0 \\ 3 & 2 \\ 2 & 0 \\ 2 & 2 \end{pmatrix} \in \mathbb{R}^{4\times 2}$$

Find

(a) the mean vector $\bar{x}$ and the centered data matrix C_4X_0;

(b) the matrix $X_0'C_4X_0$ of corrected sums of squares and products, both as $(C_4X_0)'(C_4X_0)$ and as $X_0'X_0 - 4\bar{x}\bar{x}'$.

Solution. (a) $\mathbf{1}'X_0 = (1+3+2+2,\ 0+2+0+2) = (8,\ 4)$, so $\bar{x} = \frac{1}{4}X_0'\mathbf{1} = (2,\ 1)'$. As in Example 1.17, $C_4X_0 = X_0 - \mathbf{1}\bar{x}'$:

$$C_4X_0 = \begin{pmatrix} 1-2 & 0-1 \\ 3-2 & 2-1 \\ 2-2 & 0-1 \\ 2-2 & 2-1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}$$

(b) The (j,l) entry of $(C_4X_0)'(C_4X_0)$ is $\mathrm{col}_j(C_4X_0)'\,\mathrm{col}_l(C_4X_0)$ (Theorems 1.4(d) and 1.9): $1+1+0+0 = 2$ for $(1,1)$, $1+1+0+0 = 2$ for $(1,2)$ and $(2,1)$, and $1+1+1+1 = 4$ for $(2,2)$. On the other hand

$$X_0'X_0 - 4\bar{x}\bar{x}' = \begin{pmatrix} 18 & 10 \\ 10 & 8 \end{pmatrix} - 4\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 18 & 10 \\ 10 & 8 \end{pmatrix} - \begin{pmatrix} 16 & 8 \\ 8 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$$

(for instance $18 = 1+9+4+4$ and $10 = 0+6+0+4$). So $X_0'C_4X_0 = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 4 \end{smallmatrix}\right)$: the sums of squares about the means are $\sum_i (x_{i1} - \bar{x}_1)^2 = 2$ and $\sum_i (x_{i2} - \bar{x}_2)^2 = 4$, and the corrected sum of products is $\sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) = 2$.

Exercise 1.8. Five units are measured on three variables ($n = 5$, $k = 3$), with data matrix

$$X_0 = \begin{pmatrix} 2 & 2 & 5 \\ 3 & 2 & 2 \\ 1 & 4 & 2 \\ 3 & 6 & 8 \\ 1 & 6 & 8 \end{pmatrix} \in \mathbb{R}^{5\times 3}$$

(a) Find the mean vector $\bar{x}$ and the centered data matrix C_5X_0.

(b) Find $S = X_0'C_5X_0$, both as $(C_5X_0)'(C_5X_0)$ and as $X_0'X_0 - 5\bar{x}\bar{x}'$.

(c) Let $D = \mathrm{diag}(\frac{1}{2}, \frac{1}{4}, \frac{1}{6})$. Compute DSD, and explain why its diagonal entries equal 1.

Solution. (a) $\mathbf{1}'X_0 = (10,\ 20,\ 25)$, so $\bar{x} = \frac{1}{5}X_0'\mathbf{1} = (2,\ 4,\ 5)'$. As in Example 1.17, $C_5X_0 = X_0 - \mathbf{1}\bar{x}'$:

$$C_5X_0 = \begin{pmatrix} 0 & -2 & 0 \\ 1 & -2 & -3 \\ -1 & 0 & -3 \\ 1 & 2 & 3 \\ -1 & 2 & 3 \end{pmatrix}$$

(b) Write c_1, c_2, c_3 for the columns of C_5X_0. The (j,l) entry of $(C_5X_0)'(C_5X_0)$ is $c_j'c_l$ (Theorems 1.4(d) and 1.9):

$$c_1'c_1 = 0+1+1+1+1 = 4, \quad c_2'c_2 = 4+4+0+4+4 = 16, \quad c_3'c_3 = 0+9+9+9+9 = 36$$
$$c_1'c_2 = 0-2+0+2-2 = -2, \quad c_1'c_3 = 0-3+3+3-3 = 0, \quad c_2'c_3 = 0+6+0+6+6 = 18$$

On the other hand

$$X_0'X_0 - 5\bar{x}\bar{x}' = \begin{pmatrix} 24 & 38 & 50 \\ 38 & 96 & 118 \\ 50 & 118 & 161 \end{pmatrix} - \begin{pmatrix} 20 & 40 & 50 \\ 40 & 80 & 100 \\ 50 & 100 & 125 \end{pmatrix} = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 16 & 18 \\ 0 & 18 & 36 \end{pmatrix} = S$$

(for instance $38 = 4+6+4+18+6$ and $161 = 25+4+4+64+64$).

(c) By Proposition 1.13(a), left multiplication by D scales the rows and right multiplication scales the columns, so $(DSD)_{ij} = d_i s_{ij} d_j$:

$$DSD = \begin{pmatrix} 1 & -\frac{1}{4} & 0 \\ -\frac{1}{4} & 1 & \frac{3}{4} \\ 0 & \frac{3}{4} & 1 \end{pmatrix}$$

for instance $(DSD)_{23} = \frac{1}{4}\cdot 18 \cdot \frac{1}{6} = \frac{3}{4}$. The diagonal entries are $d_i^2 s_{ii} = \frac{4}{4}, \frac{16}{16}, \frac{36}{36}$, all equal to 1, because $d_i = 1/\sqrt{s_{ii}}$ for $i = 1,2,3$.

Exercise 1.9. Four units are measured on three variables, with the data matrix $X_0 = \begin{pmatrix} 2 & 5 & 1 \\ 4 & 0 & 3 \\ 1 & 2 & 2 \\ 3 & 1 & 4 \end{pmatrix}$.

(a) Compute $e_3'X_0$, X_0e_2 and $e_3'X_0e_2$.

(b) Compute $X_0[e_3\ e_1]$, $[e_2\ e_4]'X_0$ and $[e_2\ e_4]'X_0[e_3\ e_1]$.

(c) For the matrices $E_{ij} \in \mathbb{R}^{4\times 4}$, compute $E_{12}E_{23}$ and $E_{23}E_{12}$.

Solution. (a) By Proposition 1.18(a), $e_3'X_0 = \mathrm{row}_3(X_0) = (1,2,2)$, $X_0e_2 = \mathrm{col}_2(X_0) = (5,0,2,1)'$ and $e_3'X_0e_2 = x_{32} = 2$.

(b) By Theorem 1.4(a) and (c), the first product has the columns 3 and 1 of X_0, the second the rows 2 and 4, and the third the entries of rows $2,4$ in columns $3,1$:

$$X_0[e_3\ e_1] = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 2 & 1 \\ 4 & 3 \end{pmatrix}, \qquad [e_2\ e_4]'X_0 = \begin{pmatrix} 4 & 0 & 3 \\ 3 & 1 & 4 \end{pmatrix}, \qquad [e_2\ e_4]'X_0[e_3\ e_1] = \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}$$

(c) With

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad E_{23} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Theorem 1.4(c) gives $\operatorname{row}_1(E_{12}E_{23}) = 1\ \operatorname{row}_2(E_{23}) = (0,0,1,0)$, the other rows being zero, and $\operatorname{row}_2(E_{23}E_{12}) = 1\ \operatorname{row}_3(E_{12}) = 0$, the other rows being zero as well. So

$$E_{12}E_{23} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = E_{13}, \qquad E_{23}E_{12} = 0$$

as Proposition 1.18(c) says ($\delta_{22} = 1$, $\delta_{31} = 0$).

Exercise 1.10. For $i, j \in \{1,2,3\}$ let $E_{ij} = e_i e_j' \in \mathbb{R}^{3\times 3}$, as in Proposition 1.18. Compute $\operatorname{tr}(E_{12}E_{23}E_{31})$ and $\operatorname{tr}(E_{12}E_{31}E_{23})$.

Solution. By Proposition 1.18(c), $E_{12}E_{23}E_{31} = E_{13}E_{31} = E_{11}$, with trace 1, whereas $E_{12}E_{31} = \delta_{23}E_{11} = 0$, so $E_{12}E_{31}E_{23} = 0$, with trace 0. Thus the trace of a product of three matrices is unchanged when the factors are moved cyclically (Theorem 1.11(c)), but it can change when two factors are interchanged: $\operatorname{tr}(E_{12}E_{23}E_{31}) = 1 \neq 0 = \operatorname{tr}(E_{12}E_{31}E_{23})$.

Exercise 1.11. Let

$$U = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & -1 & 3 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad N = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Compute U^2 and U^3, and compare their diagonal entries with those of U.

(b) Compute N^2, N^3 and N^4.

(c) Write down $(U')^2$ without multiplying.

Solution. (a) By rows (Theorem 1.4(c)): $\operatorname{row}_1(U^2) = 1\,(1,2,-1,0) + 2\,(0,-1,3,1) - 1\,(0,0,2,1) = (1,0,3,1)$, $\operatorname{row}_2(U^2) = -1\,(0,-1,3,1) + 3\,(0,0,2,1) + 1\,(0,0,0,-1) = (0,1,3,1)$, $\operatorname{row}_3(U^2) = 2\,(0,0,2,1) + 1\,(0,0,0,-1) = (0,0,4,1)$ and $\operatorname{row}_4(U^2) = -1\,(0,0,0,-1) = (0,0,0,1)$. Then, for $U^3 = U^2U$, $\operatorname{row}_1(U^3) = 1\,(1,2,-1,0) + 3\,(0,0,2,1) + 1\,(0,0,0,-1) = (1,2,5,2)$, $\operatorname{row}_2(U^3) = 1\,(0,-1,3,1) + 3\,(0,0,2,1) + 1\,(0,0,0,-1) = (0,-1,9,3)$, $\operatorname{row}_3(U^3) = 4\,(0,0,2,1) + 1\,(0,0,0,-1) = (0,0,8,3)$ and $\operatorname{row}_4(U^3) = 1\,(0,0,0,-1) = (0,0,0,-1)$:

$$U^2 = \begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad U^3 = \begin{pmatrix} 1 & 2 & 5 & 2 \\ 0 & -1 & 9 & 3 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Both are upper triangular; their diagonal entries $1,1,4,1$ and $1,-1,8,-1$ are the squares and the cubes of the diagonal entries $1,-1,2,-1$ of U (Proposition 1.19(e)).

(b) $\operatorname{row}_1(N^2) = 1\ \operatorname{row}_2(N) + 2\ \operatorname{row}_3(N) = (0,0,1,5)$, $\operatorname{row}_2(N^2) = 1\ \operatorname{row}_3(N) + 3\ \operatorname{row}_4(N) = (0,0,0,1)$, and rows 3, 4 of N^2 are zero. Then $\operatorname{row}_1(N^3) = 1\ \operatorname{row}_3(N) + 5\ \operatorname{row}_4(N) = (0,0,0,1)$, $\operatorname{row}_2(N^3) = 1\ \operatorname{row}_4(N) = 0$, and $\operatorname{row}_1(N^4) = 1\ \operatorname{row}_4(N) = 0$:

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^4 = 0$$

With each power one more diagonal becomes zero, as in Proposition 1.21(a), and $N^4 = 0$ (Proposition 1.21(c), $n = 4$).

(c) By Proposition 1.19(c),

$$(U')^2 = (U^2)' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 3 & 3 & 4 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Exercise 1.12. Let

$$N = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Compute N^2, N^3 and N^4.

(b) Let $G = I_4 - N + N^2 - N^3$. Show that $(I_4 + N)G = I_4 = G(I_4 + N)$, and write out G.

Solution. (a) By Theorem 1.4(c), $\operatorname{row}_i(N^{k+1}) = \operatorname{row}_i(N^k)N$ is the combination of the rows of N with coefficients $\operatorname{row}_i(N^k)$. The rows of N are $(0,1,1,1)$, $(0,0,2,1)$, $(0,0,0,3)$, 0. So

$$\operatorname{row}_1(N^2) = 1\,(0,0,2,1) + 1\,(0,0,0,3) = (0,0,2,4), \qquad \operatorname{row}_2(N^2) = 2\,(0,0,0,3) = (0,0,0,6)$$
$$\operatorname{row}_1(N^3) = 2\,(0,0,0,3) = (0,0,0,6), \qquad \operatorname{row}_2(N^3) = 6\,\operatorname{row}_4(N) = 0$$

while $\operatorname{row}_3(N^2) = 3\,\operatorname{row}_4(N) = 0$, $\operatorname{row}_4(N^2) = \operatorname{row}_4(N)N = 0$, and rows $3, 4$ of N^3 are $\operatorname{row}_3(N^2)N = 0$ and $\operatorname{row}_4(N^2)N = 0$. Hence

$$N^2 = \begin{pmatrix} 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 6\,e_1 e_4', \qquad N^4 = 6\,e_1 e_4' N = 6\,e_1\,\operatorname{row}_4(N) = 0$$

(b) By Theorem 1.3 and $NN^k = N^{k+1} = N^k N$ (Proposition 1.19(a)), the terms cancel in pairs:

$$(I + N)(I - N + N^2 - N^3) = I - N + N^2 - N^3 + N - N^2 + N^3 - N^4 = I - N^4 = I$$

and the product in the other order expands to the same sum. Adding the entries,

$$G = \begin{pmatrix} 1 & -1 & 1 & -3 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

For instance $g_{14} = -1 + 4 - 6 = -3$ and $g_{24} = -1 + 6 = 5$. As a check, $\operatorname{row}_1(I + N)\operatorname{col}_4(G) = 1 \cdot (-3) + 1 \cdot 5 + 1 \cdot (-3) + 1 \cdot 1 = 0$.

Exercise 1.13. Let

$$A = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & 0 & 1 \\ 1 & 1 & 2 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

and $f(t) = t^2 - 3t + 2 = (t - 1)(t - 2)$.

(a) Compute A^2 and $f(A)$.

(b) Compute $(A - I_4)(A - 2I_4)$ and $(A - 2I_4)(A - I_4)$.

(c) Use (a) to write A^3 as $\alpha A + \beta I_4$, and check the result by computing AA^2.

Solution. (a) By rows (Theorem 1.4(c)), $\mathrm{row}_1(A^2) = 2\,(2,-1,0,-1) - (-1,2,0,1) - (1,-1,0,0) = (4,-3,0,-3)$, $\mathrm{row}_2(A^2) = -(2,-1,0,-1) + 2\,(-1,2,0,1) + (1,-1,0,0) = (-3,4,0,3)$, $\mathrm{row}_3(A^2) = (2,-1,0,-1) + (-1,2,0,1) + 2\,(1,1,2,0) = (3,3,4,0)$ and $\mathrm{row}_4(A^2) = (2,-1,0,-1) - (-1,2,0,1) = (3,-3,0,-2)$. So

$$A^2 = \begin{pmatrix} 4 & -3 & 0 & -3 \\ -3 & 4 & 0 & 3 \\ 3 & 3 & 4 & 0 \\ 3 & -3 & 0 & -2 \end{pmatrix} = 3A - 2I_4$$

since $3A - 2I_4$ has the same entries; that is, $f(A) = A^2 - 3A + 2I_4 = 0$.

(b) Here

$$A - I_4 = \begin{pmatrix} 1 & -1 & 0 & -1 \\ -1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & -1 \end{pmatrix}, \qquad A - 2I_4 = \begin{pmatrix} 0 & -1 & 0 & -1 \\ -1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & -2 \end{pmatrix}$$

Write $r_1', \dots, r_4'$ for the rows of $A-2I_4$ and $s_1', \dots, s_4'$ for those of $A-I_4$ ($r_i, s_i \in \mathbb{R}^4$). By Theorem 1.4(c), the rows of $(A-I_4)(A-2I_4)$ are $r_1' - r_2' - r_4' = 0$, $-r_1' + r_2' + r_4' = 0$, $r_1' + r_2' + r_3' = 0$ and $r_1' - r_2' - r_4' = 0$; the rows of $(A-2I_4)(A-I_4)$ are $-s_2' - s_4' = (0,0,0,0)$, $-s_1' + s_4' = (0,0,0,0)$, $s_1' + s_2' = (0,0,0,0)$ and $s_1' - s_2' - 2s_4' = (0,0,0,0)$. Both products are 0, as Proposition 1.24(b) says: each equals $f(A)$.

(c) By (a), $A^2 = 3A - 2I_4$, so $A^3 = AA^2 = 3A^2 - 2A = 3(3A - 2I_4) - 2A = 7A - 6I_4$ (Proposition 1.19(a), Theorem 1.3):

$$A^3 = 7A - 6I_4 = \begin{pmatrix} 8 & -7 & 0 & -7 \\ -7 & 8 & 0 & 7 \\ 7 & 7 & 8 & 0 \\ 7 & -7 & 0 & -6 \end{pmatrix}$$

Check: $\mathrm{row}_1(AA^2) = 2\,(4,-3,0,-3) - (-3,4,0,3) - (3,-3,0,-2) = (8,-7,0,-7)$, $\mathrm{row}_2(AA^2) = -(4,-3,0,-3) + 2\,(-3,4,0,3) + (3,-3,0,-2) = (-7,8,0,7)$, $\mathrm{row}_3(AA^2) = (4,-3,0,-3) + (-3,4,0,3) + 2\,(3,3,4,0) = (7,7,8,0)$ and $\mathrm{row}_4(AA^2) = (4,-3,0,-3) - (-3,4,0,3) = (7,-7,0,-6)$.

Exercise 1.14. Let

$$S = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

so that $Se_1 = e_2$, $Se_2 = e_3$, $Se_3 = e_4$, $Se_4 = e_1$.

(a) Compute S^2, S^3 and S^4, and show that $S' = S^3$ and $S'S = I_4$.

(b) Let $T = I_4 + 2S + 3S^2 + 4S^3$. Write out T and T', show that $TT' = T'T$, and compute TT'.

Solution. (a) Applying S repeatedly moves each unit vector one step along the cycle $e_1 \to e_2 \to e_3 \to e_4 \to e_1$. So $S^2 e_j = e_{j+2}$ and $S^3 e_j = e_{j+3}$, indices taken cyclically ($e_5 = e_1$, $e_6 = e_2$, …), and $S^4 e_j = e_j$. With $\mathrm{col}_j(S^k) = S^k e_j$ (Theorem 1.4(b)),

$$S^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad S^3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad S^4 = I_4$$

The entries 1 of S stand in positions $(2,1)$, $(3,2)$, $(4,3)$, $(1,4)$, so by $(S')_{ij} = s_{ji}$ the entries 1 of S' stand in positions $(1,2)$, $(2,3)$, $(3,4)$, $(4,1)$, all other entries being 0. These are exactly the positions of the entries 1 of S^3, so $S' = S^3$, and then $S'S = S^3 S = S^4 = I_4$.

(b) Adding the entries of I, $2S$, $3S^2$, $4S^3$,

$$T = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}, \qquad T' = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

By Theorem 1.9, Proposition 1.19(a)–(c) and part (a), $(S^2)' = (SS)' = S'S' = (S^3)^2 = S^6 = S^4S^2 = S^2$ and $(S^3)' = (S')^3 = S^9 = (S^4)^2S = S$, so $T' = I + 4S + 3S^2 + 2S^3$. Both T and T' are polynomials in S (Definition 1.23), so $TT' = T'T$ by Proposition 1.24(b). Expanding by Theorem 1.3, and using $S^aS^b = S^{a+b}$ (Proposition 1.19(a)) and $S^4 = I$, the coefficient of S^k in $TT' = (I + 2S + 3S^2 + 4S^3)(I + 4S + 3S^2 + 2S^3)$ collects the pairs of exponents with sum k or $k + 4$:

$$\begin{aligned} I:&\ 1\cdot 1 + 2\cdot 2 + 3\cdot 3 + 4\cdot 4 = 30, & S:&\ 1\cdot 4 + 2\cdot 1 + 3\cdot 2 + 4\cdot 3 = 24 \\ S^2:&\ 1\cdot 3 + 2\cdot 4 + 3\cdot 1 + 4\cdot 2 = 22, & S^3:&\ 1\cdot 2 + 2\cdot 3 + 3\cdot 4 + 4\cdot 1 = 24 \end{aligned}$$

Hence

$$TT' = 30I + 24S + 22S^2 + 24S^3 = \begin{pmatrix} 30 & 24 & 22 & 24 \\ 24 & 30 & 24 & 22 \\ 22 & 24 & 30 & 24 \\ 24 & 22 & 24 & 30 \end{pmatrix}$$

Its $(1,1)$ entry is $\operatorname{row}_1(T)\operatorname{row}_1(T)' = 1 + 16 + 9 + 4 = 30$.

Exercise 1.15. Let

$$A = \begin{pmatrix} 2 & 3 & -1 & 1 \\ 1 & 0 & 4 & 0 \\ 5 & -2 & 1 & -1 \\ 3 & 2 & 1 & 2 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 2 \\ 1 & 0 \end{pmatrix}$$

(a) Compute $S = \frac{1}{2}(A + A')$ and $K = \frac{1}{2}(A - A')$, and check that $S' = S$, $K' = -K$ and $S + K = A$.

(b) Compute $P'SP$ and check that it is symmetric.

Solution. (a)

$$A' = \begin{pmatrix} 2 & 1 & 5 & 3 \\ 3 & 0 & -2 & 2 \\ -1 & 4 & 1 & 1 \\ 1 & 0 & -1 & 2 \end{pmatrix}, \qquad S = \tfrac{1}{2}\begin{pmatrix} 4 & 4 & 4 & 4 \\ 4 & 0 & 2 & 2 \\ 4 & 2 & 2 & 0 \\ 4 & 2 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 \\ 2 & 1 & 1 & 0 \\ 2 & 1 & 0 & 2 \end{pmatrix}$$

$$K = \tfrac{1}{2}\begin{pmatrix} 0 & 2 & -6 & -2 \\ -2 & 0 & 6 & -2 \\ 6 & -6 & 0 & -2 \\ 2 & 2 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & -3 & -1 \\ -1 & 0 & 3 & -1 \\ 3 & -3 & 0 & -1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

S equals its transpose, $K' = -K$ with zero diagonal (Proposition 1.27(c)), and $S + K = A$ entry by entry.

(b) By columns (Theorem 1.4(a)), $SP = [S(1,-1,0,1)' \;\; S(0,1,2,0)']$, and then $P'(SP)$ by rows:

$$SP = \begin{pmatrix} 2 & 6 \\ 3 & 2 \\ 1 & 3 \\ 3 & 1 \end{pmatrix}, \qquad P'SP = \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{pmatrix}\begin{pmatrix} 2 & 6 \\ 3 & 2 \\ 1 & 3 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix}$$

which is symmetric, as Proposition 1.27(d) says.

Exercise 1.16. Let

$$M_1 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 0 & 1 & 0 & 1 \\ -1 & -1 & 1 & 0 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 1 & 0 \end{pmatrix}, \qquad M_3 = \begin{pmatrix} 2 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 3 & 0 & -2 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

(a) Compute M_1^2, M_2^2, M_2^3 and M_3^2, and decide which of the three matrices is idempotent, which nilpotent and which involutory.

(b) For the idempotent one, compute $I_4 - M_1$, $(I_4 - M_1)^2$ and $M_1(I_4 - M_1)$.

Solution. (a) By rows (Theorem 1.4(c)):

- Let $B = 2M_1$. Then $\operatorname{row}_1(B^2) = 1\,(1,0,1,0) + 1\,(1,0,1,0) = (2,0,2,0)$, $\operatorname{row}_2(B^2) = 1\,(0,1,0,-1) - 1\,(0,-1,0,1) = (0,2,0,-2)$, $\operatorname{row}_3(B^2) = 1\,(1,0,1,0) + 1\,(1,0,1,0) = (2,0,2,0)$ and $\operatorname{row}_4(B^2) = -1\,(0,1,0,-1) + 1\,(0,-1,0,1) = (0,-2,0,2)$; so $B^2 = 2B$, and $M_1^2 = \frac{1}{4}B^2 = \frac{1}{2}B = M_1$ (Theorem 1.3).
- $\operatorname{row}_1(M_2^2) = \operatorname{row}_2(M_2) + \operatorname{row}_4(M_2) = (-2,-2,2,0)$, $\operatorname{row}_2(M_2^2) = -\operatorname{row}_1(M_2) - \operatorname{row}_2(M_2) + \operatorname{row}_3(M_2) = (0,0,0,0)$, $\operatorname{row}_3(M_2^2) = -\operatorname{row}_1(M_2) + \operatorname{row}_3(M_2) + \operatorname{row}_4(M_2) = (-2,-2,2,0)$ and $\operatorname{row}_4(M_2^2) = -\operatorname{row}_1(M_2) - \operatorname{row}_2(M_2) + \operatorname{row}_3(M_2) = (0,0,0,0)$. Then rows 1 and 3 of $M_2^3 = M_2^2 M_2$ are $-2\operatorname{row}_1(M_2) - 2\operatorname{row}_2(M_2) + 2\operatorname{row}_3(M_2) = (0,0,0,0)$, and rows 2 and 4 are 0.
- $\operatorname{row}_1(M_3^2) = 2\,(2,0,-1,0) - (3,0,-2,0) = (1,0,0,0)$, $\operatorname{row}_2(M_3^2) = (0,1,0,0)$, $\operatorname{row}_3(M_3^2) = 3\,(2,0,-1,0) - 2\,(3,0,-2,0) = (0,0,1,0)$ and $\operatorname{row}_4(M_3^2) = (0,1,0,0) - (0,1,0,-1) = (0,0,0,1)$.

$$M_1^2 = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}, \qquad M_2^2 = \begin{pmatrix} -2 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad M_2^3 = 0$$

$$M_3^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

So M_1 is idempotent (and symmetric), M_2 is nilpotent ($M_2^3 = 0$, $M_2^2 \neq 0$; it is not triangular), and M_3 is involutory. Each lacks the other two properties: $M_1^2 = M_1 \neq I_4$, and $M_1^k = M_1 \neq 0$ for every $k \geqslant 1$ (Proposition 1.29(c)); M_2^2 differs from M_2 and from I_4 (its second row is 0, theirs are not); and $M_3^2 = I_4 \neq M_3$, while M_3^k is I_4 for even k and M_3 for odd k (Proposition 1.19(a),(b)), never 0.

(b)

$$I_4 - M_1 = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad (I_4 - M_1)^2 = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$$M_1(I_4 - M_1) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

By rows, with $F = 2(I_4 - M_1)$: $\operatorname{row}_1(F^2) = 1\,(1,0,-1,0) - 1\,(-1,0,1,0) = (2,0,-2,0)$, $\operatorname{row}_2(F^2) = 1\,(0,1,0,1) + 1\,(0,1,0,1) = (0,2,0,2)$, $\operatorname{row}_3(F^2) = -\operatorname{row}_1(F^2)$ and $\operatorname{row}_4(F^2) = \operatorname{row}_2(F^2)$ ($\operatorname{row}_3(F) = -\operatorname{row}_1(F)$, $\operatorname{row}_4(F) = \operatorname{row}_2(F)$; Theorem 1.4(c)), so $F^2 = 2F$ and $(I_4 - M_1)^2 = \frac{1}{4}F^2 = \frac{1}{2}F = I_4 - M_1$; and $\operatorname{row}_1(BF) = 1\,(1,0,-1,0) + 1\,(-1,0,1,0) = 0$, $\operatorname{row}_2(BF) = 1\,(0,1,0,1) - 1\,(0,1,0,1) = 0$, $\operatorname{row}_3(BF) = \operatorname{row}_1(BF)$ and $\operatorname{row}_4(BF) = -\operatorname{row}_2(BF)$ ($\operatorname{row}_3(B) = \operatorname{row}_1(B)$, $\operatorname{row}_4(B) = -\operatorname{row}_2(B)$), so $M_1(I_4 - M_1) = \frac{1}{4}BF = 0$. So $I_4 - M_1$ is idempotent and $M_1(I_4 - M_1) = 0$, as Proposition 1.29(b) says.

Exercise 1.17. Let

$$H = [h_1\ h_2\ h_3\ h_4] = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

(a) Show that $H' = H$ and $H^2 = 4I_4$.

(b) Let $E = \mathrm{diag}(1, 1, 0, 0)$. Compute HE, and use the outer product expansion (Theorem 1.4(d)) to write HEH and H^2 as sums of the matrices $h_j h_j'$.

(c) Let $P = \frac{1}{4}HEH$, with E as in (b). Write out P, and show that $P^2 = P$, $\mathrm{tr}(P) = 2$ and $\frac{1}{4}H(I_4 - E)H = I_4 - P$.

Solution. (a) Row i of H equals h_i' for each i, so $H' = H$. Then $(H^2)_{ij} = \mathrm{row}_i(H)\,\mathrm{col}_j(H) = h_i' h_j$ (Theorem 1.4(d)). Each $h_i' h_i = 1 + 1 + 1 + 1 = 4$, and the six products with $i < j$ vanish:

$$h_1' h_2 = 1 - 1 + 1 - 1 = 0, \quad h_1' h_3 = 1 + 1 - 1 - 1 = 0, \quad h_1' h_4 = 1 - 1 - 1 + 1 = 0$$
$$h_2' h_3 = 1 - 1 - 1 + 1 = 0, \quad h_2' h_4 = 1 + 1 - 1 - 1 = 0, \quad h_3' h_4 = 1 - 1 + 1 - 1 = 0$$

So $H^2 = 4I_4$.

(b) By Proposition 1.13(a), right multiplication by E multiplies the columns of H by $1, 1, 0, 0$:

$$HE = [h_1\ h_2\ 0\ 0] = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

By (a), $\mathrm{row}_j(H) = h_j'$. So the outer product expansions (Theorem 1.4(d)) of $(HE)H$ and of HH are

$$HEH = \sum_{j=1}^{4} \mathrm{col}_j(HE)\,\mathrm{row}_j(H) = h_1 h_1' + h_2 h_2'$$
$$H^2 = \sum_{j=1}^{4} \mathrm{col}_j(H)\,\mathrm{row}_j(H) = h_1 h_1' + h_2 h_2' + h_3 h_3' + h_4 h_4'$$

(c) By (b), $P = \frac{1}{4}HEH = \frac{1}{4}(h_1 h_1' + h_2 h_2')$. Here $h_1 h_1' = J_4$. The ith entry of h_2 is $(-1)^{i+1}$, so the (i, j) entry of $h_2 h_2'$ is $(-1)^{i+1}(-1)^{j+1} = (-1)^{i+j}$: 1 where $i + j$ is even and -1 where $i + j$ is odd. Adding, $h_1 h_1' + h_2 h_2'$ has entry 2 where $i + j$ is even and 0 where $i + j$ is odd; so

$$P = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

By Theorem 1.3, $H^2 = 4I$ from (a), and $E^2 = E$,

$$P^2 = \tfrac{1}{16}(HEH)(HEH) = \tfrac{1}{16}HE(H^2)EH = \tfrac{1}{16}HE(4I)EH = \tfrac{1}{4}HE^2H = \tfrac{1}{4}HEH = P$$

Next $\mathrm{tr}(P) = \frac{1}{2}(1 + 1 + 1 + 1) = 2$. Finally, by Theorem 1.3 and (a), $\frac{1}{4}H(I - E)H = \frac{1}{4}H^2 - \frac{1}{4}HEH = \frac{1}{4}\,4I - P = I - P$.

Exercise 1.18. With $h_1, \dots, h_4$ as in Exercise 1.17, let

$$A = [h_2\ h_3\ h_4] = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 3}$$

(a) Compute $A'A$ and AA'.

(b) Show that $AA' = 4C_4$, $\mathbf{1}_4'A = 0$ and $C_4A = A$.

(c) Compute $\operatorname{tr}(A'A)$ and $\operatorname{tr}(AA')$.

Solution. (a) $(A'A)_{jl} = h_{j+1}'h_{l+1}$, so $A'A = 4I_3$ by the products listed in Exercise 1.17(a). By Theorem 1.4(d), $AA' = h_2h_2' + h_3h_3' + h_4h_4'$, and by Exercise 1.17(a),(b), $h_1h_1' + h_2h_2' + h_3h_3' + h_4h_4' = H^2 = 4I_4$. Since $h_1h_1' = J_4$,

$$AA' = 4I_4 - J_4 = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}$$

(b) $4I_4 - J_4 = 4(I_4 - \frac{1}{4}J_4) = 4C_4$ (Definition 1.14). Next $\mathbf{1}_4'A = (h_1'h_2,\ h_1'h_3,\ h_1'h_4) = 0$, so $C_4A = A - \frac{1}{4}\mathbf{1}_4(\mathbf{1}_4'A) = A$: each column of A already has mean 0.

(c) $\operatorname{tr}(A'A) = 12$ and $\operatorname{tr}(AA') = 4 \cdot 3 = 12$, equal by Theorem 1.11(b); both are the sum of the squares of the twelve entries ± 1 of A (Theorem 1.11(d)).

Exercise 1.19. Let $A = \begin{pmatrix} 1 & -2 \\ 0 & 3 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ -1 & 4 \end{pmatrix}$,

$$M = \begin{pmatrix} 1 & 2 & 0 & 1 \\ -1 & 3 & 1 & 0 \\ 2 & 0 & -2 & 1 \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

and $x = (1, -1, 2, 1)'$.

(a) Compute $\sum_i \sum_j a_{ij}b_{ij}$, $A'B$ and AB', and compare $\operatorname{tr}(A'B)$ and $\operatorname{tr}(AB')$ with the sum.

(b) Compute $x'Mx$, xx' and Mxx', and compare $\operatorname{tr}(Mxx')$ with $x'Mx$.

Solution. (a) $\sum_i \sum_j a_{ij}b_{ij} = 1 \cdot 2 + (-2) \cdot 1 + 0 \cdot 1 + 3 \cdot 0 + 2 \cdot (-1) + 1 \cdot 4 = 2$, and

$$A'B = \begin{pmatrix} 1 & 0 & 2 \\ -2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ -1 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 9 \\ -2 & 2 \end{pmatrix}, \qquad AB' = \begin{pmatrix} 1 & -2 \\ 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 \\ 1 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 1 & -9 \\ 3 & 0 & 12 \\ 5 & 2 & 2 \end{pmatrix}$$

So $\operatorname{tr}(A'B) = 0 + 2 = 2$ and $\operatorname{tr}(AB') = 0 + 0 + 2 = 2$, the same sum (Proposition 1.30(b)).

(b) $Mx = (1 - 2 + 0 + 1,\ -1 - 3 + 2 + 0,\ 2 + 0 - 4 + 1,\ 0 - 1 - 2 + 2)' = (0, -2, -1, -1)'$, so $x'Mx = 0 + 2 - 2 - 1 = -1$. Further

$$xx' = \begin{pmatrix} 1 & -1 & 2 & 1 \\ -1 & 1 & -2 & -1 \\ 2 & -2 & 4 & 2 \\ 1 & -1 & 2 & 1 \end{pmatrix}, \qquad Mxx' = (Mx)x' = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & 2 & -4 & -2 \\ -1 & 1 & -2 & -1 \\ -1 & 1 & -2 & -1 \end{pmatrix}$$

and $\operatorname{tr}(Mxx') = 0 + 2 - 2 - 1 = -1 = x'Mx$ (Proposition 1.30(c)).

Exercise 1.20. Five units are measured on three variables, with the data matrix

$$X_0 = \begin{pmatrix} 0 & 1 & 1 \\ 3 & 1 & 0 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{5\times 3}$$

(a) Find the mean vector $\bar{x}$ and the centered data matrix C_5X_0.

(b) Find $M = X_0'C_5X_0$, both as $(C_5X_0)'(C_5X_0)$ and as $X_0'X_0 - 5\bar{x}\bar{x}'$.

(c) For $a = (1,1,1)'$ and $b = (1,0,-1)'$, let $z = X_0a$ and $w = X_0b$. Find $\bar{z}$, $\bar{w}$, $\sum_i (z_i - \bar{z})^2$, $\sum_i (w_i - \bar{w})^2$ and $\sum_i (z_i - \bar{z})(w_i - \bar{w})$, first by Proposition 1.34(c) and then directly.

Solution. (a) $\mathbf{1}_5'X_0 = (10, 10, 5)$, so $\bar{x} = \frac{1}{5}X_0'\mathbf{1}_5 = (2, 2, 1)'$, and $C_5X_0 = X_0 - \mathbf{1}_5\bar{x}'$ (Proposition 1.33(b)):

$$C_5X_0 = \begin{pmatrix} 0-2 & 1-2 & 1-1 \\ 3-2 & 1-2 & 0-1 \\ 2-2 & 4-2 & 2-1 \\ 3-2 & 3-2 & 1-1 \\ 2-2 & 1-2 & 1-1 \end{pmatrix} = \begin{pmatrix} -2 & -1 & 0 \\ 1 & -1 & -1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \\ 0 & -1 & 0 \end{pmatrix}$$

(b) The (j,l) entry of $(C_5X_0)'(C_5X_0)$ is the product of columns j and l of C_5X_0 (Proposition 1.33(a)): $4+1+1 = 6$, $1+1+4+1+1 = 8$ and $1+1 = 2$ on the diagonal, and $2-1+1 = 2$, -1, $1+2 = 3$ off it. On the other hand, $X_0'X_0$ has the entries $\sum_i x_{ij}x_{il}$ (for instance $0+9+4+9+4 = 26$), and

$$X_0'X_0 - 5\bar{x}\bar{x}' = \begin{pmatrix} 26 & 22 & 9 \\ 22 & 28 & 13 \\ 9 & 13 & 7 \end{pmatrix} - \begin{pmatrix} 20 & 20 & 10 \\ 20 & 20 & 10 \\ 10 & 10 & 5 \end{pmatrix} = \begin{pmatrix} 6 & 2 & -1 \\ 2 & 8 & 3 \\ -1 & 3 & 2 \end{pmatrix} = M$$

(c) By Proposition 1.34(c), $\bar{z} = a'\bar{x} = 5$ and $\bar{w} = b'\bar{x} = 1$. With $Mb = \mathrm{col}_1(M) - \mathrm{col}_3(M) = (7, -1, -3)'$,

$$a'Ma = 24, \qquad b'Mb = 7 + 3 = 10, \qquad a'Mb = 7 - 1 - 3 = 3$$

($a'Ma$ is the sum of all entries of M). Directly, $z = (2, 4, 8, 7, 4)'$ and $w = (-1, 3, 0, 2, 1)'$ have the means $\frac{25}{5} = 5$ and $\frac{5}{5} = 1$, the deviations are $z - 5\,\mathbf{1}_5 = (-3, -1, 3, 2, -1)'$ and $w - \mathbf{1}_5 = (-2, 2, -1, 1, 0)'$, and

$$\sum_i (z_i - \bar{z})^2 = 9 + 1 + 9 + 4 + 1 = 24, \qquad \sum_i (w_i - \bar{w})^2 = 4 + 4 + 1 + 1 + 0 = 10$$

and $\sum_i (z_i - \bar{z})(w_i - \bar{w}) = 6 - 2 - 3 + 2 + 0 = 3$.

Problems

Problem 1.1 ([6, Exercise 2.8(a), (c), (d)])**.** Give examples of real 2×2 matrices such that: (a) $A^2 = -I$; (b) $CD = -DC$ ($CD \neq 0$); (c) $EF = 0$ (no entries of E or F are zero).

Solution. (a) No real number a has $a^2 = -1$, but $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ gives $A^2 = \left(\begin{smallmatrix} 0\cdot 0+1\cdot(-1) & 0\cdot 1+1\cdot 0 \\ (-1)\cdot 0+0\cdot(-1) & (-1)\cdot 1+0\cdot 0 \end{smallmatrix}\right) = -I$.

(b) For $C = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ and $D = \left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)$,

$$CD = \begin{pmatrix} 1-1 & -1-1 \\ 1+1 & -1+1 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad DC = \begin{pmatrix} 1-1 & 1+1 \\ -1-1 & -1+1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = -CD \neq 0$$

(c) For $E = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $F = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$, every entry of EF is $1 \cdot 1 + 1 \cdot (-1) = 0$, so $EF = 0$.

Problem 1.2 ([8, Ch. 1, Exercise 5])**.** Let $A, B \in \mathbb{R}^{n\times n}$. Show that $(A+B)(A-B) = A^2 - B^2$ if and only if A and B commute.

Solution. By Theorem 1.3, $(A+B)(A-B) = A(A-B) + B(A-B) = A^2 - AB + BA - B^2$. Thus $(A+B)(A-B) = A^2 - B^2$ if and only if $-AB + BA = 0$, that is, if and only if $AB = BA$.

Problem 1.3 ([8, Ch. 1, Exercise 6])**.** (a) Show that the product AB of two symmetric matrices $A, B \in \mathbb{R}^{n\times n}$ is symmetric if and only if A and B commute.

(b) Give an example of two symmetric matrices of the same size whose product is not symmetric.

Solution. (a) Since A and B are symmetric, $(AB)' = B'A' = BA$ (Theorem 1.9). So $AB = (AB)'$ holds if and only if $AB = BA$.

(b) Take $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $AB = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 0 \end{smallmatrix}\right) \neq \left(\begin{smallmatrix} 0 & 2 \\ 1 & 0 \end{smallmatrix}\right) = BA$, so by (a) AB is not symmetric.

Problem 1.4 ([8, Ch. 5, Exercise 1])**.** Show that for any $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$ and $C \in \mathbb{R}^{p\times m}$, $\operatorname{tr}(ABC) = \operatorname{tr}(B'A'C') = \operatorname{tr}(A'C'B')$.

Solution. By Theorem 1.11(c), then (a), then Theorem 1.9, and finally (c) again,

$$\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}\Big((CAB)'\Big) = \operatorname{tr}(B'A'C') = \operatorname{tr}(A'C'B')$$

In the last step the factors $B' \in \mathbb{R}^{p\times n}$, $A' \in \mathbb{R}^{n\times m}$, $C' \in \mathbb{R}^{m\times p}$ are moved cyclically.

Problem 1.5 ([4, Ch. III, §3, Exercise 11(e)], [7])**.** Prove that there are no matrices $A, B \in \mathbb{R}^{n\times n}$ such that $AB - BA = I_n$.

Solution. Suppose that $A, B \in \mathbb{R}^{n\times n}$ satisfy $AB - BA = I_n$. Then $\operatorname{tr}(AB - BA) = \operatorname{tr}(I_n) = n$, since the n diagonal entries of I_n are 1 (Proposition 1.30(a)). On the other hand, by Theorem 1.11(a),(b),

$$\operatorname{tr}(AB - BA) = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0$$

So $0 = n \geqslant 1$ ✘ Hence no such A and B exist.

Problem 1.6 ([6, Exercise 2.27])**.** Let $A \in \mathbb{R}^{n\times n}$ and let $D = \operatorname{diag}(a_{11}, \ldots, a_{nn})$. Show that the sum of all entries of A is

$$\mathbf{1}_n' A \mathbf{1}_n = \mathbf{1}_n' D \mathbf{1}_n + \operatorname{tr}\Big((\mathbf{1}_n \mathbf{1}_n' - I_n)A\Big)$$

Solution. By Theorem 1.11(a),(b),

$$\operatorname{tr}\Big((\mathbf{1}\mathbf{1}' - I_n)A\Big) = \operatorname{tr}(\mathbf{1}\mathbf{1}'A) - \operatorname{tr}(A) = \operatorname{tr}(\mathbf{1}'A\mathbf{1}) - \operatorname{tr}(A) = \mathbf{1}'A\mathbf{1} - \mathbf{1}'D\mathbf{1}$$

Here $\operatorname{tr}(\mathbf{1}\mathbf{1}'A) = \operatorname{tr}(\mathbf{1}'A\mathbf{1})$ moves the $n \times 1$ factor $\mathbf{1}$ past the $1 \times n$ factor $\mathbf{1}'A$; the trace of the 1×1 matrix $\mathbf{1}'A\mathbf{1}$ is that number; and $\mathbf{1}'D\mathbf{1} = \sum_i a_{ii} = \operatorname{tr}(A)$, because $D\mathbf{1} = (a_{11}, \ldots, a_{nn})'$. Rearranging gives the identity. Also $\mathbf{1}'A\mathbf{1} = \sum_i \sum_j a_{ij}$: by Theorem 1.4(b), $A\mathbf{1} = a_1 + \cdots + a_n$ has ith entry $\sum_j a_{ij}$, and by Theorem 1.9, $\mathbf{1}'(A\mathbf{1})$ is the sum of these n entries.

Problem 1.7 ([6, Exercise 2.37(a), (c)])**.** (a) Show that the only symmetric idempotent 2×2 matrices are I_2, 0, and uu' with $u \in \mathbb{R}^2$, $u'u = 1$.

(b) Give an example of an $n \times n$ idempotent matrix ($n \geqslant 2$) that is not symmetric.

Solution. (a) The symmetric matrix $A = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$ is idempotent if and only if $\left(\begin{smallmatrix} a^2+b^2 & b(a+d) \\ b(a+d) & b^2+d^2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$, that is, $a^2 + b^2 = a$, $b(a + d) = b$, $d^2 + b^2 = d$.

- If $a+d \neq 1$, the second equation gives $b = 0$; then $a^2 = a$ and $d^2 = d$, so $a, d \in \{0, 1\}$, and $a+d \neq 1$ leaves $a = d = 0$ or $a = d = 1$: $A = 0$ or $A = I_2$.
- If $a + d = 1$, then $d = 1 - a$ and the first equation gives $b^2 = a(1 - a)$, so $0 \leqslant a \leqslant 1$ and $b = \pm\sqrt{a(1-a)}$; the third equation then holds, since $(1 - a)^2 + a(1 - a) = 1 - a$. Hence

$$A = \begin{pmatrix} a & \pm\sqrt{a(1-a)} \\ \pm\sqrt{a(1-a)} & 1-a \end{pmatrix} = \begin{pmatrix} \sqrt{a} \\ \pm\sqrt{1-a} \end{pmatrix} \begin{pmatrix} \sqrt{a} & \pm\sqrt{1-a} \end{pmatrix} = uu'$$

  with the same sign throughout, and $u'u = a + (1 - a) = 1$.

Conversely, I_2 and 0 are symmetric idempotent, and if $u'u = 1$, then uu' is symmetric and $(uu')(uu') = u(u'u)u' = uu'$.

(b) Let $a, b \in \mathbb{R}^n$ with $b'a = 1$, and $A = ab'$. Then $A^2 = a(b'a)b' = ab' = A$. This A is symmetric only when b is a multiple of a: if $ab' = ba'$, multiplying on the right by a gives $a(b'a) = b(a'a)$, that is, $a = (a'a)b$; here $a \neq 0$ because $b'a = 1$, so $a'a > 0$ and $b = a/(a'a)$. Conversely, if $b = ca$, then $ab' = c\,aa'$ is symmetric. For a concrete example take $a = e_1$ and $b = e_1 + e_2$: $b'a = 1$, and $A = e_1e_1' + e_1e_2'$ has $(1, 1)$ and $(1, 2)$ entries 1 and all other entries 0; for $n = 2$, $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, with $A^2 = A \neq A'$.

Problem 1.8 ([2, Exercise 4.15])**.** Let $A, B \in \mathbb{R}^{n\times n}$ satisfy $A^2 = A$ and $B^2 = B$. Show that if $(A - B)^2 = A - B$, then $AB = BA = B$.

Solution. Expanding by Theorem 1.3, $(A - B)^2 = A - B$ reads $A^2 - AB - BA + B^2 = A - B$; with $A^2 = A$ and $B^2 = B$ this is $A - AB - BA + B = A - B$, that is,

$$2B = AB + BA$$

- Multiplying on the left by A: $2AB = A^2B + ABA = AB + ABA$, hence $AB = ABA$.
- Multiplying on the right by A: $2BA = ABA + BA^2 = ABA + BA$, hence $ABA = BA$.
- Therefore $AB = BA$, and $2B = AB + BA$ gives $2B = 2AB = 2BA$, that is, $AB = BA = B$.

Problem 1.9 ([8, Ch. 10, Exercise 6])**.** Let $A \in \mathbb{R}^{n\times n}$. Show that $\frac{1}{2}(I_n + A)$ is idempotent if and only if A is involutory.

Solution. By Theorem 1.3,

$$\left[\tfrac{1}{2}(I + A)\right]^2 = \tfrac{1}{4}I + \tfrac{1}{2}A + \tfrac{1}{4}A^2$$

So $\frac{1}{2}(I + A)$ is idempotent if and only if $\frac{1}{4}I + \frac{1}{2}A + \frac{1}{4}A^2 = \frac{1}{2}I + \frac{1}{2}A$, that is, if and only if $\frac{1}{4}A^2 = \frac{1}{4}I$, that is, if and only if $A^2 = I$: A is involutory.

Problem 1.10 ([8, Ch. 5, Exercise 3])**.** Let $A \in \mathbb{R}^{n\times n}$ satisfy $A'A = A^2$.

(a) Show that $\operatorname{tr}\big[(A-A')'(A-A')\big] = 0$.

(b) Show that A is symmetric.

Solution. (a) Since $(A-A')' = A'-A$ and $A'A' = (AA)'$ (Theorem 1.9),

$$\begin{aligned}\operatorname{tr}\big[(A-A')'(A-A')\big] &= \operatorname{tr}\big[A'A - A'A' - AA + AA'\big]\\ &= \operatorname{tr}(A'A) - \operatorname{tr}\big[(AA)'\big] - \operatorname{tr}(A^2) + \operatorname{tr}(AA')\\ &= \operatorname{tr}(AA') - \operatorname{tr}\big[(AA)'\big]\\ &= \operatorname{tr}(A'A) - \operatorname{tr}\big[(AA)'\big]\\ &= \operatorname{tr}(A'A) - \operatorname{tr}(AA) = 0\end{aligned}$$

The second line uses Theorem 1.11(a) (linearity); the third, $A'A = A^2$; the fourth, $\operatorname{tr}(AA') = \operatorname{tr}(A'A)$ (Theorem 1.11(b)); the last, $\operatorname{tr}(M') = \operatorname{tr}(M)$ (Theorem 1.11(a)) and $A'A = A^2$ again.

(b) By (a) and Theorem 1.11(d), $A - A' = 0$, that is, $A' = A$.

Problem 1.11 ([2, Exercise 5.8])**.** Let $X = [x_1\ x_2\ \cdots\ x_c] \in \mathbb{R}^{r\times c}$ and let $A \in \mathbb{R}^{r\times r}$ be symmetric. Prove that

$$\operatorname{tr}\big[(AXX')^2\big] = \sum_{j=1}^{c}\sum_{k=1}^{c}(x_j'Ax_k)^2$$

Solution. Moving factors cyclically (Theorem 1.11(c), applied twice),

$$\operatorname{tr}\big[AXX'AXX'\big] = \operatorname{tr}\big[XX'AXX'A\big] = \operatorname{tr}\big(X'AXX'AX\big)$$

Let $Y = X'AX \in \mathbb{R}^{c\times c}$; its kth column is $y_k = X'Ax_k$ (Theorem 1.4(a)).

- $Y' = X'A'X = Y$, because $A' = A$. So $\operatorname{row}_k(Y) = y_k'$, and $\operatorname{tr}(YY) = \sum_k \operatorname{row}_k(Y)\, y_k = \sum_k y_k'y_k$ (Theorem 1.4(d)).
- $y_k' = x_k'A'X = x_k'AX$, again because $A' = A$. By the outer product expansion $XX' = \sum_j x_jx_j'$ (Theorem 1.4(d)),
$$y_k'y_k = x_k'AXX'Ax_k = \sum_{j=1}^{c}(x_k'Ax_j)(x_j'Ax_k)$$
- The number $x_j'Ax_k$ equals its transpose $x_k'A'x_j = x_k'Ax_j$, so each term is $(x_k'Ax_j)^2$.

Hence $\operatorname{tr}\big[(AXX')^2\big] = \operatorname{tr}(YY) = \sum_k\sum_j(x_k'Ax_j)^2$, which is the stated double sum (the summand is symmetric in j and k).

Problem 1.12 ([10, Problems 39.1, 39.2])**.** Let $d_1, \dots, d_n$ be distinct real numbers and $D = \operatorname{diag}(d_1, \dots, d_n)$.

(a) Show that every $X \in \mathbb{R}^{n\times n}$ with $DX = XD$ is diagonal.

(b) Suppose, moreover, that $d_1, \dots, d_n$ are nonzero, and let $N \in \mathbb{R}^{n\times n}$ have the $(i, i+1)$ entries 1 $(1 \leqslant i \leqslant n-1)$ and all other entries 0. Show that every $X \in \mathbb{R}^{n\times n}$ that commutes with D and with ND is $X = cI_n$ for some $c \in \mathbb{R}$.

(c) Show that every $X \in \mathbb{R}^{n\times n}$ that commutes with all matrices in $\mathbb{R}^{n\times n}$ is $X = cI_n$ for some $c \in \mathbb{R}$.

Solution. Write $X = (x_{ij})$. By Proposition 1.13(a), left multiplication by a diagonal matrix scales the rows, and right multiplication scales the columns:

$$(DX)_{ij} = d_i x_{ij}, \qquad (XD)_{ij} = x_{ij} d_j \qquad (1 \leqslant i, j \leqslant n)$$

(a) If $DX = XD$, then $d_i x_{ij} = x_{ij} d_j$, that is, $(d_i - d_j)x_{ij} = 0$, for all i, j. For $i \neq j$, $d_i - d_j \neq 0$, so $x_{ij} = 0$: X is diagonal.

(b)

- By (a), $X = \operatorname{diag}(x_1, \ldots, x_n)$, with $x_i = x_{ii}$.
- Let $M = ND = (m_{ij})$. By the same rule, column j of M is $d_j \operatorname{col}_j(N)$, so M has the $(i, i+1)$ entries $m_{i,i+1} = d_{i+1}$ $(1 \leqslant i \leqslant n-1)$ and all other entries 0.
- By the same rule for the diagonal matrix X, $(MX)_{i,i+1} = m_{i,i+1} x_{i+1} = d_{i+1} x_{i+1}$ and $(XM)_{i,i+1} = x_i m_{i,i+1} = x_i d_{i+1}$.
- As $MX = XM$, $d_{i+1}(x_{i+1} - x_i) = 0$, and $d_{i+1} \neq 0$; so $x_{i+1} = x_i$ for $i = 1, \ldots, n-1$. Hence $x_1 = \cdots = x_n = c$, and $X = cI_n$.

(c) Let X commute with every matrix in $\mathbb{R}^{n\times n}$, and take $D = \operatorname{diag}(1, 2, \ldots, n)$, whose diagonal entries are distinct and nonzero. Then X commutes with D and with ND, and by (b), $X = cI_n$ for some $c \in \mathbb{R}$.

Problem 1.13 ([10, Problem 39.3])**.** Find all matrices $X \in \mathbb{R}^{n\times n}$ that commute with $J_n = \mathbf{1}_n \mathbf{1}_n'$, the $n \times n$ matrix with all entries 1.

Solution. Let $\mathbf{1} = \mathbf{1}_n$ and $J = J_n$, and put $p = X\mathbf{1} \in \mathbb{R}^n$ and $q' = \mathbf{1}'X \in \mathbb{R}^{1\times n}$. By Theorem 1.4(b),(c), the entries $p_i = \sum_j x_{ij}$ of p are the row sums of X, and the entries $q_j = \sum_i x_{ij}$ of q' are its column sums. By Theorem 1.3,

$$JX = \mathbf{1}(\mathbf{1}'X) = \mathbf{1}q', \qquad XJ = (X\mathbf{1})\mathbf{1}' = p\mathbf{1}'$$

so $(JX)_{ij} = q_j$ and $(XJ)_{ij} = p_i$ (Definition 1.2, with one term): every row of JX is $(q_1, \ldots, q_n)$, and every column of XJ is p.

- If $JX = XJ$, then $q_j = p_i$ for all i, j. Taking $j = 1$, every p_i equals q_1; taking $i = 1$, every q_j equals $p_1 = q_1$. So all row sums and all column sums of X equal the same number $c = q_1$.
- Conversely, if every row sum and every column sum of X equals c, then all entries of JX and of XJ are c, so $JX = XJ$.

Hence X commutes with J_n if and only if $X\mathbf{1}_n = c\mathbf{1}_n$ and $\mathbf{1}_n'X = c\mathbf{1}_n'$ for some $c \in \mathbb{R}$, that is, all row sums and all column sums of X are equal. For instance, $aI_n + bJ_n$ has all row and column sums $a + nb$, in accordance with Example 1.25.

Problem 1.14 ([10, Problems 5.1, 40.7(a)])**.** Let $A \in \mathbb{R}^{n\times n}$.

(a) Suppose that $\operatorname{tr}(AX) = 0$ for every $X \in \mathbb{R}^{n\times n}$ with $\operatorname{tr}(X) = 0$. Show that $A = cI_n$ for some $c \in \mathbb{R}$.

(b) Suppose that $\operatorname{tr}(AXY) = \operatorname{tr}(AYX)$ for all $X, Y \in \mathbb{R}^{n\times n}$. Show that $A = cI_n$ for some $c \in \mathbb{R}$.

Solution. Let $E_{ij} = e_i e_j' \in \mathbb{R}^{n\times n}$ be the matrices of Proposition 1.18. For all i, j, by Theorem 1.3, Theorem 1.11(b), applied to the $n \times 1$ matrix Ae_i and the $1 \times n$ matrix e_j', and Proposition 1.18(a),

$$\operatorname{tr}(AE_{ij}) = \operatorname{tr}\Big((Ae_i)e_j'\Big) = \operatorname{tr}\Big(e_j' Ae_i\Big) = e_j' Ae_i = a_{ji}$$

(a)

- Let $i \neq j$. The only nonzero entry of E_{ij} is its (i,j) entry (Proposition 1.18(b)), which lies off the diagonal; so $\operatorname{tr}(E_{ij}) = 0$, and by hypothesis $a_{ji} = \operatorname{tr}(AE_{ij}) = 0$. Thus A is diagonal.
- Let $i \neq j$. Then $\operatorname{tr}(E_{ii} - E_{jj}) = 1 - 1 = 0$ (Theorem 1.11(a)), and by hypothesis, with Theorems 1.3 and 1.11(a), $0 = \operatorname{tr}\Big(A(E_{ii} - E_{jj})\Big) = \operatorname{tr}(AE_{ii}) - \operatorname{tr}(AE_{jj}) = a_{ii} - a_{jj}$.
- So A is diagonal with all diagonal entries equal to a_{11}, that is, $A = a_{11}I_n$.

(b) By Theorems 1.3 and 1.11(a), the hypothesis says that $\operatorname{tr}\Big(A(XY - YX)\Big) = \operatorname{tr}(AXY) - \operatorname{tr}(AYX) = 0$ for all $X, Y \in \mathbb{R}^{n\times n}$. Let $i \neq j$.

- By Proposition 1.18(c), $E_{ij}E_{jj} = E_{ij}$ and $E_{jj}E_{ij} = 0$, as $j \neq i$. With $X = E_{ij}$ and $Y = E_{jj}$, $a_{ji} = \operatorname{tr}(AE_{ij}) = \operatorname{tr}(AE_{ij}E_{jj}) = \operatorname{tr}(AE_{jj}E_{ij}) = \operatorname{tr}(A0) = 0$.
- Likewise $E_{ij}E_{ji} = E_{ii}$ and $E_{ji}E_{ij} = E_{jj}$. With $X = E_{ij}$ and $Y = E_{ji}$, $a_{ii} = \operatorname{tr}(AE_{ii}) = \operatorname{tr}(AE_{jj}) = a_{jj}$.
- As in (a), $A = a_{11}I_n$. (So the traceless matrices $E_{ij} = E_{ij}E_{jj} - E_{jj}E_{ij}$ and $E_{ii} - E_{jj} = E_{ij}E_{ji} - E_{ji}E_{ij}$ used in (a) are of the form $XY - YX$.)

Problem 1.15 ([10, Problem 39.6(a)])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be nilpotent, with $AB = BA$. Show that $A + B$ is nilpotent.

Solution.

- $A^jB^l = B^lA^j$ for all integers $j, l \geqslant 0$. First, $AB^l = B^lA$, by induction on l: $AB^0 = A = B^0A$, and if $AB^l = B^lA$, then $AB^{l+1} = (AB^l)B = B^lAB = B^lBA = B^{l+1}A$ (Theorem 1.3). Then, by induction on j: $A^0B^l = B^l = B^lA^0$, and if $A^jB^l = B^lA^j$, then $A^{j+1}B^l = A^j(AB^l) = A^j(B^lA) = (A^jB^l)A = B^lA^jA = B^lA^{j+1}$.
- The binomial formula: for every integer $k \geqslant 0$,

$$(A+B)^k = \sum_{j=0}^{k} \binom{k}{j} A^jB^{k-j}$$

  by induction on k. For $k = 0$ both sides are I_n. If the formula holds for k, then, by Theorem 1.3 and $A^jB^{k-j}A = A^jAB^{k-j} = A^{j+1}B^{k-j}$ (first item),

$$(A+B)^{k+1} = (A+B)^k(A+B) = \sum_{j=0}^{k} \binom{k}{j} A^{j+1}B^{k-j} + \sum_{j=0}^{k} \binom{k}{j} A^jB^{k+1-j} = \sum_{j=0}^{k+1} \binom{k+1}{j} A^jB^{k+1-j}$$

  In the last step, the coefficient of A^jB^{k+1-j} is 1 for $j = 0$ and for $j = k+1$, and for $1 \leqslant j \leqslant k$ it is, by Pascal's rule,

$$\binom{k}{j-1} + \binom{k}{j} = \frac{k!\,j + k!\,(k+1-j)}{j!\,(k+1-j)!} = \frac{(k+1)!}{j!\,(k+1-j)!} = \binom{k+1}{j}$$

- Let $A^p = 0$ and $B^q = 0$ with $p, q \geqslant 1$ (Definition 1.20), and let $m = \max\{p, q\}$. For $j \geqslant m$, $A^j = A^pA^{j-p} = 0$ (Proposition 1.19(a), Theorem 1.3); likewise $B^l = 0$ for $l \geqslant m$.
- In $(A+B)^{2m-1} = \sum_{j=0}^{2m-1} \binom{2m-1}{j} A^jB^{2m-1-j}$, either $j \geqslant m$, or $j \leqslant m-1$ and then $2m-1-j \geqslant m$. So every term has a zero factor and is 0 (Theorem 1.3). Hence $(A+B)^{2m-1} = 0$, with $2m-1 \geqslant 1$: $A + B$ is nilpotent.

Problem 1.16 ([10, Problem 34.4])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric. Show that $\operatorname{tr}\big((AB)^2\big) \leqslant \operatorname{tr}(A^2B^2)$.

Solution. Let $K = AB - BA$.

- K is skew-symmetric: as $A' = A$ and $B' = B$, $K' = B'A' - A'B' = BA - AB = -K$ (Theorem 1.9).
- Hence $K'K = -K^2$, and by Theorem 1.11(a),(d), $\operatorname{tr}(K^2) = -\operatorname{tr}(K'K) = -\sum_{i,j} k_{ij}^2 \leqslant 0$.
- By Theorem 1.3, $K^2 = ABAB - ABBA - BAAB + BABA$. By Theorem 1.11(b), applied to the factors B and ABA, to A and B^2A, to B^2 and A^2, and to B and A^2B,

$$\operatorname{tr}(BABA) = \operatorname{tr}(ABAB), \qquad \operatorname{tr}(ABBA) = \operatorname{tr}(B^2A^2) = \operatorname{tr}(A^2B^2), \qquad \operatorname{tr}(BAAB) = \operatorname{tr}(A^2B^2)$$

- So, by Theorem 1.11(a),

$$0 \geqslant \operatorname{tr}(K^2) = 2\operatorname{tr}\big((AB)^2\big) - 2\operatorname{tr}(A^2B^2)$$

that is, $\operatorname{tr}\big((AB)^2\big) \leqslant \operatorname{tr}(A^2B^2)$.

Problem 1.17 ([1, §8.5, Exercise 11], [2, Exercise 5.7(a)], [8, Ch. 10, Exercise 5])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric and nilpotent: $A^k = 0$ for some integer $k \geqslant 1$ (Definition 1.20). Show that $A = 0$. (Compare the matrix X of Remark 1.7: $X \neq 0$ and $X^2 = 0$, but X is not symmetric, since $x_{12} = 1 \neq 2 = x_{21}$.)

Solution. Since $A^1 = A^0A = I_nA = A$ (Theorem 1.3), the case $k = 1$ is $A = A^1 = 0$. Let $k \geqslant 2$. It suffices to show that, for every integer j with $2 \leqslant j \leqslant k$, $A^j = 0$ implies $A^{j-1} = 0$: then $A^k = 0$ gives, in turn, $A^{k-1} = 0$, …, $A^1 = 0$, that is, $A = 0$.

So let $2 \leqslant j \leqslant k$ and $A^j = 0$. Since $A' = A$, Proposition 1.19(a) and Theorem 1.3 give

$$A'A\,A^{j-2} = A^2A^{j-2} = A^j = 0 = A'A\,0$$

where $A^{j-2} = A^0 = I_n$ if $j = 2$, and 0 is the $n \times n$ zero matrix. By Corollary 1.31(b), with $B_1 = A^{j-2}$ and $B_2 = 0$ ($m = p = n$), $A\,A^{j-2} = A\,0$, that is,

$$A^{j-1} = A\,A^{j-2} = A\,0 = 0$$

again by Proposition 1.19(a) and Theorem 1.3.

Chapter 2. Gauss elimination, RREF, and inverses

Linear systems and augmented matrices

Definition 2.1 ([1, §§1.2–1.5, Example 12], [6, Exercise 6.41])**.** Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. The system of m linear equations in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

is the matrix equation $Ax = b$, since $(Ax)_i = \sum_j a_{ij}x_j$. A is the **coefficient matrix** and $[A \mid b] \in \mathbb{R}^{m\times(n+1)}$, the matrix A with the column b appended, is the **augmented matrix**. A **solution** is an $x \in \mathbb{R}^n$ with $Ax = b$. The system is **consistent** if it has a solution and **inconsistent** otherwise. It is **homogeneous** if $b = 0$; then $x = 0$ is a solution, the **trivial solution**.

Elementary row operations

Definition 2.2 ([1, §1.3])**.** The **elementary row operations** on a matrix with m rows are, for row indices $r \neq s$:

(1) multiply row r by a number $c \neq 0$, written $R_r \leftarrow cR_r$;

(2) add c times row s to row r ($c \in \mathbb{R}$), written $R_r \leftarrow R_r + cR_s$;

(3) interchange rows r and s, written $R_r \leftrightarrow R_s$.

When several operations label one arrow, they are performed one after another, in the order written (upper label first). B is **row-equivalent** to A if B is obtained from A by finitely many elementary row operations.

Theorem 2.3 ([1, §§1.2–1.4, Theorems 1–3])**.** (a) Each elementary row operation is undone by an operation of the same type: $R_r \leftarrow cR_r$ by $R_r \leftarrow c^{-1}R_r$, $R_r \leftarrow R_r + cR_s$ by $R_r \leftarrow R_r - cR_s$, and $R_r \leftrightarrow R_s$ by itself.

(b) If row operations take $[A \mid b]$ to $[B \mid z]$, then the systems $Ax = b$ and $Bx = z$ have exactly the same solutions.

Proof. (a) is checked row by row. (b) It suffices to consider one operation. Each equation of the new system is a multiple of an equation of the old one, or a sum of an old equation and a multiple of another, so every solution of $Ax = b$ solves $Bx = z$. By (a), the old system is obtained back from the new one by a row operation, so every solution of $Bx = z$ solves $Ax = b$. □

So we solve a system by simplifying its augmented matrix with row operations; the next definition says how far to simplify.

Echelon form and RREF

Definition 2.4 ([1, §§1.3–1.4], [6, Ch. 6])**.** A matrix R is in **reduced row echelon form** (**RREF**) if

(a) the first nonzero entry of each nonzero row is 1, its **pivot**;

(b) each column that contains a pivot has all its other entries 0;

(c) every zero row lies below every nonzero row;

(d) if rows $1, \dots, r$ are the nonzero rows and the pivot of row i lies in column k_i, then $k_1 < k_2 < \cdots < k_r$.

The columns $k_1, \dots, k_r$ are the **pivot columns**; column k_i of R is the unit vector e_i. A matrix that satisfies (a), (c), (d), but not necessarily (b), is in **echelon form**; then every entry below a pivot is 0.

For example, I_n and 0 are in RREF, and so is

$$R = \begin{pmatrix} 0 & 1 & -3 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad (r = 2,\ k_1 = 2,\ k_2 = 4)$$

The matrix $\left(\begin{smallmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{smallmatrix}\right)$ is in echelon form but not in RREF (column 3 violates (b)), and $\left(\begin{smallmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ violates (a) and (d).

Definition 2.5 (Gauss–Jordan elimination; [6, Exercise 6.13, §6.3], [1, §§1.3–1.4, Theorems 4, 5])**.** Let M be any matrix (for instance an augmented matrix). Start with $i = 1$.

Step 1. Find the first column that has a nonzero entry in rows $i, i+1, \dots$; if there is none, go to Step 5. This is the next pivot column.

Step 2. If the entry of row i in this column is 0, interchange row i with a lower row that has a nonzero entry there.

Step 3. Multiply row i by the reciprocal of this entry, so that the pivot is 1.

Step 4. Add suitable multiples of row i to the rows below it, so that every entry below the pivot is 0. Increase i by 1 and return to Step 1.

Step 5. (*Backward elimination.*) For each pivot, from the last one to the first, add suitable multiples of its row to the rows above it, so that every entry above the pivot is 0.

Steps 1–4 (**forward elimination**) produce an echelon form; Step 5 produces a matrix in RREF. Indeed, when Step 1 finds the pivot column of row i, rows $i, i+1, \dots$ are 0 in every column before it, and Step 4 makes rows $i+1, i+2, \dots$ 0 in this column too; so each nonzero row begins with its pivot 1, the pivot columns are taken from left to right, and the rows left when Step 1 finds no column are zero. Thus forward elimination ends in an echelon form, with pivots in columns $k_1 < \cdots < k_r$, say. In Step 5, when the pivot of row i is processed, row i is 0 before column k_i and, the pivots of rows $i+1, \dots, r$ being done, 0 in columns $k_{i+1}, \dots, k_r$; so the multiples of row i that are added change no entry before column k_i and no other pivot column, and in the end every column k_i is e_i. The scaling in Step 3 may also be postponed, to the end of forward elimination or even to the very end (the leading entries become 1 only then); this often avoids fractions. So for every matrix M, some matrix in RREF is row-equivalent to M.

Example 2.6 (cf. [1, §1.3, Example 5])**.** Solve the homogeneous system

$$x_1 + x_2 + x_3 - 2x_4 = 0, \qquad x_1 + x_2 + 2x_3 - 3x_4 = 0, \qquad 2x_1 + x_2 + 5x_3 - 4x_4 = 0$$

Solution. Row-reduce the coefficient matrix A (the column $b = 0$ never changes, so it is not written).

$$\begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & 2 & -3 \\ 2 & 1 & 5 & -4 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & -1 & 3 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + 3R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

After the first arrow the entry of row 2 in column 2 is 0, so Step 2 interchanges rows 2 and 3, and Step 3 multiplies the new row 2 by -1. The first matrix of the second line is an echelon form (end of

forward elimination); the last matrix, R, is in RREF, with pivot columns $1, 2, 3$. The system $Rx = 0$ reads $x_1 + 2x_4 = 0$, $x_2 - 3x_4 = 0$, $x_3 - x_4 = 0$, so $x_4 = t$ is free and all solutions are

$$x = t\,(-2,\ 3,\ 1,\ 1)' \qquad (t \in \mathbb{R})$$

Check: $-2 + 3 + 1 - 2 = 0$, $-2 + 3 + 2 - 3 = 0$, $-4 + 3 + 5 - 4 = 0$.

Reading off the solutions

Proposition 2.7 ([1, §1.4])**.** Let row operations take $[A \mid b]$ ($A \in \mathbb{R}^{m\times n}$) to $[R \mid z]$ with R in RREF, nonzero rows $1, \dots, r$ and pivot columns $k_1 < \cdots < k_r$. Call $x_{k_1}, \dots, x_{k_r}$ the **pivot unknowns** and the other $n - r$ unknowns the **free unknowns**.

(a) $Ax = b$ is consistent if and only if $z_i = 0$ for every $i > r$, that is, no row of $[R \mid z]$ reads $(0 \ \cdots \ 0 \mid z_i)$ with $z_i \neq 0$.

(b) If it is consistent, all solutions are obtained by choosing arbitrary values for the free unknowns and computing each pivot unknown from its row: $x_{k_i} = z_i - \sum_{j \text{ free}} R_{ij}x_j$ $(1 \leqslant i \leqslant r)$.

Proof. By Theorem 2.3, $Ax = b$ and $Rx = z$ have the same solutions. (a) Row $i > r$ of $Rx = z$ reads $0 = z_i$. So there is no solution if one of these z_i is not 0; if all of them are 0, these rows hold for every x, and the argument for (b) gives a solution. (b) For $i \leqslant r$, column k_l of R is e_l, so the pivot unknown x_{k_i} occurs only in equation i, with coefficient 1; each choice of the free unknowns therefore gives exactly one solution. □

Theorem 2.8 ([11, One.III.2, Theorem 2.6])**.** For every matrix A, exactly one matrix in RREF is row-equivalent to A; it is called **the RREF of** A.

Proof. By Definition 2.5, for every $A \in \mathbb{R}^{m\times n}$ some matrix in RREF is row-equivalent to A. Let R and S be matrices in RREF that are row-equivalent to A; we show $R = S$ by induction on n, for fixed m.

- $n = 1$: an $m \times 1$ matrix in RREF is 0 or e_1. Row operations take 0 to 0, and a nonzero matrix to a nonzero one, since they can be undone (Theorem 2.3(a)). So $R = S = 0$ if $A = 0$, and $R = S = e_1$ otherwise.
- Let $n > 1$, and let $\hat{A}, \hat{R}, \hat{S}$ consist of the first $n - 1$ columns of A, R, S. Row operations act on each column separately: by Definition 2.2, each of them computes the new entries of a column from the entries of that column only. So the operations that take A to R take $\hat{A}$ to $\hat{R}$. Moreover $\hat{R}$ is in RREF: a row of R whose pivot lies in column n is the last nonzero row, and it becomes a zero row of $\hat{R}$. Likewise for $\hat{S}$; so $\hat{R} = \hat{S}$ by the induction hypothesis.
- Suppose $R \neq S$; then they differ in column n, say in row i. The systems $Ax = 0$, $Rx = 0$ and $Sx = 0$ have the same solutions (Theorem 2.3(b)). Subtracting equation i of $Sx = 0$ from equation i of $Rx = 0$ leaves $(r_{in} - s_{in})x_n = 0$, so $x_n = 0$ for every solution.
- Hence x_n is not a free unknown of $Rx = 0$, since a free unknown can take the value 1 (Proposition 2.7(b)): column n of R is a pivot column, and so is column n of S.
- Let r be the number of nonzero rows of $\hat{R} = \hat{S}$. In R, rows $1, \dots, r$ have their pivots before column n, and rows $r + 1, \dots, m$ are zero before column n; as zero rows come last, the pivot of column n lies in row $r+1$, and column n of R is e_{r+1} (Definition 2.4). The same holds for S. So $R = S$ ✘ □

In particular, the pivot columns, and hence the free unknowns of $Ax = b$, do not depend on the row operations used to reach the RREF.

Example 2.9 (cf. [1, §1.4, Example 8])**.** For the RREF R displayed after Definition 2.4, the system $Rx = 0$ is $x_2 - 3x_3 + 2x_5 = 0$, $x_4 + 2x_5 = 0$; the pivot unknowns are x_2, x_4 and the free unknowns x_1, x_3, x_5. With $x_1 = a$, $x_3 = b$, $x_5 = c$, all solutions are

$$x = \begin{pmatrix} a \\ 3b - 2c \\ b \\ -2c \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ -2 \\ 0 \\ -2 \\ 1 \end{pmatrix} \qquad (a, b, c \in \mathbb{R})$$

Example 2.10 (cf. [1, §1.4, Example 9])**.** Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 1 \end{pmatrix}$$

For which $y = (y_1, y_2, y_3, y_4)'$ is $Ax = y$ consistent? Solve it for $y = (4, 7, 4, 7)'$.

Solution. Row-reduce the augmented matrix, keeping $y_1, \dots, y_4$ as letters:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 2 & 3 & 1 & 1 & y_2 \\ 1 & 1 & 1 & 1 & y_3 \\ 2 & 2 & 2 & 1 & y_4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 0 & -1 & 1 & -1 & y_2 - 2y_1 \\ 0 & -1 & 1 & 0 & y_3 - y_1 \\ 0 & -2 & 2 & -1 & y_4 - 2y_1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 0 & 1 & -1 & 1 & 2y_1 - y_2 \\ 0 & -1 & 1 & 0 & y_3 - y_1 \\ 0 & -2 & 2 & -1 & y_4 - 2y_1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 0 & 1 & -1 & 1 & 2y_1 - y_2 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 1 & 2y_1 - 2y_2 + y_4 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & y_1 \\ 0 & 1 & -1 & 1 & 2y_1 - y_2 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 0 & y_1 - y_2 - y_3 + y_4 \end{array}\right) \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 0 & y_2 - y_3 \\ 0 & 1 & -1 & 0 & y_1 - y_3 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 0 & y_1 - y_2 - y_3 + y_4 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & -2y_1 + y_2 + y_3 \\ 0 & 1 & -1 & 0 & y_1 - y_3 \\ 0 & 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & 0 & y_1 - y_2 - y_3 + y_4 \end{array}\right)$$

The left part of the last matrix is the RREF of A, with $r = 3$ and pivot columns $1, 2, 4$, and $z_4 = y_1 - y_2 - y_3 + y_4$. By Proposition 2.7, the system is consistent if and only if $y_1 - y_2 - y_3 + y_4 = 0$; then $x_3 = c$ is free and

$$x_1 = -2y_1 + y_2 + y_3 - 2c, \qquad x_2 = y_1 - y_3 + c, \qquad x_3 = c, \qquad x_4 = y_1 - y_2 + y_3 \qquad (c \in \mathbb{R})$$

For $y = (4, 7, 4, 7)'$, $4 - 7 - 4 + 7 = 0$, and the solutions are $x = (3, 0, 0, 1)' + c\,(-2, 1, 1, 0)'$: the solution $(3, 0, 0, 1)'$ plus the multiples of $(-2, 1, 1, 0)'$, which solve $Ax = 0$. Check: $A(3, 0, 0, 1)' = (4, 7, 4, 7)'$, and $A(-2, 1, 1, 0)' = 0$ says $\mathrm{col}_3(A) = 2\,\mathrm{col}_1(A) - \mathrm{col}_2(A)$. For $y = (1, 0, 0, 0)'$, $y_1 - y_2 - y_3 + y_4 = 1 \neq 0$, so $Ax = y$ has no solution.

Homogeneous systems

Theorem 2.11 ([1, §1.4, Theorems 6, 7])**.** (a) If $A \in \mathbb{R}^{m \times n}$ with $m < n$ (fewer equations than unknowns), then $Ax = 0$ has a nontrivial solution.

(b) For $A \in \mathbb{R}^{n \times n}$, $Ax = 0$ has only the trivial solution if and only if the RREF of A is I_n.

Proof. Let R be the RREF of A, with r nonzero rows. (a) $r \leqslant m < n$, so there is a free unknown; giving it the value 1 and the other free unknowns the value 0 gives a solution $x \neq 0$ (Proposition 2.7(b)). (b) If $R = I$, then $Rx = x$, so $x = 0$ is the only solution (Theorem 2.3(b)). If $Ax = 0$ has only the trivial solution, there is no free unknown, so $r = n$; then $1 \leqslant k_1 < \cdots < k_n \leqslant n$ forces $k_i = i$, so column i of R is e_i and $R = I$. □

Inverses

Definition 2.12 ([1, §1.6], [4, Ch. II, §3, Exercise 34])**.** $A \in \mathbb{R}^{n\times n}$ is **invertible** if $AB = BA = I_n$ for some $B \in \mathbb{R}^{n\times n}$. Such a B is unique: if $BA = I$ and $AB_1 = I$, then $B = B(AB_1) = (BA)B_1 = B_1$. It is called the **inverse** of A, written A^{-1}. For example, $\operatorname{diag}(d_1, \ldots, d_n)$ with all $d_i \neq 0$ has inverse $\operatorname{diag}(d_1^{-1}, \ldots, d_n^{-1})$, since by Proposition 1.13(b) the products in both orders are $\operatorname{diag}(1, \ldots, 1) = I$.

Theorem 2.13 ([1, §1.6, Theorem 10], [4, Ch. II, §3, Exercise 32], [7])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be invertible.

(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(b) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$; more generally $(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$ for invertible $A_1, \ldots, A_k$.

(c) A' is invertible and $(A')^{-1} = (A^{-1})'$.

Proof. (a) $AA^{-1} = A^{-1}A = I$ says that A is the inverse of A^{-1}. (b) $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$, and similarly $(B^{-1}A^{-1})(AB) = I$; for k factors repeat this. (c) $A'(A^{-1})' = (A^{-1}A)' = I$ and $(A^{-1})'A' = (AA^{-1})' = I$ by Theorem 1.9. □

Elementary matrices

Definition 2.14 ([1, §1.5])**.** An **elementary matrix** is a matrix $E = \omega(I_m)$ obtained from I_m by one elementary row operation ω. For $r \neq s$ we write:

- $D_r(c)$ ($c \neq 0$): I_m with its (r, r) entry replaced by c (from $R_r \leftarrow cR_r$);
- $T_{rs}(c)$: I_m with its (r, s) entry replaced by c (from $R_r \leftarrow R_r + cR_s$);
- P_{rs}: I_m with rows r and s interchanged (from $R_r \leftrightarrow R_s$).

For example, for $m = 4$:

$$P_{23} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_{31}(c) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ c & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad D_3(c) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Theorem 2.15 ([1, §§1.5–1.6, Theorems 9, 11])**.** Let ω be an elementary row operation on matrices with m rows and $E = \omega(I_m)$.

(a) $\omega(A) = EA$ for every $A \in \mathbb{R}^{m\times n}$: a row operation is a left multiplication.

(b) E is invertible, and E^{-1} is the elementary matrix of the inverse operation: $D_r(c)^{-1} = D_r(c^{-1})$, $T_{rs}(c)^{-1} = T_{rs}(-c)$, $P_{rs}^{-1} = P_{rs}$.

(c) If row operations $\omega_1, \ldots, \omega_k$ (in this order) take A to B, then $B = PA$ with $P = E_k \cdots E_2E_1$, $E_j = \omega_j(I_m)$; and P is invertible.

Proof. (a) By Theorem 1.4(c), $\operatorname{row}_i(EA) = \operatorname{row}_i(E)A$. An unchanged row of E is e_i', and $e_i'A = \operatorname{row}_i(A)$ (Proposition 1.18(a)). For type (2), $\operatorname{row}_r(E) = e_r' + ce_s'$, so $\operatorname{row}_r(EA) = \operatorname{row}_r(A) + c\operatorname{row}_s(A)$; types (1) and (3) are the same computation. (b) Let $\tilde{\omega}$ be the inverse operation (Theorem 2.3(a)) and $\tilde{E} = \tilde{\omega}(I)$. By (a), $E\tilde{E} = \omega(\tilde{\omega}(I)) = I$ and $\tilde{E}E = \tilde{\omega}(\omega(I)) = I$. (c) By (a), $B = E_k(\cdots(E_1A)) = (E_k \cdots E_1)A$, and a product of invertible matrices is invertible (Theorem 2.13(b)). □

Deciding invertibility and computing inverses

Theorem 2.16 ([1, §1.6, Theorems 12, 13], [3, Result 2.11])**.** For $A \in \mathbb{R}^{n\times n}$ the following are equivalent:

(i) A is invertible;

(ii) the RREF of A is I_n;

(iii) $Ax = 0$ has only the trivial solution;

(iv) $Ax = b$ has a solution for every $b \in \mathbb{R}^n$;

(v) A is a product of elementary matrices.

Moreover, if $BA = I$ or $AB = I$ for some $B \in \mathbb{R}^{n\times n}$, then A is invertible and $B = A^{-1}$.

Proof. Let $R = PA$ be the RREF of A, with $P = E_k \cdots E_1$ (Theorem 2.15(c)).

- If $R = I$, then $A = P^{-1} = E_1^{-1} \cdots E_k^{-1}$ is a product of elementary matrices, hence invertible (Theorem 2.13(b)); $Ax = 0$ has only the trivial solution (Theorem 2.11(b)); and $x = A^{-1}b$ solves $Ax = b$.
- If $R \neq I$, then R has a zero row, hence a zero last row (by (c) of Definition 2.4); otherwise $r = n$, so $1 \leqslant k_1 < \cdots < k_n \leqslant n$ forces $k_i = i$, column i of R is e_i for every i, and $R = I$. Then $Ax = 0$ has a solution $x \neq 0$ (Theorem 2.11(b)), so A has no inverse (else $x = A^{-1}Ax = 0$ ✘), and A is not a product of elementary matrices (such a product is invertible, Theorems 2.15(b) and 2.13(b)). For $b = P^{-1}e_n$, $Ax = b$ holds if and only if $Rx = PAx = Pb = e_n$ does, as P is invertible (Theorem 2.15(c)); the last equation of $Rx = e_n$ reads $0 = 1$, so (iv) fails.

Thus each of (i)–(v) holds exactly when $R = I$. If $BA = I$, then $Ax = 0$ gives $x = BAx = 0$, so A is invertible by (iii), and $B = BAA^{-1} = A^{-1}$. If $AB = I$, then B is invertible with $B^{-1} = A$ by the case just shown, so $A = B^{-1}$ is invertible with $A^{-1} = B$. □

Corollary 2.17 (Computing A^{-1}; [1, §1.6, Corollaries to Theorem 12], [6, Exercises 6.22(b), 6.23])**.** Let $A \in \mathbb{R}^{n\times n}$, and row-reduce the $n \times 2n$ matrix $[A \mid I_n]$ by Gauss–Jordan elimination.

(a) If the left half becomes I_n, the right half is A^{-1}.

(b) If a zero row appears in the left half, A is not invertible.

Proof. (a) $[A \mid I] \to [I \mid B]$ means $E[A \mid I] = [EA \mid E] = [I \mid B]$ for the product E of the elementary matrices used (Theorem 2.15(c); E multiplies each column separately, Theorem 1.4(a)), so $EA = I$ and $B = E = A^{-1}$ by the last sentence of Theorem 2.16.

(b) The left half is then EA, with E the (invertible) product of the elementary matrices used so far (Theorem 2.15(c)). For every $X \in \mathbb{R}^{n\times n}$, EAX has the same zero row (Theorem 1.4(c)), so $EAX \neq I_n$ and EA is not invertible; if A were invertible, so would be EA (Theorem 2.13(b)). □

Proposition 2.18 ([11, Three.IV.4, Corollary 4.11, Exercise 4.21])**.** Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2\times 2}$. Then A is invertible if and only if $ad - bc \neq 0$, and in that case

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Proof. We row-reduce $[A \mid I_2]$ and read the result by Corollary 2.17. Put $\Delta = ad - bc$.

- Let $a \neq 0$. Then

$$\left(\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - \frac{c}{a}R_1} \left(\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{\Delta}{a} & -\frac{c}{a} & 1 \end{array}\right)$$

since $d - \frac{cb}{a} = \frac{\Delta}{a}$. If $\Delta = 0$, the left half has a zero row, so A is not invertible (Corollary 2.17(b)).

- Let $a \neq 0$ and $\Delta \neq 0$. Then
$$\xrightarrow[R_2 \leftarrow \frac{a}{\Delta} R_2]{R_1 \leftarrow \frac{1}{a} R_1} \left(\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 1 & -\frac{c}{\Delta} & \frac{a}{\Delta} \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - \frac{b}{a} R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{d}{\Delta} & -\frac{b}{\Delta} \\ 0 & 1 & -\frac{c}{\Delta} & \frac{a}{\Delta} \end{array}\right)$$
using $\frac{1}{a} + \frac{bc}{a\Delta} = \frac{\Delta + bc}{a\Delta} = \frac{d}{\Delta}$. The left half is I_2, so A is invertible and the right half is the stated A^{-1} (Corollary 2.17(a)).
- Let $a = 0$, so that $\Delta = -bc$. If $c = 0$, the first column of A is zero, so $Ae_1 = 0$ and A is not invertible (Theorem 2.16, (i) $\Leftrightarrow$ (iii)). Let $c \neq 0$. Then
$$\left(\begin{array}{cc|cc} 0 & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|cc} c & d & 0 & 1 \\ 0 & b & 1 & 0 \end{array}\right)$$
If $b = 0$, the left half has a zero row, and A is not invertible (Corollary 2.17(b)). So for $a = 0$, A is invertible only if $b \neq 0$ and $c \neq 0$, that is, only if $\Delta = -bc \neq 0$.
- Let $a = 0$ and $b, c \neq 0$. Then
$$\xrightarrow[R_2 \leftarrow \frac{1}{b} R_2]{R_1 \leftarrow \frac{1}{c} R_1} \left(\begin{array}{cc|cc} 1 & \frac{d}{c} & 0 & \frac{1}{c} \\ 0 & 1 & \frac{1}{b} & 0 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - \frac{d}{c} R_2} \left(\begin{array}{cc|cc} 1 & 0 & -\frac{d}{bc} & \frac{1}{c} \\ 0 & 1 & \frac{1}{b} & 0 \end{array}\right)$$
and with $\Delta = -bc$ the right half is $\frac{1}{\Delta} \left(\begin{smallmatrix} d & -b \\ -c & 0 \end{smallmatrix}\right)$, the stated A^{-1} for $a = 0$ (Corollary 2.17(a)).

In both cases A is invertible exactly when $\Delta \neq 0$, with the stated inverse. □

The formula is easy to check by multiplication: $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right) = \left(\begin{smallmatrix} ad-bc & 0 \\ 0 & ad-bc \end{smallmatrix}\right) = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right) \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$.

Example 2.19. For $A = \left(\begin{smallmatrix} 3 & 5 \\ 1 & 2 \end{smallmatrix}\right)$, $ad - bc = 6 - 5 = 1$, so Proposition 2.18 gives $A^{-1} = \left(\begin{smallmatrix} 2 & -5 \\ -1 & 3 \end{smallmatrix}\right)$. For $B = \left(\begin{smallmatrix} 2 & 6 \\ 1 & 3 \end{smallmatrix}\right)$, $ad - bc = 6 - 6 = 0$, so B is not invertible; indeed $x = (3, -1)' \neq 0$ satisfies $Bx = (6 - 6,\ 3 - 3)' = 0$ (Theorem 2.16, (i) $\Leftrightarrow$ (iii)).

Example 2.20 (cf. [1, §1.6, Example 15])**.** Find the inverse of
$$A = \begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -2 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -1 & 1 & 0 \end{pmatrix}$$
and write A and A^{-1} as products of elementary matrices.

Solution. Forward elimination of $[A \mid I_4]$:
$$\left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ -1 & 1 & -2 & 1 & 0 & 1 & 0 & 0 \\ 2 & -1 & 2 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$
$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$
After the first arrow the entry of row 2 in column 2 is 0, so Step 2 interchanges rows 2 and 3, and Step 3 multiplies the new row 3 by -1. Backward elimination, clearing columns 4, 3, 2 in turn:
$$\xrightarrow[R_2 \leftarrow R_2 - R_4]{R_1 \leftarrow R_1 + R_4} \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$
$$\xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

The left half has become I_4, so by Corollary 2.17(a)

$$A^{-1} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

Check: $\text{row}_3(A)A^{-1} = (2,-1,2,-1)A^{-1} = (0+1-2+1,\ 2+0-2+0,\ 2-1+0+0,\ 0+1+0-1) = (0,0,1,0)$, and similarly for the other rows. The nine operations have the elementary matrices

$$E_1 = T_{21}(1),\ E_2 = T_{31}(-2),\ E_3 = T_{41}(-1),\ E_4 = P_{23},\ E_5 = D_3(-1)$$
$$E_6 = T_{14}(1),\ E_7 = T_{24}(-1),\ E_8 = T_{13}(-1),\ E_9 = T_{12}(1)$$

By Theorem 2.15(c), $E_9E_8\cdots E_1A = I_4$, so $A^{-1} = E_9E_8\cdots E_1$ by the last sentence of Theorem 2.16. By Theorem 2.13(a), $A = (A^{-1})^{-1}$; so, by Theorems 2.13(b) and 2.15(b),

$$A = E_1^{-1}E_2^{-1}\cdots E_9^{-1} = T_{21}(-1)\,T_{31}(2)\,T_{41}(1)\,P_{23}\,D_3(-1)\,T_{14}(-1)\,T_{24}(1)\,T_{13}(1)\,T_{12}(-1)$$

Example 2.21. Find the inverse of

$$P = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{pmatrix}$$

(each entry outside row 1 and column 1 is the sum of the entry above it and the entry to its left; P is the 4×4 **Pascal matrix**).

Solution. Every pivot is 1, so no scaling is needed. Forward elimination:

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 0 & 1 & 0 & 0 \\ 1 & 3 & 6 & 10 & 0 & 0 & 1 & 0 \\ 1 & 4 & 10 & 20 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-R_1,\ R_3\leftarrow R_3-R_1} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & 9 & -1 & 0 & 1 & 0 \\ 0 & 3 & 9 & 19 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4\leftarrow R_4-3R_2]{R_3\leftarrow R_3-2R_2} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & 10 & 2 & -3 & 0 & 1 \end{array}\right) \xrightarrow{R_4\leftarrow R_4-3R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right)$$

Backward elimination, clearing columns 4, 3, 2 in turn:

$$\xrightarrow[R_1\leftarrow R_1-R_4]{R_3\leftarrow R_3-3R_4,\ R_2\leftarrow R_2-3R_4} \left(\begin{array}{cccc|cccc} 1 & 1 & 1 & 0 & 2 & -3 & 3 & -1 \\ 0 & 1 & 2 & 0 & 2 & -8 & 9 & -3 \\ 0 & 0 & 1 & 0 & 4 & -11 & 10 & -3 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-R_3]{R_2\leftarrow R_2-2R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & -2 & 8 & -7 & 2 \\ 0 & 1 & 0 & 0 & -6 & 14 & -11 & 3 \\ 0 & 0 & 1 & 0 & 4 & -11 & 10 & -3 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right) \xrightarrow{R_1\leftarrow R_1-R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 4 & -6 & 4 & -1 \\ 0 & 1 & 0 & 0 & -6 & 14 & -11 & 3 \\ 0 & 0 & 1 & 0 & 4 & -11 & 10 & -3 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array}\right)$$

The left half has become I_4, so by Corollary 2.17(a)

$$P^{-1} = \begin{pmatrix} 4 & -6 & 4 & -1 \\ -6 & 14 & -11 & 3 \\ 4 & -11 & 10 & -3 \\ -1 & 3 & -3 & 1 \end{pmatrix}$$

Check: $\text{row}_2(P)P^{-1} = (4-12+12-4,\ -6+28-33+12,\ 4-22+30-12,\ -1+6-9+4) = (0,1,0,0)$, and similarly for the other rows. Every operation added an integer multiple of one row to another, so the right half stayed an integer matrix throughout; this is why P^{-1} has integer entries.

Example 2.22. The matrix A of Example 2.10 is not invertible: as the left part of the last matrix there shows, its RREF is

$$\begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \neq I_4$$

Accordingly $Ax = 0$ has the nontrivial solution $x = (-2, 1, 1, 0)'$, and $Ax = y$ has no solution for $y = (1, 0, 0, 0)'$ (Theorem 2.16, (iii) and (iv)).

Corollary 2.23 ([1, §1.6], [5, §8.2])**.** (a) Let $A_1, \ldots, A_k \in \mathbb{R}^{n\times n}$. The product $A_1 A_2 \cdots A_k$ is invertible if and only if every A_j is invertible.

(b) If $A \in \mathbb{R}^{n\times n}$ is symmetric and invertible, then A^{-1} is symmetric.

Proof. (a) • If every A_j is invertible, so is the product (Theorem 2.13(b)).

- Let $M = A_1 \cdots A_k$ be invertible. If $A_k x = 0$, then $Mx = (A_1 \cdots A_{k-1})A_k x = 0$, so $x = M^{-1}Mx = 0$. Hence A_k is invertible (Theorem 2.16, (iii) $\Rightarrow$ (i)).
- Then $A_1 \cdots A_{k-1} = MA_k^{-1}$ is invertible (Theorem 2.13(b)), and the same argument, applied to these $k-1$ factors, shows that A_{k-1} is invertible. Continuing in this way, every A_j is invertible.

(b) As $A' = A$, Theorem 2.13(c) gives $(A^{-1})' = (A')^{-1} = A^{-1}$. □

For instance, the Pascal matrix P of Example 2.21 is symmetric, and so is its inverse

$$P^{-1} = \begin{pmatrix} 4 & -6 & 4 & -1 \\ -6 & 14 & -11 & 3 \\ 4 & -11 & 10 & -3 \\ -1 & 3 & -3 & 1 \end{pmatrix}$$

By (a), a product of square matrices with a factor that is not invertible is not invertible.

Row equivalence

Corollary 2.24 ([1, §1.3, §1.6, Corollaries to Theorem 12], [6, Exercise 6.21], [11, One.III.2])**.** Let $A, B, G \in \mathbb{R}^{m\times n}$.

(a) B is row-equivalent to A if and only if $B = PA$ for some invertible $P \in \mathbb{R}^{m\times m}$. Moreover, if row operations take the $m \times (n+m)$ matrix $[A \mid I_m]$ to $[B \mid P]$, then P is invertible and $B = PA$.

(b) If B is row-equivalent to A, then A is row-equivalent to B. If, in addition, G is row-equivalent to B, then G is row-equivalent to A.

(c) B is row-equivalent to A if and only if A and B have the same RREF.

Proof. (a) • If B is row-equivalent to A, then $B = PA$ with $P = E_k \cdots E_1$ invertible (Theorem 2.15(c)).

- Conversely, let $B = PA$ with P invertible. By Theorem 2.16 ((i) $\Rightarrow$ (v)), $P = F_1 F_2 \cdots F_l$ with elementary matrices F_j. Then $B = F_1(F_2(\cdots(F_l A)))$ is obtained from A by the row operations of $F_l, F_{l-1}, \ldots, F_1$, in this order (Theorem 2.15(a)).
- Let row operations take $[A \mid I_m]$ to $[B \mid P]$, and let E be the product of their elementary matrices (Theorem 2.15(c)). Since E multiplies each column separately (Theorem 1.4(a)), $[B \mid P] = E[A \mid I_m] = [EA \mid E]$. So $P = E$ is invertible and $B = EA = PA$.

(b)
- Let row operations $\omega_1, \dots, \omega_k$ take A to B, and let $\tilde{\omega}_j$ be the operation that undoes ω_j (Theorem 2.3(a)). Applying $\tilde{\omega}_k, \dots, \tilde{\omega}_1$, in this order, to B undoes $\omega_k, \dots, \omega_1$ in turn and gives A.
- If row operations $\omega'_1, \dots, \omega'_l$ take B to G, then $\omega_1, \dots, \omega_k, \omega'_1, \dots, \omega'_l$ take A to G.

(c)
- Let B be row-equivalent to A, and let S be the RREF of B. Operations taking A to B, followed by operations taking B to S, take A to S; so S is the RREF of A (Theorem 2.8).
- Let A and B have the same RREF R. Then R is row-equivalent to A, and B is row-equivalent to R (by (b)); so B is row-equivalent to A (by (b)). □

Example 2.25 (cf. [1, §1.6])**.** Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 1 \end{pmatrix}$$

be the matrix of Example 2.10, and R its RREF. Find an invertible P with $PA = R$.

Solution. Row-reduce $[A \mid I_4]$ with the operations of Example 2.10:

$$\left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 2 & 2 & 2 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & -2 & 2 & -1 & -2 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & -2 & 2 & -1 & -2 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & -2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right) \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 2 & 0 & -2 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right)$$

The left half is R. By Corollary 2.24(a),

$$P = \begin{pmatrix} -2 & 1 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 \\ 1 & -1 & -1 & 1 \end{pmatrix} \quad \text{is invertible, and} \quad PA = R = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- Check: $\text{row}_4(P)A = (1,2,0,1) - (2,3,1,1) - (1,1,1,1) + (2,2,2,1) = (0,0,0,0)$ (Theorem 1.4(c)), and likewise $\text{row}_1(P)A = (1,0,2,0)$, $\text{row}_2(P)A = (0,1,-1,0)$, $\text{row}_3(P)A = (0,0,0,1)$.
- For $y \in \mathbb{R}^4$, $Py = (-2y_1 + y_2 + y_3,\ y_1 - y_3,\ y_1 - y_2 + y_3,\ y_1 - y_2 - y_3 + y_4)'$ is the last column of the last matrix in Example 2.10: the same operations take $[A \mid y]$ to $[PA \mid Py] = [R \mid Py]$.
- As P is invertible, $Ax = y$ holds if and only if $Rx = PAx = Py$ does. Row 4 of R is zero, so by Proposition 2.7(a), $Ax = y$ is consistent exactly when $\text{row}_4(P)\, y = y_1 - y_2 - y_3 + y_4 = 0$, the condition found in Example 2.10.

Solving $Ax = b$

In practice $Ax = b$ is solved by forward elimination of $[A \mid b]$ alone: an echelon form already shows whether there is a solution, and how many.

Theorem 2.26 ([1, §1.4])**.** Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, and let row operations take $[A \mid b]$ to a matrix $[H \mid d]$ in echelon form ($H \in \mathbb{R}^{m\times n}$, $d \in \mathbb{R}^m$). Then $Ax = b$ is consistent if and only if $[H \mid d]$ has no row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$, that is, if and only if the last column of $[H \mid d]$ is not a pivot column. Moreover, if row operations take $[A \mid b]$ to any matrix, in echelon form or not, with a row $(0 \ \cdots \ 0 \mid c)$, $c \neq 0$, then $Ax = b$ has no solution. If $[H \mid d]$ has no such row, call x_j a pivot unknown when column j of H is a pivot column and a free unknown otherwise, as in Proposition 2.7; then each choice of values of the free unknowns gives, by **back substitution** (solving the equations of the nonzero rows of H, from the last one up, for their pivot unknowns), exactly one solution, and every solution arises in this way.

Proof. • By Theorem 2.3, $Ax = b$ and $Hx = d$ have the same solutions. In an echelon form, a row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ is exactly a row whose pivot lies in the last column.

- Let row operations take $[A \mid b]$ to a matrix $[B \mid z]$ with a row $(0 \ \cdots \ 0 \mid c)$, $c \neq 0$. The corresponding equation of $Bx = z$ reads $0 = c$, and no x satisfies it; as $Bx = z$ and $Ax = b$ have the same solutions (Theorem 2.3(b)), $Ax = b$ has no solution. This applies to $[H \mid d]$, and to the matrix reached at any stage of the elimination, so one may stop there.
- Suppose $[H \mid d]$ has no such row. Then every zero row of H is a zero row of $[H \mid d]$, and its equation reads $0 = 0$. Let rows $1, \dots, r$ be the nonzero rows of H, with pivots in columns $k_1 < \cdots < k_r$; so $x_{k_1}, \dots, x_{k_r}$ are the pivot unknowns, and the others are free.
- Give the free unknowns any values. Row i of $H = (h_{ij})$ is 0 before column k_i and 1 there, so for $i = r, r-1, \dots, 1$ equation i reads $x_{k_i} + \sum_{j>k_i} h_{ij}x_j = d_i$. Each x_j with $j > k_i$ is free or the pivot unknown of a lower row, already found; so the equation determines x_{k_i} (back substitution). The resulting x solves $Hx = d$, hence $Ax = b$.
- Every solution arises in this way from the values of its free unknowns, since the equations force the pivot unknowns; so each choice of values of the free unknowns gives exactly one solution. □

Theorem 2.27 ([5, Theorem 11.2.3], [3, §2.3, Ch. 2, Exercise 13], [6, Exercise 6.45])**.** Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$.

(a) If x_p is one solution of $Ax = b$ (a **particular solution**), then the solutions of $Ax = b$ are exactly the vectors $x_p + z$ with $Az = 0$.

(b) $Ax = b$ has no solution, exactly one solution, or infinitely many solutions. If it is consistent, it has exactly one solution if and only if there are no free unknowns, that is, every column of H in Theorem 2.26 is a pivot column.

(c) If A is square, $Ax = b$ has exactly one solution if and only if A is invertible; this solution is $x = A^{-1}b$.

Proof. (a) If $Az = 0$, then $A(x_p + z) = Ax_p + Az = b$. Conversely, if $Ax = b$, then $z = x - x_p$ satisfies $Az = b - b = 0$, and $x = x_p + z$.

(b) Suppose $Ax = b$ is consistent, and take $[H \mid d]$ as in Theorem 2.26.

- By that theorem, each choice of values of the free unknowns gives exactly one solution, and every solution arises in this way; different choices give different solutions (they differ in a free unknown, an entry of x).
- Without free unknowns there is therefore exactly one solution; with a free unknown, each of its values gives a different solution, so there are infinitely many.

(c)
- If A is invertible, $x = A^{-1}b$ is a solution, since $A(A^{-1}b) = b$; and every solution satisfies $x = A^{-1}(Ax) = A^{-1}b$.
- If $Ax = b$ has exactly one solution x_p, then $Az = 0$ only for $z = 0$, since otherwise $x_p + z$ would be a second solution by (a). So A is invertible by Theorem 2.16 ((iii) $\Rightarrow$ (i)). $\square$

So, to solve $Ax = b$: reduce $[A \mid b]$ by forward elimination to an echelon form; if a row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ appears, there is no solution; otherwise choose the free unknowns arbitrarily and find the pivot unknowns by back substitution (or continue to the RREF and use Proposition 2.7).

Example 2.28. Solve the system $x_1 - x_2 + x_3 - x_4 = 1$, $-x_1 + x_2 - 2x_3 + x_4 = 0$, $2x_1 - x_2 + 2x_3 - x_4 = 3$, $x_1 - x_2 + x_3 = 3$.

Solution. The coefficient matrix is the matrix A of Example 2.20, and $b = (1, 0, 3, 3)'$. Forward elimination of $[A \mid b]$ uses the operations of the forward elimination there:

$$\left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ -1 & 1 & -2 & 1 & 0 \\ 2 & -1 & 2 & -1 & 3 \\ 1 & -1 & 1 & 0 & 3 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

- The last matrix is an echelon form with no row $(0\ 0\ 0\ 0 \mid c)$, $c \neq 0$, so the system is consistent (Theorem 2.26).
- Every column of the left part is a pivot column, so there are no free unknowns and the solution is unique (Theorem 2.27(b)).
- Back substitution: $x_4 = 2$; $x_3 = -1$; $x_2 = 1 - x_4 = -1$; $x_1 = 1 + x_2 - x_3 + x_4 = 1 - 1 + 1 + 2 = 3$. So $x = (3, -1, -1, 2)'$.
- Check, with A^{-1} from Example 2.20 (Theorem 2.27(c)): $A^{-1}b = (0 + 0 + 3 + 0,\ -1 + 0 + 3 - 3,\ -1 + 0 + 0 + 0,\ -1 + 0 + 0 + 3)' = (3, -1, -1, 2)'$.

Example 2.29. Solve the system $x_1 + x_2 + x_3 - 2x_4 = 2$, $x_1 + x_2 + 2x_3 - 3x_4 = 3$, $2x_1 + x_2 + 5x_3 - 4x_4 = 7$.

Solution. The coefficient matrix is the matrix A of Example 2.6, and $b = (2, 3, 7)'$. Forward elimination of $[A \mid b]$ uses the same operations as there:

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 1 & 1 & 2 & -3 & 3 \\ 2 & 1 & 5 & -4 & 7 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & 3 & 0 & 3 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & -1 & 3 & 0 & 3 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & 1 & -3 & 0 & -3 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right)$$

- The last matrix is an echelon form with no row $(0\ 0\ 0\ 0 \mid c)$, $c \neq 0$: the system is consistent (Theorem 2.26).
- The pivot columns are $1, 2, 3$, so x_4 is free and there are infinitely many solutions (Theorem 2.27(b)).
- Back substitution with $x_4 = t$: $x_3 = 1 + t$; $x_2 = -3 + 3x_3 = 3t$; $x_1 = 2 - x_2 - x_3 + 2x_4 = 2 - 3t - (1 + t) + 2t = 1 - 2t$. So

$$x = \begin{pmatrix} 1 - 2t \\ 3t \\ 1 + t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 3 \\ 1 \\ 1 \end{pmatrix} \qquad (t \in \mathbb{R})$$

- This is the particular solution $x_p = (1,0,1,0)'$ plus the solutions $t(-2,3,1,1)'$ of $Ax = 0$ found in Example 2.6, as Theorem 2.27(a) says.

Example 2.30 (cf. [1, §1.4, Exercise 6])**.** Show that the system $x_1-2x_2+x_3+2x_4 = 1$, $x_1-x_2-x_3+x_4 = 2$, $x_1 + x_2 - 5x_3 - x_4 = 3$ has no solution.

Solution. Forward elimination of the augmented matrix $[A \mid b]$, $b = (1,2,3)'$:

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 1 & -1 & -1 & 1 & 2 \\ 1 & 1 & -5 & -1 & 3 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 3 & -6 & -3 & 2 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - 3R_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & -1 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

The last matrix is an echelon form whose last column is a pivot column: row 3 reads $0 = 1$. So the system has no solution (Theorem 2.26). Indeed, the coefficient matrix A has $\operatorname{row}_3(A) = 3\operatorname{row}_2(A) - 2\operatorname{row}_1(A)$, while $3b_2 - 2b_1 = 4 \neq 3 = b_3$: the third equation contradicts the first two.

Triangular matrices

Theorem 2.31 ([1, §1.6, Exercise 9], [5, Corollaries 8.5.6, 8.5.8, 8.5.9], [7, Ch. VI, §5, Exercise 2], [6, Exercise 7.60(a)])**.** Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ be upper triangular or lower triangular.

(a) A is invertible if and only if $a_{ii} \neq 0$ for $i = 1, \ldots, n$.

(b) If A is invertible, then A^{-1} is triangular of the same kind as A, with diagonal entries $1/a_{11}, \ldots, 1/a_{nn}$.

In particular, $\operatorname{diag}(d_1, \ldots, d_n)$ is invertible if and only if every $d_i \neq 0$, and the inverse of a **unit triangular** matrix (a triangular matrix with all diagonal entries 1) is unit triangular of the same kind.

Proof. (a) By Theorem 2.16 ((i) $\Leftrightarrow$ (iii)), A is invertible if and only if $Ax = 0$ has only the trivial solution.

- Let A be lower triangular, so that equation i of $Ax = 0$ reads $a_{i1}x_1 + \cdots + a_{ii}x_i = 0$. If every $a_{ii} \neq 0$ and $Ax = 0$, then equation 1 gives $x_1 = 0$, equation 2 then gives $a_{22}x_2 = 0$, so $x_2 = 0$, and so on: $x = 0$.
- Let A be lower triangular with some $a_{kk} = 0$, and let k be the smallest such index. Put $x_1 = \cdots = x_{k-1} = 0$. Then equations $1, \ldots, k-1$ hold, and so does equation k, which reads $a_{kk}x_k = 0$.
- Equations $k+1, \ldots, n$ are $n-k$ equations in the $n-k+1$ unknowns $x_k, \ldots, x_n$. If $k < n$, they have a nontrivial solution (Theorem 2.11(a)); if $k = n$, there are none, and we take $x_n = 1$. Either way $x \neq 0$ and $Ax = 0$, so A is not invertible.
- If A is upper triangular, then A' is lower triangular with the same diagonal entries, and A is invertible if and only if A' is (Theorem 2.13(c), with $(A')' = A$). So the lower triangular case gives the upper one.

(b)
- Let A be upper triangular and invertible, so that every $a_{ii} \neq 0$ by (a). For an upper triangular $B \in \mathbb{R}^{n\times n}$, AB is upper triangular with diagonal entries $a_{ii}b_{ii}$ (Proposition 1.13(c)), and $(AB)_{ij} = \sum_{k=i}^{j} a_{ik}b_{kj}$ for $i < j$, since $a_{ik} = 0$ for $k < i$ and $b_{kj} = 0$ for $k > j$.
- Hence an upper triangular B satisfies $AB = I_n$ if and only if

$$b_{jj} = \frac{1}{a_{jj}}, \qquad b_{ij} = -\frac{1}{a_{ii}} \sum_{k=i+1}^{j} a_{ik}b_{kj} \qquad (1 \leqslant i < j \leqslant n)$$

- For each column j, these formulas give b_{jj} and then $b_{j-1,j}, \dots, b_{1j}$ in turn, since b_{ij} uses only the b_{kj} with $k > i$. So such a B exists, and $B = A^{-1}$ (Theorem 2.16): A^{-1} is upper triangular with diagonal entries $1/a_{ii}$.
- If A is lower triangular and invertible, then A' is upper triangular and invertible, with the same diagonal; so $(A')^{-1}$ is upper triangular with diagonal entries $1/a_{ii}$, and $A^{-1} = ((A')^{-1})'$ (Theorem 2.13(c)) is lower triangular with the same diagonal.

A diagonal matrix is both upper and lower triangular, and a unit triangular matrix has $1/a_{ii} = 1$; this gives the last sentence. □

Example 2.32. Find the inverse of

$$U = \begin{pmatrix} 1 & -2 & -2 & 3 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Solution. U is upper triangular with diagonal entries $1, -1, 1, 1$, all nonzero, so it is invertible, and $B = U^{-1} = (b_{ij})$ is upper triangular with $b_{11} = 1$, $b_{22} = -1$, $b_{33} = b_{44} = 1$ (Theorem 2.31). The other entries come from $b_{ij} = -\frac{1}{u_{ii}} \sum_{k=i+1}^{j} u_{ik} b_{kj}$, column by column, from the bottom up:

- column 2: $b_{12} = -u_{12} b_{22} = -(-2)(-1) = -2$;
- column 3: $b_{23} = u_{23} b_{33} = (-1)(1) = -1$ (here $-1/u_{22} = 1$), and $b_{13} = -(u_{12} b_{23} + u_{13} b_{33}) = -\Big((-2)(-1) + (-2)(1)\Big) = 0$;
- column 4: $b_{34} = -u_{34} b_{44} = -1$, $b_{24} = u_{23} b_{34} + u_{24} b_{44} = (-1)(-1) + (1)(1) = 2$ (again $-1/u_{22} = 1$), and $b_{14} = -(u_{12} b_{24} + u_{13} b_{34} + u_{14} b_{44}) = -\Big((-2)(2) + (-2)(-1) + (3)(1)\Big) = -1$.

So

$$U^{-1} = \begin{pmatrix} 1 & -2 & 0 & -1 \\ 0 & -1 & -1 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Check: $\mathrm{row}_1(U)\, U^{-1} = (1,\ -2+2,\ 2-2,\ -1-4+2+3) = (1, 0, 0, 0)$, and similarly for the other rows.

Theorem 2.33 ([5, §8.5d, Corollary 8.5.9], [6, Exercise 7.60(a)], [4, Ch. II, §3, Exercise 18(c)])**.** Let $L = (l_{ij}) \in \mathbb{R}^{n \times n}$ be unit lower triangular, and let $N = I_n - L$; so N is strictly lower triangular, with entries $-l_{ij}$ for $i > j$. Let $M = L^{-1} = (m_{ij})$.

(a) M is unit lower triangular, and for $i > j$

$$m_{ij} = -\sum_{k=j}^{i-1} l_{ik} m_{kj}$$

so column j of M is found from the top down: $m_{jj} = 1$, then $m_{j+1,j}, \dots, m_{nj}$ in turn.

(b) $M = I_n + NM$, and repeating this recursion gives

$$L^{-1} = I_n + N + N^2 + \cdots + N^{n-1}$$

Proof. (a)

- L is invertible, and M is unit lower triangular (Theorem 2.31(a),(b)).
- Let $i > j$. The (i, j) entry of $LM = I_n$ is $\sum_k l_{ik} m_{kj} = 0$. Here $l_{ik} = 0$ for $k > i$, $l_{ii} = 1$, and $m_{kj} = 0$ for $k < j$, so the sum runs over $j \leqslant k \leqslant i$: $m_{ij} + \sum_{k=j}^{i-1} l_{ik} m_{kj} = 0$.

- The right side of the formula uses only $m_{jj}, \dots, m_{i-1,j}$, entries of column j above row i.

(b)

- $LM = I_n$ reads $(I_n - N)M = I_n$, that is, $M = I_n + NM$ (Theorem 1.3). (Entry (i, j) of this equation, $i > j$, is the formula of (a).)
- Substituting $I_n + NM$ for M on the right gives $M = I_n + N + N^2M$, and after p substitutions

$$M = I_n + N + \cdots + N^{p-1} + N^pM$$

  by induction on p (Theorem 1.3).
- N is strictly lower triangular, so $N^n = 0$ (Proposition 1.21(c)). With $p = n$, $M = I_n + N + \cdots + N^{n-1}$. □

Example 2.34. (a) Find the inverse of

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 2 & 0 & -1 & 1 \end{pmatrix}$$

(b) Let $S \in \mathbb{R}^{4\times 4}$ have the entries 1 in the positions $(2, 1), (3, 2), (4, 3)$ and 0 elsewhere. The first-difference matrix $I_4 - S$ turns $x = (x_1, x_2, x_3, x_4)'$ into $(x_1, x_2 - x_1, x_3 - x_2, x_4 - x_3)'$. Find its inverse.

Solution. (a) L is unit lower triangular. By Theorem 2.33(a), column by column from the top down, with $m_{jj} = 1$:

- column 1: $m_{21} = -l_{21}m_{11} = -1$, $m_{31} = -(l_{31}m_{11} + l_{32}m_{21}) = -(1 - 2) = 1$, $m_{41} = -(l_{41}m_{11} + l_{42}m_{21} + l_{43}m_{31}) = -(2 + 0 - 1) = -1$;
- column 2: $m_{32} = -l_{32}m_{22} = -2$, $m_{42} = -(l_{42}m_{22} + l_{43}m_{32}) = -(0 + 2) = -2$;
- column 3: $m_{43} = -l_{43}m_{33} = 1$.

So

$$L^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & -2 & 1 & 1 \end{pmatrix}$$

The formula of Theorem 2.33(b) gives the same: with

$$N = I_4 - L = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ -2 & 0 & 1 & 0 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ -1 & -2 & 0 & 0 \end{pmatrix}, \qquad N^3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix}$$

the sum $I_4 + N + N^2 + N^3$ has the entries -1, $-1 + 2 = 1$, $-2 - 1 + 2 = -1$ in column 1 below the diagonal, -2 and $0 - 2 = -2$ in column 2, and 1 in column 3: it is L^{-1}.

(b) $I_4 - S$ is unit lower triangular with $N = S$. By Theorem 2.33(b), $(I_4 - S)^{-1} = I_4 + S + S^2 + S^3$, where

$$S = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad S^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad S^3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

So

$$(I_4 - S)^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

the matrix that turns y into its partial sums $(y_1, y_1 + y_2, y_1 + y_2 + y_3, y_1 + y_2 + y_3 + y_4)'$: summing undoes differencing.

Column operations

Theorem 2.35 ([6, Exercises 6.4(a), 6.10])**.** The **elementary column operations** on matrices with n columns are, for column indices $r \neq s$: multiply column s by $c \neq 0$, written $\mathrm{col}_s \leftarrow c\,\mathrm{col}_s$; add c times column r to column s ($c \in \mathbb{R}$), written $\mathrm{col}_s \leftarrow \mathrm{col}_s + c\,\mathrm{col}_r$; and interchange columns r and s, written $\mathrm{col}_r \leftrightarrow \mathrm{col}_s$. Let γ be one of them and $F = \gamma(I_n)$.

(a) F is $D_s(c)$, $T_{rs}(c)$ or P_{rs}, respectively. Moreover $D_s(c)' = D_s(c)$, $T_{sr}(c)' = T_{rs}(c)$ and $P_{rs}' = P_{rs}$.

(b) $\gamma(A) = AF$ for every $A \in \mathbb{R}^{m \times n}$: a column operation is a right multiplication.

(c) If column operations $\gamma_1, \ldots, \gamma_l$ (in this order) take A to B, then $B = AQ$ with $Q = F_1 F_2 \cdots F_l$, $F_j = \gamma_j(I_n)$, and Q is invertible.

Proof. (a) • Column s of I_n is e_s. The three operations replace it by ce_s, by $e_s + ce_r$, or exchange it with column r. The results are I_n with its (s,s) entry replaced by c, I_n with its (r,s) entry replaced by c, and I_n with rows r and s interchanged (exchanging columns r and s of I_n, or its rows r and s, gives in both cases the matrix whose (r,s) and (s,r) entries are 1, whose (r,r) and (s,s) entries are 0, and whose other entries are those of I_n); that is, $D_s(c)$, $T_{rs}(c)$ and P_{rs} (Definition 2.14).

• $D_s(c)$ and P_{rs} are symmetric, by inspection, and $T_{sr}(c)'$ is I_n with its (r,s) entry replaced by c, that is, $T_{rs}(c)$.

(b) • The columns of $\gamma(A)$ are the transposed rows of $\gamma(A)'$ (Definition 1.8). So $\gamma(A)'$ is obtained from A' by the row operation ω with the same indices: $R_s \leftarrow cR_s$, $R_s \leftarrow R_s + cR_r$ or $R_r \leftrightarrow R_s$.

• By Theorem 2.15(a), $\gamma(A)' = EA'$ with $E = \omega(I_n)$, which is $D_s(c)$, $T_{sr}(c)$ or P_{rs}. Transposing (Theorem 1.9), $\gamma(A) = (EA')' = AE'$, and $E' = F$ by (a).

(c) By (b), $B = \gamma_l(\cdots\gamma_1(A)) = (\cdots(AF_1)\cdots)F_l = AF_1 \cdots F_l$ (Theorem 1.3). Each F_j is invertible (Theorem 2.15(b)), and so is Q (Theorem 2.13(b)). □

Example 2.36 (cf. [12, Corollary 11.5(c)])**.** Let X_0 be the data matrix of Example 1.17, and $M = [\mathbf{1}_4\ X_0] \in \mathbb{R}^{4\times 3}$. Write $[\mathbf{1}_4\ CX_0] = MQ$ with a product Q of elementary matrices, and find Q^{-1}.

Solution. Centering the columns of X_0 is a sequence of column operations on M.

• Since $\bar{x} = (1,2)'$, the columns of $CX_0 = X_0 - \mathbf{1}\bar{x}'$ are $\mathrm{col}_1(X_0) - \mathbf{1}$ and $\mathrm{col}_2(X_0) - 2 \cdot \mathbf{1}$. So $[\mathbf{1}_4\ CX_0]$ arises from M by $\mathrm{col}_2 \leftarrow \mathrm{col}_2 - \mathrm{col}_1$ and then $\mathrm{col}_3 \leftarrow \mathrm{col}_3 - 2\,\mathrm{col}_1$, and Theorem 2.35 gives $[\mathbf{1}_4\ CX_0] = MQ$ with

$$Q = T_{12}(-1)\,T_{13}(-2) = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

• Q is unit upper triangular, so it is invertible and Q^{-1} is unit upper triangular (Theorem 2.31); indeed $Q^{-1} = T_{13}(2)\,T_{12}(1)$ (Theorems 2.13(b) and 2.15(b)), that is,

$$Q^{-1} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- With CX_0 from Example 1.17,

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad MQ = [\mathbf{1}_4 \; CX_0] = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \end{pmatrix}$$

- So $M = [\mathbf{1}_4 \; CX_0]\, Q^{-1}$: by Theorem 2.35(c), the column operations of $Q^{-1} = T_{13}(2)\, T_{12}(1)$, $\mathrm{col}_3 \leftarrow \mathrm{col}_3 + 2\,\mathrm{col}_1$ and then $\mathrm{col}_2 \leftarrow \mathrm{col}_2 + \mathrm{col}_1$, undo the centering; they add $2 \cdot \mathbf{1}$ back to $\mathrm{col}_2(X_0) - 2 \cdot \mathbf{1}$ and $\mathbf{1}$ back to $\mathrm{col}_1(X_0) - \mathbf{1}$.

Exercises

Exercise 2.1. Find the RREF and the pivot columns of each matrix by Gauss–Jordan elimination (Definition 2.5).

(a) $A = \begin{pmatrix} 1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 0 \\ 2 & 3 & 1 & -3 \end{pmatrix}$

(b) $B = \begin{pmatrix} 1 & 0 & 3 & 0 & -1 \\ 1 & -1 & 2 & 1 & -2 \\ 0 & -1 & -1 & 1 & -1 \\ -2 & 3 & -3 & -2 & 2 \end{pmatrix}$

Solution. (a) After the first arrow, row 2 is 0 in column 2 and row 3 is not, so Step 2 interchanges rows 2 and 3; the matrix after the interchange is an echelon form (end of forward elimination).

$$\begin{pmatrix} 1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 0 \\ 2 & 3 & 1 & -3 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 3 & -5 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & 3 & -5 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + 5R_3} \begin{pmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & -4 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The RREF is the last matrix; its pivot columns are 1, 2 and 4.

(b) After the third arrow, column 3 has no pivot, and row 3 is zero while row 4 is not in column 4; so Step 2 interchanges rows 3 and 4.

$$\begin{pmatrix} 1 & 0 & 3 & 0 & -1 \\ 1 & -1 & 2 & 1 & -2 \\ 0 & -1 & -1 & 1 & -1 \\ -2 & 3 & -3 & -2 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 0 & 3 & 0 & -1 \\ 0 & -1 & -1 & 1 & -1 \\ 0 & -1 & -1 & 1 & -1 \\ 0 & 3 & 3 & -2 & 0 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 0 & 3 & 0 & -1 \\ 0 & 1 & 1 & -1 & 1 \\ 0 & -1 & -1 & 1 & -1 \\ 0 & 3 & 3 & -2 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 0 & 3 & 0 & -1 \\ 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 3 & 0 & -1 \\ 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_3} \begin{pmatrix} 1 & 0 & 3 & 0 & -1 \\ 0 & 1 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The RREF is the last matrix; its pivot columns are 1, 2 and 4.

Exercise 2.2. Let $A = \begin{pmatrix} 1 & -2 & -2 \\ 2 & -3 & -2 \\ 0 & 1 & 2 \\ 1 & -1 & 0 \end{pmatrix}$.

(a) For which $b = (b_1, b_2, b_3, b_4)'$ is $Ax = b$ consistent?

(b) Find all solutions of $Ax = b$ for $b = (-2, -1, 3, 1)'$.

(c) Does $Ax = b$ have a solution for $b = (1, 0, 0, 0)'$?

Solution. (a) Row-reduce $[A \mid b]$, keeping $b_1, \ldots, b_4$ as letters:

$$\left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 2 & -3 & -2 & b_2 \\ 0 & 1 & 2 & b_3 \\ 1 & -1 & 0 & b_4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & 2 & b_2 - 2b_1 \\ 0 & 1 & 2 & b_3 \\ 0 & 1 & 2 & b_4 - b_1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & -2 & -2 & b_1 \\ 0 & 1 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 2b_1 - b_2 + b_3 \\ 0 & 0 & 0 & b_1 - b_2 + b_4 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 2b_2 - 3b_1 \\ 0 & 1 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 2b_1 - b_2 + b_3 \\ 0 & 0 & 0 & b_1 - b_2 + b_4 \end{array}\right)$$

The left part of the last matrix is the RREF of A, with $r = 2$. By Proposition 2.7(a), $Ax = b$ is consistent if and only if

$$2b_1 - b_2 + b_3 = 0 \qquad \text{and} \qquad b_1 - b_2 + b_4 = 0$$

(b) Here $2(-2) - (-1) + 3 = 0$ and $-2 - (-1) + 1 = 0$, so the system is consistent, and the last matrix becomes

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

With $x_3 = t$ free, $x_1 = 4 - 2t$ and $x_2 = 3 - 2t$: $x = (4, 3, 0)' + t\,(-2, -2, 1)'$ $(t \in \mathbb{R})$. Check: $A(4, 3, 0)' = (4-6,\ 8-9,\ 3,\ 4-3)' = (-2, -1, 3, 1)'$ and $A(-2, -2, 1)' = (-2+4-2,\ -4+6-2,\ -2+2,\ -2+2)' = 0$.

(c) No: $2b_1 - b_2 + b_3 = 2 \neq 0$.

Exercise 2.3 (cf. [1, §1.4, Exercises 6–9])**.** Let A be the coefficient matrix of Example 2.30.

(a) For which $b = (b_1, b_2, b_3)'$ is $Ax = b$ consistent?

(b) Find all solutions of $Ax = b$ for $b = (1, 2, 4)'$.

Solution. (a) Row-reduce $[A \mid b]$, keeping b_1, b_2, b_3 as letters:

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 1 & -1 & -1 & 1 & b_2 \\ 1 & 1 & -5 & -1 & b_3 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 3 & -6 & -3 & b_3 - b_1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - 3R_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 - 3b_2 + 2b_1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 2b_2 - b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 - 3b_2 + 2b_1 \end{array}\right)$$

The left part of the last matrix is the RREF of A, with $r = 2$. By Proposition 2.7(a), $Ax = b$ is consistent if and only if $2b_1 - 3b_2 + b_3 = 0$. (In Example 2.30, $2 - 6 + 3 = -1 \neq 0$.)

(b) For $b = (1, 2, 4)'$, $2 - 6 + 4 = 0$, so the system is consistent, and the last matrix becomes

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 3 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The free unknowns are $x_3 = s$, $x_4 = t$, and by Proposition 2.7(b) $x_1 = 3 + 3s$, $x_2 = 1 + 2s + t$. So

$$x = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 3 \\ 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \qquad (s, t \in \mathbb{R})$$

infinitely many solutions (Theorem 2.27(b)): the particular solution $(3, 1, 0, 0)'$ plus the solutions of $Ax = 0$ (Theorem 2.27(a)). Check: $A(3, 1, 0, 0)' = (3 - 2,\ 3 - 1,\ 3 + 1)' = (1, 2, 4)'$.

Exercise 2.4. Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 1 \\ 2 & 4 & 3 & 5 & 3 & 3 \\ 1 & 2 & 2 & 3 & 3 & 1 \\ 3 & 6 & 4 & 7 & 3 & 5 \end{pmatrix} \in \mathbb{R}^{4\times 6}$$

(a) Show that $Ax = b$, $b = (b_1, b_2, b_3, b_4)'$, is consistent if and only if $2b_2 = b_3 + b_4$.

(b) Find all solutions of $Ax = b$ for $b = (2, 4, 3, 5)'$.

Solution. (a) Forward elimination of $[A \mid b]$, keeping $b_1, \ldots, b_4$ as letters:

$$\left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 2 & 4 & 3 & 5 & 3 & 3 & b_2 \\ 1 & 2 & 2 & 3 & 3 & 1 & b_3 \\ 3 & 6 & 4 & 7 & 3 & 5 & b_4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 1 & 1 & 2 & 0 & b_3 - b_1 \\ 0 & 0 & 1 & 1 & 0 & 2 & b_4 - 3b_1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & 1 & -1 & b_1 - b_2 + b_3 \\ 0 & 0 & 0 & 0 & -1 & 1 & b_4 - b_1 - b_2 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & 1 & -1 & b_1 - b_2 + b_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2b_2 + b_3 + b_4 \end{array}\right)$$

The left part of the last matrix is in echelon form, with pivots in columns $1, 3, 5$ of A, and the last row reads $0 = -2b_2 + b_3 + b_4$. By Theorem 2.26, $Ax = b$ is consistent if and only if $-2b_2 + b_3 + b_4 = 0$, that is, $2b_2 = b_3 + b_4$.

(b) For $b = (2, 4, 3, 5)'$, $2 \cdot 4 = 3 + 5$, and the last matrix has rows $(1, 2, 1, 2, 1, 1 \mid 2)$, $(0, 0, 1, 1, 1, 1 \mid 0)$, $(0, 0, 0, 0, 1, -1 \mid 1)$ and a zero row. Backward elimination:

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccccc|c} 1 & 2 & 1 & 2 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccccc|c} 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The free unknowns are $x_2 = \lambda$, $x_4 = \mu$, $x_6 = \nu$, and Proposition 2.7(b) gives $x_1 = 2 - 2\lambda - \mu$,

$x_3 = -1-\mu-2\nu$, $x_5 = 1+\nu$:

$$x = \begin{pmatrix}2\\0\\-1\\0\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}-2\\1\\0\\0\\0\\0\end{pmatrix} + \mu\begin{pmatrix}-1\\0\\-1\\1\\0\\0\end{pmatrix} + \nu\begin{pmatrix}0\\0\\-2\\0\\1\\1\end{pmatrix} \qquad (\lambda,\mu,\nu\in\mathbb{R})$$

the particular solution $(2,0,-1,0,1,0)'$ plus the solutions of $Ax=0$ (Theorem 2.27(a)). Check: $A(2,0,-1,0,1,0)' = 2\,\mathrm{col}_1(A) - \mathrm{col}_3(A) + \mathrm{col}_5(A) = (2-1+1,\ 4-3+3,\ 2-2+3,\ 6-4+3)' = (2,4,3,5)'$.

Exercise 2.5. (a) Find all solutions of $Ax=0$ for $A = \begin{pmatrix}1&1&-1&0&0\\2&1&-3&0&3\\2&1&-3&1&1\end{pmatrix}$.

(b) Does $Bx=0$ have a nontrivial solution, for $B = \begin{pmatrix}1&-2&-1&-1\\-1&3&3&1\\-1&2&1&1\\-1&2&1&0\end{pmatrix}$? Find all solutions.

Solution. (a) There are fewer equations than unknowns, so nontrivial solutions exist (Theorem 2.11(a)). Row-reduce A:

$$\begin{pmatrix}1&1&-1&0&0\\2&1&-3&0&3\\2&1&-3&1&1\end{pmatrix} \xrightarrow[R_3\leftarrow R_3-2R_1]{R_2\leftarrow R_2-2R_1} \begin{pmatrix}1&1&-1&0&0\\0&-1&-1&0&3\\0&-1&-1&1&1\end{pmatrix}$$

$$\xrightarrow{R_2\leftarrow -R_2} \begin{pmatrix}1&1&-1&0&0\\0&1&1&0&-3\\0&-1&-1&1&1\end{pmatrix} \xrightarrow{R_3\leftarrow R_3+R_2} \begin{pmatrix}1&1&-1&0&0\\0&1&1&0&-3\\0&0&0&1&-2\end{pmatrix}$$

$$\xrightarrow{R_1\leftarrow R_1-R_2} \begin{pmatrix}1&0&-2&0&3\\0&1&1&0&-3\\0&0&0&1&-2\end{pmatrix}$$

The pivot columns are $1,2,4$; the free unknowns are $x_3=s$, $x_5=t$, and the rows give $x_1 = 2s-3t$, $x_2 = -s+3t$, $x_4 = 2t$:

$$x = s\begin{pmatrix}2\\-1\\1\\0\\0\end{pmatrix} + t\begin{pmatrix}-3\\3\\0\\2\\1\end{pmatrix} \qquad (s,t\in\mathbb{R})$$

Check: $A(2,-1,1,0,0)' = (2-1-1,\ 4-1-3,\ 4-1-3)' = 0$ and $A(-3,3,0,2,1)' = (-3+3,\ -6+3+3,\ -6+3+2+1)' = 0$.

(b) Row-reduce B; after the first arrow, column 3 has no pivot and row 3 is zero, so Step 2 interchanges

rows 3 and 4:

$$\begin{pmatrix} 1 & -2 & -1 & -1 \\ -1 & 3 & 3 & 1 \\ -1 & 2 & 1 & 1 \\ -1 & 2 & 1 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & -2 & -1 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & -2 & -1 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow -R_3} \begin{pmatrix} 1 & -2 & -1 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 + R_3} \begin{pmatrix} 1 & -2 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \begin{pmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The RREF has pivot columns $1, 2, 4$, so $x_3 = t$ is free: yes, there are nontrivial solutions, and all solutions are

$$x = t\,(-3,\ -2,\ 1,\ 0)' \qquad (t \in \mathbb{R})$$

Check: $B(-3, -2, 1, 0)' = (-3 + 4 - 1,\ 3 - 6 + 3,\ 3 - 4 + 1,\ 3 - 4 + 1)' = 0$. In particular B is not invertible (Theorem 2.16).

Exercise 2.6 (cf. [6, Exercise 6.37])**.** Find the solutions of the equations

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 + 2x_5 &= 0 \\ 2x_1 + 3x_2 + x_3 + 3x_4 + 5x_5 &= 0 \\ x_1 + 2x_3 + x_4 + 2x_5 &= 0 \end{aligned}$$

Solution. Row-reduce the coefficient matrix:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 & 5 \\ 1 & 0 & 2 & 1 & 2 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & 1 & 1 \\ 0 & -1 & 1 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

The pivot columns are $1, 2, 4$, so x_3 and x_5 are free unknowns (Proposition 2.7). With $x_3 = \lambda$ and $x_5 = \mu$, the rows of the RREF give $x_1 = -2\lambda - \mu$, $x_2 = \lambda$, $x_4 = -\mu$. All solutions are

$$x = \lambda(-2, 1, 1, 0, 0)' + \mu(-1, 0, 0, -1, 1)' \qquad (\lambda, \mu \in \mathbb{R})$$

Check: $(-2, 1, 1, 0, 0)'$ gives $-2 + 1 + 1 = 0$, $-4 + 3 + 1 = 0$, $-2 + 2 = 0$, and $(-1, 0, 0, -1, 1)'$ gives $-1 - 1 + 2 = 0$, $-2 - 3 + 5 = 0$, $-1 - 1 + 2 = 0$. Three equations in five unknowns always have nontrivial solutions (Theorem 2.11(a)); here there are two free unknowns.

Exercise 2.7 (cf. [6, Exercise 6.40])**.** Solve the homogeneous system

$$\begin{aligned} x_1 - x_2 + 2x_3 + x_4 &= 0 \\ x_1 - x_3 + 3x_4 &= 0 \\ 3x_1 - 2x_2 + 3x_3 + 5x_4 &= 0 \\ x_1 + x_2 - 4x_3 + 5x_4 &= 0 \end{aligned}$$

Solution. Transform the coefficient matrix to echelon form:

$$\begin{pmatrix} 1 & -1 & 2 & 1 \\ 1 & 0 & -1 & 3 \\ 3 & -2 & 3 & 5 \\ 1 & 1 & -4 & 5 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - 3R_1} \begin{pmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 1 & -3 & 2 \\ 0 & 2 & -6 & 4 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

By Theorem 2.3, the system has the same solutions as the two equations $x_1 - x_2 + 2x_3 + x_4 = 0$ and $x_2 - 3x_3 + 2x_4 = 0$. For any values of x_3 and x_4, the second equation determines x_2 and then the first determines x_1; so x_3 and x_4 are free. Let $x_3 = \lambda$ and $x_4 = \mu$. Then $x_2 = 3x_3 - 2x_4 = 3\lambda - 2\mu$ and $x_1 = x_2 - 2x_3 - x_4 = \lambda - 3\mu$. Hence

$$x = \lambda(1, 3, 1, 0)' + \mu(-3, -2, 0, 1)' \qquad (\lambda, \mu \in \mathbb{R})$$

(One more arrow, $R_1 \leftarrow R_1 + R_2$, gives the RREF, with rows $(1, 0, -1, 3)$ and $(0, 1, -3, 2)$, from which the same solutions are read off.)

Exercise 2.8 ([6, Exercise 6.1])**.** Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$. Find an elementary matrix E with $B = EA$ when: (a) B is obtained from A by interchanging the first and third rows; (b) B is obtained from A by multiplying the second row by 7; (c) B is obtained from A by adding five times the second row to the third row.

Solution. Perform each operation on I_3 (Theorem 2.15(a)):

$$P_{13} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad D_2(7) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad T_{32}(5) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{pmatrix}$$

and

$$P_{13}A = \begin{pmatrix} 5 & 6 \\ 3 & 4 \\ 1 & 2 \end{pmatrix}, \quad D_2(7)A = \begin{pmatrix} 1 & 2 \\ 21 & 28 \\ 5 & 6 \end{pmatrix}, \quad T_{32}(5)A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 20 & 26 \end{pmatrix}$$

Exercise 2.9 (cf. [6, Exercise 6.8])**.** Let $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}$.

(a) Find the matrix E such that EA is obtained from A by first interchanging rows 1 and 2 and then multiplying row 2 by 2. Does the order of the two operations matter?

(b) Find the matrix E such that EA is obtained from A by first interchanging rows 1 and 2 and then adding 3 times row 1 to row 2. Does the order matter?

(c) Do elementary matrices commute?

Solution. By Theorem 2.15, the operation done first is the factor next to A.

(a) $E = D_2(2)P_{12} = \left(\begin{smallmatrix} 0&1&0&0\\ 2&0&0&0\\ 0&0&1&0\\ 0&0&0&1 \end{smallmatrix}\right)$, while $P_{12}D_2(2) = \left(\begin{smallmatrix} 0&2&0&0\\ 1&0&0&0\\ 0&0&1&0\\ 0&0&0&1 \end{smallmatrix}\right)$, and

$$D_2(2)P_{12}A = \begin{pmatrix} 0&1&2&3\\ 2&4&6&8\\ 1&0&1&0\\ 2&1&0&1 \end{pmatrix} \neq \begin{pmatrix} 0&2&4&6\\ 1&2&3&4\\ 1&0&1&0\\ 2&1&0&1 \end{pmatrix} = P_{12}D_2(2)A$$

The order matters.

(b) $E = T_{21}(3)P_{12} = \left(\begin{smallmatrix} 0&1&0&0\\ 1&3&0&0\\ 0&0&1&0\\ 0&0&0&1 \end{smallmatrix}\right)$, while $P_{12}T_{21}(3) = \left(\begin{smallmatrix} 3&1&0&0\\ 1&0&0&0\\ 0&0&1&0\\ 0&0&0&1 \end{smallmatrix}\right)$, and

$$T_{21}(3)P_{12}A = \begin{pmatrix} 0&1&2&3\\ 1&5&9&13\\ 1&0&1&0\\ 2&1&0&1 \end{pmatrix} \neq \begin{pmatrix} 3&7&11&15\\ 1&2&3&4\\ 1&0&1&0\\ 2&1&0&1 \end{pmatrix} = P_{12}T_{21}(3)A$$

The order matters.

(c) No: (a) and (b) give pairs of elementary matrices that do not commute.

Exercise 2.10 ([6, Exercise 6.14])**.** Reduce the matrix $A = \begin{pmatrix} 0&0&1&2&3\\ 2&4&-2&0&4\\ 2&4&-1&2&7 \end{pmatrix}$ to echelon form by elementary row operations, and record the elementary matrices used.

Solution. Follow Definition 2.5, postponing the scaling (Step 3) to the end. Column 1 is the first pivot column; since $a_{11} = 0$, interchange rows 1 and 2; then clear column 1 below the pivot, and then column 3, the next pivot column:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2&4&-2&0&4\\ 0&0&1&2&3\\ 2&4&-1&2&7 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 2&4&-2&0&4\\ 0&0&1&2&3\\ 0&0&1&2&3 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 2&4&-2&0&4\\ 0&0&1&2&3\\ 0&0&0&0&0 \end{pmatrix}$$

Finally divide row 1 by 2:

$$H = \begin{pmatrix} 1&2&-1&0&2\\ 0&0&1&2&3\\ 0&0&0&0&0 \end{pmatrix} = D_1(\tfrac{1}{2})\, T_{32}(-1)\, T_{31}(-1)\, P_{12}\, A$$

(Theorem 2.15(c)). H is in echelon form with pivot columns 1 and 3. One more operation, $R_1 \leftarrow R_1 + R_2$, clears the entry above the second pivot and gives the RREF $\left(\begin{smallmatrix} 1&2&0&2&5\\ 0&0&1&2&3\\ 0&0&0&0&0 \end{smallmatrix}\right)$.

Exercise 2.11. Row-reduce $[M \mid I_4]$ to find M^{-1}, or show that M is not invertible, for

(a) $M = \begin{pmatrix} 1&1&0&-1\\ 1&1&-1&0\\ -2&-1&3&0\\ -1&-1&2&0 \end{pmatrix}$

(b) $M = \begin{pmatrix} 1&-1&0&0\\ -3&2&2&-1\\ -2&2&1&-1\\ -1&1&1&0 \end{pmatrix}$

(c) $M = \begin{pmatrix} 1 & 2 & -1 & 1 \\ 2 & 3 & 1 & -1 \\ 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & -1 \end{pmatrix}$

Solution. (a) After the first arrow, row 2 is 0 in column 2, so Step 2 interchanges rows 2 and 3:

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 \\ -2 & -1 & 3 & 0 & 0 & 0 & 1 & 0 \\ -1 & -1 & 2 & 0 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 + 2R_1} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 2 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & -2 & 2 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 2 & -1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & -2 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 2 & -1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - 2R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & -2 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 2 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 + R_4]{R_1 \leftarrow R_1 + R_4,\ R_2 \leftarrow R_2 + 2R_4} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 1 & 3 & 0 & 0 & 4 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow R_2 - 3R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 1 & -1 & 2 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 2 & 0 & 1 \end{array}\right)$$

So

$$M^{-1} = \begin{pmatrix} 0 & 1 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ -1 & 2 & 0 & 1 \end{pmatrix}$$

Check: $\text{row}_1(M)M^{-1} = (0+0+1,\ 1+1-2,\ -1+1,\ 2-1-1) = (1,0,0,0)$, and similarly for rows $2,3,4$.

(b)

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ -3 & 2 & 2 & -1 & 0 & 1 & 0 & 0 \\ -2 & 2 & 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 + 3R_1,\ R_3 \leftarrow R_3 + 2R_1} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & -3 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & -3 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow R_3 + R_4]{R_2 \leftarrow R_2 - R_4} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 & -2 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 + 2R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & 1 \end{array}\right)$$

So

$$M^{-1} = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & -1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 \end{pmatrix}$$

Check: $\text{row}_2(M)M^{-1} = (-3+2+1,\ 3-2,\ -3+2+1,\ -3+2+2-1) = (0,1,0,0)$, and similarly for rows $1,3,4$.

(c)

$$\left(\begin{array}{cccc|cccc} 1 & 2 & -1 & 1 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 2 & -2 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|cccc} 1 & 2 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -3 & -2 & 1 & 0 & 0 \\ 0 & -1 & 3 & -3 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|cccc} 1 & 2 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & 3 & 2 & -1 & 0 & 0 \\ 0 & -1 & 3 & -3 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -2 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc|cccc} 1 & 2 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & 3 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & -1 & 0 & 0 & 1 \end{array}\right)$$

A zero row has appeared in the left half (row 3), so M is not invertible (Corollary 2.17(b)). Indeed, $Mx = 0$ has the same solutions as the system of the left half (Theorem 2.3(b)): with $x_4 = t$, $x_3 - 2x_4 = 0$ gives $x_3 = 2t$, $x_2 - 3x_3 + 3x_4 = 0$ gives $x_2 = 3t$, and then $x_1 = -2x_2 + x_3 - x_4 = -5t$. So $Mx = 0$ has the nontrivial solution $x = (-5,3,2,1)'$: $M(-5,3,2,1)' = (-5+6-2+1,\ -10+9+2-1,\ -5+3+4-2,\ -5+6+0-1)' = 0$.

Exercise 2.12 ([6, Exercise 6.23])**.** Use row reduction of $[A \mid I_3]$ to find the inverse of $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 3 & 4 & 3 \end{pmatrix}$.

Solution.

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 3 & 4 & 3 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 3 & 4 & 3 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - 3R_1} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -3 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array}\right)$$

$$\xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & -2 & 1 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array}\right)$$

Hence $A^{-1} = \left(\begin{smallmatrix} -1 & 1 & 0 \\ 0 & -3 & 1 \\ 1 & 3 & -1 \end{smallmatrix}\right)$. Check: $\text{row}_3(A)A^{-1} = (3,4,3)A^{-1} = (-3+3,\ 3-12+9,\ 4-3) = (0,0,1)$, and similarly for the other rows.

Exercise 2.13 (cf. [6, Exercise 4.19])**.** Do $A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 4 & 3 & 1 \\ 1 & 3 & 2 & 2 \\ 3 & 7 & 5 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$ have inverses?

Find the inverse when it exists.

Solution. (A) Row-reduce A; after the first arrow the entry of row 2 in column 2 is 0, so rows 2 and 3 are interchanged:

$$A \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 2 & 0 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 - R_2} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

A zero row appears, so A has no inverse (Corollary 2.17(b): the same operations give this left half in $[A \mid I_4]$). (Indeed row 4 of A is the sum of rows 2 and 3.)

(B) Row-reduce $[B \mid I_4]$. Every pivot is 1; forward elimination subtracts each row from the next one, and backward elimination each row from the one above it:

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right)$$
$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array}\right)$$
$$\xrightarrow{R_3 \leftarrow R_3 - R_4} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -3 & 3 & -2 & 1 \\ 0 & 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array}\right)$$
$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 4 & -3 & 2 & -1 \\ 0 & 1 & 0 & 0 & -3 & 3 & -2 & 1 \\ 0 & 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \end{array}\right)$$

The left block is I_4, so B is invertible and

$$B^{-1} = \begin{pmatrix} 4 & -3 & 2 & -1 \\ -3 & 3 & -2 & 1 \\ 2 & -2 & 2 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

(Corollary 2.17(a)). Check: the second row of BB^{-1} is $(4-6+2,\ -3+6-2,\ 2-4+2,\ -1+2-1) = (0, 1, 0, 0)$, and similarly for the other rows.

Exercise 2.14 (cf. [3, Ch. 2, Exercise 15])**.** Use Proposition 2.18 to decide which of $\left(\begin{smallmatrix}5&3\\3&2\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}4&6\\6&9\end{smallmatrix}\right)$ and $\begin{pmatrix}7&-3\\-5&2\end{pmatrix}$ are invertible, and find the inverses that exist. Then find the inverse of

$$\begin{pmatrix}0&1&1&1\\1&1&1&1\\0&1&1&2\\0&0&1&1\end{pmatrix}$$

Solution. (2×2) By Proposition 2.18:

- $5\cdot 2-3\cdot 3=1$, so $\left(\begin{smallmatrix}5&3\\3&2\end{smallmatrix}\right)^{-1}=\begin{pmatrix}2&-3\\-3&5\end{pmatrix}$.
- $4\cdot 9-6\cdot 6=0$, so $\left(\begin{smallmatrix}4&6\\6&9\end{smallmatrix}\right)$ is not invertible.
- $7\cdot 2-(-3)(-5)=-1$, so $\begin{pmatrix}7&-3\\-5&2\end{pmatrix}^{-1}=-\left(\begin{smallmatrix}2&3\\5&7\end{smallmatrix}\right)=\begin{pmatrix}-2&-3\\-5&-7\end{pmatrix}$.

Check: $\begin{pmatrix}7&-3\\-5&2\end{pmatrix}\begin{pmatrix}-2&-3\\-5&-7\end{pmatrix}=\begin{pmatrix}-14+15&-21+21\\10-10&15-14\end{pmatrix}=I_2$.

(4×4) Call the 4×4 matrix K and row-reduce $[K\mid I_4]$. The entry in position $(1,1)$ is 0, so Step 2 interchanges rows 1 and 2; after clearing column 2, the entry in position $(3,3)$ is 0, so rows 3 and 4 are interchanged:

$$\left(\begin{array}{cccc|cccc}0&1&1&1&1&0&0&0\\1&1&1&1&0&1&0&0\\0&1&1&2&0&0&1&0\\0&0&1&1&0&0&0&1\end{array}\right)\xrightarrow{R_1\leftrightarrow R_2}\left(\begin{array}{cccc|cccc}1&1&1&1&0&1&0&0\\0&1&1&1&1&0&0&0\\0&1&1&2&0&0&1&0\\0&0&1&1&0&0&0&1\end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3-R_2}\left(\begin{array}{cccc|cccc}1&1&1&1&0&1&0&0\\0&1&1&1&1&0&0&0\\0&0&0&1&-1&0&1&0\\0&0&1&1&0&0&0&1\end{array}\right)\xrightarrow{R_3\leftrightarrow R_4}\left(\begin{array}{cccc|cccc}1&1&1&1&0&1&0&0\\0&1&1&1&1&0&0&0\\0&0&1&1&0&0&0&1\\0&0&0&1&-1&0&1&0\end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-R_4]{R_3\leftarrow R_3-R_4,\ R_2\leftarrow R_2-R_4}\left(\begin{array}{cccc|cccc}1&1&1&0&1&1&-1&0\\0&1&1&0&2&0&-1&0\\0&0&1&0&1&0&-1&1\\0&0&0&1&-1&0&1&0\end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-R_3]{R_2\leftarrow R_2-R_3}\left(\begin{array}{cccc|cccc}1&1&0&0&0&1&0&-1\\0&1&0&0&1&0&0&-1\\0&0&1&0&1&0&-1&1\\0&0&0&1&-1&0&1&0\end{array}\right)\xrightarrow{R_1\leftarrow R_1-R_2}\left(\begin{array}{cccc|cccc}1&0&0&0&-1&1&0&0\\0&1&0&0&1&0&0&-1\\0&0&1&0&1&0&-1&1\\0&0&0&1&-1&0&1&0\end{array}\right)$$

So

$$K^{-1}=\begin{pmatrix}-1&1&0&0\\1&0&0&-1\\1&0&-1&1\\-1&0&1&0\end{pmatrix}$$

Check: $\mathrm{row}_3(K)K^{-1}=(0,1,1,2)K^{-1}=(1+1-2,\ 0,\ -1+2,\ -1+1)=(0,0,1,0)$, and similarly for the other rows.

Exercise 2.15. Find the inverse of $A = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$ by row reduction of $[A \mid I_4]$, and check $AA^{-1} = I_4$.

Solution. After the first arrow, rows 2 and 3 are 0 in column 2, so Step 2 interchanges rows 2 and 4:

$$\left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_4} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 \end{array}\right) \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 \end{array}\right)$$

$$\xrightarrow[R_2 \leftarrow R_2 + R_4]{R_3 \leftarrow R_3 - R_4} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & -2 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 \end{array}\right)$$

So

$$A^{-1} = \begin{pmatrix} 1 & -2 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 \end{pmatrix}$$

Check: $\text{row}_1(A)A^{-1} = (1,\ -2+2,\ 0,\ 2-2) = (1,0,0,0)$, $\text{row}_2(A)A^{-1} = (1-1,\ -2+2+1,\ 0,\ 2-2) = (0,1,0,0)$, $\text{row}_3(A)A^{-1} = (1-1,\ -2+2-1+1,\ 1,\ 2-2) = (0,0,1,0)$ and $\text{row}_4(A)A^{-1} = (1-1,\ -2+1+1,\ 0,\ 2-1) = (0,0,0,1)$.

Exercise 2.16. Let

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -2 & 1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

(a) Row-reduce $[A \mid I_4]$ to $[I_4 \mid A^{-1}]$, one elementary operation per arrow.

(b) Write A^{-1} as a product of elementary matrices.

(c) Write A as a product of elementary matrices, and check the product.

Solution. (a) After the first arrow, row 2 is 0 in column 2, so Step 2 interchanges rows 2 and 3:

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ -2 & 1 & 2 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 + 2R_1} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_4} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 & -2 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow R_2 + 2R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -2 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & 1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -2 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{array}\right)$$

So

$$A^{-1} = \begin{pmatrix} -1 & 1 & -1 & -1 \\ -2 & 1 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

(b) The seven operations have the elementary matrices $E_1 = T_{31}(2)$, $E_2 = P_{23}$, $E_3 = D_2(-1)$, $E_4 = T_{43}(1)$, $E_5 = T_{24}(-1)$, $E_6 = T_{23}(2)$, $E_7 = T_{12}(1)$, and $E_7E_6E_5E_4E_3E_2E_1A = I_4$ (Theorem 2.15(c)). Hence, by the last sentence of Theorem 2.16, $A^{-1} = E_7E_6E_5E_4E_3E_2E_1$, the product, in this order, of

$$E_7 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad E_6 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad E_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$E_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad E_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$E_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad E_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(c) By Theorem 2.13(a), $A = (A^{-1})^{-1}$; so, by Theorems 2.13(b) and 2.15(b),

$$\begin{aligned} A &= E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}E_6^{-1}E_7^{-1} \\ &= T_{31}(-2)\,P_{23}\,D_2(-1)\,T_{43}(-1)\,T_{24}(1)\,T_{23}(-2)\,T_{12}(-1) \end{aligned}$$

Check: the product applied to I_4 performs, from the right, the operations of $T_{12}(-1)$, $T_{23}(-2)$, $T_{24}(1)$, $T_{43}(-1)$, $D_2(-1)$, P_{23}, $T_{31}(-2)$ (Theorem 2.15(a)):

$$\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix} \xrightarrow{R_1\leftarrow R_1-R_2} \begin{pmatrix}1&-1&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix} \xrightarrow{R_2\leftarrow R_2-2R_3} \begin{pmatrix}1&-1&0&0\\0&1&-2&0\\0&0&1&0\\0&0&0&1\end{pmatrix}$$

$$\xrightarrow{R_2\leftarrow R_2+R_4} \begin{pmatrix}1&-1&0&0\\0&1&-2&1\\0&0&1&0\\0&0&0&1\end{pmatrix} \xrightarrow{R_4\leftarrow R_4-R_3} \begin{pmatrix}1&-1&0&0\\0&1&-2&1\\0&0&1&0\\0&0&-1&1\end{pmatrix} \xrightarrow{R_2\leftarrow -R_2} \begin{pmatrix}1&-1&0&0\\0&-1&2&-1\\0&0&1&0\\0&0&-1&1\end{pmatrix}$$

$$\xrightarrow{R_2\leftrightarrow R_3} \begin{pmatrix}1&-1&0&0\\0&0&1&0\\0&-1&2&-1\\0&0&-1&1\end{pmatrix} \xrightarrow{R_3\leftarrow R_3-2R_1} \begin{pmatrix}1&-1&0&0\\0&0&1&0\\-2&1&2&-1\\0&0&-1&1\end{pmatrix}$$

which is A.

Exercise 2.17. Let $A = \begin{pmatrix}1&-1&2&0\\-1&1&-1&1\\-1&1&0&2\end{pmatrix}$.

(a) Row-reduce $[A \mid I_3]$ to $[R \mid P]$, where R is the RREF of A.

(b) Check that $PA = R$.

(c) For which $y \in \mathbb{R}^3$ is $Ax = y$ consistent?

Solution. (a)

$$\left(\begin{array}{cccc|ccc}1&-1&2&0&1&0&0\\-1&1&-1&1&0&1&0\\-1&1&0&2&0&0&1\end{array}\right) \xrightarrow[R_3\leftarrow R_3+R_1]{R_2\leftarrow R_2+R_1} \left(\begin{array}{cccc|ccc}1&-1&2&0&1&0&0\\0&0&1&1&1&1&0\\0&0&2&2&1&0&1\end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3-2R_2} \left(\begin{array}{cccc|ccc}1&-1&2&0&1&0&0\\0&0&1&1&1&1&0\\0&0&0&0&-1&-2&1\end{array}\right) \xrightarrow{R_1\leftarrow R_1-2R_2} \left(\begin{array}{cccc|ccc}1&-1&0&-2&-1&-2&0\\0&0&1&1&1&1&0\\0&0&0&0&-1&-2&1\end{array}\right)$$

So $R = \left(\begin{smallmatrix}1&-1&0&-2\\0&0&1&1\\0&0&0&0\end{smallmatrix}\right)$ and $P = \left(\begin{smallmatrix}-1&-2&0\\1&1&0\\-1&-2&1\end{smallmatrix}\right)$, which is invertible (Corollary 2.24(a)).

(b) Row by row (Theorem 1.4(c)):

- $\text{row}_1(P)A = -(1,-1,2,0) - 2(-1,1,-1,1) = (1,-1,0,-2)$;
- $\text{row}_2(P)A = (1,-1,2,0) + (-1,1,-1,1) = (0,0,1,1)$;
- $\text{row}_3(P)A = -(1,-1,2,0) - 2(-1,1,-1,1) + (-1,1,0,2) = (0,0,0,0)$.

(c) As P is invertible, $Ax = y$ holds if and only if $Rx = Py$ does. Row 3 of R is zero, so $Ax = y$ is consistent if and only if $\text{row}_3(P)\,y = -y_1 - 2y_2 + y_3 = 0$ (Proposition 2.7(a)).

Exercise 2.18. (a) Solve $x_1-2x_3-x_4=2$, $2x_1-x_2-3x_3=3$, $-x_1+x_2+2x_3=-1$, $-x_1+x_2+x_3=0$ by forward elimination and back substitution.

(b) Find all solutions of $x_1+2x_2+x_3+x_4=-2$, $-2x_1-3x_2-3x_3-2x_4=3$, $-x_1-3x_2=1$, in vector form, by Gauss–Jordan elimination.

(c) Show that the system $x_1-x_3=-1$, $-x_1+x_2+x_3=1$, $x_1-x_2=-1$, $2x_1-x_2=-1$ has no solution.

Solution. (a) Forward elimination of $[A \mid b]$:

$$\left(\begin{array}{cccc|c} 1 & 0 & -2 & -1 & 2 \\ 2 & -1 & -3 & 0 & 3 \\ -1 & 1 & 2 & 0 & -1 \\ -1 & 1 & 1 & 0 & 0 \end{array}\right) \xrightarrow[R_4\leftarrow R_4+R_1]{R_2\leftarrow R_2-2R_1,\ R_3\leftarrow R_3+R_1} \left(\begin{array}{cccc|c} 1 & 0 & -2 & -1 & 2 \\ 0 & -1 & 1 & 2 & -1 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & -1 & 2 \end{array}\right)$$

$$\xrightarrow{R_2\leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 0 & -2 & -1 & 2 \\ 0 & 1 & -1 & -2 & 1 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & -1 & 2 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-R_2]{R_3\leftarrow R_3-R_2} \left(\begin{array}{cccc|c} 1 & 0 & -2 & -1 & 2 \\ 0 & 1 & -1 & -2 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

- The last matrix is an echelon form with pivots in columns $1,2,3,4$ and none in the last column: there is exactly one solution (Theorems 2.26 and 2.27(b)).
- Back substitution: $x_4=1$; $x_3=-x_4=-1$; $x_2=1+x_3+2x_4=2$; $x_1=2+2x_3+x_4=1$. So $x=(1,2,-1,1)'$.
- Check: $1+2-1=2$, $2-2+3=3$, $-1+2-2=-1$, $-1+2-1=0$.

(b) Gauss–Jordan elimination of $[A \mid b]$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & -2 \\ -2 & -3 & -3 & -2 & 3 \\ -1 & -3 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3\leftarrow R_3+R_1]{R_2\leftarrow R_2+2R_1} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & -2 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & -1 & 1 & 1 & -1 \end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3+R_2} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & -2 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right) \xrightarrow{R_1\leftarrow R_1-R_3} \left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right)$$

$$\xrightarrow{R_1\leftarrow R_1-2R_2} \left(\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right)$$

- The left part of the last matrix is the RREF of A, with pivot columns $1,2,4$, and the last column has no pivot: the system is consistent, and $x_3=t$ is free (Proposition 2.7).
- The rows give $x_1=2-3t$, $x_2=-1+t$, $x_4=-2$, so

$$x=\begin{pmatrix}2\\-1\\0\\-2\end{pmatrix}+t\begin{pmatrix}-3\\1\\1\\0\end{pmatrix} \qquad (t\in\mathbb{R})$$

- Check: $A(2,-1,0,-2)'=(2-2-2,\ -4+3+4,\ -2+3)'=(-2,3,1)'=b$ and $A(-3,1,1,0)'=(-3+2+1,\ 6-3-3,\ 3-3)'=0$.

(c) Forward elimination of $[A \mid b]$:

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 0 & -1 \\ 2 & -1 & 0 & -1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 2 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \left(\begin{array}{cccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

The last matrix is an echelon form whose last column is a pivot column: row 4 reads $0 = 1$. So the system has no solution (Theorem 2.26). Indeed, rows 1–3 give $x_3 = 0$, $x_2 = 0$, $x_1 = -1 + x_3 = -1$: the first three equations have only the solution $(-1, 0, 0)'$, and it violates the fourth, $2(-1) - 0 = -2 \neq -1$.

Exercise 2.19 (cf. [6, Exercise 6.27])**.** Solve the equations

$$\begin{pmatrix} 1 & -3 & -2 & -1 \\ -1 & 2 & 1 & 1 \\ -1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -2 \\ 1 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 2 & 3 & 1 \\ 1 & 2 & 2 & 2 \\ 0 & 1 & 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 3 \\ 1 \end{pmatrix}$$

by Gaussian elimination.

Solution. First system. Forward elimination:

$$\left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ -1 & 2 & 1 & 1 & -1 \\ -1 & 2 & 2 & 1 & -2 \\ 0 & 1 & 1 & 1 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 + R_1} \left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc|c} 1 & -3 & -2 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

Backward elimination:

$$\xrightarrow{R_1 \leftarrow R_1 + R_4} \left(\begin{array}{cccc|c} 1 & -3 & -2 & 0 & 2 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 + 2R_3} \left(\begin{array}{cccc|c} 1 & -3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 + 3R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

By Proposition 2.7, the unique solution is $(x_1, x_2, x_3, x_4) = (3, 1, -1, 1)$. Check: $3 - 3 + 2 - 1 = 1$, $-3 + 2 - 1 + 1 = -1$, $-3 + 2 - 2 + 1 = -2$, $1 - 1 + 1 = 1$.

Second system. Forward elimination; after the first arrow the entry of row 2 in column 2 is 0, so rows 2 and 3 are interchanged (Step 2):

$$\left(\begin{array}{cccc|c} 1&1&2&1&4\\ 2&2&3&1&6\\ 1&2&2&2&3\\ 0&1&1&3&1 \end{array}\right) \xrightarrow[R_3\leftarrow R_3-R_1]{R_2\leftarrow R_2-2R_1} \left(\begin{array}{cccc|c} 1&1&2&1&4\\ 0&0&-1&-1&-2\\ 0&1&0&1&-1\\ 0&1&1&3&1 \end{array}\right) \xrightarrow{R_2\leftrightarrow R_3} \left(\begin{array}{cccc|c} 1&1&2&1&4\\ 0&1&0&1&-1\\ 0&0&-1&-1&-2\\ 0&1&1&3&1 \end{array}\right)$$

$$\xrightarrow[R_3\leftarrow -R_3]{R_4\leftarrow R_4-R_2} \left(\begin{array}{cccc|c} 1&1&2&1&4\\ 0&1&0&1&-1\\ 0&0&1&1&2\\ 0&0&1&2&2 \end{array}\right) \xrightarrow{R_4\leftarrow R_4-R_3} \left(\begin{array}{cccc|c} 1&1&2&1&4\\ 0&1&0&1&-1\\ 0&0&1&1&2\\ 0&0&0&1&0 \end{array}\right)$$

Every column of the left part is a pivot column and the last column is not, so there is exactly one solution (Theorems 2.26 and 2.27(b)). Back substitution: $x_4 = 0$; $x_3 = 2 - x_4 = 2$; $x_2 = -1 - x_4 = -1$; $x_1 = 4 - x_2 - 2x_3 - x_4 = 4 + 1 - 4 - 0 = 1$. The unique solution is $(x_1, x_2, x_3, x_4) = (1, -1, 2, 0)$. Check: $1 - 1 + 4 + 0 = 4$, $2 - 2 + 6 + 0 = 6$, $1 - 2 + 4 + 0 = 3$, $0 - 1 + 2 + 0 = 1$.

Exercise 2.20. Let

$$A = \begin{pmatrix} 1&3&5&-1\\ 1&2&2&0\\ 0&1&3&-1\\ 0&0&1&-1 \end{pmatrix}, \qquad b = \begin{pmatrix} 6\\4\\2\\0 \end{pmatrix}, \qquad c = \begin{pmatrix} 6\\4\\3\\0 \end{pmatrix}$$

(a) Find all solutions of $Ax = b$.

(b) Show that $Ax = c$ has no solution.

Solution. (a) Row-reduce $[A \mid b]$:

$$\left(\begin{array}{cccc|c} 1&3&5&-1&6\\ 1&2&2&0&4\\ 0&1&3&-1&2\\ 0&0&1&-1&0 \end{array}\right) \xrightarrow{R_2\leftarrow R_2-R_1} \left(\begin{array}{cccc|c} 1&3&5&-1&6\\ 0&-1&-3&1&-2\\ 0&1&3&-1&2\\ 0&0&1&-1&0 \end{array}\right) \xrightarrow{R_2\leftarrow -R_2} \left(\begin{array}{cccc|c} 1&3&5&-1&6\\ 0&1&3&-1&2\\ 0&1&3&-1&2\\ 0&0&1&-1&0 \end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3-R_2} \left(\begin{array}{cccc|c} 1&3&5&-1&6\\ 0&1&3&-1&2\\ 0&0&0&0&0\\ 0&0&1&-1&0 \end{array}\right) \xrightarrow{R_3\leftrightarrow R_4} \left(\begin{array}{cccc|c} 1&3&5&-1&6\\ 0&1&3&-1&2\\ 0&0&1&-1&0\\ 0&0&0&0&0 \end{array}\right)$$

$$\xrightarrow[R_1\leftarrow R_1-5R_3]{R_2\leftarrow R_2-3R_3} \left(\begin{array}{cccc|c} 1&3&0&4&6\\ 0&1&0&2&2\\ 0&0&1&-1&0\\ 0&0&0&0&0 \end{array}\right) \xrightarrow{R_1\leftarrow R_1-3R_2} \left(\begin{array}{cccc|c} 1&0&0&-2&0\\ 0&1&0&2&2\\ 0&0&1&-1&0\\ 0&0&0&0&0 \end{array}\right)$$

After the third arrow the entry of row 3 in column 3 is 0, so Step 2 interchanges rows 3 and 4. The left part of the last matrix is the RREF R of A, with $r = 3$ and pivot columns $1, 2, 3$; the unknown x_4 is free. The last matrix is $[R \mid z]$ with $z = (0, 2, 0, 0)'$. As $z_4 = 0$, the system is consistent (Proposition 2.7(a)), and with $x_4 = t$, Proposition 2.7(b) gives $x_1 = 2t$, $x_2 = 2 - 2t$, $x_3 = t$:

$$x = \begin{pmatrix} 0\\2\\0\\0 \end{pmatrix} + t \begin{pmatrix} 2\\-2\\1\\1 \end{pmatrix} \qquad (t \in \mathbb{R})$$

infinitely many solutions (Theorem 2.27(a),(b)). Check: $A(0, 2, 0, 0)' = 2\,\mathrm{col}_2(A) = (6, 4, 2, 0)' = b$ and $A(2, -2, 1, 1)' = (2 - 6 + 5 - 1,\ 2 - 4 + 2,\ -2 + 3 - 1,\ 1 - 1)' = 0$.

(b) Row-reduce $[A \mid c]$ by the same first three operations:

$$\left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 1 & 2 & 2 & 0 & 4 \\ 0 & 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & -1 & -3 & 1 & -2 \\ 0 & 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right)$$

The third row of the last matrix reads $(0\ 0\ 0\ 0 \mid 1)$, that is, $0 = 1$; so $Ax = c$ has no solution (Theorem 2.26). Indeed $\mathrm{row}_3(A) = \mathrm{row}_1(A) - \mathrm{row}_2(A)$, while $c_3 = 3 \neq 2 = c_1 - c_2$.

Exercise 2.21. Consider the system

$$\begin{aligned} x_1 - 2x_3 + 3x_5 &= -1 \\ 2x_1 + x_2 - x_3 - x_4 + 3x_5 &= -4 \\ -2x_1 + 4x_3 + x_4 - 4x_5 &= 3 \\ x_1 + x_2 + x_3 + 2x_5 &= -2 \end{aligned}$$

(a) Reduce its augmented matrix $[A \mid b]$ to RREF.

(b) Write all solutions in the form $x = x_p + sv_1 + tv_2$ $(s, t \in \mathbb{R})$.

(c) Check that $Ax_p = b$ and $Av_1 = Av_2 = 0$.

Solution. (a)

$$\left(\begin{array}{ccccc|c} 1 & 0 & -2 & 0 & 3 & -1 \\ 2 & 1 & -1 & -1 & 3 & -4 \\ -2 & 0 & 4 & 1 & -4 & 3 \\ 1 & 1 & 1 & 0 & 2 & -2 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 + 2R_1} \left(\begin{array}{ccccc|c} 1 & 0 & -2 & 0 & 3 & -1 \\ 0 & 1 & 3 & -1 & -3 & -2 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 1 & 3 & 0 & -1 & -1 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_2} \left(\begin{array}{ccccc|c} 1 & 0 & -2 & 0 & 3 & -1 \\ 0 & 1 & 3 & -1 & -3 & -2 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{ccccc|c} 1 & 0 & -2 & 0 & 3 & -1 \\ 0 & 1 & 3 & -1 & -3 & -2 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow R_2 + R_3} \left(\begin{array}{ccccc|c} 1 & 0 & -2 & 0 & 3 & -1 \\ 0 & 1 & 3 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The last matrix is $[R \mid z]$, with R the RREF of A, pivot columns $1, 2, 4$ and $z = (-1, -1, 1, 0)'$.

(b) Since $z_4 = 0$, the system is consistent (Proposition 2.7(a)). The free unknowns are $x_3 = s$ and $x_5 = t$, and the rows give $x_1 = -1 + 2s - 3t$, $x_2 = -1 - 3s + t$, $x_4 = 1 - 2t$ (Proposition 2.7(b)):

$$x = \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ 1 \\ 0 \\ -2 \\ 1 \end{pmatrix} = x_p + sv_1 + tv_2$$

(c) Row by row:

- $Ax_p = (-1,\ -2 - 1 - 1,\ 2 + 1,\ -1 - 1)' = (-1, -4, 3, -2)' = b$;

- $Av_1 = (2-2,\ 4-3-1,\ -4+4,\ 2-3+1)' = 0$;
- $Av_2 = (-3+3,\ -6+1+2+3,\ 6-2-4,\ -3+1+2)' = 0$.

So x_p is a particular solution, and $sv_1 + tv_2$ runs through the solutions of $Ax = 0$ (Theorem 2.27(a)).

Exercise 2.22. For which values of k does the system

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 1 \\ 2x_1 + 5x_2 + x_3 + x_4 &= 3 \\ x_1 + 3x_2 + 3x_3 + 2x_4 &= 2 \\ x_1 + 2x_2 + k^2 x_4 &= k+2 \end{aligned}$$

have

(a) no solution,

(b) exactly one solution,

(c) infinitely many solutions?

Give the solutions when they exist.

Solution. Forward elimination:

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 1 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 3 & 3 & 2 & 2 \\ 1 & 2 & 0 & k^2 & k+2 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 1 \\ 0 & 1 & 3 & 1 & 1 \\ 0 & 1 & 4 & 2 & 1 \\ 0 & 0 & 1 & k^2 & k+1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 1 \\ 0 & 1 & 3 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & k^2 & k+1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 1 \\ 0 & 1 & 3 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & k^2-1 & k+1 \end{array}\right)$$

Row 4 reads $(k^2-1)x_4 = k+1$, where $k^2 - 1 = (k-1)(k+1)$.

(a) $k = 1$: row 4 reads $(0\ 0\ 0\ 0 \mid 2)$, that is, $0 = 2$; so there is no solution (Theorem 2.26).

(b) $k \neq \pm 1$: dividing row 4 by k^2-1 gives an echelon form with pivots in columns $1, 2, 3, 4$ and none in the last column, so there is exactly one solution (Theorems 2.26 and 2.27(b)). Back substitution: $x_4 = \frac{k+1}{k^2-1} = \frac{1}{k-1}$; $x_3 = -x_4 = -\frac{1}{k-1}$; $x_2 = 1 - 3x_3 - x_4 = \frac{k+1}{k-1}$; $x_1 = 1 - 2x_2 + x_3 = -\frac{k+4}{k-1}$. So $x = \frac{1}{k-1}(-k-4,\ k+1,\ -1,\ 1)'$; for $k = 2$, $x = (-6, 3, -1, 1)'$, and indeed $-6+6+1 = 1$, $-12+15-1+1 = 3$, $-6+9-3+2 = 2$, $-6+6+4 = 4 = k+2$.

(c) $k = -1$: row 4 is zero, and $x_4 = t$ is free, so there are infinitely many solutions (Theorems 2.26 and 2.27(b)): $x_3 = -t$, $x_2 = 1 - 3x_3 - x_4 = 1 + 2t$, $x_1 = 1 - 2x_2 + x_3 = -1 - 5t$, that is, $x = (-1, 1, 0, 0)' + t\,(-5, 2, -1, 1)'$ $(t \in \mathbb{R})$.

Exercise 2.23 (cf. [6, Exercise 6.47])**.** For which values of k and h does the system

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 1 \\ x_1 + 2x_2 + 2x_3 + 2x_4 &= 2 \\ x_1 + 2x_2 + 3x_3 + 3x_4 &= 4 \\ x_1 + 2x_2 + 3x_3 + kx_4 &= h \end{aligned}$$

have no solution, exactly one solution, or infinitely many solutions? Give the solutions when they exist.

Solution. Forward elimination; each arrow subtracts a row from all rows below it:

$$\left(\begin{array}{cccc|c} 1&1&1&1&1\\ 1&2&2&2&2\\ 1&2&3&3&4\\ 1&2&3&k&h \end{array}\right) \xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-R_1,\ R_3\leftarrow R_3-R_1} \left(\begin{array}{cccc|c} 1&1&1&1&1\\ 0&1&1&1&1\\ 0&1&2&2&3\\ 0&1&2&k-1&h-1 \end{array}\right)$$

$$\xrightarrow[R_4\leftarrow R_4-R_2]{R_3\leftarrow R_3-R_2} \left(\begin{array}{cccc|c} 1&1&1&1&1\\ 0&1&1&1&1\\ 0&0&1&1&2\\ 0&0&1&k-2&h-2 \end{array}\right) \xrightarrow{R_4\leftarrow R_4-R_3} \left(\begin{array}{cccc|c} 1&1&1&1&1\\ 0&1&1&1&1\\ 0&0&1&1&2\\ 0&0&0&k-3&h-4 \end{array}\right)$$

- $k \neq 3$: $R_4 \leftarrow \frac{1}{k-3}R_4$ gives an echelon form with pivots in columns $1,2,3,4$ and none in the last column. So there is exactly one solution (Theorems 2.26 and 2.27(b)). Back substitution with $s = \frac{h-4}{k-3}$: $x_4 = s$, $x_3 = 2 - s$, $x_2 = 1 - x_3 - x_4 = -1$, $x_1 = 1 - x_2 - x_3 - x_4 = 0$, that is, $x = (0,-1,2,0)' + s\,(0,0,-1,1)'$.
- $k = 3$, $h \neq 4$: row 4 reads $0 = h - 4 \neq 0$, so there is no solution.
- $k = 3$, $h = 4$: row 4 is 0, and the first three rows form an echelon form with pivots in columns $1,2,3$; $x_4 = t$ is free, so there are infinitely many solutions: $x = (0,-1,2,0)' + t\,(0,0,-1,1)'$ $(t \in \mathbb{R})$.

Check: $(0,-1,2,0)'$ satisfies the first three equations ($0-1+2+0 = 1$, $0-2+4+0 = 2$, $0-2+6+0 = 4$), and $(0,0,-1,1)'$ satisfies their homogeneous versions ($-1+1=0$, $-2+2=0$, $-3+3=0$). In the fourth equation, $(0,-1,2,0)'$ gives $0-2+6+k\cdot 0 = 4$ and $(0,0,-1,1)'$ gives $-3+k$; so for $k \neq 3$ its left side is $4 + s(k-3) = 4 + (h-4) = h$, and for $k = 3$, $h = 4$ it is $4 + t(3-3) = 4 = h$.

Exercise 2.24. (a) Find U^{-1} for $U = \begin{pmatrix} 1&1&2&-1\\ 0&-1&-1&1\\ 0&0&-1&1\\ 0&0&0&1 \end{pmatrix}$ by the formulas of Theorem 2.31(b).

(b) Find L^{-1} for $L = \begin{pmatrix} 1&0&0&0\\ 2&1&0&0\\ -1&2&-1&0\\ 0&1&1&1 \end{pmatrix}$ by row reduction of $[L \mid I_4]$.

(c) Show that $G = \begin{pmatrix} 1&2&-1&1\\ 0&0&3&1\\ 0&0&-1&2\\ 0&0&0&1 \end{pmatrix}$ is not invertible by finding $x \neq 0$ with $Gx = 0$.

Solution. (a) The diagonal entries of $B = U^{-1}$ are $b_{11} = 1$, $b_{22} = -1$, $b_{33} = -1$, $b_{44} = 1$, and $b_{ij} = -\frac{1}{u_{ii}} \sum_{k=i+1}^{j} u_{ik} b_{kj}$ for $i < j$, where $-1/u_{11} = -1$, $-1/u_{22} = 1$ and $-1/u_{33} = 1$:

- column 2: $b_{12} = -u_{12}b_{22} = -(1)(-1) = 1$;
- column 3: $b_{23} = u_{23}b_{33} = (-1)(-1) = 1$, $b_{13} = -(u_{12}b_{23} + u_{13}b_{33}) = -(1-2) = 1$;
- column 4: $b_{34} = u_{34}b_{44} = 1$, $b_{24} = u_{23}b_{34} + u_{24}b_{44} = -1+1 = 0$, $b_{14} = -(u_{12}b_{24} + u_{13}b_{34} + u_{14}b_{44}) = -(0+2-1) = -1$.

So

$$U^{-1} = \begin{pmatrix} 1&1&1&-1\\ 0&-1&1&0\\ 0&0&-1&1\\ 0&0&0&1 \end{pmatrix}$$

Check: $\mathrm{row}_1(U)U^{-1} = (1,\ 1-1,\ 1+1-2,\ -1+0+2-1) = (1,0,0,0)$, and similarly for the other rows.

(b) L is lower triangular, so forward elimination and one scaling already turn the left half into I_4:

$$\left(\begin{array}{cccc|cccc} 1&0&0&0&1&0&0&0\\ 2&1&0&0&0&1&0&0\\ -1&2&-1&0&0&0&1&0\\ 0&1&1&1&0&0&0&1 \end{array}\right) \xrightarrow[R_3\leftarrow R_3+R_1]{R_2\leftarrow R_2-2R_1} \left(\begin{array}{cccc|cccc} 1&0&0&0&1&0&0&0\\ 0&1&0&0&-2&1&0&0\\ 0&2&-1&0&1&0&1&0\\ 0&1&1&1&0&0&0&1 \end{array}\right)$$

$$\xrightarrow[R_4\leftarrow R_4-R_2]{R_3\leftarrow R_3-2R_2} \left(\begin{array}{cccc|cccc} 1&0&0&0&1&0&0&0\\ 0&1&0&0&-2&1&0&0\\ 0&0&-1&0&5&-2&1&0\\ 0&0&1&1&2&-1&0&1 \end{array}\right) \xrightarrow{R_4\leftarrow R_4+R_3} \left(\begin{array}{cccc|cccc} 1&0&0&0&1&0&0&0\\ 0&1&0&0&-2&1&0&0\\ 0&0&-1&0&5&-2&1&0\\ 0&0&0&1&7&-3&1&1 \end{array}\right)$$

$$\xrightarrow{R_3\leftarrow -R_3} \left(\begin{array}{cccc|cccc} 1&0&0&0&1&0&0&0\\ 0&1&0&0&-2&1&0&0\\ 0&0&1&0&-5&2&-1&0\\ 0&0&0&1&7&-3&1&1 \end{array}\right)$$

So

$$L^{-1} = \begin{pmatrix} 1&0&0&0\\ -2&1&0&0\\ -5&2&-1&0\\ 7&-3&1&1 \end{pmatrix}$$

lower triangular with diagonal entries $1, 1, -1, 1$, the reciprocals of those of L (Theorem 2.31(b)). Check: $\text{row}_4(L)L^{-1} = (-2-5+7,\ 1+2-3,\ -1+1,\ 1) = (0,0,0,1)$.

(c) $Gx = 0$ reads $x_1 + 2x_2 - x_3 + x_4 = 0$, $3x_3 + x_4 = 0$, $-x_3 + 2x_4 = 0$, $x_4 = 0$. So $x_4 = 0$, $x_3 = 0$ and $x_1 = -2x_2$; with $x_2 = 1$, $x = (-2, 1, 0, 0)' \neq 0$ and $Gx = (-2+2,\ 0,\ 0,\ 0)' = 0$. Hence G is not invertible (Theorem 2.16), as Theorem 2.31(a) predicts: $g_{22} = 0$.

Exercise 2.25. Six units are measured on two variables: $X_0 = \begin{pmatrix} 2&2&0&0&2&0\\ 2&0&2&0&1&1 \end{pmatrix}' \in \mathbb{R}^{6\times 2}$. Let $M = [\mathbf{1}_6\ X_0] \in \mathbb{R}^{6\times 3}$. Find the mean vector $\bar{x}$, the centered data CX_0, a product Q of elementary matrices with $[\mathbf{1}_6\ CX_0] = MQ$, and Q^{-1}.

Solution.

- $\mathbf{1}'X_0 = (6, 6)$, so $\bar{x} = \frac{1}{6}X_0'\mathbf{1} = (1, 1)'$, and $CX_0 = X_0 - \mathbf{1}\bar{x}'$ (Example 1.17) has the columns $(1, 1, -1, -1, 1, -1)'$ and $(1, -1, 1, -1, 0, 0)'$.
- $[\mathbf{1}_6\ CX_0]$ arises from M by $\text{col}_2 \leftarrow \text{col}_2 - \text{col}_1$ and $\text{col}_3 \leftarrow \text{col}_3 - \text{col}_1$, so (Theorem 2.35)

$$Q = T_{12}(-1)\,T_{13}(-1) = \begin{pmatrix} 1&-1&-1\\ 0&1&0\\ 0&0&1 \end{pmatrix}, \qquad Q^{-1} = T_{13}(1)\,T_{12}(1) = \begin{pmatrix} 1&1&1\\ 0&1&0\\ 0&0&1 \end{pmatrix}$$

  by Theorems 2.13(b) and 2.15(b); Q^{-1} is unit upper triangular, as Theorem 2.31 says.

Problems

Problem 2.1 ([6, Exercise 6.6])**.** Show that the product of two elementary matrices is, in general, not elementary.

Solution. For example,

$$P_{13}D_2(7) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 7 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

This is not elementary: a matrix $D_r(c)$ or $T_{rs}(c)$ differs from I_3 in one entry only, while this product differs from I_3 in five entries; and every P_{rs} has all entries 0 or 1, while this product has the entry 7.

Problem 2.2 ([6, Exercise 4.22(a), (e)])**.** Let $A \in \mathbb{R}^{n\times n}$ be invertible. Show that:

(a) $(\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$ $(\lambda \neq 0)$;

(b) if A and B commute and B is invertible, then A and B^{-1} commute.

Solution. (a) We check both products (Definition 2.12): $(\lambda A)(\frac{1}{\lambda}A^{-1}) = AA^{-1} = I$ and $(\frac{1}{\lambda}A^{-1})(\lambda A) = A^{-1}A = I$. (b) Multiply $AB = BA$ on the left and on the right by B^{-1}: $B^{-1}ABB^{-1} = B^{-1}BAB^{-1}$, that is, $B^{-1}A = AB^{-1}$.

Problem 2.3 ([4, Ch. II, §3, Exercise 18], [7])**.** Let $A \in \mathbb{R}^{n\times n}$.

(a) If $A^2 = 0$, show that $I - A$ is invertible.

(b) If $A^3 = 0$, show that $I - A$ is invertible.

(c) In general, if $A^k = 0$ for some positive integer k, show that $I - A$ is invertible.

(d) Suppose that $A^2 + 2A + I = 0$. Show that A is invertible.

(e) Suppose that $A^3 - A + I = 0$. Show that A is invertible.

Solution. (a) By Theorem 1.3, $(I-A)(I+A) = I+A-A-A^2 = I-A^2$ and $(I+A)(I-A) = I-A^2$. If $A^2 = 0$, both products equal I, so $I - A$ is invertible with $(I - A)^{-1} = I + A$.

(b) By Theorem 1.3, $(I - A)(I + A + A^2) = (I + A + A^2) - (A + A^2 + A^3) = I - A^3$ and in the same way $(I + A + A^2)(I - A) = I - A^3$. If $A^3 = 0$, both products equal I, so $(I - A)^{-1} = I + A + A^2$.

(c) For every positive integer k, by Theorem 1.3,

$$(I - A)(I + A + A^2 + \cdots + A^{k-1}) = (I + A + \cdots + A^{k-1}) - (A + A^2 + \cdots + A^k) = I - A^k$$

and in the same way $(I + A + \cdots + A^{k-1})(I - A) = I - A^k$. If $A^k = 0$, both products equal I, so $I - A$ is invertible with inverse $I + A + \cdots + A^{k-1}$.

(d) $A(-A - 2I) = (-A - 2I)A = -A^2 - 2A = I$, since $A^2 + 2A + I = 0$. So $A^{-1} = -A - 2I$.

(e) $A(-A^2 + I) = (-A^2 + I)A = -A^3 + A = I$, since $A^3 - A + I = 0$. So $A^{-1} = I - A^2$.

Problem 2.4 ([1, §1.6, Exercise 10])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$, with $n < m$. Show that AB is not invertible.

Solution. AB is an $m \times m$ matrix. By Theorem 2.16 ((i) $\Leftrightarrow$ (iii)), it suffices to find $x \neq 0$ with $(AB)x = 0$.

- The system $Bx = 0$ has n equations in the m unknowns $x_1, \ldots, x_m$, and $n < m$. So it has a nontrivial solution $x \in \mathbb{R}^m$ (Theorem 2.11(a)).
- Then $(AB)x = A(Bx) = A0 = 0$ (Theorem 1.3), with $x \neq 0$. Hence AB is not invertible.

Problem 2.5 ([10, Problem 7.3])**.** Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ satisfy $|a_{ii}| > \sum_{k\neq i}|a_{ik}|$ for $i = 1, \dots, n$. Show that A is invertible.

Solution. By Theorem 2.16 ((iii) $\Rightarrow$ (i)), it suffices to show that $Ax = 0$ only for $x = 0$. Suppose that $Ax = 0$ with $x \neq 0$, and choose an index s with $|x_s| \geqslant |x_k|$ for $k = 1, \dots, n$; then $|x_s| > 0$.

- Equation s of $Ax = 0$ reads $\sum_k a_{sk}x_k = 0$, that is, $a_{ss}x_s = -\sum_{k\neq s} a_{sk}x_k$.
- By the triangle inequality, the choice of s, and the hypothesis for $i = s$ (with $|x_s| > 0$),

$$|a_{ss}|\,|x_s| = \left|\sum_{k\neq s} a_{sk}x_k\right| \leqslant \sum_{k\neq s}|a_{sk}|\,|x_k| \leqslant |x_s|\sum_{k\neq s}|a_{sk}| < |x_s|\,|a_{ss}| \text{ ✘}$$

Hence $Ax = 0$ only for $x = 0$, and A is invertible.

Problem 2.6 ([10, Problems 21.1, 21.2])**.** Let $A \in \mathbb{R}^{n\times n}$ be skew-symmetric (Definition 1.26).

(a) Show that $I_n + A$ is invertible.

(b) Show that if A is invertible, then A^{-1} is skew-symmetric.

Solution. (a) By Theorem 2.16 ((iii) $\Rightarrow$ (i)), it suffices to show that $(I_n + A)x = 0$ only for $x = 0$.

- For every $x \in \mathbb{R}^n$, the 1×1 matrix $x'Ax$ equals its transpose, so by Theorem 1.9 and $A' = -A$, $x'Ax = (x'Ax)' = x'A'x = -x'Ax$. Hence $x'Ax = 0$.
- If $(I_n + A)x = 0$, then by Theorems 1.3 and 1.9, $0 = x'(I_n + A)x = x'x + x'Ax = x'x = \sum_i x_i^2$, so $x = 0$.

(b)

- $-A$ is invertible with $(-A)^{-1} = -A^{-1}$, since $(-A)(-A^{-1}) = AA^{-1} = I_n$ and $(-A^{-1})(-A) = A^{-1}A = I_n$ (Theorem 1.3, Definition 2.12).
- By Theorem 2.13(c) and $A' = -A$, $(A^{-1})' = (A')^{-1} = (-A)^{-1} = -A^{-1}$. So A^{-1} is skew-symmetric.

Problem 2.7 ([10, Problem 2.12])**.** Find all invertible matrices $A \in \mathbb{R}^{n\times n}$ such that all entries of A and all entries of A^{-1} are nonnegative.

Solution. These are exactly the matrices $A = (a_{ij})$ with nonnegative entries that have exactly one nonzero entry in each row and in each column; the inverse of such an A is A' with every nonzero entry replaced by its reciprocal.

- Rows. Let A be invertible, with nonnegative entries, and let $B = A^{-1} = (b_{ij})$ have nonnegative entries. Suppose that row i of A has two positive entries, $a_{ir} > 0$ and $a_{is} > 0$ with $r \neq s$. For $j \neq i$, the (i, j) entry of $AB = I_n$ is $\sum_k a_{ik}b_{kj} = 0$; its terms are $\geqslant 0$, so all of them vanish, and $a_{ir}b_{rj} = 0$, $a_{is}b_{sj} = 0$ give $b_{rj} = b_{sj} = 0$. Thus $\mathrm{row}_r(B) = b_{ri}e_i'$ and $\mathrm{row}_s(B) = b_{si}e_i'$: rows r and s of B are multiples of e_i'.
- This is impossible. By Theorem 1.4(c), $\mathrm{row}_r(BA) = \mathrm{row}_r(B)A = b_{ri}\,\mathrm{row}_i(A)$ and $\mathrm{row}_s(BA) = b_{si}\,\mathrm{row}_i(A)$, and as $BA = I_n$, these rows are e_r' and e_s'. So $b_{ri} \neq 0$, and $e_s' = b_{si}\,\mathrm{row}_i(A) = (b_{si}/b_{ri})\,e_r'$; but the sth entry of e_s' is 1, while that of $(b_{si}/b_{ri})\,e_r'$ is 0, as $r \neq s$ ✘ Hence every row of A has at most one nonzero entry.
- No row of A is zero: if $\mathrm{row}_i(A) = 0$, then $\mathrm{row}_i(AB) = \mathrm{row}_i(A)B = 0$, while $\mathrm{row}_i(I_n) = e_i' \neq 0$ (Theorem 1.4(c)). So every row of A has exactly one nonzero entry.
- Columns. A' has nonnegative entries and is invertible, with $(A')^{-1} = (A^{-1})' = B'$, which has nonnegative entries (Theorem 2.13(c)). By the first three items, applied to A', every row of A', that is, every column of A, has exactly one nonzero entry.

- Converse. Let A have nonnegative entries, with exactly one nonzero entry in each row and in each column, and let $a_{i\sigma(i)}$ be the nonzero (hence positive) entry of row i. If $i \neq l$, then $\sigma(i) \neq \sigma(l)$, as column $\sigma(i)$ has only one nonzero entry. Let $B = (b_{kj}) \in \mathbb{R}^{n\times n}$ have the entries $b_{\sigma(i)i} = 1/a_{i\sigma(i)}$ $(i = 1, \ldots, n)$ and all other entries 0; it is A' with every nonzero entry replaced by its reciprocal, and its entries are nonnegative.
- In $(AB)_{ij} = \sum_k a_{ik}b_{kj}$ only the term $k = \sigma(i)$ can be nonzero, so $(AB)_{ij} = a_{i\sigma(i)}b_{\sigma(i)j}$. Here $b_{\sigma(i)j} \neq 0$ only if $\sigma(i) = \sigma(j)$, that is, $j = i$, and $a_{i\sigma(i)}b_{\sigma(i)i} = 1$. So $AB = I_n$, and by Theorem 2.16, A is invertible with $A^{-1} = B$, whose entries are nonnegative.

Chapter 3. Vector spaces, subspaces, span, and bases

Vector spaces

Definition 3.1 ([1, §2.1], [4, Ch. I, §1], [2, Definition 1.1], [3, §1.2])**.** A (real) **vector space** is a set V, whose elements are called **vectors**, with an **addition** that gives $u+v \in V$ for $u, v \in V$ and a **scalar multiplication** that gives $cv \in V$ for $c \in \mathbb{R}$ and $v \in V$, such that for all $u, v, w \in V$ and $c, d \in \mathbb{R}$:

(V1) $u+v = v+u$ and (V2) $u+(v+w) = (u+v)+w$;

(V3) there is a **zero vector** $0 \in V$ with $v+0 = v$ for every $v \in V$;

(V4) for each $v \in V$ there is $-v \in V$ with $v+(-v) = 0$;

(V5) $1v = v$ and (V6) $(cd)v = c(dv)$;

(V7) $c(u+v) = cu+cv$ and (V8) $(c+d)v = cv+dv$.

A vector $c_1v_1 + \cdots + c_kv_k$ ($c_i \in \mathbb{R}$) is a **linear combination** of $v_1, \ldots, v_k$.

From these rules, $0v = 0$ and $(-1)v = -v$: indeed $0v = (0+0)v = 0v+0v$, and adding $-(0v)$ to both sides gives $0 = 0v$; then $v+(-1)v = (1+(-1))v = 0v = 0$, and $-v$ is the only vector w with $v+w = 0$, since such a w equals $w+(v+(-v)) = (w+v)+(-v) = 0+(-v) = -v$. Likewise $c0 = 0$, and $cv = 0$ only if $c = 0$ or $v = 0$ (if $c \neq 0$, then $v = c^{-1}(cv) = c^{-1}0 = 0$).

Example 3.2 ([1, §2.1, Example 2])**.** $\mathbb{R}^n$, $\mathbb{R}^{1\times n}$ and $\mathbb{R}^{m\times n}$, with the entrywise sum and scalar multiple of Definition 1.1, are vector spaces: (V1)–(V8) hold entry by entry, the zero vector is the zero matrix, and $-A = (-1)A$. In this course every vector space is one of these or a subspace of one of them.

Subspaces

Definition 3.3 ([1, §2.2], [2, Definition 1.2])**.** A **subspace** of a vector space V is a subset $W \subseteq V$ that is itself a vector space with the addition and scalar multiplication of V.

Theorem 3.4 ([1, §2.2, Theorem 1])**.** For a subset W of a vector space V the following are equivalent:

(a) W is a subspace of V;

(b) $0 \in W$, and $u+v \in W$ and $cu \in W$ for all $u, v \in W$ and $c \in \mathbb{R}$ (W is **closed under addition and scalar multiplication**);

(c) $0 \in W$, and $cu+v \in W$ for all $u, v \in W$ and $c \in \mathbb{R}$.

Proof. (a) $\Rightarrow$ (b): the operations of the vector space W are those of V and give vectors of W; W has a zero vector, so it contains some w, and then $0 = 0w \in W$. (b) $\Rightarrow$ (c): $cu \in W$, so $cu+v \in W$. (c) $\Rightarrow$ (a): taking $v = 0$ gives $cu \in W$, and $c = 1$ gives $u+v \in W$; so the operations of V give operations on W. The rules (V1), (V2), (V5)–(V8) hold in W because they hold in V; (V3) holds since $0 \in W$, and (V4) since $-w = (-1)w \in W$. □

So, to show that W is a subspace, check that it contains 0 and is closed under the two operations; to show that it is not, exhibit one failure.

Example 3.5 ([1, §2.2, Examples 6, 7], [2, §1.1], [6, Exercise 3.10(b)])**.** (a) $\{0\}$ and V itself are subspaces of V.

(b) In $\mathbb{R}^n$, $W = \{x : x_1 = 0\}$ is a subspace: $0 \in W$, and $(cx+y)_1 = cx_1 + y_1 = 0$ for $x, y \in W$. For $n \geqslant 2$, $\{x : x_1 = 1 + x_2\}$ is not, since it does not contain 0.

(c) In $\mathbb{R}^2$, $\{(x,y)' : 2x-5y=0\}$ is a subspace: it contains 0, and $2(cx_1+x_2)-5(cy_1+y_2) = c(2x_1-5y_1)+(2x_2-5y_2)$ for $(x_1,y_1)'$ and $(x_2,y_2)'$ in it; but for $k\neq 0$ the set $\{(x,y)' : ax+by=k\}$ is not, since it does not contain 0.

(d) The symmetric $n\times n$ matrices form a subspace of $\mathbb{R}^{n\times n}$: $0'=0$, and $(cA+B)' = cA'+B' = cA+B$ for symmetric A, B. So do the skew-symmetric matrices (Definition 1.26), those with $A'=-A$: $0'=-0$, and $(cA+B)' = cA'+B' = -(cA+B)$ if $A'=-A$ and $B'=-B$.

(e) For $A\in\mathbb{R}^{m\times n}$, the solution set $\{x\in\mathbb{R}^n : Ax=0\}$ of a homogeneous system is a subspace of $\mathbb{R}^n$: $A0=0$, and $A(cu+v) = cAu+Av = 0$ if $Au=Av=0$. For $b\neq 0$, the solutions of $Ax=b$ do not form a subspace ($A0=0\neq b$).

(f) In statistics, the **centered** vectors $W_0 = \{x\in\mathbb{R}^n : \mathbf{1}'x=0\}$, whose entries add up to 0, form a subspace: this is (e) with the $1\times n$ matrix $A=\mathbf{1}'$. For every x, the deviation vector Cx of Example 1.16 lies in W_0, since $\mathbf{1}'Cx = \mathbf{1}'C'x = (C\mathbf{1})'x = 0$ (Proposition 1.15: $C'=C$, $C\mathbf{1}=0$).

(g) The set $\{(x,y,z)'\in\mathbb{R}^3 : xy=0\}$ is not a subspace: $(1,0,0)'$ and $(0,1,0)'$ lie in it, but their sum $(1,1,0)'$ does not. It is the union of the subspaces $\{x=0\}$ and $\{y=0\}$; a union of subspaces need not be a subspace.

Span

Definition 3.6 ([4, Ch. I, §1])**.** Let $v_1,\dots,v_k$ be vectors of a vector space V. Their **span** is the set of all their linear combinations,

$$\operatorname{span}\{v_1,\dots,v_k\} = \{c_1v_1+\cdots+c_kv_k :\ c_1,\dots,c_k\in\mathbb{R}\}$$

and $\operatorname{span}\varnothing = \{0\}$. If $\operatorname{span}\{v_1,\dots,v_k\} = W$, we say that $v_1,\dots,v_k$ **span** W.

Theorem 3.7 ([1, §2.2, Theorem 3])**.** $\operatorname{span}\{v_1,\dots,v_k\}$ is a subspace of V that contains $v_1,\dots,v_k$, and it is contained in every subspace that contains $v_1,\dots,v_k$: it is the smallest subspace containing them.

Proof. $0 = 0v_1+\cdots+0v_k$ lies in the span, and $c\sum_i a_iv_i + \sum_i b_iv_i = \sum_i(ca_i+b_i)v_i$; so the span is a subspace (Theorem 3.4(c)). It contains $v_j = 1v_j + \sum_{i\neq j}0v_i$. A subspace that contains $v_1,\dots,v_k$ contains every c_iv_i and every sum of these, by closure. □

For vectors $v_1,\dots,v_k\in\mathbb{R}^m$, put $V = [v_1\ \cdots\ v_k]\in\mathbb{R}^{m\times k}$. Since $Vc = c_1v_1+\cdots+c_kv_k$ (Theorem 1.4(b)),

$$\operatorname{span}\{v_1,\dots,v_k\} = \{Vc :\ c\in\mathbb{R}^k\},\qquad b\in\operatorname{span}\{v_1,\dots,v_k\} \iff \text{the system } Vc=b \text{ is consistent}$$

So membership in a span is decided by row-reducing $[V \mid b]$ (Proposition 2.7, Theorem 2.26). For instance, by Example 2.10, a vector $y\in\mathbb{R}^4$ lies in the span of the columns of the matrix A there if and only if $y_1-y_2-y_3+y_4=0$.

Example 3.8 ([1, §2.2, Example 8])**.** Let $W = \operatorname{span}\{v_1,v_2,v_3\}\subseteq\mathbb{R}^5$, where $v_1=(1,2,0,3,0)'$, $v_2=(0,0,1,4,0)'$ and $v_3=(0,0,0,0,1)'$. Which $y=(y_1,\dots,y_5)'$ lie in W?

Solution. Row-reduce $[v_1\ v_2\ v_3 \mid y]$, keeping the y_i as letters:

$$\left(\begin{array}{ccc|c} 1&0&0&y_1\\ 2&0&0&y_2\\ 0&1&0&y_3\\ 3&4&0&y_4\\ 0&0&1&y_5 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-3R_1]{R_2\leftarrow R_2-2R_1} \left(\begin{array}{ccc|c} 1&0&0&y_1\\ 0&0&0&y_2-2y_1\\ 0&1&0&y_3\\ 0&4&0&y_4-3y_1\\ 0&0&1&y_5 \end{array}\right) \xrightarrow{R_4\leftarrow R_4-4R_3} \left(\begin{array}{ccc|c} 1&0&0&y_1\\ 0&0&0&y_2-2y_1\\ 0&1&0&y_3\\ 0&0&0&y_4-3y_1-4y_3\\ 0&0&1&y_5 \end{array}\right)$$

$$\xrightarrow{R_2\leftrightarrow R_3}\left(\begin{array}{ccc|c}1&0&0&y_1\\0&1&0&y_3\\0&0&0&y_2-2y_1\\0&0&0&y_4-3y_1-4y_3\\0&0&1&y_5\end{array}\right)\xrightarrow{R_3\leftrightarrow R_5}\left(\begin{array}{ccc|c}1&0&0&y_1\\0&1&0&y_3\\0&0&1&y_5\\0&0&0&y_4-3y_1-4y_3\\0&0&0&y_2-2y_1\end{array}\right)$$

By Proposition 2.7, $y \in W$ if and only if

$$y_2 = 2y_1, \qquad y_4 = 3y_1 + 4y_3$$

and then $y = y_1v_1 + y_3v_2 + y_5v_3$. For example, $(-3,-6,1,-5,2)' \in W$, since $-6 = 2(-3)$ and $-5 = 3(-3)+4\cdot 1$; indeed it equals $-3v_1+v_2+2v_3$. But $(2,4,6,7,8)' \notin W$, since $3\cdot 2+4\cdot 6 = 30 \neq 7$.

Example 3.9. $\operatorname{span}\{\mathbf{1}_n\} = \{c\mathbf{1}_n : c \in \mathbb{R}\}$ is the subspace of constant vectors. Every $x \in \mathbb{R}^n$ is the sum of a constant vector and a centered vector:

$$x = \bar{x}\mathbf{1} + Cx, \qquad \bar{x}\mathbf{1} \in \operatorname{span}\{\mathbf{1}\}, \quad Cx \in W_0$$

since $Cx = x - \bar{x}\mathbf{1}$ (Example 1.16) and $Cx \in W_0$ (Example 3.5(f)). For $x = (1,3,5,7,9)'$: $x = 5\mathbf{1} + (-4,-2,0,2,4)'$.

Theorem 3.10 ([5, Lemmas 4.2.1, 4.2.2, 4.3.1], [8, Ch. 4, Exercise 6])**.** Let $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m\times n}$ and $B = [b_1 \ \cdots \ b_p] \in \mathbb{R}^{m\times p}$.

(a) $\operatorname{span}\{b_1,\dots,b_p\} \subseteq \operatorname{span}\{a_1,\dots,a_n\}$ if and only if every b_j lies in $\operatorname{span}\{a_1,\dots,a_n\}$, and if and only if $B = AF$ for some $F \in \mathbb{R}^{n\times p}$.

(b) $\operatorname{span}\{b_1,\dots,b_p\} = \operatorname{span}\{a_1,\dots,a_n\}$ if and only if $B = AF$ and $A = BG$ for some $F \in \mathbb{R}^{n\times p}$ and $G \in \mathbb{R}^{p\times n}$.

(c) If every b_j lies in $\operatorname{span}\{a_1,\dots,a_n\}$, then $\operatorname{span}\{a_1,\dots,a_n,b_1,\dots,b_p\} = \operatorname{span}\{a_1,\dots,a_n\}$. In particular, deleting from a list of vectors a vector that is a linear combination of the others does not change their span.

Proof. (a)

- If $\operatorname{span}\{b_1,\dots,b_p\} \subseteq \operatorname{span}\{a_1,\dots,a_n\}$, then each b_j, which lies in $\operatorname{span}\{b_1,\dots,b_p\}$ (Theorem 3.7), lies in $\operatorname{span}\{a_1,\dots,a_n\}$.
- If every b_j lies in $\operatorname{span}\{a_1,\dots,a_n\}$, this subspace contains $b_1,\dots,b_p$, so it contains $\operatorname{span}\{b_1,\dots,b_p\}$ (Theorem 3.7).
- For $F = [f_1 \ \cdots \ f_p] \in \mathbb{R}^{n\times p}$, $\operatorname{col}_j(AF) = Af_j$ (Theorem 1.4(a)), and Af_j is the combination of $a_1,\dots,a_n$ with the coefficients f_j (Theorem 1.4(b)). So $B = AF$ for some F if and only if each b_j is a linear combination of $a_1,\dots,a_n$.

(b) The two spans are equal if and only if each is contained in the other; apply (a) to both inclusions.

(c)

- Let $b_j = k_{1j}a_1 + \cdots + k_{nj}a_n$ $(1 \leqslant j \leqslant p)$. For all scalars x_i and y_j,

$$\sum_{i=1}^{n} x_ia_i + \sum_{j=1}^{p} y_jb_j = \sum_{i=1}^{n}\Big(x_i + \sum_{j=1}^{p} k_{ij}y_j\Big)a_i$$

  so every linear combination of $a_1,\dots,a_n,b_1,\dots,b_p$ is one of $a_1,\dots,a_n$.
- Conversely, $\sum_i x_ia_i = \sum_i x_ia_i + \sum_j 0\,b_j$.
- For the last sentence, let the a_i be the remaining vectors and b_1 the deleted one ($p = 1$); the order of the vectors does not affect their span. □

Linear independence

Definition 3.11 ([1, §2.3], [4, Ch. I, §2], [2, Definition 1.3], [3, §1.3])**.** Vectors $v_1, \dots, v_k$ of a vector space are **linearly dependent** if $c_1v_1 + \cdots + c_kv_k = 0$ for some scalars $c_1, \dots, c_k$, not all 0 (a **linear relation**). Otherwise they are **linearly independent**: $c_1v_1 + \cdots + c_kv_k = 0$ only for $c_1 = \cdots = c_k = 0$. The empty list is independent.

Proposition 3.12 (cf. [1, §2.3, Example 14])**.** Let $v_1, \dots, v_k \in \mathbb{R}^m$ and $V = [v_1 \ \cdots \ v_k] \in \mathbb{R}^{m\times k}$. Then $v_1, \dots, v_k$ are linearly independent if and only if $Vc = 0$ has only the solution $c = 0$, that is, if and only if every column of the RREF of V is a pivot column (no free unknowns). In particular, more than m vectors of $\mathbb{R}^m$ are linearly dependent.

Proof. $Vc = c_1v_1 + \cdots + c_kv_k$, so the linear relations among the v_i are the solutions of $Vc = 0$. This system is consistent ($c = 0$), so by Theorem 2.27(b) $c = 0$ is its only solution exactly when there is no free unknown. If $k > m$, then $Vc = 0$ has a nontrivial solution (Theorem 2.11(a)). □

Proposition 3.13 ([1, §2.3], [5, §3.2, Lemma 3.2.1])**.** Let $v_1, \dots, v_k$ be vectors of a vector space.

(a) If some $v_i = 0$, or $v_i = v_j$ for some $i \neq j$, then $v_1, \dots, v_k$ are dependent. A single vector v is independent if and only if $v \neq 0$.

(b) A sublist of an independent list is independent.

(c) For $k \geqslant 2$, $v_1, \dots, v_k$ are dependent if and only if some v_j is a linear combination of the others. In particular, two vectors are dependent if and only if one of them is a multiple of the other.

Proof. (a) $1 \cdot 0 = 0$ and $v_i - v_j = 0$ are linear relations with a coefficient 1; and if $v \neq 0$, then $cv = 0$ forces $c = 0$. (b) A relation among some of the vectors is a relation among all of them, with coefficient 0 for the others. (c) If $v_j = \sum_{i\neq j} x_iv_i$, then $v_j - \sum_{i\neq j} x_iv_i = 0$ is a relation with coefficient 1 on v_j. Conversely, if $\sum_i x_iv_i = 0$ with $x_j \neq 0$, then $v_j = \sum_{i\neq j}(-x_i/x_j)v_i$. □

Proposition 3.14 ([5, Lemma 3.2.2], [1, §2.3, Lemma before Theorem 5])**.** Let $v_1, \dots, v_k$ be vectors of a vector space V.

(a) If $k \geqslant 2$ and $v_1 \neq 0$, then $v_1, \dots, v_k$ are linearly dependent if and only if some v_j ($2 \leqslant j \leqslant k$) is a linear combination of the preceding vectors $v_1, \dots, v_{j-1}$.

(b) If $v_1, \dots, v_k$ are linearly independent and $v \in V$ does not lie in $\operatorname{span}\{v_1, \dots, v_k\}$, then $v_1, \dots, v_k, v$ are linearly independent.

Proof. (a)

- If $v_j = \sum_{i<j} x_iv_i$, then v_j is a combination of the other vectors (with coefficient 0 for $i > j$), so the list is dependent (Proposition 3.13(c)).
- Conversely, let the list be dependent, and let j be the least index such that $v_1, \dots, v_j$ are dependent. Since $v_1 \neq 0$ is independent (Proposition 3.13(a)), $j \geqslant 2$.
- There are $x_1, \dots, x_j$, not all 0, with $x_1v_1 + \cdots + x_jv_j = 0$. Here $x_j \neq 0$, since otherwise $v_1, \dots, v_{j-1}$ would be dependent, contrary to the choice of j.
- Hence $v_j = (-x_1/x_j)v_1 + \cdots + (-x_{j-1}/x_j)v_{j-1}$.

(b)

- Let $c_1v_1 + \cdots + c_kv_k + bv = 0$. Then $b = 0$: otherwise $v = (-c_1/b)v_1 + \cdots + (-c_k/b)v_k$ would lie in $\operatorname{span}\{v_1, \dots, v_k\}$.
- So $c_1v_1 + \cdots + c_kv_k = 0$, and every $c_i = 0$, as $v_1, \dots, v_k$ are independent. □

Example 3.15. Let a_1, a_2, a_3, a_4 be the columns of the matrix A of Example 2.6:

$$a_1 = \begin{pmatrix}1\\1\\2\end{pmatrix}, \quad a_2 = \begin{pmatrix}1\\1\\1\end{pmatrix}, \quad a_3 = \begin{pmatrix}1\\2\\5\end{pmatrix}, \quad a_4 = \begin{pmatrix}-2\\-3\\-4\end{pmatrix}$$

Four vectors of $\mathbb{R}^3$ are dependent (Proposition 3.12). A relation comes from the solutions $x = t(-2,3,1,1)'$ of $Ax = 0$ found there:

$$-2a_1 + 3a_2 + a_3 + a_4 = 0, \qquad \text{so} \qquad a_4 = 2a_1 - 3a_2 - a_3$$

On the other hand a_1, a_2, a_3 are independent: the same row operations take $[a_1\ a_2\ a_3]$ to the first three columns of the RREF found there (row operations act on each column separately: Theorems 2.15(c) and 1.4(a)), which form I_3, so every column is a pivot column.

Example 3.16 ([1, §2.3, Exercises 2, 4])**.** (a) Let $u_1 = (1,1,2,4)'$, $u_2 = (2,-1,-5,2)'$, $u_3 = (1,-1,-4,0)'$ and $u_4 = (2,1,1,6)'$. Here $u_1 \neq 0$, and u_2 is not a linear combination of the preceding vector u_1, that is, not a multiple of u_1: the only multiple cu_1 with first entry 2 is $2u_1$, whose second entry is $2 \neq -1$. So u_1, u_2 are independent (Proposition 3.14(a) with $k = 2$). But u_3 is a linear combination of the preceding vectors:

$$-\tfrac{1}{3}u_1 + \tfrac{2}{3}u_2 = \tfrac{1}{3}\begin{pmatrix}-1+4\\-1-2\\-2-10\\-4+4\end{pmatrix} = \begin{pmatrix}1\\-1\\-4\\0\end{pmatrix} = u_3$$

By Proposition 3.14(a) ($j = 3$; with $k = 3$ and with $k = 4$), u_1, u_2, u_3 and $u_1, \dots, u_4$ are linearly dependent. Since the single vector $u_1 \neq 0$ is independent (Proposition 3.13(a)), and so are u_1, u_2, 3 is the least j for which $u_1, \dots, u_j$ are dependent.

(b) Let $v_1 = (1,0,-1)'$, $v_2 = (1,2,1)'$ and $v_3 = (0,-3,2)'$. The single vector $v_1 \neq 0$ is independent (Proposition 3.13(a)).

- $v_2 \notin \operatorname{span}\{v_1\}$: every multiple of v_1 has second entry 0, and v_2 has second entry 2. So v_1, v_2 are independent (Proposition 3.14(b)).
- $v_3 \notin \operatorname{span}\{v_1, v_2\}$: if $v_3 = xv_1 + yv_2$, the second entries give $2y = -3$, so $y = -\frac{3}{2}$; the first entries give $x + y = 0$, so $x = \frac{3}{2}$; but then the third entry of $xv_1 + yv_2$ is $-x + y = -3 \neq 2$. So v_1, v_2, v_3 are independent (Proposition 3.14(b) again).

Bases

Definition 3.17 ([1, §2.3], [4, Ch. I, §2])**.** Vectors $v_1, \dots, v_k$ of a vector space V form a **basis** of a subspace W if they lie in W, are linearly independent, and span W.

- The unit vectors $e_1, \dots, e_n$ form a basis of $\mathbb{R}^n$, the **standard basis**: $x = x_1e_1 + \cdots + x_ne_n$ for every $x \in \mathbb{R}^n$, and this is 0 only for $x = 0$.
- Let $E_{ij} \in \mathbb{R}^{m\times n}$ have (i,j) entry 1 and all other entries 0 (two indices; not the elementary matrices $E_1, E_2, \dots$ of Theorem 2.15(c)). The mn matrices E_{ij} form a basis of $\mathbb{R}^{m\times n}$: $A = \sum_i \sum_j a_{ij}E_{ij}$, and $\sum_i \sum_j c_{ij}E_{ij} = (c_{ij})$ is 0 only if every $c_{ij} = 0$.
- The empty list is a basis of $\{0\}$.

Theorem 3.18 ([4, Ch. I, Theorem 2.1])**.** If $v_1, \dots, v_k$ form a basis of W, every $w \in W$ can be written as $w = c_1v_1 + \cdots + c_kv_k$ in exactly one way; the numbers $c_1, \dots, c_k$ are the **coordinates** of w in this basis.

Proof. Such an expression exists because the v_i span W. If $\sum_i x_i v_i = \sum_i y_i v_i$, then $\sum_i (x_i - y_i)v_i = 0$, and independence gives $x_i = y_i$ for every i. □

Example 3.19 ([1, §2.3, Example 14])**.** If $P \in \mathbb{R}^{n\times n}$ is invertible, its columns $p_1, \dots, p_n$ form a basis of $\mathbb{R}^n$, and the coordinates of $y \in \mathbb{R}^n$ are $x = P^{-1}y$. Indeed $Px = \sum_j x_j p_j$; $Px = 0$ only for $x = 0$ (Theorem 2.16), so the columns are independent; and $y = P(P^{-1}y)$, so they span. For example, the matrix A of Example 2.20 is invertible, its inverse being computed there; so its columns $(1,-1,2,1)'$, $(-1,1,-1,-1)'$, $(1,-2,2,1)'$, $(-1,1,-1,0)'$ form a basis of $\mathbb{R}^4$. The coordinates of $y = (1,0,3,3)'$ are $A^{-1}y = (3,-1,-1,2)'$, the solution of $Ax = y$ found in Example 2.28: $y = 3(1,-1,2,1)' - (-1,1,-1,-1)' - (1,-2,2,1)' + 2(-1,1,-1,0)'$.

Example 3.20 (cf. [2, Example 1.6])**.** Let $t = (1,2,3,4)'$ and $W = \operatorname{span}\{\mathbf{1}_4, t\}$, a subspace of $\mathbb{R}^4$. Any two of the vectors $\mathbf{1}_4$, t, $\mathbf{1}_4 + t = (2,3,4,5)'$ form a basis of W.

- Each pair is independent: none of the three vectors is 0 or a multiple of another, since their first two entries are $(1,1)$, $(1,2)$, $(2,3)$, no two of them proportional (Proposition 3.13(c)).
- Each pair spans W: the pair $\mathbf{1}_4, t$ by definition, and for $w = a\mathbf{1}_4 + bt \in W$,

$$w = (a-b)\,\mathbf{1}_4 + b\,(\mathbf{1}_4 + t) = (b-a)\,t + a\,(\mathbf{1}_4 + t)$$

All three vectors together also span W, but they are dependent: $\mathbf{1}_4 + t - (\mathbf{1}_4 + t) = 0$ is a linear relation with coefficients $1, 1, -1$. So they do not form a basis. A subspace has many bases.

Example 3.21 ([5, §4.3b])**.** The symmetric 2×2 matrices (for instance 2×2 covariance matrices) form a subspace of $\mathbb{R}^{2\times 2}$ (Example 3.5(d)). The matrices E_{11}, $E_{12} + E_{21}$, E_{22} form a basis of it: they are symmetric, every symmetric matrix is

$$\begin{pmatrix} a & b \\ b & d \end{pmatrix} = a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

and this combination is 0 only if $a = b = d = 0$.

Example 3.22. In $W_0 = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$, the vectors $w_1 = (1,-1,0,0)'$, $w_2 = (0,1,-1,0)'$ and $w_3 = (0,0,1,-1)'$ form a basis. They lie in W_0. They span W_0: for $x \in W_0$, $x_4 = -x_1 - x_2 - x_3$, so

$$x_1 w_1 + (x_1 + x_2)w_2 + (x_1 + x_2 + x_3)w_3 = \big(x_1,\ x_2,\ x_3,\ -x_1 - x_2 - x_3\big)' = x$$

They are independent: $c_1 w_1 + c_2 w_2 + c_3 w_3 = (c_1,\ c_2 - c_1,\ c_3 - c_2,\ -c_3)' = 0$ forces $c_1 = 0$, then $c_2 = 0$, then $c_3 = 0$.

Coordinates and change of basis

Definition 3.23 ([1, §2.4], [4, Ch. I, §2])**.** Let W be a subspace of a vector space V. An **ordered basis** of W is a list $\mathcal{B} = (v_1, \dots, v_k)$ of vectors that form a basis of W, taken in this order. Each $w \in W$ is $w = c_1 v_1 + \cdots + c_k v_k$ for exactly one choice of $c_1, \dots, c_k$ (Theorem 3.18), and

$$[w]_{\mathcal{B}} = (c_1, \dots, c_k)' \in \mathbb{R}^k$$

is the **coordinate vector** of w relative to $\mathcal{B}$. For the **standard ordered basis** $\mathcal{E} = (e_1, \dots, e_n)$ of $\mathbb{R}^n$, $[x]_{\mathcal{E}} = x$, since $x = x_1 e_1 + \cdots + x_n e_n$.

If $W \subseteq \mathbb{R}^n$ and $\mathcal{B}$ consists of the columns of $X = [v_1\ \cdots\ v_k] \in \mathbb{R}^{n\times k}$, then $Xc = c_1 v_1 + \cdots + c_k v_k$ (Theorem 1.4(b)); so $[w]_{\mathcal{B}}$ is the unique solution c of the system $Xc = w$, found by row-reducing $[X \mid w]$.

Proposition 3.24 ([1, §2.4])**.** Let $\mathcal{B} = (v_1, \dots, v_k)$ be an ordered basis of a subspace W.

(a) $[u + w]_\mathcal{B} = [u]_\mathcal{B} + [w]_\mathcal{B}$ and $[cw]_\mathcal{B} = c[w]_\mathcal{B}$ for $u, w \in W$ and $c \in \mathbb{R}$.

(b) Every $c \in \mathbb{R}^k$ is the coordinate vector of exactly one vector of W, namely $c_1v_1 + \cdots + c_kv_k$.

Proof. (a) If $u = \sum_i x_iv_i$ and $w = \sum_i y_iv_i$, then $u + w = \sum_i (x_i + y_i)v_i$ and $cw = \sum_i (cy_i)v_i$. By the uniqueness in Theorem 3.18, these are the expressions that give $[u + w]_\mathcal{B}$ and $[cw]_\mathcal{B}$.

(b) $w = \sum_i c_iv_i$ lies in W, and $[w]_\mathcal{B} = c$ by definition. If also $[w_1]_\mathcal{B} = c$, then $w_1 = \sum_i c_iv_i = w$. □

Example 3.25 ([1, §2.2, Example 8])**.** In Example 3.8, the row operations shown take $[v_1\ v_2\ v_3]$ to the first three columns of the last matrix (row operations act on each column separately, Theorems 2.15(c) and 1.4(a)). These columns are $e_1, e_2, e_3 \in \mathbb{R}^5$; they form the RREF of $[v_1\ v_2\ v_3]$, in which every column is a pivot column. So v_1, v_2, v_3 are independent (Proposition 3.12), and $\mathcal{B} = (v_1, v_2, v_3)$ is an ordered basis of W. Every $y \in W$ is $y = y_1v_1 + y_3v_2 + y_5v_3$, so

$$[y]_\mathcal{B} = (y_1, y_3, y_5)'$$

the first, third and fifth entries of y. For instance $[(-3, -6, 1, -5, 2)']_\mathcal{B} = (-3, 1, 2)'$, and by Proposition 3.24(b), $(1, 1, 1)'$ is the coordinate vector of $v_1 + v_2 + v_3 = (1, 2, 1, 7, 1)'$.

Example 3.26 ([5, §4.3b])**.** For $1 \leqslant j \leqslant i \leqslant n$ let $U_{ii} = E_{ii}$ and $U_{ij} = E_{ij} + E_{ji}$ $(j < i)$, with the matrices $E_{ij} \in \mathbb{R}^{n\times n}$ of Definition 3.17. Each U_{ij} is symmetric. For $k \geqslant l$, U_{kl} is the only one of these matrices with a nonzero (k, l) entry, and this entry is 1; for $k < l$, the only one is U_{lk}. Hence the (k, l) entry of $\sum_{j\leqslant i} c_{ij}U_{ij}$ is c_{kl} for $k \geqslant l$, and c_{lk} for $k < l$. So every symmetric $A = (a_{ij})$ is

$$A = \sum_{j=1}^{n} \sum_{i=j}^{n} a_{ij}U_{ij}$$

and $\sum_{j\leqslant i} c_{ij}U_{ij} = 0$ only if every $c_{ij} = 0$. Thus the matrices U_{ij} $(j \leqslant i)$ form a basis of the subspace of symmetric $n \times n$ matrices (Example 3.5(d)); for column j there are $n - j + 1$ of them, $n(n + 1)/2$ in all. In the ordered basis $\mathcal{B} = (U_{11}, U_{21}, \dots, U_{n1}, U_{22}, \dots, U_{n2}, \dots, U_{nn})$, the coordinate vector of A lists the entries of A on and below the diagonal, column by column. For instance, a covariance matrix of four variables is determined by ten numbers, its variances and covariances:

$$\left[\begin{pmatrix} 4 & 2 & -1 & 1 \\ 2 & 5 & 0 & -2 \\ -1 & 0 & 3 & 1 \\ 1 & -2 & 1 & 6 \end{pmatrix}\right]_\mathcal{B} = (4, 2, -1, 1, 5, 0, -2, 3, 1, 6)'$$

For $n = 2$ this is Example 3.21.

Theorem 3.27 ([1, §2.4, Theorem 7])**.** Let $\mathcal{B} = (v_1, \dots, v_k)$ and $\tilde{\mathcal{B}} = (\tilde{v}_1, \dots, \tilde{v}_k)$ be ordered bases of a subspace W, each with k vectors, and let $P = (p_{ij}) \in \mathbb{R}^{k\times k}$ have jth column $[\tilde{v}_j]_\mathcal{B}$, that is, $\tilde{v}_j = p_{1j}v_1 + \cdots + p_{kj}v_k$ $(1 \leqslant j \leqslant k)$.

(a) $[w]_\mathcal{B} = P[w]_{\tilde{\mathcal{B}}}$ for every $w \in W$, and P is the only $k \times k$ matrix with this property.

(b) P is invertible, and $[w]_{\tilde{\mathcal{B}}} = P^{-1}[w]_\mathcal{B}$ for every $w \in W$.

P is the **change-of-basis matrix** from $\tilde{\mathcal{B}}$ to $\mathcal{B}$.

Proof. (a)

- Let $w \in W$ and $\tilde{x} = [w]_{\tilde{\mathcal{B}}}$. Then

$$w = \sum_{j=1}^{k} \tilde{x}_j\tilde{v}_j = \sum_{j=1}^{k} \tilde{x}_j \sum_{i=1}^{k} p_{ij}v_i = \sum_{i=1}^{k} \Big(\sum_{j=1}^{k} p_{ij}\tilde{x}_j\Big)v_i$$

- By the uniqueness of coordinates (Theorem 3.18), the ith coordinate of w relative to $\mathcal{B}$ is $\sum_j p_{ij}\tilde{x}_j = (P\tilde{x})_i$. So $[w]_{\mathcal{B}} = P\tilde{x}$.
- If also $[w]_{\mathcal{B}} = Q[w]_{\tilde{\mathcal{B}}}$ for every w, take $w = \tilde{v}_j$, for which $[\tilde{v}_j]_{\tilde{\mathcal{B}}} = e_j$: then $\mathrm{col}_j(Q) = Qe_j = [\tilde{v}_j]_{\mathcal{B}} = \mathrm{col}_j(P)$ (Theorem 1.4(b)) for every j, so $Q = P$.

(b)

- Let $P\tilde{x} = 0$, and let $w \in W$ be the vector with $[w]_{\tilde{\mathcal{B}}} = \tilde{x}$ (Proposition 3.24(b)). By (a), $[w]_{\mathcal{B}} = 0$, so $w = 0v_1 + \cdots + 0v_k = 0$, and $\tilde{x} = [0]_{\tilde{\mathcal{B}}} = 0$.
- So $P\tilde{x} = 0$ only for $\tilde{x} = 0$, and P is invertible (Theorem 2.16).
- Then $[w]_{\tilde{\mathcal{B}}} = P^{-1}P[w]_{\tilde{\mathcal{B}}} = P^{-1}[w]_{\mathcal{B}}$. □

If $W \subseteq \mathbb{R}^n$ and the two ordered bases are the columns of X and of $\tilde{X}$ in $\mathbb{R}^{n\times k}$, then $\mathrm{col}_j(\tilde{X}) = \tilde{v}_j = X\,\mathrm{col}_j(P)$, that is, $\tilde{X} = XP$ (Theorem 1.4(a),(b)). Hence $\tilde{X}\tilde{c} = (XP)\tilde{c} = X(P\tilde{c})$ for every $\tilde{c} \in \mathbb{R}^k$ (Theorem 1.3): the coefficients $\tilde{c}$ on the columns of $\tilde{X}$ and the coefficients $c = P\tilde{c}$ on the columns of X give the same vector, which is (a) in matrix form.

Theorem 3.28 ([1, §2.4, Theorem 8], [12, Proposition 11.4(i)])**.** Let $\mathcal{B} = (v_1, \ldots, v_k)$ be an ordered basis of a subspace W, and let $P = (p_{ij}) \in \mathbb{R}^{k\times k}$ be invertible. Put $\tilde{v}_j = p_{1j}v_1 + \cdots + p_{kj}v_k$ $(1 \leqslant j \leqslant k)$. Then $\tilde{\mathcal{B}} = (\tilde{v}_1, \ldots, \tilde{v}_k)$ is an ordered basis of W, and it is the only ordered basis of W with $[w]_{\mathcal{B}} = P[w]_{\tilde{\mathcal{B}}}$ for every $w \in W$. In particular, if the columns of $X \in \mathbb{R}^{n\times k}$ form a basis of a subspace W of $\mathbb{R}^n$, so do the columns of XP.

Proof. • At most one: if $\mathcal{Y} = (y_1, \ldots, y_k)$ is an ordered basis of W with $[w]_{\mathcal{B}} = P[w]_{\mathcal{Y}}$ for every w, then $w = y_j$, with $[y_j]_{\mathcal{Y}} = e_j$, gives $[y_j]_{\mathcal{B}} = Pe_j = \mathrm{col}_j(P)$, that is, $y_j = \sum_i p_{ij}v_i = \tilde{v}_j$.

- Spanning: the $\tilde{v}_j$ lie in W. Let $Q = (q_{jl}) = P^{-1}$. For each l,

$$\sum_{j=1}^{k} q_{jl}\tilde{v}_j = \sum_{j=1}^{k} q_{jl}\sum_{i=1}^{k} p_{ij}v_i = \sum_{i=1}^{k}\Big(\sum_{j=1}^{k} p_{ij}q_{jl}\Big)v_i = \sum_{i=1}^{k}(PQ)_{il}v_i = v_l$$

  So $\mathrm{span}\{\tilde{v}_1, \ldots, \tilde{v}_k\}$ contains $v_1, \ldots, v_k$, hence contains $W = \mathrm{span}\{v_1, \ldots, v_k\}$ (Theorem 3.7).

- Independent: if $\sum_j c_j\tilde{v}_j = 0$, the same computation gives $\sum_i (Pc)_i v_i = 0$; so $Pc = 0$, as the v_i are independent, and $c = P^{-1}Pc = 0$.
- So $\tilde{\mathcal{B}}$ is an ordered basis with $[\tilde{v}_j]_{\mathcal{B}} = \mathrm{col}_j(P)$, and Theorem 3.27(a) gives $[w]_{\mathcal{B}} = P[w]_{\tilde{\mathcal{B}}}$.
- For the last sentence take $v_j = \mathrm{col}_j(X)$; then $\tilde{v}_j = X\,\mathrm{col}_j(P) = \mathrm{col}_j(XP)$ (Theorem 1.4(a),(b)). □

Example 3.29. In $W_0 = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$ take the ordered basis $\mathcal{B} = (w_1, w_2, w_3)$ of Example 3.22. There $x = x_1w_1 + (x_1 + x_2)w_2 + (x_1 + x_2 + x_3)w_3$ for $x \in W_0$, so $[x]_{\mathcal{B}} = (x_1,\ x_1+x_2,\ x_1+x_2+x_3)'$, the partial sums of the entries of x. Let $h_1 = (1, 0, 0, -1)'$, $h_2 = (1, 1, 0, -2)'$ and $h_3 = (1, 1, 1, -3)'$, three vectors of W_0 that compare the first entry, the first two entries and the first three entries with the last.

- $[h_1]_{\mathcal{B}} = (1, 1, 1)'$, $[h_2]_{\mathcal{B}} = (1, 2, 2)'$ and $[h_3]_{\mathcal{B}} = (1, 2, 3)'$, the partial sums; that is, $h_1 = w_1 + w_2 + w_3$, $h_2 = w_1 + 2w_2 + 2w_3$ and $h_3 = w_1 + 2w_2 + 3w_3$. With these columns,

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

  since $PP^{-1} = I_3$ (Theorem 2.16).

- As P is invertible, $\tilde{\mathcal{B}} = (h_1, h_2, h_3)$ is an ordered basis of W_0 (Theorem 3.28), and P is the change-of-basis matrix from $\tilde{\mathcal{B}}$ to $\mathcal{B}$; in matrix form $[h_1\ h_2\ h_3] = [w_1\ w_2\ w_3]P$.

- By Theorem 3.27(b), for $x \in W_0$,

$$[x]_{\tilde{\mathcal{B}}} = P^{-1}[x]_{\mathcal{B}} = \begin{pmatrix} 2x_1 - (x_1 + x_2) \\ -x_1 + 2(x_1 + x_2) - (x_1 + x_2 + x_3) \\ -(x_1 + x_2) + (x_1 + x_2 + x_3) \end{pmatrix} = \begin{pmatrix} x_1 - x_2 \\ x_2 - x_3 \\ x_3 \end{pmatrix}$$

  For $x = (3, 1, 2, -6)'$: $[x]_{\mathcal{B}} = (3, 4, 6)'$ and $[x]_{\tilde{\mathcal{B}}} = (2, -1, 2)'$; indeed $3w_1 + 4w_2 + 6w_3 = 2h_1 - h_2 + 2h_3 = (3, 1, 2, -6)'$.

Sums and direct sums of subspaces

Definition 3.30 ([5, §17.1], [4, Ch. I, §4], [6, Exercise 3.12])**.** Let U and W be subspaces of a vector space V. Their **intersection** $U \cap W$ is the set of vectors that lie in both U and W, and their **sum** is

$$U + W = \{u + w : \ u \in U, \ w \in W\}$$

The sum is **direct**, written $U \oplus W$, if $U \cap W = \{0\}$, that is, if 0 is the only vector that U and W have in common. Thus $V = U \oplus W$ means that $V = U + W$ and $U \cap W = \{0\}$.

Theorem 3.31 ([5, §17.1, Lemma 17.1.1], [6, Exercise 3.12(a), (c)], [8, Ch. 17, Exercise 1(a), (b)], [1, §2.2])**.** Let U and W be subspaces of a vector space V.

(a) $U \cap W$ and $U + W$ are subspaces of V.

(b) $U + W$ contains U and W, and it is contained in every subspace of V that contains $U \cup W$: it is the smallest subspace of V containing $U \cup W$. Moreover $U + W$ is the set of all linear combinations $c_1x_1 + \cdots + c_kx_k$ of finitely many vectors $x_1, \ldots, x_k \in U \cup W$ (this set is written $\operatorname{span}(U \cup W)$).

(c) If $U = \operatorname{span}\{u_1, \ldots, u_p\}$ and $W = \operatorname{span}\{w_1, \ldots, w_q\}$, then $U + W = \operatorname{span}\{u_1, \ldots, u_p, w_1, \ldots, w_q\}$.

Proof. (a)

- Both sets contain 0: $0 \in U$, $0 \in W$, and $0 = 0 + 0$.
- Let $x, y \in U \cap W$ and $c \in \mathbb{R}$. Then $cx + y \in U$ and $cx + y \in W$, since U and W are subspaces; so $cx + y \in U \cap W$.
- Let $x = u_1 + w_1$ and $y = u_2 + w_2$ with $u_1, u_2 \in U$, $w_1, w_2 \in W$, and let $c \in \mathbb{R}$. Then $cx + y = (cu_1 + u_2) + (cw_1 + w_2)$ with $cu_1 + u_2 \in U$ and $cw_1 + w_2 \in W$; so $cx + y \in U + W$.
- By Theorem 3.4(c), $U \cap W$ and $U + W$ are subspaces.

(b)

- Every $u \in U$ is $u = u + 0$, and every $w \in W$ is $w = 0 + w$; so U and W lie in $U + W$.
- If a subspace Z of V contains $U \cup W$, then it contains u and w for all $u \in U$, $w \in W$, hence also $u + w$; so $U + W \subseteq Z$.
- In a combination $c_1x_1 + \cdots + c_kx_k$ with all $x_i \in U \cup W$, the terms with $x_i \in U$ add up to a vector $u \in U$, and the other terms, whose x_i lie in W, add up to a vector $w \in W$ (an empty sum is 0). So the combination is $u + w \in U + W$. Conversely, $u + w = 1u + 1w$ is such a combination.

(c)

- If $u = \sum_i a_iu_i \in U$ and $w = \sum_j b_jw_j \in W$, then $u + w = \sum_i a_iu_i + \sum_j b_jw_j$ is a combination of $u_1, \ldots, u_p, w_1, \ldots, w_q$.
- Conversely, $\sum_i a_iu_i + \sum_j b_jw_j = u + w$ with $u = \sum_i a_iu_i \in U$ and $w = \sum_j b_jw_j \in W$. □

For matrices: let $A \in \mathbb{R}^{m\times p}$ have columns $a_1, \ldots, a_p$, let $B \in \mathbb{R}^{m\times q}$ have columns $b_1, \ldots, b_q$, and let $[A\ B] \in \mathbb{R}^{m\times(p+q)}$ be the matrix with columns $a_1, \ldots, a_p, b_1, \ldots, b_q$. By (c), the columns of $[A\ B]$ span the sum of the subspaces spanned by the columns of A and by the columns of B,

$$\operatorname{span}\{a_1, \ldots, a_p, b_1, \ldots, b_q\} = \operatorname{span}\{a_1, \ldots, a_p\} + \operatorname{span}\{b_1, \ldots, b_q\}$$

and, by Theorem 1.4(b), the vectors of this subspace are the vectors $Ax + By$ ($x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$).

Theorem 3.32 ([6, Exercise 3.13(a), (b)])**.** Let U and W be subspaces of a vector space V. Then $U \cap W = \{0\}$ if and only if every $x \in U + W$ can be written in exactly one way as $x = u + w$ with $u \in U$ and $w \in W$, that is: if $u + w = u_1 + w_1$ with $u, u_1 \in U$ and $w, w_1 \in W$, then $u = u_1$ and $w = w_1$.

Proof.
- Let $U \cap W = \{0\}$, and let $u + w = u_1 + w_1$ as above. Then $u - u_1 = w_1 - w$. The left side lies in U and the right side in W, so this vector lies in $U \cap W = \{0\}$. Hence $u = u_1$ and $w = w_1$.
- Let $U \cap W \neq \{0\}$, and take $x \neq 0$ in $U \cap W$. Then $x = x + 0$ (with $x \in U$, $0 \in W$) and $x = 0 + x$ (with $0 \in U$, $x \in W$) are two different ways of writing $x \in U + W$. □

So, if $V = U \oplus W$, each $v \in V$ is $v = u + w$ for exactly one $u \in U$ and one $w \in W$ (the parts of v in U and in W).

Theorem 3.33 ([5, Theorem 17.1.5])**.** Let U and W be subspaces of a vector space V, let $u_1, \ldots, u_p$ form a basis of U, and let $w_1, \ldots, w_q$ form a basis of W. If $U \cap W = \{0\}$, then $u_1, \ldots, u_p, w_1, \ldots, w_q$ form a basis of $U \oplus W$. If $U \cap W \neq \{0\}$, then $u_1, \ldots, u_p, w_1, \ldots, w_q$ are linearly dependent. Hence the sum $U + W$ is direct if and only if $u_1, \ldots, u_p, w_1, \ldots, w_q$ are linearly independent.

Proof.
- The vectors $u_1, \ldots, u_p, w_1, \ldots, w_q$ lie in $U + W$ and span it, by Theorem 3.31(b),(c).
- Let $U \cap W = \{0\}$ and $\sum_i a_i u_i + \sum_j b_j w_j = 0$. Then $x = \sum_i a_i u_i = \sum_j (-b_j) w_j$ lies in U and in W, so $x = 0$. Hence $\sum_i a_i u_i = 0$ and $\sum_j b_j w_j = 0$, and since each list is independent, every a_i and every b_j is 0. So the vectors form a basis of $U + W = U \oplus W$.
- Let $U \cap W \neq \{0\}$, take $x \neq 0$ in $U \cap W$, and write $x = \sum_i a_i u_i = \sum_j b_j w_j$. Then $\sum_i a_i u_i + \sum_j (-b_j) w_j = 0$, and not every a_i is 0, since $x \neq 0$: a linear relation among $u_1, \ldots, u_p, w_1, \ldots, w_q$. □

Example 3.34. In $\mathbb{R}^4$ let $U = \{x : x_4 = 0\} = \operatorname{span}\{e_1, e_2, e_3\}$ and $W_0 = \{x : x_1 + x_2 + x_3 + x_4 = 0\} = \operatorname{span}\{w_1, w_2, w_3\}$, with $w_1 = (1, -1, 0, 0)'$, $w_2 = (0, 1, -1, 0)'$ and $w_3 = (0, 0, 1, -1)'$ (Example 3.22).

(a) $U + W_0 = \mathbb{R}^4$, but the sum is not direct.

- Every $x \in \mathbb{R}^4$ is $x = (x_1,\ x_2,\ x_3 + x_4,\ 0)' + (0,\ 0,\ -x_4,\ x_4)'$, with the first vector in U and the second, $-x_4 w_3$, in W_0. So $U + W_0 = \mathbb{R}^4$.
- $x \in U \cap W_0$ means $x_4 = 0$ and $x_1 + x_2 + x_3 = 0$, that is, $x = (a,\ b,\ -a-b,\ 0)' = a\, w_1 + (a+b)\, w_2$ for some $a, b \in \mathbb{R}$. So $U \cap W_0 = \operatorname{span}\{w_1, w_2\} \neq \{0\}$, and the sum is not direct.
- Accordingly (Theorem 3.32) the parts are not unique: for example

$$(1, 2, 3, 6)' = (1, 2, 9, 0)' + (0, 0, -6, 6)' = (7, 2, 3, 0)' + (-6, 0, 0, 6)'$$

  and the two parts in U differ by $(-6, 0, 6, 0)' = -6(w_1 + w_2) \in U \cap W_0$. Likewise $e_1, e_2, e_3, w_1, w_2, w_3$ are dependent, since $w_1 = e_1 - e_2$ (Theorem 3.33).

(b) $\mathbb{R}^4 = W_0 \oplus \operatorname{span}\{e_4\}$.

- $ce_4 = (0, 0, 0, c)'$ lies in W_0 only for $c = 0$; so $W_0 \cap \operatorname{span}\{e_4\} = \{0\}$.
- For $x \in \mathbb{R}^4$ and $c \in \mathbb{R}$, $x - ce_4 = (x_1,\ x_2,\ x_3,\ x_4 - c)'$ lies in W_0 exactly when $c = x_1 + x_2 + x_3 + x_4$. So every x is $x = (x_1,\ x_2,\ x_3,\ -x_1 - x_2 - x_3)' + (x_1 + x_2 + x_3 + x_4)\, e_4$ with the first vector in W_0, and in no other way. For example, $(1, 2, 3, 6)' = (1, 2, 3, -6)' + 12e_4$.

- The nonzero vector e_4 is a basis of $\operatorname{span}\{e_4\}$ (Proposition 3.13(a)); so, by Theorem 3.33, w_1, w_2, w_3, e_4 form a basis of $\mathbb{R}^4$.

Example 3.35. The constant vectors and the centered vectors $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$ (Example 3.5(f)) give a direct sum:

$$\mathbb{R}^n = \operatorname{span}\{\mathbf{1}_n\} \oplus W_0, \qquad x = \bar{x}\mathbf{1} + Cx$$

- Sum: every $x \in \mathbb{R}^n$ is $x = \bar{x}\mathbf{1} + Cx$ with $\bar{x}\mathbf{1} \in \operatorname{span}\{\mathbf{1}\}$ and $Cx \in W_0$ (Example 3.9).
- Intersection: if $c\mathbf{1} \in W_0$, then $0 = \mathbf{1}'(c\mathbf{1}) = cn$ (Proposition 1.15), so $c = 0$.
- Hence $\bar{x}\mathbf{1}$ is the only constant vector whose difference from x is centered (Theorem 3.32). In matrix form, $x = \bar{J}_n x + C_n x$ with $\bar{J}_n x = \frac{1}{n}\mathbf{1}(\mathbf{1}'x) = \bar{x}\mathbf{1}$, and $\bar{J}_n + C_n = I_n$ (Definition 1.14). For $x = (1, 3, 5, 7, 9)'$ the two parts are $\bar{x}\mathbf{1} = 5\mathbf{1}$ and $Cx = (-4, -2, 0, 2, 4)'$.
- For $n = 4$, the nonzero vector $\mathbf{1}_4$ is a basis of $\operatorname{span}\{\mathbf{1}_4\}$ (Proposition 3.13(a)); so $\mathbf{1}_4$ together with the basis w_1, w_2, w_3 of W_0 (Example 3.22) is a basis of $\mathbb{R}^4$ (Theorem 3.33).
- The second subspace matters: also $\mathbb{R}^4 = W_0 \oplus \operatorname{span}\{e_4\}$ (Example 3.34(b)), and the parts of $x = (1, 2, 3, 6)'$ in W_0 differ: here $\bar{x} = 3$ and $Cx = (-2, -1, 0, 3)'$, there $(1, 2, 3, -6)'$.

Example 3.36 (cf. [8, Ch. 17, Exercise 2])**.** In $\mathbb{R}^4$ let $U = \operatorname{span}\{u_1, u_2\}$ and $W = \operatorname{span}\{w_1, w_2\}$, where

$$u_1 = \begin{pmatrix}1\\0\\1\\2\end{pmatrix}, \quad u_2 = \begin{pmatrix}1\\1\\0\\1\end{pmatrix}, \quad w_1 = \begin{pmatrix}0\\1\\0\\1\end{pmatrix}, \quad w_2 = \begin{pmatrix}2\\0\\1\\2\end{pmatrix}$$

Find bases of $U + W$ and of $U \cap W$. Is the sum direct?

Solution. By Theorem 3.31(c), $U + W = \operatorname{span}\{u_1, u_2, w_1, w_2\}$. Row-reduce $M = [u_1\ u_2\ w_1\ w_2]$:

$$\begin{pmatrix}1&1&0&2\\0&1&1&0\\1&0&0&1\\2&1&1&2\end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix}1&1&0&2\\0&1&1&0\\0&-1&0&-1\\0&-1&1&-2\end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix}1&1&0&2\\0&1&1&0\\0&0&1&-1\\0&0&2&-2\end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix}1&1&0&2\\0&1&1&0\\0&0&1&-1\\0&0&0&0\end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{pmatrix}1&1&0&2\\0&1&0&1\\0&0&1&-1\\0&0&0&0\end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix}1&0&0&1\\0&1&0&1\\0&0&1&-1\\0&0&0&0\end{pmatrix} = R$$

- Relations: $Mc = 0$ has the same solutions as $Rc = 0$ (Theorem 2.3(b)), that is, $c_1 + c_4 = 0$, $c_2 + c_4 = 0$, $c_3 - c_4 = 0$, with $c_4 = t$ free. So $c = t(-1, -1, 1, 1)'$ (Proposition 2.7(b)), and $t = 1$ gives $-u_1 - u_2 + w_1 + w_2 = 0$, that is, $w_2 = u_1 + u_2 - w_1$.
- $U + W$: by Theorem 3.10(c), $U + W = \operatorname{span}\{u_1, u_2, w_1\}$. The same row operations take $[u_1\ u_2\ w_1]$ to the first three columns of R (row operations act on each column separately, Theorems 2.15(c) and 1.4(a)), which are $e_1, e_2, e_3 \in \mathbb{R}^4$; every column is a pivot column, so u_1, u_2, w_1 are independent (Proposition 3.12) and form a basis of $U + W$.
- $U \cap W$: $x \in U \cap W$ if and only if $x = a_1u_1 + a_2u_2 = b_1w_1 + b_2w_2$ for some scalars, that is, $M(a_1, a_2, -b_1, -b_2)' = 0$. By the first item, $(a_1, a_2, -b_1, -b_2)' = t(-1, -1, 1, 1)'$, so $x = -t(u_1 + u_2)$. Hence $U \cap W = \operatorname{span}\{u_1 + u_2\}$, and the nonzero vector $u_1 + u_2 = (2, 1, 1, 3)' = w_1 + w_2$ is a basis of $U \cap W$ (Proposition 3.13(a)).
- The sum is not direct, since $U \cap W \neq \{0\}$ (Definition 3.30). Here u_1, u_2 form a basis of U (a sublist of the independent u_1, u_2, w_1, Proposition 3.13(b)), and w_1, w_2 a basis of W (neither is a multiple of the other, Proposition 3.13(c)); accordingly u_1, u_2, w_1, w_2 are dependent (Theorem 3.33).

Exercises

Exercise 3.1. Decide whether each set is a subspace; for each subspace, find vectors (or matrices) that span it.

(a) $W_1 = \{x \in \mathbb{R}^4 : \ x_1 + x_2 + x_3 = 0, \ x_2 - x_4 = 0, \ x_1 + 2x_2 + x_3 - x_4 = 0\}$.

(b) $W_2 = \{x \in \mathbb{R}^3 : \ x_1 + x_2 + x_3 = 1\}$.

(c) $W_3 = \{x \in \mathbb{R}^3 : \ x_1 = x_2^2\}$.

(d) $W_4 = \{A \in \mathbb{R}^{2\times 3} : \ A\mathbf{1}_3 = 0\}$, the 2×3 matrices whose rows add up to 0.

Solution. (a) W_1 is the set of solutions of $Ax = 0$ for the coefficient matrix A below, so it is a subspace (Example 3.5(e)). Row-reduce A:

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 2 & 1 & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The RREF gives $x_1 + x_3 + x_4 = 0$ and $x_2 - x_4 = 0$, with $x_3 = s$ and $x_4 = t$ free (Proposition 2.7(b)):

$$x = \begin{pmatrix} -s - t \\ t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix} \qquad (s, t \in \mathbb{R})$$

So $W_1 = \operatorname{span}\{(-1, 0, 1, 0)', (-1, 1, 0, 1)'\}$. Check: $-1 + 0 + 1 = 0$, $0 - 0 = 0$, $-1 + 0 + 1 - 0 = 0$, and $-1 + 1 + 0 = 0$, $1 - 1 = 0$, $-1 + 2 + 0 - 1 = 0$.

(b) Not a subspace: $0 \notin W_2$, since $0 + 0 + 0 \neq 1$ (Theorem 3.4(b)).

(c) Not a subspace: $(1, 1, 0)' \in W_3$, since $1 = 1^2$, but $2(1, 1, 0)' = (2, 2, 0)' \notin W_3$, since $2 \neq 2^2$; so W_3 is not closed under scalar multiplication.

(d) A subspace: the zero matrix lies in W_4, and if $A\mathbf{1} = 0$ and $B\mathbf{1} = 0$, then $(cA+B)\mathbf{1} = cA\mathbf{1}+B\mathbf{1} = 0$ (Theorem 1.3); apply Theorem 3.4(c). A matrix lies in W_4 if and only if $a_{13} = -a_{11} - a_{12}$ and $a_{23} = -a_{21} - a_{22}$, that is,

$$\begin{pmatrix} a & b & -a-b \\ p & q & -p-q \end{pmatrix} = a \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} + p \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix} + q \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix} \qquad (a, b, p, q \in \mathbb{R})$$

So W_4 is spanned by the four matrices on the right, $E_{11} - E_{13}$, $E_{12} - E_{13}$, $E_{21} - E_{23}$ and $E_{22} - E_{23}$ (Definition 3.17).

Exercise 3.2. Which of the following subsets of $\mathbb{R}^4$ are subspaces? (a) $\{x : x_1 + x_2 = x_3 + x_4\}$; (b) $\{x : x_1x_4 = x_2x_3\}$; (c) $\{x : x_1 + x_2 + x_3 + x_4 = 1\}$; (d) $\{x : x_1 \leqslant x_2\}$; (e) $\{x : x_1 = x_2 = x_3 = x_4\}$.

Solution. (a) Yes: it is the solution set of $Ax = 0$ for the 1×4 matrix $A = (1, 1, -1, -1)$, a subspace by Example 3.5(e).

(b) No: e_1 and e_4 lie in the set ($1 \cdot 0 = 0 \cdot 0$ and $0 \cdot 1 = 0 \cdot 0$), but $e_1 + e_4 = (1, 0, 0, 1)'$ does not, since $1 \cdot 1 \neq 0 \cdot 0$.

(c) No: 0 is not in the set, since $0 + 0 + 0 + 0 \neq 1$.

(d) No: $e_2 = (0, 1, 0, 0)'$ lies in the set, but $(-1)e_2 = (0, -1, 0, 0)'$ does not, since $0 > -1$.

(e) Yes: the set is $\{c\mathbf{1}_4 : c \in \mathbb{R}\} = \operatorname{span}\{\mathbf{1}_4\}$, a subspace by Theorem 3.7.

Exercise 3.3. Let $W = \operatorname{span}\{v_1, v_2, v_3\} \subseteq \mathbb{R}^4$, where $v_1 = (1,2,1,0)'$, $v_2 = (0,1,1,1)'$ and $v_3 = (1,3,3,2)'$.

(a) Which $y = (y_1, y_2, y_3, y_4)'$ lie in W?

(b) Show that $(3,6,4,1)' \in W$, and write it as a combination of v_1, v_2, v_3.

(c) Does $(1,1,1,1)'$ lie in W?

Solution. (a) With $V = [v_1\ v_2\ v_3]$, $y \in W$ if and only if $Vc = y$ is consistent (the paragraph after Theorem 3.7). Row-reduce $[V \mid y]$, keeping the y_i as letters:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & y_1 \\ 2 & 1 & 3 & y_2 \\ 1 & 1 & 3 & y_3 \\ 0 & 1 & 2 & y_4 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & y_1 \\ 0 & 1 & 1 & y_2 - 2y_1 \\ 0 & 1 & 2 & y_3 - y_1 \\ 0 & 1 & 2 & y_4 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & y_1 \\ 0 & 1 & 1 & y_2 - 2y_1 \\ 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 1 & 2y_1 - y_2 + y_4 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & y_1 \\ 0 & 1 & 1 & y_2 - 2y_1 \\ 0 & 0 & 1 & y_1 - y_2 + y_3 \\ 0 & 0 & 0 & y_1 - y_3 + y_4 \end{array}\right)$$

The left part of the last matrix is in echelon form, and its last row is $(0\ 0\ 0 \mid y_1 - y_3 + y_4)$. If $y_1 - y_3 + y_4 \neq 0$, this row leaves no solution; if $y_1 - y_3 + y_4 = 0$, the last matrix is in echelon form without such a row. So, by Theorem 2.26, $Vc = y$ is consistent if and only if $y_1 - y_3 + y_4 = 0$:

$$y \in W \iff y_1 - y_3 + y_4 = 0$$

Then every column of the left part is a pivot column, so the coefficients are unique (Theorem 2.27(b)); back substitution gives $c_3 = y_1 - y_2 + y_3$, $c_2 = (y_2 - 2y_1) - c_3 = -3y_1 + 2y_2 - y_3$ and $c_1 = y_1 - c_3 = y_2 - y_3$.
(b) For $y = (3,6,4,1)'$, $3 - 4 + 1 = 0$, so $y \in W$, with $c_3 = 3 - 6 + 4 = 1$, $c_2 = -9 + 12 - 4 = -1$ and $c_1 = 6 - 4 = 2$: $y = 2v_1 - v_2 + v_3$. Check: $(2,4,2,0)' - (0,1,1,1)' + (1,3,3,2)' = (3,6,4,1)'$.
(c) No: $y_1 - y_3 + y_4 = 1 - 1 + 1 = 1 \neq 0$.

Exercise 3.4. Let $v_1 = (1,2,0,1)'$, $v_2 = (0,1,1,-1)'$, $v_3 = (1,4,2,-1)'$ and $b = (2,3,-1,3)'$.

(a) Show that $b \in \operatorname{span}\{v_1, v_2, v_3\}$, and find all $c \in \mathbb{R}^3$ with $b = c_1v_1 + c_2v_2 + c_3v_3$.

(b) Give the representation with $c_3 = 0$ and the one with $c_1 = 0$.

(c) Are v_1, v_2, v_3 linearly independent? If not, give a linear relation among them.

Solution. With $V = [v_1\ v_2\ v_3]$, $b = c_1v_1 + c_2v_2 + c_3v_3$ means $Vc = b$ (Theorem 1.4(b)). Row-reduce $[V \mid b]$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 1 & 4 & 3 \\ 0 & 1 & 2 & -1 \\ 1 & -1 & -1 & 3 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -1 \\ 0 & -1 & -2 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(a) The last matrix is in RREF and has no row $(0\ 0\ 0 \mid d)$ with $d \neq 0$, so $Vc = b$ is consistent (Proposition 2.7(a)) and $b \in \operatorname{span}\{v_1, v_2, v_3\}$. The pivot columns are $1, 2$, and $c_3 = t$ is free; then $c_1 = 2 - t$ and $c_2 = -1 - 2t$ (Proposition 2.7(b)):

$$c = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \qquad (t \in \mathbb{R})$$

(b) $t = 0$: $b = 2v_1 - v_2$. $t = 2$: $c = (0, -5, 2)'$, so $b = -5v_2 + 2v_3$. Check: $(2, 4, 0, 2)' - (0, 1, 1, -1)' = (2, 3, -1, 3)'$ and $-5(0, 1, 1, -1)' + 2(1, 4, 2, -1)' = (2, 3, -1, 3)'$.

(c) No: the left part of the last matrix is the RREF of V, and its column 3 is not a pivot column (Proposition 3.12). The difference of the solutions for $t = 1$ and $t = 0$, $c = (-1, -2, 1)'$, solves $Vc = 0$ (Theorem 2.27(a)): $-v_1 - 2v_2 + v_3 = 0$, that is, $v_3 = v_1 + 2v_2$.

Exercise 3.5. Let $a_1 = (1, 0, 1, 1)'$, $a_2 = (0, 1, 1, 2)'$, $b_1 = (1, 1, 2, 3)'$ and $b_2 = (1, 2, 3, 5)'$.

(a) Row-reduce $[a_1\ a_2\ b_1\ b_2]$, write b_1 and b_2 as combinations of a_1, a_2, and find $F \in \mathbb{R}^{2\times 2}$ with $[b_1\ b_2] = [a_1\ a_2]F$.

(b) Write a_1 and a_2 as combinations of b_1, b_2, and find $G \in \mathbb{R}^{2\times 2}$ with $[a_1\ a_2] = [b_1\ b_2]G$.

(c) Conclude that $\operatorname{span}\{a_1, a_2\} = \operatorname{span}\{b_1, b_2\}$, and check that $FG = I_2$.

Solution. (a)

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 3 & 5 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 2 & 4 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The last matrix is in RREF; the solutions of $c_1a_1 + c_2a_2 + c_3b_1 + c_4b_2 = 0$ are $c_1 = -s - t$, $c_2 = -s - 2t$, $c_3 = s$, $c_4 = t$ (Theorem 2.3(b), Proposition 2.7(b)). For $s = 1$, $t = 0$: $-a_1 - a_2 + b_1 = 0$; for $s = 0$, $t = 1$: $-a_1 - 2a_2 + b_2 = 0$. So

$$b_1 = a_1 + a_2, \qquad b_2 = a_1 + 2a_2, \qquad F = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

(Theorem 1.4(a),(b)). Check: $(1, 0, 1, 1)' + (0, 1, 1, 2)' = (1, 1, 2, 3)'$ and $(1, 0, 1, 1)' + 2(0, 1, 1, 2)' = (1, 2, 3, 5)'$.

(b) Subtracting, $b_2 - b_1 = a_2$, and then $a_1 = b_1 - a_2 = 2b_1 - b_2$. So $G = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

(c) By (a), (b) and Theorem 3.10(b), the two spans are equal. Finally $FG = \left(\begin{smallmatrix} 2-1 & -1+1 \\ 2-2 & -1+2 \end{smallmatrix}\right) = I_2$.

Exercise 3.6 ([6, Exercise 3.19(c)])**.** Consider $e_1 = (1, 0, 0)'$, $e_2 = (0, 1, 0)'$ and $v = (0, -2, 0)'$ in $\mathbb{R}^3$. Show that e_1, e_2, v span $W = \{(x, y, 0)' : x, y \in \mathbb{R}\}$, that e_1 and e_2 alone span the same subspace W, and that e_2 and v span only $\{(0, y, 0)' : y \in \mathbb{R}\}$.

Solution. For scalars a_1, a_2, a_3 put $\lambda = a_2 - 2a_3$. Since $v = -2e_2$,

$$a_1e_1 + a_2e_2 + a_3v = a_1e_1 + \lambda e_2 = (a_1, \lambda, 0)', \qquad a_2e_2 + a_3v = \lambda e_2 = (0, \lambda, 0)'$$

So $\operatorname{span}\{e_1, e_2, v\}$ consists of the vectors $(a_1, \lambda, 0)'$. Each lies in W, and every $(x, y, 0)' \in W$ is obtained with $a_1 = x$, $a_2 = y$, $a_3 = 0$; so $\operatorname{span}\{e_1, e_2, v\} = W$. The same computation with $a_3 = 0$ gives $\operatorname{span}\{e_1, e_2\} = W$, and $\operatorname{span}\{e_2, v\} = \{\lambda e_2 : \lambda \in \mathbb{R}\} = \{(0, y, 0)' : y \in \mathbb{R}\}$.

Exercise 3.7. Let $u_1 = (1,2,3,1)'$, $u_2 = (0,1,1,1)'$, $u_3 = (1,2,3,0)'$ and $u_4 = (0,2,2,1)'$.

(a) Are u_1, u_2, u_3 linearly independent?

(b) Are u_1, u_2, u_3, u_4 linearly independent? If not, find a linear relation among them, and write u_4 as a combination of u_1, u_2, u_3.

Solution. Row-reduce $V = [u_1\ u_2\ u_3\ u_4]$:

$$\begin{pmatrix}1&0&1&0\\2&1&2&2\\3&1&3&2\\1&1&0&1\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-2R_1,\ R_3\leftarrow R_3-3R_1} \begin{pmatrix}1&0&1&0\\0&1&0&2\\0&1&0&2\\0&1&-1&1\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-R_2]{R_3\leftarrow R_3-R_2} \begin{pmatrix}1&0&1&0\\0&1&0&2\\0&0&0&0\\0&0&-1&-1\end{pmatrix}$$

$$\xrightarrow{R_3\leftrightarrow R_4} \begin{pmatrix}1&0&1&0\\0&1&0&2\\0&0&-1&-1\\0&0&0&0\end{pmatrix} \xrightarrow{R_3\leftarrow -R_3} \begin{pmatrix}1&0&1&0\\0&1&0&2\\0&0&1&1\\0&0&0&0\end{pmatrix} \xrightarrow{R_1\leftarrow R_1-R_3} \begin{pmatrix}1&0&0&-1\\0&1&0&2\\0&0&1&1\\0&0&0&0\end{pmatrix}$$

After the second arrow the entry of row 3 in column 3 is 0, so Step 2 interchanges rows 3 and 4. The last matrix is the RREF of V, with pivot columns $1, 2, 3$.

(a) Yes. Row operations act on each column separately (Theorem 2.15(c) and Theorem 1.4(a)), so the same operations take $[u_1\ u_2\ u_3]$ to the first three columns of the last matrix, whose columns are the unit vectors e_1, e_2, e_3 of $\mathbb{R}^4$. This is the RREF of $[u_1\ u_2\ u_3]$, and every column is a pivot column; so u_1, u_2, u_3 are independent (Proposition 3.12).

(b) No. Column 4 is not a pivot column, so $Vc = 0$ has a free unknown and a solution $c \neq 0$ (Proposition 3.12). The RREF gives $c_1 - c_4 = 0$, $c_2 + 2c_4 = 0$, $c_3 + c_4 = 0$; with $c_4 = t$, $c = t\,(1,-2,-1,1)'$. For $t = 1$,

$$u_1 - 2u_2 - u_3 + u_4 = 0, \qquad \text{so} \qquad u_4 = -u_1 + 2u_2 + u_3$$

Check: $-(1,2,3,1)' + 2(0,1,1,1)' + (1,2,3,0)' = (0,2,2,1)' = u_4$.

Exercise 3.8. Let $u_1 = (1,0,1,2)'$, $u_2 = (2,1,3,k)'$ and $u_3 = (0,1,1,1)'$, where k is a number.

(a) For which k are u_1, u_2, u_3 linearly dependent?

(b) For this k, find a linear relation, and write u_2 as a combination of u_1 and u_3.

Solution. Row-reduce $U = [u_1\ u_2\ u_3]$, keeping k as a letter:

$$\begin{pmatrix}1&2&0\\0&1&1\\1&3&1\\2&k&1\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-2R_1]{R_3\leftarrow R_3-R_1} \begin{pmatrix}1&2&0\\0&1&1\\0&1&1\\0&k-4&1\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-(k-4)R_2]{R_3\leftarrow R_3-R_2} \begin{pmatrix}1&2&0\\0&1&1\\0&0&0\\0&0&5-k\end{pmatrix}$$

(a) The vectors are independent exactly when $Uc = 0$ has only the solution $c = 0$ (Proposition 3.12).

- If $k \neq 5$, interchange rows 3 and 4, and then scale row 3 by $1/(5-k)$:

$$\xrightarrow{R_3\leftrightarrow R_4} \begin{pmatrix}1&2&0\\0&1&1\\0&0&5-k\\0&0&0\end{pmatrix} \xrightarrow{R_3\leftarrow \frac{1}{5-k}R_3} \begin{pmatrix}1&2&0\\0&1&1\\0&0&1\\0&0&0\end{pmatrix}$$

  The last matrix is an echelon form in which every column is a pivot column; so $c = 0$ is the only solution (Theorem 2.27(b)), and u_1, u_2, u_3 are independent.

- If $k = 5$, the matrix after the second arrow has two zero rows; one more arrow gives the RREF:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  Column 3 is not a pivot column, so c_3 is free and $Uc = 0$ has solutions $c \neq 0$: the vectors are dependent.

So u_1, u_2, u_3 are dependent only for $k = 5$.

(b) For $k = 5$, the RREF in (a) gives $c_1 - 2c_3 = 0$ and $c_2 + c_3 = 0$, with $c_3 = t$ free: $c = t(2, -1, 1)'$. For $t = 1$, $2u_1 - u_2 + u_3 = 0$, that is, $u_2 = 2u_1 + u_3$. Check: $2(1, 0, 1, 2)' + (0, 1, 1, 1)' = (2, 1, 3, 5)'$, which is u_2 for $k = 5$.

Exercise 3.9 ([4, Ch. I, §2, Exercise 1(f)–(h)], [7])**.** Show that the following vectors are linearly independent. (a) $(1, 2)'$ and $(1, 3)'$. (b) $(1, 1, 0)'$, $(1, 1, 1)'$ and $(0, 1, -1)'$. (c) $(0, 1, 1)'$, $(0, 2, 1)'$ and $(1, 5, 3)'$.

Solution. (a) If $a(1, 2)' + b(1, 3)' = 0$, then $a + b = 0$ and $2a + 3b = 0$. The second equation minus twice the first gives $b = 0$, and then $a = 0$.

(b) If $a(1, 1, 0)' + b(1, 1, 1)' + c(0, 1, -1)' = 0$, then $a + b = 0$, $a + b + c = 0$, $b - c = 0$. The second equation minus the first gives $c = 0$; then the third gives $b = 0$, and the first $a = 0$.

(c) If $a(0, 1, 1)' + b(0, 2, 1)' + c(1, 5, 3)' = 0$, then $c = 0$, $a + 2b + 5c = 0$, $a + b + 3c = 0$. Subtracting the third equation from the second gives $b + 2c = 0$, so $b = 0$ (as $c = 0$), and then $a = 0$.

Exercise 3.10 ([6, Exercise 3.16(b)])**.** Show that the columns of the matrix $A = \begin{pmatrix} 1 & 3 & 2 & 0 \\ 1 & -5 & -3 & 1 \\ 5 & -1 & 0 & 2 \end{pmatrix}$ are linearly dependent. How about the rows?

Solution. The columns satisfy, for example, $3\,\mathrm{col}_1(A) - \mathrm{col}_2(A) - 8\,\mathrm{col}_4(A) = 0$:

$$3\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} - 8\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 - 3 - 0 \\ 3 + 5 - 8 \\ 15 + 1 - 16 \end{pmatrix} = 0$$

The coefficients $3, -1, 0, -8$ are not all 0, so the columns are dependent; equivalently, $Ax = 0$ for $x = (3, -1, 0, -8)' \neq 0$ (Theorem 1.4(b)). The rows satisfy $3\,\mathrm{row}_1(A) + 2\,\mathrm{row}_2(A) - \mathrm{row}_3(A) = 0$: $(3, 9, 6, 0) + (2, -10, -6, 2) - (5, -1, 0, 2) = (0, 0, 0, 0)$, that is, $y'A = 0$ for $y = (3, 2, -1)'$ (Theorem 1.4(c)). So the rows are dependent too.

Exercise 3.11 ([8, Ch. 3, Exercise 1])**.** For what values of the scalar k are the three row vectors $(k, 1, 0)$, $(1, k, 1)$ and $(0, 1, k)$ linearly dependent, and for what values are they linearly independent?

Solution. Let x_1, x_2, x_3 be scalars with $x_1(k, 1, 0) + x_2(1, k, 1) + x_3(0, 1, k) = 0$, that is,

$$kx_1 + x_2 = 0, \qquad x_1 + kx_2 + x_3 = 0, \qquad x_2 + kx_3 = 0$$

or equivalently (the first and third equations give (1), the second gives (2))

$$x_2 = -kx_3 = -kx_1 \quad (1), \qquad kx_2 = -x_1 - x_3 \quad (2)$$

- $k = 0$: (1) and (2) become $x_2 = 0$ and $x_3 = -x_1$; for instance $(x_1, x_2, x_3) = (1, 0, -1)$ is a solution other than 0.

- $k \neq 0$: (1) gives $kx_3 = kx_1$, so $x_3 = x_1$; then (1) and (2) give $-k^2x_1 = kx_2 = -2x_1$, so $(2-k^2)x_1 = 0$. Hence $k^2 = 2$, or else $x_1 = 0$ and then $x_3 = x_2 = 0$.
- If $k = \sqrt{2}$, conditions (1), (2) are equivalent to $x_3 = x_1$ and $x_2 = -\sqrt{2}\,x_1$; if $k = -\sqrt{2}$, to $x_3 = x_1$ and $x_2 = \sqrt{2}\,x_1$. In both cases $x_1 = 1$ gives a solution other than 0.

Thus there are solutions other than $x_1 = x_2 = x_3 = 0$ if and only if $k = 0$ or $k = \pm\sqrt{2}$. The three vectors are linearly dependent if $k \in \{0, \sqrt{2}, -\sqrt{2}\}$ and linearly independent otherwise.

Exercise 3.12. Let $W = \operatorname{span}\{v_1, \dots, v_5\} \subseteq \mathbb{R}^4$, where

$$v_1 = \begin{pmatrix}1\\1\\0\\2\end{pmatrix}, \quad v_2 = \begin{pmatrix}1\\0\\1\\1\end{pmatrix}, \quad v_3 = \begin{pmatrix}1\\2\\-1\\3\end{pmatrix}, \quad v_4 = \begin{pmatrix}1\\0\\2\\1\end{pmatrix}, \quad v_5 = \begin{pmatrix}1\\1\\-1\\2\end{pmatrix}$$

(a) Row-reduce $V = [v_1 \ \cdots \ v_5]$ to its RREF, and find all linear relations among $v_1, \dots, v_5$.

(b) Find a basis of W that consists of some of the v_j.

(c) Write the remaining vectors as combinations of this basis.

Solution. (a)

$$\begin{pmatrix}1&1&1&1&1\\1&0&2&0&1\\0&1&-1&2&-1\\2&1&3&1&2\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-2R_1]{R_2\leftarrow R_2-R_1} \begin{pmatrix}1&1&1&1&1\\0&-1&1&-1&0\\0&1&-1&2&-1\\0&-1&1&-1&0\end{pmatrix} \xrightarrow{R_2\leftarrow -R_2} \begin{pmatrix}1&1&1&1&1\\0&1&-1&1&0\\0&1&-1&2&-1\\0&-1&1&-1&0\end{pmatrix}$$

$$\xrightarrow[R_4\leftarrow R_4+R_2]{R_3\leftarrow R_3-R_2} \begin{pmatrix}1&1&1&1&1\\0&1&-1&1&0\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix} \xrightarrow[R_2\leftarrow R_2-R_3]{R_1\leftarrow R_1-R_3} \begin{pmatrix}1&1&1&0&2\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix} \xrightarrow{R_1\leftarrow R_1-R_2} \begin{pmatrix}1&0&2&0&1\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix}$$

The last matrix is the RREF R of V. The relations $c_1v_1 + \cdots + c_5v_5 = 0$ are the solutions of $Vc = 0$, that is, of $Rc = 0$ (Theorem 2.3(b)): $c_1 + 2c_3 + c_5 = 0$, $c_2 - c_3 + c_5 = 0$, $c_4 - c_5 = 0$, with $c_3 = s$ and $c_5 = t$ free (Proposition 2.7(b)):

$$c = s\begin{pmatrix}-2\\1\\1\\0\\0\end{pmatrix} + t\begin{pmatrix}-1\\-1\\0\\1\\1\end{pmatrix} \qquad (s, t \in \mathbb{R})$$

For $s = 1$, $t = 0$: $-2v_1 + v_2 + v_3 = 0$; for $s = 0$, $t = 1$: $-v_1 - v_2 + v_4 + v_5 = 0$.

(b) By these relations, v_3 and v_5 lie in $\operatorname{span}\{v_1, v_2, v_4\}$, so $W = \operatorname{span}\{v_1, v_2, v_4\}$ (Theorem 3.10(c)). The same row operations take $[v_1\ v_2\ v_4]$ to columns $1, 2, 4$ of R (Theorems 2.15(c) and 1.4(a)), which are $e_1, e_2, e_3 \in \mathbb{R}^4$; every column is a pivot column, so v_1, v_2, v_4 are independent (Proposition 3.12) and form a basis of W.

(c) $v_3 = 2v_1 - v_2$ and $v_5 = v_1 + v_2 - v_4$. Check: $2(1,1,0,2)' - (1,0,1,1)' = (1,2,-1,3)'$ and $(1,1,0,2)' + (1,0,1,1)' - (1,0,2,1)' = (1,1,-1,2)'$.

Exercise 3.13. Let $W = \{x \in \mathbb{R}^5 :\ x_1 + x_2 + x_3 + x_4 + x_5 = 0,\ x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 0\}$.

(a) Find a basis of W.

(b) Show that $w = (0, 0, 1, -2, 1)' \in W$, and find its coordinates in this basis.

Solution. (a) W is the solution set of $Ax = 0$ with $A = \left(\begin{smallmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{smallmatrix}\right)$. Row-reduce:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix}$$

The free unknowns are x_3, x_4, x_5, and $x_1 = x_3 + 2x_4 + 3x_5$, $x_2 = -2x_3 - 3x_4 - 4x_5$ (Proposition 2.7(b)). So

$$W = \{x_3 b_1 + x_4 b_2 + x_5 b_3 :\ x_3, x_4, x_5 \in \mathbb{R}\}, \qquad b_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix},\ b_2 = \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix},\ b_3 = \begin{pmatrix} 3 \\ -4 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

The b_i lie in W and span it. Entries $3, 4, 5$ of $c_1 b_1 + c_2 b_2 + c_3 b_3$ are c_1, c_2, c_3, so the combination is 0 only for $c_1 = c_2 = c_3 = 0$. Hence b_1, b_2, b_3 form a basis of W.

(b) $0 + 0 + 1 - 2 + 1 = 0$ and $0 + 0 + 3 - 8 + 5 = 0$, so $w \in W$. By (a) its coordinates are its entries $3, 4, 5$, namely $(1, -2, 1)$, and they are unique (Theorem 3.18): $w = b_1 - 2b_2 + b_3 = (1 - 4 + 3,\ -2 + 6 - 4,\ 1,\ -2,\ 1)'$.

Exercise 3.14. Let $t = (1, 2, 3, 4)'$ and $W = \operatorname{span}\{\mathbf{1}_4, t\} \subseteq \mathbb{R}^4$.

(a) Show that $\mathbf{1}_4, t$ form a basis of W.

(b) Which $y = (y_1, y_2, y_3, y_4)'$ lie in W? Give the coordinates of such y.

(c) Find the coordinates of $(3, 5, 7, 9)'$. Does $(1, 2, 4, 8)'$ lie in W?

Solution. (a) The vectors lie in W and span it. If $a\mathbf{1}_4 + bt = (a + b,\ a + 2b,\ a + 3b,\ a + 4b)' = 0$, the second entry minus the first gives $b = 0$, and then $a = 0$. So they are independent.

(b) With $c = (a, b)'$, $y \in W$ if and only if $[\mathbf{1}_4\ t]c = y$ is consistent (the paragraph after Theorem 3.7). Row-reduce $[\mathbf{1}_4\ t \mid y]$, keeping the y_i as letters:

$$\left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 1 & 2 & y_2 \\ 1 & 3 & y_3 \\ 1 & 4 & y_4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 0 & 1 & y_2 - y_1 \\ 0 & 2 & y_3 - y_1 \\ 0 & 3 & y_4 - y_1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 0 & 1 & y_2 - y_1 \\ 0 & 0 & y_1 - 2y_2 + y_3 \\ 0 & 0 & 2y_1 - 3y_2 + y_4 \end{array}\right)$$

The left part of the last matrix is in echelon form, and its last two rows are $(0\ 0 \mid y_1 - 2y_2 + y_3)$ and $(0\ 0 \mid 2y_1 - 3y_2 + y_4)$. So, by Theorem 2.26,

$$y \in W \iff y_1 - 2y_2 + y_3 = 0 \ \text{ and } \ 2y_1 - 3y_2 + y_4 = 0$$

Then back substitution gives the coordinates $b = y_2 - y_1$ and $a = y_1 - b = 2y_1 - y_2$: $y = (2y_1 - y_2)\mathbf{1}_4 + (y_2 - y_1)t$.

(c) For $(3, 5, 7, 9)'$: $3 - 10 + 7 = 0$ and $6 - 15 + 9 = 0$, so it lies in W, with $a = 1$, $b = 2$: $(3, 5, 7, 9)' = \mathbf{1}_4 + 2t$. For $(1, 2, 4, 8)'$: $1 - 4 + 4 = 1 \neq 0$, so it does not lie in W.

Exercise 3.15. Let $w_1 = (1,1,0,1)'$, $w_2 = (1,2,1,0)'$ and $W = \operatorname{span}\{w_1, w_2\} \subseteq \mathbb{R}^4$.

(a) Row-reduce $[w_1\ w_2 \mid y]$ for $y = (y_1, y_2, y_3, y_4)'$, keeping the y_i as letters. Show that $\mathcal{B} = (w_1, w_2)$ is an ordered basis of W, find the conditions for $y \in W$, and find $[y]_\mathcal{B}$ for $y \in W$.

(b) Which of $(3,4,1,2)'$ and $(1,1,1,1)'$ lie in W? Give the coordinate vector of the one that does.

(c) Find $w \in W$ with $[w]_\mathcal{B} = (3,-2)'$, and the coordinate vector of $y + w$ for $y = (3,4,1,2)'$.

Solution. (a)

$$\left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 1 & 2 & y_2 \\ 0 & 1 & y_3 \\ 1 & 0 & y_4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 0 & 1 & y_2 - y_1 \\ 0 & 1 & y_3 \\ 0 & -1 & y_4 - y_1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cc|c} 1 & 1 & y_1 \\ 0 & 1 & y_2 - y_1 \\ 0 & 0 & y_1 - y_2 + y_3 \\ 0 & 0 & -2y_1 + y_2 + y_4 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cc|c} 1 & 0 & 2y_1 - y_2 \\ 0 & 1 & y_2 - y_1 \\ 0 & 0 & y_1 - y_2 + y_3 \\ 0 & 0 & -2y_1 + y_2 + y_4 \end{array}\right)$$

The left part of the last matrix is the RREF of $[w_1\ w_2]$, with both columns pivot columns; so w_1, w_2 are independent (Proposition 3.12), and $\mathcal{B}$ is an ordered basis of W. By Proposition 2.7(a), the system $[w_1\ w_2]c = y$ is consistent, that is, $y \in W$, if and only if

$$y_1 - y_2 + y_3 = 0, \qquad -2y_1 + y_2 + y_4 = 0$$

and then its unique solution is $[y]_\mathcal{B}$ (Definition 3.23 and the paragraph after it):

$$[y]_\mathcal{B} = \begin{pmatrix} 2y_1 - y_2 \\ y_2 - y_1 \end{pmatrix}$$

(b) For $(3,4,1,2)'$: $3 - 4 + 1 = 0$ and $-6 + 4 + 2 = 0$, so it lies in W, with coordinate vector $(6-4,\ 4-3)' = (2,1)'$. Check: $2(1,1,0,1)' + (1,2,1,0)' = (3,4,1,2)'$. For $(1,1,1,1)'$: $1 - 1 + 1 = 1 \neq 0$, so it does not lie in W.

(c) $w = 3w_1 - 2w_2 = (3,3,0,3)' - (2,4,2,0)' = (1,-1,-2,3)'$ (Proposition 3.24(b)). By Proposition 3.24(a), $[y + w]_\mathcal{B} = (2,1)' + (3,-2)' = (5,-1)'$. Check: $y + w = (4,3,-1,5)' = 5(1,1,0,1)' - (1,2,1,0)'$.

Exercise 3.16. Let $u_1 = (1,0,1,1)'$, $u_2 = (1,1,1,1)'$, $u_3 = (1,1,2,1)'$, $u_4 = (0,0,1,1)'$, $U = [u_1\ u_2\ u_3\ u_4]$, and $\tilde{v}_1 = (2,1,3,2)'$, $\tilde{v}_2 = (3,2,3,2)'$, $\tilde{v}_3 = (2,2,2,1)'$, $\tilde{v}_4 = (1,1,3,2)'$, $\tilde{V} = [\tilde{v}_1\ \tilde{v}_2\ \tilde{v}_3\ \tilde{v}_4]$.

(a) Find U^{-1}, and conclude that $\mathcal{B} = (u_1, u_2, u_3, u_4)$ is an ordered basis of $\mathbb{R}^4$.

(b) Find the matrix P with columns $[\tilde{v}_1]_\mathcal{B}, \ldots, [\tilde{v}_4]_\mathcal{B}$, and check that $UP = \tilde{V}$.

(c) Check that

$$Q = \begin{pmatrix} 0 & -2 & 1 & -1 \\ 1 & 2 & -1 & 1 \\ -1 & -1 & 1 & -1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

satisfies $PQ = I_4$, and conclude that $\tilde{\mathcal{B}} = (\tilde{v}_1, \tilde{v}_2, \tilde{v}_3, \tilde{v}_4)$ is an ordered basis of $\mathbb{R}^4$. For the vector w with $[w]_\mathcal{B} = (1,1,1,1)'$, find w and $[w]_{\tilde{\mathcal{B}}}$.

Solution. (a) Row-reduce $[U \mid I_4]$:

$$\left(\begin{array}{cccc|cccc} 1&1&1&0&1&0&0&0\\ 0&1&1&0&0&1&0&0\\ 1&1&2&1&0&0&1&0\\ 1&1&1&1&0&0&0&1 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-R_1]{R_3\leftarrow R_3-R_1} \left(\begin{array}{cccc|cccc} 1&1&1&0&1&0&0&0\\ 0&1&1&0&0&1&0&0\\ 0&0&1&1&-1&0&1&0\\ 0&0&0&1&-1&0&0&1 \end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3-R_4} \left(\begin{array}{cccc|cccc} 1&1&1&0&1&0&0&0\\ 0&1&1&0&0&1&0&0\\ 0&0&1&0&0&0&1&-1\\ 0&0&0&1&-1&0&0&1 \end{array}\right) \xrightarrow[R_2\leftarrow R_2-R_3]{R_1\leftarrow R_1-R_3} \left(\begin{array}{cccc|cccc} 1&1&0&0&1&0&-1&1\\ 0&1&0&0&0&1&-1&1\\ 0&0&1&0&0&0&1&-1\\ 0&0&0&1&-1&0&0&1 \end{array}\right)$$

$$\xrightarrow{R_1\leftarrow R_1-R_2} \left(\begin{array}{cccc|cccc} 1&0&0&0&1&-1&0&0\\ 0&1&0&0&0&1&-1&1\\ 0&0&1&0&0&0&1&-1\\ 0&0&0&1&-1&0&0&1 \end{array}\right)$$

The left half has become I_4, so U is invertible and U^{-1} is the right half (Corollary 2.17(a)). By Example 3.19, $\mathcal{B}$ is an ordered basis of $\mathbb{R}^4$, and $[y]_{\mathcal{B}} = U^{-1}y$ for every $y \in \mathbb{R}^4$.

(b) Column j of P is $[\tilde{v}_j]_{\mathcal{B}} = U^{-1}\tilde{v}_j$, so $P = U^{-1}\tilde{V}$ (Theorem 1.4(a)):

$$P = \begin{pmatrix} 1&-1&0&0\\ 0&1&-1&1\\ 0&0&1&-1\\ -1&0&0&1 \end{pmatrix}\begin{pmatrix} 2&3&2&1\\ 1&2&2&1\\ 3&3&2&3\\ 2&2&1&2 \end{pmatrix} = \begin{pmatrix} 1&1&0&0\\ 0&1&1&0\\ 1&1&1&1\\ 0&-1&-1&1 \end{pmatrix}$$

Check:

$$UP = \begin{pmatrix} 1&1&1&0\\ 0&1&1&0\\ 1&1&2&1\\ 1&1&1&1 \end{pmatrix}\begin{pmatrix} 1&1&0&0\\ 0&1&1&0\\ 1&1&1&1\\ 0&-1&-1&1 \end{pmatrix} = \begin{pmatrix} 2&3&2&1\\ 1&2&2&1\\ 3&3&2&3\\ 2&2&1&2 \end{pmatrix} = \tilde{V}$$

(c) The product

$$PQ = \begin{pmatrix} 1&1&0&0\\ 0&1&1&0\\ 1&1&1&1\\ 0&-1&-1&1 \end{pmatrix}\begin{pmatrix} 0&-2&1&-1\\ 1&2&-1&1\\ -1&-1&1&-1\\ 0&1&0&1 \end{pmatrix} = \begin{pmatrix} 1&0&0&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1 \end{pmatrix}$$

shows that P is invertible with $P^{-1} = Q$ (Theorem 2.16). As $\tilde{v}_j = U\,\mathrm{col}_j(P) = p_{1j}u_1 + p_{2j}u_2 + p_{3j}u_3 + p_{4j}u_4$, Theorem 3.28 shows that $\tilde{\mathcal{B}}$ is an ordered basis of $\mathbb{R}^4$, and P is the change-of-basis matrix from $\tilde{\mathcal{B}}$ to $\mathcal{B}$. Then $w = u_1 + u_2 + u_3 + u_4 = (3,2,5,4)'$ (Proposition 3.24(b)), and by Theorem 3.27(b)

$$[w]_{\tilde{\mathcal{B}}} = Q[w]_{\mathcal{B}} = \begin{pmatrix} 0&-2&1&-1\\ 1&2&-1&1\\ -1&-1&1&-1\\ 0&1&0&1 \end{pmatrix}\begin{pmatrix} 1\\1\\1\\1 \end{pmatrix} = \begin{pmatrix} -2\\3\\-2\\2 \end{pmatrix}$$

Check: $-2\tilde{v}_1 + 3\tilde{v}_2 - 2\tilde{v}_3 + 2\tilde{v}_4 = (-4,-2,-6,-4)' + (9,6,9,6)' + (-4,-4,-4,-2)' + (2,2,6,4)' = (3,2,5,4)' = w$.

Exercise 3.17. In $\mathbb{R}^4$ let $U = \operatorname{span}\{u_1, u_2\}$ and $W = \operatorname{span}\{w_1, w_2\}$, where $u_1 = (1,0,1,1)'$, $u_2 = (0,1,1,-1)'$, $w_1 = (1,1,2,0)'$ and $w_2 = (1,0,0,1)'$.

(a) Row-reduce $M = [u_1\ u_2\ w_1\ w_2]$ and find the linear relations among its columns.

(b) Find a basis of $U + W$.

(c) Find a basis of $U \cap W$. Is the sum $U + W$ direct?

Solution. (a)

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow -R_3} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_3} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

$Rc = 0$ reads $c_1 + c_3 = 0$, $c_2 + c_3 = 0$, $c_4 = 0$, with $c_3 = t$ free; so the relations are $c = t(-1,-1,1,0)'$ (Theorem 2.3(b), Proposition 2.7(b)), the multiples of $-u_1 - u_2 + w_1 = 0$: $w_1 = u_1 + u_2$.

(b) By Theorem 3.31(c) and Theorem 3.10(c), $U + W = \operatorname{span}\{u_1, u_2, w_1, w_2\} = \operatorname{span}\{u_1, u_2, w_2\}$. The same row operations take $[u_1\ u_2\ w_2]$ to columns $1, 2, 4$ of R (Theorems 2.15(c) and 1.4(a)), which are $e_1, e_2, e_3 \in \mathbb{R}^4$; so u_1, u_2, w_2 are independent (Proposition 3.12) and form a basis of $U + W$.

(c) $x \in U \cap W$ if and only if $x = a_1u_1 + a_2u_2 = b_1w_1 + b_2w_2$ for some scalars, that is, $M(a_1, a_2, -b_1, -b_2)' = 0$. By (a), $(a_1, a_2, -b_1, -b_2)' = t(-1,-1,1,0)'$, so $b_1 = -t$, $b_2 = 0$ and $x = -tw_1$. Hence $U \cap W = \operatorname{span}\{w_1\}$, with the basis $w_1 = (1,1,2,0)'$ (Proposition 3.13(a)). The sum is not direct, since $U \cap W \neq \{0\}$ (Definition 3.30).

Exercise 3.18 ([8, Ch. 17, Exercise 2])**.** Let U and W be the subspaces of $\mathbb{R}^3$ spanned by the columns of

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad \text{and} \qquad B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \\ 2 & 3 \end{pmatrix}$$

respectively. Find (a) a basis of $U + W$; (b) a basis of $U \cap W$ (is the sum $U + W$ direct?); (c) a vector of $U + W$ that is not in $U \cup W$.

Solution. Let $b_1 = (0,1,2)'$ and $b_2 = (2,1,3)'$ be the columns of B; the columns of A are e_1, e_2.

(a)

- By Theorem 3.31(c), $U + W = \operatorname{span}\{e_1, e_2, b_1, b_2\}$.
- e_1, e_2, b_1 are independent: $c_1e_1 + c_2e_2 + c_3b_1 = (c_1,\ c_2 + c_3,\ 2c_3)' = 0$ gives $c_3 = 0$, then $c_2 = 0$ and $c_1 = 0$.
- So $M = [e_1\ e_2\ b_1] \in \mathbb{R}^{3\times 3}$ has $Mc = 0$ only for $c = 0$. By Theorem 2.16, M is invertible, and by Example 3.19 its columns form a basis of $\mathbb{R}^3$.
- These columns lie in the subspace $U + W$, so every vector of $\mathbb{R}^3$, a combination of them, lies in $U + W$. Hence $U + W = \mathbb{R}^3$, and e_1, e_2, b_1 form a basis of $U + W$.

(b)

- U consists of the vectors $(x_1, x_2, 0)'$, and W of the vectors $y_1b_1 + y_2b_2 = (2y_2,\ y_1 + y_2,\ 2y_1 + 3y_2)'$ $(x_1, x_2, y_1, y_2 \in \mathbb{R})$.

- So $U \cap W$ consists of the vectors $y_1 b_1 + y_2 b_2$ with $2y_1 + 3y_2 = 0$. This equation holds exactly when $y_1 = 3t$ and $y_2 = -2t$ for some $t \in \mathbb{R}$: these values give $6t - 6t = 0$; conversely, if $2y_1 + 3y_2 = 0$, then $t = y_1 + y_2$ has $3t = 3y_1 - 2y_1 = y_1$ and $-2t = 3y_2 - 2y_2 = y_2$.
- For $y_1 = 3t$, $y_2 = -2t$: $y_1 b_1 + y_2 b_2 = (-4t,\ t,\ 0)' = t\,(-4, 1, 0)'$. Hence $U \cap W = \operatorname{span}\{(-4, 1, 0)'\}$, and the single nonzero vector $(-4, 1, 0)'$ is a basis of it (Proposition 3.13(a)).
- Since $U \cap W \neq \{0\}$, the sum $U + W$ is not direct (Definition 3.30).

(c) Take $x = (0, 2, 2)'$; it lies in $U + W = \mathbb{R}^3$.

- $x \notin U$, since its third entry is not 0.
- If $x = (2y_2,\ y_1 + y_2,\ 2y_1 + 3y_2)'$, the first two entries give $y_2 = 0$ and $y_1 = 2$, and then the third entry would be $4 \neq 2$. So $x \notin W$.

Exercise 3.19 ([4, Ch. I, §4, Exercise 2], [7])**.** In $\mathbb{R}^3$ let $W = \operatorname{span}\{e_1\}$ and $U = \operatorname{span}\{u_1, u_2\}$, where $u_1 = (1, 1, 0)'$ and $u_2 = (0, 1, 1)'$. Show that $\mathbb{R}^3 = W \oplus U$, and write $x = (1, 2, 3)'$ as $x = w + u$ with $w \in W$ and $u \in U$.

Solution. The vectors e_1, u_1, u_2 are linearly independent: if $ae_1 + bu_1 + cu_2 = 0$, then

$$a + b = 0, \qquad b + c = 0, \qquad c = 0$$

so $c = 0$, then $b = 0$ and $a = 0$.

- $W \cap U = \{0\}$: e_1 is a basis of W, and u_1, u_2 (independent by Proposition 3.13(b)) form a basis of U. Their combined list e_1, u_1, u_2 is independent, so the sum $W + U$ is direct (Theorem 3.33).
- $W + U = \mathbb{R}^3$: the matrix $M = [e_1\ u_1\ u_2] \in \mathbb{R}^{3\times 3}$ has $Mc = 0$ only for $c = 0$, so it is invertible (Theorem 2.16), and its columns span $\mathbb{R}^3$ (Example 3.19). By Theorem 3.31(c), $W + U = \operatorname{span}\{e_1, u_1, u_2\} = \mathbb{R}^3$.

Hence $\mathbb{R}^3 = W \oplus U$. To write $x = (1, 2, 3)'$ as $ae_1 + bu_1 + cu_2$, solve the system whose augmented matrix is already in echelon form,

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

Back substitution (Theorem 2.26) gives $c = 3$, $b = 2 - 3 = -1$, $a = 1 - (-1) = 2$. So $w = 2e_1 = (2, 0, 0)'$ and $u = -u_1 + 3u_2 = (-1, 2, 3)'$; indeed $w + u = (1, 2, 3)'$. By Theorem 3.32, this is the only such decomposition.

Problems

Problem 3.1 ([6, Exercise 3.9(c)])**.** Is the set of vectors $(x, y)' \in \mathbb{R}^2$ with $x \geqslant 0$ and $y \geqslant 0$ a subspace of $\mathbb{R}^2$?

Solution. No: $(1, 1)'$ lies in the set, but $(-1)(1, 1)' = (-1, -1)'$ does not; so the set is not closed under scalar multiplication (Theorem 3.4(b)).

Problem 3.2 ([6, Exercise 3.11])**.** In the vector space $\mathbb{R}^{3\times 3}$, show that the set of lower triangular matrices is a subspace.

Solution. Let $A, B \in \mathbb{R}^{3\times 3}$ be lower triangular, that is, $a_{ij} = b_{ij} = 0$ for $i < j$, and let $c \in \mathbb{R}$. For $i < j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0$ and $(cA)_{ij} = ca_{ij} = 0$; so $A + B$ and cA are lower triangular, and so is 0. By Theorem 3.4(b), the lower triangular matrices form a subspace.

Problem 3.3 ([1, §2.2, Exercise 7])**.** Let U and W be subspaces of a vector space V such that the union $U \cup W$ is also a subspace. Show that one of the spaces U, W is contained in the other.

Solution. Suppose that neither space is contained in the other: there are $u \in U$ with $u \notin W$, and $w \in W$ with $w \notin U$. Both u and w lie in $U \cup W$.

- If $u + w \in U$, then, as U is a subspace, $w = (u + w) + (-1)u \in U$ ✘.
- If $u + w \in W$, then, as W is a subspace, $u = (u + w) + (-1)w \in W$ ✘.
- So $u + w \notin U \cup W$: the set $U \cup W$ is not closed under addition, and it is not a subspace (Theorem 3.4(b)).

Hence, if $U \cup W$ is a subspace, then $U \subseteq W$ or $W \subseteq U$.

Problem 3.4 ([8, Ch. 4, Exercise 7])**.** Suppose that $v_1, \ldots, v_k$ span a vector space V but do not form a basis of V. Show that, for any $v \in V$, the representation of v as a linear combination of $v_1, \ldots, v_k$ is not unique.

Solution. Let $v \in V$. Since $v_1, \ldots, v_k$ span V, there are scalars $x_1, \ldots, x_k$ with $v = \sum_i x_i v_i$. Since $v_1, \ldots, v_k$ span V but do not form a basis, they are linearly dependent (Definition 3.17): there are scalars $z_1, \ldots, z_k$, not all 0, with $\sum_i z_i v_i = 0$. Let $y_i = x_i + z_i$. Then $\sum_i y_i v_i = \sum_i x_i v_i + \sum_i z_i v_i = v + 0 = v$, and $(y_1, \ldots, y_k) \neq (x_1, \ldots, x_k)$ because some $z_i \neq 0$. So v has two different representations. (Compare Theorem 3.18.)

Problem 3.5 ([4, Ch. III, §3, Exercise 14], [7])**.** Show that every $A \in \mathbb{R}^{n \times n}$ can be written in exactly one way as $A = B + C$ with B symmetric and C skew-symmetric, and conclude that $\mathbb{R}^{n \times n}$ is the direct sum of the subspace of symmetric matrices and the subspace of skew-symmetric matrices.

Solution. Both sets are subspaces of $\mathbb{R}^{n \times n}$ (Example 3.5(d)).

- Existence: let $B = \frac{1}{2}(A + A')$ and $C = \frac{1}{2}(A - A')$. Then $B + C = A$, and, since $(A')' = A$ (Theorem 1.9), $B' = \frac{1}{2}(A' + A) = B$ and $C' = \frac{1}{2}(A' - A) = -C$. (This is Proposition 1.27(b).)
- Uniqueness: let also $A = B_1 + C_1$ with B_1 symmetric and C_1 skew-symmetric. Then $B_1 - B = C - C_1$, where $B_1 - B$ is symmetric and $C - C_1$ is skew-symmetric. So $B_1 - B = (B_1 - B)' = (C - C_1)' = -(C - C_1) = -(B_1 - B)$, hence $2(B_1 - B) = 0$. Thus $B_1 = B$, and then $C_1 = A - B_1 = C$.
- So the sum of the two subspaces is all of $\mathbb{R}^{n \times n}$, and by Theorem 3.32 their intersection is $\{0\}$: $\mathbb{R}^{n \times n}$ is their direct sum.

For a numerical instance, see Example 1.28, where the two parts of a 4×4 matrix are called S and K.

Problem 3.6 ([8, Ch. 17, Exercise 4])**.** Let U, W and X be subspaces of a vector space V.

(a) Show that $(U \cap W) + (U \cap X) \subseteq U \cap (W + X)$.

(b) Show by an example that $U \cap W = \{0\}$ and $U \cap X = \{0\}$ do not imply $U \cap (W + X) = \{0\}$.

(c) Show that if $W \subseteq U$, then $U + W = U$ and $U \cap (W + X) = W + (U \cap X)$.

Solution. By Theorem 3.31(a), all the intersections and sums of subspaces below are subspaces of V.

(a) Let $y \in (U \cap W) + (U \cap X)$, that is, $y = w + x$ with $w \in U \cap W$ and $x \in U \cap X$. Since w and x lie in the subspace U, so does y; since $w \in W$ and $x \in X$, $y \in W + X$. So $y \in U \cap (W + X)$.

(b) In $V = \mathbb{R}^2$ let U, W and X be the spans of $(1, 1)'$, $(1, 0)'$ and $(0, 1)'$, respectively.

- $U \cap W = \{0\}$: a vector of $U \cap W$ is $a(1, 1)' = b(1, 0)'$ for some $a, b \in \mathbb{R}$; then $(a, a)' = (b, 0)'$, so $a = 0$, and the vector is 0. Likewise $a(1, 1)' = b(0, 1)'$ gives $(a, a)' = (0, b)'$, so $a = 0$, and $U \cap X = \{0\}$.

- $W + X = \text{span}\{e_1, e_2\} = \mathbb{R}^2$ (Theorem 3.31(c) and the standard basis, Definition 3.17). So $U \cap (W + X) = U \neq \{0\}$.

(c) Let $W \subseteq U$.

- $U + W = U$: $U \subseteq U + W$ by Theorem 3.31(b). Conversely, if $y = u + w$ with $u \in U$ and $w \in W$, then $w \in U$, so $y \in U$.
- $W + (U \cap X) \subseteq U \cap (W + X)$: this is (a), since $U \cap W = W$.
- $U \cap (W + X) \subseteq W + (U \cap X)$: let $y \in U \cap (W + X)$, say $y = w + x$ with $w \in W$ and $x \in X$. Then $x = y - w$, where $y \in U$ and $w \in U$ (as $W \subseteq U$); so $x \in U$, as U is a subspace. Hence $x \in U \cap X$, and $y = w + x \in W + (U \cap X)$.

Problem 3.7 ([10, Problem 5.3])**.** Let $v \in \mathbb{R}^n$ have all entries nonzero, and let $W = \{w \in \mathbb{R}^n : v'w = 0\}$, a subspace of $\mathbb{R}^n$ (Example 3.5(e), with the $1 \times n$ matrix v'). For $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)'$ with every $\varepsilon_i \in \{1, -1\}$, the **orthant** O_ε is the set of the $w \in \mathbb{R}^n$ with $\varepsilon_i w_i > 0$ for $i = 1, \ldots, n$. Show that W contains a vector of every orthant except the two orthants that contain v and $-v$, and that W contains no vector of these two orthants.

Solution. Here $v'w = \sum_i v_i w_i$ (Theorem 1.9), and $\varepsilon_i^2 = 1$ for every i.

- Let $v \in O_\varepsilon$. For $w \in O_\varepsilon$, $v_i w_i = (\varepsilon_i v_i)(\varepsilon_i w_i) > 0$ for every i, so $v'w > 0$ and $w \notin W$. Likewise, if $-v \in O_\varepsilon$ and $w \in O_\varepsilon$, then $-v_i w_i = (-\varepsilon_i v_i)(\varepsilon_i w_i) > 0$ for every i, so $v'w < 0$ and $w \notin W$.
- Let O_ε contain neither v nor $-v$. As $v \notin O_\varepsilon$, some $\varepsilon_j v_j \leqslant 0$, hence $\varepsilon_j v_j < 0$ since $v_j \neq 0$; as $-v \notin O_\varepsilon$, some $-\varepsilon_i v_i \leqslant 0$, hence $\varepsilon_i v_i > 0$, again since $v_i \neq 0$.
- The vector ε itself lies in O_ε, since $\varepsilon_i \varepsilon_i = 1 > 0$. Let $s = v'\varepsilon$. If $s = 0$, then $\varepsilon \in W \cap O_\varepsilon$.
- If $s > 0$, enlarge entry j: let $w = \varepsilon + t\varepsilon_j e_j$ with $t = s/(-\varepsilon_j v_j) > 0$. Entry j of w is $(1 + t)\varepsilon_j$, so $\varepsilon_j w_j = 1 + t > 0$, and the other entries are those of ε; so $w \in O_\varepsilon$. Moreover $v'w = s + t\varepsilon_j v_j = s - s = 0$.
- If $s < 0$, enlarge entry i: let $w = \varepsilon + t\varepsilon_i e_i$ with $t = -s/(\varepsilon_i v_i) > 0$. As before, $w \in O_\varepsilon$, and $v'w = s + t\varepsilon_i v_i = s - s = 0$.

In each case W contains a vector of O_ε. For $v = \mathbf{1}_n$, W is the subspace W_0 of centered vectors (Example 3.5(f)): a centered vector without zero entries can have any pattern of signs except all entries positive and all entries negative.

Problem 3.8 ([10, Problems 7.1, 7.2])**.** Let $v_1, \ldots, v_m \in \mathbb{R}^n$.

(a) Show that if $m \geqslant n+2$, then there are numbers $\alpha_1, \ldots, \alpha_m$, not all 0, with $\alpha_1 v_1 + \cdots + \alpha_m v_m = 0$ and $\alpha_1 + \cdots + \alpha_m = 0$.

(b) A **convex combination** of $v_1, \ldots, v_m$ is a vector $x = t_1 v_1 + \cdots + t_m v_m$ with $t_1, \ldots, t_m \geqslant 0$ and $t_1 + \cdots + t_m = 1$. Show that every convex combination of $v_1, \ldots, v_m$ is a convex combination of at most $n + 1$ of the vectors $v_1, \ldots, v_m$.

Solution. (a) For $i = 1, \ldots, m$, let $u_i \in \mathbb{R}^{n+1}$ be v_i with the entry 1 appended: its first n entries are those of v_i, and its last entry is 1. For numbers $\alpha_1, \ldots, \alpha_m$, the first n entries of $\sum_i \alpha_i u_i$ form the vector $\sum_i \alpha_i v_i$, and its last entry is $\sum_i \alpha_i$. The m vectors $u_1, \ldots, u_m$ of $\mathbb{R}^{n+1}$ are more than $n + 1$, so they are linearly dependent (Proposition 3.12): $\sum_i \alpha_i u_i = 0$ for some $\alpha_1, \ldots, \alpha_m$, not all 0. These numbers satisfy $\sum_i \alpha_i v_i = 0$ and $\sum_i \alpha_i = 0$.

(b) We show, by induction on m, that a convex combination x of any m vectors of $\mathbb{R}^n$ is a convex combination of at most $n + 1$ of them.

- If $m \leqslant n + 1$, there is nothing to prove.

- Let $m \geqslant n+2$, assume the statement for $m-1$ vectors, and let $x = \sum_i t_i v_i$ with all $t_i \geqslant 0$ and $\sum_i t_i = 1$. Take $\alpha_1, \ldots, \alpha_m$ as in (a). Some α_i is positive: otherwise all $\alpha_i \leqslant 0$, and as $\sum_i \alpha_i = 0$, all α_i would be 0.
- For every $\lambda \in \mathbb{R}$, $x = \sum_i t_i v_i + \lambda \sum_i \alpha_i v_i = \sum_i \tilde{t}_i v_i$ with $\tilde{t}_i = t_i + \lambda\alpha_i$, and $\sum_i \tilde{t}_i = \sum_i t_i + \lambda \sum_i \alpha_i = 1$.
- Among the indices i with $\alpha_i > 0$, choose k with $t_k/\alpha_k \leqslant t_i/\alpha_i$ for all of them, and take $\lambda = -t_k/\alpha_k \leqslant 0$. Then $\tilde{t}_k = t_k - t_k = 0$. If $\alpha_i > 0$, then $\tilde{t}_i = \alpha_i\big(t_i/\alpha_i - t_k/\alpha_k\big) \geqslant 0$; if $\alpha_i \leqslant 0$, then $\lambda\alpha_i \geqslant 0$ and $\tilde{t}_i \geqslant t_i \geqslant 0$.
- So $x = \sum_{i \neq k} \tilde{t}_i v_i$ with all $\tilde{t}_i \geqslant 0$ and $\sum_{i \neq k} \tilde{t}_i = 1$: x is a convex combination of the $m-1$ vectors v_i $(i \neq k)$. By the induction hypothesis, it is a convex combination of at most $n+1$ of them.

Chapter 4. Bases, dimension, and rank

Bases and dimension of subspaces of $\mathbb{R}^n$

Theorem 4.1 (cf. [1, §2.3, Example 14])**.** Let R be the RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows. The following are equivalent:

(i) the columns $a_1, \dots, a_n$ of A are linearly independent;

(ii) the system $Ax = 0$ has only the solution $x = 0$;

(iii) $r = n$, that is, every column of R is a pivot column.

In particular, if $n > m$, then any n vectors of $\mathbb{R}^m$ are linearly dependent.

We call this the **pivot test**.

Proof. This is Proposition 3.12 with $V = A$ and $k = n$: each nonzero row of R contains exactly one pivot, and the pivots lie in different columns (Definition 2.4), so R has r pivot columns, and every column of R is a pivot column exactly when $r = n$. □

Remark 4.2 (cf. [1, §1.3, proof of Theorem 4])**.** For the pivot test one may stop at an echelon form E of A (Definitions 2.4 and 2.5). Backward elimination adds multiples of a row i, whose leading 1 is in column k_i, to rows $l < i$, whose leading 1 is in a column $k_l < k_i$. Row i is 0 before column k_i, so row l changes only in columns $\geqslant k_i$ and keeps its leading 1 in column k_l. Hence the RREF of A reached from E has the same leading columns as E. As the RREF of A is unique (Theorem 2.8), the pivot columns of A, and their number r, can be read off from any echelon form of A. In particular, (iii) of Theorem 4.1 says that every column of E contains a leading 1. The same holds before the leading entries are scaled to 1: scaling a row does not move its leading entry.

Theorem 4.3 (cf. [13, Ch. III, §6, Theorem 6.1])**.** Let R be the RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and entries R_{ij}.

(a) The columns of A and those of R satisfy the same linear relations: for $x \in \mathbb{R}^n$, $x_1a_1+\cdots+x_na_n = 0$ if and only if $x_1 \operatorname{col}_1(R) + \cdots + x_n \operatorname{col}_n(R) = 0$.

(b) If $r \geqslant 1$, the pivot columns $a_{k_1}, \dots, a_{k_r}$ of A form a basis of $\operatorname{span}\{a_1, \dots, a_n\}$, and for every $j = 1, \dots, n$

$$a_j = R_{1j}a_{k_1} + R_{2j}a_{k_2} + \cdots + R_{rj}a_{k_r}$$

If $r = 0$, then $A = 0$ and $\operatorname{span}\{a_1, \dots, a_n\} = \{0\}$.

In particular, every finite list that spans a subspace W contains a basis of W.

Proof. (a) The two sums are Ax and Rx, and $Ax = 0$ if and only if $Rx = 0$ (Theorem 2.3(b)).

(b) If $r = 0$, then $R = 0$, and $A = P^{-1}R = 0$, where $R = PA$ with $P \in \mathbb{R}^{m\times m}$ invertible (Theorem 2.15(c)). Let $r \geqslant 1$.

- Columns of R: $\operatorname{col}_{k_i}(R) = e_i$, and rows $r+1, \dots, m$ of R are zero, so $\operatorname{col}_j(R) = \sum_{i=1}^r R_{ij}e_i = \sum_{i=1}^r R_{ij} \operatorname{col}_{k_i}(R)$ for every j.
- Relations: for a column $j \notin \{k_1, \dots, k_r\}$ let $x \in \mathbb{R}^n$ have $x_j = 1$, $x_{k_i} = -R_{ij}$ $(1 \leqslant i \leqslant r)$ and all other entries 0. By the first item, $Rx = \operatorname{col}_j(R) - \sum_i R_{ij} \operatorname{col}_{k_i}(R) = 0$; by (a), $Ax = a_j - \sum_i R_{ij}a_{k_i} = 0$. For $j = k_l$ the formula reads $a_{k_l} = a_{k_l}$, since R_{ik_l} is 1 for $i = l$ and 0 otherwise.
- Spanning: each a_j lies in the subspace $U = \operatorname{span}\{a_{k_1}, \dots, a_{k_r}\}$, so $\operatorname{span}\{a_1, \dots, a_n\} \subseteq U$ (Theorem 3.7); the reverse inclusion is clear.
- Independence: if $c_1a_{k_1}+\cdots+c_ra_{k_r} = 0$, then by (a) $c_1 \operatorname{col}_{k_1}(R)+\cdots+c_r \operatorname{col}_{k_r}(R) = c_1e_1+\cdots+c_re_r = 0$, so every $c_i = 0$.

For the last sentence, write the spanning vectors as the columns of a matrix and apply (b). □

By Theorem 4.3(a), a_j is a linear combination of $a_1, \dots, a_{j-1}$ if and only if $\operatorname{col}_j(R)$ is a linear combination of $\operatorname{col}_1(R), \dots, \operatorname{col}_{j-1}(R)$ (take $x_j = -1$ and $x_l = 0$ for $l > j$). In R this happens exactly when j is not a pivot column: $\operatorname{col}_{k_i}(R) = e_i$, while the columns before k_i are 0 in rows $i, \dots, m$; and a non-pivot column j equals $\sum_{k_i<j} R_{ij} \operatorname{col}_{k_i}(R)$. So the pivot columns of A are the columns that are not combinations of the earlier ones; they do not depend on the row operations used.

Theorem 4.4 ([1, §2.3, Theorems 4, 5])**.** Every subspace W of $\mathbb{R}^n$ has a basis, and any two bases of W have the same number of elements. This number is the **dimension** of W, written $\dim W$. For example, $\dim \mathbb{R}^n = n$ (standard basis) and $\dim\{0\} = 0$ (the empty list is a basis of $\{0\}$). Conversely, $\dim W = 0$ only for $W = \{0\}$: a basis of W with no vectors is the empty list, which spans $\{0\}$ (Definition 3.6). Moreover, if W is spanned by m vectors, then any $k > m$ vectors of W are linearly dependent.

Proof. • If W is spanned by $w_1, \dots, w_m$, then any $k > m$ vectors $v_1, \dots, v_k$ of W are linearly dependent. Indeed, write $v_j = a_{1j}w_1 + \cdots + a_{mj}w_m$ $(1 \leqslant j \leqslant k)$ and $A = (a_{ij}) \in \mathbb{R}^{m\times k}$. As $k > m$, the columns of A are dependent (Theorem 4.1): $Ax = 0$ for some $x \neq 0$. Then

$$x_1v_1 + \cdots + x_kv_k = \sum_{j=1}^{k} x_j \sum_{i=1}^{m} a_{ij}w_i = \sum_{i=1}^{m} (Ax)_i w_i = 0$$

- Same size: if $v_1, \dots, v_k$ and $w_1, \dots, w_m$ are bases of W, each list spans W and is independent, so the first item gives $k \leqslant m$ and $m \leqslant k$.
- Existence: start with the empty list and, as long as the vectors chosen do not span W, adjoin a vector $v \in W$ outside their span. The list stays independent: in a relation $c_1v_1 + \cdots + c_kv_k + cv = 0$, $c \neq 0$ would put v in the span of $v_1, \dots, v_k$, so $c = 0$, and then every $c_i = 0$. More than n vectors of $\mathbb{R}^n$ are dependent (Theorem 4.1), so the process stops after at most n steps, at a basis of W. □

By Theorem 4.3, $\dim \operatorname{span}\{a_1, \dots, a_n\} = r$, the number of pivots of the RREF of $[a_1 \ \cdots \ a_n]$.

Theorem 4.5 ([1, §2.3, Theorems 4, 5], [4, Ch. I, Theorem 3.4])**.** Let W be a subspace of $\mathbb{R}^n$ with $\dim W = d$.

(a) If U is a subspace of W, then $\dim U \leqslant d$, and $\dim U = d$ only for $U = W$. In particular $d \leqslant n$, and $d = n$ only for $W = \mathbb{R}^n$.

(b) Every linearly independent list of vectors of W can be extended to a basis of W.

(c) More than d vectors of W are linearly dependent; fewer than d vectors do not span W.

(d) d vectors of W are linearly independent $\Leftrightarrow$ they span W $\Leftrightarrow$ they form a basis of W.

Proof. If $d = 0$, then $W = \{0\}$ (Theorem 4.4), and (a)–(c) are immediate: every subspace U of W is $\{0\} = W$; every vector of W is 0, so a nonempty list of vectors of W is dependent (Proposition 3.13(a)); and the empty list, the only independent one, is a basis of W (Definition 3.17). This also gives (d): the only list of 0 vectors is the empty list, which is independent, spans $\{0\} = W$ (Definition 3.6) and is a basis of W. Let $d \geqslant 1$, and let $w_1, \dots, w_d$ be a basis of W (Theorem 4.4). We prove (c) first.

(c)

- The vectors $w_1, \dots, w_d$ span W, so any $k > d$ vectors of W are linearly dependent, by Theorem 4.4 (with $m = d$).
- If $k < d$ vectors spanned W, Theorem 4.4, applied to them (with $m = k$), would make the $d > k$ vectors $w_1, \dots, w_d$ of W dependent ✘

(b)

- Let $v_1, \dots, v_k$ be linearly independent vectors of W. If they span W, they form a basis of W.
- Otherwise take $v \in W$ outside $\operatorname{span}\{v_1, \dots, v_k\}$. Then $v_1, \dots, v_k, v$ are independent: in a relation $c_1v_1 + \cdots + c_kv_k + cv = 0$, $c \neq 0$ would put v in the span, so $c = 0$, and then every $c_i = 0$.
- Repeat this. Each list obtained consists of independent vectors of W, so it has at most d vectors by (c); hence the process stops, at a basis of W that contains $v_1, \dots, v_k$.

(a)

- A basis of U (Theorem 4.4) consists of $\dim U$ independent vectors of W, so $\dim U \leqslant d$ by (c).
- If $U \neq W$, take $v \in W$ outside U. Adjoining v to a basis of U gives, as in (b), $\dim U + 1$ independent vectors of W, so $\dim U + 1 \leqslant d$ by (c). Hence $\dim U = d$ only for $U = W$.
- For the last sentence, apply this to W as a subspace of $\mathbb{R}^n$, which has dimension n (Theorem 4.4).

(d) Let $v_1, \dots, v_d \in W$. A basis of W is independent and spans W (Definition 3.17).

- Let $v_1, \dots, v_d$ be independent, and let $w \in W$. By (c), the $d+1$ vectors $w, v_1, \dots, v_d$ are dependent: $x_0w + x_1v_1 + \cdots + x_dv_d = 0$ with not all $x_i = 0$. Here $x_0 \neq 0$, since $v_1, \dots, v_d$ are independent; so $w = -(x_1/x_0)v_1 - \cdots - (x_d/x_0)v_d$. Thus $v_1, \dots, v_d$ span W, and they form a basis.
- Let $v_1, \dots, v_d$ span W. By Theorem 4.3, some of them form a basis of W, which has d vectors (Theorem 4.4); so it consists of all of $v_1, \dots, v_d$, and they form a basis. □

Transposition carries a subspace of $\mathbb{R}^{1\times n}$ onto a subspace of $\mathbb{R}^n$ and preserves linear combinations, hence independence, spanning and bases; so Theorems 4.4 and 4.5 also hold for subspaces of $\mathbb{R}^{1\times n}$, such as the row space $\mathcal{R}(A)$ below.

Example 4.6 (cf. [2, Example 1.5])**.** Let $W = \operatorname{span}\{u_1, \dots, u_5\} \subseteq \mathbb{R}^4$, where

$$u_1 = \begin{pmatrix}1\\2\\1\\1\end{pmatrix}, \quad u_2 = \begin{pmatrix}1\\3\\2\\1\end{pmatrix}, \quad u_3 = \begin{pmatrix}1\\1\\0\\1\end{pmatrix}, \quad u_4 = \begin{pmatrix}1\\2\\2\\3\end{pmatrix}, \quad u_5 = \begin{pmatrix}1\\3\\1\\-1\end{pmatrix}$$

(a) Are $u_1, \dots, u_5$ linearly independent? (b) Find a basis of W among $u_1, \dots, u_5$, and $\dim W$. (c) Express the remaining vectors in this basis.

Solution. Row-reduce $A = [u_1 \ \cdots \ u_5]$:

$$\begin{pmatrix}1&1&1&1&1\\2&3&1&2&3\\1&2&0&2&1\\1&1&1&3&-1\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-2R_1,\ R_3\leftarrow R_3-R_1} \begin{pmatrix}1&1&1&1&1\\0&1&-1&0&1\\0&1&-1&1&0\\0&0&0&2&-2\end{pmatrix}$$

$$\xrightarrow{R_3\leftarrow R_3-R_2} \begin{pmatrix}1&1&1&1&1\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&2&-2\end{pmatrix} \xrightarrow{R_4\leftarrow R_4-2R_3} \begin{pmatrix}1&1&1&1&1\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix}$$

$$\xrightarrow{R_1\leftarrow R_1-R_3} \begin{pmatrix}1&1&1&0&2\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix} \xrightarrow{R_1\leftarrow R_1-R_2} \begin{pmatrix}1&0&2&0&1\\0&1&-1&0&1\\0&0&0&1&-1\\0&0&0&0&0\end{pmatrix} = R$$

So $r = 3$, with pivot columns $k_1 = 1$, $k_2 = 2$, $k_3 = 4$.

(a) No: $r = 3 < 5$ (Theorem 4.1); indeed any five vectors of $\mathbb{R}^4$ are dependent.

(b) By Theorem 4.3, u_1, u_2, u_4 form a basis of W, and $\dim W = 3$. As $3 < 4$, $W \neq \mathbb{R}^4$ (Theorem 4.5(a)).

(c) Columns 3 and 5 of R give $u_3 = 2u_1 - u_2$ and $u_5 = u_1 + u_2 - u_4$ (Theorem 4.3(b)). Check: $2(1,2,1,1)' - (1,3,2,1)' = (1,1,0,1)' = u_3$ and $(1,2,1,1)' + (1,3,2,1)' - (1,2,2,3)' = (1,3,1,-1)' = u_5$. Equivalently, $2u_1 - u_2 - u_3 = 0$ and $u_1 + u_2 - u_4 - u_5 = 0$.

The four subspaces of a matrix

Definition 4.7 ([5, §4.1], [1, §2.2, Examples 7, 10], [3, §2.1])**.** For $A \in \mathbb{R}^{m\times n}$,

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \mathcal{R}(A) = \{y'A : y \in \mathbb{R}^m\} \subseteq \mathbb{R}^{1\times n}$$
$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n, \qquad \mathcal{N}(A') = \{y \in \mathbb{R}^m : A'y = 0\} \subseteq \mathbb{R}^m$$

are the **column space**, **row space**, **null space** and **left null space** of A. The columns of A span $\mathcal{C}(A)$, the rows of A span $\mathcal{R}(A)$ (a space of row vectors), $\mathcal{N}(A)$ is the set of solutions of $Ax = 0$, and $y \in \mathcal{N}(A')$ if and only if $y'A = 0$ (transpose both sides). All four are subspaces (Theorem 3.7 for $\mathcal{C}(A)$ and $\mathcal{R}(A)$, Example 3.5(e) for $\mathcal{N}(A)$ and $\mathcal{N}(A')$).

Let R be the RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and other columns $f_1 < \cdots < f_{n-r}$, so that $x_{f_1}, \ldots, x_{f_{n-r}}$ are the free unknowns of $Ax = 0$ (Proposition 2.7(b)). For $j = 1, \ldots, n-r$, the **special solution** $s_j \in \mathbb{R}^n$ is the solution of $Ax = 0$ with free unknowns $x_{f_j} = 1$ and $x_{f_l} = 0$ $(l \neq j)$ [1, §2.3, Example 15]. By Proposition 2.7(b), its entries are

$$(s_j)_{f_l} = \delta_{jl} \quad (1 \leqslant l \leqslant n-r), \qquad (s_j)_{k_i} = -R_{if_j} \quad (1 \leqslant i \leqslant r)$$

Theorem 4.8 ([1, §2.5, Theorems 9, 10; §2.3, Example 15])**.** Let $A \in \mathbb{R}^{m\times n}$, and let row operations take $[A \mid I_m]$ to $[R \mid P]$, where R is the RREF of A, with nonzero rows $\rho_1', \ldots, \rho_r'$ $(\rho_i \in \mathbb{R}^n)$, pivot columns $k_1 < \cdots < k_r$ and free columns $f_1 < \cdots < f_{n-r}$. Then $P \in \mathbb{R}^{m\times m}$ is invertible, $PA = R$, and:

(a) if $r \geqslant 1$, the pivot columns $a_{k_1}, \ldots, a_{k_r}$ of A form a basis of $\mathcal{C}(A)$;

(b) $\mathcal{R}(R) = \mathcal{R}(A)$, and if $r \geqslant 1$, the nonzero rows $\rho_1', \ldots, \rho_r'$ of R form a basis of $\mathcal{R}(A)$;

(c) the special solutions $s_1, \ldots, s_{n-r}$ form a basis of $\mathcal{N}(A)$ (if $r = n$, then $\mathcal{N}(A) = \{0\}$, with the empty basis);

(d) the transposed rows $\mathrm{row}_{r+1}(P)', \ldots, \mathrm{row}_m(P)'$ form a basis of $\mathcal{N}(A')$ (if $r = m$, then $\mathcal{N}(A') = \{0\}$, with the empty basis).

Hence $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, $\dim \mathcal{N}(A) = n - r$ and $\dim \mathcal{N}(A') = m - r$ (if $r = 0$, then $A = 0$ and $\mathcal{C}(A) = \mathcal{R}(A) = \{0\}$).

Proof. The row operations multiply $[A \mid I_m]$ on the left by an invertible matrix E (Theorem 2.15(c)), and $E[A \mid I_m] = [EA \mid E]$, column by column (Theorem 1.4(a)). So $P = E$ is invertible and $PA = R$.

(a) $\mathcal{C}(A) = \mathrm{span}\{a_1, \ldots, a_n\}$; apply Theorem 4.3.

(b)

- Same row space: with $P = (p_{ij})$, row i of $R = PA$ is $\mathrm{row}_i(R) = \mathrm{row}_i(P)A = p_{i1}\,\mathrm{row}_1(A) + \cdots + p_{im}\,\mathrm{row}_m(A)$ (Theorem 1.4(c)), a linear combination of the rows of A. So every row of R lies in $\mathcal{R}(A)$, and so does their span $\mathcal{R}(R)$ (Theorem 3.7): $\mathcal{R}(R) \subseteq \mathcal{R}(A)$. Likewise $A = P^{-1}R$ gives $\mathcal{R}(A) \subseteq \mathcal{R}(R)$. Hence $\mathcal{R}(R) = \mathcal{R}(A)$.
- Spanning: the rows of R span $\mathcal{R}(R)$, and rows $r+1, \ldots, m$ are zero, so $\rho_1', \ldots, \rho_r'$ span $\mathcal{R}(R) = \mathcal{R}(A)$.
- Independent: $\mathrm{col}_{k_i}(R) = e_i$, so entry k_i of $c_1\rho_1' + \cdots + c_r\rho_r'$ is c_i; if this combination is 0, every $c_i = 0$.

(c) The solutions of $Ax = 0$ are those of $Rx = 0$ (Theorem 2.3(b)).

- Independent: by the definition of the special solutions before the theorem, $(s_j)_{f_l} = \delta_{jl}$, so the f_lth entry of $c_1 s_1 + \cdots + c_{n-r} s_{n-r}$, that is, its value of the unknown x_{f_l}, is $\sum_j c_j \delta_{jl} = c_l$. If this combination is 0, all its entries are 0, so every $c_l = 0$.
- Spanning: let $t \in \mathcal{N}(A)$, with free unknowns $t_{f_1}, \ldots, t_{f_{n-r}}$, and put $\tilde{t} = t_{f_1} s_1 + \cdots + t_{f_{n-r}} s_{n-r}$. Then $\tilde{t} \in \mathcal{N}(A)$, as $\mathcal{N}(A)$ is a subspace (Definition 4.7).
- By the previous item, with $c_l = t_{f_l}$, the f_lth entry of $\tilde{t}$ is t_{f_l}: $\tilde{t}$ has the same free unknowns as t.
- By Proposition 2.7(b), with $b = 0$ and so $z = 0$, the pivot unknowns of a solution are computed from its free unknowns: $x_{k_i} = -\sum_l R_{if_l} x_{f_l}$. So two solutions with the same free unknowns are equal, and $t = \tilde{t} \in \operatorname{span}\{s_1, \ldots, s_{n-r}\}$.
- If $r = n$, there are no free unknowns, and $x = 0$ is the only solution.

(d)

- In $\mathcal{N}(A')$: for $i > r$, $\operatorname{row}_i(P)A = \operatorname{row}_i(PA) = \operatorname{row}_i(R) = 0$ (Theorem 1.4(c)), so $A' \operatorname{row}_i(P)' = (\operatorname{row}_i(P)A)' = 0$.
- Independent: if $c'P = c_1 \operatorname{row}_1(P) + \cdots + c_m \operatorname{row}_m(P) = 0$, then $c' = c'PP^{-1} = 0$; so the rows of P are independent, and so are rows $r+1, \ldots, m$ (Proposition 3.13(b)).
- Spanning: let $A'y = 0$, so $y'A = 0$, and put $z' = y'P^{-1}$, so that $y' = z'P = \sum_{i=1}^m z_i \operatorname{row}_i(P)$. Then $0 = y'A = z'PA = z'R = z_1\rho_1' + \cdots + z_r\rho_r'$, and (b) gives $z_1 = \cdots = z_r = 0$. So $y' = \sum_{i>r} z_i \operatorname{row}_i(P)$ (and $y = 0$ if $r = m$). □

Example 4.9 (cf. [2, Example 4.5])**.** Find bases of the four subspaces of

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}$$

Solution. Row-reduce $[A \mid I_3]$:

$$\left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 5 & 0 & 1 & 0 \\ 2 & 5 & 1 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & -2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|ccc} 1 & 0 & 3 & -1 & 3 & -2 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) = [R \mid P]$$

So $r = 2$, with pivot columns $1, 2$ and free unknowns x_3, x_4. (The pivots are already visible in the echelon form after the second arrow; Remark 4.2.)

- $\mathcal{C}(A)$: $a_1 = (1,1,2)'$ and $a_2 = (2,3,5)'$ form a basis; $\dim \mathcal{C}(A) = 2$. Columns 3, 4 of R give

  $$a_3 = 3a_1 - a_2, \qquad a_4 = -a_1 + 2a_2$$

  Check: $3(1,1,2)' - (2,3,5)' = (1,0,1)'$ and $-(1,1,2)' + 2(2,3,5)' = (3,5,8)'$.
- $\mathcal{R}(A)$: $\rho_1' = (1,0,3,-1)$ and $\rho_2' = (0,1,-1,2)$ form a basis; $\dim \mathcal{R}(A) = 2$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 + 3x_3 - x_4 = 0$, $x_2 - x_3 + 2x_4 = 0$. The special solutions

  $$s_1 = \begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \end{pmatrix} \ (x_3 = 1,\ x_4 = 0), \qquad s_2 = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix} \ (x_3 = 0,\ x_4 = 1)$$

  form a basis of $\mathcal{N}(A)$; $\dim \mathcal{N}(A) = 2 = 4 - 2$. Note $As_1 = -3a_1 + a_2 + a_3 = 0$ is the relation $a_3 = 3a_1 - a_2$: each special solution is a column relation.

- $\mathcal{N}(A')$: row 3 of P gives the basis $y = (-1,-1,1)'$; $\dim \mathcal{N}(A') = 1 = 3-2$. Indeed $y'A = -\operatorname{row}_1(A) - \operatorname{row}_2(A) + \operatorname{row}_3(A) = 0$, that is, $\operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$: each vector of $\mathcal{N}(A')$ is a row relation.
- Warning: every column of R has third entry 0, so every vector of $\mathcal{C}(R)$ does; but a_1 has third entry 2. Thus $\mathcal{C}(R) \neq \mathcal{C}(A)$, and the pivot columns e_1, e_2 of R do not span $\mathcal{C}(A)$.

Remark 4.10. Row operations do not change $\mathcal{R}(A)$ or $\mathcal{N}(A)$: $\mathcal{R}(R) = \mathcal{R}(A)$ (Theorem 4.8(b)) and $\mathcal{N}(R) = \mathcal{N}(A)$ (Theorem 2.3(b)), so R may be used for these two spaces. They do change $\mathcal{C}(A)$ in general: a basis of $\mathcal{C}(A)$ consists of the pivot columns of A itself, not of R (Example 4.9).

Rank

Definition 4.11 ([3, §2.1], [5, §§4.4a, 4.4c])**.** By Theorem 4.8, $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, the number of pivots of the RREF of A (also when $r = 0$). This common number is the **rank** of A, written $\operatorname{rank} A$; by Remark 4.2 it is also the number of nonzero rows of any echelon form of A. A has **full column rank** if $\operatorname{rank} A = n$, and **full row rank** if $\operatorname{rank} A = m$. The number $\dim \mathcal{N}(A)$ is the **nullity** of A.

Theorem 4.12 ([3, Result 2.10], [14, Ch. II, Corollary 2.9])**.** For every $A \in \mathbb{R}^{m\times n}$,

$$\operatorname{rank} A + \dim \mathcal{N}(A) = n, \qquad \operatorname{rank} A + \dim \mathcal{N}(A') = m$$

Proof. Each of the n columns of R is either a pivot column (r of them) or a free column ($n-r$ of them), and by Theorem 4.8(c), $\dim \mathcal{N}(A) = n - r$, the number of free unknowns. Likewise $\dim \mathcal{N}(A') = m - r$, the number of zero rows of R (Theorem 4.8(d)). Since $\operatorname{rank} A = r$, both equalities follow. □

Theorem 4.13 ([3, §2.1; Result 2.11], [5, Lemma 4.4.3], [6, Exercise 4.7])**.** Let $A \in \mathbb{R}^{m\times n}$.

(a) $\operatorname{rank} A' = \operatorname{rank} A$, and $\operatorname{rank} A \leqslant \min\{m, n\}$.

(b) A has full column rank $\Leftrightarrow$ its columns are linearly independent $\Leftrightarrow \mathcal{N}(A) = \{0\}$.

(c) A has full row rank $\Leftrightarrow$ its rows are linearly independent $\Leftrightarrow \mathcal{N}(A') = \{0\} \Leftrightarrow \mathcal{C}(A) = \mathbb{R}^m$.

(d) If $m = n$, then A is invertible $\Leftrightarrow \operatorname{rank} A = n$.

Proof. Let R be the RREF of A, with r nonzero rows and pivot columns $k_1 < \cdots < k_r$, so that $\operatorname{rank} A = r$ (Definition 4.11).

(a) For $y \in \mathbb{R}^m$, $(y'A)' = A'y$, so transposing maps $\mathcal{R}(A)$ onto $\mathcal{C}(A')$. It preserves linear combinations and is undone by transposing again, so it carries a basis of $\mathcal{R}(A)$ to a basis of $\mathcal{C}(A')$, and $\operatorname{rank} A' = \dim \mathcal{C}(A') = \dim \mathcal{R}(A) = \operatorname{rank} A$. Also $\operatorname{rank} A = r \leqslant m$, as R has m rows, and $r \leqslant n$, as $k_1 < \cdots < k_r$ lie in $\{1, \ldots, n\}$.

(b) A has full column rank if and only if $r = n$ (Definition 4.11); so (b) is Theorem 4.1: (iii) $\Leftrightarrow$ (i) $\Leftrightarrow$ (ii).

(c)

- $\operatorname{rank} A = m$ if and only if $\dim \mathcal{N}(A') = m - \operatorname{rank} A = 0$ (Theorem 4.12), that is, $\mathcal{N}(A') = \{0\}$ (Theorem 4.4).
- $y'A = y_1 \operatorname{row}_1(A) + \cdots + y_m \operatorname{row}_m(A)$ (Theorem 1.4(c)), and $y'A = 0$ if and only if $A'y = 0$. So $\mathcal{N}(A') = \{0\}$ says exactly that the rows of A are independent.
- $\operatorname{rank} A = \dim \mathcal{C}(A)$, and a subspace of $\mathbb{R}^m$ has dimension m only if it is $\mathbb{R}^m$ (Theorem 4.5(a)); conversely, $\dim \mathbb{R}^m = m$ (Theorem 4.4); so $\operatorname{rank} A = m$ if and only if $\mathcal{C}(A) = \mathbb{R}^m$.

(d) A square matrix A is invertible if and only if $Ax = 0$ has only the solution $x = 0$ (Theorem 2.16), that is, $\mathcal{N}(A) = \{0\}$; by (b), this means $\operatorname{rank} A = n$. □

Corollary 4.14 ([1, §2.6], [5, Lemma 4.5.1, Corollary 4.5.2])**.** Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and let $[A \mid b] \in \mathbb{R}^{m\times(n+1)}$ be the augmented matrix.

(a) $Ax = b$ is consistent $\Leftrightarrow b \in \mathcal{C}(A) \Leftrightarrow \operatorname{rank}[A \mid b] = \operatorname{rank} A$.

(b) If $Ax = b$ is consistent, its solution is unique $\Leftrightarrow \operatorname{rank} A = n$.

Proof. (a)

- $Ax = x_1a_1 + \cdots + x_na_n$ (Theorem 1.4(b)), so $Ax = b$ has a solution if and only if b is a linear combination of the columns of A, that is, $b \in \mathcal{C}(A)$.
- $\mathcal{C}(A) \subseteq \mathcal{C}([A \mid b])$: a combination of $a_1, \ldots, a_n$ is one of $a_1, \ldots, a_n, b$, with coefficient 0 on b.
- If $b = Ac$, then $x_1a_1 + \cdots + x_na_n + tb = A(x + tc) \in \mathcal{C}(A)$; so $\mathcal{C}([A \mid b]) = \mathcal{C}(A)$, and the ranks are equal.
- If the ranks are equal, the subspace $\mathcal{C}(A)$ of $\mathcal{C}([A \mid b])$ has the same dimension, so the two are equal (Theorem 4.5(a)), and $b \in \mathcal{C}([A \mid b]) = \mathcal{C}(A)$.

(b) By Theorem 2.27(a), the solutions are $x_p + z$, where x_p is one solution and z runs through the solutions of $Ax = 0$, that is, through $\mathcal{N}(A)$. So the solution is unique if and only if $\mathcal{N}(A) = \{0\}$, that is, $\operatorname{rank} A = n$ (Theorem 4.13(b)). □

In practice, consistency is read off from an echelon form of $[A \mid b]$ (Theorem 2.26).

Rank equalities and inequalities

Theorem 4.15 ([3, Result 2.2; §2.4], [6, Exercise 4.24])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$.

(a) $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$ and $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$; and $\operatorname{rank}(AB) \leqslant \min\{\operatorname{rank} A, \operatorname{rank} B\}$.

(b) If A has full column rank, then $\mathcal{N}(AB) = \mathcal{N}(B)$ and $\operatorname{rank}(AB) = \operatorname{rank} B$.

(c) If B has full row rank, then $\mathcal{C}(AB) = \mathcal{C}(A)$ and $\operatorname{rank}(AB) = \operatorname{rank} A$.

(d) $\operatorname{rank}(PAQ) = \operatorname{rank} A$ for all invertible $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$.

Proof. (a)

- $ABx = A(Bx) \in \mathcal{C}(A)$, and $Bx = 0$ gives $ABx = A0 = 0$.
- So $\operatorname{rank}(AB) = \dim \mathcal{C}(AB) \leqslant \dim \mathcal{C}(A) = \operatorname{rank} A$ (Theorem 4.5(a)).
- By this, $(AB)' = B'A'$ (Theorem 1.9) and Theorem 4.13(a), $\operatorname{rank}(AB) = \operatorname{rank}(B'A') \leqslant \operatorname{rank} B' = \operatorname{rank} B$.

(b)

- If $ABx = 0$, then $Bx \in \mathcal{N}(A) = \{0\}$ (Theorem 4.13(b)), so $x \in \mathcal{N}(B)$. With (a), $\mathcal{N}(AB) = \mathcal{N}(B)$.
- AB and B both have p columns, so by Theorem 4.12, $\operatorname{rank}(AB) = p - \dim \mathcal{N}(AB) = p - \dim \mathcal{N}(B) = \operatorname{rank} B$.

(c)

- $\mathcal{C}(B) = \mathbb{R}^n$ (Theorem 4.13(c)): every $x \in \mathbb{R}^n$ is Bw for some $w \in \mathbb{R}^p$, and then $Ax = (AB)w \in \mathcal{C}(AB)$.
- So $\mathcal{C}(A) \subseteq \mathcal{C}(AB)$; with (a), $\mathcal{C}(AB) = \mathcal{C}(A)$, and the ranks are equal.

(d) $Px = 0$ gives $x = P^{-1}Px = 0$, so P has full column rank (Theorem 4.13(b)); and $\operatorname{rank} Q = n$ (Theorem 4.13(d)), so Q has full row rank. By (b) and (c), $\operatorname{rank}(P(AQ)) = \operatorname{rank}(AQ) = \operatorname{rank} A$. □

Theorem 4.16 ([5, Lemma 4.5.1, Corollary 4.5.2], [15, Example 2.3.1]). (a) For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times p}$, let $[A\ B] \in \mathbb{R}^{m\times(n+p)}$ have the columns of A followed by those of B. Then

$$\max\{\operatorname{rank} A, \operatorname{rank} B\} \leqslant \operatorname{rank}[A\ B] \leqslant \operatorname{rank} A + \operatorname{rank} B$$

(b) For $A \in \mathbb{R}^{m\times n}$ and $C \in \mathbb{R}^{q\times n}$, let $\binom{A}{C} \in \mathbb{R}^{(m+q)\times n}$ have the rows of A followed by those of C. Then $\binom{A}{C}' = [A'\ C']$, and

$$\max\{\operatorname{rank} A, \operatorname{rank} C\} \leqslant \operatorname{rank}\begin{pmatrix} A \\ C \end{pmatrix} \leqslant \operatorname{rank} A + \operatorname{rank} C$$

(c) For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{q\times p}$, let $\operatorname{diag}(A, B) = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right) \in \mathbb{R}^{(m+q)\times(n+p)}$, with zero blocks of sizes $m \times p$ and $q \times n$. Then $\operatorname{rank}\operatorname{diag}(A, B) = \operatorname{rank} A + \operatorname{rank} B$.

Proof. (a)

- A combination of the columns of A, or of those of B, is a combination of the columns of $[A\ B]$ (the others with coefficient 0). So $\mathcal{C}(A)$ and $\mathcal{C}(B)$ are subspaces of $\mathcal{C}([A\ B])$, and Theorem 4.5(a) gives the lower bound.
- Let $r = \operatorname{rank} A$, $s = \operatorname{rank} B$, and take bases $v_1, \ldots, v_r$ of $\mathcal{C}(A)$ and $w_1, \ldots, w_s$ of $\mathcal{C}(B)$. Every column of $[A\ B]$ is a combination of these $r + s$ vectors of $\mathcal{C}([A\ B])$, so they span $\mathcal{C}([A\ B])$ (Theorem 3.7), and $\operatorname{rank}[A\ B] \leqslant r + s$ (Theorem 4.5(c)).

(b) Transposing turns the rows of $\binom{A}{C}$ into the columns of $[A'\ C']$, so $\binom{A}{C}' = [A'\ C']$. Apply (a) to A' and C', and use Theorem 4.13(a).

(c) Let $M = \operatorname{diag}(A, B)$, $r = \operatorname{rank} A$ and $s = \operatorname{rank} B$.

- Upper bound: $M = [M_1\ M_2]$ with $M_1 = \binom{A}{0}$ and $M_2 = \binom{0}{B}$. By (b), and as a zero matrix has rank 0, $r \leqslant \operatorname{rank} M_1 \leqslant r + 0$; so $\operatorname{rank} M_1 = r$, and likewise $\operatorname{rank} M_2 = s$. By (a), $\operatorname{rank} M \leqslant r + s$.
- Lower bound: let $a_{k_1}, \ldots, a_{k_r}$ and $b_{l_1}, \ldots, b_{l_s}$ be the pivot columns of A and of B, bases of $\mathcal{C}(A)$ and $\mathcal{C}(B)$ (Theorem 4.8(a)). The corresponding columns $\binom{a_{k_i}}{0}$ and $\binom{0}{b_{l_j}}$ of M are independent: a combination of them is $\begin{pmatrix} \sum_i \alpha_i a_{k_i} \\ \sum_j \beta_j b_{l_j} \end{pmatrix}$, which is 0 only if all α_i and β_j are 0. These $r + s$ vectors lie in $\mathcal{C}(M)$, so $\operatorname{rank} M \geqslant r + s$ (Theorem 4.5(c)). □

Corollary 4.17 ([3, Result 2.4], [15, Example 2.3.1(2)]). For $A, B \in \mathbb{R}^{m\times n}$, $\operatorname{rank}(A + B) \leqslant \operatorname{rank} A + \operatorname{rank} B$.

Proof. For $x \in \mathbb{R}^n$, $(A + B)x = Ax + Bx$ is a combination of the columns of A and of B, so it lies in $\mathcal{C}([A\ B])$. Hence $\mathcal{C}(A + B) \subseteq \mathcal{C}([A\ B])$, so $\operatorname{rank}(A + B) \leqslant \operatorname{rank}[A\ B]$ (Theorem 4.5(a)), and $\operatorname{rank}[A\ B] \leqslant \operatorname{rank} A + \operatorname{rank} B$ (Theorem 4.16(a)). □

Theorem 4.18 ([12, Theorem 2.4, Step 3]). Let $A \in \mathbb{R}^{m\times n}$.

(a) $\mathcal{N}(A'A) = \mathcal{N}(A)$.

(b) $\operatorname{rank}(A'A) = \operatorname{rank} A = \operatorname{rank}(AA')$.

(c) $\mathcal{C}(A'A) = \mathcal{C}(A')$ and $\mathcal{C}(AA') = \mathcal{C}(A)$.

Proof. (a)

- If $Ax = 0$, then $A'Ax = A'0 = 0$.
- If $A'Ax = 0$, then $(Ax)'(Ax) = x'A'Ax = 0$. But $(Ax)'(Ax)$ is the sum of the squares of the entries of Ax (Theorem 1.9), so $Ax = 0$.

(b) $A'A$ and A both have n columns, so by (a) and Theorem 4.12, $\operatorname{rank}(A'A) = n - \dim \mathcal{N}(A'A) = n - \dim \mathcal{N}(A) = \operatorname{rank} A$. Applied to A' in place of A, this gives $\operatorname{rank}(AA') = \operatorname{rank} A' = \operatorname{rank} A$ (Theorem 4.13(a)).

(c) $\mathcal{C}(A'A) \subseteq \mathcal{C}(A')$ (Theorem 4.15(a)), and both have dimension $\operatorname{rank} A$, by (b) and Theorem 4.13(a); so they are equal (Theorem 4.5(a)). Likewise $\mathcal{C}(AA') \subseteq \mathcal{C}(A)$, both of dimension $\operatorname{rank} A$. □

Theorem 4.19 (Sylvester's inequality; [6, Exercise 5.47(b); Ch. 5, Notes], [5, (17.5.9)])**.** For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$,

$$\dim \mathcal{N}(AB) \leqslant \dim \mathcal{N}(A) + \dim \mathcal{N}(B), \qquad \text{equivalently} \qquad \operatorname{rank}(AB) \geqslant \operatorname{rank} A + \operatorname{rank} B - n$$

In particular, if $AB = 0$, then $\operatorname{rank} A + \operatorname{rank} B \leqslant n$.

Proof.
- Let $z_1, \dots, z_s$ be a basis of $\mathcal{N}(B)$. Since $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$ (Theorem 4.15(a)), it extends to a basis $z_1, \dots, z_s, w_1, \dots, w_t$ of $\mathcal{N}(AB)$ (Theorem 4.5(b)); so $\dim \mathcal{N}(AB) = s + t$.
- The vectors $Bw_1, \dots, Bw_t$ lie in $\mathcal{N}(A)$, since $A(Bw_j) = (AB)w_j = 0$.
- They are independent: if $c_1Bw_1 + \cdots + c_tBw_t = 0$, then $B(c_1w_1 + \cdots + c_tw_t) = 0$, so $c_1w_1 + \cdots + c_tw_t \in \mathcal{N}(B)$ equals some $\lambda_1 z_1 + \cdots + \lambda_s z_s$; as $z_1, \dots, z_s, w_1, \dots, w_t$ are independent, every $c_j = 0$.
- Hence $t \leqslant \dim \mathcal{N}(A)$ (Theorem 4.5(c)), and $\dim \mathcal{N}(AB) = s + t \leqslant \dim \mathcal{N}(B) + \dim \mathcal{N}(A)$.
- By Theorem 4.12, $\dim \mathcal{N}(AB) = p - \operatorname{rank}(AB)$, $\dim \mathcal{N}(B) = p - \operatorname{rank} B$ and $\dim \mathcal{N}(A) = n - \operatorname{rank} A$. Substituting gives $p - \operatorname{rank}(AB) \leqslant (p - \operatorname{rank} B) + (n - \operatorname{rank} A)$, which is the rank form. If $AB = 0$, then $\operatorname{rank}(AB) = 0$. □

Example 4.20. Let $A = \operatorname{diag}(1, 1, 1, 0)$ and $B = \operatorname{diag}(0, 0, 1, 1)$ in $\mathbb{R}^{4\times 4}$. Their columns are $e_1, e_2, e_3, 0$ and $0, 0, e_3, e_4$. A zero column adds nothing to the span (Theorem 3.10(c)), so $\mathcal{C}(A) = \operatorname{span}\{e_1, e_2, e_3\}$ and $\mathcal{C}(B) = \operatorname{span}\{e_3, e_4\}$; unit vectors are independent, as part of the standard basis (Definition 3.17 and Proposition 3.13(b)). So $\operatorname{rank} A = 3$ and $\operatorname{rank} B = 2$.

- $AB = \operatorname{diag}(0, 0, 1, 0)$ (row i of AB is $a_{ii} \operatorname{row}_i(B)$, Theorem 1.4(c)) has the columns $0, 0, e_3, 0$; in the same way, $\mathcal{C}(AB) = \operatorname{span}\{e_3\}$ and $\operatorname{rank}(AB) = 1$: less than $\min\{3, 2\}$ (Theorem 4.15(a)), and equal to $\operatorname{rank} A + \operatorname{rank} B - 4 = 1$, so Sylvester's bound is attained (Theorem 4.19).
- $A + B = \operatorname{diag}(1, 1, 2, 1)$ is invertible, with inverse $\operatorname{diag}(1, 1, \frac{1}{2}, 1)$ (Definition 2.12); so $\operatorname{rank}(A + B) = 4 < 3 + 2$ (Theorem 4.13(d) and Corollary 4.17). And $A + (-A) = 0$ has rank 0, as $\mathcal{C}(0) = \{0\}$ and $\dim\{0\} = 0$ (Theorem 4.4).
- $[A\ B] \in \mathbb{R}^{4\times 8}$ has the columns $e_1, e_2, e_3, 0, 0, 0, e_3, e_4$, so $\mathcal{C}([A\ B]) = \mathbb{R}^4$ and $\operatorname{rank}[A\ B] = \dim \mathbb{R}^4 = 4$ (Theorem 4.4), strictly between $\max\{3, 2\}$ and $3 + 2$ (Theorem 4.16(a)).
- $\operatorname{diag}(A, B) = \operatorname{diag}(1, 1, 1, 0, 0, 0, 1, 1) \in \mathbb{R}^{8\times 8}$ has rank $5 = 3 + 2$ (Theorem 4.16(c)).
- $A'A = A^2 = A$ has rank $3 = \operatorname{rank} A$ (Theorem 4.18).

Theorem 4.21 ([3, Results 2.3, 2.13], [5, Theorem 4.4.8], [2, (4.42)])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank r. If $r \geqslant 1$, a product $A = BC$ with $B \in \mathbb{R}^{m\times r}$ of full column rank and $C \in \mathbb{R}^{r\times n}$ of full row rank is a **full-rank factorization** of A.

(a) If $r \geqslant 1$, let R be the RREF of A, with pivot columns $k_1 < \cdots < k_r$; let $B = [a_{k_1}\ \cdots\ a_{k_r}] \in \mathbb{R}^{m\times r}$ consist of the pivot columns of A, and $C \in \mathbb{R}^{r\times n}$ of the r nonzero rows of R. Then $A = BC$ is a full-rank factorization. If $r = 0$, then $A = 0$, and there is nothing to factor.

(b) For every full-rank factorization $A = BC$, $\mathcal{C}(A) = \mathcal{C}(B)$ and $\mathcal{N}(A) = \mathcal{N}(C)$.

(c) For every full-rank factorization $A = BC$, the $r \times r$ matrices $B'B$ and CC' are invertible, and

$$L = (B'B)^{-1}B' \in \mathbb{R}^{r\times m}, \qquad K = C'(CC')^{-1} \in \mathbb{R}^{n\times r}$$

satisfy $LB = I_r$ and $CK = I_r$: B has a left inverse and C has a right inverse.

Proof. (a)

- Column j of BC is $B\,\mathrm{col}_j(C) = R_{1j}a_{k_1} + \cdots + R_{rj}a_{k_r}$ (Theorem 1.4(a),(b)), which is a_j by Theorem 4.3(b). So $BC = A$.
- The columns of B form a basis of $\mathcal{C}(A)$, so they are independent: B has full column rank (Theorem 4.13(b)).
- The rows of C form a basis of $\mathcal{R}(A)$ (Theorem 4.8(b)), so they are independent: C has full row rank (Theorem 4.13(c)).
- If $r = 0$, then $A = 0$ (Theorem 4.8).

(b) Theorem 4.15(b) gives $\mathcal{N}(BC) = \mathcal{N}(C)$, as B has full column rank, and Theorem 4.15(c) gives $\mathcal{C}(BC) = \mathcal{C}(B)$, as C has full row rank.

(c)

- $\mathrm{rank}(B'B) = \mathrm{rank}\,B = r$ and $\mathrm{rank}(CC') = \mathrm{rank}\,C = r$ (Theorem 4.18(b)), so the $r \times r$ matrices $B'B$ and CC' are invertible (Theorem 4.13(d)).
- $LB = (B'B)^{-1}(B'B) = I_r$ and $CK = (CC')(CC')^{-1} = I_r$. □

Example 4.22. For A of Example 4.9, Theorem 4.21(a) gives

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

- Columns 1, 2 of BC are a_1, a_2, and columns 3, 4 are $3a_1 - a_2 = a_3$ and $-a_1 + 2a_2 = a_4$ (Example 4.9).
- By Theorem 4.21(b), $\mathcal{C}(A) = \mathcal{C}(B) = \mathrm{span}\{a_1, a_2\}$ and $\mathcal{N}(A) = \mathcal{N}(C) = \mathrm{span}\{s_1, s_2\}$: the system $Cx = 0$ is $Rx = 0$ without its zero row.
- The 4×2 matrix $[s_1\ s_2]$ has $A[s_1\ s_2] = 0$ and $\mathrm{rank}[s_1\ s_2] = 2$, and Sylvester's inequality $\mathrm{rank}\,A + \mathrm{rank}[s_1\ s_2] \leqslant 4$ holds with equality: $2 + 2 = 4$.

Sums and intersections of subspaces

Theorem 4.23 ([1, §2.3, Theorem 6])**.** Let U and W be subspaces of $\mathbb{R}^n$. Then

$$\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$$

Proof. $U \cap W$ and $U + W$ are subspaces of $\mathbb{R}^n$ (Theorem 3.31(a)). The lists below may be empty (the empty list is a basis of $\{0\}$, Definition 3.17).

- Bases: let $x_1, \ldots, x_k$ be a basis of $U \cap W$ (Theorem 4.4). It is an independent list of vectors of U, and also of W; so it extends to a basis $x_1, \ldots, x_k, u_1, \ldots, u_p$ of U and to a basis $x_1, \ldots, x_k, w_1, \ldots, w_q$ of W (Theorem 4.5(b)). Thus $\dim U = k + p$ and $\dim W = k + q$.
- Spanning: a vector of $U + W$ is $u + w$ with $u \in U$ and $w \in W$, that is, a combination of $x_1, \ldots, x_k, u_1, \ldots, u_p$ plus a combination of $x_1, \ldots, x_k, w_1, \ldots, w_q$; so it is a combination of the $k + p + q$ vectors
$$x_1, \ldots, x_k,\ u_1, \ldots, u_p,\ w_1, \ldots, w_q$$
These vectors lie in $U + W$ (Theorem 3.31(b)), so they span $U + W$.
- Independent: let $\sum_i a_i x_i + \sum_j b_j u_j + \sum_l c_l w_l = 0$. Then $\sum_l c_l w_l = -\sum_i a_i x_i - \sum_j b_j u_j$ lies in U as well as in W, hence in $U \cap W$; so $\sum_l c_l w_l = \sum_i \lambda_i x_i$ for some scalars λ_i. Since $x_1, \ldots, x_k, w_1, \ldots, w_q$ are independent, the relation $\sum_i \lambda_i x_i - \sum_l c_l w_l = 0$ forces every $c_l = 0$. Then $\sum_i a_i x_i + \sum_j b_j u_j = 0$, and the independence of $x_1, \ldots, x_k, u_1, \ldots, u_p$ gives every $a_i = 0$ and every $b_j = 0$.

- So these $k+p+q$ vectors form a basis of $U+W$, and

$$\dim(U+W)+\dim(U\cap W)=(k+p+q)+k=(k+p)+(k+q)=\dim U+\dim W \qquad \square$$

Example 4.24 ([1, §2.3, Exercise 7(b)])**.** Write a 2×2 matrix $\left(\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22}\end{smallmatrix}\right)$ as the column $(a_{11},a_{12},a_{21},a_{22})'$ of its entries. The matrices $\left(\begin{smallmatrix} x & -x \\ y & z\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} a & b \\ -a & c\end{smallmatrix}\right)$ $(a,b,c,x,y,z\in\mathbb{R})$ then form the subspaces

$$U=\{(x,-x,y,z)' : x,y,z\in\mathbb{R}\}, \qquad W=\{(a,b,-a,c)' : a,b,c\in\mathbb{R}\}$$

of $\mathbb{R}^4$: U consists of the vectors whose second entry is minus the first, and W of those whose third entry is minus the first; so U and W are the solution sets of $x_1+x_2=0$ and of $x_1+x_3=0$, subspaces of $\mathbb{R}^4$ by Example 3.5(e).

- $(x,-x,y,z)'=x\,u_1+y\,e_3+z\,e_4$ with $u_1=(1,-1,0,0)'$, and this vector is 0 only for $x=y=z=0$; so u_1,e_3,e_4 form a basis of U (Definition 3.17), and $\dim U=3$ (Theorem 4.4). Likewise $(a,b,-a,c)'=a\,w_1+b\,e_2+c\,e_4$ with $w_1=(1,0,-1,0)'$, so w_1,e_2,e_4 form a basis of W, and $\dim W=3$.
- A vector $(x,-x,y,z)'$ of U lies in W if and only if $y=-x$. So $U\cap W=\{(x,-x,-x,z)' : x,z\in\mathbb{R}\}$, and $(x,-x,-x,z)'=x\,(1,-1,-1,0)'+z\,e_4$ is 0 only for $x=z=0$; so $(1,-1,-1,0)'$ and e_4 form a basis of $U\cap W$, a subspace by Theorem 3.31(a), and $\dim(U\cap W)=2$ (Theorem 4.4).
- By Theorem 4.23, $\dim(U+W)=\dim U+\dim W-\dim(U\cap W)=3+3-2=4$, so $U+W=\mathbb{R}^4$ (Theorem 4.5(a)). Directly: $e_3,e_4\in U$ and $e_2\in W$ lie in $U+W$ (Theorem 3.31(b)), and $e_1=u_1+e_2\in U+W$; so the subspace $U+W$ (Theorem 3.31(a)) contains every $x=x_1e_1+x_2e_2+x_3e_3+x_4e_4$ of $\mathbb{R}^4$.

With this direct computation of $U+W$, $\dim(U+W)+\dim(U\cap W)=4+2=3+3=\dim U+\dim W$.

Corollary 4.25 ([5, Corollary 17.1.6], [3, Ch. 1, Exercise 19], [4, Ch. I, Theorem 4.2])**.** Let U and W be subspaces of $\mathbb{R}^n$.

(a) $\dim(U+W)\leqslant\dim U+\dim W$, with equality if and only if $U\cap W=\{0\}$. In particular $\dim(U\oplus W)=\dim U+\dim W$.

(b) If $\dim U+\dim W>n$, then $U\cap W\neq\{0\}$.

(c) W has a **complement** in $\mathbb{R}^n$, that is, a subspace V with $\mathbb{R}^n=W\oplus V$; and every complement V of W has $\dim V=n-\dim W$.

Proof. (a) Let $u_1,\dots,u_p$ and $w_1,\dots,w_q$ be bases of U and of W (Theorem 4.4).

- If $U\cap W=\{0\}$, then $u_1,\dots,u_p,w_1,\dots,w_q$ form a basis of $U\oplus W$ (Theorem 3.33), so $\dim(U\oplus W)=p+q$.
- If $U\cap W\neq\{0\}$, these $p+q$ vectors are dependent (Theorem 3.33). They span $U+W$ (Theorem 3.31(c)), so some of them form a basis of $U+W$ (Theorem 4.3), and not all of them, as a basis is independent. Hence $\dim(U+W)<p+q$.

(b) If $U\cap W=\{0\}$, then (a) and Theorem 4.5(a), applied to the subspace $U+W$ of $\mathbb{R}^n$, give $n<\dim U+\dim W=\dim(U+W)\leqslant n$ ✘

(c)

- Let $w_1,\dots,w_d$ be a basis of W (Theorem 4.4). By Theorem 4.5(b), applied to $\mathbb{R}^n$, it extends to a basis $w_1,\dots,w_d,v_1,\dots,v_{n-d}$ of $\mathbb{R}^n$, which has n vectors (Theorem 4.4). Let $V=\operatorname{span}\{v_1,\dots,v_{n-d}\}$ ($V=\{0\}$ if $d=n$).
- $W+V=\operatorname{span}\{w_1,\dots,w_d,v_1,\dots,v_{n-d}\}=\mathbb{R}^n$ (Theorem 3.31(c)).
- $W\cap V=\{0\}$: if $x=\sum_i a_iw_i=\sum_j b_jv_j$, then $\sum_i a_iw_i-\sum_j b_jv_j=0$, so every a_i and every b_j is 0, and $x=0$.

- Hence $\mathbb{R}^n = W \oplus V$ (Definition 3.30). If V is any complement of W, then $n = \dim(W \oplus V) = \dim W + \dim V$ by (a). □

A complement is not unique in general: in $\mathbb{R}^4$, both $\operatorname{span}\{\mathbf{1}_4\}$ and $\operatorname{span}\{e_4\}$ are complements of the centered vectors W_0 (Examples 3.35 and 3.34(b)).

Column spaces and the system $AX = B$

Proposition 4.26 ([5, Lemmas 4.2.1, 4.2.2])**.** Let $A \in \mathbb{R}^{m\times n}$.

(a) For $B \in \mathbb{R}^{m\times p}$ the following are equivalent: $\mathcal{C}(B) \subseteq \mathcal{C}(A)$; every column of B lies in $\mathcal{C}(A)$; $B = AF$ for some $F \in \mathbb{R}^{n\times p}$.

(b) For $C \in \mathbb{R}^{q\times n}$ the following are equivalent: $\mathcal{R}(C) \subseteq \mathcal{R}(A)$; every row of C lies in $\mathcal{R}(A)$; $C = LA$ for some $L \in \mathbb{R}^{q\times m}$.

Proof. (a) The columns of A span $\mathcal{C}(A)$, and those of B span $\mathcal{C}(B)$ (Definition 4.7); so (a) is Theorem 3.10(a).

(b)

- If $\mathcal{R}(C) \subseteq \mathcal{R}(A)$, then $\operatorname{row}_i(C) = e_i'C \in \mathcal{R}(C) \subseteq \mathcal{R}(A)$ (Theorem 1.4(c), with $e_i \in \mathbb{R}^q$). Conversely, if every row of C lies in the subspace $\mathcal{R}(A)$, so does every $y'C = y_1 \operatorname{row}_1(C) + \cdots + y_q \operatorname{row}_q(C)$ (Theorem 1.4(c)).
- For $L \in \mathbb{R}^{q\times m}$ with rows $l_1', \ldots, l_q'$, $\operatorname{row}_i(LA) = l_i'A$ (Theorem 1.4(c)). So $C = LA$ for some L if and only if every $\operatorname{row}_i(C)$ has the form $l'A$ with $l \in \mathbb{R}^m$, that is, lies in $\mathcal{R}(A) = \{y'A : y \in \mathbb{R}^m\}$. □

Theorem 4.27 ([5, Theorem 17.4.2, Lemma 4.5.1, Corollaries 4.5.2, 17.1.6])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times p}$, and let $[A\ B]$ be as in Theorem 4.16(a).

(a) $\mathcal{C}([A\ B]) = \mathcal{C}(A) + \mathcal{C}(B)$, and

$$\operatorname{rank}[A\ B] = \operatorname{rank} A + \operatorname{rank} B - \dim\Big(\mathcal{C}(A) \cap \mathcal{C}(B)\Big)$$

(b) $\operatorname{rank}[A\ B] = \operatorname{rank} A \Leftrightarrow \mathcal{C}([A\ B]) = \mathcal{C}(A) \Leftrightarrow \mathcal{C}(B) \subseteq \mathcal{C}(A)$.

(c) $\operatorname{rank}[A\ B] = \operatorname{rank} A + \operatorname{rank} B \Leftrightarrow \mathcal{C}(A) \cap \mathcal{C}(B) = \{0\}$.

(d) For $C \in \mathbb{R}^{q\times n}$ and $\binom{A}{C}$ as in Theorem 4.16(b),

$$\operatorname{rank}\binom{A}{C} = \operatorname{rank} A + \operatorname{rank} C - \dim\Big(\mathcal{R}(A) \cap \mathcal{R}(C)\Big)$$

and $\operatorname{rank}\binom{A}{C} = \operatorname{rank} A$ if and only if $\mathcal{R}(C) \subseteq \mathcal{R}(A)$.

Proof. (a) The columns of $[A\ B]$ are those of A followed by those of B, so $\mathcal{C}([A\ B]) = \mathcal{C}(A) + \mathcal{C}(B)$ (Theorem 3.31(c) and Definition 4.7). Theorem 4.23 with $U = \mathcal{C}(A)$ and $W = \mathcal{C}(B)$ gives the formula, since $\operatorname{rank} M = \dim \mathcal{C}(M)$ for every matrix M (Definition 4.11).

(b)

- By (a), $\mathcal{C}(A)$ and $\mathcal{C}(B)$ lie in $\mathcal{C}([A\ B])$ (Theorem 3.31(b)).
- If $\mathcal{C}(B) \subseteq \mathcal{C}(A)$, then $B = AF$ for some $F = [f_1\ \cdots\ f_p]$ (Proposition 4.26(a)), and the columns of $[A\ B]$ are $a_1, \ldots, a_n, Af_1, \ldots, Af_p$ (Theorem 1.4(a)), all in $\mathcal{C}(A)$. So $\mathcal{C}([A\ B]) \subseteq \mathcal{C}(A)$ (Proposition 4.26(a)), and with the first item the two spaces are equal. Conversely, if $\mathcal{C}([A\ B]) = \mathcal{C}(A)$, then $\mathcal{C}(B) \subseteq \mathcal{C}([A\ B]) = \mathcal{C}(A)$.

- Equal column spaces have equal dimensions, so $\mathcal{C}([A\ B]) = \mathcal{C}(A)$ gives $\operatorname{rank}[A\ B] = \operatorname{rank} A$. Conversely, if $\operatorname{rank}[A\ B] = \operatorname{rank} A$, the subspace $\mathcal{C}(A)$ of $\mathcal{C}([A\ B])$ has the same dimension, so the two are equal (Theorem 4.5(a)).

(c) This is Corollary 4.25(a) for $U = \mathcal{C}(A)$ and $W = \mathcal{C}(B)$, by (a).

(d)

- $\binom{A}{C}' = [A'\ C']$ (Theorem 4.16(b)), and a matrix and its transpose have the same rank (Theorem 4.13(a)).
- For every matrix G with n columns, $(y'G)' = G'y$, so a row vector z lies in $\mathcal{R}(G)$ if and only if $z' \in \mathcal{C}(G')$. Hence transposition maps $\mathcal{R}(A) \cap \mathcal{R}(C)$ onto $\mathcal{C}(A') \cap \mathcal{C}(C')$, and $\mathcal{R}(C) \subseteq \mathcal{R}(A)$ holds if and only if $\mathcal{C}(C') \subseteq \mathcal{C}(A')$. Transposition preserves linear combinations and is undone by transposing again, so it carries bases to bases, and the two intersections have the same dimension.
- Now apply (a) and (b) to $A' \in \mathbb{R}^{n\times m}$ and $C' \in \mathbb{R}^{n\times q}$. □

For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times p}$, the **system** $AX = B$ asks for the matrices $X \in \mathbb{R}^{n\times p}$ with $AX = B$, its **solutions**; it is **consistent** if it has a solution [5, §§7.1, 7.2]. By Theorem 1.4(a), $X = [x_1\ \cdots\ x_p]$ is a solution if and only if $Ax_j = \operatorname{col}_j(B)$ for $j = 1, \ldots, p$: the system collects p systems with the same coefficient matrix. For $p = 1$ it is the system $Ax = b$ of Definition 2.1.

Corollary 4.28 ([5, Theorems 7.2.1, 7.2.2])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times p}$. The following are equivalent:

(i) the system $AX = B$ is consistent;

(ii) $\mathcal{C}(B) \subseteq \mathcal{C}(A)$, that is, every column of B lies in $\mathcal{C}(A)$;

(iii) $\mathcal{C}([A\ B]) = \mathcal{C}(A)$;

(iv) $\operatorname{rank}[A\ B] = \operatorname{rank} A$.

If A has full row rank, then $AX = B$ is consistent for every $B \in \mathbb{R}^{m\times p}$.

Proof. $AX = B$ is consistent if and only if $B = AF$ for some $F \in \mathbb{R}^{n\times p}$, which is (ii) by Proposition 4.26(a); and (ii) ⇔ (iii) ⇔ (iv) is Theorem 4.27(b). If $\operatorname{rank} A = m$, then $m = \operatorname{rank} A \leqslant \operatorname{rank}[A\ B] \leqslant m$ (Theorems 4.16(a) and 4.13(a)), so (iv) holds. □

Theorem 4.29 ([5, Theorem 7.3.1])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times p}$. The system $AX = B$ is consistent if and only if $y'B = 0$ for every $y \in \mathcal{N}(A')$, that is, if and only if every linear relation $y_1 \operatorname{row}_1(A) + \cdots + y_m \operatorname{row}_m(A) = 0$ among the rows of A also holds among the rows of B. In particular, $Ax = b$ is consistent if and only if $y'b = 0$ for every $y \in \mathcal{N}(A')$.

Proof. If $AX_0 = B$ and $y \in \mathcal{N}(A')$, that is, $y'A = 0$ (Definition 4.7), then $y'B = y'AX_0 = 0$. Conversely, let $y'B = 0$ for every $y \in \mathcal{N}(A')$.

- If $A = 0$, every $y \in \mathbb{R}^m$ lies in $\mathcal{N}(A')$; with $y = e_i$, $\operatorname{row}_i(B) = e_i'B = 0$ for every i (Theorem 1.4(c)), so $B = 0$ and $X = 0$ is a solution. Let $A \neq 0$ and $r = \operatorname{rank} A \geqslant 1$.
- The rows of A span $\mathcal{R}(A)$, so r of them, $\operatorname{row}_{h_1}(A), \ldots, \operatorname{row}_{h_r}(A)$ with $h_1 < \cdots < h_r$, form a basis of $\mathcal{R}(A)$ (Theorem 4.3, last sentence, for row vectors by transposition, as in the paragraph after Theorem 4.5). Let $A_h \in \mathbb{R}^{r\times n}$ and $B_h \in \mathbb{R}^{r\times p}$ consist of rows $h_1, \ldots, h_r$ of A and of B. The rows of A_h are independent, so A_h has full row rank (Theorem 4.13(c)), and $A_hX = B_h$ has a solution X^* (Corollary 4.28): $\operatorname{row}_{h_j}(A)X^* = \operatorname{row}_{h_j}(B)$ for $j = 1, \ldots, r$ (Theorem 1.4(c)).
- Let $i \notin \{h_1, \ldots, h_r\}$. Then $\operatorname{row}_i(A) = \sum_j \lambda_{ij} \operatorname{row}_{h_j}(A)$ for some scalars λ_{ij}, so $y = e_i - \sum_j \lambda_{ij} e_{h_j} \in \mathbb{R}^m$ has $y'A = \operatorname{row}_i(A) - \sum_j \lambda_{ij} \operatorname{row}_{h_j}(A) = 0$, that is, $y \in \mathcal{N}(A')$. Hence $0 = y'B = \operatorname{row}_i(B) - \sum_j \lambda_{ij} \operatorname{row}_{h_j}(B)$, and

$$\operatorname{row}_i(A)X^* = \sum_j \lambda_{ij} \operatorname{row}_{h_j}(A)X^* = \sum_j \lambda_{ij} \operatorname{row}_{h_j}(B) = \operatorname{row}_i(B)$$

- Thus $\operatorname{row}_i(AX^*) = \operatorname{row}_i(A)X^* = \operatorname{row}_i(B)$ for every i (Theorem 1.4(c)), that is, $AX^* = B$. The last sentence is the case $p = 1$. □

In practice, $\mathcal{C}(A)$ is described by equations obtained from the reduction of $[A \mid I_m]$ to $[R \mid P]$ (Theorem 4.8). The same row operations take $[A \mid b]$ to $[R \mid Pb]$: by Theorem 2.15(c) they multiply on the left by an invertible E, and $E[A \mid I_m] = [EA \mid E]$, column by column (Theorem 1.4(a)), so $E = P$, $EA = R$ and $E[A \mid b] = [R \mid Pb]$. As $b \in \mathcal{C}(A)$ if and only if $Ax = b$ is consistent (Corollary 4.14(a)), Proposition 2.7(a) gives, with $r = \operatorname{rank} A$,

$$b \in \mathcal{C}(A) \iff \operatorname{row}_i(P)\, b = 0 \quad \text{for } i = r+1, \dots, m$$

(no condition if $r = m$; Example 2.25 is an instance) [1, §2.6, Example 22(h)]. The vectors $\operatorname{row}_i(P)'$ $(i > r)$ form a basis of $\mathcal{N}(A')$ (Theorem 4.8(d)), so these $m - r$ equations are the conditions $y'b = 0$ of Theorem 4.29.

Example 4.30. For A of Example 4.9, $r = 2$ and $\operatorname{row}_3(P) = (-1, -1, 1)$, so

$$b \in \mathcal{C}(A) \iff -b_1 - b_2 + b_3 = 0$$

This is the row relation $\operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$ found there, which every right side must share (Theorem 4.29).

(a) $Ax = (1, t, 4)'$ is consistent if and only if $-1 - t + 4 = 0$, that is, $t = 3$. For $t = 3$ and $b = (1, 3, 4)'$, $Pb = (3 - 6,\ -1 + 3,\ -1 - 3 + 4)' = (-3, 2, 0)'$, and $Rx = Pb$ reads $x_1 + 3x_3 - x_4 = -3$, $x_2 - x_3 + 2x_4 = 2$. The free unknowns $x_3 = x_4 = 0$ give the solution $x_p = (-3, 2, 0, 0)'$, and all solutions are $x_p + c_1 s_1 + c_2 s_2$ $(c_1, c_2 \in \mathbb{R})$, by Theorems 2.27(a) and 4.8(c). Check: $Ax_p = -3a_1 + 2a_2 = (-3 + 4,\ -3 + 6,\ -6 + 10)' = (1, 3, 4)'$.

(b) For the two right sides

$$B_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

and $y = (-1, -1, 1)'$, $y'B_1 = (-1 - 1 + 2,\ 0 - 1 + 1) = (0, 0)$ and $y'B_2 = (-1, -1) \neq (0, 0)$. Every vector of $\mathcal{N}(A')$ is a multiple of y, so $AX = B_1$ is consistent and $AX = B_2$ is not. Indeed the columns of B_1 are a_1 and $a_2 - 2a_1$, so

$$X = \begin{pmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

solves $AX = B_1$ (Theorem 1.4(a),(b)).

The normal form of a matrix

Theorem 4.31 ([3, Result 2.15], [6, Exercise 6.17], [5, Theorem 4.4.9])**.** Let $A \in \mathbb{R}^{m \times n}$ have rank r, and let $N_r \in \mathbb{R}^{m \times n}$ be the matrix whose first r columns are the unit vectors $e_1, \dots, e_r \in \mathbb{R}^m$ and whose other $n - r$ columns are 0 $(N_0 = 0)$:

$$N_r = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

There are invertible matrices $P \in \mathbb{R}^{m \times m}$ and $Q \in \mathbb{R}^{n \times n}$ such that $PAQ = N_r$. This N_r is the **normal form** of A.

Proof. If $r = 0$, then $A = 0$ (Theorem 4.8); take $P = I_m$ and $Q = I_n$. Let $r \geqslant 1$.

- By Theorem 4.21(a), $A = B_1C_1$ with $B_1 \in \mathbb{R}^{m\times r}$ of full column rank and $C_1 \in \mathbb{R}^{r\times n}$ of full row rank. So the columns $b_1, \dots, b_r$ of B_1 are independent, and so are the rows $c_1', \dots, c_r'$ of C_1 ($c_j \in \mathbb{R}^n$; Theorem 4.13(b),(c)).
- By Theorem 4.5(b), $b_1, \dots, b_r$ extend to a basis $b_1, \dots, b_m$ of $\mathbb{R}^m$, and $c_1', \dots, c_r'$ extend to a basis $c_1', \dots, c_n'$ of $\mathbb{R}^{1\times n}$ (the paragraph after Theorem 4.5; bases of $\mathbb{R}^m$ and of $\mathbb{R}^{1\times n}$ have m and n vectors, and nothing is added if $r = m$, or $r = n$).
- Let $B = [b_1 \ \cdots \ b_m] \in \mathbb{R}^{m\times m}$, and let $C \in \mathbb{R}^{n\times n}$ have the rows $c_1', \dots, c_n'$. The columns of B are independent, so $\operatorname{rank} B = m$ and B is invertible (Theorem 4.13(b),(d)); the rows of C are independent, so $\operatorname{rank} C = n$ and C is invertible (Theorem 4.13(c),(d)).
- By Theorem 1.4(a),(b), the columns of BN_r are $Be_j = b_j$ ($j \leqslant r$) and 0 ($j > r$). By the outer product expansion (Theorem 1.4(d)), applied to $(BN_r)C$ and to B_1C_1,

$$BN_rC = \sum_{j=1}^{r} b_j c_j' = B_1C_1 = A$$

- Hence $P = B^{-1}$ and $Q = C^{-1}$ are invertible (Theorem 2.13(a)), and $PAQ = B^{-1}(BN_rC)C^{-1} = N_r$. □

Remark 4.32 (cf. [6, Exercises 6.15, 6.16])**.** The matrices P and Q of Theorem 4.31 can be read off from the computation of Theorem 4.8. Let $[A \mid I_m]$ be reduced to $[R \mid P]$, with pivot columns $k_1 < \cdots < k_r$, free columns $f_1 < \cdots < f_{n-r}$ and special solutions $s_1, \dots, s_{n-r}$, and let

$$Q = [e_{k_1} \ \cdots \ e_{k_r} \ s_1 \ \cdots \ s_{n-r}] \in \mathbb{R}^{n\times n}$$

Then P and Q are invertible and $PAQ = N_r$.

- P is invertible and $PA = R$ (Theorem 4.8). The columns of RQ are $Re_{k_i} = \operatorname{col}_{k_i}(R) = e_i$ ($i \leqslant r$; Definition 2.4) and $Rs_j = 0$, as $s_j \in \mathcal{N}(A) = \mathcal{N}(R)$ (Theorem 2.3(b)). So $PAQ = RQ = N_r$ (Theorem 1.4(a)).
- Let $Qx = 0$, that is, $\sum_{i=1}^{r} x_i e_{k_i} + \sum_{j=1}^{n-r} x_{r+j} s_j = 0$. Entry f_l of the left side is x_{r+l}, since e_{k_i} has entry 0 there and s_j has entry δ_{jl}; so $x_{r+1} = \cdots = x_n = 0$, and then $\sum_i x_i e_{k_i} = 0$ gives $x_1 = \cdots = x_r = 0$. By Theorem 2.16, Q is invertible.

Example 4.33. For A of Example 4.9, the pivot columns are $1, 2$ and the special solutions are $s_1 = (-3, 1, 1, 0)'$, $s_2 = (1, -2, 0, 1)'$. Remark 4.32 gives

$$P = \begin{pmatrix} 3 & -2 & 0 \\ -1 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}, \qquad Q = [e_1 \ e_2 \ s_1 \ s_2] = \begin{pmatrix} 1 & 0 & -3 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and

$$PAQ = RQ = \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -3 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = N_2$$

These are also the matrices of the proof of Theorem 4.31 for the factorization $A = BC$ of Example 4.22, completed by e_3 and by the rows e_3', e_4':

- $\tilde{B} = [a_1 \ a_2 \ e_3]$ has $P\tilde{B} = [Pa_1 \ Pa_2 \ Pe_3] = [e_1 \ e_2 \ e_3] = I_3$, since Pa_1, Pa_2 are columns 1, 2 of $PA = R$ and $Pe_3 = \operatorname{col}_3(P) = e_3$; so $\tilde{B}^{-1} = P$ (Theorem 2.16);

- $\tilde{C} \in \mathbb{R}^{4\times 4}$, with rows $(1,0,3,-1)$, $(0,1,-1,2)$, e_3', e_4', has $\tilde{C}Q = I_4$: the first two rows of $\tilde{C}$ form the matrix C with $Ce_1 = e_1$, $Ce_2 = e_2$, $Cs_1 = Cs_2 = 0$, and $e_3'Q$, $e_4'Q$ are rows 3, 4 of Q, that is, e_3' and e_4'. So $\tilde{C}^{-1} = Q$.

One-sided inverses

Definition 4.34 ([5, §8.1])**.** Let $A \in \mathbb{R}^{m\times n}$. A **right inverse** of A is a matrix $K \in \mathbb{R}^{n\times m}$ with $AK = I_m$, and a **left inverse** of A is a matrix $L \in \mathbb{R}^{n\times m}$ with $LA = I_n$ (as for the factors of a full-rank factorization in Theorem 4.21(c)).

Theorem 4.35 ([5, Lemma 8.1.1, Corollary 8.1.2, proof of Lemma 8.1.3])**.** Let $A \in \mathbb{R}^{m\times n}$.

(a) A has a right inverse if and only if A has full row rank, $\operatorname{rank} A = m$.

(b) A has a left inverse if and only if A has full column rank, $\operatorname{rank} A = n$.

(c) A has both a right and a left inverse if and only if $m = n$ and A is invertible; then every right inverse and every left inverse of A equals A^{-1}.

Proof. (a)

- If $\operatorname{rank} A = m$, the system $AX = I_m$ is consistent (Corollary 4.28, last sentence): it has a solution $K \in \mathbb{R}^{n\times m}$, a right inverse of A.
- Conversely, if $AK = I_m$, then $m = \operatorname{rank} I_m = \operatorname{rank}(AK) \leqslant \operatorname{rank} A \leqslant m$ (Theorems 4.13(d), 4.15(a) and 4.13(a)); so $\operatorname{rank} A = m$.

(b) $LA = I_n$ if and only if $A'L' = I_n$ (Theorem 1.9), so L is a left inverse of A if and only if L' is a right inverse of $A' \in \mathbb{R}^{n\times m}$. By (a), A' has a right inverse if and only if $\operatorname{rank} A' = n$, and $\operatorname{rank} A' = \operatorname{rank} A$ (Theorem 4.13(a)).

(c)

- Let $AK = I_m$ and $LA = I_n$. By (a) and (b), $\operatorname{rank} A = m$ and $\operatorname{rank} A = n$, so $m = n$, and A is invertible (Theorem 4.13(d)).
- Then $L = LI_m = L(AK) = (LA)K = I_nK = K$ (Theorem 1.3), so K is a two-sided inverse of A, that is, $K = L = A^{-1}$ (Definition 2.12). This holds for every right inverse K and every left inverse L.
- Conversely, if $m = n$ and A is invertible, A^{-1} is a right and a left inverse of A. □

For instance, if A has full column rank, then $A'A$ is invertible (Theorems 4.18(b) and 4.13(d)), and $(A'A)^{-1}A'$ is a left inverse of A. A one-sided inverse of a matrix that is not square is never unique: if $AK_0 = I_m$ and $m \neq n$, then $\operatorname{rank} A = m$ by (a), so $m < n$ (Theorem 4.13(a)) and $\dim \mathcal{N}(A) = n - m \geqslant 1$ (Theorem 4.12), and for a nonzero $s \in \mathcal{N}(A)$ every $K = K_0 + st'$ ($t \in \mathbb{R}^m$) satisfies $AK = AK_0 + (As)t' = I_m$; for left inverses apply this to A' (L is a left inverse of A if and only if L' is a right inverse of A', Theorem 1.9). The next example finds all right inverses of a 3×4 matrix.

Example 4.36. Find all right inverses of

$$A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 3 & 3 \end{pmatrix}$$

and show that A has no left inverse.

Solution.

- Row-reduce $[A \mid I_3]$:

$$\left(\begin{array}{cccc|ccc} 1 & 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 3 & 3 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|ccc} 1 & 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & -1 & 0 & 1 \end{array}\right)$$
$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc} 1 & 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & 1 \end{array}\right) \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 2 & 2 & 1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & -1 & 1 \end{array}\right)$$

- The RREF of A, the left part, has three pivots, so $\operatorname{rank} A = 3 = m$: A has right inverses (Theorem 4.35(a)). As $\operatorname{rank} A = 3 < 4 = n$, A has no left inverse (Theorem 4.35(b)).
- Column j of a right inverse solves $Ax = e_j$, and the same row operations take $[A \mid e_j]$ to the left part and column j of the right part of the last matrix (Theorems 2.15(c) and 1.4(a)). Let $K_0 \in \mathbb{R}^{4\times 3}$ have this right part as its first three rows and a zero fourth row:

$$K_0 = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & -1 \\ -1 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad s = \begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

  The unknown x_4 is free; with $x_4 = t_j$, Proposition 2.7(b) gives $x_1 = k_{1j} - 2t_j$, $x_2 = k_{2j} - t_j$, $x_3 = k_{3j}$, where k_{ij} are the entries of K_0. So the solutions are $x = \operatorname{col}_j(K_0) + t_j s$ $(t_j \in \mathbb{R})$.
- Hence the right inverses of A are the matrices $K = K_0 + st'$ with $t = (t_1, t_2, t_3)' \in \mathbb{R}^3$, as column j of st' is $t_j s$ (Theorem 1.4(a),(d)); there are infinitely many. Check: $As = (-2+2,\ -1+1,\ -2-1+3)' = 0$, and by rows, $\operatorname{row}_3(AK_0) = 1\,(2,1,-1) + 1\,(1,2,-1) + 3\,(-1,-1,1) = (0,0,1)$; likewise $\operatorname{row}_1(AK_0) = (1,0,0)$ and $\operatorname{row}_2(AK_0) = (0,1,0)$, so $AK_0 = I_3$.

Exercises

Exercise 4.1. Let $u_1 = (1,-2,1,-1)'$ and $u_2 = (0,1,-1,1)'$ in $\mathbb{R}^4$, and let $e_1, \dots, e_4$ be the unit vectors.

(a) Row-reduce $[u_1\ u_2\ e_1\ e_2\ e_3\ e_4] \in \mathbb{R}^{4\times 6}$, and find a basis of $\mathbb{R}^4$ that consists of u_1, u_2 and two of the unit vectors.

(b) Do u_1, u_2, e_1, e_2 form a basis of $\mathbb{R}^4$?

Solution. (a) Call the matrix $A = [u_1\ u_2\ e_1\ e_2\ e_3\ e_4]$; its columns span $\mathbb{R}^4$, since they include $e_1, \dots, e_4$.

$$\left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 + 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$
$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 + R_3} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

The last matrix is an echelon form, with leading 1s in columns $1, 2, 3, 5$; these are the pivot columns of A (Remark 4.2). By Theorem 4.3(b), u_1, u_2, e_1, e_3 form a basis of the span of the columns, which is $\mathbb{R}^4$.

(b) No. Backward elimination finishes the RREF:

$$\xrightarrow{R_3\leftarrow R_3-R_4}\left(\begin{array}{cc|cccc}1&0&1&0&0&0\\0&1&2&1&0&0\\0&0&1&1&0&-1\\0&0&0&0&1&1\end{array}\right)\xrightarrow[R_2\leftarrow R_2-2R_3]{R_1\leftarrow R_1-R_3}\left(\begin{array}{cc|cccc}1&0&0&-1&0&1\\0&1&0&-1&0&2\\0&0&1&1&0&-1\\0&0&0&0&1&1\end{array}\right)$$

Column 4 of the RREF gives $e_2 = -u_1 - u_2 + e_1$ (Theorem 4.3(b)). So u_1, u_2, e_1, e_2 are linearly dependent and do not form a basis. Check: $-(1,-2,1,-1)' - (0,1,-1,1)' + (1,0,0,0)' = (0,1,0,0)' = e_2$.

Exercise 4.2 ([6, Exercise 3.23])**.** Let $a_1, \ldots, a_9$ be the columns of

$$A = \begin{pmatrix}1&1&1&1&1&1&2&2&5\\0&1&1&2&2&3&-1&3&3\\1&1&2&3&5&5&1&0&4\end{pmatrix}$$

Which of the following form a basis of $\mathbb{R}^3$? (a) a_1, a_2; (b) a_1, a_4, a_6, a_8; (c) a_2, a_4, a_7; (d) a_3, a_5, a_9.

Solution. (a) No: $\dim \mathbb{R}^3 = 3$, so two vectors cannot span $\mathbb{R}^3$ (Theorem 4.5(c)).

(b) No: $\dim \mathbb{R}^3 = 3$, so four vectors of $\mathbb{R}^3$ are dependent (Theorem 4.5(c)).

(c) Yes. Three vectors of $\mathbb{R}^3$ form a basis if and only if they are independent (Theorem 4.5(d)). Row-reduce $[a_2\ a_4\ a_7]$:

$$\begin{pmatrix}1&1&2\\1&2&-1\\1&3&1\end{pmatrix}\xrightarrow[R_3\leftarrow R_3-R_1]{R_2\leftarrow R_2-R_1}\begin{pmatrix}1&1&2\\0&1&-3\\0&2&-1\end{pmatrix}\xrightarrow{R_3\leftarrow R_3-2R_2}\begin{pmatrix}1&1&2\\0&1&-3\\0&0&5\end{pmatrix}$$

$$\xrightarrow{R_3\leftarrow \frac{1}{5}R_3}\begin{pmatrix}1&1&2\\0&1&-3\\0&0&1\end{pmatrix}\xrightarrow[R_2\leftarrow R_2+3R_3]{R_1\leftarrow R_1-2R_3}\begin{pmatrix}1&1&0\\0&1&0\\0&0&1\end{pmatrix}\xrightarrow{R_1\leftarrow R_1-R_2}\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$

All three columns are pivot columns, so a_2, a_4, a_7 are independent (Theorem 4.1).

(d) No: the three vectors are dependent, since

$$7a_3 - 2a_5 - a_9 = 7\begin{pmatrix}1\\1\\2\end{pmatrix} - 2\begin{pmatrix}1\\2\\5\end{pmatrix} - \begin{pmatrix}5\\3\\4\end{pmatrix} = \begin{pmatrix}7-2-5\\7-4-3\\14-10-4\end{pmatrix} = 0$$

Exercise 4.3 ([6, Exercise 3.31(a), (b)])**.** Let $W = \mathcal{C}(A) \subseteq \mathbb{R}^4$ for

$$A = \begin{pmatrix}1&2&3\\-2&3&8\\5&1&-3\\-3&-4&-5\end{pmatrix}$$

(a) Find a basis of W. (b) What is $\dim W$?

Solution. Let a_1, a_2, a_3 be the columns of A; they span W.

(a) a_1, a_2 are independent: otherwise, as $a_1 \neq 0$, $a_2 = ca_1$ for some c (Proposition 3.13(c)); the first entries give $c = 2$, and then the second entries give $3 = 2(-2)$ ✘. But

$$a_1 - 2a_2 + a_3 = \begin{pmatrix}1-4+3\\-2-6+8\\5-2-3\\-3+8-5\end{pmatrix} = 0$$

so $a_3 = 2a_2 - a_1$, and every combination $x_1a_1 + x_2a_2 + x_3a_3 = (x_1 - x_3)a_1 + (x_2 + 2x_3)a_2$ is a combination of a_1, a_2. Thus a_1, a_2 span W and form a basis of W.

(b) $\dim W = 2$.

Exercise 4.4. Let

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 1 & 2 & 1 & 0 & 1 \\ 2 & 4 & 1 & 1 & 0 \\ 1 & 2 & 2 & -1 & 3 \end{pmatrix}$$

(a) Find $\operatorname{rank} A$ and bases of $\mathcal{C}(A)$, $\mathcal{R}(A)$, $\mathcal{N}(A)$ and $\mathcal{N}(A')$, and write the non-pivot columns of A as combinations of the pivot columns.

(b) For which $b \in \mathbb{R}^4$ is $Ax = b$ consistent? Give all solutions for $b = (1, 2, 3, 3)'$.

Solution. (a) Row-reduce $[A \mid I_4]$ (Theorem 4.8):

$$\left(\begin{array}{ccccc|cccc} 1 & 2 & 0 & 1 & -1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 4 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 2 & -1 & 3 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{ccccc|cccc} 1 & 2 & 0 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & -2 & 0 & 1 & 0 \\ 0 & 0 & 2 & -2 & 4 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccccc|cccc} 1 & 2 & 0 & 1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 0 & 1 \end{array}\right) = [R \mid P]$$

Row 1 has entry 0 in column 3, so no backward arrow is needed. Thus $\operatorname{rank} A = 2$, with pivot columns $1, 3$ and free unknowns x_2, x_4, x_5.

- $\mathcal{C}(A)$: $a_1 = (1, 1, 2, 1)'$ and $a_3 = (0, 1, 1, 2)'$. Columns 2, 4, 5 of R give $a_2 = 2a_1$, $a_4 = a_1 - a_3$ and $a_5 = -a_1 + 2a_3$ (Theorem 4.3(b)). Check: $(1, 1, 2, 1)' - (0, 1, 1, 2)' = (1, 0, 1, -1)' = a_4$ and $-(1, 1, 2, 1)' + 2(0, 1, 1, 2)' = (-1, 1, 0, 3)' = a_5$.
- $\mathcal{R}(A)$: $\rho_1' = (1, 2, 0, 1, -1)$ and $\rho_2' = (0, 0, 1, -1, 2)$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 + 2x_2 + x_4 - x_5 = 0$, $x_3 - x_4 + 2x_5 = 0$. The special solutions

  $$s_1 = (-2, 1, 0, 0, 0)', \qquad s_2 = (-1, 0, 1, 1, 0)', \qquad s_3 = (1, 0, -2, 0, 1)'$$

  form a basis; $\dim \mathcal{N}(A) = 3 = 5 - 2$.
- $\mathcal{N}(A')$: rows 3, 4 of P give the basis $y_1 = (-1, -1, 1, 0)'$, $y_2 = (1, -2, 0, 1)'$; $\dim \mathcal{N}(A') = 2 = 4 - 2$. Indeed $\operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$ and $\operatorname{row}_4(A) = -\operatorname{row}_1(A) + 2\operatorname{row}_2(A)$.

(b) The row operations of (a) multiply on the left by the invertible matrix P of $[R \mid P]$ (Theorem 2.15(c)), so they take $[A \mid b]$ to $P[A \mid b] = [R \mid Pb]$ (Theorem 1.4(a)), where

$$Pb = (b_1,\ b_2 - b_1,\ b_3 - b_1 - b_2,\ b_4 + b_1 - 2b_2)'$$

By Proposition 2.7(a), $Ax = b$ is consistent if and only if $b_3 = b_1 + b_2$ and $b_4 = 2b_2 - b_1$, that is, $y_1'b = y_2'b = 0$. For $b = (1, 2, 3, 3)'$ both hold, and $Pb = (1, 1, 0, 0)'$. By Proposition 2.7(b), $x_1 = 1 - 2x_2 - x_4 + x_5$ and $x_3 = 1 + x_4 - 2x_5$, so the solutions are

$$x = (1, 0, 1, 0, 0)' + x_2 s_1 + x_4 s_2 + x_5 s_3 \qquad (x_2, x_4, x_5 \in \mathbb{R})$$

Check: $A(1, 0, 1, 0, 0)' = a_1 + a_3 = (1, 2, 3, 3)'$.

Exercise 4.5. Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 2 & 3 & 1 & 1 & 5 \\ 1 & 2 & 0 & 2 & 5 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix}$$

(a) Row-reduce $[A \mid I_4]$. Find $\operatorname{rank} A$, bases of $\mathcal{C}(A)$ and $\mathcal{R}(A)$, and the non-pivot columns of A as combinations of the pivot columns.

(b) Find bases of $\mathcal{N}(A)$ and $\mathcal{N}(A')$.

(c) Find an equation describing $\mathcal{C}(A)$. Which of $b = (1,3,0,1)'$ and $u = (1,1,1,1)'$ lie in $\mathcal{C}(A)$? For each one that does, find an x with Ax equal to it.

Solution. (a)

$$\left(\begin{array}{ccccc|cccc} 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 1 & 5 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 2 & 5 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 & 3 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccccc|cccc} 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 3 & -2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & 4 & -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 & 3 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccccc|cccc} 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 3 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{ccccc|cccc} 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 2 & -3 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{ccccc|cccc} 1 & 0 & 2 & 0 & -1 & 4 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 & 2 & -3 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & -1 & 0 & 1 \end{array}\right) = [R \mid P]$$

So $\operatorname{rank} A = 3$, with pivot columns $1, 2, 4$ and free unknowns x_3, x_5.

- $\mathcal{C}(A)$: $a_1 = (1,2,1,0)'$, $a_2 = (1,3,2,1)'$ and $a_4 = (0,1,2,1)'$ form a basis. Columns 3 and 5 of R give $a_3 = 2a_1 - a_2$ and $a_5 = -a_1 + 2a_2 + a_4$ (Theorem 4.3(b)). Check: $2(1,2,1,0)' - (1,3,2,1)' = (1,1,0,-1)'$ and $-(1,2,1,0)' + 2(1,3,2,1)' + (0,1,2,1)' = (1,5,5,3)'$.
- $\mathcal{R}(A)$: $\rho_1' = (1,0,2,0,-1)$, $\rho_2' = (0,1,-1,0,2)$ and $\rho_3' = (0,0,0,1,1)$ form a basis.

(b)

- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 + 2x_3 - x_5 = 0$, $x_2 - x_3 + 2x_5 = 0$, $x_4 + x_5 = 0$. The special solutions $s_1 = (-2,1,1,0,0)'$ ($x_3 = 1$, $x_5 = 0$) and $s_2 = (1,-2,0,-1,1)'$ ($x_3 = 0$, $x_5 = 1$) form a basis; $\dim \mathcal{N}(A) = 2 = 5 - 3$.
- $\mathcal{N}(A')$: row 4 of P gives the basis $y = (2,-1,0,1)'$ (Theorem 4.8(d)); $\dim \mathcal{N}(A') = 1 = 4 - 3$. Indeed $y'A = 2\,\mathrm{row}_1(A) - \mathrm{row}_2(A) + \mathrm{row}_4(A) = 0$.

(c)

- By the paragraph after Theorem 4.29, $b \in \mathcal{C}(A)$ if and only if $\mathrm{row}_4(P)\, b = 0$, that is, $2b_1 - b_2 + b_4 = 0$.
- For $b = (1,3,0,1)'$: $2 - 3 + 1 = 0$, so $b \in \mathcal{C}(A)$. Then $Pb = (4-6,\ -3+6,\ 1-3,\ 2-3+1)' = (-2,3,-2,0)'$, and $Rx = Pb$ reads $x_1 + 2x_3 - x_5 = -2$, $x_2 - x_3 + 2x_5 = 3$, $x_4 + x_5 = -2$. The free unknowns $x_3 = x_5 = 0$ give the solution $x = (-2,3,0,-2,0)'$. Check: $Ax = -2a_1 + 3a_2 - 2a_4 = (-2+3,\ -4+9-2,\ -2+6-4,\ 3-2)' = (1,3,0,1)'$.
- For $u = (1,1,1,1)'$: $2 - 1 + 1 = 2 \neq 0$, so $u \notin \mathcal{C}(A)$.

Exercise 4.6. Find $\operatorname{rank} A$, bases of $\mathcal{C}(A)$, $\mathcal{R}(A)$ and $\mathcal{N}(A)$, and $\dim \mathcal{N}(A')$ for

$$A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & 2 \\ 0 & -2 & -2 & 3 \end{pmatrix}$$

and write the non-pivot column of A as a combination of the pivot columns.

Solution. Row-reduce A:

$$A \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & -2 & -2 & 3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + R_3} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

After the second arrow column 3 has no nonzero entry in rows $3, 4$, and the next pivot column, 4, has its nonzero entry in row 4; so rows 3 and 4 are interchanged. The echelon form after the third arrow has three nonzero rows: $\operatorname{rank} A = 3$, with pivot columns $1, 2, 4$.

- $\mathcal{C}(A)$: $a_1 = (1, 0, 1, 0)'$, $a_2 = (2, 1, 1, -2)'$ and $a_4 = (1, -1, 2, 3)'$ form a basis (Theorem 4.8(a)). Column 3 of R gives $a_3 = -a_1 + a_2$; check: $-(1, 0, 1, 0)' + (2, 1, 1, -2)' = (1, 1, 0, -2)' = a_3$.
- $\mathcal{R}(A)$: $\rho_1' = (1, 0, -1, 0)$, $\rho_2' = (0, 1, 1, 0)$ and $\rho_3' = (0, 0, 0, 1)$ form a basis (Theorem 4.8(b)).
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 - x_3 = 0$, $x_2 + x_3 = 0$, $x_4 = 0$; the special solution $s = (1, -1, 1, 0)'$ ($x_3 = 1$) is a basis (Theorem 4.8(c)), and $\dim \mathcal{N}(A) = 1 = 4 - 3$.
- $\dim \mathcal{N}(A') = 4 - \operatorname{rank} A = 1$ (Theorem 4.12).

Exercise 4.7. Let $W_0 = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$, the centered vectors of $\mathbb{R}^4$, and let

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \quad w = \begin{pmatrix} 2 \\ -1 \\ 3 \\ -4 \end{pmatrix}$$

(a) Find $\dim W_0$.

(b) Show that v_1, v_2, v_3 form a basis of W_0, and find the coordinates of w in this basis.

(c) Does $(1, 2, 3, 4)'$ lie in $\operatorname{span}\{v_1, v_2, v_3\}$?

Solution. (a) $W_0 = \mathcal{N}(\mathbf{1}_4')$, where the 1×4 matrix $\mathbf{1}_4' = (1, 1, 1, 1)$ is in RREF with one nonzero row. So $\operatorname{rank} \mathbf{1}_4' = 1$ and $\dim W_0 = 4 - 1 = 3$ (Theorem 4.12).

(b) The entries of each of v_1, v_2, v_3, w add up to 0, so these vectors lie in W_0. Row-reduce $[v_1\ v_2\ v_3 \mid w]$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 3 \\ -1 & -1 & -1 & -4 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 + R_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & -1 & 0 & -2 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 + R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 1 & -3 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 1 & -3 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

- Row operations act on each column separately (Theorems 2.15(c) and 1.4(a)), so the left part shows the RREF of $[v_1\ v_2\ v_3]$: every column is a pivot column, and v_1, v_2, v_3 are independent (Theorem 4.1). Three independent vectors of the 3-dimensional space W_0 form a basis of it (Theorem 4.5(d)).
- The last column is not a pivot column, and the coordinates are read off: $c = (5, 2, -3)'$ (Proposition 2.7), that is, $w = 5v_1 + 2v_2 - 3v_3$. Check: $5(1,0,0,-1)' + 2(0,1,0,-1)' - 3(1,1,-1,-1)' = (5-3,\ 2-3,\ 3,\ -5-2+3)' = (2,-1,3,-4)'$.

(c) No. By (b), $\operatorname{span}\{v_1, v_2, v_3\} = W_0$, and the entries of $(1,2,3,4)'$ add up to $10 \neq 0$.

Exercise 4.8 ([4, Ch. V, §3, Exercise 4(b), (d)], [7])**.** Find the dimension of the space of solutions $(x, y, z)' \in \mathbb{R}^3$ of each system, and a basis of it: (a) $x - y + z = 0$; (b) $x + y + z = 0$, $x - y = 0$, $y + z = 0$.

Solution. The space of solutions is $\mathcal{N}(A)$ for the coefficient matrix A, of dimension $3 - \operatorname{rank} A$ (Theorem 4.12).

(a) $A = (1, -1, 1)$ is a nonzero RREF with one row, so $\operatorname{rank} A = 1$ and the dimension is 2. The vectors $(1,1,0)'$ and $(0,1,1)'$ are solutions ($1 - 1 + 0 = 0$, $0 - 1 + 1 = 0$) and are independent ($c_1(1,1,0)' + c_2(0,1,1)' = (c_1, c_1 + c_2, c_2)' = 0$ forces $c_1 = c_2 = 0$); so they form a basis (Theorem 4.5(d)).

(b) Row-reduce the coefficient matrix:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

This echelon form has three nonzero rows, so $\operatorname{rank} A = 3$ (Definition 4.11): the space of solutions has dimension $3 - 3 = 0$, so it is $\{0\}$ (Theorem 4.4), with the empty basis.

Exercise 4.9. For $t \in \mathbb{R}$ let

$$A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 5 & 1 & 4 \\ 1 & 3 & 1 & 3 \\ 1 & 1 & 2 & t \end{pmatrix}$$

(a) Find $\operatorname{rank} A$ for every t.

(b) For the value of t with $\operatorname{rank} A < 4$, find a basis of $\mathcal{C}(A)$, write the non-pivot column as a combination of the pivot columns, and find a basis of $\mathcal{N}(A)$.

Solution. (a) Forward elimination, keeping t as a letter:

$$\begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 5 & 1 & 4 \\ 1 & 3 & 1 & 3 \\ 1 & 1 & 2 & t \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & -1 & 1 & t-1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & t+1 \end{pmatrix}$$

The last row is $(0, 0, 0, t + 1)$. If $t \neq -1$, scaling row 4 by $1/(t+1)$ gives an echelon form with four nonzero rows, so $\operatorname{rank} A = 4$ (Remark 4.2 and Definition 4.11). If $t = -1$, the last row is zero, and the last matrix is an echelon form with three nonzero rows: $\operatorname{rank} A = 3$.

(b) Let $t = -1$. Backward elimination finishes the RREF:

$$\begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + R_3} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

- The pivot columns are $1, 2, 3$: $a_1 = (1, 2, 1, 1)'$, $a_2 = (2, 5, 3, 1)'$ and $a_3 = (1, 1, 1, 2)'$ form a basis of $\mathcal{C}(A)$. Column 4 of R gives $a_4 = -3a_1 + 2a_2$ (Theorem 4.3(b)); check: $-3(1, 2, 1, 1)' + 2(2, 5, 3, 1)' = (1, 4, 3, -1)'$, the last column of A for $t = -1$.
- $Rx = 0$ reads $x_1 - 3x_4 = 0$, $x_2 + 2x_4 = 0$, $x_3 = 0$. The special solution $s = (3, -2, 0, 1)'$ ($x_4 = 1$) is a basis of $\mathcal{N}(A)$, and $\dim \mathcal{N}(A) = 1 = 4 - 3$.

Exercise 4.10. For $t \in \mathbb{R}$ let

$$A_t = \begin{pmatrix} t & 1 & 1 & 1 \\ 1 & t & 1 & 1 \\ 1 & 1 & t & 1 \\ 1 & 1 & 1 & t \end{pmatrix}$$

Find $\operatorname{rank} A_t$ for every t, and a basis of $\mathcal{N}(A_t)$ whenever A_t is not invertible.

Solution. Interchange the rows so that the first row is $(1, 1, 1, t)$, and eliminate, keeping t as a letter:

$$A_t \xrightarrow[R_2 \leftrightarrow R_3]{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & t \\ 1 & 1 & t & 1 \\ 1 & t & 1 & 1 \\ t & 1 & 1 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - tR_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 1 & t \\ 0 & 0 & t-1 & 1-t \\ 0 & t-1 & 0 & 1-t \\ 0 & 1-t & 1-t & 1-t^2 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 1 & t \\ 0 & 0 & t-1 & 1-t \\ 0 & t-1 & 0 & 1-t \\ 0 & 1-t & 0 & -(t-1)(t+2) \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & 1 & 1 & t \\ 0 & 0 & t-1 & 1-t \\ 0 & t-1 & 0 & 1-t \\ 0 & 0 & 0 & -(t-1)(t+3) \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & t \\ 0 & t-1 & 0 & 1-t \\ 0 & 0 & t-1 & 1-t \\ 0 & 0 & 0 & -(t-1)(t+3) \end{pmatrix} = E_t$$

since $(1 - t^2) + (1 - t) = -(t - 1)(t + 2)$ and $-(t - 1)(t + 2) + (1 - t) = -(t - 1)(t + 3)$. The pivot columns can be read off before the leading entries are scaled to 1 (Remark 4.2).

- $t \neq 1$, $t \neq -3$: all four diagonal entries of E_t are nonzero, so $\operatorname{rank} A_t = 4$ (Definition 4.11) and A_t is invertible (Theorem 4.13(d)).
- $t = 1$: rows 2, 3, 4 of E_1 are zero, so $\operatorname{rank} A_1 = 1$. Here $\mathcal{N}(A_1)$ is given by $x_1 + x_2 + x_3 + x_4 = 0$, with special solutions $(-1, 1, 0, 0)'$, $(-1, 0, 1, 0)'$, $(-1, 0, 0, 1)'$ as a basis (Theorem 4.8(c)); $\dim \mathcal{N}(A_1) = 3$.
- $t = -3$: E_{-3} has nonzero rows $(1, 1, 1, -3)$, $(0, -4, 0, 4)$, $(0, 0, -4, 4)$ and a zero row, so $\operatorname{rank} A_{-3} = 3$ and $\dim \mathcal{N}(A_{-3}) = 1$ (Theorem 4.12). Every row of A_t has sum $t + 3$, so $A_{-3}\mathbf{1}_4 = 0$; the nonzero vector $\mathbf{1}_4 = (1, 1, 1, 1)'$ is a basis of $\mathcal{N}(A_{-3})$ (Proposition 3.13(a), Theorem 4.5(d)).

Exercise 4.11. Let

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 2 \\ 3 & 7 & 4 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ t \\ 2 \\ 4 \end{pmatrix} \qquad (t \in \mathbb{R})$$

(a) For which t is $Ax = b$ consistent? Compare $\operatorname{rank} A$ and $\operatorname{rank}[A \mid b]$.

(b) For this t, find all solutions. Is the solution unique?

Solution. (a) Row-reduce $[A \mid b]$, keeping t as a letter:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 5 & 3 & t \\ 1 & 3 & 2 & 2 \\ 3 & 7 & 4 & 4 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & t-2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & t-2 \\ 0 & 0 & 0 & 3-t \\ 0 & 0 & 0 & 3-t \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & t-2 \\ 0 & 0 & 0 & 3-t \\ 0 & 0 & 0 & 0 \end{array}\right)$$

- The left part of the last matrix is an echelon form with two nonzero rows, so $\operatorname{rank} A = 2$ (Remark 4.2).
- If $t \neq 3$, row 3 reads $(0\ 0\ 0 \mid 3-t)$ with $3 - t \neq 0$: the equation $0 = 3 - t$ has no solution (Theorem 2.26). Scaling row 3 by $1/(3-t)$ gives an echelon form of $[A \mid b]$ with three nonzero rows, so $\operatorname{rank}[A \mid b] = 3 > \operatorname{rank} A$.
- If $t = 3$, rows 3 and 4 are zero: $\operatorname{rank}[A \mid b] = 2 = \operatorname{rank} A$, and the system is consistent (Corollary 4.14(a)).

(b) Let $t = 3$. Backward elimination:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

With the free unknown $x_3 = c$: $x_1 = -1 + c$, $x_2 = 1 - c$, that is,

$$x = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \qquad (c \in \mathbb{R})$$

The solution is not unique, since $\operatorname{rank} A = 2 < 3$ (Corollary 4.14(b)); $(1, -1, 1)'$, the special solution of the left part, spans $\mathcal{N}(A)$ (Theorem 4.8(c)). Check: $A(-1, 1, 0)' = -a_1 + a_2 = (1, 3, 2, 4)'$ and $A(1, -1, 1)' = a_1 - a_2 + a_3 = (1 - 2 + 1,\ 2 - 5 + 3,\ 1 - 3 + 2,\ 3 - 7 + 4)' = 0$.

Exercise 4.12. Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & -1 \\ 1 & 1 & 1 & -1 \\ 0 & 1 & -1 & 1 \end{pmatrix}$$

(a) Find $\operatorname{rank} A$ and $\operatorname{rank} B$.

(b) Compute AB, BA and their ranks, and compare them with the bounds of Theorems 4.15(a) and 4.19.

(c) Compute $A + B$ and its rank, and compare it with the bound of Corollary 4.17.

Solution. (a) Row-reduce A and B. For A, Step 2 interchanges rows 3 and 4 after the second arrow; for B, Step 2 interchanges rows 1 and 3 at the first arrow and rows 3 and 4 at the fourth:

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & -1 \\ 1 & 1 & 1 & -1 \\ 0 & 1 & -1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow -R_3} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $\operatorname{rank} A = \operatorname{rank} B = 3$.

(b) By Theorem 1.4(c), $\operatorname{row}_i(AB) = \operatorname{row}_i(A)B$ and $\operatorname{row}_i(BA) = \operatorname{row}_i(B)A$: $\operatorname{row}_1(AB) = \operatorname{row}_1(B) + \operatorname{row}_2(B) = (0,0,0,0)$, $\operatorname{row}_2(AB) = \operatorname{row}_2(B) + \operatorname{row}_3(B) + \operatorname{row}_4(B) = (1,1,0,-1)$, $\operatorname{row}_3(AB) = \operatorname{row}_1(B) + 2\operatorname{row}_2(B) + \operatorname{row}_3(B) + \operatorname{row}_4(B) = (1,1,0,-1)$, $\operatorname{row}_4(AB) = \operatorname{row}_1(B) + \operatorname{row}_2(B) + \operatorname{row}_3(B) + \operatorname{row}_4(B) = (1,2,0,0)$; and $\operatorname{row}_1(BA) = \operatorname{row}_2(A) + \operatorname{row}_4(A)$, $\operatorname{row}_2(BA) = -\operatorname{row}_2(A) - \operatorname{row}_4(A)$, $\operatorname{row}_3(BA) = \operatorname{row}_1(A) + \operatorname{row}_2(A) + \operatorname{row}_3(A) - \operatorname{row}_4(A)$, $\operatorname{row}_4(BA) = \operatorname{row}_2(A) - \operatorname{row}_3(A) + \operatorname{row}_4(A)$:

$$AB = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 \\ 1 & 1 & 0 & -1 \\ 1 & 2 & 0 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 2 & 2 & 2 \\ -1 & -2 & -2 & -2 \\ 1 & 3 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

The rows of AB span $\operatorname{span}\{(1,1,0,-1),(1,2,0,0)\}$, and these two rows are independent (neither is a multiple of the other, Proposition 3.13(c)); so $\dim \mathcal{R}(AB) = 2 = \operatorname{rank}(AB)$ (Definition 4.11). For

BA:

$$\begin{pmatrix} 1 & 2 & 2 & 2 \\ -1 & -2 & -2 & -2 \\ 1 & 3 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 + R_3]{R_1 \leftarrow R_1 - 2R_3} \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $\operatorname{rank}(BA) = 3$.

- Theorem 4.15(a): $\operatorname{rank}(AB) = 2 < \min\{3, 3\}$ and $\operatorname{rank}(BA) = 3 = \min\{3, 3\}$.
- Theorem 4.19: both ranks are at least $3 + 3 - 4 = 2$; AB attains the bound. The products AB and BA have different ranks.

(c) $A + B$ is the first matrix below; row-reduce it (Step 2 interchanges rows 2 and 3):

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 2 & 3 & 2 & 0 \\ 1 & 2 & 0 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 2 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & 2 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 - 2R_4]{R_1 \leftarrow R_1 - R_4} \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + 2R_3} \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

So $\operatorname{rank}(A + B) = 4 < 3 + 3$ (Corollary 4.17).

Exercise 4.13. Let $N \in \mathbb{R}^{4\times 4}$ be the matrix of Exercise 1.2: its entries are $n_{12} = n_{23} = n_{34} = 1$ and 0 elsewhere, so that $Ne_1 = 0$ and $Ne_j = e_{j-1}$ $(j = 2, 3, 4)$.

(a) Compute N^2, N^3, N^4, and $\operatorname{rank} N^k$ for $k = 0, \ldots, 4$.

(b) Show that Sylvester's inequality for the product $N^k \cdot N$ is an equality for $k = 1, 2, 3$.

(c) Compute $N'N$, NN' and $\operatorname{rank}(N + N')$.

Solution. (a) $\operatorname{col}_j(N^k) = N^k e_j$ (Theorem 1.4(b)), and applying N repeatedly gives $N^k e_j = e_{j-k}$ for $j > k$ and $N^k e_j = 0$ for $j \leqslant k$. So

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad N^4 = 0$$

The nonzero columns of N^k are $e_1, \ldots, e_{4-k}$. They are independent, as part of the standard basis (Definition 3.17 and Proposition 3.13(b)), and they span $\mathcal{C}(N^k)$, since the zero columns add nothing to the span (Theorem 3.10(c)); so $\operatorname{rank} N^k = 4 - k$ $(k = 0, \ldots, 4)$.

(b) Theorem 4.19 gives $\operatorname{rank}(N^k \cdot N) \geqslant \operatorname{rank} N^k + \operatorname{rank} N - 4 = (4 - k) + 3 - 4 = 3 - k$, and $N^k \cdot N = N^{k+1}$ has rank $4 - (k + 1) = 3 - k$ by (a).

(c) N' has entries 1 in positions $(2,1)$, $(3,2)$, $(4,3)$ and 0 elsewhere, so $N'e_j = e_{j+1}$ $(j = 1,2,3)$ and $N'e_4 = 0$. Hence $N'Ne_1 = 0$, $N'Ne_j = N'e_{j-1} = e_j$ $(j \geqslant 2)$, and $NN'e_j = Ne_{j+1} = e_j$ $(j \leqslant 3)$, $NN'e_4 = 0$:

$$N'N = \operatorname{diag}(0,1,1,1), \qquad NN' = \operatorname{diag}(1,1,1,0)$$

both of rank $3 = \operatorname{rank} N$ (Theorem 4.18(b)). For $N + N'$:

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

an echelon form with four nonzero rows; so $\operatorname{rank}(N + N') = 4$, below the bound $\operatorname{rank} N + \operatorname{rank} N' = 6$ of Corollary 4.17.

Exercise 4.14. Four variables are measured on four units: $x_1 = (1,1,2,0)'$, $x_2 = 2x_1$ (the first variable in other units), $x_3 = (1,2,3,1)'$ and $x_4 = x_1 - x_3$. Let $A = [x_1\ x_2\ x_3\ x_4] \in \mathbb{R}^{4\times 4}$.

(a) Row-reduce A and find $\operatorname{rank} A$.

(b) Find the full-rank factorization $A = BC$ of Theorem 4.21(a), and check it.

(c) Write A as a sum of $\operatorname{rank} A$ matrices of rank 1.

Solution. (a)

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 2 & 2 & -1 \\ 2 & 4 & 3 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $\operatorname{rank} A = 2$, with pivot columns 1 and 3.

(b) B consists of the pivot columns of A, and C of the nonzero rows of the RREF:

$$B = [x_1\ x_3] = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 3 \\ 0 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

Check by columns (Theorem 1.4(a),(b)): $B\operatorname{col}_1(C) = x_1$, $B\operatorname{col}_2(C) = 2x_1 = x_2$, $B\operatorname{col}_3(C) = x_3$, $B\operatorname{col}_4(C) = x_1 - x_3 = x_4$. So $BC = A$, with $\operatorname{rank} B = \operatorname{rank} C = 2$.

(c) By the outer product expansion (Theorem 1.4(d)), $A = x_1 \operatorname{row}_1(C) + x_3 \operatorname{row}_2(C)$:

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 1 & 2 & 0 & 1 \\ 2 & 4 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 1 & -1 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

Each term is nonzero and its rows are multiples of one row, so each has rank 1. Check: the sum has rows $(1,2,1,0)$, $(1,2,2,-1)$, $(2,4,3,-1)$, $(0,0,1,-1)$, the rows of A.

Exercise 4.15. Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 \\ -1 & 0 & -1 & -1 \\ 1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

(a) Find the RREF R and rank A, and the full-rank factorization $A = BC$ of Theorem 4.21(a).

(b) Compute $B'B$, CC' and the matrices L, K of Theorem 4.21(c), and verify $LB = I_2$ and $CK = I_2$.

(c) Find a basis of $\mathcal{N}(A)$.

Solution. (a) Row-reduce A:

$$A \xrightarrow[R_5 \leftarrow R_5 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & -2 & 2 & -2 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 1 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2,\ R_5 \leftarrow R_5 + R_2]{R_1 \leftarrow R_1 - R_2,\ R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

So $\operatorname{rank} A = 2$, with pivot columns $1, 2$, and

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \end{pmatrix}$$

Columns 3, 4 of R say $a_3 = a_1 - a_2$ and $a_4 = a_1 + a_2$; check: $a_1 - a_2 = (0, -1, 2, -1, 1)'$ and $a_1 + a_2 = (2, -1, 0, 1, 1)'$.

(b) With $a_1 = (1, -1, 1, 0, 1)'$ and $a_2 = (1, 0, -1, 1, 0)'$, the entries of $B'B$ are the products $a_j' a_l$ (Theorem 1.4(d)):

$$B'B = \begin{pmatrix} a_1'a_1 & a_1'a_2 \\ a_2'a_1 & a_2'a_2 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}, \qquad CC' = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

($a_1'a_2 = 1+0-1+0+0 = 0$ and $\operatorname{row}_1(C)\operatorname{row}_2(C)' = 0+0-1+1 = 0$). Hence, as $(B'B)^{-1} = \operatorname{diag}(\frac{1}{4}, \frac{1}{3})$ and $(CC')^{-1} = \frac{1}{3} I_2$ (Definition 2.12),

$$L = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} B' = \begin{pmatrix} \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{3} & 0 & -\frac{1}{3} & \frac{1}{3} & 0 \end{pmatrix}, \qquad K = \frac{1}{3} C' = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \\ 1 & 1 \end{pmatrix}$$

Then $LB = \operatorname{diag}(\frac{1}{4}, \frac{1}{3}) B'B = \operatorname{diag}(\frac{1}{4}, \frac{1}{3}) \operatorname{diag}(4, 3) = I_2$ and $CK = \frac{1}{3} CC' = I_2$.

(c) $\mathcal{N}(A) = \mathcal{N}(C)$ (Theorem 4.21(b)); $Cx = 0$ reads $x_1 = -x_3 - x_4$, $x_2 = x_3 - x_4$, and the special solutions $(-1, 1, 1, 0)'$ and $(-1, -1, 0, 1)'$ form a basis (Theorem 4.8(c)). They are the relations $a_3 = a_1 - a_2$ and $a_4 = a_1 + a_2$.

Exercise 4.16. For $A = BC$ of Exercise 4.15:

(a) compute $A'A$ from $A'A = C'(B'B)C$;

(b) row-reduce $A'A$, and check that $\operatorname{rank}(A'A) = \operatorname{rank} A$ and $\mathcal{N}(A'A) = \mathcal{N}(A)$;

(c) find $\operatorname{rank}(AA')$ without computing the 5×5 matrix AA'.

Solution. (a) $A'A = C'B'BC$ (Theorem 1.9), with $B'B = \operatorname{diag}(4, 3)$ (Exercise 4.15(b)). The rows of C are γ_1' and γ_2', with $\gamma_1 = (1, 0, 1, 1)'$ and $\gamma_2 = (0, 1, -1, 1)'$ in $\mathbb{R}^4$. The columns of $C' \operatorname{diag}(4, 3)$ are $4\gamma_1$ and $3\gamma_2$ (Proposition 1.13(a)), so the outer product expansion (Theorem 1.4(d)) gives

$$A'A = 4\gamma_1\gamma_1' + 3\gamma_2\gamma_2' = \begin{pmatrix} 4 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 \\ 4 & 0 & 4 & 4 \\ 4 & 0 & 4 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 3 & -3 & 3 \\ 0 & -3 & 3 & -3 \\ 0 & 3 & -3 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 4 & 4 \\ 0 & 3 & -3 & 3 \\ 4 & -3 & 7 & 1 \\ 4 & 3 & 1 & 7 \end{pmatrix}$$

(b) Row-reduce $A'A$:

$$\begin{pmatrix} 4 & 0 & 4 & 4 \\ 0 & 3 & -3 & 3 \\ 4 & -3 & 7 & 1 \\ 4 & 3 & 1 & 7 \end{pmatrix} \xrightarrow[R_2 \leftarrow \frac{1}{3}R_2]{R_1 \leftarrow \frac{1}{4}R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 4 & -3 & 7 & 1 \\ 4 & 3 & 1 & 7 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 4R_1]{R_3 \leftarrow R_3 - 4R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & -3 & 3 & -3 \\ 0 & 3 & -3 & 3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 + 3R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The RREF of $A'A$ is the matrix R of Exercise 4.15 without its last row. So $\operatorname{rank}(A'A) = 2 = \operatorname{rank} A$, and $A'Ax = 0$ and $Ax = 0$ both reduce to $Cx = 0$: $\mathcal{N}(A'A) = \mathcal{N}(A)$, as Theorem 4.18(a),(b) asserts.

(c) $\operatorname{rank}(AA') = \operatorname{rank} A = 2$ (Theorem 4.18(b)).

Exercise 4.17. In $\mathbb{R}^4$ let $U = \operatorname{span}\{u_1, u_2, u_3\}$ and $W = \operatorname{span}\{w_1, w_2\}$, where

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad u_3 = \begin{pmatrix} 1 \\ 3 \\ -1 \\ 3 \end{pmatrix}, \qquad w_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}$$

(a) Find $\dim U$, $\dim W$ and $\dim(U + W)$ by one row reduction.

(b) Find $\dim(U \cap W)$ by Theorem 4.23, and a basis of $U \cap W$.

Solution. (a) $U + W = \operatorname{span}\{u_1, u_2, u_3, w_1, w_2\}$ (Theorem 3.31(c)). Row-reduce $M = [u_1\ u_2\ u_3\ w_1\ w_2]$:

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 3 & 0 & 2 \\ -1 & 0 & -1 & 1 & 1 \\ 1 & 1 & 3 & -1 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & -1 & -1 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_2} \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 & -2 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The last matrix is the RREF of M, with pivot columns $1, 2, 4$.

- $\dim U = 2$: row operations act on each column separately (Theorems 2.15(c) and 1.4(a)), so the first three columns of the last matrix form the RREF of $[u_1\ u_2\ u_3]$, with pivot columns $1, 2$. So u_1, u_2 form a basis of U, and column 3 gives $u_3 = u_1 + 2u_2$ (Theorem 4.3(b)).
- $\dim W = 2$: w_1 and w_2 are not multiples of each other (the first entry of w_1 is 0, that of w_2 is not, and $w_1 \neq 0$), so they are independent (Proposition 3.13(c)).
- $\dim(U + W) = 3$, with the basis u_1, u_2, w_1 (Theorem 4.3(b)).

(b)

- $\dim(U \cap W) = \dim U + \dim W - \dim(U + W) = 2 + 2 - 3 = 1$ (Theorem 4.23).
- Column 5 gives $w_2 = u_1 + u_2 + 2w_1$, so $v = w_2 - 2w_1 = u_1 + u_2 = (1, 2, -1, 2)'$ lies in W and in U. This nonzero vector is a basis of the one-dimensional space $U \cap W$ (Theorem 4.5(d)). Check: $w_2 - 2w_1 = (1,\ 2,\ 1-2,\ 0+2)' = (1, 2, -1, 2)'$ and $u_1 + u_2 = (1, 2, -1, 2)'$.

Exercise 4.18. Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 3 & -1 \\ 1 & 2 & 2 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Find invertible matrices P and Q with $PAQ = N_r$, where $r = \operatorname{rank} A$ (Theorem 4.31), by Remark 4.32, and check the product.

Solution. Row-reduce $[A \mid I_4]$; after the first arrow column 2 has no nonzero entry in rows 2–4, so the next pivot column is 3:

$$\left(\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 2 & 4 & 3 & -1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 & 1 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 + R_3} \left(\begin{array}{cccc|cccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 & 1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|cccc} 1 & 2 & 0 & 0 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 & 1 & 0 & 1 \end{array}\right)$$

So $\operatorname{rank} A = 3$, with pivot columns $1, 3, 4$, the free column 2, and

$$R = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad P = \begin{pmatrix} 2 & 0 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}$$

$Rx = 0$ reads $x_1 + 2x_2 = 0$, $x_3 = 0$, $x_4 = 0$, and the special solution is $s = (-2, 1, 0, 0)'$. By Remark 4.32, P and

$$Q = [e_1\ e_3\ e_4\ s] = \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

are invertible and $PAQ = N_3$. Check: the columns of AQ are a_1, a_3, a_4 and $As = -2a_1 + a_2 = 0$, so

$$PAQ = \begin{pmatrix} 2 & 0 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 3 & -1 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = N_3$$

Problems

Problem 4.1 ([8, Ch. 18, Exercise 26])**.** Show that, for all $A, B \in \mathbb{R}^{m\times n}$, $\operatorname{rank}(A+B) \geqslant |\operatorname{rank} A - \operatorname{rank} B|$.

Solution.

- $\operatorname{rank}(-B) = \operatorname{rank} B$: $-B = (-I_m)BI_n$ (Theorem 1.3), and $-I_m$ is invertible, as $(-I_m)(-I_m) = I_m$ (Definition 2.12); so Theorem 4.15(d) applies. Likewise $\operatorname{rank}(-A) = \operatorname{rank} A$.
- By Corollary 4.17, applied to $A+B$ and $-B$,

  $$\operatorname{rank} A = \operatorname{rank}\Big((A+B)+(-B)\Big) \leqslant \operatorname{rank}(A+B) + \operatorname{rank}(-B) = \operatorname{rank}(A+B) + \operatorname{rank} B$$

  so $\operatorname{rank}(A+B) \geqslant \operatorname{rank} A - \operatorname{rank} B$.
- Similarly, $\operatorname{rank} B = \operatorname{rank}\Big((A+B)+(-A)\Big) \leqslant \operatorname{rank}(A+B) + \operatorname{rank} A$, so $\operatorname{rank}(A+B) \geqslant \operatorname{rank} B - \operatorname{rank} A = -(\operatorname{rank} A - \operatorname{rank} B)$.

Together, $\operatorname{rank}(A+B) \geqslant |\operatorname{rank} A - \operatorname{rank} B|$.

Problem 4.2 ([6, Exercise 4.18])**.** Show that $\operatorname{rank}(A'AB) = \operatorname{rank}(AB) = \operatorname{rank}(ABB')$ for all $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$.

Solution. Apply Theorem 4.18(b) to AB, with $(AB)' = B'A'$ (Theorem 1.9), and Theorem 4.15(a) twice:

$$\operatorname{rank}(AB) = \operatorname{rank}\Big((AB)'AB\Big) = \operatorname{rank}\Big(B'(A'AB)\Big) \leqslant \operatorname{rank}(A'AB) = \operatorname{rank}\Big(A'(AB)\Big) \leqslant \operatorname{rank}(AB)$$
$$\operatorname{rank}(AB) = \operatorname{rank}\Big(AB(AB)'\Big) = \operatorname{rank}\Big((ABB')A'\Big) \leqslant \operatorname{rank}(ABB') = \operatorname{rank}\Big((AB)B'\Big) \leqslant \operatorname{rank}(AB)$$

So equality holds throughout.

Problem 4.3 ([3, Ch. 11, Exercise 37])**.** Let $A \in \mathbb{R}^{n\times n}$ satisfy $A^2 = I$. Show that $\operatorname{rank}(I-A) + \operatorname{rank}(I+A) = n$.

Solution.

- $2I = (I-A) + (I+A)$, and $2I = \operatorname{diag}(2, \ldots, 2)$ is invertible (Definition 2.12). By Theorem 4.13(d) and Corollary 4.17, $n = \operatorname{rank}(2I) \leqslant \operatorname{rank}(I-A) + \operatorname{rank}(I+A)$.
- $(I-A)(I+A) = I + A - A - A^2 = I - A^2 = 0$ (Theorem 1.3), so $\operatorname{rank}(I-A) + \operatorname{rank}(I+A) \leqslant n$ by Theorem 4.19.

Together, $\operatorname{rank}(I-A) + \operatorname{rank}(I+A) = n$.

Problem 4.4 ([8, Ch. 4, Exercise 10])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank r. Show that A is the sum of r matrices of rank 1.

Solution. If $r = 0$, then $A = 0$, the empty sum. Let $r \geqslant 1$, and let $A = BC$ be the full-rank factorization of Theorem 4.21(a), where $B \in \mathbb{R}^{m\times r}$ has columns $b_1, \dots, b_r$ and $C \in \mathbb{R}^{r\times n}$. By the outer product expansion (Theorem 1.4(d)),

$$A = \sum_{j=1}^{r} A_j, \qquad A_j = b_j \operatorname{row}_j(C) \in \mathbb{R}^{m\times n}$$

- The columns of B are independent, and so are the rows of C (Theorem 4.13(b),(c)); hence every b_j and every $\operatorname{row}_j(C)$ is nonzero, since a list containing 0 is dependent (Proposition 3.13(a)).
- If the lth entry c_{jl} of $\operatorname{row}_j(C)$ is nonzero, then $\operatorname{col}_l(A_j) = c_{jl}b_j \neq 0$; so $A_j \neq 0$, and $\operatorname{rank} A_j \geqslant 1$.
- $\operatorname{rank} A_j \leqslant \operatorname{rank} b_j \leqslant 1$ (Theorems 4.15(a) and 4.13(a)), as b_j has one column.

So each A_j has rank 1.

Problem 4.5 ([3, Ch. 2, Exercise 17])**.** Let $A, X, B \in \mathbb{R}^{n\times n}$. Show the following generalization of Sylvester's inequality (Theorem 4.19) for square matrices:

$$\operatorname{rank}(AXB) \geqslant \operatorname{rank}(AX) + \operatorname{rank}(XB) - \operatorname{rank} X$$

Solution. If $X = 0$, then AX, XB and AXB are zero, and both sides are 0. Let $r = \operatorname{rank} X \geqslant 1$.

- Let $X = UV$ be a full-rank factorization (Theorem 4.21(a)), with $U \in \mathbb{R}^{n\times r}$ of full column rank and $V \in \mathbb{R}^{r\times n}$ of full row rank. Then $AXB = (AU)(VB)$, $AX = (AU)V$ and $XB = U(VB)$ (Theorem 1.3).
- As V has full row rank, $\operatorname{rank}(AX) = \operatorname{rank}\big((AU)V\big) = \operatorname{rank}(AU)$ (Theorem 4.15(c)); as U has full column rank, $\operatorname{rank}(XB) = \operatorname{rank}\big(U(VB)\big) = \operatorname{rank}(VB)$ (Theorem 4.15(b)).
- Sylvester's inequality (Theorem 4.19) for $AU \in \mathbb{R}^{n\times r}$ and $VB \in \mathbb{R}^{r\times n}$, whose product has the inner size r, gives

  $$\operatorname{rank}(AXB) = \operatorname{rank}\big((AU)(VB)\big) \geqslant \operatorname{rank}(AU) + \operatorname{rank}(VB) - r = \operatorname{rank}(AX) + \operatorname{rank}(XB) - \operatorname{rank} X$$

Problem 4.6 ([3, Ch. 2, Exercise 2])**.** Let $x_1, \dots, x_p$ and $y_1, \dots, y_q$ be linearly independent lists of vectors of $\mathbb{R}^n$, where $p < q \leqslant n$. Show that there is an $i \in \{1, \dots, q\}$ such that $x_1, \dots, x_p, y_i$ are linearly independent.

Solution. Let $A = [x_1 \ \cdots \ x_p] \in \mathbb{R}^{n\times p}$ and $B = [y_1 \ \cdots \ y_q] \in \mathbb{R}^{n\times q}$, and suppose that $x_1, \dots, x_p, y_i$ are dependent for every i.

- Then each y_i lies in $\operatorname{span}\{x_1, \dots, x_p\} = \mathcal{C}(A)$: otherwise $x_1, \dots, x_p, y_i$ would be independent, by Proposition 3.14(b), as $x_1, \dots, x_p$ are independent.
- By Proposition 4.26(a), $B = AC$ for some $C \in \mathbb{R}^{p\times q}$.
- The columns of A and of B are independent, so $\operatorname{rank} A = p$ and $\operatorname{rank} B = q$ (Theorem 4.13(b)). By Theorem 4.15(a),

  $$q = \operatorname{rank} B = \operatorname{rank}(AC) \leqslant \operatorname{rank} A = p < q \ \text{✘}$$

Problem 4.7 ([8, Ch. 7, Exercise 1])**.** (a) Let $A \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{n\times q}$ and $B \in \mathbb{R}^{q\times p}$. Show that if $\operatorname{rank}(AC) = \operatorname{rank} C$, then $\mathcal{R}(ACB) = \mathcal{R}(CB)$ and $\operatorname{rank}(ACB) = \operatorname{rank}(CB)$; and that if $\operatorname{rank}(CB) = \operatorname{rank} C$, then $\mathcal{C}(ACB) = \mathcal{C}(AC)$ and $\operatorname{rank}(ACB) = \operatorname{rank}(AC)$.

(b) Let $A, B \in \mathbb{R}^{m\times n}$. Show that (1) if $C \in \mathbb{R}^{s\times q}$ and $D \in \mathbb{R}^{q\times m}$ satisfy $\operatorname{rank}(CD) = \operatorname{rank} D$, then $CDA = CDB$ implies $DA = DB$; and (2) if $C \in \mathbb{R}^{n\times q}$ and $D \in \mathbb{R}^{q\times p}$ satisfy $\operatorname{rank}(CD) = \operatorname{rank} C$, then $ACD = BCD$ implies $AC = BC$.

Solution. (a)

- $\mathcal{R}(ACB) = \mathcal{R}(A(CB)) \subseteq \mathcal{R}(CB)$ and $\mathcal{C}(ACB) = \mathcal{C}((AC)B) \subseteq \mathcal{C}(AC)$ (Proposition 4.26(b),(a)).
- Let $\operatorname{rank}(AC) = \operatorname{rank} C$. Then $\mathcal{R}(AC) \subseteq \mathcal{R}(C)$ (Proposition 4.26(b)), and both have dimension $\operatorname{rank} C$ (Definition 4.11), so $\mathcal{R}(AC) = \mathcal{R}(C)$ (Theorem 4.5(a), for row spaces). Hence $C = LAC$ for some L (Proposition 4.26(b)), and $\mathcal{R}(CB) = \mathcal{R}(L(ACB)) \subseteq \mathcal{R}(ACB)$. With the first item, $\mathcal{R}(ACB) = \mathcal{R}(CB)$, and the two ranks, the dimensions of these spaces, are equal.
- Let $\operatorname{rank}(CB) = \operatorname{rank} C$. Then $\mathcal{C}(CB) \subseteq \mathcal{C}(C)$ (Theorem 4.15(a)), both of dimension $\operatorname{rank} C$, so $\mathcal{C}(CB) = \mathcal{C}(C)$ (Theorem 4.5(a)). Hence $C = CBG$ for some G (Proposition 4.26(a)), and $\mathcal{C}(AC) = \mathcal{C}((ACB)G) \subseteq \mathcal{C}(ACB)$. With the first item, $\mathcal{C}(ACB) = \mathcal{C}(AC)$, and $\operatorname{rank}(ACB) = \operatorname{rank}(AC)$.

(b) Let $F = A - B \in \mathbb{R}^{m\times n}$.

- For (1): by (a), applied to C, D, F in place of A, C, B (so that the hypothesis is $\operatorname{rank}(CD) = \operatorname{rank} D$), $\operatorname{rank}(DF) = \operatorname{rank}(CDF) = \operatorname{rank}(CDA - CDB) = \operatorname{rank} 0 = 0$. So $DF = 0$ (Theorem 4.8), that is, $DA = DB$.
- For (2): by (a), applied to F, C, D in place of A, C, B (so that the hypothesis is $\operatorname{rank}(CD) = \operatorname{rank} C$), $\operatorname{rank}(FC) = \operatorname{rank}(FCD) = \operatorname{rank}(ACD - BCD) = 0$. So $FC = 0$, that is, $AC = BC$.

Problem 4.8 ([6, Exercise 6.19])**.** Let $A, B \in \mathbb{R}^{m\times n}$. Show that $B = PAQ$ for some invertible $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$ if and only if $\operatorname{rank} A = \operatorname{rank} B$.

Solution. If P and Q are invertible, then $\operatorname{rank}(PAQ) = \operatorname{rank} A$ (Theorem 4.15(d)). Conversely, let $\operatorname{rank} A = \operatorname{rank} B = r$. By Theorem 4.31 there are invertible $P_1, P_2 \in \mathbb{R}^{m\times m}$ and $Q_1, Q_2 \in \mathbb{R}^{n\times n}$ with $P_1AQ_1 = N_r = P_2BQ_2$. Hence

$$B = P_2^{-1}P_1AQ_1Q_2^{-1} = PAQ, \qquad P = P_2^{-1}P_1, \quad Q = Q_1Q_2^{-1}$$

and P, Q are invertible (Theorem 2.13(a),(b)).

Problem 4.9 ([10, Problem 8.1])**.** Let $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}$, and let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ have the entries $a_{ij} = x_i + y_j$. Show that $\operatorname{rank} A \leqslant 2$.

Solution. Let

$$X = \begin{pmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix} \in \mathbb{R}^{n\times 2}, \qquad Y = \begin{pmatrix} 1 & \cdots & 1 \\ y_1 & \cdots & y_n \end{pmatrix} \in \mathbb{R}^{2\times n}$$

By Theorem 1.4(d), $(XY)_{ij} = \operatorname{row}_i(X)\operatorname{col}_j(Y) = x_i \cdot 1 + 1 \cdot y_j = a_{ij}$, so $A = XY$. Hence, by Theorems 4.15(a) and 4.13(a), $\operatorname{rank} A \leqslant \operatorname{rank} X \leqslant \min\{n, 2\} \leqslant 2$.

Problem 4.10 ([10, Problem 8.7])**.** Let $A, B \in \mathbb{R}^{n\times n}$, where n is odd, satisfy $AB = 0$. Show that at least one of the matrices $A + A'$ and $B + B'$ is not invertible.

Solution. Write $n = 2m + 1$.

- By Sylvester's inequality (Theorem 4.19, the case $AB = 0$), $\operatorname{rank} A + \operatorname{rank} B \leqslant n = 2m + 1$. So $\operatorname{rank} A \leqslant m$ or $\operatorname{rank} B \leqslant m$: otherwise $\operatorname{rank} A + \operatorname{rank} B \geqslant 2(m + 1) > 2m + 1$.
- If $\operatorname{rank} A \leqslant m$, then, as $\operatorname{rank} A' = \operatorname{rank} A$ (Theorem 4.13(a)), Corollary 4.17 gives

$$\operatorname{rank}(A + A') \leqslant \operatorname{rank} A + \operatorname{rank} A' = 2 \operatorname{rank} A \leqslant 2m < n$$

  so $A + A'$ is not invertible (Theorem 4.13(d)).
- If $\operatorname{rank} B \leqslant m$, the same argument, with B in place of A, shows that $B + B'$ is not invertible.

Problem 4.11 (cf. [10, Problems 8.5, 8.6])**.** Let $A_1, A_2 \in \mathbb{R}^{m\times n}$.

(a) Show that $\operatorname{rank}(A_1 + A_2) = \operatorname{rank} A_1 + \operatorname{rank} A_2$ if and only if $\mathcal{R}(A_1) \cap \mathcal{R}(A_2) = \{0\}$ and $\mathcal{C}(A_1) \cap \mathcal{C}(A_2) = \{0\}$.

(b) Show that if $A_2'A_1 = 0$ and $A_1A_2' = 0$, then $\operatorname{rank}(A_1 + A_2) = \operatorname{rank} A_1 + \operatorname{rank} A_2$.

Solution. (a) Let $r_i = \operatorname{rank} A_i$ $(i = 1, 2)$.

- If $r_1 = 0$, then $A_1 = 0$ (Theorem 4.8), so $\operatorname{rank}(A_1 + A_2) = r_2 = r_1 + r_2$, and $\mathcal{R}(A_1) = \{0\}$ and $\mathcal{C}(A_1) = \{0\}$ meet $\mathcal{R}(A_2)$ and $\mathcal{C}(A_2)$ only in 0: both sides of the equivalence hold. The case $r_2 = 0$ is the same. Let $r_1, r_2 \geqslant 1$.
- Let $A_i = B_iC_i$ be full-rank factorizations (Theorem 4.21(a)), with $B_i \in \mathbb{R}^{m\times r_i}$ of full column rank and $C_i \in \mathbb{R}^{r_i\times n}$ of full row rank. Then $\mathcal{C}(B_i) = \mathcal{C}(A_i)$ (Theorem 4.21(b)). Also $\mathcal{R}(A_i) \subseteq \mathcal{R}(C_i)$, as $A_i = B_iC_i$ (Proposition 4.26(b)), and both spaces have dimension r_i (Definition 4.11); so $\mathcal{R}(C_i) = \mathcal{R}(A_i)$ (Theorem 4.5(a), for row spaces by the paragraph after it).
- Let $B = [B_1\ B_2] \in \mathbb{R}^{m\times(r_1+r_2)}$ and $C = \binom{C_1}{C_2} \in \mathbb{R}^{(r_1+r_2)\times n}$, as in Theorem 4.16(a),(b). The first r_1 columns of B and rows of C are those of B_1 and C_1, and the last r_2 are those of B_2 and C_2. So, by the outer product expansion (Theorem 1.4(d)), applied to BC, B_1C_1 and B_2C_2,

$$BC = \sum_{j=1}^{r_1} \operatorname{col}_j(B_1) \operatorname{row}_j(C_1) + \sum_{j=1}^{r_2} \operatorname{col}_j(B_2) \operatorname{row}_j(C_2) = B_1C_1 + B_2C_2 = A_1 + A_2$$

- By Theorem 4.27(a),(d) and the second item,

$$\operatorname{rank} B = r_1 + r_2 - \dim\Big(\mathcal{C}(A_1) \cap \mathcal{C}(A_2)\Big), \qquad \operatorname{rank} C = r_1 + r_2 - \dim\Big(\mathcal{R}(A_1) \cap \mathcal{R}(A_2)\Big)$$

- If both intersections are $\{0\}$, then $\operatorname{rank} B = r_1 + r_2$, so B has full column rank, and by Theorem 4.15(b), $\operatorname{rank}(A_1 + A_2) = \operatorname{rank}(BC) = \operatorname{rank} C = r_1 + r_2$.
- Conversely, let $\operatorname{rank}(A_1 + A_2) = r_1 + r_2$. By Theorem 4.15(a), $r_1 + r_2 = \operatorname{rank}(BC) \leqslant \min\{\operatorname{rank} B, \operatorname{rank} C\}$, so by the fourth item both intersections have dimension 0; a subspace of dimension 0 is $\{0\}$ (Theorem 4.4).

(b)

- $\mathcal{C}(A_1) \cap \mathcal{C}(A_2) = \{0\}$: if $z = A_1u = A_2w$, then $z'z = (A_2w)'(A_1u) = w'(A_2'A_1)u = 0$, so $z = 0$, as $z'z = \sum_i z_i^2$ (Theorem 1.9).
- $\mathcal{R}(A_1) \cap \mathcal{R}(A_2) = \{0\}$: if $h = y_1'A_1 = y_2'A_2$ with $y_1, y_2 \in \mathbb{R}^m$, then $hh' = (y_1'A_1)(y_2'A_2)' = y_1'(A_1A_2')y_2 = 0$, so $h = 0$, as hh' is the sum of the squares of the entries of h (Theorem 1.9, for the column h').

- By (a), $\operatorname{rank}(A_1 + A_2) = \operatorname{rank} A_1 + \operatorname{rank} A_2$.

Neither condition in (a) suffices alone, and the hypothesis $A_1 A_2' = 0$ cannot be omitted in (b). For $A_1 = E_{11} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $A_2 = E_{21} = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, $\mathcal{C}(A_1) \cap \mathcal{C}(A_2) = \operatorname{span}\{e_1\} \cap \operatorname{span}\{e_2\} = \{0\}$ and, as $A_2' = E_{12}$, $A_2' A_1 = E_{12} E_{11} = \delta_{21} E_{11} = 0$ (Proposition 1.18(b),(c)), but $\mathcal{R}(A_1) = \mathcal{R}(A_2) = \operatorname{span}\{e_1'\}$ and $\operatorname{rank}(A_1 + A_2) = \operatorname{rank}\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right) = 1 < 2$. The transposes A_1', A_2' have $\mathcal{R}(A_1') \cap \mathcal{R}(A_2') = \{0\}$, and again $\operatorname{rank}(A_1' + A_2') = 1 < 2$.

Chapter 5. Block matrices and determinants

Block matrices

Definition 5.1 ([5, §2.1], [16, §0.7.2])**.** Let $A \in \mathbb{R}^{m\times n}$. A **partition** of A splits the rows into consecutive groups of $m_1, \dots, m_r$ rows ($m_1 + \cdots + m_r = m$) and the columns into consecutive groups of $n_1, \dots, n_s$ columns ($n_1 + \cdots + n_s = n$). Put $M_0 = 0$, $M_i = m_1 + \cdots + m_i$ and $N_0 = 0$, $N_j = n_1 + \cdots + n_j$: row group i consists of the rows $M_{i-1}+1, \dots, M_i$, and column group j of the columns $N_{j-1}+1, \dots, N_j$. The **block** $A_{ij} \in \mathbb{R}^{m_i\times n_j}$ consists of the entries in row group i and column group j:

$$(A_{ij})_{xy} = a_{M_{i-1}+x,\, N_{j-1}+y} \qquad (1 \leqslant x \leqslant m_i,\ 1 \leqslant y \leqslant n_j)$$

We write

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1s} \\ \vdots & & \vdots \\ A_{r1} & \cdots & A_{rs} \end{pmatrix}$$

Lines between rows and between columns may mark the partition. The group sizes are arbitrary positive integers, so a block A_{ij} is square only when $m_i = n_j$. Two matrices of the same size, partitioned in the same way, are equal if and only if their corresponding blocks are equal (both statements say that all entries agree, Definition 1.1).

Definition 5.2 ([5, §2.2])**.** Let $A \in \mathbb{R}^{m\times n}$ be partitioned with row groups $m_1, \dots, m_r$ and column groups $n_1, \dots, n_s$.

(a) A partition of $B \in \mathbb{R}^{m\times n}$ is **conformal for addition** with that of A if it has the same row groups $m_1, \dots, m_r$ and the same column groups $n_1, \dots, n_s$.

(b) A partition of $B \in \mathbb{R}^{n\times p}$ is **conformal for multiplication** with that of A (in the product AB) if its row groups are $n_1, \dots, n_s$, the column groups of A. Its column groups $p_1, \dots, p_t$ ($p_1 + \cdots + p_t = p$) may be any.

Conformality for multiplication restricts only the column groups of A and the row groups of B; the row groups m_i of A and the column groups p_k of B are free. It means exactly that every block $A_{ij} \in \mathbb{R}^{m_i\times n_j}$ has as many columns as every block $B_{jk} \in \mathbb{R}^{n_j\times p_k}$ has rows, so that the product $A_{ij}B_{jk} \in \mathbb{R}^{m_i\times p_k}$ is defined (Definition 1.2).

Theorem 5.3 ([5, (2.2.1)–(2.2.6)])**.** Let $A \in \mathbb{R}^{m\times n}$ be partitioned into blocks $A_{ij} \in \mathbb{R}^{m_i\times n_j}$ ($1 \leqslant i \leqslant r$, $1 \leqslant j \leqslant s$).

(a) If $B \in \mathbb{R}^{m\times n}$ is partitioned conformally for addition, into blocks $B_{ij} \in \mathbb{R}^{m_i\times n_j}$, then $A + B$ has the blocks $A_{ij} + B_{ij}$, and for $c \in \mathbb{R}$ the matrix cA has the blocks cA_{ij} (same partition).

(b) Partition $A' \in \mathbb{R}^{n\times m}$ with row groups $n_1, \dots, n_s$ and column groups $m_1, \dots, m_r$. Its (j, i) block is $A'_{ij} \in \mathbb{R}^{n_j\times m_i}$: transpose the array of blocks and also each block.

(c) If $B \in \mathbb{R}^{n\times p}$ is partitioned conformally for multiplication, into blocks $B_{jk} \in \mathbb{R}^{n_j\times p_k}$ ($1 \leqslant j \leqslant s$, $1 \leqslant k \leqslant t$), then $AB \in \mathbb{R}^{m\times p}$, partitioned with row groups $m_1, \dots, m_r$ and column groups $p_1, \dots, p_t$, has the blocks

$$(AB)_{ik} = \sum_{j=1}^{s} A_{ij}B_{jk} = A_{i1}B_{1k} + A_{i2}B_{2k} + \cdots + A_{is}B_{sk} \in \mathbb{R}^{m_i\times p_k}$$

where the term $A_{ij}B_{jk}$ is the product of an $m_i \times n_j$ and an $n_j \times p_k$ matrix.

So conformal block matrices are added and multiplied as if the blocks were numbers, but the order of the factors in each $A_{ij}B_{jk}$ must be kept. No group size has to equal another, and no block has to be square. For $r = s = t = 2$,

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$$

whenever A_{11} and A_{21} have as many columns as B_{11} and B_{12} have rows (the second groups then match as well, since $n_1 + n_2 = n$ is the number of columns of A and of rows of B). The cases $r = t = 1,\ s = 2$; $r = t = 2,\ s = 1$; and $r = s = 1,\ t = 2$ read

$$[A_1\ A_2] \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = A_1B_1 + A_2B_2, \qquad \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} [B_1\ B_2] = \begin{pmatrix} A_1B_1 & A_1B_2 \\ A_2B_1 & A_2B_2 \end{pmatrix}, \qquad A[B_1\ B_2] = [AB_1\ \ AB_2]$$

Proof. Let M_i and N_j be as in Definition 5.1. Each row index u of A lies in exactly one row group i, and then $u = M_{i-1} + x$ with $1 \leqslant x \leqslant m_i$; likewise for columns.

(a) Fix i, j and $1 \leqslant x \leqslant m_i$, $1 \leqslant y \leqslant n_j$, and put $u = M_{i-1} + x$, $v = N_{j-1} + y$. Since B has the same groups as A, entry (x, y) of B_{ij} is b_{uv}. By Definition 1.1, entry (x, y) of block (i, j) of $A + B$ is

$$(A + B)_{uv} = a_{uv} + b_{uv} = (A_{ij})_{xy} + (B_{ij})_{xy} = (A_{ij} + B_{ij})_{xy}$$

and in the same way $(cA)_{uv} = ca_{uv} = (cA_{ij})_{xy}$.

(b) Row group j of A' consists of the rows $N_{j-1} + 1, \ldots, N_j$ and column group i of the columns $M_{i-1} + 1, \ldots, M_i$. By Definition 1.8, entry (y, x) of the (j, i) block of A' is

$$(A')_{N_{j-1}+y,\, M_{i-1}+x} = a_{M_{i-1}+x,\, N_{j-1}+y} = (A_{ij})_{xy} = (A'_{ij})_{yx}$$

(c) For a row group i of A and a column group k of B, let $A_{i*} \in \mathbb{R}^{m_i \times n}$ consist of the rows of A in row group i, and $B_{*k} \in \mathbb{R}^{n \times p_k}$ of the columns of B in column group k. By Definition 5.1, the columns of A_{i*} in column group j are, in order, the columns of A_{ij}; since the row groups of B are $n_1, \ldots, n_s$ (conformality), the rows of B_{*k} in row group j are, in order, the rows of B_{jk}. In blocks,

$$A_{i*} = [A_{i1}\ \cdots\ A_{is}], \qquad B_{*k} = \begin{pmatrix} B_{1k} \\ \vdots \\ B_{sk} \end{pmatrix}$$

- Columns. By Theorem 1.4(a), each column of AB is A times the column of B with the same index. The columns of B in column group k are, in order, the columns of B_{*k}, so the columns of AB in column group k are, in order, A times the columns of B_{*k}, that is, the columns of the product AB_{*k} (again Theorem 1.4(a)).
- Rows. By Theorem 1.4(c), each row of AB_{*k} is the row of A with the same index times B_{*k}. The rows of A in row group i are, in order, the rows of A_{i*}, so the rows of AB_{*k} in row group i are, in order, the rows of $A_{i*}B_{*k}$ (again Theorem 1.4(c)). These are the entries of AB in row group i and column group k, that is, the block $(AB)_{ik}$ (Definition 5.1): $(AB)_{ik} = A_{i*}B_{*k} \in \mathbb{R}^{m_i \times p_k}$.
- Inner sum. By the outer product expansion (Theorem 1.4(d)),

$$A_{i*}B_{*k} = \sum_{l=1}^{n} \operatorname{col}_l(A_{i*}) \operatorname{row}_l(B_{*k})$$

  The column groups of A cut $l = 1, \ldots, n$ into s consecutive stretches of $n_1, \ldots, n_s$ indices. In stretch j, the columns $\operatorname{col}_l(A_{i*})$ are $\operatorname{col}_1(A_{ij}), \ldots, \operatorname{col}_{n_j}(A_{ij})$ and the rows $\operatorname{row}_l(B_{*k})$ are $\operatorname{row}_1(B_{jk}), \ldots, \operatorname{row}_{n_j}(B_{jk})$, in this order (see above). So the terms of stretch j add up to

$$\sum_{y=1}^{n_j} \operatorname{col}_y(A_{ij}) \operatorname{row}_y(B_{jk}) = A_{ij}B_{jk}$$

the outer product expansion of the product of $A_{ij} \in \mathbb{R}^{m_i \times n_j}$ and $B_{jk} \in \mathbb{R}^{n_j \times p_k}$ (Theorem 1.4(d)). Adding the s stretches,

$$(AB)_{ik} = A_{i*}B_{*k} = \sum_{j=1}^{s} A_{ij}B_{jk} \qquad \square$$

In entries: let $u = M_{i-1} + x$ (row x of row group i) and let column v be the zth column of column group k of B. By Definition 1.2, $(AB)_{uv} = \sum_{l=1}^{n} a_{ul}b_{lv}$. Split the sum into the stretches of the column groups of A and write $l = N_{j-1} + y$ in stretch j; by Definition 5.1 (for A, and for B with row groups $n_1, \ldots, n_s$),

$$\sum_{l=N_{j-1}+1}^{N_j} a_{ul}b_{lv} = \sum_{y=1}^{n_j} (A_{ij})_{xy}(B_{jk})_{yz} = (A_{ij}B_{jk})_{xz}$$

so the part of stretch j is the entry (x, z) of $A_{ij}B_{jk}$: this is (c), entry by entry.

Example 5.4. Partition $A \in \mathbb{R}^{5\times 6}$ with row groups $(2, 3)$ and column groups $(4, 2)$, and $B \in \mathbb{R}^{6\times 5}$ with row groups $(4, 2)$ and column groups $(3, 2)$:

$$A = \left(\begin{array}{cccc|cc} 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 & 1 \\ \hline 1 & 1 & 1 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array}\right) = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \qquad B = \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ \hline 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 \end{array}\right) = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

The blocks are

$$A_{11} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 4}, \quad A_{12} = I_2, \quad A_{21} = \mathbf{1}_3\mathbf{1}_4' \in \mathbb{R}^{3\times 4}, \quad A_{22} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

$$B_{11} = \begin{pmatrix} I_3 \\ \mathbf{1}_3' \end{pmatrix} \in \mathbb{R}^{4\times 3}, \quad B_{12} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 2}, \quad B_{21} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 3}, \quad B_{22} = I_2$$

Only A_{12} and B_{22} are square.

(a) The column groups $(4, 2)$ of A are the row groups of B, so the partitions are conformal for multiplication (Definition 5.2). By Theorem 5.3(c), $AB \in \mathbb{R}^{5\times 5}$ has row groups $(2, 3)$, those of A, and column groups $(3, 2)$, those of B. Every term is a product of a block with n_j columns and a block with n_j rows:

$$\begin{aligned} (AB)_{11} &= A_{11}B_{11} + A_{12}B_{21}: \quad (2\times 4)(4\times 3) + (2\times 2)(2\times 3) \quad \text{is } 2\times 3 \\ (AB)_{12} &= A_{11}B_{12} + A_{12}B_{22}: \quad (2\times 4)(4\times 2) + (2\times 2)(2\times 2) \quad \text{is } 2\times 2 \\ (AB)_{21} &= A_{21}B_{11} + A_{22}B_{21}: \quad (3\times 4)(4\times 3) + (3\times 2)(2\times 3) \quad \text{is } 3\times 3 \\ (AB)_{22} &= A_{21}B_{12} + A_{22}B_{22}: \quad (3\times 4)(4\times 2) + (3\times 2)(2\times 2) \quad \text{is } 3\times 2 \end{aligned}$$

(b) The column sums of B_{11} and B_{12} are $\mathbf{1}_4'B_{11} = (2, 2, 2)$ and $\mathbf{1}_4'B_{12} = (2, 2)$, so $A_{21}B_{11} =$

$(\mathbf{1}_3\mathbf{1}_4')B_{11} = \mathbf{1}_3(\mathbf{1}_4'B_{11})$ (Theorem 1.3) has all entries 2, and so has $A_{21}B_{12}$. With $A_{12} = B_{22} = I_2$,

$$(AB)_{11} = A_{11}B_{11} + B_{21} = \begin{pmatrix} 2 & 2 & 2 \\ 0 & -2 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 1 \\ 0 & -1 & -1 \end{pmatrix}$$

$$(AB)_{12} = A_{11}B_{12} + I_2 = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix} + I_2 = \begin{pmatrix} 3 & 2 \\ 2 & -1 \end{pmatrix}$$

$$(AB)_{21} = A_{21}B_{11} + A_{22}B_{21} = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 0 & -2 \\ 0 & 1 & -1 \\ 1 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{pmatrix}$$

$$(AB)_{22} = A_{21}B_{12} + A_{22} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 3 \\ 3 & 3 \end{pmatrix}$$

So

$$AB = \left(\begin{array}{ccc|cc} 3 & 2 & 1 & 3 & 2 \\ 0 & -1 & -1 & 2 & -1 \\ \hline 4 & 2 & 0 & 4 & 2 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 0 & 3 & 3 \end{array}\right)$$

(c) Check of one entry. Entry $(4,2)$ of AB lies in row group 2 (row $u = 4 = M_1 + 2$, so $x = 2$) and column group 1 ($z = 2$). As in the paragraph after the proof of Theorem 5.3, the sum over $l = 1, \ldots, 6$ splits at $N_1 = 4$:

$$(AB)_{42} = \underbrace{1\cdot 0 + 1\cdot 1 + 1\cdot 0 + 1\cdot 1}_{l=1,\ldots,4} + \underbrace{0\cdot 0 + 1\cdot 1}_{l=5,6} = 2 + 1 = 3$$

The first part is row 2 of A_{21} times column 2 of B_{11}, entry $(2,2)$ of $A_{21}B_{11}$; the second is row 2 of A_{22} times column 2 of B_{21}, entry $(2,2)$ of $A_{22}B_{21}$.

(d) A partition of B that is not conformal. Give B the row groups $(3,3)$, with any column groups. The product AB is still defined, but $A_{11} \in \mathbb{R}^{2\times 4}$ has 4 columns while the new B_{11} has 3 rows, so $A_{11}B_{11}$ is not defined, and the blocks of AB cannot be computed from these blocks. By contrast, the column groups of B are free: with its row groups $(4,2)$ and the column groups changed, say to $(1,4)$, the partitions are still conformal (Definition 5.2).

(e) By Theorem 5.3(b), $A' \in \mathbb{R}^{6\times 5}$ with row groups $(4,2)$ and column groups $(2,3)$ is $A' = \left(\begin{smallmatrix} A'_{11} & A'_{21} \\ A'_{12} & A'_{22} \end{smallmatrix}\right)$, where $A'_{11} \in \mathbb{R}^{4\times 2}$, $A'_{21} = \mathbf{1}_4\mathbf{1}_3' \in \mathbb{R}^{4\times 3}$, $A'_{12} = I_2$ and $A'_{22} \in \mathbb{R}^{2\times 3}$.

Definition 5.5 ([5, §2.1])**.** Let $A \in \mathbb{R}^{n\times n}$ be partitioned with the same groups $n_1, \ldots, n_r$ for rows and columns. Then the **diagonal blocks** $A_{ii} \in \mathbb{R}^{n_i\times n_i}$ are square, while an off-diagonal block $A_{ij} \in \mathbb{R}^{n_i\times n_j}$ $(i \neq j)$ is rectangular when $n_i \neq n_j$. A is **block diagonal**, written $A = \operatorname{diag}(A_{11}, \ldots, A_{rr})$, if $A_{ij} = 0$ for $i \neq j$; **upper block triangular** if $A_{ij} = 0$ for $i > j$; and **lower block triangular** if $A_{ij} = 0$ for $i < j$.

Proposition 5.6 ([2, §§4.10, 4.12(f)], [5, §2.1])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be partitioned as in Definition 5.5, with the same groups.

(a) If A and B are upper (lower) block triangular, so is AB, and its diagonal blocks are $A_{ii}B_{ii}$.

(b) $\operatorname{diag}(A_{11}, \ldots, A_{rr})\operatorname{diag}(B_{11}, \ldots, B_{rr}) = \operatorname{diag}(A_{11}B_{11}, \ldots, A_{rr}B_{rr})$. If every A_{ii} is invertible, then $\operatorname{diag}(A_{11}, \ldots, A_{rr})^{-1} = \operatorname{diag}(A_{11}^{-1}, \ldots, A_{rr}^{-1})$.

(c) If $A_{11} \in \mathbb{R}^{p\times p}$ and $A_{22} \in \mathbb{R}^{q\times q}$ are invertible and $A_{12} \in \mathbb{R}^{p\times q}$, then

$$\begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}^{-1} = \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{pmatrix}$$

In (c) only the diagonal blocks must be square; for $p \neq q$ the blocks A_{12} and $-A_{11}^{-1}A_{12}A_{22}^{-1}$ are $p \times q$.

Proof. (a)

- The column groups $n_1, \dots, n_r$ of A are the row groups of B, so by Theorem 5.3(c) the (i,k) block of AB is $\sum_{j=1}^{r} A_{ij}B_{jk} \in \mathbb{R}^{n_i \times n_k}$.
- Let A and B be upper block triangular. Then $A_{ij} = 0$ unless $i \leqslant j$, and $B_{jk} = 0$ unless $j \leqslant k$. So the term $A_{ij}B_{jk}$ has a zero factor, and is 0, unless $i \leqslant j \leqslant k$.
- If $i > k$, no j satisfies $i \leqslant j \leqslant k$, so $(AB)_{ik} = 0$: AB is upper block triangular. If $i = k$, only $j = i$ remains, so $(AB)_{ii} = A_{ii}B_{ii}$.
- For lower block triangular A, B, the term j can be nonzero only if $i \geqslant j \geqslant k$; the same argument applies with the inequalities reversed.

(b) A block diagonal matrix is upper and lower block triangular, so by (a) the product is upper and lower block triangular, that is, block diagonal, with diagonal blocks $A_{ii}B_{ii}$. With $B_{ii} = A_{ii}^{-1}$ this gives $\operatorname{diag}(A_{11}, \dots, A_{rr}) \operatorname{diag}(A_{11}^{-1}, \dots, A_{rr}^{-1}) = \operatorname{diag}(I_{n_1}, \dots, I_{n_r}) = I_n$ (both have entry 1 at every (u,u) and 0 elsewhere), and likewise in the other order; so the second matrix is the inverse of the first (Definition 2.12).

(c) Call the two matrices T and X; both have row and column groups (p,q), so the 2×2 block formula of Theorem 5.3(c) applies. The four blocks of TX are

$$(TX)_{11} = A_{11}A_{11}^{-1} + A_{12} \cdot 0 = I_p, \quad (TX)_{12} = A_{11}\left(-A_{11}^{-1}A_{12}A_{22}^{-1}\right) + A_{12}A_{22}^{-1} = 0$$
$$(TX)_{21} = 0 \cdot A_{11}^{-1} + A_{22} \cdot 0 = 0, \quad (TX)_{22} = 0 \cdot \left(-A_{11}^{-1}A_{12}A_{22}^{-1}\right) + A_{22}A_{22}^{-1} = I_q$$

and those of XT are

$$(XT)_{11} = A_{11}^{-1}A_{11} + \left(-A_{11}^{-1}A_{12}A_{22}^{-1}\right) \cdot 0 = I_p, \quad (XT)_{12} = A_{11}^{-1}A_{12} - A_{11}^{-1}A_{12}A_{22}^{-1}A_{22} = 0$$
$$(XT)_{21} = 0 \cdot A_{11} + A_{22}^{-1} \cdot 0 = 0, \quad (XT)_{22} = 0 \cdot A_{12} + A_{22}^{-1}A_{22} = I_q$$

So $TX = XT = I_{p+q}$, and $X = T^{-1}$ (Definition 2.12). □

Example 5.7. Find the inverse of

$$T = \left(\begin{array}{cc|ccc} 3 & 2 & 1 & 0 & 2 \\ 1 & 1 & 1 & 1 & 1 \\ \hline 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right) = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix} \in \mathbb{R}^{5\times 5}$$

Solution. The groups are $(2,3)$: $A_{11} = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 1 \end{smallmatrix}\right)$ is 2×2, $A_{12} = \left(\begin{smallmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \end{smallmatrix}\right)$ is 2×3, and

$$A_{22} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

is 3×3. By Proposition 2.18 ($3 \cdot 1 - 2 \cdot 1 = 1$), $A_{11}^{-1} = \left(\begin{smallmatrix} 1 & -2 \\ -1 & 3 \end{smallmatrix}\right)$. For

$$Y = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A_{22}Y = \begin{pmatrix} 1 & -1+1 & 1-1 \\ 0 & 1 & -1+1 \\ 0 & 0 & 1 \end{pmatrix} = I_3$$

so $A_{22}^{-1} = Y$ (Theorem 2.16). Then

$$A_{11}^{-1}A_{12} = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -2 & 0 \\ 2 & 3 & 1 \end{pmatrix}$$

$$-A_{11}^{-1}A_{12}A_{22}^{-1} = -\begin{pmatrix}-1 & -2 & 0\\ 2 & 3 & 1\end{pmatrix}\begin{pmatrix}1 & -1 & 1\\ 0 & 1 & -1\\ 0 & 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 1 & -1\\ -2 & -1 & 0\end{pmatrix}$$

This block is 2×3. By Proposition 5.6(c) with $p=2$, $q=3$,

$$T^{-1} = \left(\begin{array}{cc|ccc} 1 & -2 & 1 & 1 & -1\\ -1 & 3 & -2 & -1 & 0\\ \hline 0 & 0 & 1 & -1 & 1\\ 0 & 0 & 0 & 1 & -1\\ 0 & 0 & 0 & 0 & 1\end{array}\right)$$

Block elimination and the Schur complement

Proposition 5.8 ([6, Exercises 5.6, 5.7], [5, Lemma 8.5.2])**.** Let $M = \begin{pmatrix}A & B\\ C & D\end{pmatrix}$ with $A\in\mathbb{R}^{m\times p}$, $B\in\mathbb{R}^{m\times q}$, $C\in\mathbb{R}^{n\times p}$, $D\in\mathbb{R}^{n\times q}$, and let $V\in\mathbb{R}^{n\times m}$, $F\in\mathbb{R}^{m\times n}$, $H\in\mathbb{R}^{p\times q}$. Then

$$\begin{pmatrix}I_m & 0\\ V & I_n\end{pmatrix}M = \begin{pmatrix}A & B\\ C+VA & D+VB\end{pmatrix}, \qquad \begin{pmatrix}I_m & F\\ 0 & I_n\end{pmatrix}M = \begin{pmatrix}A+FC & B+FD\\ C & D\end{pmatrix}$$

$$M\begin{pmatrix}I_p & H\\ 0 & I_q\end{pmatrix} = \begin{pmatrix}A & B+AH\\ C & D+CH\end{pmatrix}$$

These **block elementary matrices** are invertible: $\left(\begin{smallmatrix}I_m & 0\\ V & I_n\end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix}I_m & 0\\ -V & I_n\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}I_m & F\\ 0 & I_n\end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix}I_m & -F\\ 0 & I_n\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}I_p & H\\ 0 & I_q\end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix}I_p & -H\\ 0 & I_q\end{smallmatrix}\right)$.

None of A, B, C, D, V, F, H needs to be square.

Proof. • M has row groups (m,n) and column groups (p,q). The matrices $\left(\begin{smallmatrix}I_m & 0\\ V & I_n\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}I_m & F\\ 0 & I_n\end{smallmatrix}\right)$ have column groups (m,n), and $\left(\begin{smallmatrix}I_p & H\\ 0 & I_q\end{smallmatrix}\right)$ has row groups (p,q); so all three products are conformal, and Theorem 5.3(c) applies.

- First product: block row 1 is $(I_mA+0C,\ I_mB+0D) = (A,\ B)$, and block row 2 is $(VA+I_nC,\ VB+I_nD) = (C+VA,\ D+VB)$, where $VA\in\mathbb{R}^{n\times p}$ and $VB\in\mathbb{R}^{n\times q}$ have the sizes of C and D.
- Second product: block row 1 is $(I_mA+FC,\ I_mB+FD) = (A+FC,\ B+FD)$, and block row 2 is $(0A+I_nC,\ 0B+I_nD) = (C,\ D)$.
- Third product: block column 1 is $(AI_p+B0,\ CI_p+D0) = (A,\ C)$, and block column 2 is $(AH+BI_q,\ CH+DI_q) = (B+AH,\ D+CH)$.
- Thus left multiplication by $\left(\begin{smallmatrix}I & 0\\ V & I\end{smallmatrix}\right)$ adds V times the first block row to the second, the block analogue of the row operation $R_2\leftarrow R_2+cR_1$ of Definition 2.2; right multiplication by $\left(\begin{smallmatrix}I & H\\ 0 & I\end{smallmatrix}\right)$ adds the first block column times H to the second.
- Inverses. The first identity with $A=I_m$, $B=0$, $C=V$, $D=I_n$ (so $p=m$, $q=n$) and with $-V$ in place of V gives

$$\begin{pmatrix}I_m & 0\\ -V & I_n\end{pmatrix}\begin{pmatrix}I_m & 0\\ V & I_n\end{pmatrix} = \begin{pmatrix}I_m & 0\\ V-VI_m & I_n-V\cdot 0\end{pmatrix} = I_{m+n}$$

and with $C=-V$ and V itself it gives the product in the other order, $\left(\begin{smallmatrix}I_m & 0\\ -V+V & I_n\end{smallmatrix}\right) = I_{m+n}$. In the same way the second identity gives $\left(\begin{smallmatrix}I_m & -F\\ 0 & I_n\end{smallmatrix}\right)\left(\begin{smallmatrix}I_m & F\\ 0 & I_n\end{smallmatrix}\right) = \left(\begin{smallmatrix}I_m & F-F\\ 0 & I_n\end{smallmatrix}\right) = I_{m+n}$ and the product in the other order; the third case is the second with p, q, H in place of m, n, F. □

Theorem 5.9 ([6, Exercise 5.10])**.** Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$ invertible, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{n\times p}$, $D \in \mathbb{R}^{n\times q}$. Then

$$\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{pmatrix}\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}, \qquad \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & D - CA^{-1}B \end{pmatrix}$$

and hence $\left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{smallmatrix}\right) M \left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & 0 \\ 0 & D-CA^{-1}B \end{smallmatrix}\right)$. The matrix $D - CA^{-1}B \in \mathbb{R}^{n\times q}$ is the **Schur complement** of A in M.

Only A has to be square: B, C, D and M itself may be rectangular.

Proof. • Apply Proposition 5.8 with $m = p$. The choice $V = -CA^{-1} \in \mathbb{R}^{n\times p}$ makes the $(2,1)$ block $C + VA = C - CA^{-1}A = 0$, and then $D + VB = D - CA^{-1}B$. This is the first identity.

- The choice $H = -A^{-1}B \in \mathbb{R}^{p\times q}$ makes $B+AH = B-AA^{-1}B = 0$, and then $D+CH = D-CA^{-1}B$. This is the second identity.
- Put $S = D-CA^{-1}B$. Applying the column step to the result of the row step, by the third identity of Proposition 5.8 with C replaced by 0 and D by S, $\left(\begin{smallmatrix} A & B \\ 0 & S \end{smallmatrix}\right)\left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & B-AA^{-1}B \\ 0 & S-0\cdot A^{-1}B \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & 0 \\ 0 & S \end{smallmatrix}\right)$. □

In Theorem 5.9 the left factor adds $-CA^{-1}$ times the first block row of M to the second, and the right factor adds the first block column times $-A^{-1}B$ to the second (Proposition 5.8). When D is invertible, the matrix $A - BD^{-1}C$ is likewise called the **Schur complement** of D in M.

Let R_1, R_2 be the block rows of a matrix with row groups (p, q). The **block row operations** are $R_1 \leftarrow XR_1$ and $R_2 \leftarrow YR_2$ with $X \in \mathbb{R}^{p\times p}$ and $Y \in \mathbb{R}^{q\times q}$ invertible, $R_2 \leftarrow R_2 + VR_1$ with $V \in \mathbb{R}^{q\times p}$, and $R_1 \leftarrow R_1 + WR_2$ with $W \in \mathbb{R}^{p\times q}$. They are the left multiplications by $\operatorname{diag}(X, I_q)$, $\operatorname{diag}(I_p, Y)$, $\left(\begin{smallmatrix} I_p & 0 \\ V & I_q \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I_p & W \\ 0 & I_q \end{smallmatrix}\right)$ (Theorem 5.3(c) and Proposition 5.8), and these matrices are invertible (Propositions 5.6(b) and 5.8). **Block elimination** applies the steps of Gauss–Jordan elimination (Definition 2.5) to the block rows of $[M \mid I]$, with the inverse of a pivot block in place of the reciprocal of a pivot entry: the first pivot is a diagonal block of M, the second its Schur complement, and the result is $[I \mid M^{-1}]$, as in Corollary 2.17.

Corollary 5.10 (Block elimination; [6, Exercise 5.10], [5, §8.5])**.** Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$.

(a) If A and its Schur complement $S = D-CA^{-1}B$ are invertible, then M is invertible, the block row operations $R_2 \leftarrow R_2 - CA^{-1}R_1$, then $R_1 \leftarrow A^{-1}R_1$ and $R_2 \leftarrow S^{-1}R_2$, then $R_1 \leftarrow R_1 - A^{-1}BR_2$ take $[M \mid I_{p+q}]$ to $[I_{p+q} \mid M^{-1}]$,

$$M^{-1} = \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix}\begin{pmatrix} A^{-1} & 0 \\ 0 & S^{-1} \end{pmatrix}\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix}$$

and $M = \left(\begin{smallmatrix} I_p & 0 \\ CA^{-1} & I_q \end{smallmatrix}\right) \operatorname{diag}(A, S) \left(\begin{smallmatrix} I_p & A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$.

(b) If D and its Schur complement $F = A-BD^{-1}C$ are invertible, then M is invertible, the block row operations $R_1 \leftarrow R_1 - BD^{-1}R_2$, then $R_1 \leftarrow F^{-1}R_1$ and $R_2 \leftarrow D^{-1}R_2$, then $R_2 \leftarrow R_2 - D^{-1}CR_1$ take $[M \mid I_{p+q}]$ to $[I_{p+q} \mid M^{-1}]$,

$$M^{-1} = \begin{pmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{pmatrix}\begin{pmatrix} F^{-1} & 0 \\ 0 & D^{-1} \end{pmatrix}\begin{pmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{pmatrix}$$

and $M = \left(\begin{smallmatrix} I_p & BD^{-1} \\ 0 & I_q \end{smallmatrix}\right) \operatorname{diag}(F, D) \left(\begin{smallmatrix} I_p & 0 \\ D^{-1}C & I_q \end{smallmatrix}\right)$.

For $p \neq q$ the blocks B and C are rectangular.

Proof. (a)

- The three steps are the left multiplications by $E_1 = \left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{smallmatrix}\right)$, $E_2 = \operatorname{diag}(I_p, S^{-1})\operatorname{diag}(A^{-1}, I_q) = \operatorname{diag}(A^{-1}, S^{-1})$ (Proposition 5.6(b)) and $E_3 = \left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$.
- By Theorem 5.9 (with $n = q$), $E_1 M = \left(\begin{smallmatrix} A & B \\ 0 & S \end{smallmatrix}\right)$. By the block product formula (Theorem 5.3(c)), $E_2E_1M = \left(\begin{smallmatrix} A^{-1}A & A^{-1}B \\ S^{-1}\cdot 0 & S^{-1}S \end{smallmatrix}\right) = \left(\begin{smallmatrix} I_p & A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$. By Proposition 5.8 (second identity, with upper right block $-A^{-1}B$), $E_3E_2E_1M = \left(\begin{smallmatrix} I_p & A^{-1}B - A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = I_{p+q}$.
- Put $E = E_3E_2E_1$, the displayed product. Then $EM = I_{p+q}$, so M is invertible with $M^{-1} = E$ (Theorem 2.16), and $E[M \mid I_{p+q}] = [EM \mid E] = [I_{p+q} \mid M^{-1}]$ (Theorem 1.4(a)).
- $E_1^{-1} = \left(\begin{smallmatrix} I_p & 0 \\ CA^{-1} & I_q \end{smallmatrix}\right)$ and $E_3^{-1} = \left(\begin{smallmatrix} I_p & A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$ (Proposition 5.8), and $E_2^{-1} = \operatorname{diag}(A, S)$ (Proposition 5.6(b)). So $M = E^{-1} = E_1^{-1}E_2^{-1}E_3^{-1}$ (Theorem 2.13(a),(b)).

(b) The three steps are the left multiplications by $E_1' = \left(\begin{smallmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{smallmatrix}\right)$, $E_2' = \operatorname{diag}(F^{-1}, D^{-1})$ (as in (a)) and $E_3' = \left(\begin{smallmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{smallmatrix}\right)$. By Proposition 5.8 and Theorem 5.3(c),

$$E_1'M = \begin{pmatrix} A - BD^{-1}C & B - BD^{-1}D \\ C & D \end{pmatrix} = \begin{pmatrix} F & 0 \\ C & D \end{pmatrix}, \qquad E_2'E_1'M = \begin{pmatrix} I_p & 0 \\ D^{-1}C & I_q \end{pmatrix}$$

and $E_3'E_2'E_1'M = I_{p+q}$. The last two bullets of (a) apply to $E' = E_3'E_2'E_1'$, the displayed product, with $E_1'^{-1} = \left(\begin{smallmatrix} I_p & BD^{-1} \\ 0 & I_q \end{smallmatrix}\right)$, $E_2'^{-1} = \operatorname{diag}(F, D)$ and $E_3'^{-1} = \left(\begin{smallmatrix} I_p & 0 \\ D^{-1}C & I_q \end{smallmatrix}\right)$ (Propositions 5.8 and 5.6(b)). □

Example 5.11. Partition $M \in \mathbb{R}^{5\times 5}$ with groups $(2, 3)$ for rows and columns. Find the Schur complement S of A, the factorization of M in Corollary 5.10(a), and M^{-1}.

$$M = \left(\begin{array}{cc|ccc} 2 & 1 & 2 & 1 & 3 \\ 1 & 1 & 1 & 1 & 2 \\ \hline 2 & 1 & 3 & 2 & 3 \\ 1 & 1 & 2 & 3 & 3 \\ 3 & 2 & 3 & 3 & 7 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix}, \quad C = B', \quad D = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix}$$

Here A is 2×2, D is 3×3, B is 2×3 and C is 3×2. By Proposition 2.18 ($2 \cdot 1 - 1 \cdot 1 = 1$), $A^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Then

$$A^{-1}B = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad CA^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$$

These are 2×3 and 3×2, and the Schur complement of A is the 3×3 matrix

$$S = D - CA^{-1}B = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 3 \\ 3 & 3 & 7 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \\ 3 & 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

By Theorem 5.9,

$$\left(\begin{array}{cc|ccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 1 \end{array}\right) M \left(\begin{array}{cc|ccc} 1 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & -1 & -1 \\ \hline 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right) = \left(\begin{array}{cc|ccc} 2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

All matrices here have integer entries, because A^{-1} does. Multiplying on the left and on the right by the inverses of the two outer factors (Proposition 5.8),

$$M = \begin{pmatrix} I_2 & 0 \\ CA^{-1} & I_3 \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} I_2 & A^{-1}B \\ 0 & I_3 \end{pmatrix}$$

$$= \left(\begin{array}{cc|ccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array}\right) \left(\begin{array}{cc|ccc} 2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right) \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ \hline 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

so M is the product of a lower block triangular, a block diagonal and an upper block triangular factor, whose off-diagonal blocks $CA^{-1} \in \mathbb{R}^{3\times 2}$ and $A^{-1}B \in \mathbb{R}^{2\times 3}$ are rectangular. Since $A' = A$ and $C = B'$, $A^{-1}B = (A^{-1})'C' = (CA^{-1})'$ (Theorems 2.13(c) and 1.9), and the right factor is the transpose of the left (Theorem 5.3(b)).

Finally, M^{-1} is found by Corollary 5.10(a). The Schur complement S is invertible, with

$$S^{-1} = \begin{pmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

since $SS^{-1} = I_3$ (Theorem 2.16). By the corollary and the block product formula (Theorem 5.3(c)),

$$M^{-1} = \begin{pmatrix} I_2 & -A^{-1}B \\ 0 & I_3 \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I_2 & 0 \\ -CA^{-1} & I_3 \end{pmatrix} = \begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}$$

With $A^{-1}B$ and CA^{-1} from above,

$$-A^{-1}BS^{-1} = -\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 3 & -2 \\ 1 & -1 & 0 \end{pmatrix}$$

$$-S^{-1}CA^{-1} = -\begin{pmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 1 \\ 3 & -1 \\ -2 & 0 \end{pmatrix}$$

and

$$A^{-1} + A^{-1}BS^{-1}CA^{-1} = A^{-1} - (A^{-1}B)(-S^{-1}CA^{-1}) = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} -4 & 1 \\ 3 & -1 \\ -2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} -6 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 7 & -2 \\ -2 & 3 \end{pmatrix}$$

So

$$M^{-1} = \left(\begin{array}{cc|ccc} 7 & -2 & -4 & 3 & -2 \\ -2 & 3 & 1 & -1 & 0 \\ \hline -4 & 1 & 3 & -2 & 1 \\ 3 & -1 & -2 & 2 & -1 \\ -2 & 0 & 1 & -1 & 1 \end{array}\right)$$

Check: $\text{row}_1(M)M^{-1} = 2(7,-2,-4,3,-2) + (-2,3,1,-1,0) + 2(-4,1,3,-2,1) + (3,-1,-2,2,-1) + 3(-2,0,1,-1,1) = (1,0,0,0,0)$ (Theorem 1.4(c)).

Theorem 5.12 ([6, Exercise 5.16], [5, Theorem 8.5.11, Corollary 8.5.12], [2, §4.12(h)])**.** Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$.

(a) If A and its Schur complement $E = D - CA^{-1}B$ are invertible, then M is invertible and

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix}$$

(b) If D and its Schur complement $F = A - BD^{-1}C$ are invertible, then M is invertible and

$$M^{-1} = \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix}$$

In particular, in (a) the lower right block of M^{-1} is the inverse of the Schur complement of A.

Proof. (a)

- By Theorem 5.9, $LM = U$ with $L = \left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{smallmatrix}\right)$ and $U = \left(\begin{smallmatrix} A & B \\ 0 & E \end{smallmatrix}\right)$.
- L is invertible with inverse $\left(\begin{smallmatrix} I_p & 0 \\ CA^{-1} & I_q \end{smallmatrix}\right)$ (Proposition 5.8), so $M = L^{-1}U$; and U is invertible with $U^{-1} = \left(\begin{smallmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{smallmatrix}\right)$ (Proposition 5.6(c)).
- Hence M is invertible and $M^{-1} = U^{-1}(L^{-1})^{-1} = U^{-1}L$ (Theorem 2.13(a),(b)). By the block product formula (Theorem 5.3(c)),

$$U^{-1}L = \begin{pmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{pmatrix} \begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix} = \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix}$$

(b) Look for $X = \left(\begin{smallmatrix} P & Q \\ R & T \end{smallmatrix}\right)$ with $MX = I_{p+q}$; by the block product formula this means

$$AP + BR = I_p, \qquad CP + DR = 0, \qquad AQ + BT = 0, \qquad CQ + DT = I_q$$

- The second equation gives $R = -D^{-1}CP$; then the first becomes $(A - BD^{-1}C)P = FP = I_p$, so $P = F^{-1}$ and $R = -D^{-1}CF^{-1}$.
- The fourth equation gives $T = D^{-1} - D^{-1}CQ$; then the third becomes $AQ + BD^{-1} - BD^{-1}CQ = FQ + BD^{-1} = 0$, so $Q = -F^{-1}BD^{-1}$ and $T = D^{-1} + D^{-1}CF^{-1}BD^{-1}$.
- Conversely, these blocks satisfy the four equations: $AP+BR = (A-BD^{-1}C)F^{-1} = I_p$, $CP+DR = CF^{-1} - CF^{-1} = 0$, $AQ + BT = -(A - BD^{-1}C)F^{-1}BD^{-1} + BD^{-1} = 0$ and $CQ + DT = -CF^{-1}BD^{-1} + I_q + CF^{-1}BD^{-1} = I_q$.
- So $MX = I_{p+q}$, and since M is square, M is invertible with $M^{-1} = X$ (Theorem 2.16). □

Determinants: definition and computation

Definition 5.13 ([11, Four.I.2, Definition 2.1], [1, §5.2])**.** Let $n \geqslant 1$. A function D that assigns a number $D(A)$ to each $A \in \mathbb{R}^{n\times n}$ is a **determinant function** if, for every $A \in \mathbb{R}^{n\times n}$, all row indices $r \neq s$ and every $c \in \mathbb{R}$:

(a) multiplying row r of A by c multiplies $D(A)$ by c;

(b) adding c times row s to row r does not change $D(A)$;

(c) interchanging rows r and s changes the sign of $D(A)$;

(d) $D(I_n) = 1$.

We show that for each n there is exactly one determinant function (Theorems 5.15 and 5.21). It is written det, and $\det A$ is the **determinant** of A; many books write $|A|$, but in these notes $|\cdot|$ always means absolute value. Until its existence is proved, every statement about det holds for every determinant function. For $n = 1$, (a) and (d) give $\det(a) = a \det(1) = a$.

For $n = 2$, conditions (a)–(d) are the rules for the area of the parallelogram whose sides are the two rows u', v' of A ($u, v \in \mathbb{R}^2$), counted with a sign: stretching the side u' by c multiplies the area by c; replacing u' by $u' + cv'$ slides that side along v' and keeps both the base v' and the height; and the unit square, with sides e'_1 and e'_2, has area 1. The sign records the orientation of the pair u', v', which an interchange reverses. For $n \geqslant 3$, $|\det A|$ is accordingly taken as the volume of the box spanned by the rows [11, Four.II.1]. This is why determinants appear in the change-of-variables formula for probability densities; we do not use this here.

Theorem 5.14 ([11, Four.I.2, Lemma 2.4; Four.II.1, proof of Theorem 1.5], [2, §3.4])**.** Let $A \in \mathbb{R}^{n\times n}$.

(a) If two rows of A are equal, then $\det A = 0$.

(b) If A has a zero row, then $\det A = 0$.

(c) If A is upper triangular, then $\det A = a_{11}a_{22}\cdots a_{nn}$. In particular $\det \operatorname{diag}(d_1, \ldots, d_n) = d_1 d_2 \cdots d_n$.

(d) For the elementary matrices of Definition 2.14, $\det D_r(c) = c$, $\det T_{rs}(c) = 1$ and $\det P_{rs} = -1$; and $\det(EA) = \det E \cdot \det A$ for every elementary matrix $E \in \mathbb{R}^{n\times n}$.

Proof. (a) Interchanging the two equal rows does not change A, but it changes the sign of $\det A$ (Definition 5.13(c)). So $\det A = -\det A$, that is, $\det A = 0$.

(b) Multiplying the zero row by 2 does not change A, but it doubles $\det A$ (Definition 5.13(a)). So $\det A = 2\det A$, that is, $\det A = 0$.

(c)

- Let $a_{11}, \ldots, a_{nn}$ be nonzero. Multiplying row i by $1/a_{ii}$ for $i = 1, \ldots, n$ gives an upper triangular matrix B with diagonal entries 1, and $\det B = \det A/(a_{11}\cdots a_{nn})$ (Definition 5.13(a)).
- Backward elimination (Step 5 of Definition 2.5, with the diagonal entries of B as pivots) turns B into I_n by adding multiples of rows to other rows; so $\det B = \det I_n = 1$ (Definition 5.13(b),(d)), and $\det A = a_{11}\cdots a_{nn}$.
- Now let some a_{ii} be 0, with i the largest such index. Row i of A is 0 in columns $1, \ldots, i$. For $j = i+1, \ldots, n$ in turn, subtract a multiple of row j from row i so that entry (i, j) becomes 0 (possible as $a_{jj} \neq 0$); row j is 0 before column j, so the zeros already made stay 0. These additions do not change $\det A$ and turn row i into a zero row; so $\det A = 0$ by (b), and also $a_{11}\cdots a_{nn} = 0$.

(d)

- $D_r(c)$, $T_{rs}(c)$ and P_{rs} are obtained from I_n by $R_r \leftarrow cR_r$, $R_r \leftarrow R_r + cR_s$ and $R_r \leftrightarrow R_s$, so by Definition 5.13 their determinants are c, 1 and -1.
- EA is obtained from A by the row operation of E (Theorem 2.15(a)), so $\det(EA)$ is $c\det A$, $\det A$ or $-\det A$, that is, $\det E \cdot \det A$. □

Theorem 5.15 ([11, Four.I.2, Lemma 2.7, Examples 2.5, 2.6], [2, §3.3.5])**.** Let interchanges of two rows and additions of a multiple of one row to another take $A \in \mathbb{R}^{n\times n}$ to an upper triangular matrix U, with t interchanges. Then

$$\det A = (-1)^t u_{11}u_{22}\cdots u_{nn}$$

Forward elimination (Steps 1, 2 and 4 of Definition 2.5, without scaling the leading entries to 1) always ends with such a matrix U. In particular, for each n there is at most one determinant function.

Proof. • $\det U = u_{11}\cdots u_{nn}$ (Theorem 5.14(c)). Each addition leaves the determinant unchanged and each interchange changes its sign (Definition 5.13(b),(c)); so $\det U = (-1)^t \det A$, and the formula follows.

- In the result of forward elimination, the leading entries of the nonzero rows lie in columns $k_1 < k_2 < \cdots$, so $k_i \geqslant i$, and the zero rows lie at the bottom (Definition 2.5); hence the result is upper triangular.
- For each A, fix one forward elimination of A (Step 2 may allow several; any one will do). By the first item, every determinant function takes the value $(-1)^t u_{11} \cdots u_{nn}$ of this reduction at A; so two determinant functions are equal. □

If a row is multiplied by $c \neq 0$ on the way, the determinant is multiplied by c (Definition 5.13(a)); so divide by c at the end.

Example 5.16. Compute $\det A$ for

$$A = \begin{pmatrix} 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 6 & 4 & 4 \\ 1 & 0 & 2 & 3 \end{pmatrix}$$

Solution. Since $a_{11} = 0$, begin with an interchange:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 2 & 6 & 4 & 4 \\ 1 & 0 & 2 & 3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 2 & 2 & 4 \\ 0 & -2 & 1 & 3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & -2 \end{pmatrix} = U$$

The interchange changes the sign and the additions change nothing (Theorem 5.15, $t = 1$), so $\det A = -\det U = -\big(1 \cdot 2 \cdot 1 \cdot (-2)\big) = 4$.

Example 5.17 ([1, §5.4, Example 6])**.** Compute $\det A$ for

$$A = \begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{pmatrix}$$

Solution.

$$A \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - 4R_1} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{4} R_2} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 5R_2} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & 0 & -8 \end{pmatrix} = U$$

Only the second arrow changes the determinant: it multiplies it by $\frac{1}{4}$ (Definition 5.13(a)). Hence $\det A = 4 \det U = 4 \cdot 1 \cdot 1 \cdot (-4) \cdot (-8) = 128$.

Main properties

Theorem 5.18 ([5, Theorems 13.3.4, 13.3.7, (13.3.10), (13.3.12)], [2, Theorem 3.4, Corollary 3.3])**.** Let $A, B \in \mathbb{R}^{n\times n}$.

(a) A is invertible $\Leftrightarrow \det A \neq 0 \Leftrightarrow \operatorname{rank} A = n$.

(b) $\det(AB) = \det A \cdot \det B$.

(c) If A is invertible, then $\det A^{-1} = 1/\det A$. Moreover $\det(cA) = c^n \det A$ for $c \in \mathbb{R}$, and $\det A^k = (\det A)^k$ for $k = 1, 2, \ldots$

Proof. For $n = 1$, $\det(a) = a$: (a) holds since (a) is invertible, and $\operatorname{rank}(a) = 1$, if and only if $a \neq 0$ (Definition 4.11), and (b), (c) are identities between numbers; let $n \geqslant 2$.

(a)

- Let $R = PA$ be the RREF of A, where $P = E_k \cdots E_1$ is a product of elementary matrices (Theorem 2.15(c)). By Theorem 5.14(d), applied k times, $\det R = \det E_k \cdots \det E_1 \cdot \det A$, and each $\det E_i$ is $c \neq 0$, 1 or -1.
- If A is invertible, then $R = I_n$ (Theorem 2.16), so $\det A \neq 0$. If not, then $\operatorname{rank} A < n$ (Theorem 4.13(d)), so R has fewer than n nonzero rows (Definition 4.11), that is, a zero row; then $\det R = 0$ (Theorem 5.14(b)) and $\det A = 0$. Finally, A is invertible if and only if $\operatorname{rank} A = n$ (Theorem 4.13(d)).

(b)

- If A is not invertible, neither is AB: an inverse G of AB would give $A(BG) = I_n$, and A would be invertible (Theorem 2.16). By (a), $\det(AB) = 0 = \det A \cdot \det B$.
- If A is invertible, then $A = F_1 \cdots F_k$ with elementary matrices F_i (Theorem 2.16). By Theorem 5.14(d), applied k times, $\det(AB) = \det F_1 \cdots \det F_k \cdot \det B$; for $B = I_n$ this reads $\det A = \det F_1 \cdots \det F_k$, so $\det(AB) = \det A \cdot \det B$.

(c) By (b), $\det A \cdot \det A^{-1} = \det(AA^{-1}) = \det I_n = 1$. Next, $cA = (cI_n)A$ and $\det(cI_n) = c^n$ (Theorem 5.14(c)), so $\det(cA) = c^n \det A$ by (b). Finally $\det A^{k+1} = \det(A^k A) = \det A^k \cdot \det A$ by (b), which gives the powers by induction on k. □

Cofactor expansion and the transpose

Lemma 5.19 ([11, Four.I.3, Lemma 3.3])**.** Let $A_1, A_2, A_3 \in \mathbb{R}^{n\times n}$ agree in all rows except row i, and let $\operatorname{row}_i(A_3) = \operatorname{row}_i(A_1) + \operatorname{row}_i(A_2)$. Then $\det A_3 = \det A_1 + \det A_2$. Together with Definition 5.13(a): det is linear in each row when the other rows are held fixed.

Proof. Let ρ'_l ($l \neq i$, $\rho_l \in \mathbb{R}^n$) be the common rows, and let $u' = \operatorname{row}_i(A_1)$ and $v' = \operatorname{row}_i(A_2)$ ($u, v \in \mathbb{R}^n$).

- Let the $n-1$ vectors ρ_l be linearly dependent. The columns of A'_1 are u and the ρ_l (Definition 1.8), so they are dependent too (Proposition 3.13(b)). So A'_1 is not invertible (Theorems 4.1 and 2.16), neither is A_1 (Theorem 2.13(c)), and $\det A_1 = 0$ (Theorem 5.18(a)). Likewise $\det A_2 = \det A_3 = 0$.
- Otherwise extend the $n-1$ vectors ρ_l by one vector β to a basis of $\mathbb{R}^n$ (Theorem 4.5(b); a basis of $\mathbb{R}^n$ has n vectors, Theorem 4.4). Write $u = \sum_{l\neq i} x_l\rho_l + x_i\beta$ and $v = \sum_{l\neq i} y_l\rho_l + y_i\beta$; then $u + v = \sum_{l\neq i}(x_l + y_l)\rho_l + (x_i + y_i)\beta$.
- Let G have row i equal to β' and the rows ρ'_l elsewhere. Subtracting x_l times row l from row i for every $l \neq i$ turns A_1 into G with row i multiplied by x_i; so $\det A_1 = x_i \det G$ (Definition 5.13(b),(a)). Likewise $\det A_2 = y_i \det G$ and $\det A_3 = (x_i + y_i) \det G$. Hence $\det A_3 = \det A_1 + \det A_2$. □

Definition 5.20 ([1, §§5.2, 5.4], [11, Four.III.1, Definition 1.2], [2, §3.1.2], [13, Ch. VII, §2])**.** Let $n \geqslant 2$ and $A \in \mathbb{R}^{n\times n}$. For $1 \leqslant i, j \leqslant n$, $A(i|j) \in \mathbb{R}^{(n-1)\times(n-1)}$ is the matrix obtained from A by deleting row i and column j. The number $\det A(i|j)$ is the **minor** of a_{ij}, and $c_{ij} = (-1)^{i+j} \det A(i|j)$ is its **cofactor**.

The signs $(-1)^{i+j}$ form a checkerboard with + on the diagonal, for $n = 3$ $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$.

Theorem 5.21 ([1, §5.2, Theorem 1; §5.4], [4, Ch. VI, Theorem 2.1], [11, Four.III.1, Theorem 1.5], [2, (3.1)–(3.2)])**.** For each $n \geqslant 1$ there is exactly one determinant function on $\mathbb{R}^{n\times n}$. Let $n \geqslant 2$ and $A \in \mathbb{R}^{n\times n}$, with cofactors c_{ij}. For every column j and every row i,

$$\det A = a_{1j}c_{1j} + a_{2j}c_{2j} + \cdots + a_{nj}c_{nj} = a_{i1}c_{i1} + a_{i2}c_{i2} + \cdots + a_{in}c_{in}$$

(**expansion along column** j and **along row** i).

Proof. Existence and the expansion along a column, by induction on n.

- $n = 1$: the function $(a) \mapsto a$ satisfies Definition 5.13. Multiplying the only row by c multiplies the value by c; conditions (b) and (c) concern two different rows and say nothing for $n = 1$; and $(1) \mapsto 1$. By Theorem 5.15 it is the only determinant function.
- Let $n \geqslant 2$, and let det be the determinant function on $\mathbb{R}^{(n-1)\times(n-1)}$, which exists by the induction hypothesis. Fix a column j and let $E(A) = \sum_{i=1}^{n} a_{ij}c_{ij} = \sum_{i=1}^{n}(-1)^{i+j}a_{ij} \det A(i|j)$ for $A \in \mathbb{R}^{n\times n}$. We show that E is a determinant function on $\mathbb{R}^{n\times n}$. By Theorem 5.15 it is then the only one, so $E = \det$: a determinant function exists for n, and the expansion along column j holds.
- E is linear in each row k. For $i \neq k$, a_{ij} does not depend on row k, and $\det A(i|j)$ is linear in row k (Lemma 5.19; row k of A without its entry in column j is a row of $A(i|j)$). For $i = k$, $\det A(k|j)$ does not depend on row k, and a_{kj} is linear in it. In particular E satisfies Definition 5.13(a).
- If two adjacent rows k, $k+1$ of A are equal, then $E(A) = 0$. For $i \neq k, k+1$, $A(i|j)$ has two equal rows, so $\det A(i|j) = 0$ (Theorem 5.14(a)). The two remaining terms $(-1)^{k+j}a_{kj} \det A(k|j)$ and $(-1)^{k+1+j}a_{k+1,j} \det A(k+1|j)$ cancel, as $a_{kj} = a_{k+1,j}$ and $A(k|j) = A(k+1|j)$.
- An interchange of two adjacent rows changes the sign of E. Write $E(u, v)$ for the value of E when rows k, $k+1$ are u', v' $(u, v \in \mathbb{R}^n)$ and the other rows are fixed. By the first two items, $0 = E(u+v, u+v) = E(u, u) + E(u, v) + E(v, u) + E(v, v) = E(u, v) + E(v, u)$.
- An interchange of rows $k < l$ is a succession of $2(l-k)-1$ interchanges of adjacent rows (move row l up to position k, then the former row k down to position l), an odd number; so it changes the sign of E, which is Definition 5.13(c). In particular $E(A) = 0$ if two rows of A are equal.
- If B is A with λ times row s added to row r, then by linearity in row r, $E(B) = E(A) + \lambda E(A_s)$, where A_s is A with row r replaced by row s. As A_s has two equal rows, $E(A_s) = 0$ and $E(B) = E(A)$: Definition 5.13(b).
- Column j of I_n is e_j, and $I_n(j|j) = I_{n-1}$; so $E(I_n) = \det I_{n-1} = 1$: Definition 5.13(d).

The expansion along row i.

- Row i of A is $\sum_{j=1}^{n} a_{ij}e'_j$. As det is linear in row i (Lemma 5.19 and Definition 5.13(a)), $\det A = \sum_{j=1}^{n} a_{ij} \det G_j$, where G_j is A with row i replaced by e'_j.
- In G_j, subtract a_{kj} times row i from row k, for every $k \neq i$. As row i is e'_j, this makes entry (k, j) zero and leaves the other entries of row k unchanged; it does not change the determinant (Definition 5.13(b)). The resulting matrix H_j has column j equal to e_i, and $H_j(i|j) = A(i|j)$.
- Expanding $\det H_j$ along column j (proved above), whose only nonzero entry is the 1 in row i: $\det G_j = \det H_j = (-1)^{i+j} \det A(i|j) = c_{ij}$. Hence $\det A = \sum_{j=1}^{n} a_{ij}c_{ij}$. □

For $n = 2$, the expansion along row 1 gives $\det \left(\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{smallmatrix}\right) = a_{11}a_{22} - a_{12}a_{21}$, the number $ad - bc$ of the 2×2 inverse formula of Proposition 2.18; for $n = 3$ it gives [4, Ch. VI, §§1, 2]

$$\det A = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

In practice one expands along a row or column with many zeros; a row or column with a single nonzero entry is expanded at once. Since a determinant function exists, Theorem 5.15 shows that every reduction of A to an upper triangular matrix by interchanges and additions gives the same number $(-1)^t u_{11} \cdots u_{nn}$, namely $\det A$.

Theorem 5.22 ([11, Four.I.3, Theorem 3.15, Corollary 3.16], [1, §5.4], [2, Theorem 3.1])**.** For every $A \in \mathbb{R}^{n \times n}$, $\det A' = \det A$. Consequently Definition 5.13(a)–(c), Theorem 5.14(a),(b) and Lemma 5.19 hold with columns in place of rows, and the determinant of a lower triangular matrix is the product of its diagonal entries.

Proof. • Induction on n. For $n = 1$, $A' = A$. Let $n \geqslant 2$, and assume $\det B' = \det B$ for every $B \in \mathbb{R}^{(n-1)\times(n-1)}$.

- The $(k, 1)$ entry of A' is a_{1k}, and deleting row k and column 1 of A' leaves the transpose of $A(1|k)$: $A'(k|1) = A(1|k)'$. By the expansion of $\det A'$ along column 1 and the induction hypothesis, $\det A' = \sum_k (-1)^{k+1} a_{1k} \det A(1|k)' = \sum_k (-1)^{1+k} a_{1k} \det A(1|k)$, which is the expansion of $\det A$ along row 1 (Theorem 5.21).
- An operation on the columns of A is the same operation on the rows of A', and two equal columns or a zero column of A are two equal rows or a zero row of A'. As $\det A = \det A'$, the statements for rows give those for columns. If A is lower triangular, A' is upper triangular with the same diagonal, so $\det A = \det A' = a_{11} \cdots a_{nn}$ (Theorem 5.14(c)). □

In computing a determinant one may therefore use column operations as well as row operations.

Example 5.23 (cf. [2, §3.1.2, Example 3.4])**.** Let

$$A = \begin{pmatrix} 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

Compute $\det A$ by expansion along the first row, and check the value by expansion along the third column.

Solution. Row 1 has the nonzero entries $a_{11} = 2$ and $a_{13} = 1$, so $\det A = 2 \det A(1|1) + 1 \det A(1|3)$. Each 3×3 minor has a row with a single nonzero entry, and expanding along that row (Theorem 5.21),

$$\det A(1|1) = \det \begin{pmatrix} 3 & 0 & 2 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = (-1)^{3+3} \cdot 1 \cdot \det \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} = 6$$

$$\det A(1|3) = \det \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = (-1)^{2+2} \cdot 1 \cdot \det \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = -1$$

so $\det A = 2 \cdot 6 + 1 \cdot (-1) = 11$. Column 3 has the nonzero entries $a_{13} = 1$ and $a_{33} = 2$, with cofactors $c_{13} = \det A(1|3) = -1$ and

$$c_{33} = (-1)^{3+3} \det A(3|3) = \det \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 2 \\ 1 & 0 & 1 \end{pmatrix} = (-1)^{1+1} \cdot 2 \cdot \det \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix} = 6$$

(expansion along row 1). By Theorem 5.21, $\det A = 1 \cdot (-1) + 2 \cdot 6 = 11$.

Theorem 5.24 ([11, Four.III.1, Definition 1.8, Theorem 1.9], [2, §3.3.3], [5, Theorem 13.5.3, Corollary 13.5.4])**.** Let $A \in \mathbb{R}^{n\times n}$ with $n \geqslant 2$, and let c_{ij} be the cofactors of A. The **adjugate** of A is the transpose of the matrix of cofactors: $\operatorname{adj} A = (c_{ij})'$, that is, $(\operatorname{adj} A)_{ij} = c_{ji}$. Then

$$A \operatorname{adj} A = (\det A)\, I_n$$

Hence, if $\det A \neq 0$, then $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$.

Proof.
- The (i,k) entry of $A \operatorname{adj} A$ is $\sum_{j=1}^{n} a_{ij}(\operatorname{adj} A)_{jk} = \sum_{j=1}^{n} a_{ij}c_{kj}$.
- For $k = i$, this is the expansion of $\det A$ along row i (Theorem 5.21).
- For $k \neq i$, let $\tilde{A}$ be A with row k replaced by row i. Since $\tilde{A}(k|j) = A(k|j)$, the cofactors of row k of $\tilde{A}$ are $c_{k1}, \ldots, c_{kn}$, and the sum is the expansion of $\det \tilde{A}$ along row k. As $\tilde{A}$ has two equal rows, the sum is 0 (Theorem 5.14(a)).
- If $\det A \neq 0$, then $A\Big(\frac{1}{\det A} \operatorname{adj} A\Big) = I_n$, so $\frac{1}{\det A} \operatorname{adj} A$ is the inverse of A (Theorem 2.16). □

For $n = 2$, $\operatorname{adj} \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$, and the theorem gives again the formula of Proposition 2.18.

Remark 5.25 ([8, Ch. 13, Exercise 6])**.** Let $A \in \mathbb{R}^{n\times n}$, $n \geqslant 2$, be symmetric. Deleting row j and column i of $A = A'$ gives the transpose of $A(i|j)$: $A(j|i) = A'(j|i) = A(i|j)'$. By Theorem 5.22, $\det A(j|i) = \det A(i|j)$, so the cofactors satisfy $c_{ji} = c_{ij}$.

Example 5.26. Find A^{-1} by Theorem 5.24 for

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

Solution. The cofactors are

$$\begin{aligned}
c_{11} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{12} &= -\det \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1 \\
c_{13} &= \det \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{14} &= -\det \left(\begin{smallmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = 1 \\
c_{21} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{22} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1 \\
c_{23} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{24} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = -1 \\
c_{31} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{32} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{smallmatrix}\right) = -1 \\
c_{33} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{34} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \end{smallmatrix}\right) = 1 \\
c_{41} &= -\det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = -1, & c_{42} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = 1 \\
c_{43} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -1, & c_{44} &= \det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1
\end{aligned}$$

Each 3×3 minor here is triangular, so that its determinant is the product of its diagonal entries (Theorems 5.14(c) and 5.22), or has a row with a single nonzero entry, along which it is expanded (Theorem 5.21). Along row 1 (Theorem 5.21), $\det A = 1 \cdot c_{11} + 1 \cdot c_{12} + 0 \cdot c_{13} + 0 \cdot c_{14} = 2$. Hence

$$A^{-1} = \frac{1}{2} \operatorname{adj} A = \frac{1}{2} \begin{pmatrix} c_{11} & c_{21} & c_{31} & c_{41} \\ c_{12} & c_{22} & c_{32} & c_{42} \\ c_{13} & c_{23} & c_{33} & c_{43} \\ c_{14} & c_{24} & c_{34} & c_{44} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

Check: $\operatorname{row}_4(A) \operatorname{adj} A = (-1, 0, 0, 1) \operatorname{adj} A = (-1+1,\ 1-1,\ -1+1,\ 1+1) = (0, 0, 0, 2)$, the last row of $2I_4$.

Theorem 5.27 ([8, Ch. 13, Exercise 11], [1, §5.4], [4, Ch. VI, Theorem 4.1])**.** Let $A \in \mathbb{R}^{n\times n}$ $(n \geqslant 2)$ with $\det A \neq 0$, and let $b \in \mathbb{R}^n$. The system $Ax = b$ has exactly one solution x, and

$$x_j = \frac{\det A_j}{\det A} \qquad (1 \leqslant j \leqslant n)$$

where A_j is A with column j replaced by b (**Cramer's rule**).

Proof. • By Theorem 5.24, A is invertible with $A^{-1} = \frac{1}{\det A}\operatorname{adj} A$. If $Ax = b$, then $x = A^{-1}(Ax) = A^{-1}b$, so there is at most one solution; and $A(A^{-1}b) = b$, so $A^{-1}b$ is a solution. Hence the only solution is $x = \frac{1}{\det A}(\operatorname{adj} A)b$.

• Let c_{ij} be the cofactors of A. Entry j of $(\operatorname{adj} A)b$ is $\sum_{i=1}^n (\operatorname{adj} A)_{ji} b_i = \sum_{i=1}^n c_{ij} b_i$.

• A_j differs from A only in column j, so $A_j(i|j) = A(i|j)$, and the (i,j) entry of A_j is b_i. Expanding $\det A_j$ along column j (Theorem 5.21) gives

$$\det A_j = \sum_{i=1}^n b_i (-1)^{i+j} \det A_j(i|j) = \sum_{i=1}^n b_i (-1)^{i+j} \det A(i|j) = \sum_{i=1}^n c_{ij} b_i$$

Hence $x_j = \det A_j / \det A$. □

Example 5.28. For the matrix A of Example 5.26 and $b = (3,1,0,0)'$, find x_3 in $Ax = b$ by Cramer's rule, and check it against $x = A^{-1}b$ computed with that example.

Solution. Here $\det A = 2$, and A_3 is A with column 3 replaced by b. Row 3 of A_3 has the single nonzero entry 1, at position $(3,4)$; expanding along it and then along the last row of the 3×3 minor (Theorem 5.21),

$$\det A_3 = \det \begin{pmatrix} 1 & 1 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix} = (-1)^{3+4} \det \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \end{pmatrix} = -(-1)(-1)^{3+1} \det \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} = -2$$

So $x_3 = -2/2 = -1$ by Theorem 5.27. With $\operatorname{adj} A$ of Example 5.26 and $b = 3e_1 + e_2$, $(\operatorname{adj} A)b = 3\operatorname{col}_1(\operatorname{adj} A) + \operatorname{col}_2(\operatorname{adj} A)$ (Theorem 1.4(b)), so

$$x = \frac{1}{2}(\operatorname{adj} A)b = \frac{1}{2}\big(3(1,1,-1,1)' + (-1,1,1,-1)'\big) = \frac{1}{2}(2,4,-2,2)' = (1,2,-1,1)'$$

Its third entry is -1; indeed $Ax = (1+2,\ 2-1,\ -1+1,\ -1+1)' = b$.

Determinants of block matrices and the Schur complement

Theorem 5.29 ([6, Exercises 5.22, 5.25], [1, (5-20)], [5, Theorem 13.3.1, (13.3.5)])**.** Let $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$. Then

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \cdot \det D = \det \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$$

In particular $\det \operatorname{diag}(A, D) = \det A \cdot \det D$.

The diagonal blocks A and D are square, of possibly different sizes; for $p \neq q$ the off-diagonal blocks B and C are rectangular.

Proof. • The first row of $\left(\begin{smallmatrix} I_p & 0 \\ 0 & D \end{smallmatrix}\right)$ is $(1,0,\dots,0)$, and deleting row 1 and column 1 leaves $\left(\begin{smallmatrix} I_{p-1} & 0 \\ 0 & D \end{smallmatrix}\right)$; expanding along the first row (Theorem 5.21) p times gives $\det \left(\begin{smallmatrix} I_p & 0 \\ 0 & D \end{smallmatrix}\right) = \det D$. Likewise, expanding $\left(\begin{smallmatrix} A & 0 \\ 0 & I_q \end{smallmatrix}\right)$ q times along its last row $(0,\dots,0,1)$, whose entry 1 is a diagonal entry (sign $+1$), gives $\det A$.

- $\left(\begin{smallmatrix} I_p & B \\ 0 & I_q \end{smallmatrix}\right)$ is upper triangular with diagonal entries 1, so its determinant is 1 (Theorem 5.14(c)).
- By the block product formula (Theorem 5.3),

$$\begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix}\begin{pmatrix} I_p & B \\ 0 & I_q \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix}\begin{pmatrix} A & B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$$

  and by Theorem 5.18(b) its determinant is $\det D \cdot 1 \cdot \det A$.
- The transpose of $\left(\begin{smallmatrix} A & 0 \\ C & D \end{smallmatrix}\right)$ is $\left(\begin{smallmatrix} A' & C' \\ 0 & D' \end{smallmatrix}\right)$ (Theorem 5.3(b)); by Theorem 5.22 and the case just proved, its determinant is $\det A' \cdot \det D' = \det A \cdot \det D$. Finally, $\operatorname{diag}(A, D)$ is the case $B = 0$. □

Theorem 5.30 ([6, Exercise 5.30], [5, Theorem 13.3.8])**.** Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p\times p}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{q\times p}$ and $D \in \mathbb{R}^{q\times q}$.

(a) If A is invertible, then $\det M = \det A \cdot \det(D - CA^{-1}B)$.

(b) If D is invertible, then $\det M = \det D \cdot \det(A - BD^{-1}C)$.

In particular, if A is invertible, then M is invertible if and only if its Schur complement $D - CA^{-1}B$ is invertible.

Proof. (a) By Theorem 5.9, $\left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{smallmatrix}\right) M = \left(\begin{smallmatrix} A & B \\ 0 & D-CA^{-1}B \end{smallmatrix}\right)$. The left factor is lower triangular with diagonal entries 1, so its determinant is 1 (Theorem 5.22). By Theorems 5.18(b) and 5.29, $\det M = \det A \cdot \det(D - CA^{-1}B)$.

(b) By the block product formula, $\left(\begin{smallmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{smallmatrix}\right) M = \left(\begin{smallmatrix} A-BD^{-1}C & B-BD^{-1}D \\ C & D \end{smallmatrix}\right) = \left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ C & D \end{smallmatrix}\right)$. The left factor is upper triangular with diagonal entries 1 (Theorem 5.14(c)); as in (a), $\det M = \det(A - BD^{-1}C) \cdot \det D$.

If A is invertible, then $\det A \neq 0$, so by (a) $\det M \neq 0$ if and only if $\det(D - CA^{-1}B) \neq 0$; apply Theorem 5.18(a) to M and to $D - CA^{-1}B$. □

Here A and D are square, of possibly different sizes p and q; $B \in \mathbb{R}^{p\times q}$ and $C \in \mathbb{R}^{q\times p}$ are rectangular when $p \neq q$. For $p = q = 1$ and $a \neq 0$, (a) reads $\det\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = a\left(d - \frac{cb}{a}\right) = ad - bc$.

Theorem 5.31 ([5, Theorem 18.1.1, Corollary 18.1.3])**.** Let $A \in \mathbb{R}^{n\times n}$ be invertible and $u, v \in \mathbb{R}^n$. Then

$$\det(A + uv') = \det A \cdot (1 + v'A^{-1}u)$$

In particular $\det(I_n + uv') = 1 + v'u$. (This is the **matrix determinant lemma**.)

Proof. Apply Theorem 5.30 to $M = \left(\begin{smallmatrix} A & -u \\ v' & 1 \end{smallmatrix}\right) \in \mathbb{R}^{(n+1)\times(n+1)}$ ($p = n$, $q = 1$).

- By (a), with the invertible block A, whose Schur complement is the 1×1 matrix $1 - v'A^{-1}(-u)$: $\det M = \det A \cdot (1 + v'A^{-1}u)$.
- By (b), with the invertible block $D = (1)$: $\det M = \det(1) \cdot \det\big(A - (-u)(1)^{-1}v'\big) = \det(A + uv')$.
- Comparing gives the formula; for $A = I_n$, $A^{-1} = I_n$. □

Theorem 5.32 ([5, Theorems 18.1.1, 18.2.8, Corollary 18.2.10])**.** Let $A \in \mathbb{R}^{n\times n}$ and $T \in \mathbb{R}^{m\times m}$ be invertible, and let $U \in \mathbb{R}^{n\times m}$ and $V \in \mathbb{R}^{m\times n}$.

(a) $\det(A + UTV) = \det A \cdot \det T \cdot \det(T^{-1} + VA^{-1}U)$. Hence $A + UTV$ is invertible if and only if $T^{-1} + VA^{-1}U$ is invertible, and then

$$(A + UTV)^{-1} = A^{-1} - A^{-1}U(T^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

(the **Sherman–Morrison–Woodbury formula**).

(b) For $u, v \in \mathbb{R}^n$, $A + uv'$ is invertible if and only if $1 + v'A^{-1}u \neq 0$, and then

$$(A + uv')^{-1} = A^{-1} - \frac{1}{1 + v'A^{-1}u} A^{-1}uv'A^{-1}$$

Proof. (a) Let $M = \left(\begin{smallmatrix} A & -U \\ V & T^{-1} \end{smallmatrix}\right) \in \mathbb{R}^{(n+m)\times(n+m)}$.

- The Schur complement of A in M is $E = T^{-1} - VA^{-1}(-U) = T^{-1} + VA^{-1}U$, and that of T^{-1} is $F = A - (-U)(T^{-1})^{-1}V = A + UTV$.
- By Theorem 5.30(a),(b), $\det M = \det A \cdot \det E = \det T^{-1} \cdot \det F$. As $\det T^{-1} = 1/\det T$ (Theorem 5.18(c)), $\det F = \det A \cdot \det T \cdot \det E$.
- Since $\det A \neq 0$ and $\det T \neq 0$, $\det F \neq 0$ if and only if $\det E \neq 0$; so F is invertible if and only if E is (Theorem 5.18(a)).
- Let E and F be invertible. Theorem 5.12, with the blocks A, $-U$, V, T^{-1} of M in place of A, B, C, D, gives M^{-1} twice. Its upper left block is $A^{-1}+A^{-1}(-U)E^{-1}VA^{-1}$ by (a) of that theorem and F^{-1} by (b). An invertible matrix has only one inverse (Definition 2.12), so $F^{-1} = A^{-1} - A^{-1}UE^{-1}VA^{-1}$.

(b) Take $m = 1$, $U = u$, $T = (1)$ and $V = v'$ in (a). Then $T^{-1} + VA^{-1}U$ is the 1×1 matrix $1 + v'A^{-1}u$, which is invertible if and only if it is not 0, with inverse $1/(1 + v'A^{-1}u)$. □

For instance, let $G \in \mathbb{R}^{q\times n}$ with $G'G$ invertible, and let $x \in \mathbb{R}^n$. Appending the row x' to G changes $G'G$ into $G'G + xx'$: by the block transpose and the block product formula (Theorem 5.3(b),(c)), $\left(\begin{smallmatrix} G \\ x' \end{smallmatrix}\right)' \left(\begin{smallmatrix} G \\ x' \end{smallmatrix}\right) = [G' \;\; x] \left(\begin{smallmatrix} G \\ x' \end{smallmatrix}\right) = G'G + xx'$. So (b), with $A = G'G$ and $u = v = x$, gives $(G'G + xx')^{-1}$ from $(G'G)^{-1}$ without a new inversion. The condition of (b) holds: with $w = (G'G)^{-1}x$, $x'(G'G)^{-1}x = (G'Gw)'w = (Gw)'(Gw) \geqslant 0$ (Theorem 1.9), so $1 + x'(G'G)^{-1}x \geqslant 1$.

Example 5.33 ([2, Example 10.1])**.** Let $n \geqslant 2$ and $\rho \in \mathbb{R}$. The **equicorrelation matrix** is $R = (1 - \rho)I_n + \rho J_n \in \mathbb{R}^{n\times n}$, with diagonal entries 1 and all other entries ρ: if n variables have correlation ρ for every pair, their correlation matrix is R. Find $\det R$, the values of ρ for which R is invertible, and R^{-1} for these ρ.

Solution.

- If $\rho = 1$, then $R = J_n$ has equal rows ($n \geqslant 2$), so $\det R = 0$ (Theorem 5.14(a)).
- Let $\rho \neq 1$. As $J_n = \mathbf{1}_n\mathbf{1}_n'$, $R = A + uv'$ with $A = (1-\rho)I_n$, $u = \rho\mathbf{1}_n$ and $v = \mathbf{1}_n$. Here $\det A = (1-\rho)^n$ (Theorem 5.14(c)), $A^{-1} = (1 - \rho)^{-1}I_n$ and $v'A^{-1}u = \frac{\rho}{1-\rho}\mathbf{1}'\mathbf{1} = \frac{n\rho}{1-\rho}$ (Proposition 1.15). By Theorem 5.31,

  $$\det R = (1 - \rho)^n\Big(1 + \frac{n\rho}{1 - \rho}\Big) = (1 - \rho)^{n-1}(1 - \rho + n\rho) = (1 - \rho)^{n-1}\Big(1 + (n - 1)\rho\Big)$$

  This formula also gives 0 for $\rho = 1$.
- By Theorem 5.18(a), R is invertible if and only if $\det R \neq 0$, that is, $\rho \neq 1$ and $\rho \neq -1/(n - 1)$. For instance, for $n = 4$ and $\rho = -\frac{1}{3}$ every row of R has sum $1 + 3 \cdot (-\frac{1}{3}) = 0$, so $R\mathbf{1}_4 = 0$.
- For these ρ, $1 + v'A^{-1}u = 1 + \frac{n\rho}{1-\rho} = \frac{1+(n-1)\rho}{1-\rho} \neq 0$ and $A^{-1}uv'A^{-1} = \frac{\rho}{(1-\rho)^2}\mathbf{1}\mathbf{1}' = \frac{\rho}{(1-\rho)^2}J_n$. By Theorem 5.32(b),

  $$R^{-1} = \frac{1}{1 - \rho}I_n - \frac{1 - \rho}{1 + (n - 1)\rho} \cdot \frac{\rho}{(1 - \rho)^2}J_n = \frac{1}{1 - \rho}\Big(I_n - \frac{\rho}{1 + (n - 1)\rho}J_n\Big)$$

  For instance, for $n = 4$ and $\rho = \frac{1}{3}$, $R^{-1} = \frac{3}{2}\Big(I_4 - \frac{1}{6}J_4\Big) = \frac{3}{2}I_4 - \frac{1}{4}J_4$. Check, with $J_4^2 = 4J_4$ (Proposition 1.15): $\Big(\frac{2}{3}I_4 + \frac{1}{3}J_4\Big)\Big(\frac{3}{2}I_4 - \frac{1}{4}J_4\Big) = I_4 - \frac{1}{6}J_4 + \frac{1}{2}J_4 - \frac{1}{3}J_4 = I_4$.

Rank and the Schur complement

Theorem 5.34 ([5, Theorem 8.5.10], [6, Exercise 5.48])**.** Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$.

(a) If $A \in \mathbb{R}^{p\times p}$ is invertible, and $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{n\times p}$, $D \in \mathbb{R}^{n\times q}$, then $\operatorname{rank} M = p + \operatorname{rank}(D - CA^{-1}B)$.

(b) If $D \in \mathbb{R}^{q\times q}$ is invertible, and $A \in \mathbb{R}^{m\times p}$, $B \in \mathbb{R}^{m\times q}$, $C \in \mathbb{R}^{q\times p}$, then $\operatorname{rank} M = q + \operatorname{rank}(A - BD^{-1}C)$.

Proof. (a)

- By Theorem 5.9, $\left(\begin{smallmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{smallmatrix}\right) M \left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right) = \left(\begin{smallmatrix} A & 0 \\ 0 & D-CA^{-1}B \end{smallmatrix}\right)$, and the two outer factors are invertible (Proposition 5.8).
- Invertible factors do not change the rank (Theorem 4.15(d)), and the rank of a block-diagonal matrix is the sum of the ranks of its blocks (Theorem 4.16(c)); so $\operatorname{rank} M = \operatorname{rank} A + \operatorname{rank}(D - CA^{-1}B)$, where $\operatorname{rank} A = p$ (Theorem 4.13(d)).

(b)

- By the block product formula, $\left(\begin{smallmatrix} I_m & -BD^{-1} \\ 0 & I_q \end{smallmatrix}\right) M = \left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ C & D \end{smallmatrix}\right)$ (as in the proof of Theorem 5.30(b)), and multiplying on the right by $\left(\begin{smallmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{smallmatrix}\right)$ gives $\left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ C-DD^{-1}C & D \end{smallmatrix}\right) = \left(\begin{smallmatrix} A-BD^{-1}C & 0 \\ 0 & D \end{smallmatrix}\right)$.
- Both outer factors are invertible (Proposition 5.8); as in (a), $\operatorname{rank} M = \operatorname{rank}(A - BD^{-1}C) + \operatorname{rank} D = \operatorname{rank}(A - BD^{-1}C) + q$. □

Example 5.35. Find the rank of

$$M = \left(\begin{array}{cc|ccc} 1 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 \\ \hline 2 & 3 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 1 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbb{R}^{4\times 5}$$

Solution. Here $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has $\det A = 1$ and $A^{-1} = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ (Proposition 2.18), $B \in \mathbb{R}^{2\times 3}$, $C \in \mathbb{R}^{2\times 2}$ and $D \in \mathbb{R}^{2\times 3}$. Then

$$CA^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad CA^{-1}B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix}$$

and the Schur complement of A is

$$D - CA^{-1}B = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \end{pmatrix}$$

It is nonzero and its two rows are equal, so its row space is spanned by the nonzero row $(1, -1, 0)$ and its rank is 1 (Definition 4.11 and Proposition 3.13(a)). By Theorem 5.34(a), $\operatorname{rank} M = 2 + 1 = 3$. The equal rows of the Schur complement give a relation among the rows of M: by Theorem 5.9, the last two rows of $\left(\begin{smallmatrix} I_2 & 0 \\ -CA^{-1} & I_2 \end{smallmatrix}\right) M$ are those of $[0 \;\; D - CA^{-1}B]$, that is (Theorem 1.4(c)),

$$\operatorname{row}_3(M) - \operatorname{row}_1(M) - \operatorname{row}_2(M) = (0, 0, 1, -1, 0) = \operatorname{row}_4(M) - \operatorname{row}_2(M)$$

so $\operatorname{row}_3(M) = \operatorname{row}_1(M) + \operatorname{row}_4(M)$.

Rank and subdeterminants

Definition 5.36 ([3, §2.3])**.** Let $A \in \mathbb{R}^{m\times n}$ and $1 \leqslant k \leqslant \min\{m,n\}$. For rows $i_1 < \cdots < i_k$ and columns $j_1 < \cdots < j_k$ of A, the $k \times k$ matrix with (s,t) entry $a_{i_s j_t}$ is a $k \times k$ **submatrix** of A, and its determinant is a $k \times k$ **minor** of A.

Theorem 5.37 ([3, §2.3])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank r.

(a) Every $k \times k$ minor of A with $k > r$ is 0.

(b) If $r \geqslant 1$, then A has an $r \times r$ submatrix with nonzero determinant.

So if $A \neq 0$, then $\operatorname{rank} A$ is the largest k for which A has a nonzero $k \times k$ minor.

Proof. (a)

- Let A_* be the $k \times k$ submatrix in rows $i_1 < \cdots < i_k$ and columns $j_1 < \cdots < j_k$, with $k > r$. The k columns $a_{j_1}, \ldots, a_{j_k}$ of A lie in $\mathcal{C}(A)$, of dimension $r < k$, so they are linearly dependent (Theorem 4.5(c)): $c_1 a_{j_1} + \cdots + c_k a_{j_k} = 0$ for some $c = (c_1, \ldots, c_k)' \neq 0$.
- Entries $i_1, \ldots, i_k$ of this vector equation read $A_* c = 0$. So the columns of A_* are linearly dependent, $\operatorname{rank} A_* < k$ (Theorem 4.13(b)), and $\det A_* = 0$ (Theorem 5.18(a)).

(b)

- $\operatorname{rank} A' = r$ (Theorem 4.13(a)), so the pivot columns of A' are r linearly independent columns of A' (Theorem 4.8(a)): A has r linearly independent rows, say rows $i_1 < \cdots < i_r$.
- The matrix $B \in \mathbb{R}^{r\times n}$ formed by these rows has full row rank, $\operatorname{rank} B = r$ (Theorem 4.13(c)); so it has r linearly independent columns, say columns $j_1 < \cdots < j_r$ (its pivot columns, Theorem 4.8(a)).
- They form an $r \times r$ submatrix A_* of B, and so of A, with independent columns: $\operatorname{rank} A_* = r$ (Theorem 4.13(b)), and $\det A_* \neq 0$ (Theorem 5.18(a)).

By (a) and (b), no $k \times k$ minor with $k > r$ is nonzero, and some $r \times r$ minor is. □

Example 5.38 ([4, Ch. VI, §9, Examples 1, 2])**.** Find, by minors, the ranks of

$$B = \begin{pmatrix} 3 & 1 & 2 & 5 \\ 1 & 2 & -1 & 2 \\ 4 & 3 & 1 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 3 & 1 & 2 & 5 \\ 1 & 2 & -1 & 2 \\ 4 & 3 & 1 & 7 \end{pmatrix}$$

Solution. Both are 3×4, so their ranks are at most 3 (Theorem 4.13(a)).

- B: in the 3×3 submatrix in columns $1, 2, 3$, row 3, $(4, 3, 1)$, is the sum of rows 1 and 2. The operations $R_3 \leftarrow R_3 - R_1$, $R_3 \leftarrow R_3 - R_2$ do not change the determinant (Definition 5.13(b)) and give a zero row; so this minor is 0 (Theorem 5.14(b)). The minor in columns $2, 3, 4$ is, along row 1 (Theorem 5.21),

$$\det \begin{pmatrix} 1 & 2 & 5 \\ 2 & -1 & 2 \\ 3 & 1 & 1 \end{pmatrix} = 1 \cdot (-1-2) - 2 \cdot (2-6) + 5 \cdot (2+3) = -3 + 8 + 25 = 30 \neq 0$$

  So $\operatorname{rank} B = 3$ (Theorem 5.37).

- C: row 3 of C is the sum of rows 1 and 2, and so is row 3 of each of its four 3×3 submatrices; as for B, all four 3×3 minors are 0. The 2×2 minor in rows $1, 2$ and columns $1, 2$ is $3 \cdot 2 - 1 \cdot 1 = 5 \neq 0$. So $\operatorname{rank} C = 2$ (Theorem 5.37).

Permutations

Definition 5.39 ([1, §5.3], [11, Four.I.3, Definition 3.7, Example 3.9])**.** A **permutation** of degree n is a function φ from $\{1,\dots,n\}$ to itself that is one-to-one and onto: each of $1,\dots,n$ equals $\varphi(i)$ for exactly one i. We write $\varphi = \langle\varphi(1),\dots,\varphi(n)\rangle$; the **identity permutation** is $\mathrm{id} = \langle 1,2,\dots,n\rangle$. The **product** of permutations φ, ψ of degree n is the composite $\varphi\psi$, $(\varphi\psi)(i) = \varphi(\psi(i))$, again a permutation, and the **inverse** φ^{-1}, also a permutation, is given by $\varphi^{-1}(k) = i$ when $\varphi(i) = k$; thus $\varphi\varphi^{-1} = \varphi^{-1}\varphi = \mathrm{id}$.

There are $n! = n(n-1)\cdots 2\cdot 1$ permutations of degree n: $\varphi(1)$ can be chosen in n ways, then $\varphi(2)$ in $n-1$ ways, and so on. For $n=3$ they are $\langle 1,2,3\rangle$, $\langle 1,3,2\rangle$, $\langle 2,1,3\rangle$, $\langle 2,3,1\rangle$, $\langle 3,1,2\rangle$ and $\langle 3,2,1\rangle$.

Definition 5.40 ([11, Four.I.3, Example 3.11], [1, §5.3])**.** The **permutation matrix** of a permutation φ of degree n is the matrix $P_\varphi \in \mathbb{R}^{n\times n}$ with rows $e'_{\varphi(1)},\dots,e'_{\varphi(n)}$: its (i,j) entry is 1 if $j=\varphi(i)$ and 0 otherwise.

So P_φ has exactly one entry 1 in each row and in each column, and $P_{\mathrm{id}} = I_n$. The interchange matrix P_{rs} of Definition 2.14 is P_φ for the permutation φ that exchanges r and s and leaves the other numbers in place.

Proposition 5.41 ([1, §5.3, (5-16), Exercise 12(a), (c)], [11, Four.I.4, Exercise 4.17; Four.I.3, Exercise 3.33])**.** Let φ, ψ be permutations of degree n.

(a) For $A\in\mathbb{R}^{n\times p}$, row i of $P_\varphi A$ is row $\varphi(i)$ of A: $P_\varphi A$ lists the rows of A in the order $\varphi(1),\dots,\varphi(n)$.

(b) $P_\psi P_\varphi = P_{\varphi\psi}$.

(c) $P'_\varphi = P_{\varphi^{-1}}$, and $P'_\varphi P_\varphi = P_\varphi P'_\varphi = I_n$; so P_φ is invertible, with $P_\varphi^{-1} = P'_\varphi$.

Proof. (a) $\mathrm{row}_i(P_\varphi A) = \mathrm{row}_i(P_\varphi)A = e'_{\varphi(i)}A = \mathrm{row}_{\varphi(i)}(A)$ (Theorem 1.4(c)).

(b) By (a), row i of $P_\psi P_\varphi$ is row $\psi(i)$ of P_φ, which is $e'_{\varphi(\psi(i))} = e'_{(\varphi\psi)(i)}$, row i of $P_{\varphi\psi}$.

(c)

- The (i,j) entry of P'_φ is the (j,i) entry of P_φ. It is 1 if $i=\varphi(j)$, that is, $j=\varphi^{-1}(i)$, and 0 otherwise; this is the (i,j) entry of $P_{\varphi^{-1}}$.
- By (b), $P'_\varphi P_\varphi = P_{\varphi^{-1}}P_\varphi = P_{\varphi\varphi^{-1}} = P_{\mathrm{id}} = I_n$, and likewise $P_\varphi P'_\varphi = P_\varphi P_{\varphi^{-1}} = P_{\varphi^{-1}\varphi} = I_n$. □

If the rows of a data matrix X_0 hold the measurements on n units, $P_\varphi X_0$ lists the same units in another order, for instance sorted by one variable. By Theorem 1.9 and Proposition 5.41(c), $(P_\varphi X_0)'(P_\varphi X_0) = X_0'P'_\varphi P_\varphi X_0 = X_0'X_0$: reordering the units does not change $X_0'X_0$.

Example 5.42. (a) The data matrix X_0 below holds the measurements of two variables on four units. Sorting the units by the first variable puts them in the order $3,1,4,2$; so for $\varphi = \langle 3,1,4,2\rangle$ (Proposition 5.41(a))

$$X_0 = \begin{pmatrix} 2 & 7\\ 5 & 1\\ 1 & 4\\ 3 & 6\end{pmatrix},\qquad P_\varphi = \begin{pmatrix} 0 & 0 & 1 & 0\\ 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 1 & 0 & 0\end{pmatrix},\qquad P_\varphi X_0 = \begin{pmatrix} 1 & 4\\ 2 & 7\\ 3 & 6\\ 5 & 1\end{pmatrix}$$

(b) Let $\varphi = \langle 2,3,1\rangle$ and $\psi = \langle 2,1,3\rangle$. Then $\varphi\psi = \langle\varphi(2),\varphi(1),\varphi(3)\rangle = \langle 3,2,1\rangle$ and $\psi\varphi = \langle\psi(2),\psi(3),\psi(1)\rangle = \langle 1,3,2\rangle$, so $\varphi\psi \neq \psi\varphi$. Since $\varphi(3)=1$, $\varphi(1)=2$ and $\varphi(2)=3$, the inverse is $\varphi^{-1} = \langle 3,1,2\rangle$.

Exercises

Exercise 5.1. Let

$$A = \left(\begin{array}{cc|cc} 1 & 2 & 0 & -1 \\ 0 & 1 & 3 & 2 \\ \hline 2 & -1 & 1 & 0 \end{array}\right), \qquad B = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 3 & -1 \end{pmatrix}$$

where the rows of A are split into groups of 2 and 1 rows and its columns into groups of 2 and 2 columns.

(a) Split the columns of B into groups of 1 and 2 columns, and split its rows so that the partitions of A and B are conformal for multiplication. Write down the blocks A_{ij} and B_{jk}.

(b) Compute the blocks of AB by Theorem 5.3(c).

(c) Check the result against the ordinary product AB.

Solution. (a) The row groups of B must be the column groups of A, of 2 and 2 (Definition 5.2):

$$B = \left(\begin{array}{c|cc} 1 & 0 & 2 \\ -1 & 1 & 0 \\ \hline 2 & 0 & 1 \\ 0 & 3 & -1 \end{array}\right)$$

So $A_{11} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, $A_{12} = \left(\begin{smallmatrix} 0 & -1 \\ 3 & 2 \end{smallmatrix}\right)$, $A_{21} = (2,\ -1)$, $A_{22} = (1,\ 0)$, and $B_{11} = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$, $B_{12} = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 0 \end{smallmatrix}\right)$, $B_{21} = \left(\begin{smallmatrix} 2 \\ 0 \end{smallmatrix}\right)$, $B_{22} = \left(\begin{smallmatrix} 0 & 1 \\ 3 & -1 \end{smallmatrix}\right)$.

(b) AB is partitioned with the row groups of A (2 and 1 rows) and the column groups of B (1 and 2 columns), so its blocks are 2×1, 2×2, 1×1 and 1×2:

$$\begin{aligned} (AB)_{11} &= A_{11}B_{11} + A_{12}B_{21} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 6 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \\ (AB)_{12} &= A_{11}B_{12} + A_{12}B_{22} = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} -3 & 1 \\ 6 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 7 & 1 \end{pmatrix} \\ (AB)_{21} &= A_{21}B_{11} + A_{22}B_{21} = 3 + 2 = 5 \\ (AB)_{22} &= A_{21}B_{12} + A_{22}B_{22} = (-1,\ 4) + (0,\ 1) = (-1,\ 5) \end{aligned}$$

For instance, the first row of $A_{12}B_{22}$ is $0 \cdot (0,\ 1) + (-1)(3,\ -1) = (-3,\ 1)$ (Theorem 1.4(c)). Hence

$$AB = \left(\begin{array}{c|cc} -1 & -1 & 3 \\ 5 & 7 & 1 \\ \hline 5 & -1 & 5 \end{array}\right)$$

(c) By Theorem 1.4(c), $\text{row}_i(AB) = \text{row}_i(A)B$:

- $\text{row}_1(A)B = (1,0,2) + 2(-1,1,0) - (0,3,-1) = (-1,-1,3)$;
- $\text{row}_2(A)B = (-1,1,0) + 3(2,0,1) + 2(0,3,-1) = (5,7,1)$;
- $\text{row}_3(A)B = 2(1,0,2) - (-1,1,0) + (2,0,1) = (5,-1,5)$.

These are the rows found in (b).

Exercise 5.2 (cf. [6, Exercise 2.16])**.** Let A and B be the 3×5 matrices and C the 5×4 matrix partitioned as

$$A=\left(\begin{array}{ccc|cc} 1 & 3 & -2 & 1 & 2\\ 2 & 1 & 0 & -1 & 3\\ \hline 0 & 0 & 1 & 2 & 1 \end{array}\right),\quad B=\left(\begin{array}{ccc|cc} 1 & -3 & -2 & 2 & 1\\ 1 & 2 & 3 & 2 & 0\\ \hline 1 & 0 & 2 & 0 & 1 \end{array}\right),\quad C=\left(\begin{array}{cc|cc} 1 & 0 & 2 & 1\\ 0 & 2 & 0 & 0\\ -1 & 0 & 3 & 1\\ \hline 3 & 1 & 0 & 2\\ 2 & -1 & 1 & 1 \end{array}\right)$$

With blocks $A=(A_{st})$, $B=(B_{st})$, $C=(C_{st})$ $(s,t=1,2)$, show that

$$A+B=(A_{st}+B_{st}),\qquad A'=\begin{pmatrix}A'_{11} & A'_{21}\\ A'_{12} & A'_{22}\end{pmatrix},\qquad AC=\begin{pmatrix}A_{11}C_{11}+A_{12}C_{21} & A_{11}C_{12}+A_{12}C_{22}\\ A_{21}C_{11}+A_{22}C_{21} & A_{21}C_{12}+A_{22}C_{22}\end{pmatrix}$$

Solution. Adding entrywise,

$$A+B=\left(\begin{array}{ccc|cc} 2 & 0 & -4 & 3 & 3\\ 3 & 3 & 3 & 1 & 3\\ \hline 1 & 0 & 3 & 2 & 2 \end{array}\right)$$

and each block is the sum of the corresponding blocks of A and B. Transposing,

$$A'=\left(\begin{array}{cc|c} 1 & 2 & 0\\ 3 & 1 & 0\\ -2 & 0 & 1\\ \hline 1 & -1 & 2\\ 2 & 3 & 1 \end{array}\right)=\begin{pmatrix}A'_{11} & A'_{21}\\ A'_{12} & A'_{22}\end{pmatrix}$$

Multiplying in the ordinary way,

$$AC=\left(\begin{array}{cc|cc} 10 & 5 & -2 & 3\\ 5 & -2 & 7 & 3\\ \hline 7 & 1 & 4 & 6 \end{array}\right)$$

The blockwise products give the same four blocks:

$$\begin{aligned} A_{11}C_{11}+A_{12}C_{21}&=\begin{pmatrix}3 & 6\\ 2 & 2\end{pmatrix}+\begin{pmatrix}7 & -1\\ 3 & -4\end{pmatrix}=\begin{pmatrix}10 & 5\\ 5 & -2\end{pmatrix}\\ A_{11}C_{12}+A_{12}C_{22}&=\begin{pmatrix}-4 & -1\\ 4 & 2\end{pmatrix}+\begin{pmatrix}2 & 4\\ 3 & 1\end{pmatrix}=\begin{pmatrix}-2 & 3\\ 7 & 3\end{pmatrix}\\ A_{21}C_{11}+A_{22}C_{21}&=(-1,\ 0)+(8,\ 1)=(7,\ 1)\\ A_{21}C_{12}+A_{22}C_{22}&=(3,\ 1)+(1,\ 5)=(4,\ 6)\end{aligned}$$

For instance, $A_{12}C_{21}=\left(\begin{smallmatrix}1 & 2\\ -1 & 3\end{smallmatrix}\right)\left(\begin{smallmatrix}3 & 1\\ 2 & -1\end{smallmatrix}\right)=\left(\begin{smallmatrix}7 & -1\\ 3 & -4\end{smallmatrix}\right)$. This agrees with Theorem 5.3.

Exercise 5.3. Let

$$N=\left(\begin{array}{ccc|cc} 1 & 0 & 0 & 0 & 0\\ 1 & 1 & 0 & 0 & 0\\ 0 & 1 & 1 & 0 & 0\\ \hline 1 & 0 & 1 & 3 & 1\\ 0 & 1 & 1 & 2 & 1 \end{array}\right)=\begin{pmatrix}A & 0\\ C & D\end{pmatrix}$$

Here $A\in\mathbb{R}^{3\times 3}$, $C\in\mathbb{R}^{2\times 3}$ and $D\in\mathbb{R}^{2\times 2}$. Find N^{-1} by applying Proposition 5.6(c) to N', and check the answer by multiplication.

Solution. By Theorem 5.3(b), N' with groups $(3,2)$ is $N' = \left(\begin{smallmatrix} A' & C' \\ 0 & D' \end{smallmatrix}\right)$, upper block triangular, with

$$A' = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad C' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}, \qquad D' = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$$

The matrix A' is the block A_{22} of Example 5.7, so

$$(A')^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

and by Proposition 2.18 ($3 \cdot 1 - 2 \cdot 1 = 1$), $(D')^{-1} = \left(\begin{smallmatrix} 1 & -2 \\ -1 & 3 \end{smallmatrix}\right)$. Then

$$(A')^{-1}C' = \begin{pmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad -(A')^{-1}C'(D')^{-1} = -\begin{pmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 1 & -2 \\ 0 & -1 \end{pmatrix}$$

so by Proposition 5.6(c) with $p = 3$, $q = 2$

$$(N')^{-1} = \left(\begin{array}{ccc|cc} 1 & -1 & 1 & -2 & 4 \\ 0 & 1 & -1 & 1 & -2 \\ 0 & 0 & 1 & 0 & -1 \\ \hline 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & -1 & 3 \end{array}\right), \qquad N^{-1} = \left((N')^{-1}\right)' = \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 \\ \hline -2 & 1 & 0 & 1 & -1 \\ 4 & -2 & -1 & -2 & 3 \end{array}\right)$$

by Theorem 2.13(c). Check, with the blocks

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} -2 & 1 & 0 \\ 4 & -2 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 3}, \qquad D^{-1} = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix}$$

of N^{-1}: the $(2,1)$ block of NN^{-1} is

$$CA^{-1} + DR = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -2 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix} = 0$$

The $(1,2)$ block is $A \cdot 0 + 0 \cdot D^{-1} = 0$, and the diagonal blocks are $AA^{-1} = I_3$ and $DD^{-1} = I_2$. So $NN^{-1} = I_5$.

Exercise 5.4. Let

$$M = \left(\begin{array}{ccc|cc} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 4 & 2 \\ 0 & 1 & 2 & 2 & 3 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

Here $A \in \mathbb{R}^{3\times 3}$, $B \in \mathbb{R}^{3\times 2}$, $C \in \mathbb{R}^{2\times 3}$, $D \in \mathbb{R}^{2\times 2}$. Find the Schur complement E of A and $\det M$. Then compute M^{-1} with Corollary 5.10(a).

Solution. Here A is the block A_{22} of Example 5.7, so

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

and $\det A = 1$ (Theorem 5.14(c)). With $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{smallmatrix}\right)$ and $D = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 3 \end{smallmatrix}\right)$,

$$A^{-1}B = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad CA^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$E = D - CA^{-1}B = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

So $\det M = \det A \cdot \det E = 1 \cdot 1 = 1$ (Theorem 5.30(a)), and $E^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ by Proposition 2.18.

- With $L = \left(\begin{smallmatrix} I_3 & 0 \\ -CA^{-1} & I_2 \end{smallmatrix}\right)$ and $U = \left(\begin{smallmatrix} I_3 & -A^{-1}B \\ 0 & I_2 \end{smallmatrix}\right)$, Corollary 5.10(a), with E in place of S, gives $M^{-1} = U \operatorname{diag}(A^{-1}, E^{-1})L$.
- By the block product formula,

$$U \operatorname{diag}(A^{-1}, E^{-1}) = \begin{pmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{pmatrix} = \left(\begin{array}{ccc|cc} 1 & -1 & 1 & -2 & 2 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{array}\right)$$

  Here $-A^{-1}BE^{-1} = -\left(\begin{smallmatrix} 2 & 0 \\ -1 & 0 \\ 1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -2 & 2 \\ 1 & -1 \\ 0 & -1 \end{smallmatrix}\right) \in \mathbb{R}^{3\times 2}$. Multiplying on the right by L adds to block column 1 the block column 2 times $-CA^{-1} \in \mathbb{R}^{2\times 3}$:

$$M^{-1} = \left(\begin{array}{ccc|cc} 1 & -1 & 1 & -2 & 2 \\ 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{array}\right)\left(\begin{array}{ccc|cc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \hline -1 & 0 & -1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 1 \end{array}\right) = \left(\begin{array}{ccc|cc} 3 & -3 & 1 & -2 & 2 \\ -1 & 2 & -1 & 1 & -1 \\ 0 & 1 & 2 & 0 & -1 \\ \hline -1 & 1 & 0 & 1 & -1 \\ 1 & -2 & -1 & -1 & 2 \end{array}\right)$$

Check: $\operatorname{row}_1(M)M^{-1}$ is the sum of rows 1, 2, 4 of M^{-1}, namely $(3-1-1,\ -3+2+1,\ 1-1+0,\ -2+1+1,\ 2-1-1) = (1,0,0,0,0)$; and $\operatorname{row}_4(M)M^{-1} = (1,1,1,4,2)M^{-1} = (3-1+0-4+2,\ -3+2+1+4-4,\ 1-1+2+0-2,\ -2+1+0+4-2,\ 2-1-1-4+4) = (0,0,0,1,0)$.

Exercise 5.5. Let

$$M = \left(\begin{array}{cc|cc} 4 & 2 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ \hline 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 2 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

(a) Show that A is not invertible, so that Theorem 5.12(a) does not apply.

(b) Find D^{-1}, the Schur complement $F = A - BD^{-1}C$ of D, and $\det M$.

(c) Use Theorem 5.12(b) to find M^{-1}, and check the answer.

Solution. (a) Here $A = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$ and $D = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. Since $\det A = 4 \cdot 1 - 2 \cdot 2 = 0$, A is not invertible (Theorem 5.18(a)).

(b)

- $\det D = 1 \cdot 2 - 1 \cdot 1 = 1$, so $D^{-1} = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ (the 2×2 case of Theorem 5.24).
- Then

$$BD^{-1} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad BD^{-1}C = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

- So $F = A - BD^{-1}C = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Its determinant is $2 \cdot 0 - 1 \cdot 1 = -1$, so F is invertible, and by the same formula $F^{-1} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & -2 \end{smallmatrix}\right)$.
- By Theorem 5.30(b), $\det M = \det D \cdot \det F = 1 \cdot (-1) = -1$.

(c)

- The products in the formula of Theorem 5.12(b):

$$F^{-1}BD^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}$$
$$D^{-1}C = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$
$$D^{-1}CF^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$$
$$D^{-1}CF^{-1}BD^{-1} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix}$$

- So $D^{-1} + D^{-1}CF^{-1}BD^{-1} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and by Theorem 5.12(b)

$$M^{-1} = \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix} = \left(\begin{array}{cc|cc} 0 & 1 & 1 & -1 \\ 1 & -2 & -2 & 1 \\ \hline 1 & -2 & 0 & 0 \\ -1 & 1 & 0 & 1 \end{array}\right)$$

- Check: $\mathrm{row}_1(M)M^{-1} = 4(0,1,1,-1) + 2(1,-2,-2,1) + (1,-2,0,0) + 2(-1,1,0,1) = (1,0,0,0)$ and $\mathrm{row}_3(M)M^{-1} = (0,1,1,-1) + (1,-2,0,0) + (-1,1,0,1) = (0,0,1,0)$; rows 2 and 4 likewise give $(0,1,0,0)$ and $(0,0,0,1)$.

Exercise 5.6. Let

$$A = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 1 & 2 & 0 & -1 \\ 2 & 3 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1 & 2 & 1 \\ 2 & -1 & 3 & 4 \\ -1 & 3 & -4 & 2 \\ 3 & -2 & 5 & 5 \end{pmatrix}$$

(a) Compute $\det A$ by forward elimination (Theorem 5.15), keeping track of the interchanges.

(b) Compute $\det B$ in the same way.

(c) Which of A and B is invertible?

Solution. (a)

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 2 & 1 \\ 2 & 3 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 2 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 2 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 + 2R_2} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix} = U$$

- Since $a_{11} = 0$, Step 2 of Definition 2.5 interchanges rows 1 and 2. After the first column is cleared, the entry in position $(2,2)$ is 0, and rows 2 and 3 are interchanged.

- There are two interchanges, and the additions do not change the determinant. By Theorem 5.15, $\det A = (-1)^2 \cdot 1 \cdot (-1) \cdot 2 \cdot 3 = -6$.

(b)

$$B \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 2 & -2 & 3 \\ 0 & 1 & -1 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = U$$

There are no interchanges, and U is upper triangular with $u_{33} = u_{44} = 0$. So $\det B = 1 \cdot 1 \cdot 0 \cdot 0 = 0$ by Theorem 5.15.

(c) By Theorem 5.18(a), A is invertible, as $\det A = -6 \neq 0$, and B is not. Indeed $\mathrm{row}_4(B) = \mathrm{row}_1(B) + \mathrm{row}_2(B)$.

Exercise 5.7 ([2, Exercise 3.1])**.** Let

$$A = \begin{pmatrix} 1 & 5 & -5 \\ 3 & 2 & -5 \\ 6 & -2 & -5 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 6 & 5 \\ -2 & -3 & -5 \\ 2 & 3 & 9 \end{pmatrix}$$

Show that $\det A = -5$ and $\det B = 24$, in each case (i) by expansion along the first row and (ii) by row and column operations.

Solution. (A)

- (i) Along the first row (Theorem 5.21), $\det A = 1 \det \left(\begin{smallmatrix} 2 & -5 \\ -2 & -5 \end{smallmatrix}\right) - 5 \det \left(\begin{smallmatrix} 3 & -5 \\ 6 & -5 \end{smallmatrix}\right) - 5 \det \left(\begin{smallmatrix} 3 & 2 \\ 6 & -2 \end{smallmatrix}\right) = (-10 - 10) - 5(-15 + 30) - 5(-6 - 12) = -20 - 75 + 90 = -5$.
- (ii) Column 3 is -5 times $(1, 1, 1)'$. Taking the factor -5 out of column 3 (Definition 5.13(a) for columns, Theorem 5.22) and then subtracting row 1 from rows 2 and 3 (Definition 5.13(b)),

$$\det A = -5 \det \begin{pmatrix} 1 & 5 & 1 \\ 3 & 2 & 1 \\ 6 & -2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 5 & 1 \\ 3 & 2 & 1 \\ 6 & -2 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 5 & 1 \\ 2 & -3 & 0 \\ 5 & -7 & 0 \end{pmatrix}$$

  The last matrix has the single nonzero entry 1 in column 3, at position $(1, 3)$; expanding along column 3 (Theorem 5.21), its determinant is $(-1)^{1+3} \det \left(\begin{smallmatrix} 2 & -3 \\ 5 & -7 \end{smallmatrix}\right) = -14 + 15 = 1$. Hence $\det A = -5 \cdot 1 = -5$.

(B)

- (i) Along the first row, $\det B = 2 \det \left(\begin{smallmatrix} -3 & -5 \\ 3 & 9 \end{smallmatrix}\right) - 6 \det \left(\begin{smallmatrix} -2 & -5 \\ 2 & 9 \end{smallmatrix}\right) + 5 \det \left(\begin{smallmatrix} -2 & -3 \\ 2 & 3 \end{smallmatrix}\right) = 2(-27 + 15) - 6(-18 + 10) + 5(-6 + 6) = -24 + 48 + 0 = 24$.
- (ii)

$$B \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 2 & 6 & 5 \\ 0 & 3 & 0 \\ 0 & -3 & 4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 2 & 6 & 5 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

  No arrow changes the determinant (Theorem 5.15 with $t = 0$), so $\det B = 2 \cdot 3 \cdot 4 = 24$.

Exercise 5.8. Compute, by row and column operations,

$$\det\begin{pmatrix}1&2&-1&3\\2&1&0&3\\3&1&2&3\\1&0&1&3\end{pmatrix} \qquad \text{and} \qquad \det\begin{pmatrix}1&2&0&1\\2&5&1&2\\-1&-2&3&1\\1&2&6&7\end{pmatrix}$$

Solution. Call the two matrices A and B.

(A) Column 4 is 3 times $\mathbf{1}_4$. Taking the factor 3 out of column 4 (Definition 5.13(a) for columns, Theorem 5.22) and then subtracting row 1 from rows 2, 3, 4 (Definition 5.13(b)),

$$\det A = 3\det\begin{pmatrix}1&2&-1&1\\2&1&0&1\\3&1&2&1\\1&0&1&1\end{pmatrix} = 3\det\begin{pmatrix}1&2&-1&1\\1&-1&1&0\\2&-1&3&0\\0&-2&2&0\end{pmatrix}$$

The last matrix has the single nonzero entry 1 in column 4, at position $(1,4)$. Expanding along column 4 and then the 3×3 determinant along its first row (Theorem 5.21),

$$\det A = 3\cdot(-1)^{1+4}\det\begin{pmatrix}1&-1&1\\2&-1&3\\0&-2&2\end{pmatrix} = -3\Big(1(-2+6)+1(4-0)+1(-4-0)\Big) = -3\cdot 4 = -12$$

(B)

$$B \xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-2R_1,\ R_3\leftarrow R_3+R_1} \begin{pmatrix}1&2&0&1\\0&1&1&0\\0&0&3&2\\0&0&6&6\end{pmatrix} \xrightarrow{R_4\leftarrow R_4-2R_3} \begin{pmatrix}1&2&0&1\\0&1&1&0\\0&0&3&2\\0&0&0&2\end{pmatrix}$$

No arrow changes the determinant (Theorem 5.15 with $t=0$), so $\det B = 1\cdot 1\cdot 3\cdot 2 = 6$.

Exercise 5.9. Let $A, B \in \mathbb{R}^{4\times 4}$ with $\det A = 2$ and $\det B = -3$. Find

(a) $\det(AB)$, $\det A'$ and $\det(A'B)$;

(b) $\det(2A)$, $\det(-B)$ and $\det(-2AB)$;

(c) $\det A^{-1}$, $\det(A^{-1}B)$ and $\det(B^{-1}AB)$;

(d) $\det(A^3B^2)$, and the real numbers c with $\det(cA) = 162$.

Solution. (a) By Theorem 5.18(b), $\det(AB) = \det A\cdot\det B = 2\cdot(-3) = -6$. By Theorem 5.22, $\det A' = \det A = 2$, so $\det(A'B) = \det A'\cdot\det B = -6$.

(b) By Theorem 5.18(c) with $n=4$, $\det(2A) = 2^4\det A = 32$ and $\det(-B) = (-1)^4\det B = -3$. With (a) of this exercise, $\det(-2AB) = (-2)^4\det(AB) = 16\cdot(-6) = -96$.

(c) $\det A \neq 0$ and $\det B \neq 0$, so A and B are invertible (Theorem 5.18(a)). By Theorem 5.18(b),(c),

$$\det A^{-1} = \frac{1}{\det A} = \frac{1}{2}, \qquad \det(A^{-1}B) = \det A^{-1}\cdot\det B = \frac{1}{2}\cdot(-3) = -\frac{3}{2}$$
$$\det(B^{-1}AB) = \det B^{-1}\cdot\det A\cdot\det B = -\frac{1}{3}\cdot 2\cdot(-3) = 2$$

(d) By Theorem 5.18(b),(c), $\det(A^3B^2) = (\det A)^3(\det B)^2 = 8\cdot 9 = 72$. Also $\det(cA) = c^4\det A = 2c^4$, and $2c^4 = 162$ gives $c^4 = 81$, that is, $(c^2-9)(c^2+9) = 0$. As $c^2+9>0$, $c^2 = 9$: $c = 3$ or $c = -3$.

Exercise 5.10. Let $A, B \in \mathbb{R}^{4\times 4}$ with $\det A = 2$ and $\det B = -3$. Compute $\det(3A)$, $\det(A'B^{-1})$, $\det(A^2B')$, $\det(B^{-1}AB)$, $\det(-B)$ and $\det(\operatorname{adj} A)$.

Solution. By Theorems 5.18(b),(c) and 5.22 (B is invertible since $\det B \neq 0$):

$$\det(3A) = 3^4 \cdot 2 = 162, \qquad \det(A'B^{-1}) = \det A \cdot \frac{1}{\det B} = -\frac{2}{3}$$
$$\det(A^2B') = 2^2 \cdot (-3) = -12, \quad \det(B^{-1}AB) = \frac{1}{-3} \cdot 2 \cdot (-3) = 2$$
$$\det(-B) = (-1)^4(-3) = -3$$

By Theorem 5.24, $A \operatorname{adj} A = 2I_4$, so $\det A \cdot \det(\operatorname{adj} A) = \det(2I_4) = 2^4$ (Theorems 5.18(b) and 5.14(c)) and $\det(\operatorname{adj} A) = 16/2 = 8$.

Exercise 5.11. Let

$$A = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 2 & -1 & 4 & 1 \\ 1 & 2 & -2 & 0 \\ 0 & 1 & 0 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & -1 & 2 & 3 \\ -2 & 3 & 0 & -2 \\ 2 & 0 & 0 & -2 \\ 1 & 0 & 3 & 2 \end{pmatrix}$$

(a) Compute $\det A$ by expansion along the third column, and check the value by expansion along the first row; expand each 3×3 minor along a row or a column that contains a zero.

(b) Compute $\det B$ by expansion along the third row, expanding each 3×3 minor along a row or a column that contains a zero.

(c) Check the value of $\det B$ by expansion along the second column.

Solution. (a) Column 3 of A is $(0, 4, -2, 0)'$, so by Theorem 5.21, $\det A = 4c_{23} + (-2)c_{33}$. Expanding the first 3×3 minor along its third column $(0, 0, 2)'$ and the second along its first row $(3, 1, 0)$,

$$c_{23} = (-1)^{2+3} \det \begin{pmatrix} 3 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} = -2 \det \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} = -2 \cdot 5 = -10$$
$$c_{33} = (-1)^{3+3} \det \begin{pmatrix} 3 & 1 & 0 \\ 2 & -1 & 1 \\ 0 & 1 & 2 \end{pmatrix} = 3 \det \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = 3 \cdot (-3) - 4 = -13$$

Hence $\det A = 4 \cdot (-10) + (-2) \cdot (-13) = -40 + 26 = -14$. Along the first row $(3, 1, 0, 0)$, $\det A = 3c_{11} + 1 \cdot c_{12}$; expanding the first minor along its second row $(2, -2, 0)$ and the second along its third row $(0, 0, 2)$,

$$c_{11} = (-1)^{1+1} \det \begin{pmatrix} -1 & 4 & 1 \\ 2 & -2 & 0 \\ 1 & 0 & 2 \end{pmatrix} = -2 \det \begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix} + (-2) \det \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix} = -16 + 6 = -10$$
$$c_{12} = (-1)^{1+2} \det \begin{pmatrix} 2 & 4 & 1 \\ 1 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = -2 \det \begin{pmatrix} 2 & 4 \\ 1 & -2 \end{pmatrix} = -2 \cdot (-8) = 16$$

So $\det A = 3 \cdot (-10) + 1 \cdot 16 = -14$, as before.

(b)

- Row 3 of B is $(2, 0, 0, -2)$, so only the cofactors c_{31} and c_{34} are needed: $\det B = 2c_{31} + (-2)c_{34}$, where

$$c_{31} = (-1)^{3+1} \det \begin{pmatrix} -1 & 2 & 3 \\ 3 & 0 & -2 \\ 0 & 3 & 2 \end{pmatrix}, \qquad c_{34} = (-1)^{3+4} \det \begin{pmatrix} 2 & -1 & 2 \\ -2 & 3 & 0 \\ 1 & 0 & 3 \end{pmatrix}$$

- The first minor, along its first column $(-1,3,0)'$: $(-1)\det\left(\begin{smallmatrix}0&-2\\3&2\end{smallmatrix}\right)-3\det\left(\begin{smallmatrix}2&3\\3&2\end{smallmatrix}\right)+0 = -1\cdot 6-3\cdot(-5) = 9$. So $c_{31}=9$.
- The second minor, along its second row $(-2,3,0)$: $-(-2)\det\left(\begin{smallmatrix}-1&2\\0&3\end{smallmatrix}\right)+3\det\left(\begin{smallmatrix}2&2\\1&3\end{smallmatrix}\right)-0 = 2\cdot(-3)+3\cdot 4 = 6$. So $c_{34}=-6$.
- Hence $\det B = 2\cdot 9+(-2)\cdot(-6) = 18+12 = 30$.

(c) Column 2 of B is $(-1,3,0,0)'$, so $\det B = (-1)c_{12}+3c_{22}$.

- The two minors are

$$B(1|2)=\begin{pmatrix}-2&0&-2\\2&0&-2\\1&3&2\end{pmatrix}, \qquad B(2|2)=\begin{pmatrix}2&2&3\\2&0&-2\\1&3&2\end{pmatrix}$$

- Along the second column $(0,0,3)'$ of $B(1|2)$, $\det B(1|2) = 3\cdot(-1)^{3+2}\det\left(\begin{smallmatrix}-2&-2\\2&-2\end{smallmatrix}\right) = -3\cdot 8 = -24$, so $c_{12} = (-1)^{1+2}\cdot(-24) = 24$.
- Along the second row $(2,0,-2)$ of $B(2|2)$, $\det B(2|2) = -2\det\left(\begin{smallmatrix}2&3\\3&2\end{smallmatrix}\right)+0-(-2)\det\left(\begin{smallmatrix}2&2\\1&3\end{smallmatrix}\right) = -2\cdot(-5)+2\cdot 4 = 18$, so $c_{22}=(-1)^{2+2}\cdot 18 = 18$.
- Hence $\det B = (-1)\cdot 24+3\cdot 18 = 30$, as in (b).

Exercise 5.12 ([8, Ch. 13, Exercise 5(b)])**.** Compute the determinant of

$$A=\begin{pmatrix}0&4&0&5\\1&0&-1&2\\0&3&0&-2\\0&0&-6&0\end{pmatrix}$$

(a) by repeated cofactor expansion: expand $\det A$ in terms of determinants of 3×3 matrices, these in terms of determinants of 2×2 matrices, and finally these in terms of determinants of 1×1 matrices; (b) by row reduction to triangular form.

Solution. (a)

- Column 1 has the single nonzero entry $a_{21}=1$; expanding along column 1 (Theorem 5.21),

$$\det A = 1\cdot(-1)^{2+1}\det\begin{pmatrix}4&0&5\\3&0&-2\\0&-6&0\end{pmatrix}$$

- Row 3 of this 3×3 matrix has the single nonzero entry -6, in column 2:

$$\det\begin{pmatrix}4&0&5\\3&0&-2\\0&-6&0\end{pmatrix} = (-6)(-1)^{3+2}\det\begin{pmatrix}4&5\\3&-2\end{pmatrix}$$

- Expanding the 2×2 determinant along its first row, $4(-1)^{1+1}\det(-2)+5(-1)^{1+2}\det(3) = -8-15 = -23$.
- Hence $\det A = (-1)\cdot 6\cdot(-23) = 138$.

(b)

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 4 & 0 & 5 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 3R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & -23 \\ 0 & 0 & -6 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & -23 \end{pmatrix}$$

There are two interchanges, and the additions do not change the determinant (Theorem 5.15, $t = 2$); so $\det A = (-1)^2 \cdot 1 \cdot 1 \cdot (-6) \cdot (-23) = 138$.

Exercise 5.13. For $t \in \mathbb{R}$ let

$$A(t) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & t & 1 & 2-t \\ 1 & 1 & t & 2-t \\ 1 & t & t & 5-t \end{pmatrix}$$

(a) Compute $\det A(t)$ by row operations, and factor it.

(b) For which t is $A(t)$ invertible?

(c) Find $\operatorname{rank} A(t)$ for each t for which $A(t)$ is not invertible.

Solution. (a)

$$A(t) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & t-1 & 0 & 1-t \\ 0 & 0 & t-1 & 1-t \\ 0 & t-1 & t-1 & 4-t \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & t-1 & 0 & 1-t \\ 0 & 0 & t-1 & 1-t \\ 0 & 0 & t-1 & 3 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & t-1 & 0 & 1-t \\ 0 & 0 & t-1 & 1-t \\ 0 & 0 & 0 & t+2 \end{pmatrix}$$

using $(4-t)-(1-t) = 3$ and $3-(1-t) = t+2$. Every operation adds a multiple of a row to another row, for every value of t, and there are no interchanges; the last matrix is upper triangular. By Theorem 5.15,

$$\det A(t) = 1 \cdot (t-1) \cdot (t-1) \cdot (t+2) = (t-1)^2(t+2)$$

(b) By Theorem 5.18(a), $A(t)$ is invertible if and only if $\det A(t) \neq 0$, that is, $t \neq 1$ and $t \neq -2$.

(c) The operations of (a) do not depend on t, so for $t = 1$ and for $t = -2$ they take $A(t)$ to the last matrix of (a) with that value of t.

- For $t = 1$ its rows 2 and 3 are zero, and row 4 is not (the second and third arrows of (a) subtract these zero rows and change nothing). The next pivot column is column 4 (Step 1 of Definition 2.5), and Step 2 interchanges rows 2 and 4:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- For $t = -2$ it is

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & -3 & 0 & 3 \\ 0 & 0 & -3 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- The matrix after the interchange ($t = 1$) and this matrix ($t = -2$) are the results of forward elimination of $A(1)$ and $A(-2)$ (Definition 2.5, with the scaling of Step 3 postponed), and rank $A(t)$ is their number of nonzero rows (Definition 4.11 and Remark 4.2). So rank $A(1) = 2$ and rank $A(-2) = 3$.

Exercise 5.14 ([6, Exercise 4.31(a)])**.** Let $\alpha \in \mathbb{R}$ and

$$A = \begin{pmatrix} 1 & \alpha & 0 \\ 4 & 5 & 3 \\ 1 & 0 & 2 \end{pmatrix}$$

For which α is $\det A = 0$? Show that the columns of A are linearly dependent in that case.

Solution. Expanding along the first row (Theorem 5.21), $\det A = 1 \det \left(\begin{smallmatrix} 5 & 3 \\ 0 & 2 \end{smallmatrix}\right) - \alpha \det \left(\begin{smallmatrix} 4 & 3 \\ 1 & 2 \end{smallmatrix}\right) = 10 - 5\alpha$, so $\det A = 0$ exactly when $\alpha = 2$ (and A is invertible exactly when $\alpha \neq 2$, Theorem 5.18(a)). For $\alpha = 2$ and $x = (2, -1, -1)'$, $Ax = (2 - 2 + 0,\ 8 - 5 - 3,\ 2 - 0 - 2)' = 0$; that is, $2a_1 - a_2 - a_3 = 0$, a nontrivial relation among the columns.

Exercise 5.15. Let

$$A = \begin{pmatrix} 3 & -1 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 4 & -2 & 3 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} -1 \\ 0 \\ -3 \\ 0 \end{pmatrix}$$

(a) Compute the sixteen cofactors c_{ij} of A, and $\det A$ by expansion along the second row.

(b) Find $\operatorname{adj} A$ and A^{-1} (Theorem 5.24), and check that $AA^{-1} = I_4$.

(c) Solve $Ax = b$ by Cramer's rule (Theorem 5.27), and check the solution with $A^{-1}b$.

Solution. (a) By Definition 5.20,

$$\begin{aligned}
c_{11} &= \det \left(\begin{smallmatrix} 0 & 1 & 0 \\ -2 & 3 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = 2, & c_{12} &= -\det \left(\begin{smallmatrix} 1 & 1 & 0 \\ 4 & 3 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = 1 \\
c_{13} &= \det \left(\begin{smallmatrix} 1 & 0 & 0 \\ 4 & -2 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -2, & c_{14} &= -\det \left(\begin{smallmatrix} 1 & 0 & 1 \\ 4 & -2 & 3 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 2 \\
c_{21} &= -\det \left(\begin{smallmatrix} -1 & 2 & 0 \\ -2 & 3 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = -1, & c_{22} &= \det \left(\begin{smallmatrix} 3 & 2 & 0 \\ 4 & 3 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = 1 \\
c_{23} &= -\det \left(\begin{smallmatrix} 3 & -1 & 0 \\ 4 & -2 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 2, & c_{24} &= \det \left(\begin{smallmatrix} 3 & -1 & 2 \\ 4 & -2 & 3 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -2 \\
c_{31} &= \det \left(\begin{smallmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = -1, & c_{32} &= -\det \left(\begin{smallmatrix} 3 & 2 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) = -1 \\
c_{33} &= \det \left(\begin{smallmatrix} 3 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 1, & c_{34} &= -\det \left(\begin{smallmatrix} 3 & -1 & 2 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right) = -1 \\
c_{41} &= -\det \left(\begin{smallmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & 3 & 0 \end{smallmatrix}\right) = 0, & c_{42} &= \det \left(\begin{smallmatrix} 3 & 2 & 0 \\ 1 & 1 & 0 \\ 4 & 3 & 0 \end{smallmatrix}\right) = 0 \\
c_{43} &= -\det \left(\begin{smallmatrix} 3 & -1 & 0 \\ 1 & 0 & 0 \\ 4 & -2 & 0 \end{smallmatrix}\right) = 0, & c_{44} &= \det \left(\begin{smallmatrix} 3 & -1 & 2 \\ 1 & 0 & 1 \\ 4 & -2 & 3 \end{smallmatrix}\right) = 1
\end{aligned}$$

- Each of the twelve minors with $i \leqslant 3$ has a third row or a third column whose only nonzero entry is the 1 in position $(3, 3)$; expanding along it (Theorem 5.21) leaves the 2×2 determinant of its first two rows and columns. For instance, $\det \left(\begin{smallmatrix} 0 & 1 \\ -2 & 3 \end{smallmatrix}\right) = 0 + 2 = 2$ for c_{11}.
- The minors $\det A(4|j)$ with $j \leqslant 3$ have a zero third column, so they are 0 (Theorem 5.14(b) for columns, Theorem 5.22).
- $c_{44} = \det A(4|4)$, along its second row $(1, 0, 1)$, is $-\det \left(\begin{smallmatrix} -1 & 2 \\ -2 & 3 \end{smallmatrix}\right) - \det \left(\begin{smallmatrix} 3 & -1 \\ 4 & -2 \end{smallmatrix}\right) = -1 + 2 = 1$.
- Along row 2 of A (Theorem 5.21), $\det A = 1 \cdot c_{21} + 0 \cdot c_{22} + 1 \cdot c_{23} + 0 \cdot c_{24} = -1 + 2 = 1$.

(b) Since $\det A = 1$, Theorem 5.24 gives

$$A^{-1} = \operatorname{adj} A = \begin{pmatrix} c_{11} & c_{21} & c_{31} & c_{41} \\ c_{12} & c_{22} & c_{32} & c_{42} \\ c_{13} & c_{23} & c_{33} & c_{43} \\ c_{14} & c_{24} & c_{34} & c_{44} \end{pmatrix} = \begin{pmatrix} 2 & -1 & -1 & 0 \\ 1 & 1 & -1 & 0 \\ -2 & 2 & 1 & 0 \\ 2 & -2 & -1 & 1 \end{pmatrix}$$

Check: $\operatorname{row}_1(A)A^{-1} = 3(2,-1,-1,0) - (1,1,-1,0) + 2(-2,2,1,0) = (1,0,0,0)$, $\operatorname{row}_2(A)A^{-1} = (2,-1,-1,0)+(-2,2,1,0) = (0,1,0,0)$, $\operatorname{row}_3(A)A^{-1} = 4(2,-1,-1,0)-2(1,1,-1,0)+3(-2,2,1,0) = (0,0,1,0)$ and $\operatorname{row}_4(A)A^{-1} = (-2,2,1,0) + (2,-2,-1,1) = (0,0,0,1)$ (Theorem 1.4(c)).

(c)

- As $\det A = 1 \neq 0$, Theorem 5.27 applies. Replacing column j of A by b gives

$$A_1 = \begin{pmatrix} -1 & -1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & -2 & 3 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 3 & -1 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 4 & -3 & 3 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

$$A_3 = \begin{pmatrix} 3 & -1 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 4 & -2 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad A_4 = \begin{pmatrix} 3 & -1 & 2 & -1 \\ 1 & 0 & 1 & 0 \\ 4 & -2 & 3 & -3 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

- Each second row contains zeros; expanding along it (Theorem 5.21),

$$\det A_1 = 1 \cdot (-1)^{2+3} \det \begin{pmatrix} -1 & -1 & 0 \\ -3 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = -(2-3) = 1$$

$$\det A_2 = 1 \cdot (-1)^{2+1} \det \begin{pmatrix} -1 & 2 & 0 \\ -3 & 3 & 0 \\ 0 & 1 & 1 \end{pmatrix} + 1 \cdot (-1)^{2+3} \det \begin{pmatrix} 3 & -1 & 0 \\ 4 & -3 & 0 \\ 0 & 0 & 1 \end{pmatrix} = -(-3+6) - (-9+4) = 2$$

$$\det A_3 = 1 \cdot (-1)^{2+1} \det \begin{pmatrix} -1 & -1 & 0 \\ -2 & -3 & 0 \\ 0 & 0 & 1 \end{pmatrix} = -(3-2) = -1$$

$$\det A_4 = 1 \cdot (-1)^{2+1} \det \begin{pmatrix} -1 & 2 & -1 \\ -2 & 3 & -3 \\ 0 & 1 & 0 \end{pmatrix} + 1 \cdot (-1)^{2+3} \det \begin{pmatrix} 3 & -1 & -1 \\ 4 & -2 & -3 \\ 0 & 0 & 0 \end{pmatrix} = -(-1) - 0 = 1$$

  Here the 3×3 determinants with the third column $(0,0,1)'$ are 2×2 determinants, as in (a); the last one has a zero row, so it is 0 (Theorem 5.14(b)); and the one before it, along its third row $(0,1,0)$, is $(-1)^{3+2} \det \left(\begin{smallmatrix} -1 & -1 \\ -2 & -3 \end{smallmatrix}\right) = -(3-2) = -1$.

- Hence $x_1 = \frac{1}{1}$, $x_2 = \frac{2}{1}$, $x_3 = \frac{-1}{1}$, $x_4 = \frac{1}{1}$, that is, $x = (1,2,-1,1)'$. Check: $A^{-1}b = (-2+0+3+0,\ -1+0+3+0,\ 2+0-3+0,\ -2+0+3+0)' = (1,2,-1,1)'$.

Exercise 5.16 (cf. [8, Ch. 13, Exercise 9])**.** Let

$$T = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

(a) Compute all cofactors of T.

(b) Compute $\det T$ by expansion along the second row, and check the answer by expansion along the fourth column.

(c) Use Theorem 5.24 to show that T^{-1} is the matrix K with entries $k_{ij} = \min(i, j)$.

(d) For $b = (1, 2, 3, 4)'$, find x_2 in $Tx = b$ by Cramer's rule, and check it with $x = Kb$.

Solution. (a) T is symmetric, so $c_{ji} = c_{ij}$ (Remark 5.25). The ten cofactors with $i \leqslant j$ are (each minor is expanded along a row or column with at most two nonzero entries, Theorem 5.21)

$$c_{11} = \det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} = 2 \cdot 1 + 1 \cdot (-1) = 1, \quad c_{12} = -\det \begin{pmatrix} -1 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} = -(-1) = 1$$

$$c_{13} = \det \begin{pmatrix} -1 & 2 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = (-1)(-1)(1) = 1, \quad c_{14} = -\det \begin{pmatrix} -1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix} = -(-1) = 1$$

$$c_{22} = \det \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} = 2(2 - 1) = 2, \quad c_{23} = -\det \begin{pmatrix} 2 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = -(-2) = 2$$

$$c_{24} = \det \begin{pmatrix} 2 & -1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix} = 2(-1)(-1) = 2, \quad c_{33} = \det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 4 - 1 = 3$$

$$c_{34} = -\det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & 0 & -1 \end{pmatrix} = -(-3) = 3, \quad c_{44} = \det \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} = 2 \cdot 3 + 1 \cdot (-2) = 4$$

(the triangular minors by Theorem 5.14(c)). So $c_{ij} = \min(i, j)$ for all i, j.

(b) Along row 2: $\det T = (-1)c_{21} + 2c_{22} + (-1)c_{23} = -1 + 4 - 2 = 1$. Along column 4: $\det T = (-1)c_{34} + 1 \cdot c_{44} = -3 + 4 = 1$ (Theorem 5.21).

(c) Since $\det T = 1$, Theorem 5.24 gives $T^{-1} = \operatorname{adj} T = (c_{ij})'$, and $(c_{ij})' = (c_{ji}) = (\min(i, j)) = K$ by (a):

$$T^{-1} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

Check: $\operatorname{row}_4(T)K = -(1, 2, 3, 3) + (1, 2, 3, 4) = (0, 0, 0, 1)$ and $\operatorname{row}_2(T)K = -(1, 1, 1, 1) + 2(1, 2, 2, 2) - (1, 2, 3, 3) = (0, 1, 0, 0)$.

(d) T_2 is T with column 2 replaced by b. Expanding along column 1 (Theorem 5.21),

$$\det T_2 = \det \begin{pmatrix} 2 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & 3 & 2 & -1 \\ 0 & 4 & -1 & 1 \end{pmatrix} = 2 \det \begin{pmatrix} 2 & -1 & 0 \\ 3 & 2 & -1 \\ 4 & -1 & 1 \end{pmatrix} + \det \begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & -1 \\ 4 & -1 & 1 \end{pmatrix}$$

$$= 2\Big(2(2 - 1) + 1 \cdot (3 + 4)\Big) + 1 \cdot (2 - 1) = 19$$

(the 3×3 determinants along their first rows), and $\det T = 1$ by (b), so $x_2 = 19$ (Theorem 5.27). Check: $x_2 = \operatorname{row}_2(K)b = 1 + 4 + 6 + 8 = 19$; indeed $x = Kb = (10, 19, 26, 30)'$.

Exercise 5.17 (cf. [6, Exercise 6.29])**.** Use Cramer's rule (Theorem 5.27) to solve the first system of Exercise 2.19, $Ax = b$ with

$$A = \begin{pmatrix} 1 & -3 & -2 & -1 \\ -1 & 2 & 1 & 1 \\ -1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ -1 \\ -2 \\ 1 \end{pmatrix}$$

Solution.

- Let A_j be A with column j replaced by b. Subtracting row 2 from row 3 does not change a determinant (Definition 5.13(b)). It turns row 3 of A into $(0,0,1,0)$, and row 3 of A_j into the same row with its jth entry replaced by $b_3 - b_2 = -1$: the five matrices become

$$\begin{pmatrix} 1 & -3 & -2 & -1 \\ -1 & 2 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} (A), \quad \begin{pmatrix} 1 & -3 & -2 & -1 \\ -1 & 2 & 1 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} (A_1), \quad \begin{pmatrix} 1 & 1 & -2 & -1 \\ -1 & -1 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} (A_2)$$

$$\begin{pmatrix} 1 & -3 & 1 & -1 \\ -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} (A_3), \quad \begin{pmatrix} 1 & -3 & -2 & 1 \\ -1 & 2 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \end{pmatrix} (A_4)$$

- Expanding along the new row 3 (Theorem 5.21), and then the 3×3 determinants along their first rows,

$$\det A = \det \begin{pmatrix} 1 & -3 & -1 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} = 1(2-1) + 3(-1-0) - 1(-1-0) = -1$$

$$\det A_1 = -\det \begin{pmatrix} -3 & -2 & -1 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} + \det \begin{pmatrix} 1 & -3 & -1 \\ -1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix} = -(0+2-1) + (1-6+3) = -3$$

$$\det A_2 = \det \begin{pmatrix} 1 & -2 & -1 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} + \det \begin{pmatrix} 1 & 1 & -1 \\ -1 & -1 & 1 \\ 0 & 1 & 1 \end{pmatrix} = (0-2+1) + (-2+1+1) = -1$$

$$\det A_3 = -\det \begin{pmatrix} 1 & -3 & -1 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} = 1$$

$$\det A_4 = \det \begin{pmatrix} 1 & -3 & 1 \\ -1 & 2 & -1 \\ 0 & 1 & 1 \end{pmatrix} + \det \begin{pmatrix} 1 & -3 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} = (3-3-1) + (1-3+2) = -1$$

  (the minor of $\det A_3$ is the one of $\det A$, as columns 1, 2, 4 of A_3 are those of A).

- So $x = \left(\frac{-3}{-1}, \frac{-1}{-1}, \frac{1}{-1}, \frac{-1}{-1}\right)' = (3,1,-1,1)'$ (Theorem 5.27), the solution found by elimination in Exercise 2.19.

Exercise 5.18. Let

$$M = \left(\begin{array}{cc|ccc} 3 & 1 & 4 & -1 & 7 \\ 5 & 2 & 2 & 9 & 3 \\ \hline 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

Compute $\det M$, $\det M^3$ and $\det(2M^{-1})$.

Solution. $M = \left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right)$ with $A = \left(\begin{smallmatrix} 3 & 1 \\ 5 & 2 \end{smallmatrix}\right)$, $\det A = 6 - 5 = 1$, and

$$D = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

whose expansion along the first row (Theorem 5.21) gives $\det D = 2(4-1) - 1(2-0) = 4$. By Theorem 5.29, $\det M = 1 \cdot 4 = 4$; the block B plays no role. As $\det M \neq 0$, M is invertible, and by Theorem 5.18(a),(c), $\det M^3 = 4^3 = 64$ and $\det(2M^{-1}) = 2^5 \cdot \frac{1}{4} = 8$.

Exercise 5.19. For $t \in \mathbb{R}$ let

$$M = \left(\begin{array}{ccc|cc} 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & t & 2 \\ 0 & 1 & 1 & 2 & t+2 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

Here $A \in \mathbb{R}^{3\times 3}$, $B \in \mathbb{R}^{3\times 2}$, $C \in \mathbb{R}^{2\times 3}$, $D \in \mathbb{R}^{2\times 2}$. (a) Find $\det M$ with the Schur complement of A. (b) For which t is M invertible? (c) Find $\operatorname{rank} M$ for these exceptional values of t.

Solution. (a) Here $A = \operatorname{diag}(A_1, 1)$ with $A_1 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $C = B'$ and $D = \left(\begin{smallmatrix} t & 2 \\ 2 & t+2 \end{smallmatrix}\right)$. By Theorem 5.29, $\det A = \det A_1 \cdot 1 = 1$, as $\det A_1 = 2 \cdot 1 - 1 \cdot 1 = 1$; by Propositions 2.18 and 5.6(b),

$$A^{-1} = \operatorname{diag}(A_1^{-1}, 1) = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then

$$CA^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix}, \qquad CA^{-1}B = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$$

so the Schur complement is $S = D - CA^{-1}B = \left(\begin{smallmatrix} t-1 & 1 \\ 1 & t-1 \end{smallmatrix}\right)$. By Theorem 5.30(a), $\det M = \det A \cdot \det S = (t-1)^2 - 1 = t(t-2)$.

(b) By Theorem 5.18(a), M is invertible if and only if $t \neq 0$ and $t \neq 2$.

(c) For $t = 0$, $S = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$; for $t = 2$, $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. In both cases the second row of S is a multiple of the first and $S \neq 0$, so $\operatorname{rank} S = 1$ (the row space is spanned by the nonzero first row; Definition 4.11 and Proposition 3.13(a)). By Theorem 5.34(a) with $p = 3$, $\operatorname{rank} M = 3 + 1 = 4$.

Exercise 5.20. Let

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad M = \begin{pmatrix} I_3 & B \\ C & I_2 \end{pmatrix} \in \mathbb{R}^{5\times 5}$$

(a) Compute $\det M$ with the Schur complement of I_3, and again with the Schur complement of I_2.

(b) Compute M^{-1} with Corollary 5.10(a).

Solution. (a) The Schur complement of I_3 is $S = I_2 - CI_3^{-1}B = I_2 - CB$ and that of I_2 is $I_3 - BC$, with

$$CB = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & -1 \\ -1 & -1 \end{pmatrix}, \qquad BC = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}, \quad I_3 - BC = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ -1 & -2 & 0 \end{pmatrix}$$

By Theorem 5.30(a), $\det M = \det I_3 \cdot \det S = 0 - 1 = -1$. Expanding $\det(I_3 - BC)$ along the first row (Theorem 5.21) gives $(-1)^{1+2}(-1)\det\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right) = 0 - 1 = -1$, so Theorem 5.30(b) gives again $\det M = \det I_2 \cdot \det(I_3 - BC) = -1$.

(b) The Schur complement $S = I_2 - CB$ of I_3 is invertible ($\det S = -1$), with $S^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 0 \end{smallmatrix}\right)$ by the 2×2 formula of Proposition 2.18. Corollary 5.10(a) with $A = I_3$ gives $M^{-1} = \left(\begin{smallmatrix} I_3 & -B \\ 0 & I_2 \end{smallmatrix}\right) \operatorname{diag}(I_3, S^{-1}) \left(\begin{smallmatrix} I_3 & 0 \\ -C & I_2 \end{smallmatrix}\right)$. By the block product formula,

$$M^{-1} = \begin{pmatrix} I_3 & -BS^{-1} \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I_3 & 0 \\ -C & I_2 \end{pmatrix} = \begin{pmatrix} I_3 + BS^{-1}C & -BS^{-1} \\ -S^{-1}C & S^{-1} \end{pmatrix}$$

where

$$-BS^{-1} = \begin{pmatrix} -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad -S^{-1}C = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad BS^{-1}C = -B(-S^{-1}C) = \begin{pmatrix} 1 & 0 & -1 \\ -1 & -1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$$

Hence

$$M^{-1} = \left(\begin{array}{ccc|cc} 2 & 0 & -1 & -1 & 1 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 1 \\ \hline -1 & 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & -1 & 0 \end{array}\right)$$

Check: $\text{row}_4(M)M^{-1} = (1,1,0,1,0)M^{-1} = (2-1-1,\ 0+0+0,\ -1+0+1,\ -1+1+1,\ 1+0-1) = (0,0,0,1,0)$.

Exercise 5.21. Let

$$M = \left(\begin{array}{cc|cc} 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 1 \\ \hline 2 & 2 & 3 & 2 \\ 2 & 1 & 2 & 1 \end{array}\right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

(a) Find D^{-1} and the Schur complement $F = A - BD^{-1}C$ of D.

(b) Find $\det M$ and $\operatorname{rank} M$ from F.

(c) Find a basis of $\mathcal{N}(M)$.

Solution. (a)

- Here $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $B = C = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ and $D = \left(\begin{smallmatrix} 3 & 2 \\ 2 & 1 \end{smallmatrix}\right)$. As $\det D = 3 \cdot 1 - 2 \cdot 2 = -1$, $D^{-1} = \left(\begin{smallmatrix} -1 & 2 \\ 2 & -3 \end{smallmatrix}\right)$ (the 2×2 case of Theorem 5.24).
- Then

$$BD^{-1} = \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix}, \qquad BD^{-1}C = \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix}$$

- So $F = A - BD^{-1}C = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

(b) $\det F = 1 - 1 = 0$, so $\det M = \det D \cdot \det F = 0$ (Theorem 5.30(b)), and M is not invertible. The second row of F is -1 times the first and $F \neq 0$, so $\operatorname{rank} F = 1$ (the row space is spanned by the nonzero first row; Definition 4.11 and Proposition 3.13(a)); by Theorem 5.34(b) with $q = 2$, $\operatorname{rank} M = 2 + 1 = 3$.

(c)

- By Proposition 5.8, with $-BD^{-1}$ in the upper right block of the block elementary matrix,

$$L = \begin{pmatrix} I_2 & -BD^{-1} \\ 0 & I_2 \end{pmatrix} \text{ is invertible, and } \quad LM = \begin{pmatrix} A - BD^{-1}C & B - BD^{-1}D \\ C & D \end{pmatrix} = \begin{pmatrix} F & 0 \\ C & D \end{pmatrix}$$

- So for $x = \left(\begin{smallmatrix} y \\ z \end{smallmatrix}\right)$ with $y, z \in \mathbb{R}^2$: $Mx = 0$ if and only if $LMx = 0$, that is, $Fy = 0$ and $Cy + Dz = 0$.
- $Fy = 0$ means $y_1 - y_2 = 0$, so $y = s(1,1)'$ with $s \in \mathbb{R}$. Then $Cy = s(4,3)'$ and $z = -D^{-1}Cy = -s(-4+6,\ 8-9)' = s(-2,1)'$.
- Hence $\mathcal{N}(M) = \{s(1,1,-2,1)' : s \in \mathbb{R}\}$, and $(1,1,-2,1)'$ is a basis. This agrees with $\dim \mathcal{N}(M) = 4 - \operatorname{rank} M = 1$ (Theorem 4.12). Check: $M(1,1,-2,1)' = (1+1-4+2,\ 1+2-4+1,\ 2+2-6+2,\ 2+1-4+1)' = 0$.

Exercise 5.22. Use Theorem 5.31 to compute

$$\det\begin{pmatrix}2&1&1&1\\1&3&1&1\\1&1&4&1\\1&1&1&7\end{pmatrix} \qquad \text{and} \qquad \det\begin{pmatrix}2&-1&1&-1\\2&-1&2&-2\\3&-3&4&-3\\4&-4&4&-3\end{pmatrix}$$

Solution. The first matrix is $D + \mathbf{1}_4\mathbf{1}_4'$ with $D = \operatorname{diag}(1,2,3,6)$, $\det D = 36$ (Theorem 5.14(c)) and $D^{-1} = \operatorname{diag}(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ (Proposition 5.6(b)). By Theorem 5.31,

$$\det(D + \mathbf{1}\mathbf{1}') = \det D \cdot (1 + \mathbf{1}'D^{-1}\mathbf{1}) = 36\Big(1 + 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}\Big) = 36 \cdot 3 = 108$$

The second matrix is $I_4 + uv'$ with $u = (1,2,3,4)'$ and $v = (1,-1,1,-1)'$: its (i,j) entry is $\delta_{ij} + u_i v_j$, where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise. Hence, by Theorem 5.31, its determinant is $1 + v'u = 1 + (1 - 2 + 3 - 4) = -1$.

Exercise 5.23. Let

$$A = \begin{pmatrix}1&2&2&0\\0&1&1&0\\2&2&3&0\\0&0&1&1\end{pmatrix}, \qquad A^{-1} = \begin{pmatrix}1&-2&0&0\\2&-1&-1&0\\-2&2&1&0\\2&-2&-1&1\end{pmatrix}, \qquad u = \begin{pmatrix}0\\0\\1\\1\end{pmatrix}, \qquad v = \begin{pmatrix}1\\2\\0\\0\end{pmatrix}$$

(a) Compute $A^{-1}u$, $v'A^{-1}$ and $\det A$, and find $\det(A + uv')$ by Theorem 5.31.

(b) Find $(A + uv')^{-1}$ by Theorem 5.32(b), and check the answer by multiplication.

(c) For which $s \in \mathbb{R}$ is $A + s\,uv'$ invertible?

Solution. (a)

- By Theorem 1.4(b),(c), $A^{-1}u = \operatorname{col}_3(A^{-1}) + \operatorname{col}_4(A^{-1}) = (0,-1,1,0)'$ and $v'A^{-1} = \operatorname{row}_1(A^{-1}) + 2\operatorname{row}_2(A^{-1}) = (5,-4,-2,0)$. So $v'A^{-1}u = 5\cdot 0 + (-4)\cdot 0 + (-2)\cdot 1 + 0\cdot 1 = -2$.
- Along the fourth column, whose only nonzero entry is $a_{44} = 1$, and then along the second row $(0,1,1)$ of the 3×3 minor (Theorem 5.21),

$$\det A = \det\begin{pmatrix}1&2&2\\0&1&1\\2&2&3\end{pmatrix} = \det\begin{pmatrix}1&2\\2&3\end{pmatrix} - \det\begin{pmatrix}1&2\\2&2\end{pmatrix} = -1 + 2 = 1$$

- By Theorem 5.31, $\det(A + uv') = \det A \cdot (1 + v'A^{-1}u) = 1\cdot(1-2) = -1$.

(b)

- $1 + v'A^{-1}u = -1 \neq 0$, so Theorem 5.32(b) applies, with

$$A^{-1}uv'A^{-1} = \begin{pmatrix}0\\-1\\1\\0\end{pmatrix}(5,\ -4,\ -2,\ 0) = \begin{pmatrix}0&0&0&0\\-5&4&2&0\\5&-4&-2&0\\0&0&0&0\end{pmatrix}$$

- Hence

$$(A + uv')^{-1} = A^{-1} - \frac{1}{-1}A^{-1}uv'A^{-1} = A^{-1} + A^{-1}uv'A^{-1} = \begin{pmatrix}1&-2&0&0\\-3&3&1&0\\3&-2&-1&0\\2&-2&-1&1\end{pmatrix}$$

- Check:
$$A + uv' = A + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 3 & 4 & 3 & 0 \\ 1 & 2 & 1 & 1 \end{pmatrix}$$
and, with the rows of $(A + uv')^{-1}$, $(1, -2, 0, 0) + 2(-3, 3, 1, 0) + 2(3, -2, -1, 0) = (1, 0, 0, 0)$, $(-3, 3, 1, 0) + (3, -2, -1, 0) = (0, 1, 0, 0)$, $3(1, -2, 0, 0) + 4(-3, 3, 1, 0) + 3(3, -2, -1, 0) = (0, 0, 1, 0)$ and $(1, -2, 0, 0)+2(-3, 3, 1, 0)+(3, -2, -1, 0)+(2, -2, -1, 1) = (0, 0, 0, 1)$. So $(A+uv')(A+uv')^{-1} = I_4$.

(c) $A+s\,uv' = A+(su)v'$. By Theorem 5.31 with su in place of u, $\det(A+s\,uv') = \det A\cdot(1+s\,v'A^{-1}u) = 1 - 2s$. By Theorem 5.18(a), $A + s\,uv'$ is invertible if and only if $s \neq \frac{1}{2}$.

Exercise 5.24. Let $\varphi = \langle 2, 4, 1, 3\rangle$ and $\psi = \langle 4, 1, 3, 2\rangle$.

(a) Find $\varphi\psi$, $\psi\varphi$ and φ^{-1}.

(b) Write down P_φ and P_ψ, and compute $P_\psi P_\varphi$.

(c) Turn P_φ into I_4 by interchanges of rows, and find $\det P_\varphi$.

Solution. (a) By Definition 5.39,

- $\varphi\psi = \langle\varphi(4), \varphi(1), \varphi(3), \varphi(2)\rangle = \langle 3, 2, 1, 4\rangle$ and $\psi\varphi = \langle\psi(2), \psi(4), \psi(1), \psi(3)\rangle = \langle 1, 2, 4, 3\rangle$, so $\varphi\psi \neq \psi\varphi$;
- since $\varphi(3) = 1$, $\varphi(1) = 2$, $\varphi(4) = 3$ and $\varphi(2) = 4$, $\varphi^{-1} = \langle 3, 1, 4, 2\rangle$.

(b) P_φ has rows e'_2, e'_4, e'_1, e'_3, and P_ψ has rows e'_4, e'_1, e'_3, e'_2. By Proposition 5.41(a), the rows of $P_\psi P_\varphi$ are rows $4, 1, 3, 2$ of P_φ:
$$P_\varphi = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad P_\psi = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad P_\psi P_\varphi = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
This is $P_{\langle 3,2,1,4\rangle} = P_{\varphi\psi}$, as Proposition 5.41(b) says.

(c)
$$P_\varphi = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4$$
Each interchange changes the sign of the determinant, and $\det I_4 = 1$ (Definition 5.13(c),(d)). Three interchanges turn P_φ into I_4, so $\det P_\varphi = (-1)^3 = -1$.

Exercise 5.25. (a) The data matrix

$$X_0 = \begin{pmatrix} 3 & 8 \\ 1 & 5 \\ 4 & 2 \\ 2 & 6 \end{pmatrix}$$

holds the measurements of two variables on four units. Find the permutation φ for which $P_\varphi X_0$ lists the units in increasing order of the second variable. Write down P_φ and $P_\varphi X_0$, and find $\det P_\varphi$ by turning P_φ into I_4 by interchanges of rows.

(b) Find φ^{-1} and $P_{\varphi^{-1}}$, and check that $P_{\varphi^{-1}} = P'_\varphi$ and $P'_\varphi(P_\varphi X_0) = X_0$.

Solution. (a)

- The second variable takes the values $8, 5, 2, 6$ on units $1, 2, 3, 4$. In increasing order these are $2, 5, 6, 8$, the values of units $3, 2, 4, 1$. Row i of $P_\varphi X_0$ is row $\varphi(i)$ of X_0 (Proposition 5.41(a)), so $\varphi = \langle 3, 2, 4, 1\rangle$.
- P_φ has the rows e'_3, e'_2, e'_4, e'_1 (Definition 5.40):

$$P_\varphi = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad P_\varphi X_0 = \begin{pmatrix} 4 & 2 \\ 1 & 5 \\ 2 & 6 \\ 3 & 8 \end{pmatrix}$$

- Interchanging rows 1 and 4 of P_φ, and then rows 3 and 4, gives I_4:

$$P_\varphi \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4$$

Each interchange changes the sign of the determinant, and $\det I_4 = 1$ (Definition 5.13(c),(d)); so $\det P_\varphi = (-1)^2 = 1$.

(b) Since $\varphi(4) = 1$, $\varphi(2) = 2$, $\varphi(1) = 3$ and $\varphi(3) = 4$, the inverse is $\varphi^{-1} = \langle 4, 2, 1, 3\rangle$ (Definition 5.39), and $P_{\varphi^{-1}}$ has the rows e'_4, e'_2, e'_1, e'_3:

$$P_{\varphi^{-1}} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} = P'_\varphi, \qquad P'_\varphi(P_\varphi X_0) = \begin{pmatrix} 3 & 8 \\ 1 & 5 \\ 4 & 2 \\ 2 & 6 \end{pmatrix} = X_0$$

The last product lists rows $4, 2, 1, 3$ of $P_\varphi X_0$ (Proposition 5.41(a)): the units return to their original order, as Proposition 5.41(c) says.

Problems

Problem 5.1 ([8, Ch. 8, Exercise 4(a)])**.** Let $A \in \mathbb{R}^{n\times n}$ be invertible and partitioned by columns as $A = [A_1\ A_2]$, where A_1 has k columns, and let A^{-1} be partitioned by rows into B_1 (k rows) and B_2. Show that

$$B_1A_1 = I, \quad B_1A_2 = 0, \quad B_2A_1 = 0, \quad B_2A_2 = I, \qquad A_1B_1 = I - A_2B_2, \quad A_2B_2 = I - A_1B_1$$

Solution. With the partition $(k, n-k)$ of I_n, Theorem 5.3(c) gives

$$\begin{pmatrix} B_1A_1 & B_1A_2 \\ B_2A_1 & B_2A_2 \end{pmatrix} = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} [A_1\ A_2] = A^{-1}A = I_n = \begin{pmatrix} I_k & 0 \\ 0 & I_{n-k} \end{pmatrix}$$

and comparing blocks gives the first four identities. Also $A_1B_1 + A_2B_2 = [A_1\ A_2]\left(\begin{smallmatrix} B_1 \\ B_2 \end{smallmatrix}\right) = AA^{-1} = I_n$, which gives the last two.

Problem 5.2 ([6, Exercise 5.8])**.** Show that the matrix $Z = \begin{pmatrix} -I_m & B \\ C & I_n \end{pmatrix}$ ($B \in \mathbb{R}^{m\times n}$, $C \in \mathbb{R}^{n\times m}$) satisfies $Z^2 = I_{m+n}$ if $B = 0$ or $C = 0$. Is this condition necessary?

Solution. By the 2×2 block formula (Theorem 5.3),

$$Z^2 = \begin{pmatrix} (-I)(-I) + BC & -B + B \\ -C + C & CB + I \end{pmatrix} = \begin{pmatrix} I_m + BC & 0 \\ 0 & I_n + CB \end{pmatrix}$$

so $Z^2 = I_{m+n}$ if and only if $BC = 0$ and $CB = 0$. This holds if $B = 0$ or $C = 0$. The condition is not necessary when $m, n \geqslant 2$: for $m = n = 2$ and $B = C = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \neq 0$ we have $BC = CB = 0$ (Proposition 1.21(c)); for larger m, n put this 2×2 matrix in the top left corner of zero matrices. If $m = 1$, then $CB \in \mathbb{R}^{n\times n}$ has entries c_ib_j, so $CB = 0$ forces $B = 0$ or $C = 0$; the case $n = 1$ is the same with BC.

Problem 5.3 ([6, Exercise 5.9])**.** Show that $Z_1 = \begin{pmatrix} I_m & B_1 \\ 0 & I_n \end{pmatrix}$ and $Z_2 = \begin{pmatrix} I_m & B_2 \\ 0 & I_n \end{pmatrix}$ ($B_1, B_2 \in \mathbb{R}^{m\times n}$) commute.

Solution. Write $Z_1 = I_{m+n} + X_1$ and $Z_2 = I_{m+n} + X_2$ with $X_i = \left(\begin{smallmatrix} 0 & B_i \\ 0 & 0 \end{smallmatrix}\right)$. By the block formula, $X_1X_2 = \left(\begin{smallmatrix} 0\cdot 0 + B_1\cdot 0 & 0\cdot B_2 + B_1\cdot 0 \\ 0 & 0 \end{smallmatrix}\right) = 0$, and likewise $X_2X_1 = 0$. Hence

$$Z_1Z_2 = I + X_1 + X_2 + X_1X_2 = I + X_1 + X_2 = I + X_2 + X_1 + X_2X_1 = Z_2Z_1$$

Problem 5.4 ([1, §5.3, Exercise 2])**.** Let $a, b, c \in \mathbb{R}$. Show that the **Vandermonde matrix**

$$A = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$$

has $\det A = (b-a)(c-a)(c-b)$, so that A is invertible if and only if a, b, c are distinct.

Solution.

- Subtracting row 1 from rows 2 and 3 does not change the determinant (Definition 5.13(b)); it gives the rows $(0,\ b-a,\ b^2-a^2)$ and $(0,\ c-a,\ c^2-a^2)$.
- Expanding along column 1 (Theorem 5.21),

$$\det A = \det \begin{pmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{pmatrix} = (b-a)(c^2-a^2) - (c-a)(b^2-a^2)$$

- As $c^2 - a^2 = (c-a)(c+a)$ and $b^2 - a^2 = (b-a)(b+a)$, this is $(b-a)(c-a)\big((c+a)-(b+a)\big) = (b-a)(c-a)(c-b)$.
- The product is nonzero exactly when a, b, c are distinct, so A is invertible exactly then (Theorem 5.18(a)).

Problem 5.5 ([2, Exercise 3.13])**.** Let $K \in \mathbb{R}^{12\times 12}$ satisfy $K^5 = 3K$. What is the numerical value of $\det K$? Is there more than one possibility? If so, give all possible values.

Solution.

- By Theorem 5.18(c), $\det K^5 = (\det K)^5$ and $\det(3K) = 3^{12}\det K$. So $(\det K)^5 = 3^{12}\det K$, that is, $\det K \cdot \big((\det K)^4 - 3^{12}\big) = 0$.
- Hence $\det K = 0$ or $(\det K)^4 = 3^{12}$, that is, $\big((\det K)^2 - 3^6\big)\big((\det K)^2 + 3^6\big) = 0$. The second factor is positive, so $(\det K)^2 = 3^6$ and $\det K = \pm 3^3 = \pm 27$.
- All three values occur. Let t be the positive number with $t^4 = 3$, so $t^5 = 3t$. For $K = 0$, $\det K = 0$. For $K = tI_{12}$, $K^5 = t^5 I_{12} = 3K$ and $\det K = t^{12} = 3^3 = 27$ (Theorem 5.14(c)). For $K = \operatorname{diag}(-t, t, \ldots, t)$, $K^5 = \operatorname{diag}(-t^5, t^5, \ldots, t^5) = 3K$ and $\det K = -t^{12} = -27$.

So $\det K$ is 0, 27 or -27.

Problem 5.6 ([3, Ch. 2, Exercise 14], [8, Ch. 13, Exercise 3])**.** Let $A \in \mathbb{R}^{n\times n}$ have integer entries.

(a) Show that if A is invertible and all entries of A^{-1} are integers, then $\det A = \pm 1$.

(b) Conversely, show that if $\det A = \pm 1$, then A is invertible and all entries of A^{-1} are integers.

Solution. (a)

- A matrix with integer entries has an integer determinant, by induction on its size. For $n = 1$, $\det(a) = a$ (Definition 5.13). For $n \geqslant 2$, the expansion along the first row (Theorem 5.21) gives $\det A = \sum_{k=1}^{n} (-1)^{1+k} a_{1k} \det A(1|k)$, where every $A(1|k) \in \mathbb{R}^{(n-1)\times(n-1)}$ has integer entries; so every $\det A(1|k)$ is an integer, and so is $\det A$. Hence $d = \det A$ and $\det A^{-1}$ are integers.
- By Theorem 5.18(a),(c), $d \neq 0$ and $\det A^{-1} = 1/d$. So d and $1/d$ are nonzero integers.
- A nonzero integer has absolute value at least 1: $|d| \geqslant 1$ and $|1/d| \geqslant 1$, that is, $|d| \leqslant 1$. Hence $|d| = 1$, and $\det A = \pm 1$.

(b) As $\det A = \pm 1 \neq 0$, A is invertible (Theorem 5.18(a)). For $n = 1$, $A = (\pm 1)$ and $A^{-1} = (\pm 1)$. For $n \geqslant 2$, each cofactor $c_{ij} = (-1)^{i+j} \det A(i|j)$ is an integer by the first item of (a), applied to $A(i|j)$. By Theorem 5.24, $A^{-1} = \frac{1}{\det A} \operatorname{adj} A = \pm \operatorname{adj} A$, whose entries $\pm c_{ji}$ are integers.

Problem 5.7 ([2, Exercise 3.14])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$. Show that $\det(I_m - AB) = \det(I_n - BA)$.

Solution. By the block product formula (Theorem 5.3),

$$\begin{pmatrix} I_m - AB & A \\ 0 & I_n \end{pmatrix}\begin{pmatrix} I_m & 0 \\ B & I_n \end{pmatrix} = \begin{pmatrix} I_m - AB + AB & A \\ B & I_n \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix}$$
$$\begin{pmatrix} I_m & 0 \\ B & I_n \end{pmatrix}\begin{pmatrix} I_m & A \\ 0 & I_n - BA \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & BA + I_n - BA \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix}$$

So the two products are equal. By Theorems 5.18(b) and 5.29, the first has determinant $\det(I_m - AB)\det I_n \cdot \det I_m \det I_n = \det(I_m - AB)$, and the second $\det I_m \det I_n \cdot \det I_m \det(I_n - BA) = \det(I_n - BA)$.

Problem 5.8 ([3, Ch. 2, Exercise 18])**.** Let $A, B, C, D \in \mathbb{R}^{n\times n}$ with A invertible and $AC = CA$. Show that

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - CB)$$

Solution.

- Subtracting CA^{-1} times the first block row from the second turns the matrix into $\left(\begin{smallmatrix} A & B \\ 0 & D-CA^{-1}B \end{smallmatrix}\right)$ (Theorem 5.9). By Theorems 5.30(a) and 5.18(b),

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \cdot \det(D - CA^{-1}B) = \det\Big(A(D - CA^{-1}B)\Big) = \det(AD - ACA^{-1}B)$$

  the last equality by Theorem 1.3.

- From $AC = CA$, $ACA^{-1} = CAA^{-1} = C$, so the last determinant is $\det(AD - CB)$.

Problem 5.9 ([6, Exercise 5.45])**.** Let $A \in \mathbb{R}^{m\times p}$, $B \in \mathbb{R}^{m\times q}$, $C \in \mathbb{R}^{n\times p}$ and $D \in \mathbb{R}^{n\times q}$.

(a) Show that

$$\operatorname{rank} \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \geqslant \operatorname{rank} A + \operatorname{rank} D \qquad \text{and} \qquad \operatorname{rank} \begin{pmatrix} A & 0 \\ C & D \end{pmatrix} \geqslant \operatorname{rank} A + \operatorname{rank} D$$

(b) Show that $\operatorname{rank} \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \geqslant \operatorname{rank} A + \operatorname{rank} D$ is not true in general.

Solution. (a)

- Let $Z = \left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right)$, $r = \operatorname{rank} A$ and $s = \operatorname{rank} D$. By Theorem 4.8(a), the pivot columns of A, say columns $k_1 < \cdots < k_r$, are linearly independent, and so are the pivot columns of D, say columns $l_1 < \cdots < l_s$ (a list is empty if the rank is 0). Column k_i of Z is $\operatorname{col}_{k_i}(A)$ followed by zeros, and column $p + l_j$ of Z is $\operatorname{col}_{l_j}(B)$ followed by $\operatorname{col}_{l_j}(D)$.

- These $r + s$ columns of Z are linearly independent. Let

$$\sum_{i=1}^{r} \alpha_i \begin{pmatrix} \operatorname{col}_{k_i}(A) \\ 0 \end{pmatrix} + \sum_{j=1}^{s} \beta_j \begin{pmatrix} \operatorname{col}_{l_j}(B) \\ \operatorname{col}_{l_j}(D) \end{pmatrix} = 0$$

  The last n entries give $\sum_j \beta_j \operatorname{col}_{l_j}(D) = 0$, so every $\beta_j = 0$; then the first m entries give $\sum_i \alpha_i \operatorname{col}_{k_i}(A) = 0$, so every $\alpha_i = 0$.

- The $r + s$ columns lie in $\mathcal{C}(Z)$, whose dimension is $\operatorname{rank} Z$ (Definition 4.11); so $r + s \leqslant \operatorname{rank} Z$ (Theorem 4.5(c)).

- For the second matrix, $\left(\begin{smallmatrix} A & 0 \\ C & D \end{smallmatrix}\right)' = \left(\begin{smallmatrix} A' & C' \\ 0 & D' \end{smallmatrix}\right)$ (Theorem 5.3(b)). By Theorem 4.13(a) and the first inequality, applied to A', C', D',

$$\operatorname{rank} \begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \operatorname{rank} \begin{pmatrix} A' & C' \\ 0 & D' \end{pmatrix} \geqslant \operatorname{rank} A' + \operatorname{rank} D' = \operatorname{rank} A + \operatorname{rank} D$$

(b) Let $m = n = p = q \geqslant 1$ and $A = B = C = D = I_m$. Then $\operatorname{rank} A + \operatorname{rank} D = 2m$ (Theorem 4.13(d)). Every column of $Z = \left(\begin{smallmatrix} I_m & I_m \\ I_m & I_m \end{smallmatrix}\right)$ is a column of its first block column $G = \left(\begin{smallmatrix} I_m \\ I_m \end{smallmatrix}\right)$, so $\mathcal{C}(Z) = \mathcal{C}(G)$ (Theorem 3.10(c)). If $Gx = 0$, the first m entries give $x = I_m x = 0$; so G has full column rank (Theorem 4.13(b)), and $\operatorname{rank} Z = \dim \mathcal{C}(G) = m < 2m$ (Definition 4.11).

Problem 5.10 ([10, Problem 1.5])**.** Let $n \geqslant 1$ and $\rho \in \mathbb{R}$, and let $A \in \mathbb{R}^{n\times n}$ have the entries $a_{ij} = \rho^{|i-j|}$, with $\rho^0 = 1$ also for $\rho = 0$:

$$A = \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{n-1} \\ \rho & 1 & \rho & \cdots & \rho^{n-2} \\ \rho^2 & \rho & 1 & \cdots & \rho^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \cdots & 1 \end{pmatrix}$$

Compute $\det A$.

Solution. For $n = 1$, $A = (1)$ and $\det A = 1$. Let $n \geqslant 2$.

- For $i = n, n-1, \ldots, 2$ in turn, subtract ρ times row $i-1$ from row i. Row $i-1$ is changed only after row i, so it is still row $i-1$ of A when it is used; and no step changes the determinant (Definition 5.13(b)).
- Row 1 is unchanged, and the new row i ($i \geqslant 2$) has the entries $\rho^{|i-j|} - \rho \cdot \rho^{|i-1-j|}$. For $j \leqslant i-1$ this is $\rho^{i-j} - \rho \cdot \rho^{i-1-j} = 0$, and for $j \geqslant i$ it is $\rho^{j-i} - \rho \cdot \rho^{j-i+1} = \rho^{j-i}(1-\rho^2)$. So the result is the upper triangular matrix

$$U = \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{n-1} \\ 0 & 1-\rho^2 & \rho(1-\rho^2) & \cdots & \rho^{n-2}(1-\rho^2) \\ 0 & 0 & 1-\rho^2 & \cdots & \rho^{n-3}(1-\rho^2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1-\rho^2 \end{pmatrix}$$

- By Theorem 5.14(c), $\det A = \det U = 1 \cdot (1-\rho^2)^{n-1}$.

So $\det A = (1-\rho^2)^{n-1}$ for every $n \geqslant 1$. In particular, for $n \geqslant 2$, A is invertible if and only if $\rho \neq \pm 1$ (Theorem 5.18(a)).

Problem 5.11 ([10, Problem 39.4])**.** Let $n \geqslant 2$ and $A \in \mathbb{R}^{n\times n}$, and suppose that $AP_\varphi = P_\varphi A$ for every permutation φ of degree n (Definition 5.40). Show that $A = cI_n + dJ_n$ for some $c, d \in \mathbb{R}$.

Solution.

- Let φ be a permutation of degree n. Multiplying $AP_\varphi = P_\varphi A$ on the right by P'_φ and using $P_\varphi P'_\varphi = I_n$ (Proposition 5.41(c)) gives $A = P_\varphi A P'_\varphi$.
- Column i of P'_φ is the transpose of row i of P_φ, which is $e'_{\varphi(i)}$ (Definition 5.40); so $P'_\varphi e_i = e_{\varphi(i)}$ (Proposition 1.18(a)). Hence, by Theorem 1.9 and Proposition 1.18(a), the (i,j) entry of $P_\varphi A P'_\varphi$ is

$$e'_i P_\varphi A P'_\varphi e_j = (P'_\varphi e_i)' A (P'_\varphi e_j) = e'_{\varphi(i)} A e_{\varphi(j)} = a_{\varphi(i)\varphi(j)}$$

  and the first item gives $a_{ij} = a_{\varphi(i)\varphi(j)}$ for all i, j and every φ.
- Diagonal entries: let $p \neq q$, and let φ interchange p and q and leave the other numbers in place. Then $a_{qq} = a_{\varphi(q)\varphi(q)} = a_{pp}$. So all diagonal entries of A are equal, say to α.
- Off-diagonal entries: let $i \neq j$ and $p \neq q$. Put $\varphi(p) = i$ and $\varphi(q) = j$, and let φ map the other $n-2$ numbers of $\{1, \ldots, n\}$, in increasing order, to the $n-2$ numbers other than i and j, in increasing order. Then φ is one-to-one and onto, a permutation (Definition 5.39), and $a_{pq} = a_{\varphi(p)\varphi(q)} = a_{ij}$. So all off-diagonal entries of A are equal, say to β.
- Every entry of J_n is 1 (Definition 1.14), so $A = \alpha I_n + \beta(J_n - I_n) = (\alpha - \beta)I_n + \beta J_n$: take $c = \alpha - \beta$ and $d = \beta$.

Conversely, every row and every column of P_φ has exactly one entry 1 (Definition 5.40, as φ is one-to-one and onto), so $P_\varphi \mathbf{1}_n = \mathbf{1}_n$ and $P'_\varphi \mathbf{1}_n = \mathbf{1}_n$ (Theorem 1.4(b)); hence $P_\varphi J_n = \mathbf{1}_n \mathbf{1}'_n = J_n P_\varphi$, and every $cI_n + dJ_n$ commutes with every P_φ. In statistical terms: listing n variables in the order $\varphi(1), \dots, \varphi(n)$ replaces their correlation matrix R by $P_\varphi R P'_\varphi$, with (i,j) entry $r_{\varphi(i)\varphi(j)}$, and $P_\varphi R P'_\varphi = R$ is the same as $RP_\varphi = P_\varphi R$; so R is unchanged by every reordering exactly when $R = (1-\rho)I_n + \rho J_n$, the equicorrelation matrix of Example 5.33 (the diagonal entries 1 give $c + d = 1$).

Problem 5.12 ([10, Problem 1.23, §2.6])**.** Let $A \in \mathbb{R}^{n\times n}$, and let $B \in \mathbb{R}^{n\times n}$ have the entries $b_{ij} = (-1)^{i+j} a_{ij}$. Show that $\det B = \det A$.

Solution. Let k be the integer part of $n/2$, the number of even numbers among $1, \dots, n$.

- Multiply the rows $2, 4, \dots, 2k$ of B by -1. An entry in an odd row i stays $(-1)^{i+j} a_{ij} = (-1)^{j+1} a_{ij}$, and an entry in an even row i becomes $-(-1)^{i+j} a_{ij} = (-1)^{j+1} a_{ij}$: every entry (i,j) is now $(-1)^{j+1} a_{ij}$.
- Multiply the columns $2, 4, \dots, 2k$ of this matrix by -1. An entry in an odd column j stays $(-1)^{j+1} a_{ij} = a_{ij}$, and an entry in an even column j becomes $-(-1)^{j+1} a_{ij} = a_{ij}$. The result is A.
- Each of these $2k$ steps multiplies the determinant by -1 (Definition 5.13(a), for columns by Theorem 5.22). So $\det A = (-1)^{2k} \det B = \det B$.

For example, for $n \geqslant 2$ the matrix of the cofactors $c_{ij} = (-1)^{i+j} \det A(i|j)$ of A and the matrix of the minors $\det A(i|j)$ (Definition 5.20) have the same determinant.

Problem 5.13 ([10, Problems 1.1, 1.2])**.** Let $A \in \mathbb{R}^{n\times n}$ be skew-symmetric (Definition 1.26).

(a) Show that $\det A = 0$ if n is odd.

(b) Let n be even. Show that adding the same number $x \in \mathbb{R}$ to all entries of A does not change the determinant: $\det(A + xJ_n) = \det A$.

Solution. (a)

- $A' = -A$, so by Theorems 5.22 and 5.18(c), $\det A = \det A' = \det(-A) = (-1)^n \det A = -\det A$, as n is odd.
- Hence $2 \det A = 0$, and $\det A = 0$.

(b) Let $\mathbf{1} = \mathbf{1}_n$. The matrices below have order $n+1$ and are partitioned after the first row and column.

- Let $M = \left(\begin{smallmatrix} 0 & \mathbf{1}' \\ -\mathbf{1} & A \end{smallmatrix}\right)$. By Theorem 5.3(b), $M' = \left(\begin{smallmatrix} 0 & -\mathbf{1}' \\ \mathbf{1} & A' \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & -\mathbf{1}' \\ \mathbf{1} & -A \end{smallmatrix}\right) = -M$. So M is skew-symmetric of the odd order $n+1$, and $\det M = 0$ by (a).
- The matrix $\left(\begin{smallmatrix} 0 & \mathbf{1}' \\ -x\mathbf{1} & A \end{smallmatrix}\right)$ is M with its first column multiplied by x; so its determinant is $x \det M = 0$ (Definition 5.13(a) for columns, Theorem 5.22). Also $\det \left(\begin{smallmatrix} 1 & 0 \\ -x\mathbf{1} & A \end{smallmatrix}\right) = \det(1) \cdot \det A = \det A$ (Theorem 5.29).
- These two matrices differ only in the first row, and their first rows $(0, 1, \dots, 1)$ and $(1, 0, \dots, 0)$ add up to $(1, 1, \dots, 1)$. By Lemma 5.19,

$$\det \begin{pmatrix} 1 & \mathbf{1}' \\ -x\mathbf{1} & A \end{pmatrix} = 0 + \det A = \det A$$

- Subtract column 1 from each of the columns $2, \dots, n+1$; this does not change the determinant (Definition 5.13(b) for columns, Theorem 5.22). Column $j+1$ ($1 \leqslant j \leqslant n$) is $\left(\begin{smallmatrix} 1 \\ \mathrm{col}_j(A) \end{smallmatrix}\right)$ and column 1 is $\left(\begin{smallmatrix} 1 \\ -x\mathbf{1} \end{smallmatrix}\right)$, so column $j+1$ becomes $\left(\begin{smallmatrix} 0 \\ \mathrm{col}_j(A) + x\mathbf{1} \end{smallmatrix}\right)$. As every column of $xJ_n = x\mathbf{1}\mathbf{1}'$ is $x\mathbf{1}$ (Definition 1.14), Theorem 5.29 gives

$$\det A = \det \begin{pmatrix} 1 & 0 \\ -x\mathbf{1} & A + xJ_n \end{pmatrix} = \det(A + xJ_n)$$

Problem 5.14 ([10, Problem 1.26])**.** Let $A \in \mathbb{R}^{n\times n}$, let $s_i = a_{i1} + \cdots + a_{in}$ be the sum of row i of A, and let $B \in \mathbb{R}^{n\times n}$ have the entries $b_{ij} = s_i - a_{ij}$. Show that $\det B = (-1)^{n-1}(n-1)\det A$.

Solution. Let $s = (s_1, \dots, s_n)'$ and $\mathbf{1} = \mathbf{1}_n$. The matrices below have order $n+1$ and are partitioned after the nth row and column.

- $\det \left(\begin{smallmatrix} B & 0 \\ -\mathbf{1}' & 1 \end{smallmatrix}\right) = \det B \cdot \det(1) = \det B$ (Theorem 5.29).
- Add the columns $1, \dots, n$ to column $n+1$; this does not change the determinant (Definition 5.13(b) for columns, Theorem 5.22). Entry $i \leqslant n$ of the new last column is $0 + \sum_{j=1}^{n}(s_i - a_{ij}) = ns_i - s_i = (n-1)s_i$, and its last entry is $1 + n\cdot(-1) = -(n-1)$; so the new last column is $n-1$ times $\left(\begin{smallmatrix} s \\ -1 \end{smallmatrix}\right)$. Taking out this factor (Definition 5.13(a) for columns),

$$\det B = \det \begin{pmatrix} B & (n-1)s \\ -\mathbf{1}' & -(n-1) \end{pmatrix} = (n-1)\det \begin{pmatrix} B & s \\ -\mathbf{1}' & -1 \end{pmatrix}$$

- Subtract column $n+1$ from each of the columns $1, \dots, n$; this does not change the determinant either (Definition 5.13(b) for columns, Theorem 5.22). Entry (i,j) with $i \leqslant n$ becomes $(s_i - a_{ij}) - s_i = -a_{ij}$, and the last entry of column j becomes $-1-(-1) = 0$. So, by Theorems 5.29 and 5.18(c),

$$\det \begin{pmatrix} B & s \\ -\mathbf{1}' & -1 \end{pmatrix} = \det \begin{pmatrix} -A & s \\ 0 & -1 \end{pmatrix} = \det(-A)\cdot(-1) = (-1)^{n+1}\det A$$

- Hence $\det B = (n-1)(-1)^{n+1}\det A = (-1)^{n-1}(n-1)\det A$.

Problem 5.15 ([10, Problem 2.7])**.** Let $n \geqslant 2$ and $1 \leqslant i, j \leqslant n$. Give an example of a matrix $A \in \mathbb{R}^{n\times n}$ whose adjugate $\operatorname{adj} A$ (Theorem 5.24) has exactly one nonzero entry, in position (i,j).

Solution. Insert into I_{n-1} a column of zeros so that it becomes column i, and then a row of zeros so that it becomes row j. The result $A \in \mathbb{R}^{n\times n}$ has the zero row j and the zero column i, and deleting them gives back I_{n-1}: $A(j|i) = I_{n-1}$. For instance, for $n = 4$ and $(i,j) = (1,2)$,

$$I_3 \to \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \to A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

By Theorem 5.24, the (k,l) entry of $\operatorname{adj} A$ is the cofactor $c_{lk} = (-1)^{l+k}\det A(l|k)$ (Definition 5.20).

- $(k,l) = (i,j)$: $c_{ji} = (-1)^{j+i}\det I_{n-1} = (-1)^{i+j} \neq 0$ (Definition 5.13(d)).
- $l \neq j$: row j of A, without its entry in column k, is a zero row of $A(l|k)$; so $\det A(l|k) = 0$ (Theorem 5.14(b)).
- $l = j$ and $k \neq i$: column i of A, without its entry in row j, is a zero column of $A(j|k)$; so $\det A(j|k) = 0$ (Theorem 5.14(b) for columns, Theorem 5.22).

So $\operatorname{adj} A$ has exactly one nonzero entry, $(-1)^{i+j}$ in position (i,j). In the instance, $\operatorname{adj} A = -E_{12}$, with E_{12} as in Proposition 1.18(b).

Problem 5.16 ([10, Problem 2.9])**.** Let $n \geqslant 2$ and let $A \in \mathbb{R}^{n\times n}$ be skew-symmetric. Show that $\operatorname{adj} A$ is symmetric if n is odd and skew-symmetric if n is even.

Solution. Let c_{ij} be the cofactors of A (Definition 5.20).

- Deleting row j and column i of A and then transposing gives the same matrix as deleting row i and column j of A': $A(j|i)' = A'(i|j)$. As $A' = -A$, $A'(i|j) = -A(i|j)$.
- $A(i|j)$ has order $n-1$, so by Theorems 5.22 and 5.18(c), $\det A(j|i) = \det A(j|i)' = \det\big(-A(i|j)\big) = (-1)^{n-1}\det A(i|j)$. Hence

$$c_{ji} = (-1)^{j+i}\det A(j|i) = (-1)^{n-1}(-1)^{i+j}\det A(i|j) = (-1)^{n-1}c_{ij}$$

- By Theorem 5.24, the (i,j) entry of $\operatorname{adj} A$ is c_{ji}, and that of $(\operatorname{adj} A)'$ is c_{ij}. So $\operatorname{adj} A = (-1)^{n-1}(\operatorname{adj} A)'$, that is, $(\operatorname{adj} A)' = (-1)^{n-1}\operatorname{adj} A$.
- For odd n, $(\operatorname{adj} A)' = \operatorname{adj} A$: $\operatorname{adj} A$ is symmetric. For even n, $(\operatorname{adj} A)' = -\operatorname{adj} A$: $\operatorname{adj} A$ is skew-symmetric (Definition 1.26).

Problem 5.17 ([10, Problems 2.2, 3.1])**.** Let $n \geqslant 2$, let $A \in \mathbb{R}^{n\times n}$ have the cofactors c_{ij} (Definition 5.20), and let $x, y, u, v \in \mathbb{R}^n$.

(a) Show that

$$\det\begin{pmatrix} A & x \\ y' & 0 \end{pmatrix} = -\sum_{i=1}^{n}\sum_{j=1}^{n} x_i y_j c_{ij} = -y'(\operatorname{adj} A)x$$

(b) Show that $\det(A + uv') = \det A + v'(\operatorname{adj} A)u$, also when A is not invertible.

Solution. (a) Let M be the matrix of order $n+1$ on the left.

- Column $n+1$ of M is $\left(\begin{smallmatrix} x \\ 0 \end{smallmatrix}\right)$. Expanding along it (Theorem 5.21),

$$\det M = \sum_{i=1}^{n} x_i(-1)^{i+n+1}\det M(i|n+1)$$

- $M(i|n+1) \in \mathbb{R}^{n\times n}$ consists of the rows of A other than row i, followed by the row y'; deleting this last row and column j leaves $A(i|j)$. Expanding along the last row (Theorem 5.21), $\det M(i|n+1) = \sum_{j=1}^{n} y_j(-1)^{n+j}\det A(i|j)$.
- As $(-1)^{i+n+1}(-1)^{n+j} = -(-1)^{i+j}$,

$$\det M = -\sum_{i=1}^{n}\sum_{j=1}^{n} x_i y_j(-1)^{i+j}\det A(i|j) = -\sum_{i=1}^{n}\sum_{j=1}^{n} x_i y_j c_{ij}$$

- By Theorem 5.24, $(\operatorname{adj} A)_{ji} = c_{ij}$; so entry j of $(\operatorname{adj} A)x$ is $\sum_{i=1}^{n} c_{ij}x_i$, and $y'(\operatorname{adj} A)x = \sum_{j=1}^{n} y_j \sum_{i=1}^{n} c_{ij}x_i$ is the double sum.

(b) Let $N = \left(\begin{smallmatrix} A & -u \\ v' & 1 \end{smallmatrix}\right) \in \mathbb{R}^{(n+1)\times(n+1)}$. We compute $\det N$ in two ways.

- By Theorem 5.30(b), with the invertible block $D = (1)$, $\det N = \det(1)\cdot\det\big(A - (-u)(1)^{-1}v'\big) = \det(A + uv')$.
- The last column of N is $\left(\begin{smallmatrix} -u \\ 0 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$, and the other columns are kept. By Lemma 5.19 for columns (Theorem 5.22), then by (a) with $x = -u$, $y = v$, and by Theorem 5.29,

$$\det N = \det\begin{pmatrix} A & -u \\ v' & 0 \end{pmatrix} + \det\begin{pmatrix} A & 0 \\ v' & 1 \end{pmatrix} = -v'(\operatorname{adj} A)(-u) + \det A\cdot\det(1)$$

- Comparing the two values, $\det(A + uv') = \det A + v'(\operatorname{adj} A)u$.

If A is invertible, then $\operatorname{adj} A = (\det A)A^{-1}$ (Theorem 5.24), and (b) is Theorem 5.31. If A is not invertible, then $\det A = 0$ (Theorem 5.18(a)), and (b) reads $\det(A + uv') = v'(\operatorname{adj} A)u$.

Problem 5.18 ([10, Problem 19.5])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric with $\operatorname{rank} A = r \geqslant 1$. By Theorem 5.37(b), A has an $r \times r$ submatrix with nonzero determinant. Show that there is one whose rows and columns have the same indices: for some $i_1 < \cdots < i_r$, the $r \times r$ submatrix A_* of A in the rows $i_1, \ldots, i_r$ and the columns $i_1, \ldots, i_r$ (Definition 5.36) has $\det A_* \neq 0$.

Solution.

- Let R be the RREF of A; it has r nonzero rows (Definition 4.11). Let $i_1 < \cdots < i_r$ be its pivot columns. By Theorem 4.3(b), the columns $a_{i_1}, \ldots, a_{i_r}$ of A form a basis of $\operatorname{span}\{a_1, \ldots, a_n\}$, so they are linearly independent, and $a_j = R_{1j}a_{i_1} + \cdots + R_{rj}a_{i_r}$ for every j.
- Let $S = [e_{i_1} \ \cdots \ e_{i_r}] \in \mathbb{R}^{n\times r}$, and let $X \in \mathbb{R}^{r\times n}$ consist of the first r rows of R. Then $AS = [a_{i_1} \ \cdots \ a_{i_r}]$ (Proposition 1.18(a)), and the first item reads $A = (AS)X$ (Theorem 1.4(a),(b)).
- Multiplying on the left by S' gives $S'A = (S'AS)X$. The (s, t) entry of $S'AS$ is $e'_{i_s}Ae_{i_t} = a_{i_s i_t}$ (Proposition 1.18(a)), so $S'AS = A_*$.
- As $A' = A$, $S'A = S'A' = (AS)'$ (Theorem 1.9). The r columns of AS are linearly independent, so $\operatorname{rank}(S'A) = \operatorname{rank}\big((AS)'\big) = \operatorname{rank}(AS) = r$ (Theorem 4.13(a),(b)). So $S'A$ has full row rank, and its rows, which are the rows $i_1, \ldots, i_r$ of A (Proposition 1.18(a)), are linearly independent too (Theorem 4.13(c)).
- By Theorems 4.15(a) and 4.13(a), $r = \operatorname{rank}(S'A) = \operatorname{rank}(A_*X) \leqslant \operatorname{rank} A_* \leqslant r$. So $\operatorname{rank} A_* = r$, A_* is invertible (Theorem 4.13(d)), and $\det A_* \neq 0$ (Theorem 5.18(a)).

Chapter 6. Orthogonality and projections

Inner products, lengths, and orthogonality

Definition 6.1 ([1, §8.1], [4, Ch. V, §1], [12, §1.3]). For $x, y \in \mathbb{R}^n$, the **inner product** of x and y is the number $x'y = x_1y_1 + \cdots + x_ny_n$; it equals $y'x$ (Theorem 1.9). The **length** (or norm) of x is $\|x\| = \sqrt{x'x} = (x_1^2 + \cdots + x_n^2)^{1/2}$, and $\|x - y\|$ is the **distance** between x and y. The vectors x and y are **orthogonal**, written $x \perp y$, if $x'y = 0$.

Since $x'x$ is a sum of squares, $\|x\| \geqslant 0$, and $\|x\| = 0$ only for $x = 0$; also $\|cx\| = |c|\,\|x\|$ for $c \in \mathbb{R}$. A vector of length 1 is a **unit vector**; for $x \neq 0$, $x/\|x\|$ is one. The zero vector is orthogonal to every vector, and it is the only vector orthogonal to itself.

Proposition 6.2 ([1, §8.1], [4, Ch. V, §1], [12, proof of Theorem 2.5]). For $x, y \in \mathbb{R}^n$, $\|x + y\|^2 = \|x\|^2 + 2x'y + \|y\|^2$. In particular, if $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ (**Pythagoras**).

Proof. $(x + y)'(x + y) = x'x + x'y + y'x + y'y$, and $y'x = x'y$. □

Proposition 6.3 ([1, §8.1, (8-3)], [4, Ch. V, §1]). Let $x, y \in \mathbb{R}^n$.

(a) $\|x - y\|^2 = \|x\|^2 - 2x'y + \|y\|^2$, and $x'y = \frac{1}{4}\|x + y\|^2 - \frac{1}{4}\|x - y\|^2$ (**polarization**).

(b) (**Pythagoras for k vectors**) If $u_1, \ldots, u_k \in \mathbb{R}^n$ are pairwise orthogonal ($u_i'u_j = 0$ for $i \neq j$), then $\|u_1 + \cdots + u_k\|^2 = \|u_1\|^2 + \cdots + \|u_k\|^2$.

Proof. (a)

- Proposition 6.2 with $-y$ in place of y gives the first identity, as $x'(-y) = -x'y$ and $\|-y\| = \|y\|$.
- Subtracting it from $\|x + y\|^2 = \|x\|^2 + 2x'y + \|y\|^2$ gives $\|x + y\|^2 - \|x - y\|^2 = 4x'y$.

(b) Induction on k; for $k = 1$ there is nothing to prove.

- Let $k \geqslant 2$ and $s = u_1 + \cdots + u_{k-1}$. Then $s'u_k = u_1'u_k + \cdots + u_{k-1}'u_k = 0$, so $s \perp u_k$.
- By Proposition 6.2, $\|s + u_k\|^2 = \|s\|^2 + \|u_k\|^2$, and $\|s\|^2 = \|u_1\|^2 + \cdots + \|u_{k-1}\|^2$ by the induction hypothesis. □

Theorem 6.4 ([12, Lemma 2.16], [1, §8.2, Theorem 1(iv)], [8, Ch. 6, Exercise 1]). Let $x, y \in \mathbb{R}^n$.

(a) (**Cauchy–Schwarz inequality**) $|x'y| \leqslant \|x\|\,\|y\|$, with equality if and only if one of x, y is a multiple of the other.

(b) (**Triangle inequality**) $\|x + y\| \leqslant \|x\| + \|y\|$, with equality if and only if $y = 0$ or $x = cy$ for some $c \geqslant 0$.

Proof. (a) If $x = 0$, both sides are 0, and $x = 0y$. Let $x \neq 0$.

- Put $c = x'y/\|x\|^2$ and $w = y - cx$. Then $x'w = x'y - c\,x'x = 0$, so $cx \perp w$.
- Pythagoras for $y = cx + w$ gives $\|y\|^2 = c^2\|x\|^2 + \|w\|^2 \geqslant c^2\|x\|^2 = (x'y)^2/\|x\|^2$. Multiply by $\|x\|^2$ and take square roots.
- Equality holds if and only if $w = 0$, that is, $y = cx$. Conversely, if $y = ax$, then $c = a$ and $w = 0$; if $x = ay$ with $x \neq 0$, then $a \neq 0$ and $y = a^{-1}x$.

(b)

- By Proposition 6.2 and (a),

  $$\|x + y\|^2 = \|x\|^2 + 2x'y + \|y\|^2 \leqslant \|x\|^2 + 2|x'y| + \|y\|^2 \leqslant \|x\|^2 + 2\|x\|\,\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$$

  and both $\|x + y\|$ and $\|x\| + \|y\|$ are $\geqslant 0$.

- Equality holds if and only if both steps are equalities: $x'y = |x'y|$, that is, $x'y \geqslant 0$; and $|x'y| = \|x\|\,\|y\|$.
- If $y = 0$, both hold. Let $y \neq 0$. By (a), the second holds if and only if $x = cy$ for some c (if $y = ax$, then $a \neq 0$ and $x = a^{-1}y$); and then $x'y = c\|y\|^2 \geqslant 0$ if and only if $c \geqslant 0$. □

Definition 6.5 ([12, §1.3], [1, §8.2], [5, §12.1])**.** Let W be a subspace of $\mathbb{R}^n$. A vector $y \in \mathbb{R}^n$ is **orthogonal to** W, written $y \perp W$, if $y'w = 0$ for every $w \in W$. The **orthogonal complement** of W is $W^\perp = \{y \in \mathbb{R}^n : y \perp W\}$. Subspaces U and W of $\mathbb{R}^n$ are **orthogonal**, written $U \perp W$, if $u'w = 0$ for all $u \in U$ and $w \in W$; their sum $U + W$ is then called an **orthogonal sum**.

Theorem 6.6 ([5, Lemma 12.1.1, Corollaries 12.1.2, 12.5.3], [8, Ch. 12, Exercise 1], [6, Exercise 3.48(c)])**.** Let W be a subspace of $\mathbb{R}^n$.

(a) If $w_1, \ldots, w_k$ span W and $A = [w_1 \ \cdots \ w_k] \in \mathbb{R}^{n\times k}$, then $y \perp W$ if and only if $y'w_1 = \cdots = y'w_k = 0$, that is, $A'y = 0$. So $W^\perp = \mathcal{N}(A')$, a subspace of $\mathbb{R}^n$.

(b) $W \cap W^\perp = \{0\}$.

(c) $\dim W + \dim W^\perp = n$ and $\mathbb{R}^n = W \oplus W^\perp$: every $y \in \mathbb{R}^n$ is $y = w + v$ with $w \in W$ and $v \in W^\perp$ in exactly one way.

Proof. (a) If $y \perp W$, then $y'w_j = 0$, as $w_j \in W$. Conversely, let all $y'w_j = 0$. Every $w \in W$ is $w = Ac$ for some $c \in \mathbb{R}^k$ (Definition 3.6 and Theorem 1.4(b)), and $y'w = (A'y)'c = 0$. The jth entry of $A'y$ is $w_j'y$ (Theorem 1.4(d)).

(b) If $y \in W \cap W^\perp$, then $y \perp y$, so $y = 0$.

(c) Let $d = \dim W$. If $d = 0$, then $W = \{0\}$ and $W^\perp = \mathbb{R}^n$. Let $d \geqslant 1$.

- Let $A \in \mathbb{R}^{n\times d}$ have a basis of W as its columns, so $\operatorname{rank} A = d$ (Definition 4.11, as $\mathcal{C}(A) = W$). By (a) and Theorem 4.12, $\dim W^\perp = \dim \mathcal{N}(A') = n - \operatorname{rank} A = n - d$.
- By (b), the sum $W + W^\perp$ is direct (Definition 3.30), and a basis of W together with a basis of $W^\perp$, $d + (n - d) = n$ vectors, is a basis of $W \oplus W^\perp$ (Theorem 3.33). So $W \oplus W^\perp$ is a subspace of $\mathbb{R}^n$ (Theorem 3.31(a)) of dimension n, hence equal to $\mathbb{R}^n$ (Theorem 4.5(a)). The decomposition $y = w + v$ is unique by Theorem 3.32. □

Corollary 6.7 ([3, Result 2.9], [5, Lemma 12.5.2])**.** For $A \in \mathbb{R}^{m\times n}$,

$$\mathcal{N}(A) = \mathcal{C}(A')^\perp, \quad \mathbb{R}^n = \mathcal{C}(A') \oplus \mathcal{N}(A); \qquad \mathcal{N}(A') = \mathcal{C}(A)^\perp, \quad \mathbb{R}^m = \mathcal{C}(A) \oplus \mathcal{N}(A')$$

So $Ax = 0$ if and only if x is orthogonal to every (transposed) row of A, and the four subspaces of A (Definition 4.7) are orthogonal in pairs: the transposed row space $\mathcal{C}(A')$ and the null space in $\mathbb{R}^n$, the column space and the left null space in $\mathbb{R}^m$.

Proof. The columns of A' span $\mathcal{C}(A')$, so Theorem 6.6(a) gives $\mathcal{C}(A')^\perp = \mathcal{N}((A')') = \mathcal{N}(A)$; applied to the columns of A, it gives $\mathcal{C}(A)^\perp = \mathcal{N}(A')$. The direct sums follow from Theorem 6.6(c). □

Theorem 6.8 ([5, Theorem 12.5.4, Corollaries 12.5.5, 12.5.6])**.** Let W, W_1 and W_2 be subspaces of $\mathbb{R}^n$.

(a) $(W^\perp)^\perp = W$.

(b) $W_1 \subseteq W_2$ if and only if $W_2^\perp \subseteq W_1^\perp$.

(c) $W_1 = W_2$ if and only if $W_1^\perp = W_2^\perp$.

Proof. (a)

- If $w \in W$, then $w'v = v'w = 0$ for every $v \in W^\perp$, so $w \in (W^\perp)^\perp$. Thus $W \subseteq (W^\perp)^\perp$.

- Conversely, let $y \in (W^\perp)^\perp$, and write $y = w + v$ with $w \in W$ and $v \in W^\perp$ (Theorem 6.6(c)).
- Then $y'v = 0$, as $y \perp W^\perp$, and $w'v = 0$, as $v \perp W$; so $v'v = (y - w)'v = y'v - w'v = 0$.
- Hence $v = 0$ and $y = w \in W$.

(b)

- For subspaces Y, U of $\mathbb{R}^n$: $Y \subseteq U$ if and only if $Y \perp U^\perp$. Indeed, $Y \perp U^\perp$ says that every vector of Y lies in $(U^\perp)^\perp$, which is U by (a).
- By this, (a) for W_1, the symmetry of $\perp$ ($u'w = w'u$), and this again with $Y = W_2^\perp$ and $U = W_1^\perp$,

$$W_1 \subseteq W_2 \iff W_1 \perp W_2^\perp \iff (W_1^\perp)^\perp \perp W_2^\perp \iff W_2^\perp \perp (W_1^\perp)^\perp \iff W_2^\perp \subseteq W_1^\perp$$

(c) If $W_1^\perp = W_2^\perp$, then $W_1^\perp \subseteq W_2^\perp$ and $W_2^\perp \subseteq W_1^\perp$, so $W_2 \subseteq W_1$ and $W_1 \subseteq W_2$ by (b). Conversely, if $W_1 = W_2$, then $W_1 \subseteq W_2$ and $W_2 \subseteq W_1$, so $W_2^\perp \subseteq W_1^\perp$ and $W_1^\perp \subseteq W_2^\perp$ by (b). □

Corollary 6.9 ([5, Lemma 12.5.2, Theorem 12.5.4, Corollary 12.5.7])**.** For $A \in \mathbb{R}^{m\times n}$, $\mathcal{C}(A') = \mathcal{N}(A)^\perp$ and $\mathcal{C}(A) = \mathcal{N}(A')^\perp$. Hence, for $b \in \mathbb{R}^m$, the system $Ax = b$ has a solution if and only if $z'b = 0$ for every $z \in \mathbb{R}^m$ with $A'z = 0$; if $z_1, \dots, z_s$ span $\mathcal{N}(A')$, it suffices that $z_1'b = \cdots = z_s'b = 0$.

Proof. • By Corollary 6.7, $\mathcal{N}(A) = \mathcal{C}(A')^\perp$ and $\mathcal{N}(A') = \mathcal{C}(A)^\perp$. By Theorem 6.8(a), $\mathcal{N}(A)^\perp = (\mathcal{C}(A')^\perp)^\perp = \mathcal{C}(A')$ and $\mathcal{N}(A')^\perp = \mathcal{C}(A)$.

- $Ax = b$ has a solution if and only if $b \in \mathcal{C}(A)$ (Definition 4.7), that is, $b \perp \mathcal{N}(A')$. By Theorem 6.6(a), applied to $\mathcal{N}(A')$ and its spanning vectors $z_1, \dots, z_s$, this means $z_1'b = \cdots = z_s'b = 0$. □

Example 6.10. Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}$$

be the matrix of Example 4.9. The nonzero rows of its RREF are ρ_1' and ρ_2', with $\rho_1 = (1, 0, 3, -1)'$ and $\rho_2 = (0, 1, -1, 2)'$, and they form a basis of $\mathcal{R}(A)$; $s_1 = (-3, 1, 1, 0)'$ and $s_2 = (1, -2, 0, 1)'$ form a basis of $\mathcal{N}(A)$ (Example 4.9). Transposing maps $\mathcal{R}(A)$ onto $\mathcal{C}(A')$, as $(z'A)' = A'z$ ($z \in \mathbb{R}^3$), preserves linear combinations and is undone by transposing again; so ρ_1, ρ_2 form a basis of $\mathcal{C}(A')$. Indeed

$$\rho_1's_1 = -3+0+3+0 = 0, \quad \rho_1's_2 = 1+0+0-1 = 0, \quad \rho_2's_1 = 0+1-1+0 = 0, \quad \rho_2's_2 = 0-2+0+2 = 0$$

and $2 + 2 = 4$. In $\mathbb{R}^3$, $y = (-1, -1, 1)'$ is a basis of $\mathcal{N}(A')$, orthogonal to the basis $a_1 = (1, 1, 2)'$, $a_2 = (2, 3, 5)'$ of $\mathcal{C}(A)$: $-1 - 1 + 2 = 0$ and $-2 - 3 + 5 = 0$; and $2 + 1 = 3$. By Corollary 6.9, $Ax = b$ has a solution exactly when $y'b = -b_1 - b_2 + b_3 = 0$. For $b = (1, 2, 3)'$, $-1 - 2 + 3 = 0$, and indeed $A(-1, 1, 0, 0)' = -a_1 + a_2 = (1, 2, 3)'$ (Theorem 1.4(b)); for $b = e_1$, $y'e_1 = -1 \neq 0$, so $Ax = e_1$ has no solution.

Orthonormal bases, Gram–Schmidt, and QR

Proposition 6.11 ([1, §8.2, Theorem 2, (8-8)], [3, Result 2.5])**.** Let $u_1, \dots, u_k \in \mathbb{R}^n$ be nonzero and pairwise orthogonal ($u_i'u_j = 0$ for $i \neq j$).

(a) If $y = c_1u_1 + \cdots + c_ku_k$, then $c_j = u_j'y/\|u_j\|^2$ for $j = 1, \dots, k$.

(b) $u_1, \dots, u_k$ are linearly independent.

Proof. (a) $u_j'y = \sum_i c_iu_j'u_i = c_j\|u_j\|^2$, and $\|u_j\|^2 > 0$. (b) If $\sum_i c_iu_i = 0$, then (a) with $y = 0$ gives every $c_j = 0$. □

Definition 6.12 ([1, §8.2], [2, §5.4], [5, §8.4a]). Vectors $q_1, \dots, q_p \in \mathbb{R}^n$ are **orthonormal** if $q_i' q_j = 0$ for $i \neq j$ and $\|q_j\| = 1$ for every j. A matrix $Q = [q_1 \ \cdots \ q_p] \in \mathbb{R}^{n\times p}$ has **orthonormal columns** if $Q'Q = I_p$, since $(Q'Q)_{ij} = q_i' q_j$ (Theorem 1.4(d)). A square matrix $Q \in \mathbb{R}^{n\times n}$ with $Q'Q = I_n$ is an **orthogonal matrix**. A basis of a subspace consisting of orthonormal vectors is an **orthonormal basis**.

Proposition 6.13 ([4, Ch. VII, Theorems 3.1, 3.2], [6, Exercise 8.4], [5, §8.4a]). Let $Q \in \mathbb{R}^{n\times p}$ satisfy $Q'Q = I_p$.

(a) $(Qx)'(Qy) = x'y$ and $\|Qx\| = \|x\|$ for all $x, y \in \mathbb{R}^p$: Q preserves inner products and lengths.

(b) If $p = n$, then Q is invertible, $Q^{-1} = Q'$ and $QQ' = I_n$: the transposed rows of Q are orthonormal too.

Proof. (a) $(Qx)'(Qy) = x'Q'Qy = x'y$; take $y = x$. (b) Q' is a left inverse of the square matrix Q, so $Q^{-1} = Q'$ (Theorem 2.16), and $QQ' = QQ^{-1} = I_n$. □

Theorem 6.14 (Gram–Schmidt; [1, §8.2, Theorem 3 and its Corollary], [6, Exercise 3.49]). Let $a_1, \dots, a_p \in \mathbb{R}^n$ be linearly independent. Put $b_1 = a_1$ and, for $j = 2, \dots, p$,

$$b_j = a_j - t_{1j} b_1 - t_{2j} b_2 - \cdots - t_{j-1,j} b_{j-1}, \qquad t_{ij} = \frac{a_j' b_i}{\|b_i\|^2} \quad (i < j)$$

Then $b_1, \dots, b_p$ are nonzero and pairwise orthogonal, and $\operatorname{span}\{b_1, \dots, b_j\} = \operatorname{span}\{a_1, \dots, a_j\}$ for every j. Hence the vectors $q_j = b_j/\|b_j\|$ $(1 \leqslant j \leqslant p)$ form an orthonormal basis of $\operatorname{span}\{a_1, \dots, a_p\}$.

Proof. Induction on j: $b_1 = a_1 \neq 0$, since an independent list does not contain 0. Let $j \geqslant 2$, and let $b_1, \dots, b_{j-1}$ be nonzero, pairwise orthogonal, with span $W_{j-1} = \operatorname{span}\{a_1, \dots, a_{j-1}\}$; so b_j is defined.

- Orthogonal: for $l < j$, $b_l' b_j = b_l' a_j - \sum_{i<j} t_{ij} b_l' b_i = b_l' a_j - t_{lj}\|b_l\|^2 = 0$.
- Nonzero: otherwise $a_j = \sum_{i<j} t_{ij} b_i \in W_{j-1}$, and $a_1, \dots, a_j$ would be dependent (Proposition 3.13(c)) ✘
- Span: b_j is a combination of a_j and $b_1, \dots, b_{j-1} \in W_{j-1}$, and $a_j = b_j + \sum_{i<j} t_{ij} b_i$; so each of the lists $a_1, \dots, a_j$ and $b_1, \dots, b_j$ lies in the span of the other, and the two spans are equal (Theorem 3.7).

Finally, the q_j are pairwise orthogonal unit vectors, hence independent (Proposition 6.11), and they span $\operatorname{span}\{b_1, \dots, b_p\} = \operatorname{span}\{a_1, \dots, a_p\}$. □

Corollary 6.15 ([1, §8.2, Corollary to Theorem 3], [5, Theorem 6.4.5], [4, Ch. V, Theorems 2.1, 2.3]). Let W be a subspace of $\mathbb{R}^n$ with $\dim W = d \geqslant 1$.

(a) W has an orthonormal basis.

(b) If $q_1, \dots, q_p \in W$ are orthonormal and $p < d$, there are $q_{p+1}, \dots, q_d \in W$ such that $q_1, \dots, q_d$ form an orthonormal basis of W.

(c) If $Q \in \mathbb{R}^{n\times p}$ has orthonormal columns and $p < n$, there is $T \in \mathbb{R}^{n\times(n-p)}$ such that $[Q\ T]$ is an orthogonal matrix. For every such T, the columns of T form an orthonormal basis of $\mathcal{C}(Q)^\perp$, and $QQ' + TT' = I_n$.

Proof. (a) W has a basis $a_1, \dots, a_d$ (Theorem 4.4). By Theorem 6.14, the vectors $q_j = b_j/\|b_j\|$ obtained from it form an orthonormal basis of $\operatorname{span}\{a_1, \dots, a_d\} = W$.

(b)

- $q_1, \dots, q_p$ are independent (Proposition 6.11(b)), so they extend to a basis $q_1, \dots, q_p, a_{p+1}, \dots, a_d$ of W (Theorem 4.5(b)).

- Apply Theorem 6.14 to this list. Its first p outputs are $b_j = q_j$: $b_1 = q_1$, and if $b_i = q_i$ for all $i < j \leqslant p$, then $t_{ij} = q_j' q_i / \|q_i\|^2 = 0$, so $b_j = q_j$.
- Hence the normalized outputs $q_1, \dots, q_p, b_{p+1}/\|b_{p+1}\|, \dots, b_d/\|b_d\|$ form an orthonormal basis of W (Theorem 6.14), and its first p vectors are the given ones.

(c)

- The columns $q_1, \dots, q_p$ of Q are orthonormal (Definition 6.12). By (b) with $W = \mathbb{R}^n$, there are $q_{p+1}, \dots, q_n$ such that $q_1, \dots, q_n$ form an orthonormal basis of $\mathbb{R}^n$; then $T = [q_{p+1} \ \cdots \ q_n]$ makes $[Q \ T]$ an $n \times n$ matrix with orthonormal columns, that is, an orthogonal matrix.
- Let T be any such matrix. Each column of T is orthogonal to every column of Q, so it lies in $\mathcal{C}(Q)^\perp$ (Theorem 6.6(a)). Orthonormal vectors are independent (Proposition 6.11(b)); so $\operatorname{rank} Q = p$ (Theorem 4.13(b)), $\dim \mathcal{C}(Q)^\perp = n - p$ (Theorem 6.6(c)), and the $n - p$ independent columns of T form a basis of $\mathcal{C}(Q)^\perp$ (Theorem 4.5(d)).
- $[Q \ T][Q \ T]' = I_n$ (Proposition 6.13(b)), and by the outer product expansion (Theorem 1.4(d)), $[Q \ T][Q \ T]' = q_1 q_1' + \cdots + q_n q_n' = QQ' + TT'$. □

Theorem 6.16 (QR decomposition; [2, Theorem 5.2], [5, §6.4d], [6, Exercises 7.34, 7.35])**.** Let $A = [a_1 \ \cdots \ a_p] \in \mathbb{R}^{n \times p}$ have full column rank p. Then $A = QR$, where $Q \in \mathbb{R}^{n \times p}$ has orthonormal columns ($Q'Q = I_p$) and $R \in \mathbb{R}^{p \times p}$ is upper triangular with positive diagonal entries; moreover $R = Q'A$.

Proof.
- The columns of A are independent (Theorem 4.13(b)). Apply Theorem 6.14 to them, and let $Q = [q_1 \ \cdots \ q_p]$.
- The construction gives $a_j = b_j + \sum_{i<j} t_{ij} b_i = \sum_{i \leqslant j} r_{ij} q_i$, where $r_{jj} = \|b_j\| > 0$ and $r_{ij} = t_{ij}\|b_i\|$ for $i < j$. Let $R = (r_{ij})$, with $r_{ij} = 0$ for $i > j$. Then $\operatorname{col}_j(QR) = Q \operatorname{col}_j(R) = \sum_{i \leqslant j} r_{ij} q_i = a_j$ (Theorem 1.4(a),(b)), so $A = QR$.
- $Q'Q = I_p$, as the q_j are orthonormal; hence $Q'A = Q'QR = R$. □

Example 6.17 (cf. [1, §8.2, Example 12])**.** Find the QR decomposition of

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 0 & 2 & 4 \\ 2 & 2 & 5 \end{pmatrix}$$

Solution. Apply Gram–Schmidt (Theorem 6.14) to the columns a_1, a_2, a_3.

- $b_1 = a_1 = (1, 2, 0, 2)'$, $\|b_1\|^2 = 1 + 4 + 0 + 4 = 9$.
- $a_2' b_1 = 3 + 2 + 0 + 4 = 9$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (3, 1, 2, 2)' - (1, 2, 0, 2)' = (2, -1, 2, 0)'$, with $\|b_2\|^2 = 4 + 1 + 4 + 0 = 9$.
- $a_3' b_1 = 2 + 6 + 0 + 10 = 18$ and $a_3' b_2 = 4 - 3 + 8 + 0 = 9$, so $t_{13} = 2$, $t_{23} = 1$ and

$$b_3 = a_3 - 2b_1 - b_2 = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \qquad \|b_3\|^2 = 4 + 0 + 4 + 1 = 9$$

- Check: $b_1' b_2 = 2 - 2 + 0 + 0 = 0$, $b_1' b_3 = -2 + 0 + 0 + 2 = 0$, $b_2' b_3 = -4 + 0 + 4 + 0 = 0$.

- All three lengths are 3, so $q_j = \frac{1}{3}b_j$, $r_{jj} = 3$, $r_{12} = 3t_{12} = 3$, $r_{13} = 3t_{13} = 6$ and $r_{23} = 3t_{23} = 3$:

$$Q = \frac{1}{3}\begin{pmatrix} 1 & 2 & -2 \\ 2 & -1 & 0 \\ 0 & 2 & 2 \\ 2 & 0 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 3 & 3 & 6 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}$$

As a check, $R = Q'A$; for instance $r_{23} = q_2'a_3 = \frac{1}{3}(4 - 3 + 8 + 0) = 3$.

Example 6.18. Extend the orthonormal columns q_1, q_2, q_3 of Q in Example 6.17 to an orthonormal basis of $\mathbb{R}^4$.

Solution. By Corollary 6.15(c) we need a unit vector q_4 orthogonal to q_1, q_2, q_3, that is, a normalized $v \neq 0$ with $b_1'v = b_2'v = b_3'v = 0$:

$$v_1 + 2v_2 + 2v_4 = 0, \qquad 2v_1 - v_2 + 2v_3 = 0, \qquad -2v_1 + 2v_3 + v_4 = 0$$

- The third equation gives $v_4 = 2v_1 - 2v_3$, and the second $v_2 = 2v_1 + 2v_3$.
- Substituting both into the first: $v_1 + (4v_1 + 4v_3) + (4v_1 - 4v_3) = 9v_1 = 0$, so $v_1 = 0$, $v_2 = 2v_3$ and $v_4 = -2v_3$.
- With $v_3 = 1$: $v = (0, 2, 1, -2)'$, $v'v = 0 + 4 + 1 + 4 = 9$, so $q_4 = \frac{1}{3}v$.

Hence

$$[Q\ q_4] = \frac{1}{3}\begin{pmatrix} 1 & 2 & -2 & 0 \\ 2 & -1 & 0 & 2 \\ 0 & 2 & 2 & 1 \\ 2 & 0 & 1 & -2 \end{pmatrix}$$

is an orthogonal matrix. By Corollary 6.15(c), q_4 alone is an orthonormal basis of $\mathcal{C}(Q)^\perp = \mathcal{C}(A)^\perp$ ($\mathcal{C}(Q) = \mathcal{C}(A)$ by Theorem 6.14), and $QQ' + q_4q_4' = I_4$.

Orthogonal matrices

Theorem 6.19 ([5, §8.4], [4, Ch. VII, §3], [7, Ch. VII, §3])**.** (a) For $Q \in \mathbb{R}^{n\times n}$ the following are equivalent:

(i) $Q'Q = I_n$, that is, Q is orthogonal;
(ii) $QQ' = I_n$;
(iii) Q is invertible and $Q^{-1} = Q'$;
(iv) the columns of Q are orthonormal;
(v) the transposed rows of Q are orthonormal;
(vi) $(Qx)'(Qy) = x'y$ for all $x, y \in \mathbb{R}^n$;
(vii) $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$.

(b) If $P, Q \in \mathbb{R}^{n\times n}$ are orthogonal, so are PQ, Q' and Q^{-1}.

(c) If Q is orthogonal, then $\det Q = 1$ or $\det Q = -1$.

(d) Every permutation matrix P_φ (Definition 5.40) is orthogonal, and $\det P_\varphi = 1$ or $\det P_\varphi = -1$.

Proof. (a)

- (i)⇒(ii),(iii): Proposition 6.13(b).
- (ii)⇒(iii): Q' is a right inverse of the square matrix Q, so Q is invertible and $Q^{-1} = Q'$ (Theorem 2.16). (iii)⇒(i): $Q'Q = Q^{-1}Q = I_n$.

- (i)⇔(iv): $(Q'Q)_{ij} = q_i'q_j$ for the columns $q_1, \dots, q_n$ of Q (Theorem 1.4(d)); so $Q'Q = I_n$ exactly when $q_i'q_j$ is 1 for $i = j$ and 0 for $i \neq j$ (Definition 6.12).
- (ii)⇔(v): likewise, $(QQ')_{ij} = \mathrm{row}_i(Q)\,\mathrm{row}_j(Q)'$.
- (i)⇒(vi): Proposition 6.13(a). (vi)⇒(vii): take $y = x$.
- (vi)⇒(i): with $x = e_i$ and $y = e_j$, $(Q'Q)_{ij} = q_i'q_j = (Qe_i)'(Qe_j) = e_i'e_j$, which is 1 for $i = j$ and 0 otherwise (Theorem 1.4(b),(d)).
- (vii)⇒(vi): by polarization (Proposition 6.3(a)) and $Q(x \pm y) = Qx \pm Qy$,

$$4(Qx)'(Qy) = \|Q(x+y)\|^2 - \|Q(x-y)\|^2 = \|x+y\|^2 - \|x-y\|^2 = 4x'y$$

(b)

- By (iii), $P^{-1} = P'$ and $Q^{-1} = Q'$, so $(PQ)^{-1} = Q^{-1}P^{-1} = Q'P' = (PQ)'$ (Theorems 2.13(b) and 1.9); by (iii), PQ is orthogonal.
- $(Q')'Q' = QQ' = I_n$ by (ii), so Q' is orthogonal; and $Q^{-1} = Q'$.

(c) $1 = \det I_n = \det(Q'Q) = \det Q' \cdot \det Q = (\det Q)^2$ (Definition 5.13(d), Theorems 5.18(b) and 5.22).

(d) $P_\varphi' P_\varphi = I_n$ (Proposition 5.41(c)), so P_φ is orthogonal, and $\det P_\varphi = \pm 1$ by (c). □

So multiplying by an orthogonal matrix keeps inner products, lengths and distances: $\|Qx - Qy\| = \|Q(x-y)\| = \|x-y\|$. It only changes the coordinates in which a data vector is written.

Example 6.20 ([5, §8.4b; Ch. 8, Exercise 4(b)], [2, §5.4.1])**.** Let $n \geqslant 2$. For $k = 2, \dots, n$ let $u_k = (1, \dots, 1, -(k-1), 0, \dots, 0)' \in \mathbb{R}^n$: $k-1$ entries 1, then $-(k-1)$ in position k, then zeros. The **Helmert matrix** of order n is the $n \times n$ matrix K_n with rows

$$p_1' = \frac{1}{\sqrt{n}}\,\mathbf{1}_n', \qquad p_k' = \frac{1}{\sqrt{k(k-1)}}\,u_k' \quad (k = 2, \dots, n)$$

For instance

$$K_4 = \begin{pmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} & 0 \\ 1/\sqrt{12} & 1/\sqrt{12} & 1/\sqrt{12} & -3/\sqrt{12} \end{pmatrix}$$

(a) K_n is orthogonal. By Theorem 6.19(a) it suffices that $p_1, \dots, p_n$ are orthonormal.

- $\mathbf{1}_n'u_k = (k-1) - (k-1) = 0$ for $k \geqslant 2$.
- For $2 \leqslant j < k$: u_j is 0 outside positions $1, \dots, j$, and these positions of u_k hold 1 (as $j \leqslant k-1$); so $u_j'u_k$ is the sum of the entries of u_j, which is $\mathbf{1}_n'u_j = 0$.
- $\mathbf{1}_n'\mathbf{1}_n = n$ and $u_k'u_k = (k-1)\cdot 1 + (k-1)^2 = k(k-1)$, so every p_k has length 1.

(b) For data $y \in \mathbb{R}^n$ with mean $\bar{y}$, the first entry of $K_n y$ is $p_1'y = \frac{1}{\sqrt{n}}\mathbf{1}'y = \sqrt{n}\,\bar{y}$, and for $k \geqslant 2$ its kth entry is the **contrast** $p_k'y = \big(y_1 + \cdots + y_{k-1} - (k-1)y_k\big)/\sqrt{k(k-1)}$, a combination of $y_1, \dots, y_n$ whose coefficients add up to 0.

- By (a) and Theorem 6.19(a)(vii), $\sum_i y_i^2 = \|K_n y\|^2 = n\bar{y}^2 + \sum_{k=2}^n (p_k'y)^2$. As $\sum_i (y_i - \bar{y})^2 = \sum_i y_i^2 - n\bar{y}^2$ (Example 1.16), the sum of squared deviations is the sum of the squares of the $n-1$ contrasts:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{k=2}^n (p_k'y)^2$$

- In matrix form: $K_n'K_n = I_n$ reads $p_1p_1' + \cdots + p_np_n' = I_n$ (Theorem 1.4(d)), and $p_1p_1' = \frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$; hence $p_2p_2' + \cdots + p_np_n' = I_n - \bar{J}_n = C_n$.

(c) For $n = 4$ and $y = (5, 3, 1, 7)'$: $\bar{y} = 4$, $p_1'y = 16/2 = 8 = \sqrt{4} \cdot 4$, and

$$p_2'y = \frac{5-3}{\sqrt{2}} = \sqrt{2}, \qquad p_3'y = \frac{5+3-2}{\sqrt{6}} = \sqrt{6}, \qquad p_4'y = \frac{5+3+1-21}{\sqrt{12}} = -\sqrt{12}$$

so the contrasts give $2 + 6 + 12 = 20 = 1 + 1 + 9 + 9 = \sum_i (y_i - 4)^2$, and $\|y\|^2 = 25 + 9 + 1 + 49 = 84 = 64 + 20$.

Example 6.21 ([8, Ch. 10, Exercise 3])**.** (a) If $A \in \mathbb{R}^{n \times n}$ is symmetric and idempotent, then $I_n - 2A$ is symmetric and orthogonal: $(I - 2A)'(I - 2A) = (I - 2A)^2 = I - 4A + 4A^2 = I$.

(b) Let $h \in \mathbb{R}^n$, $h \neq 0$. The matrix $A = hh'/(h'h)$ is symmetric and idempotent, since $(hh')' = hh'$ and $hh'hh' = h(h'h)h'$. So the **reflection**

$$F_h = I_n - \frac{2}{h'h} hh'$$

is symmetric and orthogonal, and $F_h^2 = F_h' F_h = I_n$. It reverses h and fixes every vector orthogonal to h: $F_h h = h - 2h = -h$, and $F_h x = x - \frac{2h'x}{h'h} h = x$ if $h'x = 0$. By Theorem 5.31 with $u = -\frac{2}{h'h} h$ and $v = h$, $\det F_h = 1 + v'u = 1 - 2 = -1$.

(c) For $h = e_r - e_s$ ($r \neq s$), $h'h = 2$ and $F_h = I_n - (e_r - e_s)(e_r - e_s)'$ is the interchange matrix P_{rs} of Definition 2.14: its entries (r, r) and (s, s) are $1 - 1 = 0$, its entries (r, s) and (s, r) are 1, and the others are those of I_n. Interchanging two coordinates is a reflection.

(d) For $h = (1, 1, 2, 2)'$, $h'h = 10$ and

$$F_h = I_4 - \frac{1}{5} \begin{pmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 2 & 4 & 4 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 4 & -1 & -2 & -2 \\ -1 & 4 & -2 & -2 \\ -2 & -2 & 1 & -4 \\ -2 & -2 & -4 & 1 \end{pmatrix}$$

Its columns are orthonormal (for instance $16 + 1 + 4 + 4 = 25$ and $-4 - 4 + 4 + 4 = 0$), and $F_h h = \frac{1}{5}(4 - 1 - 4 - 4,\ -1 + 4 - 4 - 4,\ -2 - 2 + 2 - 8,\ -2 - 2 - 8 + 2)' = -(1, 1, 2, 2)'$.

Orthogonal projections

Definition 6.22 ([1, §8.2], [12, Remark 2.10])**.** Let W be a subspace of $\mathbb{R}^n$ and $y \in \mathbb{R}^n$. By Theorem 6.6(c), $y = \hat{w} + (y - \hat{w})$ with $\hat{w} \in W$ and $y - \hat{w} \in W^\perp$ for exactly one vector $\hat{w}$, the **orthogonal projection** of y onto W. Thus $\hat{w}$ is the only vector of W with $y - \hat{w} \perp W$.

Theorem 6.23 ([1, §8.2, Theorem 4], [12, Theorem 2.5], [5, Theorem 12.4.1])**.** Let $\hat{w}$ be the orthogonal projection of $y \in \mathbb{R}^n$ onto a subspace W of $\mathbb{R}^n$. Then

$$\|y - w\|^2 = \|y - \hat{w}\|^2 + \|\hat{w} - w\|^2 \qquad (w \in W)$$

Hence $\hat{w}$ is the vector of W closest to y: $\|y - w\| > \|y - \hat{w}\|$ for every $w \in W$ with $w \neq \hat{w}$. Equivalently, a vector $w_0 \in W$ satisfies $\|y - w_0\| \leqslant \|y - w\|$ for every $w \in W$ if and only if $y - w_0 \perp W$, and only $w_0 = \hat{w}$ does.

Proof. $y - w = (y - \hat{w}) + (\hat{w} - w)$, where $\hat{w} - w \in W$ and $y - \hat{w} \perp W$; apply Pythagoras (Proposition 6.2). If $w \neq \hat{w}$, then $\|\hat{w} - w\|^2 > 0$. So $w_0 \in W$ is a closest vector if and only if $w_0 = \hat{w}$, and by Definition 6.22 this holds if and only if $y - w_0 \perp W$. □

The number $\|y - \hat{w}\| = \min_{w \in W} \|y - w\|$ is the **distance** from y to W.

Theorem 6.24 ([1, §8.2, Theorem 4(iii), Corollary after Theorem 5], [5, Theorem 12.1.3, §12.2a, Theorem 12.5.8])**.** Let W be a subspace of $\mathbb{R}^n$, $y \in \mathbb{R}^n$, and $\hat{w}$ the orthogonal projection of y onto W.

(a) If $u_1, \dots, u_k$ ($k \geqslant 1$) form an orthogonal basis of W, that is, a basis of pairwise orthogonal vectors, then

$$\hat{w} = \frac{u_1' y}{u_1' u_1}\, u_1 + \cdots + \frac{u_k' y}{u_k' u_k}\, u_k$$

In particular, if $q_1, \dots, q_k$ form an orthonormal basis of W and $Q = [q_1 \ \cdots \ q_k]$, then $\hat{w} = (q_1' y) q_1 + \cdots + (q_k' y) q_k = QQ'y$.

(b) $\hat{w} = y$ if and only if $y \in W$.

(c) (**Bessel's inequality**) In (a), $\|\hat{w}\|^2 = \sum_{j=1}^{k} (u_j' y)^2 / u_j' u_j \leqslant \|y\|^2$, with equality if and only if $y \in W$; for an orthonormal basis, $\sum_{j=1}^{k} (q_j' y)^2 \leqslant \|y\|^2$.

(d) $y - \hat{w}$ is the orthogonal projection of y onto $W^\perp$.

Proof. By Definition 6.22, $\hat{w}$ is the only vector of W with $y - \hat{w} \perp W$.

(a)

- The u_j are nonzero (Proposition 3.13(a)); put $c_j = u_j' y / u_j' u_j$ and $w_0 = c_1 u_1 + \cdots + c_k u_k \in W$.
- For $i = 1, \dots, k$, $u_i'(y - w_0) = u_i' y - \sum_j c_j u_i' u_j = u_i' y - c_i u_i' u_i = 0$. As $u_1, \dots, u_k$ span W, $y - w_0 \perp W$ (Theorem 6.6(a)); so $w_0 = \hat{w}$.
- For an orthonormal basis, $q_j' q_j = 1$, and $\sum_j (q_j' y) q_j = Q(Q'y)$, as $Q'y$ has the entries $q_j' y$ (Theorem 1.4(b),(d)).

(b) If $\hat{w} = y$, then $y \in W$. Conversely, if $y \in W$, then $y - y = 0 \perp W$, so $\hat{w} = y$.

(c)

- $y = \hat{w} + (y - \hat{w})$ with $\hat{w} \perp y - \hat{w}$, so $\|y\|^2 = \|\hat{w}\|^2 + \|y - \hat{w}\|^2 \geqslant \|\hat{w}\|^2$ (Proposition 6.2).
- The vectors $c_j u_j$ are pairwise orthogonal, so $\|\hat{w}\|^2 = \sum_j c_j^2\, u_j' u_j = \sum_j (u_j' y)^2 / u_j' u_j$ (Proposition 6.3(b)).
- Equality holds if and only if $\|y - \hat{w}\| = 0$, that is, $y = \hat{w}$, that is, $y \in W$ by (b).

(d) $y - \hat{w} \in W^\perp$, and $y - (y - \hat{w}) = \hat{w} \in W = (W^\perp)^\perp$ (Theorem 6.8(a)), that is, $y - (y - \hat{w}) \perp W^\perp$. By Definition 6.22 for the subspace $W^\perp$, $y - \hat{w}$ is the orthogonal projection of y onto $W^\perp$. □

Example 6.25 (cf. [1, §8.2, Example 14], [5, §§12.2b, 12.5c])**.** Let $W = \operatorname{span}\{a_1, a_2\} \subseteq \mathbb{R}^4$, where $a_1 = (1, 2, 0, 2)'$ and $a_2 = (3, 1, 2, 2)'$ are the first two columns of A in Example 6.17, and let $y = (-2, 7, 1, 3)'$. Find the orthogonal projections of y onto W and onto $W^\perp$, and the distance from y to the nearest vector of W.

Solution. By Example 6.17 and Theorem 6.14, $b_1 = (1, 2, 0, 2)'$ and $b_2 = (2, -1, 2, 0)'$ form an orthogonal basis of W, with $b_1' b_1 = b_2' b_2 = 9$.

- $b_1' y = -2 + 14 + 0 + 6 = 18$ and $b_2' y = -4 - 7 + 2 + 0 = -9$. By Theorem 6.24(a),

$$\hat{w} = \frac{18}{9}\, b_1 - \frac{9}{9}\, b_2 = \begin{pmatrix} 2 \\ 4 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ -2 \\ 4 \end{pmatrix}$$

- By Theorem 6.24(d), the projection of y onto $W^\perp$ is $y - \hat{w} = (-2, 2, 3, -1)'$. Check: $b_1'(y - \hat{w}) = -2 + 4 + 0 - 2 = 0$ and $b_2'(y - \hat{w}) = -4 - 2 + 6 + 0 = 0$.
- The vector of W nearest to y is $\hat{w}$ (Theorem 6.23), at distance $\|y - \hat{w}\| = \sqrt{4 + 4 + 9 + 1} = \sqrt{18} = 3\sqrt{2}$.
- Bessel's inequality (Theorem 6.24(c)): $\|\hat{w}\|^2 = 18^2/9 + 9^2/9 = 36 + 9 = 45 \leqslant \|y\|^2 = 4 + 49 + 1 + 9 = 63$, and $63 = 45 + 18$.

Theorem 6.26 ([12, Definition 2.6, Theorem 2.9(i)–(iii), Remark 2.10], [5, Theorem 12.3.1])**.** Let $A \in \mathbb{R}^{n\times p}$ have full column rank p. Then $A'A$ is invertible. Put

$$P = A(A'A)^{-1}A' \in \mathbb{R}^{n\times n}$$

(a) For every $y \in \mathbb{R}^n$, Py is the orthogonal projection of y onto $\mathcal{C}(A)$, and $y = Py + (I_n - P)y$ is the decomposition of y along $\mathbb{R}^n = \mathcal{C}(A) \oplus \mathcal{C}(A)^{\perp}$.

(b) $P' = P = P^2$ and $(I_n - P)' = I_n - P = (I_n - P)^2$: both are symmetric and idempotent.

(c) $PA = A$, $(I_n - P)A = 0$ and $P(I_n - P) = (I_n - P)P = 0$.

Proof. $\operatorname{rank}(A'A) = \operatorname{rank} A = p$ (Theorem 4.18(b)), so the $p \times p$ matrix $A'A$ is invertible (Theorem 4.13(d)).

(a) $Py = A\Big((A'A)^{-1}A'y\Big) \in \mathcal{C}(A)$, and $A'(y - Py) = A'y - (A'A)(A'A)^{-1}A'y = 0$, so $(I_n - P)y = y - Py \in \mathcal{C}(A)^{\perp}$ (Theorem 6.6(a)). By Definition 6.22, Py is the projection.

(b) $A'A$ is symmetric, hence so is its inverse: $((A'A)^{-1})' = ((A'A)')^{-1} = (A'A)^{-1}$ (Theorem 2.13(c)). So $P' = A((A'A)^{-1})'A' = P$ and $P^2 = A(A'A)^{-1}(A'A)(A'A)^{-1}A' = P$. Then $(I_n - P)' = I_n - P' = I_n - P$ and $(I_n - P)^2 = I_n - 2P + P^2 = I_n - P$.

(c) $PA = A(A'A)^{-1}A'A = A$, $(I_n - P)A = A - PA = 0$, and $P(I_n - P) = P - P^2 = 0 = (I_n - P)P$. □

Example 6.27 ([12, Definition 6.1, Remark 6.3], [6, Exercise 8.71(a)], [1, §8.2, Example 14])**.** For $A = \mathbf{1}_n$ ($p = 1$), $A'A = \mathbf{1}'\mathbf{1} = n$, so the matrix $P = A(A'A)^{-1}A'$ of Theorem 6.26 is $\frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$, and $I_n - P = I_n - \bar{J}_n = C_n$, the centering matrix of Definition 1.14: by Theorem 6.26(a), $\bar{J}_n y = \bar{y}\mathbf{1}$ (Example 3.35) is the orthogonal projection of y onto $\operatorname{span}\{\mathbf{1}\}$, and $C_n y = y - \bar{y}\mathbf{1}$ (Example 1.16). The centered vectors form $W_0 = \{x : \mathbf{1}'x = 0\} = \operatorname{span}\{\mathbf{1}\}^{\perp}$ (Theorem 6.6(a)), so the decomposition $y = \bar{y}\mathbf{1} + C_n y$ along $\mathbb{R}^n = \operatorname{span}\{\mathbf{1}\} \oplus W_0$ of Example 3.35 is the orthogonal one, and $\bar{y}\mathbf{1}$ is the constant vector closest to y (Theorem 6.23): $\sum_i (y_i - c)^2 = \|y - c\mathbf{1}\|^2 > \|y - \bar{y}\mathbf{1}\|^2 = \sum_i (y_i - \bar{y})^2$ for every $c \neq \bar{y}$. For instance, $y = (9, 3, 5, 3)'$ has $\bar{y} = 5$ and $y = 5\,\mathbf{1}_4 + (4, -2, 0, -2)'$, with $\mathbf{1}_4'(4, -2, 0, -2)' = 0$. Likewise, for any nonzero $a \in \mathbb{R}^n$ the projection of y onto $\operatorname{span}\{a\}$ is $a(a'a)^{-1}a'y = \frac{a'y}{a'a}\,a$ (Theorem 6.26(a) with the matrix a, which has full column rank 1 as $a \neq 0$).

Example 6.28. Let $a_1 = \mathbf{1}_4$, $a_2 = (2, 0, 2, 0)'$, $a_3 = (3, 3, 1, 1)'$, $A = [a_1\ a_2\ a_3] \in \mathbb{R}^{4\times 3}$, $W = \mathcal{C}(A)$ and $y = (9, 3, 5, 3)'$. Find $(A'A)^{-1}$, the matrix $P = A(A'A)^{-1}A'$ of Theorem 6.26, the vector of W closest to y, and the distance from y to W.

Solution.

- The (j, l) entry of $A'A$ is $a_j'a_l$ (Theorem 1.4(d)), so

$$A'A = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix}$$

It is symmetric (Theorem 1.9), so its cofactors (Definition 5.20) satisfy $c_{ji} = c_{ij}$ (Remark 5.25), and six of them are needed (each 2×2 determinant along its row 1, Theorem 5.21):

$$c_{11} = \det\begin{pmatrix} 8 & 8 \\ 8 & 20 \end{pmatrix} = 96, \quad c_{12} = -\det\begin{pmatrix} 4 & 8 \\ 8 & 20 \end{pmatrix} = -16, \quad c_{13} = \det\begin{pmatrix} 4 & 8 \\ 8 & 8 \end{pmatrix} = -32$$

$$c_{22} = \det\begin{pmatrix} 4 & 8 \\ 8 & 20 \end{pmatrix} = 16, \quad c_{23} = -\det\begin{pmatrix} 4 & 4 \\ 8 & 8 \end{pmatrix} = 0, \qquad c_{33} = \det\begin{pmatrix} 4 & 4 \\ 4 & 8 \end{pmatrix} = 16$$

Along row 1 (Theorem 5.21), $\det(A'A) = 4\cdot 96 + 4\cdot(-16) + 8\cdot(-32) = 64$. Hence, by Theorem 5.24,

$$(A'A)^{-1} = \frac{1}{64}\begin{pmatrix} 96 & -16 & -32 \\ -16 & 16 & 0 \\ -32 & 0 & 16 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

As $A'A$ is invertible, $\operatorname{rank} A = \operatorname{rank}(A'A) = 3$ (Theorems 4.18(b) and 4.13(d)): A has full column rank, and Theorem 6.26 applies.

- Then
$$(A'A)^{-1}A' = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & 2 & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix}$$
$$P = A(A'A)^{-1}A' = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}\frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix}$$

- By Theorem 6.26(a), Py is the orthogonal projection of y onto W, so it is the vector of W closest to y (Theorem 6.23):
$$Py = \frac{1}{4}\begin{pmatrix} 27+3+5-3 \\ 9+9-5+3 \\ 9-3+15+3 \\ -9+3+5+9 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \\ 6 \\ 2 \end{pmatrix}, \qquad (I_4 - P)y = y - Py = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$
The distance from y to W is $\|y - Py\| = \sqrt{1+1+1+1} = 2$. Check: $A'(I_4 - P)y = (1-1-1+1,\ 2-2,\ 3-3-1+1)' = 0$, and $(Py)'(I_4 - P)y = 8 - 4 - 6 + 2 = 0$.

- Check through $W^\perp$: $h = (1, -1, -1, 1)'$ is orthogonal to a_1, a_2, a_3 ($1-1-1+1 = 0$, $2-2 = 0$, $3-3-1+1 = 0$), so $h \in W^\perp$ (Theorem 6.6(a)). As $\dim W = \operatorname{rank} A = 3$ (Definition 4.11), $\dim W^\perp = 4 - 3 = 1$ (Theorem 6.6(c)), and the nonzero h spans $W^\perp$ (Theorem 4.5(d)). By Theorem 6.24(d) and the last sentence of Example 6.27, $(I_4 - P)y = y - Py$ is the projection $\frac{1}{4}(h'y)h = \frac{1}{4}hh'y$ of y onto $W^\perp = \operatorname{span}\{h\}$, for every $y \in \mathbb{R}^4$; so (Theorem 1.4(b), with $y = e_j$)
$$I_4 - P = \frac{1}{4}\begin{pmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} = \frac{1}{4}hh'$$
which agrees with P above. Pythagoras (Proposition 6.2): $\|y\|^2 = 81 + 9 + 25 + 9 = 124$ and $\|Py\|^2 + \|(I_4 - P)y\|^2 = (64 + 16 + 36 + 4) + 4 = 124$.

Idempotent matrices: rank equals trace

Theorem 6.29 ([12, proof of Lemma 2.8], [6, Exercise 8.53], [3, Result 3.11], [5, Theorems 10.2.1, 12.3.9])**.** Let $A \in \mathbb{R}^{n\times n}$ be idempotent, $A^2 = A$.

(a) $Av = v$ for every $v \in \mathcal{C}(A)$, and $\mathbb{R}^n = \mathcal{N}(A) \oplus \mathcal{C}(A)$: every $v \in \mathbb{R}^n$ is $v = (v - Av) + Av$ with $v - Av \in \mathcal{N}(A)$ and $Av \in \mathcal{C}(A)$.

(b) $I_n - A$ is idempotent.

(c) $\operatorname{rank} A = \operatorname{tr} A$.

(d) If moreover $A' = A$, then Av is the orthogonal projection of v onto $\mathcal{C}(A)$, for every $v \in \mathbb{R}^n$.

Proof. (a)

- If $v = Aw$, then $Av = A^2w = Aw = v$. For every v, $A(v - Av) = Av - A^2v = 0$.
- If $v \in \mathcal{N}(A) \cap \mathcal{C}(A)$, then $v = Av = 0$. So the sum $\mathcal{N}(A) + \mathcal{C}(A)$ is direct (Definition 3.30), and it is $\mathbb{R}^n$ by the displayed decomposition.

(b) $(I-A)^2 = I - 2A + A^2 = I - A$.

(c) If $A = 0$, then $\operatorname{rank} A = 0 = \operatorname{tr} A$. Let $r = \operatorname{rank} A \geqslant 1$.

- Take a full-rank factorization $A = BC$, with $B \in \mathbb{R}^{n\times r}$, $C \in \mathbb{R}^{r\times n}$, and $L \in \mathbb{R}^{r\times n}$, $K \in \mathbb{R}^{n\times r}$ such that $LB = I_r$ and $CK = I_r$ (Theorem 4.21(a),(c)).
- $A^2 = A$ reads $BCBC = BC$. Multiplying on the left by L and on the right by K, $L(BCBC)K = (LB)(CB)(CK) = CB$ and $L(BC)K = (LB)(CK) = I_r$; so $CB = I_r$.
- Hence $\operatorname{tr} A = \operatorname{tr}(BC) = \operatorname{tr}(CB) = \operatorname{tr} I_r = r$ (Theorem 1.11(b)).

(d) By Corollary 6.7, $\mathcal{N}(A) = \mathcal{C}(A')^\perp = \mathcal{C}(A)^\perp$. So in (a), $Av \in \mathcal{C}(A)$ and $v - Av \in \mathcal{C}(A)^\perp$: by Definition 6.22, Av is the orthogonal projection of v onto $\mathcal{C}(A)$. □

Corollary 6.30 ([12, Theorem 2.9(iv)], [6, Exercise 8.68(c)])**.** Let $A \in \mathbb{R}^{n\times p}$ have full column rank, and let $P = A(A'A)^{-1}A'$ (Theorem 6.26). Then $\operatorname{rank} P = \operatorname{tr} P = p = \operatorname{rank} A$ and $\operatorname{rank}(I_n - P) = \operatorname{tr}(I_n - P) = n - p$. In particular $\operatorname{rank} C_n = \operatorname{tr} C_n = n - 1$.

Proof. By Theorem 1.11(a),(b), $\operatorname{tr} P = \operatorname{tr}\Big(A(A'A)^{-1}A'\Big) = \operatorname{tr}\Big((A'A)^{-1}A'A\Big) = \operatorname{tr} I_p = p$ and $\operatorname{tr}(I_n - P) = \operatorname{tr} I_n - \operatorname{tr} P = n - p$. Both matrices are idempotent (Theorem 6.26(b)), so their ranks equal their traces (Theorem 6.29(c)). For $A = \mathbf{1}_n$, $I_n - P = C_n$ (Example 6.27). □

In Example 6.28, $\operatorname{tr} P = \frac{1}{4}(3+3+3+3) = 3$ and $\operatorname{tr}(I_4 - P) = \frac{1}{4}(1+1+1+1) = 1$; indeed every column of $I_4 - P = \frac{1}{4}hh'$ is a nonzero multiple of $h = (1,-1,-1,1)'$, so $\mathcal{C}(I_4 - P) = \operatorname{span}\{h\}$ and $\operatorname{rank}(I_4 - P) = 1$ (Definition 4.11 and Proposition 3.13(a)).

Projection matrices

Theorem 6.31 ([5, Theorem 12.3.1, Corollary 12.3.2, §12.2a, Theorems 12.3.4, 12.3.8, Corollary 12.5.10])**.** Let W be a subspace of $\mathbb{R}^n$.

(a) There is exactly one matrix $P \in \mathbb{R}^{n\times n}$ such that Py is the orthogonal projection of y onto W for every $y \in \mathbb{R}^n$. It is the **(orthogonal) projection matrix** for W, written P_W. For $W = \{0\}$, $P_W = 0$.

(b) If $A \in \mathbb{R}^{n\times p}$ has full column rank and $\mathcal{C}(A) = W$, then $P_W = A(A'A)^{-1}A'$. If the columns of $Q \in \mathbb{R}^{n\times k}$ form an orthonormal basis of W, then $P_W = QQ'$.

(c) P_W is symmetric and idempotent; $P_W w = w$ for $w \in W$ and $P_W v = 0$ for $v \in W^\perp$; $\mathcal{C}(P_W) = W$ and $\operatorname{rank} P_W = \operatorname{tr} P_W = \dim W$; $\|P_W y\|^2 = y' P_W y$ for $y \in \mathbb{R}^n$; and $P_{W^\perp} = I_n - P_W$.

Proof. (a)

- Uniqueness: if $Py = \tilde{P}y$ for every y, then $\operatorname{col}_j(P) = Pe_j = \tilde{P}e_j = \operatorname{col}_j(\tilde{P})$ for every j (Theorem 1.4(b)), so $P = \tilde{P}$.
- Existence for $W = \{0\}$: $y = 0 + y$ with $0 \in W$ and $y \perp W$, so the projection of every y is $0 = 0y$, and $P = 0$ has the property.
- Existence for $W \neq \{0\}$: W has a basis $a_1, \ldots, a_p$ ($p \geqslant 1$; Theorem 4.4), so $A = [a_1 \ \cdots \ a_p]$ has full column rank (Theorem 4.13(b)) and $\mathcal{C}(A) = W$. By Theorem 6.26(a), $P = A(A'A)^{-1}A'$ has the property.

(b) By Theorem 6.26(a), every A of full column rank with $\mathcal{C}(A) = W$ has the property in (a), so $A(A'A)^{-1}A' = P_W$ by the uniqueness in (a). If the columns of Q form an orthonormal basis of W, then $\mathcal{C}(Q) = W$, Q has full column rank (Proposition 6.11(b), Theorem 4.13(b)) and $Q'Q = I_k$ (Definition 6.12), so $P_W = Q(Q'Q)^{-1}Q' = QQ'$.

(c)

- For $w \in W$, $w = w + 0$ with $0 \in W^\perp$; for $v \in W^\perp$, $v = 0 + v$ with $0 \in W$. By the uniqueness in Definition 6.22, $P_W w = w$ and $P_W v = 0$.
- If $W = \{0\}$, then $P_W = 0$ is symmetric and idempotent, $\mathcal{C}(0) = \{0\}$ and $\operatorname{rank} 0 = \operatorname{tr} 0 = 0$. Let $W \neq \{0\}$, and $P_W = A(A'A)^{-1}A'$ as in (b).
- P_W is symmetric and idempotent, and $P_W A = A$ (Theorem 6.26(b),(c)).
- $\mathcal{C}(P_W) = \mathcal{C}\big(A(A'A)^{-1}A'\big) \subseteq \mathcal{C}(A)$ and $\mathcal{C}(A) = \mathcal{C}(P_W A) \subseteq \mathcal{C}(P_W)$ (Theorem 4.15(a)); so $\mathcal{C}(P_W) = W$, and $\operatorname{rank} P_W = \dim W = \operatorname{tr} P_W$ (Definition 4.11, Theorem 6.29(c)).
- For every W and $y \in \mathbb{R}^n$, as P_W is symmetric and idempotent, $\|P_W y\|^2 = (P_W y)'(P_W y) = y' P_W' P_W y = y' P_W^2 y = y' P_W y$ (Theorem 1.9).
- For every W: $(I_n - P_W)y = y - P_W y$ is the orthogonal projection of y onto $W^\perp$ for every y (Theorem 6.24(d)); so $P_{W^\perp} = I_n - P_W$ by (a) for $W^\perp$. □

By (a) and (b), QQ' does not depend on the choice of the orthonormal basis of W.

Corollary 6.32. Let $A \in \mathbb{R}^{n \times p}$ have full column rank p, and let $A = QR$ as in Theorem 6.16. Then $\mathcal{C}(A) = \mathcal{C}(Q)$ and $P_{\mathcal{C}(A)} = QQ'$.

Proof.
- $\operatorname{col}_j(A) = Q \operatorname{col}_j(R) \in \mathcal{C}(Q)$ (Theorem 1.4(a),(b)), so $\mathcal{C}(A) \subseteq \mathcal{C}(Q)$.
- R is upper triangular with positive diagonal entries, so R is invertible (Theorem 2.31(a)). Then $Q = AR^{-1}$, so $\operatorname{col}_j(Q) = A \operatorname{col}_j(R^{-1}) \in \mathcal{C}(A)$ for every j, and $\mathcal{C}(Q) \subseteq \mathcal{C}(A)$.
- The columns of Q are orthonormal, hence linearly independent (Proposition 6.11(b)), and they span $\mathcal{C}(Q) = \mathcal{C}(A)$. So they form an orthonormal basis of $\mathcal{C}(A)$, and $P_{\mathcal{C}(A)} = QQ'$ by Theorem 6.31(b). □

Example 6.33. For $W = \operatorname{span}\{a_1, a_2\}$ of Example 6.25, find P_W in two ways, and $P_{W^\perp}$.

Solution.

- By Example 6.17, $q_1 = \frac{1}{3}b_1$ and $q_2 = \frac{1}{3}b_2$ form an orthonormal basis of W. By Theorem 6.31(b) and the outer product expansion (Theorem 1.4(d)), $P_W = q_1 q_1' + q_2 q_2' = \frac{1}{9}(b_1 b_1' + b_2 b_2')$:

$$P_W = \frac{1}{9}\left[\begin{pmatrix} 1 & 2 & 0 & 2 \\ 2 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix} + \begin{pmatrix} 4 & -2 & 4 & 0 \\ -2 & 1 & -2 & 0 \\ 4 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\right] = \frac{1}{9}\begin{pmatrix} 5 & 0 & 4 & 2 \\ 0 & 5 & -2 & 4 \\ 4 & -2 & 4 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}$$

- With $B = [a_1\ a_2]$: $B'B = \left(\begin{smallmatrix} 9 & 9 \\ 9 & 18 \end{smallmatrix}\right)$, of determinant 81, so $(B'B)^{-1} = \frac{1}{81}\left(\begin{smallmatrix} 18 & -9 \\ -9 & 9 \end{smallmatrix}\right) = \frac{1}{9}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ (Theorem 5.24), and

$$B(B'B)^{-1} = \frac{1}{9}[2a_1 - a_2 \quad -a_1 + a_2] = \frac{1}{9}\begin{pmatrix} -1 & 2 \\ 3 & -1 \\ -2 & 2 \\ 2 & 0 \end{pmatrix}, \qquad B(B'B)^{-1}B' = \frac{1}{9}\begin{pmatrix} 5 & 0 & 4 & 2 \\ 0 & 5 & -2 & 4 \\ 4 & -2 & 4 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}$$

  the same matrix, as Theorem 6.31(a),(b) asserts. Also $\operatorname{tr} P_W = \frac{1}{9}(5+5+4+4) = 2 = \dim W$, and $P_W y = (0, 5, -2, 4)' = \hat{w}$ for y of Example 6.25.

- By Theorem 6.31(c),

$$P_{W^\perp} = I_4 - P_W = \frac{1}{9}\begin{pmatrix} 4 & 0 & -4 & -2 \\ 0 & 4 & 2 & -4 \\ -4 & 2 & 5 & 0 \\ -2 & -4 & 0 & 5 \end{pmatrix}$$

  The columns $q_3 = \frac{1}{3}b_3$, $q_4 = \frac{1}{3}v$ of the orthogonal matrix of Example 6.18 form an orthonormal basis of $W^\perp$ (Corollary 6.15(c)), and indeed $\frac{1}{9}(b_3 b_3' + vv')$ is this matrix.

Theorem 6.34 ([5, Theorems 12.3.8, 12.3.9], [8, Ch. 10, Exercise 7])**.** A matrix $A \in \mathbb{R}^{n\times n}$ is the projection matrix for some subspace of $\mathbb{R}^n$ if and only if $A' = A = A^2$. In that case $A = P_{\mathcal{C}(A)}$. In particular, two symmetric idempotent matrices with the same column space are equal.

Proof. • If $A = P_W$, then A is symmetric and idempotent and $\mathcal{C}(A) = W$ (Theorem 6.31(c)).

- Conversely, let $A' = A = A^2$. By Theorem 6.29(d), Av is the orthogonal projection of v onto $\mathcal{C}(A)$ for every $v \in \mathbb{R}^n$; so $A = P_{\mathcal{C}(A)}$ (Theorem 6.31(a)).
- If A and B are symmetric and idempotent with $\mathcal{C}(A) = \mathcal{C}(B) = W$, then $A = P_W = B$. □

Corollary 6.35 ([12, Definition 6.1], [6, Exercise 8.71(a)])**.** Let $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}_n'x = 0\}$.

(a) $P_{\text{span}\{\mathbf{1}_n\}} = \bar{J}_n$ and $P_{W_0} = C_n$.

(b) $\operatorname{rank} C_n = \operatorname{tr} C_n = n - 1$.

Proof. (a) The vector $q = n^{-1/2}\mathbf{1}_n$ satisfies $q'q = n^{-1}\mathbf{1}'\mathbf{1} = 1$ (Proposition 1.15), so it is an orthonormal basis of $\text{span}\{\mathbf{1}_n\}$, and $P_{\text{span}\{\mathbf{1}_n\}} = qq' = \frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$ (Theorem 6.31(b) and Definition 1.14). By Theorem 6.6(a), $W_0 = \mathcal{N}(\mathbf{1}') = \text{span}\{\mathbf{1}_n\}^\perp$, so $P_{W_0} = I_n - \bar{J}_n = C_n$ (Theorem 6.31(c)).

(b) By (a) and Theorem 6.31(c), $\operatorname{rank} C_n = \operatorname{tr} C_n$. Every diagonal entry of $\bar{J}_n$ is $\frac{1}{n}$, so $\operatorname{tr} C_n = \operatorname{tr} I_n - \operatorname{tr} \bar{J}_n = n - 1$ (Theorem 1.11(a)). □

Projections onto orthogonal sums and nested subspaces

Theorem 6.36 ([5, Corollary 12.3.7, Theorem 12.3.5], [12, Exercise B3(b), Remark 7.7])**.** Let $U \subseteq W$ be subspaces of $\mathbb{R}^n$, with projection matrices P_U and P_W.

(a) $P_W P_U = P_U P_W = P_U$.

(b) $P_W - P_U$ is symmetric and idempotent, of rank $\dim W - \dim U$, and it is the projection matrix for $W \cap U^\perp$. Moreover $(P_W - P_U)P_U = 0$ and $(P_W - P_U)(I_n - P_W) = 0$, and W is the orthogonal sum of U and $W \cap U^\perp$.

(c) Every $y \in \mathbb{R}^n$ splits as $y = P_U y + (P_W - P_U)y + (I_n - P_W)y$ into three pairwise orthogonal parts, so that

$$\|y\|^2 = \|P_U y\|^2 + \|(P_W - P_U)y\|^2 + \|(I_n - P_W)y\|^2, \qquad \|P_W y\|^2 = \|P_U y\|^2 + \|(P_W - P_U)y\|^2$$

Proof. (a) If $U = \{0\}$, then $P_U = 0$ and (a) is clear. Otherwise let Z have full column rank with $\mathcal{C}(Z) = U$ (a basis of U as columns; Theorems 4.4 and 4.13(b)), so that $P_U = Z(Z'Z)^{-1}Z'$ (Theorem 6.31(b)).

- Each column of Z lies in $U \subseteq W$, so it is its own projection onto W (Theorem 6.24(b)): $P_W Z = Z$ (Theorem 1.4(a)).
- Hence $P_W P_U = P_W Z(Z'Z)^{-1}Z' = Z(Z'Z)^{-1}Z' = P_U$. As P_W and P_U are symmetric (Theorem 6.31(c)), $P_U P_W = P_U' P_W' = (P_W P_U)' = P_U' = P_U$.

(b)

- P_W and P_U are symmetric and idempotent (Theorem 6.31(c)), so $P_W - P_U$ is symmetric, and by (a), $(P_W - P_U)^2 = P_W^2 - P_W P_U - P_U P_W + P_U^2 = P_W - P_U - P_U + P_U = P_W - P_U$.
- Its rank is its trace (Theorem 6.29(c)): $\operatorname{tr} P_W - \operatorname{tr} P_U = \dim W - \dim U$ (Theorems 1.11(a) and 6.31(c)).
- $(P_W - P_U)P_U = P_W P_U - P_U^2 = P_U - P_U = 0$, and $(P_W - P_U)(I - P_W) = P_W - P_W^2 - P_U + P_U P_W = 0$.

- By Theorem 6.34, $P_W - P_U$ is the projection matrix for $\mathcal{C}(P_W - P_U)$. If $v = (P_W - P_U)y$, then $P_W v = (P_W^2 - P_W P_U)y = v$, so $v \in \mathcal{C}(P_W) = W$; and $P_U v = (P_U P_W - P_U^2)y = 0$, so $v \in \mathcal{N}(P_U) = \mathcal{C}(P_U')^\perp = U^\perp$ (Corollary 6.7, with $P_U' = P_U$, and $\mathcal{C}(P_W) = W$, $\mathcal{C}(P_U) = U$ by Theorem 6.31(c)). Thus $\mathcal{C}(P_W - P_U) \subseteq W \cap U^\perp$.
- Conversely, if $v \in W \cap U^\perp$, then $P_W v = v$ (Theorem 6.24(b)) and $P_U v = 0$ (as $v = 0 + v$ with $v \perp U$), so $v = (P_W - P_U)v \in \mathcal{C}(P_W - P_U)$.
- $W \cap U^\perp$ is a subspace (Theorems 6.6(a) and 3.31(a)), orthogonal to U, as it lies in $U^\perp$. For $w \in W$, $w = P_U w + (w - P_U w)$, where $P_U w \in U$, and $w - P_U w$ lies in W (as $P_U w \in U \subseteq W$) and in $U^\perp$ (Definition 6.22). So $W \subseteq U + (W \cap U^\perp)$; conversely $U + (W \cap U^\perp) \subseteq W$, as U and $W \cap U^\perp$ lie in the subspace W (Theorem 3.31(b)). Hence $W = U + (W \cap U^\perp)$, an orthogonal sum (Definition 6.5).

(c)

- $P_U + (P_W - P_U) + (I - P_W) = I$ gives the splitting.
- For symmetric A, B with $AB = 0$, $(Ay)'(By) = y'ABy = 0$. Here $P_U(P_W - P_U) = P_U P_W - P_U^2 = 0$, $P_U(I - P_W) = P_U - P_U P_W = 0$ and $(P_W - P_U)(I - P_W) = 0$, by (a) and (b); so the three parts are pairwise orthogonal.
- Proposition 6.3(b) gives the equation for $\|y\|^2$. As $P_W y = P_U y + (P_W - P_U)y$, with $P_U y \perp (P_W - P_U)y$, Pythagoras (Proposition 6.2) gives the equation for $\|P_W y\|^2$. □

Corollary 6.37 (cf. [8, Ch. 18, Exercise 23(a)], [6, Exercise 8.64])**.** Let $W_1 \perp W_2$ be orthogonal subspaces of $\mathbb{R}^n$, and $P_1 = P_{W_1}$, $P_2 = P_{W_2}$. Then $W_1 \cap W_2 = \{0\}$, so the orthogonal sum $W_1 + W_2$ (Definition 6.5) is direct, written $W_1 \oplus W_2$, and

$$P_{W_1 \oplus W_2} = P_1 + P_2, \qquad P_1 P_2 = P_2 P_1 = 0, \qquad \|P_{W_1 \oplus W_2} y\|^2 = \|P_1 y\|^2 + \|P_2 y\|^2 \quad (y \in \mathbb{R}^n)$$

Proof. • If $w \in W_1 \cap W_2$, then $w'w = 0$, as $W_1 \perp W_2$; so $w = 0$, and the sum is direct (Definition 3.30). Let $W = W_1 \oplus W_2$, a subspace that contains W_1 and W_2 (Theorem 3.31(a),(b)).

- Apply Theorem 6.36 to $U = W_1 \subseteq W$. Every vector of W_2 lies in W and is orthogonal to W_1, so $W_2 \subseteq W \cap W_1^\perp$. By Theorem 6.36(b), $P_W - P_1$ is the projection matrix for $W \cap W_1^\perp$, of rank $\dim W - \dim W_1$; so $\dim(W \cap W_1^\perp) = \dim W - \dim W_1 = \dim W_2$ (Theorem 6.31(c) and Corollary 4.25(a)). Hence $W \cap W_1^\perp = W_2$ (Theorem 4.5(a)), and $P_W - P_1 = P_2$, that is, $P_W = P_1 + P_2$.
- By Theorem 6.36(b), $P_2 P_1 = (P_W - P_1)P_1 = 0$. As P_1 and P_2 are symmetric (Theorem 6.31(c)), $P_1 P_2 = P_1' P_2' = (P_2 P_1)' = 0$ (Theorem 1.9).
- The second equation of Theorem 6.36(c), with $P_W - P_1 = P_2$, gives $\|P_W y\|^2 = \|P_1 y\|^2 + \|P_2 y\|^2$. □

Example 6.38. In Example 6.25 take $U = \text{span}\{a_1\} \subseteq W = \text{span}\{a_1, a_2\}$. Then $P_U = q_1 q_1' = \frac{1}{9} b_1 b_1'$ (Theorem 6.31(b)), and by Example 6.33, $P_W - P_U = q_2 q_2' = \frac{1}{9} b_2 b_2'$. By Theorem 6.36(b), this is the projection matrix for $W \cap U^\perp$, of dimension $2 - 1 = 1$ (Theorem 6.31(c)); as $0 \neq b_2 \in W$ and $b_2 \perp U$ ($b_2' a_1 = 0$; Theorem 6.6(a)), $W \cap U^\perp = \text{span}\{b_2\}$ (Theorem 4.5(a)). For $y = (-2, 7, 1, 3)'$,

$$P_U y = \frac{b_1' y}{9} b_1 = \begin{pmatrix} 2 \\ 4 \\ 0 \\ 4 \end{pmatrix}, \qquad (P_W - P_U)y = \frac{b_2' y}{9} b_2 = \begin{pmatrix} -2 \\ 1 \\ -2 \\ 0 \end{pmatrix}, \qquad (I_4 - P_W)y = \begin{pmatrix} -2 \\ 2 \\ 3 \\ -1 \end{pmatrix}$$

and $\|y\|^2 = 63 = 36 + 9 + 18$, as Theorem 6.36(c) asserts. Here $U \perp \text{span}\{b_2\}$ and $W = U \oplus \text{span}\{b_2\}$, so Corollary 6.37 gives $P_W = P_U + \frac{1}{9} b_2 b_2'$ and $\|P_W y\|^2 = 36 + 9 = 45$.

Exercises

Exercise 6.1. Let

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 1 & 1 & -2 \\ 1 & 2 & -2 \\ 0 & 1 & 0 \end{pmatrix}$$

with columns a_1, a_2, a_3.

(a) Apply the Gram–Schmidt process to a_1, a_2, a_3.

(b) Find the QR decomposition of A, and check that $R = Q'A$.

(c) The vector $y = (5, 0, 1, 1)'$ lies in $\mathcal{C}(A)$. Find its coordinates $q_j'y$ in the orthonormal basis q_1, q_2, q_3 of (b), and check that $\|y\|^2 = \sum_j (q_j'y)^2$.

Solution. (a) By Theorem 6.14:

- $b_1 = a_1 = (1, 1, 1, 0)'$, $b_1'b_1 = 1 + 1 + 1 + 0 = 3$.
- $a_2'b_1 = 3 + 1 + 2 + 0 = 6$, so $t_{12} = 2$ and $b_2 = a_2 - 2b_1 = (3, 1, 2, 1)' - (2, 2, 2, 0)' = (1, -1, 0, 1)'$, with $b_2'b_2 = 1 + 1 + 0 + 1 = 3$.
- $a_3'b_1 = 1 - 2 - 2 + 0 = -3$ and $a_3'b_2 = 1 + 2 + 0 + 0 = 3$, so $t_{13} = -1$, $t_{23} = 1$ and

$$b_3 = a_3 + b_1 - b_2 = \begin{pmatrix} 1 \\ -2 \\ -2 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \\ -1 \end{pmatrix}, \qquad b_3'b_3 = 1 + 0 + 1 + 1 = 3$$

- Check: $b_1'b_2 = 1 - 1 + 0 + 0 = 0$, $b_1'b_3 = 1 + 0 - 1 + 0 = 0$, $b_2'b_3 = 1 + 0 + 0 - 1 = 0$.

(b) All three lengths are $\sqrt{3}$, so $q_j = \frac{1}{\sqrt{3}} b_j$, $r_{jj} = \sqrt{3}$, $r_{12} = 2\sqrt{3}$, $r_{13} = -\sqrt{3}$ and $r_{23} = \sqrt{3}$ (Theorem 6.16):

$$Q = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}, \qquad R = \sqrt{3} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Check: the entries of $\sqrt{3}\, Q'A$ are the products $b_i'a_j$, so

$$Q'A = \frac{1}{\sqrt{3}} \begin{pmatrix} 3 & 6 & -3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix} = R$$

(c) $b_1'y = 5 + 0 + 1 + 0 = 6$, $b_2'y = 5 - 0 + 0 + 1 = 6$ and $b_3'y = 5 + 0 - 1 - 1 = 3$, so $q_1'y = 2\sqrt{3}$, $q_2'y = 2\sqrt{3}$ and $q_3'y = \sqrt{3}$: $y = 2\sqrt{3}\, q_1 + 2\sqrt{3}\, q_2 + \sqrt{3}\, q_3 = 2b_1 + 2b_2 + b_3$ (Theorem 6.24(a),(b)). Check: $2(1, 1, 1, 0)' + 2(1, -1, 0, 1)' + (1, 0, -1, -1)' = (5, 0, 1, 1)'$, and $\sum_j (q_j'y)^2 = 12 + 12 + 3 = 27 = 25 + 0 + 1 + 1 = \|y\|^2$ (Theorem 6.24(c)).

Exercise 6.2 (cf. [2, Exercise 5.13(a)])**.** Let $A = [a_1\ a_2\ a_3] \in \mathbb{R}^{4\times 3}$ have the columns $a_1 = \mathbf{1}_4$, $a_2 = (2,0,2,0)'$ and $a_3 = (3,3,1,1)'$.

(a) Find the QR decomposition of A by the Gram–Schmidt process.

(b) Show that $Q'Q = I_3$ but $QQ' \neq I_4$, and that QQ' is the matrix P of Example 6.28.

Solution. (a) By Theorem 6.14:

- $b_1 = a_1$, $\|b_1\|^2 = 4$.
- $a_2'b_1 = 4$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (1,-1,1,-1)'$, with $\|b_2\|^2 = 4$.
- $a_3'b_1 = 8$ and $a_3'b_2 = 3-3+1-1 = 0$, so $t_{13} = 2$, $t_{23} = 0$ and $b_3 = a_3 - 2b_1 = (1,1,-1,-1)'$, with $\|b_3\|^2 = 4$.
- All three lengths are 2: $q_j = \frac{1}{2}b_j$, $r_{jj} = 2$, $r_{12} = 2t_{12} = 2$, $r_{13} = 2t_{13} = 4$ and $r_{23} = 0$:

$$Q = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \end{pmatrix}, \qquad R = \begin{pmatrix} 2 & 2 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Check: the third column of QR is $Q(4,0,2)' = \frac{1}{2}(4+2,\ 4+2,\ 4-2,\ 4-2)' = (3,3,1,1)' = a_3$.

(b) The columns of Q are orthonormal, so $Q'Q = I_3$. By the outer product expansion (Theorem 1.4(d)),

$$QQ' = \tfrac{1}{4}\big(b_1b_1' + b_2b_2' + b_3b_3'\big) = \frac{1}{4}\begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix} \neq I_4$$

For instance, its $(1,1)$ entry is $\frac{1}{4}(1+1+1) = \frac{3}{4}$. The transposed rows of Q are not orthonormal. By Corollary 6.32, $QQ' = P_{\mathcal{C}(A)}$, and $P_{\mathcal{C}(A)} = A(A'A)^{-1}A'$ by Theorem 6.31(b): QQ' is the matrix P of Example 6.28.

Exercise 6.3. Let

$$Q = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

with columns $q_1, \ldots, q_4$, and let $y = (4,2,0,6)'$.

(a) Show that Q is a symmetric orthogonal matrix, and find Q^{-1}.

(b) Find $c \in \mathbb{R}^4$ with $y = c_1q_1 + \cdots + c_4q_4$, and check that $\|c\| = \|y\|$.

(c) Find the vector of $\operatorname{span}\{q_1, q_2\}$ closest to y, and its distance from y.

Solution. (a) $Q' = Q$ by inspection. Each column of $2Q$ has squared length 4, and any two of them are orthogonal (each pair of columns agrees in two entries and differs in the other two), for instance $(1,-1,1,-1)'$ and $(1,1,-1,-1)'$ have inner product $1-1-1+1 = 0$; so $Q'Q = I_4$ (Definition 6.12), and $Q^{-1} = Q' = Q$ (Proposition 6.13(b)).

(b) $y = Qc$ (Theorem 1.4(b)) gives $c = Q^{-1}y = Q'y$:

$$c = \tfrac{1}{2}\big(4+2+0+6,\ 4-2+0-6,\ 4+2-0-6,\ 4-2-0+6\big)' = (6,-2,0,4)'$$

(equivalently, $c_j = q_j'y$ by Proposition 6.11(a)). Then $\|c\|^2 = 36+4+0+16 = 56 = 16+4+0+36 = \|y\|^2$, as Proposition 6.13(a) says.

(c) q_1, q_2 are orthonormal by (a), hence linearly independent (Proposition 6.11(b)), so they form an orthonormal basis of $V = \operatorname{span}\{q_1, q_2\}$, and $P_V y = q_1(q_1'y) + q_2(q_2'y) = 6q_1 - 2q_2$ (Theorems 6.31(b) and 1.4(d)):

$$P_V y = 3\begin{pmatrix}1\\1\\1\\1\end{pmatrix} - \begin{pmatrix}1\\-1\\1\\-1\end{pmatrix} = \begin{pmatrix}2\\4\\2\\4\end{pmatrix}, \qquad y - P_V y = \begin{pmatrix}2\\-2\\-2\\2\end{pmatrix} = 4q_4$$

By Theorem 6.23, $(2,4,2,4)'$ is the closest vector, at distance $\|y - P_V y\| = \sqrt{16} = 4$; by (b), $y - P_V y = c_3 q_3 + c_4 q_4$ and $\|y - P_V y\|^2 = c_3^2 + c_4^2 = 16$.

Exercise 6.4. Let $B = [a_1\ a_2\ a_3\ a_4] \in \mathbb{R}^{4\times 4}$, with a_1, a_2, a_3 of Exercise 6.2 and $a_4 = (4,0,0,0)'$.

(a) Continue the Gram–Schmidt process of Exercise 6.2(a) to find the QR decomposition of B. Which matrix of Exercise 6.3 is its Q?

(b) Solve $Bx = b$ for $b = (9,3,5,3)'$ by solving $Rx = Q'b$.

(c) Show that $|\det B| = \det R = 16$, without expanding $\det B$.

Solution. (a) $a_4'b_1 = 4$, $a_4'b_2 = 4$, $a_4'b_3 = 4$, so $t_{14} = t_{24} = t_{34} = 1$ and

$$b_4 = a_4 - b_1 - b_2 - b_3 = (4-1-1-1,\ 0-1+1-1,\ 0-1-1+1,\ 0-1+1+1)' = (1,-1,-1,1)'$$

with $\|b_4\|^2 = 4$. So $q_4 = \frac{1}{2}b_4$, $r_{44} = 2$ and $r_{14} = r_{24} = r_{34} = 2\cdot 1 = 2$: $B = QR$ with

$$Q = \frac{1}{2}\begin{pmatrix}1&1&1&1\\1&-1&1&-1\\1&1&-1&-1\\1&-1&-1&1\end{pmatrix}, \qquad R = \begin{pmatrix}2&2&4&2\\0&2&0&2\\0&0&2&2\\0&0&0&2\end{pmatrix}$$

This Q is the orthogonal matrix of Exercise 6.3.

(b) $Bx = b$ is $QRx = b$; multiplying by Q', and back by Q, it is equivalent to $Rx = Q'b$, as $Q'Q = QQ' = I_4$. Here $Q'b = \frac{1}{2}(20,8,4,4)' = (10,4,2,2)'$, and back substitution gives

$$2x_4 = 2, \quad 2x_3 + 2x_4 = 2, \quad 2x_2 + 2x_4 = 4, \quad 2x_1 + 2x_2 + 4x_3 + 2x_4 = 10: \qquad x = (3,1,0,1)'$$

Check: $3a_1 + a_2 + a_4 = (3+2+4,\ 3,\ 3+2,\ 3)' = (9,3,5,3)'$.

(c) $\det B = \det Q \det R$ (Theorem 5.18(b)). As Q is orthogonal, $\det Q = \pm 1$ (Theorem 6.19(c)); and R is upper triangular, so $\det R = 2\cdot 2\cdot 2\cdot 2 = 16$ (Theorem 5.14(c)). Hence $|\det B| = |\det Q| \det R = 16$.

Exercise 6.5. Let $W = \operatorname{span}\{v_1, v_2, v_3\} \subseteq \mathbb{R}^4$ with $v_1 = (1,1,-2,3)'$, $v_2 = (2,1,-3,5)'$ and $v_3 = (1,2,-3,4)'$.

(a) Find a basis of $W^\perp$, and check that $\dim W + \dim W^\perp = 4$.

(b) Find an orthonormal basis of $W^\perp$.

(c) Using (b), write $y = (1,2,0,4)'$ as $y = w + z$ with $w \in W$ and $z \in W^\perp$, and check that $w'z = 0$.

Solution. (a) $W^\perp = \mathcal{N}(V')$ for $V = [v_1\ v_2\ v_3]$ (Theorem 6.6(a)). Row-reduce V':

$$V' = \begin{pmatrix}1&1&-2&3\\2&1&-3&5\\1&2&-3&4\end{pmatrix} \xrightarrow[R_3\leftarrow R_3-R_1]{R_2\leftarrow R_2-2R_1} \begin{pmatrix}1&1&-2&3\\0&-1&1&-1\\0&1&-1&1\end{pmatrix} \xrightarrow{R_2\leftarrow -R_2} \begin{pmatrix}1&1&-2&3\\0&1&-1&1\\0&1&-1&1\end{pmatrix}$$

$$\xrightarrow[R_3\leftarrow R_3-R_2]{R_1\leftarrow R_1-R_2} \begin{pmatrix}1&0&-1&2\\0&1&-1&1\\0&0&0&0\end{pmatrix}$$

So $\dim W = \operatorname{rank} V' = 2$ (Definition 4.11, Theorem 4.13(a)). With $x_3 = s$ and $x_4 = t$ free, $x_1 = s - 2t$ and $x_2 = s - t$: $s_1 = (1, 1, 1, 0)'$ and $s_2 = (-2, -1, 0, 1)'$ form a basis of $W^\perp$ (Theorem 4.8(c)), and $2 + 2 = 4$ (Theorem 6.6(c)). Check: $s_1'v_1 = 1 + 1 - 2 + 0 = 0$, $s_1'v_2 = 2 + 1 - 3 + 0 = 0$, $s_1'v_3 = 1+2-3+0 = 0$, $s_2'v_1 = -2-1+0+3 = 0$, $s_2'v_2 = -4-1+0+5 = 0$, $s_2'v_3 = -2-2+0+4 = 0$.

(b) Gram–Schmidt: $s_1's_1 = 3$ and $s_2's_1 = -2 - 1 + 0 + 0 = -3$, so $u_2 = s_2 + s_1 = (-1, 0, 1, 1)'$, with $u_2'u_2 = 3$ and $s_1'u_2 = -1+0+1+0 = 0$. An orthonormal basis of $W^\perp$ is $\frac{1}{\sqrt{3}}(1, 1, 1, 0)'$, $\frac{1}{\sqrt{3}}(-1, 0, 1, 1)'$.

(c) $s_1'y = 1+2+0+0 = 3$ and $u_2'y = -1+0+0+4 = 3$. By Theorem 6.24(a) for $W^\perp$, the projection of y onto $W^\perp$ is $z = \frac{3}{3}s_1 + \frac{3}{3}u_2 = (0, 1, 2, 1)'$. By Theorem 6.24(d) for $W^\perp$, with $(W^\perp)^\perp = W$ (Theorem 6.8(a)), $w = y - z = (1, 1, -2, 3)'$ is the projection of y onto W; indeed $w = v_1$. Check: $w'z = 0 + 1 - 4 + 3 = 0$.

Exercise 6.6 ([6, Exercise 3.44(b)], [4, Ch. V, §2, Exercise 0], [7])**.** (a) Find all vectors of $\mathbb{R}^3$ that are orthogonal to $x = (2, 2, -1)'$.

(b) What is the dimension of the subspace of $\mathbb{R}^6$ consisting of the vectors orthogonal to both $(1, 1, -2, 3, 4, 5)'$ and $(0, 0, 1, 1, 0, 7)'$?

Solution. (a) $y = (y_1, y_2, y_3)'$ is orthogonal to x if and only if $x'y = 2y_1 + 2y_2 - y_3 = 0$, that is, $y_3 = 2y_1 + 2y_2$. So these vectors are

$$y = \begin{pmatrix} y_1 \\ y_2 \\ 2y_1 + 2y_2 \end{pmatrix} = y_1 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + y_2 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \qquad (y_1, y_2 \in \mathbb{R})$$

Hence $\operatorname{span}\{x\}^\perp = \mathcal{N}(x') = \operatorname{span}\{(1, 0, 2)', (0, 1, 2)'\}$, of dimension $2 = 3 - 1$ (Theorem 6.6).

(b) Let a_1, a_2 be the two vectors and $A = [a_1\ a_2] \in \mathbb{R}^{6\times 2}$. The subspace is $\operatorname{span}\{a_1, a_2\}^\perp = \mathcal{N}(A')$ (Theorem 6.6(a)). The vectors are independent: if $c_1a_1 + c_2a_2 = 0$, the first entry gives $c_1 = 0$, and then $c_2a_2 = 0$ with $a_2 \neq 0$ gives $c_2 = 0$. So $\dim \operatorname{span}\{a_1, a_2\} = 2$, and the answer is $6 - 2 = 4$ (Theorem 6.6(c)).

Exercise 6.7. Let $b = (2, 3, 1, 5)'$, $c = \mathbf{1}_4$ and

$$A = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 3 & 5 & 1 \\ 1 & 2 & 3 & 0 \\ 3 & 4 & 7 & 2 \end{pmatrix}$$

(a) Find a basis of $\mathcal{N}(A')$.

(b) Use Corollary 6.9 to decide whether $Ax = b$ and $Ax = c$ have solutions.

(c) Solve the consistent system, and check that $\mathcal{N}(A)$ is orthogonal to the transposed rows of A.

Solution. (a) Row-reduce A':

$$A' = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 2 & 5 & 3 & 7 \\ 1 & 1 & 0 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & -1 & -1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_1 \leftarrow R_1 - 2R_2,\ R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

With $w_3 = s$ and $w_4 = t$ free, $A'w = 0$ gives $w_1 = s - t$ and $w_2 = -s - t$; so $z_1 = (1, -1, 1, 0)'$ and $z_2 = (-1, -1, 0, 1)'$ form a basis of $\mathcal{N}(A')$ (Theorem 4.8(c)), and rank A = rank $A' = 2$ (Definition 4.11, Theorem 4.13(a)).

(b) $z_1'b = 2 - 3 + 1 + 0 = 0$ and $z_2'b = -2 - 3 + 0 + 5 = 0$, while $z_1'c = 1 - 1 + 1 + 0 = 1 \neq 0$. By Corollary 6.9, $Ax = b$ has a solution and $Ax = c$ has none.

(c) Row-reduce $[A \mid b]$:

$$\left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 2 \\ 2 & 3 & 5 & 1 & 3 \\ 1 & 2 & 3 & 0 & 1 \\ 3 & 4 & 7 & 2 & 5 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & -1 & -1 \\ 0 & 1 & 1 & -1 & -1 \\ 0 & 1 & 1 & -1 & -1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_1 \leftarrow R_1 - R_2,\ R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 3 \\ 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

With $x_3 = s$ and $x_4 = t$ free:

$$x = \begin{pmatrix} 3 - s - 2t \\ -1 - s + t \\ s \\ t \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \qquad (s, t \in \mathbb{R})$$

Check: $A(3, -1, 0, 0)' = (3 - 1,\ 6 - 3,\ 3 - 2,\ 9 - 4)' = (2, 3, 1, 5)' = b$. So $\mathcal{N}(A) = \operatorname{span}\{s_1, s_2\}$ with $s_1 = (-1, -1, 1, 0)'$ and $s_2 = (-2, 1, 0, 1)'$ (Theorem 4.8(c)), and both are orthogonal to the four transposed rows of A: $-1 - 1 + 2 + 0 = 0$, $-2 - 3 + 5 + 0 = 0$, $-1 - 2 + 3 + 0 = 0$, $-3 - 4 + 7 + 0 = 0$ for s_1, and $-2 + 1 + 0 + 1 = 0$, $-4 + 3 + 0 + 1 = 0$, $-2 + 2 + 0 + 0 = 0$, $-6 + 4 + 0 + 2 = 0$ for s_2 (Corollary 6.7).

Exercise 6.8. Let $q_1 = \frac{1}{3}(1, 0, 2, 2)'$, $q_2 = \frac{1}{3}(0, 1, 2, -2)'$ and $q_3 = \frac{1}{3}(2, 2, -1, 0)'$.

(a) Check that q_1, q_2, q_3 are orthonormal, and find all vectors of $\mathbb{R}^4$ orthogonal to them.

(b) Extend q_1, q_2, q_3 to an orthonormal basis $q_1, \ldots, q_4$ of $\mathbb{R}^4$, and write down the orthogonal matrix $U = [q_1 \ \cdots \ q_4]$ and U^{-1}.

(c) Find the coordinates $q_j'y$ of $y = (3, 3, 3, 3)'$ in this basis, and check that $\|y\|^2 = \sum_j (q_j'y)^2$.

Solution. (a) Put $u_1 = (1, 0, 2, 2)'$, $u_2 = (0, 1, 2, -2)'$, $u_3 = (2, 2, -1, 0)'$, so $q_j = \frac{1}{3}u_j$. Then $u_1'u_2 = 0 + 0 + 4 - 4 = 0$, $u_1'u_3 = 2 + 0 - 2 + 0 = 0$, $u_2'u_3 = 0 + 2 - 2 + 0 = 0$, and $u_1'u_1 = 1 + 0 + 4 + 4 = 9$, $u_2'u_2 = 0 + 1 + 4 + 4 = 9$, $u_3'u_3 = 4 + 4 + 1 + 0 = 9$; so the q_j are orthonormal. The vectors orthogonal to them form $\mathcal{N}(B)$, where B has the rows u_j' (Theorem 6.6(a)):

$$B = \begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 2 & 2 & -1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 0 & 2 & -5 & -4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & -9 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow -\frac{1}{9}R_3} \begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 - 2R_3]{R_1 \leftarrow R_1 - 2R_3} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

With $x_4 = t$ free, $x_1 = -2t$, $x_2 = 2t$, $x_3 = 0$: the vectors are $t(-2, 2, 0, 1)'$ $(t \in \mathbb{R})$.

(b) $v = (-2, 2, 0, 1)'$ has $v'v = 4 + 4 + 0 + 1 = 9$, so $q_4 = \frac{1}{3}(-2, 2, 0, 1)'$ completes an orthonormal basis (Corollary 6.15(b)), and by Theorem 6.19(a),

$$U = \frac{1}{3}\begin{pmatrix} 1 & 0 & 2 & -2 \\ 0 & 1 & 2 & 2 \\ 2 & 2 & -1 & 0 \\ 2 & -2 & 0 & 1 \end{pmatrix}, \qquad U^{-1} = U' = \frac{1}{3}\begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ 2 & 2 & -1 & 0 \\ -2 & 2 & 0 & 1 \end{pmatrix}$$

(c) $u_1'y = 3(1+0+2+2) = 15$, $u_2'y = 3(0+1+2-2) = 3$, $u_3'y = 3(2+2-1+0) = 9$ and $v'y = 3(-2+2+0+1) = 3$; so $q_1'y = 5$, $q_2'y = 1$, $q_3'y = 3$, $q_4'y = 1$, and $y = 5q_1 + q_2 + 3q_3 + q_4$ (Theorem 6.24(a),(b) with $W = \mathbb{R}^4$). Check: $25 + 1 + 9 + 1 = 36 = 9 + 9 + 9 + 9 = \|y\|^2$.

Exercise 6.9. Let $Q = \frac{1}{3}G$ with

$$G = \begin{pmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & -2 & 2 \\ 2 & 2 & 1 & 0 \\ 2 & -2 & 0 & 1 \end{pmatrix}$$

(a) Show that Q is orthogonal, and write down Q^{-1}.

(b) Find $\det Q$ by row-reducing G.

(c) For $x = (3, 3, 0, 0)'$ and $y = (0, 3, 3, 0)'$, check that $\|Qx\| = \|x\|$ and $(Qx)'(Qy) = x'y$.

Solution. (a) The product of G' and G is

$$\begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & -2 \\ -2 & -2 & 1 & 0 \\ -2 & 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & -2 & 2 \\ 2 & 2 & 1 & 0 \\ 2 & -2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 9 \end{pmatrix}$$

so $Q'Q = \frac{1}{9}G'G = I_4$: Q is orthogonal, and $Q^{-1} = Q' = \frac{1}{3}G'$ (Theorem 6.19(a)).

(b)

$$G \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & -2 & 2 \\ 0 & 2 & 5 & 4 \\ 0 & -2 & 4 & 5 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 9 \end{pmatrix}$$

Adding multiples of rows does not change the determinant (Definition 5.13(b)), and the last matrix is upper triangular; so $\det G = 1 \cdot 1 \cdot 9 \cdot 9 = 81$ (Theorem 5.14(c)), and $\det Q = \left(\frac{1}{3}\right)^4 \cdot 81 = 1$ (Theorem 5.18(c)), as Theorem 6.19(c) allows.

(c) $Qx = \frac{1}{3}(3+0,\ 0+3,\ 6+6,\ 6-6)' = (1, 1, 4, 0)'$, so $\|Qx\|^2 = 1+1+16+0 = 18 = 9+9+0+0 = \|x\|^2$. Next $Qy = \frac{1}{3}(0+0-6+0,\ 0+3-6+0,\ 0+6+3+0,\ 0-6+0+0)' = (-2, -1, 3, -2)'$, and $(Qx)'(Qy) = -2 - 1 + 12 + 0 = 9 = 0 + 9 + 0 + 0 = x'y$.

Exercise 6.10. Let K_4 be the Helmert matrix of order 4 printed in Example 6.20, with rows $p_1', \ldots, p_4'$, and let $y = (7, 3, 2, 0)'$.

(a) Check that $p_1, \ldots, p_4$ are orthonormal.

(b) Compute $K_4 y$, and check that $\sum_i (y_i - \bar{y})^2$ is the sum of the squares of the three contrasts.

(c) Check that $p_2p_2' + p_3p_3' + p_4p_4' = C_4$.

Solution. (a) The rows are $p_1' = \frac{1}{2}(1,1,1,1)$, $p_2' = \frac{1}{\sqrt{2}}(1,-1,0,0)$, $p_3' = \frac{1}{\sqrt{6}}(1,1,-2,0)$ and $p_4' = \frac{1}{\sqrt{12}}(1,1,1,-3)$. Then

$$p_1'p_2 = \frac{1-1+0+0}{2\sqrt{2}} = 0, \quad p_1'p_3 = \frac{1+1-2+0}{2\sqrt{6}} = 0, \quad p_1'p_4 = \frac{1+1+1-3}{2\sqrt{12}} = 0$$
$$p_2'p_3 = \frac{1-1+0+0}{\sqrt{12}} = 0, \quad p_2'p_4 = \frac{1-1+0+0}{\sqrt{24}} = 0, \quad p_3'p_4 = \frac{1+1-2+0}{\sqrt{72}} = 0$$

and $p_1'p_1 = \frac{4}{4} = 1$, $p_2'p_2 = \frac{2}{2} = 1$, $p_3'p_3 = \frac{1+1+4}{6} = 1$, $p_4'p_4 = \frac{1+1+1+9}{12} = 1$.

(b) $K_4 y = \left(\frac{7+3+2+0}{2},\ \frac{7-3}{\sqrt{2}},\ \frac{7+3-4}{\sqrt{6}},\ \frac{7+3+2-0}{\sqrt{12}}\right)' = (6,\ 2\sqrt{2},\ \sqrt{6},\ 2\sqrt{3})'$. Here $\bar{y} = 3$, and $p_1' y = 6 = \sqrt{4}\,\bar{y}$. The contrasts give $(2\sqrt{2})^2 + (\sqrt{6})^2 + (2\sqrt{3})^2 = 8 + 6 + 12 = 26$, and $\sum_i (y_i - 3)^2 = 16 + 0 + 1 + 9 = 26$ (Example 6.20(b)).

(c)

$$p_2 p_2' + p_3 p_3' + p_4 p_4' = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 1 & 1 & -2 & 0 \\ 1 & 1 & -2 & 0 \\ -2 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \frac{1}{12}\begin{pmatrix} 1 & 1 & 1 & -3 \\ 1 & 1 & 1 & -3 \\ 1 & 1 & 1 & -3 \\ -3 & -3 & -3 & 9 \end{pmatrix}$$

$$= \frac{1}{4}\begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix} = I_4 - \bar{J}_4 = C_4$$

(the $(1,1)$ entry, for instance, is $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} = \frac{3}{4}$).

Exercise 6.11. Let $h = \mathbf{1}_4$ and $F_h = I_4 - \frac{2}{h'h} hh'$ (Example 6.21).

(a) Compute F_h, and check that $F_h^2 = I_4$.

(b) Check that $F_h h = -h$ and that $F_h x = x$ for $x = (1, -1, 0, 0)'$, and find $\det F_h$.

(c) For $y = (3, 1, 0, 4)'$, compute $F_h y$ and check that $\|F_h y\| = \|y\|$.

Solution. (a) $h'h = 4$, so $F_h = I_4 - \frac{1}{2} J_4 = \frac{1}{2} N$ with

$$N = \begin{pmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

(each diagonal entry of N^2 is $1 + 1 + 1 + 1$, each other entry is, for instance, $-1 - 1 + 1 + 1$); so $F_h^2 = \frac{1}{4} N^2 = I_4$.

(b) $F_h h = \frac{1}{2}(1 - 1 - 1 - 1,\ -1 + 1 - 1 - 1,\ -1 - 1 + 1 - 1,\ -1 - 1 - 1 + 1)' = (-1, -1, -1, -1)' = -h$, and $F_h x = \frac{1}{2}(1 + 1,\ -1 - 1,\ -1 + 1,\ -1 + 1)' = (1, -1, 0, 0)' = x$ (here $h'x = 0$). By Theorem 5.31 with $u = -\frac{1}{2} h$ and $v = h$, $\det F_h = 1 + v'u = 1 - \frac{1}{2} \cdot 4 = -1$.

(c) $h'y = 3 + 1 + 0 + 4 = 8$, so $F_h y = y - \frac{2 \cdot 8}{4} h = y - 4h = (-1, -3, -4, 0)'$, and $\|F_h y\|^2 = 1 + 9 + 16 + 0 = 26 = 9 + 1 + 0 + 16 = \|y\|^2$. (Here $F_h y = y - 2\bar{y}\mathbf{1}$: the mean $\bar{y} = 2$ is reflected to -2, and the deviations $y - \bar{y}\mathbf{1}$ are kept.)

Exercise 6.12. Let $W = \operatorname{span}\{w_1, w_2\} \subseteq \mathbb{R}^4$ with $w_1 = (1, 0, 1, 1)'$ and $w_2 = (2, 1, 1, 0)'$, and let $y = (1, 0, 1, 4)'$.

(a) Find an orthogonal basis of W by the Gram–Schmidt process.

(b) Find the orthogonal projection $\hat{w}$ of y onto W and the projection of y onto $W^{\perp}$, and check that the latter is orthogonal to w_1 and w_2.

(c) Find the distance from y to the nearest vector of W, and check Bessel's inequality.

Solution. (a) $u_1 = w_1$, $u_1' u_1 = 1 + 0 + 1 + 1 = 3$; $w_2' u_1 = 2 + 0 + 1 + 0 = 3$, so $u_2 = w_2 - u_1 = (1, 1, 0, -1)'$, with $u_2' u_2 = 3$ and $u_1' u_2 = 1 + 0 + 0 - 1 = 0$.

(b) $u_1' y = 1 + 0 + 1 + 4 = 6$ and $u_2' y = 1 + 0 + 0 - 4 = -3$. By Theorem 6.24(a),

$$\hat{w} = \frac{6}{3} u_1 - \frac{3}{3} u_2 = \begin{pmatrix} 2 \\ 0 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 3 \end{pmatrix}$$

and the projection onto $W^\perp$ is $y - \hat{w} = (0, 1, -1, 1)'$ (Theorem 6.24(d)). Check: $w_1'(y - \hat{w}) = 0 + 0 - 1 + 1 = 0$ and $w_2'(y - \hat{w}) = 0 + 1 - 1 + 0 = 0$.

(c) The distance is $\|y - \hat{w}\| = \sqrt{0 + 1 + 1 + 1} = \sqrt{3}$ (Theorem 6.23). Bessel: $\|\hat{w}\|^2 = 6^2/3 + 3^2/3 = 12 + 3 = 15 \leqslant \|y\|^2 = 1 + 0 + 1 + 16 = 18$, and $18 = 15 + 3$.

Exercise 6.13. Let $a_1 = (1, 2, 2, 0)'$, $a_2 = (3, 2, 1, 2)'$, $W = \operatorname{span}\{a_1, a_2\} \subseteq \mathbb{R}^4$ and $y = (4, 2, 5, 3)'$.

(a) Find an orthonormal basis q_1, q_2 of W by the Gram–Schmidt process.

(b) Compute P_W, and check that $\operatorname{tr} P_W = 2$.

(c) Find the vector of W closest to y, and write it as a combination of a_1 and a_2. Find the distance from y to W.

Solution. (a) $b_1 = a_1$, with $\|b_1\|^2 = 1 + 4 + 4 + 0 = 9$; $a_2'b_1 = 3 + 4 + 2 + 0 = 9$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (2, 0, -1, 2)'$, with $\|b_2\|^2 = 4 + 0 + 1 + 4 = 9$ (Theorem 6.14). Hence $q_1 = \frac{1}{3}(1, 2, 2, 0)'$ and $q_2 = \frac{1}{3}(2, 0, -1, 2)'$.

(b) With $Q = [q_1\ q_2]$, Theorems 6.31(b) and 1.4(d) give

$$P_W = QQ' = \frac{1}{9}\left(b_1b_1' + b_2b_2'\right) = \frac{1}{9}\left[\begin{pmatrix}1&2&2&0\\2&4&4&0\\2&4&4&0\\0&0&0&0\end{pmatrix} + \begin{pmatrix}4&0&-2&4\\0&0&0&0\\-2&0&1&-2\\4&0&-2&4\end{pmatrix}\right] = \frac{1}{9}\begin{pmatrix}5&2&0&4\\2&4&4&0\\0&4&5&-2\\4&0&-2&4\end{pmatrix}$$

and $\operatorname{tr} P_W = \frac{1}{9}(5 + 4 + 5 + 4) = 2 = \dim W$ (Theorem 6.31(c)).

(c) The closest vector is the projection (Theorem 6.23):

$$P_W y = \frac{1}{9}\begin{pmatrix}20 + 4 + 0 + 12\\8 + 8 + 20 + 0\\0 + 8 + 25 - 6\\16 + 0 - 10 + 12\end{pmatrix} = \begin{pmatrix}4\\4\\3\\2\end{pmatrix} = a_1 + a_2$$

The distance is $\|y - P_W y\| = \|(0, -2, 2, 1)'\| = \sqrt{0 + 4 + 4 + 1} = 3$. Check: $a_1'(y - P_W y) = 0 - 4 + 4 + 0 = 0$ and $a_2'(y - P_W y) = 0 - 4 + 2 + 2 = 0$.

Exercise 6.14. Let A be the matrix of Example 6.17, $W = \mathcal{C}(A)$ and $y = (0, 6, 3, 3)'$.

(a) Show that $h = (0, 2, 1, -2)'$ spans $W^\perp$.

(b) Compute $P_W = QQ'$, with Q of Example 6.17, and check that $P_W = I_4 - \frac{1}{9}hh'$.

(c) Find the vector of W closest to y and its distance from y, and write the closest vector as Ax by solving $Rx = Q'y$.

Solution. (a) By Example 6.17, $W = \operatorname{span}\{b_1, b_2, b_3\}$ with $b_1 = (1, 2, 0, 2)'$, $b_2 = (2, -1, 2, 0)'$, $b_3 = (-2, 0, 2, 1)'$ (Theorem 6.14). Now $b_1'h = 0 + 4 + 0 - 4 = 0$, $b_2'h = 0 - 2 + 2 + 0 = 0$ and $b_3'h = 0 + 0 + 2 - 2 = 0$, so $h \in W^\perp$ (Theorem 6.6(a)). As $\dim W = 3$ (the nonzero, pairwise orthogonal b_j are linearly independent, Proposition 6.11(b)), $\dim W^\perp = 1$ (Theorem 6.6(c)), and the nonzero vector h spans $W^\perp$ (Theorem 4.5(d)). (This h is the vector v found in Example 6.18.)

(b) By Corollary 6.32 and Theorem 1.4(d),

$$P_W = QQ' = \tfrac{1}{9}\left(b_1b_1' + b_2b_2' + b_3b_3'\right) = \frac{1}{9}\begin{pmatrix}9&0&0&0\\0&5&-2&4\\0&-2&8&2\\0&4&2&5\end{pmatrix}$$

For instance, the $(2,3)$ entry is $\frac{1}{9}(0-2+0)$ and the $(3,3)$ entry is $\frac{1}{9}(0+4+4)$. Since $h'h=9$, $\frac{1}{9}hh'=h(h'h)^{-1}h'$ is the projection matrix for $\operatorname{span}\{h\}=W^\perp$ (Theorem 6.31(b) with the matrix h), and $P_W=I_4-\frac{1}{9}hh'$ (Theorem 6.31(c)); entry by entry, $I_4-\frac{1}{9}hh'$ is the matrix above, as hh' has rows 0, $(0,4,2,-4)$, $(0,2,1,-2)$, $(0,-4,-2,4)$.

(c) $h'y=0+12+3-6=9$, so $(I_4-P_W)y=\frac{h'y}{h'h}h=h$ and $P_Wy=y-h=(0,4,2,5)'$. This is the closest vector (Theorem 6.23), at distance $\|h\|=3$. Next $Q'y=\frac{1}{3}(b_1'y,b_2'y,b_3'y)'=\frac{1}{3}(18,0,9)'=(6,0,3)'$, and with R of Example 6.17,

$$3x_3=3,\quad 3x_2+3x_3=0,\quad 3x_1+3x_2+6x_3=6:\qquad x=(1,-1,1)'$$

Then $Ax=QRx=QQ'y=P_Wy$ (Corollary 6.32); directly, $Ax=a_1-a_2+a_3=(1-3+2,\ 2-1+3,\ 0-2+4,\ 2-2+5)'=(0,4,2,5)'$.

Exercise 6.15. Let $W=\mathcal{C}(A)$ and $y=(2,1,1,4,8)'$, where

$$A=\begin{pmatrix}1&1\\1&0\\1&0\\2&0\\3&1\end{pmatrix}$$

(a) Find a basis of $W^\perp$.

(b) Find the QR decomposition of A, and compute $P_W=QQ'$. Check that $P_WA=A$ and $\operatorname{tr}P_W=2$.

(c) Write $y=P_Wy+(I_5-P_W)y$, the decomposition along $\mathbb{R}^5=W\oplus W^\perp$, check that the two parts are orthogonal, and find the distance from y to W.

Solution. (a) $W^\perp=\mathcal{N}(A')$ (Theorem 6.6(a)). Row-reduce A':

$$\begin{pmatrix}1&1&1&2&3\\1&0&0&0&1\end{pmatrix}\xrightarrow{R_1\leftrightarrow R_2}\begin{pmatrix}1&0&0&0&1\\1&1&1&2&3\end{pmatrix}\xrightarrow{R_2\leftarrow R_2-R_1}\begin{pmatrix}1&0&0&0&1\\0&1&1&2&2\end{pmatrix}$$

With x_3,x_4,x_5 free, $x_1=-x_5$ and $x_2=-x_3-2x_4-2x_5$; so $v_1=(0,-1,1,0,0)'$, $v_2=(0,-2,0,1,0)'$, $v_3=(-1,-2,0,0,1)'$ form a basis of $W^\perp$ (Theorem 4.8(c)), of dimension $5-2=3$.

(b) By Theorem 6.14, $b_1=a_1=(1,1,1,2,3)'$, $\|b_1\|^2=1+1+1+4+9=16$; $a_2'b_1=1+3=4$, so $t_{12}=\frac{1}{4}$ and

$$b_2=a_2-\tfrac{1}{4}b_1=\tfrac{1}{4}(3,-1,-1,-2,1)',\qquad \|b_2\|^2=\tfrac{1}{16}(9+1+1+4+1)=1$$

So $q_1=\frac{1}{4}b_1$, $q_2=b_2$, $r_{11}=4$, $r_{12}=t_{12}\|b_1\|=1$, $r_{22}=1$ (Theorem 6.16):

$$Q=\frac{1}{4}\begin{pmatrix}1&3\\1&-1\\1&-1\\2&-2\\3&1\end{pmatrix},\qquad R=\begin{pmatrix}4&1\\0&1\end{pmatrix},\qquad P_W=QQ'=\frac{1}{8}\begin{pmatrix}5&-1&-1&-2&3\\-1&1&1&2&1\\-1&1&1&2&1\\-2&2&2&4&2\\3&1&1&2&5\end{pmatrix}$$

(Corollary 6.32); for instance, the $(1,5)$ entry is $\frac{1}{16}(1\cdot3+3\cdot1)=\frac{3}{8}$. Row by row, $P_Wa_1=\frac{1}{8}(5-1-1-4+9,\ -1+1+1+4+3,\ -1+1+1+4+3,\ -2+2+2+8+6,\ 3+1+1+4+15)'=(1,1,1,2,3)'$ and $P_Wa_2=\frac{1}{8}(5+3,\ -1+1,\ -1+1,\ -2+2,\ 3+5)'=(1,0,0,0,1)'$, so $P_WA=A$; and $\operatorname{tr}P_W=\frac{1}{8}(5+1+1+4+5)=2=\dim W$ (Theorem 6.31(c)).

(c) $Q'y=(\frac{1}{4}(2+1+1+8+24),\ \frac{1}{4}(6-1-1-8+8))'=(9,1)'$, so $P_Wy=Q(Q'y)=9q_1+q_2=\frac{1}{4}(9+3,\ 9-1,\ 9-1,\ 18-2,\ 27+1)'=(3,2,2,4,7)'$, and

$$\begin{pmatrix}2\\1\\1\\4\\8\end{pmatrix}=\begin{pmatrix}3\\2\\2\\4\\7\end{pmatrix}+\begin{pmatrix}-1\\-1\\-1\\0\\1\end{pmatrix}$$

The second part is $v_3 - v_1 \in W^\perp$, and $(P_W y)'(I_5 - P_W)y = -3 - 2 - 2 + 0 + 7 = 0$. The distance is $\|(-1,-1,-1,0,1)'\| = 2$ (Theorem 6.23).

Exercise 6.16. Let $v = \mathbf{1}_4$ and

$$E = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & 1 & -1 \end{pmatrix}$$

(a) Show that E is idempotent but not symmetric.

(b) Find $\operatorname{rank} E$, $\operatorname{tr} E$, $\operatorname{rank}(I_4 - E)$ and $\operatorname{tr}(I_4 - E)$.

(c) Decompose v along $\mathbb{R}^4 = \mathcal{N}(E) \oplus \mathcal{C}(E)$. Are the two parts orthogonal?

Solution. Let $\rho_1', \ldots, \rho_4'$ be the rows of E ($\rho_i \in \mathbb{R}^4$); then $\rho_3 = \rho_2 - \rho_1$ and $\rho_4 = -\rho_1$. Write $E = (e_{ij})$.

(a) By Theorem 1.4(c), $\operatorname{row}_i(E^2) = \rho_i' E = e_{i1}\rho_1' + e_{i2}\rho_2' + e_{i3}\rho_3' + e_{i4}\rho_4'$. So $\rho_1' E = \rho_1' + \rho_2' - \rho_3' + \rho_4' = \rho_1' + \rho_2' - (\rho_2' - \rho_1') - \rho_1' = \rho_1'$ and $\rho_2' E = \rho_1' + \rho_3' = \rho_2'$; then $\rho_3' E = \rho_2' E - \rho_1' E = \rho_3'$ and $\rho_4' E = -\rho_1' E = \rho_4'$. Hence $E^2 = E$. It is not symmetric: $e_{13} = -1 \neq 0 = e_{31}$.

(b) Row-reduce E:

$$E \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Scaling row 2 by -1 gives an echelon form with two nonzero rows, so $\operatorname{rank} E = 2 = 1+0+2-1 = \operatorname{tr} E$ (Remark 4.2 and Definition 4.11). Next

$$I_4 - E = \begin{pmatrix} 0 & -1 & 1 & -1 \\ -1 & 1 & -1 & 0 \\ 0 & 1 & -1 & 1 \\ 1 & 1 & -1 & 2 \end{pmatrix}$$

whose rows $\sigma_1', \ldots, \sigma_4'$ ($\sigma_i \in \mathbb{R}^4$) satisfy $\sigma_3 = -\sigma_1$ and $\sigma_4 = -2\sigma_1 - \sigma_2$. Row-reduce it:

$$I_4 - E \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 0 & -1 & 1 & -1 \\ -1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 0 & -1 & 1 & -1 \\ -1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} -1 & 1 & -1 & 0 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Scaling rows 1 and 2 by -1 gives an echelon form with two nonzero rows, so $\operatorname{rank}(I_4 - E) = 2 = 0 + 1 - 1 + 2 = \operatorname{tr}(I_4 - E)$ (Remark 4.2 and Definition 4.11), as Theorem 6.29(b),(c) say.

(c) $Ev = (2,2,0,-2)' \in \mathcal{C}(E)$ (the row sums of E) and $v - Ev = (-1,-1,1,3)'$, which lies in $\mathcal{N}(E)$: $E(-1,-1,1,3)' = (-1-1-1+3,\ -1+0+1+0,\ 0+1+2-3,\ 1+1+1-3)' = 0$. So $v = (-1,-1,1,3)' + (2,2,0,-2)'$ (Theorem 6.29(a)). The parts are not orthogonal: $(v - Ev)'(Ev) = -2 - 2 + 0 - 6 = -10$. (By Theorem 6.29(d), they would be if E were symmetric.)

Exercise 6.17. Let $W = \mathcal{C}(A)$ and $y = (2, 1, 2, 0)'$, where

$$A = [a_1\ a_2] = \begin{pmatrix} 1 & 3 \\ 1 & 1 \\ 0 & 1 \\ 1 & 2 \end{pmatrix}$$

(a) Compute $P_W = A(A'A)^{-1}A'$.

(b) Find an orthonormal basis q_1, q_2 of W by the Gram–Schmidt process, and check that $q_1q_1' + q_2q_2' = P_W$.

(c) Find $P_{W^\perp}$, check that $\operatorname{tr} P_W = \operatorname{tr} P_{W^\perp} = 2$, and split y along $\mathbb{R}^4 = W \oplus W^\perp$.

Solution. (a) $A'A = \left(\begin{smallmatrix} 3 & 6 \\ 6 & 15 \end{smallmatrix}\right)$, with determinant $45 - 36 = 9$, so $(A'A)^{-1} = \frac{1}{9}\left(\begin{smallmatrix} 15 & -6 \\ -6 & 3 \end{smallmatrix}\right) = \frac{1}{3}\left(\begin{smallmatrix} 5 & -2 \\ -2 & 1 \end{smallmatrix}\right)$ (Theorem 5.24). Then

$$(A'A)^{-1}A' = \frac{1}{3}\begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 0 & 1 \\ 3 & 1 & 1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -1 & 3 & -2 & 1 \\ 1 & -1 & 1 & 0 \end{pmatrix}$$

$$P_W = A(A'A)^{-1}A' = \begin{pmatrix} 1 & 3 \\ 1 & 1 \\ 0 & 1 \\ 1 & 2 \end{pmatrix}\frac{1}{3}\begin{pmatrix} -1 & 3 & -2 & 1 \\ 1 & -1 & 1 & 0 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & -1 & 1 \\ 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

(b) $u_1 = a_1 = (1, 1, 0, 1)'$, $u_1'u_1 = 3$; $a_2'u_1 = 3 + 1 + 0 + 2 = 6$, so $u_2 = a_2 - 2u_1 = (1, -1, 1, 0)'$, with $u_2'u_2 = 3$ and $u_1'u_2 = 0$. So $q_j = \frac{1}{\sqrt{3}}u_j$, and

$$q_1q_1' + q_2q_2' = \frac{1}{3}\left[\begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\right] = \frac{1}{3}\begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & -1 & 1 \\ 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} = P_W$$

as Theorem 6.31(b) asserts.

(c) By Theorem 6.31(c),

$$P_{W^\perp} = I_4 - P_W = \frac{1}{3}\begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 1 & -1 \\ -1 & 1 & 2 & 0 \\ -1 & -1 & 0 & 2 \end{pmatrix}$$

and $\operatorname{tr} P_W = \frac{1}{3}(2 + 2 + 1 + 1) = 2$, $\operatorname{tr} P_{W^\perp} = \frac{1}{3}(1 + 1 + 2 + 2) = 2$. Next $P_Wy = \frac{1}{3}(4 + 0 + 2 + 0,\ 0 + 2 - 2 + 0,\ 2 - 1 + 2 + 0,\ 2 + 1 + 0 + 0)' = (2, 0, 1, 1)'$ and $P_{W^\perp}y = y - P_Wy = (0, 1, 1, -1)'$; check: $(2, 0, 1, 1)(0, 1, 1, -1)' = 0 + 0 + 1 - 1 = 0$.

Exercise 6.18. Let $a_1 = (0,1,1,1)'$, $a_2 = (1,2,0,1)'$, $U = \text{span}\{a_1\}$, $W = \text{span}\{a_1, a_2\}$, and $y = (2,2,1,3)'$.

(a) Compute P_U and P_W, and check that $P_W a_1 = a_1$ and hence $P_W P_U = P_U$.

(b) Compute $P_W - P_U$, check that it is idempotent of rank 1, and find a vector spanning $W \cap U^\perp$.

(c) Split $y = P_U y + (P_W - P_U)y + (I_4 - P_W)y$, and check the equation of Theorem 6.36(c).

Solution. (a) $a_1'a_1 = 3$, so $P_U = \frac{1}{3}a_1a_1'$ (Theorem 6.31(b)). For $A = [a_1\ a_2]$, $A'A = \left(\begin{smallmatrix} 3 & 3 \\ 3 & 6 \end{smallmatrix}\right)$ ($a_1'a_2 = 0+2+0+1 = 3$, $a_2'a_2 = 1+4+0+1 = 6$), with determinant 9, so $(A'A)^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ (Theorem 5.24) and $A(A'A)^{-1} = \frac{1}{3}[2a_1 - a_2 \quad -a_1 + a_2]$; by Theorem 6.31(b), $P_W = A(A'A)^{-1}A'$:

$$P_U = \frac{1}{3}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \qquad P_W = \frac{1}{3}\begin{pmatrix} -1 & 1 \\ 0 & 1 \\ 2 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 2 & 0 & 1 \\ -1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Then $P_W a_1 = \frac{1}{3}(0+1-1+0,\ 0+2+0+1,\ 0+0+2+1,\ 0+1+1+1)' = (0,1,1,1)' = a_1$, so $P_W P_U = \frac{1}{3}(P_W a_1)a_1' = P_U$.

(b)

$$P_W - P_U = \frac{1}{3}\begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 1 & -1 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \frac{1}{3}uu', \qquad u = (1,1,-1,0)'$$

As $u'u = 3$, $\left(\frac{1}{3}uu'\right)^2 = \frac{1}{9}u(u'u)u' = \frac{1}{3}uu'$: it is idempotent, of rank $\text{tr}(P_W - P_U) = \frac{1}{3}(1+1+1+0) = 1$ (Theorem 6.29(c)). By Theorem 6.36(b), it is the projection matrix for $W \cap U^\perp = \mathcal{C}(P_W - P_U) = \text{span}\{u\}$; indeed $u = a_2 - a_1 \in W$ and $u'a_1 = 0+1-1+0 = 0$.

(c) $P_U y = \frac{a_1'y}{3}\, a_1 = \frac{6}{3}\, a_1 = (0,2,2,2)'$, $(P_W - P_U)y = \frac{u'y}{3}\, u = \frac{3}{3}\, u = (1,1,-1,0)'$, and $(I_4 - P_W)y = y - (0,2,2,2)' - (1,1,-1,0)' = (1,-1,0,1)'$. Then $\|y\|^2 = 4+4+1+9 = 18 = 12+3+3$.

Problems

Problem 6.1 ([8, Ch. 17, Exercise 3])**.** Let U, W and Z be subspaces of $\mathbb{R}^n$, and $y \in \mathbb{R}^n$.

(a) Show (1) that if $y \perp W$ and $y \perp Z$, then $y \perp W + Z$; and (2) that if $U \perp W$ and $U \perp Z$, then $U \perp W + Z$.

(b) Show (1) that $(U+W)^\perp = U^\perp \cap W^\perp$; and (2) that $(U \cap W)^\perp = U^\perp + W^\perp$.

Solution. (a)

- (1) Let $t \in W + Z$, say $t = w + z$ with $w \in W$ and $z \in Z$ (Definition 3.30). Then $y't = y'w + y'z = 0 + 0 = 0$. So $y \perp W + Z$.
- (2) Every $u \in U$ satisfies $u \perp W$ and $u \perp Z$, so $u \perp W + Z$ by (1). Hence $U \perp W + Z$.

(b)

- (1) $U \subseteq U + W$ and $W \subseteq U + W$ (Theorem 3.31(b)). So, using (a)(1) for the converse, $y \in (U+W)^\perp \iff y \perp U + W \iff y \perp U$ and $y \perp W \iff y \in U^\perp \cap W^\perp$.
- (2) $U^\perp + W^\perp$ is a subspace of $\mathbb{R}^n$ (Theorems 6.6(a) and 3.31(a)). By Theorem 6.8(a), used three times, and (1) for the subspaces $U^\perp$ and $W^\perp$,

$$U^\perp + W^\perp = \left((U^\perp + W^\perp)^\perp\right)^\perp = \left((U^\perp)^\perp \cap (W^\perp)^\perp\right)^\perp = (U \cap W)^\perp$$

Problem 6.2 ([8, Ch. 12, Exercise 8])**.** Let $a_1, \dots, a_k \in \mathbb{R}^n$ $(k \geqslant 2)$ be linearly independent, and let $b_1, \dots, b_k$ be obtained from them by the Gram–Schmidt process (Theorem 6.14). Show that for $j = 2, \dots, k$ the vector b_j is the orthogonal projection of a_j onto some subspace U_j of $\mathbb{R}^n$, and describe U_j.

Solution. Let $q_i = b_i/\|b_i\|$ $(i = 1, \dots, k)$ and $W_j = \operatorname{span}\{q_1, \dots, q_j\}$.

- For $j = 2, \dots, k$, since $t_{ij} b_i = \frac{a_j' b_i}{\|b_i\|^2}\, b_i = (a_j' q_i) q_i$,

$$b_j = a_j - \sum_{i=1}^{j-1} t_{ij} b_i = a_j - \sum_{i=1}^{j-1} (a_j' q_i) q_i$$

- $q_1, \dots, q_{j-1}$ are orthonormal (Theorem 6.14), so they form an orthonormal basis of W_{j-1}, and $\sum_{i<j} (a_j' q_i) q_i$ is the orthogonal projection of a_j onto W_{j-1} (Theorem 6.24(a)).
- By Theorem 6.24(d), b_j is the orthogonal projection of a_j onto $W_{j-1}^{\perp}$.
- Each q_i is a multiple of b_i and vice versa, so $W_{j-1} = \operatorname{span}\{b_1, \dots, b_{j-1}\}$ (Theorem 3.7), and this is $\operatorname{span}\{a_1, \dots, a_{j-1}\}$ (Theorem 6.14).

Hence b_j is the orthogonal projection of a_j onto $U_j = \operatorname{span}\{a_1, \dots, a_{j-1}\}^{\perp}$, the orthogonal complement of the span of the preceding vectors.

Problem 6.3 ([5, §6.4d])**.** Let $A = [a_1 \ \cdots \ a_p] \in \mathbb{R}^{n \times p}$ have full column rank, and let $A = QR$ be a QR decomposition as in Theorem 6.16: $Q \in \mathbb{R}^{n \times p}$ has orthonormal columns and $R \in \mathbb{R}^{p \times p}$ is upper triangular with positive diagonal entries. Show that this decomposition is unique: if $Q^* \in \mathbb{R}^{n \times p}$ has orthonormal columns, $R^* \in \mathbb{R}^{p \times p}$ is upper triangular with positive diagonal entries, and $A = Q^* R^*$, then $Q^* = Q$ and $R^* = R$.

Solution. The columns $a_1, \dots, a_p$ of A are linearly independent (Theorem 4.13(b)). Let $b_1, \dots, b_p$ and t_{ij} $(i < j)$ be the vectors and coefficients obtained from them by the Gram–Schmidt process (Theorem 6.14), so that $b_1, \dots, b_p$ are nonzero and pairwise orthogonal.

- Step 1. Let $A = BX$, where the columns $c_1, \dots, c_p$ of $B \in \mathbb{R}^{n \times p}$ are nonzero and pairwise orthogonal, and $X = (x_{ij}) \in \mathbb{R}^{p \times p}$ is unit upper triangular (upper triangular with diagonal entries 1). We show that $c_j = b_j$ for every j and $x_{ij} = t_{ij}$ for $i < j$. By Theorem 1.4(a),(b), $a_j = \operatorname{col}_j(BX) = B \operatorname{col}_j(X) = c_j + \sum_{i<j} x_{ij} c_i$, that is,

$$c_j = a_j - \sum_{i<j} x_{ij} c_i \qquad (j = 1, \dots, p)$$

  We use induction on j. For $j = 1$, $c_1 = a_1 = b_1$. Let $j \geqslant 2$, and let $c_i = b_i$ for $i < j$. For $l < j$, as the c_i are pairwise orthogonal, $0 = c_l' c_j = c_l' a_j - \sum_{i<j} x_{ij} c_l' c_i = b_l' a_j - x_{lj} \|b_l\|^2$; so $x_{lj} = a_j' b_l / \|b_l\|^2 = t_{lj}$, and $c_j = a_j - \sum_{i<j} t_{ij} b_i = b_j$.

- Step 2. Let $A = Q^* R^*$ be as in the statement, with columns $q_1^*, \dots, q_p^*$ of Q^* and entries r_{ij}^* of R^*. Put $E^* = \operatorname{diag}(r_{11}^*, \dots, r_{pp}^*)$ and $D^* = \operatorname{diag}(1/r_{11}^*, \dots, 1/r_{pp}^*)$, so that $E^* D^* = I_p$, and let $B^* = Q^* E^*$ and $X^* = D^* R^*$. The columns of B^* are $r_{jj}^* q_j^*$ (Theorem 1.4(a),(b)): nonzero, as $r_{jj}^* > 0$ and $\|q_j^*\| = 1$, and pairwise orthogonal. The rows of X^* are $\operatorname{row}_i(R^*)/r_{ii}^*$ (Theorem 1.4(c)), so X^* is unit upper triangular, with entries r_{ij}^*/r_{ii}^*. And $B^* X^* = Q^* E^* D^* R^* = Q^* R^* = A$.

- By Step 1, $r_{jj}^* q_j^* = b_j$ for every j, and $r_{ij}^*/r_{ii}^* = t_{ij}$ for $i < j$. Taking lengths, $r_{jj}^* = \|b_j\|$, as $r_{jj}^* > 0$ and $\|q_j^*\| = 1$. Hence

$$q_j^* = \frac{b_j}{\|b_j\|}, \qquad r_{jj}^* = \|b_j\|, \qquad r_{ij}^* = t_{ij} \|b_i\| \quad (i < j), \qquad r_{ij}^* = 0 \quad (i > j)$$

- Step 2 applies to the decomposition $A = QR$ as well, with the same right sides. So Q and Q^* have the same columns, and R and R^* the same entries: $Q^* = Q$ and $R^* = R$.

Problem 6.4 ([6, Exercise 8.64])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be idempotent.

(a) Show that $A+B$ is idempotent if and only if $AB = BA = 0$.

(b) If $A+B$ is idempotent, show that $\operatorname{rank}(A+B) = \operatorname{rank} A + \operatorname{rank} B$.

Solution. (a)

- If $AB = BA = 0$, then $(A+B)^2 = A^2 + AB + BA + B^2 = A^2 + B^2 = A + B$.
- Conversely, if $(A+B)^2 = A+B$, expanding as above gives $AB + BA = 0$. Multiplying on the left by A gives $AB + ABA = 0$; multiplying on the right by A gives $ABA + BA = 0$.
- Subtracting, $AB - BA = 0$, so $AB = BA$; then $AB + BA = 2AB = 0$, and $AB = BA = 0$.

(b) By Theorem 6.29(c), applied to $A+B$, A and B, and Theorem 1.11(a), $\operatorname{rank}(A+B) = \operatorname{tr}(A+B) = \operatorname{tr} A + \operatorname{tr} B = \operatorname{rank} A + \operatorname{rank} B$.

Problem 6.5 ([6, Exercise 8.67])**.** Let $A = [A_1\ A_2] \in \mathbb{R}^{n\times n}$, with $A_1 \in \mathbb{R}^{n\times n_1}$ and $A_2 \in \mathbb{R}^{n\times n_2}$ ($n_1, n_2 \geqslant 1$, $n_1 + n_2 = n$). Suppose that $\operatorname{rank} A_1 + \operatorname{rank} A_2 = n$ and $A_1'A_2 = 0$. Show that A_1 and A_2 have full column rank and that

$$A_1(A_1'A_1)^{-1}A_1' + A_2(A_2'A_2)^{-1}A_2' = I_n$$

Solution.

- $\operatorname{rank} A_1 \leqslant n_1$ and $\operatorname{rank} A_2 \leqslant n_2$ (Theorem 4.13(a)), and the ranks add up to $n = n_1 + n_2$; so $\operatorname{rank} A_1 = n_1$ and $\operatorname{rank} A_2 = n_2$.
- The columns of A are independent: let $A_1c + A_2d = 0$ with $c \in \mathbb{R}^{n_1}$, $d \in \mathbb{R}^{n_2}$, and put $v = A_1c = -A_2d$. Then $v'v = -c'A_1'A_2d = 0$, so $v = 0$; hence $A_1c = 0$ and $A_2d = 0$, and $c = 0$, $d = 0$ because $\mathcal{N}(A_1) = \{0\}$ and $\mathcal{N}(A_2) = \{0\}$ (Theorem 4.13(b)). So A is invertible (Theorem 4.13(b),(d)).
- $A_2'A_1 = (A_1'A_2)' = 0$, so by Theorem 5.3(c),

$$A'A = \begin{pmatrix} A_1'A_1 & A_1'A_2 \\ A_2'A_1 & A_2'A_2 \end{pmatrix} = \begin{pmatrix} A_1'A_1 & 0 \\ 0 & A_2'A_2 \end{pmatrix}$$

  Here $A_1'A_1$ and $A_2'A_2$ are invertible (Theorems 4.18(b) and 4.13(d)), so $(A'A)^{-1}$ is block diagonal with the diagonal blocks $(A_1'A_1)^{-1}$ and $(A_2'A_2)^{-1}$ (Proposition 5.6(b)).
- As A is invertible, $(A'A)^{-1} = A^{-1}(A')^{-1}$ (Theorem 2.13(b),(c)), so $A(A'A)^{-1}A' = AA^{-1}(A')^{-1}A' = I_n$. By Theorem 5.3(c), therefore,

$$\begin{aligned} I_n &= [A_1\ A_2] \begin{pmatrix} (A_1'A_1)^{-1} & 0 \\ 0 & (A_2'A_2)^{-1} \end{pmatrix} \begin{pmatrix} A_1' \\ A_2' \end{pmatrix} \\ &= A_1(A_1'A_1)^{-1}A_1' + A_2(A_2'A_2)^{-1}A_2' \end{aligned}$$

Problem 6.6 ([2, Exercise 5.5], [8, Ch. 17, Exercise 19(a)])**.** Let U, W be subspaces of $\mathbb{R}^n$. Show that P_UP_W is symmetric if and only if $P_UP_W = P_WP_U$, and that then $P_UP_W = P_{U\cap W}$.

Solution. $(P_UP_W)' = P_W'P_U' = P_WP_U$ (Theorems 1.9 and 6.31(c)), so P_UP_W is symmetric if and only if $P_UP_W = P_WP_U$. Let them commute, and put $P = P_UP_W$.

- $P^2 = P_UP_WP_UP_W = P_UP_UP_WP_W = P_UP_W = P$. So P is symmetric idempotent, and $P = P_{\mathcal{C}(P)}$ (Theorem 6.34).
- $\mathcal{C}(P) = U \cap W$: for $y \in \mathbb{R}^n$, $Py = P_U(P_Wy) \in U$ and $Py = P_W(P_Uy) \in W$; conversely, for $x \in U \cap W$, $Px = P_U(P_Wx) = P_Ux = x$ (Theorem 6.31(c)), so $x \in \mathcal{C}(P)$. Hence $P = P_{U\cap W}$.

Problem 6.7 ([10, Problem 7.4])**.** Let $v_1, \dots, v_m \in \mathbb{R}^n$ satisfy $v_i' v_j < 0$ for all $i \neq j$.

(a) Show that if $m = n + 1$, then any n of the vectors $v_1, \dots, v_m$ form a basis of $\mathbb{R}^n$.

(b) Show that $m \leqslant n + 1$.

Solution. Both parts use the following fact.

(∗) If $c_1 v_1 + \cdots + c_m v_m = 0$, then the nonzero numbers among $c_1, \dots, c_m$ all have the same sign.

- Suppose not: the sets $S_+ = \{i : c_i > 0\}$ and $S_- = \{j : c_j < 0\}$ are both nonempty. They are disjoint, and the terms with $c_i = 0$ vanish, so

$$z = \sum_{i \in S_+} c_i v_i = \sum_{j \in S_-} (-c_j) v_j$$

- Multiplying the transpose of the first expression by the second (Theorems 1.9 and 1.3),

$$z'z = \sum_{i \in S_+} \sum_{j \in S_-} c_i (-c_j)\, v_i' v_j$$

  Every term is < 0, as $c_i > 0$, $-c_j > 0$ and $v_i' v_j < 0$ ($i \neq j$, since S_+ and S_- are disjoint), and there is at least one term. So $z'z < 0$, while $z'z = \|z\|^2 \geqslant 0$ (Definition 6.1) ✘

(a) Let $m = n + 1$, fix $r \in \{1, \dots, n + 1\}$, and let $\sum_{i \neq r} c_i v_i = 0$, the sum over the n indices $i \neq r$.

- With $c_r = 0$, this is a relation $c_1 v_1 + \cdots + c_{n+1} v_{n+1} = 0$. Suppose that some $c_i \neq 0$. By (∗), the nonzero c_i have one sign; replacing every c_i by $-c_i$ if necessary, every $c_i \geqslant 0$ and some $c_l > 0$, where $l \neq r$, as $c_r = 0$.
- Multiplying the relation on the left by v_r' gives $\sum_{i \neq r} c_i\, v_r' v_i = 0$. But every term $c_i\, v_r' v_i$ is $\leqslant 0$, as $v_r' v_i < 0$, and the term with $i = l$ is < 0; so the sum is < 0 ✘
- Hence every $c_i = 0$: the n vectors v_i ($i \neq r$) are linearly independent. As $\dim \mathbb{R}^n = n$ (Theorem 4.4), they form a basis of $\mathbb{R}^n$ (Theorem 4.5(d)).

(b) Suppose that $m \geqslant n + 2$.

- Consider the homogeneous linear equations

$$c_1 v_1 + c_2 v_2 + \cdots + c_{n+2} v_{n+2} = 0, \qquad c_1 + c_2 + \cdots + c_{n+2} = 0$$

  in the unknowns $c_1, \dots, c_{n+2}$. The first is a system of n equations, one for each entry, whose coefficient matrix is $[v_1 \ \cdots \ v_{n+2}]$ (Theorem 1.4(b)); so there are $n+1$ equations in $n+2$ unknowns, and by Theorem 2.11(a) they have a solution with some $c_i \neq 0$.
- Put $c_i = 0$ for $n + 2 < i \leqslant m$. Then $c_1 v_1 + \cdots + c_m v_m = 0$, so by (∗) the nonzero c_i all have one sign. As some $c_i \neq 0$, the sum $c_1 + \cdots + c_{n+2}$, which is the sum of the nonzero c_i, is then $\neq 0$ ✘

Hence $m \leqslant n + 1$.

Problem 6.8 ([10, Problem 9.1])**.** Let U and W be subspaces of $\mathbb{R}^n$ with $\dim U = \dim W = m$. Suppose that some $u \in U$, $u \neq 0$, is orthogonal to W. Show that some $w \in W$, $w \neq 0$, is orthogonal to U.

Solution. As $u \neq 0$, $m \geqslant 1$. Put $q_1 = u/\|u\|$, a unit vector of U.

- If $m = 1$, then q_1 alone is an orthonormal basis of U (Proposition 3.13(a), Theorem 4.5(d)). If $m \geqslant 2$, there are $q_2, \dots, q_m \in U$ such that $q_1, \dots, q_m$ form an orthonormal basis of U (Corollary 6.15(b)).
- Let $Z = \operatorname{span}\{q_2, \dots, q_m\}$, so $Z = \{0\}$ if $m = 1$ (Definition 3.6). The vectors $q_2, \dots, q_m$ are orthonormal, hence linearly independent (Proposition 6.11(b)); so they form a basis of Z, and $\dim Z = m - 1$. By Theorem 6.6(c), $\dim Z^\perp = n - m + 1$.

- Hence $\dim Z^\perp + \dim W = (n-m+1)+m = n+1 > n$, and $Z^\perp \cap W \neq \{0\}$ (Corollary 4.25(b)). Take $w \in Z^\perp \cap W$ with $w \neq 0$.
- Then $w'q_j = 0$ for $j = 2, \ldots, m$, as $w \in Z^\perp$ and $q_j \in Z$; and $w'q_1 = u'w/\|u\| = 0$, as $u \perp W$ and $w \in W$ (Definition 6.5). Since $q_1, \ldots, q_m$ span U, $w \perp U$ (Theorem 6.6(a)).

Problem 6.9 ([10, Problem 9.2])**.** Let $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ be two bases of $\mathbb{R}^n$ such that $x_i'x_j = y_i'y_j$ for all i, j. Show that there is an orthogonal matrix $U \in \mathbb{R}^{n\times n}$ with $Ux_i = y_i$ for $i = 1, \ldots, n$.

Solution. Let $A = [x_1 \ \cdots \ x_n]$ and $B = [y_1 \ \cdots \ y_n]$.

- The (i,j) entries of $A'A$ and $B'B$ are $x_i'x_j$ and $y_i'y_j$ (Theorem 1.4(d)), so $A'A = B'B$.
- The columns of A are linearly independent, so A has full column rank and is invertible (Theorem 4.13(b),(d)). By Theorem 6.16, $A = Q_1R$, where $Q_1'Q_1 = I_n$ and R is upper triangular with positive diagonal entries. So Q_1 is orthogonal (Definition 6.12), and $R = Q_1'A = Q_1^{-1}A$ is invertible (Theorems 6.19(a) and 2.13(a),(b)), with $AR^{-1} = Q_1$.
- Apply the same combinations to $y_1, \ldots, y_n$: let $Q_2 = BR^{-1}$. By Theorem 1.9 and $B'B = A'A$,

$$Q_2'Q_2 = (R^{-1})'B'BR^{-1} = (R^{-1})'A'AR^{-1} = (AR^{-1})'(AR^{-1}) = Q_1'Q_1 = I_n$$

  so Q_2 is orthogonal too.
- Let $U = Q_2Q_1'$. It is orthogonal (Theorem 6.19(b)), and $UA = Q_2(Q_1'Q_1)R = Q_2R = B$. Column i of this equation reads $Ux_i = y_i$ (Theorem 1.4(a)).

In fact $U = BR^{-1}Q_1' = B(Q_1R)^{-1} = BA^{-1}$ (Theorem 2.13(b)).

Chapter 7. Eigenvalues and diagonalization

Eigenvalues and eigenvectors

Definition 7.1 ([1, §6.2], [11, Five.II.3, Definition 3.5, Lemma 3.13], [4, Ch. VIII, §1], [5, §21.1])**.** Let $A \in \mathbb{R}^{n\times n}$. A real number λ is an **eigenvalue** of A if $Ax = \lambda x$ for some $x \in \mathbb{R}^n$ with $x \neq 0$; such a vector x is an **eigenvector** of A **belonging to** λ. Since $Ax = \lambda x$ if and only if $(A - \lambda I)x = 0$, the set

$$\mathcal{N}(A - \lambda I) = \{x \in \mathbb{R}^n : Ax = \lambda x\}$$

is called the **eigenspace** of A for λ.

Eigenvalues need not be real numbers: once the characteristic polynomial of A is defined below, the eigenvalues of A are defined as the n complex roots of that polynomial, listed with multiplicity, and the eigenvalues of this definition are exactly the real ones among them.

The eigenspace $\mathcal{N}(A - \lambda I)$ is a subspace of $\mathbb{R}^n$ (Example 3.5(e)), and its nonzero vectors are exactly the eigenvectors belonging to λ. So a nonzero multiple of an eigenvector, and a nonzero sum of eigenvectors belonging to the same λ, are again eigenvectors belonging to λ. An eigenvector belongs to only one eigenvalue: if $Ax = \lambda_1 x$ and $Ax = \lambda_2 x$, then $(\lambda_1 - \lambda_2)x = 0$ with $x \neq 0$, so $\lambda_1 = \lambda_2$ (Proposition 3.13(a)).

Example 7.2. Let

$$A = \begin{pmatrix} 1 & 2 & -2 & -2 \\ -1 & 0 & -2 & 2 \\ 0 & -1 & -1 & 1 \\ -1 & 2 & -2 & 0 \end{pmatrix}, \qquad v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \qquad v_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \qquad v_4 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

Multiplying out,

$$Av_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \qquad Av_2 = \begin{pmatrix} -1 \\ -1 \\ -1 \\ -1 \end{pmatrix}, \qquad Av_3 = \begin{pmatrix} 0 \\ 2 \\ 0 \\ 2 \end{pmatrix}, \qquad Av_4 = \begin{pmatrix} 0 \\ -2 \\ -2 \\ 0 \end{pmatrix}$$

So $Av_1 = v_1$, $Av_2 = -v_2$, $Av_3 = 2v_3$ and $Av_4 = -2v_4$: the numbers 1, -1, 2 and -2 are eigenvalues of A, and v_1, v_2, v_3, v_4 are eigenvectors belonging to them (Definition 7.1). The vector e_1 is not an eigenvector: $Ae_1 = (1, -1, 0, -1)'$ is not a multiple of e_1.

Theorem 7.3 ([4, Ch. VIII, Theorems 2.1, 2.2], [1, §6.2, Theorem 1], [5, §21.1])**.** Let $A \in \mathbb{R}^{n\times n}$ and $\lambda \in \mathbb{R}$. The following are equivalent:

(i) λ is an eigenvalue of A;

(ii) $A - \lambda I$ is not invertible;

(iii) $\det(\lambda I - A) = 0$;

(iv) $\operatorname{rank}(A - \lambda I) < n$.

Proof. • (i)⇔(ii): λ is an eigenvalue if and only if $(A - \lambda I)x = 0$ has a solution $x \neq 0$, that is, if and only if $A - \lambda I$ is not invertible (Theorem 2.16, (i)⇔(iii)).

• (ii)⇔(iii): $\det(\lambda I - A) = \det\big((-1)(A - \lambda I)\big) = (-1)^n \det(A - \lambda I)$ (Theorem 5.18(c)), and $A - \lambda I$ is invertible if and only if $\det(A - \lambda I) \neq 0$ (Theorem 5.18(a)).

• (ii)⇔(iv): Theorem 4.13(d). □

So the eigenvalues of A are the real solutions λ of $\det(\lambda I - A) = 0$, and the eigenvectors belonging to λ are the nonzero solutions of $(A - \lambda I)x = 0$, found by row reduction.

Polynomials

Complex numbers. A **complex number** is $z = a + bi$ with $a, b \in \mathbb{R}$, where $i^2 = -1$; $\mathbb{C}$ is the set of complex numbers, and a real number a is the complex number $a + 0i$. Sums and products are $(a + bi) + (c + di) = (a + c) + (b + d)i$ and $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$; they obey the rules of arithmetic of real numbers (commutative, associative and distributive laws), as one checks by multiplying out both sides [11, Five.I.1], [1, §1.1]. The **conjugate** of $z = a + bi$ is $\bar{z} = a - bi$; multiplying out likewise gives: $\bar{z} = z$ if and only if z is real, $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\,\bar{w}$, and $z\bar{z} = a^2 + b^2 > 0$ for $z \neq 0$. Hence every $z \neq 0$ has the inverse $z^{-1} = \bar{z}/(a^2 + b^2)$, and $zw = 0$ only if $z = 0$ or $w = 0$ (if $z \neq 0$, then $w = z^{-1}zw = 0$).

Definition 7.4 ([1, §§4.1, 4.2], [11, Five.I.1], [4, Ch. IX, §§1, 2])**.** A **polynomial** (in the variable t) is an expression $f = c_0 + c_1 t + \cdots + c_m t^m$ with **coefficients** $c_0, \ldots, c_m \in \mathbb{C}$; we put $c_k = 0$ for $k > m$, and f equals $g = d_0 + d_1 t + \cdots$ if $c_k = d_k$ for every $k \geqslant 0$. We write $f(t)$ for f when the variable is to be shown.

- f is **real** if every $c_k \in \mathbb{R}$; the **zero polynomial** 0 has all coefficients 0. For $f \neq 0$, the **degree** $\deg f$ is the largest k with $c_k \neq 0$, and $c_{\deg f}$ is the **leading coefficient**; f is **monic** if its leading coefficient is 1. c_0 is the **constant term**, and a polynomial with $c_k = 0$ for all $k \geqslant 1$ is written as the number c_0.
- The sum $f + g$, the multiple af $(a \in \mathbb{C})$ and the product fg have the coefficients $c_k + d_k$, ac_k and $\sum_{i+j=k} c_i d_j = c_0 d_k + c_1 d_{k-1} + \cdots + c_k d_0$; thus $(ct^i)(dt^j) = cd\,t^{i+j}$. These operations obey the rules of arithmetic of numbers ($fg = gf$, $(fg)h = f(gh)$, $f(g + h) = fg + fh$, and so on), because these hold coefficient by coefficient; for instance, with $h = \sum_l h_l t^l$, the coefficient of t^k in $(fg)h$ and in $f(gh)$ is $\sum_{i+j+l=k} c_i d_j h_l$.
- The **value** of f at $z \in \mathbb{C}$ is $f(z) = c_0 + c_1 z + \cdots + c_m z^m$, and z is a **root** of f if $f(z) = 0$. For a real polynomial f and $A \in \mathbb{R}^{n\times n}$, $f(A) = c_0 I_n + c_1 A + \cdots + c_m A^m$ (with $A^0 = I_n$; Definition 1.23); for instance, for $f = t - c$, $f(A) = A - cI$.

Proposition 7.5 ([1, §4.2, Theorems 1, 2; §4.4, Corollaries 1, 2], [11, Five.I.1, Corollaries 1.4, 1.5])**.** Let f, g, h be polynomials.

(a) If $f \neq 0$ and $g \neq 0$, then $fg \neq 0$, $\deg(fg) = \deg f + \deg g$, and the leading coefficient of fg is the product of those of f and g. If $f \neq 0$ and $fg = fh$, then $g = h$.

(b) $(af + g)(z) = af(z) + g(z)$ and $(fg)(z) = f(z)g(z)$ for $a, z \in \mathbb{C}$; the same holds for real f, g, $a \in \mathbb{R}$ and $z = A \in \mathbb{R}^{n\times n}$.

(c) (**Factor theorem**) For $c \in \mathbb{C}$, $f(c) = 0$ if and only if $f = (t - c)q$ for some polynomial q.

(d) A nonzero polynomial of degree m has at most m roots. Consequently, if $f(z) = g(z)$ for infinitely many z (for instance, for all real z, or for all real $z \neq 0$), then $f = g$.

(e) For $\mu_1, \ldots, \mu_n \in \mathbb{C}$ $(n \geqslant 1)$, $(t - \mu_1)\cdots(t - \mu_n) = t^n - (\mu_1 + \cdots + \mu_n)t^{n-1} + r$, where $r = 0$ or $\deg r \leqslant n - 2$; and the constant term of this product is $(-1)^n \mu_1 \cdots \mu_n$.

Proof. Write $f = \sum_k c_k t^k$ and $g = \sum_k d_k t^k$.

(a)

- Let $\deg f = m$ and $\deg g = l$. A term $c_i d_{m+l+j-i}$ $(j \geqslant 0)$ of the coefficient of t^{m+l+j} in fg can be nonzero only if $i \leqslant m$ and $m + l + j - i \leqslant l$, that is, $m + j \leqslant i \leqslant m$; so $j = 0$ and $i = m$. Hence the coefficient of t^{m+l} is $c_m d_l \neq 0$, and all higher coefficients are 0.
- If $fg = fh$, then $f(g - h) = 0$; if $g \neq h$, the first item would give $f(g - h) \neq 0$ ✘ So $g = h$.

(b)

- $af + g$ has the coefficients $ac_k + d_k$, so $(af + g)(z) = \sum_k (ac_k + d_k)z^k = af(z) + g(z)$.

- By the product rule of Definition 7.4 and the distributive laws, $(fg)(z) = \sum_k \sum_{i+j=k} c_i d_j z^{i+j} = \sum_{i,j} (c_i z^i)(d_j z^j) = f(z)g(z)$.
- For real f, g, a and a matrix $z = A$, both rules are Proposition 1.24(a),(b).

(c)

- First, $f = (t-c)q + r$ for a polynomial q and a number r, by induction on the degree: if $f = 0$ or $\deg f = 0$, take $q = 0$ and $r = f$; if $\deg f = m \geqslant 1$ and a is the leading coefficient, then $f - at^{m-1}(t-c)$ has coefficient 0 at t^m, so it is 0 or of degree less than m; by the induction hypothesis it equals $(t-c)q_1 + r$, and $f = (t-c)(q_1 + at^{m-1}) + r$.
- By (b), $f(c) = (c-c)q(c) + r = r$. So if $f(c) = 0$, then $r = 0$ and $f = (t-c)q$. Conversely, if $f = (t-c)q$, then $f(c) = (c-c)q(c) = 0$ by (b).

(d)

- Induction on m. If $m = 0$, f is a nonzero number and has no root. Let $m \geqslant 1$, and let f have a root a (otherwise there is nothing to prove). By (c), $f = (t-a)q$; here $q \neq 0$, as $f \neq 0$, so $\deg q = m - 1$ by (a).
- If $f(b) = 0$, then $(b-a)q(b) = 0$ by (b), so $b = a$ or $q(b) = 0$. By the induction hypothesis q has at most $m-1$ roots, so f has at most m.
- If $f \neq g$, then $f - g$ is a nonzero polynomial, and every z with $f(z) = g(z)$ is a root of it (by (b)); so there are only finitely many such z.

(e)

- Induction on n; for $n = 1$, $t - \mu_1$ has this form with $r = 0$.
- If $(t-\mu_1)\cdots(t-\mu_n) = t^n - s_n t^{n-1} + r_n$ with $s_n = \mu_1 + \cdots + \mu_n$, then multiplying by $t - \mu_{n+1}$ gives $t^{n+1} - (s_n + \mu_{n+1})t^n + r_{n+1}$ with $r_{n+1} = s_n \mu_{n+1} t^{n-1} + r_n(t - \mu_{n+1})$, which is 0 or of degree at most $n-1$ by (a).
- The constant term of a polynomial is its value at 0, here $(0-\mu_1)\cdots(0-\mu_n) = (-1)^n \mu_1 \cdots \mu_n$ by (b). □

The characteristic polynomial

Lemma 7.6 (cf. [4, Ch. VIII, §2])**.** Let M be an $m \times m$ matrix whose entries have the form $\alpha + \beta t$ with $\alpha, \beta \in \mathbb{R}$, and let $\det M$ be computed by cofactor expansion (Theorem 5.21) with t treated as a number. If at most d rows of M contain an entry with $\beta \neq 0$, then $\det M$ is 0 or a real polynomial of degree at most d, and its value at every $c \in \mathbb{R}$ is $\det M(c)$, the determinant of the real matrix $M(c)$ given by $t = c$.

Proof. Induction on m. For $m = 1$, $\det M = \alpha + \beta t$ (Definition 5.13(a),(d)), which is 0 or of degree at most d, since $\beta = 0$ if $d = 0$; its value at c is $\alpha + \beta c = \det M(c)$. Let $m \geqslant 2$.

- If no entry has $\beta \neq 0$, then $M = M(c)$ for every c, and $\det M$ is the number $\det M(c)$.
- Otherwise $d \geqslant 1$; let row i contain an entry with $\beta \neq 0$, write $m_{ij} = \alpha_{ij} + \beta_{ij} t$ for its entries, and expand along row i (Theorem 5.21):

$$\det M = \sum_{j=1}^{m} (-1)^{i+j} m_{ij} \det\Big(M(i|j)\Big)$$

- Each $M(i|j)$ is an $(m-1) \times (m-1)$ matrix with entries of the same form, and its rows are the rows of M other than row i, each shortened by one entry; so at most $d-1$ of them contain an

entry with $\beta \neq 0$. By the induction hypothesis, $q_j = \det\big(M(i|j)\big)$ is 0 or a real polynomial of degree at most $d-1$, and its value at c is $\det\big(M(c)(i|j)\big)$.

- Each term $(-1)^{i+j}(\alpha_{ij}+\beta_{ij}t)q_j$ is 0 or of degree at most $1+(d-1)=d$ (Proposition 7.5(a)), so its coefficients of t^k, $k>d$, are 0, and so are those of $\det M$ (Definition 7.4). By Proposition 7.5(b), the value of $\det M$ at c is $\sum_j (-1)^{i+j}(\alpha_{ij}+\beta_{ij}c)\det\big(M(c)(i|j)\big) = \det M(c)$ (Theorem 5.21). □

Definition 7.7 ([4, Ch. VIII, §2], [1, §6.2], [5, §21.1, (21.1.3)–(21.1.8)], [11, Five.II.3, Definition 3.9])**.** Let $A=(a_{ij}) \in \mathbb{R}^{n\times n}$. The **characteristic polynomial** of A is

$$p_A(t) = \det(tI - A)$$

computed by cofactor expansion with t treated as a number. The (i,j) entry of $tI-A$ is $\delta_{ij}t - a_{ij}$, so p_A is a real polynomial (Lemma 7.6 with $m=d=n$). The equation $p_A(t)=0$ is the **characteristic equation** of A. Moreover:

(a) $p_A(c) = \det(cI-A)$ for every $c\in\mathbb{R}$; so the eigenvalues of A are the real roots of p_A (Theorem 7.3);

(b) p_A is monic of degree n, its coefficient of t^{n-1} is $-\operatorname{tr} A$, and its constant term is $(-1)^n \det A$ (for $n=1$ both say $p_A(t)=t-a_{11}$). For $n=2$, $p_A(t) = t^2 - (\operatorname{tr} A)t + \det A$.

Proof. (a) By Lemma 7.6, $p_A(c) = \det(cI-A)$ for every real c. By Theorem 7.3, a real λ is an eigenvalue of A if and only if $\det(\lambda I - A)=0$, that is, $p_A(\lambda)=0$.

(b) Induction on n. For $n=1$, $p_A = \det(tI_1 - A) = t - a_{11}$ (Definition 5.13(a),(d)), and $-a_{11} = -\operatorname{tr} A = (-1)^1 \det A$. Let $n \geqslant 2$.

- Expand $\det(tI-A)$ along row 1, whose entries are $t-a_{11}, -a_{12}, \ldots, -a_{1n}$ (Theorem 5.21); as $(tI-A)(1|1) = tI_{n-1} - A(1|1)$,

$$p_A = (t-a_{11})\, p_{A(1|1)} + \sum_{j=2}^{n} (-1)^{1+j}(-a_{1j}) \det\Big((tI-A)(1|j)\Big)$$

- For $j \geqslant 2$, the rows of $(tI-A)(1|j)$ are rows $2,\ldots,n$ of $tI-A$ without their jth entry. Row i of $tI-A$ contains t only in column i, so the row that comes from row j does not contain t, and at most $n-2$ rows contain t. By Lemma 7.6, $q_j = \det\Big((tI-A)(1|j)\Big)$ is 0 or a real polynomial of degree at most $n-2$.

- By the induction hypothesis, $p_{A(1|1)} = t^{n-1} - st^{n-2} + r$ with $s = \operatorname{tr} A(1|1) = a_{22} + \cdots + a_{nn}$ and $r=0$ or $\deg r \leqslant n-3$ ($r=0$ if $n=2$). Multiplying out, $(t-a_{11})p_{A(1|1)} = t^n - (a_{11}+s)t^{n-1} + r_1$ with $r_1 = a_{11}st^{n-2} + (t-a_{11})r$, which is 0 or of degree at most $n-2$ (Proposition 7.5(a)); and $a_{11}+s = \operatorname{tr} A$ (Definition 1.10).

- So $p_A = (t-a_{11})p_{A(1|1)} + \sum_{j=2}^n (-1)^{1+j}(-a_{1j})q_j$ (Definition 7.4). The coefficients of t^k, $k \geqslant n-1$, of r_1 and of every q_j are 0; so p_A has the coefficient 1 at t^n, the coefficient $-\operatorname{tr} A$ at t^{n-1}, and 0 above t^n.

- The constant term is the value at 0: by (a), $p_A(0) = \det(-A) = (-1)^n \det A$ (Theorem 5.18(c)). □

In examples, $\det(tI-A)$ is also computed by row and column operations with t treated as a number (Definition 5.13, Theorem 5.22); the result is p_A, since its value at every real t is $\det(tI-A)$, by (a) (Proposition 7.5(d)). Many books call $\det(A-tI) = (-1)^n p_A(t)$ the characteristic polynomial; it has the same roots. By (a), (b) and Proposition 7.5(d), a matrix $A\in\mathbb{R}^{n\times n}$ has at most n eigenvalues; so the four eigenvalues 1, −1, 2, −2 of the 4×4 matrix of Example 7.2 are all its eigenvalues.

Example 7.8 ([11, Five.II.3, Examples 3.7, 3.14])**.** Find the eigenvalues of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 0 & -2 \\ -1 & 2 & 3 \end{pmatrix}$$

and a basis of each eigenspace.

Solution.

- For $t \in \mathbb{R}$, expanding along row 1 (Theorem 5.21),

$$\det(tI - A) = \det \begin{pmatrix} t-1 & -2 & -1 \\ -2 & t & 2 \\ 1 & -2 & t-3 \end{pmatrix} = (t-1)\Big(t(t-3)+4\Big) + 2\Big(-2(t-3)-2\Big) - (4-t)$$
$$= (t^3 - 4t^2 + 7t - 4) + (-4t + 8) + (t - 4) = t^3 - 4t^2 + 4t = t(t-2)^2$$

  So $p_A(t) = t(t-2)^2$, and the eigenvalues are 0 and 2 (Definition 7.7).

- $\lambda = 0$: row-reduce $A - 0I = A$.

$$A \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -4 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -4 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow -\frac{1}{4}R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

  The free unknown is x_3; the special solution $(1, -1, 1)'$ is a basis of $\mathcal{N}(A)$ (Theorem 4.8(c)).

- $\lambda = 2$: row-reduce $A - 2I$.

$$\begin{pmatrix} -1 & 2 & 1 \\ 2 & -2 & -2 \\ -1 & 2 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + 2R_1} \begin{pmatrix} -1 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow \frac{1}{2}R_2]{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(1, 0, 1)'$ is a basis of $\mathcal{N}(A - 2I)$.

Both eigenspaces are lines, although p_A has the factor $(t-2)^2$.

Theorem 7.9 ([1, §4.5, Theorem 9], [4, Ch. IX, §1], [11, Five.I.1])**.** (a) (**Fundamental theorem of algebra**; not proved here.) Every polynomial of degree at least 1 has a root in $\mathbb{C}$.

(b) Every monic polynomial f of degree $n \geqslant 1$ is a product

$$f = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n) \qquad (\lambda_1, \ldots, \lambda_n \in \mathbb{C})$$

and the list $\lambda_1, \ldots, \lambda_n$ is unique up to order: if also $f = (t - \mu_1) \cdots (t - \mu_n)$, then $\mu_1, \ldots, \mu_n$ are $\lambda_1, \ldots, \lambda_n$ in some order.

(c) In (b), the roots of f are exactly $\lambda_1, \ldots, \lambda_n$.

Proof. (b)

- Existence, by induction on n. For $n = 1$, $f = t + c_0 = t - (-c_0)$. Let $n \geqslant 2$. By (a), f has a root λ_n, so $f = (t - \lambda_n)g$ (Proposition 7.5(c)); here $g \neq 0$, as $f \neq 0$, so g is monic of degree $n - 1$ (Proposition 7.5(a)). By the induction hypothesis, $g = (t - \lambda_1) \cdots (t - \lambda_{n-1})$.
- Uniqueness, by induction on n. For $n = 1$, $t - \lambda_1 = t - \mu_1$ gives $\lambda_1 = \mu_1$ (constant terms). Let $n \geqslant 2$. The value of both products at λ_n (Proposition 7.5(b)) gives $0 = (\lambda_n - \mu_1) \cdots (\lambda_n - \mu_n)$, so $\lambda_n = \mu_k$ for some k.

- Renumber the μ_i so that $k = n$, and cancel the factor $t - \lambda_n \neq 0$ (Proposition 7.5(a)): $(t - \lambda_1)\cdots(t-\lambda_{n-1}) = (t-\mu_1)\cdots(t-\mu_{n-1})$. By the induction hypothesis, $\mu_1, \ldots, \mu_{n-1}$ are $\lambda_1, \ldots, \lambda_{n-1}$ in some order, and $\mu_n = \lambda_n$.

(c) By Proposition 7.5(b), $f(z) = (z - \lambda_1)\cdots(z - \lambda_n)$, which is 0 if and only if $z = \lambda_i$ for some i. □

Definition 7.10 ([3, §3.1], [11, Five.II.3, Definition 3.17], [1, §6.2, Example 1], [5, §21.1])**.** Let $A \in \mathbb{R}^{n\times n}$. By Definition 7.7(b) and Theorem 7.9(b),

$$p_A(t) = (t - \lambda_1)(t - \lambda_2)\cdots(t - \lambda_n)$$

for complex numbers $\lambda_1, \ldots, \lambda_n$, unique up to order. These n numbers are the **eigenvalues** of A, listed **with multiplicity**; by Theorem 7.9(c) they are the roots of p_A. For $c \in \mathbb{C}$, the **algebraic multiplicity** $a_A(c)$ is the number of indices i with $\lambda_i = c$ ($a_A(c) = 0$ if c is not an eigenvalue); so if $c_1, \ldots, c_k$ are the distinct eigenvalues, then $p_A = (t - c_1)^{a_A(c_1)}\cdots(t - c_k)^{a_A(c_k)}$ and $a_A(c_1) + \cdots + a_A(c_k) = n$. An eigenvalue c is **simple** if $a_A(c) = 1$ and **multiple** if $a_A(c) \geqslant 2$. A real number is an eigenvalue in this sense if and only if it is one in the sense of Definition 7.1, since both mean that it is a real root of p_A (Definition 7.7(a)); these are the **real eigenvalues**. From now on "eigenvalue" has this meaning; eigenvectors and eigenspaces belong to real eigenvalues only.

Proposition 7.11 ([1, §4.5], [6, Exercise 7.17(b)])**.** Let $A \in \mathbb{R}^{n\times n}$.

(a) If z is a non-real eigenvalue of A, then so is $\bar{z}$, and $a_A(\bar{z}) = a_A(z)$: the non-real eigenvalues come in conjugate pairs.

(b) If n is odd, then A has a real eigenvalue.

Proof. (a)

- For a polynomial $f = \sum_k c_k t^k$ put $\bar{f} = \sum_k \bar{c}_k t^k$. Conjugation respects sums and products of numbers, so the coefficient $\overline{\sum_{i+j=k} c_i d_j}$ of $\overline{fg}$ is $\sum_{i+j=k} \bar{c}_i \bar{d}_j$: $\overline{fg} = \bar{f}\,\bar{g}$. Hence $\overline{(t - \lambda_1)\cdots(t - \lambda_n)} = (t - \bar{\lambda}_1)\cdots(t - \bar{\lambda}_n)$.
- p_A is real, so $p_A = \overline{p_A}$, that is, $(t - \lambda_1)\cdots(t - \lambda_n) = (t - \bar{\lambda}_1)\cdots(t - \bar{\lambda}_n)$ (Definition 7.10). By Theorem 7.9(b), $\bar{\lambda}_1, \ldots, \bar{\lambda}_n$ are $\lambda_1, \ldots, \lambda_n$ in some order.
- So the number of indices i with $\lambda_i = \bar{z}$ equals the number of i with $\bar{\lambda}_i = \bar{z}$, that is, with $\lambda_i = z$: $a_A(\bar{z}) = a_A(z) \geqslant 1$.

(b) By (a), the non-real numbers in the list $\lambda_1, \ldots, \lambda_n$ fall into pairs $z \neq \bar{z}$ that occur equally often, so there is an even number of them. If n is odd, some λ_i is real: a real eigenvalue. □

Example 7.12 (cf. [1, §6.2, Example 1])**.** Find the eigenvalues of

$$A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

and the eigenspace of each real eigenvalue.

Solution.

- For $t \in \mathbb{R}$, column 3 of $tI - A$ has the entry $t - 1$ in position $(3, 3)$ and 0 elsewhere; expanding along it (Theorem 5.21),

$$\det(tI - A) = \det \begin{pmatrix} t-2 & 1 & 0 \\ -1 & t-2 & 0 \\ -1 & -1 & t-1 \end{pmatrix} = (t-1)\det \begin{pmatrix} t-2 & 1 \\ -1 & t-2 \end{pmatrix} = (t-1)\Big((t-2)^2 + 1\Big)$$

- Since $(t-2-i)(t-2+i) = (t-2)^2 - i^2 = (t-2)^2 + 1$, $p_A(t) = (t-1)\big(t-(2+i)\big)\big(t-(2-i)\big)$: the eigenvalues are 1, $2+i$ and $2-i$, each simple, and $2 \pm i$ is a conjugate pair (Proposition 7.11(a)). The only real eigenvalue is 1.
- Row-reduce $A - I$:

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow \frac{1}{2} R_2} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  So $x_1 = x_2 = 0$ and x_3 is free: $\mathcal{N}(A - I) = \operatorname{span}\{e_3\}$. The eigenvalues $2 \pm i$ have no eigenvectors in $\mathbb{R}^3$ (Definition 7.10).

Remark 7.13 (Stationary autoregressions)**.** In time series, the **autoregressive model** of order p is $y_k = \beta_0 + \beta_1 y_{k-1} + \cdots + \beta_p y_{k-p} + \varepsilon_k$. It describes a stationary process when every root z of the real polynomial $\Phi(t) = 1 - \beta_1 t - \cdots - \beta_p t^p$ lies outside the unit circle, that is, $|z| > 1$, where $|a + bi| = \sqrt{a^2 + b^2}$ (not proved here). The recursion $y_k = \beta_0 + \beta_1 y_{k-1} + \cdots + \beta_p y_{k-p}$, written for the vector $(y_{k-p+1}, \ldots, y_k)'$, uses the $p \times p$ **companion matrix**

$$C = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \beta_p & \beta_{p-1} & \beta_{p-2} & \cdots & \beta_1 \end{pmatrix}$$

whose eigenvalues are the roots of $f = t^p - \beta_1 t^{p-1} - \cdots - \beta_p$ [6, Appendix A.4.1]. As $\Phi(0) = 1$ and $z^p \Phi(1/z) = f(z)$ for $z \neq 0$, the nonzero roots of f are the numbers $1/z$ for the roots z of Φ, and $|1/z| = 1/|z|$ (as $1/z = \bar{z}/(a^2 + b^2)$ for $z = a + bi \neq 0$); so the condition says that every eigenvalue λ of C, real or not, has $|\lambda| < 1$.

Similar and triangular matrices

Definition 7.14 ([1, §3.4], [11, Five.II.1, Definition 1.2])**.** A matrix $B \in \mathbb{R}^{n \times n}$ is **similar** to $A \in \mathbb{R}^{n \times n}$ if $B = P^{-1}AP$ for some invertible $P \in \mathbb{R}^{n \times n}$. Then $A = PBP^{-1} = (P^{-1})^{-1} B P^{-1}$, so A is similar to B.

Theorem 7.15 ([1, §6.2], [5, Theorem 21.3.1])**.** Let $A, P \in \mathbb{R}^{n \times n}$ with P invertible, and let $B = P^{-1}AP$.

(a) $p_B = p_A$. Hence A and B have the same eigenvalues with the same algebraic multiplicities, $\operatorname{tr} B = \operatorname{tr} A$ and $\det B = \det A$.

(b) For every $c \in \mathbb{R}$, $x \mapsto P^{-1}x$ maps $\mathcal{N}(A - cI)$ onto $\mathcal{N}(B - cI)$, with inverse $y \mapsto Py$; so x is an eigenvector of A belonging to c if and only if $P^{-1}x$ is an eigenvector of B belonging to c. Moreover $\dim \mathcal{N}(B - cI) = \dim \mathcal{N}(A - cI)$, and $\operatorname{rank} B = \operatorname{rank} A$.

Proof. (a)

- For $c \in \mathbb{R}$, $cI - B = P^{-1}(cI)P - P^{-1}AP = P^{-1}(cI - A)P$ (Theorem 1.3), so $\det(cI - B) = \det P^{-1} \cdot \det(cI - A) \cdot \det P = \det(cI - A)$ (Theorem 5.18(b),(c)).
- So $p_B(c) = p_A(c)$ for every real c (Definition 7.7(a)), and $p_B = p_A$ (Proposition 7.5(d)).

- The eigenvalues of A and B come from the same factorization (Definition 7.10); $-\operatorname{tr} B$ and $-\operatorname{tr} A$ are the same coefficient of t^{n-1}, and $(-1)^n \det B$, $(-1)^n \det A$ the same constant term (Definition 7.7(b)).

(b)

- $B - cI = P^{-1}(A - cI)P$ and $A - cI = P(B - cI)P^{-1}$. If $(A - cI)x = 0$, then $(B - cI)P^{-1}x = P^{-1}(A - cI)x = 0$; if $(B - cI)y = 0$, then $(A - cI)Py = P(B - cI)y = 0$.
- The two maps are inverse to each other ($PP^{-1}x = x$, $P^{-1}Py = y$), and $P^{-1}x \neq 0$ if and only if $x \neq 0$.
- $\operatorname{rank}(B - cI) = \operatorname{rank}\Big(P^{-1}(A - cI)P\Big) = \operatorname{rank}(A - cI)$ (Theorem 4.15(d)), so the two null spaces have the same dimension (Theorem 4.12). In the same way, $\operatorname{rank} B = \operatorname{rank}(P^{-1}AP) = \operatorname{rank} A$. □

Proposition 7.16 ([1, §6.4], [5, Lemma 21.2.1, Corollary 21.2.2])**.** (a) Let $A = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix}$ or $A = \begin{pmatrix} B & 0 \\ C & D \end{pmatrix}$ with $B \in \mathbb{R}^{p\times p}$ and $D \in \mathbb{R}^{q\times q}$. Then $p_A = p_B p_D$: the eigenvalues of A, listed with multiplicity, are those of B together with those of D.

(b) If $A \in \mathbb{R}^{n\times n}$ is upper or lower triangular, then $p_A = (t - a_{11}) \cdots (t - a_{nn})$: the eigenvalues of A, listed with multiplicity, are $a_{11}, \ldots, a_{nn}$.

Proof. (a)

- For $c \in \mathbb{R}$, $cI - A$ is block triangular of the same kind, with diagonal blocks $cI_p - B$ and $cI_q - D$ (Theorem 5.3(a)). By Theorem 5.29, $\det(cI - A) = \det(cI_p - B)\det(cI_q - D) = p_B(c)p_D(c)$, the value of $p_B p_D$ at c (Proposition 7.5(b)). So $p_A = p_B p_D$ (Proposition 7.5(d)).
- If $p_B = (t - \beta_1) \cdots (t - \beta_p)$ and $p_D = (t - \delta_1) \cdots (t - \delta_q)$ (Definition 7.10), then p_A is the product of these $p + q$ linear factors, and by Theorem 7.9(b) the eigenvalues of A are $\beta_1, \ldots, \beta_p, \delta_1, \ldots, \delta_q$.

(b)

- For $c \in \mathbb{R}$, $cI - A$ is triangular of the same kind with diagonal entries $c - a_{ii}$, so $\det(cI - A) = (c - a_{11}) \cdots (c - a_{nn})$ (Theorem 5.14(c) for upper, Theorem 5.22 for lower triangular matrices), the value of $(t - a_{11}) \cdots (t - a_{nn})$ at c.
- As in (a), $p_A = (t - a_{11}) \cdots (t - a_{nn})$, and Definition 7.10 gives the eigenvalues. □

Example 7.17. Find the eigenvalues of

$$A = \begin{pmatrix} -1 & 2 & 3 & -1 \\ 2 & -1 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} B & F \\ 0 & D \end{pmatrix}, \qquad B = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

and a basis of each eigenspace.

Solution.

- By Proposition 7.16(a) and the formula for $n = 2$ in Definition 7.7(b), with $\operatorname{tr} B = -2$, $\det B = -3$, $\operatorname{tr} D = 2$ and $\det D = -3$,

$$p_A = p_B p_D = (t^2 + 2t - 3)(t^2 - 2t - 3) = (t + 3)(t - 1)(t - 3)(t + 1)$$

  So the eigenvalues are -3, -1, 1 and 3, each simple (Definition 7.10). Check: $p_A = (t^2 - 9)(t^2 - 1) = t^4 - 10t^2 + 9$; the coefficient 0 of t^3 is $-\operatorname{tr} A = -(-1 - 1 + 1 + 1)$, and the constant term 9 is $(-1)^4 \det A$, as $\det A = \det B \det D = (-3)(-3) = 9$ (Theorem 5.29, Definition 7.7(b)).

- $\lambda = -3$:

$$A+3I = \begin{pmatrix} 2 & 2 & 3 & -1 \\ 2 & 2 & 1 & 1 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 2 & 2 & 3 & -1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} 2 & 2 & 3 & -1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 6 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 2 & 2 & 3 & -1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  An echelon form with three nonzero rows: $\operatorname{rank}(A+3I) = 3$ (Definition 4.11), so $\dim \mathcal{N}(A+3I) = 4-3 = 1$ (Theorem 4.12). Back substitution gives $x_4 = 0$, $x_3 = 0$ and $x_1 = -x_2$; so $(-1,1,0,0)'$ is a basis of $\mathcal{N}(A+3I)$ (Theorem 4.5(d)).

- $\lambda = -1$:

$$A+I = \begin{pmatrix} 0 & 2 & 3 & -1 \\ 2 & 0 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_3]{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & 3 & -1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  Again $\operatorname{rank}(A+I) = 3$ and $\dim \mathcal{N}(A+I) = 1$. With $x_4 = 1$, back substitution gives $x_3 = -1$, $x_2 = 2$ and $x_1 = 0$; so $(0,2,-1,1)'$ is a basis of $\mathcal{N}(A+I)$.

- $\lambda = 1$:

$$A-I = \begin{pmatrix} -2 & 2 & 3 & -1 \\ 2 & -2 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} -2 & 2 & 3 & -1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} -2 & 2 & 3 & -1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 4 & 0 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 - 2R_2} \begin{pmatrix} -2 & 2 & 3 & -1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  So $\operatorname{rank}(A-I) = 3$, $\dim \mathcal{N}(A-I) = 1$, and back substitution gives $x_3 = x_4 = 0$ and $x_1 = x_2$: the vector $(1,1,0,0)'$ is a basis of $\mathcal{N}(A-I)$.

- $\lambda = 3$:

$$A-3I = \begin{pmatrix} -4 & 2 & 3 & -1 \\ 2 & -4 & 1 & 1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 2 & -2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & -4 & 1 & 1 \\ -4 & 2 & 3 & -1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 2 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_3]{R_2 \leftarrow R_2 + 2R_1} \begin{pmatrix} 2 & -4 & 1 & 1 \\ 0 & -6 & 5 & 1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  So $\operatorname{rank}(A-3I) = 3$, $\dim \mathcal{N}(A-3I) = 1$, and back substitution gives $x_3 = x_4$, $x_2 = x_4$ and $x_1 = x_4$: the vector $\mathbf{1}_4$ is a basis of $\mathcal{N}(A-3I)$. Indeed every row of A adds up to 3.

Check: $A(-1,1,0,0)' = (3,-3,0,0)'$ and $A(0,2,-1,1)' = (0,-2,1,-1)'$.

Eigenvalues of related matrices; trace and determinant

Proposition 7.18 ([5, Lemmas 21.1.2, 21.1.3, §21.10], [2, §§6.2.2, 6.2.3], [1, §6.2])**.** Let $A \in \mathbb{R}^{n \times n}$.

(a) If $Ax = \lambda x$ with $\lambda \in \mathbb{R}$ and $x \neq 0$, then $A^k x = \lambda^k x$ $(k = 0, 1, 2, \ldots)$ and $f(A)x = f(\lambda)x$ for every real polynomial f; in particular $(cA)x = (c\lambda)x$ and $(A + cI)x = (\lambda + c)x$ $(c \in \mathbb{R})$. So x is an eigenvector of A^k, $f(A)$, cA and $A + cI$, belonging to λ^k, $f(\lambda)$, $c\lambda$ and $\lambda + c$.

(b) If, moreover, A is invertible, then $\lambda \neq 0$ and $A^{-1}x = \lambda^{-1}x$.

(c) If A has the eigenvalues $\lambda_1, \dots, \lambda_n$ (with multiplicity) and $c \in \mathbb{R}$, then $A + cI$ has the eigenvalues $\lambda_1 + c, \dots, \lambda_n + c$, and, if $c \neq 0$, cA has the eigenvalues $c\lambda_1, \dots, c\lambda_n$.

(d) $p_{A'} = p_A$: A' and A have the same eigenvalues with the same algebraic multiplicities.

Proof. (a)

- By induction on k: $A^0x = x = \lambda^0 x$, and if $A^k x = \lambda^k x$, then $A^{k+1}x = A^k(Ax) = A^k(\lambda x) = \lambda A^k x = \lambda^{k+1}x$ (Theorem 1.3).
- If $f = \sum_k c_k t^k$, then $f(A)x = \sum_k c_k A^k x = \sum_k c_k \lambda^k x = f(\lambda)x$. For $f = ct$ and $f = t + c$ this gives $(cA)x = c\lambda x$ and $(A + cI)x = (\lambda + c)x$. In each case $x \neq 0$.

(b) If $\lambda = 0$, then $Ax = 0$ with $x \neq 0$, and A would not be invertible (Theorem 2.16) ✘ So $\lambda \neq 0$, and $x = A^{-1}(Ax) = A^{-1}(\lambda x) = \lambda A^{-1}x$, that is, $A^{-1}x = \lambda^{-1}x$.

(c)

- For $s \in \mathbb{R}$, $\det\big(sI - (A + cI)\big) = \det\big((s - c)I - A\big) = p_A(s - c) = (s - c - \lambda_1) \cdots (s - c - \lambda_n)$ (Definition 7.7(a), Proposition 7.5(b)), the value at s of $\big(t - (\lambda_1 + c)\big) \cdots \big(t - (\lambda_n + c)\big)$. By Proposition 7.5(d) this product is p_{A+cI}, and Definition 7.10 gives the eigenvalues.
- Let $c \neq 0$. For $s \in \mathbb{R}$, $\det(sI - cA) = \det\big(c(\frac{s}{c}I - A)\big) = c^n p_A(\frac{s}{c}) = c^n(\frac{s}{c} - \lambda_1) \cdots (\frac{s}{c} - \lambda_n) = (s - c\lambda_1) \cdots (s - c\lambda_n)$ (Theorem 5.18(c)); conclude as before.

(d) For $c \in \mathbb{R}$, $(cI - A)' = cI - A'$ (Theorem 1.9), so $\det(cI - A') = \det(cI - A)$ (Theorem 5.22). By Definition 7.7(a) and Proposition 7.5(d), $p_{A'} = p_A$. □

In Example 7.2, by (a), v_1 and v_2 are eigenvectors of A^2 belonging to 1, v_3 and v_4 eigenvectors of A^2 belonging to 4, and $v_1, \dots, v_4$ eigenvectors of $A - 2I$ belonging to -1, -3, 0, -4. The eigenvalues of that A are 1, -1, 2, -2 (the remark after Definition 7.7), so A is invertible (Theorem 7.3 with $\lambda = 0$), and by (b), $v_1, \dots, v_4$ are eigenvectors of A^{-1} belonging to 1, -1, $\frac{1}{2}$, $-\frac{1}{2}$.

Theorem 7.19 ([3, §3.1], [6, Exercise 7.16])**.** Let $A \in \mathbb{R}^{n \times n}$ have the eigenvalues $\lambda_1, \dots, \lambda_n$, listed with multiplicity. Then

$$\operatorname{tr} A = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \qquad \det A = \lambda_1 \lambda_2 \cdots \lambda_n$$

Hence A is invertible if and only if 0 is not an eigenvalue of A.

Proof.

- $p_A = (t - \lambda_1) \cdots (t - \lambda_n)$ (Definition 7.10). By Proposition 7.5(e), its coefficient of t^{n-1} is $-(\lambda_1 + \cdots + \lambda_n)$ and its constant term is $(-1)^n \lambda_1 \cdots \lambda_n$.
- By Definition 7.7(b), the same two coefficients are $-\operatorname{tr} A$ and $(-1)^n \det A$.
- A is invertible if and only if $\det A \neq 0$ (Theorem 5.18(a)), that is, if and only if no λ_i is 0. □

For the matrix of Example 7.8, $\operatorname{tr} A = 4 = 0 + 2 + 2$ and $\det A = 0 = 0 \cdot 2 \cdot 2$, so A is not invertible; for that of Example 7.12, $\operatorname{tr} A = 5 = 1 + (2 + i) + (2 - i)$ and $\det A = 5 = 1 \cdot (2 + i)(2 - i)$.

Eigenvectors of distinct eigenvalues; multiplicities

Theorem 7.20 ([11, Five.II.3, Theorem 3.18, Corollary 3.20], [4, Ch. VIII, §1, Remark 1])**.** Let $v_1, \dots, v_m$ be eigenvectors of $A \in \mathbb{R}^{n\times n}$ belonging to real eigenvalues $\lambda_1, \dots, \lambda_m$ that are pairwise distinct ($\lambda_i \neq \lambda_j$ for $i \neq j$). Then $v_1, \dots, v_m$ are linearly independent. In particular, if A has n distinct real eigenvalues, then eigenvectors belonging to them form a basis of $\mathbb{R}^n$.

Proof. Induction on m. For $m = 0$ the empty list is independent (Definition 3.11). Let $m \geqslant 1$, assume the statement for $m-1$ eigenvectors, and let $c_1v_1 + \cdots + c_mv_m = 0$.

- Multiplying by λ_m gives $\sum_i c_i\lambda_m v_i = 0$; multiplying by A gives $\sum_i c_i\lambda_i v_i = 0$, since $Av_i = \lambda_i v_i$ (Theorem 1.3).
- Subtracting, the v_m term vanishes: $c_1(\lambda_m - \lambda_1)v_1 + \cdots + c_{m-1}(\lambda_m - \lambda_{m-1})v_{m-1} = 0$.
- By the induction hypothesis, $c_i(\lambda_m - \lambda_i) = 0$ for $i < m$, and $c_i = 0$ since $\lambda_m \neq \lambda_i$.
- Then $c_m v_m = 0$ with $v_m \neq 0$, so $c_m = 0$ (Proposition 3.13(a)).

If A has n distinct real eigenvalues, eigenvectors belonging to them are n independent vectors of $\mathbb{R}^n$, which has dimension n (Theorem 4.4); so they form a basis (Theorem 4.5(d)). □

Definition 7.21 ([5, §21.1], [11, Five.II.3, Definition 3.17], [3, §3.2])**.** Let $A \in \mathbb{R}^{n\times n}$ and $c \in \mathbb{R}$. The **geometric multiplicity** of c is

$$g_A(c) = \dim \mathcal{N}(A - cI) = n - \operatorname{rank}(A - cI)$$

(Theorem 4.12). So $g_A(c) \geqslant 1$ if c is a real eigenvalue of A, and $g_A(c) = 0$ otherwise (Definition 7.1, Theorem 4.4).

By Theorem 7.15(a),(b), similar matrices have the same algebraic and geometric multiplicities. In Example 7.8, $g_A(0) = a_A(0) = 1$, but $g_A(2) = 1 < 2 = a_A(2)$.

Theorem 7.22 ([1, §6.4, Example 10], [5, Corollary 21.3.5], [11, Five.II.3])**.** If c is a real eigenvalue of $A \in \mathbb{R}^{n\times n}$, then $1 \leqslant g_A(c) \leqslant a_A(c)$. In particular, if c is simple, then $g_A(c) = 1$, that is, $\operatorname{rank}(A - cI) = n - 1$.

Proof. • $g = g_A(c) \geqslant 1$ (Definition 7.21). Extend a basis $x_1, \dots, x_g$ of $\mathcal{N}(A - cI)$ to a basis $x_1, \dots, x_n$ of $\mathbb{R}^n$ (Theorem 4.5(b)). Then $P = [x_1 \ \cdots \ x_n]$ has independent columns, so it is invertible (Theorem 4.13(b),(d)).

- For $j \leqslant g$, $Ax_j = cx_j$; put $m_{ij} = c\delta_{ij}$. For $j > g$, write $Ax_j = \sum_{i=1}^n m_{ij}x_i$, as $x_1, \dots, x_n$ span $\mathbb{R}^n$. With $M = (m_{ij})$, $\operatorname{col}_j(AP) = Ax_j = \sum_i m_{ij}x_i = P\operatorname{col}_j(M)$ for every j (Theorem 1.4(a),(b)); so $AP = PM$ and $M = P^{-1}AP$.
- The first g columns of M are $ce_1, \dots, ce_g$: $M = cI_n$ if $g = n$, and $M = \left(\begin{smallmatrix} cI_g & C \\ 0 & D \end{smallmatrix}\right)$ with $D \in \mathbb{R}^{(n-g)\times(n-g)}$ if $g < n$. By Theorem 7.15(a) and Proposition 7.16(a),(b), the eigenvalues of A are those of M: c (g times), together with those of D if $g < n$. Hence $a_A(c) \geqslant g$.
- If c is simple, then $1 \leqslant g_A(c) \leqslant a_A(c) = 1$, and $\operatorname{rank}(A - cI) = n - g_A(c) = n - 1$. □

Diagonalization

Definition 7.23 ([11, Five.II.2, Definition 2.1, Lemma 2.4], [1, §6.2], [5, §21.5, Theorems 21.5.1, 21.5.2])**.** A matrix $A \in \mathbb{R}^{n\times n}$ is **diagonalizable** if $P^{-1}AP$ is diagonal for some invertible $P \in \mathbb{R}^{n\times n}$, that is, if A is similar to a diagonal matrix; P is then said to **diagonalize** A. Let $P = [p_1 \ \cdots \ p_n] \in \mathbb{R}^{n\times n}$ be invertible and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_j \in \mathbb{R}$. Then

$$P^{-1}AP = \Lambda \iff AP = P\Lambda \iff A = P\Lambda P^{-1} \iff Ap_j = \lambda_j p_j \quad (1 \leqslant j \leqslant n)$$

Consequently A is diagonalizable if and only if $\mathbb{R}^n$ has a basis consisting of eigenvectors of A. If $P^{-1}AP = \Lambda$, then p_j is an eigenvector belonging to λ_j, and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A, listed with multiplicity; in particular all eigenvalues of a diagonalizable matrix are real.

Proof. • If $P^{-1}AP = \Lambda$, then $AP = P(P^{-1}AP) = P\Lambda$ and $A = P\Lambda P^{-1}$; if $AP = P\Lambda$ or $A = P\Lambda P^{-1}$, then $P^{-1}AP = \Lambda$.

- $\operatorname{col}_j(AP) = Ap_j$ and $\operatorname{col}_j(P\Lambda) = P(\lambda_j e_j) = \lambda_j p_j$ (Theorem 1.4(a),(b)); two matrices are equal if and only if their columns are.
- If $P^{-1}AP = \Lambda$, the columns of P form a basis of $\mathbb{R}^n$ (Example 3.19), so they are nonzero (Proposition 3.13(a)) and, as $Ap_j = \lambda_j p_j$, eigenvectors. Conversely, if eigenvectors $p_1, \ldots, p_n$ with $Ap_j = \lambda_j p_j$ form a basis of $\mathbb{R}^n$, then $P = [p_1 \ \cdots \ p_n]$ is invertible (Theorem 4.13(b),(d)) and $P^{-1}AP = \Lambda$.
- By Theorem 7.15(a) and Proposition 7.16(b), $p_A = p_\Lambda = (t - \lambda_1)\cdots(t - \lambda_n)$; so $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A (Definition 7.10). □

If $P^{-1}AP = \Lambda$ and $c = P^{-1}x$ is the vector of coordinates of $x \in \mathbb{R}^n$ in the basis $p_1, \ldots, p_n$ (Example 3.19), then the coordinates of Ax are $P^{-1}Ax = \Lambda P^{-1}x = \Lambda c = (\lambda_1 c_1, \ldots, \lambda_n c_n)'$ (as $P^{-1}A = \Lambda P^{-1}$; Proposition 1.13(a)): in a basis of eigenvectors, A multiplies the jth coordinate by λ_j [1, §6.2].

Lemma 7.24 ([1, §6.2])**.** Let $c_1, \ldots, c_k$ $(k \geqslant 1)$ be distinct real eigenvalues of $A \in \mathbb{R}^{n\times n}$, and $W_i = \mathcal{N}(A - c_i I)$.

(a) If $w_i \in W_i$ $(1 \leqslant i \leqslant k)$ and $w_1 + \cdots + w_k = 0$, then $w_i = 0$ for every i.

(b) If $v_{i1}, \ldots, v_{id_i}$ is a basis of W_i for each i, then the $d_1 + \cdots + d_k$ vectors $v_{11}, \ldots, v_{1d_1}, \ldots, v_{k1}, \ldots, v_{kd_k}$ are linearly independent.

Proof. (a)

- For a real polynomial f and $w \in W_i$, $f(A)w = f(c_i)w$ (Proposition 7.18(a) if $w \neq 0$; both sides are 0 if $w = 0$). Applying $f(A)$ to $w_1 + \cdots + w_k = 0$ gives $f(c_1)w_1 + \cdots + f(c_k)w_k = 0$ (Theorem 1.3).
- For a fixed i, let $f_i = \prod_{j\neq i} \frac{t-c_j}{c_i-c_j}$, the product over $j \in \{1, \ldots, k\}$, $j \neq i$ ($f_1 = 1$ if $k = 1$). By Proposition 7.5(b), $f_i(c_l) = \prod_{j\neq i} \frac{c_l-c_j}{c_i-c_j}$, which is 1 for $l = i$ and 0 for $l \neq i$ (the factor $j = l$ vanishes).
- With $f = f_i$, the relation of the first item reads $w_i = 0$.

(b) A linear relation among these vectors has the form $w_1 + \cdots + w_k = 0$ with $w_i = \sum_l a_{il} v_{il} \in W_i$. By (a), every $w_i = 0$, and then every $a_{il} = 0$, since $v_{i1}, \ldots, v_{id_i}$ are independent. □

Theorem 7.25 ([1, §6.2, Theorem 2, Exercise 7], [2, Theorem 7.1], [4, Ch. VIII, §1, Remark 1], [11, Five.II.3, Corollary 3.20])**.** Let $A \in \mathbb{R}^{n\times n}$, let $c_1, \dots, c_k$ be the distinct real eigenvalues of A ($k \geqslant 0$; an empty sum is 0, an empty product 1), and let $W_i = \mathcal{N}(A - c_i I)$. The following are equivalent:

(i) A is diagonalizable;

(ii) $p_A = (t - c_1)^{d_1} \cdots (t - c_k)^{d_k}$ for some positive integers $d_1, \dots, d_k$, and $\dim W_i = d_i$ for each i;

(iii) $\dim W_1 + \cdots + \dim W_k = n$.

In other words, A is diagonalizable if and only if all its eigenvalues are real and $g_A(c) = a_A(c)$ for each of them, that is, $\operatorname{rank}(A - c_i I) = n - a_A(c_i)$ for each i. In particular, an $n \times n$ matrix with n distinct real eigenvalues is diagonalizable.

Proof.
- (i)⇒(ii): let $P^{-1}AP = \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. By Definition 7.23, $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A; they are real, so each is one of $c_1, \dots, c_k$. Let d_i be the number of j with $\lambda_j = c_i$; $d_i \geqslant 1$, as c_i is an eigenvalue. Then $p_A = (t - \lambda_1) \cdots (t - \lambda_n) = (t - c_1)^{d_1} \cdots (t - c_k)^{d_k}$.
- Still (i)⇒(ii): $\dim W_i = \dim \mathcal{N}(\Lambda - c_i I)$ (Theorem 7.15(b)). The jth entry of $(\Lambda - c_i I)x$ is $(\lambda_j - c_i)x_j$, so $\mathcal{N}(\Lambda - c_i I)$ consists of the x with $x_j = 0$ whenever $\lambda_j \neq c_i$; it has the basis $\{e_j : \lambda_j = c_i\}$ of d_i vectors (Definition 3.17, Proposition 3.13(b)).
- (ii)⇒(iii): $\dim W_1 + \cdots + \dim W_k = d_1 + \cdots + d_k = \deg p_A = n$ (Proposition 7.5(a), Definition 7.7(b)).
- (iii)⇒(i): as $n \geqslant 1$, $k \geqslant 1$. By Lemma 7.24(b), bases of $W_1, \dots, W_k$ together give n independent eigenvectors of A; they form a basis of $\mathbb{R}^n$ (Theorem 4.5(d)), and A is diagonalizable (Definition 7.23).
- In (ii), $d_i = a_A(c_i)$ (Theorem 7.9(b)), and p_A has no other roots (Theorem 7.9(c)); so (ii) says that all eigenvalues are real and $g_A(c_i) = a_A(c_i)$. Conversely, if all eigenvalues are real, then $p_A = (t - c_1)^{a_A(c_1)} \cdots (t - c_k)^{a_A(c_k)}$ (Definition 7.10), and (ii) holds when $g_A(c_i) = a_A(c_i)$ for each i. By Definition 7.21, $g_A(c_i) = a_A(c_i)$ means $\operatorname{rank}(A - c_i I) = n - a_A(c_i)$.
- If A has n distinct real eigenvalues, eigenvectors belonging to them form a basis of $\mathbb{R}^n$ (Theorem 7.20), so A is diagonalizable (Definition 7.23). □

For example, the matrix of Example 7.8 is not diagonalizable: its eigenvalues 0, 2, 2 are real, but $g_A(2) = 1 < 2 = a_A(2)$. Nor is the matrix of Example 7.12, which has non-real eigenvalues. The matrix A of Example 7.17 has the four distinct real eigenvalues -3, -1, 1, 3, so it is diagonalizable. The eigenvectors found there are independent (Theorem 7.20), so

$$P = \begin{pmatrix} -1 & 0 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

is invertible (Theorem 4.13(b),(d)), and $P^{-1}AP = \Lambda = \operatorname{diag}(-3, -1, 1, 3)$ (Definition 7.23).

To diagonalize $A \in \mathbb{R}^{n\times n}$: find the eigenvalues, the roots of p_A; if one of them is not real, A is not diagonalizable (Definition 7.23). For each real eigenvalue c, the special solutions of $(A - cI)x = 0$ form a basis of $\mathcal{N}(A - cI)$ (Theorem 4.8(c)), and $g_A(c)$ is their number, the number of free unknowns. If $g_A(c) = a_A(c)$ for every eigenvalue c, put these bases into the columns of P and the corresponding eigenvalues, in the same order, on the diagonal of Λ; then $P^{-1}AP = \Lambda$ (Lemma 7.24(b), Theorem 4.5(d), Definition 7.23). By Theorem 7.22, only the eigenvalues with $a_A(c) \geqslant 2$ need to be checked.

Example 7.26 ([1, §6.2, Example 3])**.** Show that

$$A = \begin{pmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{pmatrix}$$

is diagonalizable, and find an invertible P and a diagonal Λ with $P^{-1}AP = \Lambda$.

Solution.

- For $t \in \mathbb{R}$, subtract column 3 from column 2 and take the factor $t-2$ out of column 2 (Definition 5.13(b),(a) for columns, Theorem 5.22):

$$\det(tI-A) = \det\begin{pmatrix} t-5 & 6 & 6 \\ 1 & t-4 & -2 \\ -3 & 6 & t+4 \end{pmatrix} = \det\begin{pmatrix} t-5 & 0 & 6 \\ 1 & t-2 & -2 \\ -3 & 2-t & t+4 \end{pmatrix}$$
$$= (t-2)\det\begin{pmatrix} t-5 & 0 & 6 \\ 1 & 1 & -2 \\ -3 & -1 & t+4 \end{pmatrix}$$

  Add row 2 to row 3 (Definition 5.13(b)) and expand along column 2, whose only nonzero entry is the $(2,2)$ entry 1 (Theorem 5.21):

$$\det(tI-A) = (t-2)\det\begin{pmatrix} t-5 & 0 & 6 \\ 1 & 1 & -2 \\ -2 & 0 & t+2 \end{pmatrix} = (t-2)\det\begin{pmatrix} t-5 & 6 \\ -2 & t+2 \end{pmatrix} = (t-2)(t^2-3t+2)$$

  As $t^2-3t+2 = (t-1)(t-2)$, $p_A = (t-1)(t-2)^2$: the eigenvalues are 1, 2, 2, with $a_A(1) = 1$ and $a_A(2) = 2$.

- $\lambda = 1$:

$$A-I = \begin{pmatrix} 4 & -6 & -6 \\ -1 & 3 & 2 \\ 3 & -6 & -5 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} -1 & 3 & 2 \\ 4 & -6 & -6 \\ 3 & -6 & -5 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3+3R_1]{R_2 \leftarrow R_2+4R_1} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 6 & 2 \\ 0 & 3 & 1 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 6 & 2 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3-2R_2} \begin{pmatrix} -1 & 3 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

  So $\operatorname{rank}(A-I) = 2$ (Definition 4.11) and $g_A(1) = 1$ (Definition 7.21). The equations $-x_1+3x_2+2x_3 = 0$, $3x_2+x_3 = 0$ with $x_3 = 3$ give $x_2 = -1$ and $x_1 = 3$: $\alpha_1 = (3,-1,3)'$ is a basis of $\mathcal{N}(A-I)$ (Theorem 4.5(d)).

- $\lambda = 2$:

$$A-2I = \begin{pmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} -1 & 2 & 2 \\ 3 & -6 & -6 \\ 3 & -6 & -6 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3+3R_1]{R_2 \leftarrow R_2+3R_1} \begin{pmatrix} -1 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & -2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  So $g_A(2) = 3-1 = 2$, and the special solutions $\alpha_2 = (2,1,0)'$, $\alpha_3 = (2,0,1)'$ form a basis of $\mathcal{N}(A-2I)$ (Theorem 4.8(c)).

- $g_A(1)+g_A(2) = 3$, so A is diagonalizable (Theorem 7.25), and $\alpha_1, \alpha_2, \alpha_3$ form a basis of $\mathbb{R}^3$ (Lemma 7.24(b), Theorem 4.5(d)). With

$$P = [\alpha_1\ \alpha_2\ \alpha_3] = \begin{pmatrix} 3 & 2 & 2 \\ -1 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(1,2,2), \qquad AP = \begin{pmatrix} 3 & 4 & 4 \\ -1 & 2 & 0 \\ 3 & 0 & 2 \end{pmatrix} = P\Lambda$$

  we have $P^{-1}AP = \Lambda$ (Definition 7.23).

Powers and rank of a diagonalizable matrix

Proposition 7.27 ([2, §6.4.2])**.** Let $A = P\Lambda P^{-1}$ with $P = [p_1 \ \cdots \ p_n] \in \mathbb{R}^{n\times n}$ invertible and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, $\lambda_i \in \mathbb{R}$.

(a) $A^k = P\Lambda^k P^{-1} = P\operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)P^{-1}$ for $k = 0, 1, 2, \ldots$; hence, for $x \in \mathbb{R}^n$ and $c = P^{-1}x$,

$$A^k x = c_1\lambda_1^k p_1 + c_2\lambda_2^k p_2 + \cdots + c_n\lambda_n^k p_n$$

(b) $f(A) = Pf(\Lambda)P^{-1} = P\operatorname{diag}\Big(f(\lambda_1), \ldots, f(\lambda_n)\Big)P^{-1}$ for every real polynomial f.

(c) A is invertible if and only if every $\lambda_i \neq 0$. Then $A^{-1} = P\Lambda^{-1}P^{-1}$, and for $k \geqslant 1$ the power $A^{-k} = (A^{-1})^k$ equals $P\operatorname{diag}(\lambda_1^{-k}, \ldots, \lambda_n^{-k})P^{-1}$.

Proof. (a)

- By induction on k: $A^0 = I = P\Lambda^0 P^{-1}$, and if $A^k = P\Lambda^k P^{-1}$, then $A^{k+1} = A^k A = P\Lambda^k P^{-1}P\Lambda P^{-1} = P\Lambda^{k+1}P^{-1}$ (Theorem 1.3).
- $\Lambda^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)$ (Proposition 1.19(d)).
- $A^k x = P\Lambda^k P^{-1}x = P\Lambda^k c$, and $\Lambda^k c = (\lambda_1^k c_1, \ldots, \lambda_n^k c_n)'$ (Proposition 1.13(a)); so $A^k x = \sum_j \lambda_j^k c_j p_j$ (Theorem 1.4(b)).

(b) If $f = \sum_k d_k t^k$, then by (a) and Theorem 1.3, $f(A) = \sum_k d_k P\Lambda^k P^{-1} = Pf(\Lambda)P^{-1}$; and $f(\Lambda) = \sum_k d_k \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k) = \operatorname{diag}\Big(f(\lambda_1), \ldots, f(\lambda_n)\Big)$, entry by entry.

(c)

- The eigenvalues of A are $\lambda_1, \ldots, \lambda_n$ (Definition 7.23), so A is invertible if and only if every $\lambda_i \neq 0$ (Theorem 7.19).
- Then $\Lambda^{-1} = \operatorname{diag}(\lambda_1^{-1}, \ldots, \lambda_n^{-1})$ (Definition 2.12), and $A^{-1} = (P^{-1})^{-1}\Lambda^{-1}P^{-1} = P\Lambda^{-1}P^{-1}$ (Theorem 2.13(a),(b)). Applying (a) to $A^{-1} = P\Lambda^{-1}P^{-1}$ gives $A^{-k} = P(\Lambda^{-1})^k P^{-1} = P\operatorname{diag}(\lambda_1^{-k}, \ldots, \lambda_n^{-k})P^{-1}$. □

Example 7.28. Let

$$B = \begin{pmatrix} 1 & 2 & -2 & 2 \\ -2 & 3 & -2 & 0 \\ 0 & -2 & 3 & -2 \\ 2 & -2 & 2 & 1 \end{pmatrix}$$

Show that 3 and 1 are eigenvalues of B, diagonalize B, and find B^k for $k = 0, 1, 2, \ldots$

Solution.

- $\lambda = 3$:

$$B - 3I = \begin{pmatrix} -2 & 2 & -2 & 2 \\ -2 & 0 & -2 & 0 \\ 0 & -2 & 0 & -2 \\ 2 & -2 & 2 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} -2 & 2 & -2 & 2 \\ 0 & -2 & 0 & -2 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} -2 & 2 & -2 & 2 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_2 \leftarrow -\frac{1}{2}R_2]{R_1 \leftarrow -\frac{1}{2}R_1} \begin{pmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $\operatorname{rank}(B-3I) = 2 < 4$ (Definition 4.11): 3 is an eigenvalue (Theorem 7.3), with $g_B(3) = 4-2 = 2$ (Definition 7.21). The special solutions $p_1 = (-1, 0, 1, 0)'$ and $p_2 = (0, -1, 0, 1)'$ form a basis of $\mathcal{N}(B - 3I)$ (Theorem 4.8(c)).

- $\lambda = 1$:

$$B - I = \begin{pmatrix} 0 & 2 & -2 & 2 \\ -2 & 2 & -2 & 0 \\ 0 & -2 & 2 & -2 \\ 2 & -2 & 2 & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} -2 & 2 & -2 & 0 \\ 0 & 2 & -2 & 2 \\ 0 & -2 & 2 & -2 \\ 2 & -2 & 2 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} -2 & 2 & -2 & 0 \\ 0 & 2 & -2 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_2 \leftarrow \frac{1}{2}R_2]{R_1 \leftarrow -\frac{1}{2}R_1} \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  So $\operatorname{rank}(B - I) = 2$: 1 is an eigenvalue with $g_B(1) = 2$, and the special solutions $p_3 = (0, 1, 1, 0)'$ and $p_4 = (-1, -1, 0, 1)'$ form a basis of $\mathcal{N}(B - I)$.

- By Lemma 7.24(b), $p_1, \dots, p_4$ are four independent eigenvectors, a basis of $\mathbb{R}^4$ (Theorem 4.5(d)). So $P^{-1}BP = \Lambda$ (Definition 7.23) with

$$P = [p_1\ p_2\ p_3\ p_4] = \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & -1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(3, 3, 1, 1), \qquad P^{-1} = \begin{pmatrix} 0 & -1 & 1 & -1 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

  Check: $P^{-1}P = I_4$, so this is the inverse (Theorem 2.16); for instance, row 1 of P^{-1} times p_4 is $0 + 1 + 0 - 1 = 0$. The eigenvalues of B, listed with multiplicity, are 3, 3, 1, 1 (Definition 7.23), so $a_B(3) = g_B(3) = 2$ and $a_B(1) = g_B(1) = 2$.

- By Proposition 7.27(a), $B^k = P \operatorname{diag}(3^k, 3^k, 1, 1)P^{-1}$. As $\operatorname{diag}(3^k, 3^k, 1, 1) = 3^k \operatorname{diag}(1, 1, 0, 0) + \Big(I_4 - \operatorname{diag}(1, 1, 0, 0)\Big)$ and $PP^{-1} = I_4$,

$$B^k = 3^k E + (I_4 - E), \qquad E = P \operatorname{diag}(1, 1, 0, 0)P^{-1}$$

  For $k = 1$, $B = 3E + I_4 - E = I_4 + 2E$, so $E = \frac{1}{2}(B - I_4)$ and

$$B^k = I_4 + \tfrac{1}{2}(3^k - 1)(B - I_4), \qquad B^2 = I_4 + 4(B - I_4) = \begin{pmatrix} 1 & 8 & -8 & 8 \\ -8 & 9 & -8 & 0 \\ 0 & -8 & 9 & -8 \\ 8 & -8 & 8 & 1 \end{pmatrix}$$

Proposition 7.29 ([2, §6.5.4], [5, Theorems 21.4.3, 21.5.1])**.** Let $A = P\Lambda P^{-1}$ with $P = [p_1\ \cdots\ p_n] \in \mathbb{R}^{n \times n}$ invertible and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_i \in \mathbb{R}$.

(a) $\operatorname{rank} A$ is the number of indices i with $\lambda_i \neq 0$, and the columns p_i with $\lambda_i \neq 0$ form a basis of $\mathcal{C}(A)$. So $\operatorname{rank} A$ is the number of nonzero eigenvalues of A, listed with multiplicity.

(b) For every real polynomial f, $f(A)$ is diagonalizable, and its eigenvalues, listed with multiplicity, are $f(\lambda_1), \dots, f(\lambda_n)$; hence $\operatorname{tr} f(A) = f(\lambda_1) + \cdots + f(\lambda_n)$ and $\det f(A) = f(\lambda_1) \cdots f(\lambda_n)$. In particular, A^k has the eigenvalues $\lambda_1^k, \dots, \lambda_n^k$ ($k = 0, 1, 2, \dots$), and, if A is invertible, A^{-1} has the eigenvalues $\lambda_1^{-1}, \dots, \lambda_n^{-1}$.

Proof. (a) Let S be the set of indices i with $\lambda_i \neq 0$, and r the number of its elements.

- $\operatorname{rank} A = \operatorname{rank}(P\Lambda P^{-1}) = \operatorname{rank} \Lambda$ (Theorem 4.15(d)).
- Column i of Λ is $\lambda_i e_i$ (Definition 1.12), which is 0 for $i \notin S$; so the r columns $\lambda_i e_i$, $i \in S$, span $\mathcal{C}(\Lambda)$ (Definition 4.7). They are independent: if $\sum_{i \in S} c_i \lambda_i e_i = 0$, then every $c_i \lambda_i = 0$, as $e_1, \dots, e_n$ are independent (Definition 3.17), and so $c_i = 0$. Hence $\operatorname{rank} \Lambda = \dim \mathcal{C}(\Lambda) = r$ (Definition 4.11).

- If $r = 0$, then $\Lambda = 0$, so $A = 0$ and $\mathcal{C}(A) = \{0\}$, with the empty basis. Let $r \geqslant 1$, let $X \in \mathbb{R}^{n\times r}$ consist of the columns p_i, $i \in S$, and let Δ be the $r \times r$ diagonal matrix of the corresponding λ_i. As $Ap_i = \lambda_i p_i$ (Definition 7.23), $AX = X\Delta$ (Theorem 1.4(a),(b)). Δ is invertible (Definition 2.12), so $X = AX\Delta^{-1} = A(X\Delta^{-1})$, and $\mathcal{C}(X) \subseteq \mathcal{C}(A)$ (Theorem 4.15(a)).
- The columns of X are columns of the invertible P, hence independent (Example 3.19, Proposition 3.13(b)). So $\dim \mathcal{C}(X) = r = \operatorname{rank} A = \dim \mathcal{C}(A)$, and $\mathcal{C}(X) = \mathcal{C}(A)$ (Theorem 4.5(a)): the columns of X form a basis of $\mathcal{C}(A)$.
- The eigenvalues of A, listed with multiplicity, are $\lambda_1, \dots, \lambda_n$ (Definition 7.23).

(b)

- By Proposition 7.27(b), $f(A) = Pf(\Lambda)P^{-1}$, that is, $P^{-1}f(A)P = f(\Lambda)$, with $f(\Lambda) = \operatorname{diag}\big(f(\lambda_1), \dots, f(\lambda_n)\big)$. So $f(A)$ is diagonalizable, and its eigenvalues, listed with multiplicity, are $f(\lambda_1), \dots, f(\lambda_n)$ (Definition 7.23).
- Theorem 7.19 gives the trace and the determinant; $f = t^k$ gives the statement on A^k.
- If A is invertible, then $P^{-1}A^{-1}P = \Lambda^{-1} = \operatorname{diag}(\lambda_1^{-1}, \dots, \lambda_n^{-1})$ (Proposition 7.27(c)), and Definition 7.23 gives the eigenvalues of A^{-1}. □

Orthogonal triangularization

Theorem 7.30 (Schur triangularization; [5, Theorem 21.5.11])**.** Let $A \in \mathbb{R}^{n\times n}$, and let all eigenvalues of A be real, listed with multiplicity in any order as $\lambda_1, \dots, \lambda_n$. Then $A = QTQ'$, where $Q \in \mathbb{R}^{n\times n}$ is orthogonal and $T \in \mathbb{R}^{n\times n}$ is upper triangular with the diagonal entries $\lambda_1, \dots, \lambda_n$, in this order.

Proof. Induction on n; for $n = 1$, $A = (\lambda_1) = I_1(\lambda_1)I_1'$. Let $n \geqslant 2$.

- λ_1 is a real eigenvalue, so $Ax = \lambda_1 x$ for some $x \neq 0$ (Definition 7.10). Then $u = x/\|x\|$ has $u'u = 1$ and $Au = \lambda_1 u$. By Corollary 6.15(c) (with $p = 1$), there is $V \in \mathbb{R}^{n\times(n-1)}$ such that $Q_1 = [u\ V]$ is orthogonal; comparing blocks in $Q_1'Q_1 = I_n$ (Theorem 5.3(b),(c)) gives $V'u = 0$.
- By Theorem 5.3(b),(c) and $Au = \lambda_1 u$,

$$Q_1'AQ_1 = \begin{pmatrix} u'Au & u'AV \\ V'Au & V'AV \end{pmatrix} = \begin{pmatrix} \lambda_1 & b' \\ 0 & A_1 \end{pmatrix}, \qquad b' = u'AV, \qquad A_1 = V'AV \in \mathbb{R}^{(n-1)\times(n-1)}$$

- $Q_1' = Q_1^{-1}$ (Proposition 6.13(b)), so A and $Q_1'AQ_1$ have the same eigenvalues (Theorem 7.15(a)); by Proposition 7.16(a) these are λ_1 together with the eigenvalues of A_1. By Theorem 7.9(b), the eigenvalues of A_1, listed with multiplicity, are $\lambda_2, \dots, \lambda_n$, all real.
- By the induction hypothesis, $A_1 = Q_2T_2Q_2'$ with $Q_2 \in \mathbb{R}^{(n-1)\times(n-1)}$ orthogonal and T_2 upper triangular with the diagonal entries $\lambda_2, \dots, \lambda_n$; so $Q_2'A_1Q_2 = T_2$. Let $S = \operatorname{diag}(1, Q_2)$. Then $S'S = \operatorname{diag}(1, Q_2'Q_2) = I_n$ (Theorem 5.3(b), Proposition 5.6(b)), so S is orthogonal, and so is $Q = Q_1S$ (Theorem 6.19(b)). As $Q' = S'Q_1'$ (Theorem 1.9), Theorem 5.3(c) gives

$$Q'AQ = S'(Q_1'AQ_1)S = \begin{pmatrix} 1 & 0 \\ 0 & Q_2' \end{pmatrix} \begin{pmatrix} \lambda_1 & b' \\ 0 & A_1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & Q_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 & b'Q_2 \\ 0 & T_2 \end{pmatrix}$$

an upper triangular matrix T with the diagonal entries $\lambda_1, \dots, \lambda_n$. As $QQ' = I_n$, $A = Q(Q'AQ)Q' = QTQ'$. □

Conversely, if $A = QTQ'$ with Q orthogonal and T upper triangular, then A is similar to T, as $Q' = Q^{-1}$, and its eigenvalues are the diagonal entries of T (Theorem 7.15(a), Proposition 7.16(b)), which are real. So such a factorization exists exactly when all eigenvalues of A are real; the matrix of Example 7.12 has none.

Example 7.31. Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{pmatrix}, \qquad G = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad Q = \tfrac{1}{2}G$$

Every row of A adds up to 4, so $A\mathbf{1}_4 = 4\mathbf{1}_4$, and $u = \frac{1}{2}\mathbf{1}_4$, the first column of Q, has $u'u = 1$ and $Au = 4u$. The columns of G are orthogonal and have squared length 4 (for instance, columns 2 and 3 give $1 - 1 - 1 + 1 = 0$), so $G'G = 4I_4$ and $Q'Q = I_4$: $Q = [u\ V]$ is an orthogonal matrix as in the first step of the proof of Theorem 7.30. As $G' = G$,

$$AG = \begin{pmatrix} 4 & 2 & 0 & -2 \\ 4 & 2 & -2 & 0 \\ 4 & -2 & 0 & 2 \\ 4 & -2 & -2 & 0 \end{pmatrix}, \qquad Q'AQ = \tfrac{1}{4}G(AG) = \begin{pmatrix} 4 & 0 & -1 & 0 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = T$$

In the notation of that proof, $\lambda_1 = 4$, $b' = (0, -1, 0)$, and $A_1 = V'AV$ is already upper triangular, so the induction stops after one step. Hence $A = QTQ'$, and the eigenvalues of A are 4, 2, 1, -1 (Theorem 7.15(a), Proposition 7.16(b)); $\operatorname{tr} T = 6 = \operatorname{tr} A$ and $\det T = -8 = \det A$. As these four eigenvalues are real and distinct, A is diagonalizable (Theorem 7.25); but no orthogonal Q_0 makes $Q_0'AQ_0$ diagonal: if $Q_0'AQ_0 = \Lambda$, then $A = Q_0\Lambda Q_0'$ (as $Q_0Q_0' = I$, Proposition 6.13(b)) and $(Q_0\Lambda Q_0')' = Q_0\Lambda' Q_0' = Q_0\Lambda Q_0'$ (Theorem 1.9), while A is not symmetric ($a_{12} = 2 \neq 1 = a_{21}$).

Corollary 7.32 ([4, Ch. X, §1, Exercise 1], [7], [6, Exercise 7.65])**.** Let $A \in \mathbb{R}^{n\times n}$, and let all eigenvalues $\lambda_1, \dots, \lambda_n$ of A, listed with multiplicity, be real.

(a) For every real polynomial f, the eigenvalues of $f(A)$, listed with multiplicity, are $f(\lambda_1), \dots, f(\lambda_n)$. In particular $\operatorname{tr} A^k = \lambda_1^k + \cdots + \lambda_n^k$ for $k = 0, 1, 2, \dots$

(b) If A is invertible, the eigenvalues of A^{-1}, listed with multiplicity, are $\lambda_1^{-1}, \dots, \lambda_n^{-1}$.

Proof. Let $A = QTQ'$ as in Theorem 7.30: $Q' = Q^{-1}$ (Proposition 6.13(b)), and T is upper triangular with the diagonal entries $\lambda_1, \dots, \lambda_n$.

(a)

- $A^k = QT^kQ'$ for $k = 0, 1, 2, \dots$, by induction: $A^0 = I = QQ'$, and $A^{k+1} = A^kA = QT^kQ'QTQ' = QT^{k+1}Q'$ (Theorem 1.3). So, for $f = \sum_k c_k t^k$, $f(A) = \sum_k c_k QT^kQ' = Qf(T)Q'$.
- T^k is upper triangular with the diagonal entries $\lambda_1^k, \dots, \lambda_n^k$ (Proposition 1.19(e)); adding multiples of these matrices entry by entry, $f(T)$ is upper triangular with the diagonal entries $f(\lambda_1), \dots, f(\lambda_n)$.
- $f(A)$ is similar to $f(T)$, so by Theorem 7.15(a) and Proposition 7.16(b) its eigenvalues are $f(\lambda_1), \dots, f(\lambda_n)$. For $f = t^k$, Theorem 7.19 gives $\operatorname{tr} A^k = \lambda_1^k + \cdots + \lambda_n^k$.

(b) 0 is not an eigenvalue of A (Theorem 7.19), so every $\lambda_i \neq 0$; hence T is invertible, and T^{-1} is upper triangular with the diagonal entries $\lambda_1^{-1}, \dots, \lambda_n^{-1}$ (Theorem 2.31). Then $A^{-1} = (Q')^{-1}T^{-1}Q^{-1} = QT^{-1}Q'$ (Theorem 2.13(a),(b)), so A^{-1} is similar to T^{-1}; conclude as in (a). □

Proposition 7.29(b) gives (a) for a diagonalizable A; here A need not be diagonalizable. For the matrix A of Example 7.17, with the eigenvalues -3, -1, 1, 3,

$$A^2 = \begin{pmatrix} 5 & -4 & 0 & 8 \\ -4 & 5 & 8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 4 & 5 \end{pmatrix}, \qquad \operatorname{tr} A^2 = 20 = (-3)^2 + (-1)^2 + 1^2 + 3^2$$

and, as $\det A = 9 \neq 0$ (Example 7.17), A^{-1} has the eigenvalues $-\frac{1}{3}$, -1, 1, $\frac{1}{3}$ (part (b)).

The products AB and BA

Theorem 7.33 ([5, Theorem 21.10.1])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$ with $m \leqslant n$.

(a) $p_{BA}(t) = t^{n-m}p_{AB}(t)$; in particular $p_{BA} = p_{AB}$ if $m = n$.

(b) The eigenvalues of BA, listed with multiplicity, are those of AB together with $n - m$ zeros. So AB and BA have the same nonzero eigenvalues, with the same algebraic multiplicities.

(c) $g_{AB}(c) = g_{BA}(c)$ for every real $c \neq 0$. More precisely, if $c \neq 0$ is a real eigenvalue of AB and the columns of $X \in \mathbb{R}^{m\times g}$ form a basis of $\mathcal{N}(AB - cI_m)$, then the g columns of BX are linearly independent eigenvectors of BA belonging to c.

Proof. (a)

- Let $t \in \mathbb{R}$, $t \neq 0$. Theorem 5.32(a), with I_n, I_m, $-t^{-1}B$ and A in place of its A, T, U and V, gives

$$\det(I_n - t^{-1}BA) = \det I_n \cdot \det I_m \cdot \det(I_m - t^{-1}AB) = \det(I_m - t^{-1}AB)$$

- By Theorem 5.18(c), $\det(tI_n - BA) = t^n \det(I_n - t^{-1}BA)$ and $\det(tI_m - AB) = t^m \det(I_m - t^{-1}AB)$. Hence $\det(tI_n - BA) = t^{n-m}\det(tI_m - AB)$.
- So p_{BA} and $t^{n-m}p_{AB}$ have the same value at every real $t \neq 0$ (Definition 7.7(a), Proposition 7.5(b)), and they are equal (Proposition 7.5(d)).

(b) If $p_{AB} = (t - \mu_1)\cdots(t - \mu_m)$ (Definition 7.10), then by (a) $p_{BA} = (t - 0)^{n-m}(t - \mu_1)\cdots(t - \mu_m)$, a product of n linear factors. By Theorem 7.9(b), the eigenvalues of BA are $\mu_1, \ldots, \mu_m$ and $n - m$ zeros; so $a_{BA}(c) = a_{AB}(c)$ for every $c \neq 0$.

(c)

- If c is not an eigenvalue of AB, then by (b) it is not one of BA, and both geometric multiplicities are 0 (Definition 7.21).
- Let $c \neq 0$ be a real eigenvalue of AB, $g = g_{AB}(c)$, and let $X \in \mathbb{R}^{m\times g}$ have a basis of $\mathcal{N}(AB - cI_m)$ as its columns. Then $\operatorname{rank} X = g$ (Theorem 4.13(b)) and $ABX = cX$ (Theorem 1.4(a)).
- By Theorem 4.15(a),(d), $g = \operatorname{rank} X \geqslant \operatorname{rank}(BX) \geqslant \operatorname{rank}(ABX) = \operatorname{rank}(cX) = g$, as $cX = X(cI_g)$ with cI_g invertible. So the g columns of BX are independent (Theorem 4.13(b)).
- $BA(BX) = B(ABX) = cBX$: the columns of BX lie in $\mathcal{N}(BA - cI_n)$. Being independent, they are nonzero (Proposition 3.13(a)), so they are eigenvectors of BA belonging to c, and $g_{BA}(c) \geqslant g$ (Theorem 4.5(c)).
- The argument does not use $m \leqslant n$; with the roles of A and B exchanged it gives $g_{AB}(c) \geqslant g_{BA}(c)$. □

Example 7.34. For $n \geqslant 2$, take $A = \mathbf{1}_n' \in \mathbb{R}^{1\times n}$ and $B = \mathbf{1}_n \in \mathbb{R}^{n\times 1}$ in Theorem 7.33 ($m = 1$): $AB = \mathbf{1}'\mathbf{1} = n$ and $BA = \mathbf{1}\mathbf{1}' = J_n$ (Definition 1.14, Proposition 1.15). So $p_{J_n}(t) = t^{n-1}(t - n)$: the eigenvalues of J_n are n (simple) and 0 ($n-1$ times). Indeed $J_n\mathbf{1} = \mathbf{1}(\mathbf{1}'\mathbf{1}) = n\mathbf{1}$, and $J_n x = \mathbf{1}(\mathbf{1}'x) = 0$ for every centered x, that is, $\mathbf{1}'x = 0$ (Example 3.5(f)).

Example 7.35. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}, \qquad A'A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \qquad AA' = \begin{pmatrix} 2 & 1 & 1 & -1 \\ 1 & 2 & 1 & 0 \\ 1 & 1 & 2 & 1 \\ -1 & 0 & 1 & 2 \end{pmatrix}$$

Find the eigenvalues of $A'A$ and AA', and an eigenvector for each.

Solution.

- For $t \in \mathbb{R}$, expanding along row 1 (Theorem 5.21),

$$\det(tI_3 - A'A) = \det\begin{pmatrix} t-3 & -1 & 0 \\ -1 & t-2 & -1 \\ 0 & -1 & t-3 \end{pmatrix} = (t-3)\Big((t-2)(t-3)-1\Big) - (t-3) = (t-3)(t^2-5t+4)$$

So $p_{A'A} = (t-1)(t-3)(t-4)$ (Definition 7.7(a), Proposition 7.5(d)). Theorem 7.33(a), with $A' \in \mathbb{R}^{3\times 4}$ and $A \in \mathbb{R}^{4\times 3}$ in place of A and B ($m = 3$, $n = 4$), gives $p_{AA'} = t^{4-3}p_{A'A} = t(t-1)(t-3)(t-4)$.

- Multiplying out, $A'A(1,1,1)' = (4,4,4)'$, $A'A(-1,0,1)' = (-3,0,3)'$ and $A'A(1,-2,1)' = (1,-2,1)'$: these are eigenvectors of $A'A$ belonging to 4, 3 and 1. Each of these eigenvalues is simple, so its eigenspace has dimension 1 (Theorem 7.22), and the vector found is a basis of it (Theorem 4.5(d)). By Theorem 7.33(c), the vectors

$$A\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}2\\2\\2\\0\end{pmatrix}, \qquad A\begin{pmatrix}-1\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\0\\1\\2\end{pmatrix}, \qquad A\begin{pmatrix}1\\-2\\1\end{pmatrix} = \begin{pmatrix}-1\\2\\-1\\0\end{pmatrix}$$

are eigenvectors of AA' belonging to 4, 3 and 1; for instance $AA'(-1,0,1,2)' = (-3,0,3,6)'$.

- The eigenvalue 0 of AA' has the eigenspace $\mathcal{N}(AA') = \mathcal{N}(A')$ (Theorem 4.18(a), applied to A'). Row-reduce A':

$$A' = \begin{pmatrix}1&1&0&-1\\1&0&1&0\\0&1&1&1\end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix}1&1&0&-1\\0&-1&1&1\\0&1&1&1\end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix}1&1&0&-1\\0&-1&1&1\\0&0&2&2\end{pmatrix}$$

So $\operatorname{rank} A' = 3$ and $\dim \mathcal{N}(A') = 1$ (Definition 4.11, Theorem 4.12). With $x_4 = 1$, back substitution gives $x_3 = -1$, $x_2 = 0$ and $x_1 = 1$; so $(1,0,-1,1)'$ is a basis of $\mathcal{N}(AA')$ (Theorem 4.5(d)).

Corollary 7.36 ([2, Lemma 7.1, §6.5.4])**.** Let $A \in \mathbb{R}^{n\times n}$ have rank r. Then $a_A(0) \geqslant n - r$: at most r of the eigenvalues of A, listed with multiplicity, are nonzero. If A is diagonalizable, exactly r of them are nonzero.

Proof. • If $r = 0$, then $A = 0$ and $p_A = t^n$ (Proposition 7.16(b)), so $a_A(0) = n$.

- Let $r \geqslant 1$, and let $A = BC$ be a full-rank factorization, with $B \in \mathbb{R}^{n\times r}$ and $C \in \mathbb{R}^{r\times n}$ (Theorem 4.21(a)); here $r \leqslant n$ (Theorem 4.13(a)). Theorem 7.33(a), applied to $C \in \mathbb{R}^{r\times n}$ and $B \in \mathbb{R}^{n\times r}$ in place of A and B, gives $p_A = p_{BC} = t^{n-r}p_{CB}$.
- By Theorem 7.33(b), the eigenvalues of A, listed with multiplicity, are the r eigenvalues of the $r \times r$ matrix CB together with $n - r$ zeros. So at most r of them are nonzero, and $a_A(0) \geqslant n - r$.
- If A is diagonalizable, exactly r of them are nonzero, by Proposition 7.29(a). □

Example 7.37 ([2, Example 7.1])**.** Let

$$A = \begin{pmatrix} -1 & 1 & 2 \\ 2 & 2 & 0 \\ 0 & 4 & 4 \end{pmatrix}$$

Find $\operatorname{rank} A$, a full-rank factorization $A = BC$, the eigenvalues of A, and compare $\operatorname{rank} A$ with the number of nonzero eigenvalues.

Solution.

- Row-reduce A to its RREF R:

$$\begin{pmatrix} -1 & 1 & 2 \\ 2 & 2 & 0 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + 2R_1} \begin{pmatrix} -1 & 1 & 2 \\ 0 & 4 & 4 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} -1 & 1 & 2 \\ 0 & 4 & 4 \\ 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow[R_2 \leftarrow \frac{1}{4} R_2]{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & -1 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

  The pivot columns are columns 1 and 2, so $\operatorname{rank} A = 2$ (Definition 4.11). By Theorem 4.21(a), $A = BC$ with

$$B = \begin{pmatrix} -1 & 1 \\ 2 & 2 \\ 0 & 4 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad CB = \begin{pmatrix} -1 & -3 \\ 2 & 6 \end{pmatrix}$$

- By Definition 7.7(b), $p_{CB} = t^2 - 5t + (-6 + 6) = t(t - 5)$, and, as in the proof of Corollary 7.36, $p_A = t^{3-2} p_{CB} = t^2(t - 5)$. So the eigenvalues of A are 0, 0, 5: $a_A(0) = 2$, and only one eigenvalue is nonzero, although $\operatorname{rank} A = 2$.
- Hence A is not diagonalizable (Proposition 7.29(a)). Indeed $g_A(0) = 3 - \operatorname{rank} A = 1 < 2 = a_A(0)$ (Definition 7.21, Theorem 7.25).
- Check: $\operatorname{tr} A = -1 + 2 + 4 = 5 = 0 + 0 + 5$ and $\det A = 0 = 0 \cdot 0 \cdot 5$ (Theorem 7.19), as $\operatorname{rank} A < 3$ (Theorem 5.18(a)).

The Cayley–Hamilton theorem

Theorem 7.38 (Cayley–Hamilton; [2, Theorem 7.7], [8, Ch. 21, Exercise 3])**.** Let $A \in \mathbb{R}^{n \times n}$ and $p_A(t) = c_0 + c_1 t + \cdots + c_n t^n$ (so $c_n = 1$). Then

$$p_A(A) = c_0 I + c_1 A + \cdots + c_n A^n = 0$$

Proof. For $n = 1$, $A = (a)$, $p_A = t - a$ and $p_A(A) = A - aI_1 = 0$. Let $n \geqslant 2$.

- For $t \in \mathbb{R}$, the (i, j) entry of $\operatorname{adj}(tI - A)$ is the cofactor $(-1)^{i+j} \det\big((tI - A)(j|i)\big)$ (Definition 5.20, Theorem 5.24). Every entry of the $(n - 1) \times (n - 1)$ matrix $(tI - A)(j|i)$ is an entry $\delta_{rs} t - a_{rs}$ of $tI - A$; by Lemma 7.6 (with $m = d = n - 1$) and Proposition 7.5(b), the cofactor is the value at t of a real polynomial h_{ij} that is 0 or of degree at most $n - 1$. Writing $h_{ij} = \sum_{s=0}^{n-1} k_{ij}^{(s)} t^s$ and $K_s = (k_{ij}^{(s)}) \in \mathbb{R}^{n \times n}$,

$$\operatorname{adj}(tI - A) = K_0 + tK_1 + \cdots + t^{n-1} K_{n-1} \qquad (t \in \mathbb{R})$$

- By Theorem 5.24, which holds for every matrix, and Definition 7.7(a), $(tI-A)\operatorname{adj}(tI-A) = p_A(t)I$ for every real t. By Theorem 1.3 the left side is

$$(tI-A)\sum_{s=0}^{n-1} t^s K_s = T_0 + tT_1 + \cdots + t^n T_n$$

with $T_0 = -AK_0$, $T_s = K_{s-1} - AK_s$ $(1 \leqslant s \leqslant n-1)$ and $T_n = K_{n-1}$. So for each (i,j) the polynomials $\sum_s (T_s)_{ij} t^s$ and $\delta_{ij} p_A$ have the same value at every real t; by Proposition 7.5(d), their coefficients agree: $T_s = c_s I$ for $0 \leqslant s \leqslant n$.

- Hence

$$\begin{aligned} p_A(A) &= \sum_{s=0}^{n} A^s (c_s I) = \sum_{s=0}^{n} A^s T_s = -AK_0 + \sum_{s=1}^{n-1} \Big(A^s K_{s-1} - A^{s+1} K_s \Big) + A^n K_{n-1} \\ &= \sum_{s=1}^{n} A^s K_{s-1} - \sum_{s=0}^{n-1} A^{s+1} K_s = 0 \end{aligned}$$

since the two sums are the same (replace s by $s-1$ in the second). □

Corollary 7.39 ([2, §7.3, Example 7.2], [8, Ch. 21, Exercise 4])**.** Let $A \in \mathbb{R}^{n\times n}$ and $p_A(t) = c_0 + c_1 t + \cdots + c_n t^n$ (so $c_n = 1$).

(a) $A^n = -(c_0 I + c_1 A + \cdots + c_{n-1} A^{n-1})$, and every power A^k $(k = 0, 1, 2, \ldots)$ is a linear combination of $I, A, \ldots, A^{n-1}$.

(b) If A is invertible, then $c_0 \neq 0$ and

$$A^{-1} = -\frac{1}{c_0}\Big(c_1 I + c_2 A + \cdots + c_n A^{n-1} \Big)$$

Proof. (a)

- By Theorem 7.38, $c_0 I + c_1 A + \cdots + c_{n-1} A^{n-1} + A^n = 0$, which is the formula for A^n.
- For $k \leqslant n-1$, A^k is one of $I, A, \ldots, A^{n-1}$. Let $k \geqslant n-1$ and $A^k = b_0 I + b_1 A + \cdots + b_{n-1} A^{n-1}$. Then $A^{k+1} = A^k A = b_0 A + \cdots + b_{n-2} A^{n-1} + b_{n-1} A^n$ (Theorem 1.3), and replacing A^n by the first item gives a linear combination of $I, A, \ldots, A^{n-1}$. By induction on k, this holds for every k.

(b)

- $c_0 = (-1)^n \det A$ (Definition 7.7(b)), which is nonzero, as A is invertible (Theorem 5.18(a)).
- Multiplying $c_0 I + c_1 A + \cdots + c_n A^n = 0$ (Theorem 7.38) by A^{-1} gives $c_0 A^{-1} + c_1 I + c_2 A + \cdots + c_n A^{n-1} = 0$; dividing by c_0 gives the formula. □

Example 7.40 (cf. [2, Example 7.2])**.** Let A be the matrix of Example 7.2,

$$A = \begin{pmatrix} 1 & 2 & -2 & -2 \\ -1 & 0 & -2 & 2 \\ 0 & -1 & -1 & 1 \\ -1 & 2 & -2 & 0 \end{pmatrix}$$

Find p_A, check Theorem 7.38, and use Corollary 7.39 to find A^{-1} and A^4.

Solution.

- By Example 7.2, 1, -1, 2 and -2 are eigenvalues of A, that is, roots of p_A (Definition 7.7(a)). The eigenvalues $\lambda_1, \ldots, \lambda_4$ of A, listed with multiplicity, are the roots of p_A (Definition 7.10, Theorem 7.9(c)), so each of the four distinct numbers 1, -1, 2, -2 occurs among them, and there is no room for more. Hence

$$p_A = (t-1)(t+1)(t-2)(t+2) = (t^2-1)(t^2-4) = t^4 - 5t^2 + 4$$

and $c_0 = 4$, $c_1 = 0$, $c_2 = -5$, $c_3 = 0$, $c_4 = 1$.

- Multiplying out,

$$A^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -3 & 4 & 0 & 0 \\ 0 & 3 & 1 & -3 \\ -3 & 0 & 0 & 4 \end{pmatrix}, \qquad A^4 = (A^2)^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -15 & 16 & 0 & 0 \\ 0 & 15 & 1 & -15 \\ -15 & 0 & 0 & 16 \end{pmatrix}$$

  and entry by entry, $p_A(A) = A^4 - 5A^2 + 4I = 0$.

- As $c_0 = (-1)^4 \det A$ (Definition 7.7(b)), $\det A = 4 \neq 0$, and A is invertible (Theorem 5.18(a)). With

$$A^3 = A^2 A = \begin{pmatrix} 1 & 2 & -2 & -2 \\ -7 & -6 & -2 & 14 \\ 0 & -7 & -1 & 7 \\ -7 & 2 & -2 & 6 \end{pmatrix}$$

  Corollary 7.39(b) gives

$$A^{-1} = -\frac{1}{4}\big(c_1 I + c_2 A + c_3 A^2 + c_4 A^3\big) = \frac{1}{4}(5A - A^3) = \frac{1}{2}\begin{pmatrix} 2 & 4 & -4 & -4 \\ 1 & 3 & -4 & -2 \\ 0 & 1 & -2 & -1 \\ 1 & 4 & -4 & -3 \end{pmatrix}$$

  Check: row 1 of AA^{-1} is $\frac{1}{2}(2+2+0-2,\ 4+6-2-8,\ -4-8+4+8,\ -4-4+2+6) = (1,0,0,0)$, and likewise rows 2, 3, 4 are e_2', e_3', e_4'.

- By Corollary 7.39(a), $A^4 = -(c_0 I + c_1 A + c_2 A^2 + c_3 A^3) = 5A^2 - 4I$, as found above; then $A^5 = 5A^3 - 4A$, and every power of A is a linear combination of I, A, A^2, A^3.

Exercises

Exercise 7.1. Find the characteristic polynomial and the eigenvalues of each matrix, and check the trace and the determinant against the eigenvalues (Theorem 7.19).

$$\text{(a)}\ \ A = \begin{pmatrix} 1 & -2 & -3 & -1 \\ 0 & -1 & -3 & -1 \\ 0 & 0 & 2 & 1 \\ 2 & -2 & 0 & 3 \end{pmatrix}, \qquad \text{(b)}\ \ B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & 3 & -1 & 1 \\ -1 & 2 & -3 & 4 \end{pmatrix}, \qquad \text{(c)}\ \ M = \begin{pmatrix} 2 & 1 & 1 & -1 \\ 4 & -1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -2 & 3 \end{pmatrix}$$

Solution. (a)

- For $t \in \mathbb{R}$, row 3 of $tI - A$ is $(0,\ 0,\ t-2,\ -1)$; expanding along it (Theorem 5.21),

$$\begin{aligned} \det(tI - A) &= \det \begin{pmatrix} t-1 & 2 & 3 & 1 \\ 0 & t+1 & 3 & 1 \\ 0 & 0 & t-2 & -1 \\ -2 & 2 & 0 & t-3 \end{pmatrix} \\ &= (t-2)\det \begin{pmatrix} t-1 & 2 & 1 \\ 0 & t+1 & 1 \\ -2 & 2 & t-3 \end{pmatrix} + \det \begin{pmatrix} t-1 & 2 & 3 \\ 0 & t+1 & 3 \\ -2 & 2 & 0 \end{pmatrix} \end{aligned}$$

- Expanding the first 3×3 determinant along column 1 and the second along row 3,

$$\det\begin{pmatrix} t-1 & 2 & 1 \\ 0 & t+1 & 1 \\ -2 & 2 & t-3 \end{pmatrix} = (t-1)\Big((t+1)(t-3)-2\Big) - 2\Big(2-(t+1)\Big)$$
$$= (t-1)(t^2-2t-5) + 2(t-1)$$
$$= (t-1)(t^2-2t-3) = (t-1)(t+1)(t-3)$$
$$\det\begin{pmatrix} t-1 & 2 & 3 \\ 0 & t+1 & 3 \\ -2 & 2 & 0 \end{pmatrix} = -2\Big(6-3(t+1)\Big) - 2\cdot 3(t-1) = (6t-6)-(6t-6) = 0$$

  So $\det(tI-A) = (t-2)(t-1)(t+1)(t-3)$, and $p_A = (t+1)(t-1)(t-2)(t-3)$ (Definition 7.7(a), Proposition 7.5(d)): the eigenvalues are -1, 1, 2, 3.

- $\operatorname{tr} A = 1-1+2+3 = 5 = -1+1+2+3$. Expanding $\det A$ along row 3 in the same way, and the two 3×3 determinants along column 1,

$$\det A = 2\det\begin{pmatrix} 1 & -2 & -1 \\ 0 & -1 & -1 \\ 2 & -2 & 3 \end{pmatrix} - \det\begin{pmatrix} 1 & -2 & -3 \\ 0 & -1 & -3 \\ 2 & -2 & 0 \end{pmatrix} = 2\Big((-3-2)+2(2-1)\Big) - \Big((0-6)+2(6-3)\Big) = -6$$

  and $-6 = (-1)\cdot 1\cdot 2\cdot 3$.

(b)

- Every row of B has the sum 2. For $t \in \mathbb{R}$, add columns 2, 3 and 4 of $tI-B$ to column 1, and take the factor $t-2$ out of column 1 (Definition 5.13(b),(a) for columns, Theorem 5.22):

$$\det(tI-B) = \det\begin{pmatrix} t & -1 & 0 & -1 \\ 1 & t-1 & -1 & -1 \\ 1 & -3 & t+1 & -1 \\ 1 & -2 & 3 & t-4 \end{pmatrix} = \det\begin{pmatrix} t-2 & -1 & 0 & -1 \\ t-2 & t-1 & -1 & -1 \\ t-2 & -3 & t+1 & -1 \\ t-2 & -2 & 3 & t-4 \end{pmatrix}$$
$$= (t-2)\det\begin{pmatrix} 1 & -1 & 0 & -1 \\ 1 & t-1 & -1 & -1 \\ 1 & -3 & t+1 & -1 \\ 1 & -2 & 3 & t-4 \end{pmatrix}$$

- Subtract row 1 from rows 2, 3 and 4 (Definition 5.13(b)), expand along column 1, and then along column 3 (Theorem 5.21):

$$\det(tI-B) = (t-2)\det\begin{pmatrix} 1 & -1 & 0 & -1 \\ 0 & t & -1 & 0 \\ 0 & -2 & t+1 & 0 \\ 0 & -1 & 3 & t-3 \end{pmatrix} = (t-2)\det\begin{pmatrix} t & -1 & 0 \\ -2 & t+1 & 0 \\ -1 & 3 & t-3 \end{pmatrix}$$
$$= (t-2)(t-3)(t^2+t-2)$$

  So $p_B = (t-2)(t-3)(t+2)(t-1)$, and the eigenvalues are -2, 1, 2, 3.

- $\operatorname{tr} B = 0+1-1+4 = 4 = -2+1+2+3$. Expanding $\det B$ along row 1, and the two 3×3 determinants along column 1,

$$\det B = -\det\begin{pmatrix} -1 & 1 & 1 \\ -1 & -1 & 1 \\ -1 & -3 & 4 \end{pmatrix} - \det\begin{pmatrix} -1 & 1 & 1 \\ -1 & 3 & -1 \\ -1 & 2 & -3 \end{pmatrix} = -(1+7-2) - (7-5+4) = -12$$

  and $-12 = (-2)\cdot 1\cdot 2\cdot 3$.

(c)

- $M = \left(\begin{smallmatrix} M_1 & M_2 \\ 0 & M_3 \end{smallmatrix}\right)$ is block upper triangular, with $M_1 = \left(\begin{smallmatrix} 2 & 1 \\ 4 & -1 \end{smallmatrix}\right)$ and $M_3 = \left(\begin{smallmatrix} 0 & 1 \\ -2 & 3 \end{smallmatrix}\right)$. By Definition 7.7(b) ($n = 2$), $p_{M_1} = t^2 - t - 6 = (t-3)(t+2)$ and $p_{M_3} = t^2 - 3t + 2 = (t-1)(t-2)$.
- By Proposition 7.16(a), $p_M = p_{M_1} p_{M_3} = (t-3)(t+2)(t-1)(t-2)$: the eigenvalues are -2, 1, 2, 3.
- $\operatorname{tr} M = 2-1+0+3 = 4 = -2+1+2+3$, and $\det M = \det M_1 \det M_3 = (-6)\cdot 2 = -12 = (-2)\cdot 1\cdot 2\cdot 3$ (Theorem 5.29).

Exercise 7.2. Find the eigenvalues of

$$B = \begin{pmatrix} 3 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ -1 & 0 & 1 & 3 \end{pmatrix}$$

and a basis of each eigenspace. Check that the coefficient of t^3 in $p_B(t)$ is $-\operatorname{tr} B$ and that the constant term is $\det B$.

Solution. For $t \in \mathbb{R}$, row 2 of $tI_4 - B$ is $(0,\ t-1,\ 0,\ 0)$. Expanding along it, and then the remaining 3×3 determinant (rows and columns 1, 3, 4) along its second row $(0,\ t-3,\ 0)$ (Theorem 5.21),

$$\begin{aligned} \det(tI_4 - B) &= (t-1)\det \begin{pmatrix} t-3 & 0 & 1 \\ 0 & t-3 & 0 \\ 1 & -1 & t-3 \end{pmatrix} = (t-1)(t-3)\det \begin{pmatrix} t-3 & 1 \\ 1 & t-3 \end{pmatrix} \\ &= (t-1)(t-3)(t-2)(t-4) \end{aligned}$$

So $p_B = (t-1)(t-2)(t-3)(t-4)$ (Definition 7.7(a), Proposition 7.5(d)), and multiplying out, $p_B = t^4 - 10t^3 + 35t^2 - 50t + 24$. Here $\operatorname{tr} B = 3 + 1 + 3 + 3 = 10$, and expanding $\det B$ in the same way, $\det B = 1 \cdot 3 \cdot (9-1) = 24$; so the coefficient of t^3 is $-\operatorname{tr} B$ and the constant term is $(-1)^4 \det B$, as Definition 7.7(b) says. The eigenvalues are 1, 2, 3, 4 (Definition 7.7(a)).

- $\lambda = 1$:

$$B - I_4 = \begin{pmatrix} 2 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 2 & 1 & 0 & -1 \end{pmatrix} \xrightarrow{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 2 & 1 & 0 & -1 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - 2R_1} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow -\frac{1}{3}R_3} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_2 \leftarrow R_2 - 3R_3]{R_1 \leftarrow R_1 + 2R_3} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The special solution $(1, -2, 1, 0)'$ is a basis of $\mathcal{N}(B - I_4)$ (Theorem 4.8(c)).

- $\lambda = 2$:

$$B - 2I_4 = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_1} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_1 \leftarrow R_1 + R_2,\ R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(1, 0, 0, 1)'$ is a basis of $\mathcal{N}(B - 2I_4)$.

- $\lambda = 3$:

$$B - 3I_4 = \begin{pmatrix} 0 & 1 & 0 & -1 \\ 0 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[R_2 \leftrightarrow R_3]{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow -R_3]{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(1, 0, 1, 0)'$ is a basis of $\mathcal{N}(B - 3I_4)$.

- $\lambda = 4$:

$$B - 4I_4 = \begin{pmatrix} -1 & 1 & 0 & -1 \\ 0 & -3 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_1} \begin{pmatrix} -1 & 1 & 0 & -1 \\ 0 & -3 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -\frac{1}{3}R_2} \begin{pmatrix} -1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_1 \leftarrow R_1 - R_2,\ R_3 \leftarrow R_3 - R_2} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow -R_3]{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(-1, 0, 0, 1)'$ is a basis of $\mathcal{N}(B - 4I_4)$.

Check: $B(1, -2, 1, 0)' = (1, -2, 1, 0)'$ and $B(-1, 0, 0, 1)' = (-4, 0, 0, 4)'$. Also $1 + 2 + 3 + 4 = 10 = \operatorname{tr} B$ and $1 \cdot 2 \cdot 3 \cdot 4 = 24 = \det B$ (Theorem 7.19).

Exercise 7.3. Let

$$A = \begin{pmatrix} 0 & 0 & 2 & -1 \\ 0 & 3 & 0 & 0 \\ 1 & 0 & 1 & -1 \\ -2 & -2 & 2 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & 1 & 2 & 2 \\ 0 & 0 & 1 & 0 \\ -2 & 2 & 0 & 1 \end{pmatrix}$$

(a) Find the eigenvalues of A and a basis of each eigenspace.

(b) Do the same for B.

(c) Find an invertible P and a diagonal Λ with $AP = P\Lambda$, and an invertible Q and a diagonal Δ with $BQ = Q\Delta$; check both equations by multiplying out.

Solution. (a)

- For $t \in \mathbb{R}$, row 2 of $tI - A$ is $(0,\ t-3,\ 0,\ 0)$; expanding along it, and then along row 1 (Theorem 5.21),

$$\begin{aligned} \det(tI - A) &= \det \begin{pmatrix} t & 0 & -2 & 1 \\ 0 & t-3 & 0 & 0 \\ -1 & 0 & t-1 & 1 \\ 2 & 2 & -2 & t-1 \end{pmatrix} = (t-3) \det \begin{pmatrix} t & -2 & 1 \\ -1 & t-1 & 1 \\ 2 & -2 & t-1 \end{pmatrix} \\ &= (t-3)\Big(t\Big((t-1)^2+2\Big) + 2\Big(-(t-1)-2\Big) + \Big(2-2(t-1)\Big)\Big) \\ &= (t-3)\Big(t^3 - 2t^2 + 3t - 2t - 2 + 4 - 2t\Big) \\ &= (t-3)(t^3 - 2t^2 - t + 2) = (t-3)(t-2)(t^2-1) \end{aligned}$$

  So $p_A = (t+1)(t-1)(t-2)(t-3)$: the eigenvalues are -1, 1, 2, 3.

- $\lambda = -1$:

$$A + I = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 4 & 0 & 0 \\ 1 & 0 & 2 & -1 \\ -2 & -2 & 2 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 6 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{4}R_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 6 & 0 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + 2R_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow \frac{1}{6}R_3} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_3} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(1, 0, 0, 1)'$ is a basis of $\mathcal{N}(A + I)$ (Theorem 4.8(c)).

- $\lambda = 1$:

$$A - I = \begin{pmatrix} -1 & 0 & 2 & -1 \\ 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ -2 & -2 & 2 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 + R_1} \begin{pmatrix} -1 & 0 & 2 & -1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & -2 & -2 & 2 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} -1 & 0 & 2 & -1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_3]{R_1 \leftarrow R_1 - R_3} \begin{pmatrix} -1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow \frac{1}{2}R_3]{R_1 \leftarrow -R_1,\ R_2 \leftarrow \frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(1, 0, 1, 1)'$ is a basis of $\mathcal{N}(A - I)$.

- $\lambda = 2$:

$$A - 2I = \begin{pmatrix} -2 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ -2 & -2 & 2 & -1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 2 & -1 \\ -2 & -2 & 2 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_3 \leftarrow R_3 + 2R_1} \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -3 \\ 0 & -2 & 0 & -3 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + 2R_2} \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & -3 \end{pmatrix} \xrightarrow{R_3 \leftarrow -\frac{1}{3} R_3} \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + 3R_3]{R_1 \leftarrow R_1 + R_3} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The special solution $(1, 0, 1, 0)'$ is a basis of $\mathcal{N}(A - 2I)$.

- $\lambda = 3$:

$$A - 3I = \begin{pmatrix} -3 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & -1 \\ -2 & -2 & 2 & -2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ -3 & 0 & 2 & -1 \\ -2 & -2 & 2 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_3 \leftarrow R_3 + 3R_1} \begin{pmatrix} 1 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & -4 \\ 0 & -2 & -2 & -4 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & -2 & -1 \\ 0 & -2 & -2 & -4 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow -\frac{1}{4} R_3]{R_2 \leftarrow -\frac{1}{2} R_2} \begin{pmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 + 2R_3} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The special solution $(-1, -1, -1, 1)'$ is a basis of $\mathcal{N}(A - 3I)$.

(b)

- For $t \in \mathbb{R}$, row 3 of $tI - B$ is $(0,\ 0,\ t-1,\ 0)$; expanding along it, and then along row 1,

$$\begin{aligned} \det(tI - B) &= \det \begin{pmatrix} t-1 & 2 & 2 & 2 \\ -2 & t-1 & -2 & -2 \\ 0 & 0 & t-1 & 0 \\ 2 & -2 & 0 & t-1 \end{pmatrix} = (t-1) \det \begin{pmatrix} t-1 & 2 & 2 \\ -2 & t-1 & -2 \\ 2 & -2 & t-1 \end{pmatrix} \\ &= (t-1)\Big((t-1)\big((t-1)^2 - 4\big) - 2\big(-2(t-1) + 4\big) + 2\big(4 - 2(t-1)\big)\Big) \\ &= (t-1)\big((t-1)(t-3)(t+1) + 4(t-1) - 8 + 8 - 4(t-1)\big) \\ &= (t-1)^2(t-3)(t+1) \end{aligned}$$

So $p_B = (t-1)^2(t-3)(t+1)$: the eigenvalues are 1, 1, 3, -1, with $a_B(1) = 2$.

- $\lambda = 1$:

$$B - I = \begin{pmatrix} 0 & -2 & -2 & -2 \\ 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & 0 & 2 & 2 \\ 0 & -2 & -2 & -2 \\ 0 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_1} \begin{pmatrix} 2 & 0 & 2 & 2 \\ 0 & -2 & -2 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 2 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 2 & 0 & 2 & 2 \\ 0 & -2 & -2 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow -\frac{1}{2}R_2]{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The free unknowns are x_3, x_4; the special solutions $(-1,-1,1,0)'$, $(-1,-1,0,1)'$ form a basis of $\mathcal{N}(B-I)$, and $g_B(1) = 2$.

- $\lambda = 3$:

$$B - 3I = \begin{pmatrix} -2 & -2 & -2 & -2 \\ 2 & -2 & 2 & 2 \\ 0 & 0 & -2 & 0 \\ -2 & 2 & 0 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} -2 & -2 & -2 & -2 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 4 & 2 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow -\frac{1}{4}R_2} \begin{pmatrix} -2 & -2 & -2 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 4 & 2 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 4R_2]{R_1 \leftarrow R_1 + 2R_2} \begin{pmatrix} -2 & 0 & -2 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}$$
$$\xrightarrow[R_4 \leftarrow R_4 + R_3]{R_1 \leftarrow R_1 - R_3} \begin{pmatrix} -2 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow -\frac{1}{2}R_3]{R_1 \leftarrow -\frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(-1,0,0,1)'$ is a basis of $\mathcal{N}(B-3I)$.

- $\lambda = -1$:

$$B + I = \begin{pmatrix} 2 & -2 & -2 & -2 \\ 2 & 2 & 2 & 2 \\ 0 & 0 & 2 & 0 \\ -2 & 2 & 0 & 2 \end{pmatrix} \xrightarrow{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & -1 & -1 & -1 \\ 2 & 2 & 2 & 2 \\ 0 & 0 & 2 & 0 \\ -2 & 2 & 0 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & -1 & -1 & -1 \\ 0 & 4 & 4 & 4 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -2 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow \frac{1}{4}R_2} \begin{pmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -2 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -2 & 0 \end{pmatrix}$$
$$\xrightarrow{R_3 \leftarrow \frac{1}{2}R_3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_3]{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $(0,-1,0,1)'$ is a basis of $\mathcal{N}(B+I)$.

(c)

- The four eigenvectors of A belong to distinct eigenvalues, so they are independent (Theorem 7.20), and P below is invertible (Theorem 4.13(b),(d)):

$$P = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & 1 \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(-1,1,2,3), \qquad AP = \begin{pmatrix} -1 & 1 & 2 & -3 \\ 0 & 0 & 0 & -3 \\ 0 & 1 & 2 & -3 \\ -1 & 1 & 0 & 3 \end{pmatrix} = P\Lambda$$

- The bases of $\mathcal{N}(B-I)$, $\mathcal{N}(B-3I)$ and $\mathcal{N}(B+I)$ together are independent (Lemma 7.24(b)), so Q below is invertible (Theorem 4.13(b),(d)):

$$Q=\begin{pmatrix}-1&-1&-1&0\\-1&-1&0&-1\\1&0&0&0\\0&1&1&1\end{pmatrix},\qquad \Delta=\operatorname{diag}(1,1,3,-1),\qquad BQ=\begin{pmatrix}-1&-1&-3&0\\-1&-1&0&1\\1&0&0&0\\0&1&3&-1\end{pmatrix}=Q\Delta$$

Exercise 7.4 ([4, Ch. VIII, §2, Exercise 4(a), (b)], [7])**.** Find the characteristic polynomial, the eigenvalues, and a basis of each eigenspace of

$$\text{(a)}\quad A=\begin{pmatrix}4&0&1\\-2&1&0\\-2&0&1\end{pmatrix},\qquad \text{(b)}\quad B=\begin{pmatrix}1&-3&3\\3&-5&3\\6&-6&4\end{pmatrix}$$

Solution. (a)

- For $t\in\mathbb{R}$, expanding along row 1 (Theorem 5.21),

$$\det(tI-A)=\det\begin{pmatrix}t-4&0&-1\\2&t-1&0\\2&0&t-1\end{pmatrix}=(t-4)(t-1)^2-\det\begin{pmatrix}2&t-1\\2&0\end{pmatrix}=(t-4)(t-1)^2+2(t-1)$$

  So $p_A=(t-1)\Big((t-4)(t-1)+2\Big)=(t-1)(t^2-5t+6)=(t-1)(t-2)(t-3)$: the eigenvalues are 1, 2, 3.

- $Ax=\lambda x$ reads $4x_1+x_3=\lambda x_1$, $-2x_1+x_2=\lambda x_2$, $-2x_1+x_3=\lambda x_3$.
- $\lambda=1$: $3x_1+x_3=0$ and $-2x_1=0$; so $x_1=x_3=0$, x_2 is free, and $(0,1,0)'$ is a basis of $\mathcal{N}(A-I)$.
- $\lambda=2$: $2x_1+x_3=0$, $-2x_1-x_2=0$, $-2x_1-x_3=0$; so $x_3=-2x_1$ and $x_2=-2x_1$, and $(1,-2,-2)'$ is a basis of $\mathcal{N}(A-2I)$.
- $\lambda=3$: $x_1+x_3=0$, $-2x_1-2x_2=0$, $-2x_1-2x_3=0$; so $x_3=-x_1$ and $x_2=-x_1$, and $(1,-1,-1)'$ is a basis of $\mathcal{N}(A-3I)$.

(b)

- For $t\in\mathbb{R}$, expanding along row 1,

$$\begin{aligned}\det(tI-B)&=\det\begin{pmatrix}t-1&3&-3\\-3&t+5&-3\\-6&6&t-4\end{pmatrix}\\&=(t-1)\Big((t+5)(t-4)+18\Big)-3\Big(-3(t-4)-18\Big)-3\Big(-18+6(t+5)\Big)\\&=(t-1)(t^2+t-2)+9(t+2)-18(t+2)=(t+2)\Big((t-1)^2-9\Big)=(t+2)^2(t-4)\end{aligned}$$

  So $p_B=t^3-12t-16=(t+2)^2(t-4)$: the eigenvalues are -2, -2, 4.

- $\lambda=-2$: $Bx=-2x$ reads $3x_1-3x_2+3x_3=0$, $3x_1-3x_2+3x_3=0$, $6x_1-6x_2+6x_3=0$, that is, $x_1-x_2+x_3=0$. Every row of $B+2I$ is a multiple of $(1,-1,1)$, so the row space of $B+2I$ is $\operatorname{span}\{(1,-1,1)\}$, $\operatorname{rank}(B+2I)=1$ (Definition 4.11) and $\dim\mathcal{N}(B+2I)=2$ (Theorem 4.12). The solutions $(1,0,-1)'$ and $(0,1,1)'$ are not multiples of each other, so they are independent (Proposition 3.13(c)) and form a basis of $\mathcal{N}(B+2I)$ (Theorem 4.5(d)).
- $\lambda=4$: $-3x_1-3x_2+3x_3=0$, $3x_1-9x_2+3x_3=0$, $6x_1-6x_2=0$; so $x_2=x_1$, then $x_3=2x_1$ by the first equation, and the second holds: $(1,1,2)'$ is a basis of $\mathcal{N}(B-4I)$.

Exercise 7.5. Let

$$A = \begin{pmatrix} 3 & 0 & 0 & 2 \\ -2 & 1 & 0 & -2 \\ 0 & -3 & 2 & 1 \\ -2 & 2 & 0 & -1 \end{pmatrix}$$

(a) Find p_A and all eigenvalues of A, and check Theorem 7.19.

(b) Find the eigenspace of each real eigenvalue. Is A diagonalizable?

Solution. (a)

- For $t \in \mathbb{R}$, column 3 of $tI - A$ has the entry $t-2$ in position $(3,3)$ and 0 elsewhere; expanding along it, and then along row 1 (Theorem 5.21),

$$\begin{aligned} \det(tI - A) &= \det \begin{pmatrix} t-3 & 0 & 0 & -2 \\ 2 & t-1 & 0 & 2 \\ 0 & 3 & t-2 & -1 \\ 2 & -2 & 0 & t+1 \end{pmatrix} = (t-2) \det \begin{pmatrix} t-3 & 0 & -2 \\ 2 & t-1 & 2 \\ 2 & -2 & t+1 \end{pmatrix} \\ &= (t-2)\Big((t-3)\big((t-1)(t+1)+4\big) - 2\big(-4-2(t-1)\big)\Big) \\ &= (t-2)(t^3 - 3t^2 + 7t - 5) \end{aligned}$$

- Let $q = t^3 - 3t^2 + 7t - 5$. Then $q(1) = 1 - 3 + 7 - 5 = 0$, so $q = (t-1)h$ for a polynomial h (Proposition 7.5(c)). Multiplying out, $(t-1)(t^2 - 2t + 5) = t^3 - 3t^2 + 7t - 5 = q$; so $h = t^2 - 2t + 5$ (Proposition 7.5(a)). Since $t^2 - 2t + 5 = (t-1)^2 + 4 = (t - 1 - 2i)(t - 1 + 2i)$, $p_A = (t-2)(t-1)\big(t-(1+2i)\big)\big(t-(1-2i)\big)$ (Definition 7.7(a), Proposition 7.5(d)): the eigenvalues are 2, 1, $1+2i$, $1-2i$.

- $\operatorname{tr} A = 3 + 1 + 2 - 1 = 5 = 2 + 1 + (1+2i) + (1-2i)$. Expanding $\det A$ along column 3, and then along row 1 (Theorem 5.21),

$$\det A = 2 \det \begin{pmatrix} 3 & 0 & 2 \\ -2 & 1 & -2 \\ -2 & 2 & -1 \end{pmatrix} = 2\big(3(-1+4) + 2(-4+2)\big) = 10$$

  and $10 = 2 \cdot 1 \cdot (1+2i)(1-2i)$ (Theorem 7.19).

(b) The real eigenvalues are 1 and 2. Row-reduce $A - I$ and $A - 2I$:

$$A - I = \begin{pmatrix} 2 & 0 & 0 & 2 \\ -2 & 0 & 0 & -2 \\ 0 & -3 & 1 & 1 \\ -2 & 2 & 0 & -2 \end{pmatrix} \xrightarrow{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & 0 & 1 \\ -2 & 0 & 0 & -2 \\ 0 & -3 & 1 & 1 \\ -2 & 2 & 0 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_2 \leftarrow R_2 + 2R_1} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -3 & 1 & 1 \\ 0 & 2 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & -3 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 + 3R_2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A-2I=\begin{pmatrix}1&0&0&2\\-2&-1&0&-2\\0&-3&0&1\\-2&2&0&-3\end{pmatrix}\xrightarrow[R_4\leftarrow R_4+2R_1]{R_2\leftarrow R_2+2R_1}\begin{pmatrix}1&0&0&2\\0&-1&0&2\\0&-3&0&1\\0&2&0&1\end{pmatrix}\xrightarrow{R_2\leftarrow -R_2}\begin{pmatrix}1&0&0&2\\0&1&0&-2\\0&-3&0&1\\0&2&0&1\end{pmatrix}$$

$$\xrightarrow[R_4\leftarrow R_4-2R_2]{R_3\leftarrow R_3+3R_2}\begin{pmatrix}1&0&0&2\\0&1&0&-2\\0&0&0&-5\\0&0&0&5\end{pmatrix}\xrightarrow{R_3\leftarrow -\frac{1}{5}R_3}\begin{pmatrix}1&0&0&2\\0&1&0&-2\\0&0&0&1\\0&0&0&5\end{pmatrix}$$

$$\xrightarrow[R_4\leftarrow R_4-5R_3]{R_1\leftarrow R_1-2R_3,\ R_2\leftarrow R_2+2R_3}\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&0&0\end{pmatrix}$$

The special solutions give $\mathcal{N}(A-I)=\operatorname{span}\{(-1,0,-1,1)'\}$ and $\mathcal{N}(A-2I)=\operatorname{span}\{e_3\}$ (Theorem 4.8(c)). A is not diagonalizable, since it has non-real eigenvalues (Definition 7.23).

Exercise 7.6. Let

$$A=\begin{pmatrix}1&1&0&-1\\0&2&0&1\\0&0&3&0\\0&0&0&4\end{pmatrix},\qquad P=\begin{pmatrix}1&0&0&0\\-1&1&0&0\\0&1&1&0\\-1&1&0&1\end{pmatrix}$$

(a) Find P^{-1} by Gauss–Jordan elimination, and compute $B=P^{-1}AP$.

(b) Find p_B, and check that $p_B=p_A$, $\operatorname{tr}B=\operatorname{tr}A$ and $\det B=\det A$.

(c) The vector $x=(1,1,0,0)'$ is an eigenvector of A belonging to 2. Find an eigenvector of B belonging to 2.

(d) Show that

$$F=\begin{pmatrix}1&1&1&1\\0&1&1&1\\0&0&1&1\\0&0&0&2\end{pmatrix},\qquad G=\begin{pmatrix}1&0&1&1\\0&1&1&1\\0&0&1&1\\0&0&0&2\end{pmatrix}$$

have the same characteristic polynomial, but are not similar.

Solution. (a) Row-reduce $[P\mid I_4]$ (Corollary 2.17(a)):

$$\left(\begin{array}{cccc|cccc}1&0&0&0&1&0&0&0\\-1&1&0&0&0&1&0&0\\0&1&1&0&0&0&1&0\\-1&1&0&1&0&0&0&1\end{array}\right)\xrightarrow[R_4\leftarrow R_4+R_1]{R_2\leftarrow R_2+R_1}\left(\begin{array}{cccc|cccc}1&0&0&0&1&0&0&0\\0&1&0&0&1&1&0&0\\0&1&1&0&0&0&1&0\\0&1&0&1&1&0&0&1\end{array}\right)$$

$$\xrightarrow[R_4\leftarrow R_4-R_2]{R_3\leftarrow R_3-R_2}\left(\begin{array}{cccc|cccc}1&0&0&0&1&0&0&0\\0&1&0&0&1&1&0&0\\0&0&1&0&-1&-1&1&0\\0&0&0&1&0&-1&0&1\end{array}\right)$$

So

$$P^{-1}=\begin{pmatrix}1&0&0&0\\1&1&0&0\\-1&-1&1&0\\0&-1&0&1\end{pmatrix},\qquad AP=\begin{pmatrix}1&0&0&-1\\-3&3&0&1\\0&3&3&0\\-4&4&0&4\end{pmatrix},\qquad B=P^{-1}(AP)=\begin{pmatrix}1&0&0&-1\\-2&3&0&0\\2&0&3&0\\-1&1&0&3\end{pmatrix}$$

(b)

- A is upper triangular, so $p_A = (t-1)(t-2)(t-3)(t-4)$ (Proposition 7.16(b)).
- For $t \in \mathbb{R}$, column 3 of $tI - B$ has the entry $t-3$ in position $(3,3)$ and 0 elsewhere; expanding along it, and then along row 1,

$$\det(tI-B) = \det\begin{pmatrix} t-1 & 0 & 0 & 1 \\ 2 & t-3 & 0 & 0 \\ -2 & 0 & t-3 & 0 \\ 1 & -1 & 0 & t-3 \end{pmatrix} = (t-3)\det\begin{pmatrix} t-1 & 0 & 1 \\ 2 & t-3 & 0 \\ 1 & -1 & t-3 \end{pmatrix}$$
$$= (t-3)\Big((t-1)(t-3)^2 + (-2-(t-3))\Big) = (t-3)\Big((t-1)(t-3)^2 - (t-1)\Big)$$

  So $p_B = (t-3)(t-1)\Big((t-3)^2 - 1\Big) = (t-1)(t-2)(t-3)(t-4) = p_A$, as Theorem 7.15(a) asserts.
- $\operatorname{tr} B = 1+3+3+3 = 10 = \operatorname{tr} A$. Expanding $\det B$ along column 3 and then along row 1,

$$\det B = 3\det\begin{pmatrix} 1 & 0 & -1 \\ -2 & 3 & 0 \\ -1 & 1 & 3 \end{pmatrix} = 3\Big(9-(-2+3)\Big) = 24$$

  and $24 = 1\cdot 2\cdot 3\cdot 4 = \det A$ (Theorem 5.14(c)).

(c) By Theorem 7.15(b), $P^{-1}x = (1,2,-2,-1)'$ is an eigenvector of B belonging to 2. Check: $B(1,2,-2,-1)' = (2,4,-4,-2)' = 2(1,2,-2,-1)'$.

(d)

- F and G are upper triangular with the diagonal entries 1, 1, 1, 2, so $p_F = p_G = (t-1)^3(t-2)$ (Proposition 7.16(b)).
- Row-reduce $F-I$ and $G-I$:

$$F - I = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$G - I = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_3]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  So $\operatorname{rank}(F-I) = 3$ and $\operatorname{rank}(G-I) = 2$ (Definition 4.11), that is, $\dim \mathcal{N}(F-I) = 1$ and $\dim \mathcal{N}(G-I) = 2$ (Theorem 4.12).
- Similar matrices have eigenspaces of equal dimension (Theorem 7.15(b)), so F and G are not similar.

Exercise 7.7. Let B be the matrix of Example 7.28, and

$$F = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 2 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad G = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 2 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(a) Show that $p_B = p_F = p_G$.

(b) Diagonalize F, and find an invertible S with $F = S^{-1}BS$.

(c) Show that G is not similar to B.

Solution. (a) By Example 7.28, the eigenvalues of B, listed with multiplicity, are 3, 3, 1, 1, so $p_B = (t-3)^2(t-1)^2$ (Definition 7.10). F and G are upper triangular with the diagonal entries 3, 3, 1, 1, so $p_F = p_G = (t-3)^2(t-1)^2$ (Proposition 7.16(b)).

(b)

- $\lambda = 3$:

$$F - 3I_4 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 0 & 0 & -2 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_2} \begin{pmatrix} 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow -\frac{1}{2}R_2]{R_1 \leftarrow -\frac{1}{2}R_1} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The free unknowns are x_1, x_2; the special solutions e_1, e_2 form a basis of $\mathcal{N}(F - 3I_4)$ (Theorem 4.8(c)).

- $\lambda = 1$:

$$F - I_4 = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow \frac{1}{2}R_2]{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The free unknowns are x_3, x_4; the special solutions $(0, -1, 1, 0)'$ and $(0, 1, 0, 1)'$ form a basis of $\mathcal{N}(F - I_4)$.

- So $g_F(3) = a_F(3) = 2$ and $g_F(1) = a_F(1) = 2$, and $P_F^{-1} F P_F = \Lambda = \operatorname{diag}(3, 3, 1, 1)$ (the paragraph after Theorem 7.25), where

$$P_F = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad P_F^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

  Check: row 2 of the second matrix times columns 1, 2, 3, 4 of P_F gives 0, 1, $-1+1=0$, $1-1=0$, and the other rows are rows of I_4 times P_F; so the product is I_4, and the second matrix is P_F^{-1} (Theorem 2.16).

- With P of Example 7.28, $P^{-1}BP = \Lambda$ with the same Λ, so $F = P_F \Lambda P_F^{-1} = P_F P^{-1} B P P_F^{-1} = S^{-1}BS$ for $S = PP_F^{-1}$ (Theorem 2.13(a),(b)):

$$S = PP_F^{-1} = \begin{pmatrix} -1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \qquad BS = SF = \begin{pmatrix} -3 & 0 & 0 & -1 \\ 0 & -3 & -2 & 2 \\ 3 & 0 & 1 & 0 \\ 0 & 3 & 3 & -2 \end{pmatrix}$$

  The equality $BS = SF$ confirms $F = S^{-1}BS$.

(c) Scaling rows 1 and 2 of

$$G - I_4 = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

by $\frac{1}{2}$ gives an echelon form with three nonzero rows, so $\operatorname{rank}(G - I_4) = 3$ (Remark 4.2, Definition 4.11) and $g_G(1) = 4 - 3 = 1$ (Definition 7.21). As $g_B(1) = 2$ (Example 7.28), Theorem 7.15(b) shows that G is not similar to B; by (a), the characteristic polynomial does not detect this.

Exercise 7.8. For each matrix, find the eigenvalues with their algebraic and geometric multiplicities, decide whether the matrix is diagonalizable, and compare its rank with the number of its nonzero eigenvalues, listed with multiplicity.

$$\text{(a)}\ \ A = \begin{pmatrix} 2 & 1 & -1 & 0 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \qquad \text{(b)}\ \ B = \begin{pmatrix} -1 & 0 & -1 & -2 \\ 1 & 3 & -2 & -1 \\ 1 & 0 & 1 & 2 \\ -1 & 0 & -1 & -1 \end{pmatrix}$$

$$\text{(c)}\ \ M = \begin{pmatrix} 3 & 1 & 1 & 2 \\ 0 & 2 & 0 & 0 \\ -1 & -1 & 1 & -2 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

Solution. (a)

- A is upper triangular, so $p_A = (t-2)^2(t+1)(t-3)$ (Proposition 7.16(b)): $a_A(2) = 2$ and $a_A(-1) = a_A(3) = 1$. As -1 and 3 are simple, $g_A(-1) = g_A(3) = 1$ (Theorem 7.22).
- Row-reduce $A - 2I$:

$$A - 2I = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  Scaling row 2 by $\frac{1}{3}$ gives an echelon form with three nonzero rows, so $\operatorname{rank}(A-2I) = 3$ (Remark 4.2, Definition 4.11) and $g_A(2) = 4 - 3 = 1 < 2 = a_A(2)$ (Definition 7.21): A is not diagonalizable (Theorem 7.25).
- $\det A = 2 \cdot 2 \cdot (-1) \cdot 3 = -12 \neq 0$ (Theorem 5.14(c)), so $\operatorname{rank} A = 4$ (Theorem 5.18(a), Theorem 4.13(d)): all four eigenvalues 2, 2, -1, 3 are nonzero, and their number is $\operatorname{rank} A$, although A is not diagonalizable.

(b)

- For $t \in \mathbb{R}$, column 2 of $tI - B$ has the entry $t-3$ in position $(2,2)$ and 0 elsewhere; expanding along it, and then along row 1,

$$\begin{aligned} \det(tI - B) &= \det \begin{pmatrix} t+1 & 0 & 1 & 2 \\ -1 & t-3 & 2 & 1 \\ -1 & 0 & t-1 & -2 \\ 1 & 0 & 1 & t+1 \end{pmatrix} = (t-3) \det \begin{pmatrix} t+1 & 1 & 2 \\ -1 & t-1 & -2 \\ 1 & 1 & t+1 \end{pmatrix} \\ &= (t-3)\Big((t+1)\big((t-1)(t+1)+2\big) - \big(-(t+1)+2\big) + 2\big(-1-(t-1)\big)\Big) \\ &= (t-3)\big(t^3 + t^2 + t + 1 - 1 + t - 2t\big) = (t-3)(t^3+t^2) = (t-3)t^2(t+1) \end{aligned}$$

  So $a_B(0) = 2$ and $a_B(3) = a_B(-1) = 1$, and $g_B(3) = g_B(-1) = 1$ (Theorem 7.22).
- Row-reduce $B - 0I = B$:

$$B = \begin{pmatrix} -1 & 0 & -1 & -2 \\ 1 & 3 & -2 & -1 \\ 1 & 0 & 1 & 2 \\ -1 & 0 & -1 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} -1 & 0 & -1 & -2 \\ 0 & 3 & -3 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} -1 & 0 & -1 & -2 \\ 0 & 3 & -3 & -3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Scaling rows 1 and 2 by -1 and $\frac{1}{3}$ gives an echelon form with three nonzero rows, so $\operatorname{rank} B = 3$ and $g_B(0) = 4 - 3 = 1 < 2 = a_B(0)$: B is not diagonalizable.

- $\operatorname{rank} B = 3$, but only two eigenvalues, 3 and -1, are nonzero: the inequality of Corollary 7.36 is strict.

(c)

- For $t \in \mathbb{R}$, rows 2 and 4 of $tI - M$ have the entry $t-2$ on the diagonal and 0 elsewhere; expanding along row 2, and then along the last row,

$$\begin{aligned}\det(tI - M) &= \det \begin{pmatrix} t-3 & -1 & -1 & -2 \\ 0 & t-2 & 0 & 0 \\ 1 & 1 & t-1 & 2 \\ 0 & 0 & 0 & t-2 \end{pmatrix} = (t-2) \det \begin{pmatrix} t-3 & -1 & -2 \\ 1 & t-1 & 2 \\ 0 & 0 & t-2 \end{pmatrix} \\ &= (t-2)^2 \det \begin{pmatrix} t-3 & -1 \\ 1 & t-1 \end{pmatrix} = (t-2)^2(t^2 - 4t + 4) = (t-2)^4\end{aligned}$$

  So 2 is the only eigenvalue, with $a_M(2) = 4$.

- Row-reduce $M - 2I$:

$$M - 2I = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ -1 & -1 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  So $\operatorname{rank}(M - 2I) = 1$ and $g_M(2) = 3 < 4 = a_M(2)$: M is not diagonalizable.

- $\det M = (-1)^4 p_M(0) = 16 \neq 0$, so $\operatorname{rank} M = 4$, the number of the nonzero eigenvalues 2, 2, 2, 2.

Exercise 7.9 ([6, Exercises 7.3, 7.11])**.** Let

$$A = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 2 & 1 & 5 \end{pmatrix}$$

(a) Find the characteristic equation and the eigenvalues of A.

(b) What is the rank of A?

(c) What are the eigenvectors of A? Are they linearly independent? Are they orthogonal?

Solution. (a) For $t \in \mathbb{R}$, column 3 of $tI - A$ has the entry $t-5$ at $(3,3)$ and 0 elsewhere; expanding along it (Theorem 5.21),

$$\det(tI - A) = \det \begin{pmatrix} t-1 & -3 & 0 \\ 0 & t-1 & 0 \\ -2 & -1 & t-5 \end{pmatrix} = (t-5) \det \begin{pmatrix} t-1 & -3 \\ 0 & t-1 \end{pmatrix} = (t-1)^2(t-5)$$

The characteristic equation is $(t-1)^2(t-5) = 0$, and the eigenvalues are 1 (twice) and 5.

(b) The constant term of p_A is $(0-1)^2(0-5) = -5 = (-1)^3 \det A$ (Definition 7.7(b)), so $\det A = 5 \neq 0$: A is invertible and $\operatorname{rank} A = 3$ (Theorem 5.18(a)).

(c)

- $\lambda = 1$: $Ax = x$ reads $x_1 + 3x_2 = x_1$, $x_2 = x_2$, $2x_1 + x_2 + 5x_3 = x_3$; so $x_2 = 0$ and $x_1 + 2x_3 = 0$: the eigenvectors are the nonzero multiples of $(2, 0, -1)'$.

- $\lambda = 5$: $Ax = 5x$ reads $x_1 + 3x_2 = 5x_1$, $x_2 = 5x_2$, $2x_1 + x_2 + 5x_3 = 5x_3$; so $x_2 = 0$, then $x_1 = 0$, and x_3 is free: the eigenvectors are the nonzero multiples of $(0, 0, 1)'$.
- $(2, 0, -1)'$ and $(0, 0, 1)'$ are independent (Theorem 7.20) but not orthogonal: their inner product (Definition 6.1) is $2 \cdot 0 + 0 \cdot 0 + (-1) \cdot 1 = -1$. The double eigenvalue 1 has only one independent eigenvector: $g_A(1) = 1 < 2 = a_A(1)$ (Definitions 7.10 and 7.21), as Theorem 7.22 allows.

Exercise 7.10. Which of the following matrices are diagonalizable?

$$A_1 = \begin{pmatrix} 3 & 1 & 0 & -1 \\ 0 & 3 & 0 & 0 \\ 0 & 1 & 3 & 0 \\ -1 & 0 & 1 & 3 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}, \qquad A_3 = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

Solution. (A_1) A_1 differs from the matrix B of Exercise 7.2 only in the $(2, 2)$ entry. The same two expansions give $p_{A_1} = (t-3) \cdot (t-3)(t-2)(t-4) = (t-3)^2(t-2)(t-4)$, so $a_{A_1}(3) = 2$. But

$$A_1 - 3I_4 = \begin{pmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

and scaling row 1 by -1 gives an echelon form with three nonzero rows; so $\operatorname{rank}(A_1 - 3I_4) = 3$ (Remark 4.2, Definition 4.11), $g_{A_1}(3) = 1 < 2$ (Definition 7.21), and A_1 is not diagonalizable (Theorem 7.25). The matrix B has four distinct real eigenvalues, so it is diagonalizable (Theorem 7.25); changing one entry has destroyed diagonalizability.

(A_2) $p_{A_2} = t^2 - 0 \cdot t + (-1 + 2) = t^2 + 1 = (t - i)(t + i)$ (Definition 7.7(b)). So the eigenvalues i and $-i$ of A_2 are not real, and A_2 is not diagonalizable (Definition 7.23).

(A_3) A_3 is upper triangular, so $p_{A_3} = (t-1)^2(t-2)^2$ (Proposition 7.16(b)).

$$A_3 - I_4 = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The free unknowns are x_1, x_3; the special solutions $(1, 0, 0, 0)'$, $(0, -1, 1, 0)'$ form a basis of $\mathcal{N}(A_3 - I_4)$ (Theorem 4.8(c)), and $g_{A_3}(1) = 2$.

$$A_3 - 2I_4 = \begin{pmatrix} -1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_2]{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} -1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The free unknowns are x_2, x_4; the special solutions $(1, 1, 0, 0)'$, $(-1, 0, -1, 1)'$ form a basis of $\mathcal{N}(A_3 - 2I_4)$, and $g_{A_3}(2) = 2$. All eigenvalues of A_3 are real and both geometric multiplicities equal the algebraic ones, so A_3 is diagonalizable (Theorem 7.25).

Exercise 7.11. Let

$$A = \begin{pmatrix} 2 & 0 & -1 & -1 \\ 0 & 3 & -2 & -2 \\ 2 & 0 & 2 & -2 \\ -2 & 0 & -1 & 3 \end{pmatrix}$$

(a) Find the eigenvalues of A, a basis of each eigenspace, and an invertible P with $P^{-1}AP$ diagonal.

(b) Find P^{-1} by Gauss–Jordan elimination.

(c) Find A^k for $k \geqslant 1$, and check the result for $k = 1$.

Solution. (a)

- For $t \in \mathbb{R}$, column 2 of $tI - A$ has the entry $t-3$ in position $(2,2)$ and 0 elsewhere; expanding along it, and then along row 1,

$$\begin{aligned} \det(tI - A) &= \det \begin{pmatrix} t-2 & 0 & 1 & 1 \\ 0 & t-3 & 2 & 2 \\ -2 & 0 & t-2 & 2 \\ 2 & 0 & 1 & t-3 \end{pmatrix} = (t-3) \det \begin{pmatrix} t-2 & 1 & 1 \\ -2 & t-2 & 2 \\ 2 & 1 & t-3 \end{pmatrix} \\ &= (t-3)\Big((t-2)\Big((t-2)(t-3)-2\Big) - \Big(-2(t-3)-4\Big) + \Big(-2-2(t-2)\Big)\Big) \\ &= (t-3)\Big((t-2)(t^2-5t+4) + (2t-2) + (2-2t)\Big) \\ &= (t-3)(t-2)(t-1)(t-4) \end{aligned}$$

  So $p_A = (t-1)(t-2)(t-3)(t-4)$: the eigenvalues 1, 2, 3, 4 are distinct, and A is diagonalizable (Theorem 7.25).

- $\lambda = 1$:

$$A - I = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 2 & -2 & -2 \\ 2 & 0 & 1 & -2 \\ -2 & 0 & -1 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 2 & -2 & -2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & -3 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow \frac{1}{3}R_3]{R_2 \leftarrow \frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -3 & 0 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + 3R_3]{R_1 \leftarrow R_1 + R_3,\ R_2 \leftarrow R_2 + R_3} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $p_1 = (1,1,0,1)'$ is a basis of $\mathcal{N}(A - I)$ (Theorem 4.8(c)).

- $\lambda = 2$:

$$A - 2I = \begin{pmatrix} 0 & 0 & -1 & -1 \\ 0 & 1 & -2 & -2 \\ 2 & 0 & 0 & -2 \\ -2 & 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 2 & 0 & 0 & -2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -1 & -1 \\ -2 & 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_1} \begin{pmatrix} 2 & 0 & 0 & -2 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & -1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_3]{R_2 \leftarrow R_2 - 2R_3} \begin{pmatrix} 2 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow -R_3]{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $p_2 = (1,0,-1,1)'$ is a basis of $\mathcal{N}(A - 2I)$.

- $\lambda = 3$:

$$A - 3I = \begin{pmatrix} -1 & 0 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 2 & 0 & -1 & -2 \\ -2 & 0 & -1 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 + 2R_1} \begin{pmatrix} -1 & 0 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \begin{pmatrix} -1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -3 & -4 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_1 \leftarrow R_1 + R_2,\ R_3 \leftarrow R_3 + 3R_2} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_3]{R_2 \leftarrow R_2 + R_3} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_3 \leftarrow -R_3]{R_1 \leftarrow -R_1} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The free unknown is x_2; the special solution $p_3 = e_2$ is a basis of $\mathcal{N}(A - 3I)$.

- $\lambda = 4$:

$$A - 4I = \begin{pmatrix} -2 & 0 & -1 & -1 \\ 0 & -1 & -2 & -2 \\ 2 & 0 & -2 & -2 \\ -2 & 0 & -1 & -1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 2 & 0 & -2 & -2 \\ 0 & -1 & -2 & -2 \\ -2 & 0 & -1 & -1 \\ -2 & 0 & -1 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 2 & 0 & -2 & -2 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & -3 & -3 \\ 0 & 0 & -3 & -3 \end{pmatrix}$$

$$\xrightarrow[R_3 \leftarrow -\frac{1}{3}R_3]{R_2 \leftarrow -R_2} \begin{pmatrix} 2 & 0 & -2 & -2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -3 & -3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 3R_3]{R_1 \leftarrow R_1 + 2R_3,\ R_2 \leftarrow R_2 - 2R_3} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  The special solution $p_4 = (0, 0, -1, 1)'$ is a basis of $\mathcal{N}(A - 4I)$.

- The eigenvectors $p_1, \ldots, p_4$ are independent (Theorem 7.20). With

$$P = [p_1\ p_2\ p_3\ p_4] = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

  $P^{-1}AP = \operatorname{diag}(1, 2, 3, 4)$ (Definition 7.23).

(b) Row-reduce $[P \mid I_4]$ (Corollary 2.17(a)):

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow R_3 - R_2]{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow R_3 - R_4]{R_1 \leftarrow R_1 + R_4,\ R_2 \leftarrow R_2 + R_4} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & -1 & 0 & 0 & -1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{array}\right)$$

So

$$P^{-1} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

(c) By Proposition 7.27(a), $A^k = P\,\mathrm{diag}(1, 2^k, 3^k, 4^k)P^{-1}$. With r_i' the ith row of P^{-1}, the outer product expansion (Theorem 1.4(d)) gives $A^k = p_1 r_1' + 2^k p_2 r_2' + 3^k p_3 r_3' + 4^k p_4 r_4'$, where

$$p_1 r_1' = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad p_2 r_2' = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 1 \\ 1 & 0 & -1 & -1 \end{pmatrix}$$

$$p_3 r_3' = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad p_4 r_4' = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

So

$$A^k = \begin{pmatrix} 2^k & 0 & 1-2^k & 1-2^k \\ 0 & 3^k & 1-3^k & 1-3^k \\ 4^k - 2^k & 0 & 2^k & 2^k - 4^k \\ 2^k - 4^k & 0 & 1-2^k & 1-2^k+4^k \end{pmatrix} \qquad (k \geqslant 1)$$

For $k = 1$ this is

$$\begin{pmatrix} 2 & 0 & -1 & -1 \\ 0 & 3 & -2 & -2 \\ 2 & 0 & 2 & -2 \\ -2 & 0 & -1 & 3 \end{pmatrix} = A$$

Exercise 7.12. Let

$$A = \begin{pmatrix} 2 & -2 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 0 & 0 & 3 & 0 \\ -1 & -2 & 1 & 2 \end{pmatrix}$$

Show that $\operatorname{rank}(A - 3I_4) = 1$, diagonalize A, find a formula for A^k ($k \geqslant 0$), and compute A^3.

Solution.

- Subtracting 3 from the diagonal entries of A,

$$A - 3I_4 = \begin{pmatrix} -1 & -2 & 1 & -1 \\ -1 & -2 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ -1 & -2 & 1 & -1 \end{pmatrix} = uv'$$

  with $u = (1, 1, 0, 1)'$ and $v' = (-1, -2, 1, -1)$: rows 1, 2 and 4 equal v', and row 3 is 0. So the row space of $A - 3I_4$ is $\operatorname{span}\{v'\}$, of dimension 1 as $v' \neq 0$ (Proposition 3.13(a)), and $\operatorname{rank}(A - 3I_4) = 1$ (Definition 4.11). Hence $g_A(3) = 4 - 1 = 3$ (Definition 7.21) and $a_A(3) \geqslant 3$ (Theorem 7.22).

- So at least three of the four eigenvalues of A, listed with multiplicity, are 3 (Definition 7.10); let μ be the fourth. By Theorem 7.19, $3 + 3 + 3 + \mu = \operatorname{tr} A = 2 + 1 + 3 + 2 = 8$, so $\mu = -1$ and $p_A = (t - 3)^3(t + 1)$.

- $\mathcal{N}(A - 3I_4) = \mathcal{N}(uv') = \mathcal{N}(v')$, as $uv'x = (v'x)u$ and $u \neq 0$. The equation $v'x = 0$, that is, $x_1 + 2x_2 - x_3 + x_4 = 0$, with x_2, x_3, x_4 free, has the special solutions $p_1 = (-2, 1, 0, 0)'$, $p_2 = (1, 0, 1, 0)'$ and $p_3 = (-1, 0, 0, 1)'$, a basis of $\mathcal{N}(A - 3I_4)$ (Theorem 4.8(c)). Next $Au = 3u + u(v'u) = 3u - 4u = -u$, so $p_4 = u$ spans $\mathcal{N}(A + I_4)$ (Theorem 7.22: $g_A(-1) \leqslant a_A(-1) = 1$).

- By Lemma 7.24(b) and Theorem 4.5(d), $p_1, \ldots, p_4$ form a basis of $\mathbb{R}^4$, so $P^{-1}AP = \Lambda$ with $P = [p_1\ p_2\ p_3\ p_4]$ and $\Lambda = \operatorname{diag}(3, 3, 3, -1)$ (Definition 7.23). The row $w' = -\frac{1}{4}v' = \frac{1}{4}(1, 2, -1, 1)$ satisfies $w'p_j = 0$ for $j \leqslant 3$ (as $v'p_j = 0$) and $w'p_4 = -\frac{1}{4}v'u = 1$; so $w'P = e_4'$, and $w' = e_4'P^{-1} = \operatorname{row}_4(P^{-1})$.

By Proposition 7.27(a),

$$A^k = P \operatorname{diag}\big(3^k, 3^k, 3^k, (-1)^k\big)P^{-1} = 3^k M_1 + (-1)^k M_2$$

with $M_1 = P \operatorname{diag}(1, 1, 1, 0)P^{-1}$ and $M_2 = P \operatorname{diag}(0, 0, 0, 1)P^{-1}$. By the outer product expansion (Theorem 1.4(d)), $M_2 = p_4 \operatorname{row}_4(P^{-1}) = -\frac{1}{4}uv'$, and $M_1 = I_4 - M_2$, as $M_1 + M_2 = PP^{-1}$. So

$$A^k = 3^k I_4 + \tfrac{1}{4}\big(3^k - (-1)^k\big)uv' \qquad (k \geqslant 0)$$

For $k = 1$ this gives back $A = 3I_4 + uv'$, and

$$A^3 = 27I_4 + 7uv' = \begin{pmatrix} 20 & -14 & 7 & -7 \\ -7 & 13 & 7 & -7 \\ 0 & 0 & 27 & 0 \\ -7 & -14 & 7 & 20 \end{pmatrix}$$

Exercise 7.13. Let

$$A = \begin{pmatrix} 0 & 2 & -1 & 0 \\ 1 & 1 & -1 & 0 \\ -2 & 2 & 1 & 0 \\ 1 & -1 & 0 & -2 \end{pmatrix}, \qquad p_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad p_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad p_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad p_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

(a) Show that $p_1, \dots, p_4$ are eigenvectors of A, and find p_A, $\operatorname{tr} A$ and $\det A$.

(b) Diagonalize A and compute A^{-1}.

(c) For $x_0 = (3, 2, 2, 1)'$, find $A^k x_0$ ($k \geqslant 0$), and check $A^3 x_0$ by computing Ax_0, $A^2 x_0$, $A^3 x_0$ in turn.

Solution. (a) Multiplying out, $Ap_1 = (2, 2, 0, 0)' = 2p_1$, $Ap_2 = (1, 1, 1, 0)' = p_2$, $Ap_3 = (-1, 0, -1, -1)' = -p_3$ and $Ap_4 = \operatorname{col}_4(A) = (0, 0, 0, -2)' = -2p_4$. The eigenvalues 2, 1, -1, -2 are distinct, so $p_1, \dots, p_4$ form a basis of $\mathbb{R}^4$ (Theorem 7.20), and $P^{-1}AP = \Lambda$ with $P = [p_1\ p_2\ p_3\ p_4]$ and $\Lambda = \operatorname{diag}(2, 1, -1, -2)$ (Definition 7.23). By the same definition, 2, 1, -1, -2 are the eigenvalues of A, listed with multiplicity; so $p_A = (t-2)(t-1)(t+1)(t+2)$ (Definition 7.10), $\operatorname{tr} A = 2 + 1 - 1 - 2 = 0$ and $\det A = 2 \cdot 1 \cdot (-1) \cdot (-2) = 4$ (Theorem 7.19). Directly, $\operatorname{tr} A = 0 + 1 + 1 - 2 = 0$.

(b)

$$P^{-1} = \begin{pmatrix} 1 & 0 & -1 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 1 \end{pmatrix}$$

Check: by Theorem 1.4(d), the (i, j) entry of the product of this matrix and P is its row i times p_j. For row 1, $(1, 0, -1, 0)p_j$ is 1, $1 - 1 = 0$, $1 - 1 = 0$, 0 ($j = 1, 2, 3, 4$); for row 2, $(-1, 1, 1, 0)p_j$ is $-1 + 1 = 0$, $-1 + 1 + 1 = 1$, $-1 + 1 = 0$, 0; for row 3, $(1, -1, 0, 0)p_j$ is $1 - 1 = 0$, $1 - 1 = 0$, 1, 0; for row 4, $(-1, 1, 0, 1)p_j$ is $-1 + 1 = 0$, $-1 + 1 = 0$, $-1 + 1 = 0$, 1. So the product is I_4, and the matrix is P^{-1} (Theorem 2.16). By Proposition 7.27(c), $A^{-1} = P\Lambda^{-1}P^{-1} = (P\Lambda^{-1})P^{-1}$ with $\Lambda^{-1} = \operatorname{diag}(\frac{1}{2}, 1, -1, -\frac{1}{2})$; the columns of $P\Lambda^{-1}$ are those of P times $\frac{1}{2}$, 1, -1, $-\frac{1}{2}$ (Proposition 1.13(a)):

$$P\Lambda^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 2 & -2 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & -2 & -1 \end{pmatrix}, \qquad A^{-1} = \frac{1}{2}\begin{pmatrix} -3 & 4 & 1 & 0 \\ -1 & 2 & 1 & 0 \\ -4 & 4 & 2 & 0 \\ -1 & 1 & 0 & -1 \end{pmatrix}$$

Check: $\operatorname{row}_1(A)\operatorname{col}_1(A^{-1}) = \frac{1}{2}(-2 + 4) = 1$ and $\operatorname{row}_1(A)\operatorname{col}_2(A^{-1}) = \frac{1}{2}(4 - 4) = 0$.

(c) $c = P^{-1}x_0 = (1, 1, 1, 0)'$, that is, $x_0 = p_1 + p_2 + p_3$. By Proposition 7.27(a),

$$A^k x_0 = 2^k p_1 + p_2 + (-1)^k p_3 = \left(2^k + 1 + (-1)^k,\ 2^k + 1,\ 1 + (-1)^k,\ (-1)^k\right)'$$

For $k = 3$ this is $(8, 9, 0, -1)'$. Directly, $Ax_0 = (2, 3, 0, -1)'$, $A^2 x_0 = A(2, 3, 0, -1)' = (6, 5, 2, 1)'$ and $A^3 x_0 = A(6, 5, 2, 1)' = (8, 9, 0, -1)'$.

Exercise 7.14. The matrix A of Exercise 7.3 has the eigenvalues -1, 1, 2, 3, with the eigenvectors $(1, 0, 0, 1)'$, $(1, 0, 1, 1)'$, $(1, 0, 1, 0)'$, $(-1, -1, -1, 1)'$. Without computing the matrices, find the eigenvalues, the trace and the determinant of

(a) $A - 2I$ and $-3A$;

(b) A^2, A^{-1} and $A^3 - 4A$;

(c) A'.

(d) Give an eigenvector of $A^3 - 4A$ for each of its eigenvalues, and check $\operatorname{tr}(A^2)$ by computing the diagonal entries of A^2.

Solution. (a) By Proposition 7.18(c), $A-2I$ has the eigenvalues -3, -1, 0, 1, so $\operatorname{tr}(A-2I)=-3$ and $\det(A-2I)=0$ (Theorem 7.19); and $-3A$ has the eigenvalues 3, -3, -6, -9, so $\operatorname{tr}(-3A)=-15$ and $\det(-3A)=-486$.

(b) $A=P\Lambda P^{-1}$ with $\Lambda=\operatorname{diag}(-1,1,2,3)$ (Exercise 7.3(c)), so by Proposition 7.29(b):

- A^2 has the eigenvalues 1, 1, 4, 9; $\operatorname{tr}(A^2)=15$, $\det(A^2)=36$.
- 0 is not an eigenvalue, so A is invertible (Theorem 7.19), and A^{-1} has the eigenvalues -1, 1, $\frac{1}{2}$, $\frac{1}{3}$; $\operatorname{tr}(A^{-1})=-1+1+\frac{1}{2}+\frac{1}{3}=\frac{5}{6}$, and $\det(A^{-1})=-\frac{1}{6}$.
- For $f(t)=t^3-4t$, $f(-1)=3$, $f(1)=-3$, $f(2)=0$, $f(3)=15$: A^3-4A has the eigenvalues 3, -3, 0, 15; $\operatorname{tr}(A^3-4A)=15$, $\det(A^3-4A)=0$.

(c) By Proposition 7.18(d), A' has the eigenvalues -1, 1, 2, 3; $\operatorname{tr}A'=5$, $\det A'=-6$.

(d)

- By Proposition 7.18(a), $f(A)x=f(\lambda)x$: $(1,0,0,1)'$ belongs to 3, $(1,0,1,1)'$ to -3, $(1,0,1,0)'$ to 0, and $(-1,-1,-1,1)'$ to 15.
- By Theorem 1.4(d), the ith diagonal entry of A^2 is $\operatorname{row}_i(A)\operatorname{col}_i(A)$: $0\cdot0+0\cdot0+2\cdot1+(-1)(-2)=4$, $0\cdot0+3\cdot3+0\cdot0+0\cdot(-2)=9$, $1\cdot2+0\cdot0+1\cdot1+(-1)\cdot2=1$ and $(-2)(-1)+(-2)\cdot0+2\cdot(-1)+1\cdot1=1$; so $\operatorname{tr}(A^2)=4+9+1+1=15$.

Exercise 7.15. Let B be the matrix of Example 7.28.

(a) Without computing these matrices, find the eigenvalues and eigenvectors of B^3, B^{-1}, $B+cI_4$ ($c\in\mathbb{R}$) and $q(B)=B^2-4B+3I_4$, and the numbers $\operatorname{tr}B^3$, $\det B^3$, $\operatorname{tr}B^{-1}$, $\det B^{-1}$.

(b) Show that $q(B)=0$. Deduce $B^{-1}=\frac{1}{3}(4I_4-B)$ and $B^3=13B-12I_4$, and check the values of $\operatorname{tr}B^{-1}$ and $\operatorname{tr}B^3$.

Solution. (a) By Example 7.28, B has the eigenvalue 3, with the eigenvectors p_1,p_2, and the eigenvalue 1, with the eigenvectors p_3,p_4, and $B=P\Lambda P^{-1}$ with $\Lambda=\operatorname{diag}(3,3,1,1)$; so B is invertible (Proposition 7.27(c)). By Proposition 7.18(a),(b), p_1,p_2 and p_3,p_4 are eigenvectors of B^3 belonging to 27 and 1, of B^{-1} belonging to $\frac{1}{3}$ and 1, of $B+cI_4$ belonging to $3+c$ and $1+c$, and of $q(B)$ belonging to $q(3)=9-12+3=0$ and $q(1)=0$. By Proposition 7.29(b), these are the eigenvalues, listed with multiplicity (each twice), and $\operatorname{tr}B^3=56$, $\det B^3=729$, $\operatorname{tr}B^{-1}=\frac{8}{3}$, $\det B^{-1}=\frac{1}{9}$. For B^3, B^{-1} and $B+cI_4$, each of the two eigenvalues has algebraic multiplicity 2, so its eigenspace has dimension at most 2 (Theorem 7.22); it contains the independent p_1,p_2, respectively p_3,p_4, so it is spanned by them (Theorem 4.5(d)). For $q(B)$, the eigenspace for 0 contains $p_1,\dots,p_4$, so it is $\mathbb{R}^4$.

(b) $q(B)$ maps the basis $p_1,\dots,p_4$ of $\mathbb{R}^4$ to 0, so $q(B)x=0$ for every $x\in\mathbb{R}^4$; with $x=e_j$, every column of $q(B)$ is 0 (Theorem 1.4(b)), and $q(B)=0$ (multiplying out $(B-I_4)(B-3I_4)=q(B)$ gives 0 as well). Hence $B(4I_4-B)=4B-B^2=3I_4$, and $B^{-1}=\frac{1}{3}(4I_4-B)$ (Theorem 2.16):

$$B^{-1}=\frac{1}{3}\begin{pmatrix}3&-2&2&-2\\2&1&2&0\\0&2&1&2\\-2&2&-2&3\end{pmatrix}$$

Also $B^2=4B-3I_4$, so $B^3=4B^2-3B=4(4B-3I_4)-3B=13B-12I_4$. With $\operatorname{tr}B=8$: $\operatorname{tr}B^{-1}=\frac{1}{3}(16-8)=\frac{8}{3}$ and $\operatorname{tr}B^3=104-48=56$.

Exercise 7.16. Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & -1 & 1 & 2 \\ -1 & 1 & 2 & 1 \end{pmatrix}$$

and let $Q = \frac{1}{2}G$ be the orthogonal matrix of Example 7.31.

(a) Show that $\frac{1}{2}\mathbf{1}_4$ is an eigenvector of A of length 1, and that $T = Q'AQ$ is upper triangular. Read off the eigenvalues of A.

(b) Use (a) to find $\operatorname{tr}(A^2 + I_4)$ and $\det(A^2 + I_4)$, and check the trace directly.

(c) Is A diagonalizable? Is there an orthogonal U with $U'AU$ diagonal?

Solution. (a) Every row of A adds up to 3, so $A\mathbf{1}_4 = 3\mathbf{1}_4$, and $\|\frac{1}{2}\mathbf{1}_4\| = 1$. As $G' = G$,

$$AG = \begin{pmatrix} 3 & 1 & 1 & -1 \\ 3 & 1 & -1 & 1 \\ 3 & -3 & 1 & 3 \\ 3 & -3 & -1 & -3 \end{pmatrix}, \qquad T = Q'AQ = \tfrac{1}{4}G(AG) = \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

As $Q' = Q^{-1}$ (Proposition 6.13(b)), T is similar to A, and by Theorem 7.15(a) and Proposition 7.16(b) the eigenvalues of A are 3, 2, 1, -2.

(b) As $Q'Q = QQ' = I_4$, $A = QTQ'$ and $A^2 + I_4 = Q(T^2 + I_4)Q'$, with

$$T^2 + I_4 = \begin{pmatrix} 10 & -5 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 5 \end{pmatrix}$$

By Theorem 7.15(a), Proposition 7.16(b) and Theorem 7.19, $\operatorname{tr}(A^2 + I_4) = 10 + 5 + 2 + 5 = 22$ and $\det(A^2 + I_4) = 10 \cdot 5 \cdot 2 \cdot 5 = 500$, as Corollary 7.32(a) predicts ($q(t) = t^2 + 1$ has $q(3) = 10$, $q(2) = 5$, $q(1) = 2$, $q(-2) = 5$). Directly,

$$A^2 + I_4 = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 1 & 4 & 3 & 2 \\ -1 & 1 & 7 & 3 \\ 1 & -1 & 3 & 7 \end{pmatrix}$$

with trace 22.

(c) A has four distinct real eigenvalues, so it is diagonalizable (Theorem 7.25). If $U'AU = \Lambda$ with U orthogonal and Λ diagonal, then $A = U\Lambda U'$ would be symmetric (Theorem 1.9); but $a_{14} = 0 \neq -1 = a_{41}$. So no such U exists.

Exercise 7.17. Let

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix} \in \mathbb{R}^{2\times 3}, \qquad B = \begin{pmatrix} -1 & 1 \\ 1 & 1 \\ 1 & -1 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

(a) Compute AB and BA. Find p_{AB}, the eigenvalues of AB, and an eigenvector for each.

(b) Use Theorem 7.33 to find p_{BA} and the eigenvalues of BA, and check p_{BA} by expanding $\det(tI - BA)$.

(c) For each eigenvector x of AB from (a), check that Bx is an eigenvector of BA. Find a basis of $\mathcal{N}(BA)$, and compare $\operatorname{rank}(BA)$ with the number of nonzero eigenvalues of BA.

Solution. (a)

- Multiplying out,
$$AB = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 2 & 2 \\ -1 & 0 & -2 \end{pmatrix}$$
- By Definition 7.7(b) ($n = 2$), $p_{AB} = t^2 - t - 2 = (t-2)(t+1)$: the eigenvalues of AB are 2 and -1.
- $(AB - 2I)x = 0$ reads $-x_1 + x_2 = 0$ and $2x_1 - 2x_2 = 0$, so $x = (1, 1)'$; $(AB + I)x = 0$ reads $2x_1 + x_2 = 0$ (twice), so $x = (-1, 2)'$.

(b)

- By Theorem 7.33(a) ($m = 2$, $n = 3$), $p_{BA} = t\,p_{AB} = t(t-2)(t+1)$, and by (b) of that theorem the eigenvalues of BA are 0, 2, -1.
- Check: for $t \in \mathbb{R}$, column 2 of $tI - BA$ has the entry $t - 2$ in position $(2, 2)$ and 0 elsewhere, so
$$\begin{aligned} \det(tI - BA) &= \det \begin{pmatrix} t-1 & 0 & -2 \\ -1 & t-2 & -2 \\ 1 & 0 & t+2 \end{pmatrix} = (t-2) \det \begin{pmatrix} t-1 & -2 \\ 1 & t+2 \end{pmatrix} \\ &= (t-2)(t^2 + t) = t(t-2)(t+1) \end{aligned}$$

(c)

- $B(1, 1)' = (0, 2, 0)'$ and $BA(0, 2, 0)' = (0, 4, 0)' = 2(0, 2, 0)'$; $B(-1, 2)' = (3, 1, -3)'$ and $BA(3, 1, -3)' = (-3, -1, 3)' = -(3, 1, -3)'$.
- Row-reduce BA:
$$\begin{pmatrix} 1 & 0 & 2 \\ 1 & 2 & 2 \\ -1 & 0 & -2 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{2} R_2} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
The special solution $(-2, 0, 1)'$ is a basis of $\mathcal{N}(BA)$, and $\operatorname{rank}(BA) = 2$.
- BA has the two nonzero eigenvalues 2, -1, as many as its rank: BA has three distinct eigenvalues, so it is diagonalizable (Theorem 7.25), and Proposition 7.29(a) applies.

Exercise 7.18. Let
$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

(a) Compute BA and AB, find p_{BA}, and deduce p_{AB}.

(b) Find eigenvectors of AB for its nonzero eigenvalues from those of BA, and a basis of $\mathcal{N}(AB)$. Diagonalize AB.

(c) Show that $\operatorname{tr}(AB)^k = 2^k + (-1)^k$ for $k \geqslant 1$.

Solution. (a)
$$BA = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \qquad AB = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}$$
By Definition 7.7(b) ($n = 2$), $p_{BA} = t^2 - t - 2 = (t-2)(t+1)$. By Theorem 7.33(a), applied with B and A in place of its A and B ($m = 2$, $n = 4$), $p_{AB} = t^2 p_{BA} = t^2(t-2)(t+1)$.

(b)

- $BA(1,1)' = (2,2)'$ and $BA(1,-2)' = (-1,2)'$: $(1,1)'$ and $(1,-2)'$ are eigenvectors of BA belonging to the simple eigenvalues 2 and -1, so each is a basis of its eigenspace (Theorem 7.22). By Theorem 7.33(c), applied as in (a),

  $$v_1 = A(1,1)' = (1,2,1,0)', \qquad v_2 = A(1,-2)' = (1,-1,-2,3)'$$

  are eigenvectors of AB belonging to 2 and -1; indeed $ABv_1 = (2,4,2,0)' = 2v_1$ and $ABv_2 = (-1,1,2,-3)' = -v_2$.
- B is in RREF, and $Bx = 0$ reads $x_1+x_3 = 0$, $x_2+x_4 = 0$, with the special solutions $n_1 = (-1,0,1,0)'$ and $n_2 = (0,-1,0,1)'$ (Theorem 4.8(c)). As $ABx = A(Bx)$, $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$, so $g_{AB}(0) \geqslant 2$ (Theorem 4.5(a)); and $g_{AB}(0) \leqslant a_{AB}(0) = 2$ (Theorem 7.22). So $\mathcal{N}(B) = \mathcal{N}(AB)$ (Theorem 4.5(a)), with the basis n_1, n_2.
- All eigenvalues 2, -1, 0, 0 of AB are real, and $g_{AB} = a_{AB}$ for each of them; so AB is diagonalizable (Theorem 7.25), and by the paragraph after that theorem, $Q^{-1}(AB)Q = \Lambda$ with $Q = [v_1\ v_2\ n_1\ n_2]$ and $\Lambda = \operatorname{diag}(2,-1,0,0)$.

(c) By (b), $AB = Q\Lambda Q^{-1}$, so by Proposition 7.29(b) (with $f = t^k$), $\operatorname{tr}(AB)^k = 2^k + (-1)^k + 0^k + 0^k = 2^k + (-1)^k$ for $k \geqslant 1$. For $k = 1$, $\operatorname{tr} AB = 1 + 1 + 0 - 1 = 1$.

Exercise 7.19. Let

$$A = \begin{pmatrix} 0 & -1 & 0 & -1 \\ 2 & -1 & 0 & -2 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 0 \end{pmatrix}$$

(a) Find p_A.

(b) Compute A^2, A^3 and A^4, and check that $p_A(A) = 0$.

(c) Use Corollary 7.39(b) to find A^{-1}, and check that $AA^{-1} = I$.

(d) Use Corollary 7.39(a) to write A^5 as a linear combination of I, A, A^2, A^3, and compute A^5.

Solution. (a) For $t \in \mathbb{R}$, column 3 of $tI - A$ has the entry $t-1$ in position $(3,3)$ and 0 elsewhere; expanding along it, and then along row 1,

$$\det(tI - A) = \det \begin{pmatrix} t & 1 & 0 & 1 \\ -2 & t+1 & 0 & 2 \\ 1 & 0 & t-1 & -1 \\ 1 & 1 & 0 & t \end{pmatrix} = (t-1)\det \begin{pmatrix} t & 1 & 1 \\ -2 & t+1 & 2 \\ 1 & 1 & t \end{pmatrix}$$

$$= (t-1)\Big(t\Big((t+1)t-2\Big) - (-2t-2) + \Big(-2-(t+1)\Big)\Big) = (t-1)(t^3+t^2-t-1)$$

So $p_A = (t-1)(t^3+t^2-t-1) = t^4 - 2t^2 + 1$: $c_0 = 1$, $c_1 = 0$, $c_2 = -2$, $c_3 = 0$, $c_4 = 1$.

(b) Multiplying out,

$$A^2 = \begin{pmatrix} -1 & 2 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 1 & 2 \\ -2 & 2 & 0 & 3 \end{pmatrix}, \qquad A^3 = A^2A = \begin{pmatrix} 2 & -3 & 0 & -3 \\ 2 & -1 & 0 & -2 \\ -3 & 0 & 1 & 3 \\ 1 & -3 & 0 & -2 \end{pmatrix}, \qquad A^4 = A^3A = \begin{pmatrix} -3 & 4 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ -4 & 0 & 1 & 4 \\ -4 & 4 & 0 & 5 \end{pmatrix}$$

Adding entry by entry, $p_A(A) = A^4 - 2A^2 + I = 0$, as Theorem 7.38 asserts.

(c) As $c_0 = (-1)^4 \det A$ (Definition 7.7(b)), $\det A = 1 \neq 0$, and A is invertible (Theorem 5.18(a)). By Corollary 7.39(b), $A^{-1} = -\frac{1}{c_0}(c_1 I + c_2 A + c_3 A^2 + c_4 A^3) = 2A - A^3$, that is,

$$A^{-1} = \begin{pmatrix} -2 & 1 & 0 & 1 \\ 2 & -1 & 0 & -2 \\ 1 & 0 & 1 & -1 \\ -3 & 1 & 0 & 2 \end{pmatrix}, \qquad AA^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(d)

- By Corollary 7.39(a), $A^4 = -(c_0 I + c_1 A + c_2 A^2 + c_3 A^3) = 2A^2 - I$, as found in (b). Then $A^5 = A^4 A = 2A^3 - A$.
- So

$$A^5 = 2\begin{pmatrix} 2 & -3 & 0 & -3 \\ 2 & -1 & 0 & -2 \\ -3 & 0 & 1 & 3 \\ 1 & -3 & 0 & -2 \end{pmatrix} - \begin{pmatrix} 0 & -1 & 0 & -1 \\ 2 & -1 & 0 & -2 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 4 & -5 & 0 & -5 \\ 2 & -1 & 0 & -2 \\ -5 & 0 & 1 & 5 \\ 3 & -5 & 0 & -4 \end{pmatrix}$$

Problems

Problem 7.1 ([6, Exercise 7.38])**.** Let $n \geqslant 2$ and $a_0, \dots, a_{n-1} \in \mathbb{R}$. Show that the companion matrix

$$C = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{pmatrix} \in \mathbb{R}^{n\times n}$$

whose entries are $c_{i,i+1} = 1$ $(i < n)$, the last row $(-a_0, \dots, -a_{n-1})$ and 0 elsewhere, has the characteristic polynomial $p_C(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0$.

Solution. Fix $t \in \mathbb{R}$. For $2 \leqslant j \leqslant n$ and $x \in \mathbb{R}$, let $A_j(x) \in \mathbb{R}^{j\times j}$ have the entries t in positions (i,i) and -1 in positions $(i, i+1)$ for $i < j$, the last row $(a_0, a_1, \dots, a_{j-2}, x)$, and 0 elsewhere. Then $tI - C = A_n(t + a_{n-1})$.

- Let $j \geqslant 3$. The last column of $A_j(x)$ has the entries -1 at $(j-1, j)$ and x at (j,j), and 0 elsewhere. Deleting row j and column j leaves an upper triangular matrix with diagonal entries t, of determinant t^{j-1} (Theorem 5.14(c)). Deleting row $j-1$ and column j leaves $A_{j-1}(a_{j-2})$. Expanding along the last column (Theorem 5.21),

$$\det A_j(x) = x\, t^{j-1} + (-1)\cdot(-1)^{(j-1)+j} \det A_{j-1}(a_{j-2}) = x\, t^{j-1} + \det A_{j-1}(a_{j-2})$$

- For $j = 2$, $\det A_2(x) = \det \left(\begin{smallmatrix} t & -1 \\ a_0 & x \end{smallmatrix}\right) = tx + a_0$.
- Hence

$$\begin{aligned} \det(tI - C) &= (t + a_{n-1})t^{n-1} + \det A_{n-1}(a_{n-2}) = t^n + a_{n-1}t^{n-1} + a_{n-2}t^{n-2} + \det A_{n-2}(a_{n-3}) \\ &= \cdots = t^n + a_{n-1}t^{n-1} + \cdots + a_2 t^2 + \det A_2(a_1) = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0 \end{aligned}$$

  (for $n = 2$, directly $\det A_2(t + a_1) = t(t + a_1) + a_0$). This holds for every real t, so p_C is this polynomial (Definition 7.7(a), Proposition 7.5(d)).

Problem 7.2 ([8, Ch. 21, Exercise 8])**.** Let $n \geqslant 2$. Show that for $B \in \mathbb{R}^{n\times n}$ to be similar to $A \in \mathbb{R}^{n\times n}$ it is not sufficient that B have the same rank, determinant, trace and characteristic polynomial as A.

Solution. Take $A = I_n$ and an upper triangular $B \neq I_n$ with all diagonal entries 1, for instance $B = I_n + E_{12}$ (Definition 3.17).

- $\det B = 1 = \det I_n$ (Theorem 5.14(c)), so B is invertible and $\operatorname{rank} B = n = \operatorname{rank} I_n$ (Theorem 5.18(a)); $\operatorname{tr} B = n = \operatorname{tr} I_n$; and $p_B = (t-1)^n = p_{I_n}$ (Proposition 7.16(b)).
- If B were similar to I_n, there would be an invertible C with $B = C^{-1} I_n C = C^{-1}C = I_n$ ✘

Problem 7.3 ([8, Ch. 21, Exercise 13])**.** Let x_1 and x_2 be eigenvectors of $A \in \mathbb{R}^{n\times n}$ belonging to λ_1 and λ_2, and let c_1, c_2 be nonzero numbers. When is $x = c_1x_1 + c_2x_2$ an eigenvector of A?

Solution.

- $Ax = c_1\lambda_1x_1 + c_2\lambda_2x_2 = \lambda_1x + (\lambda_2 - \lambda_1)c_2x_2$.
- If $\lambda_1 = \lambda_2$, then $Ax = \lambda_1x$: x is an eigenvector unless $x = 0$, that is, unless $x_2 = -(c_1/c_2)x_1$.
- If $\lambda_1 \neq \lambda_2$, then x_1 and x_2 are independent (Theorem 7.20), and so are x and x_2: if $ax + bx_2 = 0$, then $ac_1x_1 + (ac_2 + b)x_2 = 0$, so $ac_1 = 0$, $a = 0$, and then $b = 0$. If $Ax = cx$ for some c, then $(\lambda_1 - c)x + (\lambda_2 - \lambda_1)c_2x_2 = 0$, so $(\lambda_2 - \lambda_1)c_2 = 0$ ✘ Hence x is not an eigenvector.

So x is an eigenvector of A if and only if $\lambda_1 = \lambda_2$ and $x \neq 0$.

Problem 7.4 ([8, Ch. 21, Exercise 2], [5, §21.1])**.** Let $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{k\times k}$ and $X \in \mathbb{R}^{n\times k}$ satisfy $AX = XB$. A subspace U of $\mathbb{R}^n$ is **invariant** relative to A if $Au \in U$ for every $u \in U$.

(a) Show that $\mathcal{C}(X)$ is invariant relative to A.

(b) Show that if X has full column rank, then every real eigenvalue of B is an eigenvalue of A.

Solution. (a) Every $u \in \mathcal{C}(X)$ is $u = Xr$ for some $r \in \mathbb{R}^k$ (Definition 4.7), so $Au = AXr = XBr = X(Br) \in \mathcal{C}(X)$.

(b) Let $By = \lambda y$ with $\lambda \in \mathbb{R}$ and $y \neq 0$. Then $A(Xy) = XBy = X(\lambda y) = \lambda(Xy)$. If X has full column rank, then $\mathcal{N}(X) = \{0\}$ (Theorem 4.13(b)), so $Xy \neq 0$: λ is an eigenvalue of A, and Xy is an eigenvector belonging to it.

Problem 7.5 ([8, Ch. 21, Exercise 14])**.** Let $A \in \mathbb{R}^{n\times n}$, and let $Q \in \mathbb{R}^{n\times n}$ be invertible with $Q^{-1}AQ = D = \operatorname{diag}(d_1, \ldots, d_n)$. Let r_i' be row i of Q^{-1} $(1 \leqslant i \leqslant n)$. Show:

(a) $(Q^{-1})'$ diagonalizes A'.

(b) $d_1, \ldots, d_n$ are the eigenvalues of A'.

(c) r_i is an eigenvector of A' belonging to d_i.

Solution. (a) By Theorem 2.13(c),(a), $(Q^{-1})' = (Q')^{-1}$, so $\left((Q^{-1})'\right)^{-1} = Q'$. By Theorem 1.9, $D = D' = (Q^{-1}AQ)' = Q'A'(Q^{-1})' = \left((Q^{-1})'\right)^{-1}A'(Q^{-1})'$.

(b) By (a), $\left((Q^{-1})'\right)^{-1}A'(Q^{-1})' = D$; so by Definition 7.23, applied to A' and $(Q^{-1})'$, $d_1, \ldots, d_n$ are the eigenvalues of A'.

(c) Column i of $(Q^{-1})'$ is the transposed row i of Q^{-1}, that is, r_i (Definition 1.8). By Definition 7.23, applied as in (b), the columns of $(Q^{-1})'$ are eigenvectors of A': $A'r_i = d_ir_i$ with $r_i \neq 0$.

Problem 7.6 ([6, Exercise 7.37])**.** Let $A \in \mathbb{R}^{n\times n}$, and let λ be a real eigenvalue of A. A vector $w \in \mathbb{R}^n$, $w \neq 0$, with $w'A = \lambda w'$ is a **left eigenvector** of A belonging to λ.

(a) Show that a left eigenvector of A belonging to λ is an eigenvector of A' belonging to λ.

(b) Let $\lambda \neq \mu$ be real eigenvalues of A. Show that every left eigenvector of A belonging to μ is orthogonal to every eigenvector of A belonging to λ.

Solution. (a) Let $w'A = \lambda w'$ with $w \neq 0$. By Theorem 1.9, $A'w = (w'A)' = (\lambda w')' = \lambda w$; so w is an eigenvector of A' belonging to λ (Definition 7.1).

(b) Let $Ax = \lambda x$ with $x \neq 0$, and $w'A = \mu w'$ with $w \neq 0$. By Theorem 1.3,

$$w'Ax = w'(\lambda x) = \lambda(w'x), \qquad w'Ax = (\mu w')x = \mu(w'x)$$

So $(\lambda - \mu)\, w'x = 0$, and as $\lambda \neq \mu$, $w'x = 0$: w and x are orthogonal (Definition 6.1).

Problem 7.7 ([2, Exercises 7.6, 7.7]). (a) Let $A \in \mathbb{R}^{n\times n}$, $A \neq 0$, satisfy $A^n = 0$. Show that A is not diagonalizable.

(b) Find a 2×2 matrix that is not diagonalizable.

Solution. (a) Suppose $P^{-1}AP = D = \operatorname{diag}(d_1, \dots, d_n)$ with P invertible. Then $A = PDP^{-1}$, and by Proposition 7.27(a), $0 = A^n = P \operatorname{diag}(d_1^n, \dots, d_n^n)P^{-1}$. Multiplying by P^{-1} on the left and by P on the right gives $\operatorname{diag}(d_1^n, \dots, d_n^n) = 0$; so every $d_i = 0$, $D = 0$ and $A = PDP^{-1} = 0$ ✘

(b) Let $N = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Multiplying out, $N^2 = 0$, while $N \neq 0$. By (a) (with $n = 2$), N is not diagonalizable.

Problem 7.8 ([6, Exercise 7.36(a), (b)]). Let $x, y \in \mathbb{R}^n$.

(a) Show that xy' has $n - 1$ zero eigenvalues and one eigenvalue $y'x$.

(b) Show that $\det(I_n + xy') = 1 + y'x$.

Solution. (a) For $n = 1$, $xy' = (x_1y_1) = (y'x)$, and its only eigenvalue is $y'x$ (Definition 7.7(b)). Let $n \geqslant 2$, and let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of xy', listed with multiplicity (Definition 7.10).

- If $y = 0$, then $xy' = 0$, $p_{xy'} = t^n$ (Proposition 7.16(b)), and all n eigenvalues are $0 = y'x$.
- Let $y \neq 0$. The only row y' of the $1 \times n$ matrix y' is nonzero, so $\operatorname{rank} y' = 1$ (Proposition 3.13(a), Definition 4.11), and $\dim \mathcal{N}(y') = n - 1 \geqslant 1$ (Theorem 4.12): there are $n - 1$ linearly independent vectors $u_1, \dots, u_{n-1}$ with $y'u_i = 0$. Each of them satisfies $xy'u_i = x(y'u_i) = 0$ (Theorem 1.3); so $\mathcal{N}(y') \subseteq \mathcal{N}(xy')$, $g_{xy'}(0) \geqslant n-1 \geqslant 1$ (Theorem 4.5(a)), 0 is an eigenvalue of xy', and $a_{xy'}(0) \geqslant n-1$ (Theorem 7.22).
- So we may number the eigenvalues so that $\lambda_1 = \cdots = \lambda_{n-1} = 0$. By Theorem 7.19 and Theorem 1.11(b),
$$\lambda_n = \lambda_1 + \cdots + \lambda_n = \operatorname{tr}(xy') = \operatorname{tr}(y'x) = y'x$$

(b) By (a) and Proposition 7.18(c) (with $c = 1$), the eigenvalues of $I_n + xy'$ are 1 ($n - 1$ times) and $1 + y'x$. The determinant is the product of the eigenvalues (Theorem 7.19), so $\det(I_n + xy') = 1^{n-1}(1 + y'x) = 1 + y'x$. (This is also the case $u = x$, $v = y$ of Theorem 5.31.)

Problem 7.9 ([6, Exercises 7.56, 7.69]). (a) Let $A = (a_{ij})$ and $B = (b_{ij})$ be upper triangular $n \times n$ matrices, and let $1 \leqslant k < n$, such that $a_{ij} = 0$ for $1 \leqslant i \leqslant k$ and $1 \leqslant j \leqslant k$, and $b_{k+1,k+1} = 0$. Show that the matrix $C = AB$ is upper triangular with $c_{ij} = 0$ for $1 \leqslant i \leqslant k+1$ and $1 \leqslant j \leqslant k+1$.

(b) Let $A \in \mathbb{R}^{n\times n}$, and let all eigenvalues $\lambda_1, \dots, \lambda_n$ of A, listed with multiplicity, be real. Show that

$$(\lambda_1 I_n - A)(\lambda_2 I_n - A) \cdots (\lambda_n I_n - A) = 0$$

and deduce that $p_A(A) = 0$.

Solution. (a) Partition A, B and C with diagonal blocks of sizes k and $n - k$ (Definition 5.1). The blocks below the diagonal blocks of A and B are 0, as A and B are upper triangular (Definition 1.12), and the upper left $k \times k$ block of A is 0 by assumption. By Theorem 5.3(c),

$$A = \begin{pmatrix} 0 & A_{12} \\ 0 & A_{22} \end{pmatrix}, \qquad B = \begin{pmatrix} B_{11} & B_{12} \\ 0 & B_{22} \end{pmatrix}, \qquad C = AB = \begin{pmatrix} 0 & A_{12}B_{22} \\ 0 & A_{22}B_{22} \end{pmatrix}$$

- C is upper triangular (Proposition 1.13(c)), and its first k columns are 0. It remains to show that column $k + 1$ of C, which consists of the first columns of $A_{12}B_{22}$ and of $A_{22}B_{22}$, is 0.

- B_{22} is upper triangular, and its first diagonal entry is $b_{k+1,k+1} = 0$; so its first column is 0. Hence for every matrix E with $n-k$ columns, the first column of EB_{22} is $E \cdot 0 = 0$ (Theorem 1.4(a)); in particular the first columns of $A_{12}B_{22}$ and $A_{22}B_{22}$ are 0.

So the first $k+1$ columns of C are 0; in particular $c_{ij} = 0$ for $1 \leqslant i \leqslant k+1$ and $1 \leqslant j \leqslant k+1$.

(b) For $n = 1$, $A = (a_{11})$ with $\lambda_1 = a_{11}$ (Definition 7.7(b)), so $\lambda_1 I_1 - A = 0$ and $p_A(A) = A - \lambda_1 I_1 = 0$. Let $n \geqslant 2$.

- By Theorem 7.30, $A = QTQ'$, where Q is orthogonal and T is upper triangular with the diagonal entries $\lambda_1, \dots, \lambda_n$, in this order. Let $B_i = \lambda_i I_n - T$ $(1 \leqslant i \leqslant n)$: it is upper triangular, and its ith diagonal entry is 0.
- As $QQ' = Q'Q = I_n$ (Proposition 6.13(b)), $\lambda_i I_n - A = Q(\lambda_i I_n)Q' - QTQ' = QB_iQ'$ (Theorem 1.3), and
$$(\lambda_1 I_n - A) \cdots (\lambda_n I_n - A) = QB_1Q'QB_2Q' \cdots QB_nQ' = Q(B_1B_2 \cdots B_n)Q'$$
- By induction on j, $B_1 \cdots B_j$ is upper triangular, and its upper left $j \times j$ block is 0 $(1 \leqslant j \leqslant n)$. For $j = 1$, the $(1,1)$ entry of B_1 is 0. If this holds for some $j < n$, then (a), with $B_1 \cdots B_j$, B_{j+1} and j in place of A, B and k, gives it for $j+1$, as the $(j+1)$th diagonal entry of B_{j+1} is 0. For $j = n$, $B_1 \cdots B_n = 0$; so $(\lambda_1 I_n - A) \cdots (\lambda_n I_n - A) = 0$.
- $p_A = (t - \lambda_1) \cdots (t - \lambda_n)$ (Definition 7.10) is a product of the real polynomials $t - \lambda_i$, whose values at A are $A - \lambda_i I_n$ (Definition 7.4). By Proposition 7.5(b), applied $n-1$ times, and Theorem 1.3,
$$p_A(A) = (A - \lambda_1 I_n) \cdots (A - \lambda_n I_n) = (-1)^n (\lambda_1 I_n - A) \cdots (\lambda_n I_n - A) = 0$$

Problem 7.10 ([10, Problem 11.3])**.** Show that every $A \in \mathbb{R}^{n \times n}$ is the sum of two invertible matrices.

Solution.

- p_A has degree n (Definition 7.7(b)), so it has at most n roots (Proposition 7.5(d)), and the real eigenvalues of A are among them (Definition 7.7(a)). Let $c_1, \dots, c_k$ $(0 \leqslant k \leqslant n)$ be the distinct real eigenvalues of A, and put $c = 1 + |c_1| + \cdots + |c_k|$ ($c = 1$ if $k = 0$).
- Then $c > |c_i|$ for every i, so neither c nor $-c$ is an eigenvalue of A. By Theorem 7.3, (i)⇔(ii), with $\lambda = c$ and with $\lambda = -c$, the matrices $A - cI$ and $A - (-c)I = A + cI$ are invertible.
- Hence $\frac{1}{2}(A + cI)$ and $\frac{1}{2}(A - cI)$ are invertible, as $\frac{1}{2}(A \pm cI) \cdot 2(A \pm cI)^{-1} = I$ (Theorem 2.16), and
$$A = \tfrac{1}{2}(A + cI) + \tfrac{1}{2}(A - cI)$$

Problem 7.11 ([10, Problem 22.1])**.** Let $Q \in \mathbb{R}^{n \times n}$ be orthogonal, with $p_Q(t) = c_0 + c_1 t + \cdots + c_n t^n$, and let $\tilde{p}(t) = c_n + c_{n-1}t + \cdots + c_0 t^n$ be the polynomial with the coefficients of p_Q in reverse order, so that $\tilde{p}(t) = t^n p_Q(1/t)$ for real $t \neq 0$. Show that $\tilde{p} = p_Q$ or $\tilde{p} = -p_Q$.

Solution. Fix a real $t \neq 0$.

- $t^n p_Q(1/t) = t^n \sum_{k=0}^n c_k t^{-k} = \sum_{k=0}^n c_k t^{n-k} = \tilde{p}(t)$ (Definition 7.4).
- By Definition 7.7(a) and Theorem 5.18(c), $t^n p_Q(1/t) = t^n \det(t^{-1}I - Q) = \det\big(t(t^{-1}I - Q)\big) = \det(I - tQ)$.
- As $Q'Q = I$, $I - tQ = Q'Q - tQ = (Q' - tI)Q$ (Theorem 1.3), so $\det(I - tQ) = \det(Q' - tI) \cdot \det Q$ (Theorem 5.18(b)).
- By Theorem 5.18(c), Definition 7.7(a) and Proposition 7.18(d),
$$\det(Q' - tI) = \det\big((-1)(tI - Q')\big) = (-1)^n \det(tI - Q') = (-1)^n p_{Q'}(t) = (-1)^n p_Q(t)$$

So $\tilde{p}(t) = \varepsilon\, p_Q(t)$ for every real $t \neq 0$, where $\varepsilon = (-1)^n \det Q$; and $\varepsilon = 1$ or $\varepsilon = -1$, as $\det Q = \pm 1$ (Theorem 6.19(c)). The polynomials $\tilde{p}$ and εp_Q have the same value at infinitely many t (Proposition 7.5(b)), so $\tilde{p} = \varepsilon p_Q$ (Proposition 7.5(d)), that is, $c_{n-k} = \varepsilon c_k$ for $k = 0, \dots, n$.

Problem 7.12 ([10, Problem 11.2])**.** Let $n \geqslant 2$, let $d_1, \dots, d_n$ be nonzero real numbers, and let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ with $a_{ij} = d_i/d_j$. Find the eigenvalues of A and the eigenvectors belonging to each of them.

Solution. Let $D = \operatorname{diag}(d_1, \dots, d_n)$ and $d = (d_1, \dots, d_n)'$.

- D is invertible, with $D^{-1} = \operatorname{diag}(d_1^{-1}, \dots, d_n^{-1})$ (Definition 2.12). Left multiplication by D multiplies row i by d_i, and right multiplication by D^{-1} multiplies column j by d_j^{-1} (Proposition 1.13(a)); as every entry of J_n is 1, the (i, j) entry of DJ_nD^{-1} is $d_i \cdot 1 \cdot d_j^{-1} = a_{ij}$. So $A = DJ_nD^{-1}$, that is, $J_n = D^{-1}AD$: J_n is similar to A (Definition 7.14).
- By Theorem 7.15(a) and Example 7.34, $p_A = p_{J_n} = t^{n-1}(t - n)$: the eigenvalues of A are n, simple, and 0, of algebraic multiplicity $n - 1$.
- Eigenvectors of $J_n = \mathbf{1}\mathbf{1}'$: $J_n y = \mathbf{1}(\mathbf{1}'y)$ (Theorem 1.3), a multiple of $\mathbf{1}$. So $J_n y = 0$ if and only if $\mathbf{1}'y = 0$, as $\mathbf{1} \neq 0$ (Proposition 3.13(a)); and if $J_n y = ny$, then $y = \frac{1}{n}(\mathbf{1}'y)\mathbf{1}$. Hence $\mathcal{N}(J_n) = \{y : \mathbf{1}'y = 0\}$, and $\mathcal{N}(J_n - nI) = \operatorname{span}\{\mathbf{1}\}$, as $J_n\mathbf{1} = n\mathbf{1}$.
- By Theorem 7.15(b), with $P = D$ and $J_n = D^{-1}AD$, x is an eigenvector of A belonging to c if and only if $D^{-1}x$ is an eigenvector of J_n belonging to c. For $c = n$: $D^{-1}x = s\mathbf{1}$ with $s \neq 0$, that is, $x = sD\mathbf{1} = sd$. For $c = 0$: $x \neq 0$ and $\mathbf{1}'D^{-1}x = 0$, that is,

$$\frac{x_1}{d_1} + \frac{x_2}{d_2} + \cdots + \frac{x_n}{d_n} = 0$$

- So the eigenvectors belonging to n are the nonzero multiples of d, and those belonging to 0 are the nonzero vectors of $\mathcal{N}(A) = \{x : x_1/d_1 + \cdots + x_n/d_n = 0\}$. Here $\dim \mathcal{N}(A) = \dim \mathcal{N}(J_n) = n - 1$ (Theorem 7.15(b); $\mathcal{N}(J_n) = \mathcal{N}(\mathbf{1}')$, $\operatorname{rank} \mathbf{1}' = 1$ as its only row is nonzero, and Theorem 4.12), and the $n - 1$ vectors $d_1e_1 - d_je_j$ $(j = 2, \dots, n)$ of $\mathcal{N}(A)$ are linearly independent, as only the jth of them has a nonzero entry in position j; so they form a basis of $\mathcal{N}(A)$ (Theorem 4.5(d)).

Check: entry i of Ad is $\sum_j (d_i/d_j)d_j = nd_i$, and entry i of Ax is $d_i \sum_j x_j/d_j$.

Problem 7.13 ([10, Problem 25.2])**.** Find all idempotent matrices $P \in \mathbb{R}^{2\times 2}$.

Solution. Let $P \in \mathbb{R}^{2\times 2}$. First, P is idempotent if and only if $P = 0$, or $P = I_2$, or $\operatorname{tr} P = 1$ and $\det P = 0$.

- Let $P^2 = P$. Then $\operatorname{rank} P \in \{0, 1, 2\}$ (Theorem 4.13(a)). If $\operatorname{rank} P = 0$, then $\mathcal{C}(P) = \{0\}$ (Definition 4.11, Theorem 4.4), so every column of P is 0 (Definition 4.7) and $P = 0$. If $\operatorname{rank} P = 2$, then P is invertible (Theorem 4.13(d)), and multiplying $P^2 = P$ on the left by P^{-1} gives $P = I_2$. If $\operatorname{rank} P = 1$, then $\operatorname{tr} P = \operatorname{rank} P = 1$ (Theorem 6.29(c)), and $\det P = 0$, as P is not invertible (Theorem 5.18(a)).
- Conversely, 0 and I_2 are idempotent. If $\operatorname{tr} P = 1$ and $\det P = 0$, then $p_P(t) = t^2 - (\operatorname{tr} P)t + \det P = t^2 - t$ (Definition 7.7(b)), and the Cayley–Hamilton theorem (Theorem 7.38) gives $P^2 - P = 0$.

Next, the matrices with $\operatorname{tr} P = 1$ and $\det P = 0$.

- Write $P = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$. If $\operatorname{tr} P = 1$, then $\delta = 1 - \alpha$; with $a = 2\alpha - 1$, $b = 2\beta$ and $c = 2\gamma$,

$$P = \frac{1}{2}\begin{pmatrix} 1 + a & b \\ c & 1 - a \end{pmatrix}, \qquad \det P = \tfrac{1}{4}(1 + a)(1 - a) - \tfrac{1}{4}bc = \tfrac{1}{4}\big(1 - a^2 - bc\big)$$

- So $\det P = 0$ if and only if $a^2 + bc = 1$. Conversely, every matrix of this form with $a, b, c \in \mathbb{R}$ has trace $\frac{1}{2}(1+a) + \frac{1}{2}(1-a) = 1$.

Hence the idempotent 2×2 matrices are 0, I_2, and the matrices

$$P = \frac{1}{2}\begin{pmatrix} 1+a & b \\ c & 1-a \end{pmatrix} \qquad (a, b, c \in \mathbb{R},\ a^2 + bc = 1)$$

Problem 7.14 ([10, §26.1])**.** Let $A \in \mathbb{R}^{n\times n}$ satisfy $A^2 = I_n$.

(a) Show that $E = \frac{1}{2}(I_n + A)$ is idempotent and that $A = 2E - I_n$.

(b) Show that A is diagonalizable, that its eigenvalues lie in $\{1, -1\}$, that $\mathcal{N}(A - I_n) = \mathcal{C}(I_n + A)$ and $\mathcal{N}(A + I_n) = \mathcal{C}(I_n - A)$, and that $g_A(1) + g_A(-1) = n$. Compare with Problem 4.3.

Solution. (a) By Theorem 1.3 and $A^2 = I_n$, $E^2 = \frac{1}{4}(I_n + 2A + A^2) = \frac{1}{4}(2I_n + 2A) = E$, and $2E - I_n = I_n + A - I_n = A$.

(b)

- $(A - I_n)(I_n + A) = A^2 - I_n = 0$ and $(A + I_n)(I_n - A) = I_n - A^2 = 0$ (Theorem 1.3). So for every $y \in \mathbb{R}^n$, $(I_n + A)y \in \mathcal{N}(A - I_n)$ and $(I_n - A)y \in \mathcal{N}(A + I_n)$: $\mathcal{C}(I_n + A) \subseteq \mathcal{N}(A - I_n)$ and $\mathcal{C}(I_n - A) \subseteq \mathcal{N}(A + I_n)$ (Definition 4.7).
- Conversely, if $Ax = x$, then $x = \frac{1}{2}(x + Ax) = (I_n + A)(\frac{1}{2}x) \in \mathcal{C}(I_n + A)$; and if $Ax = -x$, then $x = \frac{1}{2}(x - Ax) = (I_n - A)(\frac{1}{2}x) \in \mathcal{C}(I_n - A)$. Hence $\mathcal{N}(A - I_n) = \mathcal{C}(I_n + A)$ and $\mathcal{N}(A + I_n) = \mathcal{C}(I_n - A)$.
- Every $x \in \mathbb{R}^n$ is $x = \frac{1}{2}(x + Ax) + \frac{1}{2}(x - Ax)$, where $\frac{1}{2}(x + Ax) = (I_n + A)(\frac{1}{2}x)$ lies in $\mathcal{N}(A - I_n)$ and $\frac{1}{2}(x - Ax) = (I_n - A)(\frac{1}{2}x)$ in $\mathcal{N}(A + I_n)$, by the first item. So $\mathbb{R}^n = \mathcal{N}(A - I_n) + \mathcal{N}(A + I_n)$ (Definition 3.30). If x lies in both null spaces, then $x = Ax = -x$, so $x = 0$. By Theorems 4.23 and 4.4 and Definition 7.21, $g_A(1) + g_A(-1) = \dim \mathcal{N}(A - I_n) + \dim \mathcal{N}(A + I_n) = \dim \mathbb{R}^n = n$.
- If $Ax = \lambda x$ with $\lambda \in \mathbb{R}$ and $x \neq 0$, then $x = A^2x = \lambda^2 x$ (Proposition 7.18(a)), so $\lambda^2 = 1$ (Proposition 3.13(a)): the real eigenvalues of A are among 1 and -1. So the dimensions of the eigenspaces of the distinct real eigenvalues of A add up to $g_A(1) + g_A(-1) = n$ (a null space $\mathcal{N}(A \mp I_n) = \{0\}$ adds 0), and A is diagonalizable (Theorem 7.25, (iii)⇒(i)). By Definition 7.23, all eigenvalues of A are then real; so they lie in $\{1, -1\}$.

In terms of ranks, $g_A(1) = \dim \mathcal{C}(I_n + A) = \operatorname{rank}(I_n + A)$ and $g_A(-1) = \operatorname{rank}(I_n - A)$ (Definition 4.11), so $\operatorname{rank}(I_n + A) + \operatorname{rank}(I_n - A) = n$, the identity of Problem 4.3.

Chapter 8. The spectral theorem and quadratic forms

Real eigenvalues and orthogonal eigenvectors

Theorem 8.1 ([3, Result 3.5])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then every eigenvalue of A (every complex root of p_A, Definition 7.10) is real, and for each eigenvalue λ there is a nonzero $x \in \mathbb{R}^n$ with $Ax = \lambda x$.

Proof. Let $\mu = \alpha + i\beta$ $(\alpha, \beta \in \mathbb{R})$ be a root of p_A, and put $Y = (A - \alpha I)^2 \in \mathbb{R}^{n\times n}$.

- For every real s, $\Big((\alpha + s)I - A\Big)\Big((\alpha - s)I - A\Big) = (\alpha I - A)^2 - s^2 I = Y - s^2 I$, as sI commutes with every matrix. By Theorem 5.18(b),

$$p_A(\alpha + s)\, p_A(\alpha - s) = \det(Y - s^2 I) \qquad (s \in \mathbb{R})$$

- Both sides are polynomials in s with real coefficients: p_A has real coefficients (Definition 7.7), and on the left we expand the powers of $\alpha \pm s$; on the right $\det(Y - rI) = (-1)^n p_Y(r)$ (Theorem 5.18(c)) with $r = s^2$. They agree at every real s, so they are equal as polynomials (Proposition 7.5(d)), and the identity also holds for $s = i\beta$ (Proposition 7.5(b)): $\det(Y + \beta^2 I) = p_A(\mu)\, p_A(\alpha - i\beta) = 0$.
- $A - \alpha I$ is symmetric, so $Y = (A - \alpha I)'(A - \alpha I)$, and for $x \in \mathbb{R}^n$

$$x'(Y + \beta^2 I)x = \|(A - \alpha I)x\|^2 + \beta^2 \|x\|^2$$

  If $\beta \neq 0$, this is > 0 for every $x \neq 0$, so $(Y + \beta^2 I)x \neq 0$: $Y + \beta^2 I$ is invertible (Theorem 2.16), and $\det(Y + \beta^2 I) \neq 0$ (Theorem 5.18(a)) ✘ Hence $\beta = 0$.
- So $\lambda = \mu = \alpha$ is real and $\det(\lambda I - A) = 0$; by Theorem 7.3, $Ax = \lambda x$ for some nonzero $x \in \mathbb{R}^n$. □

Most matrices of statistics in this book whose eigenvalues we use are symmetric, so their eigenvalues are real by Theorem 8.1: $A'A$ for every matrix A (Theorem 1.9), the projection matrices P_W (Theorem 6.31(c)), among them $\bar{J}_n$ and C_n (Proposition 1.15), the equicorrelation matrix of Example 5.33, and the covariance and correlation matrices of data. A matrix that is not symmetric may have eigenvalues that are not real: the matrix of Example 7.12 has the eigenvalues 1 and $2 \pm i$. The others are the matrices $A^{-1}B$ with A PD and B symmetric, which need not be symmetric, but whose eigenvalues are real too; this is shown in the section on simultaneous diagonalization.

Theorem 8.2 ([1, §8.5, Theorem 15])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $Ax = \lambda x$, $Ay = \mu y$ with $x, y \in \mathbb{R}^n$ and $\lambda \neq \mu$. Then $x'y = 0$.

Proof. $\lambda x'y = (Ax)'y = x'A'y = x'Ay = \mu x'y$, so $(\lambda - \mu)x'y = 0$, and $\lambda - \mu \neq 0$. □

The spectral theorem

Theorem 8.3 (Spectral theorem; [5, Theorem 21.5.7, Corollary 21.5.9], [4, Ch. VIII, Theorem 4.4])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric.

(a) $\mathbb{R}^n$ has an orthonormal basis $q_1, \ldots, q_n$ consisting of eigenvectors of A.

(b) Let $Aq_i = \lambda_i q_i$ $(i = 1, \ldots, n)$, $Q = [q_1 \ \cdots \ q_n]$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Then Q is orthogonal and

$$Q'AQ = \Lambda, \qquad A = Q\Lambda Q'$$

Conversely, if $B \in \mathbb{R}^{n\times n}$ and $Q'BQ$ is diagonal for an orthogonal Q, then B is symmetric. The q_i may be numbered in any order, for instance so that $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$.

(c) $p_A(t) = (t-\lambda_1)\cdots(t-\lambda_n)$: the numbers $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A, each eigenvalue λ listed $a_A(\lambda)$ times.

Proof. (a) We show by induction on n that $Q'AQ$ is diagonal for some orthogonal $Q \in \mathbb{R}^{n\times n}$.

- For $n = 1$, take $Q = (1)$.
- Let $n \geqslant 2$, and assume the claim for symmetric matrices of size $n-1$. As p_A has $n \geqslant 1$ roots, A has an eigenvalue λ (Definition 7.10); by Theorem 8.1, λ is real and $Ax = \lambda x$ for some nonzero $x \in \mathbb{R}^n$. Then $u = x/\|x\|$ has $u'u = 1$ and $Au = \lambda u$.
- By Corollary 6.15(c), with $p = 1 < n$, there is $V \in \mathbb{R}^{n\times(n-1)}$ such that $[u\ V]$ is orthogonal. Comparing blocks of $[u\ V]'[u\ V] = I_n$ (Theorem 5.3(b),(c)) gives $V'u = 0$ and $V'V = I_{n-1}$.
- $V'Au = \lambda V'u = 0$, $u'AV = (V'A'u)' = (V'Au)' = 0$ and $u'Au = \lambda u'u = \lambda$; so, by block multiplication (Theorem 5.3(b),(c)),

$$[u\ V]'A[u\ V] = \begin{pmatrix} u'Au & u'AV \\ V'Au & V'AV \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & B \end{pmatrix}, \qquad B = V'AV \in \mathbb{R}^{(n-1)\times(n-1)}$$

- $B' = V'A'V = B$. By the induction hypothesis, $R'BR = \Lambda_2$ is diagonal for some orthogonal $R \in \mathbb{R}^{(n-1)\times(n-1)}$. Let $S = \operatorname{diag}(1, R)$ and $Q = [u\ V]S$. By Theorems 1.9 and 5.3(b) and Proposition 5.6(b),

$$Q'Q = S'[u\ V]'[u\ V]S = S'S = \operatorname{diag}(1, R'R) = I_n$$
$$Q'AQ = S'\operatorname{diag}(\lambda, B)\,S = \operatorname{diag}(\lambda, R'BR) = \operatorname{diag}(\lambda, \Lambda_2)$$

  so Q is orthogonal and $Q'AQ$ is diagonal.
- Let $\Lambda = Q'AQ$, and let $q_1, \dots, q_n$ be the columns of Q. As $QQ' = I_n$ (Proposition 6.13(b)), $AQ = QQ'AQ = Q\Lambda$, whose column i reads $Aq_i = \lambda_i q_i$ (Theorem 1.4(a),(b)); and $q_i'q_j = \delta_{ij}$, as $Q'Q = I_n$. So $q_1, \dots, q_n$ are orthonormal eigenvectors of A, hence independent (Proposition 6.11(b)), and they form a basis of $\mathbb{R}^n$ (Theorem 4.5(d)).

(b)

- $(Q'Q)_{ij} = q_i'q_j = \delta_{ij}$, so $Q'Q = I_n$: Q is orthogonal, and $Q^{-1} = Q'$ (Proposition 6.13(b)).
- $AQ = [Aq_1\ \cdots\ Aq_n] = [\lambda_1 q_1\ \cdots\ \lambda_n q_n] = Q\Lambda$, since $\operatorname{col}_j(Q\Lambda) = Q(\lambda_j e_j) = \lambda_j q_j$ (Theorem 1.4(a),(b)). Multiplying by Q' on the left, or on the right, gives $Q'AQ = \Lambda$ and $A = Q\Lambda Q'$.
- Conversely, if $Q'BQ = D$ is diagonal, then $B = QDQ'$ and $B' = QD'Q' = QDQ' = B$ (Theorem 1.9).
- Any renumbering of $q_1, \dots, q_n$ is again an orthonormal basis of eigenvectors of A, so the above applies to it; numbering the q_i by decreasing λ_i gives $\lambda_1 \geqslant \cdots \geqslant \lambda_n$.

(c) As $Q^{-1} = Q'$, $Q^{-1}AQ = \Lambda$; so $p_A = p_\Lambda = (t-\lambda_1)\cdots(t-\lambda_n)$ (Theorem 7.15(a), Proposition 7.16(b)), and $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A, listed with multiplicity: this is the last sentence of Definition 7.23. □

Theorem 8.4 ([5, Lemma 14.3.1, Corollary 21.5.8], [2, §6.5.3])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric. For every real c, $\dim \mathcal{N}(A - cI) = a_A(c)$ (which is 0 if c is not an eigenvalue). Hence $g_A(c) = a_A(c)$ for every eigenvalue c; $\operatorname{rank} A$ is the number of nonzero eigenvalues of A, counted with multiplicity; and $a_A(0) = \dim \mathcal{N}(A) = n - \operatorname{rank} A$. Moreover, if $c_1, \dots, c_k$ are the distinct eigenvalues of A, then orthonormal bases of $\mathcal{N}(A - c_1 I), \dots, \mathcal{N}(A - c_k I)$, put side by side in any order, form an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A, as in Theorem 8.3(a).

Proof. Let $A = Q\Lambda Q'$ as in Theorem 8.3.

- A diagonal matrix $D = \operatorname{diag}(d_1, \dots, d_n)$ has rank equal to the number of i with $d_i \neq 0$: its nonzero columns $d_i e_i$ are nonzero and pairwise orthogonal, hence independent (Proposition 6.11(b)), and they span $\mathcal{C}(D)$.
- For real c, $A - cI = Q(\Lambda - cI)Q'$ with Q, Q' invertible, so $\operatorname{rank}(A - cI) = \operatorname{rank}(\Lambda - cI)$ (Theorem 4.15(d)), the number of i with $\lambda_i \neq c$.
- By Theorem 4.12, $\dim \mathcal{N}(A - cI) = n - \operatorname{rank}(A - cI)$ is the number of i with $\lambda_i = c$, which is $a_A(c)$ by Theorem 8.3(c). For an eigenvalue c this is $g_A(c) = a_A(c)$ (Definition 7.21), and $c = 0$ gives the statements on $\operatorname{rank} A$ and $a_A(0)$.
- The c_j are real (Theorem 8.1) and $\dim \mathcal{N}(A - c_j I) = a_A(c_j) \geqslant 1$, so each $\mathcal{N}(A - c_j I)$ has an orthonormal basis of $a_A(c_j)$ vectors (Corollary 6.15(a)). Vectors from different eigenspaces are orthogonal (Theorem 8.2), so all these vectors together are orthonormal, hence independent (Proposition 6.11(b)); there are $a_A(c_1) + \cdots + a_A(c_k) = n$ of them (Definition 7.10), so they form a basis of $\mathbb{R}^n$ (Theorem 4.5(d)). □

Example 8.5 ([2, Example 6.13])**.** Find an orthogonal Q with $Q'AQ$ diagonal for

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} = 2J_3 - I_3$$

Solution. For real t, apply $R_1 \leftarrow R_1 + R_2 + R_3$, take the factor $t - 5$ out of row 1, and apply $R_2 \leftarrow R_2 + 2R_1$, $R_3 \leftarrow R_3 + 2R_1$ (Definition 5.13(a),(b), Theorem 5.14(c)):

$$\begin{aligned} \det(tI - A) &= \det \begin{pmatrix} t-1 & -2 & -2 \\ -2 & t-1 & -2 \\ -2 & -2 & t-1 \end{pmatrix} = \det \begin{pmatrix} t-5 & t-5 & t-5 \\ -2 & t-1 & -2 \\ -2 & -2 & t-1 \end{pmatrix} \\ &= (t-5) \det \begin{pmatrix} 1 & 1 & 1 \\ -2 & t-1 & -2 \\ -2 & -2 & t-1 \end{pmatrix} = (t-5) \det \begin{pmatrix} 1 & 1 & 1 \\ 0 & t+1 & 0 \\ 0 & 0 & t+1 \end{pmatrix} = (t-5)(t+1)^2 \end{aligned}$$

This holds for every real t, hence as polynomials (Definition 7.7(a), Proposition 7.5(d)): the eigenvalues are 5 with $a_A(5) = 1$ and -1 with $a_A(-1) = 2$.

- Every row sum of A is 5, so $A\mathbf{1}_3 = 5\mathbf{1}_3$; as $\dim \mathcal{N}(A - 5I) = a_A(5) = 1$ (Theorem 8.4), $\mathcal{N}(A - 5I) = \operatorname{span}\{\mathbf{1}_3\}$.
- Reduce $A + I = 2J_3$:

$$A + I = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} \xrightarrow{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  so $\mathcal{N}(A + I) = \{x : x_1 + x_2 + x_3 = 0\}$, of dimension 2. It contains $u_2 = (-2, 1, 1)'$. A second vector of $\mathcal{N}(A + I)$ orthogonal to u_2 solves $x_1 + x_2 + x_3 = 0$ and $u_2' x = 0$:

$$\begin{pmatrix} 1 & 1 & 1 \\ -2 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + 2R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 3 & 3 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{3}R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

  so $x_1 = 0$, $x_2 = -x_3$; take $u_3 = (0, -1, 1)'$.
- $\mathbf{1}_3 \perp u_2, u_3$ (Theorem 8.2) and $u_2 \perp u_3$. Normalizing,

$$Q = \begin{pmatrix} 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{pmatrix}, \qquad Q'AQ = \operatorname{diag}(5, -1, -1)$$

  by Theorem 8.3(b). Check: $Au_2 = (2, -1, -1)' = -u_2$ and $Au_3 = (0, 1, -1)' = -u_3$.

The spectral decomposition

Theorem 8.6 (Spectral decomposition; [1, §9.5, Theorems 9, 10], [5, §21.5f, Theorem 21.4.3], [3, §3.2])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric with distinct eigenvalues $c_1, \dots, c_k$, let $W_j = \mathcal{N}(A - c_j I)$, and let $E_j = P_{W_j} \in \mathbb{R}^{n\times n}$ be the projection matrix for W_j (Theorem 6.31(a)): $E_j y$ is the orthogonal projection of y onto W_j (Definition 6.22), and $E_j = V_j(V_j'V_j)^{-1}V_j'$ whenever the columns of V_j form a basis of W_j (Theorem 6.31(b)). (Here E_j is a projection matrix, not an elementary matrix as in Theorem 2.15(c).)

(a) $W_i \perp W_j$ for $i \neq j$. Every $y \in \mathbb{R}^n$ can be written as $y = y_1 + \cdots + y_k$ with $y_j \in W_j$ in exactly one way, and then $E_j y = y_j$.

(b) $E_1 + \cdots + E_k = I_n$, $E_j' = E_j = E_j^2$, $E_i E_j = 0$ for $i \neq j$, each $E_j \neq 0$ (as $E_j w = w$ for $w \in W_j$, by (a)), and

$$A = c_1 E_1 + \cdots + c_k E_k, \qquad A^m = c_1^m E_1 + \cdots + c_k^m E_k \quad (m = 0, 1, 2, \dots)$$

If no c_j is 0, then A is invertible and $A^{-1} = c_1^{-1}E_1 + \cdots + c_k^{-1}E_k$.

(c) If $A = Q\Lambda Q'$ as in Theorem 8.3 and Q_j consists of the columns q_i of Q with $\lambda_i = c_j$, then $E_j = Q_j Q_j' = \sum_{\lambda_i = c_j} q_i q_i'$, $A = \sum_{i=1}^n \lambda_i q_i q_i'$, and $A^m = Q\Lambda^m Q'$ $(m = 0, 1, 2, \dots)$. Moreover $\mathcal{N}(A) = \operatorname{span}\{q_i : \lambda_i = 0\}$ and $\mathcal{C}(A) = \operatorname{span}\{q_i : \lambda_i \neq 0\}$.

(d) (Uniqueness) Let $A = b_1 F_1 + \cdots + b_r F_r$, where $b_1, \dots, b_r$ are distinct numbers and $F_1, \dots, F_r \in \mathbb{R}^{n\times n}$ are nonzero, symmetric and idempotent, with $F_1 + \cdots + F_r = I_n$ and $F_l F_m = 0$ for $l \neq m$. Then $r = k$, and after renumbering $b_j = c_j$ and $F_j = E_j$ for every j.

The expression $A = \sum_j c_j E_j$ is the **spectral decomposition** of A.

Proof. (a)

- $W_i \perp W_j$ by Theorem 8.2.
- Let $q_1, \dots, q_n$ be an orthonormal basis of eigenvectors (Theorem 8.3). Then $y = \sum_i (q_i' y) q_i$ (Proposition 6.11(a)), and each q_i lies in the W_j with $c_j = \lambda_i$; grouping the terms gives $y = \sum_j y_j$ with $y_j \in W_j$.
- If $z_1 + \cdots + z_k = 0$ with $z_j \in W_j$, then $0 = z_i' \sum_j z_j = \sum_j z_i' z_j = \|z_i\|^2$, so every $z_i = 0$. Applied to the differences of two such expressions of y, this gives uniqueness.
- $y - y_j = \sum_{l\neq j} y_l \perp W_j$ and $y_j \in W_j$, so y_j is the orthogonal projection of y onto W_j: $E_j y = y_j$.

(b)

- $\sum_j E_j y = \sum_j y_j = y$ for every y, so $\sum_j E_j = I_n$. $E_j' = E_j = E_j^2$ by Theorem 6.31(c).
- For $i \neq j$ and every y, $E_j y \in W_j$ is its own decomposition in (a), with zero part in W_i; so $E_i E_j y = 0$.
- $Ay = \sum_j A y_j = \sum_j c_j y_j = \sum_j c_j E_j y$ for every y, so $A = \sum_j c_j E_j$.
- $A^0 = I_n = \sum_j E_j$, and if $A^m = \sum_i c_i^m E_i$, then $A^{m+1} = \left(\sum_i c_i^m E_i\right)\left(\sum_j c_j E_j\right) = \sum_j c_j^{m+1} E_j$, since $E_i E_j = 0$ for $i \neq j$ and $E_j^2 = E_j$.
- In the same way $A \sum_j c_j^{-1} E_j = \sum_j E_j = I_n$, so $A^{-1} = \sum_j c_j^{-1} E_j$ (Theorem 2.16).

(c)

- The columns of Q_j are orthonormal eigenvectors for c_j, so they lie in W_j and are independent; their number is $a_A(c_j) = \dim W_j$ (Theorem 8.4). So they form an orthonormal basis of W_j (Theorem 4.5(d)), and $E_j = Q_j Q_j'$ (Theorem 6.31(b)).

- By the outer product expansion (Theorem 1.4(d)), $Q_jQ_j' = \sum_{\lambda_i=c_j} q_iq_i'$, and $A = (Q\Lambda)Q' = \sum_i \lambda_i q_iq_i'$, as $\mathrm{col}_i(Q\Lambda) = \lambda_i q_i$.
- $A = Q\Lambda Q^{-1}$ with $Q^{-1} = Q'$ (Proposition 6.13(b)), so $A^m = Q\Lambda^m Q^{-1} = Q\Lambda^m Q'$ (Proposition 7.27(a)).
- If $\lambda_i \neq 0$, then $q_i = A(\lambda_i^{-1}q_i) \in \mathcal{C}(A)$; if $\lambda_i = 0$, then $Aq_i = \lambda_i q_i = 0$. So $\mathrm{span}\{q_i : \lambda_i \neq 0\} \subseteq \mathcal{C}(A)$ and $\mathrm{span}\{q_i : \lambda_i = 0\} \subseteq \mathcal{N}(A)$ (Theorem 3.7). The q_i are independent (Proposition 6.11(b)), and there are $\operatorname{rank} A = \dim \mathcal{C}(A)$ of them with $\lambda_i \neq 0$ and $a_A(0) = \dim \mathcal{N}(A)$ with $\lambda_i = 0$ (Theorems 8.3(c) and 8.4, Definition 4.11); so both inclusions are equalities (Theorem 4.5(a)).

(d)

- As $F_m \neq 0$, there is y with $x = F_m y \neq 0$; then $Ax = \sum_l b_l F_l F_m y = b_m x$. So every b_m is an eigenvalue of A, and $\mathcal{C}(F_m) \subseteq \mathcal{N}(A - b_m I)$.
- Let $Ax = cx$ with $x \neq 0$. Then $x = \sum_l F_l x$ and $0 = Ax - cx = \sum_l (b_l - c)F_l x$. The vectors $F_l x$ are pairwise orthogonal ($(F_l x)'(F_m x) = x'F_l F_m x = 0$), so $0 = \|\sum_l (b_l - c)F_l x\|^2 = \sum_l (b_l - c)^2 \|F_l x\|^2$. Hence $F_l x = 0$ whenever $b_l \neq c$. As $x \neq 0$, some $F_l x \neq 0$; so c is one of the b_l, and $x = F_l x \in \mathcal{C}(F_l)$ for that l.
- So $\{b_1, \ldots, b_r\} = \{c_1, \ldots, c_k\}$, $r = k$, and after renumbering $b_j = c_j$ and $\mathcal{C}(F_j) = W_j$. F_j is symmetric and idempotent, so $F_j y$ is the orthogonal projection of y onto $\mathcal{C}(F_j) = W_j$ (Theorem 6.29(d)): $F_j = E_j$. □

Example 8.7 (cf. [2, Example 6.13])**.** For $A = 2J_3 - I_3$ of Example 8.5, with the distinct eigenvalues $c_1 = 5$ and $c_2 = -1$, Theorem 8.6(c) gives $E_1 = q_1q_1' = \frac{1}{3}\mathbf{1}\mathbf{1}' = \bar{J}_3$, and then $E_2 = I_3 - E_1 = C_3$ by (b); indeed

$$q_2q_2' + q_3q_3' = \frac{1}{6}\begin{pmatrix} 4 & -2 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = C_3$$

So $A = 5\bar{J}_3 - C_3$, $A^m = 5^m\bar{J}_3 + (-1)^m C_3$; for instance $A^2 = 25\bar{J}_3 + C_3 = I_3 + 8J_3$. As no eigenvalue is 0,

$$A^{-1} = \tfrac{1}{5}\bar{J}_3 - C_3 = \tfrac{2}{5}J_3 - I_3 = \frac{1}{5}\begin{pmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{pmatrix}$$

Example 8.8. Find an orthogonal Q with $Q'AQ$ diagonal, and the matrices E_j of Theorem 8.6, for

$$A = \begin{pmatrix} 11 & 2 & 6 & 6 \\ 2 & 11 & 6 & 6 \\ 6 & 6 & 11 & 2 \\ 6 & 6 & 2 & 11 \end{pmatrix}$$

Solution.

- For real t, in $tI_4 - A$ subtract row 2 from row 1 and row 4 from row 3; the new rows 1 and 3 are $(t-9)(1,-1,0,0)$ and $(t-9)(0,0,1,-1)$. Take the two factors $t-9$ out, add column 1 to column 2 and column 3 to column 4 (Definition 5.13(b),(a), Theorem 5.22), and expand along row 1 and

then along the row $(0,1,0)$ (Theorem 5.21):

$$\det(tI_4 - A) = \det\begin{pmatrix} t-11 & -2 & -6 & -6 \\ -2 & t-11 & -6 & -6 \\ -6 & -6 & t-11 & -2 \\ -6 & -6 & -2 & t-11 \end{pmatrix} = (t-9)^2 \det\begin{pmatrix} 1 & -1 & 0 & 0 \\ -2 & t-11 & -6 & -6 \\ 0 & 0 & 1 & -1 \\ -6 & -6 & -2 & t-11 \end{pmatrix}$$

$$= (t-9)^2 \det\begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & t-13 & -6 & -12 \\ 0 & 0 & 1 & 0 \\ -6 & -12 & -2 & t-13 \end{pmatrix} = (t-9)^2 \det\begin{pmatrix} t-13 & -6 & -12 \\ 0 & 1 & 0 \\ -12 & -2 & t-13 \end{pmatrix}$$

$$= (t-9)^2\Big((t-13)^2 - 144\Big) = (t-9)^2(t-25)(t-1)$$

This holds for every real t, hence as polynomials (Definition 7.7(a), Proposition 7.5(d)): the eigenvalues are $c_1 = 25$, $c_2 = 9$ and $c_3 = 1$, with $a_A(25) = a_A(1) = 1$ and $a_A(9) = 2$.

- The row sums of A are 25, so $A\mathbf{1}_4 = 25\,\mathbf{1}_4$; and $v = (1,1,-1,-1)'$ has $Av = (11+2-12,\ 2+11-12,\ 12-11-2,\ 12-2-11)' = v$. By Theorem 8.4, $\dim\mathcal{N}(A-25I) = \dim\mathcal{N}(A-I) = 1$, so $q_1 = \frac{1}{2}\mathbf{1}_4$ and $q_4 = \frac{1}{2}v$ are orthonormal bases of $\mathcal{N}(A-25I)$ and $\mathcal{N}(A-I)$.
- $\mathcal{N}(A-9I)$: reduce $A-9I$,

$$A - 9I = \begin{pmatrix} 2 & 2 & 6 & 6 \\ 2 & 2 & 6 & 6 \\ 6 & 6 & 2 & 2 \\ 6 & 6 & 2 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_3]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 2 & 2 & 6 & 6 \\ 0 & 0 & 0 & 0 \\ 6 & 6 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 3R_1} \begin{pmatrix} 2 & 2 & 6 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -16 & -16 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 2 & 2 & 6 & 6 \\ 0 & 0 & -16 & -16 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow -\frac{1}{16}R_2]{R_1 \leftarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 1 & 3 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 3R_2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $\mathcal{N}(A-9I) = \{x : x_1 + x_2 = 0,\ x_3 + x_4 = 0\}$, with the free unknowns x_2, x_4 and the special solutions $u_2 = (-1,1,0,0)'$ and $u_3 = (0,0,-1,1)'$. They are orthogonal, so $q_2 = \frac{1}{\sqrt{2}}u_2$ and $q_3 = \frac{1}{\sqrt{2}}u_3$ form an orthonormal basis of $\mathcal{N}(A-9I)$.

- $q_1, q_4 \perp q_2, q_3$ (Theorem 8.2), and $q_1'q_4 = 0$. So $Q = [q_1\ q_2\ q_3\ q_4]$ is orthogonal and $Q'AQ = \Lambda = \mathrm{diag}(25,9,9,1)$ (Theorem 8.3(b)).
- By Theorem 8.6(c),

$$E_1 = q_1q_1' = \tfrac{1}{4}J_4, \qquad E_2 = q_2q_2' + q_3q_3' = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \qquad E_3 = q_4q_4' = \tfrac{1}{4}vv'$$

$E_1 + E_2 + E_3 = I_4$, and $A = 25E_1 + 9E_2 + E_3$; for instance the $(1,1)$, $(1,2)$ and $(1,3)$ entries are $\frac{25}{4} + \frac{9}{2} + \frac{1}{4} = 11$, $\frac{25}{4} - \frac{9}{2} + \frac{1}{4} = 2$ and $\frac{25}{4} - \frac{1}{4} = 6$.

The vectors $\frac{1}{2}(1,-1,1,-1)'$ and $\frac{1}{2}(1,-1,-1,1)'$ also form an orthonormal basis of $\mathcal{N}(A-9I)$. They give another Q, with entries $\pm\frac{1}{2}$, but, by Theorem 8.6(c), the same projection matrix

$$E_2 = \tfrac{1}{4}\Big((1,-1,1,-1)'(1,-1,1,-1) + (1,-1,-1,1)'(1,-1,-1,1)\Big)$$

Functions of a symmetric matrix

Definition 8.9 ([1, §9.5, (9-12)])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric with spectral decomposition $A = c_1E_1 + \cdots + c_kE_k$ (Theorem 8.6), and let f be a real-valued function defined at $c_1, \ldots, c_k$. The matrix

$$f(A) = f(c_1)E_1 + \cdots + f(c_k)E_k$$

is the **function** f of A.

Proposition 8.10 ([1, §9.5, Theorems 9, 10])**.** Let $A = c_1E_1 + \cdots + c_kE_k$ be as in Definition 8.9, let $W_j = \mathcal{N}(A - c_jI)$, and let f, g be real-valued functions defined at $c_1, \ldots, c_k$.

(a) $f(A)$ is symmetric, and $f(A)x = f(c_j)x$ for every $x \in W_j$. If $A = Q\Lambda Q'$ as in Theorem 8.3, then $f(A) = Q\, f(\Lambda)\, Q'$ with $f(\Lambda) = \operatorname{diag}\big(f(\lambda_1), \ldots, f(\lambda_n)\big)$, and the eigenvalues of $f(A)$, listed with multiplicity, are $f(\lambda_1), \ldots, f(\lambda_n)$.

(b) $f(A) + g(A) = (f+g)(A)$, $af(A) = (af)(A)$ for $a \in \mathbb{R}$, and $f(A)\, g(A) = (fg)(A)$. If $f(c_j) \neq 0$ for every j, then $f(A)$ is invertible and $f(A)^{-1} = h(A)$ for $h = 1/f$.

(c) If f is a real polynomial, then $f(A)$ is the matrix $f(A)$ of Definition 7.4. With the polynomials $e_j = \prod_{i \neq j} \frac{t - c_i}{c_j - c_i}$ ($e_1 = 1$ if $k = 1$), $E_j = e_j(A)$ for every j; hence $f(A) = p(A)$ for the polynomial $p = f(c_1)e_1 + \cdots + f(c_k)e_k$: every $f(A)$ is a polynomial in A.

Proof. (a)

- $f(A)' = \sum_j f(c_j)E_j' = f(A)$, as $E_j' = E_j$ (Theorem 8.6(b)).
- Let $x \in W_j$. Then x is its own decomposition in Theorem 8.6(a), so $E_jx = x$ and $E_ix = 0$ for $i \neq j$; hence $f(A)x = f(c_j)x$.
- Each q_i lies in the W_j with $c_j = \lambda_i$, so $f(A)q_i = f(\lambda_i)q_i$, that is, $f(A)Q = Q\, f(\Lambda)$ (Theorem 1.4(a),(b)); as $QQ' = I_n$ (Proposition 6.13(b)), $f(A) = Q\, f(\Lambda)\, Q'$.
- So $q_1, \ldots, q_n$ is an orthonormal basis of eigenvectors of the symmetric matrix $f(A)$, with $f(A)q_i = f(\lambda_i)q_i$, and Theorem 8.3(b),(c), applied to $f(A)$, gives its eigenvalues.

(b)

- The first two rules hold term by term. As $E_iE_j = 0$ for $i \neq j$ and $E_j^2 = E_j$ (Theorem 8.6(b)), $f(A)\, g(A) = \sum_i \sum_j f(c_i)g(c_j)E_iE_j = \sum_j f(c_j)g(c_j)E_j = (fg)(A)$.
- If every $f(c_j) \neq 0$, then $h = 1/f$ is defined at every c_j, and $f(A)\, h(A) = (fh)(A) = \sum_j E_j = I_n$ (Theorem 8.6(b)); so $f(A)^{-1} = h(A)$ (Theorem 2.16).

(c)

- Let $f = a_0 + a_1t + \cdots + a_rt^r$. Since $\sum_j c_j^m E_j = A^m$ (Theorem 8.6(b)), the matrix of Definition 8.9 is $\sum_j \sum_m a_m c_j^m E_j = \sum_m a_m A^m$, the matrix of Definition 7.4.
- The c_i are real (Theorem 8.1), so each e_j is a real polynomial, and $e_j(c_l)$ is 1 for $l = j$ and 0 for $l \neq j$ (Proposition 7.5(b)). By the first item, $e_j(A) = \sum_l e_j(c_l)E_l = E_j$.
- Hence $f(A) = \sum_j f(c_j)E_j = \sum_j f(c_j)e_j(A) = p(A)$ (Proposition 7.5(b)). □

Example 8.11. Let $A = 2J_3 - I_3 = 5\bar{J}_3 - C_3$, with $c_1 = 5$, $c_2 = -1$, $E_1 = \bar{J}_3$ and $E_2 = C_3$ (Examples 8.5 and 8.7).

- $e_1 = \frac{t+1}{6}$ and $e_2 = \frac{5-t}{6}$, so by Proposition 8.10(c) $\bar{J}_3 = \frac{1}{6}(A + I_3)$ and $C_3 = \frac{1}{6}(5I_3 - A)$. Indeed $A + I_3 = 2J_3$ and $5I_3 - A = 6I_3 - 2J_3$.
- For $f(t) = |t|$,

$$f(A) = 5\bar{J}_3 + C_3 = I_3 + \tfrac{4}{3}J_3 = \frac{1}{3}\begin{pmatrix} 7 & 4 & 4 \\ 4 & 7 & 4 \\ 4 & 4 & 7 \end{pmatrix}$$

  By Proposition 8.10, $f(A)$ is symmetric with the eigenvalues 5, 1, 1; $f(A)^2 = 25\bar{J}_3 + C_3 = A^2$; and $f(A) = p(A)$ for $p = 5e_1 + e_2 = \frac{2t+5}{3}$: indeed $\frac{1}{3}(2A + 5I_3) = \frac{1}{3}(4J_3 + 3I_3)$.

Example 8.12. For $A = 25E_1 + 9E_2 + E_3$ of Example 8.8, Theorem 8.6(b) gives

$$A^k = 25^k E_1 + 9^k E_2 + E_3 \quad (k \geqslant 0), \qquad A^{-1} = \tfrac{1}{25}E_1 + \tfrac{1}{9}E_2 + E_3 = \frac{1}{225}\begin{pmatrix} 71 & 46 & -54 & -54 \\ 46 & 71 & -54 & -54 \\ -54 & -54 & 71 & 46 \\ -54 & -54 & 46 & 71 \end{pmatrix}$$

Check: row 1 of A times the first three columns of $225A^{-1}$ gives $781 + 92 - 324 - 324 = 225$, $506 + 142 - 324 - 324 = 0$ and $-594 - 108 + 426 + 276 = 0$; the other entries in the same way. The polynomial $f(t) = (t-25)(t-9)(t-1)$ vanishes at 25, 9 and 1, so $f(A) = f(25)E_1 + f(9)E_2 + f(1)E_3 = 0$ (Definition 8.9). As $f(t) = t^3 - 35t^2 + 259t - 225$, Proposition 8.10(c) gives $f(A) = A^3 - 35A^2 + 259A - 225I_4$, which is $(A - 25I_4)(A - 9I_4)(A - I_4)$ (Proposition 7.5(b)); hence $(A - 25I_4)(A - 9I_4)(A - I_4) = 0$.

Idempotent, orthogonal, and patterned matrices

Theorem 8.13 ([5, Theorems 21.8.2, 21.8.3], [1, §6.6], [6, Exercise 8.66(b)])**.** (a) Let $A \in \mathbb{R}^{n\times n}$ be idempotent with $\operatorname{rank} A = r$. Then $p_A(t) = t^{n-r}(t-1)^r$: the only eigenvalues of A are 0 and 1, with $g_A(0) = a_A(0) = n - r$ and $g_A(1) = a_A(1) = r$ (each an eigenvalue only when this number is positive). The eigenspaces are $\mathcal{N}(A)$ and $\mathcal{N}(A - I) = \mathcal{C}(A)$, and A is diagonalizable.

(b) A symmetric matrix whose eigenvalues all lie in $\{0, 1\}$ is idempotent.

(c) If $A \in \mathbb{R}^{n\times n}$ is symmetric and idempotent with $\operatorname{rank} A = r$, then $Q'AQ = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$ (r entries 1) for some orthogonal Q; if $r \geqslant 1$, then $A = Q_1 Q_1'$, where $Q_1 \in \mathbb{R}^{n\times r}$ consists of the first r columns of Q, and $Q_1'Q_1 = I_r$.

Proof. (a)

- If $Ax = \lambda x$ with $\lambda \in \mathbb{R}$ and $x \neq 0$, then $\lambda x = Ax = A^2x = \lambda Ax = \lambda^2 x$, so $\lambda^2 = \lambda$: λ is 0 or 1.
- $\dim \mathcal{N}(A) = n - r$ (Theorem 4.12). $I - A$ is idempotent, so $\operatorname{rank}(I - A) = \operatorname{tr}(I - A) = n - \operatorname{tr} A = n - r$ (Theorems 6.29(b),(c) and 1.11(a)), and $\dim \mathcal{N}(A - I) = n - \operatorname{rank}(I - A) = r$.
- For real c, $\dim \mathcal{N}(A - cI) \leqslant a_A(c)$: by Theorem 7.22 if c is an eigenvalue, and as $\mathcal{N}(A - cI) = \{0\}$ otherwise (Theorem 7.3). The multiplicities of all eigenvalues add up to n (Definition 7.10). Hence

  $$n = \dim \mathcal{N}(A) + \dim \mathcal{N}(A - I) \leqslant a_A(0) + a_A(1) \leqslant n$$

  so $a_A(0) = n - r$, $a_A(1) = r$, and A has no other eigenvalue, real or not: $p_A(t) = t^{n-r}(t-1)^r$ (Definition 7.10).
- $Av = v$ for every $v \in \mathcal{C}(A)$ (Theorem 6.29(a)), and $v = Av \in \mathcal{C}(A)$ if $Av = v$; so $\mathcal{N}(A - I) = \mathcal{C}(A)$. As $\dim \mathcal{N}(A) + \dim \mathcal{N}(A - I) = n$, A is diagonalizable (Theorem 7.25).

(b) By Theorem 8.3, $A = Q\Lambda Q'$ with $Q'Q = I_n$, and the diagonal entries of Λ are 0 or 1, so $\Lambda^2 = \Lambda$. Then $A^2 = Q\Lambda Q'Q\Lambda Q' = Q\Lambda^2 Q' = A$.

(c) By Theorem 8.3, $Q'AQ = \Lambda$ with Q orthogonal. By (a) every λ_i is 0 or 1, and exactly r of them are 1, as $r = \operatorname{rank} A$ is the number of nonzero λ_i (Theorem 8.4). Numbering the columns of Q so that $\lambda_1 = \cdots = \lambda_r = 1$ (Theorem 8.3(b)) gives $\Lambda = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$. If $r \geqslant 1$, the outer product expansion (Theorem 1.4(d)) gives $A = (Q\Lambda)Q' = q_1 q_1' + \cdots + q_r q_r' = Q_1 Q_1'$, as $\operatorname{col}_i(Q\Lambda) = \lambda_i q_i$; and $Q_1'Q_1 = I_r$, as the columns of Q are orthonormal. □

Without symmetry (b) fails [6, Exercise 8.56(b)]: the upper triangular matrix

$$N = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

has $p_N(t) = (t-1)^2 t$ (Proposition 7.16(b)), but $N^2 \neq N$.

Example 8.14. Let

$$E = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & 1 & -1 \end{pmatrix}$$

Show that E is idempotent, and find an invertible P with $P^{-1}EP$ diagonal.

Solution.

- Row 1 of E times the columns of E gives $1+1+0-1=1$, $1+0+1-1=1$, $-1+1-2+1=-1$ and $1+0+1-1=1$, the entries of row 1 of E; the other rows in the same way, so $E^2=E$. By Theorem 6.29(c), $\operatorname{rank} E = \operatorname{tr} E = 1+0+2-1=2$.
- The first two columns $p_1=(1,1,0,-1)'$ and $p_2=(1,0,-1,-1)'$ of E are not multiples of each other, so they are linearly independent; as $\dim \mathcal{C}(E)=2$, they form a basis of $\mathcal{C}(E)=\mathcal{N}(E-I)$ (Theorems 4.5(d) and 8.13(a)).
- The vectors $p_3=(-1,2,1,0)'$ and $p_4=(0,-1,0,1)'$ satisfy $Ep_3=Ep_4=0$ (multiply out), and they are not multiples of each other, so they are linearly independent; as $\dim \mathcal{N}(E)=4-2=2$ (Theorem 4.12), they form a basis of $\mathcal{N}(E)$ (Theorem 4.5(d)).
- By Lemma 7.24(b), $p_1,\dots,p_4$ are linearly independent, so $P=[p_1\ p_2\ p_3\ p_4]$ is invertible (Theorem 4.13(b),(d)), and by Definition 7.23

$$P^{-1}EP = \operatorname{diag}(1,1,0,0), \qquad P = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 0 & 2 & -1 \\ 0 & -1 & 1 & 0 \\ -1 & -1 & 0 & 1 \end{pmatrix}$$

  Also $p_E(t)=t^2(t-1)^2$ (Theorem 8.13(a)). Here $\det P = 1$, so P^{-1} has integer entries (Theorem 5.24).
- E is not symmetric (its $(1,3)$ entry is -1, its $(3,1)$ entry 0), so no orthogonal Q makes $Q'EQ$ diagonal (Theorem 8.3(b)): E is diagonalizable, but not by an orthogonal matrix.

Proposition 8.15 ([5, Theorem 21.8.1])**.** If $P \in \mathbb{R}^{n\times n}$ is orthogonal and λ is a real eigenvalue of P, then $\lambda = 1$ or $\lambda = -1$.

Proof. Let $Px=\lambda x$ with $x \neq 0$. Then $x'x = x'P'Px = (Px)'(Px) = \lambda^2 x'x$, and $x'x > 0$; so $\lambda^2 = 1$. □

An orthogonal matrix need not have a real eigenvalue: $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is orthogonal, with characteristic polynomial t^2+1.

Example 8.16 ([2, Exercise 6.15], [6, Exercise 8.74])**.** Let $n \geqslant 2$.

(a) $\bar{J}_n$ and C_n are symmetric and idempotent (Proposition 1.15), of ranks $\operatorname{tr}\bar{J}_n = 1$ and $n-1$ (Theorem 6.29(c), Corollary 6.30). By Theorem 8.13(a), $\bar{J}_n$ has the eigenvalues 1 (once) and 0 ($n-1$ times), and C_n the eigenvalues 1 ($n-1$ times) and 0 (once). Since $\bar{J}_n x = \bar{x}\mathbf{1}$ and $C_n x = x - \bar{x}\mathbf{1}$, the eigenspaces are $\mathcal{N}(\bar{J}_n - I_n) = \mathcal{N}(C_n) = \operatorname{span}\{\mathbf{1}\}$ and $\mathcal{N}(\bar{J}_n) = \mathcal{N}(C_n - I_n) = \{x : \mathbf{1}'x = 0\} = \operatorname{span}\{\mathbf{1}\}^\perp$ (Theorem 6.6(a)).

(b) $\bar{J}_n + C_n = I_n$ and $\bar{J}_n C_n = C_n \bar{J}_n = \bar{J}_n - \bar{J}_n^2 = 0$. So, for numbers $a \neq b$, $a\bar{J}_n + bC_n$ satisfies the hypotheses of Theorem 8.6(d): its eigenvalues are a and b. As $a\bar{J}_n + bC_n - aI_n = (b-a)C_n$ and $a\bar{J}_n + bC_n - bI_n = (a-b)\bar{J}_n$, with $a-b \neq 0$, by (a) the eigenspace of a is $\mathcal{N}(C_n) = \operatorname{span}\{\mathbf{1}\}$ and that of b is $\mathcal{N}(\bar{J}_n) = \operatorname{span}\{\mathbf{1}\}^\perp$; so a has multiplicity 1 and b multiplicity $n-1$ (Theorems 8.4 and 6.6(c)). For instance $J_n = n\bar{J}_n + 0\,C_n$ has the eigenvalues n (once) and 0 ($n-1$ times), as found in Example 7.34; and for numbers α, β with $\beta \neq 0$, $\alpha I_n + \beta J_n = (\alpha + n\beta)\bar{J}_n + \alpha C_n$ (as $I_n = \bar{J}_n + C_n$ and $J_n = n\bar{J}_n$) has the eigenvalues $\alpha + n\beta$, with eigenspace $\operatorname{span}\{\mathbf{1}\}$, and α, of multiplicity $n-1$.

(c) The equicorrelation matrix $R = (1-\rho)I_n + \rho J_n$ of Example 5.33 is, writing $I_n = \bar{J}_n + C_n$ and $J_n = n\bar{J}_n$,

$$R = \big(1 + (n-1)\rho\big)\bar{J}_n + (1-\rho)C_n$$

For $\rho \neq 0$, by (b), its eigenvalues are $1 + (n-1)\rho$, with eigenvector $\mathbf{1}$, and $1-\rho$, of multiplicity $n-1$. So $\det R = \big(1+(n-1)\rho\big)(1-\rho)^{n-1}$ (Theorem 7.19), as in Example 5.33, and for $\rho \neq 1$, $\rho \neq -1/(n-1)$, $R^{-1} = \big(1+(n-1)\rho\big)^{-1}\bar{J}_n + (1-\rho)^{-1}C_n$ (Theorem 8.6(b)). For $n = 4$ and $\rho = \frac{1}{3}$, $R = 2\bar{J}_4 + \frac{2}{3}C_4$ has the eigenvalues 2 and $\frac{2}{3}$ (three times), $\det R = 2 \cdot \left(\frac{2}{3}\right)^3 = \frac{16}{27}$, and $R^{-1} = \frac{1}{2}\bar{J}_4 + \frac{3}{2}C_4 = \frac{3}{2}I_4 - \frac{1}{4}J_4$, as in Example 5.33.

(d) Let W be a subspace of $\mathbb{R}^n$ with $\dim W = r$. The projection matrix P_W is symmetric and idempotent, of rank r (Theorem 6.31(c)). So P_W has the eigenvalue 1 with multiplicity r and 0 with multiplicity $n-r$ (Theorem 8.13(a)), and $I_n - P_W = P_{W^\perp}$ the reverse; $P_W y = y$ exactly for $y \in W$, and $P_W y = 0$ exactly for $y \in W^\perp$ (Theorem 6.31(c), Definition 6.22). For instance, the matrix P of Example 6.28 has the eigenvalues 1, 1, 1, 0, as $\operatorname{rank} P = \operatorname{tr} P = 3$ (Corollary 6.30); and for $W = \operatorname{span}\{\mathbf{1}_n\}$, $P_W = \bar{J}_n$ and $I_n - P_W = C_n$ (Corollary 6.35(a)), which is (a).

Quadratic forms and definiteness

Definition 8.17 ([2, §5.6], [5, §14.1], [6, Exercise 8.1(a)])**.** The **quadratic form** of $A \in \mathbb{R}^{n\times n}$ is the function $q_A(x) = x'Ax = \sum_{i=1}^n \sum_{j=1}^n a_{ij}x_i x_j$ $(x \in \mathbb{R}^n)$. Collecting the terms in x_i^2 and in $x_i x_j$ $(i<j)$,

$$x'Ax = \sum_{i=1}^n a_{ii}x_i^2 + \sum_{i<j}(a_{ij}+a_{ji})x_i x_j$$

As $x'Ax$ is a 1×1 matrix, $x'Ax = (x'Ax)' = x'A'x$; so $x'Ax = x'Bx$ for the symmetric matrix $B = \frac{1}{2}(A+A')$. Different matrices can give the same form, but only one symmetric matrix does.

Proposition 8.18 ([4, Ch. V, §7])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric.

(a) $x'Ay = \frac{1}{2}\big(q_A(x+y) - q_A(x) - q_A(y)\big)$ for all $x, y \in \mathbb{R}^n$.

(b) If $x'Ax = x'Bx$ for every $x \in \mathbb{R}^n$, then $A = B$.

Proof. (a) $(x+y)'A(x+y) = x'Ax + x'Ay + y'Ax + y'Ay$, and $y'Ax = (y'Ax)' = x'A'y = x'Ay$.
(b) By (a), $x'Ay = x'By$ for all x, y. For $x = e_i$ and $y = e_j$ this reads $a_{ij} = b_{ij}$, since $e_i'Ae_j$ is entry i of $Ae_j = \operatorname{col}_j(A)$ (Theorem 1.4(b)). □

So every quadratic form is q_A for exactly one symmetric A, the **matrix of the form**: a_{ii} is the coefficient of x_i^2, and $a_{ij} = a_{ji}$ is half the coefficient of $x_i x_j$ $(i \neq j)$.

Example 8.19 (cf. [2, Example 5.6, §5.6])**.** (a) For data $y \in \mathbb{R}^n$, the sum of squares about the mean $\sum_i (y_i - \bar{y})^2$ equals $y'C_n y$ (Example 1.16): it is the quadratic form of the symmetric matrix C_n.

(b) The matrices

$$T = \begin{pmatrix} 2 & 0 & 2 & -2 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \qquad T' = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 2 & 2 & 3 & 0 \\ -2 & 2 & 0 & 3 \end{pmatrix}, \qquad A = \tfrac{1}{2}(T+T') = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ -1 & 1 & 0 & 3 \end{pmatrix}$$

give the same form $2x_1^2 + 2x_2^2 + 3x_3^2 + 3x_4^2 + 2x_1x_3 - 2x_1x_4 + 2x_2x_3 + 2x_2x_4$ (Definition 8.17: $t_{12}+t_{21} = 0$, $t_{13}+t_{31} = 2$, $t_{14}+t_{41} = -2$, $t_{23}+t_{32} = 2$, $t_{24}+t_{42} = 2$, $t_{34}+t_{43} = 0$); its matrix is A.

Definition 8.20 ([6, Ch. 8], [3, §3.1])**.** A symmetric $A \in \mathbb{R}^{n\times n}$, and its quadratic form, are

- **positive definite** (PD) if $x'Ax > 0$ for every nonzero $x \in \mathbb{R}^n$;
- **positive semidefinite** (PSD) if $x'Ax \geqslant 0$ for every $x \in \mathbb{R}^n$;
- **negative definite** (ND) or **negative semidefinite** (NSD) if $-A$ is PD or PSD;
- **indefinite** if $x'Ax$ takes both positive and negative values.

A PD matrix is PSD.

Theorem 8.21 ([6, Exercises 7.53(a), 8.11, 8.13(a)], [1, §9.5, Theorem 12], [5, Corollary 14.3.12])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, with spectral decomposition $A = c_1E_1 + \cdots + c_kE_k$, and let $A = Q\Lambda Q'$ as in Theorem 8.3(b), with $\lambda_1 \geqslant \cdots \geqslant \lambda_n$. Then, for $x \in \mathbb{R}^n$ and $y = Q'x$,

$$x'Ax = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 = c_1\|E_1x\|^2 + \cdots + c_k\|E_kx\|^2$$

(a) A is PSD if and only if every eigenvalue of A is $\geqslant 0$, and PD if and only if every eigenvalue is > 0.

(b) A is NSD (ND) if and only if every eigenvalue is $\leqslant 0$ (< 0), and indefinite if and only if A has a positive and a negative eigenvalue.

(c) If A is PD, then $\det A > 0$, and A is invertible; if A is PSD, then $\det A \geqslant 0$. A PSD matrix is PD if and only if it is invertible.

(d) $\lambda_n\, x'x \leqslant x'Ax \leqslant \lambda_1\, x'x$ for every $x \in \mathbb{R}^n$, with equality on the right for $x = q_1$ and on the left for $x = q_n$.

Proof. • Let $x \in \mathbb{R}^n$ and $y = Q'x$. Then $x = QQ'x = Qy$ and $y'y = x'QQ'x = x'x$ (Proposition 6.13(b)), and, as $Q'AQ = \Lambda$,

$$x'Ax = x'QQ'AQQ'x = y'\Lambda y = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$$

- Entry i of $y = Q'x$ is $y_i = q_i'x$ (Theorem 1.4(d)). As $E_j' = E_j = E_j^2$ and $E_j = \sum_{\lambda_i=c_j} q_iq_i'$ (Theorem 8.6(b),(c)), $\|E_jx\|^2 = x'E_j'E_jx = x'E_jx = \sum_{\lambda_i=c_j}(q_i'x)^2 = \sum_{\lambda_i=c_j} y_i^2$. Grouping the terms $\lambda_i y_i^2$ by the value c_j of λ_i gives the second expression.

(a)

- If every $\lambda_i \geqslant 0$, then $x'Ax \geqslant 0$. If every $\lambda_i > 0$ and $x \neq 0$, then $y \neq 0$, as $x = Qy$; so $x'Ax > 0$.
- Conversely, for $x = q_i$, $y = Q'Qe_i = e_i$ and $x'Ax = \lambda_i$; so $\lambda_i \geqslant 0$ if A is PSD, and $\lambda_i > 0$ if A is PD. The λ_i are the eigenvalues of A (Theorem 8.3(c)).

(b)

- $-A = \sum_j(-c_j)E_j$, so the eigenvalues of $-A$ are $-c_1, \ldots, -c_k$ (Theorem 8.6(d)); apply (a) to $-A$.
- If $c_i > 0 > c_l$, unit eigenvectors x, y for c_i, c_l give $x'Ax = c_i > 0$ and $y'Ay = c_l < 0$. If no eigenvalue is negative (positive), then A is PSD (NSD), hence not indefinite.

(c) If A is PD, its eigenvalues $\lambda_1, \ldots, \lambda_n$ are > 0 by (a), so $\det A = \lambda_1 \cdots \lambda_n > 0$ (Theorem 7.19), and A is invertible (Theorem 5.18(a)). If A is PSD, its eigenvalues are $\geqslant 0$ by (a), and $\det A = \lambda_1 \cdots \lambda_n \geqslant 0$ in the same way. If A is PSD and invertible, then 0 is not an eigenvalue (Theorem 7.3), so every eigenvalue is > 0, and A is PD by (a).

(d) As $\lambda_n \leqslant \lambda_i \leqslant \lambda_1$ and $y_i^2 \geqslant 0$, $x'Ax = \sum_i \lambda_i y_i^2$ lies between $\lambda_n \sum_i y_i^2 = \lambda_n\, x'x$ and $\lambda_1 \sum_i y_i^2 = \lambda_1\, x'x$. Moreover $q_1'Aq_1 = \lambda_1 q_1'q_1 = \lambda_1$ and $q_n'Aq_n = \lambda_n$. □

Example 8.22 ([6, Exercise 8.74(c)])**.** (a) C_n has the eigenvalues 1 and 0 (Example 8.16(a)), so it is PSD but not PD. Indeed $C_n = 1 \cdot C_n + 0 \cdot \bar{J}_n$, and the identity of Theorem 8.21 gives $\sum_i (y_i - \bar{y})^2 = y'C_n y = \|C_n y\|^2 \geqslant 0$ (Example 1.16), with equality exactly when $C_n y = 0$, that is, when $y = \bar{y}\mathbf{1}$ is constant.

(b) $A = 2J_3 - I_3$ (Example 8.5) has the eigenvalues 5 and -1, so it is indefinite: for $x = \mathbf{1}_3$ and $y = (1, -1, 0)'$, $x'Ax = 15$ and $y'Ay = -2$.

(c) By Example 8.16(c), the equicorrelation matrix R is PD if and only if $1 + (n-1)\rho > 0$ and $1 - \rho > 0$, that is, $-1/(n-1) < \rho < 1$, and PSD if and only if $-1/(n-1) \leqslant \rho \leqslant 1$ (for $\rho = 0$, $R = I_n$). For $n = 4$ and $\rho = -\frac{1}{3}$, R is PSD and singular ($R\mathbf{1}_4 = 0$, Example 5.33).

(d) The matrix A of Example 8.8 has the eigenvalues 25, 9, 9, 1, all > 0; so it is PD (Theorem 8.21(a)).

Proposition 8.23 ([6, Exercise 8.6], [5, Lemmas 14.2.2, 14.2.4, 14.2.17], [4, Ch. VIII, §4, Exercise 5], [7])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric.

(a) A diagonal matrix $D = \operatorname{diag}(d_1, \ldots, d_n)$ is PD (PSD) if and only if every $d_i > 0$ ($d_i \geqslant 0$).

(b) If A and B are PSD and $\alpha, \beta \geqslant 0$, then $\alpha A + \beta B$ is PSD; if moreover A is PD and $\alpha > 0$, then $\alpha A + \beta B$ is PD.

(c) If A is PD, then A is invertible and A^{-1} is PD.

(d) If A is idempotent, then A is PSD.

(e) If A and $-A$ are PSD, then $A = 0$.

Proof. (a) $x'Dx = d_1x_1^2 + \cdots + d_nx_n^2$. This is > 0 for $x \neq 0$ if every $d_i > 0$, and $\geqslant 0$ for every x if every $d_i \geqslant 0$. Conversely $d_i = e_i'De_i$, which is > 0 if D is PD and $\geqslant 0$ if D is PSD.

(b) $\alpha A + \beta B$ is symmetric, and $x'(\alpha A + \beta B)x = \alpha\, x'Ax + \beta\, x'Bx$. Both terms are $\geqslant 0$; if A is PD, $\alpha > 0$ and $x \neq 0$, the first term is > 0.

(c)

- A is invertible (Theorem 8.21(c)), and $(A^{-1})' = (A')^{-1} = A^{-1}$ (Theorem 2.13(c)).
- Let $x \neq 0$ and $y = A^{-1}x$. Then $y \neq 0$, as $x = Ay$, and $x'A^{-1}x = (Ay)'y = y'A'y = y'Ay > 0$.

(d) As $A = A'$ and $A = A^2$, $x'Ax = x'A'Ax = (Ax)'(Ax) = \|Ax\|^2 \geqslant 0$.

(e) $x'Ax \geqslant 0$ and $-x'Ax \geqslant 0$, so $x'Ax = 0 = x'0x$ for every x. As 0 is symmetric, $A = 0$ (Proposition 8.18(b)). □

Congruence, $A'A$, and the factorization $A = P'P$

Theorem 8.24 ([5, Theorem 14.2.9, Corollary 14.2.10])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric and $P \in \mathbb{R}^{n\times m}$.

(a) If A is PSD, then $P'AP$ is PSD.

(b) If A is PD and $\operatorname{rank} P = m$, then $P'AP$ is PD.

(c) If $m = n$ and P is invertible, then A is PD (PSD) if and only if $P'AP$ is PD (PSD).

Proof. $P'AP$ is symmetric, and $x'(P'AP)x = (Px)'A(Px)$ for $x \in \mathbb{R}^m$.

(a) This is $\geqslant 0$.

(b) If $x \neq 0$, then $Px \neq 0$ (Theorem 4.13(b)), so it is > 0.

(c) $\operatorname{rank} P = n$ (Theorem 4.13(d)), so (a) and (b) give one direction; for the other, apply them to $P'AP$ and P^{-1}: $A = (P^{-1})'(P'AP)P^{-1}$ (Theorem 2.13(c)). □

Theorem 8.25 ([5, Corollary 14.2.14], [2, Theorem 5.3])**.** Let $A \in \mathbb{R}^{n\times p}$ have the columns $a_1, \dots, a_p$. Then $A'A$, the **Gram matrix** of $a_1, \dots, a_p$, with the entries $a_i'a_j$ (Theorem 1.4(d)), is PSD, and it is PD if and only if $\operatorname{rank} A = p$. Likewise AA' is PSD, and PD if and only if $\operatorname{rank} A = n$.

Proof. $A'A = A'I_nA$, and I_n is PD; so $A'A$ is PSD, and PD if $\operatorname{rank} A = p$ (Theorem 8.24(a),(b)). If $A'A$ is PD, it is invertible (Theorem 8.21(c)), so $\operatorname{rank} A = \operatorname{rank}(A'A) = p$ (Theorems 4.18(b) and 4.13(d)). For AA' apply this to A', with $\operatorname{rank} A' = \operatorname{rank} A$ (Theorem 4.13(a)). □

For a data matrix $X_0 \in \mathbb{R}^{n\times k}$, the matrix of corrected sums of squares and products is $X_0'C_nX_0 = (C_nX_0)'(C_nX_0)$ (Example 1.17). So it is PSD, and PD exactly when $\operatorname{rank}(C_nX_0) = k$ (Theorem 8.25), that is, when the centered columns are linearly independent; in Example 1.17, $X_0'C_4X_0 = 4I_2$ is PD.

Theorem 8.26 ([5, Theorem 14.3.7, Corollaries 14.3.8, 14.3.13], [1, §9.3, Theorem 5; §8.4, Example 24])**.** Let $A \in \mathbb{R}^{n\times n}$.

(a) Let $1 \leqslant r \leqslant n$. Then A is symmetric, PSD and of rank r if and only if $A = P'P$ for some $P \in \mathbb{R}^{r\times n}$ of rank r.

(b) A is symmetric and PSD if and only if $A = P'P$ for some matrix P with n columns.

(c) A is symmetric and PD if and only if $A = P'P$ for some invertible $P \in \mathbb{R}^{n\times n}$. In this case $\det A = (\det P)^2 > 0$, and $y'Ax = (Py)'(Px)$ for all $x, y \in \mathbb{R}^n$.

Proof. (a)

- Let A be symmetric, PSD and of rank r, and let $A = Q\Lambda Q' = \sum_{i=1}^n \lambda_i q_i q_i'$ (Theorems 8.3 and 8.6(c)). Exactly r of the λ_i, say $\lambda_{k_1}, \dots, \lambda_{k_r}$, are nonzero (Theorem 8.4), and they are > 0 (Theorem 8.21(a)).
- Let $P \in \mathbb{R}^{r\times n}$ have the rows $\sqrt{\lambda_{k_1}}\, q_{k_1}', \dots, \sqrt{\lambda_{k_r}}\, q_{k_r}'$. By the outer product expansion (Theorem 1.4(d)), $P'P = \sum_{l=1}^r \lambda_{k_l} q_{k_l} q_{k_l}' = A$, since the other terms of $\sum_i \lambda_i q_i q_i'$ are 0. Moreover $\operatorname{rank} P = \operatorname{rank}(P'P) = \operatorname{rank} A = r$ (Theorem 4.18(b)).
- Conversely, if $A = P'P$ with $P \in \mathbb{R}^{r\times n}$ of rank r, then A is symmetric (Theorem 1.9), PSD (Theorem 8.25), and $\operatorname{rank} A = \operatorname{rank}(P'P) = \operatorname{rank} P = r$ (Theorem 4.18(b)).

(b) Let A be symmetric and PSD. If $A = 0$, then $A = P'P$ for the zero matrix $P \in \mathbb{R}^{1\times n}$; otherwise (a) applies with $r = \operatorname{rank} A \geqslant 1$. Conversely, $P'P$ is symmetric and PSD (Theorems 1.9 and 8.25).

(c)

- If A is symmetric and PD, then A is invertible (Theorem 8.21(c)), so $\operatorname{rank} A = n$ (Theorem 4.13(d)). By (a) with $r = n$, $A = P'P$ with $P \in \mathbb{R}^{n\times n}$ of rank n, and P is invertible (Theorem 4.13(d)).
- Conversely, if $A = P'P$ with P invertible, then $\operatorname{rank} P = n$, and A is symmetric and PD (Theorems 1.9 and 8.25).
- Then $\det A = \det P' \cdot \det P = (\det P)^2$ (Theorems 5.18(b) and 5.22), and $\det P \neq 0$ (Theorem 5.18(a)). Finally $y'Ax = y'P'Px = (Py)'(Px)$. □

Corollary 8.27 ([5, Corollary 14.3.11])**.** Let $A \in \mathbb{R}^{n\times n}$ be PSD and $X \in \mathbb{R}^{n\times m}$. Then $X'AX = 0$ if and only if $AX = 0$. In particular $x'Ax = 0$ if and only if $Ax = 0$: the vectors x with $x'Ax = 0$ form the subspace $\mathcal{N}(A)$.

Proof. If $AX = 0$, then $X'AX = 0$. Conversely, let $X'AX = 0$, and write $A = P'P$ (Theorem 8.26(b)).

- $X'AX = (PX)'(PX)$, so $\operatorname{tr}\big((PX)'(PX)\big) = 0$, and $PX = 0$ (Theorem 1.11(d)).
- Hence $AX = P'(PX) = 0$. The case $m = 1$, $X = x$, gives the last sentence. □

For instance, $\sum_i (y_i - \bar{y})^2 = y'C_n y = 0$ only if $C_n y = 0$ (Example 1.16), and, for A and P as in Theorem 6.26, $y'(I_n - P)y = 0$ only if $(I_n - P)y = 0$: both C_n and $I_n - P$ are symmetric and idempotent (Proposition 1.15, Theorem 6.26(b)), hence PSD (Proposition 8.23(d)).

Example 8.28. For $x \in \mathbb{R}^4$, the sum of the squares of the two sums of three successive entries is

$$(x_1 + x_2 + x_3)^2 + (x_2 + x_3 + x_4)^2 = \|Dx\|^2 = x'Ax$$

where

$$D = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \qquad A = D'D = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Find the factor P of Theorem 8.26(a) and all x with $x'Ax = 0$.

Solution.

- A is symmetric and PSD, and $\operatorname{rank} A = \operatorname{rank} D = 2$ (Theorems 8.25 and 4.18(b)), as the two rows of D are independent.
- $A(1,2,2,1)' = (5,10,10,5)'$, $A(1,0,0,-1)' = (1,0,0,-1)'$, and $A(0,1,-1,0)' = A(2,-1,-1,2)' = 0$. So $a_A(c) \geqslant 1$ for $c = 5, 1$, and $a_A(0) = \dim \mathcal{N}(A) = 4 - \operatorname{rank} A = 2$ (Theorem 8.4); as the multiplicities add up to 4 (Definition 7.10), the eigenvalues are 5, 1, 0, 0. The unit eigenvectors for 5 and 1 are $q_1 = \frac{1}{\sqrt{10}}(1,2,2,1)'$ and $q_2 = \frac{1}{\sqrt{2}}(1,0,0,-1)'$, and the orthogonal unit vectors $q_3 = \frac{1}{\sqrt{2}}(0,1,-1,0)'$, $q_4 = \frac{1}{\sqrt{10}}(2,-1,-1,2)'$ of $\mathcal{N}(A)$ complete them to an orthonormal basis of eigenvectors (Theorem 8.2).
- The proof of Theorem 8.26(a) takes the rows $\sqrt{5}\, q_1' = \frac{1}{\sqrt{2}}(1,2,2,1)$ and $\sqrt{1}\, q_2' = \frac{1}{\sqrt{2}}(1,0,0,-1)$:

$$P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix}, \qquad P'P = \frac{1}{2} \begin{pmatrix} 2 & 2 & 2 & 0 \\ 2 & 4 & 4 & 2 \\ 2 & 4 & 4 & 2 \\ 0 & 2 & 2 & 2 \end{pmatrix} = A$$

  The matrix D is another 2×4 factor of rank 2: the factor in Theorem 8.26(a) is not unique.
- By Corollary 8.27, $x'Ax = 0$ if and only if $x \in \mathcal{N}(A) = \operatorname{span}\{(0,1,-1,0)', (2,-1,-1,2)'\}$ ($\dim \mathcal{N}(A) = 4 - 2 = 2$, Theorem 4.12, and these two vectors are independent, Theorem 4.5(d)), that is, $x_1 = x_4 = -(x_2 + x_3)$.

Principal submatrices and Schur complements

Definition 8.29 ([3, §§3.1, 3.3], [1, §9.3], [5, Corollary 14.2.12])**.** Let $A \in \mathbb{R}^{n \times n}$ and $1 \leqslant i_1 < \cdots < i_k \leqslant n$. The $k \times k$ matrix whose (s,t) entry is $a_{i_s i_t}$ (rows and columns $i_1, \ldots, i_k$ of A) is a **principal submatrix** of A; with $S = [e_{i_1} \ \cdots \ e_{i_k}] \in \mathbb{R}^{n \times k}$ it equals $S'AS$, as the (s,t) entry of $S'AS$ is $e_{i_s}' A e_{i_t} = a_{i_s i_t}$ (Theorem 1.4(b),(d)). The **leading principal submatrix** A_k uses rows and columns $1, \ldots, k$, and the numbers $\Delta_k(A) = \det A_k$ ($k = 1, \ldots, n$) are the **leading principal minors** of A; we put $\Delta_0(A) = 1$.

Theorem 8.30 ([3, Result 3.4], [5, Corollaries 14.2.12, 14.2.13, 14.5.14, Theorem 14.9.11], [6, Exercises 8.7(b),(d), 8.27(c)])**.** If A is PD (PSD), then every principal submatrix of A is PD (PSD). In particular every diagonal entry a_{ii} is > 0 ($\geqslant 0$), and if A is PD, then $\Delta_k(A) > 0$ for $k = 1, \ldots, n$. If A is PSD, then moreover $a_{ij}^2 \leqslant a_{ii} a_{jj}$ for all i, j, and if $a_{ii} = 0$, then row i and column i of A are zero.

Proof. • With S as in Definition 8.29 and $y \in \mathbb{R}^k$, the vector $x = Sy$ has the entries $y_1, \ldots, y_k$ in the positions $i_1, \ldots, i_k$ and zeros elsewhere (Theorem 1.4(b)); so $x \neq 0$ if $y \neq 0$, and $y'(S'AS)y = x'Ax$. Hence $S'AS$, which is symmetric, is PD (PSD) when A is.

- (a_{ii}) is the principal submatrix with $k = 1$ and $i_1 = i$. If A is PD, then A_k is PD and $\Delta_k(A) = \det A_k > 0$ (Theorem 8.21(c)).
- Let A be PSD and $i \neq j$. The principal submatrix with the rows and columns i and j is PSD (first item), and its determinant is $a_{ii}a_{jj} - a_{ij}a_{ji} = a_{ii}a_{jj} - a_{ij}^2$ (Theorem 5.21); so $a_{ii}a_{jj} - a_{ij}^2 \geqslant 0$ (Theorem 8.21(c)). For $i = j$ there is nothing to prove.
- If A is PSD and $a_{ii} = 0$, then $e_i'Ae_i = a_{ii} = 0$ (Theorem 1.4(b)), so $Ae_i = 0$ (Corollary 8.27): column i of A is zero, and so is row i, its transpose, as $A' = A$. □

So $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$ is not PSD ($2^2 > 1 \cdot 3$), and neither is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ ($a_{11} = 0$, but row 1 is not zero).

Theorem 8.31 ([3, Result 3.14], [5, Lemma 14.8.3, Theorems 14.8.4, 14.8.5], [6, Exercise 12.33])**.** Let $M = \begin{pmatrix} A & B \\ B' & D \end{pmatrix}$ be symmetric, with $A \in \mathbb{R}^{p\times p}$ invertible, $B \in \mathbb{R}^{p\times q}$ and $D \in \mathbb{R}^{q\times q}$, and let $M/A = D - B'A^{-1}B$ be the Schur complement of A. Then M is PD if and only if A and M/A are PD. If A is PD, then M is PSD if and only if M/A is PSD. In every case $\det M = \det A \cdot \det(M/A)$.

Proof. • A and D are symmetric, as $M' = M$ (Theorem 5.3(b)), hence so are A^{-1} (Theorem 2.13(c)) and M/A. Let $L = \left(\begin{smallmatrix} I_p & 0 \\ -B'A^{-1} & I_q \end{smallmatrix}\right)$; then $L' = \left(\begin{smallmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{smallmatrix}\right)$, L is invertible (Proposition 5.8), and by Theorem 5.9 (with $C = B'$)

$$LML' = \begin{pmatrix} A & 0 \\ 0 & M/A \end{pmatrix}$$

- If M is PD (PSD), so is LML' (Theorem 8.24(c)), and A and M/A are principal submatrices of it, hence PD (PSD) (Theorem 8.30).
- Let A be PD and M/A PSD, and let $x = \left(\begin{smallmatrix} y \\ z \end{smallmatrix}\right)$ ($y \in \mathbb{R}^p$, $z \in \mathbb{R}^q$). Then $x'(LML')x = y'Ay + z'(M/A)z \geqslant 0$, and if M/A is PD and $x \neq 0$, this is > 0, as $y \neq 0$ or $z \neq 0$. So LML' is PSD, and PD if M/A is PD; and so is M (Theorem 8.24(c)).
- The determinant is Theorem 5.30(a). □

Theorem 8.32 (Hadamard's inequality; [6, Exercise 12.26(a)], [3, Result 7.2])**.** If $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is PSD, then $\det A \leqslant a_{11}a_{22}\cdots a_{nn}$. If A is PD, equality holds if and only if A is diagonal.

Proof. First let A be PD, with the leading principal submatrices $A_1, \ldots, A_n = A$ (Definition 8.29).

- For $2 \leqslant k \leqslant n$ write

$$A_k = \begin{pmatrix} A_{k-1} & b \\ b' & a_{kk} \end{pmatrix}$$

with $b \in \mathbb{R}^{k-1}$. A_{k-1} is PD (Theorem 8.30), so $\det A_{k-1} > 0$ and A_{k-1}^{-1} is PD (Theorem 8.21(c), Proposition 8.23(c)). By Theorem 5.30(a),

$$\det A_k = \det A_{k-1} \cdot \left(a_{kk} - b'A_{k-1}^{-1}b\right) \leqslant a_{kk} \det A_{k-1}$$

with equality if and only if $b'A_{k-1}^{-1}b = 0$, that is, $b = 0$.

- Every $a_{kk} > 0$ (Theorem 8.30). Applying the first item for $k = n, n-1, \ldots, 2$, $\det A \leqslant a_{nn} \det A_{n-1} \leqslant a_{nn}a_{n-1,n-1} \det A_{n-2} \leqslant \cdots \leqslant a_{nn}\cdots a_{22} \det A_1 = a_{11}a_{22}\cdots a_{nn}$.
- Each step multiplies one of these inequalities by positive numbers, so equality holds if and only if it holds at every step, that is, if and only if the off-diagonal entries of the last row and column of A_k vanish for $k = 2, \ldots, n$: A is diagonal.
- If A is PSD but not PD, then A is not invertible (Theorem 8.21(c)), so $\det A = 0$ (Theorem 5.18(a)), while every $a_{ii} \geqslant 0$ (Theorem 8.30). □

For the equicorrelation matrix R, whose diagonal entries are 1 and which is PSD for $-1/(n-1) \leqslant \rho \leqslant 1$ (Example 8.22(c)), Hadamard's inequality says that $\det R = \big(1+(n-1)\rho\big)(1-\rho)^{n-1} \leqslant 1$ (Example 5.33), with equality for PD R only at $\rho = 0$.

Leading principal minors, LDU and Cholesky decompositions

Lemma 8.33 ([1, §9.3, Lemma])**.** Let $A \in \mathbb{R}^{n\times n}$ with $\Delta_k(A) \neq 0$ for $k = 1, \dots, n-1$. There is exactly one upper triangular $P \in \mathbb{R}^{n\times n}$ with diagonal entries 1 such that AP is lower triangular, and $\Delta_k(AP) = \Delta_k(A)$ for $k = 1, \dots, n$.

Proof. Let P be upper triangular with $p_{kk} = 1$ $(k = 1, \dots, n)$, and $B = AP$.

- $b_{jk} = \sum_r a_{jr}p_{rk} = a_{jk} + \sum_{r=1}^{k-1} a_{jr}p_{rk}$, as $p_{rk} = 0$ for $r > k$. So B is lower triangular ($b_{jk} = 0$ for $j < k$) if and only if, for $k = 2, \dots, n$,

$$\sum_{r=1}^{k-1} a_{jr}p_{rk} = -a_{jk} \qquad (j = 1, \dots, k-1)$$

- For fixed k, this is a system of $k-1$ equations in the $k-1$ unknowns $p_{1k}, \dots, p_{k-1,k}$ with coefficient matrix A_{k-1}, and $\det A_{k-1} = \Delta_{k-1}(A) \neq 0$. So it has exactly one solution (Theorems 5.18(a) and 2.16): P exists and is unique.
- By the first item, column r of B is column r of A plus $\sum_{j<r} p_{jr}$ times column j of A. So B_k arises from A_k by adding to column r a combination of columns $j < r$, for $r = k, k-1, \dots, 2$ in turn (each step uses columns not yet changed). Such column operations do not change the determinant (Definition 5.13(b) and Theorem 5.22); so $\Delta_k(B) = \Delta_k(A)$. □

Theorem 8.34 (Sylvester's criterion; [1, §9.3, Theorem 6])**.** A symmetric $A \in \mathbb{R}^{n\times n}$ is PD if and only if $\Delta_k(A) > 0$ for $k = 1, \dots, n$. In this case, with P as in Lemma 8.33 and $U = P^{-1}$,

$$P'AP = D = \operatorname{diag}(d_1, \dots, d_n), \qquad d_k = \frac{\Delta_k(A)}{\Delta_{k-1}(A)} > 0, \qquad A = U'DU$$

and U is upper triangular with diagonal entries 1.

Proof. If A is PD, every $\Delta_k(A) > 0$ by Theorem 8.30. Conversely, let every $\Delta_k(A) > 0$, and let $B = AP$ be lower triangular (Lemma 8.33).

- $D = P'AP = P'B$ is lower triangular: $(P'B)_{jk} = \sum_r p_{rj}b_{rk}$, with $p_{rj} = 0$ for $r > j$ and $b_{rk} = 0$ for $r < k$, so it is 0 for $j < k$. Also $D' = P'A'P = D$. A symmetric lower triangular matrix is diagonal.
- As $p_{jj} = 1$, row j of $P'B$ is row j of B plus $\sum_{r<j} p_{rj}$ times row r of B. So D_k arises from B_k by adding to row j a combination of the rows above it, for $j = k, \dots, 2$ in turn; this does not change the determinant (Definition 5.13(b)), and $\Delta_k(D) = \Delta_k(B) = \Delta_k(A)$.
- $\Delta_k(D) = d_1 \cdots d_k$ (Theorem 5.14(c)). So $d_1 \cdots d_k = \Delta_k(A) > 0$ for every k, and $d_k = \Delta_k(A)/\Delta_{k-1}(A) > 0$.
- $\det P = 1$ (Theorem 5.14(c)), so P is invertible (Theorem 5.18(a)), and $A = (P')^{-1}DP^{-1} = U'DU$ (Theorem 2.13(c)). For $x \neq 0$, $Ux \neq 0$ and $x'Ax = (Ux)'D(Ux) = \sum_k d_k(Ux)_k^2 > 0$: A is PD.
- P is unit upper triangular, so $U = P^{-1}$ is unit upper triangular (Theorem 2.31). □

Example 8.35. Show that the matrix

$$A = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ -1 & 1 & 0 & 3 \end{pmatrix}$$

of Example 8.19 is PD, and write $A = U'DU$.

Solution.

- $\Delta_1 = 2$, $\Delta_2 = \det(2I_2) = 4$, and along row 1 (Theorem 5.21), $\Delta_3 = 2(2 \cdot 3 - 1 \cdot 1) - 0 + 1(0 \cdot 1 - 2 \cdot 1) = 10 - 2 = 8$. For $\Delta_4 = \det A$, write $A = \left(\begin{smallmatrix} A_2 & B \\ B' & 3I_2 \end{smallmatrix}\right)$ with $A_2 = 2I_2$ and $B = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$. As $B'B = 2I_2$, the Schur complement of A_2 is $A/A_2 = 3I_2 - B'(\frac{1}{2}I_2)B = 3I_2 - I_2 = 2I_2$, and $\Delta_4 = \det A_2 \cdot \det(2I_2) = 4 \cdot 4 = 16$ (Theorems 5.30(a) and 5.14(c)). All are > 0, so A is PD (Theorem 8.34).
- P: for $k = 2$, $a_{11}p_{12} = -a_{12}$ gives $p_{12} = 0$; for $k = 3$, $A_2 \left(\begin{smallmatrix} p_{13} \\ p_{23} \end{smallmatrix}\right) = -\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ with $A_2 = 2I_2$ gives $p_{13} = p_{23} = -\frac{1}{2}$; for $k = 4$, $A_3(p_{14}, p_{24}, p_{34})' = -(a_{14}, a_{24}, a_{34})' = (1, -1, 0)'$ reads $2p_{14} + p_{34} = 1$, $2p_{24} + p_{34} = -1$ and $p_{14} + p_{24} + 3p_{34} = 0$. The sum of the first two equations gives $p_{14} + p_{24} = -p_{34}$, so the third gives $2p_{34} = 0$; hence $p_{34} = 0$, $p_{14} = \frac{1}{2}$ and $p_{24} = -\frac{1}{2}$. Then

$$P = \begin{pmatrix} 1 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad AP = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 0 \\ -1 & 1 & 0 & 2 \end{pmatrix}, \qquad U = P^{-1} = \begin{pmatrix} 1 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

  and $D = P'(AP) = \operatorname{diag}(2, 2, 2, 2)$, with $d_k = \Delta_k/\Delta_{k-1} = 2/1,\ 4/2,\ 8/4,\ 16/8$. So $A = U'DU$, that is, $q_A(x) = 2(x_1 + \frac{1}{2}x_3 - \frac{1}{2}x_4)^2 + 2(x_2 + \frac{1}{2}x_3 + \frac{1}{2}x_4)^2 + 2x_3^2 + 2x_4^2$.
- Schur complement (Theorem 8.31, $p = 2$): $A_2 = 2I_2$ and $A/A_2 = 2I_2$ (first item) are PD (Proposition 8.23(a)); so again A is PD, and $\det A = \det A_2 \cdot \det(A/A_2) = 16$.
- Eigenvalues: $A(1, 1, 2, 0)' = (4, 4, 8, 0)'$, $A(-1, 1, 0, 2)' = (-4, 4, 0, 8)'$, $A(1, 1, -1, 0)' = (1, 1, -1, 0)'$ and $A(1, -1, 0, 1)' = (1, -1, 0, 1)'$. The two vectors for 4 are nonzero and orthogonal, hence independent (Proposition 6.11(b)), and so are the two for 1. So $a_A(c) = \dim \mathcal{N}(A - cI) \geqslant 2$ for $c = 4, 1$ (Theorems 8.4 and 4.5(c)); as the multiplicities add up to 4 (Definition 7.10), the eigenvalues are 4, 4, 1, 1, all > 0, as Theorem 8.21(a) requires.

Example 8.36. For which $a \in \mathbb{R}$ is

$$A = \begin{pmatrix} 1 & a & a & 0 \\ a & 1 & 0 & a \\ a & 0 & 1 & a \\ 0 & a & a & 1 \end{pmatrix}$$

PD?

Solution.

- The first two leading principal minors are 1 and $1 - a^2$. Expanding along the first row (Theorem 5.21), $\Delta_3 = (1 - 0) - a(a - 0) + a(0 - a) = 1 - 2a^2$.
- For $\det A$, subtract row 3 from row 2, which becomes $(0, 1, -1, 0)$, and then add column 2 to column 3 (Definition 5.13(b), Theorem 5.22); then expand along row 2 and along the first row (Theorem 5.21):

$$\det A = \det \begin{pmatrix} 1 & a & 2a & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & 1 & a \\ 0 & a & 2a & 1 \end{pmatrix} = \det \begin{pmatrix} 1 & 2a & 0 \\ a & 1 & a \\ 0 & 2a & 1 \end{pmatrix} = (1 - 2a^2) - 2a(a - 0) = 1 - 4a^2$$

- By Theorem 8.34, A is PD if and only if $1-a^2$, $1-2a^2$ and $1-4a^2$ are positive, that is, $a^2 < \frac{1}{4}$, or $-\frac{1}{2} < a < \frac{1}{2}$.

Check with eigenvalues: the orthogonal vectors $h_1 = (1,1,1,1)'$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$, $h_4 = (1,-1,-1,1)'$ satisfy

$$Ah_1 = (1+2a)h_1, \qquad Ah_2 = h_2, \qquad Ah_3 = h_3, \qquad Ah_4 = (1-2a)h_4$$

(for instance, row 1 of A times h_4 is $1-a-a+0$). So $A = Q\Lambda Q'$ with $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ orthogonal and $\Lambda = \operatorname{diag}(1+2a, 1, 1, 1-2a)$ (Theorem 8.3(b)), and by Theorem 8.21(a) A is PD if and only if $1-2|a| > 0$, the same condition. For $a = \pm\frac{1}{2}$ the eigenvalues are 2, 1, 1, 0: A is PSD but not PD; for $|a| > \frac{1}{2}$, A is indefinite (Theorem 8.21(b)).

Corollary 8.37 ([5, Corollary 14.5.7, Lemma 14.5.8], [1, §9.3, (9-6)])**.** Let $A \in \mathbb{R}^{n\times n}$ with $\Delta_k(A) \neq 0$ for $k = 1, \dots, n-1$, and let P be as in Lemma 8.33.

(a) $A = LDU$, where L is unit lower triangular, $D = \operatorname{diag}(d_1, \dots, d_n)$ with $d_k = \Delta_k(A)/\Delta_{k-1}(A)$, and $U = P^{-1}$ is unit upper triangular: an **LDU decomposition** of A.

(b) It is the only one: if $A = L_1D_1U_1$ with L_1 unit lower triangular, D_1 diagonal and U_1 unit upper triangular, then $L_1 = L$, $D_1 = D$ and $U_1 = U$.

(c) If A is symmetric, then $L = U'$; so $A = U'DU$ and $P'AP = D$.

Proof. Let $B = AP$; it is lower triangular, and $\Delta_k(B) = \Delta_k(A)$ for every k (Lemma 8.33).

(a)

- As B is lower triangular, $\Delta_k(B) = b_{11}\cdots b_{kk}$ (Theorems 5.14(c) and 5.22). So $b_{11}\cdots b_{kk} = \Delta_k(A)$ for every k, and $b_{kk} = \Delta_k(A)/\Delta_{k-1}(A) = d_k$, which is $\neq 0$ for $k \leqslant n-1$.
- Let L have the columns $\frac{1}{d_k}\operatorname{col}_k(B)$ ($k \leqslant n-1$) and the last column e_n. Column k of B has zeros above the diagonal and d_k on it, so L is unit lower triangular. For every k, $\operatorname{col}_k(LD) = L(d_ke_k) = d_k\operatorname{col}_k(L) = \operatorname{col}_k(B)$ (Theorem 1.4(a),(b)); for $k = n$, both sides are d_ne_n, the last column of B. So $B = LD$.
- P is unit upper triangular, so $U = P^{-1}$ is unit upper triangular (Theorem 2.31), and $A = BP^{-1} = LDU$.

(b)

- U_1 is invertible and U_1^{-1} is unit upper triangular (Theorem 2.31), and $AU_1^{-1} = L_1D_1$ is lower triangular, as $\operatorname{col}_k(L_1D_1) = (D_1)_{kk}\operatorname{col}_k(L_1)$ (Theorem 1.4(a),(b)). By the uniqueness in Lemma 8.33, $U_1^{-1} = P$, that is, $U_1 = U$.
- Then $L_1D_1 = AP = B = LD$. The diagonal entries of L_1D_1 and of LD are those of D_1 and of D, so $D_1 = D$.
- For $k \leqslant n-1$, $\operatorname{col}_k(L_1) = \frac{1}{d_k}\operatorname{col}_k(L_1D) = \frac{1}{d_k}\operatorname{col}_k(B) = \operatorname{col}_k(L)$, and the last columns of L_1 and L are both e_n. So $L_1 = L$.

(c) If $A' = A$, then $A = A' = U'D'L' = U'DL'$, where U' is unit lower and L' unit upper triangular. By (b), $L = U'$. So $A = U'DU$, and $P'AP = (UP)'D(UP) = D$, as $UP = I_n$. □

For a symmetric A with $\Delta_k(A) \neq 0$ ($k \leqslant n-1$), (c) writes the quadratic form as a combination of squares, $q_A(x) = (Ux)'D(Ux) = d_1(Ux)_1^2 + \cdots + d_n(Ux)_n^2$; this is completing the squares, as in Example 8.35.

Proposition 8.38 ([5, §14.5b])**.** Let $A \in \mathbb{R}^{n\times n}$. A factorization $A = L_0U_0$ with L_0 unit lower triangular and U_0 upper triangular is an **LU decomposition** of A.

(a) If $\Delta_k(A) \neq 0$ for $k = 1, \ldots, n-1$, and $A = LDU$ is the LDU decomposition of Corollary 8.37, then $A = L(DU)$ is an LU decomposition, and the diagonal entries of DU are $d_1, \ldots, d_n$.

(b) If row operations $R_i \leftarrow R_i + cR_j$ with $i > j$, and no others, take A to an upper triangular matrix U_0, then $A = L_0U_0$ for a unit lower triangular L_0.

Proof. (a) DU is a product of two upper triangular matrices, so it is upper triangular, with diagonal entries $d_k \cdot 1 = d_k$ (Proposition 1.13(c)).

(b)

- By Theorem 2.15(c), $U_0 = E_N \cdots E_1 A$, where each $E_k = T_{ij}(c)$ with $i > j$ is unit lower triangular (Definition 2.14).
- Each E_k is invertible, and $E_k^{-1} = T_{ij}(-c)$ is unit lower triangular (Theorem 2.15(b)); so $A = (E_N \cdots E_1)^{-1}U_0 = E_1^{-1} \cdots E_N^{-1}U_0$ (Theorem 2.13(b)).
- $L_0 = E_1^{-1} \cdots E_N^{-1}$ is a product of unit lower triangular matrices, hence unit lower triangular (Proposition 1.13(c)). □

Example 8.39. Find an LU decomposition of

$$A = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 4 & 3 & 3 & 3 \\ 8 & 7 & 9 & 8 \\ 4 & 3 & 5 & 6 \end{pmatrix}$$

by Gauss elimination.

Solution.

- Elimination without interchanges:

$$\begin{pmatrix} 2 & 1 & 1 & 1 \\ 4 & 3 & 3 & 3 \\ 8 & 7 & 9 & 8 \\ 4 & 3 & 5 & 6 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - 4R_1} \begin{pmatrix} 2 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & 5 & 4 \\ 0 & 1 & 3 & 4 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - 3R_2} \begin{pmatrix} 2 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 3 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 2 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix} = U_0$$

- The elimination matrices are $T_{21}(-2)$, $T_{31}(-4)$, $T_{41}(-2)$, $T_{32}(-3)$, $T_{42}(-1)$, $T_{43}(-1)$, in this order. By the proof of Proposition 8.38(b), $A = L_0U_0$ with $L_0 = T_{21}(2)\,T_{31}(4)\,T_{41}(2)\,T_{32}(3)\,T_{42}(1)\,T_{43}(1)$ (Theorem 2.15(b)). Multiplying from the left, $T_{21}(2)\,T_{31}(4)\,T_{41}(2)$ is $T_{41}(2)$ with 4 times row 1 added to row 3 and then 2 times row 1 added to row 2, and $T_{32}(3)\,T_{42}(1)\,T_{43}(1)$ is $T_{43}(1)$ with row 2 added to row 4 and then 3 times row 2 added to row 3 (Theorem 2.15(a)); so

$$L_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 0 \\ 2 & 1 & 1 & 1 \end{pmatrix}$$

  Below the diagonal of L_0 stand the multipliers 2, 4, 2, 3, 1, 1 of the elimination.

- Check by rows (Theorem 1.4(c)): $\text{row}_1(L_0U_0) = (2, 1, 1, 1)$, $\text{row}_2(L_0U_0) = 2\,(2, 1, 1, 1) + (0, 1, 1, 1) = (4, 3, 3, 3)$, $\text{row}_3(L_0U_0) = 4\,(2, 1, 1, 1) + 3\,(0, 1, 1, 1) + (0, 0, 2, 1) = (8, 7, 9, 8)$ and $\text{row}_4(L_0U_0) = 2\,(2, 1, 1, 1) + (0, 1, 1, 1) + (0, 0, 2, 1) + (0, 0, 0, 2) = (4, 3, 5, 6)$.

- Let $D_0 = \operatorname{diag}(2,1,2,2)$, the diagonal of U_0. Dividing the rows of U_0 by 2, 1, 2, 2 (Proposition 1.13(a)) gives the unit upper triangular matrix

$$V_0 = D_0^{-1}U_0 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

  and $A = L_0 D_0 V_0$. As $\Delta_1 = 2$, $\Delta_2 = 2\cdot 3 - 1\cdot 4 = 2$ and, along row 1 (Theorem 5.21), $\Delta_3 = 2(3\cdot 9 - 3\cdot 7) - 1(4\cdot 9 - 3\cdot 8) + 1(4\cdot 7 - 3\cdot 8) = 12 - 12 + 4 = 4$ are nonzero, this is the LDU decomposition of Corollary 8.37, by its part (b): $d_1 = 2$, $d_2 = 1$, $d_3 = 2$, $d_4 = 2$. Indeed $d_2 = \Delta_2/\Delta_1 = 1$, $d_3 = \Delta_3/\Delta_2 = \frac{4}{2} = 2$ and $d_4 = \Delta_4/\Delta_3 = \frac{8}{4} = 2$, with $\Delta_4 = \det A = \det L_0 \det U_0 = 1\cdot(2\cdot 1\cdot 2\cdot 2) = 8$ (Theorem 5.18(b); $\det U_0$ by Theorem 5.14(c), and $\det L_0 = \det L_0' = 1$ by Theorems 5.22 and 5.14(c)).

Theorem 8.40 (Cholesky decomposition; [5, Theorem 14.5.11])**.** Let $A \in \mathbb{R}^{n\times n}$ be PD, and let $A = U'DU$ as in Theorem 8.34. There is exactly one upper triangular $T \in \mathbb{R}^{n\times n}$ with positive diagonal entries such that $A = T'T$, namely $T = D^{1/2}U$, where $D^{1/2} = \operatorname{diag}(\sqrt{d_1},\dots,\sqrt{d_n})$. This T is the **Cholesky factor** of A, and $A = T'T$ is the **Cholesky decomposition** of A.

Proof. • Every $d_k > 0$ (Theorem 8.34), so $D^{1/2}$ is defined. $D^{1/2}U$ is U with row k multiplied by $\sqrt{d_k}$: it is upper triangular with the diagonal entries $\sqrt{d_k} > 0$, and $(D^{1/2}U)'(D^{1/2}U) = U'D^{1/2}D^{1/2}U = U'DU = A$.

- Let T be upper triangular with $t_{kk} > 0$ and $A = T'T$. Put $D_* = \operatorname{diag}(t_{11}^2,\dots,t_{nn}^2)$ and $U_* = \operatorname{diag}(t_{11}^{-1},\dots,t_{nn}^{-1})\,T$, which is T with row k divided by t_{kk}, a unit upper triangular matrix. Then $T = \operatorname{diag}(t_{11},\dots,t_{nn})\,U_*$ and $A = U_*'D_*U_*$.
- This is an LDU decomposition of A, and every $\Delta_k(A) > 0$ (Theorem 8.34). By Corollary 8.37(b),(c), $U_* = U$ and $D_* = D$. So $t_{kk} = \sqrt{d_k}$, and $T = D^{1/2}U$. □

Remark 8.41 ([5, (14.5.16)–(14.5.19)])**.** For $i \leqslant j$, the (i,j) entry of $T'T$ is $\sum_{k\leqslant i} t_{ki}t_{kj}$, since $t_{ki} = 0$ for $k > i$. Comparing with a_{ij} gives T row by row ($i = 1,\dots,n$):

$$t_{ii} = \Big(a_{ii} - \sum_{k<i} t_{ki}^2\Big)^{1/2}, \qquad t_{ij} = \frac{1}{t_{ii}}\Big(a_{ij} - \sum_{k<i} t_{ki}t_{kj}\Big) \quad (j > i)$$

For a symmetric A, this succeeds with positive numbers under all square roots exactly when A is PD: if it succeeds, it gives an upper triangular T with positive diagonal entries and $T'T = A$, so A is PD (Theorems 2.31(a) and 8.26(c)); if A is PD, its Cholesky factor (Theorem 8.40) satisfies these equations. Moreover $A_k = T_k'T_k$ for the leading principal submatrices, as the first k columns of T are zero below row k; so $\Delta_k(A) = \det T_k' \det T_k = (t_{11}\cdots t_{kk})^2$ (Theorems 5.18(b), 5.22 and 5.14(c)).

Example 8.42. Find the Cholesky factor T of

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

the matrix with entries $a_{ij} = \min(i,j)$, and use it to compute A^{-1}.

Solution. By Remark 8.41:

- row 1: $t_{11} = \sqrt{1} = 1$ and $t_{1j} = a_{1j}/1 = 1$ for $j = 2,3,4$;
- row 2: $t_{22} = \sqrt{2 - 1^2} = 1$ and $t_{2j} = (2 - 1\cdot 1)/1 = 1$ for $j = 3,4$;

- row 3: $t_{33} = \sqrt{3-1-1} = 1$ and $t_{34} = (3-1-1)/1 = 1$;
- row 4: $t_{44} = \sqrt{4-1-1-1} = 1$.

So T is the upper triangular matrix with all entries $t_{ij} = 1$ for $i \leqslant j$; indeed the (i,j) entry of $T'T$ is $\operatorname{col}_i(T)' \operatorname{col}_j(T) = \min(i,j)$. Hence A is PD, and all its leading principal minors equal 1 (Remark 8.41); in particular $\det A = \Delta_4(A) = 1$, and in Theorem 8.34 $D = I_4$ and $U = T$ (Theorem 8.40: $T = D^{1/2}U$). The matrix

$$K = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

has $KT' = I_4$ (row i of K subtracts row $i-1$ of T' from row i), so $(T')^{-1} = K$ (Theorem 2.16) and, by Theorem 2.13(b),(c),

$$A^{-1} = T^{-1}(T')^{-1} = K'K = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Sylvester's law of inertia

Theorem 8.43 (Sylvester's law of inertia; [4, Ch. V, §8; Ch. VIII, §4, Exercise 25], [7, Ch. VIII, §4, Exercise 25])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $P \in \mathbb{R}^{n\times n}$ be invertible with $P'AP = \operatorname{diag}(d_1, \dots, d_n)$.

(a) The number n_0 of indices i with $d_i = 0$ is $\dim \mathcal{N}(A) = n - \operatorname{rank} A$.

(b) The number n_+ of i with $d_i > 0$ and the number n_- of i with $d_i < 0$ do not depend on the choice of P.

(c) n_+ and n_- are the numbers of positive and of negative eigenvalues of A, counted with multiplicity.

(d) There is an invertible $R \in \mathbb{R}^{n\times n}$ with $R'AR = \operatorname{diag}(I_{n_+}, -I_{n_-}, 0)$, where 0 is the $n_0 \times n_0$ zero matrix.

Proof. Let $p_1, \dots, p_n$ be the columns of P. The (i,j) entry of $P'AP$ is $p_i'Ap_j$ (Theorem 1.4(d)), so $p_i'Ap_j = 0$ for $i \neq j$ and $p_i'Ap_i = d_i$; and $p_1, \dots, p_n$ is a basis of $\mathbb{R}^n$ (Example 3.19).

(a)

- If $d_i = 0$, then $p_j'(Ap_i) = 0$ for every j, that is, $P'(Ap_i) = 0$; as P' is invertible, $Ap_i = 0$.
- Conversely, let $v \in \mathcal{N}(A)$ and $v = \sum_l x_l p_l$. For each j with $d_j \neq 0$, $0 = p_j'Av = \sum_l x_l p_j'Ap_l = x_j d_j$, so $x_j = 0$: v is a combination of the p_i with $d_i = 0$.
- These p_i are independent, so they form a basis of $\mathcal{N}(A)$, and $n_0 = \dim \mathcal{N}(A) = n - \operatorname{rank} A$ (Theorem 4.12).

(b)

- Let R be another invertible matrix with $R'AR = \operatorname{diag}(g_1, \dots, g_n)$, with columns $r_1, \dots, r_n$, and let m_+ be the number of j with $g_j > 0$. Let $\mathcal{I} = \{i : d_i > 0\}$ (n_+ indices) and $\mathcal{J} = \{j : g_j \leqslant 0\}$ ($n - m_+$ indices).
- The $n_+ + n - m_+$ vectors p_i ($i \in \mathcal{I}$) and r_j ($j \in \mathcal{J}$) are linearly independent. Indeed, let $\sum_{i\in\mathcal{I}} x_i p_i + \sum_{j\in\mathcal{J}} y_j r_j = 0$ and $u = \sum_{i\in\mathcal{I}} x_i p_i = -\sum_{j\in\mathcal{J}} y_j r_j$. Then

$$u'Au = \sum_{i\in\mathcal{I}} d_i x_i^2 \geqslant 0, \qquad u'Au = \sum_{j\in\mathcal{J}} g_j y_j^2 \leqslant 0$$

so $u'Au = 0$, and every $x_i = 0$, as each $d_i > 0$ ($i \in \mathcal{I}$). Then $\sum_{j\in\mathcal{J}} y_j r_j = 0$, and every $y_j = 0$, as the columns of R are independent.

- Hence $n_+ + n - m_+ \leqslant n$ (Theorem 4.5(c)), that is, $n_+ \leqslant m_+$; exchanging the roles of P and R gives $m_+ \leqslant n_+$.
- By (a), $n_- = n - n_+ - n_0$ does not depend on P either.

(c) By Theorem 8.3, $Q'AQ = \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ with Q orthogonal, hence invertible, and $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A, listed with multiplicity. Apply (b) with Q in place of P.

(d) Number the orthonormal eigenvectors so that $\lambda_1, \dots, \lambda_{n_+} > 0$, $\lambda_{n_+ +1}, \dots, \lambda_{n_+ + n_-} < 0$ and the remaining λ_i are 0 (by (c)). Let $R = Q \operatorname{diag}(s_1, \dots, s_n)$, with $s_i = 1/\sqrt{|\lambda_i|}$ if $\lambda_i \neq 0$ and $s_i = 1$ otherwise. R is invertible (Theorem 2.13(b)), and $R'AR = \operatorname{diag}(s_1, \dots, s_n)\,\Lambda \operatorname{diag}(s_1, \dots, s_n) = \operatorname{diag}(s_1^2\lambda_1, \dots, s_n^2\lambda_n) = \operatorname{diag}(I_{n_+}, -I_{n_-}, 0)$. □

Definition 8.44 ([4, Ch. V, §8])**.** For a symmetric $A \in \mathbb{R}^{n\times n}$, the numbers n_+, n_- and n_0 of Theorem 8.43 are the **index of positivity**, the **index of negativity** and the **index of nullity** of A (and of its quadratic form q_A); the triple (n_+, n_-, n_0) is the **inertia** of A.

By Theorems 8.43(c) and 8.21(a),(b), A is PD if and only if $n_+ = n$, PSD if and only if $n_- = 0$, and indefinite if and only if $n_+ \geqslant 1$ and $n_- \geqslant 1$. So any diagonalization $P'AP$ decides the definiteness, for instance the one of Corollary 8.37(c), found by completing the squares; no eigenvalues are needed.

Example 8.45 ([4, Ch. V, §8, Example])**.** (a) Let $C = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$, and let P have the columns $v_1 = (1,0)'$ and $v_2 = (1,1)'$. Then

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad CP = \begin{pmatrix} -1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad P'CP = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$$

so the inertia of C is $(0,1,1)$. Indeed C has the eigenvalues -2 and 0, with the eigenvectors $(1,-1)'$ and $(1,1)'$. Here P is the matrix of Lemma 8.33 ($p_{12} = -c_{12}/c_{11} = 1$), and $P'CP = \operatorname{diag}(\Delta_1, \Delta_2/\Delta_1)$ is Corollary 8.37(c).

(b) For $A = 2J_3 - I_3$ of Example 8.5, $\Delta_1 = 1$, $\Delta_2 = 1 - 4 = -3$ and $\Delta_3 = \det A = 5 \cdot (-1) \cdot (-1) = 5$ (Theorem 7.19). Solving for P as in Example 8.35 ($p_{12} = -2$, and $A_2 \left(\begin{smallmatrix} p_{13} \\ p_{23} \end{smallmatrix}\right) = -\left(\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right)$ gives $p_{13} = p_{23} = -\frac{2}{3}$),

$$P = \begin{pmatrix} 1 & -2 & -\frac{2}{3} \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 1 \end{pmatrix}, \qquad AP = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -3 & 0 \\ 2 & -2 & -\frac{5}{3} \end{pmatrix}, \qquad U = P^{-1} = \begin{pmatrix} 1 & 2 & 2 \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 1 \end{pmatrix}$$

and $P'AP = \operatorname{diag}(1, -3, -\frac{5}{3})$ (Corollary 8.37(c)). So $q_A(x) = (x_1 + 2x_2 + 2x_3)^2 - 3\left(x_2 + \frac{2}{3}x_3\right)^2 - \frac{5}{3}x_3^2$: the inertia is $(1,2,0)$, and A is indefinite, as the eigenvalues 5, -1, -1 show (Theorem 8.43(c)).

The square root of a PSD matrix

Theorem 8.46 ([1, §9.5, Theorem 13], [5, Theorem 21.9.1], [6, Exercise 8.25], [2, (6.41)])**.** Let $A \in \mathbb{R}^{n\times n}$ be PSD, with spectral decomposition $A = c_1E_1 + \cdots + c_kE_k$, and let Q, Λ be as in Theorem 8.3.

(a) There is exactly one PSD matrix B with $B^2 = A$, namely

$$A^{1/2} := \sqrt{c_1}\,E_1 + \cdots + \sqrt{c_k}\,E_k = Q\Lambda^{1/2}Q', \qquad \Lambda^{1/2} = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$$

the **square root** of A, which is $f(A)$ for $f(t) = \sqrt{t}$ (Definition 8.9).

(b) $A^{1/2}A = AA^{1/2}$, $\operatorname{rank} A^{1/2} = \operatorname{rank} A$, and $x'Ax = \|A^{1/2}x\|^2$ for $x \in \mathbb{R}^n$.

(c) If A is PD, then $A^{1/2}$ is PD, and $A^{-1/2} := (A^{1/2})^{-1} = \sum_j c_j^{-1/2}E_j = Q\Lambda^{-1/2}Q'$, with $\Lambda^{-1/2} = \operatorname{diag}(\lambda_1^{-1/2}, \ldots, \lambda_n^{-1/2})$, satisfies

$$A^{-1/2}AA^{-1/2} = I_n, \qquad A^{-1/2}A^{-1/2} = A^{-1}$$

Proof. Every $c_j \geqslant 0$ (Theorem 8.21(a)), so $\sqrt{c_j}$ is defined. Let $N = \sum_j \sqrt{c_j}\,E_j$.

(a)

- N is symmetric, $N^2 = \sum_i \sum_j \sqrt{c_i}\sqrt{c_j}\,E_iE_j = \sum_j c_jE_j = A$, and, as in the proof of Theorem 8.21, $x'Nx = \sum_j \sqrt{c_j}\|E_jx\|^2 \geqslant 0$: N is PSD. By Theorem 8.6(c), $N = \sum_i \sqrt{\lambda_i}\,q_iq_i' = Q\Lambda^{1/2}Q'$.
- Uniqueness: let B be PSD with $B^2 = A$, and let $B = \sum_m d_mF_m$ be its spectral decomposition. The d_m are distinct and $\geqslant 0$ (Theorem 8.21(a)), so the d_m^2 are distinct, and $A = B^2 = \sum_m d_m^2F_m$ (Theorem 8.6(b) for B). By Theorem 8.6(d), after renumbering $d_m^2 = c_m$ and $F_m = E_m$; so $d_m = \sqrt{c_m}$ and $B = N$.

(b)

- $AN = \sum_j c_j\sqrt{c_j}\,E_j = NA$, as $E_iE_j = 0$ for $i \neq j$.
- $N = f(A)$ for $f(t) = \sqrt{t}$, so its eigenvalues are $\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}$ (Proposition 8.10(a)), and $\sqrt{\lambda_i} \neq 0$ exactly when $\lambda_i \neq 0$; hence $\operatorname{rank} N = \operatorname{rank} A$ (Theorem 8.4, for N and for A).
- N is symmetric, so $x'Ax = x'NNx = (Nx)'(Nx) = \|Nx\|^2$.

(c)

- If A is PD, every $c_j > 0$, so $x'Nx > 0$ for $x \neq 0$ (some $E_jx \neq 0$), and $N\sum_j c_j^{-1/2}E_j = \sum_j E_j = I_n$ gives $A^{-1/2} = \sum_j c_j^{-1/2}E_j$ (Theorem 2.16). Then $A^{-1/2}AA^{-1/2} = \sum_j c_j^{-1/2}c_jc_j^{-1/2}E_j = I_n$.
- As $A = N^2$ with N invertible, $A^{-1} = (N^2)^{-1} = (N^{-1})^2 = A^{-1/2}A^{-1/2}$ (Theorem 2.13(b)); and $A^{-1/2} = f(A)$ for $f(t) = t^{-1/2}$, so $A^{-1/2} = Q\Lambda^{-1/2}Q'$ (Proposition 8.10(a)). □

Example 8.47. For $A = 25E_1 + 9E_2 + E_3$ of Example 8.8, all eigenvalues are positive, so A is PD (Theorem 8.21(a)), and Theorem 8.46(a),(c) gives

$$A^{1/2} = 5E_1 + 3E_2 + E_3 = \begin{pmatrix} 3 & 0 & 1 & 1 \\ 0 & 3 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 3 \end{pmatrix}, \qquad A^{-1/2} = \tfrac{1}{5}E_1 + \tfrac{1}{3}E_2 + E_3 = \frac{1}{15}\begin{pmatrix} 7 & 2 & -3 & -3 \\ 2 & 7 & -3 & -3 \\ -3 & -3 & 7 & 2 \\ -3 & -3 & 2 & 7 \end{pmatrix}$$

(for instance the $(1,1)$ entry of $A^{1/2}$ is $\frac{5}{4} + \frac{3}{2} + \frac{1}{4} = 3$). Check: row 1 of $A^{1/2}$ times its columns gives $9+1+1 = 11$, $0+1+1 = 2$, $3+3 = 6$, $3+3 = 6$, the first row of A. The matrix

$$G = 5E_1 - 3E_2 + E_3 = \begin{pmatrix} 0 & 3 & 1 & 1 \\ 3 & 0 & 1 & 1 \\ 1 & 1 & 0 & 3 \\ 1 & 1 & 3 & 0 \end{pmatrix}$$

also satisfies $G^2 = 25E_1 + 9E_2 + E_3 = A$, because $E_j^2 = E_j$ and $E_jE_l = 0$ for $j \neq l$ (Theorem 8.6(b)); but G is not PSD, as its diagonal entry $g_{11} = 0$ stands in a nonzero row (Theorem 8.30). Among the matrices whose square is A, Theorem 8.46(a) singles out the PSD one.

The Loewner order

Definition 8.48 ([6, Ch. 8])**.** For symmetric $A, B \in \mathbb{R}^{n\times n}$ we write $A \succcurlyeq B$ (or $B \preccurlyeq A$) if $A - B$ is PSD, and $A \succ B$ if $A - B$ is PD. This is the **Loewner order**. In particular $A \succcurlyeq 0$ means that A is PSD, and $A \succ 0$ that A is PD.

Example 8.49. (a) Unlike numbers, two symmetric matrices need not be comparable: for $A = \operatorname{diag}(1, 0)$ and $B = \operatorname{diag}(0, 1)$, neither $A-B = \operatorname{diag}(1, -1)$ nor $B-A$ is PSD (Proposition 8.23(a)), so neither $A \succcurlyeq B$ nor $B \succcurlyeq A$.

(b) The Loewner order is not the entrywise order: $I_2 - \frac{1}{2}J_2 = \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ has the eigenvalues 0 and 1, with the eigenvectors $(1, 1)'$ and $(1, -1)'$, so $I_2 \succcurlyeq \frac{1}{2}J_2$ (Theorem 8.21(a)), although the $(1, 2)$ entry of $\frac{1}{2}J_2$ is the larger one.

Proposition 8.50. Let $A, B, D \in \mathbb{R}^{n\times n}$ be symmetric.

(a) $A \succcurlyeq A$; if $A \succcurlyeq B$ and $B \succcurlyeq A$, then $A = B$; if $A \succcurlyeq B$ and $B \succcurlyeq D$, then $A \succcurlyeq D$.

(b) If $A \succcurlyeq B$, then $A + D \succcurlyeq B + D$, and $K'AK \succcurlyeq K'BK$ for every $K \in \mathbb{R}^{n\times m}$. If $A \succ B$ and $K \in \mathbb{R}^{n\times n}$ is invertible, then $K'AK \succ K'BK$.

(c) If $A \succcurlyeq B$ and $B \succ 0$, then $A \succ 0$.

(d) If $A \succcurlyeq B$, then $a_{ii} \geqslant b_{ii}$ for every i, and $\operatorname{tr} A \geqslant \operatorname{tr} B$.

Proof. (a) $A - A = 0$ is PSD. If $A - B$ and $B - A = -(A - B)$ are PSD, then $A - B = 0$ (Proposition 8.23(e)). If $A - B$ and $B - D$ are PSD, so is their sum $A - D$ (Proposition 8.23(b)).

(b) $(A + D) - (B + D) = A - B$, and $K'AK - K'BK = K'(A - B)K$, which is PSD if $A - B$ is (Theorem 8.24(a)), and PD if $A - B$ is PD and K is invertible (Theorem 8.24(c)).

(c) $A = B + (A - B)$ with B PD and $A - B$ PSD, so A is PD (Proposition 8.23(b)).

(d) $A - B$ is PSD, so its diagonal entries $a_{ii} - b_{ii}$ are $\geqslant 0$ (Theorem 8.30); adding over i, $\operatorname{tr} A - \operatorname{tr} B = \operatorname{tr}(A - B) \geqslant 0$ (Theorem 1.11(a)). □

Theorem 8.51 ([3, Result 3.16])**.** If $A \succcurlyeq B \succ 0$, then $B^{-1} \succcurlyeq A^{-1}$.

Proof. A is PD (Proposition 8.50(c)), so A^{-1} and B^{-1} exist and are PD (Proposition 8.23(c)).

- First let $B = I_n$, and let $A = c_1E_1 + \cdots + c_kE_k$ be the spectral decomposition. By Proposition 8.10(c),(a) with $f(t) = t - 1$, the eigenvalues of $A - I_n$ are $c_j - 1$, and they are $\geqslant 0$, as $A - I_n$ is PSD (Theorem 8.21(a)). So every $c_j \geqslant 1$.
- $I_n - A^{-1} = g(A)$ for $g(t) = 1 - 1/t$ (Theorem 8.6(b) and Proposition 8.10(b)), with the eigenvalues $1 - c_j^{-1} \geqslant 0$ (Proposition 8.10(a)). So $I_n - A^{-1}$ is PSD (Theorem 8.21(a)), that is, $I_n \succcurlyeq A^{-1}$.
- In general, let $B^{-1/2} = (B^{1/2})^{-1}$, which is symmetric (Theorems 8.46 and 2.13(c)), and $M = B^{-1/2}AB^{-1/2}$. By Proposition 8.50(b) with $K = B^{-1/2}$, $M \succcurlyeq B^{-1/2}BB^{-1/2} = I_n$ (Theorem 8.46).
- By the first part, $I_n \succcurlyeq M^{-1} = B^{1/2}A^{-1}B^{1/2}$ (Theorem 2.13(a),(b)).
- By Proposition 8.50(b) with $K = B^{-1/2}$ again, $B^{-1/2}B^{-1/2} \succcurlyeq B^{-1/2}B^{1/2}A^{-1}B^{1/2}B^{-1/2} = A^{-1}$, and $B^{-1/2}B^{-1/2} = (B^{1/2}B^{1/2})^{-1} = B^{-1}$ (Theorem 2.13(b)). □

Theorem 8.52 ([5, Theorem 18.1.6, Corollaries 18.1.7, 18.1.8])**.** Let $A \in \mathbb{R}^{n\times n}$ be PD and $B \in \mathbb{R}^{n\times n}$ PSD. Then $\det(A+B) \geqslant \det A$, with equality if and only if $B = 0$. Consequently, if $A \succcurlyeq B \succ 0$, then $\det A \geqslant \det B$, with equality if and only if $A = B$; and $\det A \geqslant \det B$ holds also when $A \succcurlyeq B \succcurlyeq 0$.

Proof. • By Theorem 8.26(b), $B = K'K$ for some $K \in \mathbb{R}^{r\times n}$; let $k_1', \dots, k_r'$ be the rows of K, and K_i the $i \times n$ matrix of its first i rows. Put $A_0 = A$ and $A_i = A + K_i'K_i$ $(i = 1, \dots, r)$, so that $A_r = A + B$.

- By the outer product expansion (Theorem 1.4(d)), $K_i'K_i = K_{i-1}'K_{i-1} + k_ik_i'$, that is, $A_i = A_{i-1} + k_ik_i'$ $(i \geqslant 2$; for $i = 1$, $A_1 = A + k_1k_1')$. Each $K_i'K_i$ is PSD (Theorem 8.25), so each A_i is PD, and A_{i-1}^{-1} is PD (Proposition 8.23(b),(c)).
- By Theorem 5.31, $\det A_i = \det A_{i-1} \cdot (1 + k_i'A_{i-1}^{-1}k_i)$. Here $\det A_{i-1} > 0$ (Theorem 8.21(c)) and $k_i'A_{i-1}^{-1}k_i \geqslant 0$, with equality if and only if $k_i = 0$. So $\det A_i \geqslant \det A_{i-1}$, with equality if and only if $k_i = 0$.
- Hence $\det(A + B) = \det A_r \geqslant \det A_{r-1} \geqslant \cdots \geqslant \det A_0 = \det A$, with equality if and only if equality holds at every step, that is, $k_1 = \cdots = k_r = 0$, that is, $K = 0$. And $K = 0$ if and only if $B = K'K = 0$: if $K'K = 0$, then $\operatorname{tr}(K'K) = 0$ and $K = 0$ (Theorem 1.11(d)).
- If $A \succcurlyeq B \succ 0$, apply this to the PD matrix B and the PSD matrix $A - B$: $\det A = \det\Big(B + (A - B)\Big) \geqslant \det B$, with equality if and only if $A - B = 0$.
- Let $A \succcurlyeq B \succcurlyeq 0$. Then $A \succcurlyeq 0$ (Proposition 8.50(a)), so $\det A \geqslant 0$ (Theorem 8.21(c)); if $\det B = 0$, this is $\det A \geqslant \det B$. Otherwise B is invertible (Theorem 5.18(a)), hence PD (Theorem 8.21(c)), and the previous item applies. □

Example 8.53. Let $B = (\min(i, j)) \in \mathbb{R}^{4\times 4}$ be the matrix of Example 8.42, $x = e_4$ and $A = B + xx'$. Show that $A \succcurlyeq B \succ 0$, and compare A^{-1} with B^{-1} and $\det A$ with $\det B$.

Solution.

- $xx' = e_4e_4'$ has the single nonzero entry 1 in position $(4, 4)$, so

$$B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 5 \end{pmatrix}$$

  By Example 8.42, B is PD, $\det B = 1$ and $B^{-1} = K'K$. The matrix $A - B = xx'$ is PSD (Theorem 8.25, applied to the 1×4 matrix x'); so $A \succcurlyeq B \succ 0$.

- $w = B^{-1}x = \operatorname{col}_4(B^{-1}) = (0, 0, -1, 1)'$ (Theorem 1.4(b)), and $x'B^{-1}x = x'w = 1$. As B^{-1} is symmetric (Theorem 2.13(c)), $B^{-1}xx'B^{-1} = ww'$, and Theorem 5.32(b) gives

$$B^{-1} = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \qquad A^{-1} = B^{-1} - \tfrac{1}{2}ww' = \frac{1}{2}\begin{pmatrix} 4 & -2 & 0 & 0 \\ -2 & 4 & -2 & 0 \\ 0 & -2 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

  The difference $B^{-1} - A^{-1} = \frac{1}{2}ww'$, whose only nonzero entries are $\frac{1}{2}$ in positions $(3, 3)$, $(4, 4)$ and $-\frac{1}{2}$ in positions $(3, 4)$, $(4, 3)$, is PSD (Theorem 8.25), as Theorem 8.51 predicts: $B^{-1} \succcurlyeq A^{-1}$.

- $\det A = \det B \cdot (1 + x'B^{-1}x) = 1 \cdot 2 = 2 \geqslant 1 = \det B$ (Theorems 5.31 and 8.52); the inequality is strict, as $A \neq B$.

Simultaneous diagonalization

Lemma 8.54 ([4, Ch. VIII, Theorem 4.3], [6, Exercise 7.51])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $W \neq \{0\}$ be a subspace of $\mathbb{R}^n$ that is **invariant** under A: $Aw \in W$ for every $w \in W$. Then W has an orthonormal basis consisting of eigenvectors of A.

Proof. • Let $q_1, \dots, q_m$ be an orthonormal basis of W (Corollary 6.15(a)), and $Q_W = [q_1 \ \cdots \ q_m] \in \mathbb{R}^{n\times m}$; so $Q'_W Q_W = I_m$.

- Each Aq_j lies in W, so $Aq_j = Q_W c_j$ for some $c_j \in \mathbb{R}^m$ (Theorem 1.4(b)). With $M = [c_1 \ \cdots \ c_m]$, $AQ_W = Q_W M$ (Theorem 1.4(a)); multiplying by Q'_W on the left gives $M = Q'_W A Q_W$, and $M' = Q'_W A' Q_W = M$.
- By Theorem 8.3(a),(b), applied to M, $R'MR = \Lambda$ is diagonal for an orthogonal $R \in \mathbb{R}^{m\times m}$, so $MR = RR'MR = R\Lambda$ (Proposition 6.13(b)). Let $U = Q_W R$. Then $U'U = R'Q'_W Q_W R = R'R = I_m$ and $AU = AQ_W R = Q_W MR = Q_W R\Lambda = U\Lambda$: the m columns of U are orthonormal eigenvectors of A (Theorem 1.4(a),(b)).
- They lie in W, as $\operatorname{col}_j(U) = Q_W \operatorname{col}_j(R)$, and they are independent (Proposition 6.11(b)); as $\dim W = m$, they form a basis of W (Theorem 4.5(d)). □

Theorem 8.55 ([4, Ch. VIII, §4, Exercise 20], [7], [2, Theorem 7.3])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric. There is an orthogonal Q such that $Q'AQ$ and $Q'BQ$ are both diagonal if and only if $AB = BA$.

Proof. • If $Q'AQ = D_1$ and $Q'BQ = D_2$ are diagonal, then $D_1D_2 = D_2D_1$, and $AB = QD_1Q'QD_2Q' = QD_1D_2Q' = QD_2D_1Q' = BA$.

- Conversely, let $AB = BA$, let $c_1, \dots, c_k$ be the distinct eigenvalues of A, and $W_j = \mathcal{N}(A - c_j I)$. For $x \in W_j$, $A(Bx) = B(Ax) = c_j Bx$, so $Bx \in W_j$: W_j is invariant under B.
- By Lemma 8.54, applied to B and W_j, W_j has an orthonormal basis of eigenvectors of B; they are eigenvectors of A belonging to c_j.
- Together, these vectors are orthonormal, as $W_i \perp W_j$ for $i \neq j$ (Theorem 8.2), and there are $\sum_j \dim W_j = \sum_j a_A(c_j) = n$ of them (Theorem 8.4, Definition 7.10). So they are independent (Proposition 6.11(b)) and form an orthonormal basis $q_1, \dots, q_n$ of $\mathbb{R}^n$ (Theorem 4.5(d)).
- Each q_i is an eigenvector of A and of B, so for $Q = [q_1 \ \cdots \ q_n]$, $Q'AQ$ and $Q'BQ$ are diagonal (Theorem 8.3(b), for A and for B). □

Example 8.56. Let

$$A = \begin{pmatrix} 11 & 2 & 6 & 6 \\ 2 & 11 & 6 & 6 \\ 6 & 6 & 11 & 2 \\ 6 & 6 & 2 & 11 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \\ 1 & 0 & 4 & 2 \\ 0 & 1 & 2 & 4 \end{pmatrix}$$

where A is the matrix of Example 8.8. Find an orthogonal Q such that $Q'AQ$ and $Q'BQ$ are diagonal.

Solution.

- By rows (Theorem 1.4(c)),

$$AB = \begin{pmatrix} 54 & 36 & 47 & 38 \\ 36 & 54 & 38 & 47 \\ 47 & 38 & 54 & 36 \\ 38 & 47 & 36 & 54 \end{pmatrix}$$

is symmetric, so $BA = B'A' = (AB)' = AB$ (Theorem 1.9). By Theorem 8.55, Q exists.

- The eigenspaces of A (Example 8.8) are $W_1 = \text{span}\{\mathbf{1}_4\}$ (for 25), $W_2 = \{x : x_1 + x_2 = 0,\ x_3 + x_4 = 0\} = \text{span}\{u_2, u_3\}$ with $u_2 = (-1,1,0,0)'$ and $u_3 = (0,0,-1,1)'$ (for 9), and $W_3 = \text{span}\{(1,1,-1,-1)'\}$ (for 1). Every row sum of B is 7, so $B\mathbf{1}_4 = 7\mathbf{1}_4$; and $B(1,1,-1,-1)' = (5,5,-5,-5)'$.
- In W_2, $x = au_2 + bu_3 = (-a,a,-b,b)'$ and $Bx = (-2a-b,\ 2a+b,\ -a-2b,\ a+2b)'$. If $Bx = \mu x$, entries 2 and 4 give $2a+b = \mu a$ and $a+2b = \mu b$; their sum and their difference give $(\mu-3)(a+b) = 0$ and $(\mu-1)(a-b) = 0$. So $\mu = 3$ with $x = a(-1,1,-1,1)'$, or $\mu = 1$ with $x = a(-1,1,1,-1)'$; entries 1 and 3 then agree. Indeed $B(1,-1,1,-1)' = (3,-3,3,-3)'$ and $B(1,-1,-1,1)' = (1,-1,-1,1)'$.
- The vectors $\mathbf{1}_4$, $(1,-1,1,-1)'$, $(1,-1,-1,1)'$, $(1,1,-1,-1)'$ are eigenvectors of A and of B, of length 2 and pairwise orthogonal. So

$$Q = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad Q'AQ = \text{diag}(25,9,9,1), \qquad Q'BQ = \text{diag}(7,3,1,5)$$

  by Theorem 8.3(b), for A and for B.

Theorem 8.57 ([3, Ch. 3, Exercise 17], [2, Theorem 7.4], [5, §21.14])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric, with A PD.

(a) There is an invertible $E \in \mathbb{R}^{n\times n}$ with $E'AE = I_n$ and $E'BE = \Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$, and E can be chosen so that $\lambda_1 \geqslant \cdots \geqslant \lambda_n$.

(b) For every real λ, $\det(B - \lambda A) = \det A \cdot (\lambda_1 - \lambda)\cdots(\lambda_n - \lambda)$. So $\lambda_1, \ldots, \lambda_n$ are the roots of $\det(B - \lambda A) = 0$, and they are the eigenvalues of $A^{-1}B$, listed with multiplicity.

(c) An $n \times n$ matrix $E = [f_1\ \cdots\ f_n]$ is as in (a) if and only if $Bf_i = \lambda_i A f_i$ and $f_i' A f_j = \delta_{ij}$ for all i, j. Moreover, if $Bx = \lambda Ax$ and $By = \mu Ay$ with $\lambda \neq \mu$, then $x'Ay = 0$.

Proof. (a)

- By Theorem 8.3, $P'AP = D = \text{diag}(d_1, \ldots, d_n)$ for an orthogonal P, and every $d_i > 0$ (Theorem 8.21(a)). With $D^{-1/2} = \text{diag}(d_1^{-1/2}, \ldots, d_n^{-1/2})$, $D^{-1/2}P'APD^{-1/2} = I_n$.
- $C = D^{-1/2}P'BPD^{-1/2}$ is symmetric (Theorem 1.9), so $Q'CQ = \Lambda$ is diagonal for an orthogonal Q, whose columns may be numbered so that $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ (Theorem 8.3(b)).
- $E = PD^{-1/2}Q$ is invertible (Theorem 2.13(b)), $E'AE = Q'(D^{-1/2}P'APD^{-1/2})Q = Q'Q = I_n$, and $E'BE = Q'CQ = \Lambda$.

(b)

- For real λ, $E'(B - \lambda A)E = \Lambda - \lambda I_n$, so $\det E' \cdot \det(B - \lambda A) \cdot \det E = (\lambda_1 - \lambda)\cdots(\lambda_n - \lambda)$ (Theorems 5.18(b) and 5.14(c)).
- From $E'AE = I_n$, $(\det E)^2 \det A = 1$ (Theorems 5.18(b) and 5.22); multiplying by $\det A$ gives $\det(B - \lambda A) = \det A \cdot (\lambda_1 - \lambda)\cdots(\lambda_n - \lambda)$, and $\det A \neq 0$.
- For real t, $\det(tI_n - A^{-1}B) = \det A^{-1} \cdot \det(tA - B) = (\det A)^{-1}(-1)^n \det(B - tA) = (t-\lambda_1)\cdots(t-\lambda_n)$ (Theorem 5.18(b),(c)). By Definition 7.7(a) and Proposition 7.5(d), $p_{A^{-1}B} = (t-\lambda_1)\cdots(t-\lambda_n)$, and Definition 7.10 gives the eigenvalues of $A^{-1}B$.

(c)

- The (i,j) entries of $E'AE$ and $E'BE$ are $f_i'Af_j$ and $f_i'Bf_j$ (Theorem 1.4(d)). If $Bf_j = \lambda_j Af_j$ and $f_i'Af_j = \delta_{ij}$ for all i, j, then $E'AE = I_n$, so $(\det E)^2 \det A = 1$ and E is invertible (Theorem 5.18(a)), and $f_i'Bf_j = \lambda_j f_i'Af_j = \lambda_j\delta_{ij}$: $E'BE = \Lambda$.

- Conversely, if E is as in (a), then $E'AE = I_n$ gives $AE = (E')^{-1}$, and $BE = (E')^{-1}\Lambda = AE\Lambda$, whose column i reads $Bf_i = \lambda_i Af_i$ (Theorem 1.4(a),(b)).
- If $Bx = \lambda Ax$ and $By = \mu Ay$, then $\lambda\, x'Ay = (\lambda Ax)'y = (Bx)'y = x'By = \mu\, x'Ay$, so $(\lambda - \mu)\, x'Ay = 0$. □

Corollary 8.58 ([6, Exercises 8.36, 8.37])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric, with A PD, and let E and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ be as in Theorem 8.57(a).

(a) $A^{-1}B = E\Lambda E^{-1}$: $A^{-1}B$ is diagonalizable, its eigenvalues $\lambda_1, \ldots, \lambda_n$ are real, and column i of E is an eigenvector of $A^{-1}B$ belonging to λ_i.

(b) $A^{-1}B$ and the symmetric matrix $A^{-1/2}BA^{-1/2}$ have the same eigenvalues, with the same multiplicities.

Proof. (a) By Theorem 8.57(c), $Bf_i = \lambda_i Af_i$ for the columns f_i of E, that is, $BE = AE\Lambda$ (Theorem 1.4(a),(b)). As A is invertible (Theorem 8.21(c)) and so is E, $A^{-1}BE = E\Lambda$ and $A^{-1}B = E\Lambda E^{-1}$; the rest is the last part of Definition 7.23.

(b) $A^{-1/2}$ is symmetric (Theorems 8.46(c) and 2.13(c)), so $A^{-1/2}BA^{-1/2}$ is symmetric (Theorem 1.9). As $A^{-1} = A^{-1/2}A^{-1/2}$ (Theorem 8.46(c)), $A^{1/2}(A^{-1}B)A^{-1/2} = A^{1/2}A^{-1/2}A^{-1/2}BA^{-1/2} = A^{-1/2}BA^{-1/2}$; that is, $A^{-1/2}BA^{-1/2} = P^{-1}(A^{-1}B)P$ with $P = A^{-1/2} = (A^{1/2})^{-1}$, as $P^{-1} = A^{1/2}$ (Theorem 2.13(a)), and Theorem 7.15(a) gives the claim. □

This covers the matrix $\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}'$ of canonical correlation analysis, where Σ_{11} and Σ_{22} are the covariance matrices of two random vectors and Σ_{12} is their cross-covariance matrix: when Σ_{11} and Σ_{22} are PD, $A = \Sigma_{11}$ and the symmetric matrix $B = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}'$ are as above, so its eigenvalues are real.

Example 8.59. Let $A = (\min(i,j)) \in \mathbb{R}^{4\times 4}$ be the matrix of Example 8.42, and

$$B = \begin{pmatrix} 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 6 & 6 \\ 3 & 5 & 6 & 5 \end{pmatrix}$$

the matrix with entries $b_{ij} = c_{\min(i,j)}$, where $(c_1, c_2, c_3, c_4) = (3, 5, 6, 5)$. Find E as in Theorem 8.57(a).

Solution.

- A is PD and $\det A = 1$ (Example 8.42). For real t, the (i,j) entry of $B - tA$ is $d_{\min(i,j)}$, with $d_m = c_m - tm$: $d_1 = 3 - t$, $d_2 = 5 - 2t$, $d_3 = 6 - 3t$, $d_4 = 5 - 4t$.
- Subtract row 3 from row 4, then row 2 from row 3, then row 1 from row 2 (Definition 5.13(b)). Row i minus row $i-1$ has the entry $d_j - d_j = 0$ for $j < i$ and $d_i - d_{i-1}$ for $j \geqslant i$, so the result is upper triangular, and (Theorem 5.14(c))

  $$\det(B - tA) = d_1(d_2 - d_1)(d_3 - d_2)(d_4 - d_3) = (3 - t)(2 - t)(1 - t)(-1 - t)$$

  As $\det A = 1$, Theorem 8.57(b) gives $\lambda_1 = 3$, $\lambda_2 = 2$, $\lambda_3 = 1$, $\lambda_4 = -1$.
- Let $f_k = e_k - e_{k-1}$ ($k = 1, \ldots, 4$, with $e_0 = 0$ and $d_0 = 0$). Entry i of $(B - tA)f_k$ is $d_{\min(i,k)} - d_{\min(i,k-1)}$, which is 0 for $i < k$ and $d_k - d_{k-1} = (c_k - c_{k-1}) - t$ for $i \geqslant k$. As $c_k - c_{k-1} = \lambda_k$ ($c_0 = 0$), $(B - \lambda_k A)f_k = 0$, that is, $Bf_k = \lambda_k Af_k$.
- Let $E = [f_1\ f_2\ f_3\ f_4]$. The columns of

  $$E = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad AE = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

  give $E'AE = I_4$: the (j,k) entry $f_j'Af_k$ is entry j minus entry $j-1$ of column k of AE, which is 1 for $j = k$ and 0 otherwise. Then $E'BE = \operatorname{diag}(3, 2, 1, -1)$, since $(E'BE)_{jk} = f_j'Bf_k = \lambda_k f_j'Af_k$.

Here $E = K' = T^{-1}$ for the Cholesky factor T of A and $K = (T')^{-1}$ (Example 8.42, Theorem 2.13(c)), and $B = T' \operatorname{diag}(3,2,1,-1)\, T$. In the coordinates $y = E^{-1}x = Tx$, that is, $y_k = x_k + \cdots + x_4$, $x'Ax = y_1^2 + y_2^2 + y_3^2 + y_4^2$ and $x'Bx = 3y_1^2 + 2y_2^2 + y_3^2 - y_4^2$.

Exercises

Exercise 8.1. Let

$$A = \begin{pmatrix} 2 & -1 & 2 & 2 \\ -1 & 2 & -2 & -2 \\ 2 & -2 & 5 & 4 \\ 2 & -2 & 4 & 5 \end{pmatrix}$$

(a) Find p_A and the eigenvalues of A with their algebraic multiplicities.

(b) Find an orthogonal Q and a diagonal Λ with $Q'AQ = \Lambda$.

(c) Write down the spectral decomposition of A, and check that the projection for the simple eigenvalue is the polynomial in A of Proposition 8.10(c).

(d) Find A^{-1} and A^m for $m = 1, 2, \ldots$

Solution. (a) For real t, apply $R_1 \leftarrow R_1 + R_2$ and $R_4 \leftarrow R_4 - R_3$, take the factor $t-1$ out of rows 1 and 4, apply $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 + 2R_1$, expand along column 1, add column 3 to column 2, and expand along row 3 (Definition 5.13(a),(b), Theorems 5.22 and 5.21):

$$\begin{aligned}
\det(tI - A) &= \det \begin{pmatrix} t-2 & 1 & -2 & -2 \\ 1 & t-2 & 2 & 2 \\ -2 & 2 & t-5 & -4 \\ -2 & 2 & -4 & t-5 \end{pmatrix} = \det \begin{pmatrix} t-1 & t-1 & 0 & 0 \\ 1 & t-2 & 2 & 2 \\ -2 & 2 & t-5 & -4 \\ 0 & 0 & 1-t & t-1 \end{pmatrix} \\
&= (t-1)^2 \det \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & t-2 & 2 & 2 \\ -2 & 2 & t-5 & -4 \\ 0 & 0 & -1 & 1 \end{pmatrix} = (t-1)^2 \det \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & t-3 & 2 & 2 \\ 0 & 4 & t-5 & -4 \\ 0 & 0 & -1 & 1 \end{pmatrix} \\
&= (t-1)^2 \det \begin{pmatrix} t-3 & 2 & 2 \\ 4 & t-5 & -4 \\ 0 & -1 & 1 \end{pmatrix} = (t-1)^2 \det \begin{pmatrix} t-3 & 4 & 2 \\ 4 & t-9 & -4 \\ 0 & 0 & 1 \end{pmatrix} \\
&= (t-1)^2\Big((t-3)(t-9) - 16\Big) = (t-1)^2(t^2 - 12t + 11) = (t-1)^3(t-11)
\end{aligned}$$

So $p_A(t) = (t-1)^3(t-11)$ (Definition 7.7(a), Proposition 7.5(d)): the eigenvalues are 11 with $a_A(11) = 1$ and 1 with $a_A(1) = 3$.

(b) Row-reduce $A - 11I$:

$$A - 11I = \begin{pmatrix} -9 & -1 & 2 & 2 \\ -1 & -9 & -2 & -2 \\ 2 & -2 & -6 & 4 \\ 2 & -2 & 4 & -6 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_4]{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} -10 & -10 & 0 & 0 \\ -1 & -9 & -2 & -2 \\ 0 & 0 & -10 & 10 \\ 2 & -2 & 4 & -6 \end{pmatrix}$$

$$\xrightarrow[R_3 \leftarrow -\frac{1}{10}R_3]{R_1 \leftarrow -\frac{1}{10}R_1} \begin{pmatrix} 1 & 1 & 0 & 0 \\ -1 & -9 & -2 & -2 \\ 0 & 0 & 1 & -1 \\ 2 & -2 & 4 & -6 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & -4 & 4 & -6 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 4R_3]{R_2 \leftarrow R_2 + 2R_3} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & -8 & 0 & -4 \\ 0 & 0 & 1 & -1 \\ 0 & -4 & 0 & -2 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - \frac{1}{2}R_2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & -8 & 0 & -4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $x_4 = 2$ gives $x_3 = x_4 = 2$, $x_2 = -\frac{1}{2}x_4 = -1$ and $x_1 = -x_2 = 1$: $u = (1,-1,2,2)'$ spans $\mathcal{N}(A - 11I)$. Next,

$$A - I = \begin{pmatrix} 1 & -1 & 2 & 2 \\ -1 & 1 & -2 & -2 \\ 2 & -2 & 4 & 4 \\ 2 & -2 & 4 & 4 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & -1 & 2 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $\mathcal{N}(A - I) = \{x : x_1 = x_2 - 2x_3 - 2x_4\}$ has the basis $s_1 = (1,1,0,0)'$, $s_2 = (-2,0,1,0)'$, $s_3 = (-2,0,0,1)'$. Gram–Schmidt (Theorem 6.14):

- $b_1 = s_1$, with $\|b_1\|^2 = 2$;
- $s_2' b_1 = -2$, so $b_2 = s_2 + b_1 = (-1,1,1,0)'$, with $\|b_2\|^2 = 3$;
- $s_3' b_1 = -2$ and $s_3' b_2 = 2$, so $b_3 = s_3 + b_1 - \frac{2}{3}b_2 = \frac{1}{3}(-1,1,-2,3)'$; we use $3b_3 = (-1,1,-2,3)'$, with $\|3b_3\|^2 = 15$.

The vector u is orthogonal to b_1, b_2 and b_3 (Theorem 8.2). Normalizing (Theorem 8.3(b)),

$$Q = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{3} & -1/\sqrt{15} & 1/\sqrt{10} \\ 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{15} & -1/\sqrt{10} \\ 0 & 1/\sqrt{3} & -2/\sqrt{15} & 2/\sqrt{10} \\ 0 & 0 & 3/\sqrt{15} & 2/\sqrt{10} \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(1,1,1,11)$$

Check: $A(-1,1,-2,3)' = (-1,1,-2,3)'$ and $A(1,-1,2,2)' = (11,-11,22,22)'$.

(c) With $c_1 = 11$ and $c_2 = 1$, Theorem 8.6(b),(c) gives $E_1 = q_4 q_4' = \frac{1}{10}uu'$ and $E_2 = I_4 - E_1$:

$$E_1 = \frac{1}{10}\begin{pmatrix} 1 & -1 & 2 & 2 \\ -1 & 1 & -2 & -2 \\ 2 & -2 & 4 & 4 \\ 2 & -2 & 4 & 4 \end{pmatrix}, \qquad E_2 = \frac{1}{10}\begin{pmatrix} 9 & 1 & -2 & -2 \\ 1 & 9 & 2 & 2 \\ -2 & 2 & 6 & -4 \\ -2 & 2 & -4 & 6 \end{pmatrix}, \qquad A = 11E_1 + E_2$$

The polynomial $e_1 = \frac{t-1}{11-1}$ gives $E_1 = \frac{1}{10}(A - I)$, and indeed $A - I = uu'$ is the matrix reduced in (b).

(d) No eigenvalue is 0, so by Theorem 8.6(b)

$$A^{-1} = \tfrac{1}{11}E_1 + E_2 = I_4 - \tfrac{10}{11}E_1 = \frac{1}{11}\begin{pmatrix} 10 & 1 & -2 & -2 \\ 1 & 10 & 2 & 2 \\ -2 & 2 & 7 & -4 \\ -2 & 2 & -4 & 7 \end{pmatrix}, \qquad A^m = 11^m E_1 + E_2 = I_4 + \tfrac{11^m - 1}{10}\,uu'$$

Check: $\text{row}_1(A)A^{-1} = \frac{1}{11}(20-1-4-4,\ 2-10+4+4,\ -4-2+14-8,\ -4-2-8+14) = (1,0,0,0)$, and likewise for rows 2, 3 and 4.

Exercise 8.2. Let

$$A = \begin{pmatrix} -1 & -2 & 2 & -2 \\ -2 & -1 & -2 & 2 \\ 2 & -2 & -1 & -2 \\ -2 & 2 & -2 & -1 \end{pmatrix}$$

(a) Show that $p_A(t) = (t-5)(t+3)^3$, find an orthonormal basis of each eigenspace, and an orthogonal Q with $Q'AQ$ diagonal.

(b) Find the projections E_1 and E_2 of the spectral decomposition of A ($c_1 = 5$, $c_2 = -3$), onto $\mathcal{N}(A - 5I)$ and $\mathcal{N}(A + 3I)$, and check that $A = 5E_1 - 3E_2$.

(c) Use Definition 8.9 to compute A^{-1} and A^3.

Solution. (a) For real t, subtract rows 2 and 4 of $tI_4 - A$ from row 1 and add row 3 to it; the new row 1 is $(t-5)(1,-1,1,-1)$. Take out $t-5$, and add column 1 to columns 2 and 4 and subtract it from column 3 (Definition 5.13(a),(b), Theorem 5.22):

$$\det(tI_4 - A) = \det\begin{pmatrix} t+1 & 2 & -2 & 2 \\ 2 & t+1 & 2 & -2 \\ -2 & 2 & t+1 & 2 \\ 2 & -2 & 2 & t+1 \end{pmatrix} = (t-5)\det\begin{pmatrix} 1 & -1 & 1 & -1 \\ 2 & t+1 & 2 & -2 \\ -2 & 2 & t+1 & 2 \\ 2 & -2 & 2 & t+1 \end{pmatrix}$$
$$= (t-5)\det\begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & t+3 & 0 & 0 \\ -2 & 0 & t+3 & 0 \\ 2 & 0 & 0 & t+3 \end{pmatrix} = (t-5)(t+3)^3$$

as the last matrix is lower triangular (Theorem 5.22). So $p_A(t) = (t-5)(t+3)^3$ (Definition 7.7(a), Proposition 7.5(d)).

- $v = (1,-1,1,-1)'$ has $Av = (-1+2+2+2,\ -2+1-2-2,\ 2+2-1+2,\ -2-2-2+1)' = 5v$, and $\dim\mathcal{N}(A-5I) = a_A(5) = 1$ (Theorem 8.4); so $u_1 = \frac{1}{2}v$ is an orthonormal basis of $\mathcal{N}(A-5I)$.
- $A+3I_4 = 2vv'$: row i of $A+3I_4$ is $2v_iv'$. So $\mathcal{N}(A+3I) = \{x : x_1 - x_2 + x_3 - x_4 = 0\}$, of dimension $a_A(-3) = 3$ (Theorem 8.4). The orthogonal vectors $(1,1,1,1)'$, $(1,1,-1,-1)'$, $(1,-1,-1,1)'$ lie in it, so $u_2 = \frac{1}{2}(1,1,1,1)'$, $u_3 = \frac{1}{2}(1,1,-1,-1)'$, $u_4 = \frac{1}{2}(1,-1,-1,1)'$ is an orthonormal basis of $\mathcal{N}(A+3I)$ (Proposition 6.11(b), Theorem 4.5(d)).
- $Q = [u_1\ u_2\ u_3\ u_4]$ is orthogonal ($u_1 \perp u_2, u_3, u_4$ by Theorem 8.2) and $Q'AQ = \mathrm{diag}(5,-3,-3,-3)$ (Theorem 8.3(b)).

(b) By Theorem 8.6(b),(c),

$$E_1 = u_1u_1' = \frac{1}{4}\begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}, \qquad E_2 = I_4 - E_1 = \frac{1}{4}\begin{pmatrix} 3 & 1 & -1 & 1 \\ 1 & 3 & 1 & -1 \\ -1 & 1 & 3 & 1 \\ 1 & -1 & 1 & 3 \end{pmatrix}$$

and $5E_1 - 3E_2 = 8E_1 - 3I_4 = 2vv' - 3I_4 = A$: the diagonal entries are $2-3 = -1$ and the others are $2v_iv_j = \pm 2$.

(c) By Definition 8.9 with $f(t) = 1/t$ and $f(t) = t^3$, which give A^{-1} and A^3 (Proposition 8.10(b),(c)),

$$A^{-1} = \tfrac{1}{5}E_1 - \tfrac{1}{3}E_2 = \tfrac{8}{15}E_1 - \tfrac{1}{3}I_4 = \tfrac{1}{15}(2vv' - 5I_4) = \frac{1}{15}\begin{pmatrix} -3 & -2 & 2 & -2 \\ -2 & -3 & -2 & 2 \\ 2 & -2 & -3 & -2 \\ -2 & 2 & -2 & -3 \end{pmatrix}$$

$$A^3 = 125E_1 - 27E_2 = 152E_1 - 27I_4 = 38vv' - 27I_4 = \begin{pmatrix} 11 & -38 & 38 & -38 \\ -38 & 11 & -38 & 38 \\ 38 & -38 & 11 & -38 \\ -38 & 38 & -38 & 11 \end{pmatrix}$$

Check: row 1 of A times the first two columns of $15A^{-1}$ gives $3+4+4+4 = 15$ and $2+6-4-4 = 0$. (Thus $A^{-1} = \frac{1}{15}(A - 2I_4)$, that is, $A^2 - 2A - 15I_4 = (A-5I_4)(A+3I_4) = 0$.)

Exercise 8.3. Let

$$B = \begin{pmatrix} 3 & 1 & 0 & 2 \\ 1 & 2 & 1 & 2 \\ 0 & 1 & 3 & 2 \\ 2 & 2 & 2 & 0 \end{pmatrix}$$

(a) Find p_B and the eigenvalues of B.

(b) Find an orthogonal Q and a diagonal Λ with $Q'BQ = \Lambda$.

(c) Write down the spectral decomposition of B, and use it to find B^{-1}.

Solution. (a) For real t, add rows 2, 3 and 4 to row 1, take the factor $t-6$ out of row 1, subtract column 1 from columns 2, 3 and 4, and expand along row 1 and then along row 3 (Definition 5.13(a),(b), Theorems 5.22 and 5.21):

$$\begin{aligned} \det(tI - B) &= \det \begin{pmatrix} t-3 & -1 & 0 & -2 \\ -1 & t-2 & -1 & -2 \\ 0 & -1 & t-3 & -2 \\ -2 & -2 & -2 & t \end{pmatrix} = \det \begin{pmatrix} t-6 & t-6 & t-6 & t-6 \\ -1 & t-2 & -1 & -2 \\ 0 & -1 & t-3 & -2 \\ -2 & -2 & -2 & t \end{pmatrix} \\ &= (t-6) \det \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & t-2 & -1 & -2 \\ 0 & -1 & t-3 & -2 \\ -2 & -2 & -2 & t \end{pmatrix} = (t-6) \det \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & t-1 & 0 & -1 \\ 0 & -1 & t-3 & -2 \\ -2 & 0 & 0 & t+2 \end{pmatrix} \\ &= (t-6) \det \begin{pmatrix} t-1 & 0 & -1 \\ -1 & t-3 & -2 \\ 0 & 0 & t+2 \end{pmatrix} = (t-6)(t+2)\Big((t-1)(t-3) - 0\Big) \end{aligned}$$

So $p_B(t) = (t-6)(t-3)(t-1)(t+2)$ (Definition 7.7(a), Proposition 7.5(d)), and the eigenvalues are 6, 3, 1, −2.

(b) Row-reduce $B - cI$ for $c = 6, 3, 1, -2$:

$$B - 6I = \begin{pmatrix} -3 & 1 & 0 & 2 \\ 1 & -4 & 1 & 2 \\ 0 & 1 & -3 & 2 \\ 2 & 2 & 2 & -6 \end{pmatrix} \xrightarrow{R_4 \leftarrow \frac{1}{2}R_4} \begin{pmatrix} -3 & 1 & 0 & 2 \\ 1 & -4 & 1 & 2 \\ 0 & 1 & -3 & 2 \\ 1 & 1 & 1 & -3 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 1 & -4 & 1 & 2 \\ 0 & 1 & -3 & 2 \\ -3 & 1 & 0 & 2 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + 3R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 0 & -5 & 0 & 5 \\ 0 & 1 & -3 & 2 \\ 0 & 4 & 3 & -7 \end{pmatrix} \xrightarrow{R_2 \leftarrow -\frac{1}{5}R_2} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & -3 & 2 \\ 0 & 4 & 3 & -7 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 4R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -3 & 3 \\ 0 & 0 & 3 & -3 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -3 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 1$ gives $x_3 = x_4 = 1$, $x_2 = x_4 = 1$ and $x_1 = -x_2 - x_3 + 3x_4 = 1$;

$$B - 3I = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & -1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & -3 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & -3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & -7 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - 4R_2} \begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -15 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -15 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 0$, $x_2 = -2x_4 = 0$, and $x_3 = -1$ gives $x_1 = x_2 - x_3 - 2x_4 = 1$;

$$B - I = \begin{pmatrix} 2 & 1 & 0 & 2 \\ 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 2 & 2 & 2 & -1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 2 & 2 & 2 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & -1 & -2 & -2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & -5 \end{pmatrix}$$
$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 0$, and $x_3 = 1$ gives $x_2 = -2x_3 - 2x_4 = -2$ and $x_1 = -x_2 - x_3 - 2x_4 = 1$;

$$B + 2I = \begin{pmatrix} 5 & 1 & 0 & 2 \\ 1 & 4 & 1 & 2 \\ 0 & 1 & 5 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix} \xrightarrow{R_4 \leftarrow \frac{1}{2} R_4} \begin{pmatrix} 5 & 1 & 0 & 2 \\ 1 & 4 & 1 & 2 \\ 0 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 4 & 1 & 2 \\ 0 & 1 & 5 & 2 \\ 5 & 1 & 0 & 2 \end{pmatrix}$$
$$\xrightarrow[R_4 \leftarrow R_4 - 5R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 3 & 0 & 1 \\ 0 & 1 & 5 & 2 \\ 0 & -4 & -5 & -3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 5 & 2 \\ 0 & 3 & 0 & 1 \\ 0 & -4 & -5 & -3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 4R_2]{R_3 \leftarrow R_3 - 3R_2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 5 & 2 \\ 0 & 0 & -15 & -5 \\ 0 & 0 & 15 & 5 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 5 & 2 \\ 0 & 0 & -15 & -5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = -3$ gives $x_3 = -\frac{1}{3}x_4 = 1$, $x_2 = -5x_3 - 2x_4 = 1$ and $x_1 = -x_2 - x_3 - x_4 = 1$. So the eigenspaces are spanned by $(1,1,1,1)'$, $(1,0,-1,0)'$, $(1,-2,1,0)'$ and $(1,1,1,-3)'$. They belong to distinct eigenvalues, so they are pairwise orthogonal (Theorem 8.2). Normalizing them (Theorem 8.3(b)),

$$Q = \begin{pmatrix} 1/2 & 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{12} \\ 1/2 & 0 & -2/\sqrt{6} & 1/\sqrt{12} \\ 1/2 & -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{12} \\ 1/2 & 0 & 0 & -3/\sqrt{12} \end{pmatrix}, \qquad \Lambda = \operatorname{diag}(6, 3, 1, -2)$$

(c) Each eigenvalue has multiplicity 1, so $E_j = q_j q_j'$ (Theorem 8.6(c)), and $B = 6E_1 + 3E_2 + E_3 - 2E_4$ with

$$E_1 = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad E_2 = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
$$E_3 = \frac{1}{6}\begin{pmatrix} 1 & -2 & 1 & 0 \\ -2 & 4 & -2 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad E_4 = \frac{1}{12}\begin{pmatrix} 1 & 1 & 1 & -3 \\ 1 & 1 & 1 & -3 \\ 1 & 1 & 1 & -3 \\ -3 & -3 & -3 & 9 \end{pmatrix}$$

(for instance, entry $(1,1)$ of the sum is $\frac{6}{4} + \frac{3}{2} + \frac{1}{6} - \frac{2}{12} = 3$). No eigenvalue is 0, so by Theorem 8.6(b)

$$B^{-1} = \tfrac{1}{6}E_1 + \tfrac{1}{3}E_2 + E_3 - \tfrac{1}{2}E_4 = \frac{1}{6}\begin{pmatrix} 2 & -2 & 0 & 1 \\ -2 & 4 & -2 & 1 \\ 0 & -2 & 2 & 1 \\ 1 & 1 & 1 & -2 \end{pmatrix}$$

Check: $\text{row}_1(B)B^{-1} = \frac{1}{6}(6 - 2 + 0 + 2,\ -6 + 4 + 0 + 2,\ 0 - 2 + 0 + 2,\ 3 + 1 + 0 - 4) = (1, 0, 0, 0)$, and likewise for rows 2, 3 and 4.

Exercise 8.4. Let

$$A = \begin{pmatrix} 5 & 3 & 2 & 0 \\ 3 & 5 & 0 & 2 \\ 2 & 0 & 5 & 3 \\ 0 & 2 & 3 & 5 \end{pmatrix}$$

and let $h_1 = \mathbf{1}_4$, $h_2 = (1, 1, -1, -1)'$, $h_3 = (1, -1, 1, -1)'$ and $h_4 = (1, -1, -1, 1)'$ (the vectors of Example 8.36).

(a) Show that $h_1, \dots, h_4$ are eigenvectors of A. Find an orthogonal Q and a diagonal Λ with $A = Q\Lambda Q'$ and decreasing diagonal entries, and write the spectral decomposition of A.

(b) Find $\operatorname{rank} A$ and bases of $\mathcal{C}(A)$ and $\mathcal{N}(A)$. Is A PD? PSD?

(c) Find the largest and the smallest value of $x'Ax$ on the unit sphere $x'x = 1$ of $\mathbb{R}^4$, and all points where they are attained.

Solution. (a) Multiplying the rows of A by $h_1, \dots, h_4$:

$$Ah_1 = \begin{pmatrix} 10 \\ 10 \\ 10 \\ 10 \end{pmatrix} = 10h_1, \quad Ah_2 = \begin{pmatrix} 6 \\ 6 \\ -6 \\ -6 \end{pmatrix} = 6h_2, \quad Ah_3 = \begin{pmatrix} 4 \\ -4 \\ 4 \\ -4 \end{pmatrix} = 4h_3, \quad Ah_4 = \begin{pmatrix} 5-3-2 \\ 3-5+2 \\ 2-5+3 \\ -2-3+5 \end{pmatrix} = 0$$

The h_j are orthogonal with $\|h_j\| = 2$, so $Q = \frac{1}{2}[h_1 \ h_2 \ h_3 \ h_4]$ is orthogonal, $AQ = Q\Lambda$ with $\Lambda = \operatorname{diag}(10, 6, 4, 0)$, and $A = Q\Lambda Q'$; so $10, 6, 4, 0$ are the eigenvalues of A (Theorem 8.3(b),(c)). By Theorem 8.6(c), with $q_j = \frac{1}{2}h_j$,

$$A = 10q_1q_1' + 6q_2q_2' + 4q_3q_3' = \tfrac{5}{2}J_4 + \tfrac{3}{2}h_2h_2' + h_3h_3'$$

For instance, the entries in row 1 are $\frac{5}{2} + \frac{3}{2} + 1 = 5$, $\frac{5}{2} + \frac{3}{2} - 1 = 3$, $\frac{5}{2} - \frac{3}{2} + 1 = 2$ and $\frac{5}{2} - \frac{3}{2} - 1 = 0$.

(b) $\operatorname{rank} A = 3$, the number of nonzero eigenvalues (Theorem 8.4), and $\mathcal{C}(A) = \operatorname{span}\{h_1, h_2, h_3\}$, $\mathcal{N}(A) = \operatorname{span}\{h_4\}$ (Theorem 8.6(c)); these vectors are independent (Proposition 6.11(b)), so they form bases. The eigenvalues $10, 6, 4, 0$ are $\geqslant 0$ and one is 0: A is PSD but not PD (Theorem 8.21(a)).

(c) Put $y = Q'x$. Then $x = QQ'x = Qy$ and $y'y = x'x = 1$ (Proposition 6.13(b),(a)), and $x'Ax = y'\Lambda y = 10y_1^2 + 6y_2^2 + 4y_3^2$. Hence $x'Ax = 10 - 4y_2^2 - 6y_3^2 - 10y_4^2 \leqslant 10$, with equality exactly when $y = \pm e_1$, that is, at $x = \pm q_1 = \pm\frac{1}{2}(1, 1, 1, 1)'$; and $x'Ax \geqslant 0$, with equality exactly when $y = \pm e_4$, that is, at $x = \pm q_4 = \pm\frac{1}{2}(1, -1, -1, 1)'$. The largest value is 10 and the smallest is 0 (Theorem 8.21(d)).

Exercise 8.5. Let

$$A = \begin{pmatrix} 6 & 5 & 0 & 5 \\ 5 & 7 & 1 & 4 \\ 0 & 1 & 2 & -1 \\ 5 & 4 & -1 & 7 \end{pmatrix}$$

(a) Check that $(1, 1, 0, 1)'$, $(0, 1, 1, -1)'$, $(0, 1, -2, -1)'$ and $(2, -1, 0, -1)'$ are eigenvectors of A, and find the spectral decomposition of A.

(b) Check that the projection E_3 for the eigenvalue $c_3 = 1$ (with $c_1 = 16$, $c_2 = 4$) equals $\frac{1}{45}(A - 16I)(A - 4I)$ (Proposition 8.10(c)).

(c) Find $A^{1/2}$, $A^{-1/2}$ and A^{-1}, and check $(A^{1/2})^2 = A$ and $A^{1/2}A^{-1/2} = I$.

Solution. (a) $A(1, 1, 0, 1)' = (16, 16, 0, 16)'$, $A(0, 1, 1, -1)' = (0, 4, 4, -4)'$, $A(0, 1, -2, -1)' = (0, 1, -2, -1)'$ and $A(2, -1, 0, -1)' = (2, -1, 0, -1)'$. The last two are orthogonal, hence independent (Proposition 6.11(b)), so $\dim \mathcal{N}(A - I) \geqslant 2$ (Theorem 4.5(c)). So $a_A(16) \geqslant 1$, $a_A(4) \geqslant 1$ and $a_A(1) \geqslant 2$

(Theorem 8.4); as the multiplicities add up to 4 (Definition 7.10), these are the eigenvalues, with $a_A(16) = a_A(4) = 1$ and $a_A(1) = 2$, and A is PD (Theorem 8.21(a)). The four vectors are pairwise orthogonal; with the unit vectors $q_1 = \frac{1}{\sqrt{3}}(1,1,0,1)'$, $q_2 = \frac{1}{\sqrt{3}}(0,1,1,-1)'$, $q_3 = \frac{1}{\sqrt{6}}(0,1,-2,-1)'$ and $q_4 = \frac{1}{\sqrt{6}}(2,-1,0,-1)'$, Theorem 8.6(c), with $c_1 = 16$, $c_2 = 4$, $c_3 = 1$, gives $E_1 = q_1q_1'$, $E_2 = q_2q_2'$, $E_3 = q_3q_3' + q_4q_4'$:

$$E_1 = \frac{1}{3}\begin{pmatrix}1&1&0&1\\1&1&0&1\\0&0&0&0\\1&1&0&1\end{pmatrix}, \qquad E_2 = \frac{1}{3}\begin{pmatrix}0&0&0&0\\0&1&1&-1\\0&1&1&-1\\0&-1&-1&1\end{pmatrix}, \qquad E_3 = \frac{1}{6}\begin{pmatrix}4&-2&0&-2\\-2&2&-2&0\\0&-2&4&2\\-2&0&2&2\end{pmatrix}$$

and $A = 16E_1 + 4E_2 + E_3$ (for instance, entry $(1,1)$: $\frac{16}{3} + 0 + \frac{4}{6} = 6$).

(b) $e_3 = \frac{(t-16)(t-4)}{(1-16)(1-4)} = \frac{1}{45}(t-16)(t-4)$, and

$$(A-16I)(A-4I) = \begin{pmatrix}-10&5&0&5\\5&-9&1&4\\0&1&-14&-1\\5&4&-1&-9\end{pmatrix}\begin{pmatrix}2&5&0&5\\5&3&1&4\\0&1&-2&-1\\5&4&-1&3\end{pmatrix} = \begin{pmatrix}30&-15&0&-15\\-15&15&-15&0\\0&-15&30&15\\-15&0&15&15\end{pmatrix} = 45E_3$$

(c) By Definition 8.9 with $f(t) = \sqrt{t}$, $t^{-1/2}$ and t^{-1} (Theorem 8.46, Proposition 8.10(b)),

$$A^{1/2} = 4E_1 + 2E_2 + E_3 = \frac{1}{3}\begin{pmatrix}6&3&0&3\\3&7&1&2\\0&1&4&-1\\3&2&-1&7\end{pmatrix}$$

$$A^{-1/2} = \tfrac{1}{4}E_1 + \tfrac{1}{2}E_2 + E_3 = \frac{1}{12}\begin{pmatrix}9&-3&0&-3\\-3&7&-2&-1\\0&-2&10&2\\-3&-1&2&7\end{pmatrix}$$

$$A^{-1} = \tfrac{1}{16}E_1 + \tfrac{1}{4}E_2 + E_3 = \frac{1}{16}\begin{pmatrix}11&-5&0&-5\\-5&7&-4&-1\\0&-4&12&4\\-5&-1&4&7\end{pmatrix}$$

Check: $\mathrm{row}_1(A^{1/2})A^{1/2} = \frac{1}{9}(36+9+0+9,\ 18+21+0+6,\ 0+3+0-3,\ 18+6+0+21) = (6,5,0,5) = \mathrm{row}_1(A)$ and $\mathrm{row}_1(A^{1/2})A^{-1/2} = \frac{1}{36}(54-9+0-9,\ -18+21+0-3,\ 0-6+0+6,\ -18-3+0+21) = (1,0,0,0)$, and likewise for rows 2, 3 and 4; also $A^{-1} = (A^{-1/2})^2$, as $\mathrm{row}_1(A^{-1/2})A^{-1/2} = \frac{1}{144}(81+9+0+9,\ -27-21+0+3,\ 0+6+0-6,\ -27+3+0-21) = \frac{1}{16}(11,-5,0,-5) = \mathrm{row}_1(A^{-1})$.

Exercise 8.6. Let $w = (1,-1,-1,1)'$ and

$$P = I_4 - \tfrac{1}{4}ww' = \frac{1}{4}\begin{pmatrix}3&1&1&-1\\1&3&-1&1\\1&-1&3&1\\-1&1&1&3\end{pmatrix}$$

(a) Show that P is symmetric and idempotent, and find $\operatorname{rank} P$.

(b) Find Q_1 with orthonormal columns and $P = Q_1Q_1'$, as in Theorem 8.13(c).

(c) For $y = (2,0,0,2)'$, find the orthogonal projection $\hat{y}$ of y onto $\mathcal{C}(P)$, and check that $y - \hat{y} \perp \mathcal{C}(P)$.

Solution. (a) $P' = I_4 - \frac{1}{4}(ww')' = P$, and with $w'w = 4$, $P^2 = I_4 - \frac{1}{2}ww' + \frac{1}{16}w(w'w)w' = I_4 - \frac{1}{4}ww' = P$. By Theorem 6.29(c), $\operatorname{rank} P = \operatorname{tr} P = \frac{1}{4}(3+3+3+3) = 3$.

(b) With $h_1, \dots, h_4$ of Exercise 8.4, $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ is orthogonal and $\frac{1}{2}h_4 = \frac{1}{2}w$. By Proposition 6.13(b) and the outer product expansion (Theorem 1.4(d)), $I_4 = QQ' = \frac{1}{4}\sum_j h_j h_j'$, so

$$P = I_4 - \tfrac{1}{4}h_4h_4' = \tfrac{1}{4}(h_1h_1' + h_2h_2' + h_3h_3') = Q_1Q_1', \qquad Q_1 = \frac{1}{2}\begin{pmatrix}1 & 1 & 1\\ 1 & 1 & -1\\ 1 & -1 & 1\\ 1 & -1 & -1\end{pmatrix}$$

and $Q_1'Q_1 = I_3$.

(c) $P = P_{\mathcal{C}(P)}$ (Theorem 6.34), so $\hat{y} = Py = y - \frac{1}{4}(w'y)w = (2,0,0,2)' - (1,-1,-1,1)' = (1,1,1,1)'$. Then $y - \hat{y} = w$, and $Q_1'w = 0$, as $w = h_4$ is orthogonal to h_1, h_2, h_3. Moreover $\mathcal{C}(P) = \mathcal{C}(Q_1Q_1') \subseteq \mathcal{C}(Q_1)$ (Theorem 4.15(a)), and both have dimension 3 (by (a), and as the columns of Q_1 are independent, Proposition 6.11(b)), so $\mathcal{C}(P) = \mathcal{C}(Q_1)$ (Theorem 4.5(a)). Hence $y - \hat{y} \perp \mathcal{C}(P)$ (Theorem 6.6(a)).

Exercise 8.7. For each quadratic form on $\mathbb{R}^4$, write its symmetric matrix, complete the squares with Corollary 8.37, and find the inertia (Theorem 8.43). Which forms are PD, PSD, indefinite?

(a) $q_1(x) = x_1^2 + 3x_2^2 + 7x_3^2 + 4x_4^2 - 2x_1x_2 + 4x_1x_3 + 2x_1x_4 - 2x_2x_4 + 2x_3x_4$

(b) $q_2(x) = 2x_1^2 + x_2^2 + x_3^2 + 4x_1x_2 + 4x_1x_3 - 4x_1x_4 - 2x_2x_3 - 2x_2x_4 + 2x_3x_4$

(c) $q_3(x) = 2x_1^2 + 2x_2^2 + 2x_3^2 + x_4^2 + 2x_1x_3 - 2x_1x_4 + 2x_2x_3 - 2x_2x_4 - 2x_3x_4$; for this form, find also all x with $q_3(x) = 0$.

Solution. The matrix has the coefficient of x_i^2 as entry (i,i) and half the coefficient of x_ix_j as entries (i,j) and (j,i) (Proposition 8.18 and the paragraph after it). The 2×2 inverses come from Theorem 5.24, and column 4 of P solves the system of Lemma 8.33 with the coefficient matrix of the first three rows and columns; then $D = P'(AP)$ and $\Delta_4 = d_1d_2d_3d_4$ (Corollary 8.37(a)).

(a) The matrix is

$$A = \begin{pmatrix}1 & -1 & 2 & 1\\ -1 & 3 & 0 & -1\\ 2 & 0 & 7 & 1\\ 1 & -1 & 1 & 4\end{pmatrix}$$

$\Delta_1 = 1$, $\Delta_2 = 3-1 = 2$, and along row 1, $\Delta_3 = 1(21-0)+1(-7-0)+2(0-6) = 2$. Next $p_{12} = 1$, and with $A_2^{-1} = \frac{1}{2}\left(\begin{smallmatrix}3 & 1\\ 1 & 1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}p_{13}\\ p_{23}\end{smallmatrix}\right) = -A_2^{-1}\left(\begin{smallmatrix}2\\ 0\end{smallmatrix}\right) = \left(\begin{smallmatrix}-3\\ -1\end{smallmatrix}\right)$. Column 4 of P solves $A_3(p_{14}, p_{24}, p_{34})' = -(1,-1,1)'$:

$$\left(\begin{array}{ccc|c}1 & -1 & 2 & -1\\ -1 & 3 & 0 & 1\\ 2 & 0 & 7 & -1\end{array}\right) \xrightarrow[R_3\leftarrow R_3-2R_1]{R_2\leftarrow R_2+R_1} \left(\begin{array}{ccc|c}1 & -1 & 2 & -1\\ 0 & 2 & 2 & 0\\ 0 & 2 & 3 & 1\end{array}\right) \xrightarrow{R_3\leftarrow R_3-R_2} \left(\begin{array}{ccc|c}1 & -1 & 2 & -1\\ 0 & 2 & 2 & 0\\ 0 & 0 & 1 & 1\end{array}\right)$$

so $p_{34} = 1$, $p_{24} = -p_{34} = -1$ and $p_{14} = -1 + p_{24} - 2p_{34} = -4$. Hence

$$P = \begin{pmatrix}1 & 1 & -3 & -4\\ 0 & 1 & -1 & -1\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{pmatrix}, \qquad AP = \begin{pmatrix}1 & 0 & 0 & 0\\ -1 & 2 & 0 & 0\\ 2 & 2 & 1 & 0\\ 1 & 0 & -1 & 2\end{pmatrix}, \qquad U = P^{-1} = \begin{pmatrix}1 & -1 & 2 & 1\\ 0 & 1 & 1 & 0\\ 0 & 0 & 1 & -1\\ 0 & 0 & 0 & 1\end{pmatrix}$$

and $D = \operatorname{diag}(1,2,1,2)$, so $\Delta_4 = 4$. So $q_1(x) = (x_1 - x_2 + 2x_3 + x_4)^2 + 2(x_2 + x_3)^2 + (x_3 - x_4)^2 + 2x_4^2$: the inertia is $(4,0,0)$, and q_1 is PD.

(b) The matrix is

$$B = \begin{pmatrix}2 & 2 & 2 & -2\\ 2 & 1 & -1 & -1\\ 2 & -1 & 1 & 1\\ -2 & -1 & 1 & 0\end{pmatrix}$$

$\Delta_1 = 2$, $\Delta_2 = 2-4 = -2$, and along row 1, $\Delta_3 = 2(1-1) - 2(2+2) + 2(-2-2) = -16$. Next $p_{12} = -1$, and with $B_2^{-1} = -\frac{1}{2}\begin{pmatrix} 1 & -2 \\ -2 & 2 \end{pmatrix}$, $\binom{p_{13}}{p_{23}} = -B_2^{-1}\binom{2}{-1} = \binom{2}{-3}$. Column 4 of P solves $B_3(p_{14}, p_{24}, p_{34})' = -(-2,-1,1)'$:

$$\left(\begin{array}{ccc|c} 2 & 2 & 2 & 2 \\ 2 & 1 & -1 & 1 \\ 2 & -1 & 1 & -1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|c} 2 & 2 & 2 & 2 \\ 0 & -1 & -3 & -1 \\ 0 & -3 & -1 & -3 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - 3R_2} \left(\begin{array}{ccc|c} 2 & 2 & 2 & 2 \\ 0 & -1 & -3 & -1 \\ 0 & 0 & 8 & 0 \end{array}\right)$$

so $p_{34} = 0$, $p_{24} = 1 - 3p_{34} = 1$ and $p_{14} = \frac{1}{2}(2 - 2p_{24} - 2p_{34}) = 0$. Hence

$$P = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & -3 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad BP = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 \\ 2 & -3 & 8 & 0 \\ -2 & 1 & 0 & -1 \end{pmatrix}, \qquad U = P^{-1} = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and $D = \operatorname{diag}(2, -1, 8, -1)$, with $d_k = \Delta_k/\Delta_{k-1} = 2/1,\ -2/2,\ -16/(-2)$ for $k \leqslant 3$, and $\Delta_4 = 16$. So $q_2(x) = 2(x_1 + x_2 + x_3 - x_4)^2 - (x_2 + 3x_3 - x_4)^2 + 8x_3^2 - x_4^2$: the inertia is $(2, 2, 0)$, and q_2 is indefinite; for instance $q_2(1,0,0,0) = 2$ and $q_2(-1,1,0,0) = -1$.

(c) The matrix is

$$F = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & 1 & -1 \\ 1 & 1 & 2 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix}$$

$\Delta_1 = 2$, $\Delta_2 = 4$, and along row 1, $\Delta_3 = 2(4-1) - 0 + 1(0-2) = 4$. Next $p_{12} = 0$, $\binom{p_{13}}{p_{23}} = -\frac{1}{2}I_2\binom{1}{1} = \binom{-1/2}{-1/2}$, and column 4 of P solves $F_3(p_{14}, p_{24}, p_{34})' = -(-1,-1,-1)'$:

$$\left(\begin{array}{ccc|c} 2 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \\ 2 & 0 & 1 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & -2 & -3 & -1 \end{array}\right)$$
$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \\ 0 & 0 & -2 & 0 \end{array}\right)$$

so $p_{34} = 0$, $p_{24} = \frac{1}{2}(1 - p_{34}) = \frac{1}{2}$ and $p_{14} = 1 - p_{24} - 2p_{34} = \frac{1}{2}$. Hence

$$P = \begin{pmatrix} 1 & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad FP = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix}, \qquad U = P^{-1} = \begin{pmatrix} 1 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and $D = \operatorname{diag}(2, 2, 1, 0)$, so $\Delta_4 = 0$. So $q_3(x) = 2\left(x_1 + \frac{1}{2}x_3 - \frac{1}{2}x_4\right)^2 + 2\left(x_2 + \frac{1}{2}x_3 - \frac{1}{2}x_4\right)^2 + x_3^2$: the inertia is $(3, 0, 1)$, and q_3 is PSD but not PD. By Corollary 8.27, $q_3(x) = 0$ exactly for $x \in \mathcal{N}(F)$:

$$F = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & 1 & -1 \\ 1 & 1 & 2 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_4]{R_1 \leftarrow R_1 + 2R_4} \begin{pmatrix} 0 & -2 & -1 & 1 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & 0 \\ -1 & -1 & -1 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

With $x_4 = 2$, $x_3 = 0$, $x_2 = \frac{1}{2}(x_4 - x_3) = 1$ and $x_1 = x_2 = 1$: $q_3(x) = 0$ if and only if $x = c(1, 1, 0, 2)'$ for some $c \in \mathbb{R}$.

Exercise 8.8 ([6, Exercise 8.2])**.** Write each quadratic form with a symmetric matrix, and decide whether it is positive (negative) definite, semidefinite, or indefinite: (a) $x_1^2 + x_2^2 - x_1x_2$; (b) $4x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3$; (c) $-x_1^2 + 2x_2^2 + x_1x_2$.

Solution. (a) $\begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$: $x_1^2 + x_2^2 - x_1x_2 = (x_1 - \frac{1}{2}x_2)^2 + \frac{3}{4}x_2^2 > 0$ for $x \neq 0$; positive definite.

(b) The matrix is

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 5 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

and the form is $(x_1 + x_2)^2 + (x_2 + x_3)^2 + 3x_1^2 + 3x_2^2$, which is 0 only if $x_1 = x_2 = 0$ and $x_2 + x_3 = 0$, that is, $x = 0$; positive definite.

(c) $\begin{pmatrix} -1 & \frac{1}{2} \\ \frac{1}{2} & 2 \end{pmatrix}$: $-x_1^2 + 2x_2^2 + x_1x_2 = (\frac{3}{2}x_2)^2 - (x_1 - \frac{1}{2}x_2)^2$ takes the value 9 at $x = (1, 2)'$ and -1 at $x = (1, 0)'$; indefinite.

Exercise 8.9. Write each quadratic form as $x'Ax$ with A symmetric, and decide whether A is PD, PSD, negative definite, negative semidefinite or indefinite: (a) $x_1^2+x_2^2+x_3^2+x_4^2+6x_1x_2+6x_2x_3+6x_3x_4+6x_1x_4$; (b) $x_1^2 + 2x_2^2 + 2x_3^2 + x_4^2 - 2x_1x_2 - 2x_2x_3 - 2x_3x_4$; (c) $-2x_1^2 - 2x_2^2 - 2x_3^2 - 2x_4^2 + 2x_1x_2 + 2x_2x_3 + 2x_3x_4$; (d) $2x_1^2 + 2x_2^2 + 2x_3^2 + x_4^2 - 2x_1x_2 - 2x_2x_3 - 2x_3x_4$.

Solution. (a) The matrix is

$$A = \begin{pmatrix} 1 & 3 & 0 & 3 \\ 3 & 1 & 3 & 0 \\ 0 & 3 & 1 & 3 \\ 3 & 0 & 3 & 1 \end{pmatrix}$$

It gives $4 + 24 = 28$ at $\mathbf{1}_4$ and $1 + 1 - 6 = -4$ at $(1, -1, 0, 0)'$: indefinite. (The eigenvalues are 7, 1, 1, -5: $A\mathbf{1}_4 = 7\mathbf{1}_4$, $A(1, 0, -1, 0)' = (1, 0, -1, 0)'$, $A(0, 1, 0, -1)' = (0, 1, 0, -1)'$ and $A(1, -1, 1, -1)' = -5(1, -1, 1, -1)'$.)

(b) The matrix is

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

The form equals $(x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_3 - x_4)^2 \geqslant 0$, and it is 0 at $x = \mathbf{1}_4 \neq 0$: PSD but not PD.

(c) The matrix is

$$A = -\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

Expanding along the last row (Theorem 5.21), the leading principal minors of $-A$ satisfy $\Delta_k = 2\Delta_{k-1} - \Delta_{k-2}$ for $k = 3, 4$; so they are 2, $4 - 1 = 3$, $2 \cdot 3 - 2 = 4$ and $2 \cdot 4 - 3 = 5$. They are positive, so $-A$ is PD (Theorem 8.34) and A is negative definite.

(d) The matrix is

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

the matrix $K'K$ of Example 8.42. So $x'Ax = \|Kx\|^2 = x_1^2 + (x_2 - x_1)^2 + (x_3 - x_2)^2 + (x_4 - x_3)^2$, and A is PD (Theorem 8.25, as $\operatorname{rank} K = 4$). Its leading principal minors are $2, 3, 4, 1$.

Exercise 8.10. Let

$$A = \begin{pmatrix} 2 & 2 & -1 & 1 \\ 2 & 2 & -1 & 1 \\ -1 & -1 & 5 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

(a) Find p_A, the eigenvalues, and an orthonormal basis of eigenvectors.

(b) Show that A is PSD, find $\operatorname{rank} A$, and find the matrix P of Theorem 8.26(a) with $A = P'P$.

(c) Find all x with $x'Ax = 0$.

Solution. (a) For real t, apply $R_1 \leftarrow R_1 - R_2$, take the factor t out of row 1, add column 1 to column 2, and expand along row 1 and then along row 1 of the 3×3 matrix (Definition 5.13(a),(b), Theorems 5.22 and 5.21):

$$\begin{aligned}
\det(tI - A) &= \det \begin{pmatrix} t-2 & -2 & 1 & -1 \\ -2 & t-2 & 1 & -1 \\ 1 & 1 & t-5 & -1 \\ -1 & -1 & -1 & t-1 \end{pmatrix} \\
&= \det \begin{pmatrix} t & -t & 0 & 0 \\ -2 & t-2 & 1 & -1 \\ 1 & 1 & t-5 & -1 \\ -1 & -1 & -1 & t-1 \end{pmatrix} = t \det \begin{pmatrix} 1 & -1 & 0 & 0 \\ -2 & t-2 & 1 & -1 \\ 1 & 1 & t-5 & -1 \\ -1 & -1 & -1 & t-1 \end{pmatrix} \\
&= t \det \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & t-4 & 1 & -1 \\ 1 & 2 & t-5 & -1 \\ -1 & -2 & -1 & t-1 \end{pmatrix} = t \det \begin{pmatrix} t-4 & 1 & -1 \\ 2 & t-5 & -1 \\ -2 & -1 & t-1 \end{pmatrix} \\
&= t\Big((t-4)\big((t-5)(t-1)-1\big) - \big(2(t-1)-2\big) - \big(-2+2(t-5)\big)\Big) \\
&= t(t^3 - 10t^2 + 24t)
\end{aligned}$$

So $p_A(t) = t^2(t-4)(t-6)$ (Definition 7.7(a), Proposition 7.5(d)), and the eigenvalues are 6, 4 and 0, with $a_A(0) = 2$. Row-reduce $A - cI$:

$$A - 6I = \begin{pmatrix} -4 & 2 & -1 & 1 \\ 2 & -4 & -1 & 1 \\ -1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -5 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & -5 \\ 2 & -4 & -1 & 1 \\ -1 & -1 & -1 & 1 \\ -4 & 2 & -1 & 1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + 4R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & 1 & 1 & -5 \\ 0 & -6 & -3 & 11 \\ 0 & 0 & 0 & -4 \\ 0 & 6 & 3 & -19 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 1 & -5 \\ 0 & -6 & -3 & 11 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & -8 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 1 & 1 & -5 \\ 0 & -6 & -3 & 11 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 0$, and $x_2 = 1$ gives $x_3 = -2x_2 = -2$ and $x_1 = -x_2 - x_3 = 1$;

$$A - 4I = \begin{pmatrix} -2 & 2 & -1 & 1 \\ 2 & -2 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -3 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 2 & -2 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ -2 & 2 & -1 & 1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 0 & -4 & -3 & 7 \\ 0 & 0 & 2 & -2 \\ 0 & 4 & 1 & -5 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 0 & -4 & -3 & 7 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & 1 & 1 & -3 \\ 0 & -4 & -3 & 7 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 1$ gives $x_3 = x_4 = 1$, $x_2 = \frac{1}{4}(-3x_3 + 7x_4) = 1$ and $x_1 = -x_2 - x_3 + 3x_4 = 1$;

$$A = \begin{pmatrix} 2 & 2 & -1 & 1 \\ 2 & 2 & -1 & 1 \\ -1 & -1 & 5 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & -1 & 1 \\ -1 & -1 & 5 & 1 \\ 2 & 2 & -1 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 6 & 2 \\ 0 & 0 & -3 & -1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $\mathcal{N}(A) = \{x : x_3 = -\frac{1}{3}x_4,\ x_1 = -x_2 - x_3 - x_4\}$ has the basis $s_1 = (-1, 1, 0, 0)'$ ($x_2 = 1$, $x_4 = 0$) and $s_2 = (-2, 0, -1, 3)'$ ($x_2 = 0$, $x_4 = 3$). So $\mathcal{N}(A - 6I)$ and $\mathcal{N}(A - 4I)$ are spanned by $(1, 1, -2, 0)'$ and $(1, 1, 1, 1)'$. In $\mathcal{N}(A)$, Gram–Schmidt (Theorem 6.14) gives $b_1 = s_1$, with $\|b_1\|^2 = 2$, and, as $s_2' b_1 = 2$, $b_2 = s_2 - b_1 = (-1, -1, -1, 3)'$, with $\|b_2\|^2 = 12$. The eigenspaces are pairwise orthogonal (Theorem 8.2), so an orthonormal basis of eigenvectors is $q_1 = \frac{1}{\sqrt{6}}(1, 1, -2, 0)'$, $q_2 = \frac{1}{2}(1, 1, 1, 1)'$, $q_3 = \frac{1}{\sqrt{2}}(-1, 1, 0, 0)'$, $q_4 = \frac{1}{\sqrt{12}}(-1, -1, -1, 3)'$.

(b) The eigenvalues are $\geqslant 0$, so A is PSD (Theorem 8.21(a)), and rank $A = 2$, the number of nonzero eigenvalues (Theorem 8.4). The leading minors 2, 0, 0, 0 do not decide here, as $\Delta_2 = 0$. The proof of Theorem 8.26(a) takes the rows $\sqrt{6}\, q_1' = (1, 1, -2, 0)$ and $\sqrt{4}\, q_2' = (1, 1, 1, 1)$:

$$P = \begin{pmatrix} 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad P'P = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & -1 & 1 \\ 2 & 2 & -1 & 1 \\ -1 & -1 & 5 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} = A$$

(c) By Corollary 8.27, $x'Ax = 0$ if and only if $Ax = 0$, and by the last reduction in (a) this means $x = c_1(-1, 1, 0, 0)' + c_2(-2, 0, -1, 3)'$ for some $c_1, c_2 \in \mathbb{R}$. Equivalently, $x'Ax = \|Px\|^2 = (x_1 + x_2 - 2x_3)^2 + (x_1 + x_2 + x_3 + x_4)^2$ vanishes exactly when $x_1 + x_2 - 2x_3 = 0$ and $x_1 + x_2 + x_3 + x_4 = 0$.

Exercise 8.11. Let

$$A = \begin{pmatrix} 4 & 2 & 1 & -3 \\ 2 & 4 & -1 & -3 \\ 1 & -1 & 3 & 0 \\ -3 & -3 & 0 & 9 \end{pmatrix}$$

(a) Find the leading principal minors of A, and decide whether A is PD.

(b) Find P, D and U of Theorem 8.34, and write $q_A(x)$ as a sum of squares.

(c) Check (a) with Theorem 8.31, using the Schur complement of A_2.

(d) Show that $(1,1,0,-2)'$, $(1,-1,2,0)'$, $(1,1,0,1)'$ and $(1,-1,-1,0)'$ are eigenvectors of A, and check (a) once more.

Solution. (a) $\Delta_1 = 4$, $\Delta_2 = 16-4 = 12$, and along row 1, $\Delta_3 = 4(12-1) - 2(6+1) + 1(-2-4) = 44-14-6 = 24$. For $\Delta_4 = \det A$, interchange rows 1 and 3, apply $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 4R_1$, $R_4 \leftarrow R_4 + 3R_1$, then $R_3 \leftarrow R_3 - R_2$, $R_4 \leftarrow R_4 + R_2$, then $R_4 \leftarrow R_4 + \frac{1}{2}R_3$ (Definition 5.13(b),(c), Theorem 5.14(c)):

$$\det A = -\det\begin{pmatrix} 1 & -1 & 3 & 0 \\ 2 & 4 & -1 & -3 \\ 4 & 2 & 1 & -3 \\ -3 & -3 & 0 & 9 \end{pmatrix} = -\det\begin{pmatrix} 1 & -1 & 3 & 0 \\ 0 & 6 & -7 & -3 \\ 0 & 6 & -11 & -3 \\ 0 & -6 & 9 & 9 \end{pmatrix}$$

$$= -\det\begin{pmatrix} 1 & -1 & 3 & 0 \\ 0 & 6 & -7 & -3 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 2 & 6 \end{pmatrix} = -\det\begin{pmatrix} 1 & -1 & 3 & 0 \\ 0 & 6 & -7 & -3 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix} = -1\cdot 6\cdot(-4)\cdot 6 = 144$$

All four minors are > 0, so A is PD (Theorem 8.34).

(b) For $k=2$: $4p_{12} = -2$, so $p_{12} = -\frac{1}{2}$. For $k=3$: with $A_2^{-1} = \frac{1}{12}\left(\begin{smallmatrix} 4 & -2 \\ -2 & 4 \end{smallmatrix}\right)$ (Theorem 5.24), $\left(\begin{smallmatrix} p_{13} \\ p_{23} \end{smallmatrix}\right) = -A_2^{-1}\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right) = -\frac{1}{12}\left(\begin{smallmatrix} 6 \\ -6 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -1/2 \\ 1/2 \end{smallmatrix}\right)$. For $k=4$, $A_3(p_{14},p_{24},p_{34})' = -(-3,-3,0)'$:

$$\left(\begin{array}{ccc|c} 4 & 2 & 1 & 3 \\ 2 & 4 & -1 & 3 \\ 1 & -1 & 3 & 0 \end{array}\right) \xrightarrow{R_1\leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 2 & 4 & -1 & 3 \\ 4 & 2 & 1 & 3 \end{array}\right) \xrightarrow[R_3\leftarrow R_3-4R_1]{R_2\leftarrow R_2-2R_1} \left(\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 0 & 6 & -7 & 3 \\ 0 & 6 & -11 & 3 \end{array}\right)$$

$$\xrightarrow{R_3\leftarrow R_3-R_2} \left(\begin{array}{ccc|c} 1 & -1 & 3 & 0 \\ 0 & 6 & -7 & 3 \\ 0 & 0 & -4 & 0 \end{array}\right)$$

so $p_{34} = 0$, $p_{24} = \frac{1}{6}(3+7p_{34}) = \frac{1}{2}$ and $p_{14} = p_{24} - 3p_{34} = \frac{1}{2}$. So

$$P = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad AP = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 1 & -\frac{3}{2} & 2 & 0 \\ -3 & -\frac{3}{2} & 0 & 6 \end{pmatrix}$$

$$D = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix}, \qquad U = P^{-1} = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{4} & -\frac{3}{4} \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

with $d_k = \Delta_k/\Delta_{k-1} = 4/1,\ 12/4,\ 24/12,\ 144/24$. Hence $q_A(x) = (Ux)'D(Ux) = 4(x_1 + \frac{1}{2}x_2 + \frac{1}{4}x_3 - \frac{3}{4}x_4)^2 + 3(x_2 - \frac{1}{2}x_3 - \frac{1}{2}x_4)^2 + 2x_3^2 + 6x_4^2$.

(c) A_2 is PD (its leading minors 4, 12 are positive; Theorem 8.34). With $B = \left(\begin{smallmatrix} 1 & -3 \\ -1 & -3 \end{smallmatrix}\right)$, the upper right 2×2 block of A, $A_2^{-1}B = \frac{1}{12}\left(\begin{smallmatrix} 6 & -6 \\ -6 & -6 \end{smallmatrix}\right)$ and $B'A_2^{-1}B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, so $A/A_2 = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 9 \end{smallmatrix}\right) - B'A_2^{-1}B = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 6 \end{smallmatrix}\right)$, which is PD (Proposition 8.23(a)); so A is PD. Also $\det A = 12\cdot\det(A/A_2) = 12\cdot 12 = 144$.

(d) $A(1,1,0,-2)' = (12,12,0,-24)'$, $A(1,-1,2,0)' = (4,-4,8,0)'$, $A(1,1,0,1)' = (3,3,0,3)'$ and $A(1,-1,-1,0)' = (1,-1,-1,0)'$. So $a_A(c) \geqslant 1$ for $c = 12,4,3,1$ (Theorem 8.4), and as the multiplicities add up to 4, the eigenvalues are 12, 4, 3, 1, all > 0 (Theorem 8.21(a)); also $12\cdot 4\cdot 3\cdot 1 = 144 = \det A$.

Exercise 8.12. For $a \in \mathbb{R}$ let

$$A = \begin{pmatrix} 4 & 2 & 2 & a \\ 2 & 4 & a & 2 \\ 2 & a & 4 & 2 \\ a & 2 & 2 & 4 \end{pmatrix}$$

and let $h_1, \dots, h_4$ be the vectors of Exercise 8.4.

(a) Show that $h_1, \dots, h_4$ are eigenvectors of A, and find $\det A$.

(b) For which a is A PD? Answer with Theorem 8.34, and check with (a).

(c) For which a is A PSD? Find $\mathcal{N}(A)$ at the endpoints.

Solution. (a) The row sums are $8 + a$, so $Ah_1 = (8 + a)h_1$. Multiplying the rows of A by h_2, h_3,

$$Ah_2 = \begin{pmatrix} 4+2-2-a \\ 2+4-a-2 \\ 2+a-4-2 \\ a+2-2-4 \end{pmatrix} = (4-a)h_2, \quad Ah_3 = \begin{pmatrix} 4-2+2-a \\ 2-4+a-2 \\ 2-a+4-2 \\ a-2+2-4 \end{pmatrix} = (4-a)h_3$$

In the same way $Ah_4 = (4-2-2+a,\ 2-4-a+2,\ 2-a-4+2,\ a-2-2+4)' = ah_4$. So $A = Q\Lambda Q'$ with $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ and $\Lambda = \operatorname{diag}(8+a, 4-a, 4-a, a)$, these are the eigenvalues of A, and $\det A = a(a+8)(a-4)^2$ (Theorems 8.3(b),(c) and 7.19).

(b) The leading principal minors are 4, $16 - 4 = 12$,

$$\Delta_3 = 4(16 - a^2) - 2(8 - 2a) + 2(2a - 8) = 32 + 8a - 4a^2 = -4(a-4)(a+2)$$

(expanding $\det A_3$ along the first row), and $\det A = a(a+8)(a-4)^2$. The third is positive exactly for $-2 < a < 4$, and the fourth exactly for $a \neq 4$ and $a(a+8) > 0$, that is, $a > 0$ or $a < -8$. By Theorem 8.34, A is PD if and only if $0 < a < 4$. By (a) and Theorem 8.21(a), A is PD if and only if $8 + a > 0$, $4 - a > 0$ and $a > 0$: the same condition.

(c) By (a) and Theorem 8.21(a), A is PSD if and only if $8+a \geqslant 0$, $4-a \geqslant 0$ and $a \geqslant 0$, that is, $0 \leqslant a \leqslant 4$. By Theorem 8.6(c), $\mathcal{N}(A) = \operatorname{span}\{h_4\}$ for $a = 0$ (eigenvalues $8, 4, 4, 0$) and $\mathcal{N}(A) = \operatorname{span}\{h_2, h_3\}$ for $a = 4$ (eigenvalues $12, 0, 0, 4$).

Exercise 8.13. Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 2 \\ 1 & 0 & 6 & 4 \\ 1 & 2 & 4 & 10 \end{pmatrix}$$

(a) Show that A is PD by its leading principal minors.

(b) Find P, D and U of Theorem 8.34.

(c) Find the Cholesky factor T of A, check $T'T = A$, and write $q_A(x)$ as a sum of squares.

Solution. (a) $\Delta_1 = 1$, $\Delta_2 = 2 - 1 = 1$, and along row 1, $\Delta_3 = 1(12 - 0) - 1(6 - 0) + 1(0 - 2) = 4$. For $\Delta_4 = \det A$, apply $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$, $R_4 \leftarrow R_4 - R_1$, then $R_3 \leftarrow R_3 + R_2$, $R_4 \leftarrow R_4 - R_2$, then $R_4 \leftarrow R_4 - R_3$ (Definition 5.13(b), Theorem 5.14(c)):

$$\det A = \det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 5 & 3 \\ 0 & 1 & 3 & 9 \end{pmatrix} = \det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 4 & 8 \end{pmatrix} = \det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 4 \end{pmatrix} = 16$$

All four minors are > 0, so A is PD (Theorem 8.34).

(b) For $k=2$, $p_{12}=-1$. For $k=3$, with $A_2^{-1}=\left(\begin{smallmatrix}2&-1\\-1&1\end{smallmatrix}\right)$ ($\det A_2=1$, Theorem 5.24), $\left(\begin{smallmatrix}p_{13}\\p_{23}\end{smallmatrix}\right)=-A_2^{-1}\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)=\left(\begin{smallmatrix}-2\\1\end{smallmatrix}\right)$. For $k=4$, $A_3(p_{14},p_{24},p_{34})'=-(1,2,4)'$:

$$\left(\begin{array}{ccc|c}1&1&1&-1\\1&2&0&-2\\1&0&6&-4\end{array}\right)\xrightarrow[R_3\leftarrow R_3-R_1]{R_2\leftarrow R_2-R_1}\left(\begin{array}{ccc|c}1&1&1&-1\\0&1&-1&-1\\0&-1&5&-3\end{array}\right)\xrightarrow{R_3\leftarrow R_3+R_2}\left(\begin{array}{ccc|c}1&1&1&-1\\0&1&-1&-1\\0&0&4&-4\end{array}\right)$$

so $p_{34}=-1$, $p_{24}=-1+p_{34}=-2$ and $p_{14}=-1-p_{24}-p_{34}=2$. So

$$P=\begin{pmatrix}1&-1&-2&2\\0&1&1&-2\\0&0&1&-1\\0&0&0&1\end{pmatrix},\qquad AP=\begin{pmatrix}1&0&0&0\\1&1&0&0\\1&-1&4&0\\1&1&4&4\end{pmatrix}$$

$$D=\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&4&0\\0&0&0&4\end{pmatrix},\qquad U=P^{-1}=\begin{pmatrix}1&1&1&1\\0&1&-1&1\\0&0&1&1\\0&0&0&1\end{pmatrix}$$

with $d_k=\Delta_k/\Delta_{k-1}=1/1,\ 1/1,\ 4/1,\ 16/4$.

(c) By Theorem 8.40, $T=D^{1/2}U$ is U with rows 3 and 4 doubled:

$$T=\begin{pmatrix}1&1&1&1\\0&1&-1&1\\0&0&2&2\\0&0&0&2\end{pmatrix},\qquad T'T=\begin{pmatrix}1&0&0&0\\1&1&0&0\\1&-1&2&0\\1&1&2&2\end{pmatrix}\begin{pmatrix}1&1&1&1\\0&1&-1&1\\0&0&2&2\\0&0&0&2\end{pmatrix}=\begin{pmatrix}1&1&1&1\\1&2&0&2\\1&0&6&4\\1&2&4&10\end{pmatrix}=A$$

So $q_A(x)=\|Tx\|^2=(x_1+x_2+x_3+x_4)^2+(x_2-x_3+x_4)^2+4(x_3+x_4)^2+4x_4^2$. Also $\det A=(\det T)^2=(1\cdot1\cdot2\cdot2)^2=16=\Delta_4$ (Theorem 8.26(c)).

Exercise 8.14. Let

$$A=\begin{pmatrix}9&3&-6&3\\3&5&0&-1\\-6&0&9&-1\\3&-1&-1&4\end{pmatrix}$$

(a) Find the Cholesky factor T of A. (b) Find $\det A$. (c) Solve $Ax=b$ for $b=(9,7,2,5)'$ by solving $T'y=b$ and then $Tx=y$.

Solution. (a) By Remark 8.41, row by row:

- row 1: $t_{11}=\sqrt{9}=3$, $t_{12}=\frac{3}{3}=1$, $t_{13}=-\frac{6}{3}=-2$, $t_{14}=\frac{3}{3}=1$;
- row 2: $t_{22}=\sqrt{5-1}=2$, $t_{23}=\big(0-1\cdot(-2)\big)/2=1$, $t_{24}=(-1-1\cdot1)/2=-1$;
- row 3: $t_{33}=\sqrt{9-4-1}=2$, $t_{34}=\big(-1-(-2)\cdot1-1\cdot(-1)\big)/2=1$;
- row 4: $t_{44}=\sqrt{4-1-1-1}=1$.

So

$$T=\begin{pmatrix}3&1&-2&1\\0&2&1&-1\\0&0&2&1\\0&0&0&1\end{pmatrix}$$

and $T'T=A$; in particular A is PD (Theorem 8.26(c)).

(b) $\det A=\det T'\cdot\det T=(3\cdot2\cdot2\cdot1)^2=144$ (Theorems 5.18(b), 5.22 and 5.14(c)).

(c) $Ax=T'(Tx)$, so x solves $Ax=b$ if $T'y=b$ and $Tx=y$. From the first equation downwards, $3y_1=9$, $y_1+2y_2=7$, $-2y_1+y_2+2y_3=2$, $y_1-y_2+y_3+y_4=5$ give $y=(3,2,3,1)'$. Back substitution in $Tx=y$: $x_4=1$, $2x_3+x_4=3$, $2x_2+x_3-x_4=2$, $3x_1+x_2-2x_3+x_4=3$ give $x=(1,1,1,1)'$. Check: the row sums of A are $9,7,2,5$. The solution is unique, as A is invertible.

Exercise 8.15. Let

$$A = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 2 & -1 \\ 2 & 2 & -1 & -1 \\ -1 & -1 & -1 & 5 \end{pmatrix}$$

(a) Find the leading principal minors and the matrices P, D, U with $P'AP = D$ and $A = U'DU$ (Corollary 8.37), and write $q_A(x)$ as a combination of squares.

(b) Find the inertia of A. Is A PD, PSD or indefinite?

(c) Check (b) with the eigenvectors $(1,1,1,-3)'$, $(1,1,1,1)'$, $(1,-1,0,0)'$ and $(1,1,-2,0)'$ of A.

(d) Find an invertible R with $R'AR = \operatorname{diag}(1,1,1,-1)$.

Solution. (a) $\Delta_1 = 1$, $\Delta_2 = 1$, and along row 1, $\Delta_3 = 1(-1-4) - 0 + 2(0-2) = -9$. Next $p_{12} = 0$, and as $A_2 = I_2$, $\binom{p_{13}}{p_{23}} = -\binom{2}{2}$. Column 4 of P solves $A_3(p_{14}, p_{24}, p_{34})' = -(-1,-1,-1)'$ (Lemma 8.33):

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 2 & 2 & -1 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & -5 & -1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -9 & -3 \end{array}\right)$$

so $p_{34} = \frac{1}{3}$, $p_{24} = 1 - 2p_{34} = \frac{1}{3}$ and $p_{14} = 1 - 2p_{34} = \frac{1}{3}$. So

$$P = \begin{pmatrix} 1 & 0 & -2 & \frac{1}{3} \\ 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad AP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 2 & -9 & 0 \\ -1 & -1 & 3 & 4 \end{pmatrix}, \qquad U = P^{-1} = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and $D = P'AP = \operatorname{diag}(1,1,-9,4)$, so $\Delta_4 = d_1d_2d_3d_4 = -36$ (Corollary 8.37(a)). So $q_A(x) = (x_1 + 2x_3 - x_4)^2 + (x_2 + 2x_3 - x_4)^2 - 9\left(x_3 - \frac{1}{3}x_4\right)^2 + 4x_4^2$.

(b) Three diagonal entries of D are positive and one is negative: the inertia is $(3,1,0)$ (Theorem 8.43), and A is indefinite; for instance $q_A(e_1) = 1$ and $q_A(e_3) = -1$.

(c) $A(1,1,1,-3)' = (6,6,6,-18)'$, $A(1,1,1,1)' = (2,2,2,2)'$, $A(1,-1,0,0)' = (1,-1,0,0)'$ and $A(1,1,-2,0)' = (-3,-3,6,0)'$. So $a_A(c) \geqslant 1$ for $c = 6, 2, 1, -3$ (Theorem 8.4); as the multiplicities add up to 4, the eigenvalues are 6, 2, 1, -3: three positive and one negative, as Theorem 8.43(c) requires.

(d) Let $p_1, \ldots, p_4$ be the columns of P, so that $p_i'Ap_j = 0$ for $i \neq j$ and $p_i'Ap_i = d_i$ (Theorem 1.4(d)). Scale p_3 by $\frac{1}{3}$ and p_4 by $\frac{1}{2}$, and interchange them: $R = [p_1\ p_2\ \frac{1}{2}p_4\ \frac{1}{3}p_3]$. The columns of R are nonzero multiples of those of the invertible P, in another order, so they are linearly independent and R is invertible (Theorem 4.13(b),(d)). Then

$$R = \begin{pmatrix} 1 & 0 & \frac{1}{6} & -\frac{2}{3} \\ 0 & 1 & \frac{1}{6} & -\frac{2}{3} \\ 0 & 0 & \frac{1}{6} & \frac{1}{3} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}, \qquad R'AR = \operatorname{diag}\left(1, 1, \tfrac{1}{4} \cdot 4, \tfrac{1}{9} \cdot (-9)\right) = \operatorname{diag}(1,1,1,-1)$$

Exercise 8.16. Let

$$A = \begin{pmatrix} 6 & -3 & -3 & 0 \\ -3 & 7 & -4 & 5 \\ -3 & -4 & 7 & -5 \\ 0 & 5 & -5 & 6 \end{pmatrix}$$

(a) Find p_A, the eigenvalues, and an orthonormal basis of eigenvectors.

(b) Is A PD? PSD? Find $\operatorname{rank} A$ and the leading principal minors.

(c) Find $A^{1/2}$.

Solution. (a) For real t, add rows 2 and 3 to row 1, take the factor t out of row 1, subtract column 1 from columns 2 and 3, and expand along row 1; in the 3×3 determinant, add row 2 to row 1, take the factor $t-9$ out of row 1, subtract column 1 from column 2, and expand along row 1 (Definition 5.13(a),(b), Theorems 5.22 and 5.21):

$$\begin{aligned}
\det(tI - A) &= \det \begin{pmatrix} t-6 & 3 & 3 & 0 \\ 3 & t-7 & 4 & -5 \\ 3 & 4 & t-7 & 5 \\ 0 & -5 & 5 & t-6 \end{pmatrix} = \det \begin{pmatrix} t & t & t & 0 \\ 3 & t-7 & 4 & -5 \\ 3 & 4 & t-7 & 5 \\ 0 & -5 & 5 & t-6 \end{pmatrix} \\
&= t \det \begin{pmatrix} 1 & 1 & 1 & 0 \\ 3 & t-7 & 4 & -5 \\ 3 & 4 & t-7 & 5 \\ 0 & -5 & 5 & t-6 \end{pmatrix} = t \det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 3 & t-10 & 1 & -5 \\ 3 & 1 & t-10 & 5 \\ 0 & -5 & 5 & t-6 \end{pmatrix} \\
&= t \det \begin{pmatrix} t-10 & 1 & -5 \\ 1 & t-10 & 5 \\ -5 & 5 & t-6 \end{pmatrix} = t \det \begin{pmatrix} t-9 & t-9 & 0 \\ 1 & t-10 & 5 \\ -5 & 5 & t-6 \end{pmatrix} \\
&= t(t-9) \det \begin{pmatrix} 1 & 1 & 0 \\ 1 & t-10 & 5 \\ -5 & 5 & t-6 \end{pmatrix} = t(t-9) \det \begin{pmatrix} 1 & 0 & 0 \\ 1 & t-11 & 5 \\ -5 & 10 & t-6 \end{pmatrix} \\
&= t(t-9)\Big((t-11)(t-6) - 50\Big) = t(t-9)(t^2 - 17t + 16) = t(t-1)(t-9)(t-16)
\end{aligned}$$

So $p_A(t) = t(t-1)(t-9)(t-16)$ (Definition 7.7(a), Proposition 7.5(d)), and the eigenvalues are 16, 9, 1, 0. Row-reduce $A - cI$:

$$\begin{aligned}
A - 16I &= \begin{pmatrix} -10 & -3 & -3 & 0 \\ -3 & -9 & -4 & 5 \\ -3 & -4 & -9 & -5 \\ 0 & 5 & -5 & -10 \end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - R_4} \begin{pmatrix} -10 & -8 & 2 & 10 \\ 0 & -5 & 5 & 10 \\ -3 & -4 & -9 & -5 \\ 0 & 5 & -5 & -10 \end{pmatrix} \\
&\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_1 \leftarrow R_1 + 2R_3} \begin{pmatrix} -16 & -16 & -16 & 0 \\ 0 & -5 & 5 & 10 \\ -3 & -4 & -9 & -5 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \leftarrow \frac{1}{5}R_2]{R_1 \leftarrow \frac{1}{16}R_1} \begin{pmatrix} -1 & -1 & -1 & 0 \\ 0 & -1 & 1 & 2 \\ -3 & -4 & -9 & -5 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
&\xrightarrow{R_3 \leftarrow R_3 - 3R_1} \begin{pmatrix} -1 & -1 & -1 & 0 \\ 0 & -1 & 1 & 2 \\ 0 & -1 & -6 & -5 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} -1 & -1 & -1 & 0 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & -7 & -7 \\ 0 & 0 & 0 & 0 \end{pmatrix}
\end{aligned}$$

so $x_4 = 1$ gives $x_3 = -x_4 = -1$, $x_2 = x_3 + 2x_4 = 1$ and $x_1 = -x_2 - x_3 = 0$;

$$A - 9I = \begin{pmatrix} -3 & -3 & -3 & 0 \\ -3 & -2 & -4 & 5 \\ -3 & -4 & -2 & -5 \\ 0 & 5 & -5 & -3 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_4} \begin{pmatrix} -3 & -3 & -3 & 0 \\ -3 & -7 & 1 & 8 \\ 0 & -1 & 1 & -5 \\ 0 & 5 & -5 & -3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_3]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} -3 & -3 & -3 & 0 \\ 0 & -4 & 4 & 8 \\ 0 & -1 & 1 & -5 \\ 0 & 4 & -4 & -8 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow \frac{1}{4}R_4]{R_2 \leftarrow \frac{1}{4}R_2} \begin{pmatrix} -3 & -3 & -3 & 0 \\ 0 & -1 & 1 & 2 \\ 0 & -1 & 1 & -5 \\ 0 & 1 & -1 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} -3 & -3 & -3 & 0 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & -7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 0$, and $x_3 = -1$ gives $x_2 = x_3 + 2x_4 = -1$ and $x_1 = -x_2 - x_3 = 2$;

$$A - I = \begin{pmatrix} 5 & -3 & -3 & 0 \\ -3 & 6 & -4 & 5 \\ -3 & -4 & 6 & -5 \\ 0 & 5 & -5 & 5 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_4} \begin{pmatrix} 5 & -3 & -3 & 0 \\ -3 & 1 & 1 & 0 \\ -3 & -4 & 6 & -5 \\ 0 & 5 & -5 & 5 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_2]{R_1 \leftarrow R_1 + 3R_2} \begin{pmatrix} -4 & 0 & 0 & 0 \\ -3 & 1 & 1 & 0 \\ 0 & -5 & 5 & -5 \\ 0 & 5 & -5 & 5 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_3]{R_1 \leftarrow \frac{1}{4}R_1} \begin{pmatrix} -1 & 0 & 0 & 0 \\ -3 & 1 & 1 & 0 \\ 0 & -5 & 5 & -5 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow \frac{1}{5}R_3]{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_1 = 0$, and $x_3 = -1$ gives $x_4 = 2x_3 = -2$ and $x_2 = -x_3 = 1$;

$$A = \begin{pmatrix} 6 & -3 & -3 & 0 \\ -3 & 7 & -4 & 5 \\ -3 & -4 & 7 & -5 \\ 0 & 5 & -5 & 6 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_4]{R_2 \leftarrow R_2 - R_4} \begin{pmatrix} 6 & -3 & -3 & 0 \\ -3 & 2 & 1 & -1 \\ -3 & 1 & 2 & 1 \\ 0 & 5 & -5 & 6 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_2]{R_1 \leftarrow R_1 + R_2} \begin{pmatrix} 3 & -1 & -2 & -1 \\ -3 & 2 & 1 & -1 \\ 0 & -1 & 1 & 2 \\ 0 & 5 & -5 & 6 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + 2R_3]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 3 & -1 & -2 & -1 \\ 0 & 1 & -1 & -2 \\ 0 & -1 & 1 & 2 \\ 0 & 3 & -3 & 10 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 3 & -1 & -2 & -1 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 16 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 3 & -1 & -2 & -1 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 16 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 0$, and $x_3 = 1$ gives $x_2 = x_3 + 2x_4 = 1$ and $x_1 = \frac{1}{3}(x_2 + 2x_3 + x_4) = 1$. So the eigenspaces are spanned by $(0, 1, -1, 1)'$, $(2, -1, -1, 0)'$, $(0, 1, -1, -2)'$ and $(1, 1, 1, 0)'$, pairwise orthogonal (Theorem 8.2); an orthonormal basis is $q_1 = \frac{1}{\sqrt{3}}(0, 1, -1, 1)'$, $q_2 = \frac{1}{\sqrt{6}}(2, -1, -1, 0)'$, $q_3 = \frac{1}{\sqrt{6}}(0, 1, -1, -2)'$, $q_4 = \frac{1}{\sqrt{3}}(1, 1, 1, 0)'$.

(b) The eigenvalues are $\geqslant 0$ and one is 0, so A is PSD but not PD (Theorem 8.21(a)), and rank $A = 3$ (Theorem 8.4). The leading minors are 6, $42 - 9 = 33$, along row 1 $\Delta_3 = 6(49 - 16) + 3(-21 - 12) - 3(12 + 21) = 0$, and $\det A = 16 \cdot 9 \cdot 1 \cdot 0 = 0$ (Theorem 7.19).

(c) By Theorem 8.46,

$$\begin{aligned} A^{1/2} &= 4q_1q_1' + 3q_2q_2' + 1\,q_3q_3' + 0\,q_4q_4' \\ &= \frac{4}{3}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 4 & -2 & -2 & 0 \\ -2 & 1 & 1 & 0 \\ -2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & -2 \\ 0 & -1 & 1 & 2 \\ 0 & -2 & 2 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 1 \\ -1 & -1 & 2 & -1 \\ 0 & 1 & -1 & 2 \end{pmatrix} \end{aligned}$$

Check: $\text{row}_1(A^{1/2})A^{1/2} = (4+1+1+0,\ -2-2+1+0,\ -2+1-2+0,\ 0-1+1+0) = (6,-3,-3,0) = \text{row}_1(A)$, and likewise for rows 2, 3 and 4.

Exercise 8.17. Let $A = I_4 + 6J_4$, the 4×4 matrix with diagonal entries 7 and all other entries 6.

(a) Find the eigenvalues of A, and compute $A^{1/2}$, $A^{-1/2}$ and A^{-1}. Check that $(A^{1/2})^2 = A$.

(b) Show that

$$G = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 \end{pmatrix}$$

also satisfies $G^2 = A$. Why does this not contradict Theorem 8.46(a)?

Solution. (a) By Example 8.16(b), with $\alpha = 1$ and $\beta = 6$, $A = 25\bar{J}_4 + C_4$, and the eigenvalues are 25, with $\mathcal{N}(A - 25I) = \text{span}\{\mathbf{1}_4\}$, and 1 (three times), with $\mathcal{N}(A - I) = \text{span}\{\mathbf{1}_4\}^\perp = \{x : \mathbf{1}'x = 0\}$. The projection matrices onto these eigenspaces are $\bar{J}_4 = \frac{1}{4}J_4$ and $C_4 = I_4 - \frac{1}{4}J_4$ (Corollary 6.35(a)), so $A = 25\bar{J}_4 + C_4$ is the spectral decomposition of A (Theorem 8.6(b)), and by Theorem 8.46(a),(c) and Definition 8.9

$$A^{1/2} = 5\bar{J}_4 + C_4 = I_4 + J_4, \qquad A^{-1/2} = \tfrac{1}{5}\bar{J}_4 + C_4 = I_4 - \tfrac{1}{5}J_4, \qquad A^{-1} = \tfrac{1}{25}\bar{J}_4 + C_4 = I_4 - \tfrac{6}{25}J_4$$

Indeed $(I_4 + J_4)^2 = I_4 + 2J_4 + J_4^2 = I_4 + 6J_4$, as $J_4^2 = 4J_4$ (Proposition 1.15).

(b) The $(1,1)$, $(1,2)$, $(1,3)$, $(1,4)$ entries of G^2 are $1+4+1+1 = 7$, $2+2+1+1 = 6$, $1+2+1+2 = 6$, $1+2+2+1 = 6$, and the other rows follow in the same way: $G^2 = A$. But G is not PSD: $(1,-1,0,0)G(1,-1,0,0)' = 1 - 4 + 1 = -2$. Theorem 8.46(a) asserts uniqueness only among PSD matrices. (With $h_1, \ldots, h_4$ of Exercise 8.4, $G = \frac{1}{4}(5h_1h_1' + h_2h_2' - h_3h_3' - h_4h_4')$, with the eigenvalues $5, 1, -1, -1$.)

Exercise 8.18. Let

$$B = \begin{pmatrix} 1 & -1 & 0 & 1 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ 1 & -1 & -1 & 3 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & -1 & 0 & 1 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ 1 & -1 & -1 & 4 \end{pmatrix}$$

(a) Show that B is PD and that $A \succcurlyeq B$.

(b) Find B^{-1} by Gauss–Jordan elimination, and A^{-1} by Theorem 5.32(b).

(c) Check that $B^{-1} \succcurlyeq A^{-1}$ and $\det A \geqslant \det B$ (Theorems 8.51 and 8.52).

Solution. (a) $\Delta_1(B) = 1$, $\Delta_2(B) = 2 - 1 = 1$, and along row 1, $\Delta_3(B) = 1(4-1) + 1(-2-0) = 1$. For $\det B$, apply $R_2 \leftarrow R_2 + R_1$, $R_4 \leftarrow R_4 - R_1$, then $R_3 \leftarrow R_3 + R_2$, then $R_4 \leftarrow R_4 + R_3$ (Definition 5.13(b), Theorem 5.14(c)):

$$\det B = \det \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} = \det \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} = \det \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 1$$

So B is PD (Theorem 8.34). $A - B = e_4e_4'$ is PSD (Theorem 8.25), so $A \succcurlyeq B$.

(b) Row-reduce $[B \mid I_4]$ (Corollary 2.17(a)):

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 1 & 0 & 0 & 0 \\ -1 & 2 & -1 & -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & -1 & 3 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 + R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow R_3 + R_4]{R_1 \leftarrow R_1 - R_4} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 + R_3} \left(\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & 1 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & 2 & 3 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 3 & 2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 2 & 3 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{array}\right)$$

So

$$B^{-1} = \begin{pmatrix} 3 & 2 & 1 & 0 \\ 2 & 3 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Now $A = B + e_4 e_4'$, $w = B^{-1} e_4 = (0, 1, 1, 1)'$ and $1 + e_4' B^{-1} e_4 = 2$, so by Theorem 5.32(b)

$$A^{-1} = B^{-1} - \tfrac{1}{2} w w' = \frac{1}{2} \begin{pmatrix} 6 & 4 & 2 & 0 \\ 4 & 5 & 3 & 1 \\ 2 & 3 & 3 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

Check: $\text{row}_1(A) A^{-1} = \frac{1}{2}(6 - 4 + 0 + 0,\ 4 - 5 + 0 + 1,\ 2 - 3 + 0 + 1,\ 0 - 1 + 0 + 1) = (1, 0, 0, 0)$, and likewise for rows 2, 3 and 4.

(c) $B^{-1} - A^{-1} = \frac{1}{2} w w'$ is PSD (Theorem 8.25), so $B^{-1} \succcurlyeq A^{-1}$. And $\det A = \det B \cdot (1 + e_4' B^{-1} e_4) = 2 \geqslant 1 = \det B$ (Theorem 5.31).

Exercise 8.19. Let

$$A = \begin{pmatrix} 2 & -1 & -1 & 1 \\ -1 & 2 & 1 & -1 \\ -1 & 1 & 2 & -1 \\ 1 & -1 & -1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 2 & 1 \\ 1 & 2 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

(a) Compute AB, and conclude that $AB = BA$.

(b) Find the eigenvalues and eigenspaces of A.

(c) Find an orthogonal Q such that $Q'AQ$ and $Q'BQ$ are both diagonal (Theorem 8.55).

Solution. (a) The product is

$$AB = \begin{pmatrix} 0 & 2 & 2 & -1 \\ 2 & -2 & 1 & 2 \\ 2 & 1 & -2 & 2 \\ -1 & 2 & 2 & 0 \end{pmatrix}$$

It is symmetric, so $BA = B'A' = (AB)' = AB$ (Theorem 1.9).

(b) Row-reduce $A - 5I$ and $A - I$:

$$A - 5I = \begin{pmatrix} -3 & -1 & -1 & 1 \\ -1 & -3 & 1 & -1 \\ -1 & 1 & -3 & -1 \\ 1 & -1 & -1 & -3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 + R_3} \begin{pmatrix} -3 & -1 & -1 & 1 \\ -2 & -2 & -2 & -2 \\ -1 & 1 & -3 & -1 \\ -2 & -2 & -2 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} -1 & 1 & 1 & 3 \\ -2 & -2 & -2 & -2 \\ -1 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} -1 & 1 & 1 & 3 \\ 0 & -4 & -4 & -8 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 1$ gives $x_3 = -x_4 = -1$, $x_2 = -x_3 - 2x_4 = -1$ and $x_1 = x_2 + x_3 + 3x_4 = 1$;

$$A - I = \begin{pmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 + R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $\mathcal{N}(A-5I) = \operatorname{span}\{v\}$ with $v = (1,-1,-1,1)'$, and $\mathcal{N}(A-I) = \{x : x_1 = x_2 + x_3 - x_4\}$, of dimension 3. As the dimensions add up to 4, the eigenvalues are 5 and 1, with $a_A(1) = 3$ (Theorem 8.4).

(c) $\mathcal{N}(A - I)$ is invariant under B: for x in it, $A(Bx) = B(Ax) = Bx$. For $x = (b + c - d, b, c, d)'$ in $\mathcal{N}(A - I)$, $Bx = (2b + 2c - d,\ 3c,\ 3b,\ b + c + d)'$. If $Bx = \mu x$, entries 2 and 3 give $3c = \mu b$ and $3b = \mu c$, so $9b = 3\mu c = \mu^2 b$ and likewise $9c = \mu^2 c$; entry 4 gives $b + c + d = \mu d$.

- If $\mu^2 \neq 9$, then $b = c = 0$, and $d = \mu d$ with $d \neq 0$ (as $x \neq 0$): $\mu = 1$ and $x = d(-1, 0, 0, 1)'$.
- If $\mu = 3$, then $c = b$ and $2b + d = 3d$, so $d = b$: $x = b(1, 1, 1, 1)'$.
- If $\mu = -3$, then $c = -b$ and $d = -3d$, so $d = 0$: $x = b(0, 1, -1, 0)'$.

The first entry then agrees. Also $Bv = (-1, 1, 1, -1)' = -v$. Normalizing,

$$Q = \begin{pmatrix} 1/2 & 1/2 & 0 & -1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} & 0 \\ -1/2 & 1/2 & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 0 & 1/\sqrt{2} \end{pmatrix}, \qquad Q'AQ = \operatorname{diag}(5, 1, 1, 1), \qquad Q'BQ = \operatorname{diag}(-1, 3, -3, 1)$$

Exercise 8.20. Let

$$A = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 2 & -1 \\ -1 & -1 & -1 & -1 \end{pmatrix}$$

(a) Show that A is PD, and find the roots of $\det(B - tA) = 0$.

(b) Find an invertible E with $E'AE = I_4$ and $E'BE$ diagonal (Theorem 8.57).

Solution. (a) $\Delta_1(A) = 2$, $\Delta_2(A) = 3$, and along row 1, $\Delta_3(A) = 2(4 - 1) - 1(2 - 1) + 1(1 - 2) = 4$. Applying $R_1 \leftarrow R_1 - R_4$, $R_2 \leftarrow R_2 - R_4$ and $R_3 \leftarrow R_3 - R_4$ to A gives the lower triangular matrix with rows e_1', e_2', e_3', $\mathbf{1}_4'$, so $\det A = 1$ (Definition 5.13(b), Theorem 5.22); so A is PD (Theorem 8.34).

For real t, apply the same row operations to $B - tA$; the result is lower triangular:

$$\det(B-tA) = \det\begin{pmatrix} -2t & -1-t & -1-t & -1-t \\ -1-t & 1-2t & -1-t & -1-t \\ -1-t & -1-t & 2-2t & -1-t \\ -1-t & -1-t & -1-t & -1-t \end{pmatrix}$$
$$= \det\begin{pmatrix} 1-t & 0 & 0 & 0 \\ 0 & 2-t & 0 & 0 \\ 0 & 0 & 3-t & 0 \\ -1-t & -1-t & -1-t & -1-t \end{pmatrix} = (1-t)(2-t)(3-t)(-1-t)$$

The roots are -1, 1, 2, 3.

(b) Row-reduce $B - cA$ for $c = -1, 1, 2, 3$:

$$B + A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow \frac{1}{4}R_3]{R_1 \leftarrow \frac{1}{2}R_1,\ R_2 \leftarrow \frac{1}{3}R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$B - A = \begin{pmatrix} -2 & -2 & -2 & -2 \\ -2 & -1 & -2 & -2 \\ -2 & -2 & 0 & -2 \\ -2 & -2 & -2 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} -2 & -2 & -2 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$B - 2A = \begin{pmatrix} -4 & -3 & -3 & -3 \\ -3 & -3 & -3 & -3 \\ -3 & -3 & -2 & -3 \\ -3 & -3 & -3 & -3 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_1 \leftarrow R_1 - R_2,\ R_3 \leftarrow R_3 - R_2} \begin{pmatrix} -1 & 0 & 0 & 0 \\ -3 & -3 & -3 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -3 & -3 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$B - 3A = \begin{pmatrix} -6 & -4 & -4 & -4 \\ -4 & -5 & -4 & -4 \\ -4 & -4 & -4 & -4 \\ -4 & -4 & -4 & -4 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_3]{R_1 \leftarrow R_1 - R_3,\ R_2 \leftarrow R_2 - R_3} \begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ -4 & -4 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & -4 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 4R_2} \begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $Bf = cAf$ for $f_1 = (0,0,0,1)'$ ($c = -1$), $f_2 = (1,0,0,-1)'$ ($c = 1$), $f_3 = (0,1,0,-1)'$ ($c = 2$) and $f_4 = (0,0,1,-1)'$ ($c = 3$). The roots are distinct, so $f_i' A f_j = 0$ for $i \neq j$ (Theorem 8.57(c)); and $f_1' A f_1 = 1$, $f_2' A f_2 = 2-2+1 = 1$, $f_3' A f_3 = 2-2+1 = 1$, $f_4' A f_4 = 2-2+1 = 1$. By Theorem 8.57(c),

$$E = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & -1 & -1 & -1 \end{pmatrix}, \qquad AE = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad BE = \begin{pmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 \\ -1 & 0 & 0 & 3 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

and $E'AE = I_4$, $E'BE = \operatorname{diag}(-1, 1, 2, 3)$.

Exercise 8.21. Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 4 & 0 & 0 \\ 4 & 7 & 3 & 0 \\ 0 & 3 & 5 & 2 \\ 0 & 0 & 2 & 3 \end{pmatrix}$$

(a) Show that A is PD, and find an invertible E with $E'AE = I_4$ and $E'BE$ diagonal.

(b) Show that $x'Ax \leqslant x'Bx \leqslant 4\,x'Ax$ for every $x \in \mathbb{R}^4$, that is, $A \preccurlyeq B \preccurlyeq 4A$.

Solution. (a) Expanding $\Delta_k(A) = \det A_k$ along its last row gives $\Delta_k(A) = 2\Delta_{k-1}(A) - \Delta_{k-2}(A)$ for $k = 3, 4$ (Theorem 5.21), so the leading principal minors are 1, $2-1=1$, $2-1=1$, $2-1=1$, and A is PD (Theorem 8.34). The rows of $B - \lambda A$ are $(4-\lambda,\ 4-\lambda,\ 0,\ 0)$, $(4-\lambda,\ 7-2\lambda,\ 3-\lambda,\ 0)$, $(0,\ 3-\lambda,\ 5-2\lambda,\ 2-\lambda)$, $(0,\ 0,\ 2-\lambda,\ 3-2\lambda)$. Subtract row 1 from row 2, then the new row 2 from row 3, then the new row 3 from row 4 (Definition 5.13(b)):

$$\det(B - \lambda A) = \det \begin{pmatrix} 4-\lambda & 4-\lambda & 0 & 0 \\ 0 & 3-\lambda & 3-\lambda & 0 \\ 0 & 0 & 2-\lambda & 2-\lambda \\ 0 & 0 & 0 & 1-\lambda \end{pmatrix} = (4-\lambda)(3-\lambda)(2-\lambda)(1-\lambda)$$

(Theorem 5.14(c)), so $\lambda_1 = 4$, $\lambda_2 = 3$, $\lambda_3 = 2$, $\lambda_4 = 1$ (Theorem 8.57(b), as $\det A = 1$). The null spaces of

$$B - 4A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -3 & -2 \\ 0 & 0 & -2 & -5 \end{pmatrix}, \qquad B - 3A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & -3 \end{pmatrix}$$

$$B - 2A = \begin{pmatrix} 2 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad B - A = \begin{pmatrix} 3 & 3 & 0 & 0 \\ 3 & 5 & 2 & 0 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

contain $f_1 = (1,0,0,0)'$, $f_2 = (-1,1,0,0)'$, $f_3 = (1,-1,1,0)'$ and $f_4 = (-1,1,-1,1)'$. Here $Af_1 = (1,1,0,0)'$, $Af_2 = (0,1,1,0)'$, $Af_3 = (0,0,1,1)'$ and $Af_4 = (0,0,0,1)'$, so $f_k' A f_k = 1$, and $f_j' A f_k = 0$ for $j \neq k$ (Theorem 8.57(c)). Hence

$$E = \begin{pmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

has $E'AE = I_4$ and $E'BE = \operatorname{diag}(4,3,2,1)$, since $(E'BE)_{jk} = f_j' B f_k = \lambda_k f_j' A f_k$.

(b) $\det E = 1$, so E is invertible (Theorem 5.18(a)) and every x is $x = Ey$ with $y = E^{-1}x$; here E^{-1} is the upper triangular matrix with ones on the diagonal and on the superdiagonal (multiply out E with it), so $y_k = x_k + x_{k+1}$ for $k \leqslant 3$ and $y_4 = x_4$. Then $x'Ax = y'E'AEy = y_1^2 + y_2^2 + y_3^2 + y_4^2$ and $x'Bx = y'E'BEy = 4y_1^2 + 3y_2^2 + 2y_3^2 + y_4^2$. Hence $x'Ax \leqslant x'Bx \leqslant 4\,x'Ax$.

Problems

Problem 8.1 ([2, Exercise 6.17])**.** (a) Show that a symmetric matrix whose eigenvalues are all 0 is the zero matrix.

(b) Show by a 2×2 example that (a) fails without symmetry, and that the same example shows that Theorem 8.4 needs symmetry.

Solution. (a) By Theorem 8.3, $A = Q\Lambda Q'$, and the diagonal entries of Λ are the eigenvalues of A. If they are all 0, then $\Lambda = 0$ and $A = 0$.

(b) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Then $p_A(t) = \det \left(\begin{smallmatrix} t & -1 \\ 0 & t \end{smallmatrix}\right) = t^2$, so both eigenvalues are 0, but $A \neq 0$. Also $\operatorname{rank} A = 1$ (one nonzero row, in echelon form), although A has no nonzero eigenvalue.

Problem 8.2 ([6, Exercise 8.62])**.** Show that a symmetric $A \in \mathbb{R}^{n \times n}$ is idempotent if and only if $\operatorname{rank} A + \operatorname{rank}(I_n - A) = n$.

Solution. Let $A = Q\Lambda Q'$ (Theorem 8.3), and let n_0 and n_1 be the numbers of eigenvalues λ_i equal to 0 and to 1.

- $\operatorname{rank} A = n - n_0$ (Theorem 8.4). $I_n - A = (-1)A + I_n$ is symmetric with the eigenvalues $1 - \lambda_i$ (Proposition 7.18(c), with $c = -1$ and then $c = 1$), of which n_1 are 0; so $\operatorname{rank}(I_n - A) = n - n_1$ (Theorem 8.4).
- Hence $\operatorname{rank} A + \operatorname{rank}(I_n - A) = n + (n - n_0 - n_1)$, which is n if and only if $n_0 + n_1 = n$, that is, every eigenvalue of A is 0 or 1.
- This holds if and only if A is idempotent: by Theorem 8.13(a) if A is idempotent, and by Theorem 8.13(b) conversely.

Problem 8.3 ([6, Exercise 8.75])**.** Find the matrix of the quadratic form $\sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2$ on $\mathbb{R}^n$, and its eigenvalues and eigenvectors.

Solution.

$$\begin{aligned} \sum_{i,j} (x_i - x_j)^2 &= \sum_{i,j} (x_i^2 + x_j^2 - 2x_i x_j) = 2n \sum_i x_i^2 - 2\Big(\sum_i x_i\Big)^2 \\ &= 2n\, x'x - 2(\mathbf{1}'x)^2 = 2n\, x'\Big(I_n - \tfrac{1}{n}\mathbf{1}\mathbf{1}'\Big)x = 2n\, x' C_n x \end{aligned}$$

So the matrix of the form is the symmetric matrix $2nC_n$ (Proposition 8.18). C_n has the eigenvalues 1 ($n-1$ times) and 0 (once) (Example 8.16(a)), so $2nC_n$ has the eigenvalues (Proposition 7.18(c)) $2n$ ($n-1$ times), with eigenvectors the nonzero vectors orthogonal to $\mathbf{1}$, and 0 (once), with eigenvector $\mathbf{1}$. By Example 1.16, the form equals $2n \sum_i (x_i - \bar{x})^2$.

Problem 8.4 ([8, Ch. 14, Exercise 40])**.** Show by examples that the leading principal minors of a symmetric matrix can all be $\geqslant 0$ although the matrix is not PSD: for a 2×2 matrix, and, for every $n \geqslant 3$, for a nonsingular $n \times n$ matrix.

Solution.

- $\left(\begin{smallmatrix} 0 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ has the leading minors 0, 0, but its diagonal entry -1 is negative, so it is not PSD (Theorem 8.30).
- Let

$$A_* = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

  Its leading minors are 0, $\det \left(\begin{smallmatrix} 0 & 0 \\ 0 & -1 \end{smallmatrix}\right) = 0$ and, along row 1, $\det A_* = 1 \cdot \det \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = 1$. So A_* is nonsingular, but not PSD, as $a_{22} = -1 < 0$.

- For $n \geqslant 4$ let $A = \operatorname{diag}(A_*, I_{n-3})$. Its first three leading minors are those of A_*, and for $k \geqslant 4$, $A_k = \operatorname{diag}(A_*, I_{k-3})$ has determinant $\det A_* = 1$ (Theorem 5.29). So A is nonsingular, with leading minors $\geqslant 0$, and not PSD, as $a_{22} = -1$.

Problem 8.5 ([6, Exercise 12.26(b)])**.** Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$. Use Hadamard's inequality (Theorem 8.32) to show that

$$(\det A)^2 \leqslant \prod_{i=1}^{n}\Big(\sum_{j=1}^{n} a_{ij}^2\Big)$$

Solution.

- If $\det A = 0$, the inequality holds, as the right side is a product of sums of squares, hence $\geqslant 0$.
- Let $\det A \neq 0$. Then $\operatorname{rank} A = n$ (Theorem 5.18(a)), so AA' is PD (Theorem 8.25), with $\det(AA') = \det A \cdot \det A' = (\det A)^2$ (Theorems 5.18(b) and 5.22).
- The ith diagonal entry of AA' is $\operatorname{row}_i(A)\operatorname{row}_i(A)' = \sum_j a_{ij}^2$ (Theorem 1.4(d)). By Theorem 8.32,

$$(\det A)^2 = \det(AA') \leqslant \prod_{i=1}^{n}(AA')_{ii} = \prod_{i=1}^{n}\Big(\sum_{j=1}^{n} a_{ij}^2\Big)$$

Problem 8.6 ([2, Exercise 5.19])**.** Let $V \in \mathbb{R}^{n\times n}$ be PD with $V = P'P$ for some $P \in \mathbb{R}^{n\times n}$, and let $A \in \mathbb{R}^{n\times n}$ be symmetric. Show that

(a) P is invertible;

(b) PAP' is idempotent if and only if AV is idempotent;

(c) if AV is idempotent, then $\operatorname{rank}(PAP') = \operatorname{tr}(AV)$.

Solution. (a) $\det V = \det P' \det P = (\det P)^2$ (Theorems 5.18(b) and 5.22), and $\det V \neq 0$ (Theorem 8.21(c)); so $\det P \neq 0$, and P is invertible (Theorem 5.18(a)).

(b)

- If AV is idempotent, then $(AP'P)(AP'P) = AP'P$; multiplying on the right by $(P'P)^{-1} = V^{-1}$ gives $AP'PA = A$, and then $(PAP')(PAP') = P(AP'PA)P' = PAP'$.
- If PAP' is idempotent, then $P(AP'PA)P' = PAP'$; multiplying on the left by P^{-1} and on the right by $(P')^{-1}$ gives $AP'PA = A$, that is, $AVA = A$, and multiplying on the right by V gives $AVAV = AV$.

(c) By (b), PAP' is idempotent, so $\operatorname{rank}(PAP') = \operatorname{tr}(PAP') = \operatorname{tr}(AP'P) = \operatorname{tr}(AV)$ (Theorems 6.29(c) and 1.11(b)).

Problem 8.7 ([5, Theorem 14.7.6, Corollary 14.7.7])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be PSD. Show that $\operatorname{tr}(AB) \geqslant 0$, with equality if and only if $AB = 0$.

Solution. If $B = 0$, then $\operatorname{tr}(AB) = 0$ and $AB = 0$. Let $B \neq 0$ and $r = \operatorname{rank} B \geqslant 1$.

- By Theorem 8.26(a), $B = Q'Q$ for some $Q \in \mathbb{R}^{r\times n}$ of rank r. Then $\operatorname{tr}(AB) = \operatorname{tr}(AQ'Q) = \operatorname{tr}(QAQ')$ (Theorem 1.11(b)).
- $M = QAQ' \in \mathbb{R}^{r\times r}$ is PSD (Theorem 8.24(a), with Q' in place of P), so its diagonal entries m_{ii} are $\geqslant 0$ (Theorem 8.30) and $\operatorname{tr} M = m_{11} + \cdots + m_{rr} \geqslant 0$. If $\operatorname{tr} M = 0$, then every $m_{ii} = 0$, so every row of M is zero (Theorem 8.30): $M = 0$. Hence $\operatorname{tr}(AB) \geqslant 0$, with equality if and only if $QAQ' = 0$.

- $QAQ' = 0$ if and only if $BAB = 0$. Indeed $BAB = Q'(QAQ')Q$. Conversely, Q has a right inverse K, $QK = I_r$ (Theorem 4.35(a)), so $K'Q' = I_r$ (Theorem 1.9), and if $BAB = 0$, then $QAQ' = K'Q'(QAQ')QK = K'(BAB)K = 0$.
- $BAB = 0$ if and only if $AB = 0$. If $AB = 0$, then $BAB = 0$. Conversely, let $BAB = 0$. By Theorem 8.26(b), $A = P'P$ for some matrix P with n columns, and, as $B' = B$, $0 = BAB = (PB)'(PB)$. So $\operatorname{tr}\Big((PB)'(PB)\Big) = 0$, and $PB = 0$ (Theorem 1.11(d)); hence $AB = P'(PB) = 0$.

So $\operatorname{tr}(AB) = 0$ if and only if $QAQ' = 0$, if and only if $BAB = 0$, if and only if $AB = 0$.

Problem 8.8 ([2, Exercise 9.5])**.** Let $A \in \mathbb{R}^{n\times n}$ be PSD. Show that $(x'Ay)^2 \leqslant (x'Ax)(y'Ay)$ for all $x, y \in \mathbb{R}^n$.

Solution. Let $z = A^{1/2}x$ and $w = A^{1/2}y$ (Theorem 8.46). As $A^{1/2}$ is symmetric and $A^{1/2}A^{1/2} = A$, $z'w = x'Ay$, $z'z = x'Ax$ and $w'w = y'Ay$. By the Cauchy–Schwarz inequality (Theorem 6.4(a)), $(x'Ay)^2 = (z'w)^2 \leqslant (z'z)(w'w) = (x'Ax)(y'Ay)$.

Problem 8.9 ([8, Ch. 14, Exercise 23])**.** Let $A \in \mathbb{R}^{n\times p}$ have rank p, and let $A = QR$ as in Theorem 6.16. Show that $A'A = R'R$, and that R is the Cholesky factor of $A'A$.

Solution. $Q'Q = I_p$, so $A'A = R'Q'QR = R'R$. The matrix $A'A$ is PD (Theorem 8.25), and R is upper triangular with positive diagonal entries (Theorem 6.16); by the uniqueness in Theorem 8.40, R is the Cholesky factor of $A'A$.

Problem 8.10 ([3, Ch. 11, Exercise 49])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be PSD, and let $A - B$ be PSD. Show that $\operatorname{rank} A \geqslant \operatorname{rank} B$.

Solution. It suffices to show $\mathcal{N}(A) \subseteq \mathcal{N}(B)$: then $\dim\mathcal{N}(A) \leqslant \dim\mathcal{N}(B)$ (Theorem 4.5(a)), and $\operatorname{rank} A = n - \dim\mathcal{N}(A) \geqslant n - \dim\mathcal{N}(B) = \operatorname{rank} B$ (Theorem 4.12).

- Let $x \in \mathcal{N}(A)$. Then $x'Ax = 0$, and $x'(A - B)x \geqslant 0$ gives $x'Bx \leqslant 0$; as B is PSD, $x'Bx = 0$.
- By Corollary 8.27, $Bx = 0$: $x \in \mathcal{N}(B)$.

Problem 8.11 ([6, Exercise 12.19(b)])**.** Show that for symmetric matrices A and B, $A \succcurlyeq B \succcurlyeq 0$ does not imply $A^2 \succcurlyeq B^2$.

Solution. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

- $A - B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) = \mathbf{1}_2\mathbf{1}_2'$ and $B = e_1e_1'$ are PSD (Theorem 8.25), so $A \succcurlyeq B \succcurlyeq 0$.
- $A^2 = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 2 \end{smallmatrix}\right)$ and $B^2 = B$, so $A^2 - B^2 = \left(\begin{smallmatrix} 4 & 3 \\ 3 & 2 \end{smallmatrix}\right)$, with $\det(A^2 - B^2) = 8 - 9 = -1 < 0$. A PSD matrix has a determinant $\geqslant 0$ (Theorem 8.21(c)), so $A^2 - B^2$ is not PSD: $A^2 \succcurlyeq B^2$ fails. For instance $x = (2, -3)'$ gives $x'(A^2 - B^2)x = 16 - 36 + 18 = -2$.

Problem 8.12 ([6, Exercise 12.18])**.** Let $A, B \in \mathbb{R}^{n\times k}$ with $A'B = I_k$. Show that $A'A \succcurlyeq (B'B)^{-1}$ and $B'B \succcurlyeq (A'A)^{-1}$.

Solution.

- $k = \operatorname{rank} I_k = \operatorname{rank}(A'B) \leqslant \operatorname{rank} B \leqslant k$ (Theorems 4.15(a) and 4.13(a)), so B has full column rank and $B'B$ is invertible (Theorem 6.26); likewise for A.
- $M = I_n - B(B'B)^{-1}B'$ is symmetric and idempotent (Theorem 6.26(b)), hence PSD (Proposition 8.23(d)), and $A'MA$ is PSD (Theorem 8.24(a)).

- $B'A = (A'B)' = I_k$, so $A'MA = A'A - A'B(B'B)^{-1}B'A = A'A - (B'B)^{-1}$: $A'A \succcurlyeq (B'B)^{-1}$.
- As $B'A = I_k$, the same argument with A and B exchanged, with $I_n - A(A'A)^{-1}A'$, gives $B'B \succcurlyeq (A'A)^{-1}$.

Problem 8.13 ([6, Exercise 8.58])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric and idempotent. Show that $A \succcurlyeq B$ if and only if $AB = B$.

Solution.

- Let $AB = B$. Then $BA = B'A' = (AB)' = B' = B$, and $(A-B)^2 = A^2 - AB - BA + B^2 = A - B - B + B = A - B$. So $A - B$ is symmetric and idempotent; its eigenvalues are 0 and 1 (Theorem 8.13(a)), so it is PSD (Theorem 8.21(a)): $A \succcurlyeq B$.
- Conversely, let $A \succcurlyeq B$, and let $C = (I_n - A)B$. Then $C'C \succcurlyeq 0$ (Theorem 8.25).
- $I_n - A$ is symmetric and idempotent, so $C'C = B(I_n - A)^2B = B(I_n - A)B = B^2 - BAB = -B(A-B)B$, as $B^2 = B = B^3$; and $B(A-B)B = B'(A-B)B$ is PSD (Theorem 8.24(a)). So $0 \succcurlyeq C'C$.
- Hence $C'C = 0$ (Proposition 8.50(a)), so $\operatorname{tr}(C'C) = 0$ and $C = 0$ (Theorem 1.11(d)): $B = AB$.

Problem 8.14 ([8, Ch. 21, Exercise 32])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric and $B \in \mathbb{R}^{n\times n}$ PD. Show that $A - B$ is PSD if and only if the n roots of $\det(A - \lambda B) = 0$, listed with multiplicity, are all $\geqslant 1$, and PD if and only if they are all > 1.

Solution.

- By Theorem 8.57(a),(b), applied to the PD matrix B and the symmetric matrix A, there is an invertible E with $E'BE = I_n$ and $E'AE = D = \operatorname{diag}(d_1, \dots, d_n)$, and $\det(A - \lambda B) = \det B \cdot (d_1 - \lambda) \cdots (d_n - \lambda)$: the roots are $d_1, \dots, d_n$.
- So $E'(A-B)E = D - I_n = \operatorname{diag}(d_1 - 1, \dots, d_n - 1)$.
- By Theorem 8.24(c), $A - B$ is PSD (PD) if and only if $D - I_n$ is, that is (Proposition 8.23(a)), if and only if every $d_i \geqslant 1$ ($d_i > 1$).

Problem 8.15 ([10, Problem 19.3])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, $A \neq 0$. Show that

$$\operatorname{rank} A \geqslant \frac{(\operatorname{tr} A)^2}{\operatorname{tr}(A^2)}$$

Solution. Let $A = Q\Lambda Q'$, with Q orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (Theorem 8.3).

- Let $S = \{i : \lambda_i \neq 0\}$, and let r be the number of its elements. Then $r = \operatorname{rank} A$ (Theorem 8.4), and $r \geqslant 1$, as $A \neq 0$.
- $\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n = \sum_{i\in S} \lambda_i$ (Theorem 7.19).
- $A^2 = Q\Lambda Q'Q\Lambda Q' = Q\Lambda^2 Q'$, so $\operatorname{tr}(A^2) = \operatorname{tr}(\Lambda^2 Q'Q) = \operatorname{tr}(\Lambda^2) = \sum_i \lambda_i^2 = \sum_{i\in S} \lambda_i^2$ (Theorem 1.11(b), Proposition 1.19(d)). Also $\operatorname{tr}(A^2) = \operatorname{tr}(A'A) > 0$, as $A \neq 0$ (Theorem 1.11(d)).
- Let $\mu \in \mathbb{R}^r$ have the entries λ_i, $i \in S$. By the Cauchy–Schwarz inequality (Theorem 6.4(a)) for $\mathbf{1}_r$ and μ,
$$(\operatorname{tr} A)^2 = (\mathbf{1}_r'\mu)^2 \leqslant (\mathbf{1}_r'\mathbf{1}_r)(\mu'\mu) = r\sum_{i\in S} \lambda_i^2 = \operatorname{rank} A \cdot \operatorname{tr}(A^2)$$

Dividing by $\operatorname{tr}(A^2) > 0$ gives the inequality.

Problem 8.16 ([10, Problem 33.5])**.** Let $A \in \mathbb{R}^{n\times n}$ be PD and $x \in \mathbb{R}^n$. Show that

$$\det \begin{pmatrix} 0 & x' \\ x & A \end{pmatrix} \leqslant 0$$

Solution. Let M be the $(n+1)\times(n+1)$ matrix in question, partitioned into the blocks $0 \in \mathbb{R}^{1\times 1}$, $x' \in \mathbb{R}^{1\times n}$, $x \in \mathbb{R}^{n\times 1}$ and $A \in \mathbb{R}^{n\times n}$.

- A is invertible and $\det A > 0$ (Theorem 8.21(c)). By Theorem 5.30(b), with A as the invertible lower right block,
$$\det M = \det A \cdot \det\bigl(0 - x'A^{-1}x\bigr) = -\det A \cdot x'A^{-1}x$$
as the determinant of a 1×1 matrix is its entry (Definition 5.13).
- A^{-1} is PD (Proposition 8.23(c)), so $x'A^{-1}x \geqslant 0$, with equality only for $x = 0$. Hence $\det M \leqslant 0$, with equality only for $x = 0$.

Problem 8.17 ([10, Problem 34.3])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be PSD. Show that $|\operatorname{tr}(AB)| \leqslant \operatorname{tr} A \cdot \operatorname{tr} B$.

Solution. Let $N = A^{1/2}$ and $K = B^{1/2}$ (Theorem 8.46); they are symmetric, with $N^2 = A$ and $K^2 = B$. Let $G = NK$.

- By Theorem 1.11(b), applied to the factors NNK and K, and by Theorem 1.9,
$$\operatorname{tr}(AB) = \operatorname{tr}(NNKK) = \operatorname{tr}(KNNK) = \operatorname{tr}\bigl((NK)'(NK)\bigr) = \operatorname{tr}(G'G) = \sum_{i=1}^{n}\sum_{j=1}^{n} g_{ij}^2$$
(Theorem 1.11(d)). In particular $\operatorname{tr}(AB) \geqslant 0$, so $|\operatorname{tr}(AB)| = \operatorname{tr}(AB)$.
- Let $u_i' = \operatorname{row}_i(N)$ and $v_j = \operatorname{col}_j(K)$. Then $g_{ij} = u_i'v_j$ (Theorem 1.4(d)), and $g_{ij}^2 \leqslant \|u_i\|^2\|v_j\|^2$ (Theorem 6.4(a)).
- $\sum_i \|u_i\|^2$ is the sum of the squares of all entries of N, which is $\operatorname{tr}(N'N) = \operatorname{tr}(N^2) = \operatorname{tr} A$; likewise $\sum_j \|v_j\|^2 = \operatorname{tr}(K'K) = \operatorname{tr}(K^2) = \operatorname{tr} B$ (Theorem 1.11(d)).
- Hence
$$\operatorname{tr}(AB) = \sum_{i,j} g_{ij}^2 \leqslant \sum_{i,j}\|u_i\|^2\|v_j\|^2 = \Bigl(\sum_i \|u_i\|^2\Bigr)\Bigl(\sum_j \|v_j\|^2\Bigr) = \operatorname{tr} A \cdot \operatorname{tr} B$$

Problem 8.18 ([10, Problem 33.1])**.** Let $A, B \in \mathbb{R}^{n\times n}$, $n \geqslant 2$, with A PD and B PSD. Show that $\det(A+B) \geqslant \det A + \det B$, with equality only for $B = 0$.

Solution.

- By Theorem 8.57(a), there is an invertible $E \in \mathbb{R}^{n\times n}$ with $E'AE = I_n$ and $E'BE = M = \operatorname{diag}(\mu_1, \dots, \mu_n)$. M is PSD (Theorem 8.24(a)), so every $\mu_i \geqslant 0$ (Proposition 8.23(a)).
- Let $\delta = (\det E)^2$, which is > 0 (Theorem 5.18(a)). As $E'(A+B)E = I_n + M = \operatorname{diag}(1+\mu_1, \dots, 1+\mu_n)$, Theorems 5.18(b), 5.22 and 5.14(c) give
$$\delta \det A = 1, \qquad \delta \det(A+B) = (1+\mu_1)\cdots(1+\mu_n), \qquad \delta \det B = \mu_1 \cdots \mu_n$$
- For $k \geqslant 2$ and numbers $\mu_1, \dots, \mu_k \geqslant 0$,
$$(1+\mu_1)\cdots(1+\mu_k) \geqslant 1 + (\mu_1 + \cdots + \mu_k) + \mu_1 \cdots \mu_k$$
by induction on k. For $k = 2$ both sides are $1+\mu_1+\mu_2+\mu_1\mu_2$. If it holds for k, let $s = \mu_1 + \cdots + \mu_k$ and $\pi = \mu_1 \cdots \mu_k$; multiplying by $1 + \mu_{k+1} > 0$,
$$(1+\mu_1)\cdots(1+\mu_{k+1}) \geqslant (1+s+\pi)(1+\mu_{k+1}) = 1 + (s+\mu_{k+1}) + \pi\mu_{k+1} + \pi + s\mu_{k+1}$$
which is $\geqslant 1 + (s+\mu_{k+1}) + \pi\mu_{k+1}$, as $\pi \geqslant 0$ and $s\mu_{k+1} \geqslant 0$.

- With $k = n \geqslant 2$ and the three equations above,

$$\delta \det(A + B) \geqslant 1 + (\mu_1 + \cdots + \mu_n) + \delta \det B \geqslant \delta \det A + \delta \det B$$

  as $\delta \det A = 1$ and every $\mu_i \geqslant 0$; dividing by $\delta > 0$ gives $\det(A + B) \geqslant \det A + \det B$.

- If equality holds, then $1+\delta \det B = \delta \det(A+B) \geqslant 1+(\mu_1+\cdots+\mu_n)+\delta \det B$, so $\mu_1+\cdots+\mu_n \leqslant 0$; as every $\mu_i \geqslant 0$, every $\mu_i = 0$. So $M = 0$, and $B = (E')^{-1}ME^{-1} = 0$ (Theorem 2.13(c)). Conversely, if $B = 0$, then $\det(A + B) = \det A = \det A + \det B$, as $\det 0 = 0$ (Theorem 5.14(b)).

Chapter 9. Covariance matrices and principal components

Random vectors and covariance matrices

A **random variable** is a number whose value depends on chance; every random variable in this book is assumed to have a finite variance. We use the following facts from probability without proof. For random variables u, v and numbers a, b, c, the **expectation** satisfies $\mathsf{E}(au+bv+c) = a\,\mathsf{E}(u)+b\,\mathsf{E}(v)+c$; so the expectation of a sum of finitely many terms is the sum of their expectations. The **variance** $\operatorname{var}(u) = \mathsf{E}\Big((u-\mathsf{E}(u))^2\Big)$ is $\geqslant 0$, with $\operatorname{var}(u) = 0$ if and only if $u = \mathsf{E}(u)$ with probability 1; and the **covariance** is $\operatorname{cov}(u,v) = \mathsf{E}\Big((u-\mathsf{E}(u))(v-\mathsf{E}(v))\Big) = \operatorname{cov}(v,u)$; thus $\operatorname{var}(u) = \operatorname{cov}(u,u)$ [2, §10.1], [12, Lemma 3.2].

Definition 9.1 ([3, §7.1], [2, Ch. 10, introduction, §10.1], [12, Lemma 3.2])**.** A **random vector** $z = (z_1, \dots, z_n)'$ and a **random matrix** $Z = (z_{ij})$ have random variables as entries, and their expectations are taken entry by entry: $\mathsf{E}(Z) = (\mathsf{E}(z_{ij}))$. In particular $\mu = \mathsf{E}(z) = (\mathsf{E}(z_1), \dots, \mathsf{E}(z_n))'$ is the **mean vector** of z, and

$$\Sigma = \operatorname{var}(z) = \mathsf{E}\Big((z-\mu)(z-\mu)'\Big) \in \mathbb{R}^{n\times n}$$

is its **covariance matrix**. The (i,j) entry of $(z-\mu)(z-\mu)'$ is $(z_i-\mu_i)(z_j-\mu_j)$, so the (i,j) entry of Σ is $\sigma_{ij} = \operatorname{cov}(z_i, z_j)$: the diagonal entries are the variances $\sigma_{ii} = \operatorname{var}(z_i)$, and Σ is symmetric.

Definition 9.2 ([12, Lemma 3.2], [2, §10.1], [17, Definition 4.5.1])**.** Let $u \in \mathbb{R}^p$ and $v \in \mathbb{R}^q$ be random vectors with means $\mu_u = \mathsf{E}(u)$ and $\mu_v = \mathsf{E}(v)$ (Definition 9.1). The **covariance matrix of u and v** is

$$\operatorname{cov}(u,v) = \mathsf{E}\Big((u-\mu_u)(v-\mu_v)'\Big) \in \mathbb{R}^{p\times q}$$

Its (i,j) entry is $\mathsf{E}\Big((u_i-\mathsf{E}(u_i))(v_j-\mathsf{E}(v_j))\Big) = \operatorname{cov}(u_i, v_j)$. So $\operatorname{cov}(u,u) = \operatorname{var}(u)$, and $\operatorname{cov}(v,u) = \operatorname{cov}(u,v)'$, since $(v-\mu_v)(u-\mu_u)' = \Big((u-\mu_u)(v-\mu_v)'\Big)'$ (Theorem 1.9). The vectors u and v are **uncorrelated** if $\operatorname{cov}(u,v) = 0$.

For the stacked vector $z = (u', v')' \in \mathbb{R}^{p+q}$, whose mean is $(\mu_u', \mu_v')'$, the block product (Theorem 5.3(b),(c)) gives

$$(z-\mathsf{E}(z))(z-\mathsf{E}(z))' = \begin{pmatrix} u-\mu_u \\ v-\mu_v \end{pmatrix}\Big((u-\mu_u)' \;\; (v-\mu_v)'\Big) = \begin{pmatrix} (u-\mu_u)(u-\mu_u)' & (u-\mu_u)(v-\mu_v)' \\ (v-\mu_v)(u-\mu_u)' & (v-\mu_v)(v-\mu_v)' \end{pmatrix}$$

and, as expectations are taken entry by entry,

$$\operatorname{var}(z) = \begin{pmatrix} \operatorname{var}(u) & \operatorname{cov}(u,v) \\ \operatorname{cov}(v,u) & \operatorname{var}(v) \end{pmatrix}$$

Theorem 9.3 ([3, §7.1], [12, Lemma 3.2], [2, §10.1])**.** Let $z \in \mathbb{R}^n$ be a random vector with $\mathsf{E}(z) = \mu$ and $\operatorname{var}(z) = \Sigma$.

(a) For a random matrix $Z \in \mathbb{R}^{m\times p}$ and constant matrices $A \in \mathbb{R}^{q\times m}$, $B \in \mathbb{R}^{p\times l}$ and $C \in \mathbb{R}^{q\times l}$, $\mathsf{E}(AZB + C) = A\,\mathsf{E}(Z)B + C$.

(b) For a constant matrix $A \in \mathbb{R}^{m\times n}$ and a constant vector $b \in \mathbb{R}^m$,

$$\mathsf{E}(Az+b) = A\mu + b, \qquad \operatorname{var}(Az+b) = A\Sigma A'$$

(c) Σ is positive semidefinite: $a'\Sigma a = \operatorname{var}(a'z) \geqslant 0$ for every $a \in \mathbb{R}^n$.

Proof. (a) By Definition 1.2, applied twice, the (i,j) entry of $AZB+C$ is $\sum_{r=1}^{m}\sum_{s=1}^{p} a_{ir}z_{rs}b_{sj}+c_{ij}$, a sum of the random variables z_{rs} times numbers, plus a number. Its expectation is $\sum_r\sum_s a_{ir}\,\mathsf{E}(z_{rs})b_{sj} + c_{ij}$, the (i,j) entry of $A\,\mathsf{E}(Z)B + C$.

(b)

- (a) with $Z = z$, $B = 1$ and $C = b$ gives $\mathsf{E}(Az + b) = A\mu + b$.
- Hence $Az+b-\mathsf{E}(Az+b) = A(z-\mu)$, whose transpose is $(z-\mu)'A'$ (Theorem 1.9). By Definition 9.1 and (a), $\operatorname{var}(Az + b) = \mathsf{E}\big(A(z - \mu)(z - \mu)'A'\big) = A\,\mathsf{E}\big((z - \mu)(z - \mu)'\big)A' = A\Sigma A'$.

(c) Σ is symmetric (Definition 9.1). For $a \in \mathbb{R}^n$, (b) with the $1 \times n$ matrix a' and $b = 0$ gives $\operatorname{var}(a'z) = a'\Sigma a$, and a variance is $\geqslant 0$. So Σ is positive semidefinite (Definition 8.20). □

Theorem 9.4 ([12, Lemma 3.2], [17, Theorem 4.5.6, Lemma 5.2.5, Theorem 4.5.3])**.** Let $u \in \mathbb{R}^p$ and $v \in \mathbb{R}^q$ be random vectors.

(a) For constant $A \in \mathbb{R}^{k\times p}$, $b \in \mathbb{R}^k$, $C \in \mathbb{R}^{l\times q}$ and $d \in \mathbb{R}^l$, $\operatorname{cov}(Au + b, Cv + d) = A\operatorname{cov}(u, v)C'$. In particular, if $z \in \mathbb{R}^n$ is a random vector with $\operatorname{var}(z) = \Sigma$, and $A \in \mathbb{R}^{k\times n}$ and $B \in \mathbb{R}^{l\times n}$ are constant, then $\operatorname{cov}(Az, Bz) = A\Sigma B'$.

(b) If $u_1, u_2 \in \mathbb{R}^p$ are random vectors, then $\operatorname{cov}(u_1 + u_2, v) = \operatorname{cov}(u_1, v) + \operatorname{cov}(u_2, v)$, and likewise $\operatorname{cov}(v, u_1 + u_2) = \operatorname{cov}(v, u_1) + \operatorname{cov}(v, u_2)$.

(c) If $q = p$, then $\operatorname{var}(u + v) = \operatorname{var}(u) + \operatorname{cov}(u, v) + \operatorname{cov}(v, u) + \operatorname{var}(v)$.

(d) If $x_1, \dots, x_m \in \mathbb{R}^p$ are random vectors with $\operatorname{cov}(x_i, x_j) = 0$ for $i \neq j$, then $\operatorname{var}(x_1 + \cdots + x_m) = \operatorname{var}(x_1) + \cdots + \operatorname{var}(x_m)$.

(e) $\operatorname{var}(u) = \mathsf{E}(uu') - \mu_u\mu_u'$, that is, $\mathsf{E}(uu') = \operatorname{var}(u) + \mu_u\mu_u'$, where $\mu_u = \mathsf{E}(u)$.

Proof. (a)

- By Theorem 9.3(b), $Au + b - \mathsf{E}(Au + b) = A(u - \mu_u)$ and $Cv + d - \mathsf{E}(Cv + d) = C(v - \mu_v)$, whose transpose is $(v - \mu_v)'C'$ (Theorem 1.9).
- By Theorem 9.3(a), $\operatorname{cov}(Au+b, Cv+d) = \mathsf{E}\big(A(u-\mu_u)(v-\mu_v)'C'\big) = A\operatorname{cov}(u, v)C'$. For $u = v = z$ this reads $\operatorname{cov}(Az, Bz) = A\operatorname{var}(z)B'$.

(b) The mean of $u_1 + u_2$ is $\mathsf{E}(u_1) + \mathsf{E}(u_2)$, so $(u_1 + u_2 - \mathsf{E}(u_1 + u_2))(v - \mu_v)' = (u_1 - \mathsf{E}(u_1))(v - \mu_v)' + (u_2 - \mathsf{E}(u_2))(v - \mu_v)'$; take expectations entry by entry. The second identity follows by transposing (Definition 9.2).

(c) By (b), applied in both arguments, $\operatorname{var}(u + v) = \operatorname{cov}(u + v, u + v) = \operatorname{cov}(u, u) + \operatorname{cov}(u, v) + \operatorname{cov}(v, u) + \operatorname{cov}(v, v)$.

(d) In the same way, $\operatorname{var}\big(\sum_i x_i\big) = \operatorname{cov}\big(\sum_i x_i, \sum_j x_j\big) = \sum_i \sum_j \operatorname{cov}(x_i, x_j)$, and only the m terms with $i = j$ remain.

(e) $(u - \mu_u)(u - \mu_u)' = uu' - u\mu_u' - \mu_u u' + \mu_u\mu_u'$ (Theorems 1.3 and 1.9); taking expectations entry by entry, with $\mathsf{E}(u\mu_u') = \mu_u\mu_u' = \mathsf{E}(\mu_u u')$ (Theorem 9.3(a)), $\operatorname{var}(u) = \mathsf{E}(uu') - \mu_u\mu_u' - \mu_u\mu_u' + \mu_u\mu_u' = \mathsf{E}(uu') - \mu_u\mu_u'$. □

Theorem 9.5 ([3, §7.1])**.** Let $x \in \mathbb{R}^n$ be a random vector with $\mathsf{E}(x) = \mu$ and $\operatorname{var}(x) = \Sigma$. Then Σ is singular (not invertible) if and only if $a'x = a'\mu$ with probability 1 for some nonzero $a \in \mathbb{R}^n$.

Proof.

- Σ is singular if and only if $\Sigma a = 0$ for some $a \neq 0$ (Theorem 2.16).
- Σ is PSD (Theorem 9.3(c)), so $\Sigma a = 0$ if and only if $a'\Sigma a = 0$ (Corollary 8.27).
- By Theorem 9.3(b), $a'\Sigma a = \operatorname{var}(a'x)$ and $\mathsf{E}(a'x) = a'\mu$; and a random variable has variance 0 if and only if it equals its expectation with probability 1. □

Example 9.6. Let u_1, u_2, u_3 be pairwise uncorrelated random variables with variance 1, and $x = (u_1,\ u_2,\ u_3,\ u_1+u_2+u_3)'$. Then $x = Bu$ with $u = (u_1, u_2, u_3)'$, $\operatorname{var}(u) = I_3$ (Definition 9.1) and

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \operatorname{var}(x) = BB' = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{pmatrix}$$

by Theorem 9.3(b). This matrix is singular: $\operatorname{var}(x)a = 0$ for $a = (1, 1, 1, -1)'$. Correspondingly $a'x = u_1 + u_2 + u_3 - (u_1 + u_2 + u_3) = 0$, as Theorem 9.5 says. Likewise, the shares of a budget spent on p items add up to 1, so their covariance matrix is singular.

Definition 9.7 ([17, Definitions 2.3.2, 4.5.2], [2, §10.2])**.** The **correlation** of random variables u and v with positive variances is $\rho_{uv} = \operatorname{cov}(u, v)/\sqrt{\operatorname{var}(u)\operatorname{var}(v)}$. Let $z \in \mathbb{R}^n$ be a random vector with $\mathsf{E}(z) = \mu$, $\operatorname{var}(z) = \Sigma = (\sigma_{ij})$ and $\sigma_{ii} > 0$ for every i; $\sqrt{\sigma_{ii}}$ is the **standard deviation** of z_i. The correlation of z_i and z_j is $\rho_{ij} = \sigma_{ij}/\sqrt{\sigma_{ii}\sigma_{jj}}$, and with $D^{-1/2} = \operatorname{diag}\left(\sigma_{11}^{-1/2}, \ldots, \sigma_{nn}^{-1/2}\right)$, the **correlation matrix** of z is

$$\operatorname{cor}(z) = (\rho_{ij}) = D^{-1/2}\Sigma D^{-1/2} \in \mathbb{R}^{n\times n}$$

by Definition 1.2: the (i, j) entry of $D^{-1/2}\Sigma D^{-1/2}$ is $\sigma_{ii}^{-1/2}\sigma_{ij}\sigma_{jj}^{-1/2}$.

Proposition 9.8 ([17, Theorem 4.5.7(a)], [2, §10.2], [18, §12.1])**.** In Definition 9.7:

(a) $-1 \leqslant \rho_{ij} \leqslant 1$ for all i, j;

(b) $\operatorname{cor}(z) = \operatorname{var}\left(D^{-1/2}(z-\mu)\right)$ is the covariance matrix of the **standardized variables** $(z_i-\mu_i)/\sqrt{\sigma_{ii}}$ $(i = 1, \ldots, n)$, which have mean 0 and variance 1; so $\operatorname{cor}(z)$ is symmetric and PSD, with diagonal entries 1.

Proof. (a)

- For $t \in \mathbb{R}$, Theorem 9.3(b) for $(z_i, z_j)'$, whose covariance matrix has the entries $\sigma_{ii}, \sigma_{ij}, \sigma_{ij}, \sigma_{jj}$, and the 1×2 matrix $(t, 1)$ gives $h(t) = \operatorname{var}(tz_i + z_j) = \sigma_{ii}t^2 + 2\sigma_{ij}t + \sigma_{jj} \geqslant 0$.
- A quadratic polynomial with leading coefficient $\sigma_{ii} > 0$ and two different real roots $t_1 < t_2$ equals $\sigma_{ii}(t - t_1)(t - t_2) < 0$ for $t_1 < t < t_2$. So h has no two different real roots, and its discriminant is $(2\sigma_{ij})^2 - 4\sigma_{ii}\sigma_{jj} \leqslant 0$.
- Hence $\sigma_{ij}^2 \leqslant \sigma_{ii}\sigma_{jj}$, that is, $\rho_{ij}^2 \leqslant 1$.

(b) $D^{-1/2}$ is diagonal, hence symmetric. By Theorem 9.3(b) with $A = D^{-1/2}$ and $b = -D^{-1/2}\mu$, $D^{-1/2}(z - \mu)$ has the mean $D^{-1/2}\mu - D^{-1/2}\mu = 0$ and $\operatorname{var}\left(D^{-1/2}(z - \mu)\right) = D^{-1/2}\Sigma D^{-1/2} = \operatorname{cor}(z)$. This is symmetric and PSD (Theorem 9.3(c)), and its (i, i) entry, the variance of $(z_i - \mu_i)/\sqrt{\sigma_{ii}}$, is $\sigma_{ii}/\sigma_{ii} = 1$. □

Example 9.9. Let $z \in \mathbb{R}^4$ be a random vector with

$$\operatorname{var}(z) = \Sigma = \begin{pmatrix} 4 & 0 & 4 & 0 \\ 0 & 1 & 2 & 2 \\ 4 & 2 & 9 & 6 \\ 0 & 2 & 6 & 9 \end{pmatrix}$$

- The standard deviations are $\sqrt{\sigma_{11}} = 2$, $\sqrt{\sigma_{22}} = 1$, $\sqrt{\sigma_{33}} = 3$ and $\sqrt{\sigma_{44}} = 3$, so $\rho_{12} = \rho_{14} = 0$, $\rho_{13} = \frac{4}{2\cdot 3} = \frac{2}{3}$, $\rho_{23} = \frac{2}{1\cdot 3} = \frac{2}{3}$, $\rho_{24} = \frac{2}{1\cdot 3} = \frac{2}{3}$, $\rho_{34} = \frac{6}{3\cdot 3} = \frac{2}{3}$, and

$$\operatorname{cor}(z) = \begin{pmatrix} 1 & 0 & \frac{2}{3} & 0 \\ 0 & 1 & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & 1 & \frac{2}{3} \\ 0 & \frac{2}{3} & \frac{2}{3} & 1 \end{pmatrix}$$

So z_1 is uncorrelated with z_2 and with z_4, and the other four pairs have the correlation $\frac{2}{3}$.

- Let $e = z_3 - z_1 - 2z_2 = a'z$ with $a = (-1, -2, 1, 0)'$. Then $\Sigma a = (-4+0+4+0,\ 0-2+2+0,\ -4-4+9+0,\ 0-4+6+0)' = (0, 0, 1, 2)'$, so by Theorem 9.4(a), $\operatorname{cov}(z, e) = \Sigma a = (0, 0, 1, 2)'$: e is uncorrelated with z_1 and with z_2, and $\operatorname{var}(e) = a'\Sigma a = 0+0+1+0 = 1$. Likewise $f = z_4 + 2z_1 + 2z_2 - 2z_3 = b'z$ with $b = (2, 2, -2, 1)'$ has $\Sigma b = (8+0-8+0,\ 0+2-4+2,\ 8+4-18+6,\ 0+4-12+9)' = (0, 0, 0, 1)'$: f is uncorrelated with z_1, z_2 and z_3, and $\operatorname{var}(f) = b'\Sigma b = 1$.
- By Theorem 9.4(a), $z_1 + z_2 + z_3 + z_4 = \mathbf{1}'z$ has the variance $\mathbf{1}'\Sigma\mathbf{1} = 51$, the sum of all entries of Σ. Likewise, as $(1, 1, 0, 0)\Sigma = (4, 1, 6, 2)$ and $(1, -1, 0, 0)\Sigma = (4, -1, 2, -2)$, the variables $z_1 + z_2$ and $z_1 - z_2$ have the variances $4 + 1 = 5$ and $4 + 1 = 5$, and the covariance $(4, 1, 6, 2)(1, -1, 0, 0)' = 4 - 1 = 3$.

Theorem 9.10 ([12, Lemma 3.6], [2, (10.34)])**.** Let $z \in \mathbb{R}^n$ be a random vector with $\mathsf{E}(z) = \mu$ and $\operatorname{var}(z) = \Sigma$, and let $A \in \mathbb{R}^{n\times n}$ be constant. Then

$$\mathsf{E}(z'Az) = \operatorname{tr}(A\Sigma) + \mu'A\mu$$

In particular $\mathsf{E}(z'Az) = \operatorname{tr}(A\Sigma)$ if $\mu = 0$.

Proof. • $z'Az = \sum_{i=1}^n \sum_{j=1}^n a_{ij} z_i z_j$ (Definition 8.17), so $\mathsf{E}(z'Az) = \sum_i \sum_j a_{ij}\, \mathsf{E}(z_i z_j)$.

- Expanding the product, $\sigma_{ij} = \mathsf{E}\big((z_i - \mu_i)(z_j - \mu_j)\big) = \mathsf{E}(z_i z_j) - \mu_i\mu_j - \mu_j\mu_i + \mu_i\mu_j$; so $\mathsf{E}(z_i z_j) = \sigma_{ij} + \mu_i\mu_j$, and $\mathsf{E}(z'Az) = \sum_i \sum_j a_{ij}\sigma_{ij} + \sum_i \sum_j a_{ij}\mu_i\mu_j$.
- $\operatorname{tr}(A\Sigma) = \sum_i (A\Sigma)_{ii} = \sum_i \sum_j a_{ij}\sigma_{ji} = \sum_i \sum_j a_{ij}\sigma_{ij}$, as Σ is symmetric; and $\mu'A\mu = \sum_i \sum_j a_{ij}\mu_i\mu_j$ (Definition 8.17). □

Definition 9.11 ([17, Definitions 7.1.1, 7.3.2])**.** Let observations be modeled as a random vector y whose distribution depends on unknown parameters. An **estimator** of a parameter θ is a random variable $T = g(y)$, where the function g does not involve unknown parameters; its value at the observed y is an **estimate** of θ. The estimator T is **unbiased** if $\mathsf{E}(T) = \theta$ for all values of the parameters.

Example 9.12 (cf. [17, Theorem 5.2.6(c)])**.** Let $y \in \mathbb{R}^n$ have $\mathsf{E}(y) = \mu$ and $\operatorname{var}(y) = \sigma^2 I_n$, and let $P \in \mathbb{R}^{n\times n}$ be symmetric and idempotent of rank r. Then $\operatorname{tr} P = \operatorname{rank} P = r$ (Theorem 6.29(c)), and Theorem 9.10, with $\operatorname{tr}(P\sigma^2 I_n) = \sigma^2 \operatorname{tr} P$ (Theorem 1.11(a)), gives

$$\mathsf{E}(y'Py) = \operatorname{tr}(P\sigma^2 I_n) + \mu'P\mu = \sigma^2 r + \mu'P\mu$$

- Let $n \geqslant 2$ and $P = C_n$, which is symmetric and idempotent of rank $n - 1$ (Proposition 1.15, Corollary 6.35(b)). By Example 1.16, $y'C_n y = \sum_i (y_i - \bar{y})^2$ and $\mu'C_n\mu = \sum_i (\mu_i - \bar{\mu})^2$, where $\bar{\mu} = \frac{1}{n}\mathbf{1}'\mu$. So
$$\mathsf{E}\Big(\sum_{i=1}^n (y_i - \bar{y})^2\Big) = (n-1)\sigma^2 + \sum_{i=1}^n (\mu_i - \bar{\mu})^2$$
- If the observations have a common mean, $\mu = \mu_0\mathbf{1}$, then $C_n\mu = 0$ and the last sum is 0. So $s^2 = \frac{1}{n-1}\sum_i (y_i - \bar{y})^2$ is an unbiased estimator of σ^2 (Definition 9.11): $\mathsf{E}(s^2) = \sigma^2$ for all values of μ_0 and σ^2. The estimator $\frac{1}{n}\sum_i (y_i - \bar{y})^2$ has the expectation $\frac{n-1}{n}\sigma^2$, which is smaller than σ^2 when $\sigma^2 > 0$; so it is not unbiased.
- If the means drift, s^2 overestimates σ^2 on average. For $n = 4$ and $\mu = (1, 3, 5, 7)'$, $\bar{\mu} = 4$ and $\sum_i (\mu_i - \bar{\mu})^2 = 9 + 1 + 1 + 9 = 20$, so $\mathsf{E}(s^2) = \sigma^2 + \frac{20}{3}$.

Data matrices and the sample covariance matrix

Definition 9.13 ([2, §10.3])**.** Let $X = (x_{ij}) \in \mathbb{R}^{n\times p}$ ($n \geqslant 2$) be a **data matrix**: x_{ij} is the value of variable j on unit i, and row i of X is $x_i' = (x_{i1}, \ldots, x_{ip})$, the ith observation, a point of $\mathbb{R}^p$ (as for X_0 in Example 1.17). With $\mathbf{1} = \mathbf{1}_n$ and $C_n = I_n - \bar{J}_n$ (Definition 1.14):

- the **mean vector** is $\bar{x} = \frac{1}{n}X'\mathbf{1} = (\bar{x}_1, \ldots, \bar{x}_p)'$, where $\bar{x}_j = \frac{1}{n}\sum_i x_{ij}$;
- the **centered data matrix** is $C_nX = X - \mathbf{1}\bar{x}'$, with entries $x_{ij} - \bar{x}_j$ (Example 1.17);
- the **sample covariance matrix** is $S = (s_{jl}) = \frac{1}{n-1}X'C_nX \in \mathbb{R}^{p\times p}$.

For data vectors $y, w \in \mathbb{R}^n$ with means $\bar{y}$, $\bar{w}$, the number $\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})(w_i - \bar{w})$ is their **sample covariance**, and y, w are **uncorrelated** if it is 0; for $w = y$ it is the **sample variance** $\frac{1}{n-1}\sum_i (y_i - \bar{y})^2$ of y. For $a \in \mathbb{R}^p$, the vector $Xa = (a'x_1, \ldots, a'x_n)'$ holds the values on the n units of the linear combination of the variables with coefficient vector a.

Proposition 9.14 ([2, §10.3], [18, §3.3, (3.22)–(3.24)], [9, Definition 5.19], [3, §10.1])**.** Let $X \in \mathbb{R}^{n\times p}$ be a data matrix, with $\bar{x}$, C_n and S as in Definition 9.13.

(a) $X'C_nX = (C_nX)'(C_nX) = X'X - n\bar{x}\bar{x}'$, and $S = \frac{1}{n-1}\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})'$. Hence s_{jj} is the sample variance of variable j and s_{jl} the sample covariance of variables j and l; S is symmetric and PSD; $S = 0$ only if all rows of X are equal; and S is PD if and only if $\operatorname{rank}(C_nX) = p$, which requires $n \geqslant p + 1$.

(b) For $a, b \in \mathbb{R}^p$, Xa has mean $a'\bar{x}$ and centered values $C_n(Xa) = (C_nX)a$; the sample covariance of Xa and Xb is $a'Sb$, and the sample variance of Xa is $a'Sa$.

(c) Let $A \in \mathbb{R}^{q\times p}$, $b \in \mathbb{R}^q$ and $Y = XA' + \mathbf{1}b'$, the data matrix with the rows $y_i' = (Ax_i + b)'$. Then Y has the mean vector $\bar{y} = A\bar{x} + b$, the centered data matrix $C_nY = C_nXA'$ and the sample covariance matrix $S_Y = ASA'$.

Proof. (a)

- $C_n' = C_n = C_n^2$ (Proposition 1.15), so $X'C_nX = X'C_n'C_nX = (C_nX)'(C_nX)$; and, as $X'\mathbf{1} = n\bar{x}$, $X'C_nX = X'X - \frac{1}{n}(X'\mathbf{1})(\mathbf{1}'X) = X'X - n\bar{x}\bar{x}'$.
- The (j, l) entry of $(C_nX)'(C_nX)$ is $\sum_i (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l)$ (Theorem 1.4(d)); dividing by $n - 1$ gives the sample covariance.
- Row i of $C_nX = X - \mathbf{1}\bar{x}'$ is $(x_i - \bar{x})'$, so by the outer product expansion (Theorem 1.4(d)), $(C_nX)'(C_nX) = \sum_i (x_i - \bar{x})(x_i - \bar{x})'$.
- $(C_nX)'(C_nX)$ is PSD (Theorem 8.25), hence so is S. If $S = 0$, then every $s_{jj} = \frac{1}{n-1}\sum_i (x_{ij} - \bar{x}_j)^2 = 0$, so $x_{ij} = \bar{x}_j$ for all i, j.
- By Theorem 8.25, $(C_nX)'(C_nX)$, and hence S, is PD if and only if $\operatorname{rank}(C_nX) = p$. As $\operatorname{rank}(C_nX) \leqslant \operatorname{rank} C_n = n - 1$ (Theorem 4.15(a), Corollary 6.35(b)), this requires $p \leqslant n - 1$.

(b)

- $\frac{1}{n}\mathbf{1}'(Xa) = \bar{x}'a = a'\bar{x}$, and $C_n(Xa) = (C_nX)a$ (Theorem 1.3).
- By Example 1.16, $C_n(Xa)$ and $C_n(Xb)$ hold the deviations from the means, so the sample covariance is $\frac{1}{n-1}(C_nXa)'(C_nXb) = a'\Big[\frac{1}{n-1}(C_nX)'(C_nX)\Big]b = a'Sb$, by (a). Take $b = a$ for the variance.

(c) Row i of XA' is $x_i'A' = (Ax_i)'$ (Theorems 1.4(c) and 1.9), and row i of $\mathbf{1}b'$ is b'.

- $Y' = AX' + b\mathbf{1}'$, so $\bar{y} = \frac{1}{n}Y'\mathbf{1} = A\Big(\frac{1}{n}X'\mathbf{1}\Big) + b\,\frac{1}{n}\mathbf{1}'\mathbf{1} = A\bar{x} + b$ ($\mathbf{1}'\mathbf{1} = n$, Proposition 1.15).
- $C_n\mathbf{1} = 0$ (Proposition 1.15), so $C_nY = C_nXA' + (C_n\mathbf{1})b' = C_nXA'$, and by (a) $S_Y = \frac{1}{n-1}(C_nY)'(C_nY) = A\Big[\frac{1}{n-1}(C_nX)'(C_nX)\Big]A' = ASA'$. □

Example 9.15. Five units ($n = 5$) are measured on four variables ($p = 4$), for instance five students on four tests:

$$X = \begin{pmatrix} 7 & 3 & 7 & 3 \\ 9 & 9 & 5 & 7 \\ 5 & 7 & 9 & 9 \\ 3 & 5 & 1 & 1 \\ 1 & 1 & 3 & 5 \end{pmatrix}, \qquad \mathbf{1}'X = (25,\ 25,\ 25,\ 25), \qquad \bar{x} = \tfrac{1}{5}X'\mathbf{1} = \begin{pmatrix} 5 \\ 5 \\ 5 \\ 5 \end{pmatrix}$$

Subtracting $\bar{x}' = (5, 5, 5, 5)$ from every row and using Proposition 9.14(a),

$$C_5X = \begin{pmatrix} 2 & -2 & 2 & -2 \\ 4 & 4 & 0 & 2 \\ 0 & 2 & 4 & 4 \\ -2 & 0 & -4 & -4 \\ -4 & -4 & -2 & 0 \end{pmatrix}, \qquad (C_5X)'(C_5X) = \begin{pmatrix} 40 & 28 & 20 & 12 \\ 28 & 40 & 12 & 20 \\ 20 & 12 & 40 & 28 \\ 12 & 20 & 28 & 40 \end{pmatrix}$$

and

$$S = \frac{1}{4}(C_5X)'(C_5X) = \begin{pmatrix} 10 & 7 & 5 & 3 \\ 7 & 10 & 3 & 5 \\ 5 & 3 & 10 & 7 \\ 3 & 5 & 7 & 10 \end{pmatrix}$$

For instance $40 = 4{+}16{+}0{+}4{+}16$ is the sum of squares of column 1 of C_5X, and $28 = -4{+}16{+}0{+}0{+}16$ is the inner product of columns 1 and 2. So every variable has the sample variance 10; the pairs of variables $\{1, 2\}$ and $\{3, 4\}$ have the sample covariance 7, the pairs $\{1, 3\}$ and $\{2, 4\}$ have 5, and the pairs $\{1, 4\}$ and $\{2, 3\}$ have 3.

Theorem 9.16 ([17, Theorem 5.2.6, Lemma 5.2.5], [2, §10.3])**.** Let $x_1, \dots, x_n \in \mathbb{R}^p$ ($n \geqslant 2$) be random vectors with $\mathsf{E}(x_i) = \mu$ and $\operatorname{var}(x_i) = \Sigma$ for every i, and $\operatorname{cov}(x_i, x_j) = 0$ for $i \neq j$. Let $X \in \mathbb{R}^{n\times p}$ be the random data matrix with rows $x_1', \dots, x_n'$, with mean vector $\bar{x} = \frac{1}{n}X'\mathbf{1} = \frac{1}{n}(x_1 + \cdots + x_n)$ and sample covariance matrix $S = \frac{1}{n-1}X'C_nX$ (Definition 9.13). Then

(a) $\mathsf{E}(\bar{x}) = \mu$ and $\operatorname{var}(\bar{x}) = \frac{1}{n}\Sigma$;

(b) $\mathsf{E}(S) = \Sigma$: each sample variance s_{jj} and each sample covariance s_{jl} has the expectation σ_{jj}, respectively σ_{jl}.

Proof. (a)

- The columns of X' are $x_1, \dots, x_n$, so $X'\mathbf{1} = x_1 + \cdots + x_n$ (Theorem 1.4(b)), and $\mathsf{E}(\bar{x}) = \frac{1}{n}(\mu + \cdots + \mu) = \mu$.
- By Theorem 9.3(b) with $A = \frac{1}{n}I_p$ and Theorem 9.4(d), $\operatorname{var}(\bar{x}) = \frac{1}{n^2}\operatorname{var}(x_1{+}\cdots{+}x_n) = \frac{1}{n^2}\,n\Sigma = \frac{1}{n}\Sigma$.

(b)

- By Theorem 9.4(e), $\mathsf{E}(x_ix_i') = \Sigma + \mu\mu'$ and, by (a), $\mathsf{E}(\bar{x}\bar{x}') = \frac{1}{n}\Sigma + \mu\mu'$.
- By Proposition 9.14(a), $X'C_nX = X'X - n\bar{x}\bar{x}'$, and $X'X = x_1x_1' + \cdots + x_nx_n'$ by the outer product expansion (Theorem 1.4(d)).
- Hence $\mathsf{E}(X'C_nX) = n(\Sigma + \mu\mu') - n\left(\frac{1}{n}\Sigma + \mu\mu'\right) = (n-1)\Sigma$, and $\mathsf{E}(S) = \Sigma$. □

With the divisor n in place of $n - 1$, the expectation would be $\frac{n-1}{n}\Sigma$, not Σ (if $\Sigma \neq 0$); this is why S has the divisor $n - 1$.

Definition 9.17 ([2, §10.2, Exercise 10.7], [12, Definition 2.15], [9, Proposition 5.24], [19, §12.2.4])**.** For $u, v \in \mathbb{R}^n$ with means $\bar{u}$, $\bar{v}$, put

$$S_{uv} = \sum_{i=1}^{n}(u_i - \bar{u})(v_i - \bar{v}) = (C_n u)'(C_n v), \qquad S_{uu} = \sum_{i=1}^{n}(u_i - \bar{u})^2 = \|C_n u\|^2$$

($C_n u = u - \bar{u}\mathbf{1}$ and $C_n' C_n = C_n$; Example 1.16 and Proposition 1.15). The vector u is **constant** if $u = \bar{u}\mathbf{1}_n$, that is, if $C_n u = 0$, or $S_{uu} = 0$. If u and v are not constant, their **sample correlation** is

$$\operatorname{cor}(u, v) = \frac{S_{uv}}{\sqrt{S_{uu} S_{vv}}}$$

(for random variables, the correlation is ρ_{uv} of Definition 9.7). Now let $X \in \mathbb{R}^{n \times p}$ be a data matrix whose variables all have positive sample variance, $s_{jj} > 0$ (Definition 9.13), and put

$$D_s = \operatorname{diag}(s_{11}, \ldots, s_{pp}), \qquad D_s^{1/2} = \operatorname{diag}\Big(\sqrt{s_{11}}, \ldots, \sqrt{s_{pp}}\Big), \qquad D_s^{-1/2} = \operatorname{diag}\Big(\frac{1}{\sqrt{s_{11}}}, \ldots, \frac{1}{\sqrt{s_{pp}}}\Big)$$

The number $\sqrt{s_{jj}}$ is the **sample standard deviation** of variable j.

- The number $r_{jl} = s_{jl}/\sqrt{s_{jj} s_{ll}}$ is the sample correlation of variables j and l, that is, cor of the columns j and l of X (the divisors $n - 1$ cancel), and $R = (r_{jl}) = D_s^{-1/2} S D_s^{-1/2} \in \mathbb{R}^{p \times p}$ is the **sample correlation matrix** (by Theorem 1.4(a),(c), the diagonal factors scale the rows and the columns of S).
- The **standardized data matrix** is $X_s = C_n X D_s^{-1/2}$, with the entries $(x_{ij} - \bar{x}_j)/\sqrt{s_{jj}}$: each variable is centered and measured in units of its sample standard deviation.

Thus $\operatorname{cor}(u, v)$ is the sample covariance of u and v (Definition 9.13) divided by the product of the square roots of their sample variances. The numbers ρ_{ij} and the matrix $\operatorname{cor}(z)$ of Definition 9.7 are the population counterparts of the r_{jl} and of R.

Proposition 9.18 ([9, Proposition 5.24], [12, Proposition 2.17])**.** Let X, S, D_s, R and X_s be as in Definition 9.17.

(a) $C_n X_s = X_s$, so the standardized variables have mean 0, and the sample covariance matrix of the data matrix X_s is R. Hence R is symmetric and PSD, $r_{jj} = 1$, $|r_{jl}| \leqslant 1$, and $\operatorname{tr} R = p$.

(b) Let $c_1, \ldots, c_p > 0$ and $\Gamma = \operatorname{diag}(c_1, \ldots, c_p)$. The data matrix $X\Gamma$, in which variable j is multiplied by c_j (a change of units), has the sample covariance matrix $\Gamma S \Gamma$, with the entries $c_j c_l s_{jl}$, but the same standardized data matrix X_s and the same sample correlation matrix R.

Proof. (a)

- $C_n^2 = C_n = C_n'$ and $C_n \mathbf{1} = 0$ (Proposition 1.15). So $C_n X_s = C_n^2 X D_s^{-1/2} = X_s$, and the mean vector of X_s is $\frac{1}{n} X_s' \mathbf{1} = \frac{1}{n}(C_n X_s)' \mathbf{1} = \frac{1}{n} X_s' C_n \mathbf{1} = 0$.
- By Definition 9.13 and Proposition 9.14(a), and as $(D_s^{-1/2})' = D_s^{-1/2}$, the sample covariance matrix of X_s is

$$\frac{1}{n-1} X_s' C_n X_s = \frac{1}{n-1} X_s' X_s = D_s^{-1/2}\Big(\frac{1}{n-1}(C_n X)'(C_n X)\Big) D_s^{-1/2} = D_s^{-1/2} S D_s^{-1/2} = R$$

- So R is symmetric and PSD (Proposition 9.14(a) for X_s), $r_{jj} = s_{jj}/s_{jj} = 1$, and $\operatorname{tr} R = p$.
- Let u_j be column j of $C_n X$. Then $(n-1)s_{jl} = u_j' u_l$ (Proposition 9.14(a), Theorem 1.4(d)), so $r_{jl} = u_j' u_l / (\|u_j\| \, \|u_l\|)$, and $|r_{jl}| \leqslant 1$ by the Cauchy–Schwarz inequality (Theorem 6.4(a)).

(b)

- $X\Gamma$ has the mean vector $\frac{1}{n}\Gamma X'\mathbf{1} = \Gamma\bar{x}$, the centered data matrix $C_nX\Gamma$, and the sample covariance matrix $\frac{1}{n-1}\Gamma X'C_nX\Gamma = \Gamma S\Gamma$, whose (j,l) entry is $c_j s_{jl} c_l$ (Theorem 1.4(a),(c)).
- Its diagonal entries are $c_j^2 s_{jj}$, so for $X\Gamma$ the matrices $D_s^{1/2}$ and $D_s^{-1/2}$ become $\Gamma D_s^{1/2}$ and $\Gamma^{-1}D_s^{-1/2}$ ($c_j > 0$). Hence the standardized data matrix of $X\Gamma$ is $C_nX\Gamma\Gamma^{-1}D_s^{-1/2} = X_s$, and by (a) its sample correlation matrix, the sample covariance matrix of X_s, is R. □

The Mahalanobis distance

Definition 9.19 ([18, §2.6, (2.32), (2.33), Theorem 14.2(a)])**.** Let $S \in \mathbb{R}^{p\times p}$ be PD. The **Mahalanobis distance** (with respect to S) between $x, y \in \mathbb{R}^p$ is

$$d_S(x,y) = \Big((x-y)'S^{-1}(x-y)\Big)^{1/2}$$

S^{-1} is PD (Proposition 8.23(c)), so $d_S(x,y) \geqslant 0$, with equality only for $x = y$. For $S = I_p$ this is the distance $\|x-y\|$ of Definition 6.1; for $S = \operatorname{diag}(s_{11},\dots,s_{pp})$ with every $s_{jj} > 0$, $S^{-1} = \operatorname{diag}(1/s_{11},\dots,1/s_{pp})$ (Definition 2.12) and $d_S(x,y)^2 = \sum_j (x_j-y_j)^2/s_{jj}$: each coordinate is measured in units of $\sqrt{s_{jj}}$, the sample standard deviation when S is a sample covariance matrix (Definition 9.17).

Proposition 9.20 ([18, (3.23)–(3.26), Example 6.2, Theorem 2.7])**.** Let $X \in \mathbb{R}^{n\times p}$ be a data matrix with rows $x_1',\dots,x_n'$, mean vector $\bar{x}$ and a PD sample covariance matrix S (Definition 9.13).

(a) The vectors $w_i = S^{-1/2}(x_i - \bar{x})$ $(i = 1,\dots,n)$ have the mean vector 0 and the sample covariance matrix I_p. (Passing from the x_i to the w_i is called **whitening** the data.)

(b) $d_S(x_i,\bar{x}) = \|w_i\|$ for every i, and $\sum_{i=1}^n d_S(x_i,\bar{x})^2 = (n-1)p$.

(c) Let $B \in \mathbb{R}^{p\times p}$ be invertible, $b \in \mathbb{R}^p$, and $y_i = Bx_i + b$ $(i = 1,\dots,n)$. The y_i have the mean vector $\bar{y} = B\bar{x} + b$ and the PD sample covariance matrix $S_Y = BSB'$, and $d_{S_Y}(y_i,y_l) = d_S(x_i,x_l)$ and $d_{S_Y}(y_i,\bar{y}) = d_S(x_i,\bar{x})$ for all i,l.

(d) Let $S = Q\Lambda Q'$ as in Theorem 8.3(b), with $\lambda_1 \geqslant \cdots \geqslant \lambda_p$; then $\lambda_p > 0$ (Theorem 8.21(a)). Let $c > 0$. For $x \in \mathbb{R}^p$ put $u = Q'(x-\bar{x})$, so that $x = \bar{x} + u_1q_1 + \cdots + u_pq_p$. Then

$$d_S(x,\bar{x})^2 = \frac{u_1^2}{\lambda_1} + \cdots + \frac{u_p^2}{\lambda_p}$$

So $\{x \in \mathbb{R}^p : d_S(x,\bar{x}) \leqslant c\}$ is an ellipse for $p = 2$ (an interval for $p = 1$, an **ellipsoid** for $p \geqslant 3$) with center $\bar{x}$ and axes along $q_1,\dots,q_p$: the point $\bar{x} + tq_j$ lies in it if and only if $|t| \leqslant c\sqrt{\lambda_j}$.

Proof. (a) $S^{-1/2}$ is symmetric (Theorem 8.46(a),(c) and Theorem 2.13(c)), so the matrix X_w with the rows $w_i' = (x_i-\bar{x})'S^{-1/2}$ is $X_w = C_nXS^{-1/2} = XS^{-1/2} - \mathbf{1}(S^{-1/2}\bar{x})'$. By Proposition 9.14(c) with $A = S^{-1/2}$ and $b = -S^{-1/2}\bar{x}$, its mean vector is $S^{-1/2}\bar{x} - S^{-1/2}\bar{x} = 0$ and its sample covariance matrix is $S^{-1/2}SS^{-1/2} = I_p$ (Theorem 8.46(c)).

(b)

- $\|w_i\|^2 = (x_i-\bar{x})'S^{-1/2}S^{-1/2}(x_i-\bar{x}) = (x_i-\bar{x})'S^{-1}(x_i-\bar{x})$ (Theorem 8.46(c)).
- $(x_i-\bar{x})'S^{-1}(x_i-\bar{x}) = \operatorname{tr}\Big(S^{-1}(x_i-\bar{x})(x_i-\bar{x})'\Big)$ (Theorem 1.11(b)); summing over i (Theorem 1.11(a)) and by Proposition 9.14(a), $\sum_i d_S(x_i,\bar{x})^2 = \operatorname{tr}\Big(S^{-1}(n-1)S\Big) = \operatorname{tr}\Big((n-1)I_p\Big) = (n-1)p$.

(c)

- The matrix with the rows y_i' is $Y = XB' + \mathbf{1}b'$. By Proposition 9.14(c), $\bar{y} = B\bar{x}+b$ and $S_Y = BSB'$, which is PD by Theorem 8.24(c) with $P = B'$.

- $S_Y^{-1} = (B')^{-1}S^{-1}B^{-1}$ (Theorem 2.13(b),(c)) and $y_i - y_l = B(x_i - x_l)$, so

$$(y_i - y_l)'S_Y^{-1}(y_i - y_l) = (x_i - x_l)'B'(B')^{-1}S^{-1}B^{-1}B(x_i - x_l) = (x_i - x_l)'S^{-1}(x_i - x_l)$$

  and in the same way with $y_i - \bar{y} = B(x_i - \bar{x})$.

(d)

- $QQ' = I_p$ (Proposition 6.13(b)) gives $x - \bar{x} = Qu = \sum_j u_j q_j$, and $S^{-1} = Q\Lambda^{-1}Q'$ (Proposition 8.10(a),(b) with $f(t) = t$), so $d_S(x, \bar{x})^2 = (x - \bar{x})'Q\Lambda^{-1}Q'(x - \bar{x}) = u'\Lambda^{-1}u = \sum_j u_j^2/\lambda_j$.
- For $x = \bar{x} + tq_j$, $u = tQ'q_j = te_j$, so $d_S(x, \bar{x})^2 = t^2/\lambda_j$, which is $\leqslant c^2$ if and only if $|t| \leqslant c\sqrt{\lambda_j}$. □

In the coordinates u, the set in (d) is $\left\{u : \sum_j u_j^2/(c\sqrt{\lambda_j})^2 \leqslant 1\right\}$, the standard form of an ellipse or ellipsoid with the semi-axes $c\sqrt{\lambda_1}, \ldots, c\sqrt{\lambda_p}$.

Example 9.21. Nine students take four tests. Their scores x_i' are the rows of

$$X = \begin{pmatrix} 8 & 9 & 10 & 11 \\ 5 & 4 & 5 & 6 \\ 3 & 6 & 5 & 6 \\ 3 & 4 & 7 & 6 \\ 7 & 8 & 9 & 8 \\ 6 & 5 & 6 & 9 \\ 4 & 7 & 6 & 9 \\ 4 & 5 & 8 & 9 \\ 5 & 6 & 7 & 8 \end{pmatrix}$$

Find $\bar{x}$, S and R; the sample variances of the total score and of the difference of tests 1 and 2, and their sample covariance; $S^{1/2}$, $S^{-1/2}$, the whitened data and the Mahalanobis distances of the students from $\bar{x}$.

Solution.

- $\mathbf{1}'X = (45,\ 54,\ 63,\ 72)$, so $\bar{x} = (5, 6, 7, 8)'$, and

$$C_9X = X - \mathbf{1}\bar{x}' = \begin{pmatrix} 3 & 3 & 3 & 3 \\ 0 & -2 & -2 & -2 \\ -2 & 0 & -2 & -2 \\ -2 & -2 & 0 & -2 \\ 2 & 2 & 2 & 0 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad (C_9X)'(C_9X) = \begin{pmatrix} 24 & 16 & 16 & 16 \\ 16 & 24 & 16 & 16 \\ 16 & 16 & 24 & 16 \\ 16 & 16 & 16 & 24 \end{pmatrix}$$

  for instance $24 = 9 + 0 + 4 + 4 + 4 + 1 + 1 + 1 + 0$ and $16 = 9 + 0 + 0 + 4 + 4 - 1 - 1 + 1 + 0$. Hence (Proposition 9.14(a))

$$S = \frac{1}{8}(C_9X)'(C_9X) = \begin{pmatrix} 3 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 \\ 2 & 2 & 3 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix} = I_4 + 2J_4, \qquad R = \frac{1}{3}S = \begin{pmatrix} 1 & \frac{2}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & 1 & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{2}{3} & 1 \end{pmatrix}$$

  as $D_s = 3I_4$: the four tests are equicorrelated with $r_{jl} = \frac{2}{3}$.

- The total score is Xa with $a = \mathbf{1}_4$, so its sample variance is $a'Sa = \mathbf{1}'S\mathbf{1} = 36$, the sum of the entries of S (Proposition 9.14(b)); directly, the centered totals $12, -6, -6, -6, 6, 0, 0, 0, 0$ have the squares adding up to $288 = 8 \cdot 36$. For the difference, $b = (1, -1, 0, 0)'$: $b'Sb = 3 - 2 \cdot 2 + 3 = 2$; and $a'Sb = (9, 9, 9, 9)b = 0$, as every column of S sums to 9: total and difference are uncorrelated.
- $S = I_4 + 2J_4$ has the eigenvalues $1 + 4 \cdot 2 = 9$, with eigenspace $\operatorname{span}\{\mathbf{1}\}$, and 1 (three times) (Example 8.16(b)); so S is PD (Theorem 8.21(a)). Let $B = I_4 + \frac{1}{2}J_4$, which has the eigenvalues 3 and 1 in the same way, so it is PD. As $J_4^2 = 4J_4$ (Proposition 1.15), $B^2 = I_4 + J_4 + \frac{1}{4}J_4^2 = I_4 + 2J_4 = S$, and by the uniqueness in Theorem 8.46(a), $S^{1/2} = B$. Next $\left(I_4 + \frac{1}{2}J_4\right)\left(I_4 - \frac{1}{6}J_4\right) = I_4 + \frac{1}{2}J_4 - \frac{1}{6}J_4 - \frac{1}{3}J_4 = I_4$, so (Theorem 2.16)

$$S^{-1/2} = B^{-1} = I_4 - \tfrac{1}{6}J_4, \qquad S^{-1} = \left(S^{-1/2}\right)^2 = I_4 - \tfrac{1}{3}J_4 + \tfrac{4}{36}J_4 = I_4 - \tfrac{2}{9}J_4$$

  (Theorem 8.46(c)).
- The whitened data $w_i = S^{-1/2}(x_i - \bar{x}) = (x_i - \bar{x}) - \frac{1}{6}\left(\mathbf{1}'(x_i - \bar{x})\right)\mathbf{1}$ are the rows of

$$X_w = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

  (for student 1, $(3, 3, 3, 3)' - 2\mathbf{1}$; for student 2, $(0, -2, -2, -2)' + \mathbf{1}$; for students 6, 7, 8, $\mathbf{1}'(x_i - \bar{x}) = 0$). They have the mean vector 0 and the sample covariance matrix I_4 (Proposition 9.20(a)); directly, the columns of X_w add up to 0 and are orthogonal with squared length 8, so $\frac{1}{8}X_w'X_w = I_4$.
- By Proposition 9.20(b), $d_S(x_i, \bar{x})^2 = \|w_i\|^2$, which is 4 for students $1, \dots, 8$ and 0 for student 9; the sum is $32 = (n-1)p$. The squared distances $\|x_i - \bar{x}\|^2$ are 36, 12, 12, 12, 12, 4, 4, 4, 0. Student 1, three points above average on every test, follows the strong common pattern of the tests and is no more unusual than students 6, 7, 8, who are one point above or below average on each test, against the pattern.
- S has $\lambda_1 = 9$ with $q_1 = \frac{1}{2}\mathbf{1}$ and $\lambda_2 = \lambda_3 = \lambda_4 = 1$. The ellipsoid $d_S(x, \bar{x}) \leqslant 2$ has the semi-axis $2 \cdot 3 = 6$ along q_1 and $2 \cdot 1 = 2$ in every direction orthogonal to q_1 (Proposition 9.20(d)). Student 1 lies at the end of the long axis: $x_1 = \bar{x} + 3\mathbf{1} = \bar{x} + 6q_1$.

The Rayleigh quotient

Lemma 9.22 ([5, Corollary 21.5.9], [3, §5.1])**.** Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Its eigenvalues, listed with multiplicity, are real (Theorem 8.1); we write them in decreasing order,

$$\lambda_1(A) \geqslant \lambda_2(A) \geqslant \cdots \geqslant \lambda_n(A)$$

or $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ when A is clear.

(a) There is an orthogonal $Q = [q_1 \ \cdots \ q_n] \in \mathbb{R}^{n \times n}$ with $Q'AQ = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$; then $Aq_i = \lambda_i q_i$ for every i, and $A = \lambda_1 q_1 q_1' + \cdots + \lambda_n q_n q_n'$.

(b) $\lambda_k(-A) = -\lambda_{n+1-k}(A)$ for $k = 1, \dots, n$.

(c) $\lambda_k(A + cI_n) = \lambda_k(A) + c$ for $k = 1, \dots, n$ and every $c \in \mathbb{R}$.

Proof. (a) By Theorem 8.3(a),(b), with the orthonormal eigenvectors $q_1, \dots, q_n$ numbered so that their eigenvalues decrease, $Q = [q_1 \ \cdots \ q_n]$ is orthogonal and $Q'AQ$ is the diagonal matrix of these eigenvalues, which are the eigenvalues of A listed with multiplicity (Theorem 8.3(c)), that is, $\lambda_1, \dots, \lambda_n$. The sum is Theorem 8.6(c).

(b) The eigenvalues of $-A$ are $-\lambda_1 \leqslant \cdots \leqslant -\lambda_n$ (Proposition 7.18(c) with $c = -1$); in decreasing order the kth of them is $-\lambda_{n+1-k}$.

(c) The eigenvalues of $A + cI_n$ are $\lambda_1 + c \geqslant \cdots \geqslant \lambda_n + c$ (Proposition 7.18(c)), already in decreasing order; the kth of them is $\lambda_k + c$. □

For symmetric $A \in \mathbb{R}^{n\times n}$ and $x \neq 0$, the number $x'Ax/x'x$ is the **Rayleigh quotient** of x [2, §9.7], [6, Exercise 7.53]. It does not change when x is replaced by cx ($c \neq 0$); in particular it equals the quadratic form $u'Au$ (Definition 8.17) at the unit vector $u = x/\|x\|$. So the largest and the smallest Rayleigh quotient are also the largest and the smallest value of $u'Au$ over the unit vectors u.

Theorem 9.23 ([3, Result 5.1; Ch. 5, Exercise 16], [6, Exercise 12.39(a)], [9, Theorem 5.20], [2, Lemma 9.1])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, with $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ and $Q = [q_1 \ \cdots \ q_n]$ as in Lemma 9.22.

(a) For every $x \neq 0$ in $\mathbb{R}^n$,

$$\lambda_n \leqslant \frac{x'Ax}{x'x} \leqslant \lambda_1$$

and the bounds are attained at $x = q_n$ and $x = q_1$. Hence $\max_{x\neq 0} x'Ax/x'x = \lambda_1$ and $\min_{x\neq 0} x'Ax/x'x = \lambda_n$.

(b) For $x \neq 0$: $x'Ax = \lambda_1 x'x$ if and only if $Ax = \lambda_1 x$, and $x'Ax = \lambda_n x'x$ if and only if $Ax = \lambda_n x$. So the maximum (minimum) is attained exactly at the eigenvectors belonging to λ_1 (λ_n).

(c) $\lambda_1 I_n - A$ and $A - \lambda_n I_n$ are PSD.

Proof. Let $x \in \mathbb{R}^n$ and $y = Q'x$. Then $x = Qy$ and $x'x = y'y$ (Proposition 6.13), so $x \neq 0$ if and only if $y \neq 0$, and

$$x'Ax = y'Q'AQy = y' \operatorname{diag}(\lambda_1, \dots, \lambda_n)y = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$$

(a) By Theorem 8.21(d), $\lambda_n x'x \leqslant x'Ax \leqslant \lambda_1 x'x$, with equality on the right for $x = q_1$ and on the left for $x = q_n$; divide by $x'x > 0$.

(b)

- $\lambda_1 x'x - x'Ax = \sum_i (\lambda_1 - \lambda_i)y_i^2$ is a sum of terms $\geqslant 0$. It is 0 if and only if $y_i = 0$ for every i with $\lambda_i < \lambda_1$, that is, $x = Qy = \sum_{\lambda_i=\lambda_1} y_i q_i$; and then $Ax = \sum_{\lambda_i=\lambda_1} y_i \lambda_1 q_i = \lambda_1 x$.
- Conversely, $Ax = \lambda_1 x$ gives $x'Ax = \lambda_1 x'x$.
- For λ_n, apply this to $-A$, whose largest eigenvalue is $-\lambda_n$ (Lemma 9.22(b)).

(c) Both matrices are symmetric, and by (a), $x'(\lambda_1 I - A)x = \lambda_1 x'x - x'Ax \geqslant 0$ and $x'(A - \lambda_n I)x = x'Ax - \lambda_n x'x \geqslant 0$ for every x (Definition 8.20). □

Lemma 9.24 (cf. [8, Ch. 21, Exercise 17], [6, Exercise 12.46])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, let $W \neq \{0\}$ be a subspace of $\mathbb{R}^n$ of dimension d, and let the columns of $G \in \mathbb{R}^{n\times d}$ form an orthonormal basis of W (Theorems 4.4 and 6.14). Then the nonzero vectors of W are the vectors $x = Gz$ with $z \in \mathbb{R}^d$, $z \neq 0$, and for these

$$\frac{x'Ax}{x'x} = \frac{z'(G'AG)z}{z'z}$$

Hence the maximum and the minimum of $x'Ax/x'x$ over the nonzero $x \in W$ exist: they are $\lambda_1(G'AG)$ and $\lambda_d(G'AG)$, attained at $x = Gz$ for eigenvectors z of the symmetric $d \times d$ matrix $G'AG$ belonging to these eigenvalues. Both lie between $\lambda_n(A)$ and $\lambda_1(A)$.

Proof. • The columns of G span W, so $x \in W$ if and only if $x = Gz$ for some z (Theorem 1.4(b)); they are independent, so $Gz = 0$ only for $z = 0$.

- $G'G = I_d$ (Definition 6.12), so $x'x = z'G'Gz = z'z$; also $x'Ax = z'(G'AG)z$, and $G'AG$ is symmetric.
- Apply Theorem 9.23(a),(b) to $G'AG$, and Theorem 9.23(a) to A for the last sentence. □

As before, these extrema are also the largest and the smallest value of $u'Au$ over the unit vectors $u \in W$.

Theorem 9.25 ([3, Result 5.7(ii)], [6, Exercise 12.42], [8, Ch. 21, Exercise 17])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, with $Q = [q_1 \ \cdots \ q_n]$ as in Lemma 9.22, and let $1 \leqslant k \leqslant n$. Then

$$\lambda_k = \max\left\{\frac{x'Ax}{x'x} : x \neq 0,\ x \perp q_1, \ldots, q_{k-1}\right\} = \min\left\{\frac{x'Ax}{x'x} : x \neq 0,\ x \perp q_{k+1}, \ldots, q_n\right\}$$

and both are attained at $x = q_k$. (For $k = 1$ the first set has no condition, for $k = n$ the second.)

Proof. • Let $0 \leqslant j \leqslant n$, and split $Q = [R\ T]$ with $R = [q_1 \ \cdots \ q_j]$. Then $R'T = 0$, as $q_i'q_l = 0$ for $i \neq l$, and $I_n = QQ' = RR' + TT'$ (Proposition 6.13(b) and Theorem 5.3). The entries of $R'x$ are $q_1'x, \ldots, q_j'x$. If $R'x = 0$, then $x = TT'x \in \mathcal{C}(T)$; and $R'(Tz) = 0$ for every z. So $x \perp q_1, \ldots, q_j$ if and only if $x \in \mathcal{C}(T)$; in the same way, $x \perp q_{j+1}, \ldots, q_n$ if and only if $x \in \mathcal{C}(R)$.

- First form ($j = k-1$): the $x \perp q_1, \ldots, q_{k-1}$ form $\mathcal{C}(T)$, where the columns of $T = [q_k \ \cdots \ q_n]$ are an orthonormal basis of it. $T'AT = \operatorname{diag}(\lambda_k, \ldots, \lambda_n)$, the lower right block of $Q'AQ$, has the largest eigenvalue λ_k with eigenvector e_1. By Lemma 9.24, the maximum is λ_k, attained at $Te_1 = q_k$.
- Second form ($j = k$): the $x \perp q_{k+1}, \ldots, q_n$ form $\mathcal{C}(R)$, $R = [q_1 \ \cdots \ q_k]$, and $R'AR = \operatorname{diag}(\lambda_1, \ldots, \lambda_k)$ has the smallest eigenvalue λ_k with eigenvector e_k. By Lemma 9.24, the minimum is λ_k, attained at $Re_k = q_k$. □

Definition 9.26 ([2, Definition 9.1], [3, Ch. 5, Exercise 12])**.** The **Frobenius norm** of $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ is

$$\|A\|_F = \Big(\operatorname{tr}(A'A)\Big)^{1/2} = \left(\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2\right)^{1/2}$$

(Theorem 1.11(d)). If $v \in \mathbb{R}^{mn}$ lists the entries of A column after column, then $\|A\|_F = \|v\|$ (Definition 6.1); and $\|A\|_F^2 = \|a_1\|^2 + \cdots + \|a_n\|^2$ for the columns a_j of A. (Some books write $\|A\|_2$ for $\|A\|_F$.)

Theorem 9.27 (Ky Fan; [3, Result 5.12], [6, Exercise 12.49(a)], [5, Theorem 21.12.5], [9, Theorem 5.22])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, with eigenvalues $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_n$ (listed with multiplicity) and orthonormal eigenvectors $q_1, \ldots, q_n$, $Aq_i = \lambda_i q_i$ (Lemma 9.22(a)). Let $1 \leqslant k \leqslant n$, and let $G \in \mathbb{R}^{n\times k}$ have orthonormal columns, $G'G = I_k$.

(a) $\operatorname{tr}(G'AG) \leqslant \lambda_1 + \cdots + \lambda_k$, with equality for $G = [q_1 \ \cdots \ q_k]$.

(b) $\operatorname{tr}(G'AG) \geqslant \lambda_{n-k+1} + \cdots + \lambda_n$, with equality for $G = [q_{n-k+1} \ \cdots \ q_n]$.

(c) If $k < n$ and $\lambda_k > \lambda_{k+1}$, then equality holds in (a) only if $\mathcal{C}(G) = \operatorname{span}\{q_1, \ldots, q_k\}$.

So the largest (smallest) value of $\operatorname{tr}(G'AG)$ over all $G \in \mathbb{R}^{n\times k}$ with $G'G = I_k$ is the sum of the k largest (smallest) eigenvalues of A.

Proof. (a)

- $Q = [q_1 \ \cdots \ q_n]$ is orthogonal and $Q'AQ = \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ (Theorem 8.3(b)). Put $R = Q'G = (r_{ij}) \in \mathbb{R}^{n\times k}$. As $QQ' = I_n$ (Proposition 6.13(b)), $G = QR$, and $R'R = G'QQ'G = G'G = I_k$.

- So $\operatorname{tr}(G'AG) = \operatorname{tr}(R'Q'AQR) = \operatorname{tr}(R'\Lambda R)$. The jth diagonal entry of $R'\Lambda R$ is $\operatorname{col}_j(R)'\Lambda \operatorname{col}_j(R) = \sum_i \lambda_i r_{ij}^2$ (Theorem 1.4(d)); hence

$$\operatorname{tr}(G'AG) = \sum_{j=1}^{k}\sum_{i=1}^{n} \lambda_i r_{ij}^2 = \sum_{i=1}^{n} w_i\lambda_i, \qquad w_i = \sum_{j=1}^{k} r_{ij}^2$$

- $w_i \geqslant 0$, and $\sum_i w_i = \sum_j \sum_i r_{ij}^2 = \operatorname{tr}(R'R) = \operatorname{tr} I_k = k$ (Theorem 1.11(d)).
- $w_i \leqslant 1$: w_i is the ith diagonal entry of RR' (Theorem 1.4(d)). If $k = n$, then $RR' = I_n$ (Proposition 6.13(b)) and $w_i = 1$. If $k < n$, Corollary 6.15(c) gives $T \in \mathbb{R}^{n\times(n-k)}$ with $[R\ T]$ orthogonal and $RR' + TT' = I_n$. The ith diagonal entries give $w_i + \sum_l t_{il}^2 = 1$, so $w_i \leqslant 1$.
- The right side of

$$\sum_{i\leqslant k} \lambda_i - \sum_{i=1}^{n} w_i\lambda_i = \sum_{i\leqslant k}(\lambda_i - \lambda_k)(1 - w_i) + \sum_{i>k}(\lambda_k - \lambda_i)w_i$$

  expands to $\sum_{i\leqslant k} \lambda_i - \sum_i w_i\lambda_i - \lambda_k\big(k - \sum_i w_i\big)$, which is the left side as $\sum_i w_i = k$. Every term on the right is $\geqslant 0$: $\lambda_i \geqslant \lambda_k$ and $w_i \leqslant 1$ for $i \leqslant k$; $\lambda_i \leqslant \lambda_k$ and $w_i \geqslant 0$ for $i > k$. This proves the inequality.
- For $G = [q_1\ \cdots\ q_k]$, $AG = [\lambda_1 q_1\ \cdots\ \lambda_k q_k] = G\operatorname{diag}(\lambda_1, \ldots, \lambda_k)$, so $G'AG = \operatorname{diag}(\lambda_1, \ldots, \lambda_k)$, with trace $\lambda_1 + \cdots + \lambda_k$.

(b)

- $-A$ is symmetric, with the orthonormal eigenvectors $q_n, \ldots, q_1$ for the eigenvalues $-\lambda_n \geqslant \cdots \geqslant -\lambda_1$. By (a) for $-A$, $-\operatorname{tr}(G'AG) = \operatorname{tr}(G'(-A)G) \leqslant -\lambda_n - \cdots - \lambda_{n-k+1}$, with equality for $G = [q_n\ \cdots\ q_{n-k+1}]$.
- $\operatorname{tr}(G'AG) = \sum_j g_j'Ag_j$ over the columns g_j of G (Theorem 1.4(d)), so the order of the columns does not matter.

(c)

- If $\operatorname{tr}(G'AG) = \lambda_1 + \cdots + \lambda_k$, every term on the right of the identity in (a) is 0. For $i > k$, $\lambda_k - \lambda_i \geqslant \lambda_k - \lambda_{k+1} > 0$, so $w_i = 0$. Then $\sum_{i\leqslant k} w_i = k$ with every $w_i \leqslant 1$ gives $w_i = 1$ for $i \leqslant k$.
- Row i of $R = Q'G$ is $q_i'G$, so $w_i = \|G'q_i\|^2$. As $G'G = I_k$, $\|GG'q_i\|^2 = q_i'GG'GG'q_i = w_i$, and $GG'q_i \perp q_i - GG'q_i$, since $(GG'q_i)'(q_i - GG'q_i) = w_i - w_i = 0$. By Pythagoras (Proposition 6.2), $\|q_i - GG'q_i\|^2 = \|q_i\|^2 - w_i = 0$ for $i \leqslant k$.
- Hence $q_i = G(G'q_i) \in \mathcal{C}(G)$ for $i \leqslant k$, and $\operatorname{span}\{q_1, \ldots, q_k\} \subseteq \mathcal{C}(G)$. Both spaces have dimension k (Proposition 6.11(b)), so they are equal (Theorem 4.5(a)). □

For $k = 1$ and $G = x/\|x\|$ ($x \neq 0$), (a) and (b) are the bounds $\lambda_n \leqslant x'Ax/x'x \leqslant \lambda_1$ of Theorem 9.23(a).

Example 9.28. Let

$$A = \begin{pmatrix} 15 & 5 & 10 & 2 \\ 5 & 15 & 2 & 10 \\ 10 & 2 & 15 & 5 \\ 2 & 10 & 5 & 15 \end{pmatrix}$$

Find the maximum and the minimum of $\operatorname{tr}(G'AG)$ over the matrices $G \in \mathbb{R}^{4\times 2}$ with $G'G = I_2$.

Solution.

- Let $h_1 = (1,1,1,1)'$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$ and $h_4 = (1,-1,-1,1)'$. Row by row, $Ah_1 = 32h_1$, $Ah_2 = 8h_2$, $Ah_3 = 18h_3$ and $Ah_4 = 2h_4$; for instance $15 + 5 + 10 + 2 = 32$, $15 + 5 - 10 - 2 = 8$, $15 - 5 + 10 - 2 = 18$ and $15 - 5 - 10 + 2 = 2$ in row 1. The h_m are pairwise orthogonal with $\|h_m\| = 2$, so the eigenvalues of A are $32, 18, 8, 2$, with the orthonormal eigenvectors $\frac{1}{2}h_1, \frac{1}{2}h_3, \frac{1}{2}h_2, \frac{1}{2}h_4$ (Theorem 8.3(c)).

- By Theorem 9.27, the maximum is $32 + 18 = 50$, attained at $G = \frac{1}{2}[h_1\ h_3]$, and the minimum is $8 + 2 = 10$, attained at $G = \frac{1}{2}[h_2\ h_4]$.
- Other G give values in between: $G = [e_1\ e_2]$ gives $a_{11} + a_{22} = 30$, and $G = \frac{1}{\sqrt{2}}[e_1 + e_2\ \ e_3 + e_4]$ gives $\frac{1}{2}(15 + 5 + 5 + 15) + \frac{1}{2}(15 + 5 + 5 + 15) = 40$. The matrix $G = \frac{1}{\sqrt{2}}[e_1 + e_3\ \ e_2 + e_4]$ gives $\frac{1}{2}(15 + 10 + 10 + 15) \cdot 2 = 50$: its columns span the same plane W as $h_1 = (e_1 + e_3) + (e_2 + e_4)$ and $h_3 = (e_1 + e_3) - (e_2 + e_4)$, and $\operatorname{tr}(G'AG) = \operatorname{tr}(AGG')$ (Theorem 1.11(b)) depends on G only through $GG' = P_W$ (Theorem 6.31(b)).

Corollary 9.29 ([6, Exercise 12.49(b)], [2, Exercise 6.22])**.** Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ be symmetric, with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ (Lemma 9.22), and let $1 \leqslant i_1 < i_2 < \cdots < i_k \leqslant n$. Then

$$\lambda_{n-k+1} + \cdots + \lambda_n \leqslant a_{i_1 i_1} + a_{i_2 i_2} + \cdots + a_{i_k i_k} \leqslant \lambda_1 + \cdots + \lambda_k$$

In particular $\lambda_n \leqslant a_{ii} \leqslant \lambda_1$ for every i.

Proof.
- Let $G = [e_{i_1}\ \cdots\ e_{i_k}] \in \mathbb{R}^{n\times k}$. The (s,t) entry of $G'G$ is $e'_{i_s} e_{i_t}$ (Theorem 1.4(d)), which is 1 for $s = t$ and 0 for $s \neq t$, as $i_s \neq i_t$; so $G'G = I_k$.
- The (s,t) entry of $G'AG$ is $e'_{i_s} A e_{i_t} = a_{i_s i_t}$ (Theorem 1.4(b),(d)): $G'AG$ is the submatrix of A in rows and columns $i_1, \ldots, i_k$, and $\operatorname{tr}(G'AG) = a_{i_1 i_1} + \cdots + a_{i_k i_k}$.
- Theorem 9.27(a),(b) bound $\operatorname{tr}(G'AG)$ by $\lambda_1 + \cdots + \lambda_k$ from above and by $\lambda_{n-k+1} + \cdots + \lambda_n$ from below. For $k = 1$ and $i_1 = i$ this is the last sentence. □

Principal components

Definition 9.30 ([9, Definition 5.19], [3, §10.1])**.** Let $X \in \mathbb{R}^{n\times p}$ be a data matrix with $S \neq 0$. A **first loading vector** is a unit vector $v_1 \in \mathbb{R}^p$ that maximizes $a'Sa$ over all unit vectors $a \in \mathbb{R}^p$. For $m \geqslant 2$, once $v_1, \ldots, v_{m-1}$ are chosen, an m**th loading vector** is a unit vector v_m that maximizes $a'Sa$ over the unit vectors a with $a'Sv_l = 0$ for $l = 1, \ldots, m-1$ (the m**th problem**). The linear combination of the variables with coefficient vector v_m is the m**th principal component**, and the vector of its centered values,

$$z_m = C_n X v_m \in \mathbb{R}^n, \qquad z_{im} = v'_m(x_i - \bar{x}) \quad (i = 1, \ldots, n)$$

is the m**th score vector**.

By Proposition 9.14(b), $a'Sa$ is the sample variance of Xa, and $a'Sv_l = 0$ says that Xa and Xv_l are uncorrelated. So the mth principal component has the largest sample variance among the combinations with unit coefficient vector that are uncorrelated with the first $m-1$ principal components.

Theorem 9.31 ([9, Theorem 5.20], [3, §10.1])**.** Let $X \in \mathbb{R}^{n\times p}$ be a data matrix with $S \neq 0$, and let $S = V\Lambda V'$, where $V = [v_1\ \cdots\ v_p]$ is orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ with $\lambda_1 \geqslant \cdots \geqslant \lambda_p$ (Lemma 9.22(a)). Let $r = \operatorname{rank}(C_n X)$ and $z_m = C_n X v_m$.

(a) $\lambda_p \geqslant 0$, $r \leqslant \min\{n-1, p\}$, and $\lambda_m > 0$ if and only if $m \leqslant r$.

(b) If $\lambda_l > 0$, then for every $a \in \mathbb{R}^p$, Xa and Xv_l are uncorrelated if and only if $a \perp v_l$.

(c) Let $m \leqslant r$. If $v_1, \ldots, v_{m-1}$ are taken as the first $m-1$ loading vectors, then v_m is an mth loading vector, and the maximum in the mth problem is λ_m.

(d) $z'_m z_l = (n-1)\lambda_m$ if $l = m$ and 0 if $l \neq m$: the score vectors are pairwise orthogonal and uncorrelated, and z_m has sample variance λ_m.

(e) Let $M \leqslant r$ and $\lambda_1 > \lambda_2 > \cdots > \lambda_M > \lambda_{M+1}$ (the last inequality is omitted if $M = p$). Then the first M loading vectors are $\pm v_1, \ldots, \pm v_M$, and the first M score vectors are $\pm z_1, \ldots, \pm z_M$.

Proof. (a)

- S is PSD (Proposition 9.14(a)), so $\lambda_p \geqslant 0$ (Theorem 8.21(a)).
- $\mathbf{1}'C_n = (C_n\mathbf{1})' = 0'$ (Proposition 1.15), so every column of C_nX lies in $\operatorname{span}\{\mathbf{1}\}^\perp$, which has dimension $n-1$ (Theorem 6.6(c)). Hence $r \leqslant n-1$ (Theorem 4.5(a)), and $r \leqslant p$ (Theorem 4.13(a)).
- $\operatorname{rank} S = \operatorname{rank}\Big((C_nX)'(C_nX)\Big) = r$ (Theorem 4.18(b)), and $\operatorname{rank} S = \operatorname{rank}\Lambda$, as V is invertible (Theorem 4.15(d)). The nonzero columns $\lambda_m e_m$ of Λ are pairwise orthogonal, hence independent (Proposition 6.11(b)), and span $\mathcal{C}(\Lambda)$; so $\operatorname{rank}\Lambda$ is the number of nonzero λ_m.
- As $\lambda_1 \geqslant \cdots \geqslant \lambda_p \geqslant 0$, exactly $\lambda_1, \ldots, \lambda_r$ are positive.

(b) By Proposition 9.14(b) and $Sv_l = \lambda_l v_l$, the sample covariance of Xa and Xv_l is $a'Sv_l = \lambda_l a'v_l$, which is 0 if and only if $a'v_l = 0$, as $\lambda_l > 0$.

(c)

- Let a be a unit vector and $c = V'a$. Then $a = Vc$, $c_j = v_j'a$, $\sum_j c_j^2 = \|c\|^2 = \|a\|^2 = 1$ (Proposition 6.13), and $a'Sa = c'V'SVc = c'\Lambda c = \sum_j \lambda_j c_j^2$.
- By (b) and (a) ($\lambda_l > 0$ for $l < m \leqslant r$), the constraint of the mth problem says $c_1 = \cdots = c_{m-1} = 0$. Then
$$\lambda_m - a'Sa = \sum_{j \geqslant m} \lambda_m c_j^2 - \sum_{j \geqslant m} \lambda_j c_j^2 = \sum_{j>m} (\lambda_m - \lambda_j) c_j^2 \geqslant 0$$
- v_m satisfies the constraint, as $v_m'Sv_l = \lambda_l v_m'v_l = 0$ for $l < m$, and $v_m'Sv_m = \lambda_m$. So v_m solves the mth problem, with maximum λ_m.

(d) $z_m'z_l = v_m'(C_nX)'(C_nX)v_l = (n-1)v_m'Sv_l = (n-1)\lambda_l v_m'v_l$ (Proposition 9.14(a)). The score vectors have mean 0 ($\mathbf{1}'C_n = 0'$), so $\frac{1}{n-1}z_m'z_l$ is their sample covariance.

(e) Let $m \leqslant M$, and suppose the first $m-1$ loading vectors are $\pm v_1, \ldots, \pm v_{m-1}$ (nothing for $m = 1$).

- The constraint is again $c_1 = \cdots = c_{m-1} = 0$, and a maximizer a has $a'Sa = \lambda_m$ by (c). By the display in (c), $\sum_{j>m}(\lambda_m - \lambda_j)c_j^2 = 0$, where $\lambda_m - \lambda_j \geqslant \lambda_m - \lambda_{m+1} > 0$ for $j > m$.
- So $c_j = 0$ for $j \neq m$, and $a = c_m v_m$ with $c_m^2 = 1$: $a = \pm v_m$, with score vector $\pm z_m$. By induction on m the claim follows. □

Part (c) is Theorem 9.25 for S. The loading vectors and the maxima depend on the data only through S, and the proofs of (c) and (e) use only that S is symmetric and PSD: for a PSD matrix given without the data, the principal components are found in the same way.

Proposition 9.32 ([9, Proposition 5.21], [3, §10.1])**.** Let X, S, λ_m, z_m and r be as in Theorem 9.31. The **total variance** $s_{11} + \cdots + s_{pp}$ equals $\operatorname{tr} S = \lambda_1 + \cdots + \lambda_p$, and the **proportion of variance explained** by the mth principal component is

$$\mathrm{PVE}_m = \frac{\|z_m\|^2}{\|C_nX\|_F^2} = \frac{\lambda_m}{\lambda_1 + \cdots + \lambda_p} \qquad (m = 1, \ldots, p)$$

It is positive for $m \leqslant r$ and 0 for $m > r$, and $\mathrm{PVE}_1 \geqslant \mathrm{PVE}_2 \geqslant \cdots$. The cumulative proportion $\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_M$ increases with $M \leqslant r$ and equals 1 for $M \geqslant r$.

Proof.

- $\operatorname{tr} S = s_{11} + \cdots + s_{pp}$, and $\operatorname{tr} S = \operatorname{tr}(V\Lambda V') = \operatorname{tr}(\Lambda V'V) = \operatorname{tr}\Lambda$ (Theorem 1.11(b)).
- $\|C_nX\|_F^2 = \operatorname{tr}\Big((C_nX)'(C_nX)\Big) = (n-1)\operatorname{tr} S$ (Definition 9.26, Proposition 9.14(a)) and $\|z_m\|^2 = (n-1)\lambda_m$ (Theorem 9.31(d)); this gives the second expression, and $\operatorname{tr} S \geqslant \lambda_1 > 0$ ($S \neq 0$, so $r \geqslant 1$, and Theorem 9.31(a)).
- By Theorem 9.31(a), $\lambda_m > 0$ exactly for $m \leqslant r$, and λ_m decreases with m; so $\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_r = (\lambda_1 + \cdots + \lambda_r)/\operatorname{tr} S = 1$. □

Example 9.33 ([3, Ch. 10, Exercise 1])**.** The sample covariance matrix of two variables is

$$S = \begin{pmatrix} 60 & 20 \\ 20 & 60 \end{pmatrix}$$

Find the principal components and their sample variances.

Solution. $S = 40I_2 + 20J_2$, and $J_2\mathbf{1}_2 = \mathbf{1}_2(\mathbf{1}_2'\mathbf{1}_2) = 2\mathbf{1}_2$, $J_2(1,-1)' = \mathbf{1}_2(1-1) = 0$ (Proposition 1.15). So

$$S\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}80\\80\end{pmatrix} = 80\begin{pmatrix}1\\1\end{pmatrix}, \qquad S\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}40\\-40\end{pmatrix} = 40\begin{pmatrix}1\\-1\end{pmatrix}$$

With the orthonormal eigenvectors $v_1 = \frac{1}{\sqrt{2}}(1,1)'$, $v_2 = \frac{1}{\sqrt{2}}(1,-1)'$, the eigenvalues are $\lambda_1 = 80$, $\lambda_2 = 40$ (Theorem 8.3(c)). By Theorem 9.31, the principal components are the sum of the two variables divided by $\sqrt{2}$, with sample variance 80, and their difference divided by $\sqrt{2}$, with sample variance 40; they explain $\frac{80}{120} = \frac{2}{3}$ and $\frac{40}{120} = \frac{1}{3}$ of the total variance (Proposition 9.32).

Example 9.34. Find the principal components, the score vectors and the proportions of variance explained for the data of Example 9.15.

Solution.

- Let $h_1 = (1,1,1,1)'$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$ and $h_4 = (1,-1,-1,1)'$. Row by row,

$$Sh_1 = \begin{pmatrix}25\\25\\25\\25\end{pmatrix} = 25h_1, \qquad Sh_2 = \begin{pmatrix}9\\9\\-9\\-9\end{pmatrix} = 9h_2, \qquad Sh_3 = \begin{pmatrix}5\\-5\\5\\-5\end{pmatrix} = 5h_3, \qquad Sh_4 = \begin{pmatrix}1\\-1\\-1\\1\end{pmatrix} = h_4$$

  for instance $10+7+5+3 = 25$, $10+7-5-3 = 9$, $10-7+5-3 = 5$ and $10-7-5+3 = 1$ in row 1. The h_m are pairwise orthogonal with $\|h_m\| = 2$, so $v_m = \frac{1}{2}h_m$ $(m = 1,\dots,4)$ are orthonormal eigenvectors, and the eigenvalues of S are $\lambda_1 = 25$, $\lambda_2 = 9$, $\lambda_3 = 5$, $\lambda_4 = 1$ (Theorem 8.3(c)); with $V = [v_1 \ \cdots \ v_4]$, $S = V\,\mathrm{diag}(25,9,5,1)V'$.

- The eigenvalues are positive and distinct, so $r = 4$ and the loading vectors $v_1,\dots,v_4$ are unique up to sign (Theorem 9.31(a),(e)). The first principal component, with the coefficient vector $\frac{1}{2}(1,1,1,1)'$ and sample variance 25, is half the sum of the four variables: an overall level. The second, $\frac{1}{2}(1,1,-1,-1)'$, with sample variance 9, contrasts variables 1 and 2 with variables 3 and 4; the third, $\frac{1}{2}(1,-1,1,-1)'$, with sample variance 5, contrasts variables 1 and 3 with variables 2 and 4; the fourth, $\frac{1}{2}(1,-1,-1,1)'$, has sample variance 1.

- Scores: $C_5Xh_1 = (0,10,10,-10,-10)'$, $C_5Xh_2 = (0,6,-6,6,-6)'$, $C_5Xh_3 = (8,-2,-2,-2,-2)'$ and $C_5Xh_4 = (0,2,-2,-2,2)'$, so $z_m = C_5Xv_m = \frac{1}{2}C_5Xh_m$ gives

$$z_1 = \begin{pmatrix}0\\5\\5\\-5\\-5\end{pmatrix}, \qquad z_2 = \begin{pmatrix}0\\3\\-3\\3\\-3\end{pmatrix}, \qquad z_3 = \begin{pmatrix}4\\-1\\-1\\-1\\-1\end{pmatrix}, \qquad z_4 = \begin{pmatrix}0\\1\\-1\\-1\\1\end{pmatrix}$$

  Check (Theorem 9.31(d)): $\|z_1\|^2 = 100 = 4{\cdot}25$, $\|z_2\|^2 = 36 = 4{\cdot}9$, $\|z_3\|^2 = 20 = 4{\cdot}5$, $\|z_4\|^2 = 4 = 4{\cdot}1$, and the six products $z_m'z_l$ $(m < l)$ are 0; for instance $z_1'z_2 = 0+15-15-15+15 = 0$ and $z_1'z_3 = 0-5-5+5+5 = 0$.

- $\mathrm{tr}\,S = 10+10+10+10 = 40 = 25+9+5+1$, so (Proposition 9.32) $\mathrm{PVE}_1 = \frac{25}{40} = \frac{5}{8}$, $\mathrm{PVE}_2 = \frac{9}{40}$, $\mathrm{PVE}_3 = \frac{5}{40} = \frac{1}{8}$, $\mathrm{PVE}_4 = \frac{1}{40}$: the first component explains $\frac{5}{8}$ of the total variance, the first two together $\frac{34}{40} = \frac{17}{20}$.

Remark 9.35 ([19, §12.2.4], [9, Remark after Proposition 5.24])**.** Usually only the first M principal components are kept, with M much smaller than p; their score vectors $z_1, \dots, z_M$ summarize the data, for instance in a plot of the units. By Proposition 9.32, they explain together the proportion $\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_M$ of the total variance, and $\mathrm{PVE}_1 \geqslant \mathrm{PVE}_2 \geqslant \cdots$. A common aid is the **scree plot**, the points (m, PVE_m), $m = 1, \dots, p$, joined by segments: one takes the smallest M that explains a sizable part of the total variance, typically the M after which the PVE_m drop off and stay small (an **elbow** of the plot after the Mth component). There is no objective rule; the choice depends on the data and on the purpose. Corollary 9.29, applied to S, shows what is gained: the sample variances of any M of the variables add up to at most $\lambda_1 + \cdots + \lambda_M$. So no M of the original variables explain a larger part of the total variance than the first M principal components. More generally, by Theorem 9.27(a) for S and Proposition 9.14(b), the sample variances of M combinations $Xg_1, \dots, Xg_M$ with orthonormal coefficient vectors $g_1, \dots, g_M$ add up to $\mathrm{tr}(G'SG) \leqslant \lambda_1 + \cdots + \lambda_M$, where $G = [g_1 \ \cdots \ g_M]$ ($\mathrm{tr}(G'SG) = \sum_j g_j' S g_j$, Theorem 1.4(d)): no M orthonormal coefficient vectors retain a larger total variance than the first M loading vectors.

Example 9.36 (cf. [19, §12.2.4, Figure 12.3])**.** Four tests are given to the same students: two verbal tests (variables 1 and 2) and two quantitative tests (variables 3 and 4). The sample covariance matrix of the four scores is

$$S = \begin{pmatrix} 5 & 4 & 2 & 2 \\ 4 & 5 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 2 & 4 & 5 \end{pmatrix}$$

Find the principal components and the proportions of variance explained, and decide how many components to keep.

Solution.

- Let $h_1 = (1,1,1,1)'$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$ and $h_4 = (1,-1,-1,1)'$. Row by row,

$$Sh_1 = \begin{pmatrix} 13 \\ 13 \\ 13 \\ 13 \end{pmatrix} = 13h_1, \qquad Sh_2 = \begin{pmatrix} 5 \\ 5 \\ -5 \\ -5 \end{pmatrix} = 5h_2, \qquad Sh_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} = h_3, \qquad Sh_4 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} = h_4$$

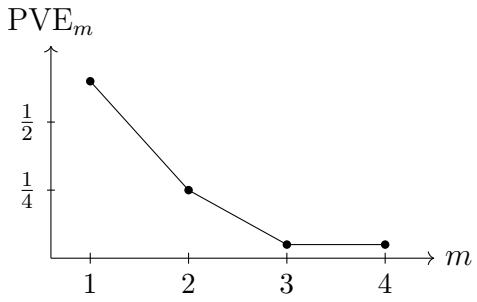
  for instance $5+4+2+2 = 13$ and $5+4-2-2 = 5$ in row 1. The h_i are pairwise orthogonal with $\|h_i\| = 2$, so $v_i = \frac{1}{2}h_i$ $(i = 1, \dots, 4)$ are orthonormal eigenvectors, and the eigenvalues of S are $13, 5, 1, 1$ (Theorem 8.3(c)).

- By Theorem 9.31(c),(e) ($13 > 5 > 1$), the first two loading vectors are $\pm v_1$ and $\pm v_2$. The first principal component, with coefficient vector $\frac{1}{2}(1,1,1,1)'$ and sample variance 13, is half the total score: an overall level. The second, with coefficient vector $\frac{1}{2}(1,1,-1,-1)'$ and sample variance 5, contrasts the verbal with the quantitative scores. As $\lambda_3 = \lambda_4 = 1$, the third loading vector is not unique: besides v_3, also $a = \frac{1}{\sqrt{2}}(v_3 + v_4) = \frac{1}{\sqrt{2}}(1,-1,0,0)'$ is a unit vector orthogonal to v_1, v_2, and $S(1,-1,0,0)' = (1,-1,0,0)'$ gives $a'Sa = 1 = \lambda_3$, the maximum of the third problem (Theorem 9.31(b),(c)).

- $\mathrm{tr}\, S = 20 = 13 + 5 + 1 + 1$, so (Proposition 9.32)

m	1	2	3	4
PVE_m	$\frac{13}{20}$	$\frac{5}{20} = \frac{1}{4}$	$\frac{1}{20}$	$\frac{1}{20}$
$\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_m$	$\frac{13}{20}$	$\frac{18}{20} = \frac{9}{10}$	$\frac{19}{20}$	1

  The scree plot (Remark 9.35)



falls steeply up to $m = 3$ and is flat afterwards: there is an elbow after the second component. Two components, which explain 90% of the total variance, are kept.

- Any two of the four scores have sample variances adding up to 10, at most $13 + 5 = 18$, as Corollary 9.29 says: the two verbal scores carry $\frac{10}{20} = 50\%$ of the total variance, the first two principal components 90%.

Standardized variables

The principal components of the standardized data matrix X_s (Definition 9.30) are obtained from R: by Proposition 9.18(a) and Theorem 9.31, the loading vectors v_m are orthonormal eigenvectors of R, the score vectors are $C_n X_s v_m = X_s v_m$, and, as $\operatorname{tr} R = p$, $\mathrm{PVE}_m = \lambda_m(R)/p$ (Proposition 9.32). By Proposition 9.18(b), they do not depend on the units of the variables, whereas those of S do: by Corollary 9.29, $\lambda_1(S) \geqslant s_{jj}$ for every j, so a variable with a much larger sample variance than the others forces a large first eigenvalue of S. Variables measured in different units are therefore usually standardized before the analysis; variables measured in the same units may be left as they are [19, §12.2.4], [9, Proposition 5.24]. For two variables with sample correlation $r = r_{12}$, for instance, $R = \left(\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}\right)$ has $R(1,1)' = (1+r)(1,1)'$ and $R(1,-1)' = (1-r)(1,-1)'$: if $r \neq 0$, the principal components of the standardized variables are their sum and their difference, divided by $\sqrt{2}$, the sum first if $r > 0$, and the first one explains the proportion $(1 + |r|)/2$.

Example 9.37 (cf. [19, §12.2.4, Figure 12.4])**.** The sample covariance matrix of four variables is

$$S = \begin{pmatrix} 4 & -2 & 0 & 0 \\ -2 & 4 & 0 & 0 \\ 0 & 0 & 16 & 4 \\ 0 & 0 & 4 & 16 \end{pmatrix}$$

Find the sample correlation matrix, and compare the principal components of the variables with those of the standardized variables.

Solution.

- The sample standard deviations are 2, 2, 4, 4, so $D_s^{1/2} = \operatorname{diag}(2,2,4,4)$, $r_{12} = \frac{-2}{2 \cdot 2} = -\frac{1}{2}$, $r_{34} = \frac{4}{4 \cdot 4} = \frac{1}{4}$, the other r_{jl} $(j \neq l)$ are 0, and

$$R = D_s^{-1/2} S D_s^{-1/2} = \begin{pmatrix} 1 & -\frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & \frac{1}{4} & 1 \end{pmatrix}$$

- The variables. Row by row, $S(0,0,1,1)' = (0,0,20,20)'$, $S(0,0,1,-1)' = (0,0,12,-12)'$, $S(1,-1,0,0)' = (6,-6,0,0)'$ and $S(1,1,0,0)' = (2,2,0,0)'$. These four vectors are pairwise orthogonal, so the eigenvalues of S are $20, 12, 6, 2$ (Theorem 8.3(c)). They are distinct, so the loading vectors are, up to sign, $\frac{1}{\sqrt{2}}(0,0,1,1)'$, $\frac{1}{\sqrt{2}}(0,0,1,-1)'$, $\frac{1}{\sqrt{2}}(1,-1,0,0)'$ and $\frac{1}{\sqrt{2}}(1,1,0,0)'$ (Theorem 9.31(c),(e)), with sample variances 20, 12, 6, 2; as $\operatorname{tr} S = 40$, the proportions of variance explained are $\frac{20}{40} = \frac{1}{2}$, $\frac{12}{40} = \frac{3}{10}$, $\frac{6}{40} = \frac{3}{20}$ and $\frac{2}{40} = \frac{1}{20}$.

- The standardized variables. In the same way $R(1,-1,0,0)' = \frac{3}{2}(1,-1,0,0)'$, $R(0,0,1,1)' = \frac{5}{4}(0,0,1,1)'$, $R(0,0,1,-1)' = \frac{3}{4}(0,0,1,-1)'$ and $R(1,1,0,0)' = \frac{1}{2}(1,1,0,0)'$. The loading vectors are, up to sign, $\frac{1}{\sqrt{2}}(1,-1,0,0)'$, $\frac{1}{\sqrt{2}}(0,0,1,1)'$, $\frac{1}{\sqrt{2}}(0,0,1,-1)'$ and $\frac{1}{\sqrt{2}}(1,1,0,0)'$, and, as $\operatorname{tr} R = 4$, the proportions are $\frac{3}{8}$, $\frac{5}{16}$, $\frac{3}{16}$ and $\frac{1}{8}$.
- Comparison. Variables 3 and 4 have the largest sample variances, 16, and are uncorrelated with variables 1 and 2; the eigenvalues 20 and 12 that belong to them exceed the eigenvalues 6 and 2 that belong to variables 1 and 2. So the first two principal components of the variables are combinations of variables 3 and 4 only, and $\lambda_1(S) = 20 \geqslant s_{33} = 16$, as Corollary 9.29 says. Of the standardized variables, the first principal component is the contrast of variables 1 and 2, which are more strongly correlated ($r_{12} = -\frac{1}{2}$) than variables 3 and 4 ($r_{34} = \frac{1}{4}$). A change of the units of variables 3 and 4 would change S, but not R (Proposition 9.18(b)).

The closest subspaces

Definition 9.38 ([9, Theorem 5.22])**.** For $a \in \mathbb{R}^p$ and a subspace W of $\mathbb{R}^p$, the set $a + W = \{a + w : w \in W\}$ is an **affine subspace** of dimension $\dim W$: a line through a if $\dim W = 1$, a plane if $\dim W = 2$. The **distance** from $y \in \mathbb{R}^p$ to $a + W$ is $\operatorname{dist}(y, a + W) = \min_{w \in W} \|y - a - w\|$. By Theorem 6.23, the minimum is attained exactly at the orthogonal projection $\hat{w}$ of $y - a$ onto W; if the columns of $G \in \mathbb{R}^{p \times M}$ form an orthonormal basis of W, then $\hat{w} = GG'(y - a)$ (Theorem 6.31(a),(b): $P_W = GG'$), and

$$\operatorname{dist}(y, a + W)^2 = \left\| (I_p - GG')(y - a) \right\|^2$$

Theorem 9.39 ([9, Theorem 5.22])**.** Let X, S, λ_m, v_m, z_m and r be as in Theorem 9.31, let $1 \leqslant M \leqslant p$, and put $V_M = [v_1 \ \cdots \ v_M] \in \mathbb{R}^{p \times M}$ and $Z_M = C_n X V_M = [z_1 \ \cdots \ z_M] \in \mathbb{R}^{n \times M}$.

(a) For all $A \in \mathbb{R}^{n \times M}$ and $B \in \mathbb{R}^{p \times M}$, $\|C_n X - AB'\|_F^2 \geqslant (n-1)(\lambda_{M+1} + \cdots + \lambda_p)$, with equality for $A = Z_M$, $B = V_M$. If $M \geqslant r$, then $C_n X = Z_M V_M'$.

(b) $\|C_n X\|_F^2 = \|z_1\|^2 + \cdots + \|z_M\|^2 + \|C_n X - Z_M V_M'\|_F^2$, so

$$\mathrm{PVE}_1 + \cdots + \mathrm{PVE}_M = 1 - \frac{\|C_n X - Z_M V_M'\|_F^2}{\|C_n X\|_F^2}$$

(c) Among all affine subspaces $a + W$ of $\mathbb{R}^p$ with $\dim W = M$, the sum of squared distances $\sum_{i=1}^n \operatorname{dist}(x_i, a + W)^2$ is smallest for $\bar{x} + \mathcal{C}(V_M)$, where it equals $(n-1)(\lambda_{M+1} + \cdots + \lambda_p)$. If $M < p$ and $\lambda_M > \lambda_{M+1}$, no other M-dimensional affine subspace gives this value.

Proof. (a)

- Equality: $V_M' V_M = I_M$, so $N = I_p - V_M V_M'$ is symmetric and idempotent, and $C_n X - Z_M V_M' = C_n X N$. By Definition 9.26, Proposition 9.14(a), Theorem 1.11(b) and $V_M' S V_M = \operatorname{diag}(\lambda_1, \ldots, \lambda_M)$,

$$\begin{aligned} \|C_n X N\|_F^2 &= \operatorname{tr}\Big(N (C_n X)' (C_n X) N\Big) = (n-1) \operatorname{tr}(S N^2) \\ &= (n-1)\Big(\operatorname{tr} S - \operatorname{tr}(V_M' S V_M)\Big) = (n-1)(\lambda_{M+1} + \cdots + \lambda_p) \end{aligned}$$

  using $\operatorname{tr} S = \lambda_1 + \cdots + \lambda_p$ (Proposition 9.32). For $M \geqslant r$ this is 0 (Theorem 9.31(a)), so $C_n X = Z_M V_M'$ (Theorem 1.11(d)).

- The bound. Let $W = \mathcal{C}(B)$, $P = P_W$ and $N_W = I_p - P$, and let $d = \dim W = \operatorname{rank} B$; then $d \leqslant M$ (Definition 4.11, Theorem 4.13(a)). Row i of AB' is $a_i' B' = (Ba_i)'$, where a_i' is row i of A (Theorems 1.4(c) and 1.9), and $Ba_i \in W$ (Theorem 1.4(b)); row i of $C_n X$ is $(x_i - \bar{x})'$. By Theorem 6.23 and Theorem 6.31(a), $\|(x_i - \bar{x}) - Ba_i\|^2 \geqslant \|N_W(x_i - \bar{x})\|^2$. Summing over i (Definition 9.26), and as row i of $C_n X N_W$ is $(N_W(x_i - \bar{x}))'$ (N_W is symmetric, Theorem 6.31(c)),

$$\|C_n X - AB'\|_F^2 \geqslant \sum_{i=1}^n \|N_W(x_i - \bar{x})\|^2 = \|C_n X N_W\|_F^2$$

- $\operatorname{tr}(PS) \leqslant \lambda_1 + \cdots + \lambda_M$ if $d = 0$ or $d = M$. If $d = 0$, then $P = 0$, and $\lambda_1 \geqslant \cdots \geqslant \lambda_M \geqslant 0$ (Theorem 9.31(a)). If $d = M$, let the columns of $G \in \mathbb{R}^{p\times M}$ form an orthonormal basis of W (Corollary 6.15(a)); then $P = GG'$ (Theorem 6.31(b)), and $\operatorname{tr}(PS) = \operatorname{tr}(G'SG) \leqslant \lambda_1 + \cdots + \lambda_M$ (Theorem 1.11(b), and Theorem 9.27(a) for S).
- $\operatorname{tr}(PS) \leqslant \lambda_1 + \cdots + \lambda_M$ if $1 \leqslant d < M$. Let the columns of $Q_1 \in \mathbb{R}^{p\times d}$ form an orthonormal basis of W (Corollary 6.15(a)), so that $P = Q_1Q_1'$ (Theorem 6.31(b)). As $d < p$, Corollary 6.15(c) gives T with $[Q_1\ T]$ orthogonal; let Q_2 consist of the first $M - d$ columns of T, and $G = [Q_1\ Q_2] \in \mathbb{R}^{p\times M}$, so that $G'G = I_M$ and $GG' = P + Q_2Q_2'$ (Theorem 5.3(c)). The diagonal entries $q'Sq$ of $Q_2'SQ_2$ are $\geqslant 0$, as S is PSD (Proposition 9.14(a)). So by Theorem 1.11(a),(b) and Theorem 9.27(a) for S,

$$\operatorname{tr}(PS) = \operatorname{tr}(GG'S) - \operatorname{tr}(Q_2Q_2'S) = \operatorname{tr}(G'SG) - \operatorname{tr}(Q_2'SQ_2) \leqslant \operatorname{tr}(G'SG) \leqslant \lambda_1 + \cdots + \lambda_M$$

- N_W is symmetric and idempotent (Theorem 6.31(c)); as in the first item, $\|C_nXN_W\|_F^2 = (n-1)\operatorname{tr}(SN_W) = (n-1)\Big(\operatorname{tr} S - \operatorname{tr}(PS)\Big) \geqslant (n-1)(\lambda_{M+1} + \cdots + \lambda_p)$.

(b) $\|C_nX\|_F^2 = (n-1)\operatorname{tr} S = (n-1)(\lambda_1 + \cdots + \lambda_p)$ (Proposition 9.14(a), Proposition 9.32) and $\|z_m\|^2 = (n-1)\lambda_m$ (Theorem 9.31(d)); with (a), both sides of the identity equal $(n-1)\operatorname{tr} S$. Divide by $\|C_nX\|_F^2$ and use the first expression of PVE_m in Proposition 9.32.

(c) Let $\dim W = M$, let the columns of $G \in \mathbb{R}^{p\times M}$ form an orthonormal basis of W (Theorems 4.4 and 6.14), and put $N = I_p - GG'$, symmetric and idempotent.

- By Definition 9.38, $\operatorname{dist}(x_i, a + W)^2 = \|N(x_i - a)\|^2$. Write $x_i - a = (x_i - \bar{x}) + (\bar{x} - a)$. As $\sum_i (x_i - \bar{x}) = X'\mathbf{1} - n\bar{x} = 0$, expanding the squares (Proposition 6.2) gives

$$\sum_{i=1}^n \operatorname{dist}(x_i, a + W)^2 = \sum_{i=1}^n \|N(x_i - \bar{x})\|^2 + n\|N(\bar{x} - a)\|^2$$

- Row i of C_nXN is $(x_i - \bar{x})'N = (N(x_i - \bar{x}))'$, so, as in (a), and by Theorem 9.27(a) for S,

$$\sum_i \|N(x_i - \bar{x})\|^2 = \|C_nXN\|_F^2 = (n-1)\operatorname{tr}(SN) = (n-1)\Big(\operatorname{tr} S - \operatorname{tr}(G'SG)\Big)$$
$$\geqslant (n-1)(\lambda_{M+1} + \cdots + \lambda_p)$$

- Hence the sum of squared distances is at least $(n-1)(\lambda_{M+1} + \cdots + \lambda_p)$. For $a = \bar{x}$ and $G = V_M$, the second term is 0 and $\operatorname{tr}(V_M'SV_M) = \lambda_1 + \cdots + \lambda_M$: the bound is attained.
- If the bound is attained, then $N(\bar{x} - a) = 0$ and $\operatorname{tr}(G'SG) = \lambda_1 + \cdots + \lambda_M$. If $M < p$ and $\lambda_M > \lambda_{M+1}$, Theorem 9.27(c) gives $W = \mathcal{C}(G) = \mathcal{C}(V_M)$. And $N(\bar{x} - a) = 0$ means $\bar{x} - a = GG'(\bar{x} - a) \in W$, so $a + W = \bar{x} + W$. □

Part (a) says that the matrix Z_MV_M', of rank at most M, built from the first M principal components is a best approximation of the centered data among the products AB' with M columns in A and B. Every matrix of rank $s \leqslant M$ is such a product ($0 = 0 \cdot 0'$, and for $s \geqslant 1$ a full-rank factorization, Theorem 4.21(a), with zero columns added), so Z_MV_M' is also a best approximation among the matrices of rank at most M. Part (b) says that the cumulative proportion of variance explained is the part of $\|C_nX\|_F^2$ that it reproduces.

Example 9.40. For the data of Example 9.15, $\|C_5X\|_F^2 = 40 + 40 + 40 + 40 = 160 = 4 \cdot 40$ (Example 9.34: $\operatorname{tr} S = 40$).

- $M = 1$: $Z_1V_1' = z_1v_1' = \frac{1}{2}(0, 5, 5, -5, -5)'(1, 1, 1, 1)$ has the rows $0'$, $\frac{5}{2}\mathbf{1}'$ twice and $-\frac{5}{2}\mathbf{1}'$ twice, and

$$C_5X - Z_1V_1' = \begin{pmatrix} 2 & -2 & 2 & -2 \\ \frac{3}{2} & \frac{3}{2} & -\frac{5}{2} & -\frac{1}{2} \\ -\frac{5}{2} & -\frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ \frac{1}{2} & \frac{5}{2} & -\frac{3}{2} & -\frac{3}{2} \\ -\frac{3}{2} & -\frac{3}{2} & \frac{1}{2} & \frac{5}{2} \end{pmatrix}$$

with the squared row lengths 16, 11, 11, 11, 11; so $\|C_5X - Z_1V_1'\|_F^2 = 16+11+11+11+11 = 60 = 4(9+5+1)$. By Theorem 9.39(c), the line $\bar{x} + \operatorname{span}\{(1,1,1,1)'\}$ is the line closest to the five points (the only one, as $25 > 9$), with the sum of squared distances 60; and $1 - \frac{60}{160} = \frac{5}{8} = \mathrm{PVE}_1$ (Theorem 9.39(b)).

- $M = 2$: $Z_2V_2' = z_1v_1' + z_2v_2'$, with $z_2v_2' = \frac{1}{2}(0,3,-3,3,-3)'(1,1,-1,-1)$. So

$$Z_2V_2' = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 4 & 4 & 1 & 1 \\ 1 & 1 & 4 & 4 \\ -1 & -1 & -4 & -4 \\ -4 & -4 & -1 & -1 \end{pmatrix}, \qquad C_5X - Z_2V_2' = \begin{pmatrix} 2 & -2 & 2 & -2 \\ 0 & 0 & -1 & 1 \\ -1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

  $\|C_5X - Z_2V_2'\|_F^2 = 16+2+2+2+2 = 24 = 4(5+1)$, and $1 - \frac{24}{160} = \frac{17}{20} = \mathrm{PVE}_1 + \mathrm{PVE}_2$. By Theorem 9.39(a) with $M = 4 \geqslant r$, $C_5X - Z_2V_2' = z_3v_3' + z_4v_4'$; for instance its first row is $4 \cdot \frac{1}{2}(1,-1,1,-1) + 0$.

- The closest plane (Theorem 9.39(c), unique as $9 > 5$) is $\bar{x} + \operatorname{span}\{v_1, v_2\}$. As $\operatorname{span}\{v_1,v_2\} = \operatorname{span}\{v_3,v_4\}^\perp$ (the first lies in the second, and both have dimension 2: Theorems 6.6(c) and 4.5(a)), it is $\{y \in \mathbb{R}^4 : h_3'(y-\bar{x}) = h_4'(y-\bar{x}) = 0\} = \{y : y_1 = y_2,\ y_3 = y_4\}$, since $h_3'u + h_4'u = 2(u_1-u_2)$ and $h_3'u - h_4'u = 2(u_3-u_4)$. The point of the plane nearest to unit i is $\bar{x} + V_2V_2'(x_i - \bar{x})$ (Definition 9.38), whose transpose is $\bar{x}'$ plus row i of Z_2V_2': unit 1, $(7,3,7,3)$, has the nearest point $(5,5,5,5)$, at squared distance 16; units 2–5 have the nearest points $(9,9,6,6)$, $(6,6,9,9)$, $(4,4,1,1)$, $(1,1,4,4)$, at squared distance 2 each.

Principal components of a random vector

Definition 9.41 ([3, §10.1])**.** Let $x = (x_1, \ldots, x_p)'$ be a random vector with covariance matrix $\Sigma = \operatorname{var}(x) \neq 0$ (Definition 9.1). A **first principal component** of x is a random variable $a_1'x$, where the unit vector $a_1 \in \mathbb{R}^p$ maximizes $\operatorname{var}(a'x)$ over all unit vectors $a \in \mathbb{R}^p$. For $m \geqslant 2$, once $a_1, \ldots, a_{m-1}$ are chosen, an m**th principal component** is $a_m'x$, where the unit vector a_m maximizes $\operatorname{var}(a'x)$ over the unit vectors a for which $a'x$ is uncorrelated with (has covariance 0 with) each of $a_1'x, \ldots, a_{m-1}'x$.

Theorem 9.42 ([3, §10.1])**.** Let x and Σ be as in Definition 9.41, and let $\Sigma = V\Lambda V'$, where $V = [v_1 \ \cdots \ v_p]$ is orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ with $\lambda_1 \geqslant \cdots \geqslant \lambda_p$ (Lemma 9.22(a)).

(a) For $a, b \in \mathbb{R}^p$, $\operatorname{var}(a'x) = a'\Sigma a$ and $\operatorname{cov}(a'x, b'x) = a'\Sigma b$; and $\lambda_p \geqslant 0$.

(b) $v_1'x$ is a first principal component, with variance λ_1. Let $m \geqslant 2$ and $\lambda_{m-1} > 0$. If $v_1'x, \ldots, v_{m-1}'x$ are taken as the first $m-1$ principal components, then $v_m'x$ is an mth principal component, with variance λ_m.

(c) The principal components $v_1'x, \ldots, v_p'x$ are pairwise uncorrelated, and the **total variance** of x is $\operatorname{var}(x_1) + \cdots + \operatorname{var}(x_p) = \operatorname{tr}\Sigma = \lambda_1 + \cdots + \lambda_p$. So the first M principal components account for the proportion $(\lambda_1 + \cdots + \lambda_M)/\operatorname{tr}\Sigma$ of the total variance.

Proof. (a)

- Let $T \in \mathbb{R}^{2\times p}$ have the rows a' and b', so that $Tx = (a'x, b'x)'$ (Theorem 1.4(c)). By Theorem 9.3(b), $\operatorname{var}(Tx) = T\Sigma T'$, whose entries are $a'\Sigma a$, $a'\Sigma b$ in the first row and $b'\Sigma a$, $b'\Sigma b$ in the second (Theorem 1.4(d)).
- By Definition 9.1, the diagonal entries of $\operatorname{var}(Tx)$ are $\operatorname{var}(a'x)$ and $\operatorname{var}(b'x)$, and its $(1,2)$ entry is $\operatorname{cov}(a'x, b'x)$. (This is also Theorem 9.4(a) for $u = v = x$ with the rows $A = a'$ and $C = b'$.)
- Σ is PSD (Theorem 9.3(c)), so $\lambda_p \geqslant 0$ (Theorem 8.21(a)).

(b)

- By (a), the first problem is to maximize $a'\Sigma a$ over the unit vectors a, and for $a \neq 0$ the Rayleigh quotient $a'\Sigma a/a'a$ is the value at the unit vector $a/\|a\|$. By Theorem 9.23(a) the maximum is λ_1, attained at $a = v_1$.
- Let $m \geqslant 2$ and $\lambda_{m-1} > 0$; then $\lambda_l > 0$ for every $l < m$. By (a) and $\Sigma v_l = \lambda_l v_l$, $\operatorname{cov}(a'x, v_l'x) = a'\Sigma v_l = \lambda_l a'v_l$, which is 0 if and only if $a \perp v_l$. So the mth problem is to maximize $a'\Sigma a$ over the unit vectors $a \perp v_1, \ldots, v_{m-1}$.
- By Theorem 9.25, the largest value of $a'\Sigma a/a'a$ over the nonzero $a \perp v_1, \ldots, v_{m-1}$ is λ_m, attained at $a = v_m$; so v_m solves the mth problem, and $\operatorname{var}(v_m'x) = v_m'\Sigma v_m = \lambda_m$.

(c)

- By (a), $\operatorname{cov}(v_m'x, v_l'x) = v_m'\Sigma v_l = \lambda_l v_m'v_l = 0$ for $l \neq m$.
- $\operatorname{var}(x_j) = \sigma_{jj}$ (Definition 9.1), so the total variance is $\operatorname{tr}\Sigma = \operatorname{tr}(V\Lambda V') = \operatorname{tr}(\Lambda V'V) = \operatorname{tr}\Lambda = \lambda_1 + \cdots + \lambda_p$ (Theorem 1.11(b)). It is positive, so the proportions are defined: every $\lambda_m \geqslant 0$ by (a), and $\Lambda \neq 0$, as $\Sigma = V\Lambda V' \neq 0$. □

Parts (b) and (c) are Theorem 9.31(c),(d) and Proposition 9.32 with Σ in place of S. In practice Σ is not known: n observations of x form the rows of a data matrix, and the principal components of the data (Definition 9.30), computed from S, take the place of those of x cf. [3, §10.1], [2, §10.3].

Example 9.43 (cf. [2, Example 10.1])**.** Let $x = (x_1, \ldots, x_p)'$ ($p \geqslant 2$) have $\operatorname{var}(x_i) = \sigma^2 > 0$ and $\operatorname{cov}(x_i, x_j) = \rho\sigma^2$ for $i \neq j$, where $0 < \rho < 1$: $\Sigma = \sigma^2\big((1-\rho)I_p + \rho J_p\big)$, and $\operatorname{cor}(x)$ is the equicorrelation matrix of Example 5.33. Find the principal components of x.

Solution.

- By Example 8.16(c), $\Sigma = \sigma^2\big(1 + (p-1)\rho\big)\bar{J}_p + \sigma^2(1-\rho)C_p$, and by Example 8.16(b) its eigenvalues are $\lambda_1 = \sigma^2\big(1 + (p-1)\rho\big)$, with eigenspace $\operatorname{span}\{\mathbf{1}_p\}$, and $\lambda_2 = \cdots = \lambda_p = \sigma^2(1-\rho)$, with eigenspace $\operatorname{span}\{\mathbf{1}_p\}^\perp$. As $0 < \rho < 1$, $\lambda_1 > \lambda_2 > 0$.
- First component. With $v_1 = \frac{1}{\sqrt{p}}\mathbf{1}_p$, $v_1'x = \frac{1}{\sqrt{p}}(x_1 + \cdots + x_p)$, a multiple of the average of the p variables, with variance $\sigma^2\big(1 + (p-1)\rho\big)$ (Theorem 9.42(b)). It is unique up to sign: a unit vector a with $a'\Sigma a = \lambda_1$ is an eigenvector for λ_1 (Theorem 9.23(b)), hence $a = \pm v_1$.
- Second component. $\operatorname{cov}(a'x, v_1'x) = \lambda_1 a'v_1$ (Theorem 9.42(a)), so the constraint of the second problem is $\mathbf{1}_p'a = 0$; and for every such unit vector a, $\Sigma a = \lambda_2 a$ and $\operatorname{var}(a'x) = \lambda_2$. So every $a'x$ with $\mathbf{1}_p'a = 0$ and $\|a\| = 1$ is a second principal component: for $p \geqslant 3$ the second component is not unique, even up to sign (for $p = 2$ it is $\pm(x_1 - x_2)/\sqrt{2}$).
- The first component accounts for the proportion $\lambda_1/\operatorname{tr}\Sigma = \big(1 + (p-1)\rho\big)/p$ of the total variance $\operatorname{tr}\Sigma = p\sigma^2$ (Theorem 9.42(c)); for $p = 4$ and $\rho = \frac{1}{2}$ it is $\frac{5}{8}$.

Exercises

Exercise 9.1. Let $z = (z_1, z_2, z_3, z_4)'$ be a random vector, and let

$$\mu = \mathsf{E}(z) = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}, \qquad \Sigma = \operatorname{var}(z) = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \qquad b = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

(a) Find $\mathsf{E}(Az + b)$ and $\operatorname{var}(Az + b)$, and $\operatorname{cov}(z_1 + z_2,\ z_3 - z_4)$.

(b) Find $\operatorname{var}(z_1 + z_2 + z_3 + z_4)$ and $\operatorname{var}(z_1 - z_4)$.

(c) Find $\mathsf{E}(z'z)$ and $\mathsf{E}(2z_1z_2 + z_3^2 + 2z_3z_4)$.

Solution. (a) By Theorem 9.3(b),

$$\mathsf{E}(Az + b) = A\mu + b = \begin{pmatrix} 1 - 1 \\ 2 - 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$$

and, with $A\Sigma$ computed row by row (its rows are $\operatorname{row}_1(\Sigma) + \operatorname{row}_2(\Sigma)$ and $\operatorname{row}_3(\Sigma) - \operatorname{row}_4(\Sigma)$, Theorem 1.4(c)),

$$A\Sigma = \begin{pmatrix} 3 & 3 & 1 & 0 \\ 0 & 1 & 1 & -1 \end{pmatrix}, \qquad \operatorname{var}(Az + b) = A\Sigma A' = \begin{pmatrix} 6 & 1 \\ 1 & 2 \end{pmatrix}$$

With $b = 0$, Theorem 9.3(b) gives $\operatorname{var}(Az) = A\Sigma A'$ as well. The entries of Az are $z_1 + z_2$ and $z_3 - z_4$, so their covariance is the $(1, 2)$ entry of $A\Sigma A'$ (Definition 9.1): $\operatorname{cov}(z_1 + z_2,\ z_3 - z_4) = 1$.

(b) By Theorem 9.3(b) with the 1×4 matrix a', $\operatorname{var}(a'z) = a'\Sigma a$. For $a = \mathbf{1}_4$ this is the sum of all entries of Σ: $\operatorname{var}(z_1 + z_2 + z_3 + z_4) = 3 + 4 + 4 + 3 = 14$, the row sums of Σ being $3, 4, 4, 3$. For $a = (1, 0, 0, -1)'$, $\operatorname{var}(z_1 - z_4) = \sigma_{11} - 2\sigma_{14} + \sigma_{44} = 2 - 0 + 2 = 4$.

(c) By Theorem 9.10 with $A = I_4$, $\mathsf{E}(z'z) = \operatorname{tr}\Sigma + \mu'\mu = (2 + 2 + 2 + 2) + (1 + 1 + 4 + 1) = 15$. Next, $2z_1z_2 + z_3^2 + 2z_3z_4 = z'Bz$ for the symmetric B below (Definition 8.17), and

$$B = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\operatorname{tr}(B\Sigma) = 2\sigma_{12} + \sigma_{33} + 2\sigma_{34} = 2 + 2 + 2 = 6, \qquad \mu'B\mu = 2\mu_1\mu_2 + \mu_3^2 + 2\mu_3\mu_4 = -2 + 4 + 4 = 6$$

so $\mathsf{E}(2z_1z_2 + z_3^2 + 2z_3z_4) = \operatorname{tr}(B\Sigma) + \mu'B\mu = 12$. Check: by Theorem 9.4(e), whose (i, j) entry reads $\mathsf{E}(z_iz_j) = \sigma_{ij} + \mu_i\mu_j$, $\mathsf{E}(z_1z_2) = 1 - 1 = 0$, $\mathsf{E}(z_3^2) = 2 + 4 = 6$ and $\mathsf{E}(z_3z_4) = 1 + 2 = 3$, and $2 \cdot 0 + 6 + 2 \cdot 3 = 12$.

Exercise 9.2. Let $x \in \mathbb{R}^4$ be a random vector with $\mathsf{E}(x) = \mu = (1, 0, 2, -1)'$ and

$$\operatorname{var}(x) = \Sigma = \begin{pmatrix} 9 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \\ 3 & 2 & 9 & 5 \\ 3 & 1 & 5 & 4 \end{pmatrix}, \qquad L = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

(a) Show that $\Sigma = LL'$, and deduce that there is no nonzero $a \in \mathbb{R}^4$ with $a'x = a'\mu$ with probability 1.

(b) Find $\mathsf{E}(y)$ and $\operatorname{var}(y)$ for $y = (x_1 - x_2,\ x_3 - x_4 - 3)'$.

(c) Find $\operatorname{cov}(x_1 + x_3,\ x_2 - x_4)$ and $\operatorname{cor}(x)$.

Solution. (a) The (i,j) entry of LL' is $\operatorname{row}_i(L)\operatorname{row}_j(L)'$ (Theorem 1.4(d)), and LL' is symmetric (Theorem 1.9). With the rows $(3,0,0,0)$, $(0,1,0,0)$, $(1,2,2,0)$, $(1,1,1,1)$ this gives $9,0,3,3$ in row 1; $1,2,1$ at $(2,2)$, $(2,3)$, $(2,4)$; $1+4+4=9$ and $1+2+2=5$ at $(3,3)$, $(3,4)$; and $1+1+1+1=4$ at $(4,4)$. So $LL' = \Sigma$. As L is lower triangular, $\det L = 3\cdot 1\cdot 2\cdot 1 = 6 \neq 0$ (Theorem 5.22), so $P = L'$ is invertible (Theorem 5.18(a)), and $\Sigma = P'P$ is PD (Theorem 8.26(c)), hence invertible (Theorem 8.21(c)). By Theorem 9.5, there is no nonzero a with $a'x = a'\mu$ with probability 1.

(b) $y = Ax + b$ with $A = \left(\begin{smallmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{smallmatrix}\right)$ and $b = (0,-3)'$. By Theorem 9.3(b), $\mathsf{E}(y) = A\mu + b = (1-0,\ 2+1-3)' = (1,0)'$, and the rows of $A\Sigma$ are $\operatorname{row}_1(\Sigma) - \operatorname{row}_2(\Sigma)$ and $\operatorname{row}_3(\Sigma) - \operatorname{row}_4(\Sigma)$:

$$A\Sigma = \begin{pmatrix} 9 & -1 & 1 & 2 \\ 0 & 1 & 4 & 1 \end{pmatrix}, \qquad \operatorname{var}(y) = A\Sigma A' = \begin{pmatrix} 10 & -1 \\ -1 & 3 \end{pmatrix}$$

(c) $\Sigma(0,1,0,-1)' = \operatorname{col}_2(\Sigma) - \operatorname{col}_4(\Sigma) = (-3,0,-3,-3)'$, so by Theorem 9.4(a) $\operatorname{cov}(x_1+x_3,\ x_2-x_4) = (1,0,1,0)(-3,0,-3,-3)' = -6$. The standard deviations are $3,1,3,2$, so $D^{-1/2} = \operatorname{diag}(\frac{1}{3}, 1, \frac{1}{3}, \frac{1}{2})$ and (Definition 9.7)

$$\operatorname{cor}(x) = D^{-1/2}\Sigma D^{-1/2} = \begin{pmatrix} 1 & 0 & \frac{1}{3} & \frac{1}{2} \\ 0 & 1 & \frac{2}{3} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} & 1 & \frac{5}{6} \\ \frac{1}{2} & \frac{1}{2} & \frac{5}{6} & 1 \end{pmatrix}$$

Exercise 9.3. Let $z \in \mathbb{R}^4$ be a random vector, and let $u = Az$ and $v = \mathbf{1}'z = z_1 + z_2 + z_3 + z_4$, where

$$\operatorname{var}(z) = \Sigma = \begin{pmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

so that $u = (z_1 - z_2,\ z_1 + z_2 + z_3 + 2z_4)'$.

(a) Find $\operatorname{cov}(u, z)$ and $\operatorname{cov}(u, v)$.

(b) Find $\operatorname{var}(u)$, $\operatorname{var}(v)$ and the covariance matrix of $(u', v)' \in \mathbb{R}^3$.

(c) Find the correlation matrix of z, and the correlation of u_2 and v.

Solution. (a) By Theorem 9.4(a) with $B = I_4$, computing $A\Sigma$ row by row,

$$\begin{aligned} \operatorname{cov}(u,z) = A\Sigma &= \begin{pmatrix} 4+1 & -1-4 & -1+1 & -1+1 \\ 4-1-1-2 & -1+4-1-2 & -1-1+4-2 & -1-1-1+8 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -5 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix} \end{aligned}$$

and, with $B = \mathbf{1}'$, $\operatorname{cov}(u,v) = A\Sigma\mathbf{1} = (5-5+0+0,\ 0+0+0+5)' = (0,5)'$. So $z_1 - z_2$ and v are uncorrelated.

(b) By Theorem 9.3(b), $\operatorname{var}(u) = A\Sigma A' = (A\Sigma)A'$, and $\operatorname{var}(v) = \mathbf{1}'\Sigma\mathbf{1}$, the sum of all entries of Σ:

$$\operatorname{var}(u) = \begin{pmatrix} 5+5+0+0 & 5-5+0+0 \\ 0+0+0+0 & 0+0+0+10 \end{pmatrix} = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}, \qquad \operatorname{var}(v) = 16 - 12 = 4$$

By the block form of the covariance matrix of a stacked vector (after Definition 9.2),

$$\operatorname{var}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \operatorname{var}(u) & \operatorname{cov}(u,v) \\ \operatorname{cov}(v,u) & \operatorname{var}(v) \end{pmatrix} = \begin{pmatrix} 10 & 0 & 0 \\ 0 & 10 & 5 \\ 0 & 5 & 4 \end{pmatrix}$$

(c) Every $\sigma_{ii} = 4$, so $D^{-1/2} = \frac{1}{2}I_4$ and (Definition 9.7)

$$\operatorname{cor}(z) = \frac{1}{4}\Sigma = \begin{pmatrix} 1 & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & 1 & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & 1 & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & 1 \end{pmatrix}$$

The correlation of u_2 and v is $\operatorname{cov}(u_2, v)/\sqrt{\operatorname{var}(u_2)\operatorname{var}(v)} = 5/\sqrt{10 \cdot 4} = 5/(2\sqrt{10}) = \sqrt{10}/4$.

Exercise 9.4. Find the correlation matrix of a random vector $z \in \mathbb{R}^4$ with

$$\operatorname{var}(z) = \begin{pmatrix} 9 & 3 & -6 & 9 \\ 3 & 4 & 2 & 6 \\ -6 & 2 & 16 & -12 \\ 9 & 6 & -12 & 36 \end{pmatrix}$$

Solution. The standard deviations are 3, 2, 4, 6, so $\rho_{12} = \frac{3}{3\cdot 2} = \frac{1}{2}$, $\rho_{13} = \frac{-6}{3\cdot 4} = -\frac{1}{2}$, $\rho_{14} = \frac{9}{3\cdot 6} = \frac{1}{2}$, $\rho_{23} = \frac{2}{2\cdot 4} = \frac{1}{4}$, $\rho_{24} = \frac{6}{2\cdot 6} = \frac{1}{2}$, $\rho_{34} = \frac{-12}{4\cdot 6} = -\frac{1}{2}$, and (Definition 9.7)

$$\operatorname{cor}(z) = \begin{pmatrix} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{4} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{4} & 1 & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 1 \end{pmatrix}$$

Exercise 9.5 ([2, Exercise 10.5(b)])**.** Let $x = (x_1, x_2, x_3)'$ be a random vector with the covariance matrix

$$\operatorname{var}(x) = V = \begin{pmatrix} 8 & -7 & 2 \\ -7 & 10 & -3 \\ 2 & -3 & 1 \end{pmatrix}$$

Show that $x_1 + 2x_2 + 3x_3$ and $x_2 + 4x_3$ have the variances 5 and 2. Are they correlated?

Solution. Let $a = (1, 2, 3)'$ and $b = (0, 1, 4)'$. By Theorem 9.42(a), $\operatorname{var}(a'x) = a'Va$, $\operatorname{var}(b'x) = b'Vb$ and $\operatorname{cov}(a'x, b'x) = a'Vb$. Here $Va = (8 - 14 + 6,\ -7 + 20 - 9,\ 2 - 6 + 3)' = (0, 4, -1)'$ and $Vb = (0 - 7 + 8,\ 0 + 10 - 12,\ 0 - 3 + 4)' = (1, -2, 1)'$, so

$$\operatorname{var}(x_1 + 2x_2 + 3x_3) = a'Va = 0 + 8 - 3 = 5, \qquad \operatorname{var}(x_2 + 4x_3) = b'Vb = 0 - 2 + 4 = 2$$

and $\operatorname{cov}(x_1 + 2x_2 + 3x_3,\ x_2 + 4x_3) = a'Vb = 1 - 4 + 3 = 0$. So the two combinations are uncorrelated.

Exercise 9.6. Let x be as in Exercise 9.2. Use Theorem 9.10 to find $\mathsf{E}(x_1x_3 + x_2^2)$ and $\mathsf{E}(\|x\|^2)$, and check the first value directly from the means, variances and covariances.

Solution. $x_1x_3 + x_2^2 = x'Ax$ with $A = e_1e_3' + e_2e_2'$. The rows of $A\Sigma$ are $\operatorname{row}_3(\Sigma)$, $\operatorname{row}_2(\Sigma)$, 0, 0 (Theorem 1.4(c)), so $\operatorname{tr}(A\Sigma) = \sigma_{31} + \sigma_{22} = 3 + 1 = 4$; and $\mu'A\mu = \mu_1\mu_3 + \mu_2^2 = 2$. Hence $\mathsf{E}(x_1x_3 + x_2^2) = 4 + 2 = 6$. Directly, by Theorem 9.4(e), whose (i, j) entry reads $\mathsf{E}(x_ix_j) = \sigma_{ij} + \mu_i\mu_j$, $\mathsf{E}(x_1x_3) = \operatorname{cov}(x_1, x_3) + \mu_1\mu_3 = 3 + 2 = 5$ and $\mathsf{E}(x_2^2) = \operatorname{var}(x_2) + \mu_2^2 = 1$, with sum 6. With $A = I_4$, $\mathsf{E}(\|x\|^2) = \operatorname{tr}\Sigma + \mu'\mu = 23 + 6 = 29$.

Exercise 9.7. Three stocks each have correlation $\frac{2}{3}$ with a market index, and any two of the stocks have correlation r. The correlation matrix of (index, stock 1, stock 2, stock 3) is

$$R_r = \begin{pmatrix} 1 & \frac{2}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & 1 & r & r \\ \frac{2}{3} & r & 1 & r \\ \frac{2}{3} & r & r & 1 \end{pmatrix}$$

(a) Show that R_r is PSD if and only if $\frac{1}{6} \leqslant r \leqslant 1$, and PD if and only if $\frac{1}{6} < r < 1$.

(b) An analyst reports the sample correlations $\frac{2}{3}, \frac{2}{3}, \frac{2}{3}$ of three series with a fourth, and reports the three series as pairwise uncorrelated, all observed over the same periods. Explain why the report must contain an error.

Solution. (a) Write $R_r = \left(\begin{smallmatrix} 1 & b' \\ b & D \end{smallmatrix}\right)$ with $b = \frac{2}{3}\mathbf{1}_3$ and $D = (1-r)I_3 + rJ_3$. The block (1) is PD, and its Schur complement is

$$R_r/(1) = D - bb' = \begin{pmatrix} 1 & r & r \\ r & 1 & r \\ r & r & 1 \end{pmatrix} - \tfrac{4}{9}J_3 = (1-r)I_3 + \left(r - \tfrac{4}{9}\right)J_3$$

For $r \neq \frac{4}{9}$, by Example 8.16(b) its eigenvalues are $(1-r) + 3\left(r - \frac{4}{9}\right) = 2r - \frac{1}{3}$ (eigenvector $\mathbf{1}_3$) and $1 - r$; for $r = \frac{4}{9}$ it is $\frac{5}{9}I_3$, and $2r - \frac{1}{3} = 1 - r = \frac{5}{9}$. By Theorem 8.31 and Theorem 8.21(a), R_r is PSD if and only if $2r - \frac{1}{3} \geqslant 0$ and $1 - r \geqslant 0$, that is, $\frac{1}{6} \leqslant r \leqslant 1$; and PD if and only if both are positive, that is, $\frac{1}{6} < r < 1$.

(b) A sample correlation matrix is PSD (Proposition 9.18(a)), and $0 < \frac{1}{6}$, so by (a) the reported matrix R_0 is not; for instance $x = (-2, 1, 1, 1)'$ gives $x'R_0x = 4 + 3 - 2 \cdot \frac{2}{3} \cdot 2 \cdot 3 = -1 < 0$. More generally, for any sample correlations r_{12}, r_{13}, r_{23} among the three series, the same x gives $0 \leqslant 7 - 8 + 2(r_{12} + r_{13} + r_{23})$: three series that each correlate $\frac{2}{3}$ with a fourth have pairwise correlations adding up to at least $\frac{1}{2}$.

Exercise 9.8. The monthly returns (in %) of four funds over five months are:

month	1	2	3	4	5
fund 1	2	2	6	10	10
fund 2	8	12	12	8	10
fund 3	9	7	5	9	10
fund 4	5	7	9	9	10

Let $X \in \mathbb{R}^{5\times 4}$ have the months as rows.

(a) Compute the mean vector $\bar{x}$, the sample covariance matrix S and the sample correlation matrix R.

(b) The equally weighted portfolio has the returns $a'x_i$ with $a = \frac{1}{4}(1,1,1,1)'$. Compute their sample variance as $a'Sa$, and directly from the five portfolio returns.

(c) Show that S is singular, find $a \neq 0$ with $a'Sa = 0$, and explain this from the data. What is $\operatorname{rank}(C_5X)$?

Solution. (a) $\bar{x} = \frac{1}{5}(30,50,40,40)' = (6,10,8,8)'$, and (Proposition 9.14(a))

$$C_5X = \begin{pmatrix} -4 & -2 & 1 & -3 \\ -4 & 2 & -1 & -1 \\ 0 & 2 & -3 & 1 \\ 4 & -2 & 1 & 1 \\ 4 & 0 & 2 & 2 \end{pmatrix}$$

$$X'C_5X = \begin{pmatrix} 64 & -8 & 12 & 28 \\ -8 & 16 & -12 & 4 \\ 12 & -12 & 16 & 0 \\ 28 & 4 & 0 & 16 \end{pmatrix}, \qquad S = \frac{1}{4}X'C_5X = \begin{pmatrix} 16 & -2 & 3 & 7 \\ -2 & 4 & -3 & 1 \\ 3 & -3 & 4 & 0 \\ 7 & 1 & 0 & 4 \end{pmatrix}$$

(for instance $64 = 16+16+0+16+16$ and $-8 = 8-8+0-8+0$). With $D_s = \operatorname{diag}(16,4,4,4)$, $r_{12} = \frac{-2}{4\cdot 2} = -\frac{1}{4}$, $r_{13} = \frac{3}{4\cdot 2} = \frac{3}{8}$, $r_{14} = \frac{7}{4\cdot 2} = \frac{7}{8}$, $r_{23} = \frac{-3}{2\cdot 2} = -\frac{3}{4}$, $r_{24} = \frac{1}{2\cdot 2} = \frac{1}{4}$ and $r_{34} = 0$ (Definition 9.17), so

$$R = \begin{pmatrix} 1 & -\frac{1}{4} & \frac{3}{8} & \frac{7}{8} \\ -\frac{1}{4} & 1 & -\frac{3}{4} & \frac{1}{4} \\ \frac{3}{8} & -\frac{3}{4} & 1 & 0 \\ \frac{7}{8} & \frac{1}{4} & 0 & 1 \end{pmatrix}$$

(b) $a'Sa = \frac{1}{16}\mathbf{1}'S\mathbf{1}$, and the entries of S add up to 40; so $a'Sa = \frac{5}{2}$ (Proposition 9.14(b)). Directly, the portfolio returns are $6,7,8,9,10$, with mean 8 and squared deviations $4,1,0,1,4$; their sample variance is $\frac{10}{4} = \frac{5}{2}$. Each fund alone has a sample variance of at least 4; the negative correlations of fund 2 with funds 1 and 3 make the portfolio less variable.

(c) In every month the return of fund 4 is the average of those of funds 1 and 2: $x_{i4} = \frac{1}{2}(x_{i1} + x_{i2})$. So for $a = (1,1,0,-2)'$, every $a'x_i$ is 0; their sample variance $a'Sa$ is 0 (Proposition 9.14(b)), and $Sa = 0$ (Corollary 8.27); directly, $S(1,1,0,-2)' = (16-2+0-14,\ -2+4+0-2,\ 3-3+0+0,\ 7+1+0-8)' = 0$. Thus S is singular and not PD, so $\operatorname{rank}(C_5X) < 4$ (Proposition 9.14(a)). The first three columns c_1, c_2, c_3 of C_5X are linearly independent: if $b_1c_1 + b_2c_2 + b_3c_3 = 0$, rows 3, 5 and 2 give $2b_2 - 3b_3 = 0$, $4b_1 + 2b_3 = 0$ and $-4b_1 + 2b_2 - b_3 = 0$, so $b_2 = \frac{3}{2}b_3$, $b_1 = -\frac{1}{2}b_3$ and $2b_3 + 3b_3 - b_3 = 4b_3 = 0$. So $\operatorname{rank}(C_5X) = 3$. (Fund 4 is a 50:50 mix of funds 1 and 2, and the combination "one unit each of funds 1 and 2, minus two units of fund 4" has the constant return 0.)

Exercise 9.9. Seven students take four quizzes (scores out of 10):

student	1	2	3	4	5	6	7
quiz 1	9	7	3	4	9	7	3
quiz 2	8	5	2	5	8	5	2
quiz 3	3	8	3	8	6	8	6
quiz 4	5	7	5	7	2	7	2

(a) Compute $\bar{x}$, S, S^{-1} and the sample correlation matrix R.

(b) Compute the squared Mahalanobis distances $d_i^2 = d_S(x_i, \bar{x})^2$ of all students, and check $\sum_i d_i^2 = (n-1)p$.

(c) A screening rule flags a student with $d_i^2 > 4$ as unusual. Which student is flagged? Compare with the standardized scores $(x_{ij} - \bar{x}_j)/\sqrt{s_{jj}}$: is the flagged student more than one standard deviation from the average on any quiz?

Solution. (a) $\bar{x} = \frac{1}{7}(42, 35, 42, 35)' = (6, 5, 6, 5)'$. The rows of C_7X are $(3, 3, -3, 0)$, $(1, 0, 2, 2)$, $(-3, -3, -3, 0)$, $(-2, 0, 2, 2)$, $(3, 3, 0, -3)$, $(1, 0, 2, 2)$, $(-3, -3, 0, -3)$, so (Proposition 9.14(a))

$$X'C_7X = \begin{pmatrix} 42 & 36 & 0 & 0 \\ 36 & 36 & 0 & 0 \\ 0 & 0 & 30 & 12 \\ 0 & 0 & 12 & 30 \end{pmatrix}, \qquad S = \frac{1}{6}X'C_7X = \begin{pmatrix} 7 & 6 & 0 & 0 \\ 6 & 6 & 0 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 2 & 5 \end{pmatrix}$$

(for instance $42 = 9+1+9+4+9+1+9$ and $30 = 9+4+9+4+0+4+0$). So $S = \operatorname{diag}(S_1, S_2)$ with $S_1 = \left(\begin{smallmatrix} 7 & 6 \\ 6 & 6 \end{smallmatrix}\right)$ and $S_2 = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 5 \end{smallmatrix}\right)$. As $\det S_1 = 42 - 36 = 6$ and $\det S_2 = 25 - 4 = 21$, the 2×2 formula of Proposition 2.18 gives $S_1^{-1} = \frac{1}{6}\left(\begin{smallmatrix} 6 & -6 \\ -6 & 7 \end{smallmatrix}\right)$ and $S_2^{-1} = \frac{1}{21}\left(\begin{smallmatrix} 5 & -2 \\ -2 & 5 \end{smallmatrix}\right)$. By Proposition 5.6(b), $S^{-1} = \operatorname{diag}(S_1^{-1}, S_2^{-1})$, that is,

$$S^{-1} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & \frac{7}{6} & 0 & 0 \\ 0 & 0 & \frac{5}{21} & -\frac{2}{21} \\ 0 & 0 & -\frac{2}{21} & \frac{5}{21} \end{pmatrix}$$

The sample standard deviations are $\sqrt{7}$, $\sqrt{6}$, $\sqrt{5}$, $\sqrt{5}$, so $r_{12} = \frac{6}{\sqrt{42}} = \sqrt{\frac{6}{7}}$, $r_{34} = \frac{2}{5}$, the other r_{jl} $(j \neq l)$ are 0, and (Definition 9.17)

$$R = \begin{pmatrix} 1 & \sqrt{\frac{6}{7}} & 0 & 0 \\ \sqrt{\frac{6}{7}} & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{2}{5} \\ 0 & 0 & \frac{2}{5} & 1 \end{pmatrix}$$

(b) For a deviation $(u, v, w, s) = (x_i - \bar{x})'$, $d_i^2 = u^2 - 2uv + \frac{7}{6}v^2 + \frac{1}{21}(5w^2 - 4ws + 5s^2) = (u-v)^2 + \frac{1}{6}v^2 + \frac{1}{21}(5w^2 - 4ws + 5s^2)$. This gives

$$d_1^2 = d_3^2 = d_5^2 = d_7^2 = \tfrac{3}{2} + \tfrac{15}{7} = \tfrac{51}{14}, \qquad d_2^2 = d_6^2 = 1 + \tfrac{8}{7} = \tfrac{15}{7}, \qquad d_4^2 = 4 + \tfrac{8}{7} = \tfrac{36}{7}$$

with sum $\frac{204}{14} + \frac{30}{7} + \frac{36}{7} = \frac{168}{7} = 24 = 6 \cdot 4$, as Proposition 9.20(b) says.

(c) Only student 4 is flagged: $d_4^2 = \frac{36}{7} > 4$, while all other $d_i^2 \leqslant \frac{51}{14} < 4$. The standardized scores of student 4 are $-2/\sqrt{7}$, 0, $2/\sqrt{5}$ and $2/\sqrt{5}$, and $2 < \sqrt{5} < \sqrt{7}$: less than one standard deviation from the average on every quiz. Students $1, 3, 5, 7$ have the standardized scores $\pm 3/\sqrt{7}$ and $\pm 3/\sqrt{6}$ on quizzes 1 and 2 and $\pm 3/\sqrt{5}$ on one of quizzes 3 and 4, more than one standard deviation from the average on three of the four quizzes $(3 > \sqrt{7} > \sqrt{6} > \sqrt{5})$, but $d_i^2 = \frac{51}{14}$. Quizzes 1 and 2 are strongly correlated $(r_{12}^2 = \frac{6}{7})$; students $1, 3, 5, 7$ follow the common pattern of these two quizzes, while student 4 is below average on quiz 1 without being below average on quiz 2. The Mahalanobis distance detects this; the coordinatewise scores do not.

Exercise 9.10. Let

$$A = \begin{pmatrix} 4 & 1 & 1 & 1 \\ 1 & 4 & 1 & -2 \\ 1 & 1 & 4 & 1 \\ 1 & -2 & 1 & 4 \end{pmatrix}$$

(a) Check that $(1,1,1,0)'$, $(0,1,0,-1)'$, $(1,0,-1,0)'$ and $(1,-2,1,-2)'$ are eigenvectors of A, and find the eigenvalues $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3 \geqslant \lambda_4$ of A.

(b) Find the maximum and the minimum of $x'Ax/x'x$ over $x \neq 0$, and all x at which they are attained.

(c) Find the eigenvalues, in decreasing order, of $-A$, $A-6I_4$ and $A+2I_4$. For which $c \in \mathbb{R}$ is $A-cI_4$ PSD, and for which is $cI_4 - A$ PSD?

Solution. (a) $A(1,1,1,0)' = (6,6,6,0)'$, $A(0,1,0,-1)' = (0,6,0,-6)'$, $A(1,0,-1,0)' = (3,0,-3,0)'$ and $A(1,-2,1,-2)' = (1,-2,1,-2)'$. The first two vectors are independent eigenvectors belonging to 6, so $a_A(6) = \dim \mathcal{N}(A-6I) \geqslant 2$, and $a_A(3) \geqslant 1$, $a_A(1) \geqslant 1$ (Theorem 8.4); as the multiplicities add up to 4 (Definition 7.10), $\lambda_1 = \lambda_2 = 6$, $\lambda_3 = 3$ and $\lambda_4 = 1$. Check: $\operatorname{tr} A = 16 = 6+6+3+1$.

(b)

- By Theorem 9.23(a), the maximum is 6 and the minimum is 1.
- By Theorem 9.23(b), the maximum is attained exactly at the $x \neq 0$ with $Ax = 6x$. As $\dim \mathcal{N}(A-6I) = a_A(6) = 2$ (Theorem 8.4), the two independent vectors $(1,1,1,0)'$ and $(0,1,0,-1)'$ form a basis of $\mathcal{N}(A-6I)$ (Theorem 4.5(d)); so these are the vectors $x = c_1(1,1,1,0)' + c_2(0,1,0,-1)' \neq 0$: a whole plane of maximizers, as $\lambda_1 = \lambda_2$.
- The minimum is attained exactly at the $x \neq 0$ with $Ax = x$: $a_A(1) = 1$, so these are the vectors $x = c(1,-2,1,-2)'$, $c \neq 0$.

(c)

- By Lemma 9.22(b), the eigenvalues of $-A$ are $-1 \geqslant -3 \geqslant -6 \geqslant -6$.
- By Lemma 9.22(c), those of $A-6I_4$ are $0 \geqslant 0 \geqslant -3 \geqslant -5$, and those of $A+2I_4$ are $8 \geqslant 8 \geqslant 5 \geqslant 3$.
- $A - cI_4$ is symmetric with the eigenvalues $6-c$, $6-c$, $3-c$, $1-c$, so it is PSD exactly when $1-c \geqslant 0$, that is, $c \leqslant 1$ (Theorem 8.21(a)). Likewise $cI_4 - A = -(A-cI_4)$ has the eigenvalues $c-1$, $c-3$, $c-6$, $c-6$ and is PSD exactly when $c \geqslant 6$. For $c = \lambda_4 = 1$ and $c = \lambda_1 = 6$ this is Theorem 9.23(c).

Exercise 9.11. Let

$$A = \begin{pmatrix} 3 & 1 & 0 & -1 \\ 1 & 4 & 2 & 2 \\ 0 & 2 & 4 & 2 \\ -1 & 2 & 2 & 4 \end{pmatrix}$$

(a) Check that $(0,1,1,1)'$, $(2,1,0,-1)'$, $(0,1,-2,1)'$ and $(1,-1,0,1)'$ are eigenvectors of A, and find the eigenvalues $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3 \geqslant \lambda_4$ of A.

(b) Find the maximum and the minimum of $x'Ax$ over the unit vectors $x \in \mathbb{R}^4$, and all unit vectors at which they are attained.

(c) Find the maximum of $x'Ax/x'x$ over the $x \neq 0$ with $x_2 + x_3 + x_4 = 0$, and the minimum over the $x \neq 0$ with $x_1 - x_2 + x_4 = 0$.

Solution. (a) $A(0,1,1,1)' = (0,8,8,8)'$, $A(2,1,0,-1)' = (8,4,0,-4)'$, $A(0,1,-2,1)' = (0,2,-4,2)'$ and $A(1,-1,0,1)' = (1,-1,0,1)'$: eigenvectors belonging to 8, 4, 2 and 1. So the algebraic multiplicities $\dim \mathcal{N}(A-cI)$ of $c = 8,4,2,1$ are at least 1 (Theorem 8.4); as all multiplicities add up to

4 (Definition 7.10), each of 8, 4, 2, 1 is simple and there are no other eigenvalues: $\lambda_1 = 8$, $\lambda_2 = 4$, $\lambda_3 = 2$, $\lambda_4 = 1$.

(b) Each eigenspace is a line, so the unit eigenvectors are $\pm q_1$, $\pm q_2$, $\pm q_3$, $\pm q_4$, with $q_1 = \frac{1}{\sqrt{3}}(0,1,1,1)'$, $q_2 = \frac{1}{\sqrt{6}}(2,1,0,-1)'$, $q_3 = \frac{1}{\sqrt{6}}(0,1,-2,1)'$ and $q_4 = \frac{1}{\sqrt{3}}(1,-1,0,1)'$. By Theorem 9.23, the maximum is 8, attained exactly at $x = \pm q_1$, and the minimum is 1, attained exactly at $x = \pm q_4$.

(c)

- q_1, q_2, q_3, q_4 are orthonormal (Theorem 8.2), so $Q = [q_1\ q_2\ q_3\ q_4]$ is as in Lemma 9.22.
- $x_2 + x_3 + x_4 = 0$ means $x \perp q_1$. By Theorem 9.25 with $k = 2$ (first form), the maximum is $\lambda_2 = 4$, attained at q_2.
- $x_1 - x_2 + x_4 = 0$ means $x \perp q_4$. By the same theorem with $k = 3$ (second form), the minimum is $\lambda_3 = 2$, attained at q_3.

Exercise 9.12. Five units are measured on four variables; row i of

$$X = \begin{pmatrix} 2 & 5 & 5 & 1 \\ 6 & 5 & 1 & 5 \\ 6 & 7 & 7 & 5 \\ 4 & 1 & 7 & 7 \\ 2 & 7 & 5 & 7 \end{pmatrix}$$

holds the values on unit i. For $a \in \mathbb{R}^4$, let $y = Xa$.

(a) Find the mean vector $\bar{x}$, the centered data matrix C_5X and $M = X'C_5X$.

(b) Check that $(0,-1,1,1)'$, $(1,0,-1,1)'$, $(0,2,1,1)'$ and $(2,0,1,-1)'$ are eigenvectors of M, and find its eigenvalues.

(c) Among the unit vectors a, find those for which $\sum_i (y_i - \bar{y})^2$ is largest and smallest, these two sums, and the corresponding sample variances.

(d) Among the unit vectors a orthogonal to the maximizers of (c), find the largest $\sum_i (y_i - \bar{y})^2$ and where it is attained.

Solution. (a) $\bar{x} = \frac{1}{5}X'\mathbf{1} = \frac{1}{5}(20,25,25,25)' = (4,5,5,5)'$, and (Proposition 9.14(a))

$$C_5X = X - \mathbf{1}\bar{x}' = \begin{pmatrix} -2 & 0 & 0 & -4 \\ 2 & 0 & -4 & 0 \\ 2 & 2 & 2 & 0 \\ 0 & -4 & 2 & 2 \\ -2 & 2 & 0 & 2 \end{pmatrix}, \qquad M = (C_5X)'(C_5X) = \begin{pmatrix} 16 & 0 & -4 & 4 \\ 0 & 24 & -4 & -4 \\ -4 & -4 & 24 & 4 \\ 4 & -4 & 4 & 24 \end{pmatrix}$$

(b) $M(0,-1,1,1)' = (0,-32,32,32)'$, $M(1,0,-1,1)' = (24,0,-24,24)'$, $M(0,2,1,1)' = (0,40,20,20)'$ and $M(2,0,1,-1)' = (24,0,12,-12)'$: eigenvectors belonging to 32, 24, 20 and 12. So $a_M(c) \geqslant 1$ for $c = 32, 24, 20, 12$ (Theorem 8.4), and as the multiplicities add up to 4, the eigenvalues are $\lambda_1 = 32$, $\lambda_2 = 24$, $\lambda_3 = 20$, $\lambda_4 = 12$. Check: $\operatorname{tr} M = 88 = 32 + 24 + 20 + 12$.

(c)

- By Proposition 9.14(b), $y = Xa$ has the sample variance $a'Sa$ with $S = \frac{1}{4}M$, so $\sum_i (y_i - \bar{y})^2 = 4a'Sa = a'Ma$.
- Each eigenspace is a line, so the unit eigenvectors are $\pm q_1, \ldots, \pm q_4$, with $q_1 = \frac{1}{\sqrt{3}}(0,-1,1,1)'$, $q_2 = \frac{1}{\sqrt{3}}(1,0,-1,1)'$, $q_3 = \frac{1}{\sqrt{6}}(0,2,1,1)'$ and $q_4 = \frac{1}{\sqrt{6}}(2,0,1,-1)'$. By Theorem 9.23, the largest sum is 32, attained exactly at $a = \pm q_1$, and the smallest is 12, attained exactly at $a = \pm q_4$.
- Check: $C_5Xq_1 = \frac{1}{\sqrt{3}}(-4,-4,0,8,0)'$, whose squared length is $\frac{96}{3} = 32$.

- Divided by $n-1=4$: the sample variances $a'Sa$ lie in $[3,8]$ for unit a, with both ends attained.

(d) $q_1,\dots,q_4$ are orthonormal (Theorem 8.2), so $Q=[q_1\ \cdots\ q_4]$ is as in Lemma 9.22. By Theorem 9.25 ($k=2$, first form), the largest $a'Ma$ over the unit vectors $a\perp q_1$ is $\lambda_2=24$, attained at $a=\pm q_2$.

Exercise 9.13. Let

$$A=\begin{pmatrix}3&1&0&1\\1&3&0&1\\0&0&4&1\\1&1&1&2\end{pmatrix}$$

(a) Find the eigenvalues of A and orthonormal eigenvectors.

(b) Find the largest and the smallest value of $\operatorname{tr}(G'AG)$ over all $G\in\mathbb{R}^{4\times 2}$ with $G'G=I_2$, and a G attaining each.

(c) For $G=[e_1\ e_2]$, compute $\operatorname{tr}(G'AG)$ and check that it lies between the two values of (b).

Solution. (a) For real t, apply $R_1\leftarrow R_1-R_2$ to tI_4-A, take the factor $t-2$ out of row 1, apply $\mathrm{col}_2\leftarrow\mathrm{col}_2+\mathrm{col}_1$, and expand along row 1 and then along row 1 again (Definition 5.13(a),(b), Theorems 5.22 and 5.21):

$$\begin{aligned}\det(tI_4-A)&=\det\begin{pmatrix}t-3&-1&0&-1\\-1&t-3&0&-1\\0&0&t-4&-1\\-1&-1&-1&t-2\end{pmatrix}=(t-2)\det\begin{pmatrix}1&-1&0&0\\-1&t-3&0&-1\\0&0&t-4&-1\\-1&-1&-1&t-2\end{pmatrix}\\&=(t-2)\det\begin{pmatrix}1&0&0&0\\-1&t-4&0&-1\\0&0&t-4&-1\\-1&-2&-1&t-2\end{pmatrix}=(t-2)\det\begin{pmatrix}t-4&0&-1\\0&t-4&-1\\-2&-1&t-2\end{pmatrix}\\&=(t-2)\Big((t-4)\Big((t-4)(t-2)-1\Big)-2(t-4)\Big)\\&=(t-2)(t-4)(t^2-6t+5)=(t-5)(t-4)(t-2)(t-1)\end{aligned}$$

so the eigenvalues are 5, 4, 2, 1 (Definition 7.7(a), Proposition 7.5(d)). Eigenvectors:

$$A-5I=\begin{pmatrix}-2&1&0&1\\1&-2&0&1\\0&0&-1&1\\1&1&1&-3\end{pmatrix}\xrightarrow{R_1\leftrightarrow R_4}\begin{pmatrix}1&1&1&-3\\1&-2&0&1\\0&0&-1&1\\-2&1&0&1\end{pmatrix}\xrightarrow[R_4\leftarrow R_4+2R_1]{R_2\leftarrow R_2-R_1}\begin{pmatrix}1&1&1&-3\\0&-3&-1&4\\0&0&-1&1\\0&3&2&-5\end{pmatrix}$$
$$\xrightarrow{R_4\leftarrow R_4+R_2}\begin{pmatrix}1&1&1&-3\\0&-3&-1&4\\0&0&-1&1\\0&0&1&-1\end{pmatrix}\xrightarrow{R_4\leftarrow R_4+R_3}\begin{pmatrix}1&1&1&-3\\0&-3&-1&4\\0&0&-1&1\\0&0&0&0\end{pmatrix}$$

so $x_4=1$ gives $x_3=x_4=1$, $x_2=\frac{1}{3}(4x_4-x_3)=1$ and $x_1=-x_2-x_3+3x_4=1$; take $(1,1,1,1)'$.

$$A-4I=\begin{pmatrix}-1&1&0&1\\1&-1&0&1\\0&0&0&1\\1&1&1&-2\end{pmatrix}\xrightarrow[R_4\leftarrow R_4+R_1]{R_2\leftarrow R_2+R_1}\begin{pmatrix}-1&1&0&1\\0&0&0&2\\0&0&0&1\\0&2&1&-1\end{pmatrix}\xrightarrow{R_2\leftrightarrow R_4}\begin{pmatrix}-1&1&0&1\\0&2&1&-1\\0&0&0&1\\0&0&0&2\end{pmatrix}$$
$$\xrightarrow{R_4\leftarrow R_4-2R_3}\begin{pmatrix}-1&1&0&1\\0&2&1&-1\\0&0&0&1\\0&0&0&0\end{pmatrix}$$

so $x_4 = 0$, and $x_3 = -2$ gives $x_2 = \frac{1}{2}(x_4 - x_3) = 1$ and $x_1 = x_2 + x_4 = 1$; take $(1, 1, -2, 0)'$.

$$A - 2I = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
$$\xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = 0$, $x_3 = x_4 = 0$, and $x_2 = -1$ gives $x_1 = -x_2 - x_4 = 1$; take $(1, -1, 0, 0)'$.

$$A - I = \begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 2 & 1 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & -1 & -2 & -1 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & -3 & -1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $x_4 = -3$ gives $x_3 = -\frac{1}{3}x_4 = 1$, $x_2 = x_3 = 1$ and $x_1 = -x_2 - x_3 - x_4 = 1$; take $(1, 1, 1, -3)'$. The four vectors are pairwise orthogonal, and $q_1 = \frac{1}{2}(1, 1, 1, 1)'$, $q_2 = \frac{1}{\sqrt{6}}(1, 1, -2, 0)'$, $q_3 = \frac{1}{\sqrt{2}}(1, -1, 0, 0)'$, $q_4 = \frac{1}{2\sqrt{3}}(1, 1, 1, -3)'$ are orthonormal eigenvectors for 5, 4, 2, 1.

(b) By Theorem 9.27 with $k = 2$, the largest value is $5 + 4 = 9$, attained at $G = [q_1\ q_2]$, and the smallest is $2 + 1 = 3$, attained at $G = [q_3\ q_4]$.

(c) $G'G = I_2$, and $G'AG$ consists of rows and columns 1, 2 of A: $G'AG = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, with trace 6. Indeed $3 \leqslant 6 \leqslant 9$, as Corollary 9.29 says for $a_{11} + a_{22}$.

Exercise 9.14 (cf. [9, Exercise 5.9], [19, §12.6, Exercise 8])**.** Five units are measured on four variables:

$$X = \begin{pmatrix} 3 & 1 & 1 & 9 \\ 3 & 4 & 8 & 4 \\ 3 & 6 & 4 & 2 \\ 5 & 6 & 6 & 4 \\ 1 & 8 & 6 & 6 \end{pmatrix}$$

(a) Find $\bar{x}$, C_5X and S.

(b) Find the loading vectors, the sample variances of the principal components and the score vectors.

(c) Find $\mathrm{PVE}_1, \ldots, \mathrm{PVE}_4$, once from the eigenvalues and once from the score vectors.

Solution. (a) $\mathbf{1}'X = (15, 25, 25, 25)$, so $\bar{x} = (3, 5, 5, 5)'$, and

$$C_5X = \begin{pmatrix} 0 & -4 & -4 & 4 \\ 0 & -1 & 3 & -1 \\ 0 & 1 & -1 & -3 \\ 2 & 1 & 1 & -1 \\ -2 & 3 & 1 & 1 \end{pmatrix}$$

$$(C_5X)'(C_5X) = \begin{pmatrix} 8 & -4 & 0 & -4 \\ -4 & 28 & 16 & -16 \\ 0 & 16 & 28 & -16 \\ -4 & -16 & -16 & 28 \end{pmatrix}, \qquad S = \frac{1}{4}(C_5X)'(C_5X) = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 7 & 4 & -4 \\ 0 & 4 & 7 & -4 \\ -1 & -4 & -4 & 7 \end{pmatrix}$$

(b) For real t, add row 3 of $tI_4 - S$ to row 2 and subtract row 4 from it; the new row 2 is $(t-15)(0,1,1,-1)$. Take out $t-15$, subtract column 2 from column 3 and add it to column 4, and expand along row 2 twice (Definition 5.13(a),(b), Theorems 5.22 and 5.21):

$$\begin{aligned}\det(tI_4 - S) &= \det\begin{pmatrix} t-2 & 1 & 0 & 1 \\ 1 & t-7 & -4 & 4 \\ 0 & -4 & t-7 & 4 \\ 1 & 4 & 4 & t-7 \end{pmatrix} = (t-15)\det\begin{pmatrix} t-2 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -4 & t-7 & 4 \\ 1 & 4 & 4 & t-7 \end{pmatrix} \\ &= (t-15)\det\begin{pmatrix} t-2 & 1 & -1 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & -4 & t-3 & 0 \\ 1 & 4 & 0 & t-3 \end{pmatrix} = (t-15)\det\begin{pmatrix} t-2 & -1 & 2 \\ 0 & t-3 & 0 \\ 1 & 0 & t-3 \end{pmatrix} \\ &= (t-15)(t-3)\Big((t-2)(t-3)-2\Big) \\ &= (t-15)(t-3)(t^2-5t+4) = (t-15)(t-4)(t-3)(t-1)\end{aligned}$$

So the eigenvalues are $\lambda_1 = 15$, $\lambda_2 = 4$, $\lambda_3 = 3$, $\lambda_4 = 1$ (Definition 7.7(a), Proposition 7.5(d)). Each is simple, so each eigenspace is a line (Theorem 8.4), spanned by the vector found in

$$S\begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \\ 15 \\ -15 \end{pmatrix}, \qquad S\begin{pmatrix} 1 \\ -1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ 0 \\ -4 \end{pmatrix}, \qquad S\begin{pmatrix} 0 \\ 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ -6 \\ -3 \end{pmatrix}, \qquad S\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

(the first is the row combination used for p_S). So the loading vectors are, up to sign (Theorem 9.31(e)), $v_1 = \frac{1}{\sqrt{3}}(0,1,1,-1)'$, $v_2 = \frac{1}{\sqrt{3}}(1,-1,0,-1)'$, $v_3 = \frac{1}{\sqrt{6}}(0,1,-2,-1)'$ and $v_4 = \frac{1}{\sqrt{6}}(2,1,0,1)'$ (Theorem 8.3(b)), and the principal components have the sample variances 15, 4, 3, 1. The first principal component contrasts variables 2 and 3 with variable 4. The score vectors are

$$z_1 = C_5 X v_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} -12 \\ 3 \\ 3 \\ 3 \\ 3 \end{pmatrix}, \qquad z_2 = C_5 X v_2 = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 \\ 2 \\ 2 \\ 2 \\ -6 \end{pmatrix}$$

$$z_3 = C_5 X v_3 = \frac{1}{\sqrt{6}}\begin{pmatrix} 0 \\ -6 \\ 6 \\ 0 \\ 0 \end{pmatrix}, \qquad z_4 = C_5 X v_4 = \frac{1}{\sqrt{6}}\begin{pmatrix} 0 \\ -2 \\ -2 \\ 4 \\ 0 \end{pmatrix}$$

with $z_m' z_l = 0$ for $m \neq l$, $\|z_1\|^2 = 60 = 4\cdot 15$, $\|z_2\|^2 = 16 = 4\cdot 4$, $\|z_3\|^2 = 12 = 4\cdot 3$ and $\|z_4\|^2 = 4 = 4\cdot 1$ (Theorem 9.31(d)).

(c) $\operatorname{tr} S = 2+7+7+7 = 23 = 15+4+3+1$, so $\mathrm{PVE}_m = \frac{15}{23}, \frac{4}{23}, \frac{3}{23}, \frac{1}{23}$. From the scores (Proposition 9.32): $\|C_5 X\|_F^2 = 8+28+28+28 = 92$, and $\frac{60}{92} = \frac{15}{23}$, $\frac{16}{92} = \frac{4}{23}$, $\frac{12}{92} = \frac{3}{23}$, $\frac{4}{92} = \frac{1}{23}$.

Exercise 9.15. Four variables are measured on five units: $(3,6,3,3)$, $(2,3,5,1)$, $(6,7,1,1)$, $(6,1,1,5)$, $(8,3,5,5)$.

(a) Find $\bar{x}$, the centered data matrix $C_5 X$ and S.

(b) Show that $(1,-1,0,1)'$, $(1,1,-1,0)'$, $(1,1,2,0)'$ and $(1,-1,0,-2)'$ are eigenvectors of S, and give the loading vectors, the variances of the components and the proportions of variance explained by the first component and by the first two.

(c) Compute the scores, and check Theorem 9.31(d) for them.

(d) Find the sum over the units of the squared errors of the approximation of C_5X by two components.

Solution. (a) $\bar{x} = \frac{1}{5}(25, 20, 15, 15)' = (5, 4, 3, 3)'$, and

$$C_5X = \begin{pmatrix} -2 & 2 & 0 & 0 \\ -3 & -1 & 2 & -2 \\ 1 & 3 & -2 & -2 \\ 1 & -3 & -2 & 2 \\ 3 & -1 & 2 & 2 \end{pmatrix}$$

$$(C_5X)'(C_5X) = \begin{pmatrix} 24 & -4 & -4 & 12 \\ -4 & 24 & -4 & -12 \\ -4 & -4 & 16 & 0 \\ 12 & -12 & 0 & 16 \end{pmatrix}, \qquad S = \frac{1}{4}(C_5X)'(C_5X) = \begin{pmatrix} 6 & -1 & -1 & 3 \\ -1 & 6 & -1 & -3 \\ -1 & -1 & 4 & 0 \\ 3 & -3 & 0 & 4 \end{pmatrix}$$

(b) $S(1, -1, 0, 1)' = (10, -10, 0, 10)'$, $S(1, 1, -1, 0)' = (6, 6, -6, 0)'$, $S(1, 1, 2, 0)' = (3, 3, 6, 0)'$ and $S(1, -1, 0, -2)' = (1, -1, 0, -2)'$. The four vectors are pairwise orthogonal, so $\lambda_1 = 10$, $\lambda_2 = 6$, $\lambda_3 = 3$, $\lambda_4 = 1$ (Theorem 8.3(c)), the sample variances of the components (Theorem 9.31(d)), with the loading vectors $v_1 = \frac{1}{\sqrt{3}}(1, -1, 0, 1)'$, $v_2 = \frac{1}{\sqrt{3}}(1, 1, -1, 0)'$, $v_3 = \frac{1}{\sqrt{6}}(1, 1, 2, 0)'$, $v_4 = \frac{1}{\sqrt{6}}(1, -1, 0, -2)'$, unique up to sign (Theorem 9.31(e)). The total variance is $\operatorname{tr} S = 20$; the first component explains $\frac{10}{20} = \frac{1}{2}$, the first two $\frac{16}{20} = \frac{4}{5}$ (Proposition 9.32).

(c) $z_{im} = v_m'(x_i - \bar{x})$:

$$(z_{i1})_i = \tfrac{1}{\sqrt{3}}(-4, -4, -4, 6, 6), \qquad (z_{i2})_i = \tfrac{1}{\sqrt{3}}(0, -6, 6, 0, 0)$$
$$(z_{i3})_i = \tfrac{1}{\sqrt{6}}(0, 0, 0, -6, 6), \qquad (z_{i4})_i = \tfrac{1}{\sqrt{6}}(-4, 2, 2, 0, 0)$$

Each has sum 0; the sums of squares are $\frac{120}{3} = 40 = 4\lambda_1$, $\frac{72}{3} = 24 = 4\lambda_2$, $\frac{72}{6} = 12 = 4\lambda_3$ and $\frac{24}{6} = 4 = 4\lambda_4$; and the sums of products of two different columns are $\frac{1}{3}(24-24) = 0$, $\frac{1}{\sqrt{18}}(-36+36) = 0$, $\frac{1}{\sqrt{18}}(16 - 8 - 8) = 0$, 0, $\frac{1}{\sqrt{18}}(-12 + 12) = 0$ and 0 for the pairs $(1, 2)$, $(1, 3)$, $(1, 4)$, $(2, 3)$, $(2, 4)$, $(3, 4)$.

(d) By Theorem 9.39(b) with $M = 2$, the sum is $\|C_5X - Z_2V_2'\|_F^2 = \|C_5X\|_F^2 - \|z_1\|^2 - \|z_2\|^2 = 80 - 40 - 24 = 16$, out of $\|C_5X\|_F^2 = (n-1)\operatorname{tr} S = 80$; by Theorem 9.39(a) it is $(n-1)(\lambda_3 + \lambda_4) = 4 \cdot 4 = 16 = \sum_i (z_{i3}^2 + z_{i4}^2)$, the smallest possible for an approximation of rank at most 2 (the paragraph after Theorem 9.39).

Exercise 9.16. (a) The sample covariance matrix of four variables is

$$S = \begin{pmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{pmatrix}$$

Find the eigenvalues of S and the proportions of variance explained. Show that the first loading vector is unique up to sign, but that every unit vector orthogonal to $\mathbf{1}_4$ is a second loading vector.

(b) For the data matrix X_0 of Example 1.17 ($n = 4$, two variables), find S and show that every unit vector is a first loading vector.

Solution. (a) $S = 2I_4 + J_4 = 2(\bar{J}_4 + C_4) + 4\bar{J}_4 = 6\bar{J}_4 + 2C_4$ (Definition 1.14). So $S\mathbf{1} = 6\mathbf{1}$, and $Sx = 2x$ whenever $\mathbf{1}'x = 0$ (then $\bar{J}_4 x = 0$, $C_4 x = x$). With $v_1 = \frac{1}{2}\mathbf{1}$ and the orthonormal vectors $\frac{1}{\sqrt{2}}(1, -1, 0, 0)'$, $\frac{1}{\sqrt{6}}(1, 1, -2, 0)'$, $\frac{1}{2\sqrt{3}}(1, 1, 1, -3)'$ orthogonal to $\mathbf{1}$, the eigenvalues are $6, 2, 2, 2$ (Theorem 8.3(c)), and $\text{PVE}_m = \frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$ (Proposition 9.32). As $6 > 2$, the first loading vector is $\pm v_1$ (Theorem 9.31(e), $M = 1$). The second problem has the constraint $a \perp v_1$ (Theorem 9.31(b)), that is, $\mathbf{1}'a = 0$; for every such unit vector $a'Sa = 2a'a = 2$, the maximum λ_2 (Theorem 9.31(c)). So every unit vector

orthogonal to $\mathbf{1}_4$ is a second loading vector: with a repeated eigenvalue the loading vectors are not unique.

(b) By Example 1.17, $X_0'C_4X_0 = 4I_2$, so $S = \frac{1}{3}\cdot 4I_2 = \frac{4}{3}I_2$. Every unit vector a gives $a'Sa = \frac{4}{3}$, so every unit vector is a first loading vector; $\lambda_1 = \lambda_2 = \frac{4}{3}$, and each component explains half of the total variance.

Exercise 9.17. Five units are measured on four variables:

$$X = \begin{pmatrix} 5 & 5 & 2 & 4 \\ 3 & 3 & 6 & 4 \\ 3 & 3 & 3 & 3 \\ 3 & 1 & 2 & 2 \\ 1 & 3 & 2 & 2 \end{pmatrix}$$

(a) Find $\bar{x}$, C_5X and S.

(b) Using Exercise 9.13, find the eigenvalues of S, orthonormal eigenvectors and $r = \operatorname{rank}(C_5X)$.

(c) Find the proportions of variance explained, and show that the five observations lie in an affine subspace of dimension 3 through $\bar{x}$.

Solution. (a) $\mathbf{1}'X = (15, 15, 15, 15)$, so $\bar{x} = (3, 3, 3, 3)'$, and

$$C_5X = \begin{pmatrix} 2 & 2 & -1 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & -1 & -1 \\ -2 & 0 & -1 & -1 \end{pmatrix}, \qquad (C_5X)'(C_5X) = \begin{pmatrix} 8 & 4 & 0 & 4 \\ 4 & 8 & 0 & 4 \\ 0 & 0 & 12 & 4 \\ 4 & 4 & 4 & 4 \end{pmatrix}, \qquad S = \begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

(b) $S = A - I_4$ with A of Exercise 9.13, so $Sq_m = Aq_m - q_m$: the vectors q_1, q_2, q_3, q_4 found there are orthonormal eigenvectors of S for $5-1=4$, $4-1=3$, $2-1=1$, $1-1=0$ (Theorem 8.3(c)). By Theorem 9.31(a), r is the number of positive eigenvalues: $r = 3$.

(c) $\operatorname{tr} S = 8$, so $\mathrm{PVE}_m = \frac{4}{8} = \frac{1}{2}, \frac{3}{8}, \frac{1}{8}, 0$ (Proposition 9.32): three components explain all the variance. The fourth score vector is $z_4 = C_5Xq_4 = \frac{1}{2\sqrt{3}}C_5X(1,1,1,-3)' = \frac{1}{2\sqrt{3}}(2+2-1-3,\ 0+0+3-3,\ 0,\ 0-2-1+3,\ -2+0-1+3)' = 0$; so $(x_i - \bar{x})'(1,1,1,-3)' = 0$ for every i, that is, all observations lie in the set $W = \{y : y_1 + y_2 + y_3 - 3y_4 = 0\}$, which contains $\bar{x} = (3,3,3,3)'$: the fourth variable is the average of the other three. W is the subspace $\operatorname{span}\{(1,1,1,-3)'\}^{\perp}$, of dimension 3 (Theorem 6.6(c)), and as $\bar{x} \in W$, it is the affine subspace $\bar{x} + W$ (Definition 9.38). (By Theorem 9.39(a) with $M = 3 \geqslant r$, $C_5X = Z_3V_3'$ exactly.)

Exercise 9.18. Five units are measured on four variables:

$$X = \begin{pmatrix} 4 & 5 & 9 & 9 \\ 7 & 9 & 1 & 9 \\ 1 & 1 & 1 & 5 \\ 1 & 9 & 5 & 1 \\ 7 & 1 & 9 & 1 \end{pmatrix}$$

(a) Find $\bar{x}$, C_5X and S.

(b) Find $D_s^{1/2}$, the sample correlation matrix R and the standardized data matrix X_s, and check that $\frac{1}{4}X_s'X_s = R$.

(c) Find the loading vectors and the proportions of variance explained, first for the variables (from S) and then for the standardized variables (from R).

Solution. (a) $\mathbf{1}'X = (20, 25, 25, 25)$, so $\bar{x} = (4, 5, 5, 5)'$, and

$$C_5X = \begin{pmatrix} 0 & 0 & 4 & 4 \\ 3 & 4 & -4 & 4 \\ -3 & -4 & -4 & 0 \\ -3 & 4 & 0 & -4 \\ 3 & -4 & 4 & -4 \end{pmatrix}$$

$$(C_5X)'(C_5X) = \begin{pmatrix} 36 & 0 & 12 & 12 \\ 0 & 64 & -16 & 16 \\ 12 & -16 & 64 & -16 \\ 12 & 16 & -16 & 64 \end{pmatrix}, \qquad S = \begin{pmatrix} 9 & 0 & 3 & 3 \\ 0 & 16 & -4 & 4 \\ 3 & -4 & 16 & -4 \\ 3 & 4 & -4 & 16 \end{pmatrix}$$

(b) $D_s^{1/2} = \operatorname{diag}(3, 4, 4, 4)$, $r_{12} = 0$, $r_{13} = r_{14} = \frac{3}{3\cdot 4} = \frac{1}{4}$, $r_{23} = r_{34} = \frac{-4}{4\cdot 4} = -\frac{1}{4}$ and $r_{24} = \frac{4}{4\cdot 4} = \frac{1}{4}$, so

$$R = \begin{pmatrix} 1 & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & -\frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} & 1 & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & 1 \end{pmatrix}$$

$$X_s = C_5XD_s^{-1/2} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ -1 & -1 & -1 & 0 \\ -1 & 1 & 0 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad X_s'X_s = \begin{pmatrix} 4 & 0 & 1 & 1 \\ 0 & 4 & -1 & 1 \\ 1 & -1 & 4 & -1 \\ 1 & 1 & -1 & 4 \end{pmatrix}$$

For instance, column 1 of X_s has the sum of squares $0 + 1 + 1 + 1 + 1 = 4$, and columns 1 and 3 the sum of products $0 - 1 + 1 + 0 + 1 = 1$. So $\frac{1}{4}X_s'X_s = R$ (Proposition 9.18(a)).

(c) From S: for real t, subtract row 3 of $tI_4 - S$ from row 2 and add row 4 to it; the new row 2 is $(t - 24)(0, 1, -1, 1)$. Take out $t - 24$, add column 2 to column 3 and subtract it from column 4, and expand along row 2 and then along row 1 (Definition 5.13(a),(b), Theorems 5.22 and 5.21):

$$\begin{aligned} \det(tI_4 - S) &= \det \begin{pmatrix} t-9 & 0 & -3 & -3 \\ 0 & t-16 & 4 & -4 \\ -3 & 4 & t-16 & 4 \\ -3 & -4 & 4 & t-16 \end{pmatrix} = (t-24)\det \begin{pmatrix} t-9 & 0 & -3 & -3 \\ 0 & 1 & -1 & 1 \\ -3 & 4 & t-16 & 4 \\ -3 & -4 & 4 & t-16 \end{pmatrix} \\ &= (t-24)\det \begin{pmatrix} t-9 & 0 & -3 & -3 \\ 0 & 1 & 0 & 0 \\ -3 & 4 & t-12 & 0 \\ -3 & -4 & 0 & t-12 \end{pmatrix} = (t-24)\det \begin{pmatrix} t-9 & -3 & -3 \\ -3 & t-12 & 0 \\ -3 & 0 & t-12 \end{pmatrix} \\ &= (t-24)\Big((t-9)(t-12)^2 - 9(t-12) - 9(t-12)\Big) \\ &= (t-24)(t-12)(t^2 - 21t + 90) = (t-24)(t-15)(t-12)(t-6) \end{aligned}$$

so the eigenvalues are 24, 15, 12, 6 (Definition 7.7(a), Proposition 7.5(d)), and $S(0, 1, -1, 1)' = 24(0, 1, -1, 1)'$, $S(1, 0, 1, 1)' = 15(1, 0, 1, 1)'$, $S(0, 2, 1, -1)' = 12(0, 2, 1, -1)'$, $S(2, 0, -1, -1)' = 6(2, 0, -1, -1)'$. The eigenvalues are simple, so the loading vectors are, up to sign, $\frac{1}{\sqrt{3}}(0, 1, -1, 1)'$, $\frac{1}{\sqrt{3}}(1, 0, 1, 1)'$, $\frac{1}{\sqrt{6}}(0, 2, 1, -1)'$ and $\frac{1}{\sqrt{6}}(2, 0, -1, -1)'$ (Theorem 9.31(c),(e)); as $\operatorname{tr} S = 57$, the proportions of variance explained (Proposition 9.32) are $\frac{24}{57} = \frac{8}{19}$, $\frac{15}{57} = \frac{5}{19}$, $\frac{12}{57} = \frac{4}{19}$ and $\frac{6}{57} = \frac{2}{19}$.

From R: $R(0, 1, -1, 1)' = \frac{3}{2}(0, 1, -1, 1)'$, $R(2, 0, 1, 1)' = \frac{5}{4}(2, 0, 1, 1)'$, $R(0, 2, 1, -1)' = \frac{3}{4}(0, 2, 1, -1)'$ and $R(1, 0, -1, -1)' = \frac{1}{2}(1, 0, -1, -1)'$. These four vectors are pairwise orthogonal, so the eigenvalues of R are $\frac{3}{2}, \frac{5}{4}, \frac{3}{4}, \frac{1}{2}$ (Theorem 8.3(c)). The loading vectors are, up to sign, $\frac{1}{\sqrt{3}}(0, 1, -1, 1)'$, $\frac{1}{\sqrt{6}}(2, 0, 1, 1)'$, $\frac{1}{\sqrt{6}}(0, 2, 1, -1)'$ and $\frac{1}{\sqrt{3}}(1, 0, -1, -1)'$; as $\operatorname{tr} R = 4$, the proportions are $\frac{3}{8}$, $\frac{5}{16}$, $\frac{3}{16}$ and $\frac{1}{8}$. The first and the third loading vectors are the same for S and for R, but the second and the fourth are not: variable

1, whose sample standard deviation 3 is the smallest, has the weight $\frac{1}{\sqrt{3}}$ in the second loading vector of the variables and the larger weight $\frac{2}{\sqrt{6}}$ in that of the standardized variables.

Exercise 9.19. Continue Exercise 9.14.

(a) Find a best approximation of C_5X by a matrix of rank at most 1, and the error.

(b) Find the line in $\mathbb{R}^4$ closest to the five observations and the sum of the squared distances. Which observations lie on it?

Solution. (a) By Theorem 9.39(a) with $M = 1$, $\|C_5X - AB'\|_F^2 \geqslant (5-1)(\lambda_2+\lambda_3+\lambda_4) = 4\cdot 8 = 32$ for all $A \in \mathbb{R}^{5\times 1}$ and $B \in \mathbb{R}^{4\times 1}$, with equality for $A = z_1$, $B = v_1$. Every $F \in \mathbb{R}^{5\times 4}$ with rank $F \leqslant 1$ is such a product: $F = 0$ is $0\cdot 0'$, and if rank $F = 1$, then F is a 5×1 times a 1×4 matrix (Theorem 4.21(a)). So $A_1 = z_1v_1'$ is a best approximation; as $z_1 = \sqrt{3}(-4,1,1,1,1)'$ and $v_1 = \frac{1}{\sqrt{3}}(0,1,1,-1)'$,

$$A_1 = \begin{pmatrix} 0 & -4 & -4 & 4 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \end{pmatrix}, \qquad C_5X - A_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & -2 & -2 \\ 2 & 0 & 0 & 0 \\ -2 & 2 & 0 & 2 \end{pmatrix}$$

$$\|C_5X - A_1\|_F^2 = 0 + 8 + 8 + 4 + 12 = 32$$

the smallest possible error, $4(\lambda_2+\lambda_3+\lambda_4)$.

(b) By Theorem 9.39(c) with $M = 1$ ($\lambda_1 > \lambda_2$), the closest line is $\bar{x} + \operatorname{span}\{(0,1,1,-1)'\} = \{(3,\ 5+s,\ 5+s,\ 5-s)' : s \in \mathbb{R}\}$, the only one, with the sum of squared distances $(5-1)(\lambda_2+\lambda_3+\lambda_4) = 32$. Row i of $C_5X - A_1$ is $\big((x_i - \bar{x}) - v_1v_1'(x_i - \bar{x})\big)'$, as $z_{i1} = v_1'(x_i - \bar{x})$; so by Definition 9.38 the squared distances of the five observations from the line are 0, 8, 8, 4, 12, with sum 32. Observation 1, $(3,1,1,9)' = \bar{x} - 4(0,1,1,-1)'$, lies on the line; the other four do not.

Exercise 9.20. A random vector $x = (x_1, x_2, x_3, x_4)'$ has the covariance matrix

$$\Sigma = \begin{pmatrix} 4 & 0 & 2 & -1 \\ 0 & 4 & -2 & -2 \\ 2 & -2 & 4 & 0 \\ -1 & -2 & 0 & 4 \end{pmatrix}$$

(a) Find the eigenvalues of Σ and orthonormal eigenvectors.

(b) Find the principal components of x, their variances and the proportions of the total variance. How many components account for at least 80% of the total variance?

(c) Check that the first two principal components are uncorrelated, and compare $\operatorname{var}(x_1) + \operatorname{var}(x_2)$ with the sum of the variances of the first two principal components.

Solution. (a) $\Sigma = \left(\begin{smallmatrix} 4I_2 & B \\ B' & 4I_2 \end{smallmatrix}\right)$ with $B = \left(\begin{smallmatrix} 2 & -1 \\ -2 & -2 \end{smallmatrix}\right)$, and $BB' = \left(\begin{smallmatrix} 5 & -2 \\ -2 & 8 \end{smallmatrix}\right)$. For real $t \neq 4$, the block $(t-4)I_2$ of $tI_4 - \Sigma = \left(\begin{smallmatrix} (t-4)I_2 & -B \\ -B' & (t-4)I_2 \end{smallmatrix}\right)$ is invertible, and Theorem 5.30(b) gives, with Definition 5.13(a) for the factor $t-4$ in both rows of the 2×2 determinant,

$$\begin{aligned} \det(tI_4 - \Sigma) &= (t-4)^2 \det\Big((t-4)I_2 - \frac{1}{t-4}BB'\Big) = \det\Big((t-4)^2I_2 - BB'\Big) \\ &= \Big((t-4)^2 - 5\Big)\Big((t-4)^2 - 8\Big) - 4 \\ &= \Big((t-4)^2 - 4\Big)\Big((t-4)^2 - 9\Big) = (t-6)(t-2)(t-7)(t-1) \end{aligned}$$

This holds for every real $t \neq 4$, hence as polynomials (Definition 7.7(a), Proposition 7.5(d)): $p_\Sigma(t) = (t-7)(t-6)(t-2)(t-1)$, and the eigenvalues are 7, 6, 2, 1. Eigenvectors: row-reduce $\Sigma - \lambda I$ and solve $(\Sigma - \lambda I)c = 0$.

$$\Sigma - 7I = \begin{pmatrix} -3 & 0 & 2 & -1 \\ 0 & -3 & -2 & -2 \\ 2 & -2 & -3 & 0 \\ -1 & -2 & 0 & -3 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & -2 & 0 & -3 \\ 0 & -3 & -2 & -2 \\ 2 & -2 & -3 & 0 \\ -3 & 0 & 2 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_3 \leftarrow R_3 + 2R_1} \begin{pmatrix} -1 & -2 & 0 & -3 \\ 0 & -3 & -2 & -2 \\ 0 & -6 & -3 & -6 \\ 0 & 6 & 2 & 8 \end{pmatrix}$$
$$\xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} -1 & -2 & 0 & -3 \\ 0 & -3 & -2 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -2 & 4 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + 2R_3} \begin{pmatrix} -1 & -2 & 0 & -3 \\ 0 & -3 & -2 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $c_4 = 1$ gives $c_3 = 2c_4 = 2$, $c_2 = -\frac{1}{3}(2c_3 + 2c_4) = -2$ and $c_1 = -2c_2 - 3c_4 = 1$; take $(1, -2, 2, 1)'$.

$$\Sigma - 6I = \begin{pmatrix} -2 & 0 & 2 & -1 \\ 0 & -2 & -2 & -2 \\ 2 & -2 & -2 & 0 \\ -1 & -2 & 0 & -2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & -2 & 0 & -2 \\ 0 & -2 & -2 & -2 \\ 2 & -2 & -2 & 0 \\ -2 & 0 & 2 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 + 2R_1} \begin{pmatrix} -1 & -2 & 0 & -2 \\ 0 & -2 & -2 & -2 \\ 0 & -6 & -2 & -4 \\ 0 & 4 & 2 & 3 \end{pmatrix}$$
$$\xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 - 3R_2} \begin{pmatrix} -1 & -2 & 0 & -2 \\ 0 & -2 & -2 & -2 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & -2 & -1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + \frac{1}{2}R_3} \begin{pmatrix} -1 & -2 & 0 & -2 \\ 0 & -2 & -2 & -2 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $c_4 = -2$ gives $c_3 = -\frac{1}{2}c_4 = 1$, $c_2 = -c_3 - c_4 = 1$ and $c_1 = -2c_2 - 2c_4 = 2$; take $(2, 1, 1, -2)'$.

$$\Sigma - 2I = \begin{pmatrix} 2 & 0 & 2 & -1 \\ 0 & 2 & -2 & -2 \\ 2 & -2 & 2 & 0 \\ -1 & -2 & 0 & 2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & -2 & 0 & 2 \\ 0 & 2 & -2 & -2 \\ 2 & -2 & 2 & 0 \\ 2 & 0 & 2 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 2R_1]{R_3 \leftarrow R_3 + 2R_1} \begin{pmatrix} -1 & -2 & 0 & 2 \\ 0 & 2 & -2 & -2 \\ 0 & -6 & 2 & 4 \\ 0 & -4 & 2 & 3 \end{pmatrix}$$
$$\xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + 3R_2} \begin{pmatrix} -1 & -2 & 0 & 2 \\ 0 & 2 & -2 & -2 \\ 0 & 0 & -4 & -2 \\ 0 & 0 & -2 & -1 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - \frac{1}{2}R_3} \begin{pmatrix} -1 & -2 & 0 & 2 \\ 0 & 2 & -2 & -2 \\ 0 & 0 & -4 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $c_4 = 2$ gives $c_3 = -\frac{1}{2}c_4 = -1$, $c_2 = c_3 + c_4 = 1$ and $c_1 = -2c_2 + 2c_4 = 2$; take $(2, 1, -1, 2)'$.

$$\Sigma - I = \begin{pmatrix} 3 & 0 & 2 & -1 \\ 0 & 3 & -2 & -2 \\ 2 & -2 & 3 & 0 \\ -1 & -2 & 0 & 3 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \begin{pmatrix} -1 & -2 & 0 & 3 \\ 0 & 3 & -2 & -2 \\ 2 & -2 & 3 & 0 \\ 3 & 0 & 2 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + 3R_1]{R_3 \leftarrow R_3 + 2R_1} \begin{pmatrix} -1 & -2 & 0 & 3 \\ 0 & 3 & -2 & -2 \\ 0 & -6 & 3 & 6 \\ 0 & -6 & 2 & 8 \end{pmatrix}$$
$$\xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} -1 & -2 & 0 & 3 \\ 0 & 3 & -2 & -2 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & -2 & 4 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} -1 & -2 & 0 & 3 \\ 0 & 3 & -2 & -2 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $c_4 = 1$ gives $c_3 = 2c_4 = 2$, $c_2 = \frac{1}{3}(2c_3 + 2c_4) = 2$ and $c_1 = -2c_2 + 3c_4 = -1$; take $(-1, 2, 2, 1)'$. The four vectors are pairwise orthogonal and have length $\sqrt{10}$, so $v_1 = \frac{1}{\sqrt{10}}(1, -2, 2, 1)'$, $v_2 = \frac{1}{\sqrt{10}}(2, 1, 1, -2)'$, $v_3 = \frac{1}{\sqrt{10}}(2, 1, -1, 2)'$ and $v_4 = \frac{1}{\sqrt{10}}(-1, 2, 2, 1)'$ are orthonormal eigenvectors for 7, 6, 2, 1. Check: $\Sigma(1, -2, 2, 1)' = (7, -14, 14, 7)'$, $\Sigma(2, 1, 1, -2)' = (12, 6, 6, -12)'$, $\Sigma(2, 1, -1, 2)' = (4, 2, -2, 4)'$, $\Sigma(-1, 2, 2, 1)' = (-1, 2, 2, 1)'$.

(b) By Theorem 9.42(b),(c), the principal components are $v_1'x = \frac{1}{\sqrt{10}}(x_1 - 2x_2 + 2x_3 + x_4)$, $v_2'x = \frac{1}{\sqrt{10}}(2x_1 + x_2 + x_3 - 2x_4)$, $v_3'x = \frac{1}{\sqrt{10}}(2x_1 + x_2 - x_3 + 2x_4)$ and $v_4'x = \frac{1}{\sqrt{10}}(-x_1 + 2x_2 + 2x_3 + x_4)$, with

variances 7, 6, 2, 1. The total variance is $\operatorname{tr}\Sigma = 16$, so they account for $\frac{7}{16}$, $\frac{6}{16} = \frac{3}{8}$, $\frac{2}{16} = \frac{1}{8}$ and $\frac{1}{16}$ of it. The cumulative proportions are $\frac{7}{16}$, $\frac{13}{16}$, $\frac{15}{16}$ and 1; as $\frac{7}{16} = \frac{35}{80} < \frac{64}{80} = \frac{4}{5} \leqslant \frac{65}{80} = \frac{13}{16}$, two components account for at least 80%, one does not.

(c) $\operatorname{cov}(v_1'x, v_2'x) = v_1'\Sigma v_2 = \frac{1}{10}(1,-2,2,1)(12,6,6,-12)' = \frac{1}{10}(12-12+12-12) = 0$ (Theorem 9.42(a)). And $\operatorname{var}(x_1) + \operatorname{var}(x_2) = 4 + 4 = 8 \leqslant 7 + 6 = 13$, as Corollary 9.29 says (every pair of the variables has the variances $4 + 4 = 8$).

Problems

Problem 9.1 ([17, Exercise 4.43], [20, Exercise 4.43])**.** Let x_1, x_2, x_3 be uncorrelated random variables, each with mean μ and variance σ^2. Find $\operatorname{cov}(x_1 + x_2,\ x_2 + x_3)$ and $\operatorname{cov}(x_1 + x_2,\ x_1 - x_2)$.

Solution. The $(1,2)$ entry of Theorem 9.4(e) for the random vector $(u,v)'$ reads $\mathsf{E}(uv) = \operatorname{cov}(u,v) + \mathsf{E}(u)\,\mathsf{E}(v)$, that is, $\operatorname{cov}(u,v) = \mathsf{E}(uv) - \mathsf{E}(u)\,\mathsf{E}(v)$. So $\mathsf{E}(x_ix_j) = \operatorname{cov}(x_i,x_j) + \mu^2 = \mu^2$ for $i \neq j$, as the x_i are uncorrelated, and $\mathsf{E}(x_i^2) = \sigma^2 + \mu^2$. Hence

$$\operatorname{cov}(x_1 + x_2,\ x_2 + x_3) = \mathsf{E}(x_1x_2 + x_1x_3 + x_2^2 + x_2x_3) - (2\mu)(2\mu) = (4\mu^2 + \sigma^2) - 4\mu^2 = \sigma^2$$

and $\operatorname{cov}(x_1 + x_2,\ x_1 - x_2) = \mathsf{E}(x_1^2 - x_2^2) - (2\mu)\cdot 0 = (\sigma^2 + \mu^2) - (\sigma^2 + \mu^2) = 0$.

Problem 9.2 ([2, Exercise 10.2])**.** Let $X \in \mathbb{R}^{n\times p}$ be a data matrix and $\theta \in \mathbb{R}$. Show that subtracting θ from every entry of X does not change $X'C_nX$, and hence does not change the sample covariance matrix S.

Solution. The new data matrix is $X^* = X - \theta\mathbf{1}_n\mathbf{1}_p'$. As $C_n\mathbf{1}_n = 0$ and $C_n' = C_n$ (Proposition 1.15), $C_n\mathbf{1}_n\mathbf{1}_p' = 0$ and $\mathbf{1}_p\mathbf{1}_n'C_n = (C_n\mathbf{1}_n\mathbf{1}_p')' = 0$. So

$$X^{*\prime}C_nX^* = X'C_nX - \theta X'C_n\mathbf{1}_n\mathbf{1}_p' - \theta\mathbf{1}_p\mathbf{1}_n'C_nX + \theta^2\mathbf{1}_p\mathbf{1}_n'C_n\mathbf{1}_n\mathbf{1}_p' = X'C_nX$$

and $S = \frac{1}{n-1}X'C_nX$ (Definition 9.13) is unchanged.

Problem 9.3 ([2, Example 10.1])**.** Let $p \geqslant 2$ and $\rho < -\frac{1}{p-1}$. Can p variables of a data matrix have all pairwise sample correlations equal to ρ?

Solution. No. Their sample correlation matrix would be the equicorrelation matrix $R = (1-\rho)I_p + \rho J_p$ of Example 5.33. A sample correlation matrix is PSD (Proposition 9.18(a)), so its determinant is $\geqslant 0$ (Theorem 8.21(c)). But for $\rho < -\frac{1}{p-1}$, $1-\rho > 0$ and $1+(p-1)\rho < 0$, so $\det R = (1-\rho)^{p-1}\big(1+(p-1)\rho\big) < 0$ (Example 5.33). For three variables, even unequal pairwise sample correlations r_{12}, r_{13}, r_{23} cannot all be below $-\frac{1}{2}$: at $x = \mathbf{1}_3$, the quadratic form of their sample correlation matrix takes the value $3 + 2(r_{12} + r_{13} + r_{23})$ (Definition 8.17), which is $\geqslant 0$; so $r_{12} + r_{13} + r_{23} \geqslant -\frac{3}{2}$.

Problem 9.4 ([2, Exercise 6.19])**.** Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ be symmetric. Show that $\lambda_n(A) \leqslant \frac{1}{n}\sum_{i=1}^n\sum_{j=1}^n a_{ij} \leqslant \lambda_1(A)$.

Solution. $\sum_i\sum_j a_{ij} = \mathbf{1}_n'A\mathbf{1}_n$ and $\mathbf{1}_n'\mathbf{1}_n = n$ (Proposition 1.15), so by Theorem 9.23(a), $\lambda_n(A) \leqslant \mathbf{1}_n'A\mathbf{1}_n/\mathbf{1}_n'\mathbf{1}_n \leqslant \lambda_1(A)$.

Problem 9.5 ([6, Exercise 12.40(a)])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric. Show that $\lambda_1(A+B) \leqslant \lambda_1(A) + \lambda_1(B)$ and $\lambda_n(A+B) \geqslant \lambda_n(A) + \lambda_n(B)$.

Solution. By Theorem 9.23(a), for every $x \neq 0$

$$\lambda_n(A) + \lambda_n(B) \leqslant \frac{x'(A+B)x}{x'x} = \frac{x'Ax}{x'x} + \frac{x'Bx}{x'x} \leqslant \lambda_1(A) + \lambda_1(B)$$

For an eigenvector x of $A+B$ belonging to $\lambda_1(A+B)$ the middle term is $\lambda_1(A+B)$, which gives the first inequality; for one belonging to $\lambda_n(A+B)$ it is $\lambda_n(A+B)$, which gives the second.

Problem 9.6 ([2, Exercise 6.21])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric, with A PSD. Show that

$$\lambda_n(B)\operatorname{tr} A \leqslant \operatorname{tr}(AB) \leqslant \lambda_1(B)\operatorname{tr} A$$

Solution.

- Let $N = A^{1/2}$ (Theorem 8.46); it is symmetric and $N^2 = A$. By Theorem 9.23(c), $B - \lambda_n(B)I$ and $\lambda_1(B)I - B$ are PSD.
- If M is PSD, so is NMN: it is symmetric, and $x'NMNx = (Nx)'M(Nx) \geqslant 0$. Hence $NBN - \lambda_n(B)A$ and $\lambda_1(B)A - NBN$ are PSD.
- A PSD matrix has eigenvalues $\geqslant 0$ (Theorem 8.21(a)), hence trace $\geqslant 0$ (Theorem 7.19). By Theorem 1.11(a), $\lambda_n(B)\operatorname{tr} A \leqslant \operatorname{tr}(NBN) \leqslant \lambda_1(B)\operatorname{tr} A$.
- $\operatorname{tr}(NBN) = \operatorname{tr}(BNN) = \operatorname{tr}(BA) = \operatorname{tr}(AB)$ (Theorem 1.11(b)).

Problem 9.7 ([2, Exercise 6.24])**.** Let $A \in \mathbb{R}^{n\times n}$ be PSD. Show that $\operatorname{tr}(A^2) \leqslant \lambda_1(A)\operatorname{tr} A$.

Solution. With Q as in Lemma 9.22, $Q'AQ = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $Q^{-1} = Q'$ (Proposition 6.13(b)), so $A = Q\operatorname{diag}(\lambda_1, \ldots, \lambda_n)Q^{-1}$. By Proposition 7.29(b) with $f(t) = t^2$, $\operatorname{tr}(A^2) = \sum_i \lambda_i^2$. As A is PSD, $0 \leqslant \lambda_i \leqslant \lambda_1$ (Theorem 8.21(a)), so $\lambda_i^2 \leqslant \lambda_1\lambda_i$. Hence $\operatorname{tr}(A^2) \leqslant \lambda_1 \sum_i \lambda_i = \lambda_1 \operatorname{tr} A$ (Theorem 7.19).

Problem 9.8 ([10, Problem 38.1])**.** A matrix $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is **doubly stochastic** if all its entries are $\geqslant 0$ and all its row sums and column sums are 1, that is, $A\mathbf{1}_n = \mathbf{1}_n$ and $\mathbf{1}_n'A = \mathbf{1}_n'$. Let A be doubly stochastic, and let $x_1 \geqslant x_2 \geqslant \cdots \geqslant x_n \geqslant 0$ and $y_1 \geqslant y_2 \geqslant \cdots \geqslant y_n \geqslant 0$. Show that

$$\sum_{r=1}^{n}\sum_{s=1}^{n} a_{rs}x_ry_s \leqslant \sum_{r=1}^{n} x_ry_r$$

Solution. All sums below run over $1, \ldots, n$ unless limits are shown, and δ_{rs} is 1 for $r = s$ and 0 for $r \neq s$.

- Put $x_{n+1} = y_{n+1} = 0$, $\xi_i = x_i - x_{i+1}$ and $\eta_j = y_j - y_{j+1}$ $(1 \leqslant i, j \leqslant n)$. By the hypotheses, $\xi_i \geqslant 0$ and $\eta_j \geqslant 0$, and the sums telescope: $x_r = \sum_{i\geqslant r} \xi_i$ and $y_s = \sum_{j\geqslant s} \eta_j$.
- As $\sum_r x_ry_r = \sum_{r,s} \delta_{rs}x_ry_s$,

  $$\sum_r x_ry_r - \sum_{r,s} a_{rs}x_ry_s = \sum_{r,s}(\delta_{rs} - a_{rs}) \sum_{i\geqslant r}\xi_i \sum_{j\geqslant s}\eta_j = \sum_{i,j} c_{ij}\xi_i\eta_j, \qquad c_{ij} = \sum_{r\leqslant i}\sum_{s\leqslant j}(\delta_{rs} - a_{rs})$$

  since the product $(\delta_{rs} - a_{rs})\xi_i\eta_j$ occurs on both sides exactly for $r \leqslant i$ and $s \leqslant j$. As every $\xi_i\eta_j \geqslant 0$, it suffices to show that every $c_{ij} \geqslant 0$.
- Let $i \leqslant j$. Among the pairs with $r \leqslant i$ and $s \leqslant j$, those with $r = s$ are $(1,1), \ldots, (i,i)$, so $\sum_{r\leqslant i}\sum_{s\leqslant j} \delta_{rs} = i$. As $a_{rs} \geqslant 0$ and the row sums of A are 1, $\sum_{r\leqslant i}\sum_{s\leqslant j} a_{rs} \leqslant \sum_{r\leqslant i}\sum_{s=1}^{n} a_{rs} = i$. So $c_{ij} \geqslant 0$.
- Let $i \geqslant j$. Likewise the pairs with $r = s$ are $(1,1), \ldots, (j,j)$, so $\sum_{r\leqslant i}\sum_{s\leqslant j} \delta_{rs} = j$. As $a_{rs} \geqslant 0$ and the column sums of A are 1, $\sum_{r\leqslant i}\sum_{s\leqslant j} a_{rs} \leqslant \sum_{s\leqslant j}\sum_{r=1}^{n} a_{rs} = j$. So $c_{ij} \geqslant 0$.

If $P = (p_{ij}) \in \mathbb{R}^{n\times n}$ is orthogonal, the matrix (p_{ij}^2) is doubly stochastic, as the rows and the columns of P are unit vectors (Theorem 6.19(a)). In the proof of Theorem 9.27(a), with $P = [R\ T]$, the weight w_i is the sum of the first k entries of row i of (p_{ij}^2).

Chapter 10. The multivariate normal distribution

The multivariate normal distribution

We write $\mathsf{P}(z \in A)$ for the probability that a random vector $z \in \mathbb{R}^n$ takes a value in a set $A \subseteq \mathbb{R}^n$. Two random vectors $y, z \in \mathbb{R}^n$ have the **same distribution** if $\mathsf{P}(y \in A) = \mathsf{P}(z \in A)$ for every set A; then also $\mathsf{E}\, g(y) = \mathsf{E}\, g(z)$ for every function g for which these expectations exist. If a random variable u takes only the values $a_1, \dots, a_m$, then $\mathsf{E}\, g(u) = g(a_1)\,\mathsf{P}(u = a_1) + \cdots + g(a_m)\,\mathsf{P}(u = a_m)$ for every function g. Random vectors $u_1, \dots, u_m$ are **independent** if $\mathsf{P}(u_1 \in A_1, \dots, u_m \in A_m) = \mathsf{P}(u_1 \in A_1) \cdots \mathsf{P}(u_m \in A_m)$ for all sets $A_1, \dots, A_m$. The **moment generating function** (MGF) of $z \in \mathbb{R}^n$ is

$$M_z(t) = \mathsf{E}\Big(\exp(t'z)\Big) \qquad (t \in \mathbb{R}^n)$$

defined wherever this expectation is finite [12, §4.1], [17, Definitions 1.5.8, 2.2.1, Theorem 4.2.10(a)]. (Here and below, the sets and functions are those for which the probabilities and expectations are defined.)

Definition 10.1 ([12, Definition 4.1])**.** Let $\mu \in \mathbb{R}^n$, and let $\Sigma \in \mathbb{R}^{n \times n}$ be symmetric and PSD. A random vector $z \in \mathbb{R}^n$ has the **multivariate normal distribution** $N_n(\mu, \Sigma)$, written $z \sim N_n(\mu, \Sigma)$, if

$$M_z(t) = \exp\Big(t'\mu + \tfrac{1}{2}t'\Sigma t\Big) \qquad \text{for all } t \in \mathbb{R}^n$$

Its mean vector is μ and its covariance matrix is Σ: $\mathsf{E}(z) = \mu$ and $\mathrm{var}(z) = \Sigma$; and $N_n(\mu, 0)$ is the distribution of the constant vector μ (both shown below). For $n = 1$ we write $N(\mu, \sigma^2)$ ($\sigma^2 \geqslant 0$), with the MGF $\exp(t\mu + \frac{1}{2}\sigma^2 t^2)$; $N(0, 1)$ is the **standard normal distribution**. Random vectors u and v are **jointly normal** if $(u', v')'$ has a multivariate normal distribution.

We use the following facts without proof [12, §4.1].

(N1) *Uniqueness.* If two random vectors have MGFs that are finite and equal on an open neighborhood of 0, they have the same distribution.

(N2) *Independence and factorization.* Random vectors $u_1 \in \mathbb{R}^{n_1}, \dots, u_m \in \mathbb{R}^{n_m}$ whose MGFs are finite near 0 are independent if and only if the MGF of $(u_1', \dots, u_m')'$ factors: $M_{(u_1', \dots, u_m')'}(t_1, \dots, t_m) = M_{u_1}(t_1) \cdots M_{u_m}(t_m)$ for all $t_k \in \mathbb{R}^{n_k}$ near 0.

(N3) *Functions of independent vectors.* If $u_1, \dots, u_m$ are independent, then so are $g_1(u_1), \dots, g_m(u_m)$, for any functions $g_1, \dots, g_m$.

(N4) *Density in the nonsingular case.* This fact is not assumed here: the density of $N_n(\mu, \Sigma)$ for PD Σ is derived below from the change-of-variables formula for integrals.

(N5) *Distributions derived from the normal.* The **chi-square distribution** χ^2_r ($r \geqslant 1$) is the distribution of $w_1^2 + \cdots + w_r^2$ for independent $w_1, \dots, w_r \sim N(0, 1)$.

(N6) *Moments from the MGF.* If M_z is finite on an open neighborhood of 0, its partial derivatives of the first and second order exist there, and at $t = 0$, $\partial M_z/\partial t_i = \mathsf{E}(z_i)$ and $\partial^2 M_z/\partial t_i \partial t_j = \mathsf{E}(z_i z_j)$.

(N7) *Independence implies uncorrelatedness.* If u and v are independent random vectors, then $\mathsf{E}(uv') = \mathsf{E}(u)\,\mathsf{E}(v)'$, that is, $\mathrm{cov}(u, v) = 0$ (Definition 9.2). In particular $\mathsf{E}(uv) = \mathsf{E}(u)\,\mathsf{E}(v)$ for independent random variables u and v.

Let $z \sim N_n(\mu, \Sigma)$. The exponent $q(t) = t'\mu + \frac{1}{2}t'\Sigma t = \sum_k t_k\mu_k + \frac{1}{2}\sum_k \sum_l \sigma_{kl} t_k t_l$ has, as $\sigma_{kl} = \sigma_{lk}$, the partial derivatives $\partial q/\partial t_i = \mu_i + \sum_l \sigma_{il} t_l = \mu_i + (\Sigma t)_i$ and $\partial^2 q/\partial t_j \partial t_i = \sigma_{ij}$. By the chain rule and the product rule,

$$\frac{\partial M_z}{\partial t_i} = \Big(\mu_i + (\Sigma t)_i\Big) M_z(t), \qquad \frac{\partial^2 M_z}{\partial t_j \partial t_i} = \Big(\sigma_{ij} + \Big(\mu_i + (\Sigma t)_i\Big)\Big(\mu_j + (\Sigma t)_j\Big)\Big) M_z(t)$$

At $t = 0$, $M_z(0) = 1$, so by (N6) $\mathsf{E}(z_i) = \mu_i$ and $\mathsf{E}(z_i z_j) = \sigma_{ij} + \mu_i\mu_j$. Hence $\operatorname{cov}(z_i, z_j) = \mathsf{E}(z_i z_j) - \mu_i\mu_j = \sigma_{ij}$ (the (i,j) entry of Theorem 9.4(e)): $\mathsf{E}(z) = \mu$ and $\operatorname{var}(z) = \Sigma$, as claimed in Definition 10.1 [12, Definition 4.1]. If $\Sigma = 0$, the constant vector μ has the MGF $\mathsf{E}(\exp(t'\mu)) = \exp(t'\mu)$; so by (N1), $z \sim N_n(\mu, 0)$ has the distribution of the constant μ, and $\mathsf{P}(z = \mu) = 1$.

Linear transformations and marginal distributions

Theorem 10.2 ([12, Theorem 4.2], [2, (10.28)–(10.32)], [3, §8.1])**.** Let $z \sim N_n(\mu, \Sigma)$, and let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ be constant. Then

$$Az + b \sim N_m(A\mu + b,\ A\Sigma A')$$

Proof. • $A\Sigma A' = (A')'\Sigma(A')$ is symmetric and PSD (Theorem 8.24(a)), so $N_m(A\mu + b, A\Sigma A')$ is defined.

- For $t \in \mathbb{R}^m$, $t'(Az + b) = t'b + (A't)'z$, so $\exp\big(t'(Az+b)\big) = \exp(t'b)\exp\big((A't)'z\big)$, and $M_{Az+b}(t) = \exp(t'b)M_z(A't)$, as $\exp(t'b)$ is a constant.
- By Definition 10.1,

$$M_{Az+b}(t) = \exp\Big(t'b + (A't)'\mu + \tfrac{1}{2}(A't)'\Sigma(A't)\Big) = \exp\Big(t'(A\mu+b) + \tfrac{1}{2}t'(A\Sigma A')t\Big)$$

for every $t \in \mathbb{R}^m$, which is the MGF of $N_m(A\mu + b, A\Sigma A')$. □

Corollary 10.3 ([3, §8.1], [17, §4.5])**.** Let $z \sim N_n(\mu, \Sigma)$.

(a) Partition $z = (z_1', z_2')'$ with $z_1 \in \mathbb{R}^p$, $z_2 \in \mathbb{R}^q$, and accordingly

$$\mu = \begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \qquad \Sigma = \begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix} \qquad (\Sigma_{11} \in \mathbb{R}^{p\times p})$$

Then $z_1 \sim N_p(\mu_1, \Sigma_{11})$ and $z_2 \sim N_q(\mu_2, \Sigma_{22})$.

(b) For $1 \leqslant i_1 < \cdots < i_k \leqslant n$, the subvector $(z_{i_1}, \ldots, z_{i_k})'$ is N_k, with the mean $(\mu_{i_1}, \ldots, \mu_{i_k})'$ and as covariance matrix the principal submatrix of Σ in the rows and columns $i_1, \ldots, i_k$. In particular $z_i \sim N(\mu_i, \sigma_{ii})$.

(c) For $a \in \mathbb{R}^n$, $a'z \sim N(a'\mu,\ a'\Sigma a)$.

Proof. (a) For $t_1 \in \mathbb{R}^p$, $t_1'z_1 = (t_1', 0')z$, so $M_{z_1}(t_1) = M_z\big((t_1', 0')'\big)$. By the block product (Theorem 5.3(c)), $(t_1', 0')\mu = t_1'\mu_1$ and $(t_1', 0')\Sigma(t_1', 0')' = t_1'\Sigma_{11}t_1$; so $M_{z_1}(t_1) = \exp(t_1'\mu_1 + \frac{1}{2}t_1'\Sigma_{11}t_1)$ for every t_1, where Σ_{11} is symmetric and PSD (Theorem 8.30). Likewise for z_2, with $t = (0', t_2')'$.

(b) With $T = [e_{i_1}\ \cdots\ e_{i_k}] \in \mathbb{R}^{n\times k}$ (the matrix S of Definition 8.29), the subvector is $T'z$ (Theorem 1.4(d)), and $T'\Sigma T$ is the principal submatrix; apply Theorem 10.2 with $A = T'$ and $b = 0$.

(c) Theorem 10.2 with $A = a'$ and $b = 0$. □

Example 10.4. Let $z \sim N_4(\mu, \Sigma)$ with $\mu = (1, 0, 2, 2)'$ and Σ of Example 9.9.

- By Corollary 10.3(b), $z_2 \sim N(0, 1)$ and

$$\begin{pmatrix}z_1\\ z_3\end{pmatrix} \sim N_2\left(\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}4 & 4\\ 4 & 9\end{pmatrix}\right)$$

- By Corollary 10.3(c) and Example 9.9, $z_1 + z_2 + z_3 + z_4 \sim N(5, 51)$, as $\mathbf{1}'\mu = 1 + 0 + 2 + 2 = 5$; $e = z_3 - z_1 - 2z_2 \sim N(1, 1)$, as $a'\mu = -1 - 0 + 2 + 0 = 1$ for $a = (-1, -2, 1, 0)'$; and $f = z_4 + 2z_1 + 2z_2 - 2z_3 \sim N(0, 1)$, as $b'\mu = 2 + 0 - 4 + 2 = 0$ for $b = (2, 2, -2, 1)'$.
- With $A = \left(\begin{smallmatrix}1 & 1 & 0 & 0\\ 1 & -1 & 0 & 0\end{smallmatrix}\right)$, Theorem 10.2 and Example 9.9 give

$$\begin{pmatrix}z_1 + z_2\\ z_1 - z_2\end{pmatrix} = Az \sim N_2\left(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}5 & 3\\ 3 & 5\end{pmatrix}\right)$$

Independence

Theorem 10.5 ([12, Theorems 4.3, 4.5], [3, Results 8.1, 8.2])**.** (a) Let $u \in \mathbb{R}^p$ and $v \in \mathbb{R}^q$ be jointly normal, $z = (u', v')' \sim N_{p+q}(\mu, \Sigma)$, with

$$\Sigma = \begin{pmatrix} \Sigma_{uu} & \Sigma_{uv} \\ \Sigma_{vu} & \Sigma_{vv} \end{pmatrix}, \qquad \Sigma_{uu} \in \mathbb{R}^{p \times p}$$

Then u and v are independent if and only if $\Sigma_{uv} = 0$, that is, if and only if u and v are uncorrelated.

(b) Let $z = (z_1', \dots, z_m')' \sim N_n(\mu, \Sigma)$, with subvectors $z_k \in \mathbb{R}^{n_k}$ and $\Sigma = (\Sigma_{kl})$ partitioned accordingly. Then $z_1, \dots, z_m$ are independent if and only if $\Sigma_{kl} = 0$ for all $k \neq l$.

(c) Let $z \sim N_n(\mu, \Sigma)$, and let $A \in \mathbb{R}^{p \times n}$ and $B \in \mathbb{R}^{q \times n}$ be constant. Then Az and Bz are independent if and only if $A\Sigma B' = 0$.

Proof. (a) Partition $\mu = (\mu_u', \mu_v')'$. By Definition 10.1 and the block form of $\operatorname{var}(z)$ after Definition 9.2, $\Sigma_{uv} = \operatorname{cov}(u, v)$.

- If u and v are independent, then $\Sigma_{uv} = \operatorname{cov}(u, v) = 0$ by (N7).
- Let $\Sigma_{uv} = 0$; then $\Sigma_{vu} = \Sigma_{uv}' = 0$. For $s \in \mathbb{R}^p$ and $t \in \mathbb{R}^q$, the block product (Theorem 5.3(c)) gives $(s', t')\mu = s'\mu_u + t'\mu_v$ and $(s', t')\Sigma(s', t')' = s'\Sigma_{uu}s + t'\Sigma_{vv}t$. So, by Corollary 10.3(a),

$$M_z(s, t) = \exp\Big(s'\mu_u + \tfrac{1}{2}s'\Sigma_{uu}s\Big) \exp\Big(t'\mu_v + \tfrac{1}{2}t'\Sigma_{vv}t\Big) = M_u(s) M_v(t)$$

and u and v are independent by (N2).

(b)

- If $\Sigma_{kl} = 0$ for $k \neq l$, then for $t = (t_1', \dots, t_m')'$ the block product gives $t'\Sigma t = \sum_k t_k'\Sigma_{kk}t_k$, so $M_z(t) = \prod_k \exp(t_k'\mu_k + \frac{1}{2}t_k'\Sigma_{kk}t_k) = \prod_k M_{z_k}(t_k)$, as $z_k \sim N_{n_k}(\mu_k, \Sigma_{kk})$ (Corollary 10.3(b)). By (N2), $z_1, \dots, z_m$ are independent.
- Conversely, let $z_1, \dots, z_m$ be independent, and $k \neq l$. By (N2), $M_z(t) = \prod_j M_{z_j}(t_j)$. Taking $t_j = 0$ for $j \neq k, l$, and as $M_{z_j}(0) = 1$, the MGF of $(z_k', z_l')'$ is $M_{z_k}(t_k)M_{z_l}(t_l)$. So z_k and z_l are independent (N2), and $\Sigma_{kl} = \operatorname{cov}(z_k, z_l) = 0$ by (N7).

(c)

- Let $G = \binom{A}{B} \in \mathbb{R}^{(p+q) \times n}$. By Theorem 10.2, $Gz = \Big((Az)', (Bz)'\Big)' \sim N_{p+q}(G\mu, G\Sigma G')$, and by the block product

$$G\Sigma G' = \begin{pmatrix} A \\ B \end{pmatrix} \Sigma \begin{pmatrix} A' & B' \end{pmatrix} = \begin{pmatrix} A\Sigma A' & A\Sigma B' \\ B\Sigma A' & B\Sigma B' \end{pmatrix}$$

- By (a) for $u = Az$ and $v = Bz$, they are independent if and only if $A\Sigma B' = 0$. □

Remark 10.6 ([12, Remark 4.4], [17, §4.5])**.** Joint normality is needed in Theorem 10.5: uncorrelated random variables need not be independent, even if each of them is normal.

- Let u take the values -1, 0, 1, each with probability $\frac{1}{3}$, and let $v = u^2$. Then $\mathsf{E}(u) = \mathsf{E}(u^3) = 0$, so $\operatorname{cov}(u, v) = \mathsf{E}(u^3) - \mathsf{E}(u)\,\mathsf{E}(u^2) = 0$; but $\mathsf{P}(u = 0,\ v = 1) = 0 \neq \frac{1}{3} \cdot \frac{2}{3} = \mathsf{P}(u = 0)\,\mathsf{P}(v = 1)$.
- Let $x \sim N(0, 1)$, let s be independent of x with $\mathsf{P}(s = 1) = \mathsf{P}(s = -1) = \frac{1}{2}$, and put $y = sx$. Let ι_+ be 1 if $s = 1$ and 0 otherwise, and $\iota_- = 1 - \iota_+$. Then $\mathsf{E}(\iota_+) = \mathsf{E}(\iota_-) = \frac{1}{2}$ and $\mathsf{E}(s) = 0$, and ι_+, ι_- and s are independent of every function of x (N3). As $e^{ty} = \iota_+ e^{tx} + \iota_- e^{-tx}$, (N7) and Definition 10.1 give $M_y(t) = \frac{1}{2}e^{t^2/2} + \frac{1}{2}e^{t^2/2} = e^{t^2/2}$ for every t: $y \sim N(0, 1)$. Also $\mathsf{E}(x) = 0$, so $\operatorname{cov}(x, y) = \mathsf{E}(xy) = \mathsf{E}(sx^2) = \mathsf{E}(s)\,\mathsf{E}(x^2) = 0$.

- But x and y are not independent. As $x+y=(1+s)x$, $e^{t(x+y)}=\iota_+e^{2tx}+\iota_-$, so $\mathsf{E}(e^{t(x+y)})=\frac{1}{2}e^{2t^2}+\frac{1}{2}$. If x and y were independent, then by (N3) and (N7), $\mathsf{E}(e^{t(x+y)})=\mathsf{E}(e^{tx})\,\mathsf{E}(e^{ty})=e^{t^2}$ for every t; but at $t=1$, $\frac{1}{2}e^2+\frac{1}{2}-e=\frac{1}{2}(e-1)^2>0$. Hence, by Theorem 10.5(a), x and y are not jointly normal either.

Example 10.7. Let $z\sim N_4(\mu,\Sigma)$ be as in Example 10.4.

- By Corollary 10.3(b),
$$\begin{pmatrix}z_1\\z_2\\z_4\end{pmatrix}\sim N_3\left(\begin{pmatrix}1\\0\\2\end{pmatrix},\begin{pmatrix}4&0&0\\0&1&2\\0&2&9\end{pmatrix}\right)$$
and $\sigma_{12}=\sigma_{14}=0$; so z_1 and $(z_2,z_4)'$ are independent (Theorem 10.5(a) with $p=1$ and $q=2$).
- Let $A=\left(\begin{smallmatrix}1&0&0&0\\0&1&0&0\end{smallmatrix}\right)$, so that $Az=(z_1,z_2)'$, and $e=a'z=z_3-z_1-2z_2$. As $\Sigma a=(0,0,1,2)'$ (Example 9.9), $A\Sigma a=(0,0)'$, and by Theorem 10.5(c), e is independent of $(z_1,z_2)'$. Thus $z_3=z_1+2z_2+e$, where $e\sim N(1,1)$ is independent of z_1 and z_2.
- Likewise, let $G=[I_3\ \ 0]\in\mathbb{R}^{3\times4}$, so that $Gz=(z_1,z_2,z_3)'$, and $f=b'z=z_4+2z_1+2z_2-2z_3$. As $\Sigma b=(0,0,0,1)'$ (Example 9.9), $G\Sigma b=(0,0,0)'$, and f is independent of $(z_1,z_2,z_3)'$. Thus $z_4=-2z_1-2z_2+2z_3+f$, where $f\sim N(0,1)$ (Example 10.4) is independent of z_1, z_2 and z_3.
- z_1+z_2 and z_1-z_2 are not independent: their covariance is $3\neq0$ (Example 10.4), and independent vectors are uncorrelated (N7).

The density of a normal vector

A random vector $z\in\mathbb{R}^n$ has the **density** $f\colon\mathbb{R}^n\to[0,\infty)$ if $\mathsf{P}(z\in A)=\int_A f(x)\,\mathrm{d}x$ for every set $A\subseteq\mathbb{R}^n$; then $\mathsf{E}\,g(z)=\int_{\mathbb{R}^n}g(x)f(x)\,\mathrm{d}x$. Every function $f\geqslant0$ with $\int_{\mathbb{R}^n}f(x)\,\mathrm{d}x=1$ is the density of some random vector. Random vectors $u_1,\dots,u_m$ with densities $f_1,\dots,f_m$ are independent if and only if $(u_1',\dots,u_m')'$ has the density $f_1(u_1)\cdots f_m(u_m)$ [17, (4.6.2), (4.6.3), Theorem 1.6.5, Definition 4.6.5]. We also use the following facts on integrals without proof:

- (change of variables) if $B\in\mathbb{R}^{n\times n}$ is invertible and $c\in\mathbb{R}^n$, then $\int_{\mathbb{R}^n}g(x)\,\mathrm{d}x=|\det B|\int_{\mathbb{R}^n}g(c+Bu)\,\mathrm{d}u$;
- (products) $\int_{\mathbb{R}^n}g_1(u_1)\cdots g_n(u_n)\,\mathrm{d}u=\int_{\mathbb{R}}g_1(s)\,\mathrm{d}s\cdots\int_{\mathbb{R}}g_n(s)\,\mathrm{d}s$ for functions $g_1,\dots,g_n\geqslant0$;
- $\int_{\mathbb{R}}e^{-s^2/2}\,\mathrm{d}s=\sqrt{2\pi}$.

Theorem 10.8 ([3, §8.1, (8.2)], [12, §4.1])**.** Let $\mu\in\mathbb{R}^n$, let $\Sigma\in\mathbb{R}^{n\times n}$ be PD, and let
$$f(x)=\frac{1}{(2\pi)^{n/2}(\det\Sigma)^{1/2}}\exp\Big(-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\Big)\qquad(x\in\mathbb{R}^n)$$
(a) $f(x)>0$ for every x, and $\int_{\mathbb{R}^n}f(x)\,\mathrm{d}x=1$.

(b) A random vector with the density f has the distribution $N_n(\mu,\Sigma)$; in particular, $N_n(\mu,\Sigma)$ exists.

(c) Every $z\sim N_n(\mu,\Sigma)$ has the density f.

For $n=1$ and $\sigma^2>0$, $N(\mu,\sigma^2)$ has the density $(2\pi\sigma^2)^{-1/2}\exp\big(-(x-\mu)^2/(2\sigma^2)\big)$.

Proof. (a)

- $\det\Sigma>0$ (Theorem 8.21(c)), so f is defined, and $f(x)>0$.
- $\Sigma^{1/2}$ is symmetric and PD, with $\Sigma^{1/2}\Sigma^{1/2}=\Sigma$ (Theorem 8.46); so $\det\Sigma^{1/2}>0$ and $(\det\Sigma^{1/2})^2=\det\Sigma$ (Theorem 5.18(b)), that is, $\det\Sigma^{1/2}=(\det\Sigma)^{1/2}$.

- $\Sigma^{-1} = (\Sigma^{1/2}\Sigma^{1/2})^{-1} = \Sigma^{-1/2}\Sigma^{-1/2}$ (Theorem 2.13(b)), so $\Sigma^{1/2}\Sigma^{-1}\Sigma^{1/2} = I_n$, and for $x = \mu + \Sigma^{1/2}u$, $(x-\mu)'\Sigma^{-1}(x-\mu) = u'\Sigma^{1/2}\Sigma^{-1}\Sigma^{1/2}u = u'u = u_1^2 + \cdots + u_n^2$.
- By the change of variables with $c = \mu$ and $B = \Sigma^{1/2}$, and then the rule for products,

$$\int_{\mathbb{R}^n} f(x)\,\mathrm{d}x = \frac{(\det\Sigma)^{1/2}}{(2\pi)^{n/2}(\det\Sigma)^{1/2}} \int_{\mathbb{R}^n} e^{-u_1^2/2}\cdots e^{-u_n^2/2}\,\mathrm{d}u = \frac{(\sqrt{2\pi})^n}{(2\pi)^{n/2}} = 1$$

(b)

- For $t, x \in \mathbb{R}^n$, expanding with $\Sigma' = \Sigma$ and $\Sigma^{-1}\Sigma = I_n$, $(x-\mu-\Sigma t)'\Sigma^{-1}(x-\mu-\Sigma t) = (x-\mu)'\Sigma^{-1}(x-\mu) - 2t'(x-\mu) + t'\Sigma t$. Hence

$$t'x - \tfrac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu) = t'\mu + \tfrac{1}{2}t'\Sigma t - \tfrac{1}{2}(x-\mu-\Sigma t)'\Sigma^{-1}(x-\mu-\Sigma t)$$

- So $e^{t'x}f(x) = \exp(t'\mu + \frac{1}{2}t'\Sigma t)\, f_t(x)$, where f_t is the function f with $\mu + \Sigma t$ in place of μ. If y has the density f, then by (a) for $\mu + \Sigma t$, $M_y(t) = \int_{\mathbb{R}^n} e^{t'x} f(x)\,\mathrm{d}x = \exp(t'\mu + \frac{1}{2}t'\Sigma t)\int_{\mathbb{R}^n} f_t(x)\,\mathrm{d}x = \exp(t'\mu + \frac{1}{2}t'\Sigma t)$ for every t: $y \sim N_n(\mu, \Sigma)$.
- By (a), f is the density of some random vector, which has the distribution $N_n(\mu, \Sigma)$.

(c) Let y be as in (b). Then $M_z = M_y$, so z and y have the same distribution (N1), and $\mathsf{P}(z \in A) = \mathsf{P}(y \in A) = \int_A f(x)\,\mathrm{d}x$ for every set A: z has the density f.

For $n = 1$, $\det(\sigma^2) = \sigma^2$ and the inverse of (σ^2) is $(1/\sigma^2)$. □

The density f of Theorem 10.8 depends on x only through $(x-\mu)'\Sigma^{-1}(x-\mu) = d_\Sigma(x,\mu)^2$, the squared Mahalanobis distance of Definition 9.19 with Σ in place of S. So f is constant on each **contour** $\{x \in \mathbb{R}^n : d_\Sigma(x,\mu) = c\}$ $(c > 0)$. Let $\Sigma = Q\Lambda Q'$ as in Theorem 8.3(b), with $Q = [q_1\ \cdots\ q_n]$, and put $u = Q'(x-\mu)$. Then $x - \mu = Qu = u_1q_1 + \cdots + u_nq_n$ (Proposition 6.13(b)), every $\lambda_j > 0$ (Theorem 8.21(a)), and $\Sigma^{-1} = Q\Lambda^{-1}Q'$ (Proposition 8.10(a),(b) with the function $t \mapsto t$), so

$$d_\Sigma(x,\mu)^2 = u'\Lambda^{-1}u = \frac{u_1^2}{\lambda_1} + \cdots + \frac{u_n^2}{\lambda_n}$$

In the coordinates u, the contour is $\left\{u : \sum_j u_j^2/(c\sqrt{\lambda_j})^2 = 1\right\}$. As $x = \mu + u_1q_1 + \cdots + u_nq_n$, the contour is an ellipse for $n = 2$ and an ellipsoid for $n \geqslant 3$, with center μ, axes along $q_1, \ldots, q_n$ and semi-axes $c\sqrt{\lambda_1}, \ldots, c\sqrt{\lambda_n}$ [18, §4.4, (4.51), Example 4.14].

Independent components and the existence of $N_n(\mu, \Sigma)$

Theorem 10.9 ([12, Theorem 4.5], [9, Definition P.113], [3, §8.1])**.** Let $\mu \in \mathbb{R}^n$.

(a) Let $\sigma^2 \geqslant 0$. Then $z \sim N_n(\mu, \sigma^2 I_n)$ if and only if $z_1, \ldots, z_n$ are independent with $z_i \sim N(\mu_i, \sigma^2)$. More generally, $z \sim N_n\big(\mu, \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)\big)$ if and only if $z_1, \ldots, z_n$ are independent with $z_i \sim N(\mu_i, \sigma_i^2)$. Let $n = n_1 + \cdots + n_m$ and $\mu = (\mu_1', \ldots, \mu_m')'$ with $\mu_k \in \mathbb{R}^{n_k}$. If random vectors $x_1 \in \mathbb{R}^{n_1}, \ldots, x_m \in \mathbb{R}^{n_m}$ are independent, with $x_k \sim N_{n_k}(\mu_k, \Sigma_k)$, then $(x_1', \ldots, x_m')' \sim N_n\big(\mu, \operatorname{diag}(\Sigma_1, \ldots, \Sigma_m)\big)$.

(b) Let $\Sigma \in \mathbb{R}^{n \times n}$ be PSD, and let $w \sim N_n(0, I_n)$, which exists by Theorem 10.8(b). Then $\mu + \Sigma^{1/2} w \sim N_n(\mu, \Sigma)$. So $N_n(\mu, \Sigma)$ exists for every PSD Σ.

(c) A random vector z has the distribution $N_n(\mu, \Sigma)$ if and only if it has the distribution of $\mu + Aw$ for some $A \in \mathbb{R}^{n \times k}$ with $AA' = \Sigma$ and some $w \sim N_k(0, I_k)$, that is (by (a)), with $w_1, \ldots, w_k$ independent and $N(0, 1)$.

Proof. (a)

- Let $z \sim N_n(\mu, D)$ with $D = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. The off-diagonal entries of D are 0, so $z_1, \ldots, z_n$ are independent by Theorem 10.5(b) (subvectors of length 1), and $z_i \sim N(\mu_i, \sigma_i^2)$ by Corollary 10.3(b).
- Conversely, let $x_1, \ldots, x_m$ be independent with $x_k \sim N_{n_k}(\mu_k, \Sigma_k)$, and let $x = (x_1', \ldots, x_m')'$ and $D = \operatorname{diag}(\Sigma_1, \ldots, \Sigma_m)$. For $t = (t_1', \ldots, t_m')'$ with $t_k \in \mathbb{R}^{n_k}$, the block product (Theorem 5.3(c)) gives $t'\mu = \sum_k t_k' \mu_k$ and $t' D t = \sum_k t_k' \Sigma_k t_k \geqslant 0$; so D, which is symmetric (Theorem 5.3(b)), is PSD. By (N2), $M_x(t) = \prod_k M_{x_k}(t_k) = \prod_k \exp(t_k' \mu_k + \frac{1}{2} t_k' \Sigma_k t_k) = \exp(t'\mu + \frac{1}{2} t' D t)$ for t near 0.
- With $w \sim N_n(0, I_n)$ (Theorem 10.8(b)) and the square root $D^{1/2}$ of Theorem 8.46, the vector $y = \mu + D^{1/2} w$ is $N_n(\mu, D^{1/2} I_n D^{1/2}) = N_n(\mu, D)$ (Theorem 10.2), so it has the same MGF as x near 0. Hence x and y have the same distribution (N1) and the same MGF $\exp(t'\mu + \frac{1}{2} t' D t)$ for every t: $x \sim N_n(\mu, D)$ (Definition 10.1).
- With $m = n$, $n_k = 1$, $x_k = z_k$ and $\Sigma_k = \sigma_k^2$, this is the converse in the second statement; and $\sigma_1^2 = \cdots = \sigma_n^2 = \sigma^2$ gives the first statement.

(b) $\Sigma^{1/2}$ is symmetric and $\Sigma^{1/2} \Sigma^{1/2} = \Sigma$ (Theorem 8.46). By Theorem 10.2 with $A = \Sigma^{1/2}$ and $b = \mu$, $\mu + \Sigma^{1/2} w \sim N_n(\mu, \Sigma^{1/2} I_n \Sigma^{1/2}) = N_n(\mu, \Sigma)$.

(c)

- If $AA' = \Sigma$ and $w \sim N_k(0, I_k)$, then $\mu + Aw \sim N_n(\mu, A I_k A') = N_n(\mu, \Sigma)$ (Theorem 10.2). A random vector with the same distribution has the same MGF, so it is $N_n(\mu, \Sigma)$.
- Conversely, let $z \sim N_n(\mu, \Sigma)$. With $k = n$, $A = \Sigma^{1/2}$ and w as in (b), $\mu + Aw \sim N_n(\mu, \Sigma)$ has the same MGF as z, hence the same distribution (N1). □

The multivariate normal distribution is often defined as the distribution of $\mu + Aw$ with $w_1, \ldots, w_k$ independent $N(0, 1)$ [3, §8.1], [9, Definition P.113]; by (c) this is the same as Definition 10.1. The matrix A is not unique: besides $A = \Sigma^{1/2}$, for PD Σ one may take $A = T'$, where $\Sigma = T'T$ is the Cholesky decomposition of Theorem 8.40; T' is lower triangular with positive diagonal entries. For instance, for Σ of Example 9.9,

$$\Sigma = AA', \qquad A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 \\ 0 & 2 & 2 & 1 \end{pmatrix}$$

where A' is upper triangular with positive diagonal entries; A' is invertible (Theorem 2.31(a)), so $\Sigma = (A')'A'$ is PD (Theorems 5.18(a) and 8.25), and A' is the Cholesky factor of Σ (Theorem 8.40),

and $z_1 = 1 + 2w_1$, $z_2 = w_2$, $z_3 = 2 + 2w_1 + 2w_2 + w_3$, $z_4 = 2 + 2w_2 + 2w_3 + w_4$ defines a vector $z \sim N_4(\mu, \Sigma)$ as in Example 10.4.

Example 10.10 ([17, Theorem 5.3.1, Example 5.2.8, Lemma 5.3.3])**.** Let $\mu \in \mathbb{R}$, $\sigma^2 > 0$ and $z \sim N_n(\mu \mathbf{1}_n, \sigma^2 I_n)$ $(n \geqslant 2)$. By Theorem 10.9(a), $z_1, \ldots, z_n$ are independent, each $N(\mu, \sigma^2)$: a **random sample** from $N(\mu, \sigma^2)$.

(a) The mean $\bar{z} = \frac{1}{n}\mathbf{1}'z$ has the distribution $N(\mu, \sigma^2/n)$, by Corollary 10.3(c) with $a = \frac{1}{n}\mathbf{1}$: $a'\mu\mathbf{1} = \mu$ and $a'(\sigma^2 I_n)a = \sigma^2/n$.

(b) The deviations $C_n z = (z_1 - \bar{z}, \ldots, z_n - \bar{z})'$ (Example 1.16) are independent of $\bar{z}$: with $A = \frac{1}{n}\mathbf{1}'$ and $B = C_n$, $A(\sigma^2 I_n)B' = \frac{\sigma^2}{n}(C_n\mathbf{1})' = 0$ (Proposition 1.15), and Theorem 10.5(c) applies. Hence, by (N3), $\bar{z}$ is independent of every function of the deviations, such as the sample variance $s^2 = \frac{1}{n-1}(C_n z)'(C_n z)$.

(c) By Theorem 10.2, $C_n z \sim N_n(0, \sigma^2 C_n)$, as $C_n\mu\mathbf{1} = 0$ and $C_n(\sigma^2 I_n)C_n' = \sigma^2 C_n^2 = \sigma^2 C_n$. This covariance matrix is singular, of rank $n - 1$ (Corollary 6.30): the deviations always add up to 0.

Standardization and singular covariance matrices

Corollary 10.11 ([3, §8.1], [12, §4.1])**.** Let $z \sim N_n(\mu, \Sigma)$.

(a) If Σ is PD, then $w = \Sigma^{-1/2}(z - \mu) \sim N_n(0, I_n)$: the w_i are independent and $N(0, 1)$, and $z = \mu + \Sigma^{1/2}w$. Moreover $(z - \mu)'\Sigma^{-1}(z - \mu) = w'w = w_1^2 + \cdots + w_n^2$ has the distribution χ_n^2.

(b) If $\Sigma \neq 0$, let the columns of $K \in \mathbb{R}^{n \times r}$ form a basis of $\mathcal{C}(\Sigma)$, so that $r = \operatorname{rank}\Sigma$. Then $\mathsf{P}\Big(z - \mu \in \mathcal{C}(\Sigma)\Big) = 1$: with probability 1, z lies in the set $\mu + \mathcal{C}(\Sigma) = \{\mu + Kc : c \in \mathbb{R}^r\}$. If $\Sigma = 0$, then $\mathsf{P}(z = \mu) = 1$.

Proof. (a)

- $\Sigma^{-1/2}$ is symmetric and $\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I_n$ (Theorem 8.46). By Theorem 10.2 with $A = \Sigma^{-1/2}$ and $b = -\Sigma^{-1/2}\mu$, $w \sim N_n(0, I_n)$, and by Theorem 10.9(a) the w_i are independent and $N(0, 1)$.
- $\Sigma^{1/2}w = \Sigma^{1/2}\Sigma^{-1/2}(z - \mu) = z - \mu$, and $w'w = (z - \mu)'\Sigma^{-1/2}\Sigma^{-1/2}(z - \mu) = (z - \mu)'\Sigma^{-1}(z - \mu)$, as in the proof of Theorem 10.8(a). By (N5), $w_1^2 + \cdots + w_n^2 \sim \chi_n^2$.

(b)

- Let $P = K(K'K)^{-1}K'$, the matrix P of Theorem 6.26 with $A = K$, which has full column rank r. Every column of Σ lies in $\mathcal{C}(K)$, so $\Sigma = KT$ for some $T \in \mathbb{R}^{r \times n}$ (Theorem 1.4(a),(b)), and $(I_n - P)\Sigma = (I_n - P)KT = 0$ (Theorem 6.26(c)).
- By Theorem 10.2, $(I_n - P)(z - \mu) \sim N_n\Big(0, (I_n - P)\Sigma(I_n - P)'\Big) = N_n(0, 0)$, the distribution of the constant 0 (Definition 10.1); so $(I_n - P)(z - \mu) = 0$ with probability 1.
- If $(I_n - P)(z - \mu) = 0$, then $z - \mu = P(z - \mu) = K\Big((K'K)^{-1}K'(z - \mu)\Big) \in \mathcal{C}(K) = \mathcal{C}(\Sigma)$.
- The case $\Sigma = 0$ is part of Definition 10.1. □

So a normal vector with a singular covariance matrix lies, with probability 1, in the affine subspace $\mu + \mathcal{C}(\Sigma)$ of dimension $\operatorname{rank}\Sigma < n$. Such vectors, like the deviations $C_n z$ of Example 10.10(c), for which $\mathcal{C}(\sigma^2 C_n) = \operatorname{span}\{\mathbf{1}\}^\perp$ (Example 6.27), are the reason why Definition 10.1 uses the MGF and not a density [12, §4.1].

Conditional distributions

Definition 10.12 ([17, Definition 4.2.3, (4.6.4), (4.6.6)])**.** Let $(u', v')'$, with $u \in \mathbb{R}^p$ and $v \in \mathbb{R}^q$, have the density f. Then u has the density $f_u(a) = \int_{\mathbb{R}^q} f(a, b)\,\mathrm{d}b$ $(a \in \mathbb{R}^p)$, the **marginal density**. For $a \in \mathbb{R}^p$ with $f_u(a) > 0$, the function

$$f(b \mid a) = \frac{f(a, b)}{f_u(a)} \qquad (b \in \mathbb{R}^q)$$

is a density on $\mathbb{R}^q$ (it is $\geqslant 0$, with integral $f_u(a)/f_u(a) = 1$), the **conditional density** of v given $u = a$; the distribution with this density is the **conditional distribution** of v given $u = a$.

Theorem 10.13 ([3, §8.1; Ch. 8, Exercise 13], [2, §10.4], [17, Example 4.5.8])**.** Let $z \sim N_n(\mu, \Sigma)$ be partitioned as in Corollary 10.3(a), with $z_1 \in \mathbb{R}^p$ and $z_2 \in \mathbb{R}^q$, and let Σ_{11} be PD. Put

$$\Sigma_{22\cdot1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

the Schur complement of Σ_{11} in Σ (Theorem 5.9). Then the **conditional distribution of z_2 given** z_1 is

$$N_q\Big(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1),\ \Sigma_{22\cdot1}\Big)$$

in the following sense.

(a) $\Sigma_{22\cdot1}$ is PSD, and $w = z_2 - \mu_2 - \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1)$ is independent of z_1, with $w \sim N_q(0, \Sigma_{22\cdot1})$. So z_2 is the sum of $\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1)$, which is known once z_1 is known, and of w, whose distribution does not depend on z_1.

(b) If Σ is PD, then $\Sigma_{22\cdot1}$ is PD and is the inverse of the lower right $q \times q$ block of Σ^{-1}; z_1 has the density of $N_p(\mu_1, \Sigma_{11})$; and for every $a \in \mathbb{R}^p$, the conditional density of z_2 given $z_1 = a$ (Definition 10.12) is the density of $N_q\Big(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(a - \mu_1), \Sigma_{22\cdot1}\Big)$.

Proof. Let

$$L = \begin{pmatrix} I_p & 0 \\ -\Sigma_{21}\Sigma_{11}^{-1} & I_q \end{pmatrix}, \qquad \text{so that} \qquad L(z - \mu) = \begin{pmatrix} z_1 - \mu_1 \\ w \end{pmatrix}$$

by the block product. Σ_{11}^{-1} is symmetric (Theorem 2.13(c)) and $\Sigma_{21}' = \Sigma_{12}$, so $L' = \left(\begin{smallmatrix} I_p & -\Sigma_{11}^{-1}\Sigma_{12} \\ 0 & I_q \end{smallmatrix}\right)$ (Theorem 5.3(b)), and Theorem 5.9, with $A = \Sigma_{11}$, $B = \Sigma_{12}$, $C = \Sigma_{21}$ and $D = \Sigma_{22}$, gives

$$L\Sigma L' = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22\cdot1} \end{pmatrix}$$

(a)

- By Theorem 10.2 with $A = L$ and $b = -L\mu$, $L(z - \mu) \sim N_n(0, L\Sigma L')$. By Corollary 10.3(a), $w \sim N_q(0, \Sigma_{22\cdot1})$; so $\Sigma_{22\cdot1} = \mathrm{var}(w)$ is PSD (Theorem 9.3(c)).
- The off-diagonal blocks of $L\Sigma L'$ are 0, so $z_1 - \mu_1$ and w are independent (Theorem 10.5(a)), and so are $z_1 = (z_1 - \mu_1) + \mu_1$ and w (N3).

(b)

- Σ_{11} and $\Sigma_{22\cdot1}$ are PD (Theorem 8.31). L is invertible (Proposition 5.8), so by Theorem 2.13(a)–(c) and Proposition 5.6(b),

$$\Sigma = L^{-1}\begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22\cdot1} \end{pmatrix}(L')^{-1}, \qquad \Sigma^{-1} = L'\begin{pmatrix} \Sigma_{11}^{-1} & 0 \\ 0 & \Sigma_{22\cdot1}^{-1} \end{pmatrix}L$$

  By Theorem 5.12(a), the lower right block of Σ^{-1} is $\Sigma_{22\cdot1}^{-1}$.

- For $x = (a', b')'$ with $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$, put $m(a) = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(a - \mu_1)$; then $L(x - \mu) = \big((a - \mu_1)', (b - m(a))'\big)'$, and

$$\begin{aligned}(x-\mu)'\Sigma^{-1}(x-\mu) &= \big(L(x-\mu)\big)' \begin{pmatrix} \Sigma_{11}^{-1} & 0 \\ 0 & \Sigma_{22\cdot1}^{-1} \end{pmatrix} L(x-\mu) \\ &= (a-\mu_1)'\Sigma_{11}^{-1}(a-\mu_1) + (b-m(a))'\Sigma_{22\cdot1}^{-1}(b-m(a))\end{aligned}$$

- $\det\Sigma = \det\Sigma_{11} \cdot \det\Sigma_{22\cdot1}$ (Theorem 5.30(a)) and $(2\pi)^{n/2} = (2\pi)^{p/2}(2\pi)^{q/2}$. So the density f of z (Theorem 10.8(c)) factors as $f(a,b) = f_1(a)\, g_a(b)$, where f_1 is the density of $N_p(\mu_1, \Sigma_{11})$ and g_a that of $N_q(m(a), \Sigma_{22\cdot1})$.
- By Theorem 10.8(a), $\int_{\mathbb{R}^q} g_a(b)\,\mathrm{d}b = 1$; so the marginal density of z_1 is $\int_{\mathbb{R}^q} f(a,b)\,\mathrm{d}b = f_1(a) > 0$, and the conditional density of z_2 given $z_1 = a$ is $f(a,b)/f_1(a) = g_a(b)$. □

The conditional mean $\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1)$ is an affine function of z_1, and the conditional covariance matrix $\Sigma_{22\cdot1}$ does not depend on z_1. It is never larger than $\Sigma_{22} = \mathrm{var}(z_2)$: the matrix $\Sigma_{11}^{-1} = (\Sigma_{11}^{-1/2})'\Sigma_{11}^{-1/2}$ is PSD (Theorems 8.46 and 8.25), so $\Sigma_{22} - \Sigma_{22\cdot1} = \Sigma_{12}'\Sigma_{11}^{-1}\Sigma_{12}$ is PSD (Theorem 8.24(a)), and for every $c \in \mathbb{R}^q$, $\mathrm{var}(c'w) = c'\Sigma_{22\cdot1}c \leqslant c'\Sigma_{22}c = \mathrm{var}(c'z_2)$: knowing z_1 can only reduce the uncertainty about z_2. If the diagonal entries of $\Sigma_{22\cdot1} = \mathrm{var}(w)$ are positive, the correlations of the entries of w (Definition 9.7), which are computed from the conditional covariance matrix $\Sigma_{22\cdot1}$, are the **partial correlations** of the entries of z_2 given z_1 [18, §5.1, Example 5.3].

Example 10.14. Let $z \sim N_4(\mu, \Sigma)$ be as in Example 10.4, with $\mu = (1, 0, 2, 2)'$. For the lower triangular A with $\Sigma = AA'$ of the paragraph after Theorem 10.9,

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & -2 & 1 & 0 \\ 2 & 2 & -2 & 1 \end{pmatrix}, \qquad \Sigma^{-1} = (A^{-1})'A^{-1} = \begin{pmatrix} \frac{21}{4} & 6 & -5 & 2 \\ 6 & 9 & -6 & 2 \\ -5 & -6 & 5 & -2 \\ 2 & 2 & -2 & 1 \end{pmatrix}$$

as $AA^{-1} = I_4$ (Theorem 2.16) and $\Sigma^{-1} = (A')^{-1}A^{-1}$ (Theorem 2.13(b),(c)); the rows 3 and 4 of A^{-1} are a' and b' of Example 9.9. As Σ is PD (the paragraph after Theorem 10.9), Σ_{11} is PD in (a) and (b) (Theorem 8.30).

(a) Given $(z_1, z_2, z_3)'$: here $p = 3$, $q = 1$, Σ_{11} is the leading 3×3 block of Σ, $\Sigma_{21} = (0, 2, 6)$ and $\Sigma_{22} = 9$. As

$$(-2,-2,2)\,\Sigma_{11} = (-2,-2,2)\begin{pmatrix} 4 & 0 & 4 \\ 0 & 1 & 2 \\ 4 & 2 & 9 \end{pmatrix} = (-8+0+8,\ 0-2+4,\ -8-4+18) = (0,2,6)$$

multiplying by Σ_{11}^{-1} on the right gives $\Sigma_{21}\Sigma_{11}^{-1} = (-2,-2,2)$, and $\Sigma_{22\cdot1} = 9 - (-2,-2,2)(0,2,6)' = 9 - 8 = 1$. Given $z_1 = a_1$, $z_2 = a_2$ and $z_3 = a_3$, the conditional distribution of z_4 is

$$N\big(2 - 2(a_1 - 1) - 2a_2 + 2(a_3 - 2),\ 1\big) = N(2a_3 - 2a_1 - 2a_2,\ 1)$$

for instance $N(4, 1)$ for $a_1 = 3$, $a_2 = 1$, $a_3 = 6$. This agrees with Example 10.7: $z_4 = -2z_1 - 2z_2 + 2z_3 + f$, with $f \sim N(0,1)$ independent of $(z_1, z_2, z_3)'$. The $(4,4)$ entry of Σ^{-1} is $1 = \Sigma_{22\cdot1}^{-1}$, as Theorem 10.13(b) says.

(b) Given $(z_1, z_2)'$: here $p = q = 2$, $\Sigma_{11} = \mathrm{diag}(4, 1)$, $\Sigma_{11}^{-1} = \mathrm{diag}(\frac{1}{4}, 1)$ and $\Sigma_{21} = \left(\begin{smallmatrix} 4 & 2 \\ 0 & 2 \end{smallmatrix}\right)$, so $\Sigma_{21}\Sigma_{11}^{-1} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 2 \end{smallmatrix}\right)$ and

$$\Sigma_{22\cdot1} = \begin{pmatrix} 9 & 6 \\ 6 & 9 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 6 & 9 \end{pmatrix} - \begin{pmatrix} 8 & 4 \\ 4 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

As $(2,2)' + \left(\begin{smallmatrix}1 & 2\\ 0 & 2\end{smallmatrix}\right)(a_1-1,\ a_2)' = (a_1+2a_2+1,\ 2a_2+2)'$, given $z_1 = a_1$ and $z_2 = a_2$ the vector $(z_3, z_4)'$ has the distribution $N_2\Big((a_1+2a_2+1,\ 2a_2+2)', \Sigma_{22\cdot 1}\Big)$. Its first entry agrees with Example 10.7: $z_3 = z_1 + 2z_2 + e$, with $e \sim N(1,1)$ independent of $(z_1, z_2)'$. The lower right 2×2 block of Σ^{-1} is $\left(\begin{smallmatrix}5 & -2\\ -2 & 1\end{smallmatrix}\right) = \Sigma_{22\cdot 1}^{-1}$, as Theorem 10.13(b) says. Knowing z_1 and z_2 reduces the variances of z_3 and z_4 from 9 and 9 to 1 and 5.

The bivariate normal distribution

Definition 10.15 ([17, Definition 4.5.10])**.** Let $\mu_u, \mu_v \in \mathbb{R}$, $\sigma_u, \sigma_v > 0$ and $-1 \leqslant \rho \leqslant 1$. A random vector $(u, v)'$ has the **bivariate normal distribution** with means μ_u, μ_v, variances σ_u^2, σ_v^2 and correlation ρ if

$$\begin{pmatrix} u \\ v \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_u \\ \mu_v \end{pmatrix},\ \Sigma\right), \qquad \Sigma = \begin{pmatrix} \sigma_u^2 & \rho\sigma_u\sigma_v \\ \rho\sigma_u\sigma_v & \sigma_v^2 \end{pmatrix}$$

This Σ is PSD: for $|\rho| < 1$ its leading principal minors σ_u^2 and $\sigma_u^2\sigma_v^2(1-\rho^2)$ are positive, so it is PD (Theorem 8.34); for $\rho = \pm 1$, $\Sigma = cc'$ with $c = (\sigma_u, \rho\sigma_v)'$, which is PSD (Theorem 8.25). Conversely, by Definition 9.7 and Proposition 9.8(a), every N_2 distribution whose covariance matrix has a positive diagonal is of this form, with $\rho = \rho_{uv}$.

Proposition 10.16 ([17, §4.5])**.** Let $(u, v)'$ have the bivariate normal distribution of Definition 10.15.

(a) $u \sim N(\mu_u, \sigma_u^2)$, $v \sim N(\mu_v, \sigma_v^2)$, $\rho_{uv} = \rho$, and for constants α, β, $\alpha u + \beta v \sim N\Big(\alpha\mu_u + \beta\mu_v,\ \alpha^2\sigma_u^2 + 2\alpha\beta\rho\sigma_u\sigma_v + \beta^2\sigma_v^2\Big)$.

(b) u and v are independent if and only if $\rho = 0$.

(c) If $|\rho| < 1$, then $(u, v)'$ has the density

$$f(s,t) = \frac{1}{2\pi\sigma_u\sigma_v\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\left(\frac{s-\mu_u}{\sigma_u}\right)^2 - 2\rho\,\frac{s-\mu_u}{\sigma_u}\cdot\frac{t-\mu_v}{\sigma_v} + \left(\frac{t-\mu_v}{\sigma_v}\right)^2\right)\right)$$

(d) The conditional distribution of v given u is $N\Big(\mu_v + \rho\frac{\sigma_v}{\sigma_u}(u-\mu_u),\ \sigma_v^2(1-\rho^2)\Big)$. If $\rho = \pm 1$, then $v = \mu_v + \rho\frac{\sigma_v}{\sigma_u}(u-\mu_u)$ with probability 1.

Proof. (a) The marginal distributions are given by Corollary 10.3(b), and the distribution of $\alpha u + \beta v$ by Corollary 10.3(c) with the vector $(\alpha, \beta)'$, as $(\alpha,\beta)\Sigma(\alpha,\beta)' = \alpha^2\sigma_u^2 + 2\alpha\beta\rho\sigma_u\sigma_v + \beta^2\sigma_v^2$. By Definition 10.1, $\operatorname{cov}(u, v) = \rho\sigma_u\sigma_v$, so $\rho_{uv} = \rho\sigma_u\sigma_v/(\sigma_u\sigma_v) = \rho$ (Definition 9.7).

(b) By Theorem 10.5(a) with $p = q = 1$, u and v are independent if and only if $\rho\sigma_u\sigma_v = 0$, that is, $\rho = 0$.

(c)

- $\det\Sigma = \sigma_u^2\sigma_v^2 - \rho^2\sigma_u^2\sigma_v^2 = \sigma_u^2\sigma_v^2(1-\rho^2) > 0$, and by Theorem 5.24, $\Sigma^{-1} = \frac{1}{\sigma_u^2\sigma_v^2(1-\rho^2)}\left(\begin{smallmatrix}\sigma_v^2 & -\rho\sigma_u\sigma_v\\ -\rho\sigma_u\sigma_v & \sigma_u^2\end{smallmatrix}\right)$.
- Hence, for $x = (s,t)'$,

$$(x-\mu)'\Sigma^{-1}(x-\mu) = \frac{\sigma_v^2(s-\mu_u)^2 - 2\rho\sigma_u\sigma_v(s-\mu_u)(t-\mu_v) + \sigma_u^2(t-\mu_v)^2}{\sigma_u^2\sigma_v^2(1-\rho^2)}$$

which is the bracket in the exponent of f divided by $1-\rho^2$; and $(2\pi)^{2/2}(\det\Sigma)^{1/2} = 2\pi\sigma_u\sigma_v\sqrt{1-\rho^2}$. Theorem 10.8(c) gives the claim.

(d) Theorem 10.13 with $p = q = 1$ and $\Sigma_{11} = \sigma_u^2 > 0$: here $\Sigma_{21}\Sigma_{11}^{-1} = \rho\sigma_u\sigma_v/\sigma_u^2 = \rho\sigma_v/\sigma_u$ and $\Sigma_{22\cdot 1} = \sigma_v^2 - \rho^2\sigma_u^2\sigma_v^2/\sigma_u^2 = \sigma_v^2(1-\rho^2)$. If $\rho = \pm 1$, then $w = v - \mu_v - \rho\frac{\sigma_v}{\sigma_u}(u-\mu_u) \sim N(0,0)$ in (a) of that theorem, so $\mathsf{P}(w = 0) = 1$ (Definition 10.1). □

Example 10.17. Let $(u,v)'$ be bivariate normal with $\mu_u=5$, $\mu_v=6$, $\sigma_u=2$, $\sigma_v=4$ and $\rho=\frac12$. Then $\rho\sigma_u\sigma_v=4$, and

$$\Sigma=\begin{pmatrix}4&4\\4&16\end{pmatrix},\qquad \det\Sigma=48,\qquad \Sigma^{-1}=\frac{1}{48}\begin{pmatrix}16&-4\\-4&4\end{pmatrix}=\frac{1}{12}\begin{pmatrix}4&-1\\-1&1\end{pmatrix}$$

- By Proposition 10.16(c), as $2\pi\cdot 2\cdot 4\sqrt{1-\frac14}=8\sqrt3\,\pi$, the density is
$$f(s,t)=\frac{1}{8\sqrt3\,\pi}\exp\Big(-\frac{1}{24}\big(4(s-5)^2-2(s-5)(t-6)+(t-6)^2\big)\Big)$$
- By Proposition 10.16(d), given u the variable v has the distribution $N\big(6+\frac12\cdot\frac42(u-5),\ 16\cdot\frac34\big)=N(u+1,\ 12)$; for instance $N(10,12)$ given $u=9$, against $v\sim N(6,16)$ without knowledge of u. With the roles exchanged, given v the variable u has the distribution $N\big(5+\frac12\cdot\frac24(v-6),\ 4\cdot\frac34\big)=N\big(5+\frac14(v-6),\ 3\big)$.
- $v-u$ is independent of u: $(1,0)\Sigma(-1,1)'=-4+4=0$ (Theorem 10.5(c)), and $v-u\sim N(1,\ 4-8+16)=N(1,12)$ by (a). So $v=u+(v-u)$ is u plus an independent $N(1,12)$ variable, in accordance with the conditional distribution found above.

Maximum likelihood estimates of μ and Σ

Lemma 10.18 ([6, Exercise 12.20])**.** Let $A\in\mathbb{R}^{p\times p}$ be PD. Then $\log\det A\leqslant \operatorname{tr}A-p$, with equality only if $A=I_p$.

Proof. • Let $\varphi(\lambda)=\lambda-1-\log\lambda$ for $\lambda>0$. Its derivative $\varphi'(\lambda)=1-1/\lambda$ is <0 for $\lambda<1$ and >0 for $\lambda>1$, so φ decreases on $(0,1]$ and increases on $[1,\infty)$, and $\varphi(\lambda)>\varphi(1)=0$ for $\lambda\neq1$: $\log\lambda\leqslant\lambda-1$, with equality only for $\lambda=1$.

- The eigenvalues $\lambda_1,\dots,\lambda_p$ of A are >0 (Theorem 8.21(a)), and $\det A=\lambda_1\cdots\lambda_p$, $\operatorname{tr}A=\lambda_1+\cdots+\lambda_p$ (Theorem 7.19). Hence
$$\log\det A=\sum_{i=1}^p\log\lambda_i\leqslant\sum_{i=1}^p(\lambda_i-1)=\operatorname{tr}A-p$$
- Equality forces $\lambda_i=1$ for every i; then $A=QI_pQ'=I_p$ for an orthogonal Q (Theorem 8.3(b)). □

Definition 10.19 ([17, Definitions 5.1.1, 7.2.4, (7.2.3)], [21, §13])**.** Let $x_1,\dots,x_n\in\mathbb{R}^p$ be independent random vectors, each $N_p(\mu,\Sigma)$ with Σ PD: a **random sample** from $N_p(\mu,\Sigma)$. Each x_i has the density $f(\cdot\,;\mu,\Sigma)$ of Theorem 10.8, and $(x_1',\dots,x_n')'$ has the density $f(x_1;\mu,\Sigma)\cdots f(x_n;\mu,\Sigma)$. For observed values $x_1,\dots,x_n$, the function

$$L(\mu,\Sigma)=\prod_{i=1}^n f(x_i;\mu,\Sigma)\qquad(\mu\in\mathbb{R}^p,\ \Sigma\in\mathbb{R}^{p\times p}\text{ PD})$$

is the **likelihood**, and values $\hat\mu$, $\hat\Sigma$ at which L is largest are **maximum likelihood estimates** of μ and Σ.

Theorem 10.20 ([6, Exercises 13.61, 12.20], [21, §13], [17, §7.2.2, Example 7.2.11])**.** Let $x_1,\dots,x_n$ ($n\geqslant2$) be a random sample from $N_p(\mu,\Sigma)$ (Definition 10.19), let X be the data matrix with rows $x_1',\dots,x_n'$, and let $\bar x$ and S be its mean vector and sample covariance matrix (Definition 9.13), so that $A=X'C_nX=(n-1)S$. If A is PD, which requires $n\geqslant p+1$, then the likelihood has exactly one maximizer,

$$\hat\mu=\bar x,\qquad\hat\Sigma=\frac1nA=\frac1n\sum_{i=1}^n(x_i-\bar x)(x_i-\bar x)'=\frac{n-1}{n}S$$

and the maximum is $L(\bar x,\hat\Sigma)=(2\pi e)^{-np/2}(\det\hat\Sigma)^{-n/2}$. Moreover $\mathsf{E}(\bar x)=\mu$ and $\mathsf{E}(\hat\Sigma)=\frac{n-1}{n}\Sigma$.

Proof. • The requirement: as $A = (n-1)S$ with $n-1 > 0$, A is PD if and only if S is; and S is PD if and only if $\operatorname{rank}(C_n X) = p$, which requires $n \geqslant p+1$ (Proposition 9.14(a)).

• By Theorem 10.8,

$$\log L(\mu, \Sigma) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log\det\Sigma - \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)'\Sigma^{-1}(x_i-\mu)$$

As log is strictly increasing, L and $\log L$ have the same maximizers.

• The step for μ. Write $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$. As $\sum_i (x_i - \bar{x}) = X'\mathbf{1} - n\bar{x} = 0$ and Σ^{-1} is symmetric, the cross terms add up to $2(\bar{x}-\mu)'\Sigma^{-1}\sum_i(x_i-\bar{x}) = 0$, and

$$\sum_{i=1}^{n}(x_i-\mu)'\Sigma^{-1}(x_i-\mu) = \sum_{i=1}^{n}(x_i-\bar{x})'\Sigma^{-1}(x_i-\bar{x}) + n(\bar{x}-\mu)'\Sigma^{-1}(\bar{x}-\mu)$$

Σ^{-1} is PD: for $y \neq 0$, $\Sigma^{-1}y \neq 0$ and $y'\Sigma^{-1}y = (\Sigma^{-1}y)'\Sigma(\Sigma^{-1}y) > 0$. So the last term is > 0 unless $\mu = \bar{x}$, and $\log L(\mu, \Sigma) \leqslant \log L(\bar{x}, \Sigma)$, with equality only if $\mu = \bar{x}$.

• The trace form. $(x_i - \bar{x})'\Sigma^{-1}(x_i - \bar{x}) = \operatorname{tr}\Big(\Sigma^{-1}(x_i-\bar{x})(x_i-\bar{x})'\Big)$ (Theorem 1.11(b)), and $\sum_i (x_i-\bar{x})(x_i-\bar{x})' = (C_nX)'(C_nX) = A$, as the rows of C_nX are the $(x_i-\bar{x})'$ (Definition 9.13, Theorem 1.4(d)). So $-2\log L(\bar{x}, \Sigma) = np\log(2\pi) + g(\Sigma)$ with $g(\Sigma) = n\log\det\Sigma + \operatorname{tr}(\Sigma^{-1}A)$.

• The step for Σ. $\hat{\Sigma} = \frac{1}{n}A$ is PD. Let $B = \hat{\Sigma}^{1/2}\Sigma^{-1}\hat{\Sigma}^{1/2}$, which is PD (Theorems 8.46 and 8.24(c)). By Theorem 1.11(b), $\operatorname{tr} B = \operatorname{tr}(\Sigma^{-1}\hat{\Sigma}^{1/2}\hat{\Sigma}^{1/2}) = \operatorname{tr}(\Sigma^{-1}\hat{\Sigma})$, and $\det B = \det\hat{\Sigma}/\det\Sigma$ (Theorem 5.18(b),(c)). As $A = n\hat{\Sigma}$ and $\operatorname{tr}(\hat{\Sigma}^{-1}A) = np$,

$$g(\Sigma) - g(\hat{\Sigma}) = n\log\frac{\det\Sigma}{\det\hat{\Sigma}} + n\operatorname{tr}(\Sigma^{-1}\hat{\Sigma}) - np = n\Big(\operatorname{tr} B - \log\det B - p\Big) \geqslant 0$$

by Lemma 10.18, with equality only if $B = I_p$, that is, $\Sigma^{-1} = \hat{\Sigma}^{-1/2}\hat{\Sigma}^{-1/2} = \hat{\Sigma}^{-1}$, or $\Sigma = \hat{\Sigma}$.

• Together, $\log L(\mu, \Sigma) \leqslant \log L(\bar{x}, \Sigma) \leqslant \log L(\bar{x}, \hat{\Sigma})$, with equality throughout only for $\mu = \bar{x}$ and $\Sigma = \hat{\Sigma}$. As $g(\hat{\Sigma}) = n\log\det\hat{\Sigma} + np$, $\log L(\bar{x}, \hat{\Sigma}) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log\det\hat{\Sigma} - \frac{np}{2}$.

• Independent vectors are uncorrelated (N7), and $\mathsf{E}(x_i) = \mu$, $\operatorname{var}(x_i) = \Sigma$ (Definition 10.1); so by Theorem 9.16, $\mathsf{E}(\bar{x}) = \mu$ and $\mathsf{E}(\hat{\Sigma}) = \frac{n-1}{n}\mathsf{E}(S) = \frac{n-1}{n}\Sigma$. □

So the maximum likelihood estimator $\hat{\Sigma}$ is not unbiased (Definition 9.11): $\mathsf{E}(\hat{\Sigma}) = \frac{n-1}{n}\Sigma \neq \Sigma$, while S is unbiased (Theorem 9.16). For $p = 1$, Theorem 10.20 says that for a random sample from $N(\mu, \sigma^2)$ whose observed values $x_1, \ldots, x_n$ are not all equal, the maximum likelihood estimates are $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n-1}{n}s^2$, where s^2 is the sample variance.

Example 10.21. Suppose that the five observations of Example 9.15 are a random sample from $N_4(\mu, \Sigma)$. There $A = (C_5X)'(C_5X) = 4S$, and S has the eigenvalues $25, 9, 5, 1$ (Example 9.34), so A is PD (Theorem 8.21(a)). By Theorem 10.20, the maximum likelihood estimates are

$$\hat{\mu} = \bar{x} = \begin{pmatrix}5\\5\\5\\5\end{pmatrix}, \qquad \hat{\Sigma} = \frac{1}{5}\begin{pmatrix}40&28&20&12\\28&40&12&20\\20&12&40&28\\12&20&28&40\end{pmatrix} = \frac{4}{5}S, \qquad S = \begin{pmatrix}10&7&5&3\\7&10&3&5\\5&3&10&7\\3&5&7&10\end{pmatrix}$$

with $\det\hat{\Sigma} = \left(\frac{4}{5}\right)^4 \det S = \frac{256}{625}\cdot 25\cdot 9\cdot 5\cdot 1 = \frac{2304}{5}$ (Theorems 5.18(c) and 7.19). So the largest value of the likelihood is $(2\pi e)^{-10}\left(\frac{2304}{5}\right)^{-5/2}$. The unbiased estimator of Σ is S (Theorem 9.16).

Exercises

Exercise 10.1. Let $z \sim N_4(\mu, \Sigma)$ with

$$\mu = \begin{pmatrix} 1 \\ -1 \\ 2 \\ 0 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 4 & 2 & 0 & 0 \\ 2 & 3 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix}$$

(a) Find the distribution of $(z_1, z_3, z_4)'$. Is z_1 independent of $(z_3, z_4)'$?

(b) Find the distribution of $e = z_1 - z_2 + z_3$. Is e independent of z_2? Of $(z_2, z_4)'$? Of z_1?

(c) Find the distribution of $(z_1 + z_2,\ z_2 + z_3,\ z_3 + z_4)'$.

Solution. (a) By Corollary 10.3(b),

$$\begin{pmatrix} z_1 \\ z_3 \\ z_4 \end{pmatrix} \sim N_3 \left(\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \right)$$

As $\sigma_{13} = \sigma_{14} = 0$, z_1 and $(z_3, z_4)'$ are independent (Theorem 10.5(a) for this vector, with $p = 1$ and $q = 2$).

(b) With $a = (1, -1, 1, 0)'$, $a'\mu = 1 + 1 + 2 + 0 = 4$, $\Sigma a = (4 - 2 + 0 + 0,\ 2 - 3 + 1 + 0,\ 0 - 1 + 2 + 0,\ 0 - 1 + 1 + 0)' = (2, 0, 1, 0)'$ and $a'\Sigma a = 2 + 0 + 1 + 0 = 3$; so $e \sim N(4, 3)$ (Corollary 10.3(c)). By Theorem 9.4(a), $\operatorname{cov}(z, e) = \Sigma a = (2, 0, 1, 0)'$. So $\operatorname{cov}(z_2, e) = \operatorname{cov}(z_4, e) = 0$, and e is independent of z_2 and of $(z_2, z_4)'$ (Theorem 10.5(c) with $B = a'$ and $A = e_2'$, resp. the matrix A with the rows e_2' and e_4'); but $\operatorname{cov}(z_1, e) = 2 \neq 0$, so e is not independent of z_1.

(c) Let

$$B = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad B\mu = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

The rows of $B\Sigma$ are $\operatorname{row}_1(\Sigma) + \operatorname{row}_2(\Sigma)$, $\operatorname{row}_2(\Sigma) + \operatorname{row}_3(\Sigma)$ and $\operatorname{row}_3(\Sigma) + \operatorname{row}_4(\Sigma)$ (Theorem 1.4(c)), and the columns of $B\Sigma B'$ are the same sums of the columns of $B\Sigma$ (Theorem 1.4(a),(b)):

$$B\Sigma = \begin{pmatrix} 6 & 5 & 1 & 1 \\ 2 & 4 & 3 & 2 \\ 0 & 2 & 3 & 3 \end{pmatrix}, \qquad B\Sigma B' = \begin{pmatrix} 6+5 & 5+1 & 1+1 \\ 2+4 & 4+3 & 3+2 \\ 0+2 & 2+3 & 3+3 \end{pmatrix} = \begin{pmatrix} 11 & 6 & 2 \\ 6 & 7 & 5 \\ 2 & 5 & 6 \end{pmatrix}$$

so $(z_1 + z_2,\ z_2 + z_3,\ z_3 + z_4)' = Bz \sim N_3(B\mu, B\Sigma B')$ (Theorem 10.2).

Exercise 10.2 ([2, Exercise 10.10])**.** The MGF of a random vector $x = (x_1, x_2, x_3)'$ is

$$M_x(t) = \exp\bigl(t_1 - t_2 + 2t_3 + t_1^2 + \tfrac{1}{2}t_2^2 + 2t_3^2 - \tfrac{1}{2}t_1t_2 - t_1t_3\bigr) \qquad (t \in \mathbb{R}^3)$$

Express $\mathsf{P}(x_1 + x_2 > x_3)$ through the distribution function Φ of $N(0, 1)$, $\Phi(c) = \mathsf{P}(y \leqslant c)$ for $y \sim N(0, 1)$.

Solution.

- The exponent is $t'\mu + \frac{1}{2}t'Vt$ with $\mu = (1, -1, 2)'$ and

$$V = \begin{pmatrix} 2 & -\frac{1}{2} & -1 \\ -\frac{1}{2} & 1 & 0 \\ -1 & 0 & 4 \end{pmatrix}$$

since $t'Vt = 2t_1^2 + t_2^2 + 4t_3^2 - t_1t_2 - 2t_1t_3$ (Definition 8.17) is twice the quadratic part. The leading principal minors 2, $2 - \frac{1}{4} = \frac{7}{4}$ and $\det V = 6$ are positive, so V is PD (Theorem 8.34), and $x \sim N_3(\mu, V)$ by Definition 10.1.

- Let $u = x_1 + x_2 - x_3 = a'x$ with $a = (1, 1, -1)'$. By Corollary 10.3(c), $u \sim N(a'\mu, a'Va)$, with $a'\mu = 1 - 1 - 2 = -2$ and $a'Va = 2 + 1 + 4 + 2\left(-\frac{1}{2}\right) - 2(-1) - 2 \cdot 0 = 8$.
- $z = (u + 2)/\sqrt{8} \sim N(0, 1)$ (Theorem 10.2 with $A = 1/\sqrt{8}$, $b = 2/\sqrt{8}$), so z has the distribution of y, and $\mathsf{P}(z > c) = 1 - \mathsf{P}(z \leqslant c) = 1 - \Phi(c)$ for every c. As $2/\sqrt{8} = \sqrt{2}/2$, $\mathsf{P}(x_1 + x_2 > x_3) = \mathsf{P}(u > 0) = \mathsf{P}\left(z > \sqrt{2}/2\right) = 1 - \Phi\left(\sqrt{2}/2\right)$.

Exercise 10.3. Let $x \sim N_4(\mu, \Sigma)$ with $\mu = (1, 2, 0, -1)'$ and

$$\Sigma = \begin{pmatrix} 3 & 1 & 1 & 0 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

(a) Find the distributions of $y = (x_1 - x_2,\ x_3 - x_4)'$ and of $(x_1, x_4)'$.

(b) Which of the six pairs $\{x_i, x_j\}$ are independent? Is x_1 independent of $(x_3, x_4)'$?

(c) Find c such that $x_1 - cx_3$ and x_3 are independent.

(d) Find the conditional distribution of $(x_1, x_2)'$ given $(x_3, x_4)' = (a_3, a_4)'$, and check that $\det \Sigma$ is the product of the determinants of the covariance matrix of $(x_3, x_4)'$ and of this conditional covariance matrix.

Solution. (a) $y = Ax$ with $A = \left(\begin{smallmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{smallmatrix}\right)$. The rows of $A\Sigma$ are $\mathrm{row}_1(\Sigma) - \mathrm{row}_2(\Sigma) = (2, -2, 0, -1)$ and $\mathrm{row}_3(\Sigma) - \mathrm{row}_4(\Sigma) = (1, 0, 1, 0)$, so by Theorem 10.2 and Corollary 10.3(b)

$$y \sim N_2\left(\begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 & 1 \\ 1 & 1 \end{pmatrix}\right), \qquad \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\right)$$

(b) Each pair forms a normal subvector of x (Corollary 10.3(b)), so Theorem 10.5(a) applies to it: x_1 and x_4 are independent, as $\sigma_{14} = 0$, and the other five pairs are not independent, as their covariances are not 0. The vector $(x_1, x_3, x_4)'$ is normal, and the cross-covariance of x_1 and $(x_3, x_4)'$ is $(\sigma_{13}, \sigma_{14}) = (1, 0) \neq 0$; so x_1 is not independent of $(x_3, x_4)'$.

(c) By Theorem 10.5(c) with $A = (1, 0, -c, 0)$ and $B = e_3'$, $A\Sigma B' = \sigma_{13} - c\sigma_{33} = 1 - 2c$, which is 0 exactly for $c = \frac{1}{2}$.

(d) Apply Theorem 10.13 to the reordered vector $Px = (x_3, x_4, x_1, x_2)'$, where P is the permutation matrix with the rows e_3', e_4', e_1', e_2'; $Px \sim N_4(P\mu, P\Sigma P')$ (Theorem 10.2), and $P\Sigma P'$ has the rows and columns of Σ in the order $3, 4, 1, 2$. The first block is $(x_3, x_4)'$ and the second $(x_1, x_2)'$:

$$\Sigma_{11} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Sigma_{11}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad \Sigma_{21} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \Sigma_{21}\Sigma_{11}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

with Σ_{11} PD (its leading principal minors are 2 and 1; Theorem 8.34) and Σ_{11}^{-1} from Proposition 2.18 ($\det \Sigma_{11} = 1$), and

$$\Sigma_{22\cdot 1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} - \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

The conditional mean is $(1, 2)' + \Sigma_{21}\Sigma_{11}^{-1}(a_3 - 0,\ a_4 + 1)' = (a_3 - a_4,\ a_4 + 3)'$, so given $(x_3, x_4)' = (a_3, a_4)'$, the vector $(x_1, x_2)'$ has the distribution $N_2\left((a_3 - a_4,\ a_4 + 3)',\ \Sigma_{22\cdot 1}\right)$. Check: subtracting row 4 of Σ from rows 2 and 3 (Definition 5.13(b)) gives the rows $(1, 2, 0, 0)$ and $(1, 0, 1, 0)$, so that column 4 becomes e_4; expanding along column 4 and then along row 1 (Theorem 5.21), $\det \Sigma = 3(2 \cdot 1 - 0) - 1(1 \cdot 1 - 0) + 1(0 - 2 \cdot 1) = 3$; and by Theorem 5.30(b), with Σ partitioned into the blocks of $(x_1, x_2)'$ and $(x_3, x_4)'$, whose lower right block Σ_{11} is invertible, $\det \Sigma = \det \Sigma_{11} \det \Sigma_{22\cdot 1} = 1 \cdot 3$.

Exercise 10.4. Let $z \sim N_4(0, \Sigma)$ with

$$\Sigma = I_4 + J_4 = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

(a) Show that $z_1 + z_2 + z_3 + z_4$ is independent of $(z_1 - z_2,\ z_2 - z_3,\ z_3 - z_4)'$.

(b) Find all $a \in \mathbb{R}^4$ for which $a'z$ is independent of $z_1 - z_2$, of $z_2 - z_3$ and of $z_3 - z_4$.

(c) Find $b \in \mathbb{R}^3$ for which $z_4 - b_1 z_1 - b_2 z_2 - b_3 z_3$ is independent of $(z_1, z_2, z_3)'$, and the variance of this variable.

Solution. (a) $\Sigma\mathbf{1} = (5, 5, 5, 5)'$, so with $A = \mathbf{1}'$ and

$$B = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$A\Sigma B' = (B\Sigma\mathbf{1})' = (5-5,\ 5-5,\ 5-5) = (0, 0, 0)$. By Theorem 10.5(c), $\mathbf{1}'z$ and Bz are independent.

(b) $\Sigma(1, -1, 0, 0)' = (2-1,\ 1-2,\ 1-1,\ 1-1)' = (1, -1, 0, 0)'$, and likewise $\Sigma(0, 1, -1, 0)' = (0, 1, -1, 0)'$ and $\Sigma(0, 0, 1, -1)' = (0, 0, 1, -1)'$, as J_4 maps these vectors to 0. By Theorem 10.5(c), $a'z$ is independent of $z_1 - z_2$ if and only if $a'\Sigma(1, -1, 0, 0)' = a_1 - a_2 = 0$, of $z_2 - z_3$ if and only if $a_2 - a_3 = 0$, and of $z_3 - z_4$ if and only if $a_3 - a_4 = 0$. So $a = c\mathbf{1}$ ($c \in \mathbb{R}$): only the multiples of the total $z_1 + z_2 + z_3 + z_4$.

(c) Let $a = (-b_1, -b_2, -b_3, 1)'$ and $A = [I_3\ \ 0] \in \mathbb{R}^{3\times 4}$. Then $A\Sigma a = (1, 1, 1)' - \Sigma_{11} b$, where $\Sigma_{11} = I_3 + J_3$ is the leading 3×3 block of Σ. Its inverse is $I_3 - \frac{1}{4}J_3$: $(I_3 + J_3)(I_3 - \frac{1}{4}J_3) = I_3 + \frac{3}{4}J_3 - \frac{1}{4}J_3^2 = I_3$, as $J_3^2 = 3J_3$ (Proposition 1.15). By Theorem 10.5(c), independence holds exactly when $b = \Sigma_{11}^{-1}(1, 1, 1)' = (1, 1, 1)' - \frac{3}{4}(1, 1, 1)' = \frac{1}{4}(1, 1, 1)'$. The variance is $a'\Sigma a = \sigma_{44} - (1, 1, 1)\Sigma_{11}^{-1}(1, 1, 1)' = 2 - \frac{3}{4} = \frac{5}{4}$, as $\Sigma a = (0, 0, 0,\ 2 - \frac{1}{4} - \frac{1}{4} - \frac{1}{4})'$.

Exercise 10.5 ([2, Exercise 10.9])**.** Let $x = (x_1, x_2, x_3)' \sim N_3(\mu, V)$ with V PD, and suppose that

$$Q = (x - \mu)'V^{-1}(x - \mu) = 2x_1^2 + 3x_2^2 + 4x_3^2 + 2x_1x_2 - 2x_1x_3 - 4x_2x_3 - 6x_1 - 6x_2 + 10x_3 + 8$$

(a) Find μ and V.

(b) Find the MGF of x.

Solution. (a)

- For a symmetric $B \in \mathbb{R}^{3\times 3}$, the partial derivatives of $(x-\mu)'B(x-\mu) = \sum_k \sum_l b_{kl}(x_k - \mu_k)(x_l - \mu_l)$ are $2\sum_l b_{il}(x_l - \mu_l)$, the entries of $2B(x - \mu)$ (as in the paragraph after the facts (N1)–(N7)). So the vector of the partial derivatives of Q is $2V^{-1}(x - \mu)$; from the given form of Q it is

$$\left(4x_1 + 2x_2 - 2x_3 - 6,\ 2x_1 + 6x_2 - 4x_3 - 6,\ -2x_1 - 4x_2 + 8x_3 + 10\right)'$$

- Comparing the coefficients of x_1, x_2, x_3, that is, the second partial derivatives,

$$2V^{-1} = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 6 & -4 \\ -2 & -4 & 8 \end{pmatrix}, \qquad V^{-1} = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 3 & -2 \\ -1 & -2 & 4 \end{pmatrix}$$

  The leading principal minors of V^{-1} are 2, 5 and 13, so V^{-1} is PD (Theorem 8.34).

- $\mathsf{E}\big(V^{-1}(x-\mu)\big) = V^{-1}(\mathsf{E}(x)-\mu) = 0$ (Theorem 9.3(b), Definition 10.1). Taking expectations of the three entries above, with $\mathsf{E}(x)=\mu$,

$$\begin{pmatrix} 4 & 2 & -2 \\ 2 & 6 & -4 \\ -2 & -4 & 8 \end{pmatrix}\mu = \begin{pmatrix} 6 \\ 6 \\ -10 \end{pmatrix}, \qquad \text{so} \qquad \mu = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

since $(4-0+2,\ 2-0+4,\ -2-0-8)' = (6,6,-10)'$ and the coefficient matrix is invertible (its determinant is $8\cdot 13$). Check: the constant term is $\mu' V^{-1}\mu = (1,0,-1)(3,3,-5)' = 8$.

- By Theorem 5.24, with $\det V^{-1} = 13$,

$$V = \frac{1}{13}\begin{pmatrix} 8 & -2 & 1 \\ -2 & 7 & 3 \\ 1 & 3 & 5 \end{pmatrix}$$

(b) By Definition 10.1, $M_x(t) = \exp\big(t'\mu + \tfrac{1}{2}t'Vt\big)$ with μ and V of (a), that is, $M_x(t) = \exp\big(t_1 - t_3 + \tfrac{1}{26}(8t_1^2 + 7t_2^2 + 5t_3^2 - 4t_1t_2 + 2t_1t_3 + 6t_2t_3)\big)$.

Exercise 10.6. A random vector $x \in \mathbb{R}^4$ has the density $f(u) = (2\pi)^{-2}\exp\big(-\tfrac{1}{2}Q(u)\big)$, where

$$Q(u) = 3u_1^2 + 3u_2^2 + 2u_3^2 + u_4^2 + 4u_1u_2 - 2u_1u_3 - 4u_2u_3 + 2u_2u_4 - 2u_3u_4 - 10u_1 - 8u_2 + 4u_3 + 9$$

Show that x is normal, and find μ, Σ, the correlation ρ_{12} of x_1 and x_2, and the conditional distribution of $(x_3,x_4)'$ given $(x_1,x_2)' = (a_1,a_2)'$.

Solution.

- The quadratic part of Q is $u'Pu$ for the symmetric matrix P below (Definition 8.17), and the linear part is $-2c'u$ with $c = (5,4,-2,0)'$. Row reduction of $[P \mid I_4]$ (Corollary 2.17(a)), with the pivots taken from the last column back:

$$\left(\begin{array}{cccc|cccc} 3 & 2 & -1 & 0 & 1 & 0 & 0 & 0 \\ 2 & 3 & -2 & 1 & 0 & 1 & 0 & 0 \\ -1 & -2 & 2 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3\leftarrow R_3+R_4]{R_2\leftarrow R_2-R_4} \left(\begin{array}{cccc|cccc} 3 & 2 & -1 & 0 & 1 & 0 & 0 & 0 \\ 2 & 2 & -1 & 0 & 0 & 1 & 0 & -1 \\ -1 & -1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4\leftarrow R_4+R_3]{R_1\leftarrow R_1+R_3,\ R_2\leftarrow R_2+R_3} \left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ -1 & -1 & 1 & 0 & 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 & 0 & 0 & 1 & 2 \end{array}\right) \xrightarrow[R_3\leftarrow R_3+R_2]{R_1\leftarrow R_1-R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & -1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 2 & 1 \\ -1 & 0 & 0 & 1 & 0 & 0 & 1 & 2 \end{array}\right)$$

$$\xrightarrow[R_4\leftarrow R_4+R_1]{R_2\leftarrow R_2-R_1} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 2 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 & -1 & 1 & 3 \end{array}\right)$$

So P is invertible, with

$$\Sigma = P^{-1} = \begin{pmatrix} 1 & -1 & 0 & 1 \\ -1 & 2 & 1 & -1 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 1 & 3 \end{pmatrix}$$

Every arrow added multiples of rows, which does not change the determinant (Definition 5.13(b)), and the left half became I_4; so $\det P = 1$ and $\det\Sigma = 1/\det P = 1$ (Theorem 5.18(c)). The leading principal minors of Σ are 1, $2-1=1$, 1 and $\det\Sigma = 1$, so Σ is PD (Theorem 8.34); for the third, adding row 1 to row 2 of the leading 3×3 block and then subtracting the new row 2 from row 3 gives an upper triangular matrix with the diagonal entries $1,1,1$ (Definition 5.13(b), Theorem 5.14(c)).

- Let $\mu = \Sigma c = (5-4+0+0,\ -5+8-2+0,\ 0+4-4+0,\ 5-4-2+0)' = (1,1,0,-1)'$, so that $P\mu = c$. Then $(u-\mu)'P(u-\mu) = u'Pu - 2c'u + \mu'c$, and $\mu'c = 5+4+0+0 = 9$ is the constant term of Q; so $Q(u) = (u-\mu)'\Sigma^{-1}(u-\mu)$ and $f(u) = (2\pi)^{-2}(\det\Sigma)^{-1/2}\exp\Big(-\frac{1}{2}(u-\mu)'\Sigma^{-1}(u-\mu)\Big)$. By Theorem 10.8(b), $x \sim N_4(\mu,\Sigma)$.
- $\rho_{12} = \sigma_{12}/\sqrt{\sigma_{11}\sigma_{22}} = -1/\sqrt{1\cdot 2} = -\sqrt{2}/2$ (Definition 9.7).
- In Theorem 10.13, the first block is $(x_1,x_2)'$: $\Sigma_{11} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ is PD as a principal submatrix of Σ (Theorem 8.30), with determinant 1 and the inverse $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ (Proposition 2.18), and $\Sigma_{21} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & -1 \end{smallmatrix}\right) = \Sigma_{12}$. So

$$\Sigma_{21}\Sigma_{11}^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \Sigma_{22\cdot 1} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

  and the conditional mean is $(0,-1)' + \Sigma_{21}\Sigma_{11}^{-1}(a_1-1,\ a_2-1)' = (a_1+a_2-2,\ a_1-2)'$. Given $(x_1,x_2)' = (a_1,a_2)'$, the vector $(x_3,x_4)'$ has the distribution $N_2\Big((a_1+a_2-2,\ a_1-2)',\ \Sigma_{22\cdot 1}\Big)$. Check: the lower right 2×2 block of $P = \Sigma^{-1}$ is $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right) = \Sigma_{22\cdot 1}^{-1}$ (Proposition 2.18), as Theorem 10.13(b) says.

Exercise 10.7. Which of the following matrices can be the covariance matrix of a random vector?

$$M_1 = \begin{pmatrix} 4 & 3 & 1 & 0 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 3 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 1 & -\frac{2}{5} & -\frac{2}{5} & -\frac{2}{5} \\ -\frac{2}{5} & 1 & -\frac{2}{5} & -\frac{2}{5} \\ -\frac{2}{5} & -\frac{2}{5} & 1 & -\frac{2}{5} \\ -\frac{2}{5} & -\frac{2}{5} & -\frac{2}{5} & 1 \end{pmatrix}, \qquad M_3 = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}$$

Solution. A covariance matrix is symmetric and PSD (Theorem 9.3(c)), and every symmetric PSD matrix is the covariance matrix of some normal vector (Theorem 10.9(b)).

- M_1: for $x = (1,-2,0,0)'$, $x'M_1x = 4-12+4 = -4 < 0$, so M_1 is not PSD and cannot be a covariance matrix. (Its entries would also give the correlation $3/\sqrt{4\cdot 1} = \frac{3}{2} > 1$, against Proposition 9.8(a).)
- M_2: $\mathbf{1}'M_2\mathbf{1}$, the sum of its entries, is $4 - 12\cdot\frac{2}{5} = -\frac{4}{5} < 0$; so M_2 is not PSD and cannot be a covariance matrix.
- $M_3 = 4I_4 - J_4 = 4C_4$ (Definition 1.14). As C_4 is symmetric and idempotent (Proposition 1.15), $x'M_3x = 4x'C_4'C_4x = 4\|C_4x\|^2 \geqslant 0$: M_3 is PSD, so it is a covariance matrix, a singular one ($M_3\mathbf{1} = 0$).

Exercise 10.8. Let $w_1,\dots,w_4$ be independent $N(0,1)$ variables and $z_k = w_1+\cdots+w_k$ ($k=1,\dots,4$), the positions of a random walk.

(a) Write $z = Lw$ and find the distribution of z.

(b) Find Σ^{-1} by row reduction of $[\Sigma \mid I_4]$, and $\det\Sigma$.

(c) Write the density of z, and show that $x'\Sigma^{-1}x = x_1^2 + (x_2-x_1)^2 + (x_3-x_2)^2 + (x_4-x_3)^2$.

(d) Show that L' is the Cholesky factor of Σ. A program produces independent $N(0,1)$ numbers; compute $\mu + Lw$ for $\mu = (1,0,-1,0)'$ and the four numbers $w = (1,-2,2,1)'$. What is the distribution of $\mu + Lw$ when w consists of four such numbers?

Solution. (a) By Theorem 10.9(a), $w \sim N_4(0,I_4)$, and $z = Lw$ with

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad \Sigma = LL' = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

So $z \sim N_4(0, \Sigma)$ (Theorem 10.2). The (i, j) entry of LL' is $\text{row}_i(L)\,\text{row}_j(L)'$ (Theorem 1.4(d)); $\text{row}_i(L)$ has ones in the positions $1, \ldots, i$, so this product counts the positions $k \leqslant i$ with $k \leqslant j$: σ_{ij} is the smaller of i and j.

(b) Row-reduce $[\Sigma \mid I_4]$ (Corollary 2.17(a)):

$$\left(\begin{array}{cccc|cccc} 1&1&1&1&1&0&0&0\\ 1&2&2&2&0&1&0&0\\ 1&2&3&3&0&0&1&0\\ 1&2&3&4&0&0&0&1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc|cccc} 1&1&1&1&1&0&0&0\\ 0&1&1&1&-1&1&0&0\\ 0&1&2&2&-1&0&1&0\\ 0&1&2&3&-1&0&0&1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|cccc} 1&1&1&1&1&0&0&0\\ 0&1&1&1&-1&1&0&0\\ 0&0&1&1&0&-1&1&0\\ 0&0&1&2&0&-1&0&1 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|cccc} 1&1&1&1&1&0&0&0\\ 0&1&1&1&-1&1&0&0\\ 0&0&1&1&0&-1&1&0\\ 0&0&0&1&0&0&-1&1 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow R_3 - R_4]{R_1 \leftarrow R_1 - R_4,\ R_2 \leftarrow R_2 - R_4} \left(\begin{array}{cccc|cccc} 1&1&1&0&1&0&1&-1\\ 0&1&1&0&-1&1&1&-1\\ 0&0&1&0&0&-1&2&-1\\ 0&0&0&1&0&0&-1&1 \end{array}\right)$$

$$\xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cccc|cccc} 1&1&0&0&1&1&-1&0\\ 0&1&0&0&-1&2&-1&0\\ 0&0&1&0&0&-1&2&-1\\ 0&0&0&1&0&0&-1&1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|cccc} 1&0&0&0&2&-1&0&0\\ 0&1&0&0&-1&2&-1&0\\ 0&0&1&0&0&-1&2&-1\\ 0&0&0&1&0&0&-1&1 \end{array}\right)$$

So

$$\Sigma^{-1} = \begin{pmatrix} 2&-1&0&0\\ -1&2&-1&0\\ 0&-1&2&-1\\ 0&0&-1&1 \end{pmatrix}$$

The first three arrows added multiples of rows, which does not change the determinant (Definition 5.13(b)), and gave an upper triangular left half with diagonal entries $1, 1, 1, 1$; so $\det \Sigma = 1$ (Theorem 5.14(c)). Also $\det \Sigma = \det L \cdot \det L' = (\det L)^2 = 1$ (Theorems 5.18(b) and 5.22).

(c) L' is upper triangular with the diagonal entries 1, so $\det L' = 1$ (Theorem 5.14(c)) and L' is invertible (Theorem 5.18(a)); hence $\Sigma = LL' = (L')'L'$ is PD (Theorem 8.26(c)). By Theorem 10.8, z has the density $f(x) = (2\pi)^{-2} \exp\left(-\frac{1}{2} x' \Sigma^{-1} x\right)$, and

$$x'\Sigma^{-1}x = 2x_1^2 + 2x_2^2 + 2x_3^2 + x_4^2 - 2x_1x_2 - 2x_2x_3 - 2x_3x_4 = x_1^2 + (x_2 - x_1)^2 + (x_3 - x_2)^2 + (x_4 - x_3)^2$$

as expanding the right side gives $x_1^2 + x_2^2 - 2x_1x_2 + x_1^2 + x_3^2 - 2x_2x_3 + x_2^2 + x_4^2 - 2x_3x_4 + x_3^2$. The four squares are the squared steps $w_1 = z_1$, $w_2 = z_2 - z_1$, $w_3 = z_3 - z_2$, $w_4 = z_4 - z_3$, that is, $L^{-1}x$.

(d) L' is upper triangular with positive diagonal entries, and $(L')'L' = LL' = \Sigma$. As Σ is PD by (c), the uniqueness in Theorem 8.40 shows that L' is the Cholesky factor of Σ. If w consists of four independent $N(0, 1)$ numbers, then $w \sim N_4(0, I_4)$ and $\mu + Lw \sim N_4(\mu, LL') = N_4(\mu, \Sigma)$ (Theorem 10.9(a),(c)): the entries of Lw are the partial sums $w_1 + \cdots + w_k$, a random walk with $N(0, 1)$ steps, shifted by μ. Here $Lw = (1,\ 1 - 2,\ 1 - 2 + 2,\ 1 - 2 + 2 + 1)' = (1, -1, 1, 2)'$, so $\mu + Lw = (2, -1, 0, 2)'$.

Exercise 10.9. Let $z \sim N_4(\mu, \Sigma)$ with

$$\mu = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 4 & 2 & -2 & 2 \\ 2 & 2 & 1 & 0 \\ -2 & 1 & 6 & -2 \\ 2 & 0 & -2 & 4 \end{pmatrix}$$

(a) Find the lower triangular A with positive diagonal entries and $AA' = \Sigma$.

(b) Write z as $\mu + Aw$ with $w \sim N_4(0, I_4)$, and express w through z.

(c) Write $(z-\mu)'\Sigma^{-1}(z-\mu)$ as a sum of four squares, and give its distribution.

Solution. (a) Comparing the entries of AA' and Σ column by column: $a_{11}^2 = 4$, so $a_{11} = 2$; $a_{21}a_{11} = 2$, $a_{31}a_{11} = -2$ and $a_{41}a_{11} = 2$, so $a_{21} = 1$, $a_{31} = -1$, $a_{41} = 1$; $a_{21}^2 + a_{22}^2 = 2$, so $a_{22} = 1$; $a_{31}a_{21} + a_{32}a_{22} = 1$ and $a_{41}a_{21} + a_{42}a_{22} = 0$, so $a_{32} = 2$, $a_{42} = -1$; $a_{31}^2 + a_{32}^2 + a_{33}^2 = 6$, so $a_{33} = 1$; $a_{41}a_{31} + a_{42}a_{32} + a_{43}a_{33} = -2$, so $a_{43} = -2 + 1 + 2 = 1$; $a_{41}^2 + a_{42}^2 + a_{43}^2 + a_{44}^2 = 4$, so $a_{44} = 1$. Thus

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 1 & -1 & 1 & 1 \end{pmatrix}, \qquad AA' = \begin{pmatrix} 4 & 2 & -2 & 2 \\ 2 & 1+1 & -1+2 & 1-1 \\ -2 & -1+2 & 1+4+1 & -1-2+1 \\ 2 & 1-1 & -1-2+1 & 1+1+1+1 \end{pmatrix} = \Sigma$$

A' is upper triangular with positive diagonal entries, so it is invertible ($\det A' = 2 \neq 0$, Theorems 5.14(c) and 5.18(a)), $\Sigma = (A')'A'$ is PD (Theorem 8.26(c)), and A' is the Cholesky factor of Σ (Theorem 8.40): A is the only such matrix.

(b) By Theorem 10.9(c), z has the distribution of $\mu + Aw$ with $w \sim N_4(0, I_4)$: $z_1 = 1 + 2w_1$, $z_2 = w_1 + w_2$, $z_3 = -1 - w_1 + 2w_2 + w_3$, $z_4 = 2 + w_1 - w_2 + w_3 + w_4$. Conversely, $w = A^{-1}(z-\mu) \sim N_4(0, A^{-1}\Sigma(A^{-1})') = N_4(0, I_4)$ (Theorem 10.2), with

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 \\ \frac{3}{2} & -2 & 1 & 0 \\ -\frac{5}{2} & 3 & -1 & 1 \end{pmatrix}, \qquad AA^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} - \frac{1}{2} & 1 & 0 & 0 \\ -\frac{1}{2} - 1 + \frac{3}{2} & 2 - 2 & 1 & 0 \\ \frac{1}{2} + \frac{1}{2} + \frac{3}{2} - \frac{5}{2} & -1 - 2 + 3 & 1 - 1 & 1 \end{pmatrix} = I_4$$

(Theorem 2.16), that is, $w_1 = \frac{1}{2}(z_1 - 1)$, $w_2 = z_2 - \frac{1}{2}(z_1 - 1)$, $w_3 = \frac{3}{2}(z_1 - 1) - 2z_2 + (z_3 + 1)$ and $w_4 = -\frac{5}{2}(z_1 - 1) + 3z_2 - (z_3 + 1) + (z_4 - 2)$.

(c) $\Sigma^{-1} = (AA')^{-1} = (A^{-1})'A^{-1}$ (Theorem 2.13(b),(c)), so $(z-\mu)'\Sigma^{-1}(z-\mu) = \|A^{-1}(z-\mu)\|^2 = w_1^2 + w_2^2 + w_3^2 + w_4^2$, with $w_1, \ldots, w_4$ as in (b). They are independent $N(0,1)$ (Theorem 10.9(a)), so the sum has the distribution χ_4^2 (N5).

Exercise 10.10. Let $z \sim N_4(\mu, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 25 & 16 & -20 & -20 \\ 16 & 25 & -20 & -20 \\ -20 & -20 & 25 & 16 \\ -20 & -20 & 16 & 25 \end{pmatrix}$$

(a) Find $\Sigma^{1/2}$, $\Sigma^{-1/2}$ and Σ^{-1}.

(b) Find the standardized vector $w = \Sigma^{-1/2}(z-\mu)$, and write z through w.

(c) Give the distribution of $(z-\mu)'\Sigma^{-1}(z-\mu)$.

Solution. (a) Let $h = (1, 1, -1, -1)'$, $E_1 = \bar{J}_4 = \frac{1}{4}J_4$, $E_2 = \frac{1}{4}hh'$ and $E_3 = I_4 - E_1 - E_2$. Here $\mathbf{1}'h = 0$ and $h'h = 4$, so $E_1E_2 = \frac{1}{16}\mathbf{1}(\mathbf{1}'h)h' = 0 = E_2E_1$ and $E_2^2 = \frac{1}{16}h(h'h)h' = E_2$; also $E_1^2 = E_1$

(Proposition 1.15). Hence $E_3^2 = E_3$ and $E_1E_3 = E_2E_3 = 0$, and E_1, E_2, E_3 are symmetric, idempotent and nonzero (the diagonal entries of E_3 are $1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$), with $E_1 + E_2 + E_3 = I_4$. Comparing entries (the diagonal $9 - 2 + 18$, the entries $(1,2)$ and $(3,4)$ $-2 + 18$, the others $-2 - 18$),

$$\Sigma = 9I_4 - 2J_4 + 18hh' = E_1 + 81E_2 + 9E_3$$

This is the spectral decomposition of Σ (Theorem 8.6(d)), with the eigenvalues 1 (on $\mathbf{1}_4$), 81 (on h) and 9 (on the plane orthogonal to $\mathbf{1}_4$ and h); they are positive, so Σ is PD (Theorem 8.21(a)). By Theorem 8.46(a),(c),

$$\Sigma^{1/2} = E_1 + 9E_2 + 3E_3 = 3I_4 - \tfrac{1}{2}J_4 + \tfrac{3}{2}hh' = \begin{pmatrix} 4 & 1 & -2 & -2 \\ 1 & 4 & -2 & -2 \\ -2 & -2 & 4 & 1 \\ -2 & -2 & 1 & 4 \end{pmatrix}$$

and

$$\Sigma^{-1/2} = E_1 + \tfrac{1}{9}E_2 + \tfrac{1}{3}E_3 = \tfrac{1}{3}I_4 + \tfrac{1}{6}J_4 - \tfrac{1}{18}hh' = \frac{1}{9}\begin{pmatrix} 4 & 1 & 2 & 2 \\ 1 & 4 & 2 & 2 \\ 2 & 2 & 4 & 1 \\ 2 & 2 & 1 & 4 \end{pmatrix}$$

(the diagonal $\frac{1}{3} + \frac{1}{6} - \frac{1}{18} = \frac{4}{9}$, the entries $(1,2)$ and $(3,4)$ $\frac{1}{6} - \frac{1}{18} = \frac{1}{9}$, the others $\frac{1}{6} + \frac{1}{18} = \frac{2}{9}$), and $\Sigma^{-1} = E_1 + \frac{1}{81}E_2 + \frac{1}{9}E_3$ (Theorem 8.6(b)), that is,

$$\Sigma^{-1} = \tfrac{1}{9}I_4 + \tfrac{2}{9}J_4 - \tfrac{2}{81}hh' = \frac{1}{81}\begin{pmatrix} 25 & 16 & 20 & 20 \\ 16 & 25 & 20 & 20 \\ 20 & 20 & 25 & 16 \\ 20 & 20 & 16 & 25 \end{pmatrix}$$

(the diagonal $\frac{1}{9} + \frac{2}{9} - \frac{2}{81} = \frac{25}{81}$, the entries $(1,2)$ and $(3,4)$ $\frac{2}{9} - \frac{2}{81} = \frac{16}{81}$, the others $\frac{2}{9} + \frac{2}{81} = \frac{20}{81}$). Check:

$$\begin{pmatrix} 4 & 1 & -2 & -2 \\ 1 & 4 & -2 & -2 \\ -2 & -2 & 4 & 1 \\ -2 & -2 & 1 & 4 \end{pmatrix}^2 = \begin{pmatrix} 25 & 16 & -20 & -20 \\ 16 & 25 & -20 & -20 \\ -20 & -20 & 25 & 16 \\ -20 & -20 & 16 & 25 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 1 & -2 & -2 \\ 1 & 4 & -2 & -2 \\ -2 & -2 & 4 & 1 \\ -2 & -2 & 1 & 4 \end{pmatrix}\begin{pmatrix} 4 & 1 & 2 & 2 \\ 1 & 4 & 2 & 2 \\ 2 & 2 & 4 & 1 \\ 2 & 2 & 1 & 4 \end{pmatrix} = 9I_4$$

and Σ times the integer matrix of Σ^{-1} is $81I_4$; for instance $25 \cdot 25 + 16 \cdot 16 - 20 \cdot 20 - 20 \cdot 20 = 81$ and $25 \cdot 16 + 16 \cdot 25 - 20 \cdot 20 - 20 \cdot 20 = 0$.

(b) By Corollary 10.11(a), $w = \Sigma^{-1/2}(z - \mu) \sim N_4(0, I_4)$: with $d = z - \mu$, $w_1 = \frac{1}{9}(4d_1 + d_2 + 2d_3 + 2d_4)$, $w_2 = \frac{1}{9}(d_1 + 4d_2 + 2d_3 + 2d_4)$, $w_3 = \frac{1}{9}(2d_1 + 2d_2 + 4d_3 + d_4)$ and $w_4 = \frac{1}{9}(2d_1 + 2d_2 + d_3 + 4d_4)$, independent $N(0,1)$; and $z = \mu + \Sigma^{1/2}w$: $z_1 = \mu_1 + 4w_1 + w_2 - 2w_3 - 2w_4$, $z_2 = \mu_2 + w_1 + 4w_2 - 2w_3 - 2w_4$, $z_3 = \mu_3 - 2w_1 - 2w_2 + 4w_3 + w_4$, $z_4 = \mu_4 - 2w_1 - 2w_2 + w_3 + 4w_4$.

(c) By Corollary 10.11(a), $(z - \mu)'\Sigma^{-1}(z - \mu) = w_1^2 + w_2^2 + w_3^2 + w_4^2 \sim \chi_4^2$.

Exercise 10.11. Let $z \sim N_4(\mu, \Sigma)$ with

$$\mu = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 1 & 0 & -1 & 2 \end{pmatrix}$$

(a) Find the RREF of Σ, its rank, and bases of $\mathcal{N}(\Sigma)$ and $\mathcal{C}(\Sigma)$.

(b) Show that $z_1 - z_2 + z_3 = 0$ and $z_2 + z_4 = 2z_1$ with probability 1.

(c) Find $A \in \mathbb{R}^{4\times 2}$ with $AA' = \Sigma$, and check (b) on the representation $z = \mu + Aw$, $w \sim N_2(0, I_2)$.

Solution. (a) Row reduction:

$$\Sigma = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 1 & 0 & -1 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $\operatorname{rank}\Sigma = 2$, and $\Sigma x = 0$ means $x_1 = x_3 - 2x_4$, $x_2 = -x_3 + x_4$: $\mathcal{N}(\Sigma)$ has the basis $a = (1, -1, 1, 0)'$ ($x_3 = 1$, $x_4 = 0$) and $b = (-2, 1, 0, 1)'$ ($x_3 = 0$, $x_4 = 1$). The pivot columns 1 and 2 of Σ, $(1, 1, 0, 1)'$ and $(1, 2, 1, 0)'$, form a basis of $\mathcal{C}(\Sigma)$ (Theorem 4.3(b)). As $\Sigma' = \Sigma$, $\mathcal{N}(\Sigma) = \mathcal{C}(\Sigma)^\perp$ (Corollary 6.7): every vector x of $\mathcal{C}(\Sigma)$ is orthogonal to a and b, that is, $x_1 - x_2 + x_3 = 0$ and $-2x_1 + x_2 + x_4 = 0$; for the basis vectors, $1 - 1 + 0 = 0$, $1 - 2 + 1 = 0$, $-2 + 1 + 1 = 0$ and $-2 + 2 + 0 = 0$.

(b) Let $G \in \mathbb{R}^{2\times 4}$ have the rows a' and b'. Then $G\Sigma = 0$, as $\Sigma a = \Sigma b = 0$ and $\Sigma' = \Sigma$, and $G\mu = (1-2+1+0,\ -2+2+0+0)' = (0, 0)'$. By Theorem 10.2, $Gz = (z_1 - z_2 + z_3,\ -2z_1 + z_2 + z_4)' \sim N_2(G\mu, G\Sigma G') = N_2(0, 0)$, the distribution of the constant 0 (Definition 10.1). So both equations hold with probability 1, in accordance with Corollary 10.11(b): with probability 1, $z - \mu \in \mathcal{C}(\Sigma)$, so $G(z - \mu) = 0$, and $G\mu = 0$.

(c)

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{gives} \quad AA' = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 1 & 0 & -1 & 2 \end{pmatrix} = \Sigma$$

so z has the distribution of $\mu + Aw$ (Theorem 10.9(c)): $z_1 = 1 + w_1$, $z_2 = 2 + w_1 + w_2$, $z_3 = 1 + w_2$, $z_4 = w_1 - w_2$; and $z_1 - z_2 + z_3 = (1 - 2 + 1) + (w_1 - w_1 - w_2 + w_2) = 0$ and $-2z_1 + z_2 + z_4 = (-2 + 2 + 0) + (-2w_1 + w_1 + w_2 + w_1 - w_2) = 0$ for every w.

Exercise 10.12. Let $z \sim N_4(\mu, \Sigma)$ with

$$\mu = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 2 & 1 & 1 & 3 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 7 & 1 \\ 3 & 1 & 1 & 8 \end{pmatrix}$$

(a) Find the conditional distribution of $(z_3, z_4)'$ given $z_1 = a_1$ and $z_2 = a_2$, and for $a_1 = 2$, $a_2 = 1$.

(b) With

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}$$

compute $L\Sigma$ and $L\Sigma L'$.

(c) Show that $(z_3 + z_1 - 3z_2,\ z_4 - 2z_1 + z_2)'$ is independent of $(z_1, z_2)'$, and find its distribution.

Solution. (a) Here $\Sigma_{11} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with determinant 1 and, by Theorem 5.24, $\Sigma_{11}^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$; it is PD (its leading principal minors are 2 and 1; Theorem 8.34). Also $\Sigma_{21} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right)$ and $\Sigma_{22} = \left(\begin{smallmatrix} 7 & 1 \\ 1 & 8 \end{smallmatrix}\right)$. So

$$\Sigma_{21}\Sigma_{11}^{-1} = \begin{pmatrix} 1-2 & -1+4 \\ 3-1 & -3+2 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 2 & -1 \end{pmatrix}$$

$$\Sigma_{22\cdot 1} = \begin{pmatrix} 7 & 1 \\ 1 & 8 \end{pmatrix} - \begin{pmatrix} -1 & 3 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 1 \\ 1 & 8 \end{pmatrix} - \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

By Theorem 10.13, given $z_1 = a_1$, $z_2 = a_2$, the vector $(z_3, z_4)'$ has the distribution

$$N_2\left(\begin{pmatrix} -1-(a_1-1)+3(a_2-0) \\ 2+2(a_1-1)-(a_2-0) \end{pmatrix},\ \Sigma_{22\cdot 1}\right) = N_2\left(\begin{pmatrix} -a_1+3a_2 \\ 2a_1-a_2 \end{pmatrix},\ \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\right)$$

For $a_1 = 2$, $a_2 = 1$ the conditional distribution is $N_2\big((1,3)', \Sigma_{22\cdot 1}\big)$.

(b) The rows 3 and 4 of $L\Sigma$ are $\text{row}_1(\Sigma) - 3\,\text{row}_2(\Sigma) + \text{row}_3(\Sigma) = (2-3+1,\ 1-3+2,\ 1-6+7,\ 3-3+1)$ and $-2\,\text{row}_1(\Sigma) + \text{row}_2(\Sigma) + \text{row}_4(\Sigma) = (-4+1+3,\ -2+1+1,\ -2+2+1,\ -6+1+8)$ (Theorem 1.4(c)), and then the columns 3 and 4 of $L\Sigma L'$ are $\text{col}_1(L\Sigma) - 3\,\text{col}_2(L\Sigma) + \text{col}_3(L\Sigma)$ and $-2\,\text{col}_1(L\Sigma) + \text{col}_2(L\Sigma) + \text{col}_4(L\Sigma)$:

$$L\Sigma = \begin{pmatrix} 2 & 1 & 1 & 3 \\ 1 & 1 & 2 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix}, \qquad L\Sigma L' = \begin{pmatrix} 2 & 1 & 2-3+1 & -4+1+3 \\ 1 & 1 & 1-3+2 & -2+1+1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

as in Theorem 5.9: the lower right block is the Schur complement $\Sigma_{22\cdot 1}$.

(c) The entries 3 and 4 of Lz are $z_1 - 3z_2 + z_3$ and $-2z_1 + z_2 + z_4$, and $Lz \sim N_4(L\mu, L\Sigma L')$ (Theorem 10.2). The off-diagonal blocks of $L\Sigma L'$ are 0, so $(z_3 + z_1 - 3z_2,\ z_4 - 2z_1 + z_2)'$ is independent of $(z_1, z_2)'$ (Theorem 10.5(a)), and it has the distribution $N_2\big((1-0-1,\ -2+0+2)',\ \Sigma_{22\cdot 1}\big) = N_2(0, \Sigma_{22\cdot 1})$ (Corollary 10.3(a)). So $(z_3, z_4)' = (-z_1 + 3z_2,\ 2z_1 - z_2)' + (z_3 + z_1 - 3z_2,\ z_4 - 2z_1 + z_2)'$: the conditional mean of (a) plus an independent $N_2(0, \Sigma_{22\cdot 1})$ vector.

Exercise 10.13. Let z be the random walk of Exercise 10.8.

(a) Find the conditional distribution of z_4 given $z_1 = a_1$, $z_2 = a_2$ and $z_3 = a_3$.

(b) Find the conditional distribution of z_2 given $z_1 = a_1$ and $z_3 = a_3$, and given $z_1 = a_1$, $z_3 = a_3$ and $z_4 = a_4$.

(c) Compare the conditional variance in (a) with the $(4,4)$ entry of Σ^{-1} found in Exercise 10.8.

Solution. (a) Here Σ_{11} is the leading 3×3 block of Σ, which is PD, as Σ is (Exercise 10.8(c), Theorem 8.30), and $\Sigma_{21} = (1,2,3)$. As

$$(0,0,1)\,\Sigma_{11} = (0,0,1)\begin{pmatrix}1&1&1\\1&2&2\\1&2&3\end{pmatrix} = (1,2,3)$$

multiplying by Σ_{11}^{-1} on the right gives $\Sigma_{21}\Sigma_{11}^{-1} = (0,0,1)$, and $\Sigma_{22\cdot 1} = 4 - (0,0,1)(1,2,3)' = 4 - 3 = 1$. By Theorem 10.13, given $z_1 = a_1$, $z_2 = a_2$, $z_3 = a_3$, the variable z_4 is $N(a_3, 1)$: the next position is the present one plus an independent $N(0,1)$ step, whatever a_1 and a_2 are.

(b) Reorder the coordinates: $y = (z_1, z_3, z_2)' = Pz$ for the matrix $P \in \mathbb{R}^{3\times 4}$ with the rows e_1', e_3', e_2', and $y \sim N_3(0, P\Sigma P')$ (Theorem 10.2), where $P\Sigma P'$ has the rows and columns $1, 3, 2$ of Σ:

$$P\Sigma P' = \begin{pmatrix}1&1&1\\1&3&2\\1&2&2\end{pmatrix}, \qquad \Sigma_{11} = \begin{pmatrix}1&1\\1&3\end{pmatrix}, \qquad \Sigma_{11}^{-1} = \frac{1}{2}\begin{pmatrix}3&-1\\-1&1\end{pmatrix}, \qquad \Sigma_{21} = (1,2)$$

(Theorem 5.24); this Σ_{11}, and the one below, are principal submatrices of Σ, hence PD, as in (a) (Theorem 8.30). So $\Sigma_{21}\Sigma_{11}^{-1} = \frac{1}{2}(3-2,\ -1+2) = (\frac{1}{2}, \frac{1}{2})$ and $\Sigma_{22\cdot 1} = 2 - (\frac{1}{2}, \frac{1}{2})(1,2)' = 2 - \frac{3}{2} = \frac{1}{2}$. Given $z_1 = a_1$, $z_3 = a_3$, the variable z_2 is $N\Big(\frac{1}{2}(a_1 + a_3),\ \frac{1}{2}\Big)$: it is centered at the midpoint of its neighbours. Given also $z_4 = a_4$, take $y = (z_1, z_3, z_4, z_2)'$ with the rows e_1', e_3', e_4', e_2': now $\Sigma_{21} = (1,2,2)$ and, as

$$\Big(\tfrac{1}{2}, \tfrac{1}{2}, 0\Big)\,\Sigma_{11} = \Big(\tfrac{1}{2}, \tfrac{1}{2}, 0\Big)\begin{pmatrix}1&1&1\\1&3&3\\1&3&4\end{pmatrix} = (1,2,2)$$

$\Sigma_{21}\Sigma_{11}^{-1} = (\frac{1}{2}, \frac{1}{2}, 0)$ and $\Sigma_{22\cdot 1} = 2 - \frac{1}{2} - 1 = \frac{1}{2}$: the same distribution $N\Big(\frac{1}{2}(a_1 + a_3),\ \frac{1}{2}\Big)$, whatever a_4 is.

(c) The $(4,4)$ entry of Σ^{-1} is 1, the inverse of the conditional variance 1 in (a), as Theorem 10.13(b) says.

Exercise 10.14 (cf. [18, Example 5.3])**.** Let $x \sim N_4(\mu, \Sigma)$ with

$$\Sigma = \begin{pmatrix}4&1&2&1\\1&4&2&1\\2&2&2&1\\1&1&1&1\end{pmatrix}$$

(a) Show that Σ is PD. Find the conditional distribution of $(x_1, x_2)'$ given $(x_3, x_4)' = (a_3, a_4)'$, and compare the partial correlation of x_1 and x_2 given x_3 and x_4 (the paragraph after Theorem 10.13) with ρ_{12}.

(b) Find the partial correlation of x_1 and x_2 given x_3 alone, and given x_4 alone.

(c) Let $u = (x_1, x_2)' - B(x_3, x_4)'$, with B as in (a). Compute $\mathrm{var}(u)$ from Theorem 9.3(b), and compare with (a).

Solution. (a) The leading principal minors of Σ are 4, 15, 6 and $\det\Sigma = 3$, all positive (Theorem 8.34). Apply Theorem 10.13 to the reordered vector $(x_3, x_4, x_1, x_2)'$, which is normal (Theorem 10.2 with the permutation matrix with the rows e_3', e_4', e_1', e_2'), with the first block $(x_3, x_4)'$: $\Sigma_{11} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, which is PD as a principal submatrix of Σ (Theorem 8.30), $\Sigma_{11}^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ (Proposition 2.18), $\Sigma_{21} = \left(\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right)$ and

$$B = \Sigma_{21}\Sigma_{11}^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \Sigma_{22\cdot 1} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

So given $(x_3, x_4)' = (a_3, a_4)'$, the vector $(x_1, x_2)'$ has the distribution $N_2\Big((\mu_1, \mu_2)' + (a_3 - \mu_3)(1, 1)',\ \Sigma_{22\cdot 1}\Big)$. The partial correlation is $-1/\sqrt{2\cdot 2} = -\frac{1}{2}$, while $\rho_{12} = 1/\sqrt{4\cdot 4} = \frac{1}{4}$: the sign changes.

(b) Given x_3 (the vector $(x_3, x_1, x_2)'$, $\Sigma_{11} = (2)$): $\left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right) - \frac{1}{2}(2, 2)'(2, 2) = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, partial correlation $-\frac{1}{2}$. Given x_4 ($\Sigma_{11} = (1)$): $\left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right) - (1, 1)'(1, 1) = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, partial correlation 0.

(c) $u = (x_1 - x_3,\ x_2 - x_3)' = Kx$ with $K = \left(\begin{smallmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{smallmatrix}\right)$. The rows of $K\Sigma$ are $(2, -1, 0, 0)$ and $(-1, 2, 0, 0)$, so $\operatorname{var}(u) = K\Sigma K' = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right) = \Sigma_{22\cdot 1}$. This agrees with Theorem 10.13(a): u differs from the vector w of that theorem (for $(x_3, x_4, x_1, x_2)'$) by the constant vector $(\mu_1, \mu_2)' - B(\mu_3, \mu_4)'$.

Exercise 10.15. Let $(u, v)'$ be bivariate normal with $\mu_u = 2$, $\mu_v = 3$, $\sigma_u = 2$, $\sigma_v = 5$ and $\rho = \frac{3}{5}$. Let Φ be the distribution function of $N(0, 1)$: $\Phi(c) = \mathsf{P}(y \leqslant c)$ for $y \sim N(0, 1)$.

(a) Find Σ, Σ^{-1} and the density of $(u, v)'$.

(b) Find the conditional distributions of v given u and of u given v.

(c) Express $\mathsf{P}(v > 10)$ and the conditional probability $\mathsf{P}(v > 10)$ given $u = 4$ through Φ. Which is larger?

Solution. (a) $\rho\sigma_u\sigma_v = \frac{3}{5}\cdot 10 = 6$, so

$$\Sigma = \begin{pmatrix} 4 & 6 \\ 6 & 25 \end{pmatrix}, \qquad \det\Sigma = 100 - 36 = 64, \qquad \Sigma^{-1} = \frac{1}{64}\begin{pmatrix} 25 & -6 \\ -6 & 4 \end{pmatrix}$$

(Theorem 5.24). As $2\pi\sigma_u\sigma_v\sqrt{1-\rho^2} = 2\pi\cdot 10\cdot\frac{4}{5} = 16\pi$, Proposition 10.16(c) gives

$$f(s, t) = \frac{1}{16\pi}\exp\Big(-\frac{1}{128}\big(25(s-2)^2 - 12(s-2)(t-3) + 4(t-3)^2\big)\Big)$$

(b) By Proposition 10.16(d), given u, $v \sim N\Big(3 + \frac{3}{5}\cdot\frac{5}{2}(u-2),\ 25\cdot\frac{16}{25}\Big) = N\Big(3 + \frac{3}{2}(u-2),\ 16\Big)$; with the roles exchanged, given v, $u \sim N\Big(2 + \frac{3}{5}\cdot\frac{2}{5}(v-3),\ 4\cdot\frac{16}{25}\Big) = N\Big(2 + \frac{6}{25}(v-3),\ \frac{64}{25}\Big)$.

(c) If $x \sim N(m, s^2)$ with $s > 0$, then $(x - m)/s \sim N(0, 1)$ (Theorem 10.2 with $A = 1/s$, $b = -m/s$), and $\mathsf{P}(x > c) = 1 - \Phi\big((c-m)/s\big)$. As $v \sim N(3, 25)$, $\mathsf{P}(v > 10) = 1 - \Phi(\frac{10-3}{5}) = 1 - \Phi(\frac{7}{5})$. Given $u = 4$, v has the distribution $N(3+3, 16) = N(6, 16)$, so the conditional probability is $1 - \Phi(\frac{10-6}{4}) = 1 - \Phi(1)$. It is the larger one: $\Phi(\frac{7}{5}) - \Phi(1) = \mathsf{P}(1 < y \leqslant \frac{7}{5}) > 0$, as $N(0, 1)$ has a positive density (Theorem 10.8). Knowing that $u = 4$, one standard deviation of u above its mean, makes the probability larger.

Exercise 10.16. The rows of the data matrix

$$X = \begin{pmatrix} 6 & 3 & 4 & 1 \\ 2 & 7 & 2 & 1 \\ 4 & 5 & 5 & 4 \\ 6 & 7 & 2 & 3 \\ 2 & 3 & 2 & 1 \end{pmatrix}$$

are a random sample of size $n = 5$ from $N_4(\mu, \Sigma)$.

(a) Find $\bar{x}$, C_5X, $A = (C_5X)'(C_5X)$ and S.

(b) Find the maximum likelihood estimates $\hat{\mu}$, $\hat{\Sigma}$, and $\det\hat{\Sigma}$.

(c) Using the proof of Theorem 10.20, compute $\log L(\bar{x}, \hat{\Sigma}) - \log L(\bar{x}, S)$.

Solution. (a) $\mathbf{1}'X = (20, 25, 15, 10)$, so $\bar{x} = (4, 5, 3, 2)'$, and, subtracting $(4, 5, 3, 2)$ from every row,

$$C_5X = \begin{pmatrix} 2 & -2 & 1 & -1 \\ -2 & 2 & -1 & -1 \\ 0 & 0 & 2 & 2 \\ 2 & 2 & -1 & 1 \\ -2 & -2 & -1 & -1 \end{pmatrix}, \qquad A = \begin{pmatrix} 16 & 0 & 4 & 4 \\ 0 & 16 & -4 & 4 \\ 4 & -4 & 8 & 4 \\ 4 & 4 & 4 & 8 \end{pmatrix}, \qquad S = \frac{1}{4}A = \begin{pmatrix} 4 & 0 & 1 & 1 \\ 0 & 4 & -1 & 1 \\ 1 & -1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

(for instance $16 = 4+4+0+4+4$, $4 = 2+2+0-2+2$ and $8 = 1+1+4+1+1$).

(b) By Theorem 5.30(a) with $S_{11} = 4I_2$, $S_{21} = \left(\begin{smallmatrix} 1 & -1 \\ 1 & -1 \end{smallmatrix}\right) = S_{12}'$ and $S_{22} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$,

$$S_{22} - S_{21}S_{11}^{-1}S_{12} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \frac{1}{4}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & 1 \\ 1 & \frac{3}{2} \end{pmatrix}, \qquad \det S = \det(4I_2)\cdot\left(\frac{9}{4} - 1\right) = 16\cdot\frac{5}{4} = 20$$

So S, which is PSD (Proposition 9.14(a)), is invertible (Theorem 5.18(a)) and hence PD (Theorem 8.21(c)), and so is $A = 4S$. By Theorem 10.20,

$$\hat{\mu} = \bar{x} = \begin{pmatrix} 4 \\ 5 \\ 3 \\ 2 \end{pmatrix}, \qquad \hat{\Sigma} = \frac{1}{5}A = \frac{4}{5}S, \qquad \det\hat{\Sigma} = \left(\frac{4}{5}\right)^4 \det S = \frac{256}{625}\cdot 20 = \frac{1024}{125}$$

(Theorem 5.18(c)).

(c) In the proof of Theorem 10.20, $\log L(\bar{x}, \hat{\Sigma}) - \log L(\bar{x}, S) = \frac{1}{2}\big(g(S) - g(\hat{\Sigma})\big) = \frac{n}{2}(\operatorname{tr} B - \log\det B - p)$ with $B = \hat{\Sigma}^{1/2}S^{-1}\hat{\Sigma}^{1/2}$. As $S = \frac{5}{4}\hat{\Sigma}$, $S^{-1} = \frac{4}{5}\hat{\Sigma}^{-1}$, and $\hat{\Sigma}^{-1} = \hat{\Sigma}^{-1/2}\hat{\Sigma}^{-1/2}$ (Theorem 8.46), so $B = \frac{4}{5}I_4$, with $\operatorname{tr} B = \frac{16}{5}$ and $\det B = \frac{256}{625}$. So the difference is

$$\frac{5}{2}\left(\frac{16}{5} - \log\frac{256}{625} - 4\right) = 10\log\frac{5}{4} - 2$$

It is positive, as Lemma 10.18 requires: $\left(\frac{5}{4}\right)^5 = \frac{3125}{1024} > 3 > e$, so $5\log\frac{5}{4} > 1$.

Exercise 10.17. The five observations $(2,4,1,1)$, $(2,2,3,1)$, $(2,6,5,3)$, $(5,4,3,4)$, $(4,4,3,1)$ are modeled as a random sample from $N_4(\mu,\Sigma)$.

(a) Find the maximum likelihood estimates $\hat\mu$, $\hat\Sigma$, the largest value of $\log L$, and the correlation matrix computed from $\hat\Sigma$.

(b) Compute $\log L(\bar x, S)$ for the unbiased estimate S, and check that it is smaller than $\log L(\bar x,\hat\Sigma)$.

(c) Find the eigenvalues $\theta_1,\dots,\theta_4$ of $B=\hat\Sigma^{1/2}S^{-1}\hat\Sigma^{1/2}$, and check the identity $g(S)-g(\hat\Sigma)=n(\operatorname{tr}B-\log\det B-p)$ of the proof of Theorem 10.20, that is, $n\sum_j(\theta_j-1-\log\theta_j)$ (Theorem 7.19), where $g(\Sigma)=n\log\det\Sigma+\operatorname{tr}(\Sigma^{-1}A)$.

Solution. (a) $\bar x=\frac15(15,20,15,10)'=(3,4,3,2)'$, and the deviations $x_i-\bar x$ are $(-1,0,-2,-1)$, $(-1,-2,0,-1)$, $(-1,2,2,1)$, $(2,0,0,2)$, $(1,0,0,-1)$. So

$$A=\sum_{i=1}^5(x_i-\bar x)(x_i-\bar x)'=\begin{pmatrix}8&0&0&4\\0&8&4&4\\0&4&8&4\\4&4&4&8\end{pmatrix},\qquad S=\frac14A=\begin{pmatrix}2&0&0&1\\0&2&1&1\\0&1&2&1\\1&1&1&2\end{pmatrix}$$

(for instance $8=1+1+1+4+1$ and $4=1+1-1+4-1$). By Theorem 5.30(a) with $S_{11}=2I_2$, $S_{21}=\left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)=S_{12}'$ and $S_{22}=\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$,

$$S_{22}-S_{21}S_{11}^{-1}S_{12}=\begin{pmatrix}2&1\\1&2\end{pmatrix}-\frac12\begin{pmatrix}1&1\\1&2\end{pmatrix}=\begin{pmatrix}\frac32&\frac12\\\frac12&1\end{pmatrix},\qquad \det S=\det(2I_2)\cdot\left(\frac32-\frac14\right)=4\cdot\frac54=5$$

so S, which is PSD (Proposition 9.14(a)), is invertible (Theorem 5.18(a)) and hence PD (Theorem 8.21(c)), and so is $A=4S$. By Theorem 10.20, $\hat\mu=\bar x=(3,4,3,2)'$, $\hat\Sigma=\frac15A=\frac45S$, $\det\hat\Sigma=\left(\frac45\right)^4\cdot5=\frac{256}{125}$ (Theorem 5.18(c)), and the largest value of the log-likelihood is

$$\log L(\bar x,\hat\Sigma)=-10\log(2\pi)-\frac52\log\frac{256}{125}-10$$

All diagonal entries of $\hat\Sigma$ are $\frac85$, so the correlation matrix computed from $\hat\Sigma$ (Definition 9.7) is $\frac58\hat\Sigma=\frac18A$:

$$\frac18A=\begin{pmatrix}1&0&0&\frac12\\0&1&\frac12&\frac12\\0&\frac12&1&\frac12\\\frac12&\frac12&\frac12&1\end{pmatrix}$$

the sample correlation matrix of the data (Definition 9.17: the diagonal entries of S are 2, and $\frac12S=\frac18A$).

(b) In the proof of Theorem 10.20, $\log L(\bar x,\Sigma)=-10\log(2\pi)-\frac12g(\Sigma)$, with $np=20$. Here $g(S)=5\log\det S+\operatorname{tr}(S^{-1}A)=5\log5+\operatorname{tr}(4I_4)=5\log5+16$, so $\log L(\bar x,S)=-10\log(2\pi)-\frac52\log5-8$. Hence

$$\log L(\bar x,\hat\Sigma)-\log L(\bar x,S)=\frac52\Big(\log5-\log\frac{256}{125}\Big)-2=\frac52\log\frac{625}{256}-2=10\log\frac54-2$$

This is positive: $\left(\frac54\right)^5=\frac{3125}{1024}>3>e$, so $5\log\frac54>1$.

(c) $S^{-1}=\frac45\hat\Sigma^{-1}$ and $\hat\Sigma^{-1}=\hat\Sigma^{-1/2}\hat\Sigma^{-1/2}$ (Theorem 8.46), so $B=\frac45\hat\Sigma^{1/2}\hat\Sigma^{-1}\hat\Sigma^{1/2}=\frac45I_4$ and $\theta_1=\cdots=\theta_4=\frac45$. Then $n\sum_j(\theta_j-1-\log\theta_j)=5\cdot4\Big(\frac45-1+\log\frac54\Big)=20\log\frac54-4$, and $g(S)-g(\hat\Sigma)=5\log5+16-\Big(5\log\frac{256}{125}+20\Big)=5\log\frac{625}{256}-4=20\log\frac54-4$, the same number, as $g(\hat\Sigma)=5\log\det\hat\Sigma+\operatorname{tr}(5I_4)$; it is $2\Big(\log L(\bar x,\hat\Sigma)-\log L(\bar x,S)\Big)$, as in the proof of Theorem 10.20. The difference of (b) is the number of Exercise 10.16(c), for other data: for all data, $S=\frac{n}{n-1}\hat\Sigma$, so $B=\frac{n-1}{n}I_p$, and the difference depends only on $n=5$ and $p=4$.

Problems

Problem 10.1 ([17, Exercise 4.42], [20, Exercise 4.42])**.** Let u and v be independent random variables with means μ_u, μ_v and positive variances σ_u^2, σ_v^2. Find the correlation of uv and v in terms of these means and variances.

Solution.

- By (N3), u and v^2 are independent, and so are u^2 and v^2; by (N7), $\mathsf{E}(uv^2) = \mu_u \mathsf{E}(v^2)$, $\mathsf{E}(uv) = \mu_u\mu_v$ and $\mathsf{E}(u^2v^2) = \mathsf{E}(u^2)\,\mathsf{E}(v^2)$, where $\mathsf{E}(u^2) = \sigma_u^2 + \mu_u^2$ and $\mathsf{E}(v^2) = \sigma_v^2 + \mu_v^2$.
- The entries $(1,2)$ and $(1,1)$ of Theorem 9.4(e) for the random vector $(uv, v)'$ read $\operatorname{cov}(uv, v) = \mathsf{E}(uv^2) - \mathsf{E}(uv)\,\mathsf{E}(v)$ and $\operatorname{var}(uv) = \mathsf{E}(u^2v^2) - \big(\mathsf{E}(uv)\big)^2$. So $\operatorname{cov}(uv, v) = \mu_u(\sigma_v^2 + \mu_v^2) - \mu_u\mu_v^2 = \mu_u\sigma_v^2$.
- $\operatorname{var}(uv) = (\sigma_u^2 + \mu_u^2)(\sigma_v^2 + \mu_v^2) - \mu_u^2\mu_v^2 = \sigma_u^2\sigma_v^2 + \sigma_u^2\mu_v^2 + \sigma_v^2\mu_u^2 > 0$.
- Hence the correlation (Definition 9.7) is

$$\frac{\operatorname{cov}(uv, v)}{\sqrt{\operatorname{var}(uv)\operatorname{var}(v)}} = \frac{\mu_u\sigma_v^2}{\sigma_v\sqrt{\sigma_u^2\sigma_v^2 + \sigma_u^2\mu_v^2 + \sigma_v^2\mu_u^2}} = \frac{\mu_u\sigma_v}{\sqrt{\mu_u^2\sigma_v^2 + \mu_v^2\sigma_u^2 + \sigma_u^2\sigma_v^2}}$$

Problem 10.2 ([17, Exercise 4.46(b),(d)], [20, Exercise 4.46])**.** Let w_1, w_2 be independent $N(0,1)$ variables, and for constants $a_u, b_u, c_u, a_v, b_v, c_v$ let $u = a_uw_1 + b_uw_2 + c_u$ and $v = a_vw_1 + b_vw_2 + c_v$. Let $\mu_u, \mu_v \in \mathbb{R}$, $\sigma_u, \sigma_v > 0$ and $-1 \leqslant \rho \leqslant 1$.

(a) Let $a_u = \sqrt{\frac{1+\rho}{2}}\,\sigma_u$, $b_u = \sqrt{\frac{1-\rho}{2}}\,\sigma_u$, $c_u = \mu_u$, $a_v = \sqrt{\frac{1+\rho}{2}}\,\sigma_v$, $b_v = -\sqrt{\frac{1-\rho}{2}}\,\sigma_v$ and $c_v = \mu_v$. Show that $\mathsf{E}(u) = \mu_u$, $\operatorname{var}(u) = \sigma_u^2$, $\mathsf{E}(v) = \mu_v$, $\operatorname{var}(v) = \sigma_v^2$, and that the correlation of u and v is ρ.

(b) Show that the constants in (a) are not the only ones with these properties, by giving another choice. How many choices are there?

Solution. Here $(u, v)' = Aw + c$ with $A = \left(\begin{smallmatrix} a_u & b_u \\ a_v & b_v \end{smallmatrix}\right)$, $w = (w_1, w_2)'$ and $c = (c_u, c_v)'$. As w has the mean 0 and the covariance matrix I_2 (Theorem 10.9(a), Definition 10.1), Theorem 9.3(b) gives the mean c and the covariance matrix

$$AA' = \begin{pmatrix} a_u^2 + b_u^2 & a_ua_v + b_ub_v \\ a_ua_v + b_ub_v & a_v^2 + b_v^2 \end{pmatrix}$$

So the requirements are $c_u = \mu_u$, $c_v = \mu_v$, $a_u^2 + b_u^2 = \sigma_u^2$, $a_v^2 + b_v^2 = \sigma_v^2$ and $a_ua_v + b_ub_v = \rho\sigma_u\sigma_v$; and then $(u, v)'$ is bivariate normal with these parameters (Theorem 10.9(c), Definition 10.15).

(a) $a_u^2 + b_u^2 = \big(\frac{1+\rho}{2} + \frac{1-\rho}{2}\big)\sigma_u^2 = \sigma_u^2$, likewise $a_v^2 + b_v^2 = \sigma_v^2$, and $a_ua_v + b_ub_v = \big(\frac{1+\rho}{2} - \frac{1-\rho}{2}\big)\sigma_u\sigma_v = \rho\sigma_u\sigma_v$; so the correlation is $\rho\sigma_u\sigma_v/(\sigma_u\sigma_v) = \rho$.

(b)

- Another choice is $a_u = \rho\sigma_u$, $b_u = \sqrt{1-\rho^2}\,\sigma_u$, $a_v = \sigma_v$, $b_v = 0$ (with $c_u = \mu_u$, $c_v = \mu_v$): $a_u^2 + b_u^2 = \sigma_u^2$, $a_v^2 + b_v^2 = \sigma_v^2$ and $a_ua_v + b_ub_v = \rho\sigma_u\sigma_v$. It differs from (a) if $\rho \neq 1$; for $\rho = 1$, take $a_u = a_v = 0$, $b_u = \sigma_u$, $b_v = \sigma_v$.
- There are infinitely many choices. If the three equations hold, then $(\rho\sigma_u\sigma_v - a_ua_v)^2 = b_u^2b_v^2 = (\sigma_u^2 - a_u^2)(\sigma_v^2 - a_v^2)$, that is,

$$\sigma_u^2a_v^2 - 2\rho\sigma_u\sigma_va_ua_v + \sigma_v^2a_u^2 = (1-\rho^2)\sigma_u^2\sigma_v^2$$

an ellipse in the (a_u, a_v)-plane if $|\rho| < 1$, and the line $\sigma_ua_v = \rho\sigma_va_u$ if $\rho = \pm 1$.

- Conversely, for every a_u with $|a_u| < \sigma_u$ this quadratic equation in a_v has a real solution, as its discriminant $4\rho^2\sigma_u^2\sigma_v^2a_u^2 - 4\sigma_u^2\big(\sigma_v^2a_u^2 - (1-\rho^2)\sigma_u^2\sigma_v^2\big) = 4(1-\rho^2)\sigma_u^2\sigma_v^2(\sigma_u^2 - a_u^2)$ is $\geqslant 0$. Then $b_u = \sqrt{\sigma_u^2 - a_u^2} > 0$ and $b_v = (\rho\sigma_u\sigma_v - a_ua_v)/b_u$ satisfy the three equations: $a_ua_v + b_ub_v = \rho\sigma_u\sigma_v$, and $b_v^2 = (\sigma_u^2 - a_u^2)(\sigma_v^2 - a_v^2)/b_u^2 = \sigma_v^2 - a_v^2$ by the displayed equation. So every $a_u \in (-\sigma_u, \sigma_u)$ gives a choice.

Problem 10.3 ([3, Ch. 8, Exercise 14])**.** Let $y \sim N_n(0, \Sigma)$, where Σ is PSD of rank $r \geqslant 1$. Show that there are $G \in \mathbb{R}^{n\times r}$ with $\Sigma = GG'$ and $x \sim N_r(0, I_r)$ such that $y = Gx$ with probability 1.

Solution.

- By Lemma 9.22(a) and Theorem 8.4, $\Sigma = P\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)P'$ with P orthogonal and $D = \operatorname{diag}(\lambda_1, \dots, \lambda_r)$, $\lambda_1 \geqslant \cdots \geqslant \lambda_r > 0$; the other eigenvalues are 0, as Σ is PSD (Theorem 8.21(a)). Write $P = [P_1\ P_2]$ with $P_1 \in \mathbb{R}^{n\times r}$, and put $G = P\left(\begin{smallmatrix} D^{1/2} \\ 0 \end{smallmatrix}\right) = P_1D^{1/2}$, where $D^{1/2} = \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_r})$.
- $GG' = P\left(\begin{smallmatrix} D^{1/2} \\ 0 \end{smallmatrix}\right)\begin{pmatrix} D^{1/2} & 0 \end{pmatrix}P' = P\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)P' = \Sigma$ (Theorem 5.3(c)). So, if $x \sim N_r(0, I_r)$, then $Gx \sim N_n(0, GG') = N_n(0, \Sigma)$ (Theorem 10.2).
- To obtain $y = Gx$, let $x = D^{-1/2}P_1'y$. The blocks of $P'\Sigma P = \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)$ are $P_1'\Sigma P_1 = D$, $P_1'\Sigma P_2 = 0$ and $P_2'\Sigma P_2 = 0$ (Theorem 5.3(b),(c)). By Theorem 10.2, $x \sim N_r(0, D^{-1/2}P_1'\Sigma P_1D^{-1/2}) = N_r(0, I_r)$.
- $Gx = P_1P_1'y = y - P_2P_2'y$, as $P_1P_1' + P_2P_2' = PP' = I_n$ (Theorem 5.3(c)). Finally $P_2'y \sim N_{n-r}(0, P_2'\Sigma P_2) = N_{n-r}(0, 0)$ (Theorem 10.2), the distribution of the constant 0 (Definition 10.1); so $y = Gx$ with probability 1. (If $r = n$, there is no P_2, and $y = Gx$.)

Problem 10.4 ([18, Theorems 4.9, 5.7(d), §7.1, (7.2)])**.** Let $x_1, \dots, x_n$ ($n \geqslant 2$) be independent random vectors, each $N_p(\mu, \Sigma)$, let X be the data matrix with the rows $x_1', \dots, x_n'$, and let $\bar{x}$ and S be its mean vector and sample covariance matrix (Definition 9.13).

(a) Show that $\operatorname{cov}(\bar{x},\ x_i - \bar{x}) = 0$ for every i.

(b) Show that $\bar{x} \sim N_p(\mu, \frac{1}{n}\Sigma)$ and that $\bar{x}$ and S are independent.

(c) Let Σ be PD. Show that $n(\bar{x} - \mu)'\Sigma^{-1}(\bar{x} - \mu) \sim \chi^2_p$, independently of S.

Solution. (a) Let $u = (x_1', \dots, x_n')' \in \mathbb{R}^{np}$. By Theorem 10.9(a), $u \sim N_{np}(\mu_u, V)$ with $\mu_u = (\mu', \dots, \mu')'$ and $V = \operatorname{diag}(\Sigma, \dots, \Sigma)$. Let $G = [I_p\ \cdots\ I_p] \in \mathbb{R}^{p\times np}$, and let $E_i \in \mathbb{R}^{p\times np}$ have the block I_p in position i and zero blocks elsewhere; so $\bar{x} = \frac{1}{n}Gu$ (Definition 9.13: $\bar{x} = \frac{1}{n}X'\mathbf{1} = \frac{1}{n}(x_1 + \cdots + x_n)$) and $x_i = E_iu$. By the block product (Theorem 5.3(b),(c)), $GVE_i' = \Sigma$ and $GVG' = n\Sigma$. By Theorem 9.4(a),

$$\operatorname{cov}\big(\bar{x},\ x_i - \bar{x}\big) = \frac{1}{n}GV\left(E_i - \frac{1}{n}G\right)' = \frac{1}{n}\Sigma - \frac{1}{n^2}n\Sigma = 0$$

(b) $G\mu_u = n\mu$, so $\bar{x} = \frac{1}{n}Gu \sim N_p(\mu, \frac{1}{n^2}GVG') = N_p(\mu, \frac{1}{n}\Sigma)$ by Theorem 10.2. Let $K \in \mathbb{R}^{np\times np}$ have the block rows $E_1 - \frac{1}{n}G, \dots, E_n - \frac{1}{n}G$, so that $Ku = \big((x_1 - \bar{x})', \dots, (x_n - \bar{x})'\big)'$. The matrix $\frac{1}{n}GVK'$ has the blocks $\frac{1}{n}GV(E_i - \frac{1}{n}G)'$, which are 0 by (a); so $\bar{x}$ and Ku are independent by Theorem 10.5(c), with $A = \frac{1}{n}G$ and $B = K$. As $S = \frac{1}{n-1}\sum_i(x_i - \bar{x})(x_i - \bar{x})'$ (Proposition 9.14(a)) is a function of Ku, $\bar{x}$ and S are independent (N3).

(c) $\frac{1}{n}\Sigma$ is PD ($y'(\frac{1}{n}\Sigma)y = \frac{1}{n}y'\Sigma y$), with the inverse $n\Sigma^{-1}$, so Corollary 10.11(a) applied to $\bar{x} \sim N_p(\mu, \frac{1}{n}\Sigma)$ gives $(\bar{x} - \mu)'(n\Sigma^{-1})(\bar{x} - \mu) \sim \chi^2_p$. The statistic is a function of $\bar{x}$, hence independent of S by (b) and (N3).

Chapter 11. Quadratic forms in normal vectors: chi-square, t, and F

The chi-square distribution

This chapter uses the facts (N1)–(N7) stated after Definition 10.1, the facts on densities stated in the paragraph before Theorem 10.8, and marginal densities as in Definition 10.12. In addition, it uses the following facts without proof: (D1) and (D2) from calculus, (D3) from probability [17, (4.3.2), (4.6.7), §3.3, Theorem 1.5.10]. Integrals over unbounded intervals are improper integrals, and a function g is **integrable** over a set Ω if $\int_\Omega |g| < \infty$.

(D1) (**Change of variables**) Let $\Omega, \Omega' \subseteq \mathbb{R}^m$ be open, and let $\psi\colon \Omega' \to \Omega$ be one-to-one and onto, with continuous partial derivatives and Jacobian determinant $J(v) = \det\bigl(\partial\psi_i(v)/\partial v_j\bigr) \neq 0$ for $v \in \Omega'$. Then $\int_\Omega g(x)\,\mathrm{d}x = \int_{\Omega'} g(\psi(v))|J(v)|\,\mathrm{d}v$ for every g integrable over Ω. Consequently, if u has a density f that is 0 outside Ω, then $\psi^{-1}(u)$ has the density $f(\psi(v))|J(v)|$ for $v \in \Omega'$, and 0 outside Ω'. For $m = 1$ this is the substitution rule of calculus; for an affine map $\psi(v) = c + Bv$ of $\mathbb{R}^m$ onto $\mathbb{R}^m$ it is the change of variables before Theorem 10.8.

(D2) (**The Gamma function**) For $c > 0$, $\Gamma(c) = \int_0^\infty x^{c-1}e^{-x}\,\mathrm{d}x$ is finite and positive, $\Gamma(c+1) = c\,\Gamma(c)$ and $\Gamma(1) = 1$. Substituting $x = by$ (D1) gives $\int_0^\infty y^{c-1}e^{-by}\,\mathrm{d}y = \Gamma(c)\,b^{-c}$ for every $b > 0$. Substituting $x = s^2/2$ ($s > 0$) gives $\Gamma(\frac{1}{2}) = \sqrt{2}\int_0^\infty e^{-s^2/2}\,\mathrm{d}s = \sqrt{\pi}$, as the integrand is even and $\int_{\mathbb{R}} e^{-s^2/2}\,\mathrm{d}s = \sqrt{2\pi}$ (the paragraph before Theorem 10.8).

(D3) (**Distribution functions**) If random variables u and v satisfy $\mathsf{P}(u \leqslant c) = \mathsf{P}(v \leqslant c)$ for every $c \in \mathbb{R}$, then u and v have the same distribution.

By Theorem 10.8 (the case $n = 1$), $N(0,1)$ has the density $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$. An identity between MGFs (defined before Definition 10.1) is needed only **near** 0, that is, for all t in some open interval (or ball) containing 0, as in (N1) and (N2).

Lemma 11.1 (cf. [17, Example 2.3.8])**.** (a) Let $w \sim N(0,1)$ and $\delta \in \mathbb{R}$. For every $t < \frac{1}{2}$,

$$\mathsf{E}\Bigl(e^{t(w+\delta)^2}\Bigr) = (1-2t)^{-1/2}\exp\Bigl(\frac{\delta^2 t}{1-2t}\Bigr)$$

(b) Let $w \sim N_r(0, I_r)$ and $\delta = (\delta_1, \ldots, \delta_r)' \in \mathbb{R}^r$, and put $\lambda = \|\delta\|^2 = \delta_1^2 + \cdots + \delta_r^2$. Then, near 0,

$$M_{\|w+\delta\|^2}(t) = (1-2t)^{-r/2}\exp\Bigl(\frac{\lambda t}{1-2t}\Bigr)$$

Hence the distribution of $\|w+\delta\|^2$ depends on δ only through λ.

Proof. (a)

- w has the density φ, so the expectation is $\int_{\mathbb{R}} e^{t(x+\delta)^2}\varphi(x)\,\mathrm{d}x$ (the paragraph before Theorem 10.8), provided this integral is finite.
- Put $c = 1 - 2t > 0$ and $m = 2t\delta/c$. Then $t(x+\delta)^2 - \frac{1}{2}x^2 = -\frac{c}{2}x^2 + 2t\delta x + t\delta^2 = -\frac{c}{2}(x-m)^2 + \frac{t\delta^2}{c}$, since $\frac{c}{2}m^2 = \frac{2t^2\delta^2}{c}$ and $t\delta^2 + \frac{2t^2\delta^2}{c} = \frac{t\delta^2(c+2t)}{c} = \frac{t\delta^2}{c}$.
- Substituting $x = m + y/\sqrt{c}$ (D1), $\int_{\mathbb{R}} e^{-c(x-m)^2/2}\,\mathrm{d}x = c^{-1/2}\int_{\mathbb{R}} e^{-y^2/2}\,\mathrm{d}y = c^{-1/2}\sqrt{2\pi}$, as φ integrates to 1 (Theorem 10.8(a)).
- Hence $\mathsf{E}\Bigl(e^{t(w+\delta)^2}\Bigr) = (2\pi)^{-1/2}e^{t\delta^2/c}\,c^{-1/2}\sqrt{2\pi} = c^{-1/2}e^{t\delta^2/c}$.

(b)

- By Theorem 10.9(a), $w_1, \dots, w_r$ are independent $N(0,1)$ variables; so the variables $u_i = (w_i + \delta_i)^2$ are independent (N3), and their MGFs are finite for $t < \frac{1}{2}$, by (a).
- $M_{u_1+\cdots+u_r}(t) = \mathsf{E}\big(e^{tu_1+\cdots+tu_r}\big)$ is the joint MGF of $(u_1, \dots, u_r)'$ at $(t, \dots, t)'$. By (N2) and (a), near 0 it equals $\prod_{i=1}^r M_{u_i}(t) = \prod_{i=1}^r (1-2t)^{-1/2} \exp\big(\delta_i^2 t/(1-2t)\big)$, the stated expression.
- If also $\|\delta^*\|^2 = \lambda$, then $\|w + \delta^*\|^2$ has the same MGF near 0, hence the same distribution (N1). □

Definition 11.2 ([12, §4.1], [9, Definition P.115])**.** Let $r \geqslant 1$, and let $w_1, \dots, w_r$ be independent $N(0,1)$ variables; equivalently, $w = (w_1, \dots, w_r)' \sim N_r(0, I_r)$ (Theorem 10.9(a)).

(a) The distribution of $w_1^2 + \cdots + w_r^2 = \|w\|^2$ is the **chi-square distribution** χ_r^2 with r **degrees of freedom**, as in (N5).

(b) For $\delta \in \mathbb{R}^r$, the distribution of $(w_1 + \delta_1)^2 + \cdots + (w_r + \delta_r)^2 = \|w + \delta\|^2$ is the **noncentral chi-square distribution** $\chi_r^2(\lambda)$ with r degrees of freedom and **noncentrality parameter** $\lambda = \delta_1^2 + \cdots + \delta_r^2$. By Lemma 11.1(b) it depends only on r and λ; every $\lambda \geqslant 0$ occurs (take $\delta_1 = \sqrt{\lambda}$ and $\delta_i = 0$ for $i \geqslant 2$), and $\chi_r^2(0) = \chi_r^2$.

We write $u \sim \chi_r^2(\lambda)$ if u has this distribution.

Lemma 11.3 ([9, Lemma P.117])**.** If $w \sim N(0,1)$, then $\mathsf{E}(w) = \mathsf{E}(w^3) = 0$, $\mathsf{E}(w^2) = 1$ and $\mathsf{E}(w^4) = 3$.

Proof. • $N(0,1)$ has mean 0 and variance 1 (Definition 10.1), so $\mathsf{E}(w) = 0$ and $\mathsf{E}(w^2) = \mathrm{var}(w) = 1$.

- $|x|^k \varphi(x)$ is integrable for $k \leqslant 4$, since $x^k e^{-x^2/4}$ is bounded; so $\mathsf{E}(w^k) = \int_{\mathbb{R}} x^k \varphi(x)\,\mathrm{d}x$ (the paragraph before Theorem 10.8). For $k = 3$ the integrand is odd, and $\mathsf{E}(w^3) = 0$.
- $\varphi'(x) = -x\varphi(x)$, so, integrating by parts on $[-R, R]$,

$$\int_{-R}^{R} x^4 \varphi(x)\,\mathrm{d}x = -\int_{-R}^{R} x^3 \varphi'(x)\,\mathrm{d}x = -2R^3\varphi(R) + 3\int_{-R}^{R} x^2 \varphi(x)\,\mathrm{d}x$$

- Let $R \to \infty$. From the exponential series, $e^{R^2/2} \geqslant \frac{1}{2}(R^2/2)^2 = R^4/8$, so $0 \leqslant R^3\varphi(R) \leqslant 8(2\pi)^{-1/2}/R \to 0$, while the integrals tend to $\mathsf{E}(w^4)$ and $\mathsf{E}(w^2) = 1$. Hence $\mathsf{E}(w^4) = 3$. □

Theorem 11.4 ([17, Lemma 5.3.2, Example 2.3.8], [9, Proposition P.118], [2, Corollary 10.2])**.** Let $r, s \geqslant 1$ and $\lambda, \kappa \geqslant 0$.

(a) If $u \sim \chi_r^2(\lambda)$, then $M_u(t) = (1-2t)^{-r/2} \exp\big(\lambda t/(1-2t)\big)$ near 0; conversely, a random variable with this MGF near 0 has the distribution $\chi_r^2(\lambda)$.

(b) χ_r^2 has the density

$$f_r(x) = \frac{x^{r/2-1} e^{-x/2}}{\Gamma(r/2)\, 2^{r/2}} \quad (x > 0), \qquad f_r(x) = 0 \quad (x \leqslant 0)$$

In particular $f_1(x) = (2\pi x)^{-1/2} e^{-x/2}$ and $f_2(x) = \frac{1}{2} e^{-x/2}$ for $x > 0$.

(c) If $u \sim \chi_r^2(\lambda)$ and $v \sim \chi_s^2(\kappa)$ are independent, then $u + v \sim \chi_{r+s}^2(\lambda + \kappa)$.

(d) If $u \sim \chi_r^2(\lambda)$, then $\mathsf{E}(u) = r + \lambda$ and $\mathrm{var}(u) = 2r + 4\lambda$.

Proof. (a) u has the distribution of $\|w + \delta\|^2$ in Definition 11.2(b), whose MGF is given by Lemma 11.1(b). The converse is (N1).

(b)

- For $t < \frac{1}{2}$, by (D2) with $c = r/2$ and $b = (1-2t)/2$,

$$\int_0^\infty e^{tx} f_r(x)\,\mathrm{d}x = \frac{1}{\Gamma(r/2)\,2^{r/2}} \int_0^\infty x^{r/2-1} e^{-(1-2t)x/2}\,\mathrm{d}x = \frac{1}{2^{r/2}} \Big(\frac{2}{1-2t}\Big)^{r/2} = (1-2t)^{-r/2}$$

- For $t = 0$ this says $\int f_r = 1$. So $f_r \geqslant 0$ is the density of some random variable U, and $M_U(t) = \int_0^\infty e^{tx} f_r(x)\,\mathrm{d}x = (1-2t)^{-r/2}$ for $t < \frac{1}{2}$ (the paragraph before Theorem 10.8). By (a) with $\lambda = 0$, $U \sim \chi^2_r$; so χ^2_r has the density f_r.
- $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(1) = 1$ (D2) give f_1 and f_2.

(c) $M_{u+v}(t) = \mathsf{E}(e^{tu+tv})$ is the joint MGF of $(u, v)'$ at $(t, t)'$. By (N2) and (a), near 0,

$$M_{u+v}(t) = M_u(t) M_v(t) = (1-2t)^{-(r+s)/2} \exp\Big(\frac{(\lambda+\kappa)t}{1-2t}\Big)$$

so $u + v \sim \chi^2_{r+s}(\lambda + \kappa)$ by (a).

(d)

- The mean and the variance depend only on the distribution, so we may take $u = u_1 + \cdots + u_r$ with $u_i = (w_i + \delta_i)^2$ as in Definition 11.2(b).
- By Lemma 11.3 and the linearity of the expectation, $\mathsf{E}(u_i) = \mathsf{E}(w_i^2) + 2\delta_i\,\mathsf{E}(w_i) + \delta_i^2 = 1 + \delta_i^2$ and

$$\mathsf{E}(u_i^2) = \mathsf{E}(w_i^4) + 4\delta_i\,\mathsf{E}(w_i^3) + 6\delta_i^2\,\mathsf{E}(w_i^2) + 4\delta_i^3\,\mathsf{E}(w_i) + \delta_i^4 = 3 + 6\delta_i^2 + \delta_i^4$$

  so $\operatorname{var}(u_i) = \mathsf{E}(u_i^2) - \big(\mathsf{E}(u_i)\big)^2 = 2 + 4\delta_i^2$.
- The u_i are independent (N3), so $\operatorname{cov}(u_i, u_j) = 0$ for $i \neq j$ (N7). By Theorem 9.4(d), $\operatorname{var}(u) = \sum_i \operatorname{var}(u_i) = 2r + 4\lambda$; and $\mathsf{E}(u) = \sum_i (1 + \delta_i^2) = r + \lambda$. □

Example 11.5. (a) If $u \sim \chi^2_2$ and $c > 0$, then $\mathsf{P}(u > c) = \int_c^\infty \frac{1}{2} e^{-x/2}\,\mathrm{d}x = e^{-c/2}$ (Theorem 11.4(b)). So $\mathsf{P}(u > c) = \frac{1}{20}$ for $c = 2\log 20$.

(b) Let $z \sim N_4(\mu, I_4)$ with $\mu = (1, 2, 2, 4)'$. Then $w = z - \mu \sim N_4(0, I_4)$ (Theorem 10.2), and $\|z\|^2 = \|w + \mu\|^2 \sim \chi^2_4(25)$, as $\lambda = 1 + 4 + 4 + 16 = 25$. By Theorem 11.4(d), $\mathsf{E}\|z\|^2 = 4 + 25 = 29$ and $\operatorname{var}\|z\|^2 = 8 + 100 = 108$. Likewise $z_1^2 \sim \chi^2_1(1)$ and $z_2^2 + z_3^2 + z_4^2 \sim \chi^2_3(24)$; these are independent by (N3), since z_1 and $(z_2, z_3, z_4)'$ are independent (Theorem 10.5(a): the off-diagonal blocks of I_4 are 0); and their sum is $\chi^2_4(25)$, as Theorem 11.4(c) says.

The t and F distributions

Theorem 11.6 ([17, §5.3.2, (5.3.6), Exercise 5.17(a)]). (a) Let $Z \sim N(0,1)$ and $V \sim \chi^2_\nu$ ($\nu \geqslant 1$) be independent. Then $V > 0$ with probability 1, and $T = Z/\sqrt{V/\nu}$ has the density

$$f_T(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\nu\pi}}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2} \qquad (x \in \mathbb{R})$$

(b) Let $U \sim \chi^2_a$ and $V \sim \chi^2_b$ ($a, b \geqslant 1$) be independent. Then $F = (U/a)/(V/b)$ has the density

$$f_F(x) = \frac{\Gamma\left(\frac{a+b}{2}\right)}{\Gamma\left(\frac{a}{2}\right)\Gamma\left(\frac{b}{2}\right)}\left(\frac{a}{b}\right)^{a/2}\frac{x^{a/2-1}}{(1+ax/b)^{(a+b)/2}} \quad (x>0), \qquad f_F(x) = 0 \quad (x \leqslant 0)$$

and F and $U+V$ are independent.

Proof. (a)

- $f_\nu = 0$ on $(-\infty, 0]$ (Theorem 11.4(b)), so $\mathsf{P}(V \leqslant 0) = \int_{-\infty}^0 f_\nu = 0$. As Z and V are independent, $(Z,V)'$ has the density $\varphi(z)f_\nu(v)$ (the paragraph before Theorem 10.8), which is 0 outside $\Omega = \mathbb{R} \times (0,\infty)$.
- The map $\psi(t,w) = \left(t\sqrt{w/\nu},\, w\right)$ of Ω onto itself is one-to-one, with inverse $(z,v) \mapsto \left(z/\sqrt{v/\nu},\, v\right)$, and
$$J(t,w) = \det\begin{pmatrix}\sqrt{w/\nu} & t/(2\sqrt{\nu w}) \\ 0 & 1\end{pmatrix} = \sqrt{w/\nu} > 0$$
By (D1), $(T,V) = \psi^{-1}(Z,V)$ has the density $\varphi\left(t\sqrt{w/\nu}\right)f_\nu(w)\sqrt{w/\nu}$ on Ω.
- So T has the marginal density (Definition 10.12)
$$f_T(t) = \int_0^\infty \frac{e^{-t^2w/(2\nu)}}{\sqrt{2\pi}} \cdot \frac{w^{\nu/2-1}e^{-w/2}}{\Gamma(\nu/2)\,2^{\nu/2}} \cdot \frac{w^{1/2}}{\nu^{1/2}}\,\mathrm{d}w = \frac{1}{\sqrt{2\pi}\,\Gamma(\nu/2)\,2^{\nu/2}\nu^{1/2}}\int_0^\infty w^{\frac{\nu+1}{2}-1}e^{-(1+t^2/\nu)w/2}\,\mathrm{d}w$$
- By (D2) with $c = \frac{\nu+1}{2}$ and $b = \frac{1}{2}(1+t^2/\nu)$, the last integral is $\Gamma\left(\frac{\nu+1}{2}\right)2^{(\nu+1)/2}(1+t^2/\nu)^{-(\nu+1)/2}$; and $2^{(\nu+1)/2}/(\sqrt{2\pi}\,2^{\nu/2}) = 1/\sqrt{\pi}$. This is $f_T(t)$.

(b) Put $k = a/b$.

- As U and V are independent, $(U,V)'$ has the density $f_a(u)f_b(v)$ (the paragraph before Theorem 10.8), which is 0 outside $\Omega = (0,\infty)^2$.
- The map $\psi(x,y) = \left(kxy/(1+kx),\ y/(1+kx)\right)$ of Ω onto itself is one-to-one, with inverse $(u,v) \mapsto \left((u/a)/(v/b),\ u+v\right)$: for $(u,v) = \psi(x,y)$, $u/v = kx$ and $u+v = y$. Its Jacobian determinant is
$$J(x,y) = \det\begin{pmatrix} ky/(1+kx)^2 & kx/(1+kx) \\ -ky/(1+kx)^2 & 1/(1+kx)\end{pmatrix} = \frac{ky(1+kx)}{(1+kx)^3} = \frac{ky}{(1+kx)^2} > 0$$
- By (D1), $(F, U+V) = \psi^{-1}(U,V)$ has, on Ω, the density
$$h(x,y) = f_a\left(\frac{kxy}{1+kx}\right)f_b\left(\frac{y}{1+kx}\right)\frac{ky}{(1+kx)^2} = \frac{k^{a/2}x^{a/2-1}y^{(a+b)/2-1}e^{-y/2}}{\Gamma(a/2)\Gamma(b/2)\,2^{(a+b)/2}(1+kx)^{(a+b)/2}}$$
collecting powers: y has the exponent $(\frac{a}{2}-1)+(\frac{b}{2}-1)+1$, and $1+kx$ the exponent $-(\frac{a}{2}-1)-(\frac{b}{2}-1)-2$.

- So $h(x,y) = f_F(x)f_{a+b}(y)$ with $f_F \geqslant 0$ as stated. As h is a density and $\int f_{a+b} = 1$, the rule for products of integrals before Theorem 10.8 gives $1 = \iint h = \int f_F$. So F has the marginal density $\int_0^\infty h(x,y)\,\mathrm{d}y = f_F(x)$ (Definition 10.12), and $U+V \sim \chi^2_{a+b}$ has the density f_{a+b} (Theorem 11.4(c),(b)). As $(F, U+V)'$ has the product density $f_F(x)f_{a+b}(y)$, F and $U+V$ are independent (the paragraph before Theorem 10.8). □

Definition 11.7 ([12, §4.1], [9, Definition P.115], [17, Definitions 5.3.4, 5.3.6])**.** Let $\nu, a, b \geqslant 1$.

(a) If $Z \sim N(0,1)$ and $V \sim \chi^2_\nu$ are independent, the distribution of $T = Z/\sqrt{V/\nu}$ is the t **distribution** t_ν with ν degrees of freedom.

(b) If $U \sim \chi^2_a$ and $V \sim \chi^2_b$ are independent, the distribution of $F = (U/a)/(V/b)$ is the F **distribution** $F_{a,b}$ with a and b degrees of freedom.

(c) For $\tau \in (0,1)$, the τ **quantiles** $t_{\nu,\tau}$, $F_{a,b,\tau}$ and $\chi^2_{\nu,\tau}$ are the numbers with $\mathsf{P}(T \leqslant t_{\nu,\tau}) = \tau$ for $T \sim t_\nu$, $\mathsf{P}(F \leqslant F_{a,b,\tau}) = \tau$ for $F \sim F_{a,b}$, and $\mathsf{P}(u \leqslant \chi^2_{\nu,\tau}) = \tau$ for $u \sim \chi^2_\nu$.

By Theorem 11.6, these distributions have the densities f_T and f_F, so they do not depend on the choice of Z, U and V. Such variables exist: if $g \sim N_{\nu+1}(0, I_{\nu+1})$ (Theorem 10.8(b)), then $Z = g_1 \sim N(0,1)$ and $V = g_2^2 + \cdots + g_{\nu+1}^2 \sim \chi^2_\nu$ (Theorem 10.9(a), Definition 11.2) are independent (Theorem 10.5(a), (N3)); U and V are obtained in the same way from $g \sim N_{a+b}(0, I_{a+b})$. The quantiles exist and are unique: $G(x) = \mathsf{P}(T \leqslant x) = \int_{-\infty}^x f_T$ is continuous; it is strictly increasing on $\mathbb{R}$, as f_T is continuous and positive there; and $G(x) \to 0$ as $x \to -\infty$, $G(x) \to 1$ as $x \to \infty$, as $\int f_T = 1$. So G takes the value τ exactly once (intermediate value theorem). The same holds for F, and for $u \sim \chi^2_\nu$, whose density f_ν (Theorem 11.4(b)) is continuous and positive on $(0,\infty)$, with $(0,\infty)$ in place of $\mathbb{R}$ and $G(0) = 0$. Hence, for $\alpha \in (0,1)$, $\mathsf{P}(T > t_{\nu,1-\alpha}) = \alpha$, $\mathsf{P}(F > F_{a,b,1-\alpha}) = \alpha$ and $\mathsf{P}(u > \chi^2_{\nu,1-\alpha}) = \alpha$.

Theorem 11.8 ([12, §4.1], [9, Corollary P.116], [17, Theorem 5.3.8(b)])**.** Let $T \sim t_\nu$ and $\alpha \in (0,1)$.

(a) $-T \sim t_\nu$: the t distribution is symmetric about 0. Hence $\mathsf{P}(T < -c) = \mathsf{P}(T > c)$ for every $c \in \mathbb{R}$, and

$$\mathsf{P}(T < -t_{\nu,1-\alpha}) = \alpha, \qquad \mathsf{P}\Big(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}\Big) = 1-\alpha$$

(b) $T^2 \sim F_{1,\nu}$, and $t^2_{\nu,1-\alpha/2} = F_{1,\nu,1-\alpha}$.

Proof. Write $T = Z/\sqrt{V/\nu}$ as in Definition 11.7.

(a)

- $-Z \sim N(0,1)$ (Theorem 10.2 with $A = -1$), and $-Z$ is independent of V (N3). So $-T = (-Z)/\sqrt{V/\nu} \sim t_\nu$ (Definition 11.7), and $\mathsf{P}(T < -c) = \mathsf{P}(-T > c) = \mathsf{P}(T > c)$.
- For $c = t_{\nu,1-\alpha}$, $\mathsf{P}(T > c) = \alpha$ (Definition 11.7); this gives the first identity.
- With $\alpha/2$ in place of α and $c = t_{\nu,1-\alpha/2}$, $\mathsf{P}(-c \leqslant T \leqslant c) = 1 - \mathsf{P}(T < -c) - \mathsf{P}(T > c) = 1 - \frac{\alpha}{2} - \frac{\alpha}{2}$.

(b)

- $T^2 = (Z^2/1)/(V/\nu)$, where $Z^2 \sim \chi^2_1$ (Definition 11.2) is independent of V (N3); so $T^2 \sim F_{1,\nu}$.
- By (a), $\mathsf{P}(T < 0) = \mathsf{P}(T > 0)$, and $\mathsf{P}(T = 0) = 0$, as T has a density; so $\mathsf{P}(T \leqslant 0) = \frac{1}{2} < 1 - \frac{\alpha}{2}$, and $c = t_{\nu,1-\alpha/2} > 0$, as G in Definition 11.7 is increasing. Then $\mathsf{P}(T^2 \leqslant c^2) = \mathsf{P}(-c \leqslant T \leqslant c) = 1-\alpha$ by (a), so c^2 is the $(1-\alpha)$ quantile of $F_{1,\nu}$, which is unique (Definition 11.7). □

Proposition 11.9 ([17, Theorem 5.3.8(a), Exercise 5.17(c)])**.** If $F \sim F_{a,b}$, then $1/F \sim F_{b,a}$. Hence $F_{b,a,\alpha} = 1/F_{a,b,1-\alpha}$ for $\alpha \in (0,1)$.

Proof. • Write $F = (U/a)/(V/b)$ as in Definition 11.7. $U, V > 0$ with probability 1 (as in the proof of Theorem 11.6(a)), and $1/F = (V/b)/(U/a)$ with $V \sim \chi^2_b$ and $U \sim \chi^2_a$ independent; so $1/F \sim F_{b,a}$.

- Let $c = F_{a,b,1-\alpha}$; $c > 0$ (Definition 11.7). As F has a density, $\mathsf{P}(F = c) = 0$, and $\mathsf{P}(1/F \leqslant 1/c) = \mathsf{P}(F \geqslant c) = \mathsf{P}(F > c) = \alpha$; so $1/c$ is the α quantile of $F_{b,a}$. □

Example 11.10 ([18, §4.4, (4.51), Theorem 4.7])**.** Let $\alpha \in (0, 1)$.

- For $T \sim t_4$, Theorem 11.8(a), with $\alpha/2$ in place of α, gives $\mathsf{P}(T < -t_{4,1-\alpha/2}) = \frac{\alpha}{2}$; as T has a density, also $\mathsf{P}(T \leqslant -t_{4,1-\alpha/2}) = \frac{\alpha}{2}$, so $t_{4,\alpha/2} = -t_{4,1-\alpha/2}$ (Definition 11.7(c)), and $\mathsf{P}(|T| \leqslant t_{4,1-\alpha/2}) = 1 - \alpha$. By Theorem 11.8(b), $F_{1,4,1-\alpha} = t^2_{4,1-\alpha/2}$; by Proposition 11.9, $F_{4,3,\alpha} = 1/F_{3,4,1-\alpha}$.
- Let $u \sim \chi^2_2$ and $\tau \in (0, 1)$. For $c > 0$, $\mathsf{P}(u \leqslant c) = 1 - \mathsf{P}(u > c) = 1 - e^{-c/2}$ (Example 11.5(a)), and $1 - e^{-c/2} = \tau$ exactly for $c = -2\log(1-\tau)$, which is > 0. So $\chi^2_{2,\tau} = -2\log(1-\tau)$ (Definition 11.7(c)); in particular $\chi^2_{2,1-\alpha} = 2\log(1/\alpha)$, and $\chi^2_{2,19/20} = 2\log 20$.
- Let $x \sim N_n(\mu, \Sigma)$ with Σ PD. By Corollary 10.11(a), $d_\Sigma(x, \mu)^2 = (x - \mu)'\Sigma^{-1}(x - \mu) \sim \chi^2_n$, with the distance d_Σ of the paragraph after Theorem 10.8. So the region $\{u \in \mathbb{R}^n : d_\Sigma(u, \mu)^2 \leqslant \chi^2_{n,1-\alpha}\}$, bounded by the contour of that paragraph with $c^2 = \chi^2_{n,1-\alpha}$, contains x with probability $1 - \alpha$. For $n = 2$ it is bounded by an ellipse, with $c^2 = 2\log(1/\alpha)$.

One normal sample

Theorem 11.11 ([17, Theorem 5.3.1], [2, Examples 10.2, 10.3])**.** Let $n \geqslant 2$, and let $x_1, \ldots, x_n$ be independent $N(\mu, \sigma^2)$ variables ($\sigma > 0$), with mean $\bar{x} = \frac{1}{n}\sum_i x_i$ and sample variance $s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2$ (Definition 9.13). Then:

(a) $\bar{x} \sim N(\mu, \sigma^2/n)$;

(b) $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$;

(c) $\bar{x}$ and s^2 are independent;

(d) $\sqrt{n}(\bar{x} - \mu)/s \sim t_{n-1}$.

Proof. Let $x = (x_1, \ldots, x_n)'$. Then $x \sim N_n(\mu\mathbf{1}, \sigma^2 I_n)$ (Theorem 10.9(a)), the deviations are $C_n x = (x_1 - \bar{x}, \ldots, x_n - \bar{x})'$ (Example 1.16), and $(n-1)s^2 = (C_n x)'(C_n x)$.

(a) This is Example 10.10(a), applied to x.

(b)

- Let $z = (x - \mu\mathbf{1})/\sigma$; then $z \sim N_n(0, I_n)$ (Theorem 10.2).
- C_n is symmetric with the eigenvalues 1 ($n - 1$ times) and 0 (once) (Example 8.16(a)). By Theorem 8.3, there is an orthogonal P with $P'C_nP = D = \mathrm{diag}(1, \ldots, 1, 0)$, so $C_n = PDP'$. (For instance $P = [p_2 \ \cdots \ p_n \ p_1]$, with the rows p_k' of the Helmert matrix K_n of Example 6.20.)
- Let $y = P'z$. Then $y \sim N_n(0, P'P) = N_n(0, I_n)$ (Theorem 10.2), so $y_1, \ldots, y_n$ are independent $N(0, 1)$ variables (Theorem 10.9(a)).
- $C_n\mathbf{1} = 0$ and $C_n'C_n = C_n$ (Proposition 1.15), so $C_n x = \sigma C_n z$ and

$$(n-1)s^2 = (C_n x)'(C_n x) = \sigma^2 z' C_n z = \sigma^2 z' PDP' z = \sigma^2 y' D y = \sigma^2(y_1^2 + \cdots + y_{n-1}^2)$$

  So $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ (Definition 11.2).

(c) By Example 10.10(b), applied to x, $\bar{x}$ is independent of the deviations $C_n x$, and hence of $s^2 = \frac{1}{n-1}(C_n x)'(C_n x)$.

(d)

- By (a) and Theorem 10.2, $Z = \sqrt{n}(\bar{x} - \mu)/\sigma \sim N(0, 1)$; by (b), $V = (n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$; and Z and V are independent by (c) and (N3).

- V has a density (Theorem 11.4(b)), so $\mathsf{P}(s^2 = 0) = \mathsf{P}(V = 0) = 0$, and $s > 0$ with probability 1. Then
$$\frac{\sqrt{n}(\bar{x} - \mu)}{s} = \frac{\sqrt{n}(\bar{x} - \mu)/\sigma}{\sqrt{(n-1)s^2/\big((n-1)\sigma^2\big)}} = \frac{Z}{\sqrt{V/(n-1)}}$$
which has the distribution t_{n-1} (Definition 11.7). □

Definition 11.12 ([17, Definitions 9.1.1, 9.1.4, 9.1.5])**.** Let observations be modelled as random variables whose distribution depends on parameters, and let $\alpha \in (0, 1)$.

(a) An interval $[L, U]$ whose endpoints $L \leqslant U$ are computed from the observations is a **confidence interval** with level $1 - \alpha$ for a parameter $\theta \in \mathbb{R}$ if $\mathsf{P}(L \leqslant \theta \leqslant U) = 1 - \alpha$ for all values of the parameters.

(b) A set $C \subseteq \mathbb{R}^p$ computed from the observations is a **confidence region** with level $1 - \alpha$ for a parameter $\theta \in \mathbb{R}^p$ if $\mathsf{P}(\theta \in C) = 1 - \alpha$ for all values of the parameters.

Example 11.13 ([17, Example 9.2.10])**.** In Theorem 11.11, let $\alpha \in (0, 1)$, $T = \sqrt{n}(\bar{x} - \mu)/s$, $V = (n-1)s^2/\sigma^2$ and $c = t_{n-1,1-\alpha/2}$; here $s > 0$ with probability 1 (the proof of Theorem 11.11(d)).

- By Theorems 11.11(d) and 11.8(a), $\mathsf{P}(-c \leqslant T \leqslant c) = 1 - \alpha$ for all values of μ and σ. The inequality $-c \leqslant T \leqslant c$ holds exactly when $\bar{x} - cs/\sqrt{n} \leqslant \mu \leqslant \bar{x} + cs/\sqrt{n}$. So $\big[\bar{x} - t_{n-1,1-\alpha/2}\, s/\sqrt{n},\ \bar{x} + t_{n-1,1-\alpha/2}\, s/\sqrt{n}\big]$ is a confidence interval for μ with level $1 - \alpha$ (Definition 11.12(a)).
- Let $c_1 = \chi^2_{n-1,\alpha/2}$ and $c_2 = \chi^2_{n-1,1-\alpha/2}$ (Definition 11.7(c)). By Theorem 11.11(b), $V \sim \chi^2_{n-1}$, which has a density (Theorem 11.4(b)); so $\mathsf{P}(V = c_1) = 0$ and $\mathsf{P}(c_1 \leqslant V \leqslant c_2) = \mathsf{P}(V \leqslant c_2) - \mathsf{P}(V < c_1) = (1 - \frac{\alpha}{2}) - \frac{\alpha}{2} = 1 - \alpha$ for all values of μ and σ. Also $0 < c_1 < c_2$, as $\mathsf{P}(V \leqslant 0) = 0 < \frac{\alpha}{2} < 1 - \frac{\alpha}{2}$. As $c_1 \leqslant V \leqslant c_2$ holds exactly when $(n-1)s^2/c_2 \leqslant \sigma^2 \leqslant (n-1)s^2/c_1$, $\big[(n-1)s^2/\chi^2_{n-1,1-\alpha/2},\ (n-1)s^2/\chi^2_{n-1,\alpha/2}\big]$ is a confidence interval for σ^2 with level $1 - \alpha$.
- Five measurements $2, 4, 5, 6, 8$ are modelled as independent $N(\mu, \sigma^2)$ variables. Here $\bar{x} = 25/5 = 5$, the deviations are $-3, -1, 0, 1, 3$, and $s^2 = (9+1+0+1+9)/4 = 5$; so $s/\sqrt{5} = 1$ and $(n-1)s^2 = 20$. The two intervals are $\big[5 - t_{4,1-\alpha/2},\ 5 + t_{4,1-\alpha/2}\big]$ and $\big[20/\chi^2_{4,1-\alpha/2},\ 20/\chi^2_{4,\alpha/2}\big]$.
- If $\mu = 3$, then $\sqrt{5}(\bar{x} - 3)/s \sim t_4$; its value here is $\sqrt{5} \cdot 2/\sqrt{5} = 2$. So 3 lies in the interval for μ exactly when $t_{4,1-\alpha/2} \geqslant 2$.

Example 11.14 (cf. [18, §7.1, (7.2), Theorem 4.9])**.** Four dimensions of each of $n = 16$ machined parts are measured; the measurement vectors $x_1, \ldots, x_{16} \in \mathbb{R}^4$ are modelled as independent $N_4(\mu, \Sigma)$, with
$$\Sigma = 2I_4 + 2J_4 = \begin{pmatrix} 4 & 2 & 2 & 2 \\ 2 & 4 & 2 & 2 \\ 2 & 2 & 4 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}$$
known from long experience and μ unknown. So each measurement has the variance 4, and any two have the correlation $\frac{1}{2}$. Let $\alpha \in (0, 1)$.

- By Theorem 10.9(a) (the block form), $u = (x_1', \ldots, x_{16}')' \sim N_{64}\big((\mu', \ldots, \mu')', \operatorname{diag}(\Sigma, \ldots, \Sigma)\big)$, and $\bar{x} = \frac{1}{16}(x_1 + \cdots + x_{16}) = \frac{1}{16}Gu$ with $G = [I_4\ \cdots\ I_4] \in \mathbb{R}^{4\times 64}$. By Theorem 10.2 and the block product (Theorem 5.3(c)), as $G \operatorname{diag}(\Sigma, \ldots, \Sigma)G' = 16\Sigma$, $\bar{x} \sim N_4\big(\mu, \frac{1}{256} \cdot 16\Sigma\big) = N_4(\mu, \frac{1}{16}\Sigma)$. Each $\bar{x}_i$ has the standard deviation $\sqrt{4/16} = \frac{1}{2}$.
- Σ has the eigenvalues $2+8 = 10$ and 2 (three times) (Example 8.16(b)), so it is PD (Theorem 8.21(a)), and so is $\frac{1}{16}\Sigma$, with the inverse $16\Sigma^{-1}$. Here $\Sigma^{-1} = \frac{1}{2}(I_4 - \frac{1}{5}J_4)$, as $(I_4 + J_4)(I_4 - \frac{1}{5}J_4) = I_4 + J_4 - \frac{1}{5}J_4 - \frac{4}{5}J_4 = I_4$ ($J_4^2 = 4J_4$, Proposition 1.15). By Corollary 10.11(a), $q(\mu) = 16(\bar{x} - \mu)'\Sigma^{-1}(\bar{x} - \mu) \sim \chi^2_4$,

so $\mathsf{P}\big(q(\mu) \leqslant \chi^2_{4,1-\alpha}\big) = 1-\alpha$ for every μ (Definition 11.7(c)). Thus the ellipsoid $\{m \in \mathbb{R}^4 : 16(\bar{x}-m)'\Sigma^{-1}(\bar{x}-m) \leqslant \chi^2_{4,1-\alpha}\}$, computed from the observations, is a confidence region for μ with level $1-\alpha$ (Definition 11.12(b)).

- By Theorem 11.4(b), χ^2_4 has the density $f_4(x) = \frac{x}{4}e^{-x/2}$ for $x > 0$, as $\Gamma(2) = \Gamma(1) = 1$ (D2). For $w \sim \chi^2_4$ and $c > 0$, integrating by parts, as $xe^{-x/2} \to 0$ for $x \to \infty$,

$$\mathsf{P}(w > c) = \int_c^\infty \frac{x}{4}e^{-x/2}\,\mathrm{d}x = \Big[-\frac{x}{2}e^{-x/2}\Big]_c^\infty + \int_c^\infty \frac{1}{2}e^{-x/2}\,\mathrm{d}x = \Big(1+\frac{c}{2}\Big)e^{-c/2}$$

- Let $\alpha = \frac{1}{20}$, and compare two possible target values m_1, m_2 with $\bar{x}-m_1 = (1,1,1,1)'$ and $\bar{x}-m_2 = (1,-1,1,-1)'$. Each coordinate of $\bar{x}$ is two of its standard deviations away from the corresponding coordinate of m_1, and also of m_2. With $d = \bar{x}-m$, $q(m) = 16d'\Sigma^{-1}d = 8\Big(d'd - \frac{1}{5}(\mathbf{1}'d)^2\Big)$, so

$$q(m_1) = 8\Big(4-\frac{16}{5}\Big) = \frac{32}{5}, \qquad q(m_2) = 8(4-0) = 32$$

- From the exponential series, $2 < e < 3$ ($e = \sum_{k\geqslant 0} 1/k!$, and $1/k! \leqslant 1/2^{k-1}$ for $k \geqslant 1$, strictly for $k \geqslant 3$). So $\mathsf{P}(w > \frac{32}{5}) = \frac{21}{5}e^{-16/5} > \frac{1}{20}$, as $e^{16} < 3^{16} = 43046721 < 84^5$; and $\mathsf{P}(w > 32) = 17e^{-16} < \frac{1}{20}$, as $e^{16} > 2^{16} = 65536 > 340$. As $\mathsf{P}(w > c)$ is strictly decreasing in c (Definition 11.7) and $\mathsf{P}(w > \chi^2_{4,19/20}) = \frac{1}{20}$, $q(m_1) < \chi^2_{4,19/20} < q(m_2)$: m_1 lies in the confidence region and m_2 does not. The measurements are positively correlated, so deviations of the same sign are common and deviations of opposite signs are unusual; only the joint region takes this into account.

Quadratic forms in normal vectors

Theorem 11.15 ([12, Theorem 4.5], [3, Result 8.3], [2, Theorem 10.2])**.** Let $z \sim N_n(\mu, \sigma^2 I_n)$ ($\sigma > 0$), and let $A \in \mathbb{R}^{n\times n}$ be symmetric and idempotent with $\operatorname{rank} A = r \geqslant 1$. Then

$$\frac{z'Az}{\sigma^2} \sim \chi^2_r\Big(\frac{\mu'A\mu}{\sigma^2}\Big)$$

In particular $z'Az/\sigma^2 \sim \chi^2_r$ if $A\mu = 0$.

Proof. • By Theorem 8.13(c), $A = P_1P_1'$ for some $P_1 \in \mathbb{R}^{n\times r}$ with $P_1'P_1 = I_r$.

- Let $u = P_1'z/\sigma$ and $\delta = P_1'\mu/\sigma$. By Theorem 10.2, $u \sim N_r\big(\delta,\ P_1'(\sigma^2 I_n)P_1/\sigma^2\big) = N_r(\delta, I_r)$, and $w = u-\delta \sim N_r(0, I_r)$.
- $z'Az/\sigma^2 = z'P_1P_1'z/\sigma^2 = \|u\|^2 = \|w+\delta\|^2$. By Definition 11.2, this has the distribution $\chi^2_r(\lambda)$ with $\lambda = \|\delta\|^2 = \mu'P_1P_1'\mu/\sigma^2 = \mu'A\mu/\sigma^2$.
- If $A\mu = 0$, then $\lambda = 0$. □

Theorem 11.16 ([2, Theorem 10.1])**.** Let $y \sim N_n(\mu, \Sigma)$ with Σ positive definite, let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $B = \Sigma^{1/2}A\Sigma^{1/2}$ (Theorem 8.46). Let $\gamma_1, \dots, \gamma_k$ be the distinct nonzero eigenvalues of B, with multiplicities $\nu_1, \dots, \nu_k$, and let the columns of $P_i \in \mathbb{R}^{n\times\nu_i}$ be orthonormal eigenvectors of B for γ_i. Then

$$y'Ay = \gamma_1 W_1 + \cdots + \gamma_k W_k, \qquad W_i = \|P_i'\Sigma^{-1/2}y\|^2$$

where $W_1, \dots, W_k$ are independent and $W_i \sim \chi^2_{\nu_i}(\theta_i)$ with $\theta_i = \mu'\Sigma^{-1/2}P_iP_i'\Sigma^{-1/2}\mu$. The γ_i and ν_i are also the distinct nonzero eigenvalues of $A\Sigma$ and their multiplicities.

Proof.
- $\Sigma^{1/2}$ is symmetric and invertible, with $\Sigma^{1/2}\Sigma^{1/2} = \Sigma$ and inverse $\Sigma^{-1/2}$ (Theorem 8.46). Let $z = \Sigma^{-1/2}y$ and $c = \Sigma^{-1/2}\mu$. By Theorem 10.2, $z \sim N_n(c, \Sigma^{-1/2}\Sigma\Sigma^{-1/2}) = N_n(c, I_n)$, and $y'Ay = z'\Sigma^{1/2}A\Sigma^{1/2}z = z'Bz$.
- B is symmetric. The ν_i columns of P_i are orthonormal vectors of the eigenspace $\mathcal{N}(B-\gamma_i I)$, whose dimension is ν_i (Theorem 8.4); so they form an orthonormal basis of it, and P_iP_i' is its projection matrix (Theorem 6.31(b)). By Theorem 8.6(b), with the eigenvalue 0, if it occurs, contributing nothing, $B = \sum_i \gamma_i P_iP_i'$, and $z'Bz = \sum_i \gamma_i\|P_i'z\|^2 = \sum_i \gamma_i W_i$.
- The $m = \nu_1 + \cdots + \nu_k$ columns of $P = [P_1 \ \cdots \ P_k]$ are orthonormal (Theorem 8.2 for different eigenvalues), so $u = P'z \sim N_m(P'c, P'P) = N_m(P'c, I_m)$ (Theorem 10.2). Its blocks are $u_i = P_i'z \sim N_{\nu_i}(P_i'c, I_{\nu_i})$, and the off-diagonal blocks of I_m are 0; so $u_1, \dots, u_k$ are independent (Theorem 10.5(b)).
- By Definition 11.2, $W_i = \|u_i\|^2 = \|(u_i - P_i'c) + P_i'c\|^2 \sim \chi^2_{\nu_i}(\theta_i)$ with $\theta_i = \|P_i'c\|^2 = c'P_iP_i'c$; and the W_i are independent (N3).
- $B = \Sigma^{1/2}(A\Sigma)\Sigma^{-1/2}$ is similar to $A\Sigma$, so the two matrices have the same characteristic polynomial (Theorem 7.15), hence the same eigenvalues with the same multiplicities. □

Theorem 11.17 ([3, Result 8.3])**.** Let $z \sim N_n(0, I_n)$, let $A \in \mathbb{R}^{n\times n}$ be symmetric, and let $r \geqslant 1$. If $z'Az \sim \chi^2_r$, then A is idempotent and $\operatorname{rank} A = r$.

Proof.
- By Theorem 8.3, $A = Q\operatorname{diag}(\gamma_1, \dots, \gamma_n)Q'$ with Q orthogonal. Then $y = Q'z \sim N_n(0, I_n)$ (Theorem 10.2), its coordinates are independent $N(0,1)$ variables (Theorem 10.9(a)), and $z'Az = y'\operatorname{diag}(\gamma_1, \dots, \gamma_n)y = \sum_j \gamma_j y_j^2$.
- Let $t_0 > 0$ with $2t_0|\gamma_j| < 1$ for every j. For $|t| < t_0$, Lemma 11.1(a) (with $\delta = 0$ and $t\gamma_j$ in place of t) gives $M_{\gamma_j y_j^2}(t) = (1-2\gamma_j t)^{-1/2}$. The $\gamma_j y_j^2$ are independent (N3), so, as in Lemma 11.1(b), $M_{z'Az}(t) = \prod_j (1-2\gamma_j t)^{-1/2}$ near 0 (N2).
- Also $M_{z'Az}(t) = (1-2t)^{-r/2}$ near 0 (Theorem 11.4(a)). Squaring and inverting, $\prod_{j=1}^n (1-2\gamma_j t) = (1-2t)^r$ for all t in an open interval; both sides are polynomials in t, so they are equal (Proposition 7.5(d)).
- The only root of the right side is $\frac{1}{2}$. If $\gamma_j \neq 0$, then $t = 1/(2\gamma_j)$ is a root of the left side (Proposition 7.5(b)); so $1/(2\gamma_j) = \frac{1}{2}$, that is, $\gamma_j = 1$. Hence the left side is $(1-2t)^m$, where m is the number of j with $\gamma_j \neq 0$, and comparing degrees (Proposition 7.5(a)) gives $m = r$.
- So the eigenvalues of A are 1 (r times) and 0 ($n-r$ times) (Theorem 8.3(c)): A is idempotent (Theorem 8.13(b)), and $\operatorname{rank} A = r$ (Theorem 8.4). □

Theorem 11.18 ([2, Theorem 10.2, Corollary 10.1], [3, Ch. 8, Exercise 15])**.** Let $y \sim N_n(\mu, \Sigma)$ with Σ positive definite, let $A \in \mathbb{R}^{n\times n}$ be symmetric, let $r \geqslant 1$, and let $B = \Sigma^{1/2}A\Sigma^{1/2}$.

(a) $A\Sigma$ is idempotent if and only if B is idempotent; and $\operatorname{rank}(A\Sigma) = \operatorname{rank} B$.

(b) If $A\Sigma$ is idempotent of rank r, then $y'Ay \sim \chi^2_r(\mu'A\mu)$.

(c) If $\mu = 0$ and $y'Ay \sim \chi^2_r$, then $A\Sigma$ is idempotent of rank r.

(d) $A\Sigma$ is idempotent of rank r if and only if $\operatorname{tr}(A\Sigma) = \operatorname{tr}\Big((A\Sigma)^2\Big) = \operatorname{rank}(A\Sigma) = r$.

Proof. (a)

- $B^2 = \Sigma^{1/2}A\Sigma A\Sigma^{1/2}$ (Theorem 8.46). If $(A\Sigma)^2 = A\Sigma$, multiplying on the right by Σ^{-1} gives $A\Sigma A = A$, and then $B^2 = \Sigma^{1/2}A\Sigma^{1/2} = B$.
- If $B^2 = B$, multiplying by $\Sigma^{-1/2}$ on both sides gives $A\Sigma A = A$, and multiplying on the right by Σ gives $(A\Sigma)^2 = A\Sigma$.
- $B = \Sigma^{1/2}(A\Sigma)\Sigma^{-1/2}$, so $\operatorname{rank} B = \operatorname{rank}(A\Sigma)$ (Theorem 4.15(d)).

(b) Let $z = \Sigma^{-1/2}y \sim N_n(\Sigma^{-1/2}\mu, I_n)$ (Theorem 10.2); then $y'Ay = z'Bz$. By (a), B is symmetric and idempotent of rank r, so by Theorem 11.15 (with $\sigma = 1$), $z'Bz \sim \chi^2_r(\mu'\Sigma^{-1/2}B\Sigma^{-1/2}\mu) = \chi^2_r(\mu'A\mu)$.

(c) Now $z \sim N_n(0, I_n)$ and $z'Bz = y'Ay \sim \chi^2_r$; by Theorem 11.17, B is idempotent of rank r, and so is $A\Sigma$, by (a).

(d)

- If $A\Sigma$ is idempotent of rank r, then $\operatorname{tr}(A\Sigma) = \operatorname{rank}(A\Sigma) = r$ (Theorem 6.29(c)), and $(A\Sigma)^2 = A\Sigma$ has the same trace.
- Conversely, $\operatorname{tr} B = \operatorname{tr}(A\Sigma)$ and $\operatorname{tr} B^2 = \operatorname{tr}\Big(\Sigma^{1/2}(A\Sigma)^2\Sigma^{-1/2}\Big) = \operatorname{tr}\Big((A\Sigma)^2\Big)$ (Theorem 1.11(b)), and $\operatorname{rank} B = \operatorname{rank}(A\Sigma)$ by (a). Write $B = Q\operatorname{diag}(\beta_1, \ldots, \beta_n)Q'$ (Theorem 8.3); then $\operatorname{tr} B = \sum_i \beta_i$, $\operatorname{tr} B^2 = \sum_i \beta_i^2$ (Theorem 1.11(b)), and $\operatorname{rank} B$ is the number of i with $\beta_i \neq 0$ (Theorem 8.4).
- If all three numbers equal r, then $\sum_{\beta_i\neq 0}(\beta_i - 1)^2 = \sum_i \beta_i^2 - 2\sum_i \beta_i + r = 0$, so every nonzero β_i is 1: B is idempotent (Theorem 8.13(b)) of rank r, and so is $A\Sigma$, by (a). □

Independence of quadratic and linear forms

Lemma 11.19. Let $A \in \mathbb{R}^{n\times n}$ be symmetric. There is a symmetric $G \in \mathbb{R}^{n\times n}$ with $AGA = A$; hence $x'Ax = (Ax)'G(Ax)$ for every $x \in \mathbb{R}^n$: the number $x'Ax$ is a function of the vector Ax.

Proof. If $A = 0$, take $G = 0$. Otherwise let $A = c_1E_1 + \cdots + c_kE_k$ be the spectral decomposition (Theorem 8.6), and let $G = \sum_{c_j\neq 0} c_j^{-1}E_j$, which is symmetric. As $E_iE_j = 0$ for $i \neq j$ and $E_j^2 = E_j$ (Theorem 8.6(b)), $AGA = \sum_{c_j\neq 0} c_jc_j^{-1}c_jE_j = \sum_j c_jE_j = A$. Then $x'Ax = x'AGAx = (Ax)'G(Ax)$, as $A' = A$. □

Theorem 11.20 ([12, Theorem 4.6], [3, Result 8.5], [2, Theorems 10.3, 10.4])**.** Let $z \sim N_n(\mu, \sigma^2 I_n)$, and let $A, B \in \mathbb{R}^{n\times n}$ be symmetric.

(a) If $AB = 0$, then $z'Az$ and $z'Bz$ are independent.

(b) If $C \in \mathbb{R}^{m\times n}$ and $CA = 0$, then Cz and $z'Az$ are independent.

Proof. (a)

- $A(\sigma^2 I_n)B' = \sigma^2 AB = 0$, so Az and Bz are independent (Theorem 10.5(c)).
- By Lemma 11.19, $z'Az$ is a function of Az, and $z'Bz$ is a function of Bz; so they are independent (N3).

(b) $C(\sigma^2 I_n)A' = \sigma^2 CA = 0$, so Cz and Az are independent (Theorem 10.5(c)); and $z'Az$ is a function of Az (Lemma 11.19). So Cz and $z'Az$ are independent (N3). □

For instance, in Theorem 11.11, $\bar{x} = \frac{1}{n}\mathbf{1}'x$ and $s^2 = x'C_n x/(n-1)$ are independent by (b), as $\frac{1}{n}\mathbf{1}'C_n = 0$.

Theorem 11.21 ([2, Theorems 10.3, 10.4])**.** Let $y \sim N_n(\mu, \Sigma)$ (Σ positive semidefinite), and let $A, B \in \mathbb{R}^{n\times n}$ be symmetric and $C \in \mathbb{R}^{m\times n}$.

(a) If $A\Sigma B = 0$, then $y'Ay$ and $y'By$ are independent.

(b) If $C\Sigma A = 0$, then Cy and $y'Ay$ are independent.

Proof. (a) $A\Sigma B' = A\Sigma B = 0$, so Ay and By are independent (Theorem 10.5(c)); by Lemma 11.19 and (N3), so are $y'Ay$ and $y'By$.

(b) $C\Sigma A' = C\Sigma A = 0$, so Cy and Ay are independent (Theorem 10.5(c)); by Lemma 11.19 and (N3), so are Cy and $y'Ay$. □

Theorem 11.22 ([3, Results 8.4, 8.5])**.** Let $z \sim N_n(0, I_n)$, and let $A_1, A_2, A \in \mathbb{R}^{n\times n}$ be symmetric and idempotent.

(a) $z'A_1 z$ and $z'A_2 z$ are independent if and only if $A_1 A_2 = 0$.

(b) For $\ell \in \mathbb{R}^n$, $z'Az$ and $\ell' z$ are independent if and only if $A\ell = 0$.

Proof. (a) If $A_1 A_2 = 0$, the two forms are independent by Theorem 11.20(a). Conversely, let them be independent. If $A_1 = 0$ or $A_2 = 0$, then $A_1 A_2 = 0$; so let $r_i = \operatorname{rank} A_i \geqslant 1$.

- $z'A_i z \sim \chi^2_{r_i}$ (Theorem 11.15), so $z'(A_1 + A_2)z \sim \chi^2_{r_1+r_2}$ (Theorem 11.4(c)). $A_1 + A_2$ is symmetric, so it is idempotent (Theorem 11.17).
- Then $A_1 + A_2 = (A_1 + A_2)^2 = A_1 + A_2 + A_1A_2 + A_2A_1$, so $A_1A_2 + A_2A_1 = 0$.
- Multiplying on the right by A_2 gives $A_1A_2 + A_2A_1A_2 = 0$; multiplying this on the left by A_2 gives $2A_2A_1A_2 = 0$. So $A_2A_1A_2 = 0$, and then $A_1A_2 = 0$.

(b) If $A\ell = 0$, then $\ell'A = (A\ell)' = 0$, and Theorem 11.20(b) (with $C = \ell'$) gives independence. Conversely, let $z'Az$ and $\ell' z$ be independent, and let $\ell \neq 0$ (for $\ell = 0$ there is nothing to prove).

- $B = \ell\ell'/\ell'\ell$ is symmetric and idempotent, and $z'Bz = (\ell' z)^2/\ell'\ell$ is a function of $\ell' z$; so $z'Az$ and $z'Bz$ are independent (N3), and $AB = 0$ by (a).
- Hence $A\ell = A\ell(\ell'\ell)/\ell'\ell = AB\ell = 0$. □

Example 11.23 ([3, Ch. 8, Exercise 3])**.** Let x_1, x_2 be independent $N(0,1)$ variables, so that $x = (x_1, x_2)' \sim N_2(0, I_2)$ (Theorem 10.9(a)). Which linear functions $c'x = c_1x_1 + c_2x_2$ are independent of $(x_1 - x_2)^2$? With $a = (1, -1)'$ and

$$A = \tfrac{1}{2}aa' = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

which is symmetric and idempotent ($A^2 = \frac{1}{4}a(a'a)a' = A$), $(x_1 - x_2)^2 = (a'x)^2 = 2\,x'Ax$. A function of $x'Ax$ is independent of $c'x$ when $x'Ax$ is, and conversely (N3); so by Theorem 11.22(b), $c'x$ is independent of $(x_1 - x_2)^2$ exactly when $Ac = \frac{1}{2}a(a'c) = 0$, that is, $c_1 = c_2$. These are the multiples of $x_1 + x_2$.

Example 11.24 (cf. [2, Example 10.1])**.** Let $n \geqslant 2$, $\sigma > 0$, $-1/(n-1) < \rho < 1$, and $x \sim N_n(\mu\mathbf{1}, \sigma^2 R)$ with $R = (1-\rho)I_n + \rho J_n$: equal means and variances, and the correlation ρ between any two observations. R is positive definite (Example 8.22(c)), and so is $\Sigma = \sigma^2 R$. Let $\bar{x}$ and s^2 be as in Theorem 11.11.

- $C_nJ_n = (C_n\mathbf{1})\mathbf{1}' = 0$, so $C_nR = (1-\rho)C_n$. With $A = C_n/\big(\sigma^2(1-\rho)\big)$, $A\Sigma = C_n$, which is idempotent of rank $n-1$ (Proposition 1.15, Corollary 6.30), and $A(\mu\mathbf{1}) = 0$. By Theorem 11.18(b), $(n-1)s^2/\big(\sigma^2(1-\rho)\big) = x'Ax \sim \chi^2_{n-1}$; in particular $\mathsf{E}(s^2) = \sigma^2(1-\rho)$ (Theorem 11.4(d)).
- $\bar{x} = c'x$ with $c = \frac{1}{n}\mathbf{1}$. As $\mathbf{1}'R = \big(1+(n-1)\rho\big)\mathbf{1}'$ and $\mathbf{1}'C_n = 0$, $c'\Sigma A = 0$; so $\bar{x}$ and s^2 are independent (Theorem 11.21(b)). Also $\bar{x} \sim N\big(\mu,\ \sigma^2 c'Rc\big) = N\big(\mu,\ \sigma^2(1+(n-1)\rho)/n\big)$ (Theorem 10.2).
- By the two steps above and Theorem 10.2, $Z = \sqrt{n}(\bar{x}-\mu)/\sqrt{\sigma^2(1+(n-1)\rho)} \sim N(0,1)$ and $V = (n-1)s^2/\big(\sigma^2(1-\rho)\big) \sim \chi^2_{n-1}$ are independent (N3). As in the proof of Theorem 11.11(d), by Definition 11.7,
$$T = \frac{\sqrt{n}(\bar{x}-\mu)}{s}\sqrt{\frac{1-\rho}{1+(n-1)\rho}} = \frac{Z}{\sqrt{V/(n-1)}} \sim t_{n-1}$$
For $\rho = 0$ this is Theorem 11.11(d). For $\rho > 0$ the usual statistic $\sqrt{n}(\bar{x}-\mu)/s$ is T times $\sqrt{(1+(n-1)\rho)/(1-\rho)}$.
- Let $n = 9$, $\rho = \frac{1}{4}$ and $\alpha \in (0,1)$. The factor is $\sqrt{(1+2)/\frac{3}{4}} = 2$. With $c = t_{8,1-\alpha/2}$, which is > 0 (the proof of Theorem 11.8(b)), the interval $\bar{x} \pm cs/\sqrt{9}$, which has the level $1-\alpha$ for independent observations (Example 11.13), contains μ exactly when $|\sqrt{9}(\bar{x}-\mu)/s| \leqslant c$, that is, when $|T| \leqslant \frac{c}{2}$. By Theorem 11.8(a), $\mathsf{P}(|T| \leqslant \frac{c}{2}) = 1 - 2\,\mathsf{P}(T > \frac{c}{2})$, and $\mathsf{P}(T > \frac{c}{2}) > \mathsf{P}(T > c) = \frac{\alpha}{2}$, as the distribution function of t_8 is strictly increasing (Definition 11.7). So the interval contains μ with probability less than $1-\alpha$: with positively correlated observations it is too short.

Cochran's theorem

Theorem 11.25 ([3, Result 8.6])**.** Let $A_1, \ldots, A_p \in \mathbb{R}^{n\times n}$ with $A_1 + \cdots + A_p = I_n$. The following are equivalent:

(a) $\operatorname{rank} A_1 + \cdots + \operatorname{rank} A_p = n$;

(b) $A_i^2 = A_i$ for $i = 1, \ldots, p$;

(c) $A_iA_j = 0$ for $i \neq j$.

Proof. (a)⇒(c)

- Products with a zero matrix vanish, and deleting the $A_i = 0$ changes neither the sum of the matrices nor the sum of the ranks; so let every $r_i = \operatorname{rank} A_i \geqslant 1$, with full-rank factorizations $A_i = B_iC_i$, $B_i \in \mathbb{R}^{n\times r_i}$, $C_i \in \mathbb{R}^{r_i\times n}$ (Theorem 4.21).

- By the block product (Theorem 5.3(c)),

$$[B_1 \ \cdots \ B_p] \begin{pmatrix} C_1 \\ \vdots \\ C_p \end{pmatrix} = B_1C_1 + \cdots + B_pC_p = I_n$$

  and the first factor is square, as $r_1 + \cdots + r_p = n$. By Theorem 2.16 the two factors are inverse to each other, so their product in the other order is I_n; its (i, j) block is C_iB_j, so $C_iB_j = 0$ for $i \neq j$.
- Hence $A_iA_j = B_i(C_iB_j)C_j = 0$ for $i \neq j$.

(c)⇒(b) $A_j = A_jI_n = A_j(A_1 + \cdots + A_p) = A_j^2$, since $A_jA_i = 0$ for $i \neq j$.
(b)⇒(a) $\operatorname{rank} A_i = \operatorname{tr} A_i$ (Theorem 6.29(c)), so $\sum_i \operatorname{rank} A_i = \sum_i \operatorname{tr} A_i = \operatorname{tr}(A_1 + \cdots + A_p) = \operatorname{tr} I_n = n$ (Theorem 1.11(a)). □

Theorem 11.26 ([2, Theorem 10.5], [3, §8.3])**.** Let $x \sim N_n(\mu, \sigma^2 I_n)$ ($\sigma > 0$), and let $A_1, \ldots, A_p \in \mathbb{R}^{n\times n}$ be symmetric with $A_1 + \cdots + A_p = I_n$ and $r_i = \operatorname{rank} A_i \geqslant 1$. Consider the conditions

(a) $r_1 + \cdots + r_p = n$;

(b) $x'A_ix/\sigma^2 \sim \chi^2_{r_i}(\mu'A_i\mu/\sigma^2)$ for $i = 1, \ldots, p$;

(c) $x'A_1x, \ldots, x'A_px$ are mutually independent.

Then (a) implies (b) and (c), and (b) implies (a).

Proof. (a)⇒(b), (c)

- By Theorem 11.25, every A_i is idempotent and $A_iA_j = 0$ for $i \neq j$. So (b) holds by Theorem 11.15.
- Let $u_i = A_ix$ and $u = (u_1', \ldots, u_p')' = Lx$, where $L \in \mathbb{R}^{pn\times n}$ has the blocks $A_1, \ldots, A_p$ stacked. By Theorem 10.2, $u \sim N_{pn}(L\mu, \sigma^2LL')$, and the (i, j) block of LL' is $A_iA_j' = A_iA_j$, which is 0 for $i \neq j$ (Theorem 5.3(c)).
- By Theorem 10.5(b), $u_1, \ldots, u_p$ are mutually independent, and so are the functions $x'A_ix = x'A_i'A_ix = \|u_i\|^2$ (N3): this is (c).

(b)⇒(a) By Theorem 11.4(d), $\mathsf{E}(x'A_ix/\sigma^2) = r_i + \mu'A_i\mu/\sigma^2$; by Theorem 9.10, $\mathsf{E}(x'A_ix) = \operatorname{tr}(A_i\sigma^2I_n) + \mu'A_i\mu$. So $\operatorname{tr} A_i = r_i$, and $\sum_i r_i = \operatorname{tr}(A_1 + \cdots + A_p) = \operatorname{tr} I_n = n$ (Theorem 1.11(a)). □

Example 11.27 ([3, Ch. 8, Exercise 5])**.** Let $x_1, \ldots, x_n$ ($n \geqslant 2$) be independent $N(0, 1)$ variables, so that $x = (x_1, \ldots, x_n)' \sim N_n(0, I_n)$ (Theorem 10.9(a)), and let $\bar{x} = \frac{1}{n}\mathbf{1}'x$. With Cochran's theorem we show that $\bar{x}$ and $\sum_i (x_i - \bar{x})^2$ are independent.

- Let $A_1 = \bar{J}_n$ and $A_2 = C_n$. They are symmetric, $A_1 + A_2 = I_n$ (Definition 1.14), and $\operatorname{rank} A_1 = \operatorname{tr} \bar{J}_n = 1$, $\operatorname{rank} A_2 = n - 1$ (Proposition 1.15, Theorem 6.29(c), Corollary 6.30); so $\operatorname{rank} A_1 + \operatorname{rank} A_2 = n$.
- $x'A_1x = \frac{1}{n}(\mathbf{1}'x)^2 = n\bar{x}^2$ and $x'A_2x = \sum_i (x_i - \bar{x})^2$ (Example 1.16). By Theorem 11.26, (a) ⇒ (b), (c), with $\mu = 0$ and $\sigma = 1$: $n\bar{x}^2 \sim \chi^2_1$ and $\sum_i (x_i - \bar{x})^2 \sim \chi^2_{n-1}$, and they are independent.
- The vectors $A_1x = \bar{x}\mathbf{1}$ and A_2x are independent (Theorem 10.5(c)), as $A_1I_nA_2' = \bar{J}_nC_n = \frac{1}{n}\mathbf{1}(C_n\mathbf{1})' = 0$ (C_n is symmetric and $C_n\mathbf{1} = 0$, Proposition 1.15). So $\bar{x}$, the first entry of A_1x, and $\sum_i (x_i - \bar{x})^2 = \|A_2x\|^2$ are independent (N3).

Moments of quadratic forms

Theorem 11.28 ([2, (10.34), (10.35), Corollary 10.2], [6, Exercises 11.21, 11.22])**.** Let $y \sim N_n(\mu, \Sigma)$ (Σ positive semidefinite), and let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then

$$\mathsf{E}(y'Ay) = \operatorname{tr}(A\Sigma) + \mu'A\mu, \qquad \operatorname{var}(y'Ay) = 2\operatorname{tr}\Big((A\Sigma)^2\Big) + 4\mu'A\Sigma A\mu$$

Proof. The mean is Theorem 9.10. For the variance:

- Let $L = \Sigma^{1/2}$ (Theorem 8.46), so that $L' = L$ and $L^2 = \Sigma$, and let $w \sim N_n(0, I_n)$. Then $\mu + Lw \sim N_n(\mu, \Sigma)$ (Theorem 10.9(b)), so $y'Ay$ has the distribution of $(\mu + Lw)'A(\mu + Lw)$, and we may take $y = \mu + Lw$. Then

  $$y'Ay = \mu'A\mu + c'w + w'Bw, \qquad c = 2LA\mu, \quad B = LAL$$

  since the number $w'LA\mu$ equals $\mu'ALw$; B is symmetric.
- Write $B = Q\operatorname{diag}(\gamma_1, \ldots, \gamma_n)Q'$ (Theorem 8.3) and $v = Q'w \sim N_n(0, I_n)$ (Theorem 10.2); the v_i are independent $N(0,1)$ variables (Theorem 10.9(a)), and $w'Bw = \sum_i \gamma_i v_i^2$. The $\gamma_i v_i^2$ are independent (N3), hence uncorrelated (N7), with $\operatorname{var}(\gamma_i v_i^2) = \gamma_i^2 \operatorname{var}(v_i^2) = 2\gamma_i^2$ (Theorems 9.3(b) and 11.4(d)). By Theorem 9.4(d), $\operatorname{var}(w'Bw) = 2\sum_i \gamma_i^2 = 2\operatorname{tr}(B^2)$, as $B^2 = Q\operatorname{diag}(\gamma_1^2, \ldots, \gamma_n^2)Q'$ (Theorem 1.11(b)).
- $\operatorname{var}(c'w) = c'c = 4\mu'ALLA\mu = 4\mu'A\Sigma A\mu$ (Theorem 9.3(b)).
- $\mathsf{E}(c'w) = 0$, so $\operatorname{cov}(w'Bw, c'w) = \mathsf{E}(w'Bw\, c'w) = \sum_{i,j,l} b_{ij}c_l\, \mathsf{E}(w_iw_jw_l)$. Each $\mathsf{E}(w_iw_jw_l) = 0$: some index h occurs once or three times among i, j, l, so $w_iw_jw_l = w_h^e\, g$ with $e \in \{1, 3\}$ and g a function of the vector Ew of the other coordinates (the rows of E are the e_k', $k \neq h$). As $e_h'I_nE' = 0$, w_h and Ew are independent (Theorem 10.5(c)), and so are w_h^e and g (N3); so the expectation is $\mathsf{E}(w_h^e)\,\mathsf{E}(g)$ (N7), and $\mathsf{E}(w_h^e) = 0$ (Lemma 11.3).
- As $\mu'A\mu$ is a constant, Theorem 9.4(a),(c) gives

  $$\operatorname{var}(y'Ay) = \operatorname{var}(w'Bw) + \operatorname{var}(c'w) + 2\operatorname{cov}(w'Bw, c'w) = 2\operatorname{tr}(B^2) + 4\mu'A\Sigma A\mu$$

  and $\operatorname{tr}(B^2) = \operatorname{tr}(LALLAL) = \operatorname{tr}(ALLALL) = \operatorname{tr}\Big((A\Sigma)^2\Big)$ (Theorem 1.11(b)). □

Corollary 11.29 ([2, Exercise 10.12])**.** Let $y \sim N_n(\mu, \Sigma)$, and let $A, B \in \mathbb{R}^{n\times n}$ be symmetric. Then

$$\operatorname{cov}(y'Ay,\, y'By) = 2\operatorname{tr}(A\Sigma B\Sigma) + 4\mu'A\Sigma B\mu$$

Proof. • By Theorem 9.4(c), $\operatorname{var}\Big(y'(A+B)y\Big) = \operatorname{var}(y'Ay) + \operatorname{var}(y'By) + 2\operatorname{cov}(y'Ay, y'By)$.

- Apply Theorem 11.28 to $A + B$, A and B. Here $\operatorname{tr}\Big(((A+B)\Sigma)^2\Big) = \operatorname{tr}\Big((A\Sigma)^2\Big) + \operatorname{tr}\Big((B\Sigma)^2\Big) + 2\operatorname{tr}(A\Sigma B\Sigma)$, as $\operatorname{tr}(B\Sigma A\Sigma) = \operatorname{tr}(A\Sigma B\Sigma)$ (Theorem 1.11(b)); and $\mu'(A+B)\Sigma(A+B)\mu = \mu'A\Sigma A\mu + \mu'B\Sigma B\mu + 2\mu'A\Sigma B\mu$, as the number $\mu'B\Sigma A\mu$ equals its transpose $\mu'A\Sigma B\mu$. So $2\operatorname{cov}(y'Ay, y'By) = 4\operatorname{tr}(A\Sigma B\Sigma) + 8\mu'A\Sigma B\mu$. □

Example 11.30 ([17, Exercise 5.10(c)], [20, Exercise 5.10(c)])**.** In Theorem 11.11, $\operatorname{var}(s^2) = 2\sigma^4/(n-1)$. Two ways:

- $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ has the variance $2(n-1)$ (Theorem 11.4(d)), so $\operatorname{var}(s^2) = \frac{\sigma^4}{(n-1)^2} \cdot 2(n-1) = \frac{2\sigma^4}{n-1}$.
- $s^2 = x'Ax$ with $A = C_n/(n-1)$ and $x \sim N_n(\mu\mathbf{1}, \sigma^2 I_n)$. Here $A\Sigma = \sigma^2 C_n/(n-1)$, so $\operatorname{tr}\Big((A\Sigma)^2\Big) = \sigma^4\operatorname{tr}(C_n)/(n-1)^2 = \sigma^4/(n-1)$ (Proposition 1.15, Corollary 6.30), and $A\mu\mathbf{1} = 0$. By Theorem 11.28, $\operatorname{var}(s^2) = 2\sigma^4/(n-1) + 0$.

Exercises

Exercise 11.1. Let u have the MGF $(1-2t)^{-3}\exp\big(4t/(1-2t)\big)$ near 0, and let $v \sim \chi^2_2(1)$ be independent of u.

(a) Identify the distribution of u, and find $\mathsf{E}(u)$ and $\mathrm{var}(u)$.

(b) Find the distribution of $u+v$, its mean and its variance, and check that the means and the variances add up.

(c) Let $z \sim N_4(\mu, I_4)$ with $\mu = (1,-1,2,1)'$. Find the distributions of $z_1^2+z_2^2$ and of $\|z\|^2$.

Solution. (a) $(1-2t)^{-3} = (1-2t)^{-6/2}$, so $u \sim \chi^2_6(4)$ (Theorem 11.4(a)). By Theorem 11.4(d), $\mathsf{E}(u) = 6+4 = 10$ and $\mathrm{var}(u) = 12+16 = 28$.

(b) By Theorem 11.4(c), $u+v \sim \chi^2_8(5)$, with mean $8+5=13$ and variance $16+20=36$. Also $\mathsf{E}(v) = 2+1 = 3$ and $\mathrm{var}(v) = 4+4 = 8$, and indeed $10+3=13$ and $28+8=36$.

(c) $w = z-\mu \sim N_4(0, I_4)$ (Theorem 10.2), so $w_1,\dots,w_4$ are independent $N(0,1)$ variables (Theorem 10.9(a)), and $z_i = w_i + \mu_i$. By Definition 11.2, $z_1^2+z_2^2 = (w_1+1)^2+(w_2-1)^2 \sim \chi^2_2(2)$ and $\|z\|^2 \sim \chi^2_4(1+1+4+1) = \chi^2_4(7)$.

Exercise 11.2. (a) Write down the densities of χ^2_2 and χ^2_4.

(b) For $u \sim \chi^2_4$ and $c>0$, show by integration by parts that $\mathsf{P}(u>c) = (1+c/2)e^{-c/2}$. Using $\frac{8}{3} < e < 3$, show that $8 < \chi^2_{4,19/20} < 10$ (Definition 11.7(c)).

(c) For $v \sim \chi^2_2$, find the median of v, the number c with $\mathsf{P}(v \leqslant c) = \frac{1}{2}$.

Solution. (a) By Theorem 11.4(b), with $\Gamma(1)=1$ and $\Gamma(2) = 1\cdot\Gamma(1) = 1$ (D2): $f_2(x) = \frac{1}{2}e^{-x/2}$ and $f_4(x) = \frac{1}{4}xe^{-x/2}$ for $x>0$, and both are 0 for $x \leqslant 0$.

(b) As u has the density f_4, integrating by parts ($xe^{-x/2} \to 0$ for $x \to \infty$),

$$\mathsf{P}(u>c) = \int_c^\infty \frac{x}{4}e^{-x/2}\,\mathrm{d}x = \Big[-\frac{x}{2}e^{-x/2}\Big]_c^\infty + \int_c^\infty \frac{1}{2}e^{-x/2}\,\mathrm{d}x = \frac{c}{2}e^{-c/2} + e^{-c/2}$$

- From the series $e = \sum_{k\geqslant 0} 1/k!$: $e > 1+1+\frac{1}{2}+\frac{1}{6} = \frac{8}{3}$, and $e < 1+1+\frac{1}{2}+\frac{1}{4}+\cdots = 3$, as $k! \geqslant 2^{k-1}$ for $k \geqslant 1$, strictly for $k \geqslant 3$.
- $\mathsf{P}(u>8) = 5e^{-4} > \frac{1}{20}$, as $e^4 < 3^4 = 81 < 100$; and $\mathsf{P}(u>10) = 6e^{-5} < \frac{1}{20}$, as $e^5 > (\frac{8}{3})^5 = \frac{32768}{243} > 120$ ($32768 > 120\cdot 243 = 29160$).
- $\mathsf{P}(u > \chi^2_{4,19/20}) = \frac{1}{20}$ (Definition 11.7), and $c \mapsto \mathsf{P}(u>c)$ is strictly decreasing, as the distribution function of u is strictly increasing on $(0,\infty)$. So $8 < \chi^2_{4,19/20} < 10$.

(c) $\mathsf{P}(v \leqslant c) = 1-e^{-c/2}$ (Example 11.5(a)), which is $\frac{1}{2}$ for $c = 2\log 2$ (as in Example 11.10, $\chi^2_{2,1/2} = -2\log\frac{1}{2}$).

Exercise 11.3. Let $T \sim t_5$, $\alpha \in (0,1)$ and $c = t_{5,1-\alpha/2}$.

(a) Find $\mathsf{P}(T>c)$, $t_{5,\alpha/2}$ and $\mathsf{P}(|T| \leqslant c)$.

(b) Express $F_{1,5,1-\alpha}$ through c.

(c) Express $F_{5,2,\alpha}$ and $F_{2,5,\alpha}$ through quantiles $F_{a,b,1-\alpha}$.

Solution. (a) $\mathsf{P}(T>c) = 1-(1-\frac{\alpha}{2}) = \frac{\alpha}{2}$ (Definition 11.7). By Theorem 11.8(a), $\mathsf{P}(T<-c) = \frac{\alpha}{2}$; as T has a density, also $\mathsf{P}(T \leqslant -c) = \frac{\alpha}{2}$, so $t_{5,\alpha/2} = -c$, and $\mathsf{P}(|T| \leqslant c) = 1-\frac{\alpha}{2}-\frac{\alpha}{2} = 1-\alpha$.

(b) $F_{1,5,1-\alpha} = t^2_{5,1-\alpha/2} = c^2$ (Theorem 11.8(b)).

(c) By Proposition 11.9, $F_{5,2,\alpha} = 1/F_{2,5,1-\alpha}$ and $F_{2,5,\alpha} = 1/F_{5,2,1-\alpha}$.

Exercise 11.4. Seven measurements $1, 3, 4, 5, 6, 7, 9$ are modelled as independent $N(\mu, \sigma^2)$ variables. Let $\alpha \in (0,1)$ and $c = t_{6,1-\alpha/2}$.

(a) Compute $\bar{x}$ and s^2.

(b) Compute $\sqrt{7}(\bar{x}-2)/s$, the statistic of Theorem 11.11(d) for $\mu = 2$, and find the condition on c under which it lies in $[-c, c]$.

(c) Compute the confidence interval $\bar{x} \pm c\,s/\sqrt{7}$ for μ with level $1-\alpha$ (Example 11.13). Does it contain 2?

Solution. (a) $\bar{x} = 35/7 = 5$; the deviations are $-4, -2, -1, 0, 1, 2, 4$, so $s^2 = (16+4+1+0+1+4+16)/6 = 42/6 = 7$.

(b) $s/\sqrt{7} = 1$, so $\sqrt{7}(\bar{x}-2)/s = (5-2)/1 = 3$, which lies in $[-c, c]$ exactly when $c \geqslant 3$. If $\mu = 2$, the statistic is a t_6 variable (Theorem 11.11(d)), and it lies outside $[-c, c]$ with probability α (Theorem 11.8(a)).

(c) $[5 - t_{6,1-\alpha/2},\ 5 + t_{6,1-\alpha/2}]$. It contains 2 exactly when $5 - c \leqslant 2$, that is, $t_{6,1-\alpha/2} \geqslant 3$, the condition of (b).

Exercise 11.5. Three measurements $4, 7, 10$ are modelled as independent $N(\mu, \sigma^2)$ variables. Let $\alpha \in (0,1)$.

(a) Write down $\bar{x}$, $(n-1)s^2$ and the confidence interval of Example 11.13 for σ^2.

(b) Write the interval of (a) in closed form with Example 11.10. Does it contain 1 for $\alpha = \frac{1}{20}$? Show that its upper end then exceeds 351.

(c) Write down the confidence interval of Example 11.13 for μ.

Solution. (a) Here $n = 3$, $\bar{x} = 7$ and $(n-1)s^2 = 9 + 0 + 9 = 18$. By Example 11.13 the interval is $\Big[18/\chi^2_{2,1-\alpha/2},\ 18/\chi^2_{2,\alpha/2}\Big]$.

(b) By Example 11.10, $\chi^2_{2,\tau} = -2\log(1-\tau)$; so $\chi^2_{2,1-\alpha/2} = 2\log(2/\alpha)$ and $\chi^2_{2,\alpha/2} = 2\log\Big(2/(2-\alpha)\Big)$, and the interval is

$$\left[\frac{9}{\log(2/\alpha)},\ \frac{9}{\log\Big(2/(2-\alpha)\Big)}\right], \qquad \text{for } \alpha = \tfrac{1}{20}: \quad \left[\frac{9}{\log 40},\ \frac{9}{\log(40/39)}\right]$$

As $e > 1 + 1 = 2$ ($e = \sum_{k\geqslant 0} 1/k!$), $e^9 > 2^9 = 512 > 40$, so $\log 40 < 9$ and the lower end exceeds 1: the interval does not contain 1. By Lemma 10.18 with $p = 1$ and $A = (\frac{40}{39}) \neq I_1$, $\log\frac{40}{39} < \frac{40}{39} - 1 = \frac{1}{39}$, so the upper end exceeds $9 \cdot 39 = 351$. Three observations say little about σ^2.

(c) $s^2 = 9$, $s = 3$ and $s/\sqrt{3} = \sqrt{3}$, so the interval is $\Big[7 - \sqrt{3}\,t_{2,1-\alpha/2},\ 7 + \sqrt{3}\,t_{2,1-\alpha/2}\Big]$.

Exercise 11.6. Let $x \sim N_4(\mu, \Sigma)$ with

$$\mu = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 4 & 3 & 1 & 1 \\ 3 & 4 & 1 & 1 \\ 1 & 1 & 4 & 3 \\ 1 & 1 & 3 & 4 \end{pmatrix}$$

and let $d^2 = (x-\mu)'\Sigma^{-1}(x-\mu)$.

(a) Check that $h_1 = (1,1,1,1)'$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$ and $h_4 = (1,-1,-1,1)'$ are eigenvectors of Σ, and conclude that Σ is PD.

(b) Show that $\mathsf{P}(d^2 \leqslant c) = 1 - (1 + c/2)e^{-c/2}$ for $c > 0$. Describe the region $E_\alpha = \{u \in \mathbb{R}^4 : (u-\mu)'\Sigma^{-1}(u-\mu) \leqslant \chi^2_{4,1-\alpha}\}$, which contains x with probability $1-\alpha$.

(c) Which of the points $a = (4,4,4,4)'$ and $b = (3,-1,3,-1)'$ lie in $E_{1/20}$? For each, find the probability that d^2 exceeds its value of $(u-\mu)'\Sigma^{-1}(u-\mu)$, and compare with the Euclidean distances from μ.

Solution. (a) Row by row,

$$\Sigma h_1 = \begin{pmatrix}9\\9\\9\\9\end{pmatrix} = 9h_1, \qquad \Sigma h_2 = \begin{pmatrix}5\\5\\-5\\-5\end{pmatrix} = 5h_2, \qquad \Sigma h_3 = \begin{pmatrix}1\\-1\\1\\-1\end{pmatrix} = h_3, \qquad \Sigma h_4 = \begin{pmatrix}1\\-1\\-1\\1\end{pmatrix} = h_4$$

The h_j are orthogonal ($h_i'h_j = 0$ for $i \neq j$) with $h_j'h_j = 4$, so $q_j = \frac{1}{2}h_j$ form an orthonormal basis of eigenvectors, and the eigenvalues of Σ are 9, 5, 1, 1 (Theorem 8.3(b),(c)). All are > 0, so Σ is PD (Theorem 8.21(a)).

(b) By Corollary 10.11(a), $d^2 \sim \chi^2_4$; so, by Exercise 11.2(b), $\mathsf{P}(d^2 \leqslant c) = 1 - \mathsf{P}(d^2 > c) = 1 - (1 + c/2)e^{-c/2}$ for $c > 0$, and $\mathsf{P}(x \in E_\alpha) = \mathsf{P}(d^2 \leqslant \chi^2_{4,1-\alpha}) = 1 - \alpha$ (Definition 11.7(c)). By the paragraph after Theorem 10.8, with $c = (\chi^2_{4,1-\alpha})^{1/2}$, E_α is bounded by an ellipsoid with center μ, axes along $q_1, \ldots, q_4$, and semi-axes $3c$, $\sqrt{5}\,c$, c, c.

(c) In the coordinates $v_j = q_j'(u-\mu)$ of that paragraph, $(u-\mu)'\Sigma^{-1}(u-\mu) = v_1^2/9 + v_2^2/5 + v_3^2 + v_4^2$.

- For a, $a - \mu = (3,3,3,3)' = 3h_1$, so $v = (6,0,0,0)'$ and the value is $36/9 = 4$. For b, $b - \mu = (2,-2,2,-2)' = 2h_3$, so $v = (0,0,4,0)'$ and the value is 16.
- By Exercise 11.2(b), $8 < \chi^2_{4,19/20} < 10$. So $4 < \chi^2_{4,19/20}$: a lies in $E_{1/20}$; and $16 > \chi^2_{4,19/20}$: b does not. By (b), $\mathsf{P}(d^2 > 4) = 3e^{-2}$ and $\mathsf{P}(d^2 > 16) = 9e^{-8}$.
- The Euclidean distances are $\|a - \mu\| = 6$ and $\|b - \mu\| = 4$: the point farther from μ is the less unusual one, because it lies in the direction h_1 of the largest variance, and b in a direction of the smallest.

Exercise 11.7. Let C_4 be the centering matrix (Definition 1.14) and $z \sim N_4(\mu, \sigma^2 I_4)$ ($\sigma > 0$), with $\bar{z} = \frac{1}{4}\mathbf{1}'z$.

(a) Find an orthonormal basis q_1, q_2, q_3 of $\mathcal{N}(C_4 - I_4)$ by row reduction and Gram–Schmidt, and compare it with the rows p_2', p_3', p_4' of the Helmert matrix K_4 (Example 6.20).

(b) Write $z'C_4z = u_1^2 + u_2^2 + u_3^2$ with $u_i = q_i'z$, and show that u_1, u_2, u_3 are independent with variance σ^2.

(c) Let $A_1 = \bar{J}_4$ and $A_{i+1} = q_iq_i'$ ($i = 1,2,3$). Show that $A_1 + A_2 + A_3 + A_4 = I_4$ and that the ranks add up to 4, and conclude with Cochran's theorem that $4\bar{z}^2$, u_1^2, u_2^2, u_3^2 are independent; give their distributions divided by σ^2.

(d) For $\mu = (2,3,7,4)'$ and $\sigma = 1$, find the distribution of $z'C_4z$, its mean and its variance, and show that $z'C_4z$ and $\bar{z}$ are independent.

Solution. (a) $C_4 - I_4 = -\frac{1}{4}J_4$ has the same null space as J_4, which reduces as follows (the three operations use row 1 and change different rows):

$$\begin{pmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix}1&1&1&1\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix}$$

So $x_1 = -x_2 - x_3 - x_4$, and the special solutions $a_1 = (-1,1,0,0)'$, $a_2 = (-1,0,1,0)'$, $a_3 = (-1,0,0,1)'$ form a basis (Theorem 4.8(c)). Gram–Schmidt (Theorem 6.14):

- $b_1 = a_1$, $\|b_1\|^2 = 2$;
- $a_2'b_1 = 1$, $b_2 = a_2 - \frac{1}{2}b_1 = (-\frac{1}{2}, -\frac{1}{2}, 1, 0)'$, $\|b_2\|^2 = \frac{3}{2}$;
- $a_3'b_1 = 1$, $a_3'b_2 = \frac{1}{2}$, $b_3 = a_3 - \frac{1}{2}b_1 - \frac{1}{3}b_2 = (-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1)'$, $\|b_3\|^2 = \frac{4}{3}$.

So $q_1 = (-1, 1, 0, 0)'/\sqrt{2}$, $q_2 = (-1, -1, 2, 0)'/\sqrt{6}$ and $q_3 = (-1, -1, -1, 3)'/\sqrt{12}$: these are $-p_2$, $-p_3$ and $-p_4$ for the rows $p_2' = (1, -1, 0, 0)/\sqrt{2}$, $p_3' = (1, 1, -2, 0)/\sqrt{6}$ and $p_4' = (1, 1, 1, -3)/\sqrt{12}$ of K_4.

(b) By Example 6.20(b), $C_4 = p_2p_2' + p_3p_3' + p_4p_4' = q_1q_1' + q_2q_2' + q_3q_3'$, so $z'C_4z = (q_1'z)^2 + (q_2'z)^2 + (q_3'z)^2$, with $u_1 = (z_2 - z_1)/\sqrt{2}$, $u_2 = (2z_3 - z_1 - z_2)/\sqrt{6}$ and $u_3 = (3z_4 - z_1 - z_2 - z_3)/\sqrt{12}$. For $P_1 = [q_1\ q_2\ q_3]$, $P_1'P_1 = I_3$, so $u = P_1'z \sim N_3(P_1'\mu, \sigma^2 I_3)$ (Theorem 10.2): u_1, u_2, u_3 are independent with variance σ^2 (Theorem 10.9(a)).

(c) By (b) and Definition 1.14, $A_2 + A_3 + A_4 = C_4$ and $A_1 + C_4 = I_4$. The A_i are symmetric and idempotent (Proposition 1.15; $(q_iq_i')^2 = q_i(q_i'q_i)q_i' = q_iq_i'$), so their ranks are their traces (Theorem 6.29(c)): $\operatorname{tr} \bar{J}_4 = 1$ and $\operatorname{tr}(q_iq_i') = q_i'q_i = 1$; they add up to 4. By Theorem 11.26, (a) $\Rightarrow$ (b), (c), the forms $z'A_1z = \frac{1}{4}(\mathbf{1}'z)^2 = 4\bar{z}^2$ and $z'A_{i+1}z = (q_i'z)^2 = u_i^2$ are independent, with $4\bar{z}^2/\sigma^2 \sim \chi_1^2(4\bar{\mu}^2/\sigma^2)$, where $\bar{\mu} = \frac{1}{4}\mathbf{1}'\mu$, and $u_i^2/\sigma^2 \sim \chi_1^2\big((q_i'\mu)^2/\sigma^2\big)$.

(d) C_4 is symmetric and idempotent of rank 3 (Proposition 1.15, Corollary 6.30). By Theorem 11.15, $z'C_4z \sim \chi_3^2(\mu'C_4\mu)$, where $\mu'C_4\mu = \sum_i(\mu_i - 4)^2 = 4 + 1 + 9 + 0 = 14$ (Example 1.16); check with (c) and Theorem 11.4(c): $(q_1'\mu)^2 + (q_2'\mu)^2 + (q_3'\mu)^2 = \frac{1}{2} + \frac{81}{6} + 0 = 14$. By Theorem 11.4(d), the mean is $3 + 14 = 17$ and the variance is $6 + 56 = 62$. As $\bar{z} = \frac{1}{4}\mathbf{1}'z$ and $\frac{1}{4}\mathbf{1}'C_4 = 0$ (Proposition 1.15), $\bar{z}$ and $z'C_4z$ are independent (Theorem 11.20(b)).

Exercise 11.8. Let $v = (1, 1, -1, -1)'$, $u = (1, -1, 0, 0)'$, $A = \frac{1}{4}vv'$ and $B = \frac{1}{2}uu'$.

(a) Write out A and B, check that $A^2 = A$, $B^2 = B$ and $AB = 0$, and find $\operatorname{rank} A$ and $\operatorname{rank} B$.

(b) Let $z \sim N_4(\mu, I_4)$ with $\mu = (3, 1, 2, 2)'$. Find the distributions of $z'Az$ and $z'Bz$, with their means and variances, and explain why $z'Az$ and $z'Bz$ are independent.

(c) Let $\mu = 0$. Find a basis of the set of $c \in \mathbb{R}^4$ such that $c'z$ is independent of $z'Az$.

Solution. (a)

$$A = \frac{1}{4}\begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{pmatrix}, \qquad B = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$A^2 = \frac{1}{16}v(v'v)v' = \frac{4}{16}vv' = A$, $B^2 = \frac{1}{4}u(u'u)u' = \frac{2}{4}uu' = B$, and $AB = \frac{1}{8}v(v'u)u' = 0$, as $v'u = 1 - 1 + 0 + 0 = 0$. Both are symmetric and idempotent, so their ranks are their traces (Theorem 6.29(c)): $\operatorname{rank} A = \frac{1}{4}v'v = 1$ and $\operatorname{rank} B = \frac{1}{2}u'u = 1$.

(b) $\mu'A\mu = \frac{1}{4}(v'\mu)^2 = \frac{1}{4}(3 + 1 - 2 - 2)^2 = 0$ and $\mu'B\mu = \frac{1}{2}(u'\mu)^2 = \frac{1}{2}(3 - 1)^2 = 2$. By Theorem 11.15, $z'Az \sim \chi_1^2$ and $z'Bz \sim \chi_1^2(2)$; by Theorem 11.4(d), the means are 1 and 3, and the variances 2 and $2 + 8 = 10$. They are independent by Theorem 11.20(a), as $AB = 0$.

(c) By Theorem 11.22(b), $c'z$ is independent of $z'Az$ exactly when $Ac = 0$. Row-reduce $4A = vv'$:

$$\begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 + R_1} \begin{pmatrix} 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

So $c_1 = -c_2 + c_3 + c_4$, and the special solutions $(-1, 1, 0, 0)'$, $(1, 0, 1, 0)'$, $(1, 0, 0, 1)'$ form a basis of $\mathcal{N}(A)$ (Theorem 4.8(c)).

Exercise 11.9. Let $y \sim N_4(\mu, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 3 \end{pmatrix}, \qquad \mu = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \end{pmatrix}$$

(a) Show that Σ is positive definite, and find Σ^{-1} by row-reducing $[\Sigma \mid I_4]$.

(b) Find the distribution of $y'\Sigma^{-1}y$, its mean and its variance.

(c) Let $A = \Sigma^{-1} - e_1e_1'$. Compute $A\Sigma$, check the conditions of Theorem 11.18(d), and find the distribution of $y'Ay$, its mean and its variance.

Solution. (a) Row reduction (the two operations of the first arrow use row 1, those of the second the new row 2, those of the third row 4; each pair changes two different rows):

$$\left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 3 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right) \xrightarrow[R_2 \leftarrow R_2 - R_4]{R_1 \leftarrow R_1 - R_4} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 1 & 1 & 0 & -1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & -2 & 3 & -1 & -1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 3 & -2 & 1 & 0 \\ 0 & 1 & 0 & 0 & -2 & 3 & -1 & -1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{array}\right)$$

So

$$\Sigma^{-1} = \begin{pmatrix} 3 & -2 & 1 & 0 \\ -2 & 3 & -1 & -1 \\ 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

Check: $\Sigma\Sigma^{-1} = I_4$; for instance row 4 of Σ times column 2 of Σ^{-1} is $-2 + 6 - 1 - 3 = 0$. The leading principal minors are $\Delta_1(\Sigma) = 1$, $\Delta_2(\Sigma) = 2 - 1 = 1$ and $\Delta_3(\Sigma) = 1 \cdot (4 - 1) - 1 \cdot (2 - 0) = 1$ (expansion along row 1, Theorem 5.21); and $\Delta_4(\Sigma) = \det \Sigma = 1$, as the first two arrows add multiples of rows to other rows (Definition 5.13(b)) and leave the left block upper triangular with diagonal entries 1 (Theorem 5.14(c)). All are positive, so Σ is positive definite (Theorem 8.34).

(b) For $A = \Sigma^{-1}$, $A\Sigma = I_4$ is idempotent of rank 4. By Theorem 11.18(b), $y'\Sigma^{-1}y \sim \chi^2_4(\mu'\Sigma^{-1}\mu)$; $\Sigma^{-1}\mu = (3-1+0,\ -2+1-1,\ 1+0-1+0,\ 0+0+0+1)' = (2, -2, 0, 1)'$, so $\mu'\Sigma^{-1}\mu = 2+0+0+1 = 3$. The mean is $4 + 3 = 7$, and the variance is $8 + 12 = 20$ (Theorem 11.4(d)).

(c)

$$A = \begin{pmatrix} 2 & -2 & 1 & 0 \\ -2 & 3 & -1 & -1 \\ 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}, \qquad A\Sigma = I_4 - e_1(e_1'\Sigma) = \begin{pmatrix} 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Squaring, $(A\Sigma)^2 = A\Sigma$ (row 1 of the square is -1 times row 2 of $A\Sigma$ plus -1 times its row 4); so $\operatorname{tr}(A\Sigma) = \operatorname{tr}\big((A\Sigma)^2\big) = 3$, and $\operatorname{rank}(A\Sigma) = 3$ (the first column is 0, the other three are independent). By Theorem 11.18(d),(b), $y'Ay \sim \chi^2_3(\mu'A\mu)$, and $A\mu = \Sigma^{-1}\mu - \mu_1 e_1 = (1, -2, 0, 1)'$, so $\mu'A\mu = 1 + 0 + 0 + 1 = 2$. So $y'Ay \sim \chi^2_3(2)$, with mean 5 and variance $6 + 8 = 14$ (Theorem 11.4(d)).

Exercise 11.10. Let $z \sim N_4(\mu, I_4)$ with $\mu = (1,1,1,1)'$, and $A = I_4 + J_4$.

(a) Write $A = 5\bar{J}_4 + C_4$, and give the eigenvalues of A with their multiplicities.

(b) Use Theorem 11.16 to write $z'Az = 5W_1 + W_2$ with independent $W_1 \sim \chi^2_1(\theta_1)$ and $W_2 \sim \chi^2_3(\theta_2)$; find W_1, W_2, θ_1 and θ_2.

(c) Compute $\mathsf{E}(z'Az)$ and $\mathrm{var}(z'Az)$ from (b), and again from Theorem 11.28.

Solution. (a) $I_4 = \bar{J}_4 + C_4$ and $J_4 = 4\bar{J}_4$, so $A = 5\bar{J}_4 + C_4$, a spectral decomposition (Example 8.16(b)): the eigenvalues are 5 (once, eigenvector $\mathbf{1}$) and 1 (three times, eigenspace $\mathrm{span}\{\mathbf{1}\}^\perp$).

(b) Here $\Sigma = I_4$ and $B = A$, with $P_1 = \mathbf{1}/2$ for $\gamma_1 = 5$ and $P_2 = [q_1\ q_2\ q_3]$ of Exercise 11.7 for $\gamma_2 = 1$ ($P_1P_1' = \bar{J}_4$, $P_2P_2' = C_4$). So $W_1 = \|P_1'z\|^2 = z'\bar{J}_4 z = 4\bar{z}^2$ and $W_2 = z'C_4z$, with $\theta_1 = \mu'\bar{J}_4\mu = 4$ and $\theta_2 = \mu'C_4\mu = 0$.

(c) From (b) and Theorem 11.4(d): $\mathsf{E}(z'Az) = 5(1+4) + (3+0) = 28$, and, as W_1 and W_2 are independent, $\mathrm{var}(z'Az) = 25(2+16) + (6+0) = 456$. From Theorem 11.28: $\mathrm{tr}\,A = 8$ and $\mu'A\mu = \mathbf{1}'(I_4 + J_4)\mathbf{1} = 4 + 16 = 20$, so the mean is 28; $A^2 = I_4 + 2J_4 + J_4^2 = I_4 + 6J_4$, so $\mathrm{tr}(A^2) = 28$ and $\mu'A^2\mu = 4 + 96 = 100$, and the variance is $56 + 400 = 456$.

Exercise 11.11. Let $y \sim N_4(\mu, \Sigma)$ with Σ and μ as in Exercise 11.9, and let

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so that $y'Ay = y_1^2 - y_4^2$ and $y'By = 2y_2y_3$.

(a) Compute $\mathsf{E}(y'Ay)$ and $\mathrm{var}(y'Ay)$.

(b) Compute $\mathsf{E}(y'By)$ and $\mathrm{var}(y'By)$.

(c) Compute $\mathrm{cov}(y'Ay, y'By)$, and find $\mathrm{var}\big(y'(A+B)y\big)$ in two ways.

Solution. (a)

$$A\Sigma = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & -2 & -1 & -3 \end{pmatrix}, \qquad (A\Sigma)^2 = \begin{pmatrix} 0 & -1 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 5 & 3 & 8 \end{pmatrix}$$

so $\mathrm{tr}(A\Sigma) = -2$ and $\mathrm{tr}\big((A\Sigma)^2\big) = 8$. Also $\mu'A\mu = 1 - 1 = 0$, $A\mu = (1,0,0,-1)'$, $A\Sigma A\mu = (0,0,0,2)'$ and $\mu'A\Sigma A\mu = 2$. By Theorem 11.28, $\mathsf{E}(y'Ay) = -2$ and $\mathrm{var}(y'Ay) = 16 + 8 = 24$.

(b)

$$B\Sigma = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad (B\Sigma)^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 5 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $\mathrm{tr}(B\Sigma) = 2$ and $\mathrm{tr}\big((B\Sigma)^2\big) = 10$. Also $\mu'B\mu = 2\mu_2\mu_3 = 0$, $B\mu = (0,-1,0,0)'$, $B\Sigma B\mu = (0,-1,-2,0)'$ and $\mu'B\Sigma B\mu = 2$. So $\mathsf{E}(y'By) = 2$ and $\mathrm{var}(y'By) = 20 + 8 = 28$.

(c)

$$A\Sigma B\Sigma = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & -4 & -5 & -4 \end{pmatrix}$$

has the trace -4, and $A\Sigma B\mu = A\Sigma(0,-1,0,0)' = (-1,0,0,2)'$, so $\mu' A\Sigma B\mu = -1+2 = 1$. By Corollary 11.29, $\operatorname{cov}(y'Ay, y'By) = -8+4 = -4$, hence $\operatorname{var}\Big(y'(A+B)y\Big) = 24+28+2\cdot(-4) = 44$. Directly,

$$(A+B)\Sigma = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ -1 & -2 & -1 & -3 \end{pmatrix}, \qquad \Big((A+B)\Sigma\Big)^2 = \begin{pmatrix} 0 & 0 & 1 & -1 \\ 1 & 3 & 3 & 2 \\ 0 & 1 & 3 & -1 \\ 1 & 1 & -2 & 4 \end{pmatrix}$$

with the trace 10, and $(A+B)\mu = (1,-1,0,-1)'$, $(A+B)\Sigma(A+B)\mu = (-1,-2,-3,4)'$, $\mu'(A+B)\Sigma(A+B)\mu = -1+3+4 = 6$; so $\operatorname{var}\Big(y'(A+B)y\Big) = 20+24 = 44$ (Theorem 11.28).

Problems

Problem 11.1 ([17, Exercise 5.13], [20, Exercise 5.13])**.** Let $x_1, \dots, x_n$ ($n \geqslant 2$) be independent $N(\mu, \sigma^2)$ variables ($\sigma > 0$) with sample variance s^2. Find a constant c such that $\mathsf{E}(c\sqrt{s^2}) = \sigma$.

Solution.

- $q = (n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$ (Theorem 11.11(b)), and $c\sqrt{s^2} = c\sqrt{\sigma^2/(n-1)}\,\sqrt{q}$.
- q has the density f_{n-1} (Theorem 11.4(b)); so, by the paragraph before Theorem 10.8 and by (D2) with $n/2$ and $\frac{1}{2}$ in place of c and b,

$$\mathsf{E}(\sqrt{q}) = \int_0^\infty \sqrt{x}\,\frac{x^{(n-1)/2-1}e^{-x/2}}{\Gamma\Big(\frac{n-1}{2}\Big)2^{(n-1)/2}}\,\mathrm{d}x = \frac{\Gamma\Big(\frac{n}{2}\Big)2^{n/2}}{\Gamma\Big(\frac{n-1}{2}\Big)2^{(n-1)/2}} = \frac{\sqrt{2}\,\Gamma\Big(\frac{n}{2}\Big)}{\Gamma\Big(\frac{n-1}{2}\Big)}$$

- So $\mathsf{E}(c\sqrt{s^2}) = c\,\sigma\sqrt{2/(n-1)}\;\Gamma\Big(\frac{n}{2}\Big)/\Gamma\Big(\frac{n-1}{2}\Big)$, which is σ for

$$c = \frac{\Gamma\Big(\frac{n-1}{2}\Big)\sqrt{n-1}}{\sqrt{2}\,\Gamma\Big(\frac{n}{2}\Big)}$$

  For instance, for $n=2$, $c = \Gamma(\frac{1}{2})/\Big(\sqrt{2}\,\Gamma(1)\Big) = \sqrt{\pi/2}$.

Problem 11.2 ([17, Exercise 5.15], [20, Exercise 5.15])**.** Let $\bar{x}_n$ and s_n^2 be the mean and the sample variance of numbers $x_1, \dots, x_n$ ($n \geqslant 2$), and let a further number x_{n+1} be added. Show that

(a) $\bar{x}_{n+1} = (x_{n+1} + n\bar{x}_n)/(n+1)$;

(b) $ns_{n+1}^2 = (n-1)s_n^2 + \frac{n}{n+1}(x_{n+1} - \bar{x}_n)^2$.

Solution. (a) $\bar{x}_{n+1} = \frac{1}{n+1}\Big(\sum_{i\leqslant n} x_i + x_{n+1}\Big) = \frac{1}{n+1}(n\bar{x}_n + x_{n+1})$.

(b)

- By (a), $\bar{x}_{n+1} = \bar{x}_n + d$ with $d = (x_{n+1} - \bar{x}_n)/(n+1)$. So $ns_{n+1}^2 = \sum_{i=1}^{n+1}(x_i - \bar{x}_{n+1})^2 = \sum_{i=1}^{n+1}\Big((x_i - \bar{x}_n) - d\Big)^2$.
- Expanding, this is $\sum_{i=1}^{n+1}(x_i - \bar{x}_n)^2 - 2d\sum_{i=1}^{n+1}(x_i - \bar{x}_n) + (n+1)d^2$. As $\sum_{i\leqslant n}(x_i - \bar{x}_n) = 0$, the middle sum is $x_{n+1} - \bar{x}_n = (n+1)d$, and the first sum is $(n-1)s_n^2 + (x_{n+1} - \bar{x}_n)^2$.
- Hence $ns_{n+1}^2 = (n-1)s_n^2 + (x_{n+1} - \bar{x}_n)^2 - (n+1)d^2 = (n-1)s_n^2 + \Big(1 - \frac{1}{n+1}\Big)(x_{n+1} - \bar{x}_n)^2$.

Problem 11.3 ([17, Exercise 5.16], [20, Exercise 5.16])**.** Let x_1, x_2, x_3 be independent, with $x_i \sim N(i, i^2)$. Use them to construct a statistic with the distribution

(a) χ^2_3;

(b) t_2;

(c) $F_{1,2}$.

Solution. Let $z_i = (x_i - i)/i$. By Theorem 10.9(a), $x \sim N_3(\mu, D)$ with $\mu = (1, 2, 3)'$ and $D = \operatorname{diag}(1, 4, 9)$, so $z = D^{-1/2}(x - \mu) \sim N_3(0, I_3)$ (Theorem 10.2): z_1, z_2, z_3 are independent $N(0, 1)$ variables (Theorem 10.9(a)), and z_1 and $(z_2, z_3)'$ are independent (Theorem 10.5(a)).

(a) $z_1^2 + z_2^2 + z_3^2 = \sum_{i=1}^3 \Big((x_i - i)/i\Big)^2 \sim \chi^2_3$ (Definition 11.2).

(b) $z_2^2 + z_3^2 \sim \chi^2_2$ is independent of z_1 (N3), so

$$\frac{z_1}{\sqrt{(z_2^2 + z_3^2)/2}} = \frac{x_1 - 1}{\sqrt{\Big(\frac{1}{4}(x_2 - 2)^2 + \frac{1}{9}(x_3 - 3)^2\Big)/2}} \sim t_2$$

(Definition 11.7).

(c) The square of the statistic in (b), $z_1^2 / \Big((z_2^2 + z_3^2)/2\Big)$, has the distribution $F_{1,2}$ (Theorem 11.8(b)).

Problem 11.4 ([17, Exercise 5.17(b)], [20, Exercise 5.17(b)])**.** Let $F \sim F_{p,q}$. Show that

$$\mathsf{E}(F) = \frac{q}{q-2} \quad (q > 2), \qquad \operatorname{var}(F) = 2\Big(\frac{q}{q-2}\Big)^2 \frac{p+q-2}{p(q-4)} \quad (q > 4)$$

Solution.

- Write $F = (U/p)/(V/q)$ with $U \sim \chi^2_p$ and $V \sim \chi^2_q$ independent; $V > 0$ with probability 1, as $f_q = 0$ on $(-\infty, 0]$. For $k, l \in \{1, 2\}$, U^k and V^{-l} are independent (N3), so $\mathsf{E}(U^k V^{-l}) = \mathsf{E}(U^k)\,\mathsf{E}(V^{-l})$ (N7), where $\mathsf{E}(V^{-l}) = \int_0^\infty v^{-l} f_q(v)\,\mathrm{d}v$ (the paragraph before Theorem 10.8), whenever these expectations are finite.
- By (D2), for $q > 2l$, $\int_0^\infty v^{-l} f_q(v)\,\mathrm{d}v = \frac{\Gamma(q/2-l)\,2^{q/2-l}}{\Gamma(q/2)\,2^{q/2}}$. With $\Gamma(q/2) = (q/2-1)\Gamma(q/2-1)$ and $\Gamma(q/2-1) = (q/2-2)\Gamma(q/2-2)$ (D2), this gives $\mathsf{E}(V^{-1}) = \frac{1}{q-2}$ for $q > 2$ and $\mathsf{E}(V^{-2}) = \frac{1}{(q-2)(q-4)}$ for $q > 4$.
- $\mathsf{E}(U) = p$ and $\mathsf{E}(U^2) = \operatorname{var}(U) + p^2 = 2p + p^2$ (Theorem 11.4(d)).
- So $\mathsf{E}(F) = \frac{q}{p} \cdot p \cdot \frac{1}{q-2} = \frac{q}{q-2}$ and $\mathsf{E}(F^2) = \frac{q^2}{p^2} \cdot p(p+2) \cdot \frac{1}{(q-2)(q-4)}$, whence

$$\operatorname{var}(F) = \frac{q^2(p+2)}{p(q-2)(q-4)} - \frac{q^2}{(q-2)^2} = \frac{q^2}{(q-2)^2} \cdot \frac{(p+2)(q-2) - p(q-4)}{p(q-4)} = 2\Big(\frac{q}{q-2}\Big)^2 \frac{p+q-2}{p(q-4)}$$

Problem 11.5 ([17, Exercise 5.18(a)], [20, Exercise 5.18(a)])**.** Let $T \sim t_p$. Show that $\mathsf{E}(T) = 0$ if $p > 1$, and $\operatorname{var}(T) = p/(p-2)$ if $p > 2$.

Solution.

- Write $T = Z/\sqrt{V/p}$ with $Z \sim N(0, 1)$ and $V \sim \chi^2_p$ independent. As in Problem 11.4, Z and $V^{-1/2}$ are independent (N3), so $\mathsf{E}(T) = \sqrt{p}\,\mathsf{E}(Z)\,\mathsf{E}(V^{-1/2})$ (N7), where $\mathsf{E}(Z) = 0$ and $\mathsf{E}(V^{-1/2}) = \int_0^\infty v^{-1/2} f_p(v)\,\mathrm{d}v = \frac{\Gamma((p-1)/2)\,2^{(p-1)/2}}{\Gamma(p/2)\,2^{p/2}}$ is finite for $p > 1$ (D2). So $\mathsf{E}(T) = 0$.
- $T^2 \sim F_{1,p}$ (Theorem 11.8(b)), so for $p > 2$, $\operatorname{var}(T) = \mathsf{E}(T^2) = p/(p-2)$, the mean of $F_{1,p}$ (Problem 11.4).

Problem 11.6 ([17, Exercise 5.19], [20, Exercise 5.19])**.** A random variable u is **stochastically larger** than v if $\mathsf{P}(u > c) \geqslant \mathsf{P}(v > c)$ for every c, with strict inequality for some c.

(a) Let $p > q \geqslant 1$, $u \sim \chi^2_p$ and $v \sim \chi^2_q$. Show that $\mathsf{P}(u > c) > \mathsf{P}(v > c)$ for every $c > 0$, so that χ^2_p is stochastically larger than χ^2_q.

(b) Let $k_1 > k_2 \geqslant 1$, $F_1 \sim F_{k_1,\nu}$ and $F_2 \sim F_{k_2,\nu}$. Show that $\mathsf{P}(k_1F_1 > c) > \mathsf{P}(k_2F_2 > c)$ for every $c > 0$, so that k_1F_1 is stochastically larger than k_2F_2.

(c) Show that $kF_{k,\nu,1-\alpha} > (k-1)F_{k-1,\nu,1-\alpha}$ for $k \geqslant 2$ and $\alpha \in (0,1)$.

Solution. (a)

- The two probabilities depend only on the distributions, so we may take $v = g_1^2 + \cdots + g_q^2$ and put $d = g_{q+1}^2 + \cdots + g_p^2$ for $g \sim N_p(0, I_p)$ (Theorem 10.8(b)): $v \sim \chi^2_q$ and $d \sim \chi^2_{p-q}$ (Definition 11.2) are independent (Theorem 10.5(a), (N3)). Then $v + d \sim \chi^2_p$ (Theorem 11.4(c)), so $\mathsf{P}(u > c) = \mathsf{P}(v + d > c)$.
- For $c > 0$, $\mathsf{P}(v + d > c) - \mathsf{P}(v > c) = \mathsf{P}(v \leqslant c < v + d)$, as $d > 0$ with probability 1. As v and d are independent, this is the integral of their joint density $f_q(x)f_{p-q}(y)$ (the paragraph before Theorem 10.8), which is positive on $(0,\infty)^2$, over the region $\{(x,y) : 0 < x \leqslant c,\ y > c - x\}$; so it is positive. For $c \leqslant 0$ both probabilities are 1.

(b)

- Let $g \sim N_m(0, I_m)$ with $m = k_1 + \nu$ (Theorem 10.8(b)), and let $v = g_1^2 + \cdots + g_{k_2}^2$, $d = g_{k_2+1}^2 + \cdots + g_{k_1}^2$ and $w = g_{k_1+1}^2 + \cdots + g_m^2$. By Definition 11.2, $v \sim \chi^2_{k_2}$, $d \sim \chi^2_{k_1-k_2}$, $v + d \sim \chi^2_{k_1}$ and $w \sim \chi^2_\nu$.
- The three blocks of g of lengths k_2, $k_1 - k_2$ and ν are independent (Theorem 10.5(b)), and so are the vectors of its first k_1 and of its last ν coordinates (Theorem 10.5(a)). So v, d, w are independent, and so are $v + d$ and w (N3). By Definition 11.7, k_1F_1 has the distribution of $(v + d)/(w/\nu)$, and k_2F_2 that of $v/(w/\nu)$.
- For $c > 0$, $\mathsf{P}\big((v + d)/(w/\nu) > c\big) - \mathsf{P}\big(v/(w/\nu) > c\big) = \mathsf{P}(v \leqslant cw/\nu < v + d) > 0$, as in (a), with the density $f_{k_2}(x)f_{k_1-k_2}(y)f_\nu(s)$ of $(v, d, w)'$ and the region $\{(x, y, s) : 0 < x \leqslant cs/\nu < x + y\}$; for $c \leqslant 0$ both probabilities are 1.

(c) Let $F_k \sim F_{k,\nu}$, $F_{k-1} \sim F_{k-1,\nu}$ and $c = (k-1)F_{k-1,\nu,1-\alpha}$; $c > 0$ (Definition 11.7). Then $\mathsf{P}(kF_k > kF_{k,\nu,1-\alpha}) = \alpha = \mathsf{P}\big((k-1)F_{k-1} > c\big)$, and by (b) with $k_1 = k$, $k_2 = k - 1$, $\mathsf{P}(kF_k > c) > \alpha$. As $x \mapsto \mathsf{P}(kF_k > x)$ is decreasing, $c < kF_{k,\nu,1-\alpha}$.

Problem 11.7 ([17, Exercise 5.22], [20, Exercise 5.22])**.** Let x and y be independent $N(0,1)$ variables, and $z = \min(x, y)$. Show that $z^2 \sim \chi^2_1$.

Solution. Let $\Phi(a) = \mathsf{P}(x \leqslant a) = \int_{-\infty}^a \varphi$. As $\varphi(-s) = \varphi(s)$, substituting $s \mapsto -s$ (D1) gives $\mathsf{P}(x \geqslant a) = \int_a^\infty \varphi = \Phi(-a)$; and y has the same distribution function. For $c > 0$, using the independence of x and y,

$$\begin{aligned}\mathsf{P}(z^2 \leqslant c) &= \mathsf{P}(z \geqslant -\sqrt{c}) - \mathsf{P}(z > \sqrt{c}) = \mathsf{P}(x \geqslant -\sqrt{c})\,\mathsf{P}(y \geqslant -\sqrt{c}) - \mathsf{P}(x > \sqrt{c})\,\mathsf{P}(y > \sqrt{c}) \\ &= \Phi(\sqrt{c})^2 - \Phi(-\sqrt{c})^2 = \Big(\Phi(\sqrt{c}) - \Phi(-\sqrt{c})\Big)\Big(\Phi(\sqrt{c}) + \Phi(-\sqrt{c})\Big) = \mathsf{P}(x^2 \leqslant c)\end{aligned}$$

since $\Phi(\sqrt{c}) + \Phi(-\sqrt{c}) = 1$ and $\Phi(\sqrt{c}) - \Phi(-\sqrt{c}) = \mathsf{P}(-\sqrt{c} \leqslant x \leqslant \sqrt{c})$. For $c \leqslant 0$ both probabilities are 0. So z^2 and x^2 have the same distribution function, and hence the same distribution (D3). As $x^2 \sim \chi^2_1$ (Definition 11.2), $z^2 \sim \chi^2_1$.

Problem 11.8 ([2, Exercise 10.11])**.** Let $u = (x', y')' \sim N_{m+n}(\mu, V)$ with $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$, partitioned accordingly as

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \qquad V = \begin{pmatrix} V_{11} & V_{12} \\ V_{12}' & V_{22} \end{pmatrix}$$

and let $A \in \mathbb{R}^{m \times n}$. Find

(a) $\mathsf{E}(x'Ay)$;

(b) $\operatorname{var}(x'Ay)$.

Solution. $x'Ay = \frac{1}{2}u'Bu$ with the symmetric matrix $B \in \mathbb{R}^{(m+n)\times(m+n)}$ below, since $u'Bu = x'Ay + y'A'x = 2x'Ay$ (Theorem 5.3(c)); by the block product,

$$B = \begin{pmatrix} 0 & A \\ A' & 0 \end{pmatrix}, \qquad BV = \begin{pmatrix} AV_{12}' & AV_{22} \\ A'V_{11} & A'V_{12} \end{pmatrix}, \qquad B\mu = \begin{pmatrix} A\mu_2 \\ A'\mu_1 \end{pmatrix}$$

(a) By Theorem 11.28, $\mathsf{E}(x'Ay) = \frac{1}{2}\big(\operatorname{tr}(BV) + \mu'B\mu\big)$. Here $\operatorname{tr}(BV) = \operatorname{tr}(AV_{12}') + \operatorname{tr}(A'V_{12}) = 2\operatorname{tr}(AV_{12}')$ (Theorem 1.11(a),(b)) and $\mu'B\mu = \mu_1'A\mu_2 + \mu_2'A'\mu_1 = 2\mu_1'A\mu_2$; so $\mathsf{E}(x'Ay) = \mu_1'A\mu_2 + \operatorname{tr}(AV_{12}')$.

(b) By Theorem 11.28, $\operatorname{var}(x'Ay) = \frac{1}{4}\big(2\operatorname{tr}((BV)^2) + 4\mu'BVB\mu\big)$.

- The diagonal blocks of $(BV)^2$ are $AV_{12}'AV_{12}' + AV_{22}A'V_{11}$ and $A'V_{11}AV_{22} + A'V_{12}A'V_{12}$. As $\operatorname{tr}(A'V_{11}AV_{22}) = \operatorname{tr}(AV_{22}A'V_{11})$ and $\operatorname{tr}(A'V_{12}A'V_{12}) = \operatorname{tr}(V_{12}'AV_{12}'A) = \operatorname{tr}(AV_{12}'AV_{12}')$ (Theorem 1.11), $\frac{1}{2}\operatorname{tr}\big((BV)^2\big) = \operatorname{tr}(AV_{12}'AV_{12}') + \operatorname{tr}(AV_{22}A'V_{11})$.
- $\mu'BVB\mu = (B\mu)'V(B\mu) = \mu_2'A'V_{11}A\mu_2 + 2\mu_2'A'V_{12}A'\mu_1 + \mu_1'AV_{22}A'\mu_1$, and the middle number equals its transpose $\mu_1'AV_{12}'A\mu_2$.
- Hence $\operatorname{var}(x'Ay) = \operatorname{tr}(AV_{12}'AV_{12}') + \operatorname{tr}(AV_{22}A'V_{11}) + \mu_1'AV_{22}A'\mu_1 + \mu_2'A'V_{11}A\mu_2 + 2\mu_1'AV_{12}'A\mu_2$.

Problem 11.9 ([3, Ch. 8, Exercise 16])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric and idempotent, with $A - B$ positive semidefinite.

(a) Show that $A - B$ is idempotent.

(b) Let $z \sim N_n(0, I_n)$ and $A \neq B$. Show that $z'Az - z'Bz \sim \chi^2_{\operatorname{rank} A - \operatorname{rank} B}$.

Solution. (a)

- Let $r = \operatorname{rank} A$. If $r = 0$, then $A = 0$, and both $-B$ and $B = B'B$ are positive semidefinite (Theorem 8.25); so $x'Bx = 0$ for every x, $B = 0$ (Proposition 8.18(b)), and $A - B = 0$. If $r = n$, then $A = I_n$ (Theorem 8.13(a), Theorem 8.3), and $A - B = I_n - B$ is idempotent (Theorem 6.29(b)). Let $1 \leqslant r < n$.
- By Theorem 8.13(c), $P'AP = \operatorname{diag}(I_r, 0)$ for an orthogonal P, so $A = P\operatorname{diag}(I_r, 0)P'$. Put $\tilde{B} = P'BP$: it is symmetric, idempotent ($\tilde{B}^2 = P'B^2P = \tilde{B}$) and positive semidefinite ($\tilde{B} = \tilde{B}'\tilde{B}$, Theorem 8.25), and $\operatorname{diag}(I_r, 0) - \tilde{B} = P'(A - B)P$ is positive semidefinite (Theorem 8.24(a)). Partition

$$\tilde{B} = \begin{pmatrix} B_{11} & B_{12} \\ B_{12}' & B_{22} \end{pmatrix}, \qquad B_{11} \in \mathbb{R}^{r\times r}$$

- B_{22} and $-B_{22}$ are principal submatrices of positive semidefinite matrices, hence positive semidefinite (Theorem 8.30); so $B_{22} = 0$ (Proposition 8.18(b)).
- Then $B_{12} = 0$: for $s \in \mathbb{R}$, $e \in \mathbb{R}^r$ and $f \in \mathbb{R}^{n-r}$, the vector $x = (se', f')'$ gives $0 \leqslant x'\tilde{B}x = s^2e'B_{11}e + 2s\,e'B_{12}f$, which is negative for small s of the sign opposite to $e'B_{12}f$ unless $e'B_{12}f = 0$; taking for e and f the columns of identity matrices gives $B_{12} = 0$.

- So $\tilde{B} = \operatorname{diag}(B_{11}, 0)$, and $\tilde{B}^2 = \tilde{B}$ gives $B_{11}^2 = B_{11}$ (Proposition 5.6(b)). Then $I_r - B_{11}$ is idempotent (Theorem 6.29(b)), hence so is $P'(A-B)P = \operatorname{diag}(I_r - B_{11}, 0)$, and $(A-B)^2 = P \operatorname{diag}(I_r - B_{11}, 0)^2 P' = A - B$.

(b) $A - B$ is symmetric and idempotent by (a), and $A - B \neq 0$; so $\operatorname{rank}(A-B) = \operatorname{tr}(A-B) = \operatorname{tr} A - \operatorname{tr} B = \operatorname{rank} A - \operatorname{rank} B \geqslant 1$ (Theorems 6.29(c) and 1.11(a)). By Theorem 11.15 (with $\mu = 0$ and $\sigma = 1$), $z'Az - z'Bz = z'(A-B)z \sim \chi^2_{\operatorname{rank} A - \operatorname{rank} B}$.

Chapter 12. Generalized inverses

Generalized inverses

Definition 12.1 ([3, §4.1], [5, §9.1, Lemma 9.1.1]). Let $A \in \mathbb{R}^{m\times n}$. A matrix $G \in \mathbb{R}^{n\times m}$ with $AGA = A$ is a **generalized inverse** (or **g-inverse**) of A; the symbol A^- denotes any g-inverse of A. If $A \in \mathbb{R}^{n\times n}$ is invertible, then A^{-1} is a g-inverse of A, and it is the only one: $AGA = A$ gives $G = A^{-1}(AGA)A^{-1} = A^{-1}AA^{-1} = A^{-1}$.

Theorem 12.2 ([3, Result 4.1], [5, Lemma 9.5.1]). Let $A \in \mathbb{R}^{m\times n}$ and $G \in \mathbb{R}^{n\times m}$.

(a) G is a g-inverse of A if and only if, for every $y \in \mathcal{C}(A)$, $x = Gy$ is a solution of $Ax = y$.

(b) If G is a g-inverse of A and $y \in \mathbb{R}^m$, then $Ax = y$ is consistent if and only if $AGy = y$.

Proof. (a)

- Let $AGA = A$ and $y \in \mathcal{C}(A)$, say $y = Az$. Then $A(Gy) = AGAz = Az = y$.
- Conversely, let $AGy = y$ for every $y \in \mathcal{C}(A)$. For $z \in \mathbb{R}^n$, $y = Az$ lies in $\mathcal{C}(A)$, so $AGAz = Az$. With $z = e_j$, column j of AGA equals column j of A (Theorem 1.4(b)), for $j = 1, \ldots, n$; so $AGA = A$.

(b) $Ax = y$ is consistent if and only if $y \in \mathcal{C}(A)$ (Corollary 4.14(a)). If $y \in \mathcal{C}(A)$, then $AGy = y$ by (a). Conversely, if $AGy = y$, then $x = Gy$ is a solution. □

Theorem 12.3 ([3, §4.1], [5, Theorem 9.1.3, Corollary 9.1.4]). Let $A \in \mathbb{R}^{m\times n}$ have rank r.

(a) Let $r \geqslant 1$, let $A = BC$ be a full-rank factorization ($B \in \mathbb{R}^{m\times r}$, $C \in \mathbb{R}^{r\times n}$), and let $L \in \mathbb{R}^{r\times m}$ and $K \in \mathbb{R}^{n\times r}$ satisfy $LB = I_r$ and $CK = I_r$. Then $G = KL$ is a g-inverse of A. Such L and K exist, for instance $L = (B'B)^{-1}B'$ and $K = C'(CC')^{-1}$.

(b) If $r = 0$, every $G \in \mathbb{R}^{n\times m}$ is a g-inverse of A.

In particular, every matrix has a g-inverse.

Proof. (a) $AGA = BC\,KL\,BC = B(CK)(LB)C = BI_rI_rC = BC = A$. The two matrices given satisfy $LB = I_r$ and $CK = I_r$ (Theorem 4.21(c)).

(b) Then $A = 0$ (Theorem 4.8), and $AGA = 0 = A$ for every G.

Every A with $r \geqslant 1$ has a full-rank factorization (Theorem 4.21(a)), so (a) and (b) give a g-inverse of every matrix. □

Theorem 12.4 ([3, §4.1, Result 5.18]). Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$, and let $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$ be invertible with $PAQ = N_r = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$ (Theorem 4.31). Then $G \in \mathbb{R}^{n\times m}$ is a g-inverse of A if and only if

$$G = Q \begin{pmatrix} I_r & U \\ V & W \end{pmatrix} P$$

for some $U \in \mathbb{R}^{r\times(m-r)}$, $V \in \mathbb{R}^{(n-r)\times r}$ and $W \in \mathbb{R}^{(n-r)\times(m-r)}$ (blocks with no rows or no columns are absent). In particular, unless $r = m = n$, the matrix A has infinitely many g-inverses.

Proof. Let $G \in \mathbb{R}^{n\times m}$, put $H = Q^{-1}GP^{-1} \in \mathbb{R}^{n\times m}$, so that $G = QHP$, and partition $H = \left(\begin{smallmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{smallmatrix}\right)$ with $H_{11} \in \mathbb{R}^{r\times r}$.

- By the block product (Theorem 5.3(c)), $N_rH = \left(\begin{smallmatrix} H_{11} & H_{12} \\ 0 & 0 \end{smallmatrix}\right)$ and

$$N_rHN_r = \begin{pmatrix} H_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

- $A = P^{-1}N_rQ^{-1}$, so $AGA = P^{-1}N_rQ^{-1}QHPP^{-1}N_rQ^{-1} = P^{-1}N_rHN_rQ^{-1}$. Multiplying by P on the left and by Q on the right, and back by P^{-1} and Q^{-1}, shows that $AGA = A$ if and only if $N_rHN_r = N_r$, that is, $H_{11} = I_r$. Then $U = H_{12}$, $V = H_{21}$ and $W = H_{22}$ are arbitrary.
- If $QH_1P = QH_2P$, then $H_1 = H_2$ (multiply by Q^{-1} on the left and by P^{-1} on the right); so different blocks U, V, W give different g-inverses. If $r < m$, the block U is present and its $(1,1)$ entry can be any number; if $r < n$, likewise V. So A has infinitely many g-inverses unless $r = m = n$. □

Corollary 12.5 (cf. [3, §4.1])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$, and let row operations take $[A \mid I_m]$ to $[R \mid P]$, where R is the RREF of A, with pivot columns $k_1 < \cdots < k_r$ (Theorem 4.8). Let $G \in \mathbb{R}^{n\times m}$ have row k_s equal to row s of P $(s = 1, \ldots, r)$ and all other rows zero. Then G is a g-inverse of A.

Proof.
- Let $B = [a_{k_1} \ \cdots \ a_{k_r}]$, and let C consist of the first r rows of R. Then $A = BC$ is a full-rank factorization (Theorem 4.21(a)).
- $PA = R$ (Theorem 4.8), so $Pa_{k_s} = \mathrm{col}_{k_s}(R) = e_s \in \mathbb{R}^m$ (Definition 2.4). Hence $PB = [e_1 \ \cdots \ e_r]$ (Theorem 1.4(a)), and the first r rows of P form a matrix $L \in \mathbb{R}^{r\times m}$ with $LB = I_r$ (Theorem 1.4(c)).
- Let $K = [e_{k_1} \ \cdots \ e_{k_r}] \in \mathbb{R}^{n\times r}$. Column s of CK is $\mathrm{col}_{k_s}(C)$, the sth column of I_r (Definition 2.4); so $CK = I_r$.
- By the outer product expansion (Theorem 1.4(d)), $KL = e_{k_1}\,\mathrm{row}_1(L) + \cdots + e_{k_r}\,\mathrm{row}_r(L)$ has row k_s equal to $\mathrm{row}_s(L) = \mathrm{row}_s(P)$ and all other rows zero: $KL = G$. By Theorem 12.3(a), G is a g-inverse of A. □

Theorem 12.6 ([5, Theorem 4.4.10], [3, §4.1], [2, §8.2.2])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$. For rows $i_1 < \cdots < i_r$ and columns $j_1 < \cdots < j_r$ of A, let $A_* \in \mathbb{R}^{r\times r}$ be the submatrix of A in these rows and columns: its (s,t) entry is $a_{i_sj_t}$.

(a) A has r linearly independent rows and r linearly independent columns. If the rows $i_1, \ldots, i_r$ and the columns $j_1, \ldots, j_r$ of A are linearly independent, then A_* is invertible.

(b) Let A_* be invertible, and let $G \in \mathbb{R}^{n\times m}$ have (j_s, i_t) entry equal to the (s,t) entry of A_*^{-1} $(s,t = 1, \ldots, r)$ and all other entries 0. Then G is a g-inverse of A. In particular, if $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ with $A_{11} \in \mathbb{R}^{r\times r}$ invertible, then $G = \left(\begin{smallmatrix} A_{11}^{-1} & 0 \\ 0 & 0 \end{smallmatrix}\right) \in \mathbb{R}^{n\times m}$ is a g-inverse of A.

Proof. (a)
- The pivot columns of A are r linearly independent columns (Theorem 4.8(a)). As $\mathrm{rank}\, A' = r$ (Theorem 4.13(a)), the pivot columns of A' are r linearly independent columns of A', that is, transposed rows of A.
- Let the rows $i_1, \ldots, i_r$ and the columns $j_1, \ldots, j_r$ be linearly independent. The vectors $\mathrm{row}_{i_1}(A)', \ldots, \mathrm{row}_{i_r}(A)'$ are r linearly independent vectors of $\mathcal{C}(A')$, of dimension r; so they form a basis of it (Theorem 4.5(d)), and $\mathrm{row}_t(A) = f_{t1}\,\mathrm{row}_{i_1}(A) + \cdots + f_{tr}\,\mathrm{row}_{i_r}(A)$ $(t = 1, \ldots, m)$ for some numbers f_{tk}.
- Let $U \in \mathbb{R}^{m\times r}$ consist of the columns $j_1, \ldots, j_r$ of A, and $F = (f_{tk}) \in \mathbb{R}^{m\times r}$. Entry j_s of the last equation reads $u_{ts} = a_{tj_s} = \sum_{k=1}^{r} f_{tk}a_{i_kj_s}$; that is, $U = FA_*$.
- The columns of U are linearly independent, so $r = \mathrm{rank}\, U = \mathrm{rank}(FA_*) \leqslant \mathrm{rank}\, A_* \leqslant r$ (Theorems 4.13(a),(b) and 4.15(a)), and A_* is invertible (Theorem 4.13(d)).

(b) First the leading case $A_* = A_{11}$ $(i_s = j_s = s)$; blocks with no rows or no columns are absent when $r = m$ or $r = n$.
- The columns of $\left(\begin{smallmatrix} A_{11} \\ A_{21} \end{smallmatrix}\right)$ are linearly independent: $\left(\begin{smallmatrix} A_{11} \\ A_{21} \end{smallmatrix}\right) z = 0$ gives $A_{11}z = 0$, so $z = 0$. They are $r = \dim \mathcal{C}(A)$ vectors of $\mathcal{C}(A)$, hence a basis of it (Theorem 4.5(d)).

- So each of the other columns is a combination of them: $\left(\begin{smallmatrix}A_{12}\\A_{22}\end{smallmatrix}\right) = \left(\begin{smallmatrix}A_{11}\\A_{21}\end{smallmatrix}\right) X$ for some $X \in \mathbb{R}^{r\times(n-r)}$ (Theorem 1.4(a),(b)), that is, $A_{12} = A_{11}X$ and $A_{22} = A_{21}X$ (Theorem 5.3(c)). Then $X = A_{11}^{-1}A_{12}$ and $A_{22} = A_{21}A_{11}^{-1}A_{12}$.
- With $F = \left(\begin{smallmatrix}A_{11}^{-1} & 0\\ 0 & 0\end{smallmatrix}\right) \in \mathbb{R}^{n\times m}$, the block product (Theorem 5.3(c)) gives

$$AF = \begin{pmatrix} I_r & 0 \\ A_{21}A_{11}^{-1} & 0 \end{pmatrix}, \qquad AFA = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{21}A_{11}^{-1}A_{12} \end{pmatrix} = A$$

Now the general case.

- Extend $i_1, \dots, i_r$ by the other row indices, in increasing order, to a permutation φ of $1, \dots, m$, and $j_1, \dots, j_r$ likewise to a permutation ψ of $1, \dots, n$; let $P = P_\varphi$ and $Q = P'_\psi$ (Definition 5.40). Then $\operatorname{row}_s(P) = e'_{i_s}$ and $\operatorname{col}_t(Q) = e_{j_t}$ for $s, t \leqslant r$, so the (s,t) entry of PAQ is $e'_{i_s}Ae_{j_t} = a_{i_s j_t}$ (Theorem 1.4(c),(d)): the leading $r \times r$ block of PAQ is A_*.
- $\operatorname{rank}(PAQ) = r$ (Theorem 4.15(d)), so by the leading case $F = \left(\begin{smallmatrix}A_*^{-1} & 0\\ 0 & 0\end{smallmatrix}\right) \in \mathbb{R}^{n\times m}$ is a g-inverse of PAQ.
- $P'P = PP' = I_m$ and $Q'Q = QQ' = I_n$ (Proposition 5.41(c)), so $A = P'(PAQ)Q'$ and

$$A(QFP)A = P'(PAQ)Q'QFPP'(PAQ)Q' = P'(PAQ)F(PAQ)Q' = P'(PAQ)Q' = A$$

- Finally $QFP = G$. Let f_{st} be the entries of F; they are 0 unless $s, t \leqslant r$. Column t of QF is $Q\operatorname{col}_t(F) = \sum_{s\leqslant r} f_{st}\operatorname{col}_s(Q) = \sum_{s\leqslant r} f_{st}e_{j_s}$ for $t \leqslant r$, and 0 for $t > r$ (Theorem 1.4(a),(b)). By the outer product expansion (Theorem 1.4(d)), $QFP = \sum_{t\leqslant r}\operatorname{col}_t(QF)\operatorname{row}_t(P) = \sum_{s,t\leqslant r} f_{st}\, e_{j_s}e'_{i_t}$, and $e_{j_s}e'_{i_t} \in \mathbb{R}^{n\times m}$ has entry 1 in position (j_s, i_t) and 0 elsewhere. □

Example 12.7 (cf. [5, §9.5])**.** Let A be the matrix of Example 4.9, with the full-rank factorization $A = BC$ of Example 4.22:

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 2 & 5 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

Find two g-inverses of A, and decide whether $Ax = y$ is consistent for $y = (1,2,3)'$ and for $y = (1,1,1)'$.

Solution.

- Columns 1, 2 of C are the unit vectors of $\mathbb{R}^2$. Rows 1, 2 of B form $B_1 = \left(\begin{smallmatrix}1&2\\1&3\end{smallmatrix}\right)$, and $B_1\left(\begin{smallmatrix}3&-2\\-1&1\end{smallmatrix}\right) = I_2$, so $B_1^{-1} = \left(\begin{smallmatrix}3&-2\\-1&1\end{smallmatrix}\right)$ (Theorem 2.16); rows 1, 3 form $B_2 = \left(\begin{smallmatrix}1&2\\2&5\end{smallmatrix}\right)$, and likewise $B_2^{-1} = \left(\begin{smallmatrix}5&-2\\-2&1\end{smallmatrix}\right)$, as $B_2\left(\begin{smallmatrix}5&-2\\-2&1\end{smallmatrix}\right) = I_2$. Put

$$K = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 3 & -2 & 0 \\ -1 & 1 & 0 \end{pmatrix}, \qquad L_2 = \begin{pmatrix} 5 & 0 & -2 \\ -2 & 0 & 1 \end{pmatrix}$$

  where L_1 and L_2 carry the columns of B_1^{-1} and B_2^{-1} in columns $1,2$ and $1,3$. Then $CK = I_2$ (Theorem 1.4(a),(b)), and by Theorem 1.4(c) the rows of L_1B are $3(1,2) - 2(1,3) = (1,0)$ and $-(1,2) + (1,3) = (0,1)$, and those of L_2B are $5(1,2) - 2(2,5) = (1,0)$ and $-2(1,2) + (2,5) = (0,1)$: $L_1B = L_2B = I_2$.
- By Theorem 12.3(a), both

$$G_1 = KL_1 = \begin{pmatrix} 3 & -2 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad G_2 = KL_2 = \begin{pmatrix} 5 & 0 & -2 \\ -2 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  are g-inverses of A: a g-inverse need not be unique.

- The same two matrices come from Theorem 12.6(b): the submatrices of A in rows $1,2$ and in rows $1,3$, with columns $1,2$, are B_1 and B_2. Also, L_1 consists of the first two rows of the matrix P of Example 4.9, so G_1 is the g-inverse of Corollary 12.5.
- Since $CK = I_2$,

$$AG_1 = BCKL_1 = BL_1 = \begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 3 & -2 & 0 \\ -1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

  so $AG_1y = (y_1,\ y_2,\ y_1+y_2)'$. By Theorem 12.2(b), $Ax = y$ is consistent if and only if $y_3 = y_1 + y_2$; this matches $\text{row}_3(A) = \text{row}_1(A) + \text{row}_2(A)$ (Example 4.9).
- $y = (1,2,3)'$: consistent, and $G_1y = (3-4,\ -1+2,\ 0,\ 0)' = (-1,1,0,0)'$ is a solution; indeed $A(-1,1,0,0)' = -a_1 + a_2 = (1,2,3)'$.
- $y = (1,1,1)'$: $AG_1y = (1,1,2)' \neq y$, so $Ax = y$ is inconsistent.

Solving linear systems with a generalized inverse

Theorem 12.8 ([3, Result 4.4], [5, Theorem 11.2.4])**.** Let $A \in \mathbb{R}^{m\times n}$, let G be a g-inverse of A, and let $y \in \mathcal{C}(A)$. The solutions of $Ax = y$ are exactly the vectors

$$x = Gy + (I_n - GA)z \qquad (z \in \mathbb{R}^n)$$

In particular ($y = 0$), $\mathcal{N}(A) = \mathcal{C}(I_n - GA)$.

Proof. • Write $y = Av$. For every z, $A\Big(Gy+(I-GA)z\Big) = AGy+(A-AGA)z = AGAv+0 = Av = y$.

- Conversely, let $Au = y$, and put $z = u - Gy$. Since $GAu = Gy$ and $GAGy = G(AGy) = Gy$ (Theorem 12.2(a)),

$$Gy + (I - GA)(u - Gy) = Gy + u - Gy - GAu + GAGy = u - Gy + Gy = u$$

- For $y = 0$, the solutions of $Ax = 0$ are the vectors $(I-GA)z$, $z \in \mathbb{R}^n$; that is, $\mathcal{N}(A) = \mathcal{C}(I-GA)$. □

Theorem 12.9 ([5, Theorem 11.10.1])**.** Let $A \in \mathbb{R}^{m\times n}$ and $y \in \mathcal{C}(A)$, let G be a g-inverse of A, and let $\lambda \in \mathbb{R}^n$. The following are equivalent:

(i) $\lambda'x$ has the same value for every solution x of $Ax = y$;

(ii) $\lambda \in \mathcal{C}(A')$, that is, $\lambda' = a'A$ for some $a \in \mathbb{R}^m$;

(iii) $\lambda'GA = \lambda'$.

If they hold, then $\lambda'x = a'y$ for every solution x.

Proof. • (ii) ⇒ (i): if $Ax = y$, then $\lambda'x = a'Ax = a'y$, whatever the solution x.

- (i) ⇒ (iii): by Theorem 12.8, $\lambda'\Big(Gy + (I - GA)z\Big) = \lambda'Gy + \lambda'(I - GA)z$ is the same for every $z \in \mathbb{R}^n$. Comparing $z = e_j$ with $z = 0$, entry j of the row vector $\lambda'(I - GA)$ is 0, for $j = 1,\ldots,n$. So $\lambda'(I - GA) = 0$, that is, $\lambda' = \lambda'GA$.
- (iii) ⇒ (ii): $\lambda' = (\lambda'G)A = a'A$ with $a = G'\lambda$ (Theorem 1.9). □

Example 12.10. For A, G_1 and $y = (1, 2, 3)'$ of Example 12.7, find all solutions of $Ax = y$, and decide whether x_1 and $x_1 + 3x_3 - x_4$ have the same value for all of them.

Solution. $G_1A = KL_1BC = KC$, as $L_1B = I_2$; so

$$G_1A = \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad I_4 - G_1A = \begin{pmatrix} 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

By Theorem 12.8, with $G_1y = (-1, 1, 0, 0)'$, the solutions are

$$x = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z_3 \begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \end{pmatrix} + z_4 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 - 3z_3 + z_4 \\ 1 + z_3 - 2z_4 \\ z_3 \\ z_4 \end{pmatrix} \qquad (z_3, z_4 \in \mathbb{R})$$

z_1 and z_2 drop out, as columns 1, 2 of $I_4 - G_1A$ are 0. Columns 3, 4 are the special solutions s_1, s_2 of Example 4.9, and $\mathcal{N}(A) = \mathcal{C}(I_4 - G_1A) = \operatorname{span}\{s_1, s_2\}$.

- $\lambda = (1, 0, 0, 0)'$: $\lambda'G_1A = \operatorname{row}_1(G_1A) = (1, 0, 3, -1) \neq \lambda'$, so $x_1 = -1 - 3z_3 + z_4$ is not the same for all solutions (Theorem 12.9).
- $\lambda = (1, 0, 3, -1)'$: $\lambda'G_1A = \operatorname{row}_1(G_1A) + 3\operatorname{row}_3(G_1A) - \operatorname{row}_4(G_1A) = (1, 0, 3, -1) = \lambda'$, so $x_1 + 3x_3 - x_4$ is the same for all solutions: $(-1 - 3z_3 + z_4) + 3z_3 - z_4 = -1$. Here λ' is the first row of the RREF of A, a vector of $\mathcal{R}(A)$.

Properties of generalized inverses

Proposition 12.11 ([5, Lemma 9.3.3, Corollary 9.3.4, Lemma 9.2.8], [2, Theorem 8.2(iv)])**.** Let $A \in \mathbb{R}^{m\times n}$, and let G be a g-inverse of A.

(a) G' is a g-inverse of A'.

(b) If A is symmetric, then G' and $\frac{1}{2}(G + G')$ are g-inverses of A, and $\frac{1}{2}(G + G')$ is symmetric. So every symmetric matrix has a symmetric g-inverse.

(c) If A has full column rank, its g-inverses are exactly its left inverses, the $L \in \mathbb{R}^{n\times m}$ with $LA = I_n$. If A has full row rank, its g-inverses are exactly its right inverses, the $K \in \mathbb{R}^{n\times m}$ with $AK = I_m$.

Proof. (a) $A'G'A' = (AGA)' = A'$ (Theorem 1.9).

(b) If $A' = A$, then (a) reads $AG'A = A$. So $A \cdot \frac{1}{2}(G + G') \cdot A = \frac{1}{2}(AGA + AG'A) = \frac{1}{2}(A + A) = A$, and $\left(\frac{1}{2}(G + G')\right)' = \frac{1}{2}(G' + G)$.

(c)

- Let $\operatorname{rank} A = n$. Then $A = AI_n$ is a full-rank factorization, so A has a left inverse L_0 (Theorem 4.21(c)).
- Every left inverse L is a g-inverse: $ALA = AI_n = A$. Every g-inverse G is a left inverse: $GA = (L_0A)GA = L_0(AGA) = L_0A = I_n$.
- Let $\operatorname{rank} A = m$. Then $A = I_mA$ is a full-rank factorization, so A has a right inverse K_0. Every right inverse K gives $AKA = I_mA = A$, and every g-inverse G gives $AG = AG(AK_0) = (AGA)K_0 = AK_0 = I_m$. □

Lemma 12.12 ([5, Lemma 9.3.5])**.** Let $A \in \mathbb{R}^{m\times n}$, and let G be a g-inverse of A.

(a) For $C \in \mathbb{R}^{m\times q}$: $\mathcal{C}(C) \subseteq \mathcal{C}(A)$ if and only if $AGC = C$.

(b) For $B \in \mathbb{R}^{p\times n}$: $\mathcal{R}(B) \subseteq \mathcal{R}(A)$ if and only if $BGA = B$.

The inclusions do not involve G; so if $AGC = C$ (or $BGA = B$) holds for one g-inverse G, it holds for every g-inverse. In particular, the system $AX = C$ (unknown $X \in \mathbb{R}^{n\times q}$) is consistent if and only if $AGC = C$, and then $X = GC$ is a solution.

Proof. (a)

- If $AGC = C$, then each column $c_j = A(Gc_j)$ of C lies in $\mathcal{C}(A)$ (Theorem 1.4(a)), and so does every combination of them: $\mathcal{C}(C) \subseteq \mathcal{C}(A)$.
- If $\mathcal{C}(C) \subseteq \mathcal{C}(A)$, then $c_j = Af_j$ for some $f_j \in \mathbb{R}^n$ $(j = 1, \ldots, q)$, that is, $C = AF$ with $F = [f_1 \ \cdots \ f_q]$ (Theorem 1.4(a)); so $AGC = AGAF = AF = C$.

(b)

- If $BGA = B$, then each row $\mathrm{row}_i(B) = (\mathrm{row}_i(B)G)A$ lies in $\mathcal{R}(A)$ (Theorem 1.4(c)), and so does every combination of the rows: $\mathcal{R}(B) \subseteq \mathcal{R}(A)$.
- If $\mathcal{R}(B) \subseteq \mathcal{R}(A)$, then $\mathrm{row}_i(B) = l_i'A$ for some $l_i \in \mathbb{R}^m$ $(i = 1, \ldots, p)$, that is, $B = LA$, where L has the rows $l_1', \ldots, l_p'$ (Theorem 1.4(c)); so $BGA = LAGA = LA = B$.

Finally, $AX = C$ is consistent if and only if every system $Ax_j = c_j$ is (Theorem 1.4(a)), that is, $c_j \in \mathcal{C}(A)$ for every j (Corollary 4.14(a)), that is, $\mathcal{C}(C) \subseteq \mathcal{C}(A)$; and then $A(GC) = C$ by (a). □

Theorem 12.13 ([5, Lemmas 10.2.5, 9.3.7, 10.2.6], [3, Result 4.2])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank r, and let G be a g-inverse of A.

(a) $AG \in \mathbb{R}^{m\times m}$ and $GA \in \mathbb{R}^{n\times n}$ are idempotent.

(b) $\mathrm{rank}(AG) = \mathrm{rank}(GA) = r$; more precisely, $\mathcal{C}(AG) = \mathcal{C}(A)$, $\mathcal{R}(GA) = \mathcal{R}(A)$ and $\mathcal{N}(GA) = \mathcal{N}(A)$.

(c) $\mathrm{tr}(AG) = \mathrm{tr}(GA) = r$, $\mathrm{rank}(I_n - GA) = n - r$ and $\mathrm{rank}(I_m - AG) = m - r$.

Proof. (a) $(AG)(AG) = (AGA)G = AG$ and $(GA)(GA) = G(AGA) = GA$.

(b)

- By Theorem 4.15(a), applied to the products $(AG)A$, AG and $A(GA)$, GA,

$$r = \mathrm{rank}(AGA) \leqslant \mathrm{rank}(AG) \leqslant \mathrm{rank}\, A = r, \qquad r = \mathrm{rank}(AGA) \leqslant \mathrm{rank}(GA) \leqslant \mathrm{rank}\, A = r$$

- $\mathcal{C}(AG) \subseteq \mathcal{C}(A)$ and $\mathcal{C}(A) = \mathcal{C}((AG)A) \subseteq \mathcal{C}(AG)$ (Theorem 4.15(a)); so $\mathcal{C}(AG) = \mathcal{C}(A)$.
- Each row $\mathrm{row}_i(GA) = \mathrm{row}_i(G)A$ lies in $\mathcal{R}(A)$, and each row $\mathrm{row}_i(A) = \mathrm{row}_i(A)(GA)$ (as $A = AGA$) lies in $\mathcal{R}(GA)$ (Theorem 1.4(c)); so $\mathcal{R}(GA) = \mathcal{R}(A)$.
- If $GAx = 0$, then $Ax = AGAx = 0$; and $Ax = 0$ gives $GAx = 0$. So $\mathcal{N}(GA) = \mathcal{N}(A)$.

(c) By (a) and (b), $\mathrm{tr}(AG) = \mathrm{rank}(AG) = r$ and $\mathrm{tr}(GA) = \mathrm{rank}(GA) = r$ (Theorem 6.29(c)). Also $I_n - GA$ and $I_m - AG$ are idempotent (Theorem 6.29(b)), so $\mathrm{rank}(I_n - GA) = \mathrm{tr}(I_n - GA) = n - \mathrm{tr}(GA) = n - r$ (Theorem 1.11(a)), and likewise $\mathrm{rank}(I_m - AG) = m - r$. □

Theorem 12.14 ([3, §4.1, Result 4.3], [5, Lemma 20.3.2, Theorem 20.3.1])**.** Let $A \in \mathbb{R}^{m\times n}$. A g-inverse G of A with $GAG = G$ is a **reflexive g-inverse** of A; then A is also a g-inverse of G.

(a) If G_1 and G_2 are g-inverses of A, then G_1AG_2 is a reflexive g-inverse of A. In particular GAG is one, for every g-inverse G.

(b) Every g-inverse G of A has $\mathrm{rank}\, G \geqslant \mathrm{rank}\, A$, and $\mathrm{rank}\, G = \mathrm{rank}\, A$ if and only if G is reflexive.

(c) Let $r = \operatorname{rank} A \geqslant 1$ and let $A = BC$ be a full-rank factorization, with $B \in \mathbb{R}^{m\times r}$ and $C \in \mathbb{R}^{r\times n}$. Then G is a reflexive g-inverse of A if and only if $G = KL$ for some $L \in \mathbb{R}^{r\times m}$ with $LB = I_r$ and some $K \in \mathbb{R}^{n\times r}$ with $CK = I_r$.

Proof. (a) $A(G_1AG_2)A = (AG_1A)G_2A = AG_2A = A$, and

$$(G_1AG_2)A(G_1AG_2) = G_1(AG_2A)G_1AG_2 = G_1(AG_1A)G_2 = G_1AG_2$$

(b)

- $\operatorname{rank} A = \operatorname{rank}(AGA) \leqslant \operatorname{rank} G$ (Theorem 4.15(a), twice). If G is reflexive, then also $\operatorname{rank} G = \operatorname{rank}(GAG) \leqslant \operatorname{rank} A$; so $\operatorname{rank} G = \operatorname{rank} A$.
- Conversely, let $\operatorname{rank} G = \operatorname{rank} A$. Then $\mathcal{C}(GA) \subseteq \mathcal{C}(G)$ (Theorem 4.15(a)), and $\operatorname{rank}(GA) = \operatorname{rank} A = \operatorname{rank} G$ (Theorem 12.13(b)); so $\mathcal{C}(GA) = \mathcal{C}(G)$ (Theorem 4.5(a)).
- So each column of G is GAx_j for some $x_j \in \mathbb{R}^n$, that is, $G = GAX$ with $X = [x_1 \ \cdots \ x_m]$ (Theorem 1.4(a)), and $GAG = GAGAX = GAX = G$.

(c)

- If $G = KL$, then G is a g-inverse of A (Theorem 12.3(a)), and $GAG = KLBCKL = K(LB)(CK)L = KL = G$.
- Conversely, let G be a reflexive g-inverse. By Theorem 4.21(c), B has a left inverse L_0 and C a right inverse K_0. Multiplying $BCGBC = AGA = A = BC$ by L_0 on the left and by K_0 on the right gives $CGB = I_r$.
- So $K = GB$ has $CK = CGB = I_r$, $L = CG$ has $LB = CGB = I_r$, and $G = GAG = (GB)(CG) = KL$. □

Example 12.15. Let A and G_1 be as in Examples 12.7 and 12.10. Is G_1 reflexive? For the matrix $U \in \mathbb{R}^{4\times 3}$ with $(3,3)$ entry 1 and all other entries 0, show that $G_3 = G_1 + (I_4 - G_1A)U$ is a g-inverse of A that is not reflexive, and find the reflexive g-inverse G_3AG_3.

Solution.

- The matrices G_1A and
$$AG_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$
are idempotent with trace $1+1 = 2 = \operatorname{rank} A$, as Theorem 12.13(a),(c) says; AG_1 is not symmetric.
- G_1 has the two linearly independent nonzero rows $(3,-2,0)$ and $(-1,1,0)$, so $\operatorname{rank} G_1 = 2 = \operatorname{rank} A$, and G_1 is reflexive (Theorem 12.14(b); also $G_1 = KL_1$, Theorem 12.14(c)).
- $(I_4 - G_1A)U$ has column 3 equal to column 3 of $I_4 - G_1A$, which is $(-3,1,1,0)'$ (Example 12.10), and its other columns 0 (Theorem 1.4(a)). So
$$G_3 = \begin{pmatrix} 3 & -2 & -3 \\ -1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$
- $A(I_4 - G_1A) = A - AG_1A = 0$, so $AG_3 = AG_1 + A(I_4 - G_1A)U = AG_1$, and $AG_3A = AG_1A = A$: G_3 is a g-inverse of A.

- The first three rows of G_3 form a matrix M with $\det M = 1 \cdot (3-2) = 1$ (expansion along its last row, Theorem 5.21), so M is invertible (Theorem 5.18(a)). If $G_3 x = 0$, then $Mx = 0$ and $x = 0$; so the three columns of G_3 are linearly independent, and $\operatorname{rank} G_3 = 3 \neq \operatorname{rank} A$ (Theorem 4.13(b)): G_3 is not reflexive (Theorem 12.14(b)).
- $G_3 A G_3 = G_3(AG_1)$. By Theorem 1.4(a),(b), its columns are $G_3(1,0,1)' = (3-3,\ -1+1,\ 1,\ 0)'$, $G_3(0,1,1)' = (-2-3,\ 1+1,\ 1,\ 0)'$ and 0:

$$G_3 A G_3 = \begin{pmatrix} 0 & -5 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  a reflexive g-inverse of A (Theorem 12.14(a)), of rank 2.

Theorem 12.16 ([5, Theorem 9.4.1], [3, Result 6.1])**.** Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{p\times n}$ and $C \in \mathbb{R}^{m\times q}$.

(a) If $\mathcal{R}(B) \subseteq \mathcal{R}(A)$ and $\mathcal{C}(C) \subseteq \mathcal{C}(A)$, then BGC is the same matrix for every g-inverse G of A.

(b) Conversely, let BGC be the same matrix for every g-inverse G of A. If $C \neq 0$, then $\mathcal{R}(B) \subseteq \mathcal{R}(A)$; if $B \neq 0$, then $\mathcal{C}(C) \subseteq \mathcal{C}(A)$.

Proof. (a) Each row of B lies in $\mathcal{R}(A)$ and each column of C in $\mathcal{C}(A)$, so $B = LA$ and $C = AF$ for some $L \in \mathbb{R}^{p\times m}$ and $F \in \mathbb{R}^{n\times q}$ (Theorem 1.4(a),(c)). Hence $BGC = LAGAF = LAF$ for every g-inverse G.

(b)

- Fix a g-inverse G of A (Theorem 12.3). For all $U, V \in \mathbb{R}^{n\times m}$, the matrix $G+(I_n-GA)U+V(I_m-AG)$ is also a g-inverse of A: as $AGA = A$,

$$A\Big(G + (I_n - GA)U + V(I_m - AG)\Big)A = AGA + (A - AGA)UA + AV(A - AGA) = A$$

- By hypothesis, $B\Big(G + (I_n - GA)U + V(I_m - AG)\Big)C = BGC$. With $V = 0$ this gives $B(I_n - GA)UC = 0$ for every U; with $U = 0$, $BV(I_m - AG)C = 0$ for every V.
- Let $C \neq 0$, with a column $c_k \neq 0$, so $c_k' c_k > 0$. For $i = 1, \ldots, n$, the matrix $U_i = \frac{1}{c_k' c_k} e_i c_k' \in \mathbb{R}^{n\times m}$ has $U_i c_k = e_i$. Column k of $B(I_n - GA)U_i C$ is $B(I_n - GA)U_i c_k = B(I_n - GA)e_i$, the ith column of $B(I_n - GA)$ (Theorem 1.4(a),(b)), and it is 0. So $B(I_n - GA) = 0$, that is, $BGA = B$, and $\mathcal{R}(B) \subseteq \mathcal{R}(A)$ (Lemma 12.12(b)).
- Let $B \neq 0$, with a row $b_k' \neq 0$ ($b_k \in \mathbb{R}^n$). For $i = 1, \ldots, m$, the matrix $V_i = \frac{1}{b_k' b_k} b_k e_i' \in \mathbb{R}^{n\times m}$ has $b_k' V_i = e_i'$. Row k of $BV_i(I_m - AG)C$ is $b_k' V_i (I_m - AG)C = e_i'(I_m - AG)C$, the ith row of $(I_m - AG)C$ (Theorem 1.4(c)), and it is 0. So $(I_m - AG)C = 0$, that is, $AGC = C$, and $\mathcal{C}(C) \subseteq \mathcal{C}(A)$ (Lemma 12.12(a)). □

Generalized inverses of $A'A$

Theorem 12.17 ([5, Theorems 12.3.4, 12.3.1], [6, Exercise 10.54])**.** Let $A \in \mathbb{R}^{m\times n}$, let G be a g-inverse of $A'A$, and put $P = AGA' = A(A'A)^- A' \in \mathbb{R}^{m\times m}$.

(a) $AGA'A = A$ and $A'AGA' = A'$. Hence GA' is a g-inverse of A.

(b) For every $y \in \mathbb{R}^m$, Py is the orthogonal projection of y onto $\mathcal{C}(A)$. Hence P does not depend on the choice of the g-inverse G, and $P = A(A'A)^{-1}A'$, the matrix of Theorem 6.26, if $\operatorname{rank} A = n$.

(c) $P' = P = P^2$, $PA = A$ and $A'P = A'$.

(d) $\operatorname{rank} P = \operatorname{tr} P = \operatorname{rank} A$, and $I_m - P$ is symmetric and idempotent, of rank $m - \operatorname{rank} A$.

Proof. (a)

- $A'A(GA'A - I_n) = A'AGA'A - A'A = 0$, so each column of $GA'A - I_n$ lies in $\mathcal{N}(A'A) = \mathcal{N}(A)$ (Theorems 1.4(a) and 4.18(a)). Hence $A(GA'A - I_n) = 0$, that is, $AGA'A = A$.
- $A'A$ is symmetric, so G' is also a g-inverse of it (Proposition 12.11(b)), and the first identity for G' reads $AG'A'A = A$. Transposing (Theorem 1.9), $A'AGA' = A'$.
- $A(GA')A = AGA'A = A$.

(b)

- $Py = A(GA'y) \in \mathcal{C}(A)$, and $A'(y - Py) = A'y - A'AGA'y = 0$ by (a). So $y - Py \perp \mathcal{C}(A)$ (Theorem 6.6(a)), and Py is the orthogonal projection of y onto $\mathcal{C}(A)$ (Definition 6.22).
- If G_1, G_2 are g-inverses of $A'A$, then $AG_1A'y = AG_2A'y$ for every $y \in \mathbb{R}^m$, as the projection is unique (Definition 6.22). Taking for y the columns of I_m, the columns of AG_1A' and AG_2A' agree (Theorem 1.4(b)): $AG_1A' = AG_2A'$.
- If $\operatorname{rank} A = n$, then $A'A$ is invertible (Theorem 6.26), its only g-inverse is $(A'A)^{-1}$ (Definition 12.1), and $P = A(A'A)^{-1}A'$.

(c)

- G' is a g-inverse of the symmetric $A'A$ (Proposition 12.11(b)), so by (b), $P = AG'A' = (AGA')' = P'$ (Theorem 1.9).
- $PA = AGA'A = A$ by (a), and $A'P = A'P' = (PA)' = A'$.
- $P^2 = (PA)GA' = AGA' = P$.

(d)

- $\mathcal{C}(P) = \mathcal{C}(A(GA')) \subseteq \mathcal{C}(A)$ and $\mathcal{C}(A) = \mathcal{C}(PA) \subseteq \mathcal{C}(P)$ (Theorem 4.15(a)); so $\operatorname{rank} P = \operatorname{rank} A$, and $\operatorname{rank} P = \operatorname{tr} P$ as P is idempotent (Theorem 6.29(c)).
- $I_m - P$ is symmetric by (c) and idempotent (Theorem 6.29(b)); so $\operatorname{rank}(I_m - P) = \operatorname{tr}(I_m - P) = m - \operatorname{tr} P = m - \operatorname{rank} A$ (Theorems 6.29(c) and 1.11(a)). □

Corollary 12.18 ([3, Result 7.1], [5, Theorem 9.4.1])**.** Let $A \in \mathbb{R}^{m\times n}$, and let $\lambda \in \mathbb{R}^n$, $\lambda \neq 0$. Then $\lambda'G\lambda$ is the same number for every g-inverse G of $A'A$ if and only if $\lambda \in \mathcal{C}(A')$. If $\lambda' = a'A$ ($a \in \mathbb{R}^m$), then for every g-inverse G of $A'A$

$$\lambda'G\lambda = a'Pa = \|Pa\|^2 \geqslant 0$$

where $P = AGA'$ is the matrix of Theorem 12.17.

Proof. • Let $\lambda' = a'A$. Then $\lambda'G\lambda = a'AGA'a = a'Pa$ (Theorem 1.9), and P does not depend on G (Theorem 12.17(b)). As P is symmetric and idempotent (Theorem 12.17(c)), $a'Pa = a'P'Pa = \|Pa\|^2$.

- Conversely, let $\lambda' G\lambda$ be the same for every g-inverse G of $A'A$. Theorem 12.16(b), for the matrix $A'A$ with $B = \lambda' \neq 0$ and $C = \lambda \neq 0$, gives $\mathcal{R}(\lambda') \subseteq \mathcal{R}(A'A)$: $\lambda' = b'A'A$ for some $b \in \mathbb{R}^n$. So $\lambda = A'(Ab) \in \mathcal{C}(A')$. □

Example 12.19. Let A be the matrix of Example 12.7. Find the g-inverses F_1 and F_2 of $A'A$ built from its submatrices in rows and columns $1, 2$ and in rows and columns $1, 3$, and the matrices AF_1A' and AF_2A'. Compare $\lambda' F_1\lambda$ and $\lambda' F_2\lambda$ for $\lambda = (1, 0, 3, -1)'$ and for $\lambda = (1, 0, 0, 0)'$.

Solution.

- With the columns $a_1, \ldots, a_4$ of A, the (i, j) entry of $A'A$ is $a_i'a_j$ (Theorem 1.4(d)):

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}, \qquad A'A = \begin{pmatrix} 6 & 15 & 3 & 24 \\ 15 & 38 & 7 & 61 \\ 3 & 7 & 2 & 11 \\ 24 & 61 & 11 & 98 \end{pmatrix}$$

  and $\operatorname{rank}(A'A) = \operatorname{rank} A = 2$ (Theorem 4.18(b), Example 4.9).

- The submatrix in rows and columns $1, 2$ is $\left(\begin{smallmatrix} 6 & 15 \\ 15 & 38 \end{smallmatrix}\right)$, with determinant $228 - 225 = 3$ and inverse $\frac{1}{3}\left(\begin{smallmatrix} 38 & -15 \\ -15 & 6 \end{smallmatrix}\right)$; the one in rows and columns $1, 3$ is $\left(\begin{smallmatrix} 6 & 3 \\ 3 & 2 \end{smallmatrix}\right)$, with determinant $12 - 9 = 3$ and inverse $\frac{1}{3}\left(\begin{smallmatrix} 2 & -3 \\ -3 & 6 \end{smallmatrix}\right)$ (Theorem 5.24). By Theorem 12.6(b),

$$F_1 = \frac{1}{3}\begin{pmatrix} 38 & -15 & 0 & 0 \\ -15 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad F_2 = \frac{1}{3}\begin{pmatrix} 2 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ -3 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- By Theorem 1.4(a),(b), the columns of AF_1 are $\frac{1}{3}(38a_1 - 15a_2) = \frac{1}{3}(8, -7, 1)'$, $\frac{1}{3}(-15a_1 + 6a_2) = \frac{1}{3}(-3, 3, 0)'$, 0 and 0. The rows of A' are $a_1', \ldots, a_4'$, so by the outer product expansion (Theorem 1.4(d))

$$AF_1A' = \frac{1}{3}\begin{pmatrix} 8 \\ -7 \\ 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 2 \end{pmatrix} + \frac{1}{3}\begin{pmatrix} -3 \\ 3 \\ 0 \end{pmatrix}\begin{pmatrix} 2 & 3 & 5 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

- Likewise the columns of AF_2 are $\frac{1}{3}(2a_1 - 3a_3) = \frac{1}{3}(-1, 2, 1)'$, 0, $\frac{1}{3}(-3a_1 + 6a_3) = \frac{1}{3}(3, -3, 0)'$ and 0, and

$$AF_2A' = \frac{1}{3}\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 2 \end{pmatrix} + \frac{1}{3}\begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

- So $AF_1A' = AF_2A' = P$, as Theorem 12.17(b) says, and $\operatorname{tr} P = 2 = \operatorname{rank} A$ (Theorem 12.17(d)). With $w = (1, 1, -1)'$, $P = I_3 - \frac{1}{3}ww'$, so $Py = y - \frac{1}{3}(w'y)w$. Here $w'y = y_1 + y_2 - y_3 = 0$ exactly when $y \in \mathcal{C}(A)$ (Example 12.7); so $Py = y$ for $y \in \mathcal{C}(A)$, and $Pw = 0$.

- $\lambda = (1, 0, 3, -1)'$: only the entries $\lambda_1 = 1$, $\lambda_2 = 0$ meet the nonzero block of F_1, so $\lambda' F_1\lambda = \frac{1}{3}\cdot 38 = \frac{38}{3}$; with $\lambda_1 = 1$, $\lambda_3 = 3$, $\lambda' F_2\lambda = \frac{1}{3}(2 - 9 - 9 + 54) = \frac{38}{3}$. Indeed $\lambda' = 3\operatorname{row}_1(A) - 2\operatorname{row}_2(A) = a'A$ with $a = (3, -2, 0)'$, so $\lambda \in \mathcal{C}(A')$, and $Pa = \frac{1}{3}(8, -7, 1)'$, $\|Pa\|^2 = \frac{64+49+1}{9} = \frac{38}{3}$, as Corollary 12.18 says.

- $\lambda = (1, 0, 0, 0)'$: $\lambda' F_1\lambda = \frac{38}{3}$, but $\lambda' F_2\lambda = \frac{2}{3}$. By Corollary 12.18, $\lambda \notin \mathcal{C}(A')$; correspondingly x_1 is not the same for all solutions in Example 12.10.

Minimum-norm and least-squares generalized inverses

Theorem 12.20 ([3, Result 4.5], [5, Theorem 20.3.6])**.** Let $A \in \mathbb{R}^{m\times n}$. A g-inverse G of A with $(GA)' = GA$ is a **minimum-norm g-inverse** of A. For $G \in \mathbb{R}^{n\times m}$, the following are equivalent:

(i) G is a minimum-norm g-inverse of A;

(ii) for every $y \in \mathcal{C}(A)$, $x = Gy$ is a solution of $Ax = y$, and $\|Gy\| \leqslant \|x\|$ for every solution x.

If they hold, then $\|x\| > \|Gy\|$ for every solution $x \neq Gy$: Gy is the only shortest solution.

Proof. For $y = Au \in \mathcal{C}(A)$ and $z \in \mathbb{R}^n$, Proposition 6.2 and Theorem 1.9 give

$$\|Gy + (I - GA)z\|^2 = \|Gy\|^2 + \|(I - GA)z\|^2 + 2c, \qquad c = y'G'(I - GA)z = u'(GA)'(I - GA)z \ (*)$$

(i) $\Rightarrow$ (ii)

- Gy is a solution, and every solution is $x = Gy + (I - GA)z$ for some z (Theorems 12.2(a) and 12.8).
- $(GA)' = GA$, and $GAGA = GA$ as $AGA = A$; so $c = u'GA(I - GA)z = u'(GA - GAGA)z = 0$.
- Hence $\|x\|^2 = \|Gy\|^2 + \|(I - GA)z\|^2 \geqslant \|Gy\|^2$, with equality if and only if $(I - GA)z = 0$, that is, $x = Gy$.

(ii) $\Rightarrow$ (i)

- By (ii) and Theorem 12.2(a), G is a g-inverse of A.
- Let $u, z \in \mathbb{R}^n$, $d = \|(I - GA)z\|^2$ and $c = u'(GA)'(I - GA)z$. Apply (ii) to $y = A(\alpha u)$ for $\alpha \in \mathbb{R}$ and to its solution $Gy + (I - GA)z$ (Theorem 12.8): by $(*)$, with αu in place of u, $0 \leqslant d + 2\alpha c$ for every $\alpha \in \mathbb{R}$.
- If $c \neq 0$, then $\alpha = -(d + 1)/(2c)$ gives $d + 2\alpha c = -1 < 0$ ✘ So $c = 0$ for all u, z.
- Taking for u and z the ith and jth columns of I_n, c is the (i, j) entry of $(GA)'(I - GA)$ (Theorem 1.4(b),(c)); so $(GA)'(I - GA) = 0$, that is, $(GA)' = (GA)'GA$. This matrix is symmetric, so $GA = ((GA)')' = (GA)'$. □

Theorem 12.21 ([3, Result 4.6], [5, Theorem 20.3.12])**.** Let $A \in \mathbb{R}^{m\times n}$. A g-inverse G of A with $(AG)' = AG$ is a **least-squares g-inverse** of A. For $G \in \mathbb{R}^{n\times m}$, the following are equivalent:

(i) G is a least-squares g-inverse of A;

(ii) $\|AGy - y\| \leqslant \|Ax - y\|$ for all $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$.

So, for every y, $x = Gy$ minimizes $\|Ax - y\|$, also when $Ax = y$ is inconsistent.

Proof. For $x = Gy + w$ ($w \in \mathbb{R}^n$), $Ax - y = (AG - I)y + Aw$, and Proposition 6.2 gives

$$\|Ax - y\|^2 = \|(AG - I)y\|^2 + \|Aw\|^2 + 2w'A'(AG - I)y \qquad (**)$$

(i) $\Rightarrow$ (ii) As $AG = (AG)'$ and $AGA = A$, $A'(AG - I) = A'(AG)' - A' = (AGA)' - A' = 0$, so $\|Ax - y\|^2 = \|AGy - y\|^2 + \|Aw\|^2 \geqslant \|AGy - y\|^2$.

(ii) $\Rightarrow$ (i)

- For $x \in \mathbb{R}^n$ put $y = Ax$ in (ii): $\|AGAx - Ax\| \leqslant \|Ax - Ax\| = 0$, so $AGAx = Ax$. Taking for x the columns of I_n, $AGA = A$ (Theorem 1.4(b)).
- Let $w \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Apply (ii) to αy ($\alpha \in \mathbb{R}$) and $x = \alpha Gy + w$: by $(**)$, $\alpha^2\|(AG - I)y\|^2 \leqslant \alpha^2\|(AG - I)y\|^2 + \|Aw\|^2 + 2\alpha w'A'(AG - I)y$, that is, $0 \leqslant d + 2\alpha c$ with $d = \|Aw\|^2$ and $c = w'A'(AG - I)y$.
- As in the proof of Theorem 12.20, $c = 0$ for all w and y; so $A'(AG - I) = 0$, that is, $A'AG = A'$.

- Transposing, $G'A'A = A$; so $AG = G'A'AG = (AG)'AG$, which is symmetric. □

For $A \in \mathbb{R}^{m\times n}$ and a g-inverse F of $A'A$, $G = FA'$ is a least-squares g-inverse of A: with $P = AFA'$, $AGA = PA = A$, and $AG = P$ is symmetric (Theorem 12.17(c)) [5, Theorem 20.3.17]. It is even reflexive (Theorem 12.14): $GAG = F(A'AFA') = FA'P = FA' = G$, as $A'P = A'$. On the other hand, for the matrix A of Example 12.7, G_1 is neither a least-squares nor a minimum-norm g-inverse: G_1A (Example 12.10) and

$$AG_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

are not symmetric.

The Moore–Penrose inverse

Definition 12.22 ([3, §4.3], [5, Theorem 20.1.1])**.** Let $A \in \mathbb{R}^{m\times n}$. A matrix $G \in \mathbb{R}^{n\times m}$ is a **Moore–Penrose inverse** of A if

(P1) $AGA = A$;

(P2) $GAG = G$;

(P3) $(AG)' = AG$;

(P4) $(GA)' = GA$.

Thus a Moore–Penrose inverse of A is a reflexive g-inverse of A (Theorem 12.14) that is both a least-squares and a minimum-norm g-inverse.

Theorem 12.23 ([3, §4.3], [5, Theorem 20.1.1])**.** A matrix $A \in \mathbb{R}^{m\times n}$ has at most one Moore–Penrose inverse. It is written A^+.

Proof. Let G_1 and G_2 both satisfy (P1)–(P4). Each step below uses the condition shown, together with $(BC)' = C'B'$ and $(B')' = B$ (Theorem 1.9):

$$\begin{aligned}
G_1 &= G_1AG_1 && \text{(P2) for } G_1 \\
&= G_1G_1'A' && AG_1 = (AG_1)' = G_1'A' \text{ by (P3)} \\
&= G_1G_1'A'G_2'A' && A' = (AG_2A)' = A'G_2'A' \text{ by (P1)} \\
&= G_1G_1'A'AG_2 && G_2'A' = (AG_2)' = AG_2 \text{ by (P3)} \\
&= G_1AG_1AG_2 && G_1'A' = (AG_1)' = AG_1 \text{ by (P3)} \\
&= G_1AG_2 && AG_1A = A \text{ by (P1)} \\
&= G_1AG_2AG_2 && G_2 = G_2AG_2 \text{ by (P2)} \\
&= G_1AA'G_2'G_2 && G_2A = (G_2A)' = A'G_2' \text{ by (P4)} \\
&= A'G_1'A'G_2'G_2 && G_1A = (G_1A)' = A'G_1' \text{ by (P4)} \\
&= A'G_2'G_2 && A'G_1'A' = (AG_1A)' = A' \text{ by (P1)} \\
&= G_2AG_2 && A'G_2' = (G_2A)' = G_2A \text{ by (P4)} \\
&= G_2 && \text{(P2) for } G_2
\end{aligned}$$

□

Theorem 12.24 ([3, §4.3], [5, Theorem 20.1.1, §20.2])**.** Every $A \in \mathbb{R}^{m\times n}$ has a Moore–Penrose inverse. More precisely, let $r = \operatorname{rank} A$.

(a) If $r = 0$, then $A^+ = 0 \in \mathbb{R}^{n\times m}$.

(b) If $r \geqslant 1$ and $A = BC$ is a full-rank factorization, then $A^+ = C'(CC')^{-1}(B'B)^{-1}B'$.

(c) If $r = n$, then $A^+ = (A'A)^{-1}A'$; if $r = m$, then $A^+ = A'(AA')^{-1}$; if A is square and invertible, then $A^+ = A^{-1}$.

Proof. (a) Then $A = 0$ (Theorem 4.8), and $G = 0$ satisfies (P1)–(P4), as every product in them is 0.
(b) Let $G = C'(CC')^{-1}(B'B)^{-1}B'$. With $L = (B'B)^{-1}B'$ and $K = C'(CC')^{-1}$, $G = KL$, $LB = I_r$ and $CK = I_r$ (Theorem 4.21(c)).

- (P1): G is a g-inverse of A (Theorem 12.3(a)).
- (P2): $GAG = KLBCKL = K(LB)(CK)L = KL = G$.
- (P3): $AG = BCKL = BL = B(B'B)^{-1}B'$, the matrix P of Theorem 6.26 for B, which has full column rank; it is symmetric (Theorem 6.26(b)).
- (P4): $GA = KLBC = KC = C'(CC')^{-1}C$, the matrix P of Theorem 6.26 for $C' \in \mathbb{R}^{n\times r}$, which has full column rank r (Theorem 4.13(a)); it is symmetric (Theorem 6.26(b)).

With (a), every matrix has a Moore–Penrose inverse, and it is unique (Theorem 12.23).
(c) If $r = n$, then $A = AI_n$ is a full-rank factorization, and (b) gives $A^+ = I_n(I_nI_n')^{-1}(A'A)^{-1}A' = (A'A)^{-1}A'$. If $r = m$, then $A = I_mA$ is one, and (b) gives $A^+ = A'(AA')^{-1}(I_m'I_m)^{-1}I_m' = A'(AA')^{-1}$. If A is invertible, A^{-1} is its only g-inverse (Definition 12.1), so $A^+ = A^{-1}$. □

Example 12.25 ([2, Example 8.1])**.** Find A^+ for

$$A = \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 2 & 2 & 2 \\ -1 & 4 & 5 & 3 \end{pmatrix}$$

Solution. Row-reduce A:

$$A \xrightarrow{R_3\leftarrow R_3+R_1} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 4 & 4 & 4 \end{pmatrix} \xrightarrow{R_3\leftarrow R_3-2R_2} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2\leftarrow \frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

So $\operatorname{rank} A = 2$, with pivot columns $1, 2$, and Theorem 4.21(a) gives $A = BC$ with

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 4 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \qquad B'B = \begin{pmatrix} 2 & -4 \\ -4 & 20 \end{pmatrix}, \qquad CC' = 3I_2$$

Hence $(B'B)(CC') = 6\begin{pmatrix} 1 & -2 \\ -2 & 10 \end{pmatrix}$, whose inverse is $\frac{1}{36}\begin{pmatrix} 10 & 2 \\ 2 & 1 \end{pmatrix}$ (Theorem 5.24; the determinant of $\begin{pmatrix} 1 & -2 \\ -2 & 10 \end{pmatrix}$ is 6). By Theorems 12.24(b) and 2.13(b),

$$A^+ = C'(CC')^{-1}(B'B)^{-1}B' = C'\Big((B'B)(CC')\Big)^{-1}B' = \frac{1}{36}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 10 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 4 \end{pmatrix}$$

$$= \frac{1}{36}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 10 & 4 & -2 \\ 2 & 2 & 2 \end{pmatrix} = \frac{1}{36}\begin{pmatrix} 10 & 4 & -2 \\ 2 & 2 & 2 \\ -8 & -2 & 4 \\ 12 & 6 & 0 \end{pmatrix} = \frac{1}{18}\begin{pmatrix} 5 & 2 & -1 \\ 1 & 1 & 1 \\ -4 & -1 & 2 \\ 6 & 3 & 0 \end{pmatrix}$$

Check of (P3):

$$AA^+ = B(B'B)^{-1}B' = \frac{1}{6}\begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix}$$

is symmetric; for instance, its first row is $\text{row}_1(A)A^+ = \frac{1}{18}(5+0+4+6,\ 2+0+1+3,\ -1+0-2+0) = \frac{1}{6}(5, 2, -1)$.

Proposition 12.26 ([5, Theorems 20.5.1, 20.5.3], [3, Result 6.3], [6, Exercises 10.25, 10.26, 10.28])**.** Let $A \in \mathbb{R}^{m\times n}$.

(a) $\text{rank}\, A^+ = \text{rank}\, A$, $\mathcal{C}(A^+) = \mathcal{C}(A')$, $\mathcal{R}(A^+) = \mathcal{R}(A')$ and $\mathcal{N}(A^+) = \mathcal{N}(A')$.

(b) AA^+ and A^+A are symmetric and idempotent, with $\mathcal{C}(AA^+) = \mathcal{C}(A)$ and $\mathcal{C}(A^+A) = \mathcal{C}(A')$. Hence AA^+y is the orthogonal projection of $y \in \mathbb{R}^m$ onto $\mathcal{C}(A)$, and A^+Ax is that of $x \in \mathbb{R}^n$ onto $\mathcal{C}(A')$.

(c) $(A')^+ = (A^+)'$, $(A^+)^+ = A$, and $(cA)^+ = c^{-1}A^+$ for every number $c \neq 0$.

(d) If A is symmetric, then A^+ is symmetric and $AA^+ = A^+A$. If A is PSD, then A^+ is PSD.

Proof. (a)

- By (P1) and (P2), A^+ is a reflexive g-inverse of A; so $\text{rank}\, A^+ = \text{rank}\, A$ (Theorem 12.14(b)).
- By (P2), (P4) and Theorem 1.9, $A^+ = A^+AA^+ = (A^+A)'A^+ = A'(A^+)'A^+$; so $\mathcal{C}(A^+) \subseteq \mathcal{C}(A')$ (Theorem 4.15(a)). Both spaces have dimension $\text{rank}\, A^+ = \text{rank}\, A = \text{rank}\, A'$ (Theorem 4.13(a)), so they are equal (Theorem 4.5(a)).
- Likewise, by (P2) and (P3), $A^+ = A^+(AA^+)' = A^+(A^+)'A'$: each row of A^+ is $\text{row}_i(A^+(A^+)')A' \in \mathcal{R}(A')$ (Theorem 1.4(c)). So $\mathcal{R}(A^+) \subseteq \mathcal{R}(A')$, and the dimensions agree as before; the transposed spaces $\mathcal{C}((A^+)') \subseteq \mathcal{C}(A)$ are therefore equal (Theorem 4.5(a)).
- Hence $\mathcal{N}(A^+) = \mathcal{C}((A^+)')^\perp = \mathcal{C}(A)^\perp = \mathcal{N}(A')$ (Corollary 6.7).

(b)

- AA^+ and A^+A are symmetric by (P3) and (P4), and idempotent by Theorem 12.13(a), as A^+ is a g-inverse of A.
- $\mathcal{C}(AA^+) = \mathcal{C}(A)$ (Theorem 12.13(b)). Also $\mathcal{C}(A^+A) \subseteq \mathcal{C}(A^+)$ (Theorem 4.15(a)) and $\text{rank}(A^+A) = \text{rank}\, A = \text{rank}\, A^+$ (Theorem 12.13(b) and part (a)); so $\mathcal{C}(A^+A) = \mathcal{C}(A^+) = \mathcal{C}(A')$ (Theorem 4.5(a)).
- The projections: Theorem 6.29(d).

(c) Each claim is checked through (P1)–(P4) and the uniqueness of Theorem 12.23.

- $(A^+)'$ for A': $A'(A^+)'A' = (AA^+A)' = A'$, $(A^+)'A'(A^+)' = (A^+AA^+)' = (A^+)'$, and $A'(A^+)' = (A^+A)' = A^+A$ and $(A^+)'A' = (AA^+)' = AA^+$ are symmetric.
- A for A^+: the four conditions $A^+AA^+ = A^+$, $AA^+A = A$, $(A^+A)' = A^+A$ and $(AA^+)' = AA^+$ are (P2), (P1), (P4) and (P3).
- $c^{-1}A^+$ for cA: $(cA)(c^{-1}A^+)(cA) = cAA^+A = cA$, $(c^{-1}A^+)(cA)(c^{-1}A^+) = c^{-1}A^+AA^+ = c^{-1}A^+$, and $(cA)(c^{-1}A^+) = AA^+$ and $(c^{-1}A^+)(cA) = A^+A$ are symmetric.

(d)

- If $A' = A$, then $(A^+)' = (A')^+ = A^+$ by (c), and $AA^+ = (AA^+)' = (A^+)'A' = A^+A$ by (P3).
- If A is PSD (so symmetric), then A^+ is symmetric and $A^+ = A^+AA^+ = (A^+)'AA^+$; so A^+ is PSD (Theorem 8.24(a), with $P = A^+$). □

Proposition 12.27 ([6, Exercise 10.35])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, with $A = Q\Lambda Q'$ as in Theorem 8.3(b): Q is orthogonal, with columns $q_1, \ldots, q_n$, and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$. For a number λ let $\lambda^+ = 1/\lambda$ if $\lambda \neq 0$ and $0^+ = 0$, and let $\Lambda^+ = \text{diag}(\lambda_1^+, \ldots, \lambda_n^+)$. Then

$$A^+ = Q\Lambda^+Q' = \sum_{\lambda_i \neq 0} \lambda_i^{-1} q_i q_i'$$

Proof. • For every number λ, $\lambda\lambda^+\lambda = \lambda$ and $\lambda^+\lambda\lambda^+ = \lambda^+$ (both sides are 0 if $\lambda = 0$). A product of diagonal matrices is the diagonal matrix of the products of their diagonal entries; so $\Lambda\Lambda^+\Lambda = \Lambda$, $\Lambda^+\Lambda\Lambda^+ = \Lambda^+$, and $\Lambda\Lambda^+ = \Lambda^+\Lambda$ is diagonal.

- Let $G = Q\Lambda^+Q'$. As $Q'Q = I_n$, $AGA = Q\Lambda\Lambda^+\Lambda Q' = A$ and $GAG = Q\Lambda^+\Lambda\Lambda^+Q' = G$: (P1), (P2). Also $AG = Q\Lambda\Lambda^+Q' = GA$, which is symmetric (Theorem 1.9): (P3), (P4). So $G = A^+$ (Theorem 12.23).
- Column i of $Q\Lambda^+$ is $\lambda_i^+ q_i$, so by the outer product expansion (Theorem 1.4(d)) $Q\Lambda^+Q' = \sum_i \lambda_i^+ q_i q_i'$, in which the terms with $\lambda_i = 0$ are 0. □

For a symmetric $A \neq 0$, the symmetric g-inverse $G = \sum_{c_j \neq 0} c_j^{-1} E_j$ of Lemma 11.19 is A^+: by Theorem 8.6(c), $E_j = \sum_{\lambda_i = c_j} q_i q_i'$, so $G = \sum_{\lambda_i \neq 0} \lambda_i^{-1} q_i q_i'$, which is A^+ by Proposition 12.27.

The minimum-norm least-squares solution

Theorem 12.28 ([5, Corollary 20.5.5, Theorem 20.6.1], [6, Exercise 10.29(e)])**.** Let $A \in \mathbb{R}^{m\times n}$ and $y \in \mathbb{R}^m$.

(a) $A^+ = (A'A)^+A'$, and $AA^+ = A(A'A)^-A'$, the matrix P of Theorem 12.17.

(b) A vector $x \in \mathbb{R}^n$ minimizes $\|Ax - y\|$ if and only if $Ax = AA^+y$. The vector A^+y is one of them, and every other one has $\|x\| > \|A^+y\|$: A^+y is the only shortest minimizer of $\|Ax - y\|$, the **minimum-norm least-squares solution** of $Ax = y$. It is also the only minimizer in $\mathcal{C}(A')$.

(c) If $G \in \mathbb{R}^{n\times m}$ and Gy is the shortest minimizer of $\|Ax - y\|$ for every $y \in \mathbb{R}^m$, then $G = A^+$.

Proof. (a) Let $F = (A'A)^+$ (Theorem 12.24) and $H = FA'$.

- F is a g-inverse of $A'A$ by (P1), so $AH = AFA' = P$ (Theorem 12.17(b)).
- (P1): $AHA = PA = A$; (P3): $AH = P$ is symmetric (Theorem 12.17(c)).
- (P2): $HAH = (FA'AF)A' = FA' = H$, by (P2) for F.
- (P4): $HA = FA'A$ is symmetric, by (P4) for F, a Moore–Penrose inverse of $A'A$.
- So $H = A^+$ (Theorem 12.23), and $AA^+ = AH = P$.

(b)

- By Proposition 12.26(b), AA^+y is the orthogonal projection of y onto $\mathcal{C}(A)$. Every Ax lies in $\mathcal{C}(A)$, so by Theorem 6.23, $\|Ax - y\| \geqslant \|AA^+y - y\|$, with equality exactly when $Ax = AA^+y$. In particular $x = A^+y$ is a minimizer.
- The minimizers are also the solutions of $A'Ax = A'y$. If $Ax = AA^+y$, then $A'Ax = A'(AA^+)'y = (AA^+A)'y = A'y$, by (P3), Theorem 1.9 and (P1). Conversely, if $A'Ax = A'y$, then multiplying by $(A^+)'$ on the left gives $(AA^+)'Ax = (AA^+)'y$, that is, $AA^+Ax = AA^+y$ by (P3), and $Ax = AA^+y$ by (P1).
- $A'y \in \mathcal{C}(A') = \mathcal{C}(A'A)$ (Theorem 4.18(c)), and $F = (A'A)^+$ satisfies (P1) and (P4) for $A'A$: it is a minimum-norm g-inverse of $A'A$. By Theorem 12.20, for the matrix $A'A$ and the right side $A'y$, $FA'y$ is a solution of $A'Ax = A'y$, and every other solution x has $\|x\| > \|FA'y\|$. Here $FA'y = A^+y$ by (a).
- If x is a minimizer, then $A(x - A^+y) = AA^+y - AA^+y = 0$, so $x - A^+y \in \mathcal{N}(A) = \mathcal{C}(A')^\perp$ (Corollary 6.7); and $A^+y \in \mathcal{C}(A^+) = \mathcal{C}(A')$ (Proposition 12.26(a)). If moreover $x \in \mathcal{C}(A')$, then $x - A^+y \in \mathcal{C}(A') \cap \mathcal{C}(A')^\perp = \{0\}$ (Theorem 6.6(b)), that is, $x = A^+y$.

(c) For every y, Gy and A^+y are shortest minimizers of $\|Ax - y\|$, and by (b) there is only one; so $Gy = A^+y$. Taking for y the columns of I_m gives $G = A^+$ (Theorem 1.4(b)). □

Example 12.29 (cf. [2, Example 8.1])**.** For A of Example 12.25 and $y = (0, 0, 6)'$, decide whether $Ax = y$ is consistent, and find the minimizers of $\|Ax - y\|$ and the minimum-norm least-squares solution.

Solution.

- By Example 12.25,

$$A^+ = \frac{1}{18}\begin{pmatrix} 5 & 2 & -1 \\ 1 & 1 & 1 \\ -4 & -1 & 2 \\ 6 & 3 & 0 \end{pmatrix}, \qquad AA^+ = \frac{1}{6}\begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix}$$

  so $AA^+y = (-1, 2, 5)' \neq y$, and $Ax = y$ is inconsistent (Theorem 12.2(b); A^+ is a g-inverse of A by (P1)).

- $y - AA^+y = (1, -2, 1)'$, and $A'(1, -2, 1)' = (1 - 1,\ -4 + 4,\ -1 - 4 + 5,\ 1 - 4 + 3)' = 0$: it is orthogonal to $\mathcal{C}(A)$. So the smallest value of $\|Ax - y\|^2$ is $\|(1, -2, 1)'\|^2 = 6$ (Theorem 12.28(b)).

- $A^+y = \frac{6}{18}(-1, 1, 2, 0)' = \frac{1}{3}(-1, 1, 2, 0)'$, with $\|A^+y\|^2 = \frac{6}{9} = \frac{2}{3}$; check: $A(-1, 1, 2, 0)' = -a_1 + a_2 + 2a_3 = (-3, 6, 15)' = 3(-1, 2, 5)'$.

- By Theorem 12.28(b), the minimizers are the solutions of $Ax = (-1, 2, 5)'$, that is, $A^+y + z$ with $z \in \mathcal{N}(A)$ (Theorem 2.27(a)). The RREF R of Example 12.25 gives the special solutions $s_1 = (1, -1, 1, 0)'$ and $s_2 = (-1, -1, 0, 1)'$, a basis of $\mathcal{N}(A)$ (Theorem 4.8(c)); so the minimizers are

$$x = \frac{1}{3}\begin{pmatrix} -1 \\ 1 \\ 2 \\ 0 \end{pmatrix} + t_1\begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t_2\begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \qquad (t_1, t_2 \in \mathbb{R})$$

- For instance $x_0 = (-1, 1, 0, 0)'$ ($t_1 = -\frac{2}{3}$, $t_2 = 0$) has $Ax_0 = -a_1 + a_2 = (-1, 2, 5)'$, and $\|x_0\|^2 = 2 > \frac{2}{3}$. Here $x_0 - A^+y = -\frac{2}{3}s_1$ is orthogonal to A^+y, as $(-1, 1, 2, 0)s_1 = -1 - 1 + 2 = 0$, and $\|x_0\|^2 = \frac{2}{3} + \frac{4}{9} \cdot 3 = 2$ (Proposition 6.2).

- $(-1, 1, 2, 0) = -(1, 0, -1, 1) + (0, 1, 1, 1)$ is a combination of the nonzero rows of R, so $A^+y \in \mathcal{C}(A')$ (Theorem 4.8(b)), as Theorem 12.28(b) says.

Pearson's statistic

Example 12.30 ([17, Exercise 4.39])**.** In N independent trials, each trial results in exactly one of $k \geqslant 2$ categories, category j with probability $p_j > 0$ ($p_1 + \cdots + p_k = 1$). Let $y_i \in \mathbb{R}^k$ record trial i: $y_i = e_j$ if trial i results in category j. Let $c = y_1 + \cdots + y_N$ be the vector of counts, $p = (p_1, \ldots, p_k)'$, $D = \operatorname{diag}(p_1, \ldots, p_k)$, $V = D - pp'$ and $u = (c - Np)/\sqrt{N}$. **Pearson's statistic** is

$$X^2 = \sum_{j=1}^{k} \frac{(c_j - Np_j)^2}{Np_j}$$

where $Np_j = \mathsf{E}(c_j)$ is the expected count of cell j. We show:

(a) $\mathsf{E}(c) = Np$ and $\operatorname{var}(c) = NV$; in particular $\operatorname{cov}(c_i, c_j) = -Np_ip_j$ for $i \neq j$;

(b) $V\mathbf{1} = 0$, $\mathcal{N}(V) = \operatorname{span}\{\mathbf{1}\}$ and $\operatorname{rank} V = k - 1$, so $\operatorname{var}(c)$ is singular;

(c) D^{-1} is a g-inverse of V;

(d) $X^2 = u'D^{-1}u$, and $u'Gu = X^2$ for every g-inverse G of V;

(e) let $u_* \sim N_k(0, V)$, and take u_* as the normal approximation to u for large N (assumed, not proved here), so that $X^2 = u'D^{-1}u$ is approximated by $u_*'D^{-1}u_*$; then $u_*'D^{-1}u_* \sim \chi^2_{k-1}$.

Solution. (a)

- Entry j of y_i is 1 with probability p_j and 0 otherwise, so its expectation is p_j: $\mathsf{E}(y_i) = p$. As y_i is one of $e_1, \dots, e_k$, $y_i y_i' = \operatorname{diag}(y_i)$, so $\mathsf{E}(y_i y_i') = D$ and $\operatorname{var}(y_i) = D - pp' = V$ (Theorem 9.4(e)).
- With $Z = [y_1 \ \cdots \ y_N] \in \mathbb{R}^{k\times N}$, $c = Z\mathbf{1}_N$, and $\mathsf{E}(c) = \mathsf{E}(Z)\mathbf{1}_N = Np$ (Theorem 9.3(a)).
- The trials are independent, so $y_1, \dots, y_N$ are independent and $\operatorname{cov}(y_i, y_j) = 0$ for $i \neq j$ (N7). By Theorem 9.4(d), $\operatorname{var}(c) = \operatorname{var}(y_1) + \cdots + \operatorname{var}(y_N) = NV$. Its (i, j) entry for $i \neq j$ is $\operatorname{cov}(c_i, c_j) = -Np_ip_j$ (Definition 9.1).

(b) $V\mathbf{1} = p - p(p'\mathbf{1}) = p - p = 0$. If $Vx = 0$, then $Dx = p(p'x)$, so $p_jx_j = p_j(p'x)$ and $x_j = p'x$ for every j (as $p_j > 0$): $x = (p'x)\mathbf{1}$. Hence $\mathcal{N}(V) = \operatorname{span}\{\mathbf{1}\}$, and $\operatorname{rank} V = k - 1$ (Theorem 4.12). So $\operatorname{var}(c) = NV$ is singular; correspondingly $\mathbf{1}'c = N$ for every outcome of the trials, a constant, as Theorem 9.5 requires.

(c) As $p'D^{-1} = \mathbf{1}'$, $VD^{-1} = I_k - p\mathbf{1}'$; as $\mathbf{1}'D = p'$ and $\mathbf{1}'p = 1$,

$$VD^{-1}V = (I_k - p\mathbf{1}')(D - pp') = D - pp' - pp' + p(\mathbf{1}'p)p' = D - pp' = V$$

So D^{-1} is a g-inverse of V (Definition 12.1).

(d)

- $u'D^{-1}u = \sum_j u_j^2/p_j = \sum_j (c_j - Np_j)^2/(Np_j) = X^2$.
- $\mathbf{1}'u = (\mathbf{1}'c - N\mathbf{1}'p)/\sqrt{N} = 0$, so $u \in W = \operatorname{span}\{\mathbf{1}\}^\perp$, a subspace of dimension $k-1$ (Theorem 6.6(c)). W contains $\mathcal{C}(V)$, since $\mathbf{1}'Vw = (V\mathbf{1})'w = 0$ for every $w \in \mathbb{R}^k$, and $\dim \mathcal{C}(V) = \operatorname{rank} V = k - 1$ by (b); so $\mathcal{C}(V) = W$ (Theorem 4.5(a)), and $u = Vw$ for some $w \in \mathbb{R}^k$.
- For every g-inverse G of V, $u'Gu = w'VGVw = w'Vw$, the same number for all G (this is Theorem 12.16(a) with $B = u'$, $C = u$). By (c) it equals $u'D^{-1}u = X^2$.

(e)

- Let $q = (\sqrt{p_1}, \dots, \sqrt{p_k})'$, so $q'q = 1$, and $D^{-1/2} = \operatorname{diag}(p_1^{-1/2}, \dots, p_k^{-1/2})$. Then $D^{-1/2}p = q$ and $D^{-1/2}VD^{-1/2} = I_k - qq' = P$, which is symmetric and idempotent ($P^2 = I_k - 2qq' + q(q'q)q' = P$), of rank $\operatorname{tr} P = k - q'q = k - 1$ (Theorems 6.29(c) and 1.11).
- By Theorem 10.2, $v = D^{-1/2}u_* \sim N_k(0, P)$, and for $z \sim N_k(0, I_k)$ also $Pz \sim N_k(0, PP') = N_k(0, P)$. So v and Pz have the same MGF, hence the same distribution (N1), and $v'v$ has the distribution of $(Pz)'(Pz) = z'Pz$.
- $u_*'D^{-1}u_* = v'v$, and $z'Pz \sim \chi^2_{k-1}$ (Theorem 11.15 with $\mu = 0$, $\sigma = 1$). Hence $u_*'D^{-1}u_* \sim \chi^2_{k-1}$.

Exercises

Exercise 12.1. Let

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ -1 & -2 & 0 & 1 \\ 1 & 2 & 2 & 1 \end{pmatrix}$$

(a) Find a full-rank factorization $A = BC$ from the RREF of A, a left inverse L of B and a right inverse K of C with integer entries, and the g-inverse $G = KL$ of A.

(b) For which $y \in \mathbb{R}^3$ is $Ax = y$ consistent? Decide for $y = (1, 1, 3)'$ and $y = (1, 0, 1)'$.

(c) Find all solutions of $Ax = (1, 1, 3)'$.

(d) Which of $x_1 + 2x_2 - x_4$, $x_3 + x_4$ and $x_1 + 2x_2$ have the same value for all these solutions?

Solution. (a) Row-reduce A:

$$A \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

The pivot columns are $1, 3$, so (Theorem 4.21(a))

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 1 & 2 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

- Columns 1, 3 of C are the unit vectors of $\mathbb{R}^2$, and rows 1, 2 of B form $B_1 = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with $B_1 \left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right) = I_2$, so $B_1^{-1} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ (Theorem 2.16). So

$$K = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad L = [B_1^{-1} \;\; 0] = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

  give $CK = I_2$ and $LB = B_1^{-1} B_1 = I_2$ (Theorems 1.4(a),(b) and 5.3(c)).

- By Theorem 12.3(a),

$$G = KL = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

  is a g-inverse of A.

(b) Since $CK = I_2$,

$$AG = BCKL = BL = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 1 & 0 \end{pmatrix}$$

so $AGy = (y_1,\ y_2,\ 2y_1 + y_2)'$. By Theorem 12.2(b), $Ax = y$ is consistent if and only if $y_3 = 2y_1 + y_2$. For $y = (1, 1, 3)'$, $3 = 2 + 1$: consistent. For $y = (1, 0, 1)'$, $AGy = (1, 0, 2)' \neq y$: inconsistent.

(c) $Gy = (-1,\ 0,\ 1 + 1,\ 0)' = (-1, 0, 2, 0)'$, and $GA = KLBC = KC$, so

$$GA = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad I_4 - GA = \begin{pmatrix} 0 & -2 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

By Theorem 12.8, the solutions are

$$x = \begin{pmatrix}-1\\0\\2\\0\end{pmatrix} + z_2\begin{pmatrix}-2\\1\\0\\0\end{pmatrix} + z_4\begin{pmatrix}1\\0\\-1\\1\end{pmatrix} = \begin{pmatrix}-1-2z_2+z_4\\z_2\\2-z_4\\z_4\end{pmatrix} \qquad (z_2, z_4 \in \mathbb{R})$$

Check: $A(-1,0,2,0)' = -a_1 + 2a_3 = (-1+2,\ 1+0,\ -1+4)' = (1,1,3)'$.

(d) Use Theorem 12.9(iii) with the rows of GA (Theorem 1.4(c)).

- $\lambda = (1,2,0,-1)'$: $\lambda'GA = \operatorname{row}_1(GA) + 2\operatorname{row}_2(GA) - \operatorname{row}_4(GA) = (1,2,0,-1) = \lambda'$; indeed $x_1 + 2x_2 - x_4 = (-1-2z_2+z_4) + 2z_2 - z_4 = -1$ for all solutions.
- $\lambda = (0,0,1,1)'$: $\lambda'GA = \operatorname{row}_3(GA) + \operatorname{row}_4(GA) = (0,0,1,1) = \lambda'$; indeed $x_3 + x_4 = 2$.
- $\lambda = (1,2,0,0)'$: $\lambda'GA = \operatorname{row}_1(GA) + 2\operatorname{row}_2(GA) = (1,2,0,-1) \neq \lambda'$; indeed $x_1 + 2x_2 = -1 + z_4$ changes with z_4.

Exercise 12.2. Let

$$A = \begin{pmatrix}1 & -1 & 2 & 1\\2 & -1 & 3 & 4\\1 & 0 & 1 & 3\end{pmatrix}$$

(a) Row-reduce $[A \mid I_3]$ to $[R \mid P]$, and write down the g-inverse G of Corollary 12.5.

(b) Compute AG, and check that $AGA = A$.

(c) Decide whether $Ax = y$ is consistent for $y = (1,3,2)'$ and for $y = (2,1,0)'$, and find all solutions of the consistent system.

Solution. (a)

$$\left(\begin{array}{cccc|ccc}1 & -1 & 2 & 1 & 1 & 0 & 0\\2 & -1 & 3 & 4 & 0 & 1 & 0\\1 & 0 & 1 & 3 & 0 & 0 & 1\end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{cccc|ccc}1 & -1 & 2 & 1 & 1 & 0 & 0\\0 & 1 & -1 & 2 & -2 & 1 & 0\\0 & 1 & -1 & 2 & -1 & 0 & 1\end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc}1 & -1 & 2 & 1 & 1 & 0 & 0\\0 & 1 & -1 & 2 & -2 & 1 & 0\\0 & 0 & 0 & 0 & 1 & -1 & 1\end{array}\right) \xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{cccc|ccc}1 & 0 & 1 & 3 & -1 & 1 & 0\\0 & 1 & -1 & 2 & -2 & 1 & 0\\0 & 0 & 0 & 0 & 1 & -1 & 1\end{array}\right) = [R \mid P]$$

R is the RREF of A, with pivot columns 1, 2; so $\operatorname{rank} A = 2$. By Corollary 12.5, rows 1, 2 of G are rows 1, 2 of P, and rows 3, 4 are zero:

$$G = \begin{pmatrix}-1 & 1 & 0\\-2 & 1 & 0\\0 & 0 & 0\\0 & 0 & 0\end{pmatrix}$$

(b) By the outer product expansion (Theorem 1.4(d)), with $a_1 = (1,2,1)'$ and $a_2 = (-1,-1,0)'$,

$$AG = a_1\operatorname{row}_1(G) + a_2\operatorname{row}_2(G) = \begin{pmatrix}-1 & 1 & 0\\-2 & 2 & 0\\-1 & 1 & 0\end{pmatrix} + \begin{pmatrix}2 & -1 & 0\\2 & -1 & 0\\0 & 0 & 0\end{pmatrix} = \begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\-1 & 1 & 0\end{pmatrix}$$

By Theorem 1.4(c), AGA has the rows $\operatorname{row}_1(A)$, $\operatorname{row}_2(A)$ and $-\operatorname{row}_1(A) + \operatorname{row}_2(A) = (1,0,1,3) = \operatorname{row}_3(A)$: $AGA = A$.

(c) $AGy = (y_1,\ y_2,\ -y_1 + y_2)'$, so $Ax = y$ is consistent if and only if $y_3 = y_2 - y_1$ (Theorem 12.2(b)).

- $y = (2,1,0)'$: $AGy = (2,1,-1)' \neq y$, so $Ax = y$ is inconsistent.

- $y = (1,3,2)'$: $AGy = (1,3,2)' = y$, so $Ax = y$ is consistent, and $Gy = (-1+3,\ -2+3,\ 0,\ 0)' = (2,1,0,0)'$ is a solution.
- Rows 1, 2 of GA are rows 1, 2 of $PA = R$, and rows 3, 4 are zero (Theorem 1.4(c)):

$$GA = \begin{pmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad I_4 - GA = \begin{pmatrix} 0 & 0 & -1 & -3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- By Theorem 12.8, the solutions are

$$x = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z_3 \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + z_4 \begin{pmatrix} -3 \\ -2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 - z_3 - 3z_4 \\ 1 + z_3 - 2z_4 \\ z_3 \\ z_4 \end{pmatrix} \qquad (z_3, z_4 \in \mathbb{R})$$

(columns 1, 2 of $I_4 - GA$ are 0). Check: $A(2,1,0,0)' = 2a_1 + a_2 = (2-1,\ 4-1,\ 2+0)' = (1,3,2)'$.

Exercise 12.3 (cf. [3, Ch. 4, Exercise 2])**.** Let

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{pmatrix}$$

(a) Find $\operatorname{rank} A$ by row reduction.

(b) Find the g-inverse G_1 of Theorem 12.6(b) built from the leading 2×2 submatrix, and check that $AG_1A = A$.

(c) Find the g-inverse G_2 built from the submatrix in rows $2, 3$ and columns $1, 3$, and check that $AG_2A = A$.

Solution. (a)

$$A \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

So $\operatorname{rank} A = 2$; the steps show $\operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$.

(b) The leading block $A_{11} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ is invertible, with $A_{11}^{-1} = \left(\begin{smallmatrix} 1 & -2 \\ 0 & 1 \end{smallmatrix}\right)$ (the product is I_2). By Theorem 12.6(b),

$$G_1 = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad AG_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

where $AG_1 = a_1 \operatorname{row}_1(G_1) + a_2 \operatorname{row}_2(G_1)$ (Theorem 1.4(d)). By Theorem 1.4(c), AG_1A has the rows $\operatorname{row}_1(A)$, $\operatorname{row}_2(A)$ and $\operatorname{row}_1(A) + \operatorname{row}_2(A) = \operatorname{row}_3(A)$: $AG_1A = A$.

(c) The submatrix in rows $i_1 = 2$, $i_2 = 3$ and columns $j_1 = 1$, $j_2 = 3$ is $A_* = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, with $\det A_* = -1$ and $A_*^{-1} = \left(\begin{smallmatrix} -2 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ (Theorem 5.24). By Theorem 12.6(b), the (s,t) entry of A_*^{-1} goes to position (j_s, i_t) of G_2: the entries -2, 1, 1, 0 go to positions $(1,2)$, $(1,3)$, $(3,2)$, $(3,3)$. So

$$G_2 = \begin{pmatrix} 0 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

and, by Theorem 1.4(d),

$$AG_2 = a_1 \operatorname{row}_1(G_2) + a_3 \operatorname{row}_3(G_2) = \begin{pmatrix} 0 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

AG_2A has the rows $-\operatorname{row}_2(A) + \operatorname{row}_3(A) = (1, 2, 1) = \operatorname{row}_1(A)$, $\operatorname{row}_2(A)$ and $\operatorname{row}_3(A)$: $AG_2A = A$. ($G_2 \neq G_1$: a g-inverse is not unique.)

Exercise 12.4. Let

$$A = \begin{pmatrix} 2 & 2 & 0 & 1 \\ 0 & -2 & 2 & 1 \\ 1 & 0 & 1 & 1 \\ 2 & 4 & -2 & 0 \end{pmatrix}, \qquad p_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 2 \end{pmatrix}, \quad p_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad q_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \quad q_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

(a) Show that $A = p_1q_1' + p_2q_2'$, that this is a full-rank factorization $A = BC$ with $B = [p_1\ p_2]$ and the rows q_1', q_2' of C, and that $\operatorname{rank} A = 2$.

(b) Find the g-inverses G_1 and G_2 of A of Theorem 12.6(b) built from the leading 2×2 submatrix and from the submatrix in rows $1, 3$ and columns $3, 4$.

(c) Use G_2 to find the condition on $y \in \mathbb{R}^4$ for $Ax = y$ to be consistent, and find all solutions of $Ax = y$ for $y = (2, 0, 1, 2)'$.

Solution. (a) By the outer product expansion (Theorem 1.4(d)), $p_1q_1' + p_2q_2' = BC$, and

$$p_1q_1' + p_2q_2' = \begin{pmatrix} 1 & 2 & -1 & 0 \\ -1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & -2 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = A$$

Since $p_1'p_2 = 0$ and $q_1'q_2 = 0$, the nonzero vectors p_1, p_2 are linearly independent, and so are q_1, q_2 (Proposition 6.11(b)): B has full column rank and C full row rank (Theorem 4.13(b),(c)). By Theorem 4.15(b), $\operatorname{rank} A = \operatorname{rank} C = 2$, and $A = BC$ is a full-rank factorization.

(b)

- The leading submatrix $M_1 = \left(\begin{smallmatrix} 2 & 2 \\ 0 & -2 \end{smallmatrix}\right)$ has $M_1^{-1} = \frac{1}{4}\left(\begin{smallmatrix} 2 & 2 \\ 0 & -2 \end{smallmatrix}\right)$ (the product is I_2). By Theorem 12.6(b),

$$G_1 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- The submatrix in rows $i_1 = 1$, $i_2 = 3$ and columns $j_1 = 3$, $j_2 = 4$ is $M_2 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with $M_2^{-1} = \left(\begin{smallmatrix} -1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ (the product is I_2). By Theorem 12.6(b), the (s, t) entry of M_2^{-1} goes to position (j_s, i_t): the entries -1, 1, 1, 0 go to the positions $(3, 1)$, $(3, 3)$, $(4, 1)$, $(4, 3)$:

$$G_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

(c)

- With the columns $a_1, \ldots, a_4$ of A, $AG_2 = [-a_3 + a_4 \ \ 0 \ \ a_3 \ \ 0]$ (Theorem 1.4(a),(b)), so $AG_2y = y_1(1, -1, 0, 2)' + y_3(0, 2, 1, -2)' = (y_1,\ 2y_3 - y_1,\ y_3,\ 2y_1 - 2y_3)'$. By Theorem 12.2(b), $Ax = y$ is consistent if and only if $y_2 = 2y_3 - y_1$ and $y_4 = 2y_1 - 2y_3$.

- For $y = (2,0,1,2)'$ both conditions hold. The rows of G_2A are 0, 0, $\text{row}_3(A) - \text{row}_1(A) = (-1,-2,1,0)$ and $\text{row}_1(A) = (2,2,0,1)$ (Theorem 1.4(c)), so

$$G_2y = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \end{pmatrix}, \qquad I_4 - G_2A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{pmatrix}$$

- By Theorem 12.8, $\mathcal{N}(A) = \mathcal{C}(I_4 - G_2A) = \text{span}\{(1,0,1,-2)', (0,1,2,-2)'\}$, and the solutions are

$$x = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \end{pmatrix} + z_1 \begin{pmatrix} 1 \\ 0 \\ 1 \\ -2 \end{pmatrix} + z_2 \begin{pmatrix} 0 \\ 1 \\ 2 \\ -2 \end{pmatrix} \qquad (z_1, z_2 \in \mathbb{R})$$

- The solution $G_1y = (1,0,0,0)' = e_1$ ($z_1 = 1$, $z_2 = 0$) is visible directly: y is the first column of A.

Exercise 12.5. Let

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{pmatrix}, \qquad G_1 = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

where $\text{rank}\, A = 2$ and G_1 is a g-inverse of A (Exercise 12.3).

(a) Compute G_1A, check that it is idempotent, and find $\text{tr}(G_1A)$ and $\text{rank}(I_3 - G_1A)$.

(b) Is G_1 reflexive?

(c) Let $U \in \mathbb{R}^{3\times 3}$ have $(3,3)$ entry 1 and all other entries 0. Show that $G_3 = G_1 + (I_3 - G_1A)U$ is an invertible g-inverse of A, and find the reflexive g-inverse G_3AG_3.

Solution. (a) By Theorem 1.4(c), the rows of G_1A are $\text{row}_1(A) - 2\,\text{row}_2(A) = (1,0,-1)$, $\text{row}_2(A) = (0,1,1)$ and 0:

$$G_1A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad I_3 - G_1A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

The rows of $(G_1A)(G_1A)$ are $\text{row}_1(G_1A) - \text{row}_3(G_1A) = (1,0,-1)$, $\text{row}_2(G_1A) + \text{row}_3(G_1A) = (0,1,1)$ and 0: so G_1A is idempotent. $\text{tr}(G_1A) = 1 + 1 + 0 = 2 = \text{rank}\, A$, and $I_3 - G_1A$ has rank $1 = 3 - 2$ (its columns are multiples of $(1,-1,1)'$), as Theorem 12.13(c) says.

(b) The rows $(1,-2,0)$ and $(0,1,0)$ of G_1 are linearly independent, so $\text{rank}\, G_1 = 2 = \text{rank}\, A$, and G_1 is reflexive (Theorem 12.14(b)). Check: $G_1AG_1 = (G_1A)G_1$ has the rows $\text{row}_1(G_1) - \text{row}_3(G_1) = (1,-2,0)$, $\text{row}_2(G_1) + \text{row}_3(G_1) = (0,1,0)$ and 0, so $G_1AG_1 = G_1$.

(c)

- $(I_3 - G_1A)U$ has column 3 equal to column 3 of $I_3 - G_1A$, that is, $(1,-1,1)'$, and its other columns 0 (Theorem 1.4(a)). So

$$G_3 = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

- $A(I_3 - G_1A) = A - AG_1A = 0$, so $AG_3 = AG_1$ and $AG_3A = AG_1A = A$: G_3 is a g-inverse of A. It is upper triangular with diagonal $1,1,1$, so $\det G_3 = 1$ (Theorem 5.14(c)) and G_3 is invertible (Theorem 5.18(a)); $\text{rank}\, G_3 = 3 > \text{rank}\, A$, so G_3 is not reflexive (Theorem 12.14(b)).

- By Theorem 1.4(d),
$$AG_1 = a_1 \operatorname{row}_1(G_1) + a_2 \operatorname{row}_2(G_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$
and $G_3AG_3 = G_3(AG_1)$ has the columns $G_3(1,0,1)' = (1+1,\ 0-1,\ 0+1)'$, $G_3(0,1,1)' = (-2+1,\ 1-1,\ 0+1)'$ and 0 (Theorem 1.4(a),(b)):
$$G_3AG_3 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$
a reflexive g-inverse of A (Theorem 12.14(a)), of rank 2.

Exercise 12.6. Let
$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 1 & 2 & -1 \\ 1 & -2 & 3 \end{pmatrix}$$

(a) Compute $A'A$, show that $\operatorname{rank} A = 2$, and find the g-inverse G of $A'A$ of Theorem 12.6(b) built from its leading 2×2 submatrix.

(b) Check that $AGA'A = A$, and write down the g-inverse GA' of A.

(c) Use GA' to decide whether $Ax = y$ is consistent for $y = (2,2,2,2)'$ and for $y = e_1$. In the consistent case find all solutions and the shortest one.

Solution. (a) The columns are $a_1 = (1,1,1,1)'$, $a_2 = (1,-1,2,-2)'$ and $a_3 = (0,2,-1,3)' = a_1 - a_2$, and, with the entries $a_i'a_j$ (Theorem 1.4(d)),
$$A'A = \begin{pmatrix} 4 & 0 & 4 \\ 0 & 10 & -10 \\ 4 & -10 & 14 \end{pmatrix}$$
a_1 and a_2 are nonzero and orthogonal, hence linearly independent (Proposition 6.11(b)), and $a_3 \in \operatorname{span}\{a_1, a_2\}$; so $\operatorname{rank} A = 2$, and $\operatorname{rank}(A'A) = 2$ (Theorem 4.18(b)). The leading submatrix $\operatorname{diag}(4,10)$ is invertible, so $G = \operatorname{diag}(\frac{1}{4}, \frac{1}{10}, 0)$ is a g-inverse of $A'A$ (Theorem 12.6(b)).

(b) Multiplying the rows of $A'A$ by $\frac{1}{4}, \frac{1}{10}, 0$ (Theorem 1.4(c)),
$$GA'A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$
so the columns of $A(GA'A)$ are a_1, a_2 and $a_1 - a_2 = a_3$ (Theorem 1.4(a),(b)): $AGA'A = A$, as Theorem 12.17(a) says. By Theorem 12.17(a),
$$GA' = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{10} & -\frac{1}{10} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
is a g-inverse of A.

(c)

- $A(GA')y = \frac{1}{4}(a_1'y)a_1 + \frac{1}{10}(a_2'y)a_2$ (Theorem 1.4(d)). For $y = (2,2,2,2)' = 2a_1$, $a_1'y = 8$ and $a_2'y = 0$, so $AGA'y = 2a_1 = y$: consistent. For $y = e_1$, $AGA'e_1 = \frac{1}{4}a_1 + \frac{1}{10}a_2 = \frac{1}{20}(7,3,9,1)' \neq e_1$: inconsistent (Theorem 12.2(b)).

- For $y = 2a_1$, $GA'y = (2,0,0)'$, and

$$I_3 - GA'A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

so the solutions are $x = (2,0,0)' + z(-1,1,1)'$, $z \in \mathbb{R}$ (Theorem 12.8).

- As $Ax = y$ is consistent, its solutions are the minimizers of $\|Ax - y\|$, and the shortest one is the only one in $\mathcal{C}(A') = \mathcal{N}(A)^\perp$ (Theorem 12.28(b); Corollary 6.7 and Theorem 6.8(a)), where $\mathcal{N}(A) = \text{span}\{(-1,1,1)'\}$. The condition $(2-z)(-1) + z + z = 0$ gives $z = \frac{2}{3}$ and $x^+ = \frac{1}{3}(4,2,2)'$, with $\|x^+\|^2 = \frac{24}{9} = \frac{8}{3}$; the solution $(2,0,0)'$ has $\|(2,0,0)'\|^2 = 4 = \frac{8}{3} + \frac{4}{3}$.

Exercise 12.7. Let

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix}$$

(a) Find A^+ from a full-rank factorization, and check that AA^+ and A^+A are symmetric.

(b) Find the shortest solution of $Ax = (0,3,3,3)'$, and check it against all solutions.

Solution. (a) Row-reduce A:

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 1 & 3 & 4 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The last matrix is the RREF; the pivot columns are $1, 2$, so $A = BC$ with (Theorem 4.21(a))

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix}$$

By Theorem 5.24,

$$B'B = \begin{pmatrix} 6 & 3 \\ 3 & 3 \end{pmatrix}, \quad (B'B)^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad CC' = \begin{pmatrix} 3 & 3 \\ 3 & 6 \end{pmatrix}, \quad (CC')^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

(both determinants are 9). Then

$$L = (B'B)^{-1}B' = \frac{1}{3}\begin{pmatrix} 1 & -1 & 0 & 1 \\ -1 & 2 & 1 & 0 \end{pmatrix}, \qquad K = C'(CC')^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and by Theorem 12.24(b)

$$A^+ = KL = \frac{1}{9}\begin{pmatrix} 2 & -1 \\ -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 & 0 & 1 \\ -1 & 2 & 1 & 0 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 3 & -4 & -1 & 2 \\ -2 & 3 & 1 & -1 \\ 1 & -1 & 0 & 1 \\ -1 & 2 & 1 & 0 \end{pmatrix}$$

Since $CK = I_2$ and $LB = I_2$,

$$AA^+ = BCKL = BL = \frac{1}{3}\begin{pmatrix} 1 & -1 & 0 & 1 \\ -1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 \end{pmatrix}, \qquad A^+A = KLBC = KC = \frac{1}{3}\begin{pmatrix} 2 & -1 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix}$$

both symmetric.

(b)

- Let $y = (0,3,3,3)'$. A^+ is a g-inverse of A, and $AA^+y = \frac{1}{3}(0-3+0+3,\ 0+6+3+0,\ 0+3+3+3,\ 0+0+3+6)' = (0,3,3,3)' = y$, so $Ax = y$ is consistent (Theorem 12.2(b)).
- By (P1) and (P4), A^+ is a minimum-norm g-inverse of A, so the shortest solution is (Theorem 12.20)

$$A^+y = \frac{1}{9}\begin{pmatrix} 0-12-3+6 \\ 0+9+3-3 \\ 0-3+0+3 \\ 0+6+3+0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

  check: $A(-1,1,0,1)' = -a_1 + a_2 + a_4 = (0,3,3,3)'$.
- By Theorem 12.8, the solutions are $A^+y + (I_4 - A^+A)z$, where

$$I_4 - A^+A = \frac{1}{3}\begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ 0 & -1 & -1 & 1 \end{pmatrix}$$

  has the columns $-\frac{1}{3}s_1$, $-\frac{1}{3}s_2$, $\frac{1}{3}(2s_1 - s_2)$ and $\frac{1}{3}(s_2 - s_1)$, with the special solutions $s_1 = (-1,-1,1,0)'$ and $s_2 = (-1,-2,0,1)'$ of the RREF (Theorem 4.8(c)). So the solutions are $x = (-1-t_1-t_2,\ 1-t_1-2t_2,\ t_1,\ 1+t_2)'$ $(t_1, t_2 \in \mathbb{R})$, with

$$\|x\|^2 = (1+t_1+t_2)^2 + (1-t_1-2t_2)^2 + t_1^2 + (1+t_2)^2 = 3 + 3(t_1+t_2)^2 + 3t_2^2$$

  smallest exactly for $t_1 = t_2 = 0$.

Exercise 12.8. Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}, \qquad f_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},\ f_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},\ g_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix},\ g_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

(a) Show that $A = f_1 g_1' + f_2 g_2'$, that this is a full-rank factorization $A = BC$ with $B = [f_1\ f_2]$ and the rows g_1', g_2' of C, and that $B'B = CC' = \operatorname{diag}(3,2)$.

(b) Find A^+ and AA^+.

(c) Find the condition on $y \in \mathbb{R}^3$ for $Ax = y$ to be consistent. For $y = (8,4,6)'$ find the shortest solution, and compare it with the solution $(6,2,0,0,0)'$.

Solution. (a) $f_1 g_1'$ has the three rows $(1,0,1,0,1)$, and $f_2 g_2'$ the rows $(0,1,0,1,0)$, $(0,-1,0,-1,0)$ and 0; their sum is A, and it is BC by the outer product expansion (Theorem 1.4(d)). The nonzero vectors $f_1 \perp f_2$ are linearly independent, and so are $g_1 \perp g_2$ (Proposition 6.11(b)); so B has full column rank, C has full row rank (Theorem 4.13(b),(c)), and $\operatorname{rank} A = \operatorname{rank} C = 2$ (Theorem 4.15(b)). With the entries $f_i'f_j$ and $g_i'g_j$ (Theorem 1.4(d)), $B'B = \operatorname{diag}(3,2)$ and $CC' = \operatorname{diag}(3,2)$.

(b) By Theorem 12.24(b) and the outer product expansion,

$$A^+ = C' \operatorname{diag}(\tfrac{1}{3}, \tfrac{1}{2}) \operatorname{diag}(\tfrac{1}{3}, \tfrac{1}{2}) B' = \frac{1}{9} g_1 f_1' + \frac{1}{4} g_2 f_2' = \frac{1}{36} \begin{pmatrix} 4 & 4 & 4 \\ 9 & -9 & 0 \\ 4 & 4 & 4 \\ 9 & -9 & 0 \\ 4 & 4 & 4 \end{pmatrix}$$

and, by the proof of Theorem 12.24(b),

$$AA^+ = B(B'B)^{-1}B' = \frac{1}{3} f_1 f_1' + \frac{1}{2} f_2 f_2' = \frac{1}{6} \begin{pmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{pmatrix}$$

(c)

- A^+ is a g-inverse of A, so $Ax = y$ is consistent if and only if $AA^+y = y$ (Theorem 12.2(b)), that is, $(I_3 - AA^+)y = 0$. With $w = (1, 1, -2)'$,

$$I_3 - AA^+ = \frac{1}{6} \begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ -2 & -2 & 4 \end{pmatrix} = \frac{1}{6} ww'$$

  so $(I_3 - AA^+)y = \frac{1}{6}(w'y)w$, and $Ax = y$ is consistent if and only if $y_1 + y_2 - 2y_3 = 0$.
- For $y = (8, 4, 6)'$, $8 + 4 - 12 = 0$: consistent. By (P1) and (P4), A^+ is a minimum-norm g-inverse of A, so the shortest solution is (Theorem 12.20) $x^+ = A^+y = \frac{1}{36}(72, 36, 72, 36, 72)' = (2, 1, 2, 1, 2)'$, with $\|x^+\|^2 = 14$.
- The solution $x = (6, 2, 0, 0, 0)'$ ($Ax = 6a_1 + 2a_2 = (8, 4, 6)'$) has $\|x\|^2 = 40 = 14 + \|(4, 1, -2, -1, -2)'\|^2 = 14 + 26$.

Exercise 12.9. For the matrix $A = p_1 q_1' + p_2 q_2' = BC$ of Exercise 12.4, find A^+, AA^+ and A^+A, and the shortest solution of $Ax = (2, 0, 1, 2)'$.

Solution.

- $p_1'p_1 = q_1'q_1 = 6$, $p_2'p_2 = q_2'q_2 = 3$ and $p_1'p_2 = q_1'q_2 = 0$, so $B'B = CC' = \operatorname{diag}(6, 3)$ (Theorem 1.4(d)). By Theorem 12.24(b) and the outer product expansion,

$$A^+ = C' \operatorname{diag}(\tfrac{1}{6}, \tfrac{1}{3}) \operatorname{diag}(\tfrac{1}{6}, \tfrac{1}{3}) B' = \frac{1}{36} q_1 p_1' + \frac{1}{9} q_2 p_2' = \frac{1}{36} \begin{pmatrix} 5 & 3 & 4 & 2 \\ 2 & -2 & 0 & 4 \\ 3 & 5 & 4 & -2 \\ 4 & 4 & 4 & 0 \end{pmatrix}$$

- By the proof of Theorem 12.24(b), $AA^+ = B(B'B)^{-1}B'$ and $A^+A = C'(CC')^{-1}C$:

$$AA^+ = \frac{1}{6} p_1 p_1' + \frac{1}{3} p_2 p_2' = \frac{1}{6} \begin{pmatrix} 3 & 1 & 2 & 2 \\ 1 & 3 & 2 & -2 \\ 2 & 2 & 2 & 0 \\ 2 & -2 & 0 & 4 \end{pmatrix}, \qquad A^+A = \frac{1}{6} q_1 q_1' + \frac{1}{3} q_2 q_2' = \frac{1}{6} \begin{pmatrix} 3 & 2 & 1 & 2 \\ 2 & 4 & -2 & 0 \\ 1 & -2 & 3 & 2 \\ 2 & 0 & 2 & 2 \end{pmatrix}$$

  They are the projection matrices onto $\mathcal{C}(A)$ and $\mathcal{C}(A')$ (Proposition 12.26(b)); both have trace $2 = \operatorname{rank} A$ (Theorem 12.13(c)).
- The system is consistent (Exercise 12.4). By (P1) and (P4), A^+ is a minimum-norm g-inverse of A, so by Theorem 12.20 the shortest solution is $x^+ = A^+(2, 0, 1, 2)' = \frac{1}{36}(18, 12, 6, 12)' = \frac{1}{6}(3, 2, 1, 2)'$, with $\|x^+\|^2 = \frac{18}{36} = \frac{1}{2}$. It is the first column of A^+A: $A^+(2, 0, 1, 2)' = A^+Ae_1$, the projection onto $\mathcal{C}(A')$ of the solution e_1.

- For the solutions e_1 and $(0,0,-1,2)'$ of Exercise 12.4, $\|e_1 - x^+\|^2 = \frac{9+4+1+4}{36} = \frac{1}{2}$ and $\|(0,0,-1,2)' - x^+\|^2 = \frac{9+4+49+100}{36} = \frac{9}{2}$, and indeed $1 = \frac{1}{2} + \frac{1}{2}$ and $5 = \frac{1}{2} + \frac{9}{2}$.

Exercise 12.10. In an experiment of Mendel's type, $N = 160$ pea seeds fall into the four categories round yellow, round green, wrinkled yellow and wrinkled green, with the counts $c = (93, 27, 31, 9)'$. Mendel's theory gives the probabilities $p = \frac{1}{16}(9,3,3,1)'$. Let D, V, u and X^2 be as in Example 12.30.

(a) Compute the expected counts Np_j, the contributions $(c_j - Np_j)^2/(Np_j)$ and X^2.

(b) Under the approximation of Example 12.30(e), the distribution of X^2 is approximated by χ^2_3. Show that $\mathsf{P}(Y > \frac{8}{15}) > \frac{1}{2}$ for $Y \sim \chi^2_3$.

(c) Compute V, its leading 3×3 block V_{11} and V_{11}^{-1}, and check that $u'Gu = X^2$ for the g-inverse $G = \begin{pmatrix} V_{11}^{-1} & 0 \\ 0 & 0 \end{pmatrix}$ of V.

Solution. (a) The expected counts are $160 \cdot \frac{9}{16} = 90$, $160 \cdot \frac{3}{16} = 30$ (twice) and $160 \cdot \frac{1}{16} = 10$, and $c - Np = (3, -3, 1, -1)'$. The contributions are

$$\frac{9}{90} = \frac{1}{10}, \qquad \frac{9}{30} = \frac{3}{10}, \qquad \frac{1}{30}, \qquad \frac{1}{10}$$

and $X^2 = \frac{3+9+1+3}{30} = \frac{8}{15}$.

(b) Let z_1, z_2, z_3 be independent $N(0,1)$ variables. By Definition 11.2(a), $v_3 = z_1^2 + z_2^2 + z_3^2 \sim \chi^2_3$ and $v_2 = z_1^2 + z_2^2 \sim \chi^2_2$, and $v_3 \geqslant v_2$. So the event $\{v_2 > c\}$ is contained in $\{v_3 > c\}$, and $\mathsf{P}(Y > c) = \mathsf{P}(v_3 > c) \geqslant \mathsf{P}(v_2 > c) = e^{-c/2}$ for $c > 0$ (Example 11.5(a)). For $c = \frac{8}{15}$, $e^{-4/15} > \frac{1}{2}$, since $e^4 < 3^4 = 81 < 2^{15}$ ($e < 3$, Example 11.14). So under the approximation a value of X^2 larger than the one computed in (a) has probability more than $\frac{1}{2}$: the counts agree well with Mendel's $9:3:3:1$ law.

(c)

- $256D = \operatorname{diag}(144, 48, 48, 16)$ and $256pp' = (9,3,3,1)'(9,3,3,1)$, so

$$V = D - pp' = \frac{1}{256}\begin{pmatrix} 63 & -27 & -27 & -9 \\ -27 & 39 & -9 & -3 \\ -27 & -9 & 39 & -3 \\ -9 & -3 & -3 & 15 \end{pmatrix}, \qquad V_{11} = \frac{1}{256}\begin{pmatrix} 63 & -27 & -27 \\ -27 & 39 & -9 \\ -27 & -9 & 39 \end{pmatrix}$$

- With $D_3 = \operatorname{diag}(p_1, p_2, p_3)$ and $p_4 = \frac{1}{16}$, the matrix $D_3^{-1} + \frac{1}{p_4}\mathbf{1}_3\mathbf{1}_3' = \operatorname{diag}(\frac{16}{9}, \frac{16}{3}, \frac{16}{3}) + 16J_3$ is

$$V_{11}^{-1} = \frac{16}{9}\begin{pmatrix} 10 & 9 & 9 \\ 9 & 12 & 9 \\ 9 & 9 & 12 \end{pmatrix}$$

Check:

$$\begin{pmatrix} 63 & -27 & -27 \\ -27 & 39 & -9 \\ -27 & -9 & 39 \end{pmatrix}\begin{pmatrix} 10 & 9 & 9 \\ 9 & 12 & 9 \\ 9 & 9 & 12 \end{pmatrix} = 144 I_3$$

and $\frac{1}{256} \cdot \frac{16}{9} \cdot 144 = 1$; so V_{11} is invertible with this inverse (Theorem 2.16).

- $\operatorname{rank} V = 3$ (Example 12.30(b)), so G is a g-inverse of V (Theorem 12.6(b)).
- $u = (c - Np)/\sqrt{160}$, so with $u_{(3)} = (u_1, u_2, u_3)' = (3, -3, 1)'/\sqrt{160}$,

$$u'Gu = u'_{(3)}V_{11}^{-1}u_{(3)} = \frac{1}{160} \cdot \frac{16}{9}\begin{pmatrix} 3 & -3 & 1 \end{pmatrix}\begin{pmatrix} 12 \\ 0 \\ 12 \end{pmatrix} = \frac{1}{160} \cdot \frac{16}{9} \cdot 48 = \frac{8}{15} = X^2$$

where $(12, 0, 12)'$ is the product of the matrix of V_{11}^{-1} (without the factor $\frac{16}{9}$) with $(3, -3, 1)'$. In the form $D_3^{-1} + 16J_3$: $u'_{(3)}D_3^{-1}u_{(3)} + 16(u_1 + u_2 + u_3)^2 = \frac{1}{160}\Big(16 + 48 + \frac{16}{3} + 16\Big) = \frac{8}{15}$, as $u_1 + u_2 + u_3 = -u_4 = \frac{1}{\sqrt{160}}$.

Problems

Problem 12.1 ([3, Ch. 4, Exercise 3], [5, Theorem 9.2.7])**.** Let G be a g-inverse of $A \in \mathbb{R}^{m\times n}$. Show that the g-inverses of A are exactly the matrices

$$G + (I_n - GA)U + V(I_m - AG) \qquad (U, V \in \mathbb{R}^{n\times m})$$

Solution.

- Each such matrix is a g-inverse: since $AGA = A$, $A\Big(G + (I - GA)U + V(I - AG)\Big)A = AGA + (A - AGA)UA + AV(A - AGA) = A$.
- Conversely, let H be a g-inverse of A, and put $U = HAG$, $V = H - G$. Then

$$\begin{aligned} G + (I - GA)HAG + (H - G)(I - AG) &= G + HAG - GAHAG + H - HAG - G + GAG \\ &= H - G(AHA)G + GAG = H - GAG + GAG = H \end{aligned}$$

Problem 12.2 ([8, Ch. 14, Exercise 44])**.** Let $A \in \mathbb{R}^{q\times p}$, $B \in \mathbb{R}^{p\times n}$ and $C \in \mathbb{R}^{m\times q}$, and let G be a g-inverse of CAB. Show that

(a) $CABGC = C$ if and only if $\operatorname{rank}(CAB) = \operatorname{rank} C$;

(b) $BGCAB = B$ if and only if $\operatorname{rank}(CAB) = \operatorname{rank} B$.

Solution. (a)

- Let $\operatorname{rank}(CAB) = \operatorname{rank} C$. As $\mathcal{C}(CAB) \subseteq \mathcal{C}(C)$ (Theorem 4.15(a)) and the two dimensions agree, $\mathcal{C}(CAB) = \mathcal{C}(C)$ (Theorem 4.5(a)). So each column of C is $CABr_j$ for some $r_j \in \mathbb{R}^n$, that is, $C = CABR$ with $R = [r_1 \ \cdots \ r_q]$ (Theorem 1.4(a)). Hence $CABGC = CABGCABR = CABR = C$.
- Conversely, if $CABGC = C$, then $\operatorname{rank} C = \operatorname{rank}\Big((CAB)(GC)\Big) \leqslant \operatorname{rank}(CAB) \leqslant \operatorname{rank} C$ (Theorem 4.15(a)), so $\operatorname{rank}(CAB) = \operatorname{rank} C$.

(b)

- Let $\operatorname{rank}(CAB) = \operatorname{rank} B$. Each row $\operatorname{row}_i(CAB) = \operatorname{row}_i(CA)B$ lies in $\mathcal{R}(B)$ (Theorem 1.4(c)), so $\mathcal{R}(CAB) \subseteq \mathcal{R}(B)$; both have dimension $\operatorname{rank} B$, so the spaces of the transposed vectors are equal (Theorem 4.5(a)), and $\mathcal{R}(CAB) = \mathcal{R}(B)$. So each row of B is $l_i'CAB$ for some $l_i \in \mathbb{R}^m$, that is, $B = LCAB$, where L has the rows $l_1', \ldots, l_p'$ (Theorem 1.4(c)). Hence $BGCAB = LCABGCAB = LCAB = B$.
- Conversely, if $BGCAB = B$, then $\operatorname{rank} B = \operatorname{rank}\Big((BG)(CAB)\Big) \leqslant \operatorname{rank}(CAB) \leqslant \operatorname{rank} B$, so $\operatorname{rank}(CAB) = \operatorname{rank} B$.

Problem 12.3 ([3, Ch. 4, Exercise 12])**.** Let $A \in \mathbb{R}^{m\times n}$ and $G \in \mathbb{R}^{n\times m}$. Show that G is a minimum-norm g-inverse of A if and only if $GAA' = A'$.

Solution.

- If $AGA = A$ and $(GA)' = GA$, then $GAA' = (GA)'A' = A'G'A' = (AGA)' = A'$.
- Conversely, let $GAA' = A'$. Then $AGAA' = AA'$, that is, $(AG - I_m)AA' = 0$. Transposing, $AA'(AG - I_m)' = 0$: the columns of $(AG - I_m)'$ lie in $\mathcal{N}(AA') = \mathcal{N}(A')$ (Theorem 4.18(a), applied to A'). So $A'(AG - I_m)' = 0$, and transposing again, $(AG - I_m)A = 0$: $AGA = A$.
- Multiplying $GAA' = A'$ by G' on the right, $A'G' = GAA'G' = (GA)(GA)'$, which is symmetric. So $(GA)' = A'G'$ is symmetric, and $GA = ((GA)')' = (GA)'$.

Problem 12.4 ([6, Exercise 10.27])**.** (a) Let $A \in \mathbb{R}^{n\times n}$ be symmetric and idempotent. Show that $A^+ = A$.

(b) Does the converse hold?

(c) Show that $(AA^+)^+ = AA^+$ and $(A^+A)^+ = A^+A$ for every $A \in \mathbb{R}^{m\times n}$.

Solution. (a) If $A = A' = A^2$, then $A^3 = A \cdot A^2 = A^2 = A$ and $(A^2)' = A^2$. So $G = A$ satisfies (P1)–(P4): $AAA = A$, $AAA = A$, $(AA)' = AA$, $(AA)' = AA$. Hence $A^+ = A$ (Theorem 12.23).

(b) No. For $A = -I_n$, $A^+ = A^{-1} = -I_n = A$ (Theorem 12.24(c)), but $A^2 = I_n \neq A$: A is not idempotent.

(c) AA^+ and A^+A are symmetric and idempotent (Proposition 12.26(b)). Apply (a).

Problem 12.5 ([8, Ch. 20, Exercise 2])**.** Let $A \in \mathbb{R}^{m\times n}$. Show that $A^+ = A'$ if and only if $A'A$ is idempotent.

Solution.

- Let $A'A$ be idempotent: $A'A = A'AA'A$. Multiply on the left by $A^+(A^+)'$ and on the right by A^+. By (P1)–(P4) and Theorem 1.9,

$$\begin{aligned} A^+(A^+)'A'AA^+ &= A^+(AA^+)'AA^+ = A^+AA^+AA^+ = A^+AA^+ = A^+ \\ A^+(A^+)'A'AA'AA^+ &= A^+(AA^+)'AA'(AA^+)' = A^+AA^+A(AA^+A)' \\ &= A^+AA' = (A^+A)'A' = (AA^+A)' = A' \end{aligned}$$

  So $A^+ = A'$.

- Conversely, let $A^+ = A'$. Then $A'A = A^+A$, and $(A^+A)(A^+A) = A^+(AA^+A) = A^+A$: $A'A$ is idempotent.

Problem 12.6 ([5, §20.2], [6, Exercise 10.33(a)])**.** Verify the conditions (P1)–(P4) to show:

(a) $\operatorname{diag}(d_1, \ldots, d_n)^+ = \operatorname{diag}(d_1^+, \ldots, d_n^+)$, where $d^+ = 1/d$ for a number $d \neq 0$ and $0^+ = 0$;

(b) $(uv')^+ = vu'/(\|u\|^2\|v\|^2)$ for nonzero $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$; that is, $A^+ = A'/\operatorname{tr}(AA')$ for $A = uv'$.

Solution. (a) For $D = \operatorname{diag}(d_1, \ldots, d_n)$ and $G = \operatorname{diag}(d_1^+, \ldots, d_n^+)$, $DG = GD = \operatorname{diag}(d_1d_1^+, \ldots, d_nd_n^+)$ is symmetric, and $dd^+d = d$, $d^+dd^+ = d^+$ for every number d (both sides are 0 if $d = 0$); so $DGD = D$ and $GDG = G$. Hence $G = D^+$ (Theorem 12.23).

(b) Let $A = uv'$, $c = \|u\|^2\|v\|^2$ and $G = vu'/c$.

- $AG = uv'vu'/c = uu'/\|u\|^2$ and $GA = vu'uv'/c = vv'/\|v\|^2$ are symmetric: (P3), (P4).
- $AGA = uu'uv'/\|u\|^2 = uv' = A$ and $GAG = vv'vu'/(\|v\|^2c) = vu'/c = G$: (P1), (P2). So $G = A^+$ (Theorem 12.23).
- $A' = vu'$, and $\operatorname{tr}(AA') = \operatorname{tr}(uv'vu') = \|v\|^2 \operatorname{tr}(uu') = \|v\|^2u'u = c$ (Theorem 1.11(a),(b)); so $A^+ = A'/\operatorname{tr}(AA')$.

Problem 12.7 ([6, Exercise 10.29(c)–(e)])**.** Let $A \in \mathbb{R}^{m\times n}$. Show that

(a) $(A'A)^+ = A^+(A^+)'$;

(b) $(AA')^+ = (A^+)'A^+$;

(c) $A^+ = A'(AA')^+$.

Solution. First, by (P1)–(P4) and Theorem 1.9,

$$A'AA^+ = A'(AA^+)' = A'(A^+)'A' = (AA^+A)' = A', \qquad A^+(A^+)'A' = A^+(AA^+)' = A^+AA^+ = A^+$$

and likewise $A'(A^+)'A^+ = (A^+A)'A^+ = A^+AA^+ = A^+$.

(a) Let $G = A^+(A^+)'$. We check (P1)–(P4) for the matrix $A'A$, using the identities above.

- (P1): $A'A\,G\,A'A = (A'AA^+)(A^+)'A'A = A'(A^+)'A'A = (AA^+A)'A = A'A$.
- (P2): $G\,A'A\,G = A^+(A^+)'(A'AA^+)(A^+)' = A^+(A^+)'A'(A^+)' = A^+(AA^+)'(A^+)' = A^+AA^+(A^+)' = A^+(A^+)' = G$.
- (P3): $A'AG = (A'AA^+)(A^+)' = A'(A^+)' = (A^+A)' = A^+A$, which is symmetric by (P4).
- (P4): $GA'A = (A^+(A^+)'A')A = A^+A$, which is symmetric.

So $(A'A)^+ = G$ (Theorem 12.23).

(b) By Proposition 12.26(c), $(A')^+ = (A^+)'$. Part (a) for A' in place of A, as $(A')'A' = AA'$, gives $(AA')^+ = (A')^+((A')^+)' = (A^+)'A^+$.

(c) By the identity $A'(A^+)'A^+ = A^+$ and (b), $A^+ = A'\big((A^+)'A^+\big) = A'(AA')^+$.

Problem 12.8 ([10, Problem 44.1])**.** Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{p\times q}$ and $C \in \mathbb{R}^{m\times q}$, and let $C = AX = YB$ for some $X \in \mathbb{R}^{n\times q}$ and $Y \in \mathbb{R}^{m\times p}$. Show that $C = AZB$ for some $Z \in \mathbb{R}^{n\times p}$.

Solution. Let G and H be g-inverses of A and B (Theorem 12.3).

- The system $AX = C$ is consistent, so $AGC = C$ (Lemma 12.12); directly, $AGC = AGAX = AX = C$, as $AGA = A$.
- Likewise, as $BHB = B$, $CHB = YBHB = YB = C$.
- Hence $C = AGC = AG(CHB) = A(GCH)B$, that is, $C = AZB$ with $Z = GCH \in \mathbb{R}^{n\times p}$.

Problem 12.9 ([10, Problem 44.2])**.** Let $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$, and let G and H be g-inverses of A and B. Show that a matrix $X \in \mathbb{R}^{n\times p}$ satisfies $AX = 0$ and $XB = 0$ if and only if

$$X = (I_n - GA)Y(I_p - BH) \qquad \text{for some } Y \in \mathbb{R}^{n\times p}$$

Solution.

- If $X = (I_n - GA)Y(I_p - BH)$, then, as $AGA = A$ and $BHB = B$, $AX = (A - AGA)Y(I_p - BH) = 0$ and $XB = (I_n - GA)Y(B - BHB) = 0$.
- Conversely, let $AX = 0$ and $XB = 0$, and let $r = \operatorname{rank} X$. If $r = 0$, then $X = 0$ (Theorem 4.8), and $Y = 0$ will do. Let $r \geqslant 1$, and let $X = PQ$ be a full-rank factorization, with $P \in \mathbb{R}^{n\times r}$ and $Q \in \mathbb{R}^{r\times p}$ (Theorem 4.21(a)).
- $\mathcal{C}(P) = \mathcal{C}(X)$ and $\mathcal{N}(Q) = \mathcal{N}(X)$ (Theorem 4.21(b)). So each column of P is Xk_j for some $k_j \in \mathbb{R}^p$, that is, $P = XK$ with $K = [k_1 \ \cdots \ k_r]$ (Theorem 1.4(a)), and $AP = AXK = 0$. Also $XB = 0$ says that each column of B lies in $\mathcal{N}(X) = \mathcal{N}(Q)$ (Theorem 1.4(a)), so $QB = 0$.
- As $AP = 0$, each column of P lies in $\mathcal{N}(A) = \mathcal{C}(I_n - GA)$ (Theorem 12.8), so $P = (I_n - GA)Y_1$ for some $Y_1 \in \mathbb{R}^{n\times r}$ (Theorem 1.4(a)).

- $QB = 0$ gives $B'Q' = 0$ (Theorem 1.9). H' is a g-inverse of B' (Proposition 12.11(a)), so each column of Q' lies in $\mathcal{N}(B') = \mathcal{C}(I_p - H'B')$ (Theorem 12.8), and $Q' = (I_p - H'B')Z$ for some $Z \in \mathbb{R}^{p\times r}$. Transposing, $Q = Z'(I_p - BH) = Y_2(I_p - BH)$ with $Y_2 = Z'$.
- Hence $X = PQ = (I_n - GA)Y_1Y_2(I_p - BH)$, which has the required form with $Y = Y_1Y_2 \in \mathbb{R}^{n\times p}$.

Problem 12.10 ([10, Problem 44.3])**.** Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{m\times p}$ and $D \in \mathbb{R}^{n\times q}$, and let G and H be g-inverses of A and B. Show that the equations $AX = C$ and $XB = D$ have a common solution $X \in \mathbb{R}^{n\times p}$ if and only if each of them has a solution and $AD = CB$; and that then $X_0 = GC + DH - GADH$ is a common solution.

Solution.

- If $AX = C$ and $XB = D$, then each equation has a solution, and $AD = A(XB) = (AX)B = CB$ (Theorem 1.3).
- Conversely, let $AX_1 = C$, $X_2B = D$ and $AD = CB$. As $AGA = A$ and $BHB = B$, $AGC = AGAX_1 = AX_1 = C$ (Lemma 12.12) and $DHB = X_2BHB = X_2B = D$.
- Using $AGA = A$,

$$AX_0 = AGC + ADH - AGADH = C + ADH - ADH = C$$

- Using $DHB = D$ and $AD = CB$,

$$X_0B = GCB + DHB - GADHB = GCB + D - GAD = GCB + D - GCB = D$$

So X_0 is a common solution.

Chapter 13. Linear models: least squares and the Gauss–Markov theorem

The linear model and least squares

Definition 13.1 ([12, Definitions 1.1, 1.2, 2.1], [3, §7.1], [2, (12.12)])**.** In the **linear model** $y = X\beta+\varepsilon$, the **design matrix** $X \in \mathbb{R}^{n\times(k+1)}$ is a known constant matrix, $\beta = (\beta_0, \dots, \beta_k)' \in \mathbb{R}^{k+1}$ is an unknown constant vector of **parameters**, the **errors** $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)'$ form a random vector, and the **response** $y = (y_1, \dots, y_n)'$ is observed. In **multiple regression**, x_{ij} is the value of the jth explanatory variable on observation i $(1 \leqslant j \leqslant k)$, $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i$, and the ith row of X is $x_i' = (1, x_{i1}, \dots, x_{ik})$: the first column of X is $\mathbf{1}_n$, the column of the **intercept** β_0. The **classical assumptions** are:

(A1) $\mathsf{E}(y) = X\beta$;

(A2) $\mathsf{E}(\varepsilon) = 0$;

(A3) $\operatorname{var}(\varepsilon) = \sigma^2 I_n$ for a number $\sigma^2 > 0$: the errors have the same variance σ^2 and are uncorrelated, $\operatorname{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$;

(A4) $\operatorname{rank} X = k + 1$: X has full column rank.

(A1)–(A4) are the **Gauss–Markov assumptions**. Results stated for $X \in \mathbb{R}^{n\times(k+1)}$ hold for every such X, with or without the column $\mathbf{1}_n$ and of any rank, unless (A4) or an intercept is assumed.

A **least-squares estimate** of β is a vector $\hat\beta \in \mathbb{R}^{k+1}$ that minimizes

$$S(\beta) = \|y - X\beta\|^2 = \sum_{i=1}^{n} (y_i - x_i'\beta)^2 \qquad (\beta \in \mathbb{R}^{k+1})$$

and the $k + 1$ equations $X'X\beta = X'y$ are the **normal equations**.

By Theorem 9.3(b), with $A = I_n$ and the constant vector $b = X\beta$, $\mathsf{E}(y) = X\beta + \mathsf{E}(\varepsilon)$ and $\operatorname{var}(y) = \operatorname{var}(\varepsilon)$. So (A1) and (A2) are equivalent, and under (A3), $\operatorname{var}(y) = \sigma^2 I_n$. Since $\operatorname{rank} X \leqslant n$ (Theorem 4.13(a)), (A4) needs at least $k + 1$ observations. Estimators of the parameters are functions of y that involve no unknown parameter (Definition 9.11).

Theorem 13.2 ([5, Theorem 7.4.1], [12, Theorem 2.4], [2, §12.3])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$ and $y \in \mathbb{R}^n$.

(a) The normal equations $X'X\beta = X'y$ (unknown $\beta \in \mathbb{R}^{k+1}$) are consistent.

(b) $X'X$ is invertible if and only if X has full column rank, $\operatorname{rank} X = k + 1$. Then the normal equations have exactly one solution, $\hat\beta = (X'X)^{-1}X'y$.

(c) Let G be a g-inverse of $X'X$. Then $GX'y$ is a solution, and the solutions are the vectors $GX'y + (I_{k+1} - GX'X)z$ with $z \in \mathbb{R}^{k+1}$; equivalently, $\hat\beta + z$ with $z \in \mathcal{N}(X)$, for any one solution $\hat\beta$. All of them give the same vector $X\hat\beta$. If $\operatorname{rank} X < k + 1$, there are infinitely many solutions.

Proof. (a) $X'y \in \mathcal{C}(X') = \mathcal{C}(X'X)$ (Theorem 4.18(c)), so the system is consistent (Corollary 4.14(a)).

(b) $X'X \in \mathbb{R}^{(k+1)\times(k+1)}$ is invertible if and only if $\operatorname{rank}(X'X) = k + 1$ (Theorem 4.13(d)), that is, $\operatorname{rank} X = k+1$ (Theorem 4.18(b)). Then multiplying $X'X\beta = X'y$ by $(X'X)^{-1}$ gives $\beta = (X'X)^{-1}X'y$, and this vector is a solution.

(c)

- By (a), $X'y \in \mathcal{C}(X'X)$. Theorems 12.2(a) and 12.8, applied to the matrix $X'X$ and the right side $X'y$, show that $GX'y$ is a solution and that the solutions are the vectors $GX'y + (I - GX'X)z$.

- By Theorem 2.27(a), the solutions are $\hat{\beta} + z$ with $z \in \mathcal{N}(X'X) = \mathcal{N}(X)$ (Theorem 4.18(a)), and $X(\hat{\beta} + z) = X\hat{\beta} + Xz = X\hat{\beta}$.
- If $\operatorname{rank} X < k+1$, some $z \neq 0$ lies in $\mathcal{N}(X)$ (Theorem 4.13(b)), and the solutions $\hat{\beta} + tz$ ($t \in \mathbb{R}$) are pairwise distinct, since $t_1 z = t_2 z$ forces $t_1 = t_2$. □

Let $X \in \mathbb{R}^{n\times p}$ and $c > 0$. The matrix $X'X$ is PSD (Theorem 8.25), so its eigenvalues are $\geqslant 0$ (Theorem 8.21(a)). By Lemma 9.22(c), every eigenvalue of the symmetric matrix $X'X + cI_p$ is $\geqslant c > 0$. So $X'X + cI_p$, the matrix used in ridge regression, is PD and invertible (Theorem 8.21(a),(c)), even when $\operatorname{rank} X < p$ and $X'X$ is singular.

Theorem 13.3 ([12, Theorems 2.4, 2.5], [5, Theorem 12.4.3])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$ and $y \in \mathbb{R}^n$.

(a) $\hat{\beta}$ is a least-squares estimate if and only if $X'X\hat{\beta} = X'y$. For every solution $\hat{\beta}$ of the normal equations,

$$S(\beta) = S(\hat{\beta}) + \|X(\hat{\beta} - \beta)\|^2 \qquad (\beta \in \mathbb{R}^{k+1})$$

(b) All least-squares estimates give the same vector of **fitted values** $\hat{y} = X\hat{\beta}$, the orthogonal projection of y onto $\mathcal{C}(X)$; the vector of **residuals** $e = y - \hat{y}$ lies in $\mathcal{C}(X)^\perp$.

(c) If $\operatorname{rank} X = k+1$, the least-squares estimate is unique, $\hat{\beta} = (X'X)^{-1}X'y$, and $\hat{y} = Hy$, $e = My$, with the **hat matrix** $H = X(X'X)^{-1}X'$ (the matrix P of Theorem 6.26 for $A = X$) and the **residual maker** $M = I_n - H$.

Proof. (a)

- Let $X'X\hat{\beta} = X'y$. Then $X'(y - X\hat{\beta}) = 0$, so $y - X\hat{\beta} \perp \mathcal{C}(X)$ (Theorem 6.6(a)); as $X\hat{\beta} \in \mathcal{C}(X)$, it is the orthogonal projection of y onto $\mathcal{C}(X)$ (Definition 6.22).
- For $\beta \in \mathbb{R}^{k+1}$, $X\beta \in \mathcal{C}(X)$, and Theorem 6.23 gives $\|y - X\beta\|^2 = \|y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\beta\|^2$, the identity. Hence $S(\beta) \geqslant S(\hat{\beta})$: $\hat{\beta}$ is a least-squares estimate.
- Solutions exist (Theorem 13.2(a)). If β^* is a least-squares estimate and $\hat{\beta}$ a solution, then $S(\beta^*) \leqslant S(\hat{\beta})$, so the identity gives $\|X(\hat{\beta} - \beta^*)\|^2 = 0$, that is, $X\beta^* = X\hat{\beta}$; then $X'X\beta^* = X'X\hat{\beta} = X'y$.

(b) By the last step, every least-squares estimate β^* gives $X\beta^* = X\hat{\beta}$, the projection found in the first step; and $e = y - X\hat{\beta} \perp \mathcal{C}(X)$.

(c) By Theorem 13.2(b) and (a), $(X'X)^{-1}X'y$ is the only least-squares estimate, and then $\hat{y} = X(X'X)^{-1}X'y = Hy$ and $e = y - Hy = My$. □

So least squares needs no calculus: $\hat{y}$ is the projection of y onto $\mathcal{C}(X)$. The formula $\hat{\beta} = (X'X)^{-1}X'y$ needs a design matrix of full column rank, $\operatorname{rank} X = k+1$. Then Corollary 6.30 with $p = k+1$ gives $\operatorname{tr} H = k+1 = \operatorname{rank} X$ and $\operatorname{tr} M = n - k - 1$ [12, Theorem 2.9].

Theorem 13.4 ([5, Corollary 12.2.2, Theorems 12.3.1, 12.3.4], [2, (12.13), (12.25)–(12.26)], [12, Definition 2.6])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$ have rank r, let G be a g-inverse of $X'X$ (Definition 12.1), and put

$$H = XGX' = X(X'X)^{-}X', \qquad M = I_n - H$$

the **hat matrix** and the **residual maker** of X. For every $y \in \mathbb{R}^n$, Hy is the orthogonal projection of y onto $\mathcal{C}(X)$, whatever g-inverse is used; so H does not depend on the choice of G.

(a) H and M are symmetric and idempotent, $HX = X$, $X'H = X'$, $MX = 0$ and $HM = MH = 0$.

(b) $\operatorname{rank} H = \operatorname{tr} H = r$ and $\operatorname{rank} M = \operatorname{tr} M = n - r$.

(c) If $r = k+1$, then $X'X$ is invertible, its only g-inverse is $(X'X)^{-1}$, and $H = X(X'X)^{-1}X'$ and M are the matrices of Theorem 13.3(c).

(d) For every $y \in \mathbb{R}^n$ and every least-squares estimate $\hat{\beta}$, the fitted values and the residuals are $\hat{y} = X\hat{\beta} = Hy$ and $e = y - \hat{y} = My$, and

$$S(\hat{\beta}) = e'e = y'My = y'y - \hat{\beta}'X'y$$

Proof. The first statement and the independence of G are Theorem 12.17(b) with $A = X$ ($m = n$).

(a) By Theorem 12.17(c),(d) with $A = X$, $H' = H = H^2$, $HX = X$, $X'H = X'$, and $M = I_n - H$ is symmetric and idempotent. Then $MX = X - HX = 0$ and $HM = H - H^2 = 0 = MH$.

(b) This is Theorem 12.17(d) with $A = X$, and $\operatorname{tr} M = \operatorname{tr} I_n - \operatorname{tr} H = n - r$ (Theorem 1.11(a)).

(c) $X'X$ is invertible by Theorem 13.2(b), and its only g-inverse is $(X'X)^{-1}$ (Definition 12.1); so $H = X(X'X)^{-1}X'$ (Theorem 12.17(b) with $A = X$).

(d) Every least-squares estimate gives the fitted values $X\hat{\beta}$, the orthogonal projection of y onto $\mathcal{C}(X)$ (Theorem 13.3(b)), which is Hy; so $e = y - Hy = My$. As M is symmetric and idempotent, $S(\hat{\beta}) = e'e = y'M'My = y'My$, and $y'Hy = y'X\hat{\beta} = \hat{\beta}'X'y$ (Theorem 1.9); hence $y'My = y'y - y'Hy = y'y - \hat{\beta}'X'y$. □

By Theorems 13.3(a),(b) and 13.2(c), the least-squares estimates $\hat{\beta}$ of β are the solutions of the normal equations $X'X\hat{\beta} = X'y$, for instance $\hat{\beta} = (X'X)^- X'y$ for any g-inverse. All of them give the same fitted values $\hat{y} = X\hat{\beta} = Hy$ and residuals $e = y - \hat{y} = My$, with H and M as in Theorem 13.4(d). The minimum of $\|y - X\beta\|^2$ is the **error sum of squares** $\mathrm{SSE} = e'e = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ [12, Definition 3.1], [2, (12.25)]. Under (A4), $\hat{\beta} = (X'X)^{-1}X'y$ is the only least-squares estimate.

By Theorem 13.4(a),(b) and Theorem 8.13(a), H has the eigenvalue 1 with multiplicity r and 0 with multiplicity $n - r$, and M the reverse (each an eigenvalue only when this number is positive; both are eigenvalues when $1 \leqslant r < n$); as Hy is the projection of y onto $\mathcal{C}(X)$, $Hy = y$ exactly for $y \in \mathcal{C}(X)$ and $Hy = 0$ exactly for $y \in \mathcal{C}(X)^\perp$. For $X = \mathbf{1}_n$, $H = \bar{J}_n$ and $M = C_n$ (Example 6.27). Under (A4), $X'X$ is PD (Theorem 8.25); and if $X = QR$ as in Theorem 6.16, then $X'X = R'Q'QR = R'R$ with R invertible (it is triangular with positive diagonal entries, Theorem 2.31(a)), so

$$H = QR(R'R)^{-1}R'Q' = QRR^{-1}(R')^{-1}R'Q' = QQ'$$

Proposition 13.5 ([12, Theorem 2.12], [2, §11.1.2])**.** Let $\hat{\beta}$ be a least-squares estimate, $\hat{y} = X\hat{\beta}$ and $e = y - \hat{y}$.

(a) $X'e = 0$: the residuals are orthogonal to every column of X.

(b) If $\mathbf{1}_n \in \mathcal{C}(X)$ (for instance, if the model has an intercept), then $\mathbf{1}'e = e_1 + \cdots + e_n = 0$; hence $\hat{y}$ and y have the same mean $\bar{y}$.

(c) $\hat{y}'e = 0$.

Proof. (a) This is $X'(y - X\hat{\beta}) = 0$, the normal equations (Theorem 13.3(a)).

(b) If $\mathbf{1} = Xc$, then $\mathbf{1}'e = c'X'e = 0$ by (a), and $\mathbf{1}'\hat{y} = \mathbf{1}'y - \mathbf{1}'e = \mathbf{1}'y$.

(c) $\hat{y}'e = \hat{\beta}'X'e = 0$. □

Proposition 13.6 ([3, §8.5])**.** Let $X = [\mathbf{1}_n\ X_0] \in \mathbb{R}^{n \times p}$ with $X_0 \in \mathbb{R}^{n \times k}$ and $p = k + 1$, let $\bar{x} = \frac{1}{n}X_0'\mathbf{1}_n$ be the mean vector of X_0, and let $S_c = X_0'C_nX_0$.

(a) $X'X = \begin{pmatrix} n & n\bar{x}' \\ n\bar{x} & X_0'X_0 \end{pmatrix}$, the Schur complement of the entry n in $X'X$ is S_c, and $\det(X'X) = n \det S_c$. Hence X has full column rank $p = k + 1$ if and only if S_c is invertible. For $n = 4$ and the data matrix X_0 of Example 1.17, $\bar{x} = (1, 2)'$ and $S_c = 4I_2$.

(b) If S_c is invertible, then

$$(X'X)^{-1} = \begin{pmatrix} \frac{1}{n} + \bar{x}'S_c^{-1}\bar{x} & -\bar{x}'S_c^{-1} \\ -S_c^{-1}\bar{x} & S_c^{-1} \end{pmatrix}$$

Proof. (a) By the block product formula (Theorem 5.3(c)) and Theorems 1.4(d) and 1.9, $X'X = \begin{pmatrix} \mathbf{1}'\mathbf{1} & \mathbf{1}'X_0 \\ X_0'\mathbf{1} & X_0'X_0 \end{pmatrix}$, with $\mathbf{1}'\mathbf{1} = n$ and $X_0'\mathbf{1} = n\bar{x}$. The Schur complement of $A = (n)$ is $X_0'X_0 - (n\bar{x})\frac{1}{n}(n\bar{x}') = X_0'X_0 - n\bar{x}\bar{x}' = X_0'C_nX_0$ (Example 1.17). Theorem 5.30(a) gives $\det(X'X) = n \det S_c$, so $X'X$ is invertible if and only if S_c is (Theorem 5.18(a)); and $X'X$ is invertible if and only if $\operatorname{rank} X = p$ (Theorem 13.2(b)). The numbers for Example 1.17 are computed there.

(b) Apply Theorem 5.12(a) with $A = (n)$, $B = n\bar{x}'$, $C = n\bar{x}$, $D = X_0'X_0$ and $E = S_c$: here $A^{-1}B = \bar{x}'$ and $CA^{-1} = \bar{x}$, so $A^{-1} + A^{-1}BE^{-1}CA^{-1} = \frac{1}{n} + \bar{x}'S_c^{-1}\bar{x}$, $-A^{-1}BE^{-1} = -\bar{x}'S_c^{-1}$ and $-E^{-1}CA^{-1} = -S_c^{-1}\bar{x}$. □

Example 13.7 (cf. [2, Example 11.1])**.** Let $X_0 \in \mathbb{R}^{4\times 2}$ be the data matrix of Example 1.17 ($n = 4$, $k = 2$), and let $y = (9, 3, 5, 3)'$ be a response measured on the four units. The design matrix $X = [\mathbf{1}_4\ X_0] \in \mathbb{R}^{4\times 3}$ has ith row $x_i' = (1, x_{i1}, x_{i2})$; write $X_{\cdot 1}$ and $X_{\cdot 2}$ for its other two columns, the columns of X_0. By Theorems 1.4(d) and 1.9, the (j, l) entry of $X'X$ is $\operatorname{col}_j(X)'\operatorname{col}_l(X)$, so

$$X = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 9+3+5+3 \\ 18+10 \\ 27+9+5+3 \end{pmatrix} = \begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix}$$

- By Example 1.17, $\bar{x} = (1, 2)'$ and $S_c = X_0'C_4X_0 = 4I_2$. So $\det(X'X) = 4 \det S_c = 64 \neq 0$, and X has full column rank 3 (Proposition 13.6(a)). With $S_c^{-1} = \frac{1}{4}I_2$, $\frac{1}{4} + \bar{x}'S_c^{-1}\bar{x} = \frac{1}{4} + \frac{1+4}{4} = \frac{3}{2}$ and $-\bar{x}'S_c^{-1} = (-\frac{1}{4},\ -\frac{1}{2})$, Proposition 13.6(b) gives

$$(X'X)^{-1} = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

- Then

$$(X'X)^{-1}X' = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & 2 & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix}$$

$$H = X(X'X)^{-1}X' = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}\frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix}$$

and

$$M = I_4 - H = \frac{1}{4}\begin{pmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

Hence

$$Hy = \frac{1}{4}\begin{pmatrix} 27+3+5-3 \\ 9+9-5+3 \\ 9-3+15+3 \\ -9+3+5+9 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \\ 6 \\ 2 \end{pmatrix}, \qquad My = y - Hy = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

- The least-squares estimate is unique (Theorem 13.3(c)):

$$\hat{\beta} = (X'X)^{-1}X'y = \frac{1}{4}\begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix}\begin{pmatrix} 9 \\ 3 \\ 5 \\ 3 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} -18+0+10+12 \\ 9-3+5-3 \\ 9+3-5-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

and indeed $X'X\hat{\beta} = (4+8+8,\ 4+16+8,\ 8+16+20)' = (20,28,44)' = X'y$. The fitted values and residuals are $\hat{y} = X\hat{\beta} = Hy = (8,4,6,2)'$ and $e = My = (1,-1,-1,1)'$, and the smallest value of S is $\text{SSE} = S(\hat{\beta}) = e'e = 1+1+1+1 = 4$. As it is positive, no β gives $X\beta = y$: the system $X\beta = y$ has no solution.

- As Proposition 13.5 says, $X'e = (1-1-1+1,\ 2-2,\ 3-3-1+1)' = 0$, $\mathbf{1}'e = 1-1-1+1 = 0$ and $\hat{y}'e = 8-4-6+2 = 0$. By the identity of Theorem 13.3(a), for instance $\beta = (1,2,2)'$ gives $S(\beta) = 4 + \|X(0,0,-1)'\|^2 = 4 + (9+9+1+1) = 24$.

Rank in regression

Proposition 13.8 ([12, Remark 10.2])**.** Let $X \in \mathbb{R}^{n\times p}$ with $p \geqslant 2$. Then X has full column rank if and only if no column of X is a linear combination of the other columns. If some column is such a combination (**exact collinearity**), then $Xz = 0$ for some $z \neq 0$, the matrix $X'X$ is singular, and β and $\beta + tz$ give the same $X\beta$ for every $t \in \mathbb{R}$.

Proof. By Theorem 4.13(b), X has full column rank if and only if its columns are independent, and $p \geqslant 2$ vectors are dependent if and only if one of them is a linear combination of the others (Proposition 3.13(c)). If they are dependent, $Xz = 0$ for some $z \neq 0$ (Theorem 4.13(b)), $X'X$ is singular, as $\operatorname{rank}(X'X) = \operatorname{rank} X < p$ (Theorems 4.18(b) and 4.13(d)), and $X(\beta + tz) = X\beta + tXz = X\beta$. □

For example, if a regression uses the scores x_{i1}, x_{i2} of two tests and also their total $x_{i3} = x_{i1} + x_{i2}$, then column 4 of X is the sum of columns 2 and 3, and $X(0,1,1,-1)' = 0$.

Proposition 13.9 ([12, Proposition 11.4])**.** Let $X \in \mathbb{R}^{n\times p}$, $T \in \mathbb{R}^{p\times q}$ and $X_1 = XT \in \mathbb{R}^{n\times q}$, with $\operatorname{rank} X_1 = \operatorname{rank} X$. Then $\mathcal{C}(X_1) = \mathcal{C}(X)$, the least-squares fitted values (and residuals) of the designs X_1 and X are the same, and if $\hat{\beta}_1$ is a least-squares estimate for X_1, then $T\hat{\beta}_1$ is one for X.

Proof. $\mathcal{C}(X_1) \subseteq \mathcal{C}(X)$ (Theorem 4.15(a)), and both have dimension $\operatorname{rank} X$, so they are equal (Theorem 4.5(a)). For either design, the fitted values are the projection $\hat{y}$ of y onto this subspace (Theorem 13.3(b)). Finally $X(T\hat{\beta}_1) = X_1\hat{\beta}_1 = \hat{y}$, so $\|y - X(T\hat{\beta}_1)\|^2 = \|y - \hat{y}\|^2 \leqslant \|y - X\beta\|^2$ for every β (Theorem 6.23). □

For instance, adding to the design X of Example 13.7 the total $X_{\cdot 1} + X_{\cdot 2}$ of its two variables gives $\tilde{X} = [X \ \ X_{\cdot 1} + X_{\cdot 2}] \in \mathbb{R}^{4\times 4}$, with $X = \tilde{X}T$ for T the first three columns of I_4, and $\operatorname{rank} \tilde{X} = \operatorname{rank} X = 3$. So $\tilde{X}$ has the same fitted values $(8,4,6,2)'$, and $T(1,2,1)' = (1,2,1,0)'$ is a least-squares estimate for $\tilde{X}$.

Theorem 13.10 ([2, §12.1, §12.2.1], [12, Remark 10.2, Proposition 11.2])**.** Let $n_1, \dots, n_g \geqslant 1$ and $n = n_1 + \cdots + n_g$. Arrange n observations in g groups (first the n_1 observations of group 1, then the n_2 of group 2, and so on), and let the **indicator** $d_l \in \mathbb{R}^n$ of group l have entry 1 in the rows of group l and 0 elsewhere. The model $y_i = \mu + \alpha_l + \varepsilon_i$ for an observation i of group l reads $y = X\beta + \varepsilon$ with $\beta = (\mu, \alpha_1, \dots, \alpha_g)'$ and

$$X = [\mathbf{1}_n \; d_1 \; \cdots \; d_g] \in \mathbb{R}^{n \times (g+1)}$$

Let $X_1 = [d_1 \; \cdots \; d_g]$ (intercept dropped) and $X_2 = [\mathbf{1}_n \; d_2 \; \cdots \; d_g]$ (indicator d_1 dropped), both in $\mathbb{R}^{n \times g}$.

(a) X_1 has full column rank, and $X = X_1 S$ with $S = [\mathbf{1}_g \; I_g] \in \mathbb{R}^{g \times (g+1)}$ of full row rank.

(b) $\operatorname{rank} X = g$ and $\mathcal{N}(X) = \operatorname{span}\{(-1, 1, \dots, 1)'\}$: X does not have full column rank.

(c) $X_1 = XT_1$ and $X_2 = XT_2$, where $T_1, T_2 \in \mathbb{R}^{(g+1) \times g}$ are I_{g+1} with its first, respectively second, column deleted. Both X_1 and X_2 have full column rank g, and $\mathcal{C}(X_1) = \mathcal{C}(X_2) = \mathcal{C}(X)$.

Proof. (a)

- No observation lies in two groups, so $d_l' d_m = 0$ for $l \neq m$, and $d_l' d_l = n_l$. If $c_1 d_1 + \cdots + c_g d_g = 0$, multiplying by d_m' gives $c_m n_m = 0$, so $c_m = 0$: X_1 has full column rank (Theorem 4.13(b)).
- Every observation lies in exactly one group, so $d_1 + \cdots + d_g = \mathbf{1}_n$, that is, $X_1 \mathbf{1}_g = \mathbf{1}_n$. By the block product (Theorem 5.3), $X_1 S = [X_1 \mathbf{1}_g \;\; X_1 I_g] = [\mathbf{1}_n \; d_1 \; \cdots \; d_g] = X$.
- The columns of S include those of I_g, so $\mathcal{C}(S) = \mathbb{R}^g$: S has full row rank (Theorem 4.13(c)).

(b) By (a) and Theorem 4.15(b), $\operatorname{rank} X = \operatorname{rank} S = g$ and $\mathcal{N}(X) = \mathcal{N}(S)$. Now $S(c_0, c_1, \dots, c_g)' = c_0 \mathbf{1}_g + (c_1, \dots, c_g)'$ is 0 if and only if $c_l = -c_0$ for every l, that is, $(c_0, c_1, \dots, c_g)' = -c_0(-1, 1, \dots, 1)'$.

(c)

- The columns of XT_1 are the columns of X other than the first (Theorem 1.4(a),(b)), so $XT_1 = X_1$; likewise $XT_2 = X_2$.
- X_1 has full column rank by (a). For X_2: as $\mathbf{1}_n = d_1 + \cdots + d_g$ and $d_1 = \mathbf{1}_n - d_2 - \cdots - d_g$, every column of X_2 lies in $\mathcal{C}(X_1)$ and every column of X_1 lies in $\mathcal{C}(X_2)$. So $\mathcal{C}(X_2) = \mathcal{C}(X_1)$, which has dimension g by (a); as X_2 has g columns, it has full column rank.
- So $\operatorname{rank} X_1 = \operatorname{rank} X_2 = g = \operatorname{rank} X$, and Proposition 13.9 gives $\mathcal{C}(X_1) = \mathcal{C}(X_2) = \mathcal{C}(X)$. □

Using the intercept together with all g indicators (the **dummy variable trap**) [12, Remark 10.2] thus makes $X'X$ singular (Theorems 4.18(b) and 4.13(d)); dropping the intercept or one indicator gives a design of full column rank with the same column space, hence the same fitted values (Proposition 13.9).

Example 13.11 (cf. [2, Examples 12.1–12.3])**.** Six plants of three types have weights $5, 7, 6$ (type 1), $3, 5$ (type 2) and 2 (type 3). With $y = (5, 7, 6, 3, 5, 2)'$, the model of Theorem 13.10 ($g = 3$, $n_1 = 3$, $n_2 = 2$, $n_3 = 1$) has

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 6 & 3 & 2 & 1 \\ 3 & 3 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 28 \\ 18 \\ 8 \\ 2 \end{pmatrix}$$

The (j, l) entry of $X'X$ is $\operatorname{col}_j(X)' \operatorname{col}_l(X)$ (Theorems 1.4(d) and 1.9), and $X'y$ holds the total and the three group totals.

- $\operatorname{rank} X = 3$ and $\mathcal{N}(X) = \operatorname{span}\{(-1,1,1,1)'\}$ (Theorem 13.10), so $X'X$ is singular (Theorems 4.18(b) and 4.13(d)). The same follows from a Schur complement: partition $X'X$ after its first row and column, with $A = (6)$, $B = (3,2,1)$, $C = B'$ and $D = \operatorname{diag}(3,2,1)$. Then $D^{-1} = \operatorname{diag}(\frac{1}{3},\frac{1}{2},1)$, $BD^{-1} = (1,1,1)$, and the Schur complement of D is $A - BD^{-1}C = 6-(3+2+1) = 0$; so $\operatorname{rank}(X'X) = 3 + \operatorname{rank}(0) = 3$ (Theorem 5.34(b)) and $\det(X'X) = \det D \cdot 0 = 0$ (Theorem 5.30(b)).
- $\hat\beta = (0,6,4,2)'$ solves the normal equations: $X'X\hat\beta = (18+8+2,\ 18,\ 8,\ 2)' = X'y$. By Theorem 2.27(a) and $\mathcal{N}(X'X) = \mathcal{N}(X)$ (Theorem 4.18(a)), all solutions, that is, all least-squares estimates (Theorem 13.3(a)), are

$$\begin{pmatrix}0\\6\\4\\2\end{pmatrix} + t\begin{pmatrix}-1\\1\\1\\1\end{pmatrix} = \begin{pmatrix}-t\\6+t\\4+t\\2+t\end{pmatrix} \qquad (t \in \mathbb{R})$$

  and all give $X\hat\beta = (6,6,6,4,4,2)'$: the fitted value $\hat\mu + \hat\alpha_l$ is the mean of group l ($18/3 = 6$, $8/2 = 4$, $2/1 = 2$).
- $G = \operatorname{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$ is a g-inverse of $X'X$. Multiplying the columns of $X'X$ by $0, \frac{1}{3}, \frac{1}{2}, 1$ (Theorem 1.4(a),(b)) gives

$$X'XG = \begin{pmatrix}0&1&1&1\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}$$

  By Theorem 1.4(c), row 1 of $(X'XG)X'X$ is the sum of rows $2,3,4$ of $X'X$, which is $(6,3,2,1)$, and rows $2,3,4$ are those of $X'X$; so $X'XGX'X = X'X$. The solution $GX'y = (0,\ 18/3,\ 8/2,\ 2)' = (0,6,4,2)'$ of Theorem 13.2(c) is the one above with $t = 0$. Multiplying the rows of $X'X$ by $0, \frac{1}{3}, \frac{1}{2}, 1$,

$$GX'X = \begin{pmatrix}0&0&0&0\\1&1&0&0\\1&0&1&0\\1&0&0&1\end{pmatrix}$$

- By Theorem 12.9, for the consistent system $X'X\beta = X'y$, the value $\lambda'\hat\beta$ is the same for all least-squares estimates exactly when $\lambda'GX'X = \lambda'$. For $\lambda = (0,1,-1,0)'$, $\operatorname{row}_2(GX'X) - \operatorname{row}_3(GX'X) = (0,1,-1,0) = \lambda'$, so $\hat\alpha_1 - \hat\alpha_2 = (6+t) - (4+t) = 2$ for all of them. For $\lambda = (1,1,0,0)'$, $\operatorname{row}_1(GX'X) + \operatorname{row}_2(GX'X) = (1,1,0,0) = \lambda'$, so $\hat\mu + \hat\alpha_1 = 6$. For $\lambda = (0,1,0,0)'$, $\operatorname{row}_2(GX'X) = (1,1,0,0) \neq \lambda'$, and indeed $\hat\alpha_1 = 6 + t$ changes with t.
- By Theorem 13.4(d), $S(\hat\beta) = y'y - \hat\beta' X'y = 148 - (0 + 108 + 32 + 4) = 4$, with $y'y = 25 + 49 + 36 + 9 + 25 + 4 = 148$ and $\hat\beta = (0,6,4,2)'$; indeed $e = y - \hat y = (-1,1,0,-1,1,0)'$ and $e'e = 4$.
- Intercept dropped: $X_1'X_1 = \operatorname{diag}(3,2,1)$ and $X_1'y = (18,8,2)'$, so $\hat\beta = (X_1'X_1)^{-1}X_1'y = (6,4,2)'$, the three group means.
- Indicator d_1 dropped:

$$X_2'X_2 = \begin{pmatrix}6&2&1\\2&2&0\\1&0&1\end{pmatrix}, \qquad X_2'y = \begin{pmatrix}28\\8\\2\end{pmatrix}, \qquad X_2'X_2\begin{pmatrix}6\\-2\\-4\end{pmatrix} = \begin{pmatrix}36-4-4\\12-4\\6-4\end{pmatrix} = X_2'y$$

  so $\hat\beta = (6,-2,-4)'$, the unique solution ($X_2'X_2$ is invertible, Theorem 6.26): the mean 6 of group 1 and the differences $4-6$, $2-6$ of the other group means from it.
- In all three designs $\hat y = (6,6,6,4,4,2)'$, as Proposition 13.9 says; moreover $T_1(6,4,2)' = (0,6,4,2)'$ and $T_2(6,-2,-4)' = (6,0,-2,-4)'$ are the solutions above with $t = 0$ and $t = -6$.

- The columns d_1, d_2, d_3 of X_1 form a basis of $\mathcal{C}(X)$, the vectors that are constant within each group, and so do the columns $\mathbf{1}_6, d_2, d_3$ of X_2 (Theorem 13.10(a),(c)). The estimates $(6, 4, 2)'$ and $(6, -2, -4)'$ are the coordinates of $\hat{y}$ in these two bases: $\hat{y} = 6d_1 + 4d_2 + 2d_3 = 6\,\mathbf{1}_6 - 2d_2 - 4d_3$. In this **baseline coding**, the coefficient of $\mathbf{1}_6$ is the value of group 1, and the coefficients of d_2 and d_3 are the differences between the other groups and group 1.
- A covariate $x = (1, 2, 3, 1, 2, 1)'$ measured on the six plants is not constant within group 1, so $x \notin \mathcal{C}(X)$, and the four columns of $[\mathbf{1}_6\ d_2\ d_3\ x]$ are linearly independent (Proposition 3.14(b)): adding the covariate gives a design of full column rank 4.

Theorem 13.12 ([5, Corollary 20.5.5, Theorem 20.6.1])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$ and $y \in \mathbb{R}^n$.

(a) $X^+ = (X'X)^+X'$, and $XX^+ = H$, the hat matrix of Theorem 13.4.

(b) X^+y is a least-squares estimate, and every other least-squares estimate $\hat{\beta}$ has $\|\hat{\beta}\| > \|X^+y\|$: X^+y is the shortest least-squares estimate. It is also the only least-squares estimate in $\mathcal{C}(X')$.

(c) If $G \in \mathbb{R}^{(k+1)\times n}$ and Gy is the shortest least-squares estimate for every $y \in \mathbb{R}^n$, then $G = X^+$.

Proof. The least-squares estimates are the minimizers of $S(\beta) = \|X\beta - y\|^2$, that is, of $\|X\beta - y\|$ (Definition 13.1). So (a)–(c) are Theorem 12.28(a)–(c) with $A = X$, $m = n$ and $x = \beta$; in (a), $X(X'X)^-X' = H$ (Theorem 13.4). □

Example 13.13 (cf. [2, Example 12.3])**.** In Example 13.11, find the g-inverse $\dot{G}$ of $X'X$ built from its submatrix in rows and columns $1, 2, 4$, and the least-squares estimate $\dot{G}X'y$. Compare $\lambda'G\lambda$ and $\lambda'\dot{G}\lambda$ for $\alpha_1 - \alpha_2$, $\mu + \alpha_1$ and α_1, and find the shortest least-squares estimate X^+y.

Solution.

- $\operatorname{rank}(X'X) = \operatorname{rank} X = 3$ (Theorem 4.18(b)). The submatrix in rows and columns $1, 2, 4$ and its adjugate are
$$W = \begin{pmatrix} 6 & 3 & 1 \\ 3 & 3 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \operatorname{adj} W = \begin{pmatrix} 3 & -3 & -3 \\ -3 & 5 & 3 \\ -3 & 3 & 9 \end{pmatrix}$$
and $\det W = 1\cdot(0-3) + 1\cdot(18-9) = 6$ (expansion along row 3, Theorem 5.21); so $W^{-1} = \frac{1}{6}\operatorname{adj} W$ (Theorem 5.24). By Theorem 12.6(b), with $i_s = j_s$ running through $1, 2, 4$,
$$\dot{G} = \frac{1}{6}\begin{pmatrix} 3 & -3 & 0 & -3 \\ -3 & 5 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ -3 & 3 & 0 & 9 \end{pmatrix}, \qquad \dot{G}X'y = \frac{1}{6}\begin{pmatrix} 84 - 54 - 6 \\ -84 + 90 + 6 \\ 0 \\ -84 + 54 + 18 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 0 \\ -2 \end{pmatrix}$$
the estimate with $t = -4$ in Example 13.11.
- $\alpha_1 - \alpha_2$, $\lambda = (0, 1, -1, 0)'$: with $G = \operatorname{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$, $\lambda'G\lambda = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$, and $\lambda'\dot{G}\lambda = \frac{1}{6}(5 - 0 - 0 + 0) = \frac{5}{6}$.
- $\mu + \alpha_1$, $\lambda = (1, 1, 0, 0)'$: $\lambda'G\lambda = 0 + \frac{1}{3} = \frac{1}{3}$, and $\lambda'\dot{G}\lambda = \frac{1}{6}(3 - 3 - 3 + 5) = \frac{1}{3}$.
- α_1, $\lambda = (0, 1, 0, 0)'$: $\lambda'G\lambda = \frac{1}{3}$, but $\lambda'\dot{G}\lambda = \frac{5}{6}$.
- This agrees with Corollary 12.18 for $A = X$: the first two λ lie in $\mathcal{C}(X')$, the third does not (Example 13.11 and Theorem 12.9, with $\mathcal{C}(X'X) = \mathcal{C}(X')$ by Theorem 4.18(c)). For instance $(0, 1, -1, 0) = a'X$ with $a = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -\frac{1}{2}, -\frac{1}{2}, 0)' = \frac{1}{3}d_1 - \frac{1}{2}d_2$; as $a \in \mathcal{C}(X)$, its projection onto $\mathcal{C}(X)$ is $Ha = a$ (Theorem 13.4, Definition 6.22), and $\lambda'G\lambda = \|a\|^2 = 3\cdot\frac{1}{9} + 2\cdot\frac{1}{4} = \frac{5}{6}$.

- The least-squares estimates are $\hat{\beta} = (-t,\ 6+t,\ 4+t,\ 2+t)'$ $(t \in \mathbb{R})$, and

$$\|\hat{\beta}\|^2 = t^2 + (6+t)^2 + (4+t)^2 + (2+t)^2 = 4t^2 + 24t + 56 = 4(t+3)^2 + 20$$

  This is smallest exactly for $t = -3$. So, by Theorem 13.12(b), $X^+y = (3,3,1,-1)'$, of squared length 20.

Example 13.14. Four observations are classified by two factors with two levels each; observations $1,2,3,4$ have the levels $(1,1)$, $(2,1)$, $(1,2)$, $(2,2)$ of (first factor, second factor). The indicators of the two levels of the first factor, and of the second factor, are

$$d_1 = \begin{pmatrix}1\\0\\1\\0\end{pmatrix}, \quad d_2 = \begin{pmatrix}0\\1\\0\\1\end{pmatrix}, \qquad g_1 = \begin{pmatrix}1\\1\\0\\0\end{pmatrix}, \quad g_2 = \begin{pmatrix}0\\0\\1\\1\end{pmatrix}$$

Let $U = \operatorname{span}\{d_1, d_2\}$ and $W = \operatorname{span}\{g_1, g_2\}$. Find the dimensions of U, W, $U+W$ and $U \cap W$, and a basis of $U \cap W$.

Solution. $U + W = \operatorname{span}\{d_1, d_2, g_1, g_2\}$ (Theorem 3.31(c)). Row-reduce $[d_1\ d_2\ g_1\ g_2]$:

$$\begin{pmatrix}1&0&1&0\\0&1&1&0\\1&0&0&1\\0&1&0&1\end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix}1&0&1&0\\0&1&1&0\\0&0&-1&1\\0&0&-1&1\end{pmatrix} \xrightarrow{R_3 \leftarrow -R_3} \begin{pmatrix}1&0&1&0\\0&1&1&0\\0&0&1&-1\\0&0&-1&1\end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix}1&0&1&0\\0&1&1&0\\0&0&1&-1\\0&0&0&0\end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 - R_3]{R_1 \leftarrow R_1 - R_3} \begin{pmatrix}1&0&0&1\\0&1&0&1\\0&0&1&-1\\0&0&0&0\end{pmatrix}$$

- The pivot columns are $1,2,3$, so d_1, d_2, g_1 form a basis of $U+W$ and $\dim(U+W) = 3$ (Theorem 4.3(b)). Column 4 gives $g_2 = d_1 + d_2 - g_1$; check: $(1,0,1,0)' + (0,1,0,1)' - (1,1,0,0)' = (0,0,1,1)'$.
- Neither of d_1, d_2 is a multiple of the other (each has a 1 where the other has a 0), and the same holds for g_1, g_2; so $\dim U = \dim W = 2$ (Proposition 3.13(c)).
- By Theorem 4.23, $\dim(U \cap W) = 2 + 2 - 3 = 1$. The relation of column 4 reads $d_1 + d_2 = g_1 + g_2$, and both sides equal $\mathbf{1}_4$. So $\mathbf{1}_4$ lies in U and in W, and this nonzero vector is a basis of the one-dimensional space $U \cap W$ (Theorem 4.5(d)): $U \cap W = \operatorname{span}\{\mathbf{1}_4\}$. A combination of the indicators of the first factor that is also a combination of the indicators of the second factor is a constant vector.

In particular, in the additive model $y_i = \alpha_l + \gamma_m + \varepsilon_i$ for an observation i at level l of the first factor and level m of the second, the design matrix $X = [d_1\ d_2\ g_1\ g_2] \in \mathbb{R}^{4\times 4}$ has rank 3, because the indicators of each factor add up to $\mathbf{1}_4$: $X(1,1,-1,-1)' = 0$, and $X'X$ is singular (Theorems 4.18(b) and 4.13(d)), as for the one-way layout of Theorem 13.10.

Proposition 13.15 ([12, Definition 11.6])**.** Let $x = (x_1, \ldots, x_n)' \in \mathbb{R}^n$, and let $X \in \mathbb{R}^{n\times 3}$ have ith row $(1, x_i, x_i^2)$: the design matrix of the quadratic regression $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i$ ($k = 2$, with $x_{i1} = x_i$ and $x_{i2} = x_i^2$). Then X has full column rank if and only if $x_1, \ldots, x_n$ take at least three different values.

Proof. • If every x_i is one of two numbers a, b (possibly $a = b$), then $(x_i - a)(x_i - b) = ab - (a+b)x_i + x_i^2 = 0$ for every i; so $X(ab, -(a+b), 1)' = 0$, and X does not have full column rank (Theorem 4.13(b)).

- Let $x_i = a$, $x_j = b$, $x_l = c$ be three different values. If $Xv = 0$, rows i, j, l give $Av = 0$ for

$$A = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$$

The operations $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$ give the rows $(0,\ b-a,\ b^2-a^2)$ and $(0,\ c-a,\ c^2-a^2)$; dividing them by $b - a \neq 0$ and $c - a \neq 0$ gives $(0, 1, a + b)$ and $(0, 1, a + c)$; then $R_3 \leftarrow R_3 - R_2$ gives $(0, 0, c - b)$, and dividing by $c - b \neq 0$ gives an echelon form with a leading 1 in every column. So $Av = 0$ only for $v = 0$ (Theorem 4.1 and Remark 4.2), and hence $Xv = 0$ only for $v = 0$. □

Example 13.16. (a) For $x = (0, 1, 2, 3)'$, four different values, X has full column rank by Proposition 13.15. Directly:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 3 & 9 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 6 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - 3R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow \frac{1}{2} R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

an echelon form with three nonzero rows, so $\operatorname{rank} X = 3$.

(b) If a dose is given at only two levels, 1 and 3, say $x = (1, 3, 3, 1)'$, then $x_i^2 = 4x_i - 3$ for every i ($1 = 4 - 3$, $9 = 12 - 3$). So column 3 of X is 4 times column 2 minus 3 times column 1, and $X(3, -4, 1)' = 0$. Columns 1 and 2, $(1, 1, 1, 1)'$ and $(1, 3, 3, 1)'$, are not multiples of each other, so $\operatorname{rank} X = 2$. Thus β and $\beta + t(3, -4, 1)'$ give the same $X\beta$: two dose levels cannot determine the curvature β_2.

The Gauss–Markov theorem

Theorem 13.17 ([12, Theorems 3.3, 3.4, 3.8, Corollary 3.10], [2, §11.2.1])**.** In (a) and (b), assume (A4), and let $\hat{\beta} = (X'X)^{-1}X'y$.

(a) $\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$; under (A2), $\mathsf{E}(\hat{\beta}) = \beta$.

(b) Under (A3), $\operatorname{var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$.

(c) For X of any rank, with H and M of Theorem 13.4, under (A2) and (A3): $e = M\varepsilon$, $\mathsf{E}(e) = 0$ and $\operatorname{var}(e) = \sigma^2 M$; $\mathsf{E}(\hat{y}) = X\beta$ and $\operatorname{var}(\hat{y}) = \sigma^2 H$; and $\operatorname{cov}(\hat{y}, e) = 0$. Under (A4) also $\operatorname{cov}(\hat{\beta}, e) = 0$, with $\hat{\beta}$ of (a).

Only (A2)–(A4) are used; no distribution of ε is assumed.

Proof. (a) $\hat{\beta} = (X'X)^{-1}X'(X\beta + \varepsilon) = (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon = \beta + (X'X)^{-1}X'\varepsilon$. By Theorem 9.3(b) and (A2), $\mathsf{E}(\hat{\beta}) = \beta + (X'X)^{-1}X'\,\mathsf{E}(\varepsilon) = \beta$.

(b) $(X'X)^{-1}$ is symmetric (Theorem 2.13(c)). By (a), Theorem 9.3(b) and (A3),

$$\operatorname{var}(\hat{\beta}) = (X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1} = \sigma^2(X'X)^{-1}(X'X)(X'X)^{-1} = \sigma^2(X'X)^{-1}$$

(c)

- $e = My$ and $\hat{y} = Hy$ (Theorem 13.4(d)), and $MX = 0$, $HX = X$ (Theorem 13.4(a)), so $e = MX\beta + M\varepsilon = M\varepsilon$ and $\hat{y} = X\beta + H\varepsilon$. As H and M are symmetric and idempotent (Theorem 13.4(a)), Theorem 9.3(b) with (A2) and (A3) gives $\mathsf{E}(e) = 0$, $\operatorname{var}(e) = M(\sigma^2 I_n)M' = \sigma^2 M$, $\mathsf{E}(\hat{y}) = X\beta$ and $\operatorname{var}(\hat{y}) = \sigma^2 H$.

- By Theorem 9.4(a) and (A3), $\operatorname{cov}(\hat{y}, e) = \operatorname{cov}(H\varepsilon, M\varepsilon) = \sigma^2 HM' = \sigma^2 HM = 0$ (Theorem 13.4(a)).
- Under (A4), with $A = (X'X)^{-1}X'$, (a) gives $\hat{\beta} = \beta + A\varepsilon$, and by Theorem 9.4(a) and (A3), $\operatorname{cov}(\hat{\beta}, e) = \operatorname{cov}(A\varepsilon, M\varepsilon) = A(\sigma^2 I_n)M' = \sigma^2(X'X)^{-1}(MX)' = 0$. □

Theorem 13.18 (Gauss–Markov; [12, Theorem 3.5], [2, Theorem 11.1])**.** Assume (A1)–(A4), and let $\hat{\beta} = (X'X)^{-1}X'y$. Let $C \in \mathbb{R}^{(k+1)\times n}$ be constant, and let $\tilde{\beta} = Cy$ be a **linear unbiased estimator** of β: $\mathsf{E}(Cy) = \beta$ for every $\beta \in \mathbb{R}^{k+1}$. Then $\operatorname{var}(\tilde{\beta}) - \operatorname{var}(\hat{\beta})$ is positive semidefinite, and it is 0 only if $\tilde{\beta} = \hat{\beta}$. Consequently $\operatorname{var}(a'\tilde{\beta}) \geqslant \operatorname{var}(a'\hat{\beta})$ for every $a \in \mathbb{R}^{k+1}$; in particular $\operatorname{var}(\tilde{\beta}_j) \geqslant \operatorname{var}(\hat{\beta}_j)$ for $j = 0, \dots, k$. We say that $\hat{\beta}$ is the BLUE of β.

Proof. • By Theorem 9.3(b) and (A1), $\mathsf{E}(Cy) = CX\beta$. As $CX\beta = \beta$ for every β, taking for β the columns of I_{k+1} gives $CX = I_{k+1}$.

- Write $C = (X'X)^{-1}X' + D$. Then $DX = CX - (X'X)^{-1}X'X = I_{k+1} - I_{k+1} = 0$, and $X'D' = (DX)' = 0$.
- By Theorem 9.3(b) and (A3), $\operatorname{var}(\tilde{\beta}) = C(\sigma^2 I_n)C' = \sigma^2 CC'$, and

$$CC' = (X'X)^{-1}X'X(X'X)^{-1} + (X'X)^{-1}X'D' + DX(X'X)^{-1} + DD' = (X'X)^{-1} + DD'$$

- By Theorem 13.17(b), $\operatorname{var}(\tilde{\beta}) - \operatorname{var}(\hat{\beta}) = \sigma^2 DD'$, which is positive semidefinite (Theorem 8.25, as $DD' = (D')'D'$). If it is 0, then $\operatorname{tr}\big((D')'D'\big) = 0$, so $D' = 0$ (Theorem 1.11(d)), $C = (X'X)^{-1}X'$ and $\tilde{\beta} = \hat{\beta}$.
- For $a \in \mathbb{R}^{k+1}$, $\operatorname{var}(a'\tilde{\beta}) = a' \operatorname{var}(\tilde{\beta})a$ and likewise for $\hat{\beta}$ (Theorem 9.3(b)); so $\operatorname{var}(a'\tilde{\beta}) - \operatorname{var}(a'\hat{\beta}) = \sigma^2 a'DD'a \geqslant 0$. For a the column of I_{k+1} that picks out β_j, this is $\operatorname{var}(\tilde{\beta}_j) \geqslant \operatorname{var}(\hat{\beta}_j)$. □

Example 13.19. In Example 13.7, $\mathsf{E}(y_1 - y_2) = x_1'\beta - x_2'\beta = (0, 2, 0)\beta = 2\beta_1$ by (A1). So $c'y = \frac{1}{2}(y_1 - y_2)$, with $c = \frac{1}{2}(1, -1, 0, 0)'$, is a linear unbiased estimator of β_1 ($a = (0, 1, 0)'$); its variance is $\sigma^2 c'c = \sigma^2/2$. The BLUE is $\hat{\beta}_1 = \frac{1}{4}(y_1 - y_2 + y_3 - y_4)$, row 2 of $(X'X)^{-1}X'y$ (Example 13.7), with variance $\sigma^2 \cdot \frac{1}{4}$, the $(2,2)$ entry of $\sigma^2(X'X)^{-1}$ (Theorem 13.17(b)). As in the proof of Theorem 13.18, write $c' = a'(X'X)^{-1}X' + d'$ with $d'X = 0$; then $\operatorname{var}(c'y) - \operatorname{var}(\hat{\beta}_1) = \sigma^2 d'd$. Here $X'c = a$ (unbiasedness, as in the first step of that proof), so $X(X'X)^{-1}a = X(X'X)^{-1}X'c = Hc$ and $d = c - Hc = Mc$:

$$Hc = \tfrac{1}{4}(1, -1, 1, -1)', \qquad d = Mc = c - Hc = \tfrac{1}{4}(1, -1, -1, 1)', \qquad \frac{\sigma^2}{2} - \frac{\sigma^2}{4} = \sigma^2\|d\|^2 = \sigma^2 \cdot \tfrac{4}{16}$$

For $y = (9, 3, 5, 3)'$ the two estimates of β_1 are $\frac{1}{2}(9 - 3) = 3$ and $\hat{\beta}_1 = 2$.

Example 13.20. For an invertible $X'X$ and $u = v = x$, Theorem 5.32(b) updates $(X'X)^{-1}$ when an observation with row x' is added to the design matrix X, since then $X'X$ becomes $X'X + xx'$ (Theorem 5.3(c)): no new inversion is needed. The condition of (b) holds: with $w = (X'X)^{-1}x$, $x'(X'X)^{-1}x = (X'Xw)'w = (Xw)'(Xw) \geqslant 0$ (Theorem 1.9), so $1 + x'(X'X)^{-1}x \geqslant 1$.

For instance, let four observations fall into a baseline group (observations $1, 2$) and groups A (observation 3) and B (observation 4), with the baseline coding $y_i = \beta_0 + \beta_1 a_i + \beta_2 b_i + \varepsilon_i$, where a_i, b_i are the indicators of A and B; add a fifth observation, in group A, with row $x' = (1, 1, 0)$:

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 & -1 \\ -1 & 3 & 1 \\ -1 & 1 & 3 \end{pmatrix}$$

($\det(X'X) = 4 - 1 - 1 = 2$, and $X'X \cdot (X'X)^{-1} = I_3$, Theorem 2.16).

- $w = (X'X)^{-1}x = \frac{1}{2}(1-1,\ -1+3,\ -1+1)' = (0,1,0)'$ and $x'(X'X)^{-1}x = x'w = 1$. The new design $\tilde{X}$ has $\tilde{X}'\tilde{X} = X'X + xx'$, and Theorem 5.32(b) gives

$$\tilde{X}'\tilde{X} = \begin{pmatrix} 5 & 2 & 1 \\ 2 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad (\tilde{X}'\tilde{X})^{-1} = (X'X)^{-1} - \frac{1}{1+1}ww' = \frac{1}{2}\begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 3 \end{pmatrix}$$

- The difference $(X'X)^{-1} - (\tilde{X}'\tilde{X})^{-1} = \frac{1}{2}ww'$ is PSD (Theorem 8.25), as Theorem 8.51 predicts from $\tilde{X}'\tilde{X} \succcurlyeq X'X \succ 0$; and $\det(\tilde{X}'\tilde{X}) = \det(X'X)\left(1 + x'(X'X)^{-1}x\right) = 2 \cdot 2 = 4$ (Theorem 5.31).

- Under (A3), $\operatorname{var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$ (Theorem 13.17(b)). So the added observation lowers $\operatorname{var}(\hat{\beta}_1)$, the variance of the estimated difference between group A and the baseline, from $\frac{3}{2}\sigma^2$ to σ^2, and leaves the other variances and covariances unchanged, as ww' has only one nonzero entry.

Proposition 13.21 ([5, Theorem 14.12.20], [12, Exercise B5])**.** Let $X \in \mathbb{R}^{n\times p}$, let $W \in \mathbb{R}^{n\times n}$ be PD, and let $y \in \mathbb{R}^n$. The minimizers of $Q(\beta) = (y - X\beta)'W(y - X\beta)$ ($\beta \in \mathbb{R}^p$) are the solutions of $X'WX\beta = X'Wy$. If $\operatorname{rank} X = p$, then $X'WX$ is PD, and the unique minimizer of Q is the **weighted least-squares estimate**

$$\hat{\beta}_W = (X'WX)^{-1}X'Wy$$

For $W = \operatorname{diag}(w_1, \ldots, w_n)$ with all $w_i > 0$, $Q(\beta) = \sum_i w_i(y_i - x_i'\beta)^2$.

Proof. • Let $T = W^{1/2}$ (Theorem 8.46): T is PD, hence symmetric and invertible, and $T^2 = W$. So $Q(\beta) = (y - X\beta)'T'T(y - X\beta) = \|Ty - TX\beta\|^2$: the weighted sum of squares of the response y and the design X is the sum of squares of the response Ty and the design TX.

- By Theorem 13.3(a) for the design TX and the response Ty, the minimizers of Q are the solutions of $(TX)'TX\beta = (TX)'Ty$, that is, of $X'WX\beta = X'Wy$, as $T'T = T^2 = W$.

- If $\operatorname{rank} X = p$, then $\operatorname{rank}(TX) = \operatorname{rank} X = p$ (Theorem 4.15(d)), so $X'WX = (TX)'(TX)$ is PD (Theorem 8.25), and by Theorem 13.3(c) the unique minimizer of Q is $\left((TX)'TX\right)^{-1}(TX)'Ty = (X'WX)^{-1}X'Wy = \hat{\beta}_W$.

- A diagonal W with positive diagonal entries is PD (Proposition 8.23(a)), and $Q(\beta) = \sum_i w_i u_i^2$ for $u = y - X\beta$. □

Theorem 13.22 (Aitken; [12, Exercise B5], [2, §11.6.1])**.** Let $y = X\beta + \varepsilon$, where $X \in \mathbb{R}^{n\times p}$ is known with $\operatorname{rank} X = p$, $\mathsf{E}(\varepsilon) = 0$ and $\operatorname{var}(\varepsilon) = \sigma^2 V$ with $V \in \mathbb{R}^{n\times n}$ known and PD, and β and $\sigma^2 > 0$ are unknown. Then $X'V^{-1}X$ is invertible. Put

$$\tilde{\beta} = (X'V^{-1}X)^{-1}X'V^{-1}y \qquad \text{(the } \textbf{generalized least-squares estimator}\text{)}$$

(a) $\tilde{\beta}$ is the unique minimizer of $(y - X\beta)'V^{-1}(y - X\beta)$, $\mathsf{E}(\tilde{\beta}) = \beta$ and $\operatorname{var}(\tilde{\beta}) = \sigma^2(X'V^{-1}X)^{-1}$.

(b) $\tilde{\beta}$ is the BLUE of β: for every linear unbiased estimator $\check{\beta} = Cy$ of β, with $C \in \mathbb{R}^{p\times n}$ constant, $\operatorname{var}(\check{\beta}) - \operatorname{var}(\tilde{\beta})$ is PSD, and it is 0 only if $\check{\beta} = \tilde{\beta}$; so $\operatorname{var}(c'\check{\beta}) \geqslant \operatorname{var}(c'\tilde{\beta})$ for every $c \in \mathbb{R}^p$.

(c) $\hat{\beta} = (X'X)^{-1}X'y$ is unbiased, and $\operatorname{var}(\hat{\beta}) = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}$; the matrix $\operatorname{var}(\hat{\beta}) - \operatorname{var}(\tilde{\beta})$ is PSD.

Proof. • Let $T = V^{-1/2}$, so that $TVT = I_n$ and $T^2 = V^{-1}$ (Theorem 8.46(c)). Also $T = (V^{1/2})^{-1}$ is symmetric, as $V^{1/2}$ is PD, hence symmetric (Theorem 2.13(c)).

- Put $y^* = Ty$, $X^* = TX$, $\varepsilon^* = T\varepsilon$. Then $y^* = X^*\beta + \varepsilon^*$, $\mathsf{E}(\varepsilon^*) = 0$ and $\operatorname{var}(\varepsilon^*) = T(\sigma^2 V)T' = \sigma^2 I_n$ (Theorem 9.3(b)), and $\operatorname{rank} X^* = \operatorname{rank} X = p$ (Theorem 4.15(d)): the transformed model is a linear model with full column rank.

- $X^{*\prime}X^* = X'T^2X = X'V^{-1}X$, which is therefore invertible (Theorem 13.2(b)), and $X^{*\prime}y^* = X'V^{-1}y$. So $\tilde{\beta}$ is the least-squares estimator of the transformed model.

(a)

- $\|y^* - X^*\beta\|^2 = (y - X\beta)'T^2(y - X\beta) = (y - X\beta)'V^{-1}(y - X\beta)$, whose unique minimizer is $\tilde{\beta}$ (Theorem 13.3(c)).
- Theorem 13.17(a),(b) for the transformed model gives $\mathsf{E}(\tilde{\beta}) = \beta$ and $\operatorname{var}(\tilde{\beta}) = \sigma^2(X^{*\prime}X^*)^{-1}$.

(b)

- For constant $C \in \mathbb{R}^{p\times n}$, $Cy = (CT^{-1})y^*$, and CT^{-1} runs through all constant $p \times n$ matrices as C does. So the linear estimators of β of the two descriptions are the same random vectors, with the same means and variances, and the linear unbiased estimators of β are the same.
- Theorem 13.18 for the transformed model, whose least-squares estimator is $\tilde{\beta}$, gives the claim.

(c)

- $\hat{\beta} = Ay$ with $A = (X'X)^{-1}X'$ and $AX = I_p$, so $\mathsf{E}(\hat{\beta}) = AX\beta = \beta$ and $\operatorname{var}(\hat{\beta}) = A(\sigma^2 V)A'$ (Theorem 9.3(b)).
- So $\hat{\beta}$ is a linear unbiased estimator of β, and (b) gives that $\operatorname{var}(\hat{\beta}) - \operatorname{var}(\tilde{\beta})$ is PSD. □

If the errors are uncorrelated with $\operatorname{var}(\varepsilon_i) = \sigma^2 v_i$ and known $v_i > 0$ (**heteroscedastic** errors), then $V = \operatorname{diag}(v_1, \ldots, v_n)$ and $V^{-1} = \operatorname{diag}(1/v_1, \ldots, 1/v_n)$, and $\tilde{\beta}$ is the weighted least-squares estimate with weights $w_i = 1/v_i$ (Proposition 13.21).

Estimating σ^2

Theorem 13.23 ([2, §12.6], [3, Result 7.5], [12, Theorem 3.7])**.** Assume (A2) and (A3), and let $\operatorname{rank} X = r < n$. Then $\mathsf{E}(\mathrm{SSE}) = (n - r)\sigma^2$. Hence

$$\hat{\sigma}^2 = \frac{\mathrm{SSE}}{n - r}$$

is an unbiased estimator of σ^2. Under (A4), $r = k + 1$, and $\hat{\sigma}^2$ is the **mean squared error** $s^2 = \mathrm{MSE} = \mathrm{SSE}/(n - k - 1)$.

Proof.
- $e = M\varepsilon$ (Theorem 13.17(c)).
- M is symmetric and idempotent (Theorem 13.4(a)), so $\mathrm{SSE} = e'e = \varepsilon'M'M\varepsilon = \varepsilon'M\varepsilon$.
- By Theorem 9.10 with $z = \varepsilon$, $\mu = 0$ (A2) and $\Sigma = \sigma^2 I_n$ (A3), $\mathsf{E}(\mathrm{SSE}) = \operatorname{tr}(M\sigma^2 I_n) = \sigma^2 \operatorname{tr} M = \sigma^2(n - r)$ (Theorem 13.4(b)).
- Dividing by $n - r > 0$ gives $\mathsf{E}(\hat{\sigma}^2) = \sigma^2$. □

The diagonal entries $h_{11}, \ldots, h_{nn}$ of the hat matrix H are the **leverages** of the n observations. Under (A4), with s^2 of Theorem 13.23 (so $n > k + 1$), the numbers $e_i/(s\sqrt{1 - h_{ii}})$, defined when $s > 0$ and $h_{ii} < 1$, are the **standardized residuals** [12, Definitions 2.13, 3.11].

Example 13.24 (cf. [2, Example 12.6])**.** (a) In Example 13.7, $\mathrm{SSE} = 4$, $n = 4$ and $r = k + 1 = 3$, so $s^2 = 4/(4 - 3) = 4$. Replacing σ^2 by s^2 in Theorem 13.17(b) estimates $\operatorname{var}(\hat{\beta})$ by

$$s^2(X'X)^{-1} = 4 \cdot \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

with $(X'X)^{-1}$ of Example 13.7: for instance $\operatorname{var}(\hat{\beta}_1)$ is estimated by 1.

(b) In Example 13.11, $e = y - \hat{y} = (5 - 6,\ 7 - 6,\ 6 - 6,\ 3 - 4,\ 5 - 4,\ 2 - 2)' = (-1, 1, 0, -1, 1, 0)'$, so $\mathrm{SSE} = 4$, and $r = 3$: $\hat{\sigma}^2 = 4/(6 - 3) = \frac{4}{3}$.

Sums of squares and R^2

Definition 13.25 ([12, Definitions 3.1, 6.6], [2, §11.1.2])**.** For a least-squares fit, with fitted values $\hat{y}$ and residuals $e = y - \hat{y}$, the **total**, **regression** and **error sums of squares** are

$$\mathrm{SST} = \sum_{i=1}^{n}(y_i - \bar{y})^2, \qquad \mathrm{SSR} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2, \qquad \mathrm{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = e'e$$

If $\mathbf{1}_n \in \mathcal{C}(X)$ and $\mathrm{SST} > 0$, the **coefficient of determination** is $R^2 = \mathrm{SSR}/\mathrm{SST}$.

Theorem 13.26 ([12, Theorem 6.4, Proposition 6.7])**.** If $\mathbf{1}_n \in \mathcal{C}(X)$ (for instance, if the model has an intercept), then $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$, where $\mathrm{SST} = \|C_n y\|^2 = y'C_n y$ and $\mathrm{SSR} = \|C_n\hat{y}\|^2$. Hence, if $\mathrm{SST} > 0$, then $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$ and $0 \leqslant R^2 \leqslant 1$.

Proof. • For $v \in \mathbb{R}^n$, $C_n v = v - \bar{v}\mathbf{1}$, so $\|C_n v\|^2 = \sum_i (v_i - \bar{v})^2$, and $\|C_n v\|^2 = v'C_n'C_n v = v'C_n v$ (Proposition 1.15). With $v = y$ this is SST; with $v = \hat{y}$, whose mean is $\bar{y}$ because $\mathbf{1}'\hat{y} = \mathbf{1}'y - \mathbf{1}'e = \mathbf{1}'y$ (Proposition 13.5(b)), it is SSR.

- $C_n e = e - \bar{e}\mathbf{1} = e$, as $\bar{e} = 0$ (Proposition 13.5(b)). So $C_n y = C_n\hat{y} + C_n e = C_n\hat{y} + e$.
- $(C_n\hat{y})'e = \hat{y}'C_n e = \hat{y}'e = 0$ (Proposition 13.5(c)). By Pythagoras (Proposition 6.2), $\|C_n y\|^2 = \|C_n\hat{y}\|^2 + \|e\|^2$, that is, $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$.
- Dividing by $\mathrm{SST} > 0$ gives $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$; and $0 \leqslant \mathrm{SSR} \leqslant \mathrm{SST}$, as $\mathrm{SSE} \geqslant 0$. □

In terms of projections: $y = \bar{y}\mathbf{1} + C_n y$ along $\mathrm{span}\{\mathbf{1}\} \oplus \mathrm{span}\{\mathbf{1}\}^\perp$ (Example 6.27), and $C_n y = C_n\hat{y} + e$ splits the centered response into $C_n\hat{y} = \hat{y} - \bar{y}\mathbf{1} \in \mathcal{C}(X)$ and $e \in \mathcal{C}(X)^\perp$; $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ is Pythagoras for this split [12, Remark 6.5]. For the design $X = \mathbf{1}_n$ alone, $\hat{y} = \bar{y}\mathbf{1}$, so $\mathrm{SSR} = 0$ and $\mathrm{SSE} = \mathrm{SST}$; R^2 is the fraction of SST that the other columns of X explain.

For instance, $\mathrm{SST} = y'C_n y = 0$ only if $C_n y = 0$, and $\mathrm{SSE} = y'My = 0$ for the residual maker M of Theorem 13.4 only if $My = e = 0$ (Corollary 8.27): both matrices are symmetric and idempotent (Proposition 1.15, Theorem 13.4(a)), hence PSD (Proposition 8.23(d)). Without centering, $y = \bar{y}\mathbf{1}_n + C_n y$ with $\mathbf{1}_n'C_n y = 0$ (Proposition 1.15), so Pythagoras (Proposition 6.2) and Theorem 13.26 give the partition of the uncorrected sum of squares [2, §12.7]

$$y'y = \|\bar{y}\mathbf{1}_n\|^2 + \|C_n y\|^2 = n\bar{y}^2 + \mathrm{SSR} + \mathrm{SSE}$$

Example 13.27. In Example 13.7, $\bar{y} = \frac{1}{4}(9 + 3 + 5 + 3) = 5$, so

$$C_4 y = \begin{pmatrix} 4 \\ -2 \\ 0 \\ -2 \end{pmatrix}, \qquad C_4\hat{y} = \begin{pmatrix} 3 \\ -1 \\ 1 \\ -3 \end{pmatrix}, \qquad e = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \qquad C_4 y = C_4\hat{y} + e$$

Hence $\mathrm{SST} = 16 + 4 + 0 + 4 = 24$, $\mathrm{SSR} = 9 + 1 + 1 + 9 = 20$ and $\mathrm{SSE} = 1 + 1 + 1 + 1 = 4$; indeed $24 = 20 + 4$, and $R^2 = \frac{20}{24} = \frac{5}{6}$. Without centering, $y'y = 81 + 9 + 25 + 9 = 124$ and $n\bar{y}^2 = 4 \cdot 25 = 100$, so $124 = 100 + 20 + 4$.

Theorem 13.28 ([12, Theorem 6.8])**.** Let $X \in \mathbb{R}^{n\times p}$ with $\mathbf{1}_n \in \mathcal{C}(X)$, let $x \in \mathbb{R}^n$, let $X_+ = [X\ x]$ be X with the column x added, and let $y \in \mathbb{R}^n$ with $\mathrm{SST} > 0$. Let SSE and R^2 belong to the least-squares fit with design X, and SSE_+ and R^2_+ to the fit with design X_+. Then $\mathrm{SSE}_+ \leqslant \mathrm{SSE}$ and $R^2_+ \geqslant R^2$: adding a column cannot decrease R^2.

Proof. • Every column of X is a column of X_+, so $\mathcal{C}(X) \subseteq \mathcal{C}(X_+)$, and both contain $\mathbf{1}_n$.

- The fitted values $\hat{y}_+$ of X_+ form the vector of $\mathcal{C}(X_+)$ closest to y (Theorems 13.3(b) and 6.23), and the fitted values $\hat{y}$ of X lie in $\mathcal{C}(X) \subseteq \mathcal{C}(X_+)$. Hence $\mathrm{SSE}_+ = \|y - \hat{y}_+\|^2 \leqslant \|y - \hat{y}\|^2 = \mathrm{SSE}$.

- SST does not depend on the design, so by Theorem 13.26, $R^2_+ = 1 - \text{SSE}_+/\text{SST} \geqslant 1 - \text{SSE}/\text{SST} = R^2$. □

So R^2 always favours the design with more columns; the adjusted R^2 below charges for the number of columns.

Definition 13.29 ([12, Definition 6.6])**.** Let $\mathbf{1}_n \in \mathcal{C}(X)$, $\operatorname{rank} X = k+1 < n$ and $\text{SST} > 0$. The **adjusted** R^2 is

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n-k-1)}{\text{SST}/(n-1)} = 1 - \frac{n-1}{n-k-1}(1-R^2)$$

The second form follows from $\text{SSE}/\text{SST} = 1 - R^2$ (Theorem 13.26). The numerator $\text{SSE}/(n-k-1)$ is the unbiased estimate s^2 of σ^2 (Theorem 13.23), and $\text{SST}/(n-1)$ is the sample variance of y. For $k \geqslant 1$, $\frac{n-1}{n-k-1} > 1$, so $\bar{R}^2 < R^2$ unless $R^2 = 1$, and $\bar{R}^2$ can be negative: $R^2 = 0$ gives $\bar{R}^2 = 1 - \frac{n-1}{n-k-1} < 0$. In Example 13.27, $n = 4$ and $k = 2$, so $\bar{R}^2 = 1 - \frac{4/1}{24/3} = \frac{1}{2}$, while $R^2 = \frac{5}{6}$.

Proposition 13.30 ([12, Proposition 6.12])**.** Let $\mathbf{1}_n \in \mathcal{C}(X)$ and $\text{SST} > 0$, with $\hat{y}$, SSR and R^2 as in Definition 13.25, and S_{uv}, constant vectors and $\operatorname{cor}(u,v)$ as in Definition 9.17.

(a) If $\text{SSR} > 0$, then $\hat{y}$ is not constant and $R^2 = \operatorname{cor}(y,\hat{y})^2$.

(b) $\operatorname{cor}(y,v)^2 \leqslant R^2$ for every $v \in \mathcal{C}(X)$ that is not constant.

Proof. First let $v \in \mathcal{C}(X)$. Then $C_n v = v - \bar{v}\mathbf{1} \in \mathcal{C}(X)$, as $\mathbf{1} \in \mathcal{C}(X)$, so $e'C_n v = 0$ (Theorem 13.3(b)). Moreover $C_n e = e$, as $\bar{e} = 0$ (Proposition 13.5(b)), so $C_n y = C_n\hat{y} + C_n e = C_n\hat{y} + e$. Hence

$$S_{yv} = (C_n y)'(C_n v) = (C_n\hat{y})'(C_n v) + e'C_n v = (C_n\hat{y})'(C_n v) = S_{\hat{y}v}$$

(a)

- By Theorem 13.26, $S_{\hat{y}\hat{y}} = \|C_n\hat{y}\|^2 = \text{SSR} > 0$, so $\hat{y}$ is not constant; and $S_{yy} = \|C_n y\|^2 = \text{SST}$.
- The first step with $v = \hat{y} \in \mathcal{C}(X)$ gives $S_{y\hat{y}} = S_{\hat{y}\hat{y}} = \text{SSR}$. So $\operatorname{cor}(y,\hat{y})^2 = \text{SSR}^2/(\text{SST}\cdot\text{SSR}) = \text{SSR}/\text{SST} = R^2$.

(b) By the first step and the Cauchy–Schwarz inequality (Theorem 6.4(a)) for $C_n\hat{y}$ and $C_n v$, $S^2_{yv} = \left((C_n\hat{y})'(C_n v)\right)^2 \leqslant \|C_n\hat{y}\|^2\|C_n v\|^2 = \text{SSR}\cdot S_{vv}$; so $\operatorname{cor}(y,v)^2 = S^2_{yv}/(\text{SST}\cdot S_{vv}) \leqslant \text{SSR}/\text{SST} = R^2$. □

Thus, when $\text{SSR} > 0$, $\hat{y}$ is the vector of $\mathcal{C}(X)$, that is, the linear combination of the columns of X, most highly correlated with y, and R^2 is the square of that correlation.

Example 13.31. In Example 13.27, $C_4 y = (4,-2,0,-2)'$, $C_4\hat{y} = (3,-1,1,-3)'$, $\text{SST} = 24$ and $\text{SSR} = 20$.

- $S_{y\hat{y}} = 12 + 2 + 0 + 6 = 20 = \text{SSR}$, so $\operatorname{cor}(y,\hat{y})^2 = \frac{20^2}{24\cdot 20} = \frac{5}{6} = R^2$.
- For the column $v = X_{\cdot 1}$ of X, $C_4 v = (1,-1,1,-1)'$ (Example 1.17), $S_{yv} = 4 + 2 + 0 + 2 = 8$ and $S_{vv} = 4$, so $\operatorname{cor}(y,v)^2 = \frac{8^2}{24\cdot 4} = \frac{2}{3}$. For $v = X_{\cdot 2}$, $C_4 v = (1,1,-1,-1)'$, $S_{yv} = 4 - 2 + 0 + 2 = 4$ and $S_{vv} = 4$, so $\operatorname{cor}(y,v)^2 = \frac{4^2}{24\cdot 4} = \frac{1}{6}$. Both are at most $R^2 = \frac{5}{6}$, as Proposition 13.30(b) says.

Partitioned regression

Theorem 13.32 (Frisch–Waugh–Lovell; [12, Theorem 8.1])**.** Assume (A4). Partition $X = [X_1\ X_2]$ with $X_1 \in \mathbb{R}^{n\times p}$ and $X_2 \in \mathbb{R}^{n\times q}$ $(p+q=k+1)$, and accordingly $\beta = (\gamma', \delta')'$ and $\hat\beta = (X'X)^{-1}X'y = (\hat\gamma', \hat\delta')'$ with $\gamma, \hat\gamma \in \mathbb{R}^p$, $\delta, \hat\delta \in \mathbb{R}^q$; so $X\beta = X_1\gamma + X_2\delta$. Let $H_2 = X_2(X_2'X_2)^{-1}X_2'$ and $M_2 = I_n - H_2$ be the hat matrix and the residual maker of X_2.

(a) $X_1'M_2X_1$ is invertible and $\hat\gamma = (X_1'M_2X_1)^{-1}X_1'M_2y$.

(b) The least-squares fit of the response M_2y on the design M_2X_1 has the estimate $\hat\gamma$, and its residual vector is the residual vector e of the full fit.

With the roles of X_1 and X_2 exchanged, (a) and (b) hold for $\hat\delta$ and the residual maker M_1 of X_1; in particular $\hat\delta = (X_2'M_1X_2)^{-1}X_2'M_1y$.

Proof. The columns of X_1 and of X_2 are columns of X, hence independent (Proposition 3.13(b)); so H_2, M_2 and the hat matrix and the residual maker of X_1 are as in Theorem 6.26.

(a)

- By the block product (Theorem 5.3), the normal equations $X'X\hat\beta = X'y$ read
$$X_1'X_1\hat\gamma + X_1'X_2\hat\delta = X_1'y, \qquad X_2'X_1\hat\gamma + X_2'X_2\hat\delta = X_2'y$$
- The second gives $\hat\delta = (X_2'X_2)^{-1}X_2'(y - X_1\hat\gamma)$. Substituting into the first, with $X_1'X_2(X_2'X_2)^{-1}X_2' = X_1'H_2 = X_1' - X_1'M_2$, $X_1'X_1\hat\gamma + X_1'(y - X_1\hat\gamma) - X_1'M_2(y - X_1\hat\gamma) = X_1'y$. The first two terms add up to $X_1'y$, so $X_1'M_2(y - X_1\hat\gamma) = 0$, that is, $X_1'M_2X_1\hat\gamma = X_1'M_2y$.
- M_2X_1 has full column rank: if $M_2X_1v = 0$, then $X_1v = H_2X_1v = X_2w$ with $w = (X_2'X_2)^{-1}X_2'X_1v$, so $X(v', -w')' = X_1v - X_2w = 0$, and (A4) gives $v = 0$ (Theorem 4.13(b)). As M_2 is symmetric and idempotent (Theorem 6.26(b)), $X_1'M_2X_1 = (M_2X_1)'(M_2X_1)$, which is therefore invertible (Theorem 6.26); multiplying by its inverse gives the formula for $\hat\gamma$.

(b)

- By (a), $(M_2X_1)'M_2X_1 = X_1'M_2X_1$ is invertible, so the fit of M_2y on M_2X_1 has exactly one least-squares estimate (Theorem 13.3(c)); as $M_2'M_2 = M_2$, it is
$$\big((M_2X_1)'M_2X_1\big)^{-1}(M_2X_1)'M_2y = (X_1'M_2X_1)^{-1}X_1'M_2y = \hat\gamma$$
- Its residual vector is $M_2y - M_2X_1\hat\gamma = M_2(y - X_1\hat\gamma) = M_2(X_2\hat\delta + e) = M_2e$, since $y = X_1\hat\gamma + X_2\hat\delta + e$ and $M_2X_2 = 0$.
- $X_2'e = 0$ (Proposition 13.5(a)), so $H_2e = 0$ and $M_2e = e$.

The normal equations are the same two block equations with $(X_1, \hat\gamma)$ and $(X_2, \hat\delta)$ exchanged, so the same steps give the last statement. □

Here $X_1'M_2X_1 = X_1'X_1 - X_1'X_2(X_2'X_2)^{-1}X_2'X_1$ is the Schur complement of $X_2'X_2$ in $X'X = \left(\begin{smallmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{smallmatrix}\right)$ (the paragraph after Theorem 5.9): solving the second block equation for $\hat\delta$ and substituting it into the first is block elimination. The vectors M_2y and M_2X_1 are the residuals of y and of the columns of X_1 after the least-squares fit on X_2 (Theorem 13.3(c)); so, by (b), $\hat\gamma$ measures the effect of X_1 on y after both are adjusted for X_2 [12, Remark 8.5].

Example 13.33. Let $X = [\mathbf{1}_n\ X_0] \in \mathbb{R}^{n\times(k+1)}$ satisfy (A4), and let $\bar x = \frac{1}{n}X_0'\mathbf{1}$ be the mean vector of the columns of X_0 (Example 1.17). In Theorem 13.32 take $X_1 = \mathbf{1}_n$ and $X_2 = X_0$, so that $\gamma = \beta_0$ and $\delta = (\beta_1, \dots, \beta_k)'$ holds the slopes. The residual maker of $\mathbf{1}_n$ is $M_1 = C_n$ (Example 6.27), so
$$\hat\delta = (X_0'C_nX_0)^{-1}X_0'C_ny$$

the least-squares estimate of the fit of the centered response $C_n y$ on the centered data $C_n X_0$; and $X_0' C_n X_0$ is the Schur complement of n in $X'X$ (Proposition 13.6(a)). The first normal equation, $n\hat{\beta}_0 + \mathbf{1}'X_0\hat{\delta} = \mathbf{1}'y$, gives $\hat{\beta}_0 = \bar{y} - \bar{x}'\hat{\delta}$.

In Example 13.7, $X_0'C_4X_0 = 4I_2$ and C_4X_0 has the columns $(1,-1,1,-1)'$ and $(1,1,-1,-1)'$ (Example 1.17), and $C_4y = (4,-2,0,-2)'$ (Example 13.27). So

$$X_0'C_4y = (C_4X_0)'(C_4y) = \begin{pmatrix} 4+2+0+2 \\ 4-2+0+2 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}, \qquad \hat{\delta} = \frac{1}{4}\begin{pmatrix} 8 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

and $\hat{\beta}_0 = 5 - (1 \cdot 2 + 2 \cdot 1) = 1$, with $\bar{y} = 5$ and $\bar{x} = (1,2)'$: this is $\hat{\beta} = (1,2,1)'$ of Example 13.7. The residuals of the fit of C_4y on C_4X_0 are $C_4y - C_4X_0\hat{\delta} = (4,-2,0,-2)' - (3,-1,1,-3)' = (1,-1,-1,1)' = e$, as Theorem 13.32(b) says.

The centered design $\tilde{X} = [\mathbf{1}_n \; C_nX_0]$ gives the same fit: $C_nX_0 = X_0 - \mathbf{1}\bar{x}'$ (Example 1.17), so $\tilde{X} = XA$ with the invertible $A = \left(\begin{smallmatrix} 1 & -\bar{x}' \\ 0 & I_k \end{smallmatrix}\right)$ (Theorem 5.3), and $\tilde{X}$ has the fitted values of X (Proposition 13.9). As $\mathbf{1}'C_n = 0$ and $C_n'C_n = C_n$ (Proposition 1.15), $\tilde{X}'\tilde{X} = \left(\begin{smallmatrix} n & 0 \\ 0 & X_0'C_nX_0 \end{smallmatrix}\right)$. In Example 13.7, $\tilde{X}'\tilde{X} = 4I_3$ and $\tilde{X}'y = (20,8,4)'$, so the estimate for $\tilde{X}$ is $\frac{1}{4}(20,8,4)' = (5,2,1)' = (\bar{y}, \hat{\delta}')'$, and $A(5,2,1)' = (5 - (1 \cdot 2 + 2 \cdot 1),\; 2,\; 1)' = (1,2,1)' = \hat{\beta}$.

Corollary 13.34 ([12, Corollaries 8.10, 8.12])**.** Assume (A3) and (A4).

(a) In Theorem 13.32, $\operatorname{var}(\hat{\gamma}) = \sigma^2(X_1'M_2X_1)^{-1}$ and $\operatorname{var}(\hat{\delta}) = \sigma^2(X_2'M_1X_2)^{-1}$.

(b) Let $X = [\mathbf{1}_n \; X_0]$ with $k \geqslant 2$, and let $1 \leqslant j \leqslant k$. Let $X_{(j)}$ be X with the column $X_{\cdot j}$ deleted, $M_{(j)}$ its residual maker, $\tilde{x}_j = M_{(j)}X_{\cdot j}$, $S_{jj} = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$, and R_j^2 the R^2 of the least-squares fit of the response $X_{\cdot j}$ on the design $X_{(j)}$. Then $\tilde{x}_j \neq 0$, $S_{jj} > 0$, $0 \leqslant R_j^2 < 1$, and

$$\operatorname{var}(\hat{\beta}_j) = \frac{\sigma^2}{\|\tilde{x}_j\|^2} = \frac{\sigma^2}{S_{jj}(1-R_j^2)} = \frac{\sigma^2}{S_{jj}}\,\mathrm{VIF}_j, \qquad \text{where} \qquad \mathrm{VIF}_j = \frac{1}{1-R_j^2} \geqslant 1$$

The number VIF_j is the **variance inflation factor** of the jth variable.

Proof. (a) By Theorem 13.32(a), $\hat{\gamma} = Ly$ with $L = (X_1'M_2X_1)^{-1}X_1'M_2$. The matrix $(X_1'M_2X_1)^{-1}$ is symmetric (Theorem 2.13(c)), and $M_2M_2' = M_2$ (Theorem 6.26(b)); so by Theorem 9.3(b) and (A3),

$$\operatorname{var}(\hat{\gamma}) = \sigma^2 LL' = \sigma^2(X_1'M_2X_1)^{-1}X_1'M_2X_1(X_1'M_2X_1)^{-1} = \sigma^2(X_1'M_2X_1)^{-1}$$

With the roles of the blocks exchanged, the same steps give $\operatorname{var}(\hat{\delta})$.

(b)

- Let $P \in \mathbb{R}^{(k+1)\times(k+1)}$ have as its columns the columns of I_{k+1}, column $j+1$ first and the others in their order, so that $XP = [X_{\cdot j} \; X_{(j)}]$ (Theorem 1.4(a),(b)). The columns of P are distinct unit vectors, so $P'P = I_{k+1}$ (Theorem 1.4(d)) and $PP' = I_{k+1}$ (Theorem 2.16).
- XP has full column rank ($\operatorname{rank}(XP) = \operatorname{rank} X$, Theorem 4.15(d)), and $(XP)'(XP)P'\hat{\beta} = P'X'X\hat{\beta} = P'X'y = (XP)'y$. So $P'\hat{\beta}$ is the least-squares estimate for XP (Theorem 13.2(b)); its first entry is $\hat{\beta}_j$.
- Theorem 13.32(a) for XP, with $X_1 = X_{\cdot j}$ and $X_2 = X_{(j)}$, gives $X_{\cdot j}'M_{(j)}X_{\cdot j} \neq 0$ and $\hat{\beta}_j = (X_{\cdot j}'M_{(j)}X_{\cdot j})^{-1}X_{\cdot j}'M_{(j)}y$. As $M_{(j)}$ is symmetric and idempotent, $X_{\cdot j}'M_{(j)}X_{\cdot j} = \|\tilde{x}_j\|^2$. So $\tilde{x}_j \neq 0$, and (a) gives $\operatorname{var}(\hat{\beta}_j) = \sigma^2/\|\tilde{x}_j\|^2$.
- If $X_{\cdot j}$ were constant, $X_{\cdot j} - \bar{x}_j\mathbf{1}_n = 0$ would be a linear relation among the columns of X, which (A4) excludes (Theorem 4.13(b)). So $S_{jj} = \|C_nX_{\cdot j}\|^2 > 0$ (Definition 9.17).

- $X_{(j)}$ has full column rank (Proposition 3.13(b)) and contains the column $\mathbf{1}_n$. In the fit of $X_{\cdot j}$ on $X_{(j)}$, the total sum of squares is S_{jj} and the residual vector is $M_{(j)}X_{\cdot j} = \tilde{x}_j$ (Theorem 13.3(c)). By Theorem 13.26, $0 \leqslant R_j^2 = 1 - \|\tilde{x}_j\|^2/S_{jj}$, which is < 1 as $\tilde{x}_j \neq 0$. Hence $\|\tilde{x}_j\|^2 = S_{jj}(1 - R_j^2)$, and $\mathrm{VIF}_j \geqslant 1$. □

So $\mathrm{var}(\hat{\beta}_j)$ is large when σ^2 is large, when the jth variable varies little (small S_{jj}), or when it is close to a linear combination of the other columns (R_j^2 close to 1); $\mathrm{VIF}_j = 1$ exactly when $R_j^2 = 0$.

Example 13.35. Let $X = [\mathbf{1}_4\ X_0]$, where the columns of X_0 are $X_{\cdot 1} = (4, 4, 2, 2)'$ and $X_{\cdot 2} = (3, 2, 2, 1)'$. Then $\bar{x} = (3, 2)'$ and, by Example 1.16 and Theorem 5.24,

$$C_4X_0 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ -1 & 0 \\ -1 & -1 \end{pmatrix}, \qquad X_0'C_4X_0 = (C_4X_0)'(C_4X_0) = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}, \qquad (X_0'C_4X_0)^{-1} = \frac{1}{4}\begin{pmatrix} 2 & -2 \\ -2 & 4 \end{pmatrix}$$

As $X_0'C_4X_0$ is invertible, so is $X'X$ (Proposition 13.6(a)), and X satisfies (A4) (Theorems 4.18(b) and 4.13(d)).

- With $X_1 = \mathbf{1}_4$, $X_2 = X_0$ as in Example 13.33, Corollary 13.34(a) gives $\mathrm{var}\big((\hat{\beta}_1, \hat{\beta}_2)'\big) = \sigma^2(X_0'C_4X_0)^{-1}$: $\mathrm{var}(\hat{\beta}_1) = \frac{1}{2}\sigma^2$ and $\mathrm{var}(\hat{\beta}_2) = \sigma^2$.
- The diagonal of $X_0'C_4X_0$ holds $S_{11} = 4$ and $S_{22} = 2$. By Corollary 13.34(b), $\mathrm{VIF}_j = S_{jj}\,\mathrm{var}(\hat{\beta}_j)/\sigma^2$: $\mathrm{VIF}_1 = 4 \cdot \frac{1}{2} = 2$ and $\mathrm{VIF}_2 = 2 \cdot 1 = 2$, so $R_1^2 = R_2^2 = 1 - \frac{1}{2} = \frac{1}{2}$. The correlation between the two variables doubles the variances σ^2/S_{jj}.
- In Example 13.7, by contrast, $X_0'C_4X_0 = 4I_2$ (Example 1.17), so $\mathrm{var}(\hat{\beta}_1) = \mathrm{var}(\hat{\beta}_2) = \frac{1}{4}\sigma^2 = \sigma^2/S_{jj}$: there $\mathrm{VIF}_1 = \mathrm{VIF}_2 = 1$ and $R_1^2 = R_2^2 = 0$.

Theorem 13.36 ([12, Theorem 8.3])**.** Let $X_1 \in \mathbb{R}^{n\times p}$ have full column rank, let $X_2 \in \mathbb{R}^{n\times q}$, and suppose that $y = X_1\gamma + X_2\delta + \varepsilon$ with $\mathsf{E}(\varepsilon) = 0$. Let $\tilde{\gamma} = (X_1'X_1)^{-1}X_1'y$ be the least-squares estimate for the design X_1 alone, which omits X_2. Then

$$\mathsf{E}(\tilde{\gamma}) = \gamma + (X_1'X_1)^{-1}X_1'X_2\delta$$

The **omitted variable bias** $(X_1'X_1)^{-1}X_1'X_2\delta$ is 0 if and only if $X_1'X_2\delta = 0$, for instance when $X_1'X_2 = 0$ or $\delta = 0$; and it is 0 for every $\delta \in \mathbb{R}^q$ if and only if $X_1'X_2 = 0$.

Proof.
- $X_1'X_1$ is invertible (Theorem 6.26). Substituting y, $\tilde{\gamma} = (X_1'X_1)^{-1}X_1'(X_1\gamma + X_2\delta + \varepsilon) = \gamma + (X_1'X_1)^{-1}X_1'X_2\delta + (X_1'X_1)^{-1}X_1'\varepsilon$.
- By Theorem 9.3(b) and $\mathsf{E}(\varepsilon) = 0$, $\mathsf{E}(\tilde{\gamma}) = \gamma + (X_1'X_1)^{-1}X_1'X_2\delta$.
- As $(X_1'X_1)^{-1}$ is invertible, the bias is 0 if and only if $X_1'X_2\delta = 0$.
- If $X_1'X_2\delta = 0$ for every δ, then taking for δ the columns of I_q gives $X_1'X_2 = 0$; conversely, $X_1'X_2 = 0$ gives $X_1'X_2\delta = 0$ for every δ. □

Example 13.37. In Example 13.7, fit the two variables without the intercept: $X_1 = X_0$, $X_2 = \mathbf{1}_4$, $\gamma = (\beta_1, \beta_2)'$ and $\delta = \beta_0$. By Example 1.17, $X_0'\mathbf{1} = (4, 8)'$ and

$$X_0'X_0 = \begin{pmatrix} 8 & 8 \\ 8 & 20 \end{pmatrix}, \qquad (X_0'X_0)^{-1} = \frac{1}{96}\begin{pmatrix} 20 & -8 \\ -8 & 8 \end{pmatrix}, \qquad (X_0'X_0)^{-1}X_0'\mathbf{1} = \frac{1}{96}\begin{pmatrix} 80 - 64 \\ -32 + 64 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} \\ \frac{1}{3} \end{pmatrix}$$

($\det X_0'X_0 = 160 - 64 = 96$; Theorem 5.24). By Theorem 13.36, the slopes of the fit without the intercept have the expectations $\beta_1 + \frac{1}{6}\beta_0$ and $\beta_2 + \frac{1}{3}\beta_0$: they are biased unless $\beta_0 = 0$. With centered variables, $X_0'\mathbf{1} = 0$, there would be no bias for any β_0.

Best linear prediction

Theorem 13.38 (Best linear predictor)**.** Let y be a random variable and $x \in \mathbb{R}^k$ a random vector, with $\mathsf{E}\, y = \mu_y$, $\mathsf{E}\, x = \mu_x$, $\operatorname{var}(y) = \sigma_{yy}$, $\operatorname{cov}(x, y) = \sigma_{xy} \in \mathbb{R}^k$, $\sigma_{yx} = \sigma_{xy}'$, and $\operatorname{var}(x) = \Sigma_{xx}$ PD. Put

$$b^* = \Sigma_{xx}^{-1}\sigma_{xy}, \qquad a^* = \mu_y - b^{*\prime}\mu_x$$

Then for all $a \in \mathbb{R}$ and $b \in \mathbb{R}^k$,

$$\mathsf{E}(y - a - b'x)^2 = \left(\sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}\right) + (b - b^*)'\Sigma_{xx}(b - b^*) + (\mu_y - a - b'\mu_x)^2$$

So the **mean squared prediction error** $\mathsf{E}(y - a - b'x)^2$ is smallest exactly for $a = a^*$, $b = b^*$, with minimum $\sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}$: the Schur complement of Σ_{xx} in $\operatorname{var}\left((x', y)'\right) = \left(\begin{smallmatrix} \Sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{smallmatrix}\right)$ (Theorem 5.9). The random variable $a^* + b^{*\prime}x$ is the **best linear predictor** of y from x, and the prediction error $u = y - a^* - b^{*\prime}x$ has mean 0 and is uncorrelated with x.

Proof. • Let $v = y - a - b'x$. By Theorem 9.4(e) for $p = 1$, $\mathsf{E}\, v^2 = \operatorname{var}(v) + (\mathsf{E}\, v)^2$, and $\mathsf{E}\, v = \mu_y - a - b'\mu_x$ (Theorem 9.3(b)).

- $v = (1, -b')(y, x')' - a$, and $\operatorname{var}\left((y, x')'\right)$ has the blocks σ_{yy}, σ_{yx}, σ_{xy}, Σ_{xx}; by Theorem 9.3(b), $\operatorname{var}(v) = \sigma_{yy} - 2b'\sigma_{xy} + b'\Sigma_{xx}b$.
- As $\Sigma_{xx}b^* = \sigma_{xy}$, $(b - b^*)'\Sigma_{xx}(b - b^*) = b'\Sigma_{xx}b - 2b'\sigma_{xy} + \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}$; substituting gives the identity.
- Σ_{xx} is PD, so the middle term is $\geqslant 0$, with equality only for $b = b^*$; the last term is $\geqslant 0$, with equality only for $a = \mu_y - b'\mu_x$. Hence the minimum is attained exactly at $b = b^*$, $a = a^*$.
- As $a^* = \mu_y - b^{*\prime}\mu_x$, $u = y - a^* - b^{*\prime}x = (y - \mu_y) - b^{*\prime}(x - \mu_x)$, so $\mathsf{E}\, u = 0$ (Theorem 9.3(b)). By Theorem 9.4(a),(b), $\operatorname{cov}(x, u) = \operatorname{cov}(x, y) - \operatorname{cov}(x, x)b^* = \sigma_{xy} - \Sigma_{xx}b^* = 0$. □

Under joint normality the best linear predictor is the conditional mean. If $(x', y)' \sim N_{k+1}(\mu, \Sigma)$ with Σ_{xx} PD, Theorem 10.13 with $z_1 = x$ and $z_2 = y$ gives the conditional distribution of y given x as

$$N\left(\mu_y + \sigma_{yx}\Sigma_{xx}^{-1}(x - \mu_x),\ \sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}\right) = N\left(a^* + b^{*\prime}x,\ \sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy}\right)$$

in the sense of that theorem: the conditional mean is the best linear predictor of Theorem 13.38, and the conditional variance is its mean squared prediction error. Likewise, for $z \sim N_n(\mu, \Sigma)$ partitioned as in Theorem 10.13, entry j of $w = z_2 - \mu_2 - \Sigma_{21}\Sigma_{11}^{-1}(z_1 - \mu_1)$ is the prediction error of the best linear predictor of entry j of z_2 from z_1 (row j of $\Sigma_{21}\Sigma_{11}^{-1}$ is $b^{*\prime}$); so the partial correlations of the entries of z_2 given z_1 (the paragraph after Theorem 10.13) are the correlations of these prediction errors. This uses only the covariance matrix.

Example 13.39. Let $(y, x_1, x_2, x_3)'$ have mean $(8, 1, 2, 3)'$ and covariance matrix

$$\begin{pmatrix} \sigma_{yy} & \sigma_{yx} \\ \sigma_{xy} & \Sigma_{xx} \end{pmatrix} = \left(\begin{array}{c|ccc} 12 & 3 & 4 & 3 \\ \hline 3 & 2 & 1 & 0 \\ 4 & 1 & 2 & 1 \\ 3 & 0 & 1 & 2 \end{array}\right)$$

which is PD (leading principal minors 12, 15, 10, 8; Theorem 8.34); so is Σ_{xx} (leading principal minors 2, 3, 4), which is therefore invertible (Proposition 8.23(c)). So $b^* = \Sigma_{xx}^{-1}\sigma_{xy}$ is the unique solution of $\Sigma_{xx}b = \sigma_{xy}$. As

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix}$$

$b^* = (1, 1, 1)'$ and $a^* = 8 - (1 + 2 + 3) = 2$. By Theorem 13.38 the best linear predictor of y is $2 + x_1 + x_2 + x_3$, with mean squared prediction error $12 - (3, 4, 3)(1, 1, 1)' = 2$. With fewer predictors the minimum is larger:

- with $(x_1, x_3)'$, whose covariance matrix is $2I_2$, it is $12 - (3, 3)\frac{1}{2}I_2(3, 3)' = 3$;
- with no predictor, $\mathsf{E}(y - a)^2 = \operatorname{var}(y) + (\mu_y - a)^2$ is smallest for $a = \mu_y = 8$, with value 12.

Exercises

Exercise 13.1. Four observations of two variables and a response are

x_{i1}	2	0	1	1
x_{i2}	1	1	0	2
y_i	7	5	3	5

Fit $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$.

(a) Compute $X'X$, $(X'X)^{-1}$ by row reduction of $[X'X \mid I_3]$, $(X'X)^{-1}X'$ and $\hat{\beta}$.

(b) Compute the hat matrix H and its trace, $\hat{y} = Hy$ and $e = y - \hat{y}$, and check that $X'e = 0$.

(c) Compute SST, SSR, SSE and R^2. Check that $R^2 = \operatorname{cor}(y, \hat{y})^2$, and compare R^2 with $\operatorname{cor}(y, X_{\cdot 1})^2$ and $\operatorname{cor}(y, X_{\cdot 2})^2$.

Solution. (a) The ith row of X is $(1, x_{i1}, x_{i2})$, and the (j,l) entry of $X'X$ is $\operatorname{col}_j(X)'\operatorname{col}_l(X)$ (Theorems 1.4(d) and 1.9):

$$X = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 4 & 4 & 4 \\ 4 & 6 & 4 \\ 4 & 4 & 6 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 7+5+3+5 \\ 14+0+3+5 \\ 7+5+0+10 \end{pmatrix} = \begin{pmatrix} 20 \\ 22 \\ 22 \end{pmatrix}$$

Row-reduce $[X'X \mid I_3]$, postponing the scaling to the end:

$$\left(\begin{array}{ccc|ccc} 4 & 4 & 4 & 1 & 0 & 0 \\ 4 & 6 & 4 & 0 & 1 & 0 \\ 4 & 4 & 6 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{ccc|ccc} 4 & 4 & 4 & 1 & 0 & 0 \\ 0 & 2 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2,\ R_1 \leftarrow R_1 - 2R_3} \left(\begin{array}{ccc|ccc} 4 & 0 & 0 & 5 & -2 & -2 \\ 0 & 2 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow \frac{1}{2}R_3]{R_1 \leftarrow \frac{1}{4}R_1,\ R_2 \leftarrow \frac{1}{2}R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array}\right)$$

So

$$(X'X)^{-1} = \frac{1}{4}\begin{pmatrix} 5 & -2 & -2 \\ -2 & 2 & 0 \\ -2 & 0 & 2 \end{pmatrix}, \qquad (X'X)^{-1}X' = \frac{1}{4}\begin{pmatrix} -1 & 3 & 3 & -1 \\ 2 & -2 & 0 & 0 \\ 0 & 0 & -2 & 2 \end{pmatrix}$$

and

$$\hat{\beta} = (X'X)^{-1}X'y = \frac{1}{4}\begin{pmatrix} -7+15+9-5 \\ 14-10 \\ -6+10 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

(b) $H = X(X'X)^{-1}X'$ (Theorem 6.26); its trace is $\frac{1}{4}(3+3+3+3) = 3 = \operatorname{rank} X$ (Corollary 6.30):

$$H = \frac{1}{4}\begin{pmatrix} 3 & -1 & 1 & 1 \\ -1 & 3 & 1 & 1 \\ 1 & 1 & 3 & -1 \\ 1 & 1 & -1 & 3 \end{pmatrix}, \qquad \hat{y} = Hy = \frac{1}{4}\begin{pmatrix} 21-5+3+5 \\ -7+15+3+5 \\ 7+5+9-5 \\ 7+5-3+15 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 4 \\ 6 \end{pmatrix}, \qquad e = y - \hat{y} = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

Also $\hat{y} = X\hat{\beta} = (3+2+1,\ 3+0+1,\ 3+1+0,\ 3+1+2)'$. Check: $X'e = (1+1-1-1,\ 2+0-1-1,\ 1+1+0-2)' = 0$.

(c) $\bar{y} = 5$, and $\hat{y}$ also has mean 5, so $C_4 y = (2, 0, -2, 0)'$ and $C_4\hat{y} = (1, -1, -1, 1)'$. Hence SST $= 4+0+4+0 = 8$, SSR $= 1+1+1+1 = 4$, SSE $= e'e = 1+1+1+1 = 4$; indeed $8 = 4+4$ (Theorem 13.26), and $R^2 = \frac{4}{8} = \frac{1}{2}$.

- $S_{y\hat{y}} = (C_4 y)'(C_4\hat{y}) = 2+0+2+0 = 4$, so $\operatorname{cor}(y,\hat{y})^2 = \frac{4^2}{8\cdot 4} = \frac{1}{2} = R^2$ (Proposition 13.30(a)).
- $v = X_{\cdot 1} = (2,0,1,1)'$ has mean 1 and $C_4 v = (1,-1,0,0)'$, so $S_{yv} = 2$, $S_{vv} = 2$ and $\operatorname{cor}(y,v)^2 = \frac{2^2}{8\cdot 2} = \frac{1}{4}$. Likewise $v = X_{\cdot 2} = (1,1,0,2)'$ has $C_4 v = (0,0,-1,1)'$, $S_{yv} = 2$, $S_{vv} = 2$ and $\operatorname{cor}(y,v)^2 = \frac{1}{4}$. Both are below $R^2 = \frac{1}{2}$ (Proposition 13.30(b)).

Exercise 13.2. Let $X = [\mathbf{1}_5\ X_0]$, where the columns of X_0 are $x_1 = (1,2,3,4,5)'$ and $x_2 = (3,0,2,4,1)'$, and let $y = (9,4,4,14,9)'$.

(a) Compute $\bar{x}$, $S_c = X_0' C_5 X_0$ and $(X'X)^{-1}$ (Proposition 13.6).

(b) Compute the diagonal entries $h_{11},\dots,h_{55}$ of the hat matrix H, the leverages, and check that their sum is $\operatorname{tr} H = 3$.

(c) Compute $\hat{\beta}$, e, s^2, and the standardized residuals $e_i/(s\sqrt{1-h_{ii}})$.

Solution. (a) $\bar{x} = \frac{1}{5}X_0'\mathbf{1}_5 = \frac{1}{5}(15,10)' = (3,2)'$, and the columns of $C_5X_0 = X_0 - \mathbf{1}\bar{x}'$ (Example 1.17) are $a = (-2,-1,0,1,2)'$ and $b = (1,-2,0,2,-1)'$, with $a'a = b'b = 10$ and $a'b = -2+2+0+2-2 = 0$. So $S_c = (C_5X_0)'(C_5X_0) = 10I_2$ (Proposition 1.15). Proposition 13.6(b), with $\frac{1}{5} + \bar{x}'S_c^{-1}\bar{x} = \frac{1}{5} + \frac{9+4}{10} = \frac{3}{2}$ and $S_c^{-1}\bar{x} = \frac{1}{10}(3,2)'$, gives

$$(X'X)^{-1} = \frac{1}{10}\begin{pmatrix} 15 & -3 & -2 \\ -3 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

(b) $h_{ii} = \operatorname{row}_i(X)(X'X)^{-1}\operatorname{row}_i(X)'$ (Theorem 1.4(d)). For a row $(1,u,v)$,

$$(1,u,v)(X'X)^{-1}(1,u,v)' = \tfrac{1}{10}\big(15 - 6u - 4v + u^2 + v^2\big) = \tfrac{1}{10}\big((u-3)^2 + (v-2)^2 + 2\big)$$

and $(u-3, v-2)$ is row i of C_5X_0, that is, (a_i, b_i). As $a_i^2 + b_i^2 = 5$ for $i \neq 3$ and 0 for $i = 3$, $h_{ii} = \frac{7}{10}$ for $i \neq 3$ and $h_{33} = \frac{1}{5}$. Their sum is $4\cdot\frac{7}{10} + \frac{1}{5} = 3$, which is $\operatorname{tr} H = \operatorname{rank} X$ (Theorem 13.4(b)), as $\operatorname{rank} X = 3$ (Proposition 13.6(a), $\det S_c = 100$).

(c) $X'y = (9+4+4+14+9,\ 9+8+12+56+45,\ 27+0+8+56+9)' = (40,130,100)'$, so

$$\hat{\beta} = (X'X)^{-1}X'y = \frac{1}{10}\begin{pmatrix} 600-390-200 \\ -120+130 \\ -80+100 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

Then $\hat{y} = X\hat{\beta} = (1+1+6,\ 1+2+0,\ 1+3+4,\ 1+4+8,\ 1+5+2)' = (8,3,8,13,8)'$, $e = y - \hat{y} = (1,1,-4,1,1)'$, $\mathrm{SSE} = 1+1+16+1+1 = 20$ and $s^2 = 20/(5-3) = 10$ (Theorem 13.23). The standardized residuals are $1/\sqrt{10\cdot\frac{3}{10}} = \frac{1}{\sqrt{3}}$ for $i \neq 3$ and $-4/\sqrt{10\cdot\frac{4}{5}} = -\sqrt{2}$ for $i = 3$: the observation of smallest leverage has the largest residual.

Exercise 13.3. Add to the design matrix X of Example 13.7 the sum of its two variables as a fourth column: $\tilde{X} = [\mathbf{1}_4\ X_{\cdot 1}\ X_{\cdot 2}\ X_{\cdot 1} + X_{\cdot 2}] \in \mathbb{R}^{4\times 4}$, and let $y = (9,3,5,3)'$.

(a) Find $\operatorname{rank}\tilde{X}$ and a basis of $\mathcal{N}(\tilde{X})$.

(b) Solve the normal equations $\tilde{X}'\tilde{X}\beta = \tilde{X}'y$ by row reduction.

(c) Show that all least-squares estimates give the fitted values $\hat{y} = (8,4,6,2)'$ of Example 13.7, and find $S(\hat{\beta})$.

Solution. (a) The columns are $\mathbf{1}_4$, $(2,0,2,0)'$, $(3,3,1,1)'$ and $(5,3,3,1)'$. The fourth is the sum of the second and the third, so $\tilde{X}(0,1,1,-1)' = 0$; the first three are the columns of X, which are independent (Example 13.7). So $\operatorname{rank}\tilde{X} = 3$, $\dim\mathcal{N}(\tilde{X}) = 4-3 = 1$ (Theorem 4.12), and $(0,1,1,-1)'$ is a basis of $\mathcal{N}(\tilde{X})$.

(b) The entries of $\tilde{X}'\tilde{X}$ and $\tilde{X}'y$ are inner products of columns (Theorems 1.4(d) and 1.9):

$$\left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 4 & 8 & 8 & 16 & 28 \\ 8 & 8 & 20 & 28 & 44 \\ 12 & 16 & 28 & 44 & 72 \end{array}\right) \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 0 & 4 & 0 & 4 & 8 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 4 & 4 & 8 & 12 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_2} \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 0 & 4 & 0 & 4 & 8 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 4 & 4 & 4 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 0 & 4 & 0 & 4 & 8 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow \frac{1}{4}R_3]{R_1 \leftarrow \frac{1}{4}R_1,\ R_2 \leftarrow \frac{1}{4}R_2} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 5 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The last matrix is an echelon form with pivots in columns $1, 2, 3$ and none in the last column, so the system is consistent and $\beta_3 = t$ is free (Theorem 2.26). Back substitution: $\beta_2 = 1 - t$, $\beta_1 = 2 - t$ and $\beta_0 = 5 - \beta_1 - 2\beta_2 - 3\beta_3 = 5 - (2 - t) - 2(1 - t) - 3t = 1$. So the solutions are

$$\beta = \begin{pmatrix} 1 \\ 2 - t \\ 1 - t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} \qquad (t \in \mathbb{R})$$

a particular solution plus $\mathcal{N}(\tilde{X}'\tilde{X}) = \mathcal{N}(\tilde{X})$, as Theorem 13.2(c) says.

(c) These solutions are the least-squares estimates (Theorem 13.3(a)), and each gives $\tilde{X}\beta = X(1, 2, 1)' + t\tilde{X}(0, -1, -1, 1)' = (8, 4, 6, 2)'$. So $\hat{y}$ and $e = (1, -1, -1, 1)'$ are those of Example 13.7, and $S(\hat{\beta}) = e'e = 4$.

Exercise 13.4. Let

$$\tilde{X} = \begin{pmatrix} 1 & 2 & 3 & 5 \\ 1 & 0 & 3 & 3 \\ 1 & 2 & 1 & 3 \\ 1 & 0 & 1 & 1 \end{pmatrix}, \qquad y = \begin{pmatrix} 9 \\ 3 \\ 5 \\ 3 \end{pmatrix}$$

as in Exercise 13.3, where the least-squares estimates are found to be $\beta = (1,\ 2 - t,\ 1 - t,\ t)'$ $(t \in \mathbb{R})$.

(a) Row-reduce $\tilde{X}$ to its RREF, and write down the full-rank factorization $\tilde{X} = BC$ of Theorem 4.21(a).

(b) Here B is the design matrix X of Example 13.7, so $(B'B)^{-1}B'y = (1, 2, 1)'$. Compute $\tilde{X}^+ y$ by Theorem 12.24(b).

(c) Check that $\tilde{X}^+ y$ is the shortest least-squares estimate.

Solution. (a)

$$\tilde{X} \xrightarrow[R_3 \leftarrow R_3 - R_1,\ R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 2 & 3 & 5 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & -2 & -2 \\ 0 & -2 & -2 & -4 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - R_2} \begin{pmatrix} 1 & 2 & 3 & 5 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & -2 & -2 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 1 & 2 & 3 & 5 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_3 \leftarrow -\frac{1}{2}R_3]{R_2 \leftarrow -\frac{1}{2}R_2} \begin{pmatrix} 1 & 2 & 3 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_1 \leftarrow R_1 - 3R_3]{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The pivot columns are $1, 2, 3$, so $\tilde{X} = BC$ with B the first three columns of $\tilde{X}$, that is, the design matrix X of Example 13.7, and

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad CC' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

(b) $\det(CC') = 1 \cdot (4 - 1) = 3$ and, by Theorem 5.24,

$$(CC')^{-1} = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad (CC')^{-1}\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 \\ 4-1 \\ -2+2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

By Theorem 12.24(b), $\tilde{X}^+ y = C'(CC')^{-1}(B'B)^{-1}B'y = C'(1, 1, 0)' = \mathrm{col}_1(C') + \mathrm{col}_2(C') = (1, 1, 0, 1)'$ (Theorem 1.4(b)).

(c) $\|\beta\|^2 = 1 + (2 - t)^2 + (1 - t)^2 + t^2 = 3t^2 - 6t + 6 = 3(t - 1)^2 + 3$ is smallest exactly for $t = 1$, where $\beta = (1, 1, 0, 1)' = \tilde{X}^+ y$, of squared length 3, as Theorem 13.12 says.

Exercise 13.5. Let $X_0 \in \mathbb{R}^{6\times 2}$ be the data of Exercise 2.25, with the columns $(2, 2, 0, 0, 2, 0)'$ and $(2, 0, 2, 0, 1, 1)'$; let $X = [\mathbf{1}_6\ X_0]$, and let $\tilde{X} = [\mathbf{1}_6\ C_6 X_0]$ be the design with the centered columns. By Exercise 2.25, $\tilde{X} = XQ$ for the product Q of elementary matrices found there.

(a) Compute $\tilde{X}'\tilde{X}$ and its inverse.

(b) Find $(X'X)^{-1}$ from Q and (a), and check the first row of $(X'X)(X'X)^{-1} = I_3$.

Solution. (a) By Exercise 2.25, the columns of $\tilde{X}$ are $\mathbf{1}_6$, $u = (1, 1, -1, -1, 1, -1)'$ and $v = (1, -1, 1, -1, 0, 0)'$, with $\mathbf{1}'\mathbf{1} = 6$, $\mathbf{1}'u = 0$, $\mathbf{1}'v = 0$, $u'u = 6$, $u'v = 1 - 1 - 1 + 1 = 0$ and $v'v = 4$. So (Theorems 1.4(d) and 1.9, Definition 2.12)

$$\tilde{X}'\tilde{X} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad (\tilde{X}'\tilde{X})^{-1} = \begin{pmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{6} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$$

(b) $\tilde{X}'\tilde{X} = (XQ)'XQ = Q'(X'X)Q$ (Theorem 1.9), and Q is invertible (Exercise 2.25). So $X'X = (Q')^{-1}(\tilde{X}'\tilde{X})Q^{-1}$, and by Theorem 2.13(a)–(c), $(X'X)^{-1} = Q(\tilde{X}'\tilde{X})^{-1}Q'$:

$$(X'X)^{-1} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{6} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \frac{1}{12}\begin{pmatrix} 7 & -2 & -3 \\ -2 & 2 & 0 \\ -3 & 0 & 3 \end{pmatrix}$$

Here the entries of $X'X$ are n, the column sums and the sums of squares and products of the columns (Example 1.17):

$$X'X = \begin{pmatrix} 6 & 6 & 6 \\ 6 & 12 & 6 \\ 6 & 6 & 10 \end{pmatrix}$$

and the first row of $(X'X)(X'X)^{-1}$ is $\frac{1}{12}(42 - 12 - 18,\ -12 + 12,\ -18 + 18) = (1, 0, 0)$.

Exercise 13.6. A dose is given at the three coded levels $-1, 0, 1$, two observations at each level:

x_i	-1	-1	0	0	1	1
y_i	5	3	3	1	7	5

Fit the quadratic regression $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i$ under (A1)–(A3).

(a) Explain why X has full column rank, and compute $X'X$, $X'y$, $(X'X)^{-1}$ and $\hat\beta$.

(b) Compute $\hat y$, e, SSE and s^2, and the estimate $s^2(X'X)^{-1}$ of $\operatorname{var}(\hat\beta)$.

(c) Compute SST, SSR, R^2 and $\bar R^2$.

Solution. (a) The x_i take three different values, so X, with rows $(1, x_i, x_i^2)$, has full column rank (Proposition 13.15). The columns of X are $\mathbf{1}_6$, $x = (-1,-1,0,0,1,1)'$ and $(1,1,0,0,1,1)'$, and

$$X'X = \begin{pmatrix} 6 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 4 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 5+3+3+1+7+5 \\ -5-3+7+5 \\ 5+3+7+5 \end{pmatrix} = \begin{pmatrix} 24 \\ 4 \\ 20 \end{pmatrix}$$

Row-reduce $[X'X \mid I_3]$, postponing the scaling to the end:

$$\left(\begin{array}{ccc|ccc} 6 & 0 & 4 & 1 & 0 & 0 \\ 0 & 4 & 0 & 0 & 1 & 0 \\ 4 & 0 & 4 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & -1 \\ 0 & 4 & 0 & 0 & 1 & 0 \\ 4 & 0 & 4 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & -1 \\ 0 & 4 & 0 & 0 & 1 & 0 \\ 0 & 0 & 4 & -2 & 0 & 3 \end{array}\right)$$

$$\xrightarrow[R_3 \leftarrow \frac14 R_3]{R_1 \leftarrow \frac12 R_1,\ R_2 \leftarrow \frac14 R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac12 & 0 & -\frac12 \\ 0 & 1 & 0 & 0 & \frac14 & 0 \\ 0 & 0 & 1 & -\frac12 & 0 & \frac34 \end{array}\right)$$

So

$$(X'X)^{-1} = \frac14 \begin{pmatrix} 2 & 0 & -2 \\ 0 & 1 & 0 \\ -2 & 0 & 3 \end{pmatrix}, \qquad \hat\beta = (X'X)^{-1}X'y = \frac14 \begin{pmatrix} 48-40 \\ 4 \\ -48+60 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$$

(b) $\hat y_i = 2 + x_i + 3x_i^2$ is 4, 2, 6 at $x_i = -1, 0, 1$: $\hat y = (4,4,2,2,6,6)'$ and $e = y - \hat y = (1,-1,1,-1,1,-1)'$. Check: $X'e = (0,\ -1+1+1-1,\ 1-1+1-1)' = 0$. Then SSE $= 6$, and with $n = 6$, $k = 2$, $s^2 = 6/(6-3) = 2$ (Theorem 13.23), so

$$s^2(X'X)^{-1} = \frac12 \begin{pmatrix} 2 & 0 & -2 \\ 0 & 1 & 0 \\ -2 & 0 & 3 \end{pmatrix}$$

for instance $\operatorname{var}(\hat\beta_1)$ is estimated by $\frac12$.

(c) $\bar y = 24/6 = 4$, so SST $= 1+1+1+9+9+1 = 22$ and SSR $= 0+0+4+4+4+4 = 16$; indeed $22 = 16 + 6$ (Theorem 13.26). So $R^2 = \frac{16}{22} = \frac{8}{11}$ and $\bar R^2 = 1 - \frac{6/3}{22/5} = 1 - \frac{10}{22} = \frac{6}{11}$ (Definition 13.29).

Exercise 13.7. Let $a_1 = \mathbf{1}_5$, $a_2 = (1,2,3,4,5)'$, $a_3 = (1,4,9,16,25)'$ and $a_4 = (1,8,27,64,125)'$, the values of $1, t, t^2, t^3$ at $t = 1, \dots, 5$, and let $U_3 = \operatorname{span}\{a_1, a_2, a_3\}$ and $U_4 = \operatorname{span}\{a_1, \dots, a_4\}$.

(a) Apply the Gram–Schmidt process to $a_1, \dots, a_4$, and show that b_2, b_3 and $\frac56 b_4$ are

$$c_1 = (-2,-1,0,1,2)', \qquad c_2 = (2,-1,-2,-1,2)', \qquad c_3 = (-1,2,0,-2,1)'$$

(b) Show that b_2, b_3, b_4 are the values at $t = 1, \dots, 5$ of the polynomials $t - 3$, $(t-3)^2 - 2$ and $(t-3)^3 - \frac{17}{5}(t-3)$, and that $c_4 = (1,-4,6,-4,1)'$ spans $U_4^{\perp}$. (The vectors $c_1, \dots, c_4$ are the tabulated coefficients of the orthogonal polynomials for five equally spaced levels.)

(c) For $y = (0, 7, 7, 5, 6)'$, find the vector of U_3 closest to y, the polynomial of degree at most 2 whose values at $t = 1, \dots, 5$ it lists, and the distance from y to U_3. Read the answer as the least-squares fit of the quadratic regression $y_i = \beta_0 + \beta_1 t_i + \beta_2 t_i^2 + \varepsilon_i$ with $t_i = i$: find $\hat\beta$ and SSE.

Solution. (a) By Theorem 6.14:

- $b_1 = \mathbf{1}_5$, $\|b_1\|^2 = 5$; $a_2'b_1 = 15$, so $t_{12} = 3$ and $b_2 = a_2 - 3b_1 = (-2, -1, 0, 1, 2)' = c_1$, with $\|b_2\|^2 = 10$.
- $a_3'b_1 = 55$ and $a_3'b_2 = -2 - 4 + 0 + 16 + 50 = 60$, so $t_{13} = 11$, $t_{23} = 6$ and $b_3 = a_3 - 11b_1 - 6b_2 = (1 - 11 + 12,\ 4 - 11 + 6,\ 9 - 11 + 0,\ 16 - 11 - 6,\ 25 - 11 - 12)' = c_2$, with $\|b_3\|^2 = 14$.
- $a_4'b_1 = 225$, $a_4'b_2 = -2 - 8 + 0 + 64 + 250 = 304$ and $a_4'b_3 = 2 - 8 - 54 - 64 + 250 = 126$, so $t_{14} = 45$, $t_{24} = \frac{152}{5}$, $t_{34} = 9$, and $b_4 = a_4 - 45b_1 - \frac{152}{5}b_2 - 9b_3 = \frac{1}{5}(-6, 12, 0, -12, 6)' = \frac{6}{5}c_3$.

(b) The values of $s = t - 3$ at $t = 1, \dots, 5$ are $b_2 = (-2, -1, 0, 1, 2)'$; those of $s^2 - 2$ are $(4 - 2,\ 1 - 2,\ 0 - 2,\ 1 - 2,\ 4 - 2)' = b_3$; and those of $s^3 - \frac{17}{5}s$ are $(-8 + \frac{34}{5},\ -1 + \frac{17}{5},\ 0,\ 1 - \frac{17}{5},\ 8 - \frac{34}{5})' = \frac{1}{5}(-6, 12, 0, -12, 6)' = b_4$. Next $c_4'a_1 = 1 - 4 + 6 - 4 + 1 = 0$, $c_4'a_2 = 1 - 8 + 18 - 16 + 5 = 0$, $c_4'a_3 = 1 - 16 + 54 - 64 + 25 = 0$ and $c_4'a_4 = 1 - 32 + 162 - 256 + 125 = 0$, so $c_4 \in U_4^\perp$ (Theorem 6.6(a)). The vectors $b_1, \dots, b_4$ are nonzero, pairwise orthogonal and span U_4 (Theorem 6.14), so they form a basis of U_4 (Proposition 6.11(b)); hence $\dim U_4^\perp = 5 - 4 = 1$ (Theorem 6.6(c)), and $c_4 \neq 0$ spans $U_4^\perp$ (Theorem 4.5(d)).

(c) $b_1'y = 0 + 7 + 7 + 5 + 6 = 25$, $b_2'y = c_1'y = 0 - 7 + 0 + 5 + 12 = 10$ and $b_3'y = c_2'y = 0 - 7 - 14 - 5 + 12 = -14$, with $b_1'b_1 = 5$, $b_2'b_2 = 10$ and $b_3'b_3 = 14$ by (a). By (a), b_1, b_2, b_3 are nonzero, pairwise orthogonal and span U_3 (Theorem 6.14), so $b_j/\|b_j\|$ $(j = 1, 2, 3)$ form an orthonormal basis of U_3. By Theorems 6.23 and 6.31(b), the vector of U_3 closest to y is

$$P_{U_3}y = \sum_{j=1}^{3} \frac{b_j'y}{b_j'b_j}\, b_j = \frac{25}{5}\,\mathbf{1}_5 + \frac{10}{10}\,c_1 + \frac{-14}{14}\,c_2 = 5\,\mathbf{1}_5 + c_1 - c_2 = (1, 5, 7, 7, 5)'$$

and $y - P_{U_3}y = (-1, 2, 0, -2, 1)' = c_3$, so the distance from y to U_3 is $\sqrt{10}$. By (a) and (b), $P_{U_3}y = 5b_1 + b_2 - b_3$ lists the values at $t = 1, \dots, 5$ of $5 + (t-3) - \big((t-3)^2 - 2\big) = 7 + (t-3) - (t-3)^2$; for instance $7 - 2 - 4 = 1$ at $t = 1$. The columns of $X = [a_1\ a_2\ a_3]$, with rows $(1, t_i, t_i^2)$, span U_3, so $P_{U_3}y$ is the vector $\hat y$ of fitted values (Theorem 13.3(b)). As $7 + (t-3) - (t-3)^2 = -5 + 7t - t^2$, $\hat y = X(-5, 7, -1)'$; the t_i take five different values, so X has full column rank (Proposition 13.15), and $\hat\beta = (-5, 7, -1)'$ is the only least-squares estimate (Theorems 13.3(a) and 13.2(b)). Finally $\mathrm{SSE} = \|y - \hat y\|^2 = \|c_3\|^2 = 10$.

Exercise 13.8. Let X and $y = (7, 5, 3, 5)'$ be as in Exercise 13.1, under (A1)–(A4). Let X_s consist of the first three rows of X, and $C = [X_s^{-1}\ \ 0] \in \mathbb{R}^{3\times 4}$: $\tilde\beta = Cy$ solves $X_s\beta = (y_1, y_2, y_3)'$, so it uses the first three observations only.

(a) Find X_s^{-1} by Theorem 5.24, check that $CX = I_3$, and compute $\tilde\beta$.

(b) Find $\mathrm{var}(\tilde\beta)$, $\mathrm{var}(\hat\beta)$ and $D = C - (X'X)^{-1}X'$, and check that $\mathrm{var}(\tilde\beta) - \mathrm{var}(\hat\beta) = \sigma^2 DD'$.

(c) For which j is $\mathrm{var}(\tilde\beta_j) = \mathrm{var}(\hat\beta_j)$? Compare $\tilde\beta_j$ with $\hat\beta_j$ for it.

Solution. (a) The first three rows of X are

$$X_s = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Expanding along row 2, $\det X_s = -(2\cdot 0 - 1\cdot 1) + 0 - (1\cdot 1 - 2\cdot 1) = 1 + 1 = 2$, and the nine cofactors give

$$\operatorname{adj} X_s = \begin{pmatrix} -1 & 1 & 2 \\ 1 & -1 & 0 \\ 1 & 1 & -2 \end{pmatrix}, \qquad X_s^{-1} = \frac{1}{2}\begin{pmatrix} -1 & 1 & 2 \\ 1 & -1 & 0 \\ 1 & 1 & -2 \end{pmatrix}, \qquad C = \frac{1}{2}\begin{pmatrix} -1 & 1 & 2 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \end{pmatrix}$$

(Theorem 5.24). By the block product, $CX = X_s^{-1}X_s + 0 = I_3$, so $\mathsf{E}(Cy) = CX\beta = \beta$ by Theorem 9.3(b) and (A1): $\tilde\beta$ is a linear unbiased estimator of β. For the data, $\tilde\beta = X_s^{-1}(7,5,3)' = \frac12(-7+5+6,\ 7-5+0,\ 7+5-6)' = (2,1,3)'$, whereas $\hat\beta = (3,1,1)'$.

(b) By Theorem 9.3(b) and (A3), $\mathrm{var}(\tilde\beta) = \sigma^2 CC' = \sigma^2 X_s^{-1}(X_s^{-1})'$, and $\mathrm{var}(\hat\beta) = \sigma^2(X'X)^{-1}$ (Theorem 13.17(b)), with $(X'X)^{-1}$ and $(X'X)^{-1}X'$ of Exercise 13.1(a):

$$\mathrm{var}(\tilde\beta) = \frac{\sigma^2}{4}\begin{pmatrix}6&-2&-4\\-2&2&0\\-4&0&6\end{pmatrix}, \qquad \mathrm{var}(\hat\beta) = \frac{\sigma^2}{4}\begin{pmatrix}5&-2&-2\\-2&2&0\\-2&0&2\end{pmatrix}, \qquad D = \frac14\begin{pmatrix}-1&-1&1&1\\0&0&0&0\\2&2&-2&-2\end{pmatrix}$$

Then

$$DD' = \frac{1}{16}\begin{pmatrix}4&0&-8\\0&0&0\\-8&0&16\end{pmatrix} = \frac14\begin{pmatrix}1&0&-2\\0&0&0\\-2&0&4\end{pmatrix}, \qquad \mathrm{var}(\tilde\beta) - \mathrm{var}(\hat\beta) = \frac{\sigma^2}{4}\begin{pmatrix}1&0&-2\\0&0&0\\-2&0&4\end{pmatrix} = \sigma^2 DD'$$

as in the proof of Theorem 13.18. The rows of D are multiples of $e = (1,1,-1,-1)'$ of Exercise 13.1(b), and $X'e = 0$; so $DX = 0$.

(c) Only for $j = 1$: the diagonal of DD' is $\frac14(1,0,4)$, so $\mathrm{var}(\tilde\beta_1) = \mathrm{var}(\hat\beta_1) = \frac12\sigma^2$, while $\mathrm{var}(\tilde\beta_0) = \frac64\sigma^2 > \frac54\sigma^2$ and $\mathrm{var}(\tilde\beta_2) = \frac64\sigma^2 > \frac24\sigma^2$. Row 2 of D is 0, so $\tilde\beta_1 = \hat\beta_1 = \frac12(y_1 - y_2)$ for every y (here 1): the same estimator.

Exercise 13.9. Each of six responses y_i is the mean of m_i uncorrelated determinations with variance σ^2, where $m = (1,1,2,2,1,1)'$. So in the model $y = X\beta + \varepsilon$, with $\mathsf{E}(\varepsilon) = 0$, the errors are uncorrelated with $\mathrm{var}(\varepsilon_i) = \sigma^2/m_i$: $\mathrm{var}(\varepsilon) = \sigma^2 V$ with $V = \mathrm{diag}\Big(1,1,\frac12,\frac12,1,1\Big)$. Let $X = [\mathbf{1}_6\ x_1\ x_2]$ and

$$x_1 = \begin{pmatrix}-1\\-1\\1\\1\\-1\\-1\end{pmatrix}, \qquad x_2 = \begin{pmatrix}-1\\0\\-1\\1\\0\\1\end{pmatrix}, \qquad y = \begin{pmatrix}0\\3\\7\\7\\3\\6\end{pmatrix}$$

(a) Compute $X'V^{-1}X$, $X'V^{-1}y$ and the generalized least-squares estimate $\tilde\beta$ of Theorem 13.22.

(b) Compute $X'X$, $(X'X)^{-1}$, $X'y$ and the least-squares estimate $\hat\beta$.

(c) Find $\mathrm{var}(\tilde\beta)$ and $\mathrm{var}(\hat\beta)$, and check that $\mathrm{var}(\hat\beta) - \mathrm{var}(\tilde\beta)$ is PSD.

(d) Compute $Q(\tilde\beta) = (y - X\tilde\beta)'V^{-1}(y - X\tilde\beta)$ and the unbiased estimate $Q(\tilde\beta)/(6-3)$ of σ^2.

Solution. (a) $V^{-1} = \mathrm{diag}(1,1,2,2,1,1)$, so $V^{-1}X$ is X with rows 3 and 4 doubled (Theorem 1.4(c)), $X'V^{-1} = (V^{-1}X)'$, and

$$X'V^{-1} = \begin{pmatrix}1&1&2&2&1&1\\-1&-1&2&2&-1&-1\\-1&0&-2&2&0&1\end{pmatrix}, \qquad X'V^{-1}X = \begin{pmatrix}8&0&0\\0&8&0\\0&0&6\end{pmatrix}, \qquad X'V^{-1}y = \begin{pmatrix}40\\16\\6\end{pmatrix}$$

for instance $\mathbf{1}'V^{-1}y = 0+3+14+14+3+6 = 40$ and $x_1'V^{-1}y = 0-3+14+14-3-6 = 16$. By Theorem 13.22, $\tilde\beta = (X'V^{-1}X)^{-1}X'V^{-1}y = (\frac{40}{8}, \frac{16}{8}, \frac{6}{6})' = (5,2,1)'$.

(b) The (j,l) entry of $X'X$ is the product of columns j and l of X (Theorems 1.4(d) and 1.9):

$$X'X = \begin{pmatrix}6&-2&0\\-2&6&0\\0&0&4\end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{16}\begin{pmatrix}3&1&0\\1&3&0\\0&0&4\end{pmatrix}, \qquad X'y = \begin{pmatrix}26\\2\\6\end{pmatrix}$$

where the product of $X'X$ with the second matrix is I_3: its first row is $\frac{1}{16}(18-2,\ 6-6,\ 0) = (1,0,0)$, its second row $\frac{1}{16}(-6+6,\ -2+18,\ 0) = (0,1,0)$ and its third row $\frac{1}{16}(0,0,16)$ (Definition 2.12). So

$$\hat{\beta} = (X'X)^{-1}X'y = \frac{1}{16}\begin{pmatrix} 78+2 \\ 26+6 \\ 24 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ \frac{3}{2} \end{pmatrix}$$

(c) By Theorem 13.22(a), $\operatorname{var}(\tilde{\beta}) = \sigma^2(X'V^{-1}X)^{-1} = \sigma^2 \operatorname{diag}\left(\frac{1}{8}, \frac{1}{8}, \frac{1}{6}\right)$. By Theorem 13.22(c), $\operatorname{var}(\hat{\beta}) = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}$, with

$$X'VX = \begin{pmatrix} 5 & -3 & 0 \\ -3 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad (X'X)^{-1}X'VX = \frac{1}{4}\begin{pmatrix} 3 & -1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \operatorname{var}(\hat{\beta}) = \sigma^2 \begin{pmatrix} \frac{1}{8} & 0 & 0 \\ 0 & \frac{1}{8} & 0 \\ 0 & 0 & \frac{3}{16} \end{pmatrix}$$

So $\operatorname{var}(\hat{\beta}) - \operatorname{var}(\tilde{\beta}) = \sigma^2 \operatorname{diag}\left(0, 0, \frac{3}{16} - \frac{1}{6}\right) = \frac{\sigma^2}{48}\operatorname{diag}(0,0,1)$, which is PSD (Proposition 8.23(a)): the two estimators have the same variances for β_0 and β_1, and $\operatorname{var}(\hat{\beta}_2) = \frac{3}{16}\sigma^2 > \frac{1}{6}\sigma^2 = \operatorname{var}(\tilde{\beta}_2)$.

(d) $X\tilde{\beta} = (5-2-1,\ 5-2,\ 5+2-1,\ 5+2+1,\ 5-2,\ 5-2+1)' = (2,3,6,8,3,4)'$, so $y - X\tilde{\beta} = (-2,0,1,-1,0,2)'$ and, with the weights $1,1,2,2,1,1$ of V^{-1}, $Q(\tilde{\beta}) = 4+0+2+2+0+4 = 12$. Let $T = \operatorname{diag}(1,1,\sqrt{2},\sqrt{2},1,1)$, so that $T^2 = V^{-1}$ and $TVT = I_6$. The model $Ty = (TX)\beta + T\varepsilon$ has $\mathsf{E}(T\varepsilon) = 0$ and $\operatorname{var}(T\varepsilon) = \sigma^2 TVT = \sigma^2 I_6$ (Theorem 9.3(b)), and $\operatorname{rank}(TX) = \operatorname{rank} X = 3$ (T is invertible; Theorem 4.15(d)). As $\|Ty - TX\beta\|^2 = (y-X\beta)'V^{-1}(y-X\beta)$, its least-squares estimate is $\tilde{\beta}$ (Theorem 13.22(a)) and its error sum of squares is $Q(\tilde{\beta})$. By Theorem 13.23, $Q(\tilde{\beta})/(6-3) = 12/3 = 4$ is an unbiased estimate of σ^2.

Exercise 13.10. Let $X = [\mathbf{1}_6\ x_1\ x_2\ x_3] \in \mathbb{R}^{6\times 4}$ and $y \in \mathbb{R}^6$, with

$$x_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 3 \\ 3 \\ 0 \\ -1 \\ -1 \\ 2 \end{pmatrix}, \quad y = \begin{pmatrix} 9 \\ 5 \\ 4 \\ -1 \\ -1 \\ 2 \end{pmatrix}$$

In Theorem 13.32 take $X_1 = x_3$ and $X_2 = [\mathbf{1}_6\ x_1\ x_2]$; the design $[X_1\ X_2]$ is X with its columns reordered, and $\hat{\gamma}$ is the coefficient of x_3.

(a) Compute $X_2'X_2$, $(X_2'X_2)^{-1}$, $X_2'x_3$ and $X_2'y$, and then H_2x_3, H_2y, M_2X_1 and M_2y.

(b) Use Theorem 13.32 to find $\hat{\gamma}$ and the residual vector e of the least-squares fit of y on X, and check that $X'e = 0$.

(c) Check that $X_1'M_2X_1 = X_1'X_1 - X_1'X_2(X_2'X_2)^{-1}X_2'X_1$, the Schur complement of $X_2'X_2$ in $X'X$ (Theorem 5.9, with the blocks of $X'X$ in the order X_2, X_1).

Solution. (a) The (j,l) entry of $X_2'X_2$ is the product of columns j and l of X_2 (Theorems 1.4(d) and 1.9); these columns are orthogonal, so $X_2'X_2$ is diagonal, and its inverse is diagonal (Theorem 2.31):

$$X_2'X_2 = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad (X_2'X_2)^{-1} = \begin{pmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{6} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}, \qquad X_2'x_3 = \begin{pmatrix} 6 \\ 6 \\ 0 \end{pmatrix}, \qquad X_2'y = \begin{pmatrix} 18 \\ 18 \\ 4 \end{pmatrix}$$

for instance $\mathbf{1}'x_3 = 3+3+0-1-1+2 = 6$, $x_1'x_3 = (3+3+0) - (-1-1+2) = 6$ and $x_2'y = 9-5+0-1+1+0 = 4$. The vector $c = (X_2'X_2)^{-1}X_2'v$ holds the coefficients of $H_2v = X_2c$ on

the columns of X_2 (Theorem 1.4(b)): $c = (1,1,0)'$ for $v = x_3$ and $c = (3,3,1)'$ for $v = y$. So

$$H_2x_3 = \mathbf{1} + x_1 = \begin{pmatrix}2\\2\\2\\0\\0\\0\end{pmatrix}, \qquad H_2y = 3\cdot\mathbf{1} + 3x_1 + x_2 = \begin{pmatrix}7\\5\\6\\1\\-1\\0\end{pmatrix}$$

$$M_2X_1 = x_3 - H_2x_3 = \begin{pmatrix}1\\1\\-2\\-1\\-1\\2\end{pmatrix}, \qquad M_2y = y - H_2y = \begin{pmatrix}2\\0\\-2\\-2\\0\\2\end{pmatrix}$$

(b) First, X satisfies (A4): $\mathbf{1}_6$, x_1, x_2 are nonzero and pairwise orthogonal, hence independent (Proposition 6.11(b)); $x_3 \notin \mathcal{C}(X_2)$, since otherwise $H_2x_3 = x_3$ (Theorem 6.26(a)), while $M_2x_3 \neq 0$; so the four columns of X are independent (Proposition 3.14(b)). The design $[X_1\ X_2] = [x_3\ \mathbf{1}_6\ x_1\ x_2]$ has the same columns as X, so its least-squares fit has the same fitted values $\hat{y}$ and residual vector e (Theorem 13.3(b)); as these four columns are independent, they form a basis of $\mathcal{C}(X)$, and the coordinates of $\hat{y}$ in it are unique (Theorem 3.18), so the coefficient of x_3 is the same in both fits. M_2 is symmetric and idempotent (Theorem 6.26(b)), so $X_1'M_2X_1 = (M_2X_1)'(M_2X_1) = 1{+}1{+}4{+}1{+}1{+}4 = 12$ and $X_1'M_2y = (M_2X_1)'(M_2y) = 2+0+4+2+0+4 = 12$. By Theorem 13.32(a), $\hat{\gamma} = 12/12 = 1$. By Theorem 13.32(b), e is the residual vector of the fit of M_2y on M_2X_1:

$$e = M_2y - \hat{\gamma}M_2X_1 = \begin{pmatrix}2\\0\\-2\\-2\\0\\2\end{pmatrix} - \begin{pmatrix}1\\1\\-2\\-1\\-1\\2\end{pmatrix} = \begin{pmatrix}1\\-1\\0\\-1\\1\\0\end{pmatrix}$$

Check: $\mathbf{1}'e = 0$, $x_1'e = (1-1+0) - (-1+1+0) = 0$, $x_2'e = 1+1+0-1-1+0 = 0$ and $x_3'e = 3-3+0+1-1+0 = 0$; so $X'e = 0$, as Proposition 13.5(a) says.

(c) $X_1'X_1 = 9{+}9{+}0{+}1{+}1{+}4 = 24$ and $X_1'X_2 = (X_2'x_3)' = (6,6,0)$, so $X_1'X_2(X_2'X_2)^{-1}X_2'X_1 = 6\cdot\frac{1}{6}\cdot 6{+}6\cdot\frac{1}{6}\cdot 6 = 12$, as the third entry of $X_2'X_1$ is 0 (Theorem 1.4(d)), and $X_1'X_1 - X_1'X_2(X_2'X_2)^{-1}X_2'X_1 = 24 - 12 = 12 = X_1'M_2X_1$.

Exercise 13.11. Five observations of two variables and a response are

x_{i1}	1	2	3	4	5
x_{i2}	2	3	3	3	4
y_i	1	3	2	7	7

Fit $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ under (A1)–(A4), with $X = [\mathbf{1}_5\ X_0]$.

(a) Compute C_5X_0, $X_0'C_5X_0$ and its inverse, C_5y and $X_0'C_5y$, and find $\hat{\beta}_1$, $\hat{\beta}_2$ and $\hat{\beta}_0$ as in Example 13.33.

(b) Find $\text{var}(\hat{\beta}_1)$ and $\text{var}(\hat{\beta}_2)$, the variance inflation factors VIF_1, VIF_2, and R_1^2, R_2^2 (Corollary 13.34).

(c) Compute e, SSE and s^2, and the estimates of $\text{var}(\hat{\beta}_1)$ and $\text{var}(\hat{\beta}_2)$ with s^2 in place of σ^2.

Solution. (a) $\bar{x} = (3,3)'$ and $\bar{y} = 4$, so (Example 1.16)

$$C_5X_0 = \begin{pmatrix} -2 & -1 \\ -1 & 0 \\ 0 & 0 \\ 1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad X_0'C_5X_0 = (C_5X_0)'(C_5X_0) = \begin{pmatrix} 10 & 4 \\ 4 & 2 \end{pmatrix}, \qquad C_5y = \begin{pmatrix} -3 \\ -1 \\ -2 \\ 3 \\ 3 \end{pmatrix}$$

$\det X_0'C_5X_0 = 20 - 16 = 4$, so $(X_0'C_5X_0)^{-1} = \frac{1}{4}\left(\begin{smallmatrix} 2 & -4 \\ -4 & 10 \end{smallmatrix}\right)$ (Theorem 5.24). Then

$$X_0'C_5y = (C_5X_0)'(C_5y) = \begin{pmatrix} 6+1+0+3+6 \\ 3+0+0+0+3 \end{pmatrix} = \begin{pmatrix} 16 \\ 6 \end{pmatrix}, \qquad \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 32-24 \\ -64+60 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

and $\hat{\beta}_0 = \bar{y} - \bar{x}'(\hat{\beta}_1, \hat{\beta}_2)' = 4 - (6-3) = 1$.

(b) By Corollary 13.34(a), with $X_1 = \mathbf{1}_5$ and $X_2 = X_0$, $\operatorname{var}\big((\hat{\beta}_1, \hat{\beta}_2)'\big) = \sigma^2(X_0'C_5X_0)^{-1}$: $\operatorname{var}(\hat{\beta}_1) = \frac{1}{2}\sigma^2$ and $\operatorname{var}(\hat{\beta}_2) = \frac{10}{4}\sigma^2 = \frac{5}{2}\sigma^2$. The diagonal of $X_0'C_5X_0$ holds $S_{11} = 10$ and $S_{22} = 2$; by Corollary 13.34(b), $\mathrm{VIF}_j = S_{jj}\operatorname{var}(\hat{\beta}_j)/\sigma^2$: $\mathrm{VIF}_1 = 10 \cdot \frac{1}{2} = 5$, $\mathrm{VIF}_2 = 2 \cdot \frac{5}{2} = 5$, and $R_1^2 = R_2^2 = 1 - \frac{1}{5} = \frac{4}{5}$. Without the correlation of the two variables ($S_{12} = 0$), the variances would be σ^2/S_{jj}: $\frac{1}{10}\sigma^2$ and $\frac{1}{2}\sigma^2$.

(c) $\hat{y}_i = 1 + 2x_{i1} - x_{i2}$ gives $\hat{y} = (1,2,4,6,7)'$ and $e = y - \hat{y} = (0,1,-2,1,0)'$. Check: $X'e = (0+1-2+1+0,\ 0+2-6+4+0,\ 0+3-6+3+0)' = 0$. So $\mathrm{SSE} = 0+1+4+1+0 = 6$ and $s^2 = 6/(5-3) = 3$ (Theorem 13.23); the estimates of the two variances are $\frac{1}{2}s^2 = \frac{3}{2}$ and $\frac{5}{2}s^2 = \frac{15}{2}$.

Exercise 13.12. Let $x = (x_1, x_2, x_3, x_4)'$ be the random vector of Exercise 9.2, with $\mathsf{E}(x) = \mu = (1,0,2,-1)'$ and

$$\operatorname{var}(x) = \Sigma = \begin{pmatrix} 9 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \\ 3 & 2 & 9 & 5 \\ 3 & 1 & 5 & 4 \end{pmatrix}$$

(a) Find the best linear predictor of x_1 from $(x_2, x_3, x_4)'$ and its mean squared prediction error.

(b) Check directly that the prediction error u of (a) has mean 0, that $\operatorname{cov}(u, x_2) = \operatorname{cov}(u, x_3) = \operatorname{cov}(u, x_4) = 0$, and that $\operatorname{var}(u)$ is the mean squared prediction error.

Solution. (a) In Theorem 13.38 take $y = x_1$, with $\mu_y = 1$ and $\sigma_{yy} = 9$, and $(x_2, x_3, x_4)'$ as the predictors:

$$\Sigma_{xx} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 9 & 5 \\ 1 & 5 & 4 \end{pmatrix}, \qquad \sigma_{xy} = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}, \qquad \Sigma_{xx}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}$$

By Exercise 9.2(a), $\Sigma = LL'$ with $\det L = 6 \neq 0$, so Σ is PD (Theorem 8.26(c) with $P = L'$). Its principal submatrix Σ_{xx} is PD (Theorem 8.30), hence invertible (Proposition 8.23(c)), and $b^* = \Sigma_{xx}^{-1}\sigma_{xy} = (-1,0,1)'$. Then $a^* = \mu_1 - b^{*\prime}(\mu_2, \mu_3, \mu_4)' = 1 - (0+0-1) = 2$, the best linear predictor is $2 - x_2 + x_4$, and its mean squared prediction error is $\sigma_{yy} - \sigma_{yx}\Sigma_{xx}^{-1}\sigma_{xy} = 9 - (0,3,3)(-1,0,1)' = 6$.

(b) $u = x_1 - 2 + x_2 - x_4 = c'x - 2$ with $c = (1,1,0,-1)'$. By Theorem 9.3(b), $\mathsf{E}(u) = c'\mu - 2 = 1+0+0+1-2 = 0$. By Theorem 1.4(b), $\Sigma c = \mathrm{col}_1(\Sigma) + \mathrm{col}_2(\Sigma) - \mathrm{col}_4(\Sigma) = (9+0-3,\ 0+1-1,\ 3+2-5,\ 3+1-4)' = (6,0,0,0)'$. By Theorem 9.4(a), with $A = c'$ and B the row j of I_4, $\operatorname{cov}(u, x_j) = \operatorname{cov}(c'x, x_j)$ is entry j of Σc, which is 0 for $j = 2,3,4$; and $\operatorname{var}(u) = c'\Sigma c = 6$ (Theorem 9.3(b)), the mean squared prediction error of (a).

Problems

Problem 13.1 ([12, Exercise D2])**.** Wages y_i of n workers are regressed on years of schooling x_i, an intercept, and the indicators $d_1, d_2, d_3, d_4 \in \mathbb{R}^n$ of all four regions of a country (each worker lives in exactly one region): $X = [\mathbf{1}_n\ x\ d_1\ d_2\ d_3\ d_4] \in \mathbb{R}^{n\times 6}$. (a) Show that X does not have full column rank, so that $\hat{\beta} = (X'X)^{-1}X'y$ is not defined. (b) Let $X_2 = [\mathbf{1}_n\ x\ d_2\ d_3\ d_4]$ (indicator d_1 dropped). Show that $\mathcal{C}(X_2) = \mathcal{C}(X)$, so that $\operatorname{rank} X_2 = \operatorname{rank} X$.

Solution. (a) Each worker lives in exactly one region, so $d_1 + d_2 + d_3 + d_4 = \mathbf{1}_n$, that is, $X(-1, 0, 1, 1, 1, 1)' = 0$. So X does not have full column rank (Theorem 4.13(b)), and $X'X$ is singular (Theorems 4.18(b) and 4.13(d)): the dummy variable trap.

(b) Every column of X_2 is a column of X, so $\mathcal{C}(X_2) \subseteq \mathcal{C}(X)$. Conversely $d_1 = \mathbf{1}_n - d_2 - d_3 - d_4 \in \mathcal{C}(X_2)$, so every column of X lies in the subspace $\mathcal{C}(X_2)$, and therefore $\mathcal{C}(X) \subseteq \mathcal{C}(X_2)$ (Theorem 3.7). Hence $\mathcal{C}(X_2) = \mathcal{C}(X)$, and $\operatorname{rank} X_2 = \operatorname{rank} X$; by Proposition 13.9 (with $X_2 = XT$, T the columns $1, 2, 4, 5, 6$ of I_6), the two designs give the same fitted values.

Problem 13.2 ([12, Remark 2.11, Theorem 3.7, Lemma 7.2, Remark 6.2])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$ have full column rank, with hat matrix H and residual maker M.

(a) In the model $y = X\beta + \varepsilon$, show that $\text{SSE} = \varepsilon' M \varepsilon$.

(b) Let $\mathbf{1}_n \in \mathcal{C}(X)$. Show that $H\mathbf{1}_n = \mathbf{1}_n$, $H\bar{J}_n = \bar{J}_n H = \bar{J}_n$ and $MC_n = C_n M = M$.

(c) Under (b), show that $H - \bar{J}_n$ is symmetric and idempotent of rank k, that $(H - \bar{J}_n)M = 0$, and that $\text{SSR} = y'(H - \bar{J}_n)y$.

Solution. (a) $MX = 0$ (Theorem 13.4(a)), so $e = My = M(X\beta + \varepsilon) = MX\beta + M\varepsilon = M\varepsilon$ (Theorem 13.4(d)). As M is symmetric and idempotent, $\text{SSE} = e'e = \varepsilon' M' M \varepsilon = \varepsilon' M \varepsilon$.

(b)

- $\mathbf{1}_n = Xc$ for some c, so $H\mathbf{1}_n = HXc = Xc = \mathbf{1}_n$ (Theorem 6.26(c)).
- $H\bar{J}_n = \frac{1}{n}H\mathbf{1}\mathbf{1}' = \frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$, and, as H and $\bar{J}_n$ are symmetric, $\bar{J}_n H = (H\bar{J}_n)' = \bar{J}_n$.
- $M\bar{J}_n = \bar{J}_n - H\bar{J}_n = 0$, so $MC_n = M(I_n - \bar{J}_n) = M$, and $C_n M = (MC_n)' = M$.

(c)

- $H - \bar{J}_n$ is symmetric, and by (b) and $\bar{J}_n^2 = \bar{J}_n$ (Proposition 1.15), $(H - \bar{J}_n)^2 = H^2 - H\bar{J}_n - \bar{J}_n H + \bar{J}_n^2 = H - \bar{J}_n - \bar{J}_n + \bar{J}_n = H - \bar{J}_n$.
- Its rank is its trace (Theorem 6.29(c)): $\operatorname{tr} H - \operatorname{tr} \bar{J}_n = (k+1) - 1 = k$ (Corollary 6.30; the diagonal entries of $\bar{J}_n$ are $\frac{1}{n}$).
- $(H - \bar{J}_n)M = HM - \bar{J}_n M = 0 - (M\bar{J}_n)' = 0$, by Theorem 6.26(c) and (b).
- $\hat{y} - \bar{y}\mathbf{1} = Hy - \bar{J}_n y = (H - \bar{J}_n)y$ (Example 6.27), so $\text{SSR} = \|(H - \bar{J}_n)y\|^2 = y'(H - \bar{J}_n)'(H - \bar{J}_n)y = y'(H - \bar{J}_n)y$.

Problem 13.3 ([12, Theorem 2.14, Exercises B1, D3])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$ have full column rank, and let $h_{11}, \dots, h_{nn}$ be the diagonal entries of its hat matrix H, the leverages of the n observations.

(a) Show that $0 \leqslant h_{ii} \leqslant 1$.

(b) If $\mathbf{1}_n \in \mathcal{C}(X)$, show that $h_{ii} \geqslant \frac{1}{n}$.

(c) In the model $y = X\beta + \varepsilon$ under (A2) and (A3), show that $\operatorname{var}(e_i) = \sigma^2(1 - h_{ii})$.

Solution. (a) H is symmetric and idempotent (Theorem 6.26(b)), so $H = H^2 = H'H$. Comparing the (i,i) entries, with $h_{ji} = h_{ij}$, $h_{ii} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2$. So $h_{ii} \geqslant 0$, as a sum of squares, and $h_{ii} \geqslant h_{ii}^2$, that is, $h_{ii}(1 - h_{ii}) \geqslant 0$; together, $h_{ii} \leqslant 1$.

(b) By Problem 13.2(c), $H - \bar{J}_n$ is symmetric and idempotent, and its (i,j) entry is $h_{ij} - \frac{1}{n}$. As in (a), $h_{ii} - \frac{1}{n} = \sum_{j=1}^n \left(h_{ij} - \frac{1}{n}\right)^2 \geqslant 0$.

(c) By Theorem 13.17(c), $\operatorname{var}(e) = \sigma^2 M$. Its (i,i) entry is $\operatorname{var}(e_i) = \sigma^2(1 - h_{ii})$, as $M = I_n - H$.

Problem 13.4 ([12, Exercise D4])**.** Let $X_1 \in \mathbb{R}^{n\times p}$ have full column rank, let $x \in \mathbb{R}^n$ be nonzero with $X_1'x = 0$, and let $X_+ = [X_1\ x]$. For $y \in \mathbb{R}^n$, let $\tilde{\gamma} = (X_1'X_1)^{-1}X_1'y$ and SSE_1 belong to the least-squares fit with design X_1, and $(\hat{\gamma}', \hat{\theta})'$ ($\hat{\gamma} \in \mathbb{R}^p$) and SSE_+ to the fit with design X_+.

(a) Show that $\hat{\theta} = x'y/x'x$.

(b) Show that $\hat{\gamma} = \tilde{\gamma}$: the coefficients of X_1 do not change when x is added.

(c) Show that $\mathrm{SSE}_1 - \mathrm{SSE}_+ = (x'y)^2/x'x$.

(d) Relate (b) to Theorem 13.36.

Solution. X_+ has full column rank: if $X_1a + cx = 0$, then multiplying by x' gives $c\,x'x = 0$, as $x'X_1 = 0$, so $c = 0$; then $X_1a = 0$ gives $a = 0$ (Theorem 4.13(b)). Let H_1, M_1 be the hat matrix and the residual maker of X_1 (Theorem 6.26). As $X_1'x = 0$, $H_1x = X_1(X_1'X_1)^{-1}X_1'x = 0$, so $M_1x = x$, $x'M_1x = x'x > 0$ and $x'M_1y = (M_1x)'y = x'y$.

(a) By Theorem 13.32 for the design X_+, with the roles of the blocks exchanged ($X_2 = x$), $\hat{\theta} = (x'M_1x)^{-1}x'M_1y = x'y/x'x$.

(b) By the block product (Theorem 5.3), the first block of the normal equations of X_+ reads $X_1'X_1\hat{\gamma} + X_1'x\hat{\theta} = X_1'y$. As $X_1'x = 0$, $\hat{\gamma} = (X_1'X_1)^{-1}X_1'y = \tilde{\gamma}$.

(c)

- By (b), the fitted values of X_+ are $\hat{y}_+ = X_1\tilde{\gamma} + x\hat{\theta} = \hat{y}_1 + x\hat{\theta}$, with $\hat{y}_1 = X_1\tilde{\gamma}$; so the residual vectors satisfy $e_+ = e_1 - x\hat{\theta}$.
- $x'e_1 = x'(y - X_1\tilde{\gamma}) = x'y$, as $x'X_1 = 0$. Hence

$$\mathrm{SSE}_+ = e_1'e_1 - 2\hat{\theta}\,x'e_1 + \hat{\theta}^2x'x = \mathrm{SSE}_1 - 2\frac{(x'y)^2}{x'x} + \frac{(x'y)^2}{x'x} = \mathrm{SSE}_1 - \frac{(x'y)^2}{x'x}$$

(d) Read the fit with X_+ as the true model $y = X_1\gamma + x\theta + \varepsilon$, $\mathsf{E}(\varepsilon) = 0$, and the fit with X_1 as the one that omits x. By Theorem 13.36 with $X_2 = x$, the omitted variable bias $(X_1'X_1)^{-1}X_1'x\,\theta$ is 0 for every θ, as $X_1'x = 0$: omitting x does not bias $\tilde{\gamma}$. Part (b) is the exact statement about the estimates behind this statement about expectations.

Problem 13.5 ([9, Exercise 5.8], [19, §12.6, Exercise 6])**.** Let X, S, λ_m, $v_m = (v_{1m}, \dots, v_{pm})'$, z_m and r be as in Theorem 9.31, let $M \leqslant r$, let V_M and Z_M be as in Theorem 9.39, and let c_j be column j of C_nX. Suppose that the first M score vectors are known, and take $A = Z_M$ in the problem of Theorem 9.39(a): minimize $\|C_nX - Z_MB'\|_F^2$ over $B \in \mathbb{R}^{p\times M}$. Show that this problem splits into p separate least-squares problems, one for each column c_j as the response, with the design Z_M; and that their solutions are the rows of V_M, so that $B = V_M$. So the first M loading vectors can be recovered from the first M score vectors.

Solution.

- Let b_j' be row j of B. Column j of Z_MB' is Z_Mb_j (Theorem 1.4(a)), so column j of $C_nX - Z_MB'$ is $c_j - Z_Mb_j$, and $\|C_nX - Z_MB'\|_F^2 = \sum_{j=1}^p \|c_j - Z_Mb_j\|^2$ (Definition 9.26). The jth term involves only row j of B; so B minimizes the sum if and only if each b_j minimizes $\|c_j - Z_Mb_j\|^2$, the least-squares criterion for the design Z_M and the response c_j.
- $Z_M'Z_M = (n-1)\operatorname{diag}(\lambda_1, \dots, \lambda_M)$ with $\lambda_1, \dots, \lambda_M > 0$ (Theorem 9.31(a),(d)) is invertible, so $\operatorname{rank} Z_M = M$ (Theorems 4.18(b) and 4.13(d)), and the jth problem has exactly one solution, $\hat{b}_j = (Z_M'Z_M)^{-1}Z_M'c_j$ (Theorem 13.3(c)).
- $SV_M = V_M \operatorname{diag}(\lambda_1, \dots, \lambda_M)$ and $S' = S$, so $Z_M'C_nX = V_M'(C_nX)'(C_nX) = (n-1)V_M'S = (n-1)\operatorname{diag}(\lambda_1, \dots, \lambda_M)V_M'$. Column j of this reads $Z_M'c_j = (n-1)\operatorname{diag}(\lambda_1, \dots, \lambda_M)(v_{j1}, \dots, v_{jM})'$.
- Hence $\hat{b}_j = \Big((n-1)\operatorname{diag}(\lambda_1, \dots, \lambda_M)\Big)^{-1}(n-1)\operatorname{diag}(\lambda_1, \dots, \lambda_M)(v_{j1}, \dots, v_{jM})' = (v_{j1}, \dots, v_{jM})'$, row j of V_M transposed: the minimizing B is V_M.

Problem 13.6 ([12, Proposition 7.14], [2, (11.40)–(11.41)])**.** Assume (A1)–(A4) with $X = [\mathbf{1}_n\ X_0] \in \mathbb{R}^{n\times(k+1)}$, $k \geqslant 1$ and $n > k+1$, and write $\beta = (\beta_0, \beta_*')'$ with $\beta_* = (\beta_1, \dots, \beta_k)'$. Show that

$$\mathsf{E}(\mathrm{SSR}) = k\sigma^2 + \|C_nX_0\beta_*\|^2$$

and that $\mathsf{E}(\mathrm{SSR}/k) \geqslant \sigma^2 = \mathsf{E}(s^2)$, with equality if and only if $\beta_* = 0$.

Solution.

- By Problem 13.2(c), $\mathrm{SSR} = y'Py$ with $P = H - \bar{J}_n$ symmetric and idempotent of rank k.
- By (A1) and (A3), $\mathsf{E}(y) = X\beta$ and $\operatorname{var}(y) = \sigma^2 I_n$ (the paragraph after Definition 13.1). So Example 9.12 with $\mu = X\beta$ gives $\mathsf{E}(\mathrm{SSR}) = k\sigma^2 + \beta'X'PX\beta = k\sigma^2 + \|PX\beta\|^2$, as $P = P'P$.
- $HX = X$ (Theorem 13.4(a)), so $(H - \bar{J}_n)X\beta = C_nX\beta = C_n(\beta_0\mathbf{1}_n + X_0\beta_*) = C_nX_0\beta_*$, as $C_n\mathbf{1}_n = 0$ (Proposition 1.15).
- Hence $\mathsf{E}(\mathrm{SSR}/k) = \sigma^2 + \|C_nX_0\beta_*\|^2/k \geqslant \sigma^2 = \mathsf{E}(s^2)$ (Theorem 13.23). Equality holds if and only if $C_nX_0\beta_* = X_0\beta_* - (\bar{x}'\beta_*)\mathbf{1}_n = X(-\bar{x}'\beta_*,\ \beta_*')'$ vanishes, where $\bar{x} = \frac{1}{n}X_0'\mathbf{1}_n$; as $\operatorname{rank} X = k+1$, this means $\beta_* = 0$ (Theorem 4.13(b)).

So SSR/k estimates σ^2 plus a nonnegative term that vanishes exactly when $\beta_* = 0$.

Chapter 14. Estimability, inference, and the analysis of variance

Estimable functions

Definition 14.1 ([2, Definition 12.1], [3, §7.2])**.** Consider the linear model $y = X\beta + \varepsilon$ under (A1), with $X \in \mathbb{R}^{n\times(k+1)}$ of any rank, and let $a \in \mathbb{R}^{k+1}$. A **linear estimator** is a random variable $c'y$ with a constant $c \in \mathbb{R}^n$; it is a **linear unbiased estimator** of $a'\beta$ if $\mathsf{E}(c'y) = a'\beta$ for every $\beta \in \mathbb{R}^{k+1}$. The function $a'\beta$ is **estimable** if it has a linear unbiased estimator. A linear unbiased estimator of $a'\beta$ whose variance is the smallest among all linear unbiased estimators of $a'\beta$ is a **best linear unbiased estimator** (**BLUE**) of $a'\beta$.

Theorem 14.2 ([2, Theorems 12.1, 12.2, Corollary 12.1], [3, §7.2])**.** Assume (A1), and let $a \in \mathbb{R}^{k+1}$.

(a) A linear estimator $c'y$ is unbiased for $a'\beta$ if and only if $X'c = a$. More generally, for constant $A \in \mathbb{R}^{m\times n}$ and $C \in \mathbb{R}^{m\times(k+1)}$, $\mathsf{E}(Ay) = C\beta$ for every $\beta \in \mathbb{R}^{k+1}$ if and only if $AX = C$. Hence

$$a'\beta \text{ is estimable} \iff a \in \mathcal{C}(X') \iff a \perp \mathcal{N}(X) \iff \operatorname{rank}\begin{pmatrix} X \\ a' \end{pmatrix} = \operatorname{rank} X$$

In particular, if $\operatorname{rank} X = k+1$, every $a'\beta$ is estimable.

(b) If $a = X'c$, then $a'\hat\beta = c'\hat y = c'Hy$ for every least-squares estimate $\hat\beta$: it is the same linear estimator for all of them (for instance for $\hat\beta = (X'X)^- X'y$ with any g-inverse), and $\mathsf{E}(a'\hat\beta) = a'\beta$.

(c) If $a'\beta$ is not estimable, then $a'\hat\beta$ takes every real value as $\hat\beta$ runs through the least-squares estimates.

Proof. (a)

- By Theorem 9.3(b) and (A1), $\mathsf{E}(c'y) = c'X\beta$. If $c'X\beta = a'\beta$ for every β, then taking for β the columns of I_{k+1} gives $c'X = a'$, that is, $X'c = a$. Conversely, if $X'c = a$, then $\mathsf{E}(c'y) = c'X\beta = a'\beta$ for every β. In the same way, $\mathsf{E}(Ay) = AX\beta$, and taking for β the columns of I_{k+1} shows that $AX\beta = C\beta$ for every β if and only if $AX = C$ (Theorem 1.4(a),(b)). So $a'\beta$ is estimable if and only if $a = X'c$ for some c, that is, $a \in \mathcal{C}(X')$.
- $\mathcal{N}(X) = \mathcal{C}(X')^\perp$ (Corollary 6.7), so every vector of $\mathcal{C}(X')$ is orthogonal to $\mathcal{N}(X)$: $\mathcal{C}(X') \subseteq \mathcal{N}(X)^\perp$. Both have dimension $\operatorname{rank} X$: $\dim \mathcal{C}(X') = \operatorname{rank} X' = \operatorname{rank} X$ (Theorem 4.13(a)), and $\dim \mathcal{N}(X)^\perp = (k+1) - \dim \mathcal{N}(X) = \operatorname{rank} X$ (Theorems 6.6(c) and 4.12). So $\mathcal{C}(X') = \mathcal{N}(X)^\perp$ (Theorem 4.5(a)).
- $a \in \mathcal{C}(X')$ if and only if $\operatorname{rank}[X'\ a] = \operatorname{rank} X'$ (Corollary 4.14(a)). Transposing (Theorems 5.3(b) and 4.13(a)) gives the rank condition.
- If $\operatorname{rank} X = k+1$, then $\operatorname{rank} X' = k+1$, the number of rows of X', so $\mathcal{C}(X') = \mathbb{R}^{k+1}$ (Theorem 4.13(a),(c)).

(b) Every least-squares estimate gives $X\hat\beta = \hat y = Hy$ (Theorem 13.4(d)), so $a'\hat\beta = c'X\hat\beta = c'Hy$. By Theorem 9.3(b), (A1) and $HX = X$ (Theorem 13.4(a)), $\mathsf{E}(c'Hy) = c'HX\beta = c'X\beta = a'\beta$.

(c) By (a), a is not orthogonal to $\mathcal{N}(X)$: $a'z \neq 0$ for some $z \in \mathcal{N}(X)$. If $\hat\beta$ is a least-squares estimate, so is $\hat\beta + tz$ for every $t \in \mathbb{R}$, since $Xz = 0$ gives $X'X(\hat\beta + tz) = X'X\hat\beta = X'y$ (Theorem 13.3(a)); and $a'(\hat\beta + tz) = a'\hat\beta + t\,a'z$ takes every real value. □

Corollary 14.3 ([5, Corollary 11.10.2], [2, Theorems 12.2, 12.4])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$ and $y \in \mathbb{R}^n$, let G be a g-inverse of $X'X$, and let $\lambda \in \mathbb{R}^{k+1}$. The following are equivalent:

(i) $\lambda'\hat\beta$ has the same value for every least-squares estimate $\hat\beta$;

(ii) $\lambda \in \mathcal{C}(X')$, that is, $\lambda' = c'X$ for some $c \in \mathbb{R}^n$;

(iii) $\lambda' G X' X = \lambda'$.

If they hold, then $\lambda'\hat{\beta} = c'\hat{y} = c'Hy$ for every least-squares estimate $\hat{\beta}$. By Theorem 14.2(a), (ii) says that $\lambda'\beta$ is estimable.

Proof. • The least-squares estimates are the solutions of the consistent system $X'X\beta = X'y$ (Theorems 13.3(a) and 13.2(a)); so $X'y \in \mathcal{C}(X'X)$ (Corollary 4.14(a)).

- Theorem 12.9, for the matrix $X'X$ and the right side $X'y$, shows that (i) holds if and only if $\lambda \in \mathcal{C}((X'X)') = \mathcal{C}(X'X)$, and if and only if $\lambda' G X' X = \lambda'$. Here $\mathcal{C}(X'X) = \mathcal{C}(X')$ (Theorem 4.18(c)).
- If $\lambda' = c'X$, then $\lambda'\hat{\beta} = c'X\hat{\beta} = c'\hat{y} = c'Hy$ (Theorem 13.4(d)). □

Example 14.4 (cf. [2, Example 12.1], [3, Ch. 7, Exercise 11])**.** In the one-way model of Example 13.11 ($g = 3$; $n_1, n_2, n_3 = 3, 2, 1$), $\mathcal{N}(X) = \text{span}\{(-1, 1, 1, 1)'\}$. By Theorem 14.2(a), $a'\beta = a_0\mu + a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$ is estimable if and only if $-a_0 + a_1 + a_2 + a_3 = 0$.

- $\mu + \alpha_1$ ($a = (1, 1, 0, 0)'$) and $\alpha_1 - \alpha_2$ ($a = (0, 1, -1, 0)'$) are estimable. On the least-squares estimates $(-t, 6 + t, 4 + t, 2 + t)'$ they take the values 6 and 2 for every t.
- μ ($a = (1, 0, 0, 0)'$) and α_1 ($a = (0, 1, 0, 0)'$) are not estimable: $-1 \neq 0$ and $1 \neq 0$. Their values $-t$ and $6 + t$ change with t, as Theorem 14.2(c) says.
- For every y, with the group means $\bar{y}_1, \bar{y}_2, \bar{y}_3$, the vector $\hat{\beta} = (0, \bar{y}_1, \bar{y}_2, \bar{y}_3)'$ is a least-squares estimate (Theorem 13.3(a)): $X'X\hat{\beta} = (3\bar{y}_1 + 2\bar{y}_2 + \bar{y}_3,\ 3\bar{y}_1,\ 2\bar{y}_2,\ \bar{y}_3)'$ is the vector of the total and the group totals, which is $X'y$. So the estimator of $\alpha_1 - \alpha_2$ is

$$\bar{y}_1 - \bar{y}_2 = \tfrac{1}{3}(y_1 + y_2 + y_3) - \tfrac{1}{2}(y_4 + y_5) = c'y, \qquad c = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, -\tfrac{1}{2}, -\tfrac{1}{2}, 0\right)'$$

and indeed $X'c = (1 - 1,\ 1,\ -1,\ 0)' = (0, 1, -1, 0)'$.

Proposition 14.5 ([2, §12.4])**.** Let G be a g-inverse of $X'X$, and let $\beta^0 = GX'y$, a least-squares estimate (Theorems 13.2(c) and 13.3(a)).

(a) Under (A1), $\mathsf{E}(\beta^0) = GX'X\beta$. If $\operatorname{rank} X < k + 1$, then $GX'X\beta \neq \beta$ for some $\beta \in \mathbb{R}^{k+1}$: β^0 is not an unbiased estimator of β.

(b) Under (A3), $\text{var}(\beta^0) = \sigma^2 GX'XG'$.

Proof. (a)

- By Theorem 9.3(b) with $A = GX'$ and (A1), $\mathsf{E}(\beta^0) = GX'\,\mathsf{E}(y) = GX'X\beta$.
- If $GX'X\beta = \beta$ for every β, then taking for β the columns of I_{k+1} gives $GX'X = I_{k+1}$, so $k + 1 = \text{rank}(GX'X) \leqslant \text{rank}(X'X) = \operatorname{rank} X$ (Theorems 4.15(a) and 4.18(b)). Hence $\operatorname{rank} X < k + 1$ gives $GX'X\beta \neq \beta$ for some β.

(b) By Theorem 9.3(b) and (A3), $\text{var}(\beta^0) = GX'(\sigma^2 I_n)(GX')' = \sigma^2 GX'XG'$ (Theorem 1.9). □

For X of full column rank, $G = (X'X)^{-1}$ is the only g-inverse of $X'X$ (Theorem 13.4(c)), $GX'X = I_{k+1}$ and $\beta^0 = \hat{\beta}$, and Proposition 14.5 is Theorem 13.17(a),(b).

Example 14.6 (cf. [2, Example 12.1])**.** In the one-way model of Example 13.11 ($g = 3$; $n_1, n_2, n_3 = 3, 2, 1$), $G = \text{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$ is a g-inverse of $X'X$, with

$$GX'X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

- $X'y$ holds the total and the group totals $3\bar{y}_1$, $2\bar{y}_2$, $\bar{y}_3$, so $\beta^0 = GX'y = (0, \bar{y}_1, \bar{y}_2, \bar{y}_3)'$, the least-squares estimate of Example 14.4. By Proposition 14.5(a),

$$\mathsf{E}(\beta^0) = GX'X\beta = (0,\ \mu+\alpha_1,\ \mu+\alpha_2,\ \mu+\alpha_3)'$$

  So $\bar{y}_l$ estimates the mean $\mu+\alpha_l$ of group l, not α_l; for the estimable $\alpha_1-\alpha_2$, $\mathsf{E}(\bar{y}_1-\bar{y}_2) = \alpha_1-\alpha_2$, as Theorem 14.2(b) says.

- Multiplying the columns of $GX'X$ by $0, \frac{1}{3}, \frac{1}{2}, 1$ (Theorem 1.4(a),(b)) gives $GX'XG' = GX'XG = \operatorname{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$. So $\operatorname{var}(\beta^0) = \sigma^2 \operatorname{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$ by Proposition 14.5(b): the group means are uncorrelated, with $\operatorname{var}(\bar{y}_l) = \sigma^2/n_l$. For the estimable $\alpha_1-\alpha_2$, with $a = (0, 1, -1, 0)'$, Theorem 9.3(b) gives $\operatorname{var}(\bar{y}_1-\bar{y}_2) = \operatorname{var}(a'\beta^0) = a' \operatorname{var}(\beta^0) a = \sigma^2(\frac{1}{3}+\frac{1}{2}) = \frac{5}{6}\sigma^2$, which is $\sigma^2 a'Ga$.

Theorem 14.7 ([2, Theorem 12.3, Exercise 12.4], [3, Result 7.1])**.** Assume (A1)–(A3), with X of any rank, and let $a'\beta$ be estimable. Then $a'\hat{\beta}$, the same for every least-squares estimate $\hat{\beta}$, is the BLUE of $a'\beta$: it is a linear unbiased estimator of $a'\beta$, and every linear unbiased estimator $c'y$ of $a'\beta$ has $\operatorname{var}(c'y) \geqslant \operatorname{var}(a'\hat{\beta})$, with equality only if $c'y = a'\hat{\beta}$ for every y. Moreover $\operatorname{var}(a'\hat{\beta}) = \sigma^2 a'(X'X)^- a$ for every g-inverse $(X'X)^-$ of $X'X$.

Proof.
- Let $c'y$ be a linear unbiased estimator of $a'\beta$. Then $X'c = a$, and $a'\hat{\beta} = c'Hy$ is a linear unbiased estimator of $a'\beta$ (Theorem 14.2(a),(b)).
- By Theorem 9.3(b) and (A3), $\operatorname{var}(c'y) = \sigma^2 c'c$ and $\operatorname{var}(c'Hy) = \sigma^2 c'HH'c = \sigma^2 c'Hc$, as H is symmetric and idempotent (Theorem 13.4(a)). So

$$\operatorname{var}(c'y) - \operatorname{var}(a'\hat{\beta}) = \sigma^2 c'(I_n - H)c = \sigma^2 c'M'Mc = \sigma^2 \|Mc\|^2 \geqslant 0$$

- Equality holds only if $Mc = 0$, that is, $c = Hc$; then $c'y = c'Hy = a'\hat{\beta}$ for every y.
- For every g-inverse, $H = X(X'X)^-X'$ (Theorem 13.4), so $c'Hc = c'X(X'X)^-X'c = a'(X'X)^-a$, as in Corollary 12.18 with $A = X$. □

The proof shows that $\operatorname{var}(c'y) = \operatorname{var}(a'\hat{\beta}) + \sigma^2\|Mc\|^2$ for every linear unbiased estimator $c'y$ of an estimable $a'\beta$. Under (A4), every $a'\beta$ is estimable (Theorem 14.2(a)), $a'\hat{\beta} = a'(X'X)^{-1}X'y$, and this is the scalar form of the Gauss–Markov theorem [2, Theorem 11.1]: for X of full column rank, Theorem 14.7 is Theorem 13.18 for one function $a'\beta$ at a time. For $a'\beta = \beta_j$, $\operatorname{var}(c'y) \geqslant \operatorname{var}(\hat{\beta}_j)$, with equality only if $c'y = \hat{\beta}_j$ for every y.

The normal linear model

Definition 14.8 ([12, Definition 1.2], [17, Definitions 8.1.1–8.1.3, 8.3.6, 11.3.5])**.** The **normal linear model** is the linear model $y = X\beta + \varepsilon$ of Definition 13.1, with $X \in \mathbb{R}^{n\times(k+1)}$ of any rank, under the assumption

(A5) $\varepsilon \sim N_n(0, \sigma^2 I_n)$ for a number $\sigma^2 > 0$.

Then (A2) and (A3) hold (Definition 10.1), $y \sim N_n(X\beta, \sigma^2 I_n)$ (Theorem 10.2), so (A1) holds, and $y_1, \ldots, y_n$ are independent with $y_i \sim N(x_i'\beta, \sigma^2)$ (Theorem 10.9(a)). In general, let $\alpha \in (0, 1)$, and let the distribution of a random vector y depend on unknown parameters ϑ.

- The **likelihood** is the density $f(y;\vartheta)$ of y at the observed y, as a function of ϑ, and the **log-likelihood** is its logarithm; values of ϑ at which the likelihood is largest are **maximum likelihood estimates** (MLE), as in Definition 10.19.
- Confidence intervals of level $1-\alpha$ for real functions of the parameters, such as $a'\beta$, are those of Definition 11.12(a). For a further random variable y_0 (a future observation), an interval $[L, U]$ whose endpoints are computed from y is a **prediction interval** of level $1-\alpha$ for y_0 if $\mathsf{P}(L \leqslant y_0 \leqslant U) = 1-\alpha$ for all values of the parameters.

- A **hypothesis** H_0 is a condition on the parameters, such as $K\beta = m$. A **test** of H_0 is given by a **rejection region** $\mathcal{R}$, a set of possible values of y: H_0 is rejected if the observed y lies in $\mathcal{R}$. The test has **level** α if $\mathsf{P}(y \in \mathcal{R}) = \alpha$ for all parameter values that satisfy H_0. Usually $\mathcal{R}$ is described through a **test statistic** T computed from y, as $\{T > c\}$ or $\{|T| > c\}$. When H_0 is rejected at level α, the departure from H_0 is called **significant** at level α.
- The quantiles $t_{\nu,\tau}$, $F_{a,b,\tau}$ and $\chi^2_{\nu,\tau}$ $(0 < \tau < 1)$ are those of Definition 11.7(c).

Theorem 14.9 ([2, (12.46)–(12.47)], [3, Result 8.7], [12, Theorems 5.1–5.3])**.** In the normal linear model let $\operatorname{rank} X = r < n$, let G be a g-inverse of $X'X$, and let $\beta^0 = GX'y$, a least-squares estimate (Theorems 13.2(c) and 13.3(a)). Let $K \in \mathbb{R}^{q\times(k+1)}$ be such that every entry of $K\beta$ is estimable.

(a) $K\beta^0 \sim N_q(K\beta, \sigma^2 KGK')$, and $K\beta^0$ and KGK' are the same for every g-inverse G of $X'X$. Under (A4), $\hat\beta = (X'X)^{-1}X'y \sim N_{k+1}(\beta, \sigma^2(X'X)^{-1})$.

(b) $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r}$; in particular $\mathsf{P}(\mathrm{SSE} = 0) = 0$.

(c) $K\beta^0$ and SSE are independent.

(d) The log-likelihood is

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|y - X\beta\|^2 \qquad (\beta \in \mathbb{R}^{k+1},\ \sigma^2 > 0)$$

If $\mathrm{SSE} > 0$, its maximizers are the pairs $(\hat\beta, \mathrm{SSE}/n)$ with $\hat\beta$ a least-squares estimate.

Proof. (a)

- Each row k_i' of K lies in $\mathcal{C}(X')$ (Theorem 14.2(a)): $k_i' = t_i'X$ with $t_i \in \mathbb{R}^n$. So $K = TX$, where $T \in \mathbb{R}^{q\times n}$ has the rows t_i' (Theorem 1.4(c)).
- With $H = XGX'$, $K\beta^0 = TXGX'y = THy$. By (A5), $y \sim N_n(X\beta, \sigma^2 I_n)$ (Definition 14.8), so Theorem 10.2 gives $K\beta^0 \sim N_q(THX\beta,\ \sigma^2 THH'T')$. Here $HX = X$ and $HH' = H$ (Theorem 13.4(a)), so $THX\beta = TX\beta = K\beta$ and $THH'T' = THT' = TXGX'T' = KGK'$.
- H does not depend on G (Theorem 13.4), so neither do $K\beta^0 = THy$ and $KGK' = THT'$.
- Under (A4), $G = (X'X)^{-1}$ (Theorem 13.4(c)) and every $a'\beta$ is estimable (Theorem 14.2(a)); take $K = I_{k+1}$.

(b) By Theorem 13.4(a),(b),(d), $\mathrm{SSE} = y'My$ with M symmetric and idempotent of rank $n - r \geqslant 1$, and $MX = 0$. Theorem 11.15, for $y \sim N_n(X\beta, \sigma^2 I_n)$ and $A = M$, gives $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r}$, as $MX\beta = 0$. This distribution has a density (Theorem 11.4(b)), so $\mathsf{P}(\mathrm{SSE} = 0) = 0$.

(c) By (a), $K\beta^0 = Cy$ with $C = TH$, and $CM = T(HM) = 0$ (Theorem 13.4(a)). By Theorem 11.20(b), Cy and $y'My = \mathrm{SSE}$ are independent.

(d)

- $\sigma^2 I_n$ is PD with determinant σ^{2n} and inverse $\sigma^{-2}I_n$, so by Theorem 10.8(c) the density of y is $(2\pi\sigma^2)^{-n/2}\exp\bigl(-\|y - X\beta\|^2/(2\sigma^2)\bigr)$, whose logarithm is ℓ.
- For fixed σ^2, ℓ is largest exactly when $\|y - X\beta\|^2$ is smallest, that is, at the least-squares estimates $\hat\beta$; and $\|y - X\hat\beta\|^2 = \mathrm{SSE}$ for all of them (Theorem 13.3(a), Definition 13.25).
- So it remains to maximize $h(u) = \ell(\hat\beta, u) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log u - \mathrm{SSE}/(2u)$ over $u > 0$. Its derivative $h'(u) = -\frac{n}{2u} + \frac{\mathrm{SSE}}{2u^2} = \frac{\mathrm{SSE} - nu}{2u^2}$ vanishes only at $\hat u = \mathrm{SSE}/n$. As $\mathrm{SSE} > 0$, $h(u) \to -\infty$ both as $u \to 0^+$ (the term $-\mathrm{SSE}/(2u)$ dominates) and as $u \to \infty$ (the term $-\frac{n}{2}\log u$ dominates); a differentiable function on $(0, \infty)$ with these limits attains its largest value at a zero of its derivative, which is $\hat u$, and nowhere else.

- Hence $\ell(\beta,\sigma^2) \leqslant \ell(\hat\beta,\sigma^2) \leqslant \ell(\hat\beta,\hat u)$, with equality throughout only if β is a least-squares estimate and $\sigma^2 = \hat u$. □

By Theorem 13.23, $\mathsf{E}(\mathrm{SSE}/n) = \sigma^2(n-r)/n < \sigma^2$: the maximum likelihood estimator of σ^2 is biased [12, Theorem 5.3], and the unbiased $\hat\sigma^2 = \mathrm{SSE}/(n-r)$ is used instead. We write $s^2 = \hat\sigma^2$ and $s = \sqrt{s^2}$; by Theorem 14.9(b), $s > 0$ with probability 1.

Theorem 14.10 ([2, §12.11, (12.51)–(12.52)], [12, Theorems 7.1, 7.9, 9.1, 9.2, Proposition 7.13])**.** In the normal linear model let $\operatorname{rank} X = r < n$, let G be a g-inverse of $X'X$ and $\beta^0 = GX'y$, and let $a \in \mathbb{R}^{k+1}$ be nonzero with $a'\beta$ estimable. Then $a'Ga > 0$, and we put $\mathrm{SE}(a'\beta^0) = s\sqrt{a'Ga}$, the **standard error** of $a'\beta^0$, and $t^* = t_{n-r,1-\alpha/2}$.

(a) $T = (a'\beta^0 - a'\beta)/\mathrm{SE}(a'\beta^0) \sim t_{n-r}$.

(b) $\left[a'\beta^0 - t^*\,\mathrm{SE}(a'\beta^0),\ a'\beta^0 + t^*\,\mathrm{SE}(a'\beta^0)\right]$ is a confidence interval of level $1-\alpha$ for $a'\beta$. For $m_0 \in \mathbb{R}$, the test of $H_0\colon a'\beta = m_0$ that rejects when $|T_0| > t^*$, where $T_0 = (a'\beta^0 - m_0)/\mathrm{SE}(a'\beta^0)$, has level α; it rejects exactly when m_0 lies outside this interval.

(c) Let $x_0 \in \mathbb{R}^{k+1}$ with $x_0'\beta$ estimable, and let $y_0 = x_0'\beta + \varepsilon_0$ be a future observation with $\varepsilon_0 \sim N(0,\sigma^2)$ independent of ε. Put $w_0 = s\sqrt{1 + x_0'Gx_0}$. Then $(y_0 - x_0'\beta^0)/w_0 \sim t_{n-r}$, and $[x_0'\beta^0 - t^*w_0,\ x_0'\beta^0 + t^*w_0]$ is a prediction interval of level $1-\alpha$ for y_0.

$\mathrm{SE}(a'\beta^0)$ and the intervals are the same for every g-inverse G of $X'X$ (Theorem 14.9(a) with $K = a'$). Under (A4), $G = (X'X)^{-1}$, $\beta^0 = \hat\beta$ and $r = k+1$.

Proof. (a)

- $a = X'c$ for some $c \in \mathbb{R}^n$ (Theorem 14.2(a)), so, with $H = XGX'$ symmetric and idempotent (Theorem 13.4(a)), $a'Ga = c'Hc = c'H'Hc = \|Hc\|^2$. As $X'H = X'$, $X'Hc = a \neq 0$, so $Hc \neq 0$ and $a'Ga > 0$.
- By Theorem 14.9(a) with $K = a'$ and Theorem 10.2, $z = (a'\beta^0 - a'\beta)/(\sigma\sqrt{a'Ga}) \sim N(0,1)$. Next $v = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r}$ (Theorem 14.9(b)), and z and v are independent, as functions of $a'\beta^0$ and of SSE (Theorem 14.9(c), (N3)). Since $s/\sigma = \sqrt{v/(n-r)}$, $T = z/\sqrt{v/(n-r)} \sim t_{n-r}$ (Definition 11.7(a)).

(b) By (a) and Theorem 11.8(a), $\mathsf{P}(|T| \leqslant t^*) = 1-\alpha$, and $|T| \leqslant t^*$ holds exactly when $a'\beta$ lies in the interval. If H_0 holds, then $T_0 = T$, so $\mathsf{P}(|T_0| > t^*) = 1 - (1-\alpha) = \alpha$. Finally $|T_0| > t^*$ means $|a'\beta^0 - m_0| > t^*\,\mathrm{SE}(a'\beta^0)$, that is, m_0 lies outside the interval.

(c)

- As $\varepsilon \sim N_n(0,\sigma^2 I_n)$ and $\varepsilon_0 \sim N(0,\sigma^2)$ are independent, $\tilde\varepsilon = (\varepsilon', \varepsilon_0)' \sim N_{n+1}(0,\sigma^2 I_{n+1})$ (Theorem 10.9(a)).
- $x_0 = X'c_0$ for some $c_0 \in \mathbb{R}^n$ (Theorem 14.2(a)), so $x_0'\beta^0 = c_0'XGX'y = c_0'H(X\beta+\varepsilon) = x_0'\beta + c_0'H\varepsilon$, as $HX = X$. Hence $y_0 - x_0'\beta^0 = \varepsilon_0 - c_0'H\varepsilon = b'\tilde\varepsilon$, where $b = (-c_0'H,\ 1)'$ has $b'b = c_0'HH'c_0 + 1 = 1 + x_0'Gx_0$ ($c_0'Hc_0 = c_0'XGX'c_0 = x_0'Gx_0$). So $y_0 - x_0'\beta^0 \sim N\big(0, \sigma^2(1 + x_0'Gx_0)\big)$ (Theorem 10.2).
- As $MX = 0$, $\mathrm{SSE} = y'My = \varepsilon'M\varepsilon = \tilde\varepsilon'P\tilde\varepsilon$ with $P = \operatorname{diag}(M, 0)$, which is symmetric and idempotent, and $b'P = (-c_0'HM,\ 0) = 0$. By Theorem 11.20(b), $y_0 - x_0'\beta^0$ and SSE are independent. As in (a), $(y_0 - x_0'\beta^0)/w_0 \sim t_{n-r}$, and as in (b), $\mathsf{P}(|y_0 - x_0'\beta^0| \leqslant t^*w_0) = 1-\alpha$. □

For $x_0 \neq 0$, the confidence interval of (b) for $a = x_0$ has half-width $t^*s\sqrt{x_0'Gx_0}$; the prediction interval of (c) is wider, because the new error ε_0 adds σ^2 to the variance.

Definition 14.11 ([2, Definition 12.2, Lemma 12.1, (12.46)–(12.47)])**.** Let $\operatorname{rank} X = r$, let $K \in \mathbb{R}^{q\times(k+1)}$ and $m \in \mathbb{R}^q$. The hypothesis $H_0\colon K\beta = m$ is **testable** if every entry of $K\beta$ is estimable and $\operatorname{rank} K = q$ (the rows of K are linearly independent).

Let G be a g-inverse of $X'X$.

- $H_0\colon K\beta = m$ is testable if and only if $\operatorname{rank} K = q$ and $K = WX'X$ for some $W \in \mathbb{R}^{q\times(k+1)}$. Indeed, if the entries of $K\beta$ are estimable, then each row k_i' of K lies in $\mathcal{C}(X')$ (Theorem 14.2(a)): $k_i' = t_i'X$ with $t_i \in \mathbb{R}^n$. So $K = TX$, where $T \in \mathbb{R}^{q\times n}$ has the rows t_i' (Theorem 1.4(c)), and $X = XGX'X$ (Theorem 13.4(a)) gives $K = TXGX'X = WX'X$ with $W = TXG$. Conversely, if $K = WX'X$, then $K = TX$ with $T = WX'$, so every row of K lies in $\mathcal{C}(X')$, and the entries of $K\beta$ are estimable (Theorem 14.2(a)).
- Let H_0 be testable, $K = WX'X$ and $T = WX'$, so that $K = TX$. Then $q = \operatorname{rank} K \leqslant \operatorname{rank} T \leqslant q$ (Theorems 4.15(a) and 4.13(a); T has q rows), so $\operatorname{rank} T = q$ and TT' is PD (Theorem 8.25). Also $q = \operatorname{rank} K \leqslant \operatorname{rank} X = r$ (Theorem 4.15(a)). By $X'XGX'X = X'X$,

$$V = KGK' = WX'XGX'XW' = WX'XW' = TT'$$

  so V is PD, hence invertible (Proposition 8.23(c)), and it does not depend on G.

Under (A4), $H_0\colon K\beta = m$ is testable for every K of rank q (Theorem 14.2(a)).

Theorem 14.12 ([3, Result 7.6], [12, Theorem 10.3, Corollary 7.6])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$, $y \in \mathbb{R}^n$, let G be a g-inverse of $X'X$ and $\beta^0 = GX'y$, and let $H_0\colon K\beta = m$ be testable, with $K \in \mathbb{R}^{q\times(k+1)}$, so that $V = KGK'$ is invertible (the paragraph after Definition 14.11). Put

$$\tilde\beta = \beta^0 - GK'V^{-1}(K\beta^0 - m)$$

(a) $K\tilde\beta = m$, and for every β with $K\beta = m$, $\|y - X\beta\|^2 = \|y - X\tilde\beta\|^2 + \|X(\tilde\beta - \beta)\|^2$.

(b) The minimum of $S(\beta) = \|y - X\beta\|^2$ subject to $K\beta = m$ is attained at $\tilde\beta$, the **constrained least-squares estimate**; the minimizers are the β with $K\beta = m$ and $X\beta = X\tilde\beta$, and under (A4) $\tilde\beta$ is the only one.

(c) $\mathrm{SSE}_K - \mathrm{SSE} = (K\beta^0 - m)'V^{-1}(K\beta^0 - m)$, where $\mathrm{SSE}_K = \|y - X\tilde\beta\|^2$ and $\mathrm{SSE} = \|y - X\beta^0\|^2$.

Proof. Let $K = WX'X$ and $T = WX'$, so that $K = TX$, $K' = X'XW'$ and $V = TT'$ (the paragraph after Definition 14.11).

(a)

- $K\tilde\beta = K\beta^0 - KGK'V^{-1}(K\beta^0 - m) = K\beta^0 - (K\beta^0 - m) = m$.
- $X'XGK' = X'XGX'XW' = X'XW' = K'$ and $X'X\beta^0 = X'y$ (Theorem 13.2(c)), so $X'X\tilde\beta = X'y - K'V^{-1}(K\beta^0 - m)$, that is, $X'(y - X\tilde\beta) = K'V^{-1}(K\beta^0 - m)$.
- Let $K\beta = m$. Then $(\tilde\beta - \beta)'X'(y - X\tilde\beta) = (K\tilde\beta - K\beta)'V^{-1}(K\beta^0 - m) = 0$, as $K\tilde\beta = m = K\beta$. So, with $y - X\beta = (y - X\tilde\beta) + X(\tilde\beta - \beta)$,

$$\|y - X\beta\|^2 = \|y - X\tilde\beta\|^2 + 2(\tilde\beta - \beta)'X'(y - X\tilde\beta) + \|X(\tilde\beta - \beta)\|^2 = \|y - X\tilde\beta\|^2 + \|X(\tilde\beta - \beta)\|^2$$

(b) By (a), $S(\beta) \geqslant S(\tilde\beta)$ for every β with $K\beta = m$, with equality exactly when $X(\tilde\beta - \beta) = 0$. Under (A4), $X(\tilde\beta - \beta) = 0$ only for $\beta = \tilde\beta$ (Theorem 4.13(b)).

(c) By Theorem 13.3(a), $S(\tilde\beta) = S(\beta^0) + \|X(\beta^0 - \tilde\beta)\|^2$. Here $X(\beta^0 - \tilde\beta) = XGK'V^{-1}d$ with $d = K\beta^0 - m$, and $XGK' = XGX'XW' = XW' = T'$ (Theorem 13.4(a)); so, as $V = TT'$ and V^{-1} are symmetric, $\|X(\beta^0 - \tilde\beta)\|^2 = d'V^{-1}TT'V^{-1}d = d'V^{-1}d$. □

Theorem 14.13 ([2, (12.48)], [3, §8.4], [12, Theorems 7.3, 7.4, Corollaries 7.6, 7.8, 7.15, 7.16])**.** In the normal linear model let $\operatorname{rank} X = r < n$, let G be a g-inverse of $X'X$ and $\beta^0 = GX'y$, let $H_0\colon K\beta = m$ be testable with $K \in \mathbb{R}^{q\times(k+1)}$, $V = KGK'$, and let $\tilde\beta$ and SSE_K be as in Theorem 14.12.

(a) For every y with $\mathrm{SSE} > 0$,

$$F = \frac{(K\beta^0 - m)'V^{-1}(K\beta^0 - m)/q}{s^2} = \frac{(\mathrm{SSE}_K - \mathrm{SSE})/q}{\mathrm{SSE}/(n-r)}$$

(b) If H_0 holds, then F, which is defined with probability 1 (Theorem 14.9(b)), has the distribution $F_{q,n-r}$; so the test that rejects H_0 when $F > F_{q,n-r,1-\alpha}$ has level α.

(c) Let $X = [X_1\ X_2]$ with $X_1 \in \mathbb{R}^{n\times q}$, $\beta = (\beta_1', \beta_2')'$ accordingly, and let $H_0\colon \beta_1 = 0$ ($K = [I_q\ 0]$, $m = 0$) be testable. Then SSE_K is the error sum of squares of the least-squares fit with design X_2. The order of the columns does not matter: reordering the columns of X and the entries of β alike changes neither $\mathcal{C}(X)$, $\mathcal{C}(X_2)$ nor the sums of squares, so X_1 may be any q of the columns of X, and X_2 the others. If $\mathbf{1}_n \in \mathcal{C}(X_2)$ and $\mathrm{SSE} > 0$, then, with R^2 and R_2^2 the coefficients of determination of the fits with designs X and X_2,

$$F = \frac{(R^2 - R_2^2)/q}{(1-R^2)/(n-r)}$$

In particular, under (A4), for the **overall F test** of $H_0\colon \beta_1 = \cdots = \beta_k = 0$, where $X = [\mathbf{1}_n\ X_0]$ and $k \geqslant 1$, $\mathrm{SSE}_K = \mathrm{SST}$, $\mathrm{SSE}_K - \mathrm{SSE} = \mathrm{SSR}$ and

$$F = \frac{\mathrm{SSR}/k}{\mathrm{SSE}/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)}$$

(d) If $q = 1$ and $K = a'$, then $F = T_0^2$, where T_0 is the statistic of Theorem 14.10(b) for $m_0 = m$; the test of (b) rejects exactly when that of Theorem 14.10(b) does.

Part (a) and the identities in (c) and (d) are algebraic: they hold for every y with $\mathrm{SSE} > 0$. Under (A4), $G = (X'X)^{-1}$, $\beta^0 = \hat{\beta}$ and $r = k + 1$.

Proof. (a) By Theorem 14.12(c), the numerator quadratic form is $\mathrm{SSE}_K - \mathrm{SSE}$, and $s^2 = \mathrm{SSE}/(n-r)$.

(b)

- Let $K = WX'X$, $T = WX'$, so that $K = TX$ and $V = TT'$ (the paragraph after Definition 14.11); put $H = XGX'$, $M = I_n - H$ and $\varepsilon = y - X\beta$. Then $K\beta^0 = TXGX'y = THy = Ty$, as $TH = WX'XGX' = WX' = T$ (Theorem 13.4(a)), and $XGK' = XGX'XW' = XW' = T'$.
- So $X\tilde{\beta} = Hy - T'V^{-1}(Ty - m)$. Under H_0, $m = K\beta = TX\beta$, so $Ty - m = T\varepsilon$, and $y - X\tilde{\beta} = My + U\varepsilon$ with $U = T'V^{-1}T$. Also $My = M\varepsilon$, as $MX = 0$.
- U is symmetric, $U^2 = T'V^{-1}(TT')V^{-1}T = U$, and $\operatorname{rank} U = \operatorname{tr} U = \operatorname{tr}(V^{-1}TT') = \operatorname{tr} I_q = q$ (Theorems 6.29(c) and 1.11(b)). $MU = MXW'V^{-1}T = 0$ and $UM = (MU)' = 0$.
- Hence $\mathrm{SSE}_K = \|M\varepsilon + U\varepsilon\|^2 = \varepsilon'M\varepsilon + \varepsilon'U\varepsilon$, so $\mathrm{SSE}_K - \mathrm{SSE} = \varepsilon'U\varepsilon$, with $\mathrm{SSE} = \varepsilon'M\varepsilon$.
- By (A5) and Theorem 11.15, $u = \varepsilon'U\varepsilon/\sigma^2 \sim \chi^2_q$ and $v = \varepsilon'M\varepsilon/\sigma^2 \sim \chi^2_{n-r}$; by Theorem 11.20(a), u and v are independent. So $F = (u/q)/(v/(n-r)) \sim F_{q,n-r}$ (Definition 11.7(b)), and $\mathsf{P}(F > F_{q,n-r,1-\alpha}) = 1 - (1-\alpha) = \alpha$ (Definition 11.7(c)).

(c)

- The β with $\beta_1 = 0$ give exactly the vectors $X\beta = X_2\beta_2$, β_2 arbitrary. By Theorem 14.12(b), SSE_K is the minimum of $\|y - X\beta\|^2$ over these β, that is, the minimum of $\|y - X_2\beta_2\|^2$: the error sum of squares of the design X_2.
- Reordering the columns of X and the entries of β alike leaves $X\beta = \sum_j \beta_j \operatorname{col}_j(X)$ (Theorem 1.4(b)) and the hypothesis $\beta_1 = 0$ as they are, and also $\mathcal{C}(X)$ and $\mathcal{C}(X_2)$, which are spanned by the same columns as before (Definition 4.7). So $r = \dim \mathcal{C}(X)$ (Definition 4.11) does not change, and neither do the fitted values Hy and H_2y, where H and H_2 are the hat matrices of X and X_2: they are the orthogonal projections of y onto $\mathcal{C}(X)$ and $\mathcal{C}(X_2)$ (Theorem 13.4). Hence SSE, SSE_K (by the first point), R^2 and R_2^2 (Definition 13.25) do not change.

- If $\mathbf{1}_n \in \mathcal{C}(X_2)$, then also $\mathbf{1}_n \in \mathcal{C}(X)$, SST $\geqslant$ SSE > 0, and Theorem 13.26 gives SSE $=$ SST$(1-R^2)$ and SSE$_K =$ SST$(1-R_2^2)$; substitute into (a) and cancel SST.
- For the overall test, $X_2 = \mathbf{1}_n$, whose fitted values are $\bar{y}\mathbf{1}_n$ (Example 6.27). So SSE$_K =$ SST, $R_2^2 = 0$ and SSE$_K -$ SSE $=$ SSR (Theorem 13.26); under (A4), $r = k+1$ and the hypothesis is testable (Definition 14.11).

(d) For $K = a'$, $V = a'Ga$, and the numerator of F is $(a'\beta^0 - m)^2/(a'Ga)$, so $F = (a'\beta^0 - m)^2/(s^2 a'Ga) = T_0^2$. Hence $F > F_{1,n-r,1-\alpha} \iff T_0^2 > (t^*)^2 \iff |T_0| > t^*$, as $(t^*)^2 = F_{1,n-r,1-\alpha}$ and $t^* > 0$ (Theorem 11.8(b) and its proof). □

Definition 14.14 ([2, §11.9, §12.9, §12.13])**.** Let $X = [X_1\ X_2]$, with the columns in any order as in Theorem 14.13(c), $\beta = (\beta_1', \beta_2')'$ accordingly, and let H and H_2 be the hat matrices of X and X_2 (Theorem 13.4). The **reduction in the sum of squares** due to fitting $X\beta$ is $R(\beta_1, \beta_2) = y'Hy = \|Hy\|^2$, and $R(\beta_2) = y'H_2y$ is that due to fitting $X_2\beta_2$ alone; the reduction due to β_1 after β_2 is

$$R(\beta_1 \mid \beta_2) = R(\beta_1, \beta_2) - R(\beta_2)$$

By Theorem 13.4(d), $R(\beta_1, \beta_2) = \hat{\beta}'X'y = y'y -$ SSE for every least-squares estimate $\hat{\beta}$, and $R(\beta_2) = y'y -$ SSE$_2$, where SSE$_2$ is the error sum of squares of the fit with design X_2. So $R(\beta_1 \mid \beta_2) =$ SSE$_2 -$ SSE: by Theorem 14.13(c), if $H_0\colon \beta_1 = 0$ is testable, then $F = \big(R(\beta_1 \mid \beta_2)/q\big)/s^2$. For a model with an intercept μ ($X_2 = \mathbf{1}_n$), $R(\mu) = n\bar{y}^2$ (Example 6.27), and $R(\beta_1 \mid \mu) =$ SST $-$ SSE $=$ SSR (Theorem 13.26).

Example 14.15. Eight plots receive a fertilizer at two levels, coded $x_{i1} = -1, 1$, and irrigation at two levels, $x_{i2} = -1, 1$, two plots for each combination; the yields are

x_{i1}	-1	-1	-1	-1	1	1	1	1
x_{i2}	-1	-1	1	1	-1	-1	1	1
y_i	9	3	9	7	11	13	11	17

In the normal linear model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ ($n = 8$, $k = 2$), find the confidence interval of level $1-\alpha$ for β_1, the tests of $\beta_2 = 0$ and of $\beta_1 = \beta_2 = 0$ at level α, and the prediction interval of level $1-\alpha$ for a plot with $x_1 = x_2 = 1$. Put $t^* = t_{5,1-\alpha/2}$.

Solution.

- The three columns of X are orthogonal with squared length 8, so $X'X = 8I_3$, rank $X = 3$ (A4), $G = (X'X)^{-1} = \frac{1}{8}I_3$, $X'y = (80, 24, 8)'$ and $\hat{\beta} = (10, 3, 1)'$. The fitted values are $6, 8, 12, 14$ for the four combinations, the residuals $e = (3, -3, 1, -1, -1, 1, -3, 3)'$, SSE $= 40$ and $s^2 = 40/5 = 8$. With $\bar{y} = 10$, SSR $= 2(16 + 4 + 4 + 16) = 80$, SST $= 120$ and $R^2 = \frac{2}{3}$.
- SE$(\hat{\beta}_j) = s\sqrt{1/8} = 1$ for every j (Theorem 14.10). By Theorem 14.10(b) the interval for β_1 is $[3 - t^*,\ 3 + t^*]$, and the test of $\beta_2 = 0$ has $T_0 = 1$: it rejects if and only if $t^* < 1$; equivalently $F = T_0^2 = 1$ (Theorem 14.13(d)).
- The overall F statistic is $F = (80/2)/(40/5) = 5 = \frac{(2/3)/2}{(1/3)/5}$ (Theorem 14.13(c)); the test rejects if and only if $F_{2,5,1-\alpha} < 5$.
- For $x_0 = (1, 1, 1)'$, $x_0'\hat{\beta} = 14$ and $x_0'(X'X)^{-1}x_0 = \frac{3}{8}$, so $w_0 = s\sqrt{1 + \frac{3}{8}} = \sqrt{11}$: the prediction interval is $[14 - t^*\sqrt{11},\ 14 + t^*\sqrt{11}]$ (Theorem 14.10(c)), while the confidence interval for $x_0'\beta$ is $[14 - t^*\sqrt{3},\ 14 + t^*\sqrt{3}]$.

The one-way layout and the analysis of variance

Theorem 14.16 ([12, Proposition 11.2], [2, §12.12.1], [3, §8.3])**.** In the one-way layout of Theorem 13.10, let $\mathcal{I}_l$ be the set of the observations (row indices) of group l, let $\bar{y}_l = n_l^{-1}\sum_{i\in\mathcal{I}_l} y_i$ be the **group mean** and $\bar{y}$ the **grand mean** of all n observations, and let H and M be the hat matrix and the residual maker of X (Theorem 13.4).

(a) $H = \sum_l n_l^{-1} d_l d_l' = \operatorname{diag}(\bar{J}_{n_1}, \dots, \bar{J}_{n_g})$, and $\hat{y} = Hy = \bar{y}_1 d_1 + \cdots + \bar{y}_g d_g$: the fitted value of each observation is the mean of its group.

(b) With $\bar{y}$, $\bar{y}_l - \bar{y}$ and $y_i - \bar{y}_l$ in a row $i \in \mathcal{I}_l$, the three vectors $\bar{y}\mathbf{1}_n$, $Hy - \bar{y}\mathbf{1}_n$ and My are orthogonal, lie in subspaces of dimensions 1, $g-1$ and $n-g$, and add up to y; and

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{l=1}^{g} n_l(\bar{y}_l - \bar{y})^2 + \sum_{l=1}^{g}\sum_{i\in\mathcal{I}_l}(y_i - \bar{y}_l)^2$$

that is, SST = SSR + SSE, with the **between-group sum of squares** $\mathrm{SSR} = \sum_l n_l(\bar{y}_l - \bar{y})^2$ and the **within-group sum of squares** $\mathrm{SSE} = \sum_l \sum_{i\in\mathcal{I}_l}(y_i - \bar{y}_l)^2$.

Proof. (a)

- No observation lies in two groups, so $d_l' d_m = 0$ for $l \neq m$, and $d_l' d_l = n_l$; these are the entries of $X_1'X_1$ (Theorems 1.4(d) and 1.9), so $X_1'X_1 = \operatorname{diag}(n_1, \dots, n_g)$.
- $\mathcal{C}(X_1) = \mathcal{C}(X)$ and X_1 has full column rank (Theorem 13.10(c)). For every y, Hy and the hat matrix $H_1 = X_1(X_1'X_1)^{-1}X_1'$ of X_1 give the orthogonal projection of y onto the same subspace (Theorem 13.4, and (c) there for X_1), so $H = H_1$ (Theorem 1.4(b), with the columns of I_n for y).
- By the outer product expansion (Theorem 1.4(d)), $H_1 = X_1 \operatorname{diag}(n_1^{-1}, \dots, n_g^{-1})X_1' = \sum_l n_l^{-1} d_l d_l'$. The (i,j) entry of $d_l d_l'$ is 1 if $i, j \in \mathcal{I}_l$ and 0 otherwise, which gives the block diagonal form; and $n_l^{-1} d_l d_l' y = n_l^{-1}(d_l' y) d_l = \bar{y}_l d_l$.

(b) Let $U = \operatorname{span}\{\mathbf{1}_n\} \subseteq W = \mathcal{C}(X)$, so that $P_W = H$. Then $P_U y = \bar{y}\mathbf{1}_n$ (Corollary 6.35(a)), and $P_W y = Hy$ is given by (a). By Theorem 6.36(b),(c), $y = P_U y + (P_W - P_U)y + (I_n - P_W)y$, with entries $\bar{y}$, $\bar{y}_l - \bar{y}$ and $y_i - \bar{y}_l$ in row $i \in \mathcal{I}_l$, is a sum of three orthogonal vectors in U, $W \cap U^\perp$ and $W^\perp$, of dimensions 1, $g-1$ and $n-g$ (Theorems 13.10(b) and 6.6(c)), and the squared lengths add up. Subtract $\|P_U y\|^2 = n\bar{y}^2$ and use $\|y\|^2 - n\bar{y}^2 = \sum_i (y_i - \bar{y})^2$ (Example 1.16). The first sum on the right is $\|Hy - \bar{y}\mathbf{1}_n\|^2 = \mathrm{SSR}$ and the second is $\|My\|^2 = \mathrm{SSE}$ (Definition 13.25). □

Example 14.17 ([12, Proposition 11.2])**.** In Example 13.11 ($g = 3$; $n_1, n_2, n_3 = 3, 2, 1$), let $X_1 = [d_1\ d_2\ d_3]$, so that $\mathcal{C}(X_1) = \mathcal{C}(X)$ is the set of vectors that are constant within each group, and let $W = \{x \in \mathbb{R}^6 : d_1'x = d_2'x = d_3'x = 0\}$, the vectors whose entries add up to 0 within each group. Then

$$\mathbb{R}^6 = \mathcal{C}(X_1) \oplus W$$

- W is the set of solutions of $X_1'x = 0$, a subspace (Example 3.5(e)). The coefficient matrix

$$X_1' = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

is in RREF, with free unknowns x_2, x_3, x_5. By Proposition 2.7(b), $x \in W$ if and only if $x_1 = -x_2 - x_3$, $x_4 = -x_5$ and $x_6 = 0$, that is,

$$x = x_2 f_1 + x_3 f_2 + x_5 f_3, \qquad f_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad f_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad f_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}$$

Entries $2, 3, 5$ of $x_2 f_1 + x_3 f_2 + x_5 f_3$ are x_2, x_3, x_5, so f_1, f_2, f_3 are independent: they form a basis of W.

- $\mathcal{C}(X_1) \cap W = \{0\}$: if $x = c_1 d_1 + c_2 d_2 + c_3 d_3$ lies in W, then $0 = d_m' x = c_m d_m' d_m = c_m n_m$ for each m, as $d_m' d_l = 0$ for $l \neq m$ and $d_m' d_m = n_m$; so every $c_m = 0$ and $x = 0$.
- $\mathcal{C}(X_1) + W = \mathbb{R}^6$: for $x \in \mathbb{R}^6$ let $\bar{x}_l = d_l' x / n_l$, the mean of the entries of x in group l, and $u = \bar{x}_1 d_1 + \bar{x}_2 d_2 + \bar{x}_3 d_3 \in \mathcal{C}(X_1)$. Then $d_m'(x - u) = d_m' x - \bar{x}_m n_m = 0$ for each m, so $x - u \in W$, and $x = u + (x - u)$.
- Hence $\mathbb{R}^6 = \mathcal{C}(X_1) \oplus W$ (Definition 3.30), and $d_1, d_2, d_3, f_1, f_2, f_3$ form a basis of $\mathbb{R}^6$ (Theorem 3.33). By Theorem 3.32, each x is in exactly one way the sum of a vector of $\mathcal{C}(X_1)$ and a vector of W: its **between-group part** u, which replaces each entry by the mean of its group, and its **within-group part** $x - u$, the deviations from the group means. By Theorem 14.16(a), $u = Hx$ and $x - u = Mx$. Also $W = \mathcal{N}(X_1') = \mathcal{C}(X_1)^\perp$ by Theorem 6.6(a), as the columns d_1, d_2, d_3 of X_1 span $\mathcal{C}(X_1)$; and $\mathcal{C}(X_1) = \mathcal{C}(X)$ (Theorem 13.10(c)), so $W = \mathcal{C}(X)^\perp$.

For the responses $y = (5, 7, 6, 3, 5, 2)'$, the group means are $6, 4, 2$, and $y = (6, 6, 6, 4, 4, 2)' + (-1, 1, 0, -1, 1, 0)'$, the fitted values and the residuals of Example 13.11. With a single group, $\mathcal{C}(X_1) = \text{span}\{\mathbf{1}\}$ and $W = W_0$: this is Example 3.35.

Corollary 14.18 ([18, §3.6, (3.55)–(3.56), Exercise 3.22], [3, §8.3])**.** In the one-way layout of Theorem 13.10, use the design $X_1 = [d_1 \ \cdots \ d_g]$, with $\beta = (\mu_1, \ldots, \mu_g)'$, the group means, and assume the normal linear model with $n > g \geqslant 2$. Then H_0: $\mu_1 = \cdots = \mu_g$ is $K\beta = 0$ for the $(g-1) \times g$ matrix K whose row l is $\text{row}_l(I_g) - \text{row}_{l+1}(I_g)$, the hypothesis is testable, and

$$F = \frac{\sum_l n_l (\bar{y}_l - \bar{y})^2 / (g-1)}{\sum_l \sum_{i \in \mathcal{I}_l} (y_i - \bar{y}_l)^2 / (n-g)} \sim F_{g-1, n-g} \quad \text{under } H_0$$

with $\bar{y}_l$, $\bar{y}$ and $\mathcal{I}_l$ as in Theorem 14.16.

Proof.
- $K\beta = 0$ says $\mu_l = \mu_{l+1}$ for $l = 1, \ldots, g-1$, that is, H_0. The first $g-1$ columns of K form an upper triangular matrix with diagonal entries 1, which is invertible (Theorems 5.14(c) and 5.18(a)); so these $g-1$ columns are linearly independent, and $\text{rank}\, K = g - 1$, as K has $g-1$ rows (Theorem 4.13(a)). X_1 has full column rank (Theorem 13.10(a)), so H_0 is testable (Definition 14.11).
- No observation lies in two groups, so $d_l' d_m = 0$ for $l \neq m$, and $d_l' d_l = n_l$; these are the entries of $X_1' X_1$ (Theorems 1.4(d) and 1.9), so $X_1' X_1 = \text{diag}(n_1, \ldots, n_g)$. And $X_1' y$ holds the group totals $d_l' y$ (Theorem 1.4(d)); so $\hat{\beta} = (\bar{y}_1, \ldots, \bar{y}_g)'$, each fitted value is the mean of its group, and $\text{SSE} = \sum_l \sum_{i \in \mathcal{I}_l} (y_i - \bar{y}_l)^2$.
- The β with $K\beta = 0$ are $\beta = \mu \mathbf{1}_g$, with $X_1 \beta = \mu \mathbf{1}_n$. By Theorem 14.12(b), $\text{SSE}_K = \min_\mu \|y - \mu \mathbf{1}_n\|^2 = \text{SST}$ (Example 6.27). As $\mathbf{1}_n \in \mathcal{C}(X_1)$, Theorem 14.16(b) gives $\text{SSE}_K - \text{SSE} = \text{SSR} = \sum_l n_l (\bar{y}_l - \bar{y})^2$. Theorem 14.13(a),(b) with $q = g - 1$ and $r = g$ gives the formula and the distribution. □

The computations of Corollary 14.18 are arranged in the **analysis-of-variance table** [2, Tables 12.5, 12.6], [3, §8.3]; the mean squares are the sums of squares divided by their degrees of freedom, and F is the ratio of the two mean squares:

source	degrees of freedom	sum of squares	mean square
between groups	$g-1$	$\text{SSR} = \sum_l n_l (\bar{y}_l - \bar{y})^2$	$\text{SSR}/(g-1)$
within groups	$n-g$	$\text{SSE} = \sum_l \sum_{i \in \mathcal{I}_l} (y_i - \bar{y}_l)^2$	$s^2 = \text{SSE}/(n-g)$
total	$n-1$	$\text{SST} = \sum_i (y_i - \bar{y})^2$	

In the notation of Definition 14.14, with the intercept μ and the effects $\alpha = (\alpha_1, \ldots, \alpha_g)'$ of Theorem 13.10, $R(\mu) = n\bar{y}^2$, $R(\mu, \alpha) = \|Hy\|^2 = \sum_l n_l \bar{y}_l^2$ and $R(\alpha \mid \mu) = \text{SSR}$; so $y'y = R(\mu) + R(\alpha \mid \mu) + \text{SSE}$, the uncentered split of the paragraph after Theorem 13.26.

Example 14.19. In Example 13.11, the six weights $y = (5, 7, 6, 3, 5, 2)'$ of three types of plants, assume the normal linear model. Test $H_0\colon \alpha_1 = \alpha_2 = \alpha_3$ at level α, and find the confidence interval of level $1-\alpha$ for $\alpha_1 - \alpha_2$.

Solution.

- The group means are $6, 4, 2$ and $\bar{y} = 28/6 = 14/3$; the residuals are $e = (-1, 1, 0, -1, 1, 0)'$, so $\mathrm{SSE} = 4$, $r = \operatorname{rank} X = 3$ and $s^2 = 4/3$ (Example 13.11). By Theorem 14.16(b),

$$\mathrm{SSR} = 3\Big(6 - \frac{14}{3}\Big)^2 + 2\Big(4 - \frac{14}{3}\Big)^2 + \Big(2 - \frac{14}{3}\Big)^2 = \frac{48 + 8 + 64}{9} = \frac{40}{3}, \qquad \mathrm{SST} = \frac{40}{3} + 4 = \frac{52}{3}$$

- In the design X_1, H_0 is $\mu_1 = \mu_2 = \mu_3$, and Corollary 14.18 gives $F = \frac{40/3}{2}\Big/\frac{4}{3} = 5$, with 2 and 3 degrees of freedom: the test of level α rejects H_0 if and only if $F_{2,3,1-\alpha} < 5$. The analysis-of-variance table:

source	degrees of freedom	sum of squares	mean square
between groups	2	40/3	20/3
within groups	3	4	4/3
total	5	52/3	

- The same F from Theorem 14.13(a) in the design X of rank 3: H_0 is $K\beta = 0$ with the rows $(0, 1, -1, 0)$ and $(0, 0, 1, -1)$ of K, and with $G = \operatorname{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$ and $\beta^0 = (0, 6, 4, 2)'$ (Example 14.6), $K\beta^0 = (2, 2)'$ and

$$V = KGK' = \begin{pmatrix} \frac{5}{6} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}, \qquad V^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{5}{6} \end{pmatrix}$$

  ($\det V = \frac{5}{4} - \frac{1}{4} = 1$). So $(K\beta^0)'V^{-1}K\beta^0 = 4\big(\frac{3}{2} + \frac{1}{2} + \frac{1}{2} + \frac{5}{6}\big) = \frac{40}{3}$, the same $\mathrm{SSE}_K - \mathrm{SSE}$, and $F = \frac{40}{3}\big/(2 \cdot \frac{4}{3}) = 5$.

- $\alpha_1 - \alpha_2$ is estimable with BLUE $\bar{y}_1 - \bar{y}_2 = 2$ and $a'Ga = \frac{5}{6}$ for $a = (0, 1, -1, 0)'$ (Examples 14.4 and 14.6); so $\mathrm{SE} = s\sqrt{5/6} = \sqrt{10/9} = \sqrt{10}/3$, and by Theorem 14.10(b) the interval is $\big[2 - t_{3,1-\alpha/2}\sqrt{10}/3,\ 2 + t_{3,1-\alpha/2}\sqrt{10}/3\big]$. It contains 0 exactly when $t_{3,1-\alpha/2} \geqslant 3 \cdot 2/\sqrt{10} = 6/\sqrt{10}$.

- Uncentered: $n\bar{y}^2 = 6 \cdot \frac{196}{9} = \frac{392}{3}$, and $y'y = 148 = \frac{392}{3} + \frac{40}{3} + 4 = R(\mu) + R(\alpha \mid \mu) + \mathrm{SSE}$.

Example 14.20 ([2, Exercise 10.3], [3, §8.3])**.** In a balanced one-way layout, $g \geqslant 2$ groups have $m \geqslant 2$ observations each, $n = gm$; x_{ij} is observation j of group i, and $x \in \mathbb{R}^n$ lists group 1 first, then group 2, and so on. With the indicators $d_1, \dots, d_g$ of the groups (Theorem 13.10), the group means are $\bar{x}_i = d_i'x/m$ and the overall mean is $\bar{x} = \mathbf{1}'x/n$. Each observation lies in exactly one group, so $d_i'd_l = 0$ for $i \neq l$, $d_i'd_i = m$ and $d_1 + \dots + d_g = \mathbf{1}_n$. Let $D = d_1d_1' + \dots + d_gd_g' = \operatorname{diag}(J_m, \dots, J_m)$ and

$$A_1 = \bar{J}_n, \qquad A_2 = \tfrac{1}{m}D - \bar{J}_n, \qquad A_3 = I_n - \tfrac{1}{m}D$$

- $\frac{1}{m}Dx = \sum_i \bar{x}_i d_i$ has $\bar{x}_i$ in the rows of group i: it is the between-group part of x, and $A_3x = x - \frac{1}{m}Dx$ is the within-group part (Example 14.17); $\frac{1}{m}D$ is the hat matrix H of Theorem 14.16(a) with $n_1 = \dots = n_g = m$. As $\sum_i \bar{x}_i = \mathbf{1}'x/m = g\bar{x}$,

$$x'A_1x = n\bar{x}^2, \qquad x'A_2x = m\sum_{i=1}^{g}\bar{x}_i^2 - n\bar{x}^2 = m\sum_{i=1}^{g}(\bar{x}_i - \bar{x})^2 = \mathrm{SS}_A$$

  and

$$x'A_3x = x'x - m\sum_{i=1}^{g}\bar{x}_i^2 = \sum_{i=1}^{g}\sum_{j=1}^{m}(x_{ij} - \bar{x}_i)^2 = \mathrm{SS}_E$$

  using Example 1.16 within each group. SS_A and SS_E are the sums of squares between and within the groups, and $A_1 + A_2 + A_3 = I_n$.

- $D^2 = \sum_{i,l} d_i(d_i'd_l)d_l' = mD$ and $D\mathbf{1} = m\mathbf{1}$, so $(\frac{1}{m}D)^2 = \frac{1}{m}D$ and $\frac{1}{m}D\bar{J}_n = \bar{J}_n = \bar{J}_n\frac{1}{m}D$. With $\bar{J}_n^2 = \bar{J}_n$ (Proposition 1.15), $A_2^2 = \frac{1}{m}D - 2\bar{J}_n + \bar{J}_n = A_2$ and $A_3^2 = I_n - \frac{2}{m}D + \frac{1}{m}D = A_3$: all three matrices are symmetric and idempotent. By Theorem 11.25, their ranks add up to n and $A_iA_j = 0$ for $i \neq j$; the ranks are the traces 1, $\frac{n}{m} - 1 = g - 1$ and $n - g$ (Theorem 6.29(c)).
- Let $x_{ij} = \mu_i + \varepsilon_{ij}$ with independent $\varepsilon_{ij} \sim N(0, \sigma^2)$ ($\sigma > 0$): $x \sim N_n(\mu, \sigma^2 I_n)$ with $\mu = \sum_i \mu_i d_i$ (Theorem 10.9(a)). By Theorem 11.26(a), $n\bar{x}^2$, SS_A and SS_E are independent; $\frac{1}{m}D\mu = \mu$ gives $A_3\mu = 0$, so $\mathrm{SS}_E/\sigma^2 \sim \chi^2_{n-g}$; and, computing $\mu'A_2\mu$ as $x'A_2x$ above, $\mathrm{SS}_A/\sigma^2 \sim \chi^2_{g-1}\Big(m\sum_i(\mu_i - \bar{\mu})^2/\sigma^2\Big)$ with $\bar{\mu} = (\mu_1 + \cdots + \mu_g)/g$.
- If $\mu_1 = \cdots = \mu_g$, then $\mathrm{SS}_A/\sigma^2 \sim \chi^2_{g-1}$, and by Definition 11.7

$$F = \frac{\mathrm{SS}_A/(g-1)}{\mathrm{SS}_E/(n-g)} \sim F_{g-1,\,n-g}$$

Exercises

Exercise 14.1 (cf. [2, §12.12.2])**.** Six plants of three varieties are grown under two fertilizer treatments, one plant for each combination, and follow the model $y_{ij} = \mu+\alpha_i+\gamma_j+\varepsilon_{ij}$ (variety i, treatment j) under (A1)–(A3), with $\beta = (\mu, \alpha_1, \alpha_2, \alpha_3, \gamma_1, \gamma_2)'$. With the observations ordered $y_{11}, y_{12}, y_{21}, y_{22}, y_{31}, y_{32}$,

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}, \qquad y = \begin{pmatrix} 7 \\ 3 \\ 3 \\ 3 \\ 2 \\ 0 \end{pmatrix}$$

(a) Row-reduce X, and find $\operatorname{rank} X$ and a basis of $\mathcal{N}(X)$. Check the basis against the linear relations among the columns of X.

(b) Which of $\alpha_1 - \alpha_2$, $\gamma_1 - \gamma_2$, $\mu + \alpha_1 + \gamma_1$, α_1 and μ are estimable?

(c) Check that $\hat{\beta} = (0, 5, 3, 1, 1, -1)'$ is a least-squares estimate, and find the estimates of the estimable functions in (b).

(d) Find $\hat{y}$, e, SSE and the unbiased estimate of σ^2 of Theorem 13.23.

Solution. (a) Row operations do not change the solutions of $Xw = 0$:

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \xrightarrow[R_5 \leftarrow R_5 - R_1]{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow[R_6 \leftarrow R_6 - R_1,\ R_6 \leftarrow R_6 - R_5]{R_4 \leftarrow R_4 - R_1,\ R_4 \leftarrow R_4 - R_3} \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix} \xrightarrow[R_6 \leftarrow R_6 - R_4]{R_2 \leftarrow R_2 + R_3} \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_5} \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_5 \leftarrow R_5 + R_4]{R_5 \leftarrow R_5 - R_2,\ R_5 \leftarrow R_5 + R_3} \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Multiplying rows 2 and 4 of the last matrix by -1 gives an echelon form with leading entries in columns $1, 2, 3, 5$; so $\operatorname{rank} X = 4$ (Remark 4.2), and in $Xw = 0$, $w = (w_1, \ldots, w_6)'$, the unknowns w_4 and w_6 are free. Back substitution: row 4 gives $w_5 = w_6$; row 3 gives $w_3 = w_4$; row 2 gives $w_2 = w_3 - w_5 + w_6 = w_4$; row 1 gives $w_1 = -w_2 - w_5 = -w_4 - w_6$. So the solutions are

$$w = w_4 z_1 + w_6 z_2, \qquad z_1 = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad z_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

and z_1, z_2, independent (entries 4 and 6 of $t_1 z_1 + t_2 z_2$ are t_1 and t_2), form a basis of $\mathcal{N}(X)$. Check: with the columns $c_1, \ldots, c_6$ of X, each plant has one variety and one treatment, so $c_2 + c_3 + c_4 = c_1$ and $c_5 + c_6 = c_1$, that is, $Xz_1 = 0$ and $Xz_2 = 0$ (Theorem 1.4(b)).

(b) By (a), $a'\beta$ is estimable if and only if $a'z_1 = -a_0 + a_1 + a_2 + a_3 = 0$ and $a'z_2 = -a_0 + a_4 + a_5 = 0$ (Theorem 14.2(a)), with $a = (a_0, a_1, \dots, a_5)'$.

- $\alpha_1 - \alpha_2$, $a = (0,1,-1,0,0,0)'$: $1 - 1 = 0$ and $0 = 0$; estimable.
- $\gamma_1 - \gamma_2$, $a = (0,0,0,0,1,-1)'$: $0 = 0$ and $1 - 1 = 0$; estimable.
- $\mu + \alpha_1 + \gamma_1$, $a = (1,1,0,0,1,0)'$: $-1 + 1 = 0$ and $-1 + 1 = 0$; estimable.
- α_1, $a = (0,1,0,0,0,0)'$: $a'z_1 = 1 \neq 0$; not estimable. μ, $a = (1,0,0,0,0,0)'$: $a'z_1 = -1 \neq 0$; not estimable.

(c) The (j,l) entry of $X'X$ is $c_j'c_l$ for the columns $c_1, \dots, c_6$ of X (Theorems 1.4(d) and 1.9), and $X'y$ holds the total, the three variety totals and the two treatment totals:

$$X'X = \begin{pmatrix} 6 & 2 & 2 & 2 & 3 & 3 \\ 2 & 2 & 0 & 0 & 1 & 1 \\ 2 & 0 & 2 & 0 & 1 & 1 \\ 2 & 0 & 0 & 2 & 1 & 1 \\ 3 & 1 & 1 & 1 & 3 & 0 \\ 3 & 1 & 1 & 1 & 0 & 3 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 18 \\ 10 \\ 6 \\ 2 \\ 12 \\ 6 \end{pmatrix}, \qquad X'X\hat\beta = \begin{pmatrix} 10+6+2+3-3 \\ 10+1-1 \\ 6+1-1 \\ 2+1-1 \\ 5+3+1+3 \\ 5+3+1-3 \end{pmatrix} = X'y$$

So $\hat\beta$ solves the normal equations and is a least-squares estimate (Theorem 13.3(a)). By Theorem 14.2(b), the estimates are $5 - 3 = 2$ for $\alpha_1 - \alpha_2$, $1 - (-1) = 2$ for $\gamma_1 - \gamma_2$, and $0 + 5 + 1 = 6$ for $\mu + \alpha_1 + \gamma_1$.

(d) Row by row, $\hat y = X\hat\beta = (5+1,\ 5-1,\ 3+1,\ 3-1,\ 1+1,\ 1-1)' = (6,4,4,2,2,0)'$, so $e = y - \hat y = (1,-1,-1,1,0,0)'$ and $\mathrm{SSE} = 1+1+1+1 = 4$. As $\operatorname{rank} X = 4$ by (a), $\hat\sigma^2 = 4/(6-4) = 2$. Check: $X'e = (1-1-1+1+0+0,\ 1-1,\ -1+1,\ 0+0,\ 1-1+0,\ -1+1+0)' = 0$.

Exercise 14.2. Four observations in $g = 3$ groups of sizes $2, 1, 1$ follow the one-way model of Theorem 13.10, $y_i = \mu + \alpha_l + \varepsilon_i$ for an observation i of group l, with $\beta = (\mu, \alpha_1, \alpha_2, \alpha_3)'$, under (A1)–(A3). Let $y = (3,5,7,2)'$ and

$$G_1 = \operatorname{diag}\!\left(0, \tfrac12, 1, 1\right), \qquad G_2 = \frac12 \begin{pmatrix} 2 & -2 & -2 & 0 \\ -2 & 3 & 2 & 0 \\ -2 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Write down X, $X'X$ and $X'y$, and check that G_1 and G_2 are g-inverses of $X'X$.

(b) Compute the least-squares estimates $b_1 = G_1X'y$ and $b_2 = G_2X'y$ and their expectations. Which functions of β do their entries estimate without bias?

(c) Find the estimate of $\alpha_1 - \alpha_2$ from b_1 and from b_2, and its variance $\sigma^2 a'Ga$ with $G = G_1$ and with $G = G_2$.

Solution. (a) The rows of X are $(1,1,0,0)$ twice, $(1,0,1,0)$ and $(1,0,0,1)$; as in Example 13.11,

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 4 & 2 & 1 & 1 \\ 2 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 17 \\ 8 \\ 7 \\ 2 \end{pmatrix}$$

(the total and the group totals). Multiplying the columns of $X'X$ by $0, \frac12, 1, 1$ (Theorem 1.4(a),(b)), and computing $X'XG_2$ column by column,

$$X'XG_1 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad X'XG_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & -1 & -1 & 0 \end{pmatrix}$$

By Theorem 1.4(c), row 1 of $(X'XG_1)X'X$ is the sum of rows $2,3,4$ of $X'X$, which is $(4,2,1,1)$, and rows $2,3,4$ are those of $X'X$; rows $1,2,3$ of $(X'XG_2)X'X$ are those of $X'X$, and row 4 is $(4,2,1,1)-(2,2,0,0)-(1,0,1,0)=(1,0,0,1)$. So $X'XG_1X'X=X'XG_2X'X=X'X$.

(b) By Theorems 13.2(c) and 13.3(a), both are least-squares estimates:

$$b_1=G_1X'y=\begin{pmatrix}0\\4\\7\\2\end{pmatrix},\qquad b_2=G_2X'y=\frac{1}{2}\begin{pmatrix}34-16-14\\-34+24+14\\-34+16+28\\0\end{pmatrix}=\begin{pmatrix}2\\2\\5\\0\end{pmatrix}$$

and $b_1-b_2=(-2,2,2,2)'=2(-1,1,1,1)'\in\mathcal{N}(X)$ (Theorem 13.10(b)). Multiplying the rows of $X'X$ by $0,\frac{1}{2},1,1$ (Theorem 1.4(c)), and computing $G_2X'X$ row by row,

$$G_1X'X=\begin{pmatrix}0&0&0&0\\1&1&0&0\\1&0&1&0\\1&0&0&1\end{pmatrix},\qquad G_2X'X=\begin{pmatrix}1&0&0&1\\0&1&0&-1\\0&0&1&-1\\0&0&0&0\end{pmatrix}$$

So, by Proposition 14.5(a), $\mathsf{E}(b_1)=(0,\ \mu+\alpha_1,\ \mu+\alpha_2,\ \mu+\alpha_3)'$: entries $2,3,4$ of b_1, the group means $4,7,2$, estimate the group means $\mu+\alpha_l$. And $\mathsf{E}(b_2)=(\mu+\alpha_3,\ \alpha_1-\alpha_3,\ \alpha_2-\alpha_3,\ 0)'$: b_2 estimates the mean of group 3 and the differences of groups 1, 2 from group 3.

(c) For $a=(0,1,-1,0)'$, $a'b_1=4-7=-3$ and $a'b_2=2-5=-3$, as Theorem 14.2(b) says. Further $a'G_1a=\frac{1}{2}+1=\frac{3}{2}$ and $a'G_2a=\frac{3}{2}-2\cdot1+2=\frac{3}{2}$: in both cases the estimate of $\alpha_1-\alpha_2$ has the variance $\frac{3}{2}\sigma^2$ (Theorem 14.7). The estimate is $\bar{y}_1-\bar{y}_2=c'y$ with $c=(\frac{1}{2},\frac{1}{2},-1,0)'$, and indeed $\sigma^2c'c=\sigma^2\left(\frac{1}{4}+\frac{1}{4}+1\right)=\frac{3}{2}\sigma^2$.

Exercise 14.3. Add to the design X of Example 13.7 the total of its two variables:

$$\tilde{X}=[X\ \ X_{\cdot1}+X_{\cdot2}]=\begin{pmatrix}1&2&3&5\\1&0&3&3\\1&2&1&3\\1&0&1&1\end{pmatrix},\qquad y=\begin{pmatrix}9\\3\\5\\3\end{pmatrix}$$

and consider the model $y=\tilde{X}\beta+\varepsilon$, $\beta=(\beta_0,\beta_1,\beta_2,\beta_3)'$, under (A1)–(A3).

(a) Row-reduce $\tilde{X}$ to its RREF, and find a basis of $\mathcal{N}(\tilde{X})$.

(b) Which of $\beta_1+\beta_3$, $\beta_1-\beta_2$, β_3 and β_0 are estimable?

(c) Check that $\hat{\beta}=(1,2,1,0)'$ is a least-squares estimate, and find the BLUEs of the estimable functions in (b).

(d) Check that $G=\left(\begin{smallmatrix}(X'X)^{-1}&0\\0&0\end{smallmatrix}\right)\in\mathbb{R}^{4\times4}$ is a g-inverse of $\tilde{X}'\tilde{X}$, and find the variances of these BLUEs. Compare with the linear unbiased estimator $\frac{1}{2}(y_1-y_2)$ of $\beta_1+\beta_3$.

Solution. (a)

$$\begin{pmatrix}1&2&3&5\\1&0&3&3\\1&2&1&3\\1&0&1&1\end{pmatrix}\xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-R_1,\ R_3\leftarrow R_3-R_1}\begin{pmatrix}1&2&3&5\\0&-2&0&-2\\0&0&-2&-2\\0&-2&-2&-4\end{pmatrix}\xrightarrow{R_4\leftarrow R_4-R_2,\ R_4\leftarrow R_4-R_3}\begin{pmatrix}1&2&3&5\\0&-2&0&-2\\0&0&-2&-2\\0&0&0&0\end{pmatrix}$$

$$\xrightarrow{R_2\leftarrow-\frac{1}{2}R_2,\ R_3\leftarrow-\frac{1}{2}R_3}\begin{pmatrix}1&2&3&5\\0&1&0&1\\0&0&1&1\\0&0&0&0\end{pmatrix}\xrightarrow{R_1\leftarrow R_1-2R_2,\ R_1\leftarrow R_1-3R_3}\begin{pmatrix}1&0&0&0\\0&1&0&1\\0&0&1&1\\0&0&0&0\end{pmatrix}$$

In $\tilde{X}w = 0$ the unknown w_4 is free, and $w_1 = 0$, $w_2 = -w_4$, $w_3 = -w_4$. So $z = (0, -1, -1, 1)'$ is a basis of $\mathcal{N}(\tilde{X})$, and $\operatorname{rank} \tilde{X} = 3$.

(b) By Theorem 14.2(a), $a'\beta$ is estimable if and only if $a'z = -a_1 - a_2 + a_3 = 0$, with $a = (a_0, a_1, a_2, a_3)'$. So $\beta_1 + \beta_3$ $(-1 + 1 = 0)$, $\beta_1 - \beta_2$ $(-1 + 1 = 0)$ and β_0 $(0 = 0)$ are estimable, and β_3 $(a'z = 1)$ is not.

(c) By Example 13.7 (the matrix $X'X$) and $\tilde{X}_{\cdot 3} = X(0, 1, 1)'$,

$$\tilde{X}'\tilde{X} = \begin{pmatrix} 4 & 4 & 8 & 12 \\ 4 & 8 & 8 & 16 \\ 8 & 8 & 20 & 28 \\ 12 & 16 & 28 & 44 \end{pmatrix}, \qquad \tilde{X}'y = \begin{pmatrix} 20 \\ 28 \\ 44 \\ 72 \end{pmatrix}, \qquad \tilde{X}'\tilde{X}\hat{\beta} = \begin{pmatrix} 4+8+8 \\ 4+16+8 \\ 8+16+20 \\ 12+32+28 \end{pmatrix} = \tilde{X}'y$$

so $\hat{\beta}$ is a least-squares estimate (Theorem 13.3(a)). By Theorem 14.7, the BLUEs have the values $\hat{\beta}_1 + \hat{\beta}_3 = 2$, $\hat{\beta}_1 - \hat{\beta}_2 = 1$ and $\hat{\beta}_0 = 1$.

(d) With

$$(X'X)^{-1} = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

(Example 13.7), rows $1, 2, 3$ of $\tilde{X}'\tilde{X}G$ are $(I_3\ 0)$, and row 4 is $\big((12, 16, 28)(X'X)^{-1},\ 0\big) = (0, 1, 1, 0)$:

$$G = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 & 0 \\ -1 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \tilde{X}'\tilde{X}G = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

So rows $1, 2, 3$ of $(\tilde{X}'\tilde{X}G)\tilde{X}'\tilde{X}$ are those of $\tilde{X}'\tilde{X}$, and row 4 is the sum of rows 2 and 3, $(12, 16, 28, 44)$ (Theorem 1.4(c)): $\tilde{X}'\tilde{X}G\tilde{X}'\tilde{X} = \tilde{X}'\tilde{X}$. By Theorem 14.7, the variances are $\sigma^2 a'Ga$:

- $\beta_1 + \beta_3$, $a = (0, 1, 0, 1)'$: $a'Ga = \frac{1}{4}$, variance $\frac{1}{4}\sigma^2$;
- $\beta_1 - \beta_2$, $a = (0, 1, -1, 0)'$: $a'Ga = \frac{1}{4}(1 - 0 + 1) = \frac{1}{2}$, variance $\frac{1}{2}\sigma^2$;
- β_0, $a = (1, 0, 0, 0)'$: $a'Ga = \frac{6}{4}$, variance $\frac{3}{2}\sigma^2$.

By (A1), $\mathsf{E}(y_1 - y_2) = (\tilde{x}_1 - \tilde{x}_2)'\beta = (0, 2, 0, 2)\beta = 2(\beta_1 + \beta_3)$, where $\tilde{x}_i'$ is row i of $\tilde{X}$; so $\frac{1}{2}(y_1 - y_2)$ is a linear unbiased estimator of $\beta_1 + \beta_3$, with variance $\frac{1}{4}(\sigma^2 + \sigma^2) = \frac{1}{2}\sigma^2 > \frac{1}{4}\sigma^2$ (Theorem 9.3(b), (A3)). Its value is $\frac{1}{2}(9 - 3) = 3$, against the BLUE 2.

Exercise 14.4 (cf. [3, §7.2])**.** In a two-way layout with one observation in each of four cells, the additive model $y_{ij} = \alpha_i + \gamma_j + \varepsilon_{ij}$ $(i, j = 1, 2)$ reads $y = X\beta + \varepsilon$ with $y = (y_{11}, y_{12}, y_{21}, y_{22})'$, $\beta = (\alpha_1, \alpha_2, \gamma_1, \gamma_2)'$ and

$$X = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad y = \begin{pmatrix} 5 \\ 3 \\ 7 \\ 1 \end{pmatrix}$$

(a) Find $\operatorname{rank} X$ by row reduction, $X'X$, and the g-inverse G of $X'X$ of Theorem 12.6(b) built from its submatrix in rows and columns $2, 3, 4$.

(b) Find $H = XGX'$ and check that $\operatorname{tr} H = \operatorname{rank} X$. Find the least-squares estimate $GX'y$, the fitted values, the residuals and $S(\hat{\beta})$.

(c) Which of $\alpha_1 - \alpha_2$, $\gamma_1 - \gamma_2$, $\alpha_1 + \gamma_1$ and α_1 have the same value $\lambda'\hat{\beta}$ for all least-squares estimates $\hat{\beta}$? Give these values, and $\lambda'G\lambda$ for $\alpha_1 - \alpha_2$ and $\gamma_1 - \gamma_2$.

Solution. (a)

$$X \xrightarrow{R_2\leftrightarrow R_3} \begin{pmatrix}1&0&1&0\\0&1&1&0\\1&0&0&1\\0&1&0&1\end{pmatrix} \xrightarrow[R_4\leftarrow R_4-R_2]{R_3\leftarrow R_3-R_1} \begin{pmatrix}1&0&1&0\\0&1&1&0\\0&0&-1&1\\0&0&-1&1\end{pmatrix}$$

$$\xrightarrow{R_4\leftarrow R_4-R_3} \begin{pmatrix}1&0&1&0\\0&1&1&0\\0&0&-1&1\\0&0&0&0\end{pmatrix} \xrightarrow{R_3\leftarrow -R_3} \begin{pmatrix}1&0&1&0\\0&1&1&0\\0&0&1&-1\\0&0&0&0\end{pmatrix}$$

The last matrix is an echelon form with 3 pivots, so $\operatorname{rank} X = 3$ (Definition 4.11). The entries of $X'X$ are the inner products of the columns of X, and $\operatorname{rank}(X'X) = 3$ (Theorem 4.18(b)):

$$X'X = \begin{pmatrix}2&0&1&1\\0&2&1&1\\1&1&2&0\\1&1&0&2\end{pmatrix}, \qquad W = \begin{pmatrix}2&1&1\\1&2&0\\1&0&2\end{pmatrix}, \qquad \operatorname{adj} W = \begin{pmatrix}4&-2&-2\\-2&3&1\\-2&1&3\end{pmatrix}$$

where W is the submatrix in rows and columns $2,3,4$, with $\det W = 1\cdot(0-2)+2\cdot(4-1) = 4$ (expansion along row 3, Theorem 5.21). So $W^{-1} = \frac{1}{4}\operatorname{adj} W$ (Theorem 5.24), and by Theorem 12.6(b)

$$G = \frac{1}{4}\begin{pmatrix}0&0&0&0\\0&4&-2&-2\\0&-2&3&1\\0&-2&1&3\end{pmatrix}$$

(b)

- The rows of $4XG$ are sums of two rows of $4G$ (Theorem 1.4(c)), and then

$$4XG = \begin{pmatrix}0&-2&3&1\\0&-2&1&3\\0&2&1&-1\\0&2&-1&1\end{pmatrix}, \qquad H = XGX' = \frac{1}{4}\begin{pmatrix}3&1&1&-1\\1&3&-1&1\\1&-1&3&1\\-1&1&1&3\end{pmatrix}$$

  and $\operatorname{tr} H = \frac{1}{4}(3+3+3+3) = 3 = \operatorname{rank} X$ (Theorem 13.4(b)).

- $X'y = (5+3,\ 7+1,\ 5+7,\ 3+1)' = (8,8,12,4)'$, and

$$\hat\beta = GX'y = \frac{1}{4}\begin{pmatrix}0\\32-24-8\\-16+36+4\\-16+12+12\end{pmatrix} = \begin{pmatrix}0\\0\\6\\2\end{pmatrix}$$

  is a least-squares estimate (Theorems 13.2(c) and 13.3(a)).

- $\hat y = X\hat\beta = (\hat\alpha_1+\hat\gamma_1,\ \hat\alpha_1+\hat\gamma_2,\ \hat\alpha_2+\hat\gamma_1,\ \hat\alpha_2+\hat\gamma_2)' = (6,2,6,2)'$; indeed $Hy = \frac{1}{4}(15+3+7-1,\ 5+9-7+1,\ 5-3+21+1,\ -5+3+7+3)' = (6,2,6,2)'$.

- $e = y-\hat y = (-1,1,1,-1)'$ and $S(\hat\beta) = e'e = 4$; check (Theorem 13.4(d)): $y'y - \hat\beta' X'y = (25+9+49+1) - (0+0+72+8) = 84-80 = 4$.

(c)

- By Theorem 1.4(c), $4\operatorname{row}_2(GX'X) = 4(0,2,1,1) - 2(1,1,2,0) - 2(1,1,0,2) = (-4,4,0,0)$, and likewise for rows 3, 4:

$$GX'X = \begin{pmatrix}0&0&0&0\\-1&1&0&0\\1&0&1&0\\1&0&0&1\end{pmatrix}$$

- By Corollary 14.3(iii): $\alpha_1-\alpha_2$ passes, as $\mathrm{row}_1(GX'X)-\mathrm{row}_2(GX'X)=(1,-1,0,0)$, with value $\hat\alpha_1-\hat\alpha_2=0$; $\gamma_1-\gamma_2$ passes ($\mathrm{row}_3-\mathrm{row}_4=(0,0,1,-1)$), with value $6-2=4$; $\alpha_1+\gamma_1$ passes ($\mathrm{row}_1+\mathrm{row}_3=(1,0,1,0)$), with value $0+6=6$; α_1 fails, as $\mathrm{row}_1(GX'X)=0\neq(1,0,0,0)$.
- Indeed $I_4-GX'X$ has first column $(1,1,-1,-1)'$ and its other columns 0, so the least-squares estimates are $(t,\ t,\ 6-t,\ 2-t)'$ ($t\in\mathbb{R}$) (Theorem 13.2(c)): $\hat\alpha_1=t$ changes, the three other values do not.
- $\alpha_1-\alpha_2$, $\lambda=(1,-1,0,0)'$: $\lambda'G\lambda=0-0-0+1=1$; $\gamma_1-\gamma_2$, $\lambda=(0,0,1,-1)'$: $\lambda'G\lambda=\frac14(3-1-1+3)=1$. Both values are the same for every g-inverse of $X'X$ (Corollary 12.18 with $A=X$).

Exercise 14.5 ([3, Ch. 7, Exercise 2])**.** Consider the model $\mathsf{E}(y_1)=2\beta_1-\beta_2-\beta_3$, $\mathsf{E}(y_2)=\beta_2-\beta_4$, $\mathsf{E}(y_3)=\beta_2+\beta_3-2\beta_4$ under (A1)–(A3), with $\beta=(\beta_1,\beta_2,\beta_3,\beta_4)'$. Determine the estimable functions.

Solution. The model is $\mathsf{E}(y)=X\beta$ with

$$X=\begin{pmatrix}2&-1&-1&0\\0&1&0&-1\\0&1&1&-2\end{pmatrix}\xrightarrow{R_3\leftarrow R_3-R_2}\begin{pmatrix}2&-1&-1&0\\0&1&0&-1\\0&0&1&-1\end{pmatrix}$$
$$\xrightarrow{R_1\leftarrow R_1+R_2,\ R_1\leftarrow R_1+R_3}\begin{pmatrix}2&0&0&-2\\0&1&0&-1\\0&0&1&-1\end{pmatrix}\xrightarrow{R_1\leftarrow\frac12R_1}\begin{pmatrix}1&0&0&-1\\0&1&0&-1\\0&0&1&-1\end{pmatrix}$$

The last matrix is the RREF of X: $\operatorname{rank}X=3$, and in $Xw=0$ the unknown w_4 is free, with $w_1=w_2=w_3=w_4$. So $\mathcal{N}(X)=\operatorname{span}\{(1,1,1,1)'\}$; check: $X(1,1,1,1)'=(2-1-1,\ 1-1,\ 1+1-2)'=0$. By Theorem 14.2(a), $a'\beta=a_1\beta_1+a_2\beta_2+a_3\beta_3+a_4\beta_4$ is estimable if and only if $a\perp(1,1,1,1)'$, that is, $a_1+a_2+a_3+a_4=0$. For instance $\beta_1-\beta_2$ is estimable, and β_1 is not.

Exercise 14.6. For the four units of Example 13.7, consider $H_0\colon\beta_1=\beta_2$, that is, $K\beta=0$ with $K=(0,1,-1)$.

(a) Use Theorem 14.12 to compute V, $\tilde\beta$ and $\mathrm{SSE}_K-\mathrm{SSE}$, and check SSE_K from the residuals $y-X\tilde\beta$.

(b) Assuming the normal linear model, compute the statistic F of Theorem 14.13 and check $F=T_0^2$.

Solution. (a) X has full column rank, so H_0 is testable (Definition 14.11) and $G=(X'X)^{-1}$. With $(X'X)^{-1}$ and $\hat\beta=(1,2,1)'$ of Example 13.7, $(X'X)^{-1}K'=\frac14(1,1,-1)'$, $V=K(X'X)^{-1}K'=\frac14(1+1)=\frac12$ and $K\hat\beta=2-1=1$. So

$$\tilde\beta=\begin{pmatrix}1\\2\\1\end{pmatrix}-\frac14\begin{pmatrix}1\\1\\-1\end{pmatrix}\cdot2\cdot1=\begin{pmatrix}1/2\\3/2\\3/2\end{pmatrix},\qquad\mathrm{SSE}_K-\mathrm{SSE}=1\cdot2\cdot1=2$$

and $\mathrm{SSE}_K=4+2=6$. Check: $K\tilde\beta=\frac32-\frac32=0$, $X\tilde\beta=(8,5,5,2)'$, $y-X\tilde\beta=(1,-2,0,1)'$ and $1+4+0+1=6$.

(b) $n-r=1$, so $F=(2/1)/(4/1)=\frac12$. With $a=(0,1,-1)'$, $\mathrm{SE}(a'\hat\beta)=s\sqrt{a'(X'X)^{-1}a}=2\sqrt{1/2}=\sqrt2$ and $T_0=1/\sqrt2$, so $T_0^2=\frac12=F$ (Theorem 14.13(d)).

Exercise 14.7 ([3, Ch. 7, Exercise 12])**.** Consider the model $\mathsf{E}(y_1) = \beta_1 + 2\beta_2$, $\mathsf{E}(y_2) = 2\beta_1$, $\mathsf{E}(y_3) = \beta_1 + \beta_2$ under (A1)–(A3), with $\beta = (\beta_1, \beta_2)'$. Find the residual sum of squares subject to the restriction $\beta_1 = \beta_2$, that is, SSE_K of Theorem 14.12 for $K = (1, -1)$ and $m = 0$, as a function of y.

Solution.

- The design and the matrices of Theorem 14.12 are

$$X = \begin{pmatrix} 1 & 2 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 6 & 3 \\ 3 & 5 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{21}\begin{pmatrix} 5 & -3 \\ -3 & 6 \end{pmatrix}, \qquad X'y = \begin{pmatrix} y_1 + 2y_2 + y_3 \\ 2y_1 + y_3 \end{pmatrix}$$

  with $\det(X'X) = 30 - 9 = 21$; indeed $X'X \cdot \frac{1}{21}\begin{pmatrix} 5 & -3 \\ -3 & 6 \end{pmatrix} = \frac{1}{21}\begin{pmatrix} 30-9 & -18+18 \\ 15-15 & -9+30 \end{pmatrix} = I_2$. So $\operatorname{rank} X = 2$, $H_0\colon \beta_1 = \beta_2$ is testable (Definition 14.11) and $G = (X'X)^{-1}$.

- $\hat\beta = (X'X)^{-1}X'y = \frac{1}{21}(-y_1 + 10y_2 + 2y_3,\ 9y_1 - 6y_2 + 3y_3)'$, so $K\hat\beta = \frac{1}{21}(-10y_1 + 16y_2 - y_3)$, and $V = K(X'X)^{-1}K' = \frac{1}{21}(5 + 3 + 3 + 6) = \frac{17}{21}$.
- $w = (2, 1, -4)'$ has $X'w = (2 + 2 - 4,\ 4 - 4)' = 0$, and $\dim \mathcal{C}(X)^\perp = 3 - 2 = 1$ (Theorem 6.6(c)); so $\mathcal{C}(X)^\perp = \operatorname{span}\{w\}$. The residual vector e lies in it (Theorem 13.3(b)), $e = tw$, and $w'y = w'(\hat y + e) = w'e = 21t$. Hence $\mathrm{SSE} = e'e = 21t^2 = (2y_1 + y_2 - 4y_3)^2/21$.
- By Theorem 14.12(c),

$$\begin{aligned} \mathrm{SSE}_K &= \mathrm{SSE} + \frac{(K\hat\beta)^2}{V} = \frac{17(2y_1 + y_2 - 4y_3)^2 + (-10y_1 + 16y_2 - y_3)^2}{357} \\ &= \frac{8y_1^2 + 13y_2^2 + 13y_3^2 - 12y_1y_2 - 12y_1y_3 - 8y_2y_3}{17} \end{aligned}$$

  since the numerator of the middle fraction is $168y_1^2 + 273y_2^2 + 273y_3^2 - 252y_1y_2 - 252y_1y_3 - 168y_2y_3$, which is 21 times that of the last.

Exercise 14.8. (a) A regression with intercept and $k = 4$ further columns, of full column rank, on $n = 105$ observations has overall $F = 20$. Find R^2.

(b) A regression with intercept and 6 further columns, of full column rank, on $n = 40$ observations has $R^2 = \frac{1}{2}$; without three of these columns, $R^2 = \frac{2}{5}$. Compute the statistic F for the hypothesis that the three coefficients are 0, and state when the test of level α rejects it.

Solution. (a) By Theorem 14.13(c) (overall F test), $F(1 - R^2)/(n - k - 1) = R^2/k$, so $kF(1 - R^2) = (n - k - 1)R^2$ and $R^2 = kF/(kF + n - k - 1) = 80/180 = \frac{4}{9}$.

(b) $r = 7$, $n - r = 33$, $q = 3$; under (A4) the hypothesis is testable, and by Theorem 14.13(c), $F = \frac{(1/2 - 2/5)/3}{(1-1/2)/33} = \frac{1/30}{1/66} = \frac{11}{5}$. The test rejects if and only if $\frac{11}{5} > F_{3,33,1-\alpha}$ (Theorem 14.13(b)).

Exercise 14.9. Four plants given a new fertilizer have heights $12, 15, 13, 16$, and five control plants $10, 9, 12, 11, 8$. Model them as the one-way layout of Theorem 13.10 with $g = 2$ groups ($n_1 = 4$, $n_2 = 5$), the design $X_1 = [d_1\ d_2]$ and $\beta = (\mu_1, \mu_2)'$, in the normal linear model. Compute the statistic T_0 of Theorem 14.10(b) for $H_0\colon \mu_1 = \mu_2$ and the confidence interval of level $1 - \alpha$ for $\mu_1 - \mu_2$, and state when the test of level α rejects H_0.

Solution.

- $X_1'X_1 = \operatorname{diag}(4, 5)$ and $X_1'y = (56, 50)'$ (the group totals), so $\hat\beta = (14, 10)'$, the group means. The residuals are the deviations from the group means, $-2, 1, -1, 2$ and $0, -1, 2, 1, -2$, so $\mathrm{SSE} = (4 + 1 + 1 + 4) + (0 + 1 + 4 + 1 + 4) = 20$. Here $r = 2$, $n - r = 7$ and $s^2 = \frac{20}{7}$.

- For $a = (1,-1)'$, $a'\hat\beta = 4$ and $a'(X_1'X_1)^{-1}a = \frac14 + \frac15 = \frac{9}{20}$, so $\mathrm{SE}(a'\hat\beta) = \sqrt{\frac{20}{7}\cdot\frac{9}{20}} = \frac{3}{\sqrt7}$, and $T_0 = 4\Big/\frac{3}{\sqrt7} = \frac{4\sqrt7}{3}$.
- By Theorem 14.10(b), the interval is $\left[4 - \frac{3}{\sqrt7}t_{7,1-\alpha/2},\ 4 + \frac{3}{\sqrt7}t_{7,1-\alpha/2}\right]$, and the test rejects H_0 if and only if $t_{7,1-\alpha/2} < \frac{4\sqrt7}{3}$, that is, exactly when 0 lies outside the interval.

Exercise 14.10. Six units form three groups, $\{1,2\}$, $\{3,4\}$ and $\{5,6\}$, and $y = (1,3,6,8,4,2)'$ is measured on them. Let $H \in \mathbb{R}^{6\times6}$ be the matrix that replaces each y_i by the mean of its group (the hat matrix of the one-way design, Theorem 14.16(a)).

(a) Write down H and compute Hy and $\bar J_6 y$.

(b) Compute C_6y, $(H-\bar J_6)y$ and $(I_6-H)y$, and check that $C_6y = (H-\bar J_6)y + (I_6-H)y$.

(c) Compute $y'C_6y$, $y'(H-\bar J_6)y$ and $y'(I_6-H)y$, and check that the first is the sum of the other two.

Solution. (a) Row i of H averages the two entries of the group of unit i:

$$H = \tfrac12\begin{pmatrix}1&1&0&0&0&0\\1&1&0&0&0&0\\0&0&1&1&0&0\\0&0&1&1&0&0\\0&0&0&0&1&1\\0&0&0&0&1&1\end{pmatrix}, \qquad Hy = \begin{pmatrix}2\\2\\7\\7\\3\\3\end{pmatrix}, \qquad \bar J_6 y = \bar y\mathbf{1}_6 = 4\,\mathbf{1}_6$$

with the group means 2, 7, 3 and the mean $\bar y = \frac{24}{6} = 4$.

(b) By Example 1.16, $C_6y = y - 4\,\mathbf{1}_6$; further $(H-\bar J_6)y = Hy - 4\,\mathbf{1}_6$ and $(I_6-H)y = y - Hy$:

$$C_6y = \begin{pmatrix}-3\\-1\\2\\4\\0\\-2\end{pmatrix}, \qquad (H-\bar J_6)y = \begin{pmatrix}-2\\-2\\3\\3\\-1\\-1\end{pmatrix}, \qquad (I_6-H)y = \begin{pmatrix}-1\\1\\-1\\1\\1\\-1\end{pmatrix}$$

and the last two add up to C_6y.

(c) H is symmetric, every row and every column of H adds up to 1, and $H^2 = H$, since each row of H^2 is the average of two equal rows of H (Theorem 1.4(c)). So $H\bar J_6 = \frac16(H\mathbf{1})\mathbf{1}' = \bar J_6 = \bar J_6 H$, and with $\bar J_6^2 = \bar J_6$ (Proposition 1.15), $(H-\bar J_6)^2 = H - \bar J_6$; C_6 and $I_6 - H$ are symmetric and idempotent (Propositions 1.15 and 1.29(b)). For a symmetric idempotent P, $y'Py = (Py)'(Py)$ is the sum of the squares of the entries of Py:

$$y'C_6y = 9+1+4+16+0+4 = 34, \qquad y'(H-\bar J_6)y = 4+4+9+9+1+1 = 28, \qquad y'(I_6-H)y = 6$$

and $34 = 28 + 6$, as Theorem 14.16(b) says.

Exercise 14.11. Five observations form three groups: observations 1, 2 (group 1), observations 3, 4 (group 2) and observation 5 (group 3), with the indicators $d_1 = (1,1,0,0,0)'$, $d_2 = (0,0,1,1,0)'$ and $d_3 = (0,0,0,0,1)'$; let $U = \text{span}\{d_1, d_2, d_3\}$ and $y = (3,5,4,8,2)'$.

(a) Find the between-group part u and the within-group part $y - u$ of y, as in Example 14.17, and check that the entries of $y - u$ add up to 0 within each group.

(b) Find the coordinate vectors of u relative to $\mathcal{B} = (d_1, d_2, d_3)$ and to $\tilde{\mathcal{B}} = (\mathbf{1}, d_2, d_3)$.

(c) Find the change-of-basis matrix $\hat{P}$ from $\hat{\mathcal{B}} = (\mathbf{1}, d_1, d_2)$, in which group 3 is the baseline, to $\mathcal{B}$; check that

$$\hat{Q} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}$$

is its inverse, and find $[u]_{\hat{\mathcal{B}}}$.

Solution. (a) The group means of y are $(3+5)/2 = 4$, $(4+8)/2 = 6$ and $2/1 = 2$. So $u = 4d_1 + 6d_2 + 2d_3 = (4,4,6,6,2)'$ and $y - u = (-1,1,-2,2,0)'$; within the groups, $-1+1 = 0$, $-2+2 = 0$ and $0 = 0$.

(b) d_1, d_2, d_3 are the columns of the design X_1 of Theorem 13.10 for this layout, which are linearly independent (Theorem 13.10(a)), so $\mathcal{B}$ is an ordered basis of U, and $[u]_{\mathcal{B}} = (4,6,2)'$. Since $\mathbf{1} = d_1 + d_2 + d_3$, the matrix with the columns $[\mathbf{1}]_{\mathcal{B}} = (1,1,1)'$, $[d_2]_{\mathcal{B}} = (0,1,0)'$, $[d_3]_{\mathcal{B}} = (0,0,1)'$ and its inverse are

$$T = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

as $\text{row}_1(TS) = \text{row}_1(S)$, $\text{row}_2(TS) = \text{row}_1(S) + \text{row}_2(S)$ and $\text{row}_3(TS) = \text{row}_1(S) + \text{row}_3(S)$ (Theorem 1.4(c)) give $TS = I_3$ (Theorem 2.16). So $\tilde{\mathcal{B}}$ is an ordered basis of U, and T is the change-of-basis matrix from $\tilde{\mathcal{B}}$ to $\mathcal{B}$ (Theorem 3.28); and

$$[u]_{\tilde{\mathcal{B}}} = S[u]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}$$

(Theorem 3.27(b)). Check: $4\mathbf{1} + 2d_2 - 2d_3 = (4,4,6,6,2)'$.

(c) $[\mathbf{1}]_{\mathcal{B}} = (1,1,1)'$, $[d_1]_{\mathcal{B}} = (1,0,0)'$ and $[d_2]_{\mathcal{B}} = (0,1,0)'$, so

$$\hat{P} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \hat{P}\hat{Q} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

So $\hat{P}$ is invertible with $\hat{P}^{-1} = \hat{Q}$ (Theorem 2.16), $\hat{\mathcal{B}}$ is an ordered basis of U (Theorem 3.28), and

$$[u]_{\hat{\mathcal{B}}} = \hat{Q}[u]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$$

the value 2 of group 3 and the differences $4-2$, $6-2$ of groups 1 and 2 from it. Check: $2\mathbf{1} + 2d_1 + 4d_2 = (4,4,6,6,2)'$.

Exercise 14.12. In $\mathbb{R}^4$ let $U = \text{span}\{\mathbf{1}_4\}$, $W = \{x : x_1 + x_2 = x_3 + x_4\}$ and $y = (5, 1, 2, 0)'$.

(a) Show that $U \subseteq W$, and find $W^\perp$, P_W and P_U.

(b) Check that $P_W P_U = P_U$.

(c) Find an orthonormal basis of $W \cap U^\perp$, and check that the projection matrix onto $W \cap U^\perp$ is $P_W - P_U$.

(d) Write $y = P_U y + (P_W - P_U)y + (I_4 - P_W)y$, and check that the squared lengths of the three terms add up to $\|y\|^2$.

Solution. Let $h = (1, 1, -1, -1)'$, so that $W = \{x : h'x = 0\}$.

(a) $h'\mathbf{1}_4 = 0$, so $\mathbf{1}_4 \in W$ and $U \subseteq W$. By Theorem 6.6(a), $W = \mathcal{N}(h') = \text{span}\{h\}^\perp$, so $W^\perp = \text{span}\{h\}$ (Theorem 6.8(a)). By Theorem 6.31(c) for $\text{span}\{h\}$, and by Example 6.27 and Theorem 6.31(b) for $P_{\text{span}\{h\}}$, $P_W = I_4 - P_{\text{span}\{h\}} = I_4 - \frac{1}{4}hh'$; and $P_U = \bar{J}_4$ (Corollary 6.35(a)):

$$P_W = \frac{1}{4}\begin{pmatrix} 3 & -1 & 1 & 1 \\ -1 & 3 & 1 & 1 \\ 1 & 1 & 3 & -1 \\ 1 & 1 & -1 & 3 \end{pmatrix}, \qquad P_U = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

(b) Every row of P_W has sum 1, that is, $P_W\mathbf{1} = \mathbf{1}$ (as $\mathbf{1} \in W$). So $P_W P_U = \frac{1}{4}(P_W\mathbf{1})\mathbf{1}' = \frac{1}{4}\mathbf{1}\mathbf{1}' = P_U$, as Theorem 6.36(a) says.

(c) $x \in W \cap U^\perp$ means $x_1 + x_2 = x_3 + x_4$ and $x_1 + x_2 + x_3 + x_4 = 0$, that is, $x_1 + x_2 = 0 = x_3 + x_4$. So $W \cap U^\perp = \{(s, -s, t, -t)' : s, t \in \mathbb{R}\}$, with the orthonormal basis $u_1 = \frac{1}{\sqrt{2}}(1, -1, 0, 0)'$, $u_2 = \frac{1}{\sqrt{2}}(0, 0, 1, -1)'$. By Theorem 6.31(a),

$$P_{W\cap U^\perp} = u_1u_1' + u_2u_2' = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

which is $P_W - P_U$, as Theorem 6.36(b) says; for instance, row 1 of $P_W - P_U$ is $\frac{1}{4}(3-1,\ -1-1,\ 1-1,\ 1-1) = \frac{1}{2}(1, -1, 0, 0)$. The dimension of $W \cap U^\perp$ is $2 = 3 - 1$.

(d) $P_U y = \bar{y}\mathbf{1}_4 = 2\,\mathbf{1}_4$; $h'y = 5 + 1 - 2 - 0 = 4$, so $(I_4 - P_W)y = P_{W^\perp}y = \frac{h'y}{h'h}h = h$ and $P_W y = y - h = (4, 0, 3, 1)'$; and $(P_W - P_U)y = (2, -2, 1, -1)'$. So

$$\begin{pmatrix} 5 \\ 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ -2 \\ 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \qquad 30 = 16 + 10 + 4$$

as Theorem 6.36(c) says; also $\|P_W y\|^2 = 16 + 0 + 9 + 1 = 26 = 16 + 10$.

Exercise 14.13. Let $y \sim N_4(\mu, \sigma^2 I_4)$ with $\sigma > 0$, and

$$P = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

(a) Show that P and $I_4 - P$ are symmetric and idempotent of rank 2, that $y'Py = \frac{1}{2}(y_1 + y_2)^2 + \frac{1}{2}(y_3 + y_4)^2$, and that $y'(I_4 - P)y = \frac{1}{2}(y_1 - y_2)^2 + \frac{1}{2}(y_3 - y_4)^2$.

(b) Find $\mathsf{E}(y'Py)$ and $\mathsf{E}\big(y'(I_4-P)y\big)$ for $\mu=(1,3,5,7)'$.

(c) Let $\mu=(m_1,m_1,m_2,m_2)'$. Show that $y'(I_4-P)y/\sigma^2\sim\chi^2_2$, that $y'(I_4-P)y$ is independent of Py, and that $\frac12 y'(I_4-P)y$ is an unbiased estimator of σ^2.

Solution. (a) $P=\frac12\operatorname{diag}(J_2,J_2)$ is symmetric, and $J_2^2=2J_2$ (Proposition 1.15), so $P^2=\frac14\operatorname{diag}(J_2^2,J_2^2)=\frac12\operatorname{diag}(J_2,J_2)=P$ (Theorem 5.3(c)). By Theorem 6.29(b),(c), I_4-P is idempotent, $\operatorname{rank}P=\operatorname{tr}P=2$ and $\operatorname{rank}(I_4-P)=\operatorname{tr}(I_4-P)=2$. As $Py=\frac12(y_1+y_2,\ y_1+y_2,\ y_3+y_4,\ y_3+y_4)'$, $y'Py=\frac12(y_1+y_2)^2+\frac12(y_3+y_4)^2$; and $y'(I_4-P)y=y'y-y'Py$, where $y_1^2+y_2^2-\frac12(y_1+y_2)^2=\frac12(y_1-y_2)^2$, and likewise for y_3,y_4.

(b) By Theorem 9.10, with $\operatorname{tr}(P\sigma^2I_4)=\sigma^2\operatorname{tr}P=2\sigma^2$, $\mathsf{E}(y'Py)=2\sigma^2+\mu'P\mu=2\sigma^2+\frac12\cdot4^2+\frac12\cdot12^2=2\sigma^2+80$ and $\mathsf{E}\big(y'(I_4-P)y\big)=2\sigma^2+\frac12\cdot2^2+\frac12\cdot2^2=2\sigma^2+4$.

(c) Let $\varepsilon=y-\mu\sim N_4(0,\sigma^2I_4)$ (Theorem 10.2). Here $(I_4-P)\mu=0$, so $(I_4-P)y=(I_4-P)\varepsilon$ and, as I_4-P is symmetric and idempotent, $y'(I_4-P)y=\|(I_4-P)y\|^2=\varepsilon'(I_4-P)\varepsilon$. Theorem 11.15 gives $y'(I_4-P)y/\sigma^2\sim\chi^2_2$. As $P(I_4-P)=P-P^2=0$, Theorem 11.20(b) gives that $P\varepsilon$ and $\varepsilon'(I_4-P)\varepsilon$ are independent, and $Py=P\mu+P\varepsilon$ is a function of $P\varepsilon$ (N3). $\mathsf{E}\big(y'(I_4-P)y\big)=2\sigma^2+0$ (Theorem 9.10), so $\frac12y'(I_4-P)y$ is unbiased for σ^2 for all values of m_1,m_2,σ^2 (Definition 9.11).

Exercise 14.14. In the one-way layout of Example 14.20 with $g=3$ groups of $m=2$ observations, let $x=(5,7,2,4,8,10)'$.

(a) Compute the group means, $\bar x$, SS_A and SS_E.

(b) Write out A_2 and A_3, compute A_2x and A_3x, and check that $x'A_2x=\mathrm{SS}_A$ and $x'A_3x=\mathrm{SS}_E$.

(c) Compute $F=\big(\mathrm{SS}_A/(g-1)\big)/\big(\mathrm{SS}_E/(n-g)\big)$, and state when the test of level α that rejects $H_0\colon\mu_1=\mu_2=\mu_3$ for large F rejects it.

Solution. (a) The group means are 6, 3 and 9, and $\bar x=36/6=6$. So $\mathrm{SS}_A=2\big(0^2+(-3)^2+3^2\big)=36$ and $\mathrm{SS}_E=1+1+1+1+1+1=6$.

(b) Here $\frac12D=\operatorname{diag}(\frac12J_2,\frac12J_2,\frac12J_2)$, and

$$A_2=\frac16\begin{pmatrix}2&2&-1&-1&-1&-1\\2&2&-1&-1&-1&-1\\-1&-1&2&2&-1&-1\\-1&-1&2&2&-1&-1\\-1&-1&-1&-1&2&2\\-1&-1&-1&-1&2&2\end{pmatrix},\qquad A_3=\frac12\begin{pmatrix}1&-1&0&0&0&0\\-1&1&0&0&0&0\\0&0&1&-1&0&0\\0&0&-1&1&0&0\\0&0&0&0&1&-1\\0&0&0&0&-1&1\end{pmatrix}$$

$\frac12Dx=(6,6,3,3,9,9)'$ and $\bar J_6x=6\mathbf{1}$, so $A_2x=(0,0,-3,-3,3,3)'$ and $A_3x=x-\frac12Dx=(-1,1,-1,1,-1,1)'$. Then $x'A_2x=0+0-6-12+24+30=36$ and $x'A_3x=-5+7-2+4-8+10=6$.

(c) $F=(36/2)/(6/3)=9$. If the three group means μ_i are equal, then $F\sim F_{2,3}$ (Example 14.20), and the test that rejects when $F>F_{2,3,1-\alpha}$ has level α (Definition 11.7(c)): it rejects H_0 if and only if $F_{2,3,1-\alpha}<9$.

Exercise 14.15 (cf. [3, §8.3])**.** Four observations y_{ij} $(i,j=1,2)$ are ordered $y=(y_{11},y_{12},y_{21},y_{22})'$. Let $a=(1,1,-1,-1)'$, $b=(1,-1,1,-1)'$, $c=(1,-1,-1,1)'$, and $A_1=\bar J_4$, $A_2=\frac14aa'$, $A_3=\frac14bb'$, $A_4=\frac14cc'$.

(a) Check that $A_1+A_2+A_3+A_4=I_4$ and that the ranks add up to 4. What does Theorem 11.25 give?

(b) For $y=(7,3,4,2)'$, compute $y'A_iy$ $(i=1,\dots,4)$, and check that they add up to $y'y$.

(c) In the model $y_{ij} = \mu+\alpha_i+\gamma_j+\varepsilon_{ij}$ with independent $\varepsilon_{ij} \sim N(0,\sigma^2)$ ($\sigma > 0$), find the distributions of the $y'A_iy/\sigma^2$, and show that $(y'A_2y)/(y'A_4y) \sim F_{1,1}$ when $\alpha_1 = \alpha_2$. Compute its value for the data of (b).

Solution. (a) The vectors $\mathbf{1}, a, b, c$ have squared length 4, so each A_i has the form $\frac{1}{4}qq'$ with $q'q = 4$: it is symmetric, idempotent ($\frac{1}{16}q(q'q)q' = \frac{1}{4}qq'$) and of rank $\operatorname{tr} A_i = 1$ (Theorem 6.29(c)). The (k,l) entry of $4(A_1+A_2+A_3+A_4)$ is $1 + a_ka_l + b_kb_l + c_kc_l$, which is 4 for $k = l$; for $k \neq l$ it is 0: for $(1,2)$, $1+1-1-1$; for $(1,3)$, $1-1+1-1$; for $(1,4)$, $1-1-1+1$; for $(2,3)$, $1-1-1+1$; for $(2,4)$, $1-1+1-1$; for $(3,4)$, $1+1-1-1$. So the sum is I_4, and the ranks add up to 4. By Theorem 11.25, $A_iA_j = 0$ for $i \neq j$ (directly: $a'b = a'c = b'c = 0$, and $\mathbf{1}$ is orthogonal to a, b, c).

(b) $\mathbf{1}'y = 16$, $a'y = 7+3-4-2 = 4$, $b'y = 7-3+4-2 = 6$ and $c'y = 7-3-4+2 = 2$, so $y'A_1y = \frac{256}{4} = 64$, $y'A_2y = \frac{16}{4} = 4$, $y'A_3y = \frac{36}{4} = 9$ and $y'A_4y = \frac{4}{4} = 1$. Their sum is $78 = 49+9+16+4 = y'y$.

(c) $y \sim N_4(m, \sigma^2 I_4)$ with $m = (\mu+\alpha_1+\gamma_1,\ \mu+\alpha_1+\gamma_2,\ \mu+\alpha_2+\gamma_1,\ \mu+\alpha_2+\gamma_2)'$ (Theorem 10.9(a)). Here $a'm = 2(\alpha_1-\alpha_2)$, $b'm = 2(\gamma_1-\gamma_2)$ and $c'm = 0$. By Theorem 11.26(a), the four forms are independent, and $y'A_1y/\sigma^2 \sim \chi^2_1\big((\mathbf{1}'m)^2/(4\sigma^2)\big)$, $y'A_2y/\sigma^2 \sim \chi^2_1\big((\alpha_1-\alpha_2)^2/\sigma^2\big)$, $y'A_3y/\sigma^2 \sim \chi^2_1\big((\gamma_1-\gamma_2)^2/\sigma^2\big)$ and $y'A_4y/\sigma^2 \sim \chi^2_1$. If $\alpha_1 = \alpha_2$, then $y'A_2y/\sigma^2 \sim \chi^2_1$, and $(y'A_2y)/(y'A_4y) = (y'A_2y/\sigma^2)/(y'A_4y/\sigma^2) \sim F_{1,1}$ (Definition 11.7). For the data of (b) it is $4/1 = 4$.

Exercise 14.16. Let $\mathbf{1} = (1,1,1,1)'$, $x_A = (0,0,1,1)'$, $x_B = (0,1,0,1)'$, $x_{AB} = (0,0,0,1)'$ and $X = [\mathbf{1}\ x_A\ x_B\ x_{AB}]$.

(a) Row-reduce $[X \mid y]$ for $y = (y_1,y_2,y_3,y_4)'$, keeping the y_i as letters. Conclude that $\mathcal{B} = (\mathbf{1}, x_A, x_B, x_{AB})$ is an ordered basis of $\mathbb{R}^4$, and find $[y]_{\mathcal{B}}$.

(b) Find $[\mu]_{\mathcal{B}}$ for $\mu = (2,5,4,9)'$, and check the answer.

(c) Show that $\mathbb{R}^4 = \operatorname{span}\{\mathbf{1}, x_A\} \oplus \operatorname{span}\{x_B, x_{AB}\}$, and find the parts of μ in the two subspaces.

Solution. (a)

$$\left(\begin{array}{cccc|c} 1&0&0&0&y_1\\ 1&0&1&0&y_2\\ 1&1&0&0&y_3\\ 1&1&1&1&y_4 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-R_1]{R_2\leftarrow R_2-R_1,\ R_3\leftarrow R_3-R_1} \left(\begin{array}{cccc|c} 1&0&0&0&y_1\\ 0&0&1&0&y_2-y_1\\ 0&1&0&0&y_3-y_1\\ 0&1&1&1&y_4-y_1 \end{array}\right) \xrightarrow{R_2\leftrightarrow R_3} \left(\begin{array}{cccc|c} 1&0&0&0&y_1\\ 0&1&0&0&y_3-y_1\\ 0&0&1&0&y_2-y_1\\ 0&1&1&1&y_4-y_1 \end{array}\right)$$

$$\xrightarrow{R_4\leftarrow R_4-R_2} \left(\begin{array}{cccc|c} 1&0&0&0&y_1\\ 0&1&0&0&y_3-y_1\\ 0&0&1&0&y_2-y_1\\ 0&0&1&1&y_4-y_3 \end{array}\right) \xrightarrow{R_4\leftarrow R_4-R_3} \left(\begin{array}{cccc|c} 1&0&0&0&y_1\\ 0&1&0&0&y_3-y_1\\ 0&0&1&0&y_2-y_1\\ 0&0&0&1&y_1-y_2-y_3+y_4 \end{array}\right)$$

After the first step the entry of row 2 in column 2 is 0, so Step 2 interchanges rows 2 and 3. The left part has become I_4: X is invertible (Theorem 2.16), its columns form an ordered basis $\mathcal{B}$ of $\mathbb{R}^4$ (Example 3.19), and $Xc = y$ has the unique solution $c = [y]_{\mathcal{B}}$, read off the last column:

$$[y]_{\mathcal{B}} = \begin{pmatrix} y_1\\ y_3-y_1\\ y_2-y_1\\ y_1-y_2-y_3+y_4 \end{pmatrix}$$

(b) For $\mu = (2,5,4,9)'$: $[\mu]_{\mathcal{B}} = (2,\ 4-2,\ 5-2,\ 2-5-4+9)' = (2,2,3,2)'$. Check: $2\mathbf{1}+2x_A+3x_B+2x_{AB} = (2,2,2,2)'+(0,0,2,2)'+(0,3,0,3)'+(0,0,0,2)' = (2,5,4,9)'$.

(c) The four vectors are independent, so $\mathbf{1}, x_A$ and x_B, x_{AB} form bases of $U = \operatorname{span}\{\mathbf{1}, x_A\}$ and $W = \operatorname{span}\{x_B, x_{AB}\}$ (Proposition 3.13(b)), and the sum $U+W$ is direct (Theorem 3.33). Also $U+W = \operatorname{span}\{\mathbf{1}, x_A, x_B, x_{AB}\} = \mathbb{R}^4$ (Theorem 3.31(c)). So $\mathbb{R}^4 = U \oplus W$, and $\mu = u+w$ with

$$u = 2\mathbf{1}+2x_A = (2,2,4,4)' \in U, \qquad w = 3x_B+2x_{AB} = (0,3,0,5)' \in W$$

which is the only such decomposition (Theorem 3.32).

Exercise 14.17 ([2, §12.12.3, Example 12.10])**.** Six plants of three varieties are grown under two treatments: variety 1 has two plants under treatment 1, with yields $3, 7$, and one under treatment 2, with yield 8; variety 2 has one plant under each treatment, with yields 34 and 2; variety 3 has one plant, under treatment 1, with yield 12. The yield y_{ijk} of plant k of variety i under treatment j follows $y_{ijk} = \mu + \alpha_i + \gamma_j + \varepsilon_{ijk}$. In the order given, $y = (y_{111}, y_{112}, y_{121}, y_{211}, y_{221}, y_{311})' = (3, 7, 8, 34, 2, 12)'$, and $y = X\beta + \varepsilon$ with $\beta = (\mu, \alpha_1, \alpha_2, \alpha_3, \gamma_1, \gamma_2)'$ under (A1)–(A3).

(a) Write down X, $X'X$ and $X'y$, check that

$$G = \frac{1}{7}\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 2 & 4 & -4 & 0 \\ 0 & 2 & 5 & 3 & -3 & 0 \\ 0 & 4 & 3 & 13 & -6 & 0 \\ 0 & -4 & -3 & -6 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is a g-inverse of $X'X$, and compute $\beta^0 = GX'y$.

(b) Compute $GX'X$, and decide with Corollary 14.3(iii) which of $\alpha_1 - \alpha_2$, $\gamma_1 - \gamma_2$ and α_1 are estimable; give the BLUEs of the estimable ones.

(c) Compute $R(\mu)$, $R(\mu, \alpha)$, $R(\mu, \gamma)$, $R(\mu, \alpha, \gamma)$ and SSE (Definition 14.14), and the two splits of $y'y$: fitting the varieties before the treatments, and the treatments before the varieties.

(d) Assuming the normal linear model, compute the statistic F of Theorem 14.13 for $H_0\colon \gamma_1 = \gamma_2$, and state when the test of level α rejects it.

Solution. (a) The rows of X have a 1 in column 1, in the column of the variety and in the column of the treatment; the (j, l) entry of $X'X$ is $\mathrm{col}_j(X)'\,\mathrm{col}_l(X)$ (Theorems 1.4(d) and 1.9), and $X'y$ holds the total, the variety totals and the treatment totals:

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 6 & 3 & 2 & 1 & 4 & 2 \\ 3 & 3 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 4 & 2 & 1 & 1 & 4 & 0 \\ 2 & 1 & 1 & 0 & 0 & 2 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 66 \\ 18 \\ 36 \\ 12 \\ 56 \\ 10 \end{pmatrix}$$

Multiplying,

$$X'XG = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \end{pmatrix}$$

so, by Theorem 1.4(c), rows 2–5 of $X'XGX'X$ are rows 2–5 of $X'X$, row 1 is the sum of rows $2, 3, 4$, $(6, 3, 2, 1, 4, 2)$, and row 6 is the sum of rows $2, 3, 4$ minus row 5, $(6, 3, 2, 1, 4, 2) - (4, 2, 1, 1, 4, 0) = (2, 1, 1, 0, 0, 2)$: $X'XGX'X = X'X$. Then

$$\beta^0 = GX'y = \frac{1}{7}\begin{pmatrix} 0 \\ 90 + 72 + 48 - 224 \\ 36 + 180 + 36 - 168 \\ 72 + 108 + 156 - 336 \\ -72 - 108 - 72 + 336 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ 12 \\ 0 \\ 12 \\ 0 \end{pmatrix}$$

a least-squares estimate (Theorems 13.2(c) and 13.3(a)).

(b) Multiplying the rows of G by $X'X$,

$$GX'X = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

By Corollary 14.3(iii), $\lambda'\beta$ is estimable exactly when $\lambda'GX'X = \lambda'$, and $\lambda'GX'X$ is the combination of the rows of $GX'X$ with the entries of λ (Theorem 1.4(c)).

- $\alpha_1 - \alpha_2$, $\lambda = (0, 1, -1, 0, 0, 0)'$: $\text{row}_2(GX'X) - \text{row}_3(GX'X) = (0, 1, -1, 0, 0, 0) = \lambda'$; estimable, with BLUE $\lambda'\beta^0 = -2 - 12 = -14$ (Theorem 14.7).
- $\gamma_1 - \gamma_2$, $\lambda = (0, 0, 0, 0, 1, -1)'$: $\text{row}_5(GX'X) - \text{row}_6(GX'X) = (0, 0, 0, 0, 1, -1) = \lambda'$; estimable, with BLUE $\lambda'\beta^0 = 12 - 0 = 12$.
- α_1, $\lambda = (0, 1, 0, 0, 0, 0)'$: $\text{row}_2(GX'X) = (1, 1, 0, 0, 0, 1) \neq \lambda'$; not estimable.

(c)

- $\bar{y} = 66/6 = 11$, so $R(\mu) = n\bar{y}^2 = 6 \cdot 121 = 726$ (the paragraph after Definition 14.14).
- Fitting $\mu + \alpha_i$ is the one-way layout of the varieties, with group sizes $3, 2, 1$ and totals $18, 36, 12$; by Theorem 14.16(a), $R(\mu, \alpha) = \|Hy\|^2 = \sum_l n_l \bar{y}_l^2 = \frac{18^2}{3} + \frac{36^2}{2} + \frac{12^2}{1} = 108 + 648 + 144 = 900$.
- Fitting $\mu + \gamma_j$ is the one-way layout of the treatments, with design Z (columns $1, 5, 6$ of X), but Theorem 14.16(a) lists the observations group by group. Relabel the plants so that the four under treatment 1 come first: this replaces Z and y by PZ and Py for a permutation matrix P (Definition 5.40), and $P'P = I_6$ (Theorem 6.19(d)), so $(PZ)'PZ = Z'Z$ and $(PZ)'Py = Z'y$. So the normal equations $Z'Zb = Z'y$, their solutions $\hat{b}$ and the reduction $R(\mu, \gamma) = \hat{b}'Z'y$ (the paragraph after Definition 14.14) do not change. By Theorem 14.16(a) for the relabelled plants, with group sizes $4, 2$ and totals $56, 10$, $R(\mu, \gamma) = \frac{56^2}{4} + \frac{10^2}{2} = 784 + 50 = 834$.
- $R(\mu, \alpha, \gamma) = \beta^{0\prime}X'y = -2 \cdot 18 + 12 \cdot 36 + 12 \cdot 56 = -36 + 432 + 672 = 1068$ (Theorem 13.4(d)), and $y'y = 9 + 49 + 64 + 1156 + 4 + 144 = 1426$, so $\text{SSE} = 1426 - 1068 = 358$.
- Varieties first: $R(\alpha \mid \mu) = 900 - 726 = 174$, $R(\gamma \mid \mu, \alpha) = 1068 - 900 = 168$, and $1426 = 726 + 174 + 168 + 358$. Treatments first: $R(\gamma \mid \mu) = 834 - 726 = 108$, $R(\alpha \mid \mu, \gamma) = 1068 - 834 = 234$, and $1426 = 726 + 108 + 234 + 358$. The two splits differ because the numbers of plants in the cells are unequal.

(d)

- H_0 is $K\beta = 0$ with $K = (0, 0, 0, 0, 1, -1)$, estimable by (b) and of rank 1: testable (Definition 14.11).
- The β with $\gamma_1 = \gamma_2 = \gamma$ give $X\beta = (\mu + \gamma)\mathbf{1}_6 + \alpha_1 d_1 + \alpha_2 d_2 + \alpha_3 d_3$, where d_i is the indicator of variety i (the two treatment indicators add up to $\mathbf{1}_6$); these are all vectors of $\mathcal{C}([\mathbf{1}_6\ d_1\ d_2\ d_3])$. So, by Theorem 14.12(b), SSE_K is the error sum of squares of the varieties alone, $1426 - 900 = 526$, and $\text{SSE}_K - \text{SSE} = R(\gamma \mid \mu, \alpha) = 168$.
- $r = \text{rank}\, X = 4$. Column 1 of X is the sum of columns $2, 3, 4$ and also the sum of columns $5, 6$, so $(-1, 1, 1, 1, 0, 0)'$ and $(-1, 0, 0, 0, 1, 1)'$ lie in $\mathcal{N}(X)$ (Theorem 1.4(b)); they are independent (look at entries 2 and 5), so $\dim \mathcal{N}(X) \geqslant 2$ (Theorem 4.5(c)) and $r \leqslant 4$ (Theorem 4.12). Rows 2–5 of $GX'X$ in (b) lie in $\mathcal{R}(GX'X)$ and are independent (their entries 2–5 form I_4), so $\text{rank}(GX'X) \geqslant 4$ (Theorem 4.5(c), Definition 4.11); and $\text{rank}(GX'X) \leqslant \text{rank}\, X$ (Theorem 4.15(a), as $GX'X = (GX')X$). So $r = 4$.

- Theorem 14.13(a) gives
$$F = \frac{168/1}{358/2} = \frac{168}{179}$$
Check: $K\beta^0 = 12$ and $V = KGK' = \frac{6}{7}$, so $(K\beta^0)^2/V = 144 \cdot \frac{7}{6} = 168$. The test of level α rejects H_0 if and only if $F_{1,2,1-\alpha} < \frac{168}{179}$ (Theorem 14.13(b)).

Problems

Problem 14.1 ([2, Theorems 12.4, 12.5])**.** Assume (A1), and let G be a g-inverse of $X'X$. Show that for $a \in \mathbb{R}^{k+1}$ the following are equivalent: (a) $a'\beta$ is estimable; (b) $a' = a'GX'X$; (c) $a' = w'GX'X$ for some $w \in \mathbb{R}^{k+1}$.

Solution.

- (a)⇒(b): By Theorem 14.2(a), $a' = c'X$ for some $c \in \mathbb{R}^n$. By Theorem 13.4(a), $XGX'X = HX = X$, so $a' = c'XGX'X = a'GX'X$.
- (b)⇒(c): take $w = a$.
- (c)⇒(a): $a' = w'GX'X = (w'GX')X$, so $a = X'(XG'w) \in \mathcal{C}(X')$, and $a'\beta$ is estimable (Theorem 14.2(a)).

Problem 14.2 ([12, Corollary 11.3])**.** Let $x_1, \dots, x_m$ be $N(\mu_1, \sigma^2)$ and $y_1, \dots, y_n$ be $N(\mu_2, \sigma^2)$, all $m+n$ independent, with $m, n \geqslant 2$ and $\sigma > 0$, and let $s_p^2 = \left(\sum_i (x_i - \bar{x})^2 + \sum_j (y_j - \bar{y})^2\right)/(m+n-2)$.

(a) Show that $T = \dfrac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)}{s_p\sqrt{1/m + 1/n}} \sim t_{m+n-2}$.

(b) Deduce a confidence interval with level $1-\alpha$ for $\mu_1 - \mu_2$.

Solution. (a)

- By Theorem 10.9(a), $v = (x_1, \dots, x_m, y_1, \dots, y_n)'$ is normal with mean $(\mu_1 \mathbf{1}_m', \mu_2 \mathbf{1}_n')'$ and covariance matrix $\sigma^2 I_{m+n}$; so $\varepsilon = v - \mathsf{E}\, v \sim N_{m+n}(0, \sigma^2 I_{m+n})$ (Theorem 10.2).
- $\bar{x} - \bar{y} - (\mu_1 - \mu_2) = a'\varepsilon$ with $a = (\frac{1}{m}\mathbf{1}_m', -\frac{1}{n}\mathbf{1}_n')'$, $a'a = \frac{1}{m} + \frac{1}{n}$.
- Let $x = (x_1, \dots, x_m)'$ and $y = (y_1, \dots, y_n)'$, so that $\varepsilon = (\varepsilon_1', \varepsilon_2')'$ with $\varepsilon_1 = x - \mu_1 \mathbf{1}_m$ and $\varepsilon_2 = y - \mu_2 \mathbf{1}_n$. By Example 1.16, $\sum_{i=1}^m (x_i - \bar{x})^2 = (C_m x)'(C_m x)$; as $C_m \mathbf{1}_m = 0$ and $C_m' C_m = C_m$ (Proposition 1.15), $C_m x = C_m \varepsilon_1$ and $\sum_{i=1}^m (x_i - \bar{x})^2 = \varepsilon_1' C_m \varepsilon_1$. Likewise $\sum_{j=1}^n (y_j - \bar{y})^2 = \varepsilon_2' C_n \varepsilon_2$. So $(m+n-2)s_p^2 = \varepsilon' P \varepsilon$ with $P = \operatorname{diag}(C_m, C_n)$ (Theorem 5.3(c)).
- P is symmetric and idempotent (Proposition 5.6(b)), of rank $(m-1) + (n-1)$ (Theorem 4.16(c), Corollary 6.35(b)).
- $a'P = (\frac{1}{m}\mathbf{1}_m' C_m,\ -\frac{1}{n}\mathbf{1}_n' C_n) = 0$, so $a'\varepsilon$ and $\varepsilon' P \varepsilon$ are independent (Theorem 11.20(b)). Moreover $u = a'\varepsilon/(\sigma\sqrt{a'a}) \sim N(0,1)$ and $w = \varepsilon' P \varepsilon/\sigma^2 \sim \chi^2_{m+n-2}$ (Theorems 10.2 and 11.15), and $T = u/\sqrt{w/(m+n-2)} \sim t_{m+n-2}$ by (N3) and Definition 11.7(a).

(b) With $c = t_{m+n-2,1-\alpha/2}$, Theorem 11.8(a) gives $\mathsf{P}(-c \leqslant T \leqslant c) = 1 - \alpha$ for all values of the parameters; as in Example 11.13, $-c \leqslant T \leqslant c$ holds exactly when $\mu_1 - \mu_2$ lies in the interval below, so $\bar{x} - \bar{y} \pm c\, s_p \sqrt{1/m + 1/n}$ is a confidence interval with level $1-\alpha$ for $\mu_1 - \mu_2$ (Definition 11.12(a)).

Chapter 15. Extremal principles and the singular value decomposition

The generalized Rayleigh quotient

Theorem 15.1 ([3, Result 5.2], [2, Corollary 9.3])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric, with B PD, and let $\mu_1 \geqslant \cdots \geqslant \mu_n$ be the eigenvalues of the symmetric matrix $C = B^{-1/2}AB^{-1/2}$ (Theorem 8.46). Then for every $x \neq 0$

$$\mu_n \leqslant \frac{x'Ax}{x'Bx} \leqslant \mu_1$$

The maximum μ_1 is attained at $x = B^{-1/2}y$ for every eigenvector y of C belonging to μ_1, and the minimum μ_n likewise. The numbers $\mu_1, \ldots, \mu_n$ are also the eigenvalues of $B^{-1}A$ and of AB^{-1}, listed with multiplicity.

Proof. • $B^{1/2}$ is symmetric and invertible, and so is $B^{-1/2} = (B^{1/2})^{-1}$ (Theorems 8.46 and 2.13(c)); hence $C' = C$.

- Put $y = B^{1/2}x$. Then $x = B^{-1/2}y$, $x \neq 0$ if and only if $y \neq 0$, $x'Bx = x'B^{1/2}B^{1/2}x = y'y$ and $x'Ax = y'B^{-1/2}AB^{-1/2}y = y'Cy$; so
$$\frac{x'Ax}{x'Bx} = \frac{y'Cy}{y'y}$$
- Theorem 9.23(a),(b) for C gives the bounds, attained at the eigenvectors y of C belonging to μ_1 and to μ_n, that is, at $x = B^{-1/2}y$.
- $B^{-1/2}B^{-1/2} = (B^{1/2}B^{1/2})^{-1} = B^{-1}$ (Theorem 2.13(b)). So $C = B^{-1/2}(AB^{-1/2})$ with $(AB^{-1/2})B^{-1/2} = AB^{-1}$, and $B^{-1}A = B^{-1/2}(B^{-1/2}A)$ with $(B^{-1/2}A)B^{-1/2} = C$. By Theorem 7.33(a) ($m = n$), $p_{AB^{-1}} = p_C = p_{B^{-1}A}$: both matrices have the eigenvalues of C with the same multiplicities. □

Theorem 15.2 ([3, Result 5.3])**.** Let $B \in \mathbb{R}^{n\times n}$ be PD and $y \in \mathbb{R}^n$. Then

$$(x'y)^2 \leqslant (x'Bx)(y'B^{-1}y) \qquad (x \in \mathbb{R}^n)$$

If $y \neq 0$, then $\max_{x\neq 0}(x'y)^2/x'Bx = y'B^{-1}y$, attained at $x = B^{-1}y$.

Proof. If $y = 0$ or $x = 0$, both sides of the inequality are 0. Let $y \neq 0$.

- $(x'y)^2 = x'yy'x$, so $(x'y)^2/x'Bx$ is $x'Ax/x'Bx$ with the symmetric matrix $A = yy'$. By Theorem 15.1, its maximum over $x \neq 0$ is the largest eigenvalue of $C = B^{-1/2}yy'B^{-1/2} = ww'$, where $w = B^{-1/2}y \neq 0$. Here $B^{-1/2} = (B^{1/2})^{-1}$ is symmetric and $B^{-1/2}B^{-1/2} = (B^{1/2}B^{1/2})^{-1} = B^{-1}$ (Theorems 8.46 and 2.13(b),(c)), so $w'w = y'B^{-1}y$ and $B^{-1/2}w = B^{-1}y$.
- $\operatorname{rank}(ww') \leqslant \operatorname{rank} w = 1$ (Theorem 4.15(a)) and $Cw = w(w'w) \neq 0$, so C has rank 1. By Theorem 8.4, exactly one eigenvalue of C, counted with multiplicity, is nonzero, and it equals $\operatorname{tr} C = \operatorname{tr}(w'w) = w'w = y'B^{-1}y > 0$ (Theorems 7.19 and 1.11(b)). So the largest eigenvalue of C is $y'B^{-1}y$; this gives the inequality for $x \neq 0$.
- $Cw = (w'w)w$: w is an eigenvector of C belonging to $y'B^{-1}y$, and the maximum is attained at $x = B^{-1/2}w = B^{-1}y$ (Theorem 15.1). □

In statistics, let the rows $x_1', \ldots, x_n'$ of a data matrix $X \in \mathbb{R}^{n\times p}$ fall into g groups: group l has the set $\mathcal{I}_l$ of row indices, n_l rows and the mean vector $\bar{x}_l$, and $\bar{x}$ is the mean vector of all n rows. For $a \in \mathbb{R}^p$, the values $Xa = (a'x_1, \ldots, a'x_n)'$ of the combination $a'x$ have the group means $a'\bar{x}_l$ and the grand mean $a'\bar{x}$. By Theorem 14.16(b) for $y = Xa$ (after the rows are arranged by group; the sums

of squares do not depend on the order of the rows), their total sum of squares is the sum of the between-group and the within-group sums of squares

$$\sum_{l=1}^{g} n_l (a'\bar{x}_l - a'\bar{x})^2 = a'Ba, \qquad \sum_{l=1}^{g} \sum_{i \in \mathcal{I}_l} (a'x_i - a'\bar{x}_l)^2 = a'Wa$$

where, as $\big(a'(x_i - \bar{x}_l)\big)^2 = a'(x_i - \bar{x}_l)(x_i - \bar{x}_l)'a$,

$$B = \sum_{l=1}^{g} n_l (\bar{x}_l - \bar{x})(\bar{x}_l - \bar{x})', \qquad W = \sum_{l=1}^{g} \sum_{i \in \mathcal{I}_l} (x_i - \bar{x}_l)(x_i - \bar{x}_l)'$$

Fisher's linear discriminant function is a combination $a'x$ that maximizes $a'Ba/a'Wa$ [18, §14.2, Theorem 14.4]. Let W be PD. By Theorem 15.1 for the symmetric B and the PD W (in the roles of A and B there), the maximum is the largest eigenvalue μ_1 of $W^{-1}B$, attained at $a = W^{-1/2}y$ for an eigenvector y of $C = W^{-1/2}BW^{-1/2}$ belonging to μ_1; this a is an eigenvector of $W^{-1}B$, as $W^{-1}Ba = W^{-1/2}(W^{-1/2}BW^{-1/2})y = \mu_1 a$.

For two groups, let $d = \bar{x}_1 - \bar{x}_2$. As $\bar{x} = (n_1\bar{x}_1 + n_2\bar{x}_2)/n$ and $n = n_1 + n_2$,

$$\bar{x}_1 - \bar{x} = \frac{n_2}{n}\, d, \qquad \bar{x}_2 - \bar{x} = -\frac{n_1}{n}\, d, \qquad B = \frac{n_1 n_2^2 + n_2 n_1^2}{n^2}\, dd' = \frac{n_1 n_2}{n}\, dd'$$

so that

$$\frac{a'Ba}{a'Wa} = \frac{n_1 n_2}{n} \cdot \frac{(a'd)^2}{a'Wa}$$

If $d \neq 0$, Theorem 15.2 gives the maximum $(n_1 n_2/n)\, d'W^{-1}d$, attained at $a = W^{-1}d$. With the **pooled sample covariance matrix** $S_p = W/(n-2)$, the ratio $(a'd)^2/a'S_p a$ is the squared difference of the two group means of $a'x$ divided by its pooled sample variance, and it is largest for $a = S_p^{-1}d = (n-2)W^{-1}d$.

Proposition 15.3 ([3, §8.5])**.** Let y be a random variable and $x \in \mathbb{R}^p$ a random vector with

$$\operatorname{var}\begin{pmatrix} y \\ x \end{pmatrix} = V = \begin{pmatrix} \sigma^2 & u' \\ u & \Sigma \end{pmatrix}$$

where $\sigma^2 > 0$, $u \in \mathbb{R}^p$ and $\Sigma \in \mathbb{R}^{p \times p}$ is PD. For $\alpha \neq 0$ in $\mathbb{R}^p$, let $\rho(\alpha)$ be the correlation of y and $\alpha'x$ (Definition 9.7).

(a) $\rho(\alpha)^2 \leqslant u'\Sigma^{-1}u/\sigma^2$ for every $\alpha \neq 0$, with equality at $\alpha = \Sigma^{-1}u$ if $u \neq 0$ (and at every α if $u = 0$). The number $r_{y(x)} \geqslant 0$ with

$$r_{y(x)}^2 = \max_{\alpha \neq 0} \rho(\alpha)^2 = \frac{u'\Sigma^{-1}u}{\sigma^2}$$

is the **multiple correlation coefficient** of y and $x_1, \ldots, x_p$.

(b) If V is PD and $Z = V^{-1} = (z_{ij})$, then $r_{y(x)}^2 = 1 - 1/(\sigma^2 z_{11})$.

Proof. (a)

- The matrix $T = \left(\begin{smallmatrix} 1 & 0' \\ 0 & \alpha' \end{smallmatrix}\right) \in \mathbb{R}^{2 \times (p+1)}$ maps $(y, x')'$ to $(y, \alpha'x)'$, so by Theorems 9.3(b) and 5.3

$$\operatorname{var}\begin{pmatrix} y \\ \alpha'x \end{pmatrix} = TVT' = \begin{pmatrix} \sigma^2 & u'\alpha \\ \alpha'u & \alpha'\Sigma\alpha \end{pmatrix}$$

  Here $\operatorname{var}(\alpha'x) = \alpha'\Sigma\alpha > 0$, as Σ is PD and $\alpha \neq 0$; so $\rho(\alpha)^2 = (\alpha'u)^2/(\sigma^2\, \alpha'\Sigma\alpha)$.

- By Theorem 15.2 for the PD Σ and the vector u, $(\alpha'u)^2 \leqslant (\alpha'\Sigma\alpha)(u'\Sigma^{-1}u)$; divide by $\sigma^2\, \alpha'\Sigma\alpha > 0$. If $u \neq 0$, the same theorem gives equality at $\alpha = \Sigma^{-1}u \neq 0$; if $u = 0$, both sides are 0.

(b)

- Σ is invertible, so by Theorem 5.30(b) (the Schur complement of the lower right block), $\det V = \det\Sigma \cdot (\sigma^2 - u'\Sigma^{-1}u)$.
- $V(1|1) = \Sigma$ (Definition 5.20), so the $(1,1)$ entry of $\operatorname{adj} V$ is $\det\Sigma$, and $z_{11} = \det\Sigma/\det V$ (Theorem 5.24). Both determinants are positive, as V and Σ are PD (Theorem 8.21(c)).
- Hence $1/z_{11} = \det V/\det\Sigma = \sigma^2 - u'\Sigma^{-1}u$, and by (a) $r^2_{y(x)} = (\sigma^2 - 1/z_{11})/\sigma^2 = 1 - 1/(\sigma^2 z_{11})$. □

If $(x', y)'$ is normal with mean $(\mu', \tau)'$ and the PD covariance matrix $\left(\begin{smallmatrix}\Sigma & u\\ u' & \sigma^2\end{smallmatrix}\right)$ (the entries of V rearranged), then by Theorem 10.13 with $z_1 = x$ and $z_2 = y$, the conditional distribution of y given x is $N\Big(\tau + u'\Sigma^{-1}(x-\mu),\ \sigma^2 - u'\Sigma^{-1}u\Big)$. Its variance is $\sigma^2(1 - r^2_{y(x)})$, and its mean contains $u'\Sigma^{-1}x$, the combination $\alpha' x$ of maximum correlation in (a); this is also the best linear predictor of Theorem 13.38 up to the constant term ($b^* = \Sigma^{-1}u$). So $r^2_{y(x)}$ is the population counterpart of $R^2 = \operatorname{cor}(y, \hat{y})^2$ (Proposition 13.30).

The minimax principle

Lemma 15.4 ([3, Ch. 1, Exercise 19])**.** If U and W are subspaces of $\mathbb{R}^n$ with $\dim U + \dim W > n$, then $U \cap W$ contains a unit vector.

Proof. If $U \cap W = \{0\}$, then a basis of U together with a basis of W is linearly independent (Theorem 3.33); these are $\dim U + \dim W > n$ vectors of $\mathbb{R}^n$ ✘ (Theorem 4.1). So there is $x \neq 0$ in the subspace $U \cap W$ (Theorem 3.31(a)), and $x/\|x\|$ is a unit vector in it. □

Theorem 15.5 (Courant–Fischer; [3, Result 5.8], [6, Exercises 12.43, 12.44])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, with $Q = [q_1\ \cdots\ q_n]$ as in Lemma 9.22.

(a) (Min–max form.) Let $2 \leqslant k \leqslant n$. For $w_1, \dots, w_{k-1} \in \mathbb{R}^n$, the vectors u with $u \perp w_1, \dots, w_{k-1}$ form a subspace W of dimension at least $n - k + 1$, and

$$m(w_1, \dots, w_{k-1}) = \max\{u'Au : u \in W,\ \|u\| = 1\} \geqslant \lambda_k$$

with equality for $w_i = q_i$ $(i = 1, \dots, k-1)$. That is,

$$\lambda_k = \min_{w_1,\dots,w_{k-1}\in\mathbb{R}^n} \max\{u'Au : u \perp w_1, \dots, w_{k-1},\ \|u\| = 1\}$$

(b) (Max–min form.) Let $1 \leqslant k \leqslant n-1$. For $w_1, \dots, w_{n-k} \in \mathbb{R}^n$, the minimum of $u'Au$ over the unit vectors $u \perp w_1, \dots, w_{n-k}$ exists and is $\leqslant \lambda_k$, with equality for $w_i = q_{k+i}$ $(i = 1, \dots, n-k)$. That is,

$$\lambda_k = \max_{w_1,\dots,w_{n-k}\in\mathbb{R}^n} \min\{u'Au : u \perp w_1, \dots, w_{n-k},\ \|u\| = 1\}$$

Proof. (a)

- $W = \mathcal{N}(M')$ with $M = [w_1\ \cdots\ w_{k-1}] \in \mathbb{R}^{n\times(k-1)}$, since the entries of $M'u$ are $w_i'u$. As $\operatorname{rank} M' \leqslant k-1$ (Theorem 4.13(a)), $\dim W = n - \operatorname{rank} M' \geqslant n - k + 1 \geqslant 1$ (Theorem 4.12); so the maximum exists (Lemma 9.24).
- For $w_i = q_i$ the maximum is λ_k (Theorem 9.25).
- $T = \operatorname{span}\{q_1, \dots, q_k\}$ has dimension k (orthonormal vectors are independent, Proposition 6.11(b)), and $\dim W + \dim T \geqslant n + 1$; so there is a unit vector $u \in W \cap T$ (Lemma 15.4). Write $u = c_1 q_1 + \cdots + c_k q_k$. As $q_i'q_j$ is 1 for $i = j$ and 0 otherwise, $c_1^2 + \cdots + c_k^2 = u'u = 1$ and

$$u'Au = \Big(\sum_{i\leqslant k} c_i q_i\Big)' \sum_{j\leqslant k} c_j \lambda_j q_j = \sum_{i\leqslant k} \lambda_i c_i^2 \geqslant \lambda_k \sum_{i\leqslant k} c_i^2 = \lambda_k$$

So $m(w_1, \dots, w_{k-1}) \geqslant u'Au \geqslant \lambda_k$.

(b)

- $-A$ is symmetric, and $\lambda_j(-A) = -\lambda_{n+1-j}(A)$ (Lemma 9.22(b)). Put $j = n+1-k$; then $2 \leqslant j \leqslant n$, $j-1 = n-k$ and $\lambda_j(-A) = -\lambda_k$.
- Let $w_1, \dots, w_{n-k} \in \mathbb{R}^n$. By (a) for $-A$ and j, the maximum of $u'(-A)u = -u'Au$ over the unit vectors $u \perp w_1, \dots, w_{n-k}$ exists and is $\geqslant \lambda_j(-A) = -\lambda_k$. So the minimum of $u'Au$ over these u exists, it is minus that maximum, and it is $\leqslant \lambda_k$.
- For $w_i = q_{k+i}$ $(i = 1, \dots, n-k)$ the minimum is λ_k (Theorem 9.25, second form). □

Theorem 15.6 (Interlacing; [3, Result 5.9])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric ($n \geqslant 2$), let $1 \leqslant i \leqslant n$, and let $B = A(i|i) \in \mathbb{R}^{(n-1)\times(n-1)}$ be A with row i and column i deleted (Definition 5.20). Then B is symmetric and

$$\lambda_1(A) \geqslant \lambda_1(B) \geqslant \lambda_2(A) \geqslant \lambda_2(B) \geqslant \cdots \geqslant \lambda_{n-1}(A) \geqslant \lambda_{n-1}(B) \geqslant \lambda_n(A)$$

that is, $\lambda_k(A) \geqslant \lambda_k(B) \geqslant \lambda_{k+1}(A)$ for $k = 1, \dots, n-1$.

Proof. Let $S \in \mathbb{R}^{n\times(n-1)}$ be I_n with column i deleted. Then $S'S = I_{n-1}$ and $S'AS = B$: entry (s,t) of $S'AS$ is a_{jl}, where j and l are the sth and the tth of the indices $\neq i$. So B is symmetric. For $\tilde{u} \in \mathbb{R}^{n-1}$, $u = S\tilde{u}$ has $u_i = 0$, $\|u\| = \|\tilde{u}\|$ and $u'Au = \tilde{u}'B\tilde{u}$.

- Left inequality. Let $q_1, \dots, q_n$ be as in Lemma 9.22. By Theorem 9.25, $u'Au \leqslant \lambda_k(A)$ for every unit vector $u \perp q_1, \dots, q_{k-1}$. For $u = S\tilde{u}$, $u'q_j = \tilde{u}'(S'q_j)$; so $\tilde{u}'B\tilde{u} \leqslant \lambda_k(A)$ for every unit $\tilde{u} \in \mathbb{R}^{n-1}$ with $\tilde{u} \perp S'q_1, \dots, S'q_{k-1}$. The largest such $\tilde{u}'B\tilde{u}$ is $\geqslant \lambda_k(B)$: for $k = 1$ it is $\lambda_1(B)$ (Theorem 9.23), and for $k \geqslant 2$ apply Theorem 15.5(a) to B with $w_j = S'q_j$. Hence $\lambda_k(A) \geqslant \lambda_k(B)$.
- Right inequality. $-B$ is $-A$ with row and column i deleted. The left inequality for $-A$ at the index $n-k \in \{1, \dots, n-1\}$ gives $\lambda_{n-k}(-A) \geqslant \lambda_{n-k}(-B)$, that is, $-\lambda_{k+1}(A) \geqslant -\lambda_k(B)$ (Lemma 9.22(b), for A of order n and B of order $n-1$). □

For a data matrix X_0 with n rows, the entries of $X_0'C_nX_0$ are $\mathrm{col}_j(X_0)'C_n\,\mathrm{col}_l(X_0)$ (Example 1.17); deleting variable i deletes row i and column i of this matrix, so its eigenvalues interlace those of the full matrix.

Theorem 15.7 ([3, Result 5.10])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric with $A - B$ PSD. Then $\lambda_k(A) \geqslant \lambda_k(B)$ for $k = 1, \dots, n$.

Proof. Let $u_1, \dots, u_n$ and $v_1, \dots, v_n$ be the columns of orthogonal matrices as in Lemma 9.22 for A and for B, and fix k.

- $T_1 = \mathrm{span}\{u_k, \dots, u_n\}$ and $T_2 = \mathrm{span}\{v_1, \dots, v_k\}$ have dimensions $n-k+1$ and k (orthonormal vectors are independent, Proposition 6.11(b)). By Lemma 15.4 there is a unit vector $z \in T_1 \cap T_2$.
- $z \in T_1$, so $z = \sum_{i\geqslant k} c_iu_i$, with $c_i = u_i'z$, $u_i'z = 0$ for $i < k$, and $\sum_{i\geqslant k} c_i^2 = z'z = 1$. By Lemma 9.22(a),

$$z'Az = \sum_{i=1}^{n} \lambda_i(A)(u_i'z)^2 = \sum_{i\geqslant k} \lambda_i(A)c_i^2 \leqslant \lambda_k(A)\sum_{i\geqslant k} c_i^2 = \lambda_k(A)$$

- Likewise $z = \sum_{j\leqslant k} d_jv_j$ with $\sum_{j\leqslant k} d_j^2 = 1$, and $z'Bz = \sum_{j\leqslant k} \lambda_j(B)d_j^2 \geqslant \lambda_k(B)$.
- $z'(A-B)z \geqslant 0$, so $\lambda_k(A) \geqslant z'Az \geqslant z'Bz \geqslant \lambda_k(B)$. □

If $X \in \mathbb{R}^{n\times p}$ has the rows $x_1', \dots, x_n'$, then $X'X = x_1x_1' + \cdots + x_nx_n'$ (outer product expansion, Theorem 1.4(d)). So adding an observation x' as a new row turns $X'X$ into $X'X + xx'$ (as in Example 13.20), and xx' is PSD ($v'xx'v = (x'v)^2 \geqslant 0$). By Theorem 15.7, no eigenvalue of $X'X$ decreases when an observation is added.

The singular value decomposition

Proposition 15.8 ([2, Definition 9.1], [5, Lemmas 18.2.21, 8.4.2])**.** Let $A \in \mathbb{R}^{m\times n}$, with the Frobenius norm $\|A\|_F$ of Definition 9.26.

(a) $\|A\|_F \geqslant 0$, with equality only for $A = 0$; $\|cA\|_F = |c|\,\|A\|_F$ for $c \in \mathbb{R}$; and $\|A + B\|_F \leqslant \|A\|_F + \|B\|_F$ for $B \in \mathbb{R}^{m\times n}$.

(b) $\|AB\|_F \leqslant \|A\|_F\|B\|_F$ for $B \in \mathbb{R}^{n\times p}$.

(c) $\|PAQ\|_F = \|A\|_F$ for orthogonal $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$.

Proof. (a) The entries of $A+B$ and cA are $a_{ij}+b_{ij}$ and ca_{ij} (Definition 1.1), so if v, w list the entries of A, B, then $v+w$ and cv list those of $A+B$ and cA. Apply the remarks after Definition 6.1 and Theorem 6.4(b) in $\mathbb{R}^{mn}$.

(b)

- Column j of AB is Ab_j, where $b_j = \mathrm{col}_j(B)$ (Theorem 1.4(a)), and entry i of Ab_j is $\mathrm{row}_i(A)b_j = \sum_l a_{il}b_{lj}$.
- By Cauchy–Schwarz (Theorem 6.4(a)), $\|Ab_j\|^2 = \sum_i(\mathrm{row}_i(A)b_j)^2 \leqslant \sum_i\Big(\sum_l a_{il}^2\Big)\|b_j\|^2 = \|A\|_F^2\|b_j\|^2$.
- Summing over j: $\|AB\|_F^2 = \sum_j\|Ab_j\|^2 \leqslant \|A\|_F^2\sum_j\|b_j\|^2 = \|A\|_F^2\|B\|_F^2$.

(c) $P'P = I_m$ and $QQ' = I_n$ (Proposition 6.13(b)), so by Theorem 1.11(b)

$$\|PAQ\|_F^2 = \mathrm{tr}(Q'A'P'PAQ) = \mathrm{tr}(Q'A'AQ) = \mathrm{tr}(A'AQQ') = \mathrm{tr}(A'A) = \|A\|_F^2$$

□

Theorem 15.9 (Singular value decomposition; [5, Theorem 21.12.1, Corollary 21.12.2])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$. Below, $\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)$ denotes the matrix of the size required by the product, with D in the top left $r \times r$ block and zeros elsewhere; blocks with no rows or no columns are absent (when $r = n$ there is no Q_2, and when $r = m$ no P_2).

(a) Let $Q = [Q_1\ Q_2] \in \mathbb{R}^{n\times n}$ be orthogonal, with $Q_1 \in \mathbb{R}^{n\times r}$, and let $D \in \mathbb{R}^{r\times r}$ be an invertible diagonal matrix with

$$Q'A'AQ = \begin{pmatrix} D^2 & 0 \\ 0 & 0 \end{pmatrix}$$

Put $P_1 = AQ_1D^{-1} \in \mathbb{R}^{m\times r}$, and let $P = [P_1\ P_2]$, where $P_2 \in \mathbb{R}^{m\times(m-r)}$ is any matrix with $P_1'P_2 = 0$. Then

$$P'AQ = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$$

(b) Such Q and D exist with $D = \mathrm{diag}(\sigma_1, \dots, \sigma_r)$ and $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$: take $\sigma_i^2 = \lambda_i(A'A)$, the nonzero eigenvalues of $A'A$, and Q as in Lemma 9.22 for $A'A$. Then $P_1'P_1 = I_r$, P_2 can be chosen so that P is orthogonal, and, with the columns p_i of P and q_i of Q,

$$A = P\begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}Q' = P_1DQ_1' = \sigma_1p_1q_1' + \cdots + \sigma_rp_rq_r'$$

Proof. (a)

- By the block product (Theorem 5.3), $Q'A'AQ$ has the blocks $Q_1'A'AQ_1$, $Q_1'A'AQ_2$, $Q_2'A'AQ_1$, $Q_2'A'AQ_2$. So $Q_1'A'AQ_1 = D^2$, and $(AQ_2)'(AQ_2) = Q_2'A'AQ_2 = 0$; its trace is 0, so $AQ_2 = 0$ (Theorem 1.11(d)).

- $AQ_1 = P_1 D$, and $P_1' = D^{-1} Q_1' A'$, as D is diagonal. Hence

$$P'AQ = \begin{pmatrix} P_1'AQ_1 & P_1'AQ_2 \\ P_2'AQ_1 & P_2'AQ_2 \end{pmatrix} = \begin{pmatrix} D^{-1}Q_1'A'AQ_1 & 0 \\ P_2'P_1 D & 0 \end{pmatrix} = \begin{pmatrix} D^{-1}D^2 & 0 \\ (P_1'P_2)'D & 0 \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$$

(b)

- $A'A$ is symmetric and PSD (Theorem 8.25), so its eigenvalues are $\geqslant 0$ (Theorem 8.21(a)). As $\operatorname{rank}(A'A) = \operatorname{rank} A = r$ (Theorem 4.18(b)), exactly r of them are nonzero (Theorem 8.4): $\lambda_1(A'A) \geqslant \cdots \geqslant \lambda_r(A'A) > 0 = \lambda_{r+1}(A'A) = \cdots = \lambda_n(A'A)$. So Q from Lemma 9.22 and D satisfy the condition of (a).
- $P_1'P_1 = D^{-1}Q_1'A'AQ_1D^{-1} = D^{-1}D^2D^{-1} = I_r$: the columns of P_1 are orthonormal, hence independent (Proposition 6.11(b)), and $\operatorname{rank} P_1' = \operatorname{rank} P_1 = r$ (Theorem 4.13(a),(b)). So $\dim \mathcal{N}(P_1') = m - r$ (Theorem 4.12).
- If $r < m$, let the columns of P_2 form an orthonormal basis of $\mathcal{N}(P_1')$ (Theorems 4.4 and 6.14). Then $P_1'P_2 = 0$ and $P_2'P_2 = I_{m-r}$, so $P'P = I_m$ by blocks: P is orthogonal.
- As $PP' = I_m$ and $QQ' = I_n$ (Proposition 6.13(b)), $A = P(P'AQ)Q'$, the first form by (a). By blocks it is $[P_1 D \;\; 0] \left(\begin{smallmatrix} Q_1' \\ Q_2' \end{smallmatrix}\right) = P_1 D Q_1'$, and the outer product expansion (Theorem 1.4(d)), with $\operatorname{col}_i(P_1 D) = \sigma_i p_i$ and $\operatorname{row}_i(Q_1') = q_i'$, gives the sum. □

A factorization $A = P \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right) Q'$ with orthogonal $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$ and $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$, is a **singular value decomposition** (SVD) of A, and $A = P_1 D Q_1'$ is the corresponding **compact SVD**. The numbers $\sigma_1, \ldots, \sigma_r$ are the **singular values** of A, and the columns of P and Q are **singular vectors** of A. We put $\sigma_k = 0$ for $r < k \leqslant \min\{m, n\}$, and $\sigma_1 = 0$ for $A = 0$. (Theorem 15.10(a) shows that $r = \operatorname{rank} A$ and that $\sigma_1, \ldots, \sigma_r$ do not depend on the choice of the SVD.)

Theorem 15.10 ([5, Theorem 21.12.3, §21.12a])**.** Let $A \in \mathbb{R}^{m\times n}$, let $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$ be orthogonal, and let $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$ with $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$ be such that $P'AQ = \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Split $P = [P_1 \; P_2]$ and $Q = [Q_1 \; Q_2]$, where P_1 and Q_1 have the r columns $p_1, \ldots, p_r$ and $q_1, \ldots, q_r$.

(a) $r = \operatorname{rank} A$, and $\sigma_1^2, \ldots, \sigma_r^2$ are the nonzero eigenvalues of $A'A$ and of AA', listed with multiplicity. So the singular values of A depend only on A, not on the choice of P and Q.

(b) $Q'A'AQ = \left(\begin{smallmatrix} D^2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $P'AA'P = \left(\begin{smallmatrix} D^2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$: the columns of Q are orthonormal eigenvectors of $A'A$, and those of P of AA'; q_i and p_i belong to σ_i^2 $(i \leqslant r)$, the other columns to 0.

(c) $P_1 = AQ_1D^{-1}$ and $Q_1 = A'P_1D^{-1}$; that is, $Aq_i = \sigma_i p_i$ and $A'p_i = \sigma_i q_i$ $(i = 1, \ldots, r)$.

(d) $\|A\|_F^2 = \sigma_1^2 + \cdots + \sigma_r^2$.

(e) If A is symmetric, its singular values are the absolute values of its nonzero eigenvalues; if A is PSD, they are its nonzero eigenvalues. For orthogonal $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$, the matrices UAV and A' have the same singular values as A.

Proof. (a)

- $\operatorname{rank} A = \operatorname{rank}(P'AQ)$ (Theorem 4.15(d)). The nonzero columns of $P'AQ$ are $\sigma_1 e_1, \ldots, \sigma_r e_r$, which are independent (Proposition 6.11(b)) and span its column space; so the rank is r.
- As $PP' = I_m$, $Q'A'AQ = Q'A'PP'AQ = (P'AQ)'(P'AQ)$, the product of an $n \times m$ and an $m \times n$ block matrix, which is $\left(\begin{smallmatrix} D^2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ (Theorem 5.3); likewise $P'AA'P = (P'AQ)(P'AQ)' = \left(\begin{smallmatrix} D^2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.
- As $Q^{-1} = Q'$ (Proposition 6.13(b)), $Q^{-1}(A'A)Q$ is diagonal with the diagonal entries $\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0$; these are the eigenvalues of $A'A$ listed with multiplicity (Definition 7.23), and likewise for AA' with P. Hence $\sigma_1 \geqslant \cdots \geqslant \sigma_r$ are the positive square roots of the nonzero eigenvalues of $A'A$, in decreasing order.

(b) The two identities were shown in (a). Multiplying the first by Q on the left, $A'AQ = Q\left(\begin{smallmatrix} D^2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, whose columns say $A'Aq_i = \sigma_i^2 q_i$ $(i \leqslant r)$ and $A'Aq_i = 0$ $(i > r)$; likewise for P.

(c)

- $PP' = P_1P_1' + P_2P_2' = I_m$, and $P_1'AQ_1 = D$ and $P_2'AQ_1 = 0$ are blocks of $P'AQ$. So

$$P_1 = P_1(P_1'AQ_1)D^{-1} = (I_m - P_2P_2')AQ_1D^{-1} = AQ_1D^{-1} - P_2(P_2'AQ_1)D^{-1} = AQ_1D^{-1}$$

- In the same way, $QQ' = Q_1Q_1' + Q_2Q_2' = I_n$, $Q_1'A'P_1 = (P_1'AQ_1)' = D$ and $Q_2'A'P_1 = (P_1'AQ_2)' = 0$, so

$$Q_1 = Q_1(Q_1'A'P_1)D^{-1} = (I_n - Q_2Q_2')A'P_1D^{-1} = A'P_1D^{-1} - Q_2(Q_2'A'P_1)D^{-1} = A'P_1D^{-1}$$

(d) By Proposition 15.8(c), $\|A\|_F = \|P'AQ\|_F$, and the squares of the entries of $P'AQ$ add up to $\sigma_1^2 + \cdots + \sigma_r^2$.

(e)

- Let A be symmetric; we choose a new pair P, Q. Number an orthonormal basis of eigenvectors $q_1, \ldots, q_n$ (Theorem 8.3(a)), with eigenvalues $d_1, \ldots, d_n$, so that $|d_1| \geqslant \cdots \geqslant |d_n|$. Then $Q'AQ = \operatorname{diag}(d_1, \ldots, d_n)$ for $Q = [q_1 \ \cdots \ q_n]$, and $d_1, \ldots, d_n$ are the eigenvalues of A, listed with multiplicity (Theorem 8.3(b),(c)). By Theorem 8.4, exactly $\operatorname{rank} A$ of the d_i are nonzero, namely $d_1, \ldots, d_{\operatorname{rank} A}$.
- Let $\Delta = \operatorname{diag}(\delta_1, \ldots, \delta_n)$, with $\delta_i = 1$ if $d_i \geqslant 0$ and $\delta_i = -1$ if $d_i < 0$, and $P = Q\Delta$. Then $P'P = \Delta Q'Q\Delta = \Delta^2 = I_n$, and

$$P'AQ = \Delta Q'AQ = \operatorname{diag}(\delta_1 d_1, \ldots, \delta_n d_n) = \operatorname{diag}(|d_1|, \ldots, |d_n|)$$

  is an SVD of A. By (a), the singular values are the nonzero $|d_i|$. If A is PSD, every $d_i \geqslant 0$ (Theorem 8.21(a)), so $|d_i| = d_i$ (and $P = Q$).
- In the setting of the theorem, UP and $V'Q$ are orthogonal ($(UP)'(UP) = P'U'UP = I_m$, and likewise $(V'Q)'(V'Q) = I_n$), and $(UP)'(UAV)(V'Q) = P'AQ$: an SVD of UAV with the same D. Also $Q'A'P = (P'AQ)'$ is the $n \times m$ matrix with $D' = D$ in the top left corner: an SVD of A' with Q and P exchanged. □

Corollary 15.11 ([8, Ch. 21, Exercise 28])**.** In Theorem 15.10 let $r \geqslant 1$. Then

$$\mathcal{C}(A) = \mathcal{C}(P_1), \qquad \mathcal{N}(A') = \mathcal{C}(P_2), \qquad \mathcal{C}(A') = \mathcal{C}(Q_1), \qquad \mathcal{N}(A) = \mathcal{C}(Q_2)$$

so the columns of P_1, P_2, Q_1, Q_2 are orthonormal bases of these four subspaces (an absent P_2 or Q_2 means that the subspace is $\{0\}$). Moreover, $P_1P_1'y$ is the orthogonal projection of $y \in \mathbb{R}^m$ onto $\mathcal{C}(A)$, and if A has full column rank, then $A(A'A)^{-1}A' = P_1P_1'$.

Proof. • As $PP' = I_m$ and $QQ' = I_n$, $A = P(P'AQ)Q' = P\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)Q' = P_1DQ_1'$ by blocks, so $Ax = P_1(DQ_1'x) \in \mathcal{C}(P_1)$; and $P_1 = AQ_1D^{-1}$ (Theorem 15.10(c)) gives $\mathcal{C}(P_1) \subseteq \mathcal{C}(A)$.

- $Q'Q = I_n$ gives $Q_1'Q_2 = 0$, so $AQ_2 = P_1DQ_1'Q_2 = 0$ and $\mathcal{C}(Q_2) \subseteq \mathcal{N}(A)$. The $n - r$ columns of Q_2 are orthonormal, hence independent (Proposition 6.11(b)), so $\dim \mathcal{C}(Q_2) = n - r = \dim \mathcal{N}(A)$ (Theorems 4.12 and 15.10(a)). Hence $\mathcal{C}(Q_2) = \mathcal{N}(A)$ (Theorem 4.5(a)).
- $Q'A'P = (P'AQ)'$ is an SVD of A' with Q and P exchanged (Theorem 15.10(e)); the two items above, for A', give $\mathcal{C}(A') = \mathcal{C}(Q_1)$ and $\mathcal{N}(A') = \mathcal{C}(P_2)$.
- P_1 has full column rank and $P_1'P_1 = I_r$, so by Theorem 6.26(a), $P_1(P_1'P_1)^{-1}P_1'y = P_1P_1'y$ is the orthogonal projection of y onto $\mathcal{C}(P_1) = \mathcal{C}(A)$. If A has full column rank, $A(A'A)^{-1}A'y$ is also this projection (Theorem 6.26(a)), which is unique (Definition 6.22); so $A(A'A)^{-1}A'y = P_1P_1'y$ for every y, and with $y = e_1, \ldots, e_m$ the two matrices have the same columns. □

Example 15.12 ([2, Example 7.3], [6, Exercise 8.39])**.** Find an SVD and the compact SVD of

$$A = \begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

and orthonormal bases of its four subspaces.

Solution.

- The columns satisfy $a_3 = a_4 = \frac{1}{2}(a_1 + a_2)$, and a_1, a_2 are independent; so $\operatorname{rank} A = 2 = r$. Hence $A(0,0,1,-1)' = a_3 - a_4 = 0$ and $A(1,1,-1,-1)' = a_1 + a_2 - a_3 - a_4 = 0$.
- The matrix

$$A'A = \begin{pmatrix} 5 & 1 & 3 & 3 \\ 1 & 5 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}$$

  has every row sum 12, so $A'A\mathbf{1}_4 = 12\mathbf{1}_4$; and $A'A(1,-1,0,0)' = (4,-4,0,0)'$. The two vectors of the first item lie in $\mathcal{N}(A'A) = \mathcal{N}(A)$ (Theorem 4.18(a)), of dimension $4 - 2 = 2$ (Theorem 4.12). So the algebraic multiplicities $\dim \mathcal{N}(A'A - cI)$ (Theorem 8.4) are at least 1, 1, 2 for $c = 12, 4, 0$. All multiplicities add up to 4 (Definition 7.10); so these are exactly 1, 1, 2, and the eigenvalues of $A'A$ are 12, 4, 0, 0. Check: $\operatorname{tr}(A'A) = 16 = 12 + 4$.
- The four vectors are pairwise orthogonal (Theorem 8.2, and $(1,1,-1,-1)(0,0,1,-1)' = 0$). Normalized: $q_1 = \frac{1}{2}(1,1,1,1)'$, $q_2 = \frac{1}{\sqrt{2}}(1,-1,0,0)'$, $q_3 = \frac{1}{2}(1,1,-1,-1)'$, $q_4 = \frac{1}{\sqrt{2}}(0,0,1,-1)'$; and $D = \operatorname{diag}(\sqrt{12}, 2) = \operatorname{diag}(2\sqrt{3}, 2)$.
- $P_1 = AQ_1D^{-1}$: $Aq_1 = (2,2,2)'$ and $Aq_2 = \frac{1}{\sqrt{2}}(2,-2,0)'$, so $p_1 = \frac{1}{2\sqrt{3}}(2,2,2)' = \frac{1}{\sqrt{3}}(1,1,1)'$ and $p_2 = \frac{1}{2}Aq_2 = \frac{1}{\sqrt{2}}(1,-1,0)'$.
- $\mathcal{N}(P_1')$: $y_1 + y_2 + y_3 = 0$ and $y_1 - y_2 = 0$ give $y = t(1,1,-2)'$; so $p_3 = \frac{1}{\sqrt{6}}(1,1,-2)'$.
- Hence $A = P \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right) Q'$ with

$$P = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}, \qquad \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2\sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

  and the compact SVD is $A = 2\sqrt{3}\, p_1 q_1' + 2\, p_2 q_2'$. Check:

$$2\sqrt{3}\, p_1 q_1' + 2\, p_2 q_2' = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} (1,1,1,1) + \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} (1,-1,0,0) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = A$$

- By Corollary 15.11: p_1, p_2 is an orthonormal basis of $\mathcal{C}(A)$, p_3 of $\mathcal{N}(A')$, q_1, q_2 of $\mathcal{C}(A')$, and q_3, q_4 of $\mathcal{N}(A)$.

The vector $-p_2$ is also a unit eigenvector of AA' belonging to 4, but $2\sqrt{3}\, p_1 q_1' - 2\, p_2 q_2' \neq A$. So P_1 must be computed from Q_1 as AQ_1D^{-1}; it cannot be chosen separately among the eigenvectors of AA'.

Singular values as maxima; the spectral norm

Theorem 15.13 ([3, Results 5.5, 5.6])**.** Let $A \in \mathbb{R}^{m \times n}$, $A \neq 0$, have an SVD as in Theorem 15.10, with the columns $p_1, \ldots, p_m$ of P and $q_1, \ldots, q_n$ of Q.

(a) $|u'Av| \leqslant \sigma_1$ for all unit vectors $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$, and $p_1'Aq_1 = \sigma_1$. Hence

$$\sigma_1 = \max\{u'Av : \|u\| = \|v\| = 1\}$$

(b) Let $2 \leqslant k \leqslant \min\{m, n\}$. If moreover $u \perp p_1, \dots, p_{k-1}$ and $v \perp q_1, \dots, q_{k-1}$, then $|u'Av| \leqslant \sigma_k$; and $p_k'Aq_k = \sigma_k$. So σ_k is the maximum of $u'Av$ under these conditions.

Proof. Let $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$ be unit vectors, $w = P'u$ and $z = Q'v$. Then $\|w\| = \|z\| = 1$ (Proposition 6.13(a); P' and Q' are orthogonal), and $A = P \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right) Q'$ gives

$$u'Av = w' \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} z = \sigma_1 w_1 z_1 + \cdots + \sigma_r w_r z_r$$

(a)

- As $0 < \sigma_i \leqslant \sigma_1$, $|u'Av| \leqslant \sigma_1 \sum_{i \leqslant r} |w_i||z_i| \leqslant \sigma_1 \|w\|\|z\| = \sigma_1$, by Cauchy–Schwarz (Theorem 6.4(a)) for the vectors $(|w_1|, \dots, |w_r|)'$ and $(|z_1|, \dots, |z_r|)'$, whose lengths are at most $\|w\|$ and $\|z\|$.
- $p_1'Aq_1 = p_1'(\sigma_1 p_1) = \sigma_1$ (Theorem 15.10(c)). As $\sigma_1 > 0$, the maximum of $u'Av$ is σ_1.

(b)

- The entries of $w = P'u$ are $w_i = p_i'u$, so $u \perp p_1, \dots, p_{k-1}$ means $w_1 = \cdots = w_{k-1} = 0$; likewise $z_1 = \cdots = z_{k-1} = 0$.
- So $u'Av = \sum_{i=k}^{r} \sigma_i w_i z_i$ (0 if $k > r$), and as in (a) $|u'Av| \leqslant \sigma_k$.
- p_k and q_k satisfy the conditions, and $p_k'Aq_k = \sigma_k$: for $k \leqslant r$, $Aq_k = \sigma_k p_k$; for $k > r$, $Aq_k = 0$ (Corollary 15.11). □

Definition 15.14 ([2, (9.3)], [3, Result 5.5])**.** The **spectral norm** of $A \in \mathbb{R}^{m \times n}$ is $\|A\|_2 = \sigma_1$, its largest singular value; and $\|0\|_2 = 0$. It is not the Frobenius norm of Definition 9.26, which some books also write $\|A\|_2$.

Proposition 15.15 ([6, Exercise 7.53(b)], [3, Result 5.5])**.** Let $A \in \mathbb{R}^{m \times n}$.

(a) $\|A\|_2 = \big(\lambda_1(A'A)\big)^{1/2} = \max_{x \neq 0} \|Ax\|/\|x\|$; so $\|Ax\| \leqslant \|A\|_2 \|x\|$ for every $x \in \mathbb{R}^n$, with equality at $x = q_1$ if $A \neq 0$, where q_1 is the first column of Q in an SVD of A as in Theorem 15.10. If $\operatorname{rank} A = n$, then also $\|Ax\| \geqslant \sigma_n \|x\|$ for every $x \in \mathbb{R}^n$.

(b) $\|A\|_2 = \max\{u'Av : \|u\| = \|v\| = 1\}$.

(c) $\|UAV\|_2 = \|A\|_2 = \|A'\|_2$ for orthogonal $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$.

(d) $\|A\|_2 \leqslant \|A\|_F$.

(e) If A is symmetric, then $\|A\|_2 = \max_i |\lambda_i(A)| = \max\{\lambda_1(A), -\lambda_n(A)\}$; if A is PSD, then $\|A\|_2 = \lambda_1(A)$.

Proof. For $A = 0$ every statement reads $0 = 0$ or $0 \leqslant 0$. Let $A \neq 0$, with an SVD as in Theorem 15.10.

(a)

- $\lambda_1(A'A) = \sigma_1^2$ (Theorem 15.10(a)).
- For $x \neq 0$, $\|Ax\|^2/\|x\|^2 = x'A'Ax/x'x \leqslant \lambda_1(A'A) = \sigma_1^2$ (Theorem 9.23(a) for $A'A$); take square roots.
- $\|Aq_1\| = \|\sigma_1 p_1\| = \sigma_1 = \sigma_1 \|q_1\|$ (Theorem 15.10(c)).
- Let $\operatorname{rank} A = n$. Then $A'A \in \mathbb{R}^{n \times n}$ has the n nonzero eigenvalues $\sigma_1^2 \geqslant \cdots \geqslant \sigma_n^2$ (Theorem 15.10(a), with $r = n$), so $\lambda_n(A'A) = \sigma_n^2$. For $x \neq 0$, $\|Ax\|^2 = x'A'Ax \geqslant \lambda_n(A'A)\, x'x = \sigma_n^2 \|x\|^2$ (Theorem 9.23(a) for $A'A$); take square roots. For $x = 0$ both sides are 0.

(b) This is Theorem 15.13(a). (c) By Theorem 15.10(e), the three matrices have the same largest singular value. (d) $\sigma_1^2 \leqslant \sigma_1^2 + \cdots + \sigma_r^2 = \|A\|_F^2$ (Theorem 15.10(d)).

(e) By Theorem 15.10(e), σ_1 is the largest $|\lambda_i|$ with $\lambda_i \neq 0$, which is $\max_i|\lambda_i|$, as $A \neq 0$ has a nonzero eigenvalue (Theorem 8.4). As $\lambda_n \leqslant \lambda_i \leqslant \lambda_1$, $|\lambda_i| \leqslant \max\{\lambda_1, -\lambda_n\}$, with equality for $i = 1$ or $i = n$. If A is PSD, then $\lambda_n \geqslant 0$ (Theorem 8.21(a)), and the maximum is λ_1. □

Proposition 15.16 ([22, Ch. III, Proposition 3.2])**.** Let $A, B \in \mathbb{R}^{m\times n}$.

(a) $\|A\|_2 \geqslant 0$, with equality only for $A = 0$; $\|cA\|_2 = |c|\,\|A\|_2$ for $c \in \mathbb{R}$; and $\|A+B\|_2 \leqslant \|A\|_2+\|B\|_2$.

(b) $\|AC\|_2 \leqslant \|A\|_2\|C\|_2$ for $C \in \mathbb{R}^{n\times p}$.

Proof. By Proposition 15.15(a), every matrix $M \in \mathbb{R}^{s\times t}$ has $\|M\|_2 = \max_{x\neq 0}\|Mx\|/\|x\|$ and $\|Mx\| \leqslant \|M\|_2\|x\|$ for every $x \in \mathbb{R}^t$.

(a)

- $\|A\|_2$ is a maximum of numbers $\geqslant 0$. If $\|A\|_2 = 0$, then $Ax = 0$ for every x, so $A = [Ae_1 \cdots Ae_n] = 0$ (Theorem 1.4(b)).
- $\|cAx\| = |c|\,\|Ax\|$ for every x (the remarks after Definition 6.1), so the maxima for cA and for A differ by the factor $|c|$.
- For every $x \neq 0$, by Theorem 6.4(b),

$$\|(A + B)x\| = \|Ax + Bx\| \leqslant \|Ax\| + \|Bx\| \leqslant \big(\|A\|_2 + \|B\|_2\big)\|x\|$$

  Divide by $\|x\|$ and take the maximum over $x \neq 0$.

(b) For every $x \neq 0$ in $\mathbb{R}^p$, $\|ACx\| = \|A(Cx)\| \leqslant \|A\|_2\|Cx\| \leqslant \|A\|_2\|C\|_2\|x\|$. Divide by $\|x\|$ and take the maximum over $x \neq 0$. □

Approximation by matrices of lower rank

Theorem 15.17 (Eckart–Young; [5, Theorem 21.12.4])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank r and singular value decomposition

$$A = P\begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}Q' = \sigma_1 p_1 q_1' + \cdots + \sigma_r p_r q_r', \qquad D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r), \quad \sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$$

(Theorem 15.9), let $1 \leqslant k < r$, and let $A_k = \sigma_1 p_1 q_1' + \cdots + \sigma_k p_k q_k'$, which has rank k. For every $B \in \mathbb{R}^{m\times n}$ with $\operatorname{rank} B \leqslant k$:

(a) $\|A - B\|_F^2 \geqslant \sigma_{k+1}^2 + \cdots + \sigma_r^2$ (Definition 9.26), with equality for $B = A_k$;

(b) $\|A - B\|_2 \geqslant \sigma_{k+1}$ (Definition 15.14), with equality for $B = A_k$.

Proof. (a)

- Let $s = \operatorname{rank} B \leqslant k$. Take an orthonormal basis $u_1, \ldots, u_s$ of $\mathcal{C}(B)$ and, as $\dim \mathcal{C}(B)^\perp = m - s \geqslant k - s$ (Theorem 6.6(c); $k < r \leqslant m$), $k - s$ vectors $u_{s+1}, \ldots, u_k$ of an orthonormal basis of $\mathcal{C}(B)^\perp$ (Theorems 4.4 and 6.14). Then $U = [u_1 \cdots u_k]$ has $U'U = I_k$, as $\mathcal{C}(B) \perp \mathcal{C}(B)^\perp$, and $\mathcal{C}(B) \subseteq \mathcal{C}(U)$: column j of B is $U\ell_j$ for some $\ell_j \in \mathbb{R}^k$, so $B = UL$ with $L = [\ell_1 \cdots \ell_n]$ (Theorem 1.4(a)). It therefore suffices to show $\|UL - A\|_F^2 \geqslant \sigma_{k+1}^2 + \cdots + \sigma_r^2$ for all $U \in \mathbb{R}^{m\times k}$ with $U'U = I_k$ and all $L \in \mathbb{R}^{k\times n}$.
- As $U'U = I_k$, $(UL-A)'(UL-A) = L'L-A'UL-L'U'A+A'A$, and $L'L-A'UL-L'U'A+A'UU'A = (L - U'A)'(L - U'A)$. By Definition 9.26 and Theorem 1.11(d), (b),

$$\|UL - A\|_F^2 = \operatorname{tr}\big[(L - U'A)'(L - U'A)\big] - \operatorname{tr}(A'UU'A) + \operatorname{tr}(A'A) \geqslant \operatorname{tr}(A'A) - \operatorname{tr}(U'AA'U)$$

- $Q'Q = I_n$, so $AA' = P\,\mathrm{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0)P'$ (Theorem 5.3(c)). Thus $\mathrm{tr}(A'A) = \mathrm{tr}(AA') = \sigma_1^2 + \cdots + \sigma_r^2$ (Theorem 1.11(b), $P'P = I_m$), and the columns of P are orthonormal eigenvectors of the symmetric matrix AA' for the eigenvalues $\sigma_1^2 \geqslant \cdots \geqslant \sigma_r^2 \geqslant 0 = \cdots = 0$ (Theorem 8.3(c)).
- By Theorem 9.27(a) for $AA' \in \mathbb{R}^{m\times m}$, $\mathrm{tr}(U'AA'U) \leqslant \sigma_1^2 + \cdots + \sigma_k^2$. Hence $\|UL - A\|_F^2 \geqslant \sum_{i\leqslant r} \sigma_i^2 - \sum_{i\leqslant k} \sigma_i^2 = \sigma_{k+1}^2 + \cdots + \sigma_r^2$.
- $A - A_k = \sum_{i=k+1}^r \sigma_i p_i q_i' = PEQ'$ (Theorem 1.4(d)), where $E \in \mathbb{R}^{m\times n}$ has the entries $\sigma_{k+1}, \dots, \sigma_r$ in positions $(k+1, k+1), \dots, (r, r)$ and zeros elsewhere. For orthogonal P, Q, $\|PEQ'\|_F^2 = \mathrm{tr}(QE'P'PEQ') = \mathrm{tr}(E'EQ'Q) = \mathrm{tr}(E'E) = \sigma_{k+1}^2 + \cdots + \sigma_r^2$ (Theorem 1.11(b)): equality holds.
- Likewise $A_k = PE_kQ'$, where E_k has $\sigma_1, \dots, \sigma_k$ in positions $(1,1), \dots, (k,k)$; its nonzero columns $\sigma_i e_i$ are independent, so $\mathrm{rank}\, A_k = \mathrm{rank}\, E_k = k$ (Theorem 4.15(d)).

(b)

- Let $\mathrm{rank}\, B \leqslant k$. Then $\dim \mathcal{N}(B) = n - \mathrm{rank}\, B \geqslant n-k$ (Theorem 4.12), and $T = \mathrm{span}\{q_1, \dots, q_{k+1}\}$ has dimension $k+1$ (orthonormal vectors are independent, Proposition 6.11(b)). As $(n-k)+(k+1) > n$, there is a unit vector $x \in \mathcal{N}(B) \cap T$ (Lemma 15.4).
- Write $x = c_1q_1 + \cdots + c_{k+1}q_{k+1}$; then $c_1^2 + \cdots + c_{k+1}^2 = x'x = 1$, as the q_i are orthonormal. Since $k+1 \leqslant r$, $Aq_i = \sigma_i p_i$ (Theorem 15.10(c)), so $Ax = \sum_{i\leqslant k+1} c_i\sigma_i p_i$, and, the p_i being orthonormal,

$$\|Ax\|^2 = \sum_{i\leqslant k+1} c_i^2\sigma_i^2 \geqslant \sigma_{k+1}^2 \sum_{i\leqslant k+1} c_i^2 = \sigma_{k+1}^2$$

- $Bx = 0$, so $(A-B)x = Ax$, and $\|A - B\|_2 \geqslant \|(A-B)x\| = \|Ax\| \geqslant \sigma_{k+1}$ (Proposition 15.15(a)).
- Equality: $A - A_k = PEQ'$ with E as in (a), so $\|A - A_k\|_2 = \|E\|_2$ (Proposition 15.15(c)). For $y \in \mathbb{R}^n$, $\|Ey\|^2 = \sum_{i=k+1}^r \sigma_i^2 y_i^2 \leqslant \sigma_{k+1}^2\|y\|^2$, with equality for $y = e_{k+1}$; so $\|E\|_2 = \sigma_{k+1}$ (Proposition 15.15(a)). □

So A_k keeps the part $\sigma_1^2 + \cdots + \sigma_k^2$ of $\|A\|_F^2 = \sigma_1^2 + \cdots + \sigma_r^2$ and loses $\sigma_{k+1}^2 + \cdots + \sigma_r^2$; no matrix B of rank at most k does better: $\|A\|_F^2 - \|A - B\|_F^2 \leqslant \sigma_1^2 + \cdots + \sigma_k^2$. As $A_k = PE_kQ'$ in the proof of (a), $\|A_k\|_F^2 = \|E_k\|_F^2 = \sigma_1^2 + \cdots + \sigma_k^2$ (Proposition 15.8(c)): A_k reproduces the proportion

$$\frac{\|A_k\|_F^2}{\|A\|_F^2} = \frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_r^2}$$

of the sum of squares $\|A\|_F^2 = \sum_{i,j} a_{ij}^2$ of the entries of A.

Corollary 15.18 ([5, Corollary 21.12.6])**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric of rank $r \geqslant 2$. Number an orthonormal basis $q_1, \dots, q_n$ of eigenvectors of A, $Aq_i = d_iq_i$, so that $|d_1| \geqslant \cdots \geqslant |d_n|$ (Theorem 8.3(a)). Then $d_1, \dots, d_r$ are the nonzero eigenvalues of A, listed with multiplicity, and $d_{r+1} = \cdots = d_n = 0$ (Theorems 8.3(c) and 8.4). Let $1 \leqslant k < r$. For every $B \in \mathbb{R}^{n\times n}$ with $\mathrm{rank}\, B \leqslant k$,

$$\|A - B\|_F^2 \geqslant d_{k+1}^2 + \cdots + d_r^2$$

with equality for $B = d_1q_1q_1' + \cdots + d_kq_kq_k'$. If A is PSD, then $d_1 \geqslant \cdots \geqslant d_r > 0$ are its positive eigenvalues.

Proof. • $Q = [q_1 \ \cdots \ q_n]$ is orthogonal and $Q'AQ = \mathrm{diag}(d_1, \dots, d_n)$ (Theorem 8.3(b)). Let $\delta_i = -1$ if $d_i < 0$ and $\delta_i = 1$ otherwise, $\Delta = \mathrm{diag}(\delta_1, \dots, \delta_n)$ and $P = Q\Delta$, whose columns are $p_i = \delta_i q_i$ (Theorem 1.4(a)). Then $P'P = \Delta Q'Q\Delta = \Delta^2 = I_n$, and, as Δ scales the rows (Theorem 1.4(c)),

$$P'AQ = \Delta Q'AQ = \mathrm{diag}(\delta_1 d_1, \dots, \delta_n d_n) = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}, \qquad D = \mathrm{diag}(|d_1|, \dots, |d_r|)$$

- As $PP' = I_n$ and $QQ' = I_n$ (Proposition 6.13(b)), $A = P(P'AQ)Q'$ is a singular value decomposition of A, with the singular values $\sigma_i = |d_i|$ $(i \leqslant r)$.
- By Theorem 15.17(a), $\|A - B\|_F^2 \geqslant \sigma_{k+1}^2 + \cdots + \sigma_r^2 = d_{k+1}^2 + \cdots + d_r^2$, with equality for $B = \sum_{i\leqslant k} \sigma_i p_i q_i' = \sum_{i\leqslant k} |d_i|\delta_i q_i q_i' = \sum_{i\leqslant k} d_i q_i q_i'$, as $|d_i|\delta_i = d_i$.
- If A is PSD, then every $d_i \geqslant 0$ (Theorem 8.21(a)), so the nonzero ones are positive and $|d_i| = d_i$. □

The SVD in statistics

Proposition 15.19 ([9, Proposition 5.23])**.** Let $X \in \mathbb{R}^{n\times p}$ be a data matrix with $r = \operatorname{rank}(C_n X) \geqslant 1$.

(a) Let $C_n X = P \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} Q' = \sigma_1 p_1 q_1' + \cdots + \sigma_r p_r q_r'$ be a singular value decomposition (Theorem 15.9). Then $S = \frac{1}{n-1} Q \operatorname{diag}\big(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0\big) Q'$. Hence $\lambda_m = \sigma_m^2/(n-1)$ for $m \leqslant r$ and $\lambda_m = 0$ for $m > r$; the columns $q_1, \ldots, q_p$ of Q can serve as $v_1, \ldots, v_p$ in Theorem 9.31; with this choice $z_m = \sigma_m p_m$ for $m \leqslant r$; and $\mathrm{PVE}_m = \sigma_m^2/(\sigma_1^2 + \cdots + \sigma_r^2)$.

(b) Conversely, let v_m, λ_m, z_m be as in Theorem 9.31. Then $C_n X = \sigma_1 p_1 v_1' + \cdots + \sigma_r p_r v_r'$ with $\sigma_m = \|z_m\| = \sqrt{(n-1)\lambda_m}$ and $p_m = z_m/\sigma_m$. The vectors $p_1, \ldots, p_r$ extend to an orthogonal $P \in \mathbb{R}^{n\times n}$, and $C_n X = P \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right) V'$, $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, is a singular value decomposition, with the sum above as its compact SVD.

Proof. (a)

- $P'P = I_n$, so $(C_n X)'(C_n X) = Q \operatorname{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0) Q'$ (Theorem 5.3(c)); divide by $n-1$ (Proposition 9.14(a)).
- So Q is orthogonal and its columns are orthonormal eigenvectors of S for $\sigma_1^2/(n-1) \geqslant \cdots \geqslant \sigma_r^2/(n-1) > 0 = \cdots = 0$. By Theorem 8.3(c) these are the eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_p$ of S, and $V = Q$ is allowed in Theorem 9.31.
- $z_m = C_n X q_m = \sum_i \sigma_i p_i (q_i' q_m) = \sigma_m p_m$, as $q_i' q_m = 0$ for $i \neq m$ and $q_m' q_m = 1$.
- By Proposition 9.32, $\mathrm{PVE}_m = \lambda_m / \sum_l \lambda_l = \sigma_m^2 / \sum_l \sigma_l^2$.

(b)

- $VV' = I_p$ (Proposition 6.13(b)), so $C_n X = (C_n X V)V' = \sum_{m=1}^p z_m v_m'$ (Theorem 1.4(d)), and $z_m = 0$ for $m > r$, as $\|z_m\|^2 = (n-1)\lambda_m = 0$ (Theorem 9.31(a),(d)).
- $(C_n X)'(C_n X) = (n-1)S = V \operatorname{diag}\big((n-1)\lambda_1, \ldots, (n-1)\lambda_p\big) V'$ (Proposition 9.14(a)), with $(n-1)\lambda_1 \geqslant \cdots \geqslant (n-1)\lambda_p$, and its nonzero eigenvalues are $\sigma_1^2 \geqslant \cdots \geqslant \sigma_r^2$ (Theorem 9.31(a)). So $Q = V$ and D are as in Theorem 15.9(b) for $A = C_n X$, and there $P_1 = C_n X[v_1 \ \cdots \ v_r] D^{-1} = [p_1 \ \cdots \ p_r]$, as $C_n X v_m/\sigma_m = z_m/\sigma_m = p_m$. Theorem 15.9(b) extends P_1 to an orthogonal P and gives $C_n X = P \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right) V' = \sum_{m\leqslant r} \sigma_m p_m v_m'$. □

For $A = C_n X$ and $1 \leqslant k < r$, the choice $v_m = q_m$ in Proposition 15.19(a) gives, in Theorem 15.17, $A_k = \sigma_1 p_1 q_1' + \cdots + \sigma_k p_k q_k' = z_1 v_1' + \cdots + z_k v_k' = Z_k V_k'$ (Theorem 1.4(d); Z_k and V_k as in Theorem 9.39 with $M = k$) and $\sigma_{k+1}^2 + \cdots + \sigma_r^2 = (n-1)(\lambda_{k+1} + \cdots + \lambda_p)$. So Theorem 15.17(a) for $C_n X$ is Theorem 9.39(a) again, in the form of the paragraph after it (a best approximation among the matrices of rank at most $M = k$): the same bound, attained at the same matrix $Z_k V_k'$.

Example 15.20. For the data of Example 9.15, Example 9.34 and Proposition 15.19(b) give $\sigma_1 = \sqrt{4 \cdot 25} = 10$, $\sigma_2 = \sqrt{4 \cdot 9} = 6$, $\sigma_3 = \sqrt{4 \cdot 5} = 2\sqrt{5}$, $\sigma_4 = \sqrt{4 \cdot 1} = 2$ and

$$p_1 = \frac{z_1}{\sigma_1} = \frac{1}{2}\begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \qquad p_2 = \frac{z_2}{\sigma_2} = \frac{1}{2}\begin{pmatrix} 0 \\ 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \qquad p_3 = \frac{z_3}{\sigma_3} = \frac{1}{2\sqrt{5}}\begin{pmatrix} 4 \\ -1 \\ -1 \\ -1 \\ -1 \end{pmatrix}, \qquad p_4 = \frac{z_4}{\sigma_4} = \frac{1}{2}\begin{pmatrix} 0 \\ 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

so $C_5X = \sigma_1 p_1 v_1' + \sigma_2 p_2 v_2' + \sigma_3 p_3 v_3' + \sigma_4 p_4 v_4'$, with the loading vectors $v_m = \frac{1}{2}h_m$ of Example 9.34. For instance $\sigma_1 p_1 v_1' = z_1 v_1' = \frac{1}{2}(0,5,5,-5,-5)'(1,1,1,1)$. Check: $\sigma_1^2+\sigma_2^2+\sigma_3^2+\sigma_4^2 = 100+36+20+4 = 160$, which is $\|C_5X\|_F^2 = \operatorname{tr}\big((C_5X)'(C_5X)\big) = 4 \cdot 40$ (Theorem 15.10(d)).

Theorem 15.21 ([5, §21.12b], [6, Exercise 10.23])**.** Let $A \in \mathbb{R}^{m\times n}$ have rank $r \geqslant 1$, and let $A = P_1 D Q_1'$, where $P_1 \in \mathbb{R}^{m\times r}$ and $Q_1 \in \mathbb{R}^{n\times r}$ have orthonormal columns ($P_1'P_1 = Q_1'Q_1 = I_r$) and $D = \operatorname{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_1, \dots, \sigma_r > 0$ (a compact SVD, Theorem 15.9). Then

$$A^+ = Q_1 D^{-1} P_1' = \sigma_1^{-1} q_1 p_1' + \cdots + \sigma_r^{-1} q_r p_r'$$

where p_i and q_i are the columns of P_1 and Q_1; and $AA^+ = P_1P_1'$, $A^+A = Q_1Q_1'$. If moreover $P = [P_1\ P_2] \in \mathbb{R}^{m\times m}$ and $Q = [Q_1\ Q_2] \in \mathbb{R}^{n\times n}$ are orthogonal, so that $A = P\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right) Q'$, then $A^+ = Q\left(\begin{smallmatrix} D^{-1} & 0 \\ 0 & 0 \end{smallmatrix}\right) P'$.

Proof. Put $G = Q_1 D^{-1} P_1'$.

- $AG = P_1 D Q_1' Q_1 D^{-1} P_1' = P_1 P_1'$ and $GA = Q_1 D^{-1} P_1' P_1 D Q_1' = Q_1 Q_1'$ are symmetric: (P3), (P4) of Definition 12.22.
- $AGA = P_1 P_1' P_1 D Q_1' = P_1 D Q_1' = A$ and $GAG = Q_1 Q_1' Q_1 D^{-1} P_1' = G$: (P1), (P2).
- So $G = A^+$ (Theorem 12.23), and $AA^+ = AG = P_1P_1'$, $A^+A = GA = Q_1Q_1'$. Row i of $D^{-1}P_1'$ is $\sigma_i^{-1} p_i'$, so the outer product expansion (Theorem 1.4(d)) gives the sum.
- By the block product (Theorem 5.3), $Q\left(\begin{smallmatrix} D^{-1} & 0 \\ 0 & 0 \end{smallmatrix}\right) P' = [Q_1\ Q_2]\left(\begin{smallmatrix} D^{-1}P_1' \\ 0 \end{smallmatrix}\right) = Q_1 D^{-1} P_1'$. □

Definition 15.22 ([3, §10.2])**.** Let the $p = p_1 + p_2$ variables of a data matrix X be split into two groups, $X = [X_1\ X_2]$ with $X_1 \in \mathbb{R}^{n\times p_1}$, and partition S accordingly:

$$S = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix}, \qquad S_{11} \in \mathbb{R}^{p_1\times p_1}, \quad S_{21} = S_{12}'$$

Assume that S_{11} and S_{22} are PD. For $a \in \mathbb{R}^{p_1}$ and $b \in \mathbb{R}^{p_2}$, Proposition 9.14(b), applied to $(a', 0')'$ and $(0', b')'$, shows that X_1a and X_2b have sample variances $a'S_{11}a$, $b'S_{22}b$ and sample covariance $a'S_{12}b$ (Theorem 5.3(c)).

- A **first canonical pair** is a pair (a_1, b_1) that maximizes $a'S_{12}b$ subject to $a'S_{11}a = b'S_{22}b = 1$; the maximum ρ_1 is the **first canonical correlation**.
- For $k \geqslant 2$, once $(a_1, b_1), \dots, (a_{k-1}, b_{k-1})$ are chosen, a **kth canonical pair** maximizes $a'S_{12}b$ subject to $a'S_{11}a = b'S_{22}b = 1$ and $a'S_{11}a_i = b'S_{22}b_i = 0$ for $i < k$; the maximum ρ_k is the **kth canonical correlation**.

When both variances are 1, $a'S_{12}b$ is the sample correlation (Definition 9.17) $a'S_{12}b/(a'S_{11}a \cdot b'S_{22}b)^{1/2}$ of X_1a and X_2b; the side conditions make X_1a uncorrelated with X_1a_i, and X_2b with X_2b_i ($i < k$).

Theorem 15.23 ([3, §10.2])**.** In Definition 15.22, let $K = S_{11}^{-1/2} S_{12} S_{22}^{-1/2} \in \mathbb{R}^{p_1\times p_2}$ (Theorem 8.46) have rank $s \geqslant 1$ and compact SVD $K = \rho_1 u_1 w_1' + \cdots + \rho_s u_s w_s'$, $\rho_1 \geqslant \cdots \geqslant \rho_s > 0$ (Theorem 15.9), and put $a_k = S_{11}^{-1/2} u_k$, $b_k = S_{22}^{-1/2} w_k$. For $k = 1, \dots, s$: if (a_i, b_i) ($i < k$) are taken as the first $k-1$ canonical pairs, then (a_k, b_k) is a kth canonical pair, and the kth canonical correlation is ρ_k.

Proof. • $S_{11}^{1/2}$ is symmetric and invertible, with inverse $S_{11}^{-1/2}$ (Theorem 8.46); so $a \mapsto u = S_{11}^{1/2}a$ maps $\mathbb{R}^{p_1}$ onto $\mathbb{R}^{p_1}$, with inverse $u \mapsto S_{11}^{-1/2}u$. Likewise $b \mapsto w = S_{22}^{1/2}b$, and $S_{11}^{1/2}a_i = u_i$, $S_{22}^{1/2}b_i = w_i$.

- With $u = S_{11}^{1/2}a$ and $w = S_{22}^{1/2}b$: $a'S_{11}a = a'S_{11}^{1/2}S_{11}^{1/2}a = u'u$, $b'S_{22}b = w'w$, $a'S_{12}b = u'S_{11}^{-1/2}S_{12}S_{22}^{-1/2}w = u'Kw$, $a'S_{11}a_i = u'u_i$ and $b'S_{22}b_i = w'w_i$.
- So the kth problem is to maximize $u'Kw$ over unit vectors $u \in \mathbb{R}^{p_1}$, $w \in \mathbb{R}^{p_2}$ with $u \perp u_1, \ldots, u_{k-1}$ and $w \perp w_1, \ldots, w_{k-1}$. By Theorem 15.13(a),(b), the maximum is ρ_k, attained at $u = u_k$, $w = w_k$, that is, at $a = a_k$, $b = b_k$. □

Example 15.24. For the data of Example 9.15, take variables 1, 2 as the first group and variables 3, 4 as the second:

$$S_{11} = S_{22} = \begin{pmatrix} 10 & 7 \\ 7 & 10 \end{pmatrix}, \qquad S_{12} = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$$

- $I_2 = \bar{J}_2 + C_2$, where $\bar{J}_2$ and C_2 are symmetric and idempotent and $\bar{J}_2C_2 = \bar{J}_2 - \bar{J}_2^2 = 0$ (Definition 1.14, Proposition 1.15); also $C_2\bar{J}_2 = (\bar{J}_2C_2)' = 0$. As $\bar{J}_2 = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $C_2 = \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$,

$$S_{11} = S_{22} = 17\bar{J}_2 + 3C_2, \qquad S_{12} = 8\bar{J}_2 + 2C_2$$

- So $(\sqrt{17}\bar{J}_2 + \sqrt{3}C_2)^2 = 17\bar{J}_2 + 3C_2 = S_{11}$, and $\sqrt{17}\bar{J}_2 + \sqrt{3}C_2$ is PSD, as $x'(\sqrt{17}\bar{J}_2 + \sqrt{3}C_2)x = \sqrt{17}\|\bar{J}_2x\|^2 + \sqrt{3}\|C_2x\|^2$. By Theorem 8.46, $S_{11}^{1/2} = S_{22}^{1/2} = \sqrt{17}\bar{J}_2 + \sqrt{3}C_2$, and $S_{11}^{-1/2} = S_{22}^{-1/2} = \frac{1}{\sqrt{17}}\bar{J}_2 + \frac{1}{\sqrt{3}}C_2$, since the product of the two is $\bar{J}_2 + C_2 = I_2$.
- Multiplying out, with $\bar{J}_2^2 = \bar{J}_2$, $C_2^2 = C_2$ and $\bar{J}_2C_2 = C_2\bar{J}_2 = 0$,

$$K = S_{11}^{-1/2}S_{12}S_{22}^{-1/2} = \frac{8}{17}\bar{J}_2 + \frac{2}{3}C_2 = \frac{1}{51}\begin{pmatrix} 29 & -5 \\ -5 & 29 \end{pmatrix}$$

  With $u_1 = w_1 = \frac{1}{\sqrt{2}}(1, -1)'$ and $u_2 = w_2 = \frac{1}{\sqrt{2}}(1, 1)'$, $C_2 = u_1u_1'$ and $\bar{J}_2 = u_2u_2'$, so $K = \frac{2}{3}u_1w_1' + \frac{8}{17}u_2w_2'$. As $\frac{2}{3} > \frac{8}{17} > 0$ and u_1, u_2 are orthonormal, this is a compact SVD with $s = 2$, $\rho_1 = \frac{2}{3}$ and $\rho_2 = \frac{8}{17}$.
- By Theorem 15.23, the canonical correlations are $\frac{2}{3}$ and $\frac{8}{17}$, with

$$a_1 = b_1 = S_{11}^{-1/2}u_1 = \frac{1}{\sqrt{3}}u_1 = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad a_2 = b_2 = S_{11}^{-1/2}u_2 = \frac{1}{\sqrt{17}}u_2 = \frac{1}{\sqrt{34}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

  as $C_2u_1 = u_1$, $\bar{J}_2u_1 = 0$, $\bar{J}_2u_2 = u_2$ and $C_2u_2 = 0$. The first pair contrasts variable 1 with variable 2, and variable 3 with variable 4; the second pair adds them. Check: $a_1'S_{11}a_1 = \frac{1}{6} \cdot 6 = 1$, $a_1'S_{12}b_1 = \frac{1}{6} \cdot 4 = \frac{2}{3}$, $a_2'S_{11}a_2 = \frac{1}{34} \cdot 34 = 1$, $a_2'S_{12}b_2 = \frac{1}{34} \cdot 16 = \frac{8}{17}$ and $a_2'S_{11}a_1 = 0$.

Remark 15.25 ([3, §10.2]). Canonical correlations are also defined for two random vectors $x \in \mathbb{R}^{p_1}$ and $y \in \mathbb{R}^{p_2}$. Let

$$\operatorname{var}\begin{pmatrix} x \\ y \end{pmatrix} = \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}, \qquad \Sigma_{11} \in \mathbb{R}^{p_1 \times p_1}$$

with $\Sigma_{11} = \operatorname{var}(x)$ and $\Sigma_{22} = \operatorname{var}(y)$ PD (Definition 9.1). For $a \in \mathbb{R}^{p_1}$ and $b \in \mathbb{R}^{p_2}$, the matrix $T = \left(\begin{smallmatrix} a' & 0' \\ 0' & b' \end{smallmatrix}\right) \in \mathbb{R}^{2\times(p_1+p_2)}$ maps $(x', y')'$ to $(a'x, b'y)'$, and by Theorems 9.3(b) and 5.3(b),(c)

$$\operatorname{var}\begin{pmatrix} a'x \\ b'y \end{pmatrix} = T\Sigma T' = \begin{pmatrix} a'\Sigma_{11}a & a'\Sigma_{12}b \\ b'\Sigma_{21}a & b'\Sigma_{22}b \end{pmatrix}$$

So $a'x$ and $b'y$ have the variances $a'\Sigma_{11}a$, $b'\Sigma_{22}b$ and the covariance $a'\Sigma_{12}b$; and, by Theorem 9.42(a) for x and for y, $\operatorname{cov}(a'x, a_i'x) = a'\Sigma_{11}a_i$ and $\operatorname{cov}(b'y, b_i'y) = b'\Sigma_{22}b_i$. These are the expressions of Definition 15.22 with Σ_{jl} in place of S_{jl}. Canonical pairs (a_k, b_k) and canonical correlations of x and y are defined as there, with the variances and covariances of $a'x$ and $b'y$. The proof of Theorem 15.23 uses only that S_{11} and S_{22} are PD. Hence the canonical correlations of x and y are the singular values $\rho_1 \geqslant \cdots \geqslant \rho_s > 0$ of $\Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}$, attained at $a_k = \Sigma_{11}^{-1/2}u_k$, $b_k = \Sigma_{22}^{-1/2}w_k$ for its singular vectors u_k, w_k. With $K = \Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}$, $KK' = \Sigma_{11}^{-1/2}B\Sigma_{11}^{-1/2}$ for the symmetric $B = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$, so $\rho_1^2, \ldots, \rho_s^2$ are the nonzero eigenvalues of KK' (Theorem 15.10(a)) and hence of $\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ (Corollary 8.58(b) with $A = \Sigma_{11}$), the matrix named after Corollary 8.58. If $p_1 = 1$ and $\Sigma_{12} \neq 0$, then $s = 1$ and $\rho_1^2 = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}/\Sigma_{11}$: the canonical correlation of the random variable x and y is the multiple correlation coefficient $r_{x(y)}$ of Proposition 15.3, with Σ_{11}, Σ_{21}, Σ_{22} in the roles of σ^2, u, Σ.

The condition number

Definition 15.26 ([9, Definition P.52]). The **condition number** of an invertible $A \in \mathbb{R}^{n\times n}$ is $\kappa(A) = \|A\|_2\|A^{-1}\|_2$. For $X \in \mathbb{R}^{n\times p}$ of full column rank with $p < n$, with singular values $\sigma_1 \geqslant \cdots \geqslant \sigma_p > 0$, put $\kappa(X) = \sigma_1/\sigma_p$ (for $p = n$ the first definition gives the same number, by Proposition 15.27(a)).

Proposition 15.27 ([9, Proposition P.53, Corollary P.55]). (a) Let $A \in \mathbb{R}^{n\times n}$ be invertible. Then A has n singular values $\sigma_1 \geqslant \cdots \geqslant \sigma_n > 0$, $\|A^{-1}\|_2 = 1/\sigma_n$, and $\kappa(A) = \sigma_1/\sigma_n \geqslant 1$. If A is PD, with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_n > 0$, then $\kappa(A) = \lambda_1/\lambda_n$.

(b) If $X \in \mathbb{R}^{n\times p}$ has full column rank, then $X'X$ is PD and $\kappa(X'X) = \kappa(X)^2$.

Proof. (a)

- $\operatorname{rank} A = n$ (Theorem 4.13(d)), so $A = PDQ'$ with P, Q orthogonal and $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$, $\sigma_1 \geqslant \cdots \geqslant \sigma_n > 0$ (Theorem 15.9(b), with $r = n$).
- $P^{-1} = P'$ and $Q^{-1} = Q'$ (Proposition 6.13(b)), so $A^{-1} = QD^{-1}P'$ (Theorem 2.13(b)), and $\|A^{-1}\|_2 = \|D^{-1}\|_2$ (Proposition 15.15(c)).
- For $x \in \mathbb{R}^n$, $\|D^{-1}x\|^2 = \sum_i x_i^2/\sigma_i^2 \leqslant \|x\|^2/\sigma_n^2$, with equality for $x = e_n$; so $\|D^{-1}\|_2 = 1/\sigma_n$ (Proposition 15.15(a)).
- Hence $\kappa(A) = \sigma_1 \cdot \frac{1}{\sigma_n} \geqslant 1$ (Definition 15.14). If A is PD, its singular values are its eigenvalues (Theorem 15.10(e)), so $\kappa(A) = \lambda_1/\lambda_n$.

(b) $X'X$ is PD (Theorem 8.25). Its eigenvalues are $\sigma_1^2 \geqslant \cdots \geqslant \sigma_p^2 > 0$ (Theorem 15.10(a), with $r = p$), so by (a), $\kappa(X'X) = \sigma_1^2/\sigma_p^2 = \kappa(X)^2$. □

Theorem 15.28 ([9, Theorem P.54]). Let $A \in \mathbb{R}^{n\times n}$ be invertible, $Ax = b \neq 0$, and $A(x+\delta x) = b+\delta b$. Then

$$\frac{\|\delta x\|}{\|x\|} \leqslant \kappa(A)\,\frac{\|\delta b\|}{\|b\|}$$

Proof. • $\|My\| \leqslant \|M\|_2\|y\|$ for every matrix M and vector y (Proposition 15.15(a)).

- Subtracting $Ax = b$ from $A(x+\delta x) = b+\delta b$ gives $A\,\delta x = \delta b$, so $\delta x = A^{-1}\delta b$ and $\|\delta x\| \leqslant \|A^{-1}\|_2\|\delta b\|$.
- $x \neq 0$, as $b \neq 0$; and $\|b\| = \|Ax\| \leqslant \|A\|_2\|x\|$, so $1/\|x\| \leqslant \|A\|_2/\|b\|$.
- Multiplying the two bounds, both sides of each being $\geqslant 0$, $\|\delta x\|/\|x\| \leqslant \|A^{-1}\|_2\|A\|_2\,\|\delta b\|/\|b\|$. □

A large $\kappa(A)$ means that a small relative change of b may change the solution of $Ax = b$ by a large relative amount. By Proposition 15.27(b), forming $X'X$ squares the condition number of X. A QR decomposition avoids this: if X has full column rank and $X = QR$ (Theorem 6.16), then $X'X = R'Q'QR = R'R$, so the normal equations $X'Xb = X'y$ (Theorem 13.2) read $R'Rb = R'Q'y$, that is, $Rb = Q'y$, as R' is invertible (Theorem 2.31(a)); and $R'R = X'X$ shows that R has the same singular values as X (Theorem 15.10(a)), so $\kappa(R) = \kappa(X)$ (Proposition 15.27(a)). This is why least-squares estimates (Theorem 13.3) are computed in practice from $Rb = Q'y$ rather than from $X'X$.

A large $\kappa(X)$ also means that the columns of X are nearly linearly dependent, a near version of the exact collinearity of Proposition 13.8. Indeed, let $X \in \mathbb{R}^{n\times p}$ have full column rank $p \geqslant 2$. A matrix $B \in \mathbb{R}^{n\times p}$ has linearly dependent columns if and only if $\operatorname{rank} B < p$ (Theorem 4.13(b)), and by Theorem 15.17(b) for $A = X$ (so $r = p$) and $k = p-1$, every such B has $\|X - B\|_2 \geqslant \sigma_p$, with equality for the matrix $B = A_{p-1}$ of that theorem. So σ_p is the smallest spectral norm $\|E\|_2$ of a change E for which $X + E$ has linearly dependent columns, and relative to $\|X\|_2 = \sigma_1$ this distance is $\sigma_p/\sigma_1 = 1/\kappa(X)$.

Example 15.29. The matrix

$$A = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ -1 & 1 & 0 & 3 \end{pmatrix}$$

of Example 8.35 is PD, with the eigenvalues 4, 4, 1, 1 and the eigenvectors $(1,1,2,0)'$, $(-1,1,0,2)'$ for 4 and $(1,1,-1,0)'$, $(1,-1,0,1)'$ for 1 (Example 8.35). So $\kappa(A) = 4/1 = 4$ (Proposition 15.27(a)). Take $x = (1,1,2,0)'$, so that $b = Ax = 4x = (4,4,8,0)'$, and $\delta b = (1,1,-1,0)'$. As $A\,\delta b = \delta b$, the change of the solution is $\delta x = A^{-1}\delta b = \delta b = (1,1,-1,0)'$. Then

$$\frac{\|\delta x\|}{\|x\|} = \frac{\sqrt{3}}{\sqrt{6}}, \qquad \frac{\|\delta b\|}{\|b\|} = \frac{\sqrt{3}}{\sqrt{96}} = \frac{1}{4}\cdot\frac{\sqrt{3}}{\sqrt{6}}$$

so the bound of Theorem 15.28 holds with equality: the relative change of b is magnified four times in the solution.

The orthogonal Procrustes problem

Theorem 15.30 (Orthogonal Procrustes; [16, §7.4.5], [10, Problem 34.2])**.** Let $M \in \mathbb{R}^{n\times n}$, with an SVD $M = P\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)Q'$ (Theorem 15.9; for $M = 0$ take $P = Q = I_n$) and the singular values $\sigma_1, \dots, \sigma_n$, where $\sigma_j = 0$ for $j > \operatorname{rank} M$.

(a) For every orthogonal $U \in \mathbb{R}^{n\times n}$, $\operatorname{tr}(U'M) \leqslant \sigma_1 + \cdots + \sigma_n$, with equality for $U = PQ'$.

(b) Let $B, C \in \mathbb{R}^{n\times k}$ and $M = CB'$. For every orthogonal $U \in \mathbb{R}^{n\times n}$,

$$\|UB - C\|_F^2 \geqslant \|B\|_F^2 + \|C\|_F^2 - 2(\sigma_1 + \cdots + \sigma_n)$$

with equality for $U = PQ'$.

Proof. (a)

- $\Delta = \left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right) = \operatorname{diag}(\sigma_1, \dots, \sigma_n)$, and P, Q, U are orthogonal: $P'P = PP' = I_n$, and likewise for Q, U (Proposition 6.13(b)). So $Z = Q'U'P$ is orthogonal, as $Z'Z = P'UQQ'U'P = I_n$.
- By Theorem 1.11(b),

$$\operatorname{tr}(U'M) = \operatorname{tr}(U'P\Delta Q') = \operatorname{tr}(Q'U'P\Delta) = \operatorname{tr}(Z\Delta) = z_{11}\sigma_1 + \cdots + z_{nn}\sigma_n$$

  since Δ is diagonal.
- Column j of Z is a unit vector ($Z'Z = I_n$), so $z_{jj} \leqslant 1$; and $\sigma_j \geqslant 0$. Hence $\operatorname{tr}(U'M) \leqslant \sigma_1 + \cdots + \sigma_n$.
- For $U = PQ'$, $Z = Q'QP'P = I_n$, and equality holds.

(b)

- $(UB)'(UB) = B'U'UB = B'B$, and $\operatorname{tr}(C'UB) = \operatorname{tr}\Big((C'UB)'\Big) = \operatorname{tr}(B'U'C) = \operatorname{tr}(U'CB')$ (Theorem 1.11(a),(b)). So

$$\|UB - C\|_F^2 = \operatorname{tr}\Big((UB - C)'(UB - C)\Big) = \operatorname{tr}(B'B) - 2\operatorname{tr}(U'CB') + \operatorname{tr}(C'C)$$

- By Definition 9.26, $\operatorname{tr}(B'B) = \|B\|_F^2$ and $\operatorname{tr}(C'C) = \|C\|_F^2$; and (a) for $M = CB'$ gives $\operatorname{tr}(U'CB') \leqslant \sigma_1 + \cdots + \sigma_n$, with equality for $U = PQ'$. □

In (b) the columns of B and C are two configurations of k points in $\mathbb{R}^n$, and PQ' is an orthogonal transformation (a rotation, possibly composed with a reflection) that moves the first configuration as close as possible to the second; this is the **orthogonal Procrustes problem** of statistics (Procrustes analysis). For $B = I_n$, (b) says that PQ' is an orthogonal matrix nearest to $C = M$ in the Frobenius norm.

Exercises

Exercise 15.1. Let

$$A = \begin{pmatrix} 2 & 4 & 0 & 2 \\ 4 & 8 & 2 & 0 \\ 0 & 2 & 2 & 4 \\ 2 & 0 & 4 & 8 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}, \qquad y = \begin{pmatrix} 2 \\ 4 \\ 1 \\ -4 \end{pmatrix}$$

and $h_1 = (1, 1, 1, 1)'$, $h_2 = (1, -1, 1, -1)'$, $h_3 = (1, 1, -1, -1)'$, $h_4 = (1, -1, -1, 1)'$.

(a) Find $C = B^{-1/2}AB^{-1/2}$, and check that $h_1, \dots, h_4$ are eigenvectors of C. Find the maximum and the minimum of $x'Ax/x'Bx$ over $x \neq 0$, and vectors at which they are attained.

(b) Find the maximum of $(x'y)^2/x'Bx$ over $x \neq 0$, and a vector at which it is attained.

Solution. (a)

- B is PD (its eigenvalues 1, 4, 1, 4 are positive, Theorem 8.21(a)). $N = \operatorname{diag}(1, 2, 1, 2)$ is PD with $N^2 = B$, so $B^{1/2} = N$ (Theorem 8.46) and $B^{-1/2} = \operatorname{diag}(1, \frac{1}{2}, 1, \frac{1}{2})$.
- Row i and column j of A are multiplied by the ith and the jth diagonal entry of $B^{-1/2}$:

$$C = B^{-1/2}AB^{-1/2} = \begin{pmatrix} 2 & 2 & 0 & 1 \\ 2 & 2 & 1 & 0 \\ 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \end{pmatrix}$$

- $Ch_1 = (5,5,5,5)' = 5h_1$, $Ch_2 = (-1,1,-1,1)' = -h_2$, $Ch_3 = (3,3,-3,-3)' = 3h_3$ and $Ch_4 = (1,-1,-1,1)' = h_4$. So the algebraic multiplicities $\dim \mathcal{N}(C - cI)$ of $c = 5, 3, 1, -1$ are at least 1 (Theorem 8.4); as all multiplicities add up to 4 (Definition 7.10), the eigenvalues of C are $\mu_1 = 5$, $\mu_2 = 3$, $\mu_3 = 1$ and $\mu_4 = -1$.
- By Theorem 15.1, the maximum is 5, attained at $x = B^{-1/2}h_1 = (1, \frac{1}{2}, 1, \frac{1}{2})'$, and so also at $(2,1,2,1)'$; the minimum is -1, attained at $B^{-1/2}h_2 = (1, -\frac{1}{2}, 1, -\frac{1}{2})'$ and at $(2,-1,2,-1)'$. Check: at $x = (2,1,2,1)'$, $Ax = (10,20,10,20)' = 5Bx$, so $x'Ax = 5x'Bx = 5 \cdot 16 = 80$; at $x = (2,-1,2,-1)'$, $Ax = (-2,4,-2,4)' = -Bx$, so $x'Ax = -x'Bx = -16$.

(b) By Theorem 15.2, the maximum is $y'B^{-1}y = 4 + \frac{16}{4} + 1 + \frac{16}{4} = 13$, attained at $x = B^{-1}y = (2,1,1,-1)'$. Check: $(x'y)^2 = (4+4+1+4)^2 = 13^2$ and $x'Bx = 4+4+1+4 = 13$.

Exercise 15.2. Let

$$S_p = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

be the pooled sample covariance matrix of four variables in two groups of units (the paragraph after Theorem 15.2), and $d = (1,2,1,1)'$ the difference of their mean vectors.

(a) Show that S_p is PD.

(b) Solve $S_p a = d$ by Gauss–Jordan elimination. Find the maximum of $(a'd)^2/a'S_pa$ over $a \neq 0$ and a vector at which it is attained.

(c) Compute $(a'd)^2/a'S_pa$ for $a = d$ and for each variable alone ($a = e_1, e_2, e_3, e_4$).

Solution. (a) The leading principal minors are $\Delta_1 = 1$, $\Delta_2 = 2-1 = 1$, $\Delta_3 = 1(4-1) - 1(2-0) + 0 = 1$ along row 1 (Theorem 5.21), and, along row 1 again,

$$\Delta_4 = 1 \cdot \det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \big(2(4-1) - 1(2-0)\big) - 1(4-1) = 4 - 3 = 1$$

All are > 0, so S_p is PD (Theorem 8.34).

(b) Row-reduce the augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 1 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_4} \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

So $a = S_p^{-1}d = (-1,2,-1,1)'$; check: $S_pa = (-1+2,\ -1+4-1,\ 2-2+1,\ -1+2)' = (1,2,1,1)'$. By Theorem 15.2, the maximum of $(a'd)^2/a'S_pa$ is $d'S_p^{-1}d = d'a = -1+4-1+1 = 3$, attained at $a = (-1,2,-1,1)'$: there $a'd = 3$ and $a'S_pa = a'd = 3$, and $3^2/3 = 3$. The best combination gives the first and the third variable negative weights, although their mean differences are positive.

(c) For $a = d$: $d'd = 1+4+1+1 = 7$, $S_p d = (3,6,5,3)'$ and $d'S_p d = 3+12+5+3 = 23$; so the ratio is $7^2/23 = \frac{49}{23}$. For $a = e_j$ the ratio is d_j^2 divided by the jth diagonal entry of S_p: $1/1 = 1$, $4/2 = 2$, $1/2$ and $1/2$. All are below the maximum 3.

Exercise 15.3 ([3, Ch. 8, Exercise 10])**.** The random vector $(x_1, x_2, x_3)'$ has the covariance matrix

$$V = \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}$$

Find the multiple correlation coefficient between x_1 and x_2, x_3, and a combination $\alpha_1 x_2 + \alpha_2 x_3$ with the largest correlation with x_1. Check the result with the second expression of Proposition 15.3.

Solution.

- Partition V with $y = x_1$ and $x = (x_2, x_3)'$: $\sigma^2 = 3$, $u = (-1,-1)'$ and $\Sigma = \left(\begin{smallmatrix} 3 & -1 \\ -1 & 3 \end{smallmatrix}\right)$. Σ is PD: its leading principal minors are 3 and $9-1 = 8$ (Theorem 8.34).
- $\det \Sigma = 8$, so $\Sigma^{-1} = \frac{1}{8}\left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ (Theorem 5.24), and $\alpha = \Sigma^{-1}u = \frac{1}{8}(-4,-4)' = -\frac{1}{2}(1,1)'$.
- $u'\Sigma^{-1}u = u'\alpha = \frac{1}{2} + \frac{1}{2} = 1$. By Proposition 15.3, the largest squared correlation of x_1 with a combination of x_2, x_3 is $u'\Sigma^{-1}u/\sigma^2 = \frac{1}{3}$, attained at $\alpha = -\frac{1}{2}(1,1)'$, that is, for $-\frac{1}{2}(x_2 + x_3)$ (or any nonzero multiple of it). The multiple correlation coefficient is $\frac{1}{\sqrt{3}}$; the covariance $\alpha' u = 1$ is positive, so $\operatorname{cor}\left(x_1, -\frac{1}{2}(x_2+x_3)\right) = \frac{1}{\sqrt{3}}$.
- Second expression: along row 1 (Theorem 5.21), $\det V = 3(9-1) + 1(-3-1) - 1(1+3) = 24 - 4 - 4 = 16$. So the leading principal minors of V are 3, $9-1 = 8$ and $\det V = 16$, all positive, and V is PD (Theorem 8.34), as Proposition 15.3(b) requires. The $(1,1)$ entry of $Z = V^{-1}$ is $z_{11} = \det\Sigma / \det V = \frac{8}{16} = \frac{1}{2}$ (Theorem 5.24; the $(1,1)$ cofactor of V is $\det\Sigma$), and $1 - 1/(\sigma^2 z_{11}) = 1 - \frac{1}{3\cdot\frac{1}{2}} = 1 - \frac{2}{3} = \frac{1}{3}$.

Exercise 15.4. Let

$$A = \begin{pmatrix} 3 & 3 & 1 & -1 \\ 3 & 3 & -1 & 1 \\ 1 & -1 & -2 & -2 \\ -1 & 1 & -2 & -2 \end{pmatrix}$$

and $w = (1,-1,0,0)'$.

(a) Check that $(1,1,0,0)'$, $(1,-1,1,-1)'$, $(1,-1,-1,1)'$ and $(0,0,1,1)'$ are eigenvectors of A, and find the eigenvalues $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3 \geqslant \lambda_4$ of A.

(b) Find an orthonormal basis of the subspace of the vectors $u \perp w$. Use it to find the maximum and the minimum of $u'Au$ over the unit vectors $u \perp w$, and unit vectors at which they are attained.

(c) Check the inequality of Theorem 15.5(a) for $k = 2$ and that of Theorem 15.5(b) for $k = 3$, both with $w_1 = w$, and give a vector w_1 for which each of them is an equality.

Solution. (a) $A(1,1,0,0)' = (6,6,0,0)'$, $A(1,-1,1,-1)' = (2,-2,2,-2)'$, $A(1,-1,-1,1)' = (-2,2,2,-2)'$ and $A(0,0,1,1)' = (0,0,-4,-4)'$: eigenvectors belonging to 6, 2, -2 and -4. So $\dim \mathcal{N}(A - cI) \geqslant 1$ for $c = 6, 2, -2, -4$ (Theorem 8.4), and as the multiplicities add up to 4 (Definition 7.10), these are all the eigenvalues, each simple: $\lambda_1 = 6$, $\lambda_2 = 2$, $\lambda_3 = -2$, $\lambda_4 = -4$. The unit eigenvectors are $\pm q_i$, with $q_1 = \frac{1}{\sqrt{2}}(1,1,0,0)'$, $q_2 = \frac{1}{2}(1,-1,1,-1)'$, $q_3 = \frac{1}{2}(1,-1,-1,1)'$ and $q_4 = \frac{1}{\sqrt{2}}(0,0,1,1)'$. Check: $\operatorname{tr} A = 2 = 6 + 2 - 2 - 4$.

(b)

- $u \perp w$ means $u_1 = u_2$. The vectors $g_1 = \frac{1}{\sqrt{2}}(1,1,0,0)'$, $g_2 = \frac{1}{\sqrt{2}}(0,0,1,1)'$ and $g_3 = \frac{1}{\sqrt{2}}(0,0,1,-1)'$ satisfy this and are orthonormal, hence linearly independent (Proposition 6.11(b)); the subspace has dimension 3 (Theorem 6.6(c)), so they form an orthonormal basis of it (Theorem 4.5(d)).
- $g_1 = q_1$ and $g_2 = q_4$ are eigenvectors, so $Ag_1 = 6g_1$ and $Ag_2 = -4g_2$; and $Ag_3 = \frac{1}{\sqrt{2}}(2,-2,0,0)'$, so $g_3'Ag_3 = \frac{1}{2}(0,0,1,-1)(2,-2,0,0)' = 0$. As A is symmetric, $g_i'Ag_j = (Ag_i)'g_j$: so $g_1'Ag_1 = 6$, $g_2'Ag_2 = -4$, $g_1'Ag_j = 6g_1'g_j = 0$ for $j = 2,3$, and $g_2'Ag_3 = -4g_2'g_3 = 0$. With $G = [g_1\ g_2\ g_3]$, therefore, $G'AG = \operatorname{diag}(6,-4,0)$.
- By Lemma 9.24, the maximum is $\lambda_1(G'AG) = 6$, attained at $u = Ge_1 = g_1$, and the minimum is $\lambda_3(G'AG) = -4$, attained at $u = Ge_2 = g_2$.

(c)

- Theorem 15.5(a) with $k = 2$, $w_1 = w$: $m(w) = 6 \geqslant \lambda_2 = 2$.
- Theorem 15.5(b) with $k = 3$, $n - k = 1$, $w_1 = w$: the minimum $-4 \leqslant \lambda_3 = -2$.
- Equality holds in (a) for $w_1 = q_1$ and in (b) for $w_1 = q_4$: the maximum of $u'Au$ over the unit vectors $u \perp q_1$ is $\lambda_2 = 2$, attained at q_2, and the minimum over the unit vectors $u \perp q_4$ is $\lambda_3 = -2$, attained at q_3 (Theorem 9.25).

Exercise 15.5. The symmetric matrix

$$A = \begin{pmatrix} 3 & 1 & 0 & -1 \\ 1 & 4 & 2 & 2 \\ 0 & 2 & 4 & 2 \\ -1 & 2 & 2 & 4 \end{pmatrix}$$

has the eigenvalues 8, 4, 2, 1 (Exercise 9.11). Find the eigenvalues of $B = A(1|1)$, and check the inequalities of Theorem 15.6 for A and B.

Solution.

- Deleting row 1 and column 1 of A leaves

$$B = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix} = 2I_3 + 2J_3$$

- $2J_3 = 6\bar{J}_3$ and $I_3 = \bar{J}_3 + C_3$ (Definition 1.14), so $B = 8\bar{J}_3 + 2C_3$. As $\bar{J}_3\mathbf{1} = \mathbf{1}$ and $C_3\mathbf{1} = 0$, $B\mathbf{1} = 8\mathbf{1}$; if $\mathbf{1}'x = 0$, then $\bar{J}_3x = 0$ and $C_3x = x$, so $Bx = 2x$. Hence $\frac{1}{\sqrt{3}}(1,1,1)'$, $\frac{1}{\sqrt{2}}(1,-1,0)'$, $\frac{1}{\sqrt{6}}(1,1,-2)'$ are orthonormal eigenvectors for 8, 2, 2, and these are the eigenvalues of B (Theorem 8.3(c)): $\lambda_1(B) = 8$, $\lambda_2(B) = \lambda_3(B) = 2$.
- With $\lambda_1(A) = 8$, $\lambda_2(A) = 4$, $\lambda_3(A) = 2$, $\lambda_4(A) = 1$, the inequalities $\lambda_k(A) \geqslant \lambda_k(B) \geqslant \lambda_{k+1}(A)$ $(k = 1,2,3)$ read

$$8 \geqslant 8 \geqslant 4 \geqslant 2 \geqslant 2 \geqslant 2 \geqslant 1$$

and they hold.

Exercise 15.6. Five units are measured on four variables; the rows of

$$X = \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 2 & 1 & 1 & 0 \end{pmatrix}$$

hold the values.

(a) Find $X'X$, check that $(1,1,1,2)'$, $(-2,1,1,0)'$, $(-2,-2,-2,3)'$ and $(0,-1,1,0)'$ are eigenvectors of it, and find its eigenvalues.

(b) A sixth unit with the values $x' = (2,2,2,2)$ is added as a new row of X. Find the new matrix $X'X + xx'$, check that $(2,2,2,3)'$, $(-2,1,1,0)'$, $(-1,-1,-1,2)'$ and $(0,-1,1,0)'$ are eigenvectors of it, find its eigenvalues, and check Theorem 15.7.

(c) Check Theorem 15.6 for $X'X$ and its 3×3 submatrix $(X'X)(4|4)$, which belongs to the first three variables.

Solution. (a) The entries of $X'X$ are the inner products of the columns of X (Theorem 1.4(d)):

$$X'X = \begin{pmatrix} 9 & 2 & 2 & 6 \\ 2 & 6 & 5 & 6 \\ 2 & 5 & 6 & 6 \\ 6 & 6 & 6 & 16 \end{pmatrix}$$

Then $X'X(1,1,1,2)' = (25,25,25,50)'$, $X'X(-2,1,1,0)' = (-14,7,7,0)'$, $X'X(-2,-2,-2,3)' = (-8,-8,-8,12)'$ and $X'X(0,-1,1,0)' = (0,-1,1,0)'$: eigenvectors belonging to 25, 7, 4 and 1. So $\dim \mathcal{N}(X'X - cI) \geqslant 1$ for these four c (Theorem 8.4); as the multiplicities add up to 4 (Definition 7.10), the eigenvalues are 25, 7, 4 and 1.

(b) $xx' = 4J_4$, the 4×4 matrix with every entry 4, and

$$X'X + xx' = \begin{pmatrix} 13 & 6 & 6 & 10 \\ 6 & 10 & 9 & 10 \\ 6 & 9 & 10 & 10 \\ 10 & 10 & 10 & 20 \end{pmatrix}$$

which is $X'X$ for the 6×4 data matrix (Theorem 1.4(d)). Then

$$(X'X + xx')(2,2,2,3)' = (80,80,80,120)', \qquad (X'X + xx')(-2,1,1,0)' = (-14,7,7,0)'$$
$$(X'X + xx')(-1,-1,-1,2)' = (-5,-5,-5,10)', \qquad (X'X + xx')(0,-1,1,0)' = (0,-1,1,0)'$$

so, as in (a), the eigenvalues are 40, 7, 5 and 1. As xx' is PSD ($v'xx'v = (x'v)^2 \geqslant 0$), Theorem 15.7 applies, and indeed $40 \geqslant 25$, $7 \geqslant 7$, $5 \geqslant 4$ and $1 \geqslant 1$.

(c) Deleting row 4 and column 4 of $X'X$ leaves

$$B = (X'X)(4|4) = \begin{pmatrix} 9 & 2 & 2 \\ 2 & 6 & 5 \\ 2 & 5 & 6 \end{pmatrix}$$

the matrix of the inner products of the first three columns of X. $B(1,1,1)' = (13,13,13)'$, $B(-2,1,1)' = (-14,7,7)'$ and $B(0,-1,1)' = (0,-1,1)'$, so, as in (a), the eigenvalues of B are 13, 7 and 1. By Theorem 15.6 ($n = 4$), $\lambda_k(X'X) \geqslant \lambda_k(B) \geqslant \lambda_{k+1}(X'X)$ for $k = 1,2,3$; indeed

$$25 \geqslant 13 \geqslant 7 \geqslant 7 \geqslant 4 \geqslant 1 \geqslant 1$$

Exercise 15.7. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 1 & -1 \end{pmatrix}$$

(a) Find an SVD and the compact SVD of A.

(b) Find $\|A\|_F$ and $\|A\|_2$.

(c) Give orthonormal bases of the four subspaces of A, and the matrix P_1P_1' of the projection onto $\mathcal{C}(A)$; check that it equals $A(A'A)^{-1}A'$.

Solution. (a)

- $A'A = \left(\begin{smallmatrix} 6 & 3 \\ 3 & 6 \end{smallmatrix}\right)$, with characteristic polynomial $t^2 - 12t + 27 = (t-9)(t-3)$ (Definition 7.7(b)) and the unit eigenvectors $q_1 = \frac{1}{\sqrt{2}}(1,1)'$ for 9 and $q_2 = \frac{1}{\sqrt{2}}(1,-1)'$ for 3. So $r = 2$, $\sigma_1 = 3$, $\sigma_2 = \sqrt{3}$, and $Q = [q_1\ q_2]$.
- $Aq_1 = \frac{1}{\sqrt{2}}(3,3,0)'$ and $Aq_2 = \frac{1}{\sqrt{2}}(1,-1,2)'$, so $p_1 = \frac{1}{3}Aq_1 = \frac{1}{\sqrt{2}}(1,1,0)'$ and $p_2 = \frac{1}{\sqrt{3}}Aq_2 = \frac{1}{\sqrt{6}}(1,-1,2)'$ (Theorem 15.9).
- $\mathcal{N}(P_1')$: $y_1 + y_2 = 0$ and $y_1 - y_2 + 2y_3 = 0$ give $y = t(1,-1,-1)'$; so $p_3 = \frac{1}{\sqrt{3}}(1,-1,-1)'$.
- Hence, with $Q' = Q = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$,

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & \sqrt{3} \\ 0 & 0 \end{pmatrix} Q', \qquad A = 3p_1q_1' + \sqrt{3}\,p_2q_2' = \frac{3}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 2 & -2 \end{pmatrix}$$

(b) $\|A\|_F^2 = 4 + 1 + 1 + 4 + 1 + 1 = 12 = 9 + 3$ (Theorem 15.10(d)), so $\|A\|_F = 2\sqrt{3}$; and $\|A\|_2 = \sigma_1 = 3 \leqslant 2\sqrt{3}$ (Proposition 15.15(d)).

(c)

- By Corollary 15.11: p_1, p_2 is an orthonormal basis of $\mathcal{C}(A)$, p_3 of $\mathcal{N}(A')$, q_1, q_2 of $\mathcal{C}(A') = \mathbb{R}^2$, and $\mathcal{N}(A) = \{0\}$ ($r = n = 2$, no Q_2).
- $P_1P_1' = p_1p_1' + p_2p_2'$ (Theorem 1.4(d)):

$$P_1P_1' = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

- $\det(A'A) = 27$, so by Theorem 5.24

$$(A'A)^{-1} = \frac{1}{27}\begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad A(A'A)^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}$$

and

$$A(A'A)^{-1}A' = \frac{1}{3}\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} = P_1P_1'$$

Exercise 15.8 (cf. [5, Ch. 21, Exercise 26], [8, Ch. 21, Exercise 26])**.** Find an SVD and the compact SVD of $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ by applying Theorem 15.9 to A', starting from the 2×2 matrix AA'.

Solution.

- $A' \in \mathbb{R}^{3\times 2}$ has rank 2, and $(A')'A' = AA' = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, with characteristic polynomial $t^2 - 4t + 3 = (t-3)(t-1)$ and the unit eigenvectors $p_1 = \frac{1}{\sqrt{2}}(1,1)'$ for 3 and $p_2 = \frac{1}{\sqrt{2}}(1,-1)'$ for 1. So $D = \operatorname{diag}(\sqrt{3}, 1)$.
- Theorem 15.9 for A', with $P = [p_1\ p_2]$ in the role of Q: the columns of $A'PD^{-1}$ are $q_1 = \frac{1}{\sqrt{3}}A'p_1 = \frac{1}{\sqrt{6}}(1,1,2)'$ and $q_2 = A'p_2 = \frac{1}{\sqrt{2}}(1,-1,0)'$. A unit vector orthogonal to both: $y_1 + y_2 + 2y_3 = 0$ and $y_1 - y_2 = 0$ give $q_3 = \frac{1}{\sqrt{3}}(1,1,-1)'$.
- So $Q'A'P = \left(\begin{smallmatrix} D \\ 0 \end{smallmatrix}\right)$ with $Q = [q_1\ q_2\ q_3]$, and transposing, $P'AQ = [D\ \ 0]$:

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{pmatrix}$$

- The compact SVD is $A = \sqrt{3}\, p_1 q_1' + p_2 q_2' = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \end{smallmatrix}\right) + \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \end{smallmatrix}\right)$. Also $Aq_3 = 0$, so q_3 spans $\mathcal{N}(A)$.

Exercise 15.9. Let

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{pmatrix}$$

(a) Find $\operatorname{rank} A$, the eigenvalues of $A'A$, and an SVD of A.

(b) Write down the compact SVD of A and orthonormal bases of the four subspaces of A.

(c) Find $\|A\|_F$, $\|A\|_2$ and the matrix P_1P_1' of the projection onto $\mathcal{C}(A)$.

Solution. (a)

- The two columns are equal and nonzero, so $\operatorname{rank} A = 1 = r$.
- $A'A = \left(\begin{smallmatrix} 9 & 9 \\ 9 & 9 \end{smallmatrix}\right)$, with characteristic polynomial $t^2 - 18t = t(t-18)$ (Definition 7.7(b)), and $A'A(1,1)' = (18,18)'$, $A'A(1,-1)' = 0$. So the eigenvalues are 18 and 0, with the unit eigenvectors $q_1 = \frac{1}{\sqrt{2}}(1,1)'$ and $q_2 = \frac{1}{\sqrt{2}}(1,-1)'$; $\sigma_1 = \sqrt{18} = 3\sqrt{2}$ and $D = (3\sqrt{2})$.
- $P_1 = AQ_1D^{-1}$ (Theorem 15.9): $Aq_1 = \frac{1}{\sqrt{2}}(2,4,4)'$, so $p_1 = \frac{1}{3\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(2,4,4)' = \frac{1}{3}(1,2,2)'$.
- $\mathcal{N}(P_1')$ consists of the y with $y_1 + 2y_2 + 2y_3 = 0$. The vectors $(2,1,-2)'$ and $(2,-2,1)'$ lie in it and are orthogonal ($4 - 2 - 2 = 0$); normalized, $p_2 = \frac{1}{3}(2,1,-2)'$ and $p_3 = \frac{1}{3}(2,-2,1)'$.
- Hence $A = P\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)Q'$ with

$$P = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

(b) The compact SVD is $A = 3\sqrt{2}\, p_1 q_1' = 3\sqrt{2} \cdot \frac{1}{3}(1,2,2)' \cdot \frac{1}{\sqrt{2}}(1,1) = (1,2,2)'(1,1)$, which is A. By Corollary 15.11: p_1 is an orthonormal basis of $\mathcal{C}(A)$, p_2, p_3 of $\mathcal{N}(A')$, q_1 of $\mathcal{C}(A')$, and q_2 of $\mathcal{N}(A)$.

(c) $\|A\|_F^2 = 1+1+4+4+4+4 = 18 = \sigma_1^2$ (Theorem 15.10(d)), so $\|A\|_F = 3\sqrt{2} = \sigma_1 = \|A\|_2$: the two norms agree for a matrix of rank 1. By Corollary 15.11,

$$P_1 P_1' = p_1 p_1' = \frac{1}{9}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}$$

Exercise 15.10. Let

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \\ 3 & 3 \\ 1 & 1 \end{pmatrix}$$

(a) Find a singular value decomposition $A = P\left(\begin{smallmatrix} D \\ 0 \end{smallmatrix}\right) Q'$ with $P \in \mathbb{R}^{4\times 4}$, and the compact form.

(b) Give orthonormal bases of $\mathcal{C}(A)$, $\mathcal{N}(A')$, $\mathcal{C}(A')$ and $\mathcal{N}(A)$.

(c) Find $\|A\|_2$, $\|A\|_F$, $\kappa(A)$ and $\kappa(A'A)$.

(d) Find A^+ from the singular value decomposition, and check it with $(A'A)^{-1}A'$.

Solution. (a)

- $A'A = \left(\begin{smallmatrix} 20 & 16 \\ 16 & 20 \end{smallmatrix}\right)$, with $A'A(1,1)' = 36(1,1)'$ and $A'A(1,-1)' = 4(1,-1)'$. By Theorem 15.9(b): $\sigma_1 = 6$, $\sigma_2 = 2$, $q_1 = \frac{1}{\sqrt{2}}(1,1)'$, $q_2 = \frac{1}{\sqrt{2}}(1,-1)'$, $p_1 = \frac{1}{6}Aq_1 = \frac{1}{6\sqrt{2}}(4,4,6,2)' = \frac{1}{3\sqrt{2}}(2,2,3,1)'$ and $p_2 = \frac{1}{2}Aq_2 = \frac{1}{\sqrt{2}}(-1,1,0,0)'$.
- A vector y orthogonal to p_1 and p_2 satisfies $y_1 = y_2$ and $4y_1 + 3y_3 + y_4 = 0$; the vectors $(0,0,1,-3)'$ and $(5,5,-6,-2)'$ satisfy both and are orthogonal, with lengths $\sqrt{10}$ and $3\sqrt{10}$. Take $p_3 = \frac{1}{\sqrt{10}}(0,0,1,-3)'$ and $p_4 = \frac{1}{3\sqrt{10}}(5,5,-6,-2)'$.
- Then $A = P\left(\begin{smallmatrix} D \\ 0 \end{smallmatrix}\right) Q'$ with $P = [p_1\ p_2\ p_3\ p_4]$, $D = \operatorname{diag}(6,2)$, $Q = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, and

$$A = 6p_1q_1' + 2p_2q_2' = \begin{pmatrix} 2 & 2 \\ 2 & 2 \\ 3 & 3 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

(b) By Corollary 15.11: p_1, p_2 is an orthonormal basis of $\mathcal{C}(A)$, p_3, p_4 of $\mathcal{N}(A')$, and q_1, q_2 of $\mathcal{C}(A') = \mathbb{R}^2$; $\mathcal{N}(A) = \{0\}$ ($r = n = 2$, no Q_2).

(c) $\|A\|_2 = \sigma_1 = 6$ (Definition 15.14); $\|A\|_F^2 = 10+10+18+2 = 40 = 6^2 + 2^2$ (Theorem 15.10(d)), so $\|A\|_F = 2\sqrt{10}$; $\kappa(A) = 6/2 = 3$ (Definition 15.26) and $\kappa(A'A) = \kappa(A)^2 = 9$ (Proposition 15.27(b)).

(d) By Theorem 15.21,

$$A^+ = \frac{1}{6}q_1p_1' + \frac{1}{2}q_2p_2' = \frac{1}{36}\begin{pmatrix} 2 & 2 & 3 & 1 \\ 2 & 2 & 3 & 1 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix} = \frac{1}{36}\begin{pmatrix} -7 & 11 & 3 & 1 \\ 11 & -7 & 3 & 1 \end{pmatrix}$$

As $\operatorname{rank} A = 2$, the number of columns, $A^+ = (A'A)^{-1}A'$ (Theorem 12.24(c)). Here $\det(A'A) = 400 - 256 = 144$, so $(A'A)^{-1} = \frac{1}{144}\left(\begin{smallmatrix} 20 & -16 \\ -16 & 20 \end{smallmatrix}\right) = \frac{1}{36}\left(\begin{smallmatrix} 5 & -4 \\ -4 & 5 \end{smallmatrix}\right)$ (Theorem 5.24), and

$$(A'A)^{-1}A' = \frac{1}{36}\begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}\begin{pmatrix} 1 & 3 & 3 & 1 \\ 3 & 1 & 3 & 1 \end{pmatrix} = \frac{1}{36}\begin{pmatrix} -7 & 11 & 3 & 1 \\ 11 & -7 & 3 & 1 \end{pmatrix}$$

Exercise 15.11. Let

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 4 & 2 & 4 \\ 8 & 4 & 8 \end{pmatrix}$$

(a) Write $A = uv'$, and find a compact singular value decomposition, $\|A\|_2$ and $\|A\|_F$.

(b) Find A^+, AA^+ and A^+A.

(c) Decide whether $Ax = b$ is consistent for $b = (3, 6, 6, 12)'$ and for $b = (1, 0, 0, 0)'$. Find the solution of minimum length in the consistent case, and compare it with the solution $(0, 3, 0)'$.

Solution. (a)

- The rows of A are 1, 2, 2, 4 times $(2, 1, 2)$, so $A = uv'$ with $u = (1, 2, 2, 4)'$, $\|u\| = 5$, and $v = (2, 1, 2)'$, $\|v\| = 3$; $\operatorname{rank} A = 1$.
- $A'A = v(u'u)v' = 25vv'$ has rank $\operatorname{rank} A = 1$ (Theorem 4.18(b)) and $A'Av = 25v(v'v) = 225v$. By Theorem 8.4, 225 is its only nonzero eigenvalue, and the others are 0.
- By Theorem 15.9(b): $\sigma_1 = \sqrt{225} = 15$, $q_1 = \frac{1}{3}v$, and $p_1 = \frac{1}{15}Aq_1 = \frac{1}{45}u(v'v) = \frac{1}{5}u$. So $A = 15p_1q_1'$ is a compact SVD.
- $\|A\|_2 = \sigma_1 = 15$ (Definition 15.14) and $\|A\|_F^2 = \sigma_1^2 = 225$ (Theorem 15.10(d)), so $\|A\|_F = 15$; indeed $\|A\|_F^2 = 9 + 36 + 36 + 144 = 225$.

(b) By Theorem 15.21,

$$A^+ = \frac{1}{15}q_1p_1' = \frac{1}{225}vu' = \frac{1}{225}\begin{pmatrix} 2 & 4 & 4 & 8 \\ 1 & 2 & 2 & 4 \\ 2 & 4 & 4 & 8 \end{pmatrix}, \qquad AA^+ = p_1p_1' = \frac{1}{25}uu', \qquad A^+A = q_1q_1' = \frac{1}{9}vv'$$

(c)

- A^+ is a g-inverse of A ((P1) of Definition 12.22), so by Theorem 12.2(b), $Ax = b$ is consistent if and only if $AA^+b = \frac{u'b}{25}u$ equals b. For $b = (3, 6, 6, 12)' = 3u$, $u'b = 75$ and $AA^+b = 3u = b$: consistent. For $b = (1, 0, 0, 0)'$, $AA^+b = \frac{1}{25}u \neq b$: there is no solution.
- A^+ is a minimum-norm g-inverse ((P1) and (P4)), so by Theorem 12.20 the shortest solution for $b = 3u$ is $x^+ = A^+b = \frac{u'b}{225}v = \frac{75}{225}v = \frac{1}{3}(2, 1, 2)'$, with $\|x^+\|^2 = 1$.
- The solution $x = (0, 3, 0)'$ ($Ax = 3\,\mathrm{col}_2(A) = 3u$) has $\|x\|^2 = 9$; and $x - x^+ = \frac{1}{3}(-2, 8, -2)'$ is orthogonal to x^+, with $\|x - x^+\|^2 = \frac{72}{9} = 8$, so $9 = 1 + 8$.

Exercise 15.12. Let

$$A = \begin{pmatrix} 6 & 0 & 0 \\ 4 & 4 & 6 \\ 5 & 2 & 6 \\ 2 & -4 & 0 \end{pmatrix}$$

(a) Find a singular value decomposition of A and its compact form. (Write $A'A = 9M$ and use the vectors $w_1 = (2, 1, 2)'$, $w_2 = (-2, 2, 1)'$, $w_3 = (1, 2, -2)'$.)

(b) Give orthonormal bases of the four subspaces of A, and the eigenvalues of AA'.

(c) Find $\|A\|_2$, $\|A\|_F$, $\kappa(A)$ and $\kappa(A'A)$.

(d) Find A^+, once from $(A'A)^{-1}A'$ and once from the singular value decomposition, and AA^+.

(e) Find a matrix A_2 of rank at most 2 nearest to A in the spectral norm, and $\|A - A_2\|_2$.

Solution. (a)

- The entries of $A'A$ are the inner products of the columns of A (Theorem 1.4(d)):

$$A'A = \begin{pmatrix} 81 & 18 & 54 \\ 18 & 36 & 36 \\ 54 & 36 & 72 \end{pmatrix} = 9M, \qquad M = \begin{pmatrix} 9 & 2 & 6 \\ 2 & 4 & 4 \\ 6 & 4 & 8 \end{pmatrix}$$

  $Mw_1 = (32, 16, 32)' = 16w_1$, $Mw_2 = (-8, 8, 4)' = 4w_2$ and $Mw_3 = (1, 2, -2)' = w_3$. So $A'Aw_k = \lambda_k w_k$ with $\lambda_1 = 144$, $\lambda_2 = 36$, $\lambda_3 = 9$; as $\dim \mathcal{N}(A'A - cI) \geqslant 1$ for these three values and the multiplicities add up to 3 (Theorem 8.4, Definition 7.10), they are the eigenvalues of $A'A$. The w_k are orthogonal with length 3, so $q_k = \frac{1}{3}w_k$ gives an orthogonal $Q = [q_1\ q_2\ q_3]$. $A'A$ has three nonzero eigenvalues, so $\operatorname{rank} A = \operatorname{rank}(A'A) = 3$ (Theorems 8.4 and 4.18(b)), and by Theorem 15.9(b), $\sigma_1 = 12$, $\sigma_2 = 6$, $\sigma_3 = 3$.

- $p_k = \sigma_k^{-1} A q_k = (3\sigma_k)^{-1} A w_k$: $Aw_1 = (12, 24, 24, 0)'$, $Aw_2 = (-12, 6, 0, -12)'$ and $Aw_3 = (6, 0, -3, -6)'$ give $p_1 = \frac{1}{3}(1, 2, 2, 0)'$, $p_2 = \frac{1}{3}(-2, 1, 0, -2)'$ and $p_3 = \frac{1}{3}(2, 0, -1, -2)'$.

- A vector y orthogonal to p_1, p_2, p_3 satisfies $y_1 + 2y_2 + 2y_3 = 0$, $-2y_1 + y_2 - 2y_4 = 0$ and $2y_1 - y_3 - 2y_4 = 0$, so $y = c\,(0, 2, -2, 1)'$; take $p_4 = \frac{1}{3}(0, 2, -2, 1)'$.

Hence $A = P \begin{pmatrix} D \\ 0 \end{pmatrix} Q'$ with $D = \operatorname{diag}(12, 6, 3)$,

$$P = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 & 0 \\ 2 & 1 & 0 & 2 \\ 2 & 0 & -1 & -2 \\ 0 & -2 & -2 & 1 \end{pmatrix}, \qquad Q = \frac{1}{3}\begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}$$

and the compact form is

$$A = 12p_1q_1' + 6p_2q_2' + 3p_3q_3' = \frac{4}{3}\begin{pmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 4 & 2 & 4 \\ 0 & 0 & 0 \end{pmatrix} + \frac{2}{3}\begin{pmatrix} 4 & -4 & -2 \\ -2 & 2 & 1 \\ 0 & 0 & 0 \\ 4 & -4 & -2 \end{pmatrix} + \frac{1}{3}\begin{pmatrix} 2 & 4 & -4 \\ 0 & 0 & 0 \\ -1 & -2 & 2 \\ -2 & -4 & 4 \end{pmatrix}$$

(b) By Corollary 15.11: p_1, p_2, p_3 is an orthonormal basis of $\mathcal{C}(A)$, p_4 of $\mathcal{N}(A')$ (indeed $A'(0, 2, -2, 1)' = 0$), q_1, q_2, q_3 of $\mathcal{C}(A') = \mathbb{R}^3$, and $\mathcal{N}(A) = \{0\}$. By Theorem 15.10(a),(b), the eigenvalues of the 4×4 matrix AA' are 144, 36, 9, 0.

(c) $\|A\|_2 = 12$; $\|A\|_F^2 = 36 + 68 + 65 + 20 = 189 = 12^2 + 6^2 + 3^2$ (the four terms are the squared lengths of the rows; Theorem 15.10(d)), so $\|A\|_F = 3\sqrt{21}$; $\kappa(A) = 12/3 = 4$ (Definition 15.26) and $\kappa(A'A) = 16$ (Proposition 15.27(b)).

(d)

- $\operatorname{rank} A = 3$, the number of columns, so $A^+ = (A'A)^{-1}A' = \frac{1}{9}M^{-1}A'$ (Theorem 12.24(c)). Along row 1, $\det M = 9 \cdot 16 - 2 \cdot (-8) + 6 \cdot (-16) = 64$ (Theorem 5.21), and by Theorem 5.24

$$M^{-1} = \frac{1}{64}\begin{pmatrix} 16 & 8 & -16 \\ 8 & 36 & -24 \\ -16 & -24 & 32 \end{pmatrix} = \frac{1}{16}\begin{pmatrix} 4 & 2 & -4 \\ 2 & 9 & -6 \\ -4 & -6 & 8 \end{pmatrix}$$

  so

$$A^+ = \frac{1}{144}\begin{pmatrix} 4 & 2 & -4 \\ 2 & 9 & -6 \\ -4 & -6 & 8 \end{pmatrix}\begin{pmatrix} 6 & 4 & 5 & 2 \\ 0 & 4 & 2 & -4 \\ 0 & 6 & 6 & 0 \end{pmatrix} = \frac{1}{36}\begin{pmatrix} 6 & 0 & 0 & 0 \\ 3 & 2 & -2 & -8 \\ -6 & 2 & 4 & 4 \end{pmatrix}$$

- By Theorem 15.21, $A^+ = \frac{1}{12}q_1p_1' + \frac{1}{6}q_2p_2' + \frac{1}{3}q_3p_3'$, the same matrix:

$$\frac{1}{108}\begin{pmatrix} 2 & 4 & 4 & 0 \\ 1 & 2 & 2 & 0 \\ 2 & 4 & 4 & 0 \end{pmatrix} + \frac{1}{54}\begin{pmatrix} 4 & -2 & 0 & 4 \\ -4 & 2 & 0 & -4 \\ -2 & 1 & 0 & -2 \end{pmatrix} + \frac{1}{27}\begin{pmatrix} 2 & 0 & -1 & -2 \\ 4 & 0 & -2 & -4 \\ -4 & 0 & 2 & 4 \end{pmatrix} = \frac{1}{36}\begin{pmatrix} 6 & 0 & 0 & 0 \\ 3 & 2 & -2 & -8 \\ -6 & 2 & 4 & 4 \end{pmatrix}$$

- By Theorem 15.21, $AA^+ = P_1P_1'$, where $P_1 = [p_1\ p_2\ p_3]$; as $PP' = I_4$ (Proposition 6.13(b)), $P_1P_1' = I_4 - p_4p_4'$:

$$AA^+ = I_4 - \frac{1}{9}\begin{pmatrix}0&0&0&0\\0&4&-4&2\\0&-4&4&-2\\0&2&-2&1\end{pmatrix} = \frac{1}{9}\begin{pmatrix}9&0&0&0\\0&5&4&-2\\0&4&5&2\\0&-2&2&8\end{pmatrix}$$

  the matrix of the orthogonal projection onto $\mathcal{C}(A)$ (Corollary 15.11).

(e) By Theorem 15.17(b) with $k = 2 < r = 3$, every B of rank at most 2 has $\|A - B\|_2 \geqslant \sigma_3 = 3$, with equality for

$$A_2 = 12p_1q_1' + 6p_2q_2' = A - 3p_3q_3' = \frac{1}{3}\begin{pmatrix}16&-4&4\\12&12&18\\16&8&16\\8&-8&-4\end{pmatrix}$$

So $\|A - A_2\|_2 = 3 = \|A\|_2/\kappa(A)$: the matrix $E = A_2 - A = -3p_3q_3'$, of spectral norm 3, makes the columns of $A + E = A_2$ linearly dependent ($A_2w_3 = 0$), and no E of smaller spectral norm does.

Exercise 15.13. Let

$$A = \begin{pmatrix}1&4&0&2\\4&1&2&0\\0&2&1&4\\2&0&4&1\end{pmatrix}, \qquad h_1 = \begin{pmatrix}1\\1\\1\\1\end{pmatrix},\ h_2 = \begin{pmatrix}1\\-1\\1\\-1\end{pmatrix},\ h_3 = \begin{pmatrix}1\\1\\-1\\-1\end{pmatrix},\ h_4 = \begin{pmatrix}1\\-1\\-1\\1\end{pmatrix}$$

(a) Show that $Ah_1 = 7h_1$, $Ah_2 = -5h_2$, $Ah_3 = 3h_3$ and $Ah_4 = -h_4$.

(b) Find a singular value decomposition $A = PDQ'$ of A. Why is $P \neq Q$?

(c) Find $\|A\|_2$, $\|A\|_F$ and $\kappa(A)$.

Solution. (a) $Ah_1 = (7, 7, 7, 7)'$ (every row sum is 7), $Ah_2 = (-5, 5, -5, 5)'$, $Ah_3 = (3, 3, -3, -3)'$ and $Ah_4 = (-1, 1, 1, -1)'$.

(b)

- The h_k are orthogonal with length 2, so $Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$ is orthogonal, and $Q'AQ = \Lambda = \operatorname{diag}(7, -5, 3, -1)$ (from $AQ = Q\Lambda$).
- $A'A = A^2 = Q\Lambda^2Q'$, so $A'Aq_k = \lambda_k^2q_k$ for $q_k = \frac{1}{2}h_k$, with $\lambda_k^2 = 49, 25, 9, 1$ in decreasing order. By Theorem 15.9(b), $\sigma_1 = 7$, $\sigma_2 = 5$, $\sigma_3 = 3$, $\sigma_4 = 1$, $D = \operatorname{diag}(7, 5, 3, 1)$, and $p_k = \sigma_k^{-1}Aq_k = (\lambda_k/|\lambda_k|)q_k$:

$$P = \frac{1}{2}[h_1\ \ -h_2\ \ h_3\ \ -h_4], \qquad D = \operatorname{diag}(7, 5, 3, 1), \qquad Q = \frac{1}{2}[h_1\ h_2\ h_3\ h_4]$$

- The columns of P and Q differ in sign exactly where the eigenvalue is negative: $P = Q\Delta$ with $\Delta = \operatorname{diag}(1, -1, 1, -1)$. As A has negative eigenvalues, $P \neq Q$; the singular values are the absolute values of the eigenvalues (Theorem 15.10(e)).

(c) $\|A\|_2 = \sigma_1 = 7$ (Definition 15.14); $\|A\|_F^2 = 4 \cdot 21 = 84 = 49 + 25 + 9 + 1$ (Theorem 15.10(d)), so $\|A\|_F = 2\sqrt{21}$; $\operatorname{rank} A = 4$ and $\kappa(A) = 7/1 = 7$ (Proposition 15.27(a)).

Exercise 15.14. Let

$$A = \begin{pmatrix} 4 & 2 & 1 & 0 \\ 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 1 \\ 0 & 1 & 2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} -4 & -2 & -1 & 0 \\ 2 & 4 & 0 & 1 \\ 1 & 0 & 4 & 2 \\ 0 & -1 & -2 & -4 \end{pmatrix}$$

and $h_1 = (1,1,1,1)'$, $h_2 = (1,-1,1,-1)'$, $h_3 = (1,1,-1,-1)'$, $h_4 = (1,-1,-1,1)'$.

(a) Find $A'A$ and $A'Ah_1, \dots, A'Ah_4$. Find the singular values, $\|\cdot\|_2$ and $\|\cdot\|_F$ of A and of B.

(b) Find $Ah_1, \dots, Ah_4$ and the eigenvalues of A, and compare their absolute values with the singular values of A.

(c) Find $\|A+B\|_2$ and $\|AB\|_2$, and check the inequalities of Proposition 15.16.

(d) Find unit vectors u, v with $u'Av = \|A\|_2$.

Solution. (a)

- The entries of $A'A$ are the inner products of the columns of A (Theorem 1.4(d)):

$$A'A = \begin{pmatrix} 21 & 16 & 8 & 4 \\ 16 & 21 & 4 & 8 \\ 8 & 4 & 21 & 16 \\ 4 & 8 & 16 & 21 \end{pmatrix}$$

  and $A'Ah_1 = (49,49,49,49)' = 49h_1$, $A'Ah_2 = (9,-9,9,-9)' = 9h_2$, $A'Ah_3 = (25,25,-25,-25)' = 25h_3$, $A'Ah_4 = (1,-1,-1,1)' = h_4$. So $\dim \mathcal{N}(A'A - cI) \geqslant 1$ for $c = 49, 25, 9, 1$ (Theorem 8.4), and as the multiplicities add up to 4 (Definition 7.10), these are the eigenvalues of $A'A$. By Theorem 15.10(a), A has the singular values 7, 5, 3, 1.

- $B = UA$ with

$$U = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

  (row 1 of B is minus row 1 of A, rows 2 and 3 are rows 3 and 2 of A, row 4 is minus row 4 of A), and $U'U = I_4$; so $B'B = A'U'UA = A'A$, and B has the same singular values.

- $\|A\|_2 = \|B\|_2 = 7$, and $\|A\|_F^2 = 49 + 25 + 9 + 1 = 84 = \|B\|_F^2$ (Theorem 15.10(d); also the sum of the squared entries, $4 \cdot 21$), so $\|A\|_F = \|B\|_F = 2\sqrt{21}$.

(b)

- $Ah_1 = (7,7,7,7)' = 7h_1$, $Ah_2 = (3,3,-3,-3)' = 3h_3$, $Ah_3 = (5,-5,5,-5)' = 5h_2$ and $Ah_4 = (1,-1,-1,1)' = h_4$. With $H = [h_1\ h_2\ h_3\ h_4]$, invertible as $H'H = 4I_4$, this says $AH = HM$, that is, $H^{-1}AH = M$, where

$$M = \begin{pmatrix} 7 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- By Theorem 7.15(a), $p_A = p_M$; and M is block diagonal, so $p_M(t) = (t-7)(t^2-15)(t-1)$ (Proposition 7.16(a), applied twice; the middle block has the characteristic polynomial $t^2 - 15$). So the eigenvalues of A are 7, $\sqrt{15}$, $-\sqrt{15}$, 1 (Definition 7.10).

- Their absolute values 7, $\sqrt{15}$, $\sqrt{15}$, 1 are not the singular values 7, 5, 3, 1: Theorem 15.10(e) needs a symmetric matrix, and A is not symmetric. (The products agree: $7 \cdot 15 \cdot 1 = 105 = 7 \cdot 5 \cdot 3 \cdot 1$, the absolute value of $\det A$.)

(c)

- $A+B = (0,1,1,0)'(3,4,4,3)$, the matrix with rows 0, $(3,4,4,3)$, $(3,4,4,3)$, 0. With $w = (3,4,4,3)'$, $(A+B)'(A+B) = w(0,1,1,0)(0,1,1,0)'w' = 2ww'$, of rank 1, and $2ww'w = 2(w'w)w = 100w$. By Theorem 8.4, its eigenvalues are 100, 0, 0, 0; so $\|A+B\|_2 = 10 \leqslant \|A\|_2 + \|B\|_2 = 14$.
- $$AB = \begin{pmatrix} -11 & 0 & 0 & 4 \\ 0 & -4 & 11 & 0 \\ 0 & 11 & -4 & 0 \\ 4 & 0 & 0 & -11 \end{pmatrix}$$
  is symmetric. $AB(1,0,0,1)' = -7(1,0,0,1)'$, $AB(1,0,0,-1)' = -15(1,0,0,-1)'$, $AB(0,1,1,0)' = 7(0,1,1,0)'$ and $AB(0,1,-1,0)' = -15(0,1,-1,0)'$; as in (a), its eigenvalues are 7, -7, -15, -15, and $\|AB\|_2 = 15$ (Proposition 15.15(e)); $15 \leqslant \|A\|_2\|B\|_2 = 49$.

(d) $A'Ah_1 = 49h_1$, so $q_1 = \frac{1}{2}h_1$, and $p_1 = \frac{1}{7}Aq_1 = \frac{1}{14}(7,7,7,7)' = \frac{1}{2}(1,1,1,1)'$ (Theorem 15.9). By Theorem 15.13(a), $u = p_1$ and $v = q_1$ give $u'Av = \sigma_1 = 7$. Check: every row sum of A is 7, so $Av = \frac{1}{2}(7,7,7,7)'$ and $u'Av = \frac{1}{4}\cdot 28 = 7$.

Exercise 15.15. Let

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 2 & -2 \end{pmatrix}$$

(a) Find a compact SVD $A = \sigma_1 p_1 q_1' + \sigma_2 p_2 q_2'$.

(b) Find a best approximation A_1 of A by a matrix of rank at most 1, and $\|A - A_1\|_F^2$.

(c) The matrix

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$$

also has rank 1. Compute $\|A - B\|_F^2$.

Solution. (a) $A'A = \begin{pmatrix} 6 & -2 \\ -2 & 6 \end{pmatrix}$, and $\det(tI_2 - A'A) = (t-6)^2 - 4 = (t-8)(t-4)$. Indeed $A'A(1,-1)' = (8,-8)'$ and $A'A(1,1)' = (4,4)'$, so $q_1 = \frac{1}{\sqrt{2}}(1,-1)'$, $q_2 = \frac{1}{\sqrt{2}}(1,1)'$, $\sigma_1 = \sqrt{8} = 2\sqrt{2}$ and $\sigma_2 = 2$. Then $p_m = Aq_m/\sigma_m$, the columns of $P_1 = AQ_1D^{-1}$ (Theorem 15.9(a),(b)):

$$p_1 = \frac{1}{2\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}0\\0\\4\end{pmatrix} = \begin{pmatrix}0\\0\\1\end{pmatrix}, \qquad p_2 = \frac{1}{2}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}2\\2\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}$$

p_1, p_2 are orthonormal, and

$$\sigma_1 p_1 q_1' + \sigma_2 p_2 q_2' = \begin{pmatrix}0&0\\0&0\\2&-2\end{pmatrix} + \begin{pmatrix}1&1\\1&1\\0&0\end{pmatrix} = A$$

With $p_3 = \frac{1}{\sqrt{2}}(1,-1,0)'$, $P = [p_1\ p_2\ p_3]$ is orthogonal, and this is the compact form of the SVD $A = P\left(\begin{smallmatrix} D & 0 \\ 0 & 0 \end{smallmatrix}\right)Q'$ of Theorem 15.9(b).

(b) By Theorem 15.17(a) ($k = 1$, $r = 2$),

$$A_1 = \sigma_1 p_1 q_1' = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 2 & -2 \end{pmatrix}$$

and $\|A - A_1\|_F^2 = \|\sigma_2 p_2 q_2'\|_F^2 = 1+1+1+1 = 4 = \sigma_2^2$.

(c)

$$A - B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 2 & -2 \end{pmatrix}$$

so $\|A - B\|_F^2 = 8 = \sigma_1^2 > 4$: $B = \sigma_2 p_2 q_2'$ keeps the smaller singular value and loses the larger one.

Exercise 15.16. Let $A = 2J_4 - I_4$.

(a) Show that $A = 7\bar{J}_4 - C_4$, and find the eigenvalues of A and orthonormal eigenvectors.

(b) Write $A = PDQ'$ with P, Q orthogonal and $D = \operatorname{diag}(7,1,1,1)$, a singular value decomposition.

(c) Find a best approximation of A by a matrix of rank at most 1, and the error $\|A - A_1\|_F^2$.

Solution. (a) $2J_4 = 8\bar{J}_4$ and $I_4 = \bar{J}_4 + C_4$ (Definition 1.14), so $A = 8\bar{J}_4 - \bar{J}_4 - C_4 = 7\bar{J}_4 - C_4$. As $\bar{J}_4\mathbf{1} = \mathbf{1}$ and $C_4\mathbf{1} = 0$, $A\mathbf{1} = 7\mathbf{1}$; if $\mathbf{1}'x = 0$, then $\bar{J}_4 x = 0$ and $C_4 x = x$, so $Ax = -x$. Hence $q_1 = \frac{1}{2}(1,1,1,1)'$, $q_2 = \frac{1}{2}(1,-1,1,-1)'$, $q_3 = \frac{1}{2}(1,1,-1,-1)'$, $q_4 = \frac{1}{2}(1,-1,-1,1)'$ are orthonormal eigenvectors for 7, -1, -1, -1, and these are the eigenvalues (Theorem 8.3(c)).

(b) $A = 7q_1q_1' - q_2q_2' - q_3q_3' - q_4q_4' = 7q_1q_1' + 1\cdot(-q_2)q_2' + 1\cdot(-q_3)q_3' + 1\cdot(-q_4)q_4'$ (Theorem 8.3(b), Theorem 1.4(d)). So $A = PDQ'$ with $Q = [q_1\ q_2\ q_3\ q_4]$, $P = [q_1\ -q_2\ -q_3\ -q_4]$ (both orthogonal) and $D = \operatorname{diag}(7,1,1,1)$: the singular values are $7,1,1,1$, the absolute values of the eigenvalues.

(c) By Theorem 15.17(a) ($k = 1$), or by Corollary 15.18 with $d_1 = 7$, $d_2 = d_3 = d_4 = -1$, $A_1 = 7q_1q_1' = 7\bar{J}_4 = \frac{7}{4}J_4$, and $\|A - A_1\|_F^2 = \|C_4\|_F^2 = \operatorname{tr}(C_4'C_4) = \operatorname{tr} C_4 = 4 - 1 = 3 = 1^2 + 1^2 + 1^2$ (C_4 is symmetric and idempotent, $\operatorname{tr}\bar{J}_4 = 1$).

Exercise 15.17. Four stores sell three products; the sales (in hundreds) are

$$B = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 6 & 2 \\ 0 & 2 & 7 \\ 4 & 8 & 8 \end{pmatrix}$$

(row i: store i; column j: product j). Let $w_1 = (1,2,2)'$, $w_2 = (2,1,-2)'$ and $w_3 = (2,-2,1)'$.

(a) Compute Bw_1, Bw_2, Bw_3, and deduce the singular values and a compact singular value decomposition of B.

(b) Find the best approximation B_1 of B by a matrix of rank at most 1, the errors $\|B - B_1\|_2$ and $\|B - B_1\|_F$, and the fraction of $\|B\|_F^2$ that B_1 reproduces. Interpret B_1.

(c) Find the best approximation B_2 of rank at most 2 and its errors.

Solution. (a)

- $Bw_1 = (9,18,18,36)' = 9(1,2,2,4)'$, $Bw_2 = (12,6,-12,0)' = 6(2,1,-2,0)'$ and $Bw_3 = (6,-6,3,0)' = 3(2,-2,1,0)'$.
- The w_j are orthogonal with length 3, so the $q_j = \frac{1}{3}w_j$ form an orthogonal matrix Q. The vectors $p_1 = \frac{1}{5}(1,2,2,4)'$, $p_2 = \frac{1}{3}(2,1,-2,0)'$, $p_3 = \frac{1}{3}(2,-2,1,0)'$ are orthonormal (lengths 1; inner products $\frac{1}{15}(2+2-4)$, $\frac{1}{15}(2-4+2)$, $\frac{1}{9}(4-2-2)$, all 0), and $Bq_1 = 15p_1$, $Bq_2 = 6p_2$, $Bq_3 = 3p_3$.
- So $BQ = P_1D$ with $P_1 = [p_1\ p_2\ p_3]$, $D = \operatorname{diag}(15,6,3)$, and $Q'B'BQ = (P_1D)'(P_1D) = DP_1'P_1D = D^2$. By Theorem 15.9(a) (with $Q_1 = Q$, no Q_2, and $P_1 = BQD^{-1}$), $P'BQ = \binom{D}{0}$ for $P = [p_1\ p_2\ p_3\ p_4]$, with any unit vector p_4 orthogonal to p_1, p_2, p_3. By Theorem 15.10(a) the singular values are 15, 6, 3, and $B = BQQ' = 15p_1q_1' + 6p_2q_2' + 3p_3q_3'$ (Theorem 1.4(d)) is a compact SVD.

(b) By Theorem 15.17 ($k = 1$),

$$B_1 = 15p_1q_1' = (1,2,2,4)'(1,2,2) = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \\ 4 & 8 & 8 \end{pmatrix}, \qquad B - B_1 = \begin{pmatrix} 4 & 0 & -2 \\ 0 & 2 & -2 \\ -2 & -2 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

$\|B - B_1\|_2 = \sigma_2 = 6$ and $\|B - B_1\|_F^2 = 36 + 9 = 45$, so $\|B - B_1\|_F = 3\sqrt{5}$; no matrix of rank at most 1 comes closer in either norm. As $\|B\|_F^2 = 225 + 36 + 9 = 270$ (the sum of the squared entries), B_1 reproduces $\frac{225}{270} = \frac{5}{6}$ of it. The entries of B_1 are products a_ib_j with $a = (1,2,2,4)'$ and $b = (1,2,2)'$: among all tables of the form "store effect times product effect" ab', B_1 has the smallest sum of squared differences from B; store 4 follows it exactly.

(c) By Theorem 15.17 ($k = 2$),

$$B_2 = B_1 + 6p_2q_2' = B_1 + \frac{2}{3}\begin{pmatrix} 2 \\ 1 \\ -2 \\ 0 \end{pmatrix}(2,1,-2) = \frac{1}{3}\begin{pmatrix} 11 & 10 & -2 \\ 10 & 14 & 8 \\ -2 & 8 & 20 \\ 12 & 24 & 24 \end{pmatrix}$$

with $B - B_2 = 3p_3q_3'$, $\|B - B_2\|_2 = 3$ and $\|B - B_2\|_F^2 = 9$; B_2 reproduces $\frac{261}{270} = \frac{29}{30}$ of $\|B\|_F^2$.

Exercise 15.18. Write a compact SVD of the centered data matrix C_5X, and give its singular values:

(a) for the data of Exercise 9.14;

(b) for the data of Exercise 9.15.

Solution. (a) In Exercise 9.14, S has the eigenvalues 15, 4, 3, 1, with the loading vectors $v_1, \dots, v_4$ and the score vectors $z_1, \dots, z_4$ found there. By Proposition 15.19(b), $r = 4$, $\sigma_m = \|z_m\| = \sqrt{4\lambda_m}$: $\sigma_1 = \sqrt{60} = 2\sqrt{15}$, $\sigma_2 = \sqrt{16} = 4$, $\sigma_3 = \sqrt{12} = 2\sqrt{3}$, $\sigma_4 = \sqrt{4} = 2$, and $p_m = z_m/\sigma_m$:

$$p_1 = \frac{1}{2\sqrt{5}}\begin{pmatrix} -4 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \qquad p_2 = \frac{1}{2\sqrt{3}}\begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \\ -3 \end{pmatrix}, \qquad p_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad p_4 = \frac{1}{\sqrt{6}}\begin{pmatrix} 0 \\ -1 \\ -1 \\ 2 \\ 0 \end{pmatrix}$$

So $C_5X = 2\sqrt{15}\,p_1v_1' + 4p_2v_2' + 2\sqrt{3}\,p_3v_3' + 2p_4v_4'$, a compact SVD; for instance $2\sqrt{15}\,p_1v_1' = z_1v_1' = (-4,1,1,1,1)'(0,1,1,-1)$. Check: $\sigma_1^2 + \cdots + \sigma_4^2 = 60 + 16 + 12 + 4 = 92 = \|C_5X\|_F^2$ (Theorem 15.10(d)).

(b) In Exercise 9.15, S has the eigenvalues 10, 6, 3, 1. In the same way, $r = 4$, $\sigma_1 = \sqrt{40} = 2\sqrt{10}$, $\sigma_2 = \sqrt{24} = 2\sqrt{6}$, $\sigma_3 = \sqrt{12} = 2\sqrt{3}$, $\sigma_4 = \sqrt{4} = 2$, and

$$p_1 = \frac{1}{\sqrt{30}}\begin{pmatrix} -2 \\ -2 \\ -2 \\ 3 \\ 3 \end{pmatrix}, \qquad p_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad p_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \qquad p_4 = \frac{1}{\sqrt{6}}\begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

from the scores of that exercise ($z_{im} = v_m'(x_i - \bar{x})$); so $C_5X = 2\sqrt{10}\,p_1v_1' + 2\sqrt{6}\,p_2v_2' + 2\sqrt{3}\,p_3v_3' + 2p_4v_4'$, for instance $2\sqrt{10}\,p_1v_1' = z_1v_1' = \frac{1}{3}(-4,-4,-4,6,6)'(1,-1,0,1)$. Check: $40 + 24 + 12 + 4 = 80 = \|C_5X\|_F^2$.

Exercise 15.19. The matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 1 & -1 \end{pmatrix}$$

has the compact SVD $A = 3p_1q_1' + \sqrt{3}\,p_2q_2'$, with $p_1 = \frac{1}{\sqrt{2}}(1,1,0)'$, $p_2 = \frac{1}{\sqrt{6}}(1,-1,2)'$, $q_1 = \frac{1}{\sqrt{2}}(1,1)'$ and $q_2 = \frac{1}{\sqrt{2}}(1,-1)'$ (Exercise 15.7).

(a) Find A^+ from the compact SVD.

(b) Check the result with the formula for a matrix of full column rank.

(c) For $y = (3,0,0)'$, find A^+y, the orthogonal projection AA^+y of y onto $\mathcal{C}(A)$ and the residual $y - AA^+y$. Find also $(A')^+$ and $(2A)^+$.

Solution. (a) By Theorem 15.21, $A^+ = \frac{1}{3}q_1p_1' + \frac{1}{\sqrt{3}}q_2p_2'$, where

$$q_1p_1' = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \qquad q_2p_2' = \frac{1}{\sqrt{12}}\begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \end{pmatrix}$$

As $\sqrt{3}\sqrt{12} = 6$,

$$A^+ = \frac{1}{6}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & -2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

(b) $\operatorname{rank} A = 2$, the number of columns, so $A^+ = (A'A)^{-1}A'$ (Theorem 12.24(c)). Here $A'A = \left(\begin{smallmatrix} 6 & 3 \\ 3 & 6 \end{smallmatrix}\right)$, with determinant 27, so $(A'A)^{-1} = \frac{1}{27}\left(\begin{smallmatrix} 6 & -3 \\ -3 & 6 \end{smallmatrix}\right) = \frac{1}{9}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ (Theorem 5.24), and

$$(A'A)^{-1}A' = \frac{1}{9}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 3 & 0 & 3 \\ 0 & 3 & -3 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

(c)

- $A^+y = \frac{1}{3}(3,0)' = (1,0)'$, and $AA^+y = A(1,0)' = (2,1,1)'$, the orthogonal projection of y onto $\mathcal{C}(A)$ (Proposition 12.26(b)).
- The residual is $y - AA^+y = (1,-1,-1)'$; check: $A'(1,-1,-1)' = (2-1-1,\ 1-2+1)' = 0$, so it is orthogonal to $\mathcal{C}(A)$.
- By Proposition 12.26(c),

$$(A')^+ = (A^+)' = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}, \qquad (2A)^+ = \frac{1}{2}A^+ = \frac{1}{6}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

Exercise 15.20. The sample covariance matrix of four variables is

$$S = \begin{pmatrix} 4 & 0 & 2 & 1 \\ 0 & 4 & 1 & 2 \\ 2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 4 \end{pmatrix}$$

Find the canonical correlations and canonical pairs between the first two and the last two variables.

Solution. $S_{11} = S_{22} = 4I_2$, so $S_{11}^{-1/2} = S_{22}^{-1/2} = \frac{1}{2}I_2$ (Theorem 8.46: $2I_2$ is PSD and $(2I_2)^2 = 4I_2$). Then

$$K = \tfrac{1}{2}S_{12} \cdot \tfrac{1}{2} = \frac{1}{4}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \frac{1}{4}(3\bar{J}_2 + C_2) = \frac{3}{4}u_1u_1' + \frac{1}{4}u_2u_2', \qquad u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix},\ u_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

since $\bar{J}_2 = u_1u_1'$ and $C_2 = I_2 - \bar{J}_2 = u_2u_2'$. This is a singular value decomposition with $\rho_1 = \frac{3}{4}$, $\rho_2 = \frac{1}{4}$ and $w_k = u_k$. By Theorem 15.23, the canonical correlations are $\frac{3}{4}$ and $\frac{1}{4}$, with $a_1 = b_1 = \frac{1}{2}u_1 = \frac{1}{2\sqrt{2}}(1,1)'$ and $a_2 = b_2 = \frac{1}{2\sqrt{2}}(1,-1)'$. Check: $a_1'S_{11}a_1 = 4 \cdot \frac{2}{8} = 1$ and $a_1'S_{12}b_1 = \frac{1}{8}(1,1)\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{6}{8} = \frac{3}{4}$.

Exercise 15.21. Let A be the matrix of Exercise 15.13.

(a) Find the largest value of $\operatorname{tr}(U'A)$ over the orthogonal $U \in \mathbb{R}^{4\times 4}$, and an orthogonal matrix U_0 at which it is attained. Compare with $U = I_4$.

(b) Find an orthogonal matrix nearest to A in the Frobenius norm, and the distance. Compare with $\|I_4 - A\|_F$.

Solution. (a) By Theorem 15.30(a) and the singular value decomposition $A = PDQ'$ of Exercise 15.13, the largest value is $7 + 5 + 3 + 1 = 16$, attained at

$$U_0 = PQ' = \sum_{k=1}^{4} p_k q_k' = \frac{1}{4}\big(h_1h_1' - h_2h_2' + h_3h_3' - h_4h_4'\big) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

(the (i,j) entry is $\frac{1}{4}$ times the sum of the products of the entries i and j of the h_k, with signs $+,-,+,-$). Indeed $\operatorname{tr}(U_0'A) = a_{21} + a_{12} + a_{43} + a_{34} = 16$, whereas $\operatorname{tr}(I_4'A) = \operatorname{tr} A = 4$.

(b) Theorem 15.30(b) with $B = I_4$ and $C = A$ (so $CB' = A$) gives $\|U - A\|_F^2 \geqslant \|I_4\|_F^2 + \|A\|_F^2 - 2 \cdot 16 = 4 + 84 - 32 = 56$ for every orthogonal U, with equality at $U = U_0$. So U_0 is nearest, at distance $\sqrt{56} = 2\sqrt{14}$; directly, $U_0 - A$ has four entries -1, four entries -3 and four entries -2, and $4 + 36 + 16 = 56$. By comparison, $\|I_4 - A\|_F^2 = 4 \cdot 16 + 4 \cdot 4 = 80$.

Problems

Problem 15.1 ([6, Exercise 12.45])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric, with B PD. Show that $\lambda_k(A+B) > \lambda_k(A)$ for $k = 1, \dots, n$.

Solution.

- B is PD, so $\lambda_n(B) > 0$ (Theorem 8.21(a)), and $u'Bu \geqslant \lambda_n(B)$ for every unit vector u (Theorem 9.23(a)).
- $k = n$: for every unit vector u, $u'(A+B)u = u'Au + u'Bu \geqslant \lambda_n(A) + \lambda_n(B)$ (Theorem 9.23(a)); at a unit eigenvector of $A+B$ belonging to $\lambda_n(A+B)$ this gives $\lambda_n(A+B) \geqslant \lambda_n(A) + \lambda_n(B)$.
- $k \leqslant n-1$: let $w_1, \dots, w_{n-k} \in \mathbb{R}^n$. For every unit vector $u \perp w_1, \dots, w_{n-k}$, $u'(A+B)u \geqslant u'Au + \lambda_n(B)$, so the minimum of $u'(A+B)u$ over these u is at least the minimum of $u'Au$ plus $\lambda_n(B)$. Taking the maximum over $w_1, \dots, w_{n-k}$ on both sides, Theorem 15.5(b) for $A+B$ and for A gives $\lambda_k(A+B) \geqslant \lambda_k(A) + \lambda_n(B)$.
- In both cases $\lambda_k(A+B) \geqslant \lambda_k(A) + \lambda_n(B) > \lambda_k(A)$.

Problem 15.2 ([3, Ch. 5, Exercise 17])**.** Let $A \in \mathbb{R}^{n\times n}$ ($n \geqslant 2$) be symmetric with $\det A = 0$. Show that the n minors $\det A(1|1), \dots, \det A(n|n)$ are either all $\geqslant 0$ or all $\leqslant 0$.

Solution.

- $\det A = \lambda_1 \cdots \lambda_n = 0$ (Theorem 7.19), so 0 is an eigenvalue of A. Let p be the number of positive eigenvalues of A; then $p \leqslant n-1$ and $\lambda_{p+1}(A) = 0$.
- Let $B = A(i|i)$. By Theorem 15.6, $\lambda_k(B) \geqslant \lambda_{k+1}(A) \geqslant \lambda_{p+1}(A) = 0$ for $k \leqslant p$, and $\lambda_k(B) \leqslant \lambda_k(A) \leqslant \lambda_{p+1}(A) = 0$ for $p+1 \leqslant k \leqslant n-1$.
- If 0 has multiplicity at least 2, then also $\lambda_{p+2}(A) = 0$, and $0 = \lambda_{p+1}(A) \geqslant \lambda_{p+1}(B) \geqslant \lambda_{p+2}(A) = 0$. So every $A(i|i)$ has the eigenvalue 0, and $\det A(i|i) = 0$ for every i.
- Let 0 be a simple eigenvalue, and let $\det B \neq 0$. Then no eigenvalue of B is 0, so by the second item B has exactly p positive and $n-1-p$ negative eigenvalues, and $\det B$ has the sign $(-1)^{n-1-p}$ (Theorem 7.19). So all nonzero minors $\det A(i|i)$ have this same sign.

Problem 15.3 ([3, Ch. 5, Exercise 4])**.** Let n be even, and let $A \in \mathbb{R}^{n\times n}$ have $n^2/2$ entries equal to 2 and the other entries equal to 4. Show that the sum of the squares of the singular values of A is $10n^2$.

Solution. By Theorem 15.10(d) and Definition 9.26, $\sigma_1^2 + \cdots + \sigma_r^2 = \|A\|_F^2 = \sum_i \sum_j a_{ij}^2 = \frac{n^2}{2}\cdot 2^2 + \frac{n^2}{2}\cdot 4^2 = 2n^2 + 8n^2 = 10n^2$.

Problem 15.4 ([10, Problem 34.6])**.** Let $A \in \mathbb{R}^{n\times n}$, and let all singular values $\sigma_1, \dots, \sigma_n$ of A be equal to the same number c. Show that $A = cU$ for an orthogonal matrix $U \in \mathbb{R}^{n\times n}$.

Solution.

- The singular values are $\geqslant 0$ (the definitions after Theorem 15.9), so $c \geqslant 0$.
- Let $c = 0$. Then $\sigma_1 = 0$, so $A = 0$: if $A \neq 0$, then $r = \operatorname{rank} A \geqslant 1$ and $\sigma_1 \geqslant \sigma_r > 0$ (Theorem 15.9(b) and the definitions after it). So $A = 0 \cdot I_n$.
- Let $c > 0$. As $\sigma_k = 0$ for $\operatorname{rank} A < k \leqslant n$ and no σ_k is 0, $\operatorname{rank} A = n$. By Theorem 15.10(a), applied to an SVD of A (Theorem 15.9(b)), the n eigenvalues of $A'A$, listed with multiplicity, are $\sigma_1^2, \dots, \sigma_n^2$, that is, c^2 (n times).
- $A'A$ is symmetric (Theorem 1.9). By Theorem 8.3, $A'A = V\Lambda V'$ with V orthogonal and Λ the diagonal matrix of the eigenvalues of $A'A$, here $\Lambda = c^2 I_n$; so $A'A = c^2 VV' = c^2 I_n$ (Proposition 6.13(b)).
- Put $U = c^{-1}A$. Then $U'U = c^{-2}A'A = I_n$, so U is orthogonal (Definition 6.12), and $A = cU$.

Problem 15.5 ([10, Problem 34.8])**.** Let $A \in \mathbb{R}^{n\times n}$ have the singular values $\sigma_1 \geqslant \cdots \geqslant \sigma_n$ (with $\sigma_k = 0$ for $k > \operatorname{rank} A$, as in the definitions after Theorem 15.9). Show that the eigenvalues of

$$M = \begin{pmatrix} 0 & A \\ A' & 0 \end{pmatrix} \in \mathbb{R}^{2n\times 2n}$$

listed with multiplicity, are $\sigma_1, \ldots, \sigma_n, -\sigma_1, \ldots, -\sigma_n$.

Solution.

- For $t \in \mathbb{R}$, $p_M(t) = \det(tI_{2n} - M)$ (Definition 7.7(a)), and $tI_{2n} - M = \left(\begin{smallmatrix} tI_n & -A \\ -A' & tI_n \end{smallmatrix}\right)$.
- Let $t \neq 0$. Then tI_n is invertible with inverse $t^{-1}I_n$, and $\det(tI_n) = t^n$ (Theorem 5.14(c)). By Theorem 5.30(a), and $\det(cB) = c^n \det B$ (Theorem 5.18(c)) with $c = t^{-1}$,

  $$p_M(t) = t^n \det\Big(tI_n - (-A')(t^{-1}I_n)(-A)\Big) = t^n \det\Big(t^{-1}(t^2 I_n - A'A)\Big) = t^n t^{-n} \det(t^2 I_n - A'A)$$

  that is, $p_M(t) = \det(t^2 I_n - A'A)$ for every $t \neq 0$.
- The eigenvalues of $A'A$, listed with multiplicity, are $\sigma_1^2, \ldots, \sigma_n^2$. If $A = 0$, then $A'A = 0$, $p_{A'A}(t) = \det(tI_n) = t^n$ (Theorem 5.14(c)), and all of them, like all σ_i, are 0. If $A \neq 0$, then by Theorem 15.10(a), applied to an SVD of A (Theorem 15.9(b)), $\sigma_1^2, \ldots, \sigma_r^2$ ($r = \operatorname{rank} A$) are the nonzero ones, so the other $n - r$ are $0 = \sigma_{r+1}^2 = \cdots = \sigma_n^2$. Hence, for every $s \in \mathbb{R}$ (Definitions 7.7(a) and 7.10, Proposition 7.5(b)),

  $$\det(sI_n - A'A) = p_{A'A}(s) = (s - \sigma_1^2)(s - \sigma_2^2)\cdots(s - \sigma_n^2)$$

- Take $s = t^2$, and write $t^2 - \sigma_i^2 = (t - \sigma_i)(t + \sigma_i)$. Then, for every $t \neq 0$, $p_M(t) = g(t)$, where

  $$g = (t - \sigma_1)\cdots(t - \sigma_n)\Big(t - (-\sigma_1)\Big)\cdots\Big(t - (-\sigma_n)\Big)$$

  (Proposition 7.5(b)). The polynomials p_M and g have the same value at every real $t \neq 0$, so $p_M = g$ (Proposition 7.5(d)). By Definition 7.10, the eigenvalues of M, listed with multiplicity, are $\sigma_1, \ldots, \sigma_n, -\sigma_1, \ldots, -\sigma_n$.

Problem 15.6 ([10, Problem 35.1])**.** Let $A \in \mathbb{R}^{n\times n}$ be invertible, and let c be a real eigenvalue of A. Show that

$$\frac{1}{\|A^{-1}\|_2} \leqslant |c| \leqslant \|A\|_2$$

Solution.

- Let $Ax = cx$ with $x \neq 0$ (Definition 7.1). As A is invertible, $c \neq 0$ and $A^{-1}x = c^{-1}x$ (Proposition 7.18(b)).
- By Proposition 15.15(a) and $\|cx\| = |c|\,\|x\|$ (the remarks after Definition 6.1), $\|A\|_2 = \max_{y\neq 0}\|Ay\|/\|y\| \geqslant \|Ax\|/\|x\| = \|cx\|/\|x\| = |c|$.
- Likewise $\|A^{-1}\|_2 \geqslant \|A^{-1}x\|/\|x\| = \|c^{-1}x\|/\|x\| = 1/|c| > 0$. Taking reciprocals, $1/\|A^{-1}\|_2 \leqslant |c|$.

As $\|A\|_2 = \sigma_1$ and $\|A^{-1}\|_2 = 1/\sigma_n$ (Definition 15.14, Proposition 15.27(a)), this says $\sigma_n \leqslant |c| \leqslant \sigma_1$; and for two real eigenvalues c, d of A, $|c|/|d| \leqslant \|A\|_2\|A^{-1}\|_2 = \kappa(A)$ (Definition 15.26).

Problem 15.7 ([3, Ch. 5, Exercise 15])**.** Let $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{q\times q}$ be PD and $C \in \mathbb{R}^{m\times q}$. Show that

$$F = \begin{pmatrix} A & C \\ C' & B \end{pmatrix}$$

is PD if and only if the largest singular value σ_1 of $K = A^{-1/2}CB^{-1/2}$ is less than 1 (with $\sigma_1 = 0$ if $C = 0$).

Solution.

- $A^{-1/2}$, $B^{-1/2}$ are symmetric and $B^{-1/2}B^{-1/2} = (B^{1/2}B^{1/2})^{-1} = B^{-1}$ (Theorem 8.46), so $KK' = A^{-1/2}CB^{-1}C'A^{-1/2}$, and σ_1^2 is the largest eigenvalue of KK' (Theorem 15.10(a); 0 if $C = 0$).
- With $KK' = Q\Lambda Q'$ (Theorem 8.3(b)), $I_m - KK' = Q(I_m - \Lambda)Q'$ has the eigenvalues $1 - \lambda_i$; so $\sigma_1 < 1$ if and only if $I_m - KK'$ is PD (Theorem 8.21(a)).
- $A^{1/2}(I_m - KK')A^{1/2} = A - CB^{-1}C'$, and $x \mapsto y = A^{1/2}x$ maps the nonzero vectors onto the nonzero vectors ($A^{1/2}$ is invertible), with $x'(A - CB^{-1}C')x = y'(I_m - KK')y$. So $I_m - KK'$ is PD if and only if $A - CB^{-1}C'$ is PD.
- Let $L = \begin{pmatrix} I_m & -CB^{-1} \\ 0 & I_q \end{pmatrix}$, invertible (Proposition 5.8). By Theorem 5.3(c), with $(B^{-1})' = B^{-1}$,

$$LF = \begin{pmatrix} A - CB^{-1}C' & 0 \\ C' & B \end{pmatrix}, \qquad LFL' = \begin{pmatrix} A - CB^{-1}C' & 0 \\ 0 & B \end{pmatrix}$$

  As $x'(LFL')x = (L'x)'F(L'x)$ and L' is invertible, F is PD if and only if LFL' is PD. For $x = (y', z')'$, $x'(LFL')x = y'(A - CB^{-1}C')y + z'Bz$; as B is PD, this is positive for every $x \neq 0$ if and only if $A - CB^{-1}C'$ is PD (take $z = 0$ for the converse).
- Combining the three equivalences: F is PD if and only if $\sigma_1 < 1$.

Problem 15.8 ([10, Problem 34.2])**.** Let $U \in \mathbb{R}^{n\times n}$ be orthogonal and $S \in \mathbb{R}^{n\times n}$ PSD.

(a) Show that $|\operatorname{tr}(US)| \leqslant \operatorname{tr} S$.

(b) Let S be PD. Show that $\operatorname{tr}(US) = \operatorname{tr} S$ only for $U = I_n$, and $|\operatorname{tr}(US)| = \operatorname{tr} S$ only for $U = I_n$ and $U = -I_n$.

Solution. (a)

- By Theorem 8.3, $S = V\Lambda V'$ with $V \in \mathbb{R}^{n\times n}$ orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of S; they are $\geqslant 0$, as S is PSD (Theorem 8.21(a)).
- Let $W = V'UV = (w_{ij})$. By Theorem 1.11(b), $\operatorname{tr}(US) = \operatorname{tr}\big((UV\Lambda)V'\big) = \operatorname{tr}(V'UV\Lambda) = \operatorname{tr}(W\Lambda)$. Column i of $W\Lambda$ is $\lambda_i \operatorname{col}_i(W)$ (Theorem 1.4(a),(b)), whose ith entry is $\lambda_i w_{ii}$; so

$$\operatorname{tr}(US) = \lambda_1 w_{11} + \lambda_2 w_{22} + \cdots + \lambda_n w_{nn}$$

- W is orthogonal, as V', U and V are (Theorem 6.19(b)), so its columns are unit vectors (Theorem 6.19(a)): $w_{1i}^2 + \cdots + w_{ni}^2 = 1$. Hence $w_{ii}^2 \leqslant 1$, that is, $|w_{ii}| \leqslant 1$, for $i = 1, \dots, n$.
- $\operatorname{tr} S = \operatorname{tr}\big((V\Lambda)V'\big) = \operatorname{tr}(V'V\Lambda) = \operatorname{tr}\Lambda = \lambda_1 + \cdots + \lambda_n$ (Theorem 1.11(b), $V'V = I_n$). As every $\lambda_i \geqslant 0$,

$$|\operatorname{tr}(US)| \leqslant \lambda_1|w_{11}| + \cdots + \lambda_n|w_{nn}| \leqslant \lambda_1 + \cdots + \lambda_n = \operatorname{tr} S$$

(b) Now every $\lambda_i > 0$ (Theorem 8.21(a)).

- Let $\operatorname{tr}(US) = \operatorname{tr} S$. By (a), $0 = \operatorname{tr} S - \operatorname{tr}(US) = \lambda_1(1 - w_{11}) + \cdots + \lambda_n(1 - w_{nn})$, a sum of terms $\geqslant 0$, as $w_{ii} \leqslant 1$. So every term is 0, and as $\lambda_i > 0$, $w_{ii} = 1$ for $i = 1, \dots, n$.

- Column i of W is a unit vector with ith entry 1, so its other entries are 0: $\mathrm{col}_i(W) = e_i$, and $W = I_n$. As $VV' = I_n$ (Proposition 6.13(b)), $U = VV'UVV' = VWV' = VV' = I_n$.
- Let $|\mathrm{tr}(US)| = \mathrm{tr}\, S$. If $\mathrm{tr}(US) = \mathrm{tr}\, S$, then $U = I_n$. If $\mathrm{tr}(US) = -\mathrm{tr}\, S$, then $-U$ is orthogonal, as $(-U)'(-U) = U'U = I_n$, and $\mathrm{tr}\big((-U)S\big) = -\mathrm{tr}(US) = \mathrm{tr}\, S$ (Theorem 1.11(a)); so $-U = I_n$, that is, $U = -I_n$.

Problem 15.9 ([6, Exercise 8.40])**.** Let $A \in \mathbb{R}^{m\times n}$.

(a) If A has full column rank n, show that $A = UV$ for some $U \in \mathbb{R}^{m\times n}$ with $U'U = I_n$ and some PD $V \in \mathbb{R}^{n\times n}$.

(b) If A has full row rank m, show that $A = VU'$ for some $U \in \mathbb{R}^{n\times m}$ with $U'U = I_m$ and some PD $V \in \mathbb{R}^{m\times m}$.

Solution. (a)

- Here $r = n$, so in the compact SVD $A = P_1 D Q_1'$ of Theorem 15.9, $Q_1 = Q$ is an orthogonal $n \times n$ matrix, $P_1'P_1 = I_n$, and $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_n)$ with every $\sigma_i > 0$.
- As $Q'Q = I_n$, $A = P_1(Q'Q)DQ' = (P_1Q')(QDQ') = UV$ with $U = P_1Q'$ and $V = QDQ'$.
- $U'U = QP_1'P_1Q' = QQ' = I_n$ (Proposition 6.13(b)). V is symmetric and $Q'VQ = D$, so its eigenvalues are $\sigma_1, \ldots, \sigma_n > 0$ (Definition 7.23), and V is PD (Theorem 8.21(a)).

(b) $A' \in \mathbb{R}^{n\times m}$ has full column rank m (Theorem 4.13(a)). By (a), $A' = UV$ with $U'U = I_m$ and V PD, so $A = V'U' = VU'$.

Chapter 16. Kronecker products, vec, and matrix derivatives

The Kronecker product

Definition 16.1 ([21, §2.3], [5, §16.1], [2, §4.11])**.** Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$. The **Kronecker product** of A and B is the $mp \times nq$ matrix

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}$$

partitioned into m rows and n columns of $p \times q$ blocks, the (i, j) block being $a_{ij}B$. Thus the entry of $A \otimes B$ in row $(i-1)p + r$ and column $(j-1)q + s$ is $a_{ij}b_{rs}$ $(1 \leqslant r \leqslant p,\ 1 \leqslant s \leqslant q)$. For column vectors $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^p$, $a \otimes b = (a_1 b',\ \ldots,\ a_m b')' \in \mathbb{R}^{mp}$. For a number c (a 1×1 matrix), $c \otimes B = B \otimes c = cB$.

Unlike AB, the product $A \otimes B$ is defined for matrices of any sizes.

Example 16.2 ([2, §4.11])**.** With $A = (1,\ 2,\ 3)$ and $B = \begin{pmatrix} 6 & 7 \\ 8 & 9 \end{pmatrix}$,

$$A \otimes B = \begin{pmatrix} 6 & 7 & 12 & 14 & 18 & 21 \\ 8 & 9 & 16 & 18 & 24 & 27 \end{pmatrix}, \qquad B \otimes A = \begin{pmatrix} 6 & 12 & 18 & 7 & 14 & 21 \\ 8 & 16 & 24 & 9 & 18 & 27 \end{pmatrix}$$

Both contain the twelve products $a_j b_{rs}$, in different orders, so $A \otimes B \neq B \otimes A$. Two Kronecker products with the summing vector (Definition 1.14) occur in designs:

$$I_2 \otimes \mathbf{1}_3 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{1}_3 \otimes I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Theorem 16.3 ([21, Propositions 3, 4, Exercise 1], [6, Exercises 10.2–10.9], [8, Ch. 16, Exercise 10], [5, Lemma 16.1.2])**.** Let the sizes of the matrices be such that the operations below are defined, and let $c \in \mathbb{R}$.

(a) $(cA) \otimes B = A \otimes (cB) = c(A \otimes B)$, and $A \otimes B = 0$ if $A = 0$ or $B = 0$. For column vectors $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$, $a \otimes b' = ab' = b' \otimes a$.

(b) $(A_1 + A_2) \otimes B = A_1 \otimes B + A_2 \otimes B$ and $A \otimes (B_1 + B_2) = A \otimes B_1 + A \otimes B_2$.

(c) If A is partitioned into blocks A_{kl} (Definition 5.1), then $A \otimes B$ is partitioned into the blocks $A_{kl} \otimes B$, in the same arrangement; for instance $[A_1\ A_2] \otimes B = [A_1 \otimes B\ \ A_2 \otimes B]$.

(d) $(A \otimes B) \otimes C = A \otimes (B \otimes C)$; this matrix is written $A \otimes B \otimes C$.

(e) $(A \otimes B)' = A' \otimes B'$.

(f) (**Mixed product**) For $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{p\times q}$, $C \in \mathbb{R}^{n\times s}$ and $D \in \mathbb{R}^{q\times t}$,

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$$

Proof. Throughout, $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$, and we compare matrices block by block (Definition 5.1).

(a)

- The (i,j) blocks of $(cA)\otimes B$, $A\otimes(cB)$ and $c(A\otimes B)$ are $(ca_{ij})B$, $a_{ij}(cB)$ and $c(a_{ij}B)$, which are equal. If $A=0$ or $B=0$, every block $a_{ij}B$ is 0.
- $a\otimes b'$ has the rows $a_1b',\dots,a_mb'$, which are the rows of ab' (Theorem 1.4(c)). $b'\otimes a=(b_1a,\ \dots,\ b_na)$ has the columns b_ja, which are the columns of ab' (Theorem 1.4(a)).

(b) The (i,j) block of $(A_1+A_2)\otimes B$ is $(a_{ij}^{(1)}+a_{ij}^{(2)})B=a_{ij}^{(1)}B+a_{ij}^{(2)}B$, the sum of the (i,j) blocks of $A_1\otimes B$ and $A_2\otimes B$; and $a_{ij}(B_1+B_2)=a_{ij}B_1+a_{ij}B_2$.

(c)

- Let the row groups of A have $m_1,\dots,m_r$ rows and the column groups $n_1,\dots,n_c$ columns. Row i of A gives the block row of $A\otimes B$ formed by rows $(i-1)p+1,\dots,ip$, and column j the block column formed by columns $(j-1)q+1,\dots,jq$. So the rows of A in group k give m_kp consecutive rows of $A\otimes B$, and the columns in group l give n_lq consecutive columns.
- The submatrix of $A\otimes B$ in these rows and columns consists of the blocks $a_{ij}B$ with row i in group k and column j in group l, arranged as the entries a_{ij} are arranged in A_{kl}. By Definition 16.1, it is $A_{kl}\otimes B$.

(d) Partition $A\otimes B$ into its blocks $a_{ij}B$. By (c), $(A\otimes B)\otimes C$ is partitioned into the blocks $(a_{ij}B)\otimes C$, which equal $a_{ij}(B\otimes C)$ by (a). These are the blocks of $A\otimes(B\otimes C)$ (Definition 16.1).

(e) By Theorem 5.3(b), $(A\otimes B)'$ has n block rows and m block columns, and its (j,i) block is $(a_{ij}B)'=a_{ij}B'$. This is the (j,i) block of $A'\otimes B'$, since the (j,i) entry of A' is a_{ij}.

(f)

- $A\otimes B$ has n block columns of width q, and $C\otimes D$ has n block rows of height q: the partitions are conformal for multiplication. By Theorem 5.3(c), the (i,k) block of the product is

$$\sum_{j=1}^{n}(a_{ij}B)(c_{jk}D)=\Big(\sum_{j=1}^{n}a_{ij}c_{jk}\Big)BD=(AC)_{ik}\,BD$$

- This is the (i,k) block of $(AC)\otimes(BD)$, for $i=1,\dots,m$ and $k=1,\dots,s$. □

Corollary 16.4 ([5, §16.1], [21, Proposition 4], [6, Exercise 10.8], [8, Ch. 16, Exercise 4])**.** Let $A\in\mathbb{R}^{m\times n}$ and $B\in\mathbb{R}^{p\times q}$.

(a) $A\otimes B=(A\otimes I_p)(I_n\otimes B)=(I_m\otimes B)(A\otimes I_q)$; $I_m\otimes I_p=I_{mp}$; $I_m\otimes B$ is block diagonal with m diagonal blocks B; and for a diagonal $D=\operatorname{diag}(d_1,\dots,d_m)$, $D\otimes B=\operatorname{diag}(d_1B,\dots,d_mB)$.

(b) If A and B are square and invertible, then $A\otimes B$ is invertible and $(A\otimes B)^{-1}=A^{-1}\otimes B^{-1}$.

(c) If A and B are square, then $\operatorname{tr}(A\otimes B)=\operatorname{tr}A\cdot\operatorname{tr}B$.

(d) If G is a g-inverse of A and F a g-inverse of B (Definition 12.1), then $G\otimes F$ is a g-inverse of $A\otimes B$.

(e) If A and B are square and both symmetric, both orthogonal, or both idempotent, then $A\otimes B$ is symmetric, orthogonal, or idempotent, respectively.

Proof. (a) By Theorem 16.3(f), $(A\otimes I_p)(I_n\otimes B)=(AI_n)\otimes(I_pB)=A\otimes B$ and $(I_m\otimes B)(A\otimes I_q)=(I_mA)\otimes(BI_q)=A\otimes B$. The (i,j) block of $D\otimes B$ is d_iB for $i=j$ and 0 for $i\neq j$; for $D=I_m$ this gives $I_m\otimes B$, and for $B=I_p$ it gives $I_m\otimes I_p=I_{mp}$.

(b) Let $A\in\mathbb{R}^{m\times m}$, $B\in\mathbb{R}^{p\times p}$. By Theorem 16.3(f) and (a), $(A\otimes B)(A^{-1}\otimes B^{-1})=(AA^{-1})\otimes(BB^{-1})=I_m\otimes I_p=I_{mp}$; so $A\otimes B$ is invertible with this inverse (Theorem 2.16).

(c) The diagonal blocks of $A\otimes B$ are $a_{11}B,\dots,a_{mm}B$, so $\operatorname{tr}(A\otimes B)=\sum_i\operatorname{tr}(a_{ii}B)=\big(\sum_i a_{ii}\big)\operatorname{tr}B=\operatorname{tr}A\cdot\operatorname{tr}B$ (Theorem 1.11(a)).

(d) By Theorem 16.3(f), applied twice, $(A\otimes B)(G\otimes F)(A\otimes B)=(AGA)\otimes(BFB)=A\otimes B$.

(e) By Theorem 16.3(e),(f): if $A' = A$ and $B' = B$, then $(A \otimes B)' = A' \otimes B' = A \otimes B$; if $A'A = I$ and $B'B = I$, then $(A \otimes B)'(A \otimes B) = (A'A) \otimes (B'B) = I$; if $A^2 = A$ and $B^2 = B$, then $(A \otimes B)^2 = A^2 \otimes B^2 = A \otimes B$. □

Example 16.5 (cf. [5, §16.1], [3, §8.3])**.** (a) In the one-way model of Theorem 13.10 with g groups of m observations each ($n = gm$), the indicator of group l is $e_l \otimes \mathbf{1}_m$, so $X_1 = [d_1 \ \cdots \ d_g] = I_g \otimes \mathbf{1}_m$ and $X = [\mathbf{1}_n \ \ X_1] = [\mathbf{1}_g \otimes \mathbf{1}_m \ \ I_g \otimes \mathbf{1}_m]$. By Theorem 16.3(e),(f) and Corollary 16.4(a),(b),

$$X_1'X_1 = (I_g \otimes \mathbf{1}_m')(I_g \otimes \mathbf{1}_m) = I_g \otimes (\mathbf{1}_m'\mathbf{1}_m) = mI_g$$

and the hat matrix of X (Theorem 13.4), the orthogonal projection matrix onto $\mathcal{C}(X) = \mathcal{C}(X_1)$ (Theorem 13.10(c)), is (Theorem 6.26)

$$H = X_1(X_1'X_1)^{-1}X_1' = (I_g \otimes \mathbf{1}_m)\left(\tfrac{1}{m}I_g\right)(I_g \otimes \mathbf{1}_m') = I_g \otimes \bar{J}_m$$

So the hat matrix of the balanced one-way design is block diagonal with g blocks $\bar{J}_m$, and, as $I_n = I_g \otimes I_m = I_g \otimes (\bar{J}_m + C_m)$, the residual maker is $M = I_n - H = I_g \otimes C_m$ (Definition 1.14, Theorem 16.3(b)).

(b) In a **two-way layout** a response y_{ij} is observed once for each level i of a factor with a levels and each level j of a second factor with b levels; the model without interaction is $y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$. List the observations as $y = (y_{11}, \ldots, y_{1b}, y_{21}, \ldots, y_{ab})'$. The observation (i, j) is entry $(i-1)b + j$, so the column of α_i is $e_i \otimes \mathbf{1}_b$ and that of β_j is $\mathbf{1}_a \otimes e_j$ (Definition 16.1), and

$$X = [\mathbf{1}_a \otimes \mathbf{1}_b \ \ I_a \otimes \mathbf{1}_b \ \ \mathbf{1}_a \otimes I_b] \in \mathbb{R}^{ab \times (1+a+b)}, \qquad \beta = (\mu, \alpha_1, \ldots, \alpha_a, \beta_1, \ldots, \beta_b)'$$

By Theorem 16.3(e),(f), $(\mathbf{1}_a \otimes \mathbf{1}_b)'(I_a \otimes \mathbf{1}_b) = \mathbf{1}_a' \otimes (\mathbf{1}_b'\mathbf{1}_b) = b\mathbf{1}_a'$, $(I_a \otimes \mathbf{1}_b)'(\mathbf{1}_a \otimes I_b) = \mathbf{1}_a \otimes \mathbf{1}_b' = \mathbf{1}_a\mathbf{1}_b'$, $(\mathbf{1}_a \otimes I_b)'(\mathbf{1}_a \otimes I_b) = aI_b$, and so on; with the block product (Theorem 5.3(c)),

$$X'X = \begin{pmatrix} ab & b\mathbf{1}_a' & a\mathbf{1}_b' \\ b\mathbf{1}_a & bI_a & \mathbf{1}_a\mathbf{1}_b' \\ a\mathbf{1}_b & \mathbf{1}_b\mathbf{1}_a' & aI_b \end{pmatrix}$$

For $a = 2$ and $b = 3$,

$$X = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 6 & 3 & 3 & 2 & 2 & 2 \\ 3 & 3 & 0 & 1 & 1 & 1 \\ 3 & 0 & 3 & 1 & 1 & 1 \\ 2 & 1 & 1 & 2 & 0 & 0 \\ 2 & 1 & 1 & 0 & 2 & 0 \\ 2 & 1 & 1 & 0 & 0 & 2 \end{pmatrix}$$

The vec operator

Definition 16.6 ([21, §2.4, Exercise 2], [5, §16.2], [2, §9.4.1], [6, Exercise 10.17])**.** For $A \in \mathbb{R}^{m \times n}$ with columns $a_1, \ldots, a_n$, $\operatorname{vec} A \in \mathbb{R}^{mn}$ is the vector obtained by stacking the columns:

$$\operatorname{vec} A = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

Thus a_{ij} is entry $(j-1)m + i$ of $\operatorname{vec} A$. For instance, for $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, $\operatorname{vec} A = (1, 4, 2, 5, 3, 6)'$ and $\operatorname{vec} A' = (1, 2, 3, 4, 5, 6)'$.

Since sums and multiples are taken entrywise, $\operatorname{vec}(cA + dB) = c \operatorname{vec} A + d \operatorname{vec} B$ for $A, B \in \mathbb{R}^{m \times n}$ and $c, d \in \mathbb{R}$; and for $A, B \in \mathbb{R}^{m \times n}$, $\operatorname{vec} A = \operatorname{vec} B$ if and only if $A = B$. For a column vector a, $\operatorname{vec} a = \operatorname{vec} a' = a$. The squared length of $\operatorname{vec} A$ is $\sum_{i,j} a_{ij}^2 = \|A\|_F^2$ (Definition 9.26).

Theorem 16.7 ([21, Propositions 5, 6], [5, §16.2], [6, Exercises 10.18–10.20])**.** (a) For column vectors $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$, $\operatorname{vec}(ab') = b \otimes a$.

(b) For $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$ and $C \in \mathbb{R}^{p\times q}$,

$$\operatorname{vec}(ABC) = (C' \otimes A)\operatorname{vec} B$$

(c) In particular, $\operatorname{vec}(AB) = (I_p \otimes A)\operatorname{vec} B = (B' \otimes I_m)\operatorname{vec} A$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, and $ABx = (x' \otimes A)\operatorname{vec} B$ for $x \in \mathbb{R}^p$.

(d) For $A, B \in \mathbb{R}^{m\times n}$, $(\operatorname{vec} A)'\operatorname{vec} B = \operatorname{tr}(A'B)$. More generally, for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{m\times p}$, $C \in \mathbb{R}^{p\times q}$ and $D \in \mathbb{R}^{n\times q}$,

$$\operatorname{tr}(A'BCD') = (\operatorname{vec} A)'(D \otimes B)\operatorname{vec} C$$

Proof. (a) Column j of ab' is $b_j a$ (Theorem 1.4(a)), so $\operatorname{vec}(ab') = (b_1 a',\ \ldots,\ b_n a')' = b \otimes a$ (Definition 16.1).

(b)

- Let b_j be column j of B and u_j column j of I_p. By the outer product expansion of $B = BI_p$ (Theorem 1.4(d)), $B = \sum_{j=1}^{p} b_j u_j'$, so $ABC = \sum_j (Ab_j)(C'u_j)'$ (Theorems 1.3 and 1.9).
- By (a) and the mixed product (Theorem 16.3(f)), $\operatorname{vec}\big((Ab_j)(C'u_j)'\big) = (C'u_j) \otimes (Ab_j) = (C' \otimes A)(u_j \otimes b_j) = (C' \otimes A)\operatorname{vec}(b_j u_j')$.
- Summing over j (vec is linear): $\operatorname{vec}(ABC) = (C' \otimes A)\operatorname{vec}\big(\sum_j b_j u_j'\big) = (C' \otimes A)\operatorname{vec} B$.

(c) Apply (b) to $AB\,I_p$, to $I_m\,AB$, and to ABx (with $C = x \in \mathbb{R}^{p\times 1}$), using $\operatorname{vec}(ABx) = ABx$.

(d)

- The jth diagonal entry of $A'B$ is $\operatorname{col}_j(A)'\operatorname{col}_j(B) = \sum_i a_{ij}b_{ij}$ (Theorem 1.4(d)), so $\operatorname{tr}(A'B) = \sum_{i,j} a_{ij}b_{ij}$; and $(\operatorname{vec} A)'\operatorname{vec} B$ is the same sum, as both vectors list the entries in the same order.
- Hence, by (b), $\operatorname{tr}(A'BCD') = (\operatorname{vec} A)'\operatorname{vec}(BCD') = (\operatorname{vec} A)'(D \otimes B)\operatorname{vec} C$. □

Applied to A' and B, (d) also gives $\operatorname{tr}(AB) = (\operatorname{vec} A')'\operatorname{vec} B$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$.

Example 16.8 (cf. [3, §8.3], [2, §12.12.2])**.** In the two-way layout of Example 16.5(b), arrange the data in $Y = (y_{ij}) \in \mathbb{R}^{a\times b}$; then $y = \operatorname{vec} Y'$, since row i of Y is column i of Y'. Let

$$P = I_a \otimes \bar{J}_b + \bar{J}_a \otimes C_b \in \mathbb{R}^{ab\times ab}$$

- P is symmetric (Corollary 16.4(e)), and $P^2 = P$: by Theorem 16.3(b),(f) and $\bar{J}^2 = \bar{J}$, $C^2 = C$, $C\bar{J} = \bar{J}C = 0$ (Proposition 1.15; $\bar{J}_b\mathbf{1}_b = \mathbf{1}_b$), $P^2 = I_a \otimes \bar{J}_b + \bar{J}_a \otimes (\bar{J}_b C_b) + \bar{J}_a \otimes (C_b\bar{J}_b) + \bar{J}_a \otimes C_b = P$.
- $PX = X$: $P(\mathbf{1}_a \otimes \mathbf{1}_b) = \mathbf{1}_a \otimes \mathbf{1}_b + \mathbf{1}_a \otimes C_b\mathbf{1}_b = \mathbf{1}_a \otimes \mathbf{1}_b$, $P(I_a \otimes \mathbf{1}_b) = I_a \otimes \mathbf{1}_b + \bar{J}_a \otimes C_b\mathbf{1}_b = I_a \otimes \mathbf{1}_b$, and $P(\mathbf{1}_a \otimes I_b) = \mathbf{1}_a \otimes \bar{J}_b + \mathbf{1}_a \otimes C_b = \mathbf{1}_a \otimes I_b$. So $\mathcal{C}(X) \subseteq \mathcal{C}(P)$.
- $P = XT$ for a matrix T: $I_a \otimes \bar{J}_b = (I_a \otimes \mathbf{1}_b)(I_a \otimes \frac{1}{b}\mathbf{1}_b')$ and $\bar{J}_a \otimes C_b = (\mathbf{1}_a \otimes I_b)(\frac{1}{a}\mathbf{1}_a' \otimes C_b)$. So $\mathcal{C}(P) \subseteq \mathcal{C}(X)$, and $\mathcal{C}(P) = \mathcal{C}(X)$.
- By Theorem 6.29(d), Py is the orthogonal projection of y onto $\mathcal{C}(X)$, that is, the vector $\hat{y}$ of fitted values (Theorem 13.3(b)).
- By Theorem 16.7(b), $(I_a \otimes \bar{J}_b)\operatorname{vec} Y' = \operatorname{vec}(\bar{J}_b Y')$, whose entry (i,j) is the row mean $\bar{y}_{i\cdot} = \frac{1}{b}\sum_l y_{il}$; and $(\bar{J}_a \otimes C_b)\operatorname{vec} Y' = \operatorname{vec}(C_b Y'\bar{J}_a)$, whose entry (i,j) is $\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, the column mean minus the overall mean (Example 1.16). Hence

$$\hat{y}_{ij} = \bar{y}_{i\cdot} + \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$$

For $a = 2$, $b = 3$ and $Y = \begin{pmatrix} 6 & 8 & 4 \\ 2 & 4 & 6 \end{pmatrix}$: the row means are 6, 4, the column means 4, 6, 5, and $\bar{y}_{\cdot\cdot} = 5$. So $\hat{y} = (5, 7, 6, 3, 5, 4)'$ and $e = y - \hat{y} = (1, 1, -2, -1, -1, 2)'$; indeed $X'e = (0,\ 0,\ 0,\ 0,\ 0,\ 0)'$: the entries of e add up to 0 along each row and each column of the layout.

Example 16.9. In **multivariate regression**, p responses are measured on each of n units, $Y \in \mathbb{R}^{n\times p}$ holds them (column j is response j), and $Y = XB + E$ with a design $X \in \mathbb{R}^{n\times(k+1)}$ of full column rank, an unknown $B \in \mathbb{R}^{(k+1)\times p}$ and errors E. The least-squares estimate minimizes $\|Y - XB\|_F^2 = \|\operatorname{vec} Y - \operatorname{vec}(XB)\|^2$. By Theorem 16.7(c), $\operatorname{vec}(XB) = (I_p \otimes X)\operatorname{vec} B$: this is the least-squares problem with response $\operatorname{vec} Y$ and design $I_p \otimes X$. As $(I_p \otimes X)'(I_p \otimes X) = I_p \otimes X'X$ is invertible with inverse $I_p \otimes (X'X)^{-1}$ (Corollary 16.4(b)), $I_p \otimes X$ has full column rank (Theorems 4.18(b) and 4.13(d)), and by Theorem 13.3(c) the unique estimate is

$$\operatorname{vec}\hat{B} = \Big(I_p \otimes (X'X)^{-1}\Big)(I_p \otimes X')\operatorname{vec} Y = \Big(I_p \otimes (X'X)^{-1}X'\Big)\operatorname{vec} Y = \operatorname{vec}\Big((X'X)^{-1}X'Y\Big)$$

that is, $\hat{B} = (X'X)^{-1}X'Y$: column j of $\hat{B}$ is the least-squares estimate for response j alone. If X has any rank, let G be a g-inverse of $X'X$; then $I_p \otimes G$ is a g-inverse of $I_p \otimes X'X$ (Corollary 16.4(d)), and by Theorems 13.2(c) and 13.3(a) the same computation gives the least-squares estimate $\operatorname{vec} B^0 = \Big(I_p \otimes GX'\Big)\operatorname{vec} Y = \operatorname{vec}(GX'Y)$, that is, $B^0 = GX'Y$: column j of B^0 is a least-squares estimate for response j alone.

Theorem 16.10 ([6, Exercise 13.65], [21, §13])**.** Let $Y \in \mathbb{R}^{n\times p}$ be a random matrix with rows $y_1', \dots, y_n'$ such that $\mathsf{E}(y_i) = \mu$ and $\operatorname{var}(y_i) = \Sigma = (\sigma_{jl})$ for every i, and $\operatorname{cov}(y_{ij}, y_{kl}) = 0$ for $i \neq k$: the observations on different units are uncorrelated. Then $\mathsf{E}(\operatorname{vec} Y) = \mu \otimes \mathbf{1}_n$ and $\operatorname{var}(\operatorname{vec} Y) = \Sigma \otimes I_n$; and $\operatorname{vec} Y' = (y_1', \dots, y_n')'$ has $\mathsf{E}(\operatorname{vec} Y') = \mathbf{1}_n \otimes \mu$ and $\operatorname{var}(\operatorname{vec} Y') = I_n \otimes \Sigma$.

Proof.
- Block j of $\operatorname{vec} Y$ is column j of Y, $(y_{1j}, \dots, y_{nj})'$, with mean $\mu_j \mathbf{1}_n$. So $\mathsf{E}(\operatorname{vec} Y)$ has the blocks $\mu_1\mathbf{1}_n, \dots, \mu_p\mathbf{1}_n$: it is $\mu \otimes \mathbf{1}_n$ (Definitions 9.1 and 16.1).
- The entry of $\operatorname{var}(\operatorname{vec} Y)$ in block (j, l), position (i, k), is $\operatorname{cov}(y_{ij}, y_{kl})$ (Definition 9.1). It is σ_{jl} for $i = k$ (entry (j, l) of $\operatorname{var}(y_i)$) and 0 for $i \neq k$. So block (j, l) is $\sigma_{jl} I_n$, the (j, l) block of $\Sigma \otimes I_n$.
- Likewise block i of $\operatorname{vec} Y'$ is y_i, with mean μ, and block (i, k) of its covariance matrix, with entries $\operatorname{cov}(y_{ij}, y_{kl})$, is Σ for $i = k$ and 0 for $i \neq k$: the (i, k) block of $I_n \otimes \Sigma$. □

For $n \geqslant 2$ the rows of Y satisfy the hypotheses of Theorem 9.16, which gives $\mathsf{E}(\bar{y}) = \mu$ and $\operatorname{var}(\bar{y}) = \frac{1}{n}\Sigma$ for the mean vector $\bar{y} = \frac{1}{n}Y'\mathbf{1}_n$, and $\mathsf{E}(S) = \Sigma$ for the sample covariance matrix $S = \frac{1}{n-1}Y'C_nY$ (Definition 9.13). The covariance matrix of $\bar{y}$ also follows from the theorem with the mixed product: by Theorem 16.7(b), $\bar{y} = I_pY'(\frac{1}{n}\mathbf{1}_n) = (\frac{1}{n}\mathbf{1}_n' \otimes I_p)\operatorname{vec} Y'$, and by Theorem 9.3(b) and Theorem 16.3(e),(f),

$$\operatorname{var}(\bar{y}) = \Big(\tfrac{1}{n}\mathbf{1}_n' \otimes I_p\Big)(I_n \otimes \Sigma)\Big(\tfrac{1}{n}\mathbf{1}_n \otimes I_p\Big) = \tfrac{1}{n^2}(\mathbf{1}_n'\mathbf{1}_n) \otimes \Sigma = \tfrac{1}{n}\Sigma$$

The commutation matrix

Definition 16.11 ([21, §2.5], [5, §16.3a], [6, Exercise 11.9(a)])**.** Let $m, n \geqslant 1$. The **commutation matrix** $K_{mn} \in \mathbb{R}^{mn\times mn}$ is the matrix with

$$K_{mn}\operatorname{vec} A = \operatorname{vec} A' \qquad \text{for every } A \in \mathbb{R}^{m\times n}$$

(the next theorem shows that there is exactly one such matrix). We write $K_n = K_{nn}$ (not the Helmert matrix K_n of Example 6.20). For example, for $A = (a_{ij}) \in \mathbb{R}^{2\times 3}$, $\operatorname{vec} A = (a_{11}, a_{21}, a_{12}, a_{22}, a_{13}, a_{23})'$ and $\operatorname{vec} A' = (a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23})'$, so

$$K_{23} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Theorem 16.12 ([5, §16.3, Theorem 16.3.2], [21, Propositions 7, 8, Exercise 4], [6, Exercises 11.1–11.5]). Let $m, n, p, q \geqslant 1$.

(a) There is exactly one matrix K_{mn} as in Definition 16.11: row $(i-1)n+j$ of K_{mn} is row $(j-1)m+i$ of I_{mn} ($1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$). It is a permutation matrix.

(b) $K'_{mn} = K_{mn}^{-1} = K_{nm}$; $K_{m1} = K_{1m} = I_m$; and K_n is symmetric.

(c) $\operatorname{tr} K_n = n$.

(d) For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$,

$$K_{pm}(A \otimes B) = (B \otimes A)K_{qn}, \qquad B \otimes A = K_{pm}(A \otimes B)K_{nq}$$

and for column vectors $a \in \mathbb{R}^m$, $b \in \mathbb{R}^p$, $K_{pm}(a \otimes b) = b \otimes a$.

Proof. (a)

- For $A \in \mathbb{R}^{m\times n}$, a_{ij} is entry $(j-1)m+i$ of $\operatorname{vec} A$; in $A' \in \mathbb{R}^{n\times m}$ it is the (j,i) entry, so it is entry $(i-1)n+j$ of $\operatorname{vec} A'$ (Definition 16.6).
- As (i,j) runs over the mn pairs, both $(i-1)n+j$ and $(j-1)m+i$ run over $1, \dots, mn$, each value once. So $\varphi\big((i-1)n+j\big) = (j-1)m+i$ defines a permutation φ of degree mn (Definition 5.39), and its permutation matrix P_φ has row $(i-1)n+j$ equal to row $(j-1)m+i$ of I_{mn} (Definition 5.40).
- By Proposition 5.41(a), entry $(i-1)n+j$ of $P_\varphi \operatorname{vec} A$ is entry $(j-1)m+i$ of $\operatorname{vec} A$, that is, a_{ij}, which is entry $(i-1)n+j$ of $\operatorname{vec} A'$. So $P_\varphi \operatorname{vec} A = \operatorname{vec} A'$ for every A.
- If also $K \operatorname{vec} A = \operatorname{vec} A'$ for every A, then $Kv = P_\varphi v$ for every $v \in \mathbb{R}^{mn}$, since every v is $\operatorname{vec} A$ for some A; taking $v = e_r$ gives equal columns (Theorem 1.4(b)), so $K = P_\varphi$.

(b)

- By (a) and Proposition 5.41(c), K_{mn} is invertible with $K_{mn}^{-1} = K'_{mn}$.
- For every $A \in \mathbb{R}^{m\times n}$, $K_{nm}K_{mn} \operatorname{vec} A = K_{nm} \operatorname{vec} A' = \operatorname{vec}(A')' = \operatorname{vec} A$; as in (a), $K_{nm}K_{mn} = I_{mn}$, so $K_{nm} = K_{mn}^{-1}$ (Theorem 2.16).
- For $a \in \mathbb{R}^m$, $\operatorname{vec} a = \operatorname{vec} a' = a$, so $K_{m1}a = a$ and $K_{1m}a = a$ for every a: $K_{m1} = K_{1m} = I_m$. Finally $K'_n = K_n^{-1} = K_n$.

(c) For $n = 1$, $K_1 = I_1$. Let $n \geqslant 2$. By (a), the diagonal entry of K_n in position $(i-1)n+j$ is 1 if $(j-1)n+i = (i-1)n+j$, that is, $(j-i)(n-1) = 0$, that is, $i = j$, and 0 otherwise. So exactly n diagonal entries are 1, and $\operatorname{tr} K_n = n$.

(d)

- Let $X \in \mathbb{R}^{q\times n}$. By Definition 16.11 and Theorem 16.7(b), applied three times,

$$K_{pm}(A \otimes B) \operatorname{vec} X = K_{pm} \operatorname{vec}(BXA') = \operatorname{vec}(AX'B') = (B \otimes A) \operatorname{vec} X' = (B \otimes A)K_{qn} \operatorname{vec} X$$

  As $\operatorname{vec} X$ runs over $\mathbb{R}^{nq}$, this gives $K_{pm}(A \otimes B) = (B \otimes A)K_{qn}$, as in (a).
- Multiplying on the right by $K_{nq} = K_{qn}^{-1}$ (by (b)) gives $K_{pm}(A \otimes B)K_{nq} = B \otimes A$.
- By Theorem 16.7(a), $K_{pm}(a \otimes b) = K_{pm} \operatorname{vec}(ba') = \operatorname{vec}(ab') = b \otimes a$. □

For the data matrix of Theorem 16.10, $\operatorname{vec} Y' = K_{np} \operatorname{vec} Y$, and (d) with (b) gives again $\operatorname{var}(\operatorname{vec} Y') = K_{np}(\Sigma \otimes I_n)K'_{np} = (I_n \otimes \Sigma)K_{np}K_{pn} = I_n \otimes \Sigma$ (Theorem 9.3(b)).

Theorem 16.13 ([5, (16.3.18), (16.1.26)], [6, Exercise 10.12(b)]). (a) For $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{n\times n}$, $\det(A \otimes B) = (\det A)^n (\det B)^m$.

(b) For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$, $\operatorname{rank}(A \otimes B) = \operatorname{rank} A \cdot \operatorname{rank} B$.

(c) For square A and B, $A \otimes B$ is invertible if and only if A and B are invertible.

Proof. (a)

- By Corollary 16.4(a), $A \otimes B = (A \otimes I_n)(I_m \otimes B)$, and $I_m \otimes B$ is block diagonal with m blocks B. Applying Theorem 5.29 ($\det \operatorname{diag}(A_1, D) = \det A_1 \cdot \det D$) $m-1$ times, $\det(I_m \otimes B) = (\det B)^m$; likewise $\det(I_n \otimes A) = (\det A)^n$.
- By Theorem 16.12(d) with $B = I_n$ ($p = q = n$), $K_{nm}(A \otimes I_n) = (I_n \otimes A)K_{nm}$, so $A \otimes I_n = K_{mn}(I_n \otimes A)K_{nm}$ (Theorem 16.12(b)).
- By Theorem 5.18(b), $\det(A \otimes I_n) = \det K_{mn} \cdot (\det A)^n \cdot \det K_{nm}$, and $\det K_{mn} \cdot \det K_{nm} = \det(K_{mn}K_{nm}) = \det I_{mn} = 1$. Hence $\det(A \otimes B) = (\det A)^n(\det B)^m$.

(b)

- For any matrix M with a g-inverse N, $\operatorname{rank} M = \operatorname{tr}(MN)$ (Theorem 12.13(c)).
- Let G and F be g-inverses of A and B (Theorem 12.3). By Corollary 16.4(d), $G \otimes F$ is a g-inverse of $A \otimes B$. By the first step (for $A \otimes B$, A and B), the mixed product and Corollary 16.4(c),

$$\operatorname{rank}(A \otimes B) = \operatorname{tr}\Big((A \otimes B)(G \otimes F)\Big) = \operatorname{tr}(AG \otimes BF) = \operatorname{tr}(AG)\operatorname{tr}(BF) = \operatorname{rank} A \cdot \operatorname{rank} B$$

(c) By (a), $\det(A \otimes B) \neq 0$ if and only if $\det A \neq 0$ and $\det B \neq 0$; apply Theorem 5.18(a). □

Theorem 16.14 ([2, Theorem 6.10], [6, Exercises 10.11(a), 10.12(a)])**.** Let $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{n\times n}$.

(a) If $Ax = \lambda x$ and $By = \mu y$ with $x \neq 0$, $y \neq 0$, then $(A \otimes B)(x \otimes y) = \lambda\mu(x \otimes y)$ and $x \otimes y \neq 0$: $\lambda\mu$ is an eigenvalue of $A \otimes B$ with eigenvector $x \otimes y$.

(b) If $P^{-1}AP = \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$ and $Q^{-1}BQ = M = \operatorname{diag}(\mu_1, \ldots, \mu_n)$ (Definition 7.23), then $(P \otimes Q)^{-1}(A \otimes B)(P \otimes Q) = \Lambda \otimes M$. So $A \otimes B$ is diagonalizable, its eigenvalues, listed with multiplicity, are the mn products $\lambda_i\mu_j$, and $p_i \otimes q_j$ is an eigenvector for $\lambda_i\mu_j$, where $p_i = \operatorname{col}_i(P)$, $q_j = \operatorname{col}_j(Q)$.

(c) If A and B are symmetric, with $A = P\Lambda P'$, $B = QMQ'$ as in Theorem 8.3(b), then $P \otimes Q$ is orthogonal and $A \otimes B = (P \otimes Q)(\Lambda \otimes M)(P \otimes Q)'$. If A and B are PSD (PD), then $A \otimes B$ is PSD (PD).

Proof. (a) By the mixed product, $(A \otimes B)(x \otimes y) = (Ax) \otimes (By) = (\lambda x) \otimes (\mu y) = \lambda\mu(x \otimes y)$ (Theorem 16.3(a),(f)). If $x_i \neq 0$ and $y_j \neq 0$, the entry x_iy_j of $x \otimes y$ is not 0.

(b)

- $P \otimes Q$ is invertible with inverse $P^{-1} \otimes Q^{-1}$ (Corollary 16.4(b)), and by the mixed product $(P^{-1} \otimes Q^{-1})(A \otimes B)(P \otimes Q) = (P^{-1}AP) \otimes (Q^{-1}BQ) = \Lambda \otimes M$.
- $\Lambda \otimes M = \operatorname{diag}(\lambda_1 M, \ldots, \lambda_m M)$ (Corollary 16.4(a)) is diagonal, with $\lambda_i\mu_j$ in position $(i-1)n + j$; and column $(i-1)n+j$ of $P \otimes Q$ is $p_i \otimes q_j$ (Definition 16.1). By Definition 7.23, these mn numbers are the eigenvalues of $A \otimes B$, listed with multiplicity, and $p_i \otimes q_j$ belongs to $\lambda_i\mu_j$.

(c)

- $P \otimes Q$ is orthogonal and $A \otimes B$ symmetric (Corollary 16.4(e)); as $(P \otimes Q)^{-1} = (P \otimes Q)'$, (b) gives $A \otimes B = (P \otimes Q)(\Lambda \otimes M)(P \otimes Q)'$.
- If A, B are PSD (PD), all $\lambda_i, \mu_j \geqslant 0$ (> 0) (Theorem 8.21(a)), so all eigenvalues $\lambda_i\mu_j$ of $A \otimes B$ are $\geqslant 0$ (> 0), and $A \otimes B$ is PSD (PD) by the same theorem. □

In (b), $\det(A \otimes B) = \prod_{i,j} \lambda_i\mu_j = \prod_i \Big(\lambda_i^n \prod_j \mu_j\Big) = (\det A)^n(\det B)^m$ (Theorem 7.19), in agreement with Theorem 16.13(a); and $\operatorname{tr}(A \otimes B) = \sum_{i,j} \lambda_i\mu_j = \operatorname{tr} A \cdot \operatorname{tr} B$.

The duplication matrix

Definition 16.15 ([21, §2.6], [5, §16.4], [2, §9.4.1])**.** For $A \in \mathbb{R}^{n\times n}$, $\operatorname{vech} A \in \mathbb{R}^{n(n+1)/2}$ is the vector obtained from $\operatorname{vec} A$ by deleting the entries above the diagonal:

$$\operatorname{vech} A = (a_{11}, \dots, a_{n1}, a_{22}, \dots, a_{n2}, \dots, a_{nn})'$$

For symmetric A it lists each distinct entry once. The **duplication matrix** $D_n \in \mathbb{R}^{n^2\times n(n+1)/2}$ is the matrix with $D_n \operatorname{vech} A = \operatorname{vec} A$ for every symmetric $A \in \mathbb{R}^{n\times n}$. For example,

$$\operatorname{vech}\begin{pmatrix}1&7&6\\7&3&8\\6&8&2\end{pmatrix} = (1,7,6,3,8,2)', \qquad D_2 = \begin{pmatrix}1&0&0\\0&1&0\\0&1&0\\0&0&1\end{pmatrix}$$

Proposition 16.16 ([5, §16.4b, (16.4.6), (16.4.14)], [21, Proposition 9, Exercise 5])**.** Let $n \geqslant 1$.

(a) There is exactly one duplication matrix D_n. Its column for the position of a_{ij} $(i \geqslant j)$ in $\operatorname{vech} A$ has the entry 1 in rows $(j-1)n+i$ and $(i-1)n+j$ (one row if $i=j$) and 0 elsewhere. The columns of D_n are nonzero and pairwise orthogonal, $D_n'D_n$ is diagonal with the entries 1 $(i=j)$ and 2 $(i>j)$, and D_n has full column rank $n(n+1)/2$.

(b) For symmetric A, $\operatorname{vech} A = (D_n'D_n)^{-1}D_n' \operatorname{vec} A$.

(c) $K_nD_n = D_n$, and $D_n(D_n'D_n)^{-1}D_n' = \frac{1}{2}(I_{n^2} + K_n)$.

Proof. (a)

- For symmetric A and $i \geqslant j$, the entries $(j-1)n+i$ and $(i-1)n+j$ of $\operatorname{vec} A$ are a_{ij} and $a_{ji} = a_{ij}$, the entry of $\operatorname{vech} A$ for (i,j). So the matrix described has $D_n \operatorname{vech} A = \operatorname{vec} A$. As A runs over the symmetric matrices, $\operatorname{vech} A$ runs over all of $\mathbb{R}^{n(n+1)/2}$; as in Theorem 16.12(a), D_n is unique.
- Every row of D_n has exactly one entry 1, so two different columns have no nonzero entry in a common row: they are orthogonal, and $D_n'D_n$ is diagonal with the squared lengths 1 or 2 of the columns (Theorem 1.4(d)).
- $D_n'D_n$ is invertible, so $\operatorname{rank} D_n = \operatorname{rank}(D_n'D_n) = n(n+1)/2$ (Theorems 4.18(b) and 4.13(d)).

(b) Multiply $D_n \operatorname{vech} A = \operatorname{vec} A$ on the left by $(D_n'D_n)^{-1}D_n'$.

(c)

- For symmetric A, $K_nD_n \operatorname{vech} A = K_n \operatorname{vec} A = \operatorname{vec} A' = \operatorname{vec} A = D_n \operatorname{vech} A$; as $\operatorname{vech} A$ runs over $\mathbb{R}^{n(n+1)/2}$, $K_nD_n = D_n$.
- Let $M = D_n(D_n'D_n)^{-1}D_n'$ and $N = \frac{1}{2}(I_{n^2}+K_n)$. M is symmetric and idempotent. K_n is symmetric with $K_n^2 = K_n'K_n = I$ (Theorem 16.12(b)), so $N' = N$ and $N^2 = \frac{1}{4}(I + 2K_n + K_n^2) = N$.
- $ND_n = \frac{1}{2}(D_n + K_nD_n) = D_n$, so $NM = M$ and $MN = (NM)' = M$. Hence $\Delta = N - M$ is symmetric with $\Delta^2 = N - NM - MN + M = N - M = \Delta$.
- $\operatorname{tr} M = \operatorname{tr}\Big((D_n'D_n)^{-1}D_n'D_n\Big) = \frac{n(n+1)}{2}$ and $\operatorname{tr} N = \frac{1}{2}(n^2+n)$ by Theorem 16.12(c) (Theorem 1.11). So $\operatorname{rank}\Delta = \operatorname{tr}\Delta = 0$ (Theorem 6.29(c)), and $\Delta = 0$. □

Derivatives and differentials

Let $S \subseteq \mathbb{R}^n$ and $c \in S$. The **ball** $B(c,\delta)$ of radius $\delta > 0$ is the set of $x \in \mathbb{R}^n$ with $\|x - c\| < \delta$, and S is **open** if every $c \in S$ has a ball $B(c,\delta) \subseteq S$. For matrix variables $X \in \mathbb{R}^{n\times q}$ the distance is $\|X - C\|_F = \|\operatorname{vec} X - \operatorname{vec} C\|$ (Definition 9.26), so a set of matrices is open if the corresponding set of vectors $\operatorname{vec} X$ is. For a function f on an open set S and $x \in S$, the **partial derivative** $\partial f(x)/\partial x_j$ is the derivative at $t = 0$ of the function $t \mapsto f(x + te_j)$ of one variable, when it exists [5, §15.1].

Definition 16.17 ([21, §3], [5, §15.1], [2, (9.11)]). Let $S \subseteq \mathbb{R}^n$ be open and $f = (f_1, \dots, f_m)' : S \to \mathbb{R}^m$. If all partial derivatives of $f_1, \dots, f_m$ exist at $x \in S$, the **derivative** (or **Jacobian matrix**) of f at x is the $m \times n$ matrix

$$\mathrm{D}f(x) = \frac{\partial f(x)}{\partial x'} \quad \text{with } (i,j) \text{ entry } \frac{\partial f_i(x)}{\partial x_j}$$

Row i holds the partial derivatives of f_i, column j the partial derivatives with respect to x_j. For a scalar function ($m = 1$), $\mathrm{D}f(x)$ is a row vector, and the column $\partial f(x)/\partial x = (\mathrm{D}f(x))'$ is the **gradient**. A matrix function F, with values $F(X) \in \mathbb{R}^{m\times p}$, of a matrix X in an open set of $\mathbb{R}^{n\times q}$ is regarded as the vector function $\operatorname{vec} X \mapsto \operatorname{vec} F(X)$; its derivative is the $mp \times nq$ matrix

$$\mathrm{D}F(X) = \frac{\partial \operatorname{vec} F(X)}{\partial (\operatorname{vec} X)'}$$

For a scalar function $\varphi(X)$ we also write $\partial\varphi/\partial X$ for the $n \times q$ matrix with (i,j) entry $\partial\varphi/\partial x_{ij}$; thus $\mathrm{D}\varphi(X) = (\operatorname{vec}(\partial\varphi/\partial X))'$.

Some books write $\partial y/\partial x$ for the *transposed* derivative of a vector function: there $\partial(Ax)/\partial x = A'$. In these notes D always has one row for each entry of the function and one column for each variable.

We use the following facts from calculus without proof [5, Lemmas 15.1.1, 15.2.2, Theorem 15.7.1]. A function on an open set is **continuously differentiable** (C^1) if its partial derivatives exist and are continuous there, and **twice continuously differentiable** (C^2) if moreover the partial derivatives of the partial derivatives exist and are continuous; vector and matrix functions are C^1 (C^2) if all their entries are.

- (**Rules**) Sums, products and quotients (with nonzero denominator) of C^1 (C^2) functions are C^1 (C^2), and $\partial(f+g)/\partial x_j = \partial f/\partial x_j + \partial g/\partial x_j$, $\partial(fg)/\partial x_j = (\partial f/\partial x_j)g + f\,\partial g/\partial x_j$, $\partial(cf)/\partial x_j = c\,\partial f/\partial x_j$ (c constant); constants and the variables are C^2, so polynomials in the variables are C^2. A C^1 function is continuous, and a continuous function that is not 0 (or is positive) at c is not 0 (positive) on some ball around c.
- (**Chain rule**) If g is C^1 on an open $S \subseteq \mathbb{R}^n$ with values in an open $T \subseteq \mathbb{R}^m$, and f is C^1 (C^2) on T with values in $\mathbb{R}^p$, then $f \circ g$ is C^1 on S (C^2 if g is too), and $\mathrm{D}(f\circ g)(x) = \mathrm{D}f(g(x))\,\mathrm{D}g(x)$. The functions $\log t$ and $t^{-1/2}$ of $t > 0$ are C^2, with derivatives $1/t$ and $-\frac{1}{2}t^{-3/2}$.

Example 16.18 ([2, Example 9.1, §9.3.1, §9.5.2]). (a) For $f(x) = Ax$ with $A \in \mathbb{R}^{m\times n}$ constant, $f_i(x) = \sum_j a_{ij}x_j$, so $\partial f_i/\partial x_j = a_{ij}$ and $\mathrm{D}f(x) = A$. For

$$A = \begin{pmatrix} 2 & 6 & -1 \\ 3 & -2 & 4 \\ 3 & 4 & 7 \end{pmatrix}, \qquad Ax = \begin{pmatrix} 2x_1 + 6x_2 - x_3 \\ 3x_1 - 2x_2 + 4x_3 \\ 3x_1 + 4x_2 + 7x_3 \end{pmatrix}$$

the partial derivatives of the first entry are $2, 6, -1$, the first row of $\mathrm{D}(Ax) = A$; in the transposed notation, $\partial(Ax)/\partial x = A'$ has these numbers in its first column.

(b) For $\lambda = 3x_1 + 4x_2 + 9x_3 = a'x$ with $a = (3,4,9)'$, $\mathrm{D}\lambda = (3,\ 4,\ 9) = a'$, and the gradient is a.

(c) For $X \in \mathbb{R}^{n\times q}$ and constant $A \in \mathbb{R}^{q\times n}$, $\operatorname{tr}(XA) = \sum_{r,s} x_{rs}a_{sr}$, so $\partial \operatorname{tr}(XA)/\partial x_{ij} = a_{ji}$: $\partial \operatorname{tr}(XA)/\partial X = A'$, and $\mathrm{D}\operatorname{tr}(XA) = (\operatorname{vec} A')'$.

Lemma 16.19 ([21, Proposition 2]). (a) If $A, B \in \mathbb{R}^{m\times n}$ and $Ax = Bx$ for every $x \in \mathbb{R}^n$, then $A = B$.

(b) For $A \in \mathbb{R}^{n\times n}$, $x'Ax = 0$ for every $x \in \mathbb{R}^n$ if and only if $A' = -A$ (A is skew-symmetric).

(c) For $A, B \in \mathbb{R}^{n\times n}$, $x'Ax = x'Bx$ for every x if and only if $A + A' = B + B'$. In particular, for symmetric A and B, $x'Ax = x'Bx$ for every x only if $A = B$.

Proof. (a) With $x = e_j$, $\operatorname{col}_j(A) = Ae_j = Be_j = \operatorname{col}_j(B)$ for every j (Theorem 1.4(b)).

(b)

- If $A' = -A$, then $x'Ax = (x'Ax)' = x'A'x = -x'Ax$, as a number is its own transpose; so $x'Ax = 0$.
- Conversely, let $x'Ax = 0$ for every x. With $x = e_i$, $a_{ii} = 0$. With $x = e_i + e_j$ $(i \neq j)$, $0 = a_{ii} + a_{ij} + a_{ji} + a_{jj} = a_{ij} + a_{ji}$. So $A' = -A$.

(c) $x'Ax = x'Bx$ for every x means $x'(A - B)x = 0$ for every x, that is, by (b), $(A - B)' = -(A - B)$, or $A + A' = B + B'$. For symmetric A, B this reads $2A = 2B$. □

Definition 16.20 ([21, §4], [6, Ch. 13])**.** Let f be C^1 on an open $S \subseteq \mathbb{R}^n$ with values in $\mathbb{R}^m$, and $x \in S$. The **differential** of f at x is the linear function of $\mathrm{d}x \in \mathbb{R}^n$

$$\mathrm{d}f(x) = \mathrm{D}f(x)\,\mathrm{d}x$$

where $\mathrm{d}x$ is an arbitrary vector (an increment of x). For a C^1 matrix function F of $X \in \mathbb{R}^{n\times q}$, the differential $\mathrm{d}F(X)$ is the matrix of the size of $F(X)$ with $\operatorname{vec}\mathrm{d}F(X) = \mathrm{D}F(X)\operatorname{vec}\mathrm{d}X$, $\mathrm{d}X \in \mathbb{R}^{n\times q}$ arbitrary; its (k,l) entry is $\sum_{i,j}(\partial f_{kl}/\partial x_{ij})\,\mathrm{d}x_{ij}$.

So $\mathrm{d}f(x)$ has the size of f, whatever the number of variables, while $\mathrm{D}f(x)$ has one column for each variable; this makes differentials easier to compute with. For the identity function $f(x) = x$, $\mathrm{D}f(x) = I_n$ and the differential is $\mathrm{d}x$ itself.

Theorem 16.21 (First identification theorem; [21, Proposition 14])**.** (a) Let f be C^1 on an open $S \subseteq \mathbb{R}^n$ with values in $\mathbb{R}^m$, $x \in S$ and $A \in \mathbb{R}^{m\times n}$. Then $\mathrm{d}f(x) = A\,\mathrm{d}x$ for every $\mathrm{d}x$ if and only if $\mathrm{D}f(x) = A$.

(b) For a C^1 matrix function F of $X \in \mathbb{R}^{n\times q}$ with values in $\mathbb{R}^{m\times p}$ and $A \in \mathbb{R}^{mp\times nq}$: $\operatorname{vec}\mathrm{d}F(X) = A\operatorname{vec}\mathrm{d}X$ for every $\mathrm{d}X$ if and only if $\mathrm{D}F(X) = A$.

(c) For a C^1 scalar function φ of $X \in \mathbb{R}^{n\times q}$ and $C \in \mathbb{R}^{n\times q}$: $\mathrm{d}\varphi(X) = \operatorname{tr}(C'\mathrm{d}X)$ for every $\mathrm{d}X$ if and only if $\mathrm{D}\varphi(X) = (\operatorname{vec}C)'$, that is, $\partial\varphi/\partial X = C$.

Proof. (a) If $\mathrm{D}f(x) = A$, then $\mathrm{d}f(x) = A\,\mathrm{d}x$ by Definition 16.20. Conversely, $A\,\mathrm{d}x = \mathrm{D}f(x)\,\mathrm{d}x$ for every $\mathrm{d}x$ gives $\mathrm{D}f(x) = A$ by Lemma 16.19(a).

(b) This is (a) for the vector function $\operatorname{vec}X \mapsto \operatorname{vec}F(X)$ (Definitions 16.17 and 16.20).

(c) By Theorem 16.7(d), $\operatorname{tr}(C'\mathrm{d}X) = (\operatorname{vec}C)'\operatorname{vec}\mathrm{d}X$; so this is (b) with $m = p = 1$ and $A = (\operatorname{vec}C)'$, and $\mathrm{D}\varphi(X) = (\operatorname{vec}C)'$ means $\partial\varphi/\partial x_{ij} = c_{ij}$ for all i, j. □

So, to find a derivative, we compute the differential and bring it to one of these forms.

Theorem 16.22 ([21, §4.1, §7, Proposition 15], [6, Exercises 13.1–13.3])**.** Let F and G be C^1 matrix functions (vector and scalar functions included) of the same variable on an open set, and let A be a constant matrix and α a number.

(a) $\mathrm{d}A = 0$, $\mathrm{d}(\alpha F) = \alpha\,\mathrm{d}F$, $\mathrm{d}(F + G) = \mathrm{d}F + \mathrm{d}G$, $\mathrm{d}(F') = (\mathrm{d}F)'$, $\mathrm{d}\operatorname{vec}F = \operatorname{vec}\mathrm{d}F$ and, for square F, $\mathrm{d}\operatorname{tr}F = \operatorname{tr}\mathrm{d}F$.

(b) $\mathrm{d}(FG) = (\mathrm{d}F)G + F\,\mathrm{d}G$ and $\mathrm{d}(F \otimes G) = (\mathrm{d}F) \otimes G + F \otimes \mathrm{d}G$, whenever the products are defined.

(c) (**Chain rule**) Let $y = g(x)$ be C^1 on an open $S \subseteq \mathbb{R}^n$ with values in an open $T \subseteq \mathbb{R}^m$, and let $z = f(y)$ be C^1 on T, with differentials $\mathrm{d}z = A(y)\,\mathrm{d}y$ and $\mathrm{d}y = B(x)\,\mathrm{d}x$. Then the differential of $x \mapsto f(g(x))$ is $\mathrm{d}z = A(g(x))B(x)\,\mathrm{d}x$: we may substitute $\mathrm{d}y = B(x)\,\mathrm{d}x$ into $\mathrm{d}z = A(y)\,\mathrm{d}y$. The same holds for matrix functions, through vec.

Proof. (a) By Definition 16.20, the (k,l) entry of $\mathrm{d}F$ is $\sum_{i,j}(\partial f_{kl}/\partial x_{ij})\,\mathrm{d}x_{ij}$. The partial derivatives of a constant are 0, and those of αf_{kl} and of $f_{kl}+g_{kl}$ are $\alpha\,\partial f_{kl}/\partial x_{ij}$ and $\partial f_{kl}/\partial x_{ij}+\partial g_{kl}/\partial x_{ij}$ (rules of calculus); this gives the first three rules. F', $\operatorname{vec} F$ and $\operatorname{tr} F$ rearrange entries of F or add them up, so their differentials rearrange or add up the differentials of these entries.

(b)

- For C^1 scalar functions u, v, the product rule gives $\mathrm{d}(uv)=\sum_{i,j}\Big((\partial u/\partial x_{ij})v+u\,\partial v/\partial x_{ij}\Big)\mathrm{d}x_{ij}=(\mathrm{d}u)v+u\,\mathrm{d}v$.
- The (k,l) entry of FG is $\sum_r f_{kr}g_{rl}$, so by (a) its differential is $\sum_r\Big((\mathrm{d}f_{kr})g_{rl}+f_{kr}\,\mathrm{d}g_{rl}\Big)$, the (k,l) entry of $(\mathrm{d}F)G+F\,\mathrm{d}G$.
- The entries of $F\otimes G$ are the products $f_{ij}g_{st}$ (Definition 16.1), with differentials $(\mathrm{d}f_{ij})g_{st}+f_{ij}\,\mathrm{d}g_{st}$, the corresponding entries of $(\mathrm{d}F)\otimes G+F\otimes\mathrm{d}G$.

(c) By the chain rule of calculus, $\mathrm{D}(f\circ g)(x)=\mathrm{D}f(g(x))\,\mathrm{D}g(x)$, and by Theorem 16.21(a), $\mathrm{D}f(y)=A(y)$ and $\mathrm{D}g(x)=B(x)$. So the differential of $f\circ g$ at x is $A(g(x))B(x)\,\mathrm{d}x$ (Definition 16.20). □

In practice we write $f(y)$ and $f(x)$ for the same quantity seen as a function of y or of x. The rules let us compute differentials without writing down partial derivatives.

Example 16.23 ([21, Exercises 10, 12])**.** (a) For $f(x)=x_1^2x_2$, Theorem 16.22(b) gives $\mathrm{d}(x_1^2)=2x_1\,\mathrm{d}x_1$ and

$$\mathrm{d}f=(\mathrm{d}x_1^2)x_2+x_1^2\,\mathrm{d}x_2=2x_1x_2\,\mathrm{d}x_1+x_1^2\,\mathrm{d}x_2=(2x_1x_2,\ x_1^2)\,\mathrm{d}x$$

so $\mathrm{D}f(x)=(2x_1x_2,\ x_1^2)$ (Theorem 16.21(a)).

(b) For $f(x)=(x_1^2-x_2^2,\ x_1x_2x_3)'$ with $x=(x_1,x_2,x_3)'$,

$$\mathrm{d}f=\begin{pmatrix}2x_1\,\mathrm{d}x_1-2x_2\,\mathrm{d}x_2\\ x_2x_3\,\mathrm{d}x_1+x_1x_3\,\mathrm{d}x_2+x_1x_2\,\mathrm{d}x_3\end{pmatrix}=\begin{pmatrix}2x_1 & -2x_2 & 0\\ x_2x_3 & x_1x_3 & x_1x_2\end{pmatrix}\mathrm{d}x$$

and $\mathrm{D}f(x)$ is the 2×3 matrix on the right.

Theorem 16.24 ([21, Exercise 11], [6, Exercises 13.4, 13.6(a)])**.** Let $a\in\mathbb{R}^n$ and $A\in\mathbb{R}^{m\times n}$ be constant.

(a) $\mathrm{d}(a'x)=a'\,\mathrm{d}x$: $\mathrm{D}(a'x)=a'$, and the gradient of $a'x$ is a.

(b) $\mathrm{d}(Ax)=A\,\mathrm{d}x$: $\mathrm{D}(Ax)=A$.

(c) For $A\in\mathbb{R}^{n\times n}$, $\mathrm{d}(x'Ax)=x'(A+A')\,\mathrm{d}x$: $\mathrm{D}(x'Ax)=x'(A+A')$, and the gradient is $(A+A')x$, which is $2Ax$ if A is symmetric.

(d) For $X\in\mathbb{R}^{n\times q}$, $a\in\mathbb{R}^n$ and $b\in\mathbb{R}^q$, $\mathrm{d}(a'Xb)=a'(\mathrm{d}X)b$: $\partial(a'Xb)/\partial X=ab'$, and $\mathrm{D}(a'Xb)=(b\otimes a)'$.

Proof. (a), (b) By Theorem 16.22(a),(b), $\mathrm{d}(Ax)=(\mathrm{d}A)x+A\,\mathrm{d}x=A\,\mathrm{d}x$; take $A=a'$ for (a), and apply Theorem 16.21(a).

(c) By the product rule, $\mathrm{d}(x'Ax)=(\mathrm{d}x)'Ax+x'A\,\mathrm{d}x$. The number $(\mathrm{d}x)'Ax$ equals its transpose $x'A'\,\mathrm{d}x$; so $\mathrm{d}(x'Ax)=x'(A+A')\,\mathrm{d}x$, and Theorem 16.21(a) gives the derivative.

(d) $\mathrm{d}(a'Xb)=a'(\mathrm{d}X)b=\operatorname{tr}\Big(b\,a'\,\mathrm{d}X\Big)=\operatorname{tr}\Big((ab')'\mathrm{d}X\Big)$ (Theorem 1.11(b)), so $\partial(a'Xb)/\partial X=ab'$ by Theorem 16.21(c), and $\operatorname{vec}(ab')=b\otimes a$ (Theorem 16.7(a)). □

Example 16.25 ([2, Example 9.2])**.** For

$$A=\begin{pmatrix}1 & 3 & 5\\ 3 & 4 & 7\\ 5 & 7 & 9\end{pmatrix},\qquad q=x'Ax=x_1^2+6x_1x_2+10x_1x_3+4x_2^2+14x_2x_3+9x_3^2$$

(Definition 8.17), and by Theorem 16.24(c) the gradient is

$$2Ax = \begin{pmatrix} 2x_1 + 6x_2 + 10x_3 \\ 6x_1 + 8x_2 + 14x_3 \\ 10x_1 + 14x_2 + 18x_3 \end{pmatrix}$$

the partial derivatives of q read off its terms. The upper triangular

$$B = \begin{pmatrix} 1 & 6 & 10 \\ 0 & 4 & 14 \\ 0 & 0 & 9 \end{pmatrix}$$

gives the same $q = x'Bx$, and indeed $B + B' = 2A$ (Lemma 16.19(c)): the derivative $x'(B + B')$ is the same.

Traces, determinants and inverses

Theorem 16.26 ([21, §7, (30)–(32)], [6, Exercises 13.10(a), 13.21(a), 13.32, 13.36(a)], [2, (9.26), (9.33)–(9.40)], [3, Result 9.5])**.** Let X be a variable in $\mathbb{R}^{n\times n}$ ($n \geqslant 2$) and A a constant matrix; c_{ij} are the cofactors of X (Definition 5.20).

(a) $\mathrm{d}\,\mathrm{tr}(AX) = \mathrm{tr}(A\,\mathrm{d}X)$, so $\partial\,\mathrm{tr}(AX)/\partial X = A'$. For $X \in \mathbb{R}^{n\times q}$ and $A \in \mathbb{R}^{n\times n}$, $\mathrm{d}\,\mathrm{tr}(X'AX) = \mathrm{tr}\Big(X'(A + A')\,\mathrm{d}X\Big)$, so $\partial\,\mathrm{tr}(X'AX)/\partial X = (A + A')X$.

(b) $\det X$ is a polynomial in the entries of X, $\partial \det X/\partial x_{ij} = c_{ij}$, and $\mathrm{d}\det X = \mathrm{tr}(\mathrm{adj}\,X\,\mathrm{d}X)$. If X is invertible, then

$$\mathrm{d}\det X = \det X\ \mathrm{tr}(X^{-1}\mathrm{d}X) \qquad (\textbf{Jacobi's formula}), \qquad \frac{\partial \det X}{\partial X} = \det X\,(X^{-1})'$$

(c) The invertible matrices form an open set, on which $X \mapsto X^{-1}$ is C^2, and

$$\mathrm{d}X^{-1} = -X^{-1}(\mathrm{d}X)X^{-1}, \qquad \mathrm{D}(X^{-1}) = -(X^{-1})' \otimes X^{-1}$$

(d) On the open set $\{X : \det X \neq 0\}$, $\mathrm{d}\log|\det X| = \mathrm{tr}(X^{-1}\mathrm{d}X)$; so $\partial \log\det X/\partial X = (X^{-1})'$ where $\det X > 0$.

(e) (**Matrices depending on one parameter**) Let the entries of $A(t) \in \mathbb{R}^{n\times n}$ be C^1 functions of t on an open interval, and let $\frac{\mathrm{d}A}{\mathrm{d}t}$ be the matrix of their derivatives. Then $\frac{\mathrm{d}}{\mathrm{d}t}\det A(t) = \mathrm{tr}\Big(\mathrm{adj}\,A(t)\,\frac{\mathrm{d}A}{\mathrm{d}t}\Big)$, and where $\det A(t) > 0$, $\frac{\mathrm{d}}{\mathrm{d}t}\log\det A(t) = \mathrm{tr}\Big(A(t)^{-1}\frac{\mathrm{d}A}{\mathrm{d}t}\Big)$.

Proof. (a)

- By Theorem 16.22(a),(b), $\mathrm{d}\,\mathrm{tr}(AX) = \mathrm{tr}(A\,\mathrm{d}X) = \mathrm{tr}((A')'\mathrm{d}X)$; apply Theorem 16.21(c).
- By the product rule, $\mathrm{d}\,\mathrm{tr}(X'AX) = \mathrm{tr}((\mathrm{d}X)'AX) + \mathrm{tr}(X'A\,\mathrm{d}X)$. As $\mathrm{tr}\,M = \mathrm{tr}\,M'$ (Theorem 1.11(a)), $\mathrm{tr}((\mathrm{d}X)'AX) = \mathrm{tr}(X'A'\,\mathrm{d}X)$. So $\mathrm{d}\,\mathrm{tr}(X'AX) = \mathrm{tr}\Big(((A + A')X)'\mathrm{d}X\Big)$; apply Theorem 16.21(c).

(b)

- $\det X$ is a polynomial in the entries, by induction on the size: $\det(x_{11}) = x_{11}$ for a 1×1 matrix, and the expansion along row 1 (Theorem 5.21), $\det X = \sum_k x_{1k}(-1)^{1+k}\det X(1|k)$, is a sum of products of entries of X with determinants of size $n - 1$.

- By the expansion along row i (Theorem 5.21), $\det X = \sum_k x_{ik} c_{ik}$, where $c_{ik} = (-1)^{i+k} \det X(i|k)$ does not involve row i of X. With all entries other than x_{ij} fixed, $\det X$ is $c_{ij} x_{ij}$ plus a constant, so $\partial \det X / \partial x_{ij} = c_{ij}$.
- Hence $\mathrm{d} \det X = \sum_{i,j} c_{ij}\, \mathrm{d}x_{ij} = \sum_j \sum_i (\operatorname{adj} X)_{ji}\, \mathrm{d}x_{ij} = \sum_j (\operatorname{adj} X\, \mathrm{d}X)_{jj} = \operatorname{tr}(\operatorname{adj} X\, \mathrm{d}X)$, as $(\operatorname{adj} X)_{ji} = c_{ij}$ (Theorem 5.24).
- If X is invertible, $\operatorname{adj} X = \det X \cdot X^{-1}$ (Theorem 5.24), which gives Jacobi's formula; and $\partial \det X / \partial X = (c_{ij}) = (\operatorname{adj} X)' = \det X\, (X^{-1})'$.

(c)

- det is a polynomial, hence continuous; if $\det C \neq 0$, then $\det X \neq 0$ on a ball around C (rules of calculus): the set of invertible matrices is open.
- On it, $X^{-1} = \frac{1}{\det X} \operatorname{adj} X$ (Theorem 5.24), whose entries are polynomials divided by the nonzero polynomial $\det X$; so $X \mapsto X^{-1}$ is C^2.
- Differentiating $X^{-1}X = I_n$ by the product rule, $(\mathrm{d}X^{-1})X + X^{-1}\mathrm{d}X = \mathrm{d}I_n = 0$; multiplying on the right by X^{-1} gives $\mathrm{d}X^{-1} = -X^{-1}(\mathrm{d}X)X^{-1}$.
- By Theorem 16.7(b), $\operatorname{vec} \mathrm{d}X^{-1} = -\Big((X^{-1})' \otimes X^{-1}\Big) \operatorname{vec} \mathrm{d}X$; apply Theorem 16.21(b).

(d)

- As det is continuous, $\det C > 0$ ($\det C < 0$) gives $\det X > 0$ ($\det X < 0$) on a ball around C, as in (c): the sets $\{\det X > 0\}$ and $\{\det X < 0\}$ are open, so the chain rule applies on them.
- On $\{\det X > 0\}$, $\log \det X = h(\det X)$ with $h(t) = \log t$; by the chain rule (Theorem 16.22(c)) and (b), $\mathrm{d} \log \det X = \frac{1}{\det X} \mathrm{d} \det X = \operatorname{tr}(X^{-1}\mathrm{d}X)$.
- On $\{\det X < 0\}$, $\log|\det X| = h(-\det X)$, and $\mathrm{d} \log|\det X| = \frac{1}{-\det X}(-\mathrm{d} \det X) = \operatorname{tr}(X^{-1}\mathrm{d}X)$.
- Where $\det X > 0$, Theorem 16.21(c) with $C = (X^{-1})'$ gives $\partial \log \det X / \partial X = (X^{-1})'$.

(e)

- By the chain rule of calculus applied to $t \mapsto A(t) \mapsto \det A(t)$, and (b), $\frac{\mathrm{d}}{\mathrm{d}t} \det A = \sum_{i,j} \frac{\partial \det A}{\partial a_{ij}} \frac{\mathrm{d}a_{ij}}{\mathrm{d}t} = \sum_{i,j} c_{ij} \frac{\mathrm{d}a_{ij}}{\mathrm{d}t} = \operatorname{tr}\Big(\operatorname{adj} A \frac{\mathrm{d}A}{\mathrm{d}t}\Big)$, as in (b).
- Where $\det A > 0$, $\frac{\mathrm{d}}{\mathrm{d}t} \log \det A = \frac{1}{\det A} \frac{\mathrm{d}}{\mathrm{d}t} \det A = \operatorname{tr}\Big(\frac{\operatorname{adj} A}{\det A} \frac{\mathrm{d}A}{\mathrm{d}t}\Big) = \operatorname{tr}\Big(A^{-1} \frac{\mathrm{d}A}{\mathrm{d}t}\Big)$. □

Example 16.27. Let $f(X) = \log \det X$ on the set of $X \in \mathbb{R}^{4\times 4}$ with $\det X > 0$, which is open, as det is a polynomial in the entries (Theorem 16.26(b)) and hence continuous. Evaluate $\partial f / \partial X$ at

$$X_0 = \Big(\min(i,j)\Big) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

and check it along a direction in which $f(X_0 + t\,\mathrm{d}X)$ is computed exactly.

Solution. By Theorem 16.26(d), $\partial f / \partial X = (X^{-1})'$ where $\det X > 0$.

- Let $L \in \mathbb{R}^{4\times 4}$ have $l_{ik} = 1$ for $k \leqslant i$ and $l_{ik} = 0$ for $k > i$. Then $(LL')_{ij} = \sum_k l_{ik} l_{jk}$ counts the k with $k \leqslant i$ and $k \leqslant j$, so $LL' = X_0$. As L' is upper triangular with unit diagonal, $\det L = \det L' = 1$ (Theorems 5.22 and 5.14(c)), and $\det X_0 = 1$ (Theorem 5.18(b)). So $\det X_0 > 0$ and $f(X_0) = 0$.
- Let

$$T = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Column j of X_0T is $-c_{j-1}+2c_j-c_{j+1}$ for $j\leqslant 3$ (with $c_0=0$) and c_4-c_3 for $j=4$, where c_j is column j of X_0; these are $e_1,\dots,e_4$. So $X_0T=I_4$ and $X_0^{-1}=T$ (Theorem 2.16), and, as $T'=T$,

$$\frac{\partial f}{\partial X}(X_0)=(X_0^{-1})'=T,\qquad \mathrm{d}f=\operatorname{tr}(T\,\mathrm{d}X)\text{ at }X_0$$

- Check along the rank-one direction $\mathrm{d}X=uv'$ with $u=e_1-e_2$ and $v=e_1$ (adding t to x_{11} and $-t$ to x_{21}): by the matrix determinant lemma (Theorem 5.31), $\det(X_0+tuv')=\det X_0\,(1+t\,v'X_0^{-1}u)=1+t\big(2-(-1)\big)=1+3t$. So $f(X_0+tuv')=\log(1+3t)$ for $t>-\frac{1}{3}$, with derivative 3 at $t=0$; and $\operatorname{tr}(Tuv')=v'Tu=t_{11}-t_{12}=3$ (Theorem 1.11(b)).

Example 16.28 ([21, Exercise 22], [6, Exercises 13.22(a), 13.25(a)])**.** Let X range over the invertible $n\times n$ matrices, and let A, B be constant.

(a) For $F(X)=AX^{-1}B$, $\mathrm{d}F=A(\mathrm{d}X^{-1})B=-AX^{-1}(\mathrm{d}X)X^{-1}B$ (Theorems 16.22 and 16.26(c)). By Theorem 16.7(b), $\operatorname{vec}\mathrm{d}F=-\big((X^{-1}B)'\otimes AX^{-1}\big)\operatorname{vec}\mathrm{d}X$, so $\mathrm{D}F(X)=-(X^{-1}B)'\otimes AX^{-1}$.

(b) For $\varphi(X)=\operatorname{tr}(AX^{-1})$, $\mathrm{d}\varphi=-\operatorname{tr}(AX^{-1}(\mathrm{d}X)X^{-1})=-\operatorname{tr}(X^{-1}AX^{-1}\mathrm{d}X)$ (Theorem 1.11(c)), so $\partial\varphi/\partial X=-(X^{-1}AX^{-1})'$ (Theorem 16.21(c)).

Example 16.29 ([3, Results 9.1, 9.2, 9.5])**.** In the **one-way random effects model** the m observations of a group are $y_j=\mu+a+\varepsilon_j$ $(j=1,\dots,m)$, where a, $\varepsilon_1,\dots,\varepsilon_m$ are uncorrelated random variables with means 0, $\operatorname{var}(a)=\sigma_1^2$ and $\operatorname{var}(\varepsilon_j)=\sigma^2>0$ (the variance components). With $y=\mu\mathbf{1}_m+[\mathbf{1}_m\ I_m](a,\varepsilon')'$, Theorem 9.3(b) gives

$$V=\operatorname{var}(y)=[\mathbf{1}_m\ I_m]\begin{pmatrix}\sigma_1^2 & 0\\ 0 & \sigma^2 I_m\end{pmatrix}\begin{pmatrix}\mathbf{1}_m'\\ I_m\end{pmatrix}=\sigma_1^2J_m+\sigma^2I_m$$

Maximum-likelihood methods for σ_1^2, σ^2 need the derivatives of $\log\det V$. As $\mathrm{d}V/\mathrm{d}\sigma_1^2=J_m$ and $\mathrm{d}V/\mathrm{d}\sigma^2=I_m$, Theorem 16.26(e) gives

$$\frac{\partial\log\det V}{\partial\sigma_1^2}=\operatorname{tr}(V^{-1}J_m),\qquad \frac{\partial\log\det V}{\partial\sigma^2}=\operatorname{tr}(V^{-1})$$

By Theorem 5.32(b) ($A=\sigma^2I_m$, $u=\sigma_1^2\mathbf{1}_m$, $v=\mathbf{1}_m$), $V^{-1}=\frac{1}{\sigma^2}\Big(I_m-\frac{\sigma_1^2}{\sigma^2+m\sigma_1^2}J_m\Big)$, so $\operatorname{tr}(V^{-1}J_m)=\frac{m}{\sigma^2+m\sigma_1^2}$. This agrees with $\det V=\sigma^{2(m-1)}(\sigma^2+m\sigma_1^2)$ (Theorem 5.31), whose logarithm has the derivative $\frac{m}{\sigma^2+m\sigma_1^2}$ in σ_1^2. For $m=4$, $\sigma_1^2=1$, $\sigma^2=2$:

$$V=J_4+2I_4=\begin{pmatrix}3&1&1&1\\1&3&1&1\\1&1&3&1\\1&1&1&3\end{pmatrix},\qquad V^{-1}=\frac{1}{2}\Big(I_4-\frac{1}{6}J_4\Big)=\frac{1}{12}\begin{pmatrix}5&-1&-1&-1\\-1&5&-1&-1\\-1&-1&5&-1\\-1&-1&-1&5\end{pmatrix}$$

so $\operatorname{tr}(V^{-1}J_4)=\mathbf{1}_4'V^{-1}\mathbf{1}_4=\frac{4\cdot2}{12}=\frac{2}{3}=\frac{m}{\sigma^2+m\sigma_1^2}$ and $\operatorname{tr}(V^{-1})=\frac{4\cdot5}{12}=\frac{5}{3}$; and $\det V=2^3\cdot6=48$. Indeed $\frac{m-1}{\sigma^2}+\frac{1}{\sigma^2+m\sigma_1^2}=\frac{3}{2}+\frac{1}{6}=\frac{5}{3}$.

Theorem 16.30 ([2, §§9.5.2–9.5.4], [6, Exercises 13.25(b), 13.26(b)])**.** Let X range over an open set of symmetric $n \times n$ matrices, the variables being the entries x_{ij} with $i \geqslant j$ (that is, $\operatorname{vech} X$). For a function φ of these variables, write $\partial\varphi/\partial X$ for the symmetric matrix with (i,j) and (j,i) entries $\partial\varphi/\partial x_{ij}$; and write $\operatorname{diag}(M)$ for the diagonal part of a square matrix M.

(a) Let φ be C^1, with differential $\mathrm{d}\varphi = \operatorname{tr}(C\,\mathrm{d}X)$ for symmetric $\mathrm{d}X$, where $C = (c_{ij})$ is symmetric (and may depend on X). Then

$$\frac{\partial\varphi}{\partial x_{ij}} = 2c_{ij}\ (i > j), \qquad \frac{\partial\varphi}{\partial x_{ii}} = c_{ii}, \qquad \frac{\partial\varphi}{\partial(\operatorname{vech} X)'} = (\operatorname{vec} C)'D_n$$

that is, $\partial\varphi/\partial X = 2C - \operatorname{diag}(C)$.

(b) If X is PD, $\partial \log\det X/\partial X = 2X^{-1} - \operatorname{diag}(X^{-1})$.

(c) $\partial \operatorname{tr}(AX)/\partial X = A + A' - \operatorname{diag}(A)$ for constant $A \in \mathbb{R}^{n\times n}$.

(d) $\partial(a'Xa)/\partial X = 2aa' - \operatorname{diag}(aa')$ for constant $a \in \mathbb{R}^n$.

Proof. (a)

- Let $E_{ij} = e_i e_j'$. The differential $\mathrm{d}\varphi = \mathrm{D}\varphi\,\mathrm{d}\operatorname{vech} X$ at the unit vector of the variable x_{ij} is $\partial\varphi/\partial x_{ij}$ (Definitions 16.17 and 16.20). This increment changes only x_{ij}, by 1, that is, the (i,j) and (j,i) entries of X: $\mathrm{d}X = E_{ij} + E_{ji}$ for $i > j$ and $\mathrm{d}X = E_{ii}$ for $i = j$.
- Hence $\partial\varphi/\partial x_{ij} = \operatorname{tr}(CE_{ij}) + \operatorname{tr}(CE_{ji}) = c_{ji} + c_{ij} = 2c_{ij}$ for $i > j$ ($\operatorname{tr}(Ce_ie_j') = e_j'Ce_i$, Theorem 1.11(b)), and $\partial\varphi/\partial x_{ii} = \operatorname{tr}(CE_{ii}) = c_{ii}$.
- Vector form: $\operatorname{vec}\mathrm{d}X = D_n\,\mathrm{d}\operatorname{vech} X$ (Definition 16.15), so by Theorem 16.7(d) ($C' = C$)

$$\mathrm{d}\varphi = (\operatorname{vec} C)'\operatorname{vec}\mathrm{d}X = (\operatorname{vec} C)'D_n\,\mathrm{d}\operatorname{vech} X$$

and Theorem 16.21(a) applies. The entries of $(\operatorname{vec} C)'D_n$ are c_{ii} and $c_{ij} + c_{ji} = 2c_{ij}$ (Proposition 16.16(a)), as above.

(b) $\log\det X$ is the composite of the linear map $\operatorname{vech} X \mapsto X$ and $\log\det$; by the chain rule (Theorem 16.22(c)) and Theorem 16.26(d), $\mathrm{d}\log\det X = \operatorname{tr}(X^{-1}\mathrm{d}X)$ for symmetric $\mathrm{d}X$, and $C = X^{-1}$ is symmetric. Apply (a).

(c) For symmetric $\mathrm{d}X$, $\operatorname{tr}(A\,\mathrm{d}X) = \operatorname{tr}((A\,\mathrm{d}X)') = \operatorname{tr}(\mathrm{d}X\,A') = \operatorname{tr}(A'\mathrm{d}X)$ (Theorem 1.11(a),(b)), so $\mathrm{d}\operatorname{tr}(AX) = \operatorname{tr}\left(\frac{1}{2}(A + A')\mathrm{d}X\right)$ with the symmetric $C = \frac{1}{2}(A + A')$. Apply (a).

(d) $a'Xa = \operatorname{tr}(Xaa') = \operatorname{tr}(aa'X)$ (Theorem 1.11(b)), so this is (c) with $A = aa'$. □

So the derivative with respect to a symmetric matrix is not simply the unrestricted one: for $\log\det X$ the unrestricted derivative is $(X^{-1})' = X^{-1}$, while each off-diagonal variable x_{ij} $(i > j)$ enters X twice and gets the derivative $2(X^{-1})_{ij}$.

Second differentials and the Hessian matrix

We also use without proof [21, §9], [5, §15.1]: if f is C^2 on an open set, then $\partial^2 f/\partial x_i\partial x_j = \partial^2 f/\partial x_j\partial x_i$, where $\partial^2 f/\partial x_i\partial x_j$ is the partial derivative with respect to x_i of $\partial f/\partial x_j$ (**symmetry of second derivatives**).

Definition 16.31 ([21, §9], [2, §9.5.7], [5, §15.1])**.** Let f be a C^2 scalar function on an open $S \subseteq \mathbb{R}^n$. The **Hessian matrix** of f at $x \in S$ is the $n \times n$ matrix

$$\mathrm{H}f(x) = \frac{\partial^2 f(x)}{\partial x\,\partial x'} \quad \text{with } (i,j) \text{ entry } \frac{\partial^2 f(x)}{\partial x_i\,\partial x_j}$$

It is the derivative of the gradient, $\mathrm{H}f(x) = \mathrm{D}\big((\mathrm{D}f)'\big)(x)$, and it is symmetric. The **second differential** $\mathrm{d}^2 f(x)$ is the differential of $x \mapsto \mathrm{d}f(x) = \mathrm{D}f(x)\,\mathrm{d}x$ with $\mathrm{d}x$ held fixed. For a scalar function of a matrix X, $\mathrm{H}f(X) = \partial^2 f/\partial \operatorname{vec} X\, \partial(\operatorname{vec} X)'$.

With $a_j = \partial f/\partial x_j$, $\mathrm{d}f(x) = \sum_j a_j(x)\,\mathrm{d}x_j$, and by Theorem 16.22, $\mathrm{d}^2 f(x) = \sum_j (\mathrm{d}a_j)\,\mathrm{d}x_j = \sum_j \sum_i (\partial a_j/\partial x_i)\,\mathrm{d}x_i\,\mathrm{d}x_j$, that is,

$$\mathrm{d}^2 f(x) = (\mathrm{d}x)'\,\mathrm{H}f(x)\,\mathrm{d}x$$

To compute $\mathrm{d}^2 f$ we apply the rules of Theorem 16.22 to $\mathrm{d}f$, treating $\mathrm{d}x$ (or $\mathrm{d}X$) as a constant: $\mathrm{d}(\mathrm{d}x) = 0$.

Theorem 16.32 (Second identification theorem; [21, Propositions 16, 18], [6, Exercises 13.47(c), 13.48(c)])**.** Let f be a C^2 scalar function on an open set.

(a) For a vector variable $x \in \mathbb{R}^n$ and $B \in \mathbb{R}^{n\times n}$: $\mathrm{d}^2 f(x) = (\mathrm{d}x)'B\,\mathrm{d}x$ for every $\mathrm{d}x$ if and only if $\mathrm{H}f(x) = \frac{1}{2}(B + B')$.

(b) For a matrix variable $X \in \mathbb{R}^{n\times q}$, $A \in \mathbb{R}^{q\times q}$ and $B \in \mathbb{R}^{n\times n}$: $\mathrm{d}^2 f(X) = \operatorname{tr}\big(A(\mathrm{d}X)'B\,\mathrm{d}X\big)$ for every $\mathrm{d}X$ if and only if $\mathrm{H}f(X) = \frac{1}{2}(A' \otimes B + A \otimes B')$.

(c) For $X \in \mathbb{R}^{n\times q}$ and $A, B \in \mathbb{R}^{q\times n}$: $\mathrm{d}^2 f(X) = \operatorname{tr}\big(A(\mathrm{d}X)B\,\mathrm{d}X\big)$ for every $\mathrm{d}X$ if and only if $\mathrm{H}f(X) = \frac{1}{2}K_{qn}(A' \otimes B + B' \otimes A)$.

Proof. (a) Both $\mathrm{H}f(x)$ and $\frac{1}{2}(B + B')$ are symmetric. If $\mathrm{d}^2 f(x) = (\mathrm{d}x)'B\,\mathrm{d}x$ for every $\mathrm{d}x$, then $(\mathrm{d}x)'\mathrm{H}f(x)\,\mathrm{d}x = (\mathrm{d}x)'B\,\mathrm{d}x$, and Lemma 16.19(c) gives $2\mathrm{H}f(x) = B + B'$. Conversely, $(\mathrm{d}x)'\frac{1}{2}(B + B')\,\mathrm{d}x = (\mathrm{d}x)'B\,\mathrm{d}x$, as $(\mathrm{d}x)'B'\,\mathrm{d}x$ is a number equal to its transpose.

(b) By Theorems 1.11(b) and 16.7(d),(b),

$$\operatorname{tr}(A(\mathrm{d}X)'B\,\mathrm{d}X) = \operatorname{tr}((\mathrm{d}X)'B(\mathrm{d}X)A) = (\operatorname{vec}\mathrm{d}X)'\operatorname{vec}(B(\mathrm{d}X)A) = (\operatorname{vec}\mathrm{d}X)'(A' \otimes B)\operatorname{vec}\mathrm{d}X$$

By (a) for the variable $\operatorname{vec} X$, $\mathrm{H}f(X) = \frac{1}{2}\big(A' \otimes B + (A' \otimes B)'\big) = \frac{1}{2}(A' \otimes B + A \otimes B')$ (Theorem 16.3(e)).

(c)

- As $\operatorname{tr} M = \operatorname{tr} M'$, $\operatorname{tr}(A(\mathrm{d}X)B\,\mathrm{d}X) = \operatorname{tr}((\mathrm{d}X)'B'(\mathrm{d}X)'A') = (\operatorname{vec}\mathrm{d}X)'\operatorname{vec}(B'(\mathrm{d}X)'A') = (\operatorname{vec}\mathrm{d}X)'(A \otimes B')\operatorname{vec}(\mathrm{d}X)'$, and $\operatorname{vec}(\mathrm{d}X)' = K_{nq}\operatorname{vec}\mathrm{d}X$ (Definition 16.11). So $\mathrm{d}^2 f = (\operatorname{vec}\mathrm{d}X)'(A \otimes B')K_{nq}\operatorname{vec}\mathrm{d}X$.
- By (a), $\mathrm{H}f(X) = \frac{1}{2}\big((A \otimes B')K_{nq} + K'_{nq}(A' \otimes B)\big)$, with $K'_{nq} = K_{qn}$ (Theorem 16.12(b)). By Theorem 16.12(d) for $B' \in \mathbb{R}^{n\times q}$ and $A \in \mathbb{R}^{q\times n}$, $K_{qn}(B' \otimes A) = (A \otimes B')K_{nq}$. Hence $\mathrm{H}f(X) = \frac{1}{2}K_{qn}(A' \otimes B + B' \otimes A)$. □

Example 16.33 ([21, Exercises 23, 25, 29], [6, Exercises 13.46, 13.47(b), 13.52])**.** (a) For $f(x) = x'Ax$ ($A \in \mathbb{R}^{n\times n}$), $\mathrm{d}f = (\mathrm{d}x)'Ax + x'A\,\mathrm{d}x$ and $\mathrm{d}^2 f = 2(\mathrm{d}x)'A\,\mathrm{d}x$, so $\mathrm{H}f(x) = A + A'$ (Theorem 16.32(a)); for symmetric A, $\mathrm{H}f = 2A$.

(b) For $f(X) = \operatorname{tr}(X'AX)$ with $X \in \mathbb{R}^{n\times q}$, $\mathrm{d}f = \operatorname{tr}(X'(A + A')\,\mathrm{d}X)$ (Theorem 16.26(a)) and $\mathrm{d}^2 f = \operatorname{tr}((\mathrm{d}X)'(A + A')\,\mathrm{d}X) = \operatorname{tr}\big(I_q(\mathrm{d}X)'(A + A')\,\mathrm{d}X\big)$; by Theorem 16.32(b), $\mathrm{H}f(X) = I_q \otimes (A + A')$.

(c) For $f(X) = \log\det X$ ($\det X > 0$), $\mathrm{d}f = \operatorname{tr}(X^{-1}\mathrm{d}X)$ and, by Theorem 16.26(c), $\mathrm{d}^2 f = \operatorname{tr}((\mathrm{d}X^{-1})\,\mathrm{d}X) = -\operatorname{tr}\big(X^{-1}(\mathrm{d}X)X^{-1}\mathrm{d}X\big)$. Theorem 16.32(c) with $q = n$, $A = -X^{-1}$, $B = X^{-1}$ gives $\mathrm{H}f(X) = -K_n\big((X^{-1})' \otimes X^{-1}\big)$.

Conditions for extrema

For functions of one variable we use without proof [6, §A.3.4]: (**Fermat's rule**) if h is differentiable on an open interval and has a local minimum or maximum at a point t_0 of it, then $h'(t_0) = 0$; and (**Taylor's formula**) if h is C^2 on an open interval that contains $[0, 1]$, then $h(1) = h(0) + h'(0) + \frac{1}{2}h''(\tau)$ for some τ with $0 < \tau < 1$.

Definition 16.34 ([23, §7.2, §7.7], [6, §A.4.3])**.** Let f be a scalar function on $S \subseteq \mathbb{R}^n$ and $c \in S$. f has a **local minimum** at c if $f(x) \geqslant f(c)$ for all $x \in S$ in some ball $B(c, \delta)$, a **strict local minimum** if $f(x) > f(c)$ there for $x \neq c$, and a **global minimum** (**strict global minimum**) if $f(x) \geqslant f(c)$ for all $x \in S$ ($f(x) > f(c)$ for $x \neq c$); maxima likewise with the inequalities reversed. If S is open and f is C^1, c is a **stationary point** of f if $\mathrm{D}f(c) = 0$. A set S is **convex** if $tx + (1-t)y \in S$ for all $x, y \in S$ and $0 \leqslant t \leqslant 1$.

Lemma 16.35 ([23, Theorems 7.2, 7.4, 7.7])**.** Let $S \subseteq \mathbb{R}^n$ be open, f a C^2 scalar function on S, $c \in S$ and $u \in \mathbb{R}^n$. Let $I = \{t \in \mathbb{R} : c + tu \in S\}$ and $h(t) = f(c + tu)$ for $t \in I$.

(a) I is open and contains 0, and h is C^2 on I, with $h'(t) = \mathrm{D}f(c + tu)u$ and $h''(t) = u'\mathrm{H}f(c + tu)u$.

(b) Suppose that $\|u\| < \delta$ for a ball $B(c, \delta) \subseteq S$, or that S is convex and $c + u \in S$. Then, for some $0 < \tau < 1$,

$$f(c + u) = f(c) + \mathrm{D}f(c)u + \tfrac{1}{2}u'\mathrm{H}f(c + \tau u)u$$

Proof. (a)

- $0 \in I$ as $c \in S$. I is open: if $c + t_0 u \in S$, some ball $B(c + t_0 u, \varepsilon)$ lies in S, and it contains $c + tu$ for $|t - t_0| \, \|u\| < \varepsilon$.
- By the chain rule of calculus ($t \mapsto c + tu$ has the derivative u), $h'(t) = \mathrm{D}f(c + tu)u = \sum_j u_j \, \partial f / \partial x_j (c + tu)$.
- The same rule for each $t \mapsto \partial f / \partial x_j (c + tu)$ gives $h''(t) = \sum_j \sum_i u_j u_i \, \partial^2 f / \partial x_i \partial x_j (c + tu) = u'\mathrm{H}f(c + tu)u$ (Definition 16.31), which is continuous in t: h is C^2 on I.

(b)

- In the first case, $c + tu \in B(c, \delta) \subseteq S$ for $|t| < \delta / \|u\|$ (for every t if $u = 0$), and $\delta / \|u\| > 1$: I contains an open interval that contains $[0, 1]$.
- In the second case, I is an interval: for t between two points t_1, t_2 of I, $c + tu$ is a convex combination of $c + t_1 u$ and $c + t_2 u$, so it lies in S. By (a) I is open, and it contains 0 and 1.
- By Taylor's formula and (a), for some $0 < \tau < 1$, $f(c + u) = h(1) = h(0) + h'(0) + \frac{1}{2}h''(\tau) = f(c) + \mathrm{D}f(c)u + \frac{1}{2}u'\mathrm{H}f(c + \tau u)u$. □

Theorem 16.36 ([23, Theorems 7.2, 7.4, 7.7, 7.8, 7.13])**.** Let $S \subseteq \mathbb{R}^n$ be open, f a C^2 scalar function on S, and $c \in S$.

(a) (**First-order condition**) If f has a local minimum or maximum at c, then $\mathrm{D}f(c) = 0$.

(b) If f has a local minimum (maximum) at c, then $\mathrm{H}f(c)$ is PSD (NSD).

(c) (**Second-order condition**) If $\mathrm{D}f(c) = 0$ and $\mathrm{H}f(c)$ is PD (ND), then f has a strict local minimum (maximum) at c. If $\mathrm{D}f(c) = 0$ and $\mathrm{H}f(c)$ is indefinite, then f has neither a local minimum nor a local maximum at c.

(d) (**Global minimum**) Let S also be convex. If $\mathrm{D}f(c) = 0$ and $\mathrm{H}f(x)$ is PSD for every $x \in S$, then f has a global minimum at c; if $\mathrm{H}f(x)$ is PD for every $x \in S$, the global minimum at c is strict.

(e) (**Constraints**) Let S be convex, let g be C^2 on S with values in $\mathbb{R}^m$, let $l \in \mathbb{R}^m$, and let $\psi(x) = f(x) - l'g(x)$. If $\mathrm{D}\psi(c) = 0$, $g(c) = 0$, and $\mathrm{H}\psi(x)$ is PSD (PD) for every $x \in S$, then $f(x) \geqslant f(c)$ ($f(x) > f(c)$ for $x \neq c$) for every $x \in S$ with $g(x) = 0$.

Proof. Throughout, for a given $u \in \mathbb{R}^n$, I and $h(t) = f(c + tu)$ are as in Lemma 16.35.

(a) Let $f(x) \geqslant f(c)$ for $x \in B(c, \delta) \subseteq S$ (a local minimum; δ small, as S is open). For $u = e_j$, h has a local minimum at $t = 0$, as $c + te_j \in B(c, \delta)$ for $|t| < \delta$. By Fermat's rule and Lemma 16.35(a), $0 = h'(0) = \mathrm{D}f(c)e_j = \partial f/\partial x_j(c)$, for every j: $\mathrm{D}f(c) = 0$. For a maximum apply this to $-f$.

(b)

- Let f have a local minimum at c, say $f(x) \geqslant f(c)$ for $x \in B(c, \delta) \subseteq S$ (δ small, as in (a)), and suppose $u'\mathrm{H}f(c)u < 0$ for some u. By Lemma 16.35(a), $h''(0) = u'\mathrm{H}f(c)u < 0$. As h'' is continuous and I is open, $h''(t) < 0$ for $|t| \leqslant \varepsilon$, for some $\varepsilon > 0$ with $[-\varepsilon, \varepsilon] \subseteq I$.
- Choose $0 < s_0 \leqslant \varepsilon$ with $s_0\|u\| < \delta$. By Lemma 16.35(b) for the vector s_0u (first case) and $\mathrm{D}f(c) = 0$ (by (a)), for some $0 < \tau < 1$,

$$f(c + s_0u) = f(c) + \tfrac{1}{2}s_0^2u'\mathrm{H}f(c + \tau s_0u)u = f(c) + \tfrac{1}{2}s_0^2h''(\tau s_0) < f(c)$$

  as $0 < \tau s_0 < \varepsilon$. But $c + s_0u \in B(c, \delta)$, so $f(c + s_0u) \geqslant f(c)$ ✘
- Hence $\mathrm{H}f(c)$ is PSD. For a maximum apply this to $-f$, whose Hessian is $-\mathrm{H}f$.

(c)

- Let $\mathrm{H}f(c)$ be PD with smallest eigenvalue $\lambda > 0$ (Theorem 8.21(a)); then $u'\mathrm{H}f(c)u \geqslant \lambda\|u\|^2$ for every u (Theorem 9.23(a)).
- The second partial derivatives are continuous, so the $2n^2$ functions $\frac{\lambda}{2n} \pm \left(\partial^2 f/\partial x_i \partial x_j(z) - \partial^2 f/\partial x_i \partial x_j(c)\right)$ of z, positive at $z = c$, are positive on a ball $B(c, \delta) \subseteq S$ (rules of calculus). For $z \in B(c, \delta)$, every entry of $E = \mathrm{H}f(z) - \mathrm{H}f(c)$ is less than $\frac{\lambda}{2n}$ in absolute value, so $\|E\|_F^2 < n^2\frac{\lambda^2}{4n^2}$, and by Theorem 6.4(a) and Proposition 15.8(b), $|u'Eu| \leqslant \|u\|\,\|Eu\| \leqslant \|E\|_F\|u\|^2 < \frac{\lambda}{2}\|u\|^2$ for $u \neq 0$.
- For $0 < \|u\| < \delta$, Lemma 16.35(b) (first case) with $\mathrm{D}f(c) = 0$ gives $f(c + u) - f(c) = \frac{1}{2}u'\mathrm{H}f(z)u$ with $z = c + \tau u \in B(c, \delta)$; so $f(c + u) - f(c) = \frac{1}{2}u'\mathrm{H}f(c)u + \frac{1}{2}u'Eu > \frac{1}{2}\lambda\|u\|^2 - \frac{1}{4}\lambda\|u\|^2 > 0$: a strict local minimum. For ND apply this to $-f$.
- If $\mathrm{H}f(c)$ is indefinite, it is neither PSD nor NSD (Definition 8.20), so by (b) c is neither a local minimum nor a local maximum.

(d) Let $x \in S$ and $u = x - c$. By Lemma 16.35(b) (S convex, $c + u = x \in S$) and $\mathrm{D}f(c) = 0$, $f(x) = f(c) + \frac{1}{2}u'\mathrm{H}f(c + \tau u)u$, where $c + \tau u = (1 - \tau)c + \tau x \in S$. This is $\geqslant f(c)$ if $\mathrm{H}f(c + \tau u)$ is PSD, and $> f(c)$ for $x \neq c$ if it is PD.

(e) ψ is C^2 on S (rules of calculus), so by (d) applied to ψ, $\psi(x) \geqslant \psi(c)$ for every $x \in S$ ($>$ for $x \neq c$ in the PD case). If $g(x) = 0$, then $f(x) = \psi(x)$, and $\psi(c) = f(c)$ as $g(c) = 0$. □

To use (e) one solves $\mathrm{D}\psi(x) = 0$, $g(x) = 0$ for x and the **Lagrange multipliers** l (**Lagrange's method**), and then checks the Hessian of the **Lagrangian** ψ. For linear constraints $g(x) = Rx - r$, $\mathrm{H}\psi = \mathrm{H}f$.

We also use without proof **Lagrange's theorem** [23, Theorem 7.10]: let $S \subseteq \mathbb{R}^n$ be open, let f be a C^1 scalar function and g a C^1 function with values in $\mathbb{R}^m$ ($m < n$) on S, and let $c \in S$ with $g(c) = 0$ and $\operatorname{rank} \mathrm{D}g(c) = m$. If $f(x) \geqslant f(c)$ for every x with $g(x) = 0$ in some ball $B(c, \delta) \subseteq S$ (a local minimum under the constraint $g(x) = 0$), or $f(x) \leqslant f(c)$ for all these x (a local maximum), then there is exactly one $l \in \mathbb{R}^m$ with

$$\mathrm{D}f(c) = l' \,\mathrm{D}g(c)$$

that is, $\mathrm{D}\psi(c) = 0$ for the Lagrangian $\psi = f - l'g$. For S, f and g as here but with $m \leqslant n$, a point $c \in S$ with $g(c) = 0$, $\operatorname{rank} \mathrm{D}g(c) = m$ and $\mathrm{D}f(c) = l' \,\mathrm{D}g(c)$ for some $l \in \mathbb{R}^m$ is a **stationary point of f under the constraint** $g(x) = 0$. So, for $m < n$, Lagrange's method finds all the points where f has a local minimum or maximum under the constraint and $\operatorname{rank} \mathrm{D}g = m$; Theorem 16.36(e) shows that a stationary point is a minimum when the Hessian of ψ is PSD on a convex S.

Example 16.37 ([21, Exercises 14, 15], [5, Theorem 19.1.1])**.** Let $A \in \mathbb{R}^{n\times n}$ be PD, $b \in \mathbb{R}^n$, and $f(x) = \frac{1}{2}x'Ax - b'x$.

(a) By Theorem 16.24, $\mathrm{d}f = x'A\,\mathrm{d}x - b'\,\mathrm{d}x = (Ax - b)'\mathrm{d}x$, and $\mathrm{d}^2 f = (\mathrm{d}x)'A\,\mathrm{d}x$, so $\mathrm{H}f = A$ is PD at every x. The only stationary point is $\hat{x} = A^{-1}b$, and by Theorem 16.36(d) it is the strict global minimizer. Directly: $(x - \hat{x})'A(x - \hat{x}) = x'Ax - 2b'x + b'A^{-1}b = 2f(x) + b'A^{-1}b$, which is smallest, 0, at $x = \hat{x}$; the minimum is $f(\hat{x}) = -\frac{1}{2}b'A^{-1}b$.

(b) Minimize f subject to $c'x = 0$ ($c \neq 0$). The Lagrangian $\psi(x) = \frac{1}{2}x'Ax - b'x - \lambda c'x$ has $\mathrm{d}\psi = (Ax - b - \lambda c)'\mathrm{d}x$, so the conditions are $Ax = b + \lambda c$ and $c'x = 0$, or, in block form,

$$\begin{pmatrix} A & c \\ c' & 0 \end{pmatrix} \begin{pmatrix} x \\ -\lambda \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}$$

From $x = A^{-1}(b + \lambda c)$ and $0 = c'x = c'A^{-1}b + \lambda c'A^{-1}c$, with $c'A^{-1}c > 0$ (A^{-1} is PD, Proposition 8.23(c)), we get $\tilde{\lambda} = -c'A^{-1}b/c'A^{-1}c$ and $\tilde{x} = A^{-1}b - \frac{c'A^{-1}b}{c'A^{-1}c}A^{-1}c$. As $\mathrm{H}\psi = A$ is PD, $\tilde{x}$ is the strict minimizer under the constraint (Theorem 16.36(e)).

Proposition 16.38 ([23, Example 7.3])**.** Let $A \in \mathbb{R}^{n\times n}$ ($n \geqslant 2$) be symmetric, $f(x) = x'Ax$ and $g(x) = x'x - 1$ on $\mathbb{R}^n$. A vector x is a stationary point of f under the constraint $g(x) = 0$ (on the unit sphere) if and only if x is a unit eigenvector of A; then $f(x)$ is its eigenvalue. The maximum λ_1 and the minimum λ_n of f on the unit sphere (the largest and the smallest eigenvalue of A) are attained exactly at the unit eigenvectors belonging to λ_1 and to λ_n.

Proof. • By Theorem 16.24(c), $\mathrm{D}g(x) = 2x'$ and $\mathrm{D}f(x) = 2x'A$, as I_n and A are symmetric. For $x'x = 1$, $x \neq 0$ and $\operatorname{rank} \mathrm{D}g(x) = 1$.

- So a unit vector x is a stationary point of f under the constraint if and only if $2x'A = l \cdot 2x'$ for some $l \in \mathbb{R}$, that is (transposing, $A' = A$), $Ax = lx$: x is an eigenvector belonging to l. Then $f(x) = x'(lx) = l\,x'x = l$.
- On the unit sphere $f(x) = x'Ax/x'x$, so the last statement is Theorem 9.23(a),(b). □

By Lagrange's theorem, the points where f is largest or smallest on the sphere must be stationary points under the constraint, and the proposition finds them all; Theorem 9.23 gives the same answer without derivatives.

Theorem 16.39 (Quadratic minimization under linear constraints; [5, Theorem 19.2.1, §19.3a])**.** Let $A \in \mathbb{R}^{p\times p}$ be PD, $b \in \mathbb{R}^p$, $C \in \mathbb{R}^{m\times p}$ of rank m and $d \in \mathbb{R}^m$. Let $q(x) = \frac{1}{2}x'Ax - b'x$ and $\hat{x} = A^{-1}b$, its unconstrained minimizer (Example 16.37(a)). Then $V = CA^{-1}C'$ is PD, hence invertible, and with $\lambda = V^{-1}(C\hat{x} - d)$:

(a) the unique minimizer of q on $\{x \in \mathbb{R}^p : Cx = d\}$ is

$$x^* = \hat{x} - A^{-1}C'V^{-1}(C\hat{x} - d) = \hat{x} - A^{-1}C'\lambda$$

(b) $q(x) = q(x^*) + \frac{1}{2}(x - x^*)'A(x - x^*)$ for every x with $Cx = d$;

(c) $q(x^*) - q(\hat{x}) = \frac{1}{2}(C\hat{x} - d)'V^{-1}(C\hat{x} - d)$;

(d) (x^*, λ) is the unique solution of the $(p + m) \times (p + m)$ linear system

$$\begin{pmatrix} A & C' \\ C & 0 \end{pmatrix} \begin{pmatrix} x \\ \lambda \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix}$$

Its first block row says $\mathrm{D}q(x^*) = (Ax^* - b)' = (-\lambda)'C$: x^* is a stationary point of q under the constraint $Cx - d = 0$ (the paragraph after Theorem 16.36; $\mathrm{D}(Cx - d) = C$), with multiplier vector $-\lambda$.

Proof. A^{-1} is PD (Proposition 8.23(c)) and $C' \in \mathbb{R}^{p \times m}$ has rank m (Theorem 4.13(a)), so $V = (C')'A^{-1}C'$ is PD (Theorem 8.24(b)), hence invertible (Proposition 8.23(c)).

(d) Let K be the block matrix. The Schur complement of A in K is $0 - CA^{-1}C' = -V$, so $\det K = \det A \cdot \det(-V) = (-1)^m \det A \det V \neq 0$ (Theorems 5.30(a) and 5.18(c)), and K is invertible (Theorem 5.18(a)): the system has exactly one solution. The pair (x^*, λ) is a solution: $Ax^* + C'\lambda = A\hat{x} - C'\lambda + C'\lambda = b$ and $Cx^* = C\hat{x} - V\lambda = C\hat{x} - (C\hat{x} - d) = d$. By Theorem 16.24(a),(c), $\mathrm{D}q(x) = x'A - b' = (Ax - b)'$.

(b) For $h \in \mathbb{R}^p$, as A is symmetric,

$$q(x^* + h) = \tfrac{1}{2}(x^*)'Ax^* + h'Ax^* + \tfrac{1}{2}h'Ah - b'x^* - b'h = q(x^*) + h'(Ax^* - b) + \tfrac{1}{2}h'Ah$$

By (d), $Ax^* - b = -C'\lambda$. If $Cx = d$, then $h = x - x^*$ satisfies $Ch = d - d = 0$, so $h'(Ax^* - b) = -(Ch)'\lambda = 0$, and $q(x) = q(x^*) + \frac{1}{2}h'Ah$.

(a) $Cx^* = d$ by (d). By (b), $q(x) \geqslant q(x^*)$ whenever $Cx = d$, with equality only if $(x - x^*)'A(x - x^*) = 0$, that is, $x = x^*$, as A is PD.

(c) By Example 16.37(a), $2q(x) + b'A^{-1}b = (x - \hat{x})'A(x - \hat{x})$ for every x, and $2q(\hat{x}) + b'A^{-1}b = 0$; so $q(x^*) = q(\hat{x}) + \frac{1}{2}(x^* - \hat{x})'A(x^* - \hat{x})$, and $x^* - \hat{x} = -A^{-1}C'\lambda$. So $q(x^*) - q(\hat{x}) = \frac{1}{2}\lambda'CA^{-1}AA^{-1}C'\lambda = \frac{1}{2}\lambda'V\lambda = \frac{1}{2}(C\hat{x} - d)'V^{-1}(C\hat{x} - d)$, as V^{-1} is symmetric. □

For $A = I_p$ and $b = a$, $q(x) = \frac{1}{2}\|x - a\|^2 - \frac{1}{2}\|a\|^2$, and (a) gives the point of the set $\{x : Cx = d\}$ closest to a, namely $x^* = a - C'(CC')^{-1}(Ca - d)$. For $a = 0$ this is $C'(CC')^{-1}d$, the shortest solution of $Cx = d$, as in Theorem 12.20 with the minimum-norm g-inverse $G = C'(CC')^{-1}$ of C ($CGC = C$, and GC is symmetric).

Example 16.40. Minimize

$$q(x) = x_1^2 + x_2^2 + x_3^2 + x_4^2 - x_1x_2 - x_2x_3 - x_3x_4 - 2(x_1 + x_2 + x_3 + x_4)$$

subject to $x_1 + x_2 + x_3 + x_4 = 10$ and $x_1 - x_2 + x_3 - x_4 = 6$.

Solution. Here $q(x) = \frac{1}{2}x'Ax - b'x$ with $b = 2\mathbf{1}_4$ and

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}, \qquad A^{-1} = \frac{1}{5}\begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

(multiply to check $AA^{-1} = I_4$, Theorem 2.16). The leading principal minors of A are $2, 3, 4, 5$, so A is PD (Theorem 8.34). The constraints are $Cx = d$ with

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad d = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$$

and $\operatorname{rank} C = 2$ (the rows are not multiples of each other). The unconstrained minimizer is $\hat{x} = A^{-1}b = \frac{2}{5}(10, 15, 15, 10)' = (4, 6, 6, 4)'$, with $q(\hat{x}) = -\frac{1}{2}b'\hat{x} = -20$ (Example 16.37(a)). Next

$$A^{-1}C' = \frac{1}{5}\begin{pmatrix} 10 & 2 \\ 15 & -1 \\ 15 & 1 \\ 10 & -2 \end{pmatrix}, \qquad V = CA^{-1}C' = \begin{pmatrix} 10 & 0 \\ 0 & \frac{6}{5} \end{pmatrix}, \qquad C\hat{x} - d = \begin{pmatrix} 20 \\ 0 \end{pmatrix} - \begin{pmatrix} 10 \\ 6 \end{pmatrix} = \begin{pmatrix} 10 \\ -6 \end{pmatrix}$$

so $\lambda = V^{-1}(C\hat{x} - d) = (1, -5)'$. By Theorem 16.39(a),(c),

$$x^* = \begin{pmatrix} 4 \\ 6 \\ 6 \\ 4 \end{pmatrix} - \frac{1}{5}\begin{pmatrix} 10 & 2 \\ 15 & -1 \\ 15 & 1 \\ 10 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ -5 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 6 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 4 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 4 \\ 0 \end{pmatrix}, \qquad q(x^*) - q(\hat{x}) = \tfrac{1}{2}\big(10 \cdot 1 + (-6)(-5)\big) = 20$$

so the constrained minimum is $q(x^*) = 0$. Checks: x^* satisfies both constraints; $q(x^*) = 16 + 4 + 16 - 8 - 8 - 20 = 0$; and $Ax^* - b = (6, -4, 6, -4)' - (2, 2, 2, 2)' = (4, -6, 4, -6)' = -C'\lambda$, as in Theorem 16.39(d).

Least squares by differentiation

Theorem 16.41 ([12, Theorem 2.4], [2, Example 9.3], [6, Exercise 13.54])**.** Let $X \in \mathbb{R}^{n\times(k+1)}$, $y \in \mathbb{R}^n$, and $S(\beta) = (y - X\beta)'(y - X\beta)$ for $\beta \in \mathbb{R}^{k+1}$.

(a) $\mathrm{d}S = -2(y - X\beta)'X\,\mathrm{d}\beta$, so $\mathrm{D}S(\beta) = -2(y - X\beta)'X$ and the gradient is $2X'X\beta - 2X'y$; and $\mathrm{H}S(\beta) = 2X'X$, which is PSD.

(b) $\hat{\beta}$ minimizes S if and only if $X'X\hat{\beta} = X'y$ (the normal equations).

(c) If $\operatorname{rank} X = k + 1$, then $\mathrm{H}S$ is PD and $\hat{\beta} = (X'X)^{-1}X'y$ is the strict global minimizer.

Proof. (a)

- With $e = y - X\beta$, $\mathrm{d}e = -X\,\mathrm{d}\beta$, and by the product rule $\mathrm{d}S = (\mathrm{d}e)'e + e'\,\mathrm{d}e = 2e'\,\mathrm{d}e = -2e'X\,\mathrm{d}\beta$ (Theorems 16.22 and 16.24); so $\mathrm{D}S(\beta) = -2(y - X\beta)'X$ (Theorem 16.21(a)). Equivalently, $S = y'y - 2\beta'X'y + \beta'X'X\beta$ has the derivative $-2y'X + 2\beta'X'X$ (Theorem 16.24).
- With $\mathrm{d}\beta$ fixed, $\mathrm{d}^2S = -2(\mathrm{d}e)'X\,\mathrm{d}\beta = 2(\mathrm{d}\beta)'X'X\,\mathrm{d}\beta$, so $\mathrm{H}S(\beta) = 2X'X$ (Theorem 16.32(a)), which is PSD (Theorem 8.25).

(b) If $\hat{\beta}$ minimizes S on the open set $\mathbb{R}^{k+1}$, then $\mathrm{D}S(\hat{\beta}) = 0$ (Theorem 16.36(a)), that is, $X'(y - X\hat{\beta}) = 0$. Conversely, if $X'X\hat{\beta} = X'y$, then $\hat{\beta}$ is a stationary point, and as $\mathbb{R}^{k+1}$ is convex and $\mathrm{H}S$ is PSD everywhere, $\hat{\beta}$ is a global minimizer (Theorem 16.36(d)).

(c) If $\operatorname{rank} X = k + 1$, $X'X$ is PD (Theorem 8.25) and invertible, the normal equations have the single solution $(X'X)^{-1}X'y$ (Theorem 13.2(b)), and the minimum is strict (Theorem 16.36(d)). □

The projection route of Chapter 13 (least squares needs no calculus: the paragraph after Theorem 13.3) gives the same normal equations: by Theorem 13.3(a), $S(\beta) = S(\hat{\beta}) + \|X(\hat{\beta} - \beta)\|^2$. This identity is Lemma 16.35(b) written out for the quadratic function S: with $u = \beta - \hat{\beta}$, $\mathrm{D}S(\hat{\beta})u = 0$ and $\frac{1}{2}u'(2X'X)u = \|Xu\|^2$, with no remainder because the Hessian $2X'X$ does not depend on β. Differentiation finds the normal equations as the condition for a stationary point; the projection explains them as $y - X\hat{\beta} \perp \mathcal{C}(X)$ (Proposition 13.5(a)).

Example 16.42. For the design $X = [\mathbf{1}_4\ X_0] \in \mathbb{R}^{4\times 3}$ ($k = 2$) and $y = (9, 3, 5, 3)'$ of Example 13.7,

$$X'X = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix}$$

(computed there). At $\beta = 0$ the gradient $2X'X\beta - 2X'y$ is $-2(20, 28, 44)' \neq 0$. At $\hat\beta = (1, 2, 1)'$, $X'X\hat\beta = (4+8+8,\ 4+16+8,\ 8+16+20)' = X'y$, so the gradient is 0. The leading principal minors of $X'X$ are 4, $4\cdot 8 - 4\cdot 4 = 16$ and $\det(X'X) = 64$, all positive, so $\mathrm{H}S = 2X'X$ is PD (Theorem 8.34) and $\hat\beta = (1, 2, 1)'$ is the strict global minimizer, with $S(\hat\beta) = 4$ (Theorem 16.41(c)).

Further statistical applications

Example 16.43 (Hazel's selection index; [2, Example 9.4], [3, Result 5.3])**.** An animal is measured on p traits, $x \in \mathbb{R}^p$ (the phenotypes), with $x = g + e$, where g holds the unobserved genetic values and e environmental effects uncorrelated with g. Let $G = \mathrm{var}(g)$, $P = \mathrm{var}(x)$ (PD), and let $a \in \mathbb{R}^p$ weigh the traits by their economic value, with $a'Ga > 0$. The breeder ranks animals by an **index** $I = b'x$ and wants it to be as highly correlated as possible with the genetic worth $H = a'g$.

- By Theorem 9.3(b) for $z = (g', e')'$, with $\mathrm{var}(z) = \mathrm{diag}(G, E)$, $E = \mathrm{var}(e)$: $P = G + E$, and the 2×2 covariance matrix of $I = (b', b')z$ and $H = (a', 0)z$ has the off-diagonal entry $\mathrm{cov}(I, H) = b'Ga$. So the correlation is $r(b) = \dfrac{b'Ga}{\sqrt{b'Pb}\,\sqrt{a'Ga}}$ for $b \neq 0$.
- The derivative of $\varphi(b) = b'Ga/\sqrt{b'Pb}$ is found from $\mathrm{d}\varphi = \frac{a'G\,\mathrm{d}b}{(b'Pb)^{1/2}} - \frac{(b'Ga)\,b'P\,\mathrm{d}b}{(b'Pb)^{3/2}}$ (Theorem 16.24, chain rule); it is 0 if and only if $(b'Pb)\,Ga = (b'Ga)\,Pb$. So at a stationary point with $b'Ga \neq 0$, b is a multiple of $P^{-1}Ga$, and r does not change when b is multiplied by a positive number.
- The stationary condition alone does not show a maximum. By Theorem 15.2 with $B = P$ and the vector Ga, $(b'Ga)^2 \leqslant (b'Pb)(a'GP^{-1}Ga)$, with equality at $b = P^{-1}Ga$. Hence $r(b) \leqslant \sqrt{a'GP^{-1}Ga/a'Ga}$ for every $b \neq 0$, and the bound is attained at $b = P^{-1}Ga$, where $b'Ga = a'GP^{-1}Ga$ and $r = \sqrt{b'Ga/a'Ga}$.

For four traits with $E = I_4$,

$$G = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix}, \qquad P = G + I_4 = \begin{pmatrix} 3 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \\ 1 & 0 & 3 & 1 \\ 0 & 1 & 1 & 3 \end{pmatrix}, \qquad a = \begin{pmatrix} 3 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

- G is PSD, as $x'Gx = (x_1 + x_2)^2 + (x_1 + x_3)^2 + (x_2 + x_4)^2 + (x_3 + x_4)^2 \geqslant 0$, so $P = G + I_4$ is PD (Proposition 8.23(b)); and $a'Ga = 4^2 + 4^2 + 1^2 + 1^2 = 34 > 0$.
- $a = Pe_1$ is the first column of P, and $GP = G^2 + G = PG$. So $Ga = GPe_1 = PGe_1$, and

$$b = P^{-1}Ga = Ge_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \qquad Ga = \begin{pmatrix} 8 \\ 5 \\ 5 \\ 2 \end{pmatrix} = Pb$$

- Then $b'Ga = 16 + 5 + 5 + 0 = 26$, and the best index $I = 2x_1 + x_2 + x_3$ has $r = \sqrt{26/34} = \sqrt{13/17}$.
- The index $I = a'x$ has $a'Pa = a'Ga + a'a = 34 + 11 = 45$ and only $r = 34/\sqrt{45\cdot 34} = \sqrt{34/45}$, less than $\sqrt{13/17}$ as $34 \cdot 17 = 578 < 585 = 13 \cdot 45$.

Example 16.44 (Newton–Raphson; [2, §9.2])**.** To solve n nonlinear equations $f(x) = 0$ in n unknowns, with f C^1 and derivative $G(x) = \mathrm{D}f(x)$, one uses the first-order approximation $f(x_r + \Delta) \approx f(x_r) + G(x_r)\Delta$, the differential at the current guess x_r. Setting the right side equal to 0 gives the **Newton–Raphson** step

$$x_{r+1} = x_r - G(x_r)^{-1} f(x_r) \qquad (r = 0, 1, 2, \dots)$$

provided $G(x_r)$ is invertible; one stops when the step is negligible. To find a stationary point of a C^2 scalar function φ, apply it to the gradient: $f = (\mathrm{D}\varphi)'$, $G = \mathrm{H}\varphi$, so $x_{r+1} = x_r - \mathrm{H}\varphi(x_r)^{-1}(\mathrm{D}\varphi(x_r))'$. If $\varphi(x) = \frac{1}{2}x'Ax - b'x$ with A symmetric and invertible, the gradient $Ax - b$ is linear, the approximation is exact, and one step from any x_0 gives $x_1 = x_0 - A^{-1}(Ax_0 - b) = A^{-1}b$, the stationary point. For least squares with $\operatorname{rank} X = k+1$ (Theorem 16.41(c)), $x_1 = \beta_0 - (2X'X)^{-1}(2X'X\beta_0 - 2X'y) = (X'X)^{-1}X'y$: in Example 16.42, one step from $\beta_0 = 0$ gives $\hat{\beta} = (1, 2, 1)'$. For other functions several steps are needed.

For the stationary points of a C^2 scalar function φ, the step has a second reading. Let

$$q_r(y) = \varphi(x_r) + \mathrm{D}\varphi(x_r)(y - x_r) + \tfrac{1}{2}(y - x_r)'\mathrm{H}\varphi(x_r)(y - x_r) \qquad (y \in \mathbb{R}^n)$$

the second-order Taylor polynomial of φ at x_r. By Theorem 16.24 and the symmetry of $\mathrm{H}\varphi(x_r)$, the gradient of q_r is $(\mathrm{D}\varphi(x_r))' + \mathrm{H}\varphi(x_r)(y - x_r)$ and its Hessian is $\mathrm{H}\varphi(x_r)$: q_r has the same value, gradient and Hessian as φ at x_r, and its gradient is 0 exactly at $y = x_{r+1}$. So the Newton–Raphson step jumps to the stationary point of this quadratic approximation of φ. If $\mathrm{H}\varphi(x_r)$ is PD, x_{r+1} is the unique minimizer of q_r: with $z = y - x_r$, $q_r(y) = \varphi(x_r) + \frac{1}{2}z'\mathrm{H}\varphi(x_r)z - \big(-(\mathrm{D}\varphi(x_r))'\big)'z$, and Example 16.37(a) gives the unique minimizer $z = -\mathrm{H}\varphi(x_r)^{-1}(\mathrm{D}\varphi(x_r))' = x_{r+1} - x_r$; the constant $\varphi(x_r)$ does not change it.

Theorem 16.45 ([2, §9.5.5], [6, Exercises 13.37(a), 13.38(a)])**.** (a) If h is a one-to-one C^1 map of an open $U \subseteq \mathbb{R}^n$ onto an open W with a C^1 inverse, then $\mathrm{D}(h^{-1})(h(x))\,\mathrm{D}h(x) = I_n$; so $|\det \mathrm{D}(h^{-1})(h(x))| = 1/|\det \mathrm{D}h(x)|$. (The determinant $\det \mathrm{D}h(x)$ is the **Jacobian determinant** of h at x.)

(b) Let $A \in \mathbb{R}^{n\times n}$ be invertible and $b \in \mathbb{R}^n$. If x has the density f_x, then $y = Ax + b$ has the density

$$f_y(y) = \frac{f_x\big(A^{-1}(y - b)\big)}{|\det A|}$$

(c) Let $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{n\times n}$ be invertible. The map $X \mapsto Y = AXB$ of $\mathbb{R}^{m\times n}$ onto itself reads $\operatorname{vec} Y = (B' \otimes A)\operatorname{vec} X$, and $\det(B' \otimes A) = (\det A)^n(\det B)^m$. So if the random matrix X has the density f_X (of $\operatorname{vec} X$), then $Y = AXB$ has the density $f_Y(Y) = f_X(A^{-1}YB^{-1})\,|\det A|^{-n}|\det B|^{-m}$.

(d) On the open set of invertible $X \in \mathbb{R}^{n\times n}$, the map $X \mapsto X^{-1}$ has the Jacobian determinant

$$\det \mathrm{D}(X^{-1}) = (-1)^n(\det X)^{-2n}$$

Proof. (a) $h^{-1}(h(x)) = x$ on U; by the chain rule of calculus, $\mathrm{D}(h^{-1})(h(x))\,\mathrm{D}h(x) = I_n$, and taking determinants (Theorem 5.18(b)) gives the second statement.

(b) $h(x) = Ax + b$ is one-to-one from $\mathbb{R}^n$ onto $\mathbb{R}^n$, with $\mathrm{D}h(x) = A$ (Theorem 16.24(b)) and the C^1 inverse $h^{-1}(y) = A^{-1}(y - b)$. By the change of variables (D1), stated before Lemma 11.1, with $\psi = h^{-1}$, whose Jacobian determinant is $\det \mathrm{D}(h^{-1})(y)$, and (a), $f_y(y) = f_x(A^{-1}(y - b))/|\det A|$.

(c) By Theorem 16.7(b), $\operatorname{vec}(AXB) = (B' \otimes A)\operatorname{vec} X$. $B' \otimes A$ is invertible (Theorem 16.13(c)), and $\det(B' \otimes A) = (\det B')^m(\det A)^n = (\det A)^n(\det B)^m$ (Theorems 16.13(a) and 5.22). Apply (b) to the vector $\operatorname{vec} X$, with $(B' \otimes A)^{-1}\operatorname{vec} Y = \operatorname{vec}(A^{-1}YB^{-1})$ (Corollary 16.4(b), Theorem 16.7(b)).

(d) By Theorem 16.26(c), $\mathrm{D}(X^{-1}) = -(X^{-1})' \otimes X^{-1}$, an $n^2 \times n^2$ matrix. By Theorem 5.18(c) ($\det(cM) = c^{n^2} \det M$ for $M \in \mathbb{R}^{n^2 \times n^2}$), Theorem 16.13(a) and Theorems 5.22 and 5.18(c),

$$\det \mathrm{D}(X^{-1}) = (-1)^{n^2} \Big(\det(X^{-1})'\Big)^n (\det X^{-1})^n = (-1)^{n^2} (\det X)^{-2n} = (-1)^n (\det X)^{-2n}$$

as $n^2 - n = n(n-1)$ is even. □

Example 16.46 (Aitken's integral; [2, §9.5.6])**.** Let $A \in \mathbb{R}^{n \times n}$ be PD. Then

$$\int_{\mathbb{R}^n} e^{-\frac{1}{2} x'Ax} \, \mathrm{d}x = (2\pi)^{n/2} (\det A)^{-1/2}$$

- Let $P = A^{-1/2}$ (Theorem 8.46): P is symmetric, PD as the inverse of the PD matrix $A^{1/2}$ (Proposition 8.23(c)), $P'AP = A^{-1/2} A A^{-1/2} = I_n$, and $(\det P)^2 = \det(P^2) = \det A^{-1} = 1/\det A$ (Theorem 5.18(b),(c)); as $\det P > 0$ (Theorem 8.21(c)), $\det P = (\det A)^{-1/2}$.
- Change variables by $x = Py$ (the change of variables stated before Theorem 10.8, with $c = 0$ and $B = P$): the integral is

$$\begin{aligned} \int e^{-\frac{1}{2} y'P'APy} \det P \, \mathrm{d}y &= (\det A)^{-1/2} \int e^{-\frac{1}{2}(y_1^2 + \cdots + y_n^2)} \, \mathrm{d}y \\ &= (\det A)^{-1/2} \prod_{i=1}^{n} \int_{\mathbb{R}} e^{-y_i^2/2} \, \mathrm{d}y_i = (\det A)^{-1/2} (2\pi)^{n/2} \end{aligned}$$

(The second line uses the rule for products and $\int_{\mathbb{R}} e^{-s^2/2} \, \mathrm{d}s = \sqrt{2\pi}$, stated at the same place.) With $A = \Sigma^{-1}$ and the further change $x \mapsto x - \mu$, this is the computation in the proof of Theorem 10.8 that the normal density integrates to 1.

Example 16.47 ([21, §13], [6, Exercise 13.61])**.** Let $S \in \mathbb{R}^{p \times p}$ be PD (for instance a sample covariance matrix, Definition 9.13), and minimize

$$f(\Sigma) = \log \det \Sigma + \operatorname{tr}(\Sigma^{-1} S)$$

over the PD matrices $\Sigma \in \mathbb{R}^{p \times p}$, with the variables $\operatorname{vech} \Sigma$. The PD matrices form an open set (they are the symmetric matrices whose leading principal minors, polynomials in the entries, are positive, Theorem 8.34) and a convex set (Proposition 8.23(b)); $\det \Sigma > 0$ on it, and f is C^2 (Theorem 16.26).

(a) By Theorems 16.22 and 16.26(c),(d) and Theorem 1.11(c),

$$\mathrm{d}f = \operatorname{tr}(\Sigma^{-1} \mathrm{d}\Sigma) - \operatorname{tr}\Big(\Sigma^{-1} (\mathrm{d}\Sigma) \Sigma^{-1} S\Big) = \operatorname{tr}(M \, \mathrm{d}\Sigma), \qquad M = \Sigma^{-1} - \Sigma^{-1} S \Sigma^{-1}$$

At a stationary point $\operatorname{tr}(M \, \mathrm{d}\Sigma) = 0$ for every symmetric $\mathrm{d}\Sigma$; as M is symmetric, $\mathrm{d}\Sigma = M$ gives $\operatorname{tr}(M'M) = 0$, so $M = 0$ (Theorem 1.11(d)), that is, $\Sigma^{-1} = \Sigma^{-1} S \Sigma^{-1}$, or $\Sigma = S$. So S is the only stationary point.

(b) With $\mathrm{d}\Sigma$ fixed, $\mathrm{d}^2 f = \operatorname{tr}\Big((\mathrm{d}M) \, \mathrm{d}\Sigma\Big)$, where $\mathrm{d}M = -\Sigma^{-1}(\mathrm{d}\Sigma)\Sigma^{-1} + \Sigma^{-1}(\mathrm{d}\Sigma)\Sigma^{-1} S \Sigma^{-1} + \Sigma^{-1} S \Sigma^{-1} (\mathrm{d}\Sigma) \Sigma^{-1}$; the first term of $\mathrm{d}M$ contributes $-\operatorname{tr}\Big(\Sigma^{-1}(\mathrm{d}\Sigma)\Sigma^{-1}\mathrm{d}\Sigma\Big)$, the second differential of $\log \det \Sigma$ (Example 16.33(c)). At $\Sigma = S$, $\mathrm{d}M = S^{-1}(\mathrm{d}\Sigma)S^{-1}$, and with $W = S^{-1/2}(\mathrm{d}\Sigma)S^{-1/2}$ (Theorem 8.46), $\mathrm{d}^2 f = \operatorname{tr}(S^{-1}\mathrm{d}\Sigma \, S^{-1}\mathrm{d}\Sigma) = \operatorname{tr}(W^2) = \|W\|_F^2 > 0$ for $\mathrm{d}\Sigma \neq 0$. So the Hessian at S (with respect to $\operatorname{vech} \Sigma$) is PD (Theorem 16.32(a)), and S is a strict local minimizer (Theorem 16.36(c)).

The minimum at S is global, $f(\Sigma) > f(S) = \log \det S + p$ for every PD $\Sigma \neq S$: this is the step for Σ in the proof of Theorem 10.20, with S in place of $\hat{\Sigma}$ (that step uses only that $\hat{\Sigma}$ is PD), by Lemma 10.18, without derivatives.

In multivariate statistics this criterion is used to fit a covariance matrix: among all PD matrices Σ, the one that makes $\log \det \Sigma + \operatorname{tr}(\Sigma^{-1} S)$ smallest is the matrix S that is put in, whether it is the sample covariance matrix $\frac{1}{n-1} Y'C_n Y$ of Definition 9.13 or the maximum-likelihood estimate $\frac{1}{n} Y'C_n Y$ of Theorem 10.20.

Exercises

Exercise 16.1. Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $C = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

(a) Compute $A \otimes B$ and $B \otimes A$, and find $\operatorname{rank}(A \otimes B)$.

(b) Find A^{-1}, C^{-1} and $(A \otimes C)^{-1}$.

(c) Find $\operatorname{tr}(A \otimes C)$ and $\det(A \otimes C)$.

Solution. (a) By Definition 16.1,

$$A \otimes B = \begin{pmatrix} B & 2B \\ 0 & -B \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 & 4 & 2 & 0 \\ 1 & 0 & -1 & 2 & 0 & -2 \\ 0 & 0 & 0 & -2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix}, \qquad B \otimes A = \begin{pmatrix} 2 & 4 & 1 & 2 & 0 & 0 \\ 0 & -2 & 0 & -1 & 0 & 0 \\ 1 & 2 & 0 & 0 & -1 & -2 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

A is invertible ($\det A = -1$) and the rows of B are not proportional, so $\operatorname{rank} A = \operatorname{rank} B = 2$ and $\operatorname{rank}(A \otimes B) = 2 \cdot 2 = 4$ (Theorem 16.13(b)).

(b) $A \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, so $A^{-1} = A$; and $C \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = I_2$, so $C^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ (Theorem 2.16). By Corollary 16.4(b),

$$A \otimes C = \begin{pmatrix} 2 & 1 & 4 & 2 \\ 1 & 1 & 2 & 2 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & -1 & -1 \end{pmatrix}, \qquad (A \otimes C)^{-1} = A^{-1} \otimes C^{-1} = \begin{pmatrix} 1 & -1 & 2 & -2 \\ -1 & 2 & -2 & 4 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 \end{pmatrix}$$

(c) $\operatorname{tr}(A \otimes C) = \operatorname{tr} A \cdot \operatorname{tr} C = 0 \cdot 3 = 0$ (Corollary 16.4(c)); directly, $2+1-2-1 = 0$. By Theorem 16.13(a) with $m = n = 2$, $\det(A \otimes C) = (\det A)^2 (\det C)^2 = (-1)^2 \cdot 1^2 = 1$.

Exercise 16.2. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

(a) Find the eigenvalues of A and of B, with eigenvectors.

(b) Find the eigenvalues of $A \otimes B$ with their multiplicities, and $\operatorname{tr}(A \otimes B)$, $\det(A \otimes B)$ and $\operatorname{rank}(A \otimes B)$. Is $A \otimes B$ PSD?

(c) Check directly that $(1,1)' \otimes (0,0,1)'$ and $(1,-1)' \otimes (1,1,0)'$ are eigenvectors of $A \otimes B$.

Solution. (a) $A(1,1)' = (3,3)'$ and $A(1,-1)' = (1,-1)'$: the eigenvalues of A are 3 and 1. $B(1,1,0)' = (2,2,0)'$, $B(1,-1,0)' = 0$ and $B(0,0,1)' = (0,0,2)'$: the eigenvalues of B are 2, 0, 2. The eigenvectors found are orthogonal, so (after normalizing) they give $A = P\Lambda P'$, $B = QMQ'$ as in Theorem 8.3.

(b) By Theorem 16.14(b),(c), the eigenvalues of $A \otimes B$ are the products $3 \cdot 2, 3 \cdot 0, 3 \cdot 2, 1 \cdot 2, 1 \cdot 0, 1 \cdot 2$: 6 (twice), 2 (twice) and 0 (twice). Hence $\operatorname{tr}(A \otimes B) = 6 + 6 + 2 + 2 = 16 = \operatorname{tr} A \cdot \operatorname{tr} B = 4 \cdot 4$, $\det(A \otimes B) = 0 = (\det A)^3 (\det B)^2$, and, $A \otimes B$ being symmetric, $\operatorname{rank}(A \otimes B) = 4$, the number of nonzero eigenvalues (Theorem 8.4), which is $\operatorname{rank} A \cdot \operatorname{rank} B = 2 \cdot 2$. All eigenvalues are $\geqslant 0$, so $A \otimes B$ is PSD (Theorem 8.21(a)), but not PD.

(c)

$$A \otimes B = \begin{pmatrix} 2 & 2 & 0 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 4 & 0 & 0 & 2 \\ 1 & 1 & 0 & 2 & 2 & 0 \\ 1 & 1 & 0 & 2 & 2 & 0 \\ 0 & 0 & 2 & 0 & 0 & 4 \end{pmatrix}$$

With $u = (1,1)' \otimes (0,0,1)' = (0,0,1,0,0,1)'$, $(A \otimes B)u = (0,0,6,0,0,6)' = 6u$; with $w = (1,-1)' \otimes (1,1,0)' = (1,1,0,-1,-1,0)'$, $(A \otimes B)w = (2,2,0,-2,-2,0)' = 2w$, as Theorem 16.14(a) predicts $(3 \cdot 2$ and $1 \cdot 2)$.

Exercise 16.3. Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

(a) Compute ABC and $\operatorname{vec}(ABC)$.

(b) Compute $C' \otimes A$ and check that $(C' \otimes A) \operatorname{vec} B = \operatorname{vec}(ABC)$.

(c) Compute $\operatorname{tr}(A'B)$ as $(\operatorname{vec} A)' \operatorname{vec} B$, and check it directly.

Solution. (a) $AB = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$ and $ABC = \begin{pmatrix} 5 & 9 \\ 2 & 4 \end{pmatrix}$, so $\operatorname{vec}(ABC) = (5,2,9,4)'$ (Definition 16.6).

(b) $C' = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$, so

$$C' \otimes A = \begin{pmatrix} A & 0 \\ A & 2A \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 2 & 4 \\ 0 & 1 & 0 & 2 \end{pmatrix}, \qquad (C' \otimes A) \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+4 \\ 2 \\ 1+4+0+4 \\ 2+2 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 9 \\ 4 \end{pmatrix}$$

with $\operatorname{vec} B = (1,2,0,1)'$, as Theorem 16.7(b) says.

(c) $\operatorname{vec} A = (1,0,2,1)'$, so $(\operatorname{vec} A)' \operatorname{vec} B = 1+0+0+1 = 2$. Directly, $A'B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$, with trace 2 (Theorem 16.7(d)).

Exercise 16.4. (a) Write down K_{32} and check $K_{32} \operatorname{vec} A = \operatorname{vec} A'$ for

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

(b) Check that $K_{32} = K'_{23}$, with K_{23} of Definition 16.11.

(c) For $a = (1,2)'$ and $b = (3,0,1)'$, compute $a \otimes b$, $b \otimes a$ and $K_{32}(a \otimes b)$.

Solution. (a) By Theorem 16.12(a) ($m = 3$, $n = 2$), row $(i-1)2 + j$ of K_{32} is row $(j-1)3 + i$ of I_6: rows $1, 2, \ldots, 6$ of K_{32} are rows $1, 4, 2, 5, 3, 6$ of I_6,

$$K_{32} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

For the given A, $\operatorname{vec} A = (1,2,3,4,5,6)'$ and $K_{32} \operatorname{vec} A = (1,4,2,5,3,6)' = \operatorname{vec} A'$, as $A' = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$.

(b) The rows of K_{23} are rows $1, 3, 5, 2, 4, 6$ of I_6; transposing, the columns of K'_{23} are $e_1, e_3, e_5, e_2, e_4, e_6$, so the 1 of column j of K'_{23} stands in the row listed jth. Hence row 1 of K'_{23} has its 1 in column 1, row 2 in column 4, row 3 in column 2, row 4 in column 5, row 5 in column 3, row 6 in column 6: this is K_{32} (Theorem 16.12(b)).

(c) $a \otimes b = (3,0,1,6,0,2)'$ and $b \otimes a = (3,6,0,0,1,2)'$ (Definition 16.1). Taking entries $1, 4, 2, 5, 3, 6$ of $a \otimes b$, $K_{32}(a \otimes b) = (3,6,0,0,1,2)' = b \otimes a$ (Theorem 16.12(d)).

Exercise 16.5. Three groups of two observations each give $y = (5, 7,\ 2, 4,\ 6, 6)'$. In the one-way model without intercept, $y = X_1\beta + \varepsilon$ with $X_1 = I_3 \otimes \mathbf{1}_2$ (Example 16.5(a)).

(a) Compute $X_1'X_1$, $X_1'y$ and the least-squares estimate $\hat{\beta}$.

(b) Compute the hat matrix $H = I_3 \otimes \bar{J}_2$, the fitted values Hy, the residuals $e = (I_3 \otimes C_2)y$ and $\mathrm{SSE} = e'e$.

Solution. (a) By Example 16.5(a), $X_1'X_1 = 2I_3 = \mathrm{diag}(2, 2, 2)$, and $X_1'y = (I_3 \otimes \mathbf{1}_2')y$ holds the three group totals, $(12, 6, 12)'$. So $\hat{\beta} = (X_1'X_1)^{-1}X_1'y = \frac{1}{2}(12, 6, 12)' = (6, 3, 6)'$ (Theorem 13.3(c)): the group means.

(b)

$$H = I_3 \otimes \bar{J}_2 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}, \qquad Hy = \begin{pmatrix} 6 \\ 6 \\ 3 \\ 3 \\ 6 \\ 6 \end{pmatrix}, \qquad e = y - Hy = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

and $(I_3 \otimes C_2)y$ gives the same e (deviations from the group means). Hence $\mathrm{SSE} = 1 + 1 + 1 + 1 = 4$.

Exercise 16.6. Three units are measured on two variables; the rows of the random data matrix $Y \in \mathbb{R}^{3\times 2}$ are uncorrelated and have the same mean vector, each with covariance matrix $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$.

(a) Write down $\mathrm{var}(\mathrm{vec}\, Y)$ and $\mathrm{var}(\mathrm{vec}\, Y')$.

(b) Find $\mathrm{var}(\bar{y})$ and the variance of the difference $\bar{y}_1 - \bar{y}_2$ of the two variable means.

(c) Find $\mathrm{var}(y_{11} + y_{21})$ and $\mathrm{var}(y_{11} + y_{12})$.

Solution. (a) By Theorem 16.10,

$$\mathrm{var}(\mathrm{vec}\, Y) = \Sigma \otimes I_3 = \begin{pmatrix} 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 3 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & 0 & 3 \end{pmatrix}, \qquad \mathrm{var}(\mathrm{vec}\, Y') = I_3 \otimes \Sigma = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{pmatrix}$$

(b) $\mathrm{var}(\bar{y}) = \frac{1}{3}\Sigma = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 1 \end{pmatrix}$ (Theorem 9.16(a), with the rows of Y as x_1, x_2, x_3). With $c = (1, -1)'$, $\mathrm{var}(c'\bar{y}) = c'\,\mathrm{var}(\bar{y})c = \frac{1}{3}(2 - 1 - 1 + 3) = 1$ (Theorem 9.3(b)).

(c) y_{11} and y_{21} are entries 1 and 2 of $\mathrm{vec}\, Y$; with $d = (1, 1, 0, 0, 0, 0)'$, $\mathrm{var}(d'\,\mathrm{vec}\, Y) = d'(\Sigma \otimes I_3)d = 2 + 2 = 4$: different units are uncorrelated. y_{11} and y_{12} are entries 1 and 2 of $\mathrm{vec}\, Y'$, so $\mathrm{var}(y_{11} + y_{12}) = (1, 1)\Sigma(1, 1)' = 2 + 1 + 1 + 3 = 7$.

Exercise 16.7. Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 3}, \qquad B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \\ 1 & -1 \end{pmatrix} \in \mathbb{R}^{3\times 2}$$

(a) Write out $A \otimes B \in \mathbb{R}^{6\times 6}$ and find its rank.

(b) Compute AB and BA, show that $(A \otimes B)(B \otimes A) = AB \otimes BA$, and find the eigenvalues and the trace of this 6×6 matrix.

(c) Write the equation $AXB = C$ ($X \in \mathbb{R}^{3\times 3}$, $C \in \mathbb{R}^{2\times 2}$) as a linear system $M \operatorname{vec} X = \operatorname{vec} C$ with $M \in \mathbb{R}^{4\times 9}$. Show that $\operatorname{rank} M = 4$ and $MM' = B'B \otimes AA'$.

(d) Show that the solution of $AXB = C$ of least Frobenius norm is $X^* = A'(AA')^{-1}C(B'B)^{-1}B'$. Find X^* for $C = \operatorname{diag}(18, 6)$.

Solution. (a) By Definition 16.1, the blocks of $A \otimes B$ are B, B, B in the first block row and $B, 0, -B$ in the second:

$$A \otimes B = \begin{pmatrix} 2 & 1 & 2 & 1 & 2 & 1 \\ -1 & 1 & -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 2 & 1 & 0 & 0 & -2 & -1 \\ -1 & 1 & 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 & -1 & 1 \end{pmatrix}$$

The two rows of A are not multiples of each other, nor are the two columns of B, so $\operatorname{rank} A = \operatorname{rank} B = 2$, and $\operatorname{rank}(A \otimes B) = 2 \cdot 2 = 4$ by Theorem 16.13(b). Indeed rows 3 and 6 are the negatives of rows 2 and 5.

(b) Multiplying,

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad BA = \begin{pmatrix} 3 & 2 & 1 \\ 0 & -1 & -2 \\ 0 & 1 & 2 \end{pmatrix}$$

By the mixed product (Theorem 16.3(f)), with the 2×3 and 3×2 factors A, B and the 3×2 and 2×3 factors B, A, $(A \otimes B)(B \otimes A) = AB \otimes BA$.

- $AB(1, 1)' = 3(1, 1)'$ and $AB(1, -1)' = (1, -1)'$: the eigenvalues of AB are 3 and 1, and $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ gives $P^{-1}(AB)P = \operatorname{diag}(3, 1)$ (Definition 7.23).
- Expanding along the first column (Theorem 5.21), $p_{BA}(t) = \det(tI_3 - BA) = (t-3)\Big((t+1)(t-2)+2\Big) = (t-3)(t^2 - t) = t(t-1)(t-3)$. The three eigenvalues 3, 1, 0 are distinct, so eigenvectors belonging to them form a basis of $\mathbb{R}^3$ (Theorem 7.20); the matrix Q with these columns is invertible and $Q^{-1}(BA)Q = \operatorname{diag}(3, 1, 0)$ (Definition 7.23).
- By Theorem 16.14(b), the eigenvalues of $AB \otimes BA$, with multiplicity, are the products $3 \cdot 3, 3 \cdot 1, 3 \cdot 0, 1 \cdot 3, 1 \cdot 1, 1 \cdot 0$, that is, $9, 3, 3, 1, 0, 0$; and the trace is $16 = \operatorname{tr}(AB)\operatorname{tr}(BA) = 4 \cdot 4$ (Corollary 16.4(c)).

(c) By Theorem 16.7(b), $\operatorname{vec}(AXB) = (B' \otimes A)\operatorname{vec} X$, so $M = B' \otimes A \in \mathbb{R}^{4\times 9}$, and $\operatorname{rank} M = \operatorname{rank} B' \cdot \operatorname{rank} A = 2 \cdot 2 = 4$ (Theorems 16.13(b) and 4.13(a)). By Theorem 16.3(e),(f), $MM' = (B' \otimes A)(B \otimes A') = B'B \otimes AA'$. The rows of A and the columns of B are orthogonal, so $AA' = \operatorname{diag}(3, 2)$, $B'B = \operatorname{diag}(6, 3)$ and $MM' = \operatorname{diag}(18, 12, 9, 6)$ (Corollary 16.4(a)).

(d)

- $\|X\|_F = \|\operatorname{vec} X\|$ (Definition 16.6). M has rank 4, its number of rows, so $\mathcal{C}(M) = \mathbb{R}^4$ (Theorem 4.13(c)): the system $M \operatorname{vec} X = \operatorname{vec} C$ has solutions, and they are the minimizers of $\|M \operatorname{vec} X - \operatorname{vec} C\|$ (minimum 0). By Theorem 12.28(b), the shortest of them is $M^+ \operatorname{vec} C$, and $M^+ = M'(MM')^{-1}$ (Theorem 12.24(c), $\operatorname{rank} M = 4$).
- By Corollary 16.4(b) and Theorem 16.7(b), as $(B'B)^{-1}$ is symmetric,

$$(MM')^{-1} \operatorname{vec} C = \Big((B'B)^{-1} \otimes (AA')^{-1}\Big) \operatorname{vec} C = \operatorname{vec}\Big((AA')^{-1}C(B'B)^{-1}\Big)$$

  and $M' = B \otimes A'$ (Theorem 16.3(e)) maps $\operatorname{vec} Y$ to $\operatorname{vec}(A'YB')$ (Theorem 16.7(b)). Hence $X^* = A'(AA')^{-1}C(B'B)^{-1}B'$.
- For $C = \operatorname{diag}(18, 6)$, $(AA')^{-1}C(B'B)^{-1} = \operatorname{diag}(\tfrac{1}{3}, \tfrac{1}{2})\operatorname{diag}(18, 6)\operatorname{diag}(\tfrac{1}{6}, \tfrac{1}{3}) = I_2$, and

$$X^* = A'B' = (BA)' = \begin{pmatrix} 3 & 0 & 0 \\ 2 & -1 & 1 \\ 1 & -2 & 2 \end{pmatrix}, \qquad \|X^*\|_F^2 = 24$$

- Check: $AX^*B = (AA')(B'B) = \operatorname{diag}(18, 6)$. Other solutions are longer: $A(1,-2,1)' = 0$, so $X = X^* + (1,-2,1)'e_1'$ also solves $AXB = C$, and

$$X = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -1 & 1 \\ 2 & -2 & 2 \end{pmatrix}, \qquad \|X\|_F^2 = 16 + 1 + 1 + 4 + 4 + 4 = 30 > 24$$

Exercise 16.8. Let $f(x) = x'Ax + b'x$ for $x \in \mathbb{R}^4$, with

$$A = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} -2 \\ -4 \\ -4 \\ 2 \end{pmatrix}$$

(a) Find $\mathrm{d}f$, the gradient and the Hessian of f.

(b) Find the stationary point by Gauss elimination.

(c) Show that it is the strict global minimizer, and find the minimum value.

Solution. (a) By Theorem 16.24, $\mathrm{d}f = \big(x'(A + A') + b'\big)\mathrm{d}x$; the gradient is $(A + A')x + b$, and by Example 16.33(a) the Hessian is

$$A + A' = \begin{pmatrix} 2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 \\ 0 & 2 & 6 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix}$$

(b) Solve $(A + A')x = -b = (2, 4, 4, -2)'$:

$$\left(\begin{array}{cccc|c} 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 0 & 4 \\ 0 & 2 & 6 & 2 & 4 \\ 0 & 0 & 2 & 4 & -2 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 2 & 2 & 0 & 0 & 2 \\ 0 & 2 & 2 & 0 & 2 \\ 0 & 2 & 6 & 2 & 4 \\ 0 & 0 & 2 & 4 & -2 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 2 & 2 & 0 & 0 & 2 \\ 0 & 2 & 2 & 0 & 2 \\ 0 & 0 & 4 & 2 & 2 \\ 0 & 0 & 2 & 4 & -2 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 2 & 2 & 0 & 0 & 2 \\ 0 & 2 & 2 & 0 & 2 \\ 0 & 0 & 4 & 2 & 2 \\ 0 & 0 & 2 & 4 & -2 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 - \frac{1}{2}R_3} \left(\begin{array}{cccc|c} 2 & 2 & 0 & 0 & 2 \\ 0 & 2 & 2 & 0 & 2 \\ 0 & 0 & 4 & 2 & 2 \\ 0 & 0 & 0 & 3 & -3 \end{array}\right)$$

so $x_4 = -1$; $4x_3 - 2 = 2$ gives $x_3 = 1$; $2x_2 + 2 = 2$ gives $x_2 = 0$; and $2x_1 = 2$ gives $x_1 = 1$: the stationary point is $\hat{x} = (1, 0, 1, -1)'$.

(c) The operations add multiples of a row to a later row, so the leading principal minors of $A + A'$ are the products of the pivots: 2, $2 \cdot 2 = 4$, $2 \cdot 2 \cdot 4 = 16$ and $\det(A + A') = 2 \cdot 2 \cdot 4 \cdot 3 = 48$ (Theorem 5.15), all positive. So the Hessian is PD at every x (Theorem 8.34), and $\hat{x}$ is the strict global minimizer (Theorem 16.36(d)). As $\hat{x}'(A+A')\hat{x} = -b'\hat{x}$, $f(\hat{x}) = \frac{1}{2}\hat{x}'(A+A')\hat{x} + b'\hat{x} = \frac{1}{2}b'\hat{x} = \frac{1}{2}(-2+0-4-2) = -4$.

Exercise 16.9. Let $A = J_4$ and

$$X_0 = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

(a) Find $\partial \det X/\partial X$ and $\partial \log\det X/\partial X$ at $X = X_0$ (X an unrestricted 4×4 variable).

(b) Write $\mathrm{D}(X^{-1})$ at X_0 as a Kronecker product, and find the differential of X^{-1} at X_0 for the increment $\mathrm{d}X = E_{41}$.

(c) Find $\partial \operatorname{tr}(AX^{-1})/\partial X$ at X_0.

(d) Now let X be a symmetric variable. Find $\partial \log\det X/\partial X$ at X_0 in the sense of Theorem 16.30.

Solution. X_0 is the matrix T of Example 16.27, the inverse of the matrix $M = \big(\min(i,j)\big)$ there, and $\det M = 1$. So X_0 is invertible, $\det X_0 = 1/\det M = 1$ (Theorem 5.18(c)), and

$$X_0^{-1} = M = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

which is symmetric (check: $X_0 M = I_4$).

(a) By Theorem 16.26(b),(d), $\partial \det X/\partial X = \det X_0\,(X_0^{-1})' = M$, and $\partial \log\det X/\partial X = (X_0^{-1})' = M$ is the same matrix, as $\det X_0 = 1 > 0$.

(b) By Theorem 16.26(c), $\mathrm{D}(X^{-1}) = -(X_0^{-1})' \otimes X_0^{-1} = -M \otimes M$, a 16×16 matrix, and $\mathrm{d}X^{-1} = -X_0^{-1}(\mathrm{d}X)X_0^{-1}$. With $\mathrm{d}X = E_{41} = e_4 e_1'$ (Proposition 1.18), $Me_4 = (1,2,3,4)'$ is column 4 of M and $e_1'M = (1,1,1,1)$ is row 1, so

$$\mathrm{d}X^{-1} = -(Me_4)(e_1'M) = -\begin{pmatrix}1\\2\\3\\4\end{pmatrix}\begin{pmatrix}1 & 1 & 1 & 1\end{pmatrix} = -\begin{pmatrix}1&1&1&1\\2&2&2&2\\3&3&3&3\\4&4&4&4\end{pmatrix}$$

(c) By Example 16.28(b), $\partial \operatorname{tr}(AX^{-1})/\partial X = -(X_0^{-1}AX_0^{-1})'$. Here $A = J_4 = \mathbf{1}_4\mathbf{1}_4'$, so $X_0^{-1}AX_0^{-1} = (M\mathbf{1}_4)(\mathbf{1}_4'M) = vv'$ with $v = M\mathbf{1}_4 = (4,7,9,10)'$, the row sums of M ($\mathbf{1}_4'M = v'$, as M is symmetric). The derivative is

$$-vv' = -\begin{pmatrix}16&28&36&40\\28&49&63&70\\36&63&81&90\\40&70&90&100\end{pmatrix}$$

(d) The leading principal minors of X_0 are 2, $4-1=3$, $2(4-1)+(-1)\cdot 2 = 4$ (along row 1, Theorem 5.21) and $\det X_0 = 1$, all positive, so X_0 is PD (Theorem 8.34). By Theorem 16.30(b),

$$\frac{\partial \log\det X}{\partial X} = 2X_0^{-1} - \operatorname{diag}(X_0^{-1}) = 2M - \operatorname{diag}(1,2,3,4) = \begin{pmatrix}1&2&2&2\\2&2&4&4\\2&4&3&6\\2&4&6&4\end{pmatrix}$$

the off-diagonal entries double, since $x_{ij} = x_{ji}$ is one variable.

Exercise 16.10. Let

$$X_0 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$$

(a) Show that

$$X_0^{-1} = \begin{pmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and find $\det X_0$, $\partial \det X/\partial X$ and $\partial \log\det X/\partial X$ at X_0.

(b) Find the differential of X^{-1} at X_0 for the increment $\mathrm{d}X = E_{41}$, and show that it equals vv'.

Solution. (a) Multiplying, X_0 times the given matrix is I_4, so it is X_0^{-1} (Theorem 2.16). X_0 is upper triangular with diagonal entries 1, so $\det X_0 = 1$ (Theorem 5.14(c)). By Theorem 16.26(b),(d),

$$\frac{\partial \det X}{\partial X}(X_0) = \det X_0\,(X_0^{-1})' = \frac{\partial \log\det X}{\partial X}(X_0) = (X_0^{-1})' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

(b) By Theorem 16.26(c), $\mathrm{d}(X^{-1}) = -X_0^{-1}e_4e_1'X_0^{-1}$ ($E_{41} = e_4e_1'$, Proposition 1.18). Here $X_0^{-1}e_4 = (-1,1,-1,1)' = -v$ (the fourth column) and $e_1'X_0^{-1} = v'$ (the first row), so

$$\mathrm{d}(X^{-1}) = vv' = \begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}$$

Exercise 16.11. Let $f(x) = x_1^3 + x_2^3 - 3x_1x_2 + x_3^2 + x_3x_4 + x_4^2 - 3x_3$ for $x \in \mathbb{R}^4$.

(a) Find the gradient, the Hessian and the stationary points of f.

(b) Classify each stationary point by Theorem 16.36(c).

(c) Is the local minimum a global minimum?

Solution. (a) The gradient is $(3x_1^2 - 3x_2,\ 3x_2^2 - 3x_1,\ 2x_3 + x_4 - 3,\ x_3 + 2x_4)'$ and

$$\mathrm{H}f(x) = \begin{pmatrix} 6x_1 & -3 & 0 & 0 \\ -3 & 6x_2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

The gradient is 0 when $x_2 = x_1^2$ and $x_1 = x_2^2 = x_1^4$, that is, $x_1(x_1^3 - 1) = 0$: $x_1 = 0$ or $x_1 = 1$; and when $x_3 = -2x_4$ and $-4x_4 + x_4 = 3$, that is, $x_4 = -1$, $x_3 = 2$. The stationary points are $(0,0,2,-1)'$ and $(1,1,2,-1)'$.

(b) The lower right block $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ of $\mathrm{H}f$ has the eigenvalues 3 and 1, with the eigenvectors $(0,0,1,1)'$ and $(0,0,1,-1)'$ of $\mathrm{H}f$.

- At $(0,0,2,-1)'$, the upper left block is $\left(\begin{smallmatrix} 0 & -3 \\ -3 & 0 \end{smallmatrix}\right)$, with the eigenvalues 3 and -3 (eigenvectors $(1,-1,0,0)'$, $(1,1,0,0)'$). So $\mathrm{H}f(0,0,2,-1)$ has the eigenvalues 3, -3, 3, 1: it is indefinite, and $(0,0,2,-1)'$ is neither a local minimum nor a local maximum.

- At $(1,1,2,-1)'$, the upper left block is $\left(\begin{smallmatrix}6 & -3\\ -3 & 6\end{smallmatrix}\right)$, with the eigenvalues 9 and 3 (eigenvectors $(1,-1,0,0)'$, $(1,1,0,0)'$). So $\mathrm{H}f(1,1,2,-1)$ has the eigenvalues 9, 3, 3, 1: it is PD (Theorem 8.21(a)), and f has a strict local minimum at $(1,1,2,-1)'$, with $f(1,1,2,-1) = 1+1-3+4-2+1-6 = -4$.

(c) No: $f(x_1,0,2,-1) = x_1^3 + 4 - 2 + 1 - 6 = x_1^3 - 3$ is less than -4 for $x_1 < -1$. The Hessian is not PSD on all of $\mathbb{R}^4$ (for instance at $(0,0,2,-1)'$), so Theorem 16.36(d) does not apply.

Exercise 16.12. Let $M = 4I_4 + 3J_4$ and $f(x) = \frac{1}{4}(x'x)^2 - \frac{1}{2}x'Mx$ on $\mathbb{R}^4$.

(a) Find the gradient and the Hessian of f, and the eigenvalues and eigenvectors of M.

(b) Find all stationary points of f, and classify them by the eigenvalues of the Hessian. Treat $x = (1,1,-1,-1)'$ explicitly.

(c) Find the global minimum of f.

Solution. (a)

- By Theorems 16.24(c) and 16.22, $\mathrm{d}(x'x) = 2x'\mathrm{d}x$ and $\mathrm{d}f = \frac{1}{2}(x'x)\,2x'\mathrm{d}x - x'M\,\mathrm{d}x = \big((x'x)x - Mx\big)'\mathrm{d}x$: the gradient is $g(x) = (x'x)x - Mx$.
- $\mathrm{d}g = \big(\mathrm{d}(x'x)\big)x + (x'x)\,\mathrm{d}x - M\,\mathrm{d}x = \big(2xx' + (x'x)I_4 - M\big)\mathrm{d}x$, so by Definition 16.31, $\mathrm{H}f(x) = \mathrm{D}g(x) = (x'x)I_4 + 2xx' - M$.
- $J_4\mathbf{1}_4 = 4\mathbf{1}_4$ and $J_4 v = \mathbf{1}_4(\mathbf{1}_4'v) = 0$ for $v \perp \mathbf{1}_4$, so $M\mathbf{1}_4 = 16\mathbf{1}_4$ and $Mv = 4v$ for every $v \neq 0$ in the 3-dimensional subspace $\mathbf{1}_4^{\perp}$: the eigenvalues of M are 16 and 4 (three times).

(b) $g(x) = 0$ means $Mx = (x'x)x$: $x = 0$, or x is an eigenvector of M with $x'x$ equal to its eigenvalue. The stationary points are 0, $\pm 2\mathbf{1}_4$ ($x'x = 16$), and every $x \perp \mathbf{1}_4$ with $x'x = 4$, infinitely many.

- At $\pm 2\mathbf{1}_4$, $\mathrm{H}f = 16I_4 + 8J_4 - 4I_4 - 3J_4 = 12I_4 + 5J_4$, with the eigenvalues 32 (on $\mathbf{1}_4$) and 12 (three times): a strict local minimum (Theorem 16.36(c)), with $f = 64 - 128 = -64$.
- At $x \perp \mathbf{1}_4$ with $x'x = 4$, $\mathrm{H}f = 2xx' - 3J_4$. Then $\mathrm{H}f\,x = 8x$, $\mathrm{H}f\,\mathbf{1}_4 = -12\mathbf{1}_4$, and $\mathrm{H}f\,v = 0$ for $v \perp x, \mathbf{1}_4$: the eigenvalues are $8, -12, 0, 0$. As $8 > 0 > -12$, $\mathrm{H}f$ is indefinite, and x is neither a local minimum nor a local maximum (Theorem 16.36(c)), with $f = 4 - 8 = -4$. For $x = (1,1,-1,-1)'$,

$$\mathrm{H}f(x) = \begin{pmatrix} -1 & -1 & -5 & -5 \\ -1 & -1 & -5 & -5 \\ -5 & -5 & -1 & -1 \\ -5 & -5 & -1 & -1 \end{pmatrix}$$

  and $\mathrm{H}f(x)(1,-1,0,0)' = \mathrm{H}f(x)(0,0,1,-1)' = 0$, $\mathrm{H}f(x)\,x = 8x$, $\mathrm{H}f(x)\mathbf{1}_4 = -12\mathbf{1}_4$.
- At 0, $\mathrm{H}f = -M$ has the eigenvalues $-16, -4, -4, -4$: it is ND, a strict local maximum, $f(0) = 0$.

(c) By Theorem 9.23(a) for M, whose largest eigenvalue is 16, $x'Mx \leqslant 16s$ with $s = x'x$, so $f(x) \geqslant \frac{1}{4}s^2 - 8s = \frac{1}{4}(s-16)^2 - 64 \geqslant -64$. The global minimum is -64, attained at $\pm 2\mathbf{1}_4$.

Exercise 16.13. Four unbiased estimators $t = (t_1, t_2, t_3, t_4)'$ of a number θ have the covariance matrix

$$\Sigma = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

A combination $a't$ is unbiased when $\mathbf{1}'a = 1$; we seek the one of least variance (as in a minimum-variance portfolio).

(a) Write down the Lagrangian for minimizing $\operatorname{var}(a't) = a'\Sigma a$ subject to $\mathbf{1}'a = 1$, and the conditions $\mathrm{D}\psi = 0$.

(b) Solve them: find $w = \Sigma^{-1}\mathbf{1}$ by Gauss elimination, then a and the multiplier.

(c) Show that this a gives the smallest variance, and compare with the equal weights $\frac{1}{4}\mathbf{1}$.

Solution. (a) $\psi(a) = a'\Sigma a - 2\lambda(\mathbf{1}'a - 1)$ (multiplier 2λ); by Theorem 16.24, $\mathrm{d}\psi = 2(\Sigma a - \lambda\mathbf{1})'\mathrm{d}a$, so the conditions are $\Sigma a = \lambda\mathbf{1}$ and $\mathbf{1}'a = 1$.

(b) Solve $\Sigma w = \mathbf{1}$:

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ -1 & 4 & -1 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 + R_1} \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ 0 & 3 & -1 & 0 & 2 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 3 & -1 & 0 & 2 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 3 & -1 & 0 & 2 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 + 3R_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & 5 & -3 & 5 \\ 0 & 0 & -1 & 1 & 1 \end{array}\right) \xrightarrow{R_3 \leftrightarrow R_4} \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 5 & -3 & 5 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 5 & -3 & 5 \end{array}\right) \xrightarrow{R_4 \leftarrow R_4 + 5R_3} \left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 1 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 2 & 10 \end{array}\right)$$

so $w_4 = 5$; $-w_3 + 5 = 1$ gives $w_3 = 4$; $-w_2 + 8 - 5 = 1$ gives $w_2 = 2$; and $w_1 - 2 = 1$ gives $w_1 = 3$: $w = (3, 2, 4, 5)'$. Then $a = \lambda w$ with $1 = \mathbf{1}'a = 14\lambda$, so $\lambda = \frac{1}{14}$ and $a = \frac{1}{14}(3, 2, 4, 5)'$. Check: $\Sigma w = (3 - 2,\ -3 + 8 - 4,\ -2 + 8 - 5,\ -4 + 5)' = \mathbf{1}$, so $\Sigma a = \frac{1}{14}\mathbf{1}$.

(c) The leading principal minors of Σ are 1, $4 - 1 = 3$, $1 \cdot (8 - 1) + (-1) \cdot 2 = 5$ (along row 1, Theorem 5.21) and $\det\Sigma = (-1)^2 \cdot 1 \cdot (-1) \cdot (-1) \cdot 2 = 2$ (Theorem 5.15, two interchanges), all positive, so Σ is PD (Theorem 8.34), and $\mathrm{H}\psi = 2\Sigma$ is PD everywhere: by Theorem 16.36(e), a is the strict minimizer under the constraint. Its variance is $a'\Sigma a = \lambda\, a'\mathbf{1} = \frac{1}{14}$. The equal weights give $\frac{1}{16}\mathbf{1}'\Sigma\mathbf{1} = \frac{1}{16}(0 + 2 + 0 + 0) = \frac{1}{8} > \frac{1}{14}$, where $0, 2, 0, 0$ are the row sums of Σ.

Exercise 16.14. Let $a = (3, 1, 4, 1)'$ and $L = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 1,\ x_1 + x_2 - x_3 - x_4 = 3\}$.

(a) Find the point x^* of L closest to a, and the distance from a to L, by Theorem 16.39 with $A = I_4$ and $b = a$.

(b) Check Theorem 16.39(b) for the point $x = (2, 0, 0, -1)'$ of L.

Solution. (a) With $A = I_4$ and $b = a$, $q(x) = \frac{1}{2}x'x - a'x = \frac{1}{2}\|x - a\|^2 - \frac{1}{2}\|a\|^2$, so minimizing q on L is minimizing the distance to a, and $\hat{x} = a$. Here

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \end{pmatrix}, \qquad d = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad V = CC' = 4I_2, \qquad Ca - d = \begin{pmatrix} 9 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \end{pmatrix}$$

$\operatorname{rank} C = 2$ (the rows are not multiples of each other), and $\lambda = V^{-1}(Ca - d) = (2, -1)'$. By Theorem 16.39(a),

$$x^* = a - C'\lambda = \begin{pmatrix} 3 \\ 1 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \\ -2 \end{pmatrix}$$

which satisfies $2 + 0 + 1 - 2 = 1$ and $2 + 0 - 1 + 2 = 3$. By Theorem 16.39(c), $\frac{1}{2}\|x^* - a\|^2 = q(x^*) - q(a) = \frac{1}{2}(Ca - d)'V^{-1}(Ca - d) = \frac{1}{2} \cdot \frac{64 + 16}{4} = 10$; so the distance is $\sqrt{20} = 2\sqrt{5}$, and directly $\|a - x^*\|^2 = 1 + 1 + 9 + 9 = 20$.

(b) $x = (2, 0, 0, -1)'$ lies in L ($2 + 0 + 0 - 1 = 1$, $2 + 0 - 0 + 1 = 3$). Here $q(x) = \frac{1}{2} \cdot 5 - (6 - 1) = -\frac{5}{2}$, $q(x^*) = \frac{1}{2} \cdot 9 - (6 + 4 - 2) = -\frac{7}{2}$, and $\frac{1}{2}\|x - x^*\|^2 = \frac{1}{2}\|(0, 0, -1, 1)'\|^2 = 1$; indeed $-\frac{5}{2} = -\frac{7}{2} + 1$.

Exercise 16.15. Five units have the values $x_{i1} = 2, 2, 3, 3, 5$ and $x_{i2} = 1, 5, 2, 2, 0$ of two explanatory variables and the responses $y = (4, 8, 5, 7, 6)'$. In the model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ $(k = 2)$:

(a) Write down X, $X'X$, $X'y$ and the gradient of $S(\beta)$ at $\beta = 0$.

(b) Solve the normal equations by Gauss elimination.

(c) Show that the solution minimizes S, and find the residuals and $S(\hat{\beta})$.

Solution. (a)

$$X = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 5 \\ 1 & 3 & 2 \\ 1 & 3 & 2 \\ 1 & 5 & 0 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 5 & 15 & 10 \\ 15 & 51 & 24 \\ 10 & 24 & 34 \end{pmatrix}, \qquad X'y = \begin{pmatrix} 30 \\ 90 \\ 68 \end{pmatrix}$$

By Theorem 16.41(a), the gradient at $\beta = 0$ is $-2X'y = (-60, -180, -136)'$.

(b)

$$\left(\begin{array}{ccc|c} 5 & 15 & 10 & 30 \\ 15 & 51 & 24 & 90 \\ 10 & 24 & 34 & 68 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 3R_1} \left(\begin{array}{ccc|c} 5 & 15 & 10 & 30 \\ 0 & 6 & -6 & 0 \\ 0 & -6 & 14 & 8 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|c} 5 & 15 & 10 & 30 \\ 0 & 6 & -6 & 0 \\ 0 & 0 & 8 & 8 \end{array}\right)$$

so $\beta_2 = 1$, $6\beta_1 - 6 = 0$ gives $\beta_1 = 1$, and $5\beta_0 + 15 + 10 = 30$ gives $\beta_0 = 1$: $\hat{\beta} = (1, 1, 1)'$.

(c) The pivots 5, 6, 8 are positive, so the leading principal minors 5, 30, 240 of $X'X$ are positive and $\mathrm{H}S = 2X'X$ is PD (Theorem 8.34): $\hat{\beta}$ is the strict global minimizer (Theorem 16.41(c)). $X\hat{\beta} = (4, 8, 6, 6, 6)'$, so $e = y - X\hat{\beta} = (0, 0, -1, 1, 0)'$ and $S(\hat{\beta}) = e'e = 2$; indeed $X'e = (0,\ 0,\ 0)'$.

Exercise 16.16. Let $f(x) = \begin{pmatrix} x_1^2 - x_1 x_2 - 2x_2 + 2 \\ 2x_1 + x_2^2 - 3 \end{pmatrix}$, which is 0 at $x = (1, 1)'$.

(a) Find the derivative $G(x) = \mathrm{D}f(x)$, and check that $G(1, 1)$ is invertible.

(b) Starting from $x_0 = (-1, -1)'$, compute two Newton–Raphson steps x_1 and x_2 (Example 16.44).

(c) Compare the squared distances of x_0, x_1, x_2 from the solution $(1, 1)'$.

Solution. (a) $G(x) = \begin{pmatrix} 2x_1 - x_2 & -x_1 - 2 \\ 2 & 2x_2 \end{pmatrix}$, and $G(1, 1) = \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix}$ has determinant $8 \neq 0$.

(b) At $x_0 = (-1, -1)'$: $f(x_0) = (4, -4)'$ and $G(x_0) = \begin{pmatrix} -1 & -1 \\ 2 & -2 \end{pmatrix}$, with determinant 4 and $G(x_0)^{-1} = \frac{1}{4}\begin{pmatrix} -2 & 1 \\ -2 & -1 \end{pmatrix}$ (Theorem 5.24). So $x_1 = x_0 - G(x_0)^{-1} f(x_0) = (-1, -1)' - \frac{1}{4}(-12, -4)' = (2, 0)'$. At x_1: $f(x_1) = (6, 1)'$ and $G(x_1) = \begin{pmatrix} 4 & -4 \\ 2 & 0 \end{pmatrix}$, with determinant 8 and $G(x_1)^{-1} = \frac{1}{8}\begin{pmatrix} 0 & 4 \\ -2 & 4 \end{pmatrix}$. So $x_2 = (2, 0)' - \frac{1}{8}(4, -8)' = (\frac{3}{2}, 1)'$.

(c) The squared distances from $(1, 1)'$ are $\|(-2, -2)'\|^2 = 8$, $\|(1, -1)'\|^2 = 2$ and $\|(\frac{1}{2}, 0)'\|^2 = \frac{1}{4}$. A third step gives $x_3 = (\frac{23}{22}, \frac{21}{22})'$, at squared distance $\|(\frac{1}{22}, -\frac{1}{22})'\|^2 = \frac{2}{484} = \frac{1}{242}$: near the solution each step roughly squares the error.

Exercise 16.17. Let $A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, and consider $Y = AXB$ for $X \in \mathbb{R}^{2\times 2}$.

(a) Write down the matrix J with $\operatorname{vec} Y = J \operatorname{vec} X$, and find $\det J$ in two ways. Check $J \operatorname{vec} X$ for $X = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$.

(b) If the four entries of X have the joint density f_X, what is the density of Y?

(c) Compute $\int_{\mathbb{R}^4} e^{-\frac{1}{2}x'Px}\,\mathrm{d}x$ for the matrix

$$P = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ -1 & 1 & 0 & 3 \end{pmatrix}$$

of Example 8.35.

Solution. (a) By Theorem 16.45(c), $J = B' \otimes A$ with $B' = \begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix}$:

$$J = \begin{pmatrix} A & 0 \\ A & 3A \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 0 & 6 & 0 \\ 1 & 1 & 3 & 3 \end{pmatrix}$$

J is lower triangular, so $\det J = 2 \cdot 1 \cdot 6 \cdot 3 = 36$ (Theorems 5.14(c) and 5.22); and $\det J = (\det A)^2(\det B)^2 = 2^2 \cdot 3^2 = 36$ (Theorem 16.13(a)). For the given X, $AXB = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix} B = \begin{pmatrix} 2 & 2 \\ 3 & 6 \end{pmatrix}$, and $J(1, 2, 0, 1)' = (2, 3, 2, 6)' = \operatorname{vec} Y$.

(b) $f_Y(Y) = f_X(A^{-1}YB^{-1})/36$ (Theorem 16.45(c)), with $A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$ and $B^{-1} = \frac{1}{3}\begin{pmatrix} 3 & -1 \\ 0 & 1 \end{pmatrix}$.

(c) By Example 8.35, P is PD with $\det P = \Delta_4 = 16$, so by Example 16.46 with $n = 4$ the integral is $(2\pi)^{4/2} \cdot 16^{-1/2} = \frac{4\pi^2}{4} = \pi^2$.

Exercise 16.18. (a) Find the Jacobian determinant of $X \mapsto AXB$ on $\mathbb{R}^{2\times 3}$ for

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

(b) Find X_0^{-1} and the Jacobian determinant of $X \mapsto X^{-1}$ at

$$X_0 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

(c) Find the Jacobian matrix of $X \mapsto X'$ on $\mathbb{R}^{2\times 2}$ and its determinant.

Solution. (a) $\det A = 2$ and, along row 1 (Theorem 5.21), $\det B = 2(4-1) - (2-1) + (1-2) = 4$. By Theorem 16.45(c) with $m = 2$ and $n = 3$, the Jacobian matrix is $B' \otimes A \in \mathbb{R}^{6\times 6}$, with determinant $(\det A)^3(\det B)^2 = 8 \cdot 16 = 128$.

(b) Along column 1 (Theorem 5.21),

$$\det X_0 = 1 \cdot \det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} + (-1) \cdot (-1)^{4+1} \det \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = 1 + 1 = 2$$

the two minors being triangular with diagonal entries 1 (Theorems 5.14(c) and 5.22). Multiplying checks $X_0 X_0^{-1} = I_4$ for

$$X_0^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

(Theorem 2.16). By Theorem 16.45(d) with $n = 4$, the Jacobian determinant is $(-1)^4 (\det X_0)^{-8} = 2^{-8} = \frac{1}{256}$.

(c) The map is linear, with $\operatorname{vec}(X') = (x_{11}, x_{12}, x_{21}, x_{22})'$ for $\operatorname{vec} X = (x_{11}, x_{21}, x_{12}, x_{22})'$: $\operatorname{vec}(X') = K_2 \operatorname{vec} X$ (Definition 16.11), and $\mathrm{d}(X') = (\mathrm{d}X)'$ gives $\operatorname{vec} \mathrm{d}(X') = K_2 \operatorname{vec} \mathrm{d}X$. So the Jacobian matrix is (Theorem 16.21(b))

$$K_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

obtained from I_4 by interchanging rows 2 and 3, and $\det K_2 = -1$ (Theorem 5.14(d)).

Exercise 16.19. Let

$$A = \begin{pmatrix} 5 & 1 & 2 & 0 \\ 1 & 5 & 0 & 2 \\ 2 & 0 & 5 & 1 \\ 0 & 2 & 1 & 5 \end{pmatrix}$$

and $h_1 = (1, 1, 1, 1)'$, $h_2 = (1, -1, 1, -1)'$, $h_3 = (1, 1, -1, -1)'$, $h_4 = (1, -1, -1, 1)'$.

(a) Compute $Ah_1, \ldots, Ah_4$, and show that every eigenvector of A is a multiple of one of $h_1, \ldots, h_4$.

(b) Find all stationary points of $f(x) = x'Ax$ on the unit sphere of $\mathbb{R}^4$, and the maximum and the minimum of f on the sphere.

Solution. (a)

- Row by row, $Ah_1 = (8, 8, 8, 8)' = 8h_1$, $Ah_2 = (6, -6, 6, -6)' = 6h_2$, $Ah_3 = (4, 4, -4, -4)' = 4h_3$ and $Ah_4 = (2, -2, -2, 2)' = 2h_4$.
- The h_j are nonzero and pairwise orthogonal, hence linearly independent (Proposition 6.11(b)), and a basis of $\mathbb{R}^4$ (Theorem 4.5(d)). For $x = \sum_j c_j h_j$, $Ax = 8c_1h_1 + 6c_2h_2 + 4c_3h_3 + 2c_4h_4$, so $Ax = \lambda x$ means $(8 - \lambda)c_1 = (6 - \lambda)c_2 = (4 - \lambda)c_3 = (2 - \lambda)c_4 = 0$.
- If $x \neq 0$, some $c_j \neq 0$, so $\lambda \in \{8, 6, 4, 2\}$, and then the other three coefficients vanish: x is a multiple of one h_j.

(b) Each $\|h_j\| = 2$. By Proposition 16.38 and (a), the stationary points are the unit eigenvectors $\pm\frac{1}{2}h_1$ ($f = 8$), $\pm\frac{1}{2}h_2$ ($f = 6$), $\pm\frac{1}{2}h_3$ ($f = 4$) and $\pm\frac{1}{2}h_4$ ($f = 2$). On the sphere $f(x) = x'Ax/x'x$, so the maximum of f on the sphere is 8 and the minimum is 2 (Theorem 9.23(a)).

Exercise 16.20. Let

$$S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \otimes \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 2 & 3 & 1 \\ 2 & 6 & 1 & 3 \\ 3 & 1 & 6 & 2 \\ 1 & 3 & 2 & 6 \end{pmatrix}$$

and $g(X) = \log\det X - \operatorname{tr}(SX)$ for PD $X \in \mathbb{R}^{4\times 4}$.

(a) Maximize $g(cI_4)$ over $c > 0$.

(b) Find the maximizer and the maximum of g over the PD $X \in \mathbb{R}^{4\times 4}$, and compare with (a).

Solution. Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$. Their leading principal minors are $2, 3$ and $3, 8$, so A and B are PD (Theorem 8.34), and so is $S = A \otimes B$ (Theorem 16.14(c)).

(a) $\operatorname{tr} S = 24$, so $h(c) = g(cI_4) = 4\log c - 24c$ on the open convex set $(0, \infty)$, with derivative $4/c - 24$ and second derivative $-4/c^2 < 0$. By Theorem 16.36(d) applied to $-h$ ($n = 1$), the stationary point $c = \frac{1}{6}$ is the strict global maximizer, and $g(\frac{1}{6}I_4) = -4\log 6 - 4 = -2\log 36 - 4$.

(b)

- The PD matrices form an open set, and we take the variables $\operatorname{vech} X$, as in Example 16.47. By Theorems 16.22 and 16.26(a),(d), $\mathrm{d}g = \operatorname{tr}(X^{-1}\mathrm{d}X) - \operatorname{tr}(S\,\mathrm{d}X) = \operatorname{tr}\Big((X^{-1} - S)\,\mathrm{d}X\Big)$. At a stationary point this is 0 for every symmetric $\mathrm{d}X$; as $X^{-1} - S$ is symmetric, $\mathrm{d}X = X^{-1} - S$ gives $\operatorname{tr}\Big((X^{-1} - S)'(X^{-1} - S)\Big) = 0$, so $X^{-1} = S$ (Theorem 1.11(d)), as in Example 16.47(a). So the only stationary point is $X = S^{-1}$, which is PD (Proposition 8.23(c)).
- It is the global maximizer. Let $C = S^{1/2}XS^{1/2}$ (Theorem 8.46), PD (Theorem 8.24(c)). Then $\operatorname{tr} C = \operatorname{tr}(SX)$ (Theorem 1.11(b)) and $\log\det C = \log\det S + \log\det X$ (Theorem 5.18(b)), so by Lemma 10.18
$$g(X) = \log\det C - \operatorname{tr} C - \log\det S \leqslant -4 - \log\det S$$
with equality only if $C = I_4$, that is, $X = S^{-1}$.
- By Corollary 16.4(b), with $A^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ and $B^{-1} = \frac{1}{8}\left(\begin{smallmatrix} 3 & -1 \\ -1 & 3 \end{smallmatrix}\right)$,
$$S^{-1} = A^{-1} \otimes B^{-1} = \frac{1}{24}\begin{pmatrix} 6 & -2 & -3 & 1 \\ -2 & 6 & 1 & -3 \\ -3 & 1 & 6 & -2 \\ 1 & -3 & -2 & 6 \end{pmatrix}$$
and $\det S = (\det A)^2(\det B)^2 = 9\cdot 64 = 576$ (Theorem 16.13(a)). The maximum is $-\log\det S - 4 = -\log 576 - 4 = -2\log 24 - 4$, larger than the value $-2\log 36 - 4$ in (a).

Problems

Problem 16.1 ([6, Exercise 10.13])**.** Let $A \in \mathbb{R}^{m\times p}$ and $B \in \mathbb{R}^{n\times q}$. If A and B are not square, $A \otimes B$ may still be square. Show that $A \otimes B$ is singular unless both A and B are square and invertible.

Solution.

- $A \otimes B \in \mathbb{R}^{mn\times pq}$ is square when $mn = pq$. By Theorem 16.13(b) and Theorem 4.13(a), $\operatorname{rank}(A \otimes B) = \operatorname{rank} A \cdot \operatorname{rank} B \leqslant \min\{m, p\}\min\{n, q\}$.
- If $A \otimes B$ is invertible, its rank is $mn = pq$ (Theorem 4.13(d)). From $mn \leqslant \min\{m, p\}\min\{n, q\}$ we get $m \leqslant p$ and $n \leqslant q$, and from $pq \leqslant \min\{m, p\}\min\{n, q\}$ we get $p \leqslant m$ and $q \leqslant n$. So $p = m$, $q = n$: A and B are square.
- Then $\operatorname{rank} A \cdot \operatorname{rank} B = mn$ with $\operatorname{rank} A \leqslant m$ and $\operatorname{rank} B \leqslant n$ forces $\operatorname{rank} A = m$ and $\operatorname{rank} B = n$: A and B are invertible.

Problem 16.2 ([6, Exercise 10.14])**.** Let $A, B \in \mathbb{R}^{n\times n}$ be PSD. Show that $A\otimes A - B\otimes B$ is PSD if and only if $A - B$ is PSD.

Solution.

- Let $A-B$ be PSD. By Theorem 16.3(b), $A\otimes A - B\otimes B = A\otimes A - A\otimes B + A\otimes B - B\otimes B = A\otimes(A-B)+(A-B)\otimes B$. Both terms are PSD by Theorem 16.14(c), and a sum of PSD matrices is PSD (Proposition 8.23(b)).
- Conversely, let $A\otimes A - B\otimes B$ be PSD and $x\in\mathbb{R}^n$. By the mixed product, $(x\otimes x)'(A\otimes A)(x\otimes x) = (x'Ax)(x'Ax)$, so

$$0 \leqslant (x\otimes x)'(A\otimes A - B\otimes B)(x\otimes x) = (x'Ax)^2 - (x'Bx)^2 = x'(A+B)x\cdot x'(A-B)x$$

- $x'(A+B)x \geqslant 0$. If it is >0, the display gives $x'(A-B)x\geqslant 0$. If it is 0, then $x'Ax = x'Bx = 0$ (both are $\geqslant 0$), and $x'(A-B)x = 0$. So $x'(A-B)x\geqslant 0$ for every x: $A-B$ is PSD.

Problem 16.3 ([10, Problem 27.1])**.** Let $a, c\in\mathbb{R}^m$ and $b, d\in\mathbb{R}^n$ be nonzero vectors. Show that $a\otimes b = c\otimes d$ if and only if $a=\alpha c$ and $d=\alpha b$ for some number α.

Solution.

- If $a=\alpha c$ and $d=\alpha b$, then $a\otimes b = (\alpha c)\otimes b = \alpha(c\otimes b) = c\otimes(\alpha b) = c\otimes d$ (Theorem 16.3(a)).
- Conversely, let $a\otimes b = c\otimes d$. By Definition 16.1, $a\otimes b = (a_1b', \dots, a_mb')'$ and $c\otimes d = (c_1d', \dots, c_md')'$; comparing the ith blocks of n entries, $a_ib = c_id$ for $i = 1, \dots, m$.
- As $a\neq 0$, some $a_k\neq 0$. Then $c_kd = a_kb \neq 0$, as $b\neq 0$; so $c_k\neq 0$, and $d=\alpha b$ with $\alpha = a_k/c_k$.
- For every i, $a_ib = c_id = \alpha c_ib$, that is, $(a_i - \alpha c_i)b = 0$. As b has a nonzero entry, $a_i = \alpha c_i$. So $a=\alpha c$.

As $a\otimes b = \operatorname{vec}(ba')$ (Theorem 16.7(a)), this also says: for nonzero vectors, $ba' = dc'$ if and only if $a=\alpha c$ and $d=\alpha b$ for some α.

Problem 16.4 ([6, Exercise 10.21])**.** Let $n\geqslant 1$, $\alpha\in\mathbb{R}$, $v = \operatorname{vec} I_n$, and let $A\in\mathbb{R}^{n\times n}$ be PD. Consider the $n^2\times n^2$ matrices

$$B = I_n\otimes I_n - \alpha vv', \qquad C = A\otimes A - \alpha(\operatorname{vec} A)(\operatorname{vec} A)'$$

Show that:

(a) the eigenvalues of B are $1-n\alpha$ (once) and 1 (n^2-1 times);

(b) $\det B = 1 - n\alpha$;

(c) B is PD if and only if $\alpha < 1/n$, and PSD if and only if $\alpha\leqslant 1/n$;

(d) the same holds for C.

Solution. (a)

- $v'v = n$, as v has n entries 1 and the others 0. $vv'v = nv$, and $vv'x = 0$ for $x\perp v$; an orthonormal basis of $\mathbb{R}^{n^2}$ made of $v/\sqrt{n}$ and n^2-1 vectors orthogonal to v (Theorems 6.6(c) and 6.14) consists of eigenvectors of the symmetric vv', so its eigenvalues are n (once) and 0 (n^2-1 times) (Theorem 8.3(c)).
- $B = I_{n^2} - \alpha vv'$ (Corollary 16.4(a)), so its eigenvalues are $1-\alpha n$ and $1-\alpha\cdot 0 = 1$ (Proposition 7.18(c)).

(b) $\det B$ is the product of the eigenvalues (Theorem 7.19): $1 - n\alpha$.

(c) B is symmetric; it is PD (PSD) if and only if all eigenvalues are > 0 ($\geqslant 0$) (Theorem 8.21(a)), that is, $1 - n\alpha > 0$ ($\geqslant 0$).

(d)

- Let $R = A^{1/2} \otimes A^{1/2}$ (Theorem 8.46), symmetric and invertible (Corollary 16.4(b),(e)). Then $R(I_n \otimes I_n)R = A^{1/2}A^{1/2} \otimes A^{1/2}A^{1/2} = A \otimes A$ (mixed product), and $Rv = \operatorname{vec}(A^{1/2}I_nA^{1/2}) = \operatorname{vec} A$ (Theorem 16.7(b)), so $R(vv')R = (\operatorname{vec} A)(\operatorname{vec} A)'$. Hence $C = RBR = R'BR$.
- By Theorem 8.24(c), C is PD (PSD) if and only if B is; apply (c).

Problem 16.5 ([5, Ch. 16, Exercise 6], [8, Ch. 16, Exercise 6])**.** For a matrix A let $P_A = A(A'A)^-A'$, the matrix of the orthogonal projection onto $\mathcal{C}(A)$ (Theorem 12.17(b)). Let $X = A \otimes B$, where $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$. Show that $P_X = P_A \otimes P_B$.

Solution.

- By Theorem 16.3(e),(f), $X'X = (A' \otimes B')(A \otimes B) = (A'A) \otimes (B'B)$, so $(A'A)^- \otimes (B'B)^-$ is a g-inverse of $X'X$ (Corollary 16.4(d)).
- As P_X does not depend on the g-inverse (Theorem 12.17(b)), the mixed product gives

$$P_X = (A \otimes B)\big((A'A)^- \otimes (B'B)^-\big)(A' \otimes B') = \big(A(A'A)^-A'\big) \otimes \big(B(B'B)^-B'\big) = P_A \otimes P_B$$

For instance, for the balanced one-way design $X_1 = I_g \otimes \mathbf{1}_m$, $P_{X_1} = I_g \otimes \bar{J}_m$, as in Example 16.5(a).

Problem 16.6 ([6, Exercise 11.14])**.** (a) Show that $\frac{1}{2}(I_{n^2} - K_n)$ is symmetric and idempotent.

(b) Show that $\operatorname{tr}(A^2) \leqslant \operatorname{tr}(A'A)$ for every $A \in \mathbb{R}^{n\times n}$, with equality if and only if A is symmetric (**Schur's inequality**).

Solution. (a) K_n is symmetric and $K_n^2 = K_n'K_n = I_{n^2}$ (Theorem 16.12(b)), so $N = \frac{1}{2}(I - K_n)$ is symmetric and $N^2 = \frac{1}{4}(I - 2K_n + K_n^2) = \frac{1}{2}(I - K_n) = N$.

(b)

- By Theorem 16.7(d), $\operatorname{tr}(A'A) = (\operatorname{vec} A)'\operatorname{vec} A$ and $\operatorname{tr}(A^2) = \operatorname{tr}((A')'A) = (\operatorname{vec} A')'\operatorname{vec} A = (\operatorname{vec} A)'K_n'\operatorname{vec} A = (\operatorname{vec} A)'K_n \operatorname{vec} A$.
- So $\operatorname{tr}(A'A) - \operatorname{tr}(A^2) = (\operatorname{vec} A)'(I - K_n)\operatorname{vec} A = 2x'Nx$ with $x = \operatorname{vec} A$, and $x'Nx = x'N'Nx = \|Nx\|^2 \geqslant 0$.
- Equality holds if and only if $Nx = 0$, that is, $K_n \operatorname{vec} A = \operatorname{vec} A$, that is, $\operatorname{vec} A' = \operatorname{vec} A$, or $A' = A$.

Problem 16.7 ([10, Problem 32.1])**.** Let $m, n \geqslant 1$ with $mn > 1$. Show that

(a) there are no matrices A, B with $AXB = X'$ for every $X \in \mathbb{R}^{m\times n}$;

(b) there are no matrices A, B with $AX'B = X$ for every $X \in \mathbb{R}^{m\times n}$.

Solution. (a)

- Suppose $AXB = X'$ for every $X \in \mathbb{R}^{m\times n}$. As AXB is defined and has the size $n \times m$ of X', $A = (a_{ij}) \in \mathbb{R}^{n\times m}$ and $B = (b_{ij}) \in \mathbb{R}^{n\times m}$.
- Let $1 \leqslant k \leqslant m$ and $1 \leqslant s \leqslant n$, and take $X = E_{ks} = e_ke_s' \in \mathbb{R}^{m\times n}$ (Proposition 1.18). Then $AXB = (Ae_k)(e_s'B) = \operatorname{col}_k(A)\operatorname{row}_s(B)$ (Theorem 1.3, Proposition 1.18(a)), whose (i, j) entry is $a_{ik}b_{sj}$ (Theorem 1.4(d)); and $X' = e_se_k' = E_{sk} \in \mathbb{R}^{n\times m}$ (Theorem 1.9), whose (s, k) entry is 1 and whose other entries are 0 (Proposition 1.18(b)). So, for $1 \leqslant i, s \leqslant n$ and $1 \leqslant j, k \leqslant m$,

$$a_{ik}b_{sj} = 1 \quad \text{if } (i, j) = (s, k), \qquad a_{ik}b_{sj} = 0 \quad \text{if } (i, j) \neq (s, k)$$

- With $i = s$ and $j = k$: $a_{sk}b_{sk} = 1$ for all s and k. So every entry of A and every entry of B is nonzero.
- As $mn > 1$, $n \geqslant 2$ or $m \geqslant 2$. If $n \geqslant 2$, then $i = j = k = 1$ and $s = 2$ give $a_{11}b_{21} = 0$ ✘ If $m \geqslant 2$, then $i = j = s = 1$ and $k = 2$ give $a_{12}b_{11} = 0$ ✘

(b) If $AX'B = X$ for every $X \in \mathbb{R}^{m\times n}$, then transposing gives $B'XA' = X'$ for every $X \in \mathbb{R}^{m\times n}$ (Theorem 1.9), which (a) excludes.

By Theorem 16.7(b) and Definition 16.11, $AXB = X'$ for every X means $(B' \otimes A)\operatorname{vec} X = K_{mn}\operatorname{vec} X$ for every X, that is, $B' \otimes A = K_{mn}$, as $\operatorname{vec} X$ runs over $\mathbb{R}^{mn}$ (Proposition 1.18(d)). So (a) says that for $mn > 1$ the commutation matrix K_{mn} is not of the form $C \otimes A$ with $C \in \mathbb{R}^{m\times n}$ and $A \in \mathbb{R}^{n\times m}$.

Problem 16.8 ([2, Exercise 9.1])**.** Show that the derivative of $f(x) = (y_1, y_2)'$, where $y_1 = 6x_1^2x_2 + 2x_1x_2 + x_2^2$ and $y_2 = 2x_1^3 + x_1^2 + 2x_1x_2$, with respect to $x = (x_1, x_2)'$ is the same as the Hessian matrix of $y = 2x_1^3x_2 + x_1^2x_2 + x_1x_2^2$.

Solution.

- Differentiating the entries,

$$\mathrm{D}f(x) = \begin{pmatrix} 12x_1x_2 + 2x_2 & 6x_1^2 + 2x_1 + 2x_2 \\ 6x_1^2 + 2x_1 + 2x_2 & 2x_1 \end{pmatrix}$$

- For y, $\partial y/\partial x_1 = 6x_1^2x_2 + 2x_1x_2 + x_2^2$ and $\partial y/\partial x_2 = 2x_1^3 + x_1^2 + 2x_1x_2$, and the second partial derivatives are $\partial^2 y/\partial x_1^2 = 12x_1x_2 + 2x_2$, $\partial^2 y/\partial x_1\partial x_2 = \partial^2 y/\partial x_2\partial x_1 = 6x_1^2 + 2x_1 + 2x_2$, $\partial^2 y/\partial x_2^2 = 2x_1$. So $\mathrm{H}y(x)$ is the matrix above.
- The reason: y_1 and y_2 are the partial derivatives of y, so $f = (\mathrm{D}y)'$ is the gradient of y, and $\mathrm{H}y = \mathrm{D}((\mathrm{D}y)') = \mathrm{D}f$ (Definition 16.31).

Problem 16.9 ([6, Exercise 13.24(a)])**.** Let $X \in \mathbb{R}^{n\times k}$ have rank k, and let $M = I_n - X(X'X)^{-1}X'$ (the residual maker). Find the differential and the derivative of M as a function of X.

Solution.

- Let $F = X(X'X)^{-1} \in \mathbb{R}^{n\times k}$ and $H = I_n - M = X(X'X)^{-1}X'$. By the product rule and $\mathrm{d}(X'X)^{-1} = -(X'X)^{-1}\mathrm{d}(X'X)(X'X)^{-1}$ with $\mathrm{d}(X'X) = (\mathrm{d}X)'X + X'\mathrm{d}X$ (Theorems 16.22 and 16.26(c)),

$$\mathrm{d}M = -(\mathrm{d}X)F' + F(\mathrm{d}X)'H + H(\mathrm{d}X)F' - F(\mathrm{d}X)'$$

- The first and third terms give $-(I_n - H)(\mathrm{d}X)F' = -M(\mathrm{d}X)F'$, the second and fourth $-F(\mathrm{d}X)'(I_n - H) = -F(\mathrm{d}X)'M$: $\mathrm{d}M = -M(\mathrm{d}X)(X'X)^{-1}X' - X(X'X)^{-1}(\mathrm{d}X)'M$.
- By Theorem 16.7(b), $\operatorname{vec}\big(M(\mathrm{d}X)F'\big) = (F \otimes M)\operatorname{vec}\mathrm{d}X$ and $\operatorname{vec}\big(F(\mathrm{d}X)'M\big) = (M \otimes F)\operatorname{vec}(\mathrm{d}X)' = (M \otimes F)K_{nk}\operatorname{vec}\mathrm{d}X$.
- By Theorem 16.12(d), $(M \otimes F)K_{nk} = K_n(F \otimes M)$. Hence $\operatorname{vec}\mathrm{d}M = -(I_{n^2} + K_n)(F \otimes M)\operatorname{vec}\mathrm{d}X$, and

$$\mathrm{D}M(X) = -(I_{n^2} + K_n)\big(X(X'X)^{-1} \otimes M\big)$$

Problem 16.10 ([21, Exercise 16])**.** Show that optimizing a function subject to a constraint is not the same as optimizing its Lagrangian: maximize xy subject to $x + y = 2$, and examine the Lagrangian $\psi(x, y) = xy - \lambda(x + y - 2)$ at the solution.

Solution.

- On the line $x + y = 2$, $xy = x(2 - x) = 1 - (x - 1)^2 \leqslant 1$, with equality only at $x = 1$: the constrained maximum is attained at $x = y = 1$.
- $\mathrm{d}\psi = (y - \lambda)\,\mathrm{d}x + (x - \lambda)\,\mathrm{d}y$, so $\mathrm{D}\psi = 0$ and $x + y = 2$ give $x = y = \lambda = 1$: Lagrange's method finds the point $(1, 1)$ with $\lambda = 1$.
- But $\mathrm{H}\psi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with the eigenvalues 1 and -1 (eigenvectors $(1, 1)'$, $(1, -1)'$), is indefinite; by Theorem 16.36(c), ψ has neither a maximum nor a minimum at $(1, 1)$: it is a saddle point ($\psi = 1 + st$ at $(1 + s, 1 + t)$ for $\lambda = 1$, which is > 1 for $s = t \neq 0$ and < 1 for $s = -t \neq 0$). So the solution of the constrained problem is not a maximum of the Lagrangian, and Theorem 16.36(e) does not apply here, since $\mathrm{H}\psi$ is not PSD.

Problem 16.11 ([6, Exercise 13.56])**.** Let $X \in \mathbb{R}^{n\times k}$ have rank k, $y \in \mathbb{R}^n$, $\Omega \in \mathbb{R}^{n\times n}$ PD, $R \in \mathbb{R}^{k\times r}$ of rank r, and $c \in \mathbb{R}^r$. Minimize $(y - X\beta)'\Omega^{-1}(y - X\beta)$ subject to $R'\beta = c$.

Solution.

- Take $\psi(\beta) = \frac{1}{2}(y - X\beta)'\Omega^{-1}(y - X\beta) - l'(R'\beta - c)$ with $l \in \mathbb{R}^r$. Then $\mathrm{d}\psi = -\Big((y - X\beta)'\Omega^{-1}X + l'R'\Big)\mathrm{d}\beta$, so the conditions are $X'\Omega^{-1}X\beta - Rl = X'\Omega^{-1}y$ and $R'\beta = c$.
- Ω^{-1} is PD (Proposition 8.23(c)), so $W = X'\Omega^{-1}X$ is PD (Theorem 8.24(b)), so is W^{-1} (the same proposition), and $R'W^{-1}R$ is PD (Theorem 8.24(b), $\operatorname{rank} R = r$), hence invertible. Let $\hat\beta = W^{-1}X'\Omega^{-1}y$ (the unconstrained solution, the generalized least-squares estimator of Theorem 13.22).
- Multiplying the first condition by $R'W^{-1}$ and using $R'\beta = c$: $c - R'W^{-1}Rl = R'\hat\beta$, so $\tilde l = (R'W^{-1}R)^{-1}(c - R'\hat\beta)$, and then

$$\tilde\beta = \hat\beta + W^{-1}R\tilde l = \hat\beta + W^{-1}R(R'W^{-1}R)^{-1}(c - R'\hat\beta)$$

- $\mathrm{H}\psi = X'\Omega^{-1}X = W$ is PD at every β, so $\tilde\beta$ is the strict minimizer under the constraint (Theorem 16.36(e)).

For $\Omega = I_n$, $W = X'X$ and $\tilde\beta$ is the constrained least-squares estimate of Theorem 14.12, with $K = R'$, $m = c$ and $G = (X'X)^{-1}$, found there without derivatives.

Problem 16.12 ([6, Exercises 8.46, 12.34(a)])**.** (a) Let

$$M = \begin{pmatrix} A & B \\ B' & D \end{pmatrix}$$

be PD, with square blocks A and D. Show that $\det M \leqslant \det A \cdot \det D$ (**Fischer's inequality**), with equality if and only if $B = 0$.

(b) Let M be PD and partitioned into k row and column groups, with square diagonal blocks $M_{11}, \ldots, M_{kk}$. Show that $\det M \leqslant \det M_{11} \cdots \det M_{kk}$. For blocks of size 1 this is Hadamard's inequality (Theorem 8.32) for PD M.

Solution. (a)

- A and D are principal submatrices of M, hence PD (Theorem 8.30); in particular A is invertible and $\det A > 0$ (Theorem 8.21(c)).

- By Theorem 8.31, the Schur complement $S = D - B'A^{-1}B$ is PD and $\det M = \det A \cdot \det S$.
- A^{-1} is PD (Proposition 8.23(c)), so $B'A^{-1}B$ is PSD (Theorem 8.24(a)). As $D = S + B'A^{-1}B$ with S PD, Theorem 8.52 gives $\det D \geqslant \det S$, with equality if and only if $B'A^{-1}B = 0$. Multiplying by $\det A > 0$: $\det M = \det A \cdot \det S \leqslant \det A \cdot \det D$, with equality if and only if $B'A^{-1}B = 0$.
- If $B = 0$, then $B'A^{-1}B = 0$. Conversely, if $B'A^{-1}B = 0$, then $(Bx)'A^{-1}(Bx) = 0$ for every x, so $Bx = 0$ for every x, as A^{-1} is PD; hence $B = 0$ (Proposition 1.18(a)).

(b) Induction on k; for $k = 1$ there is nothing to prove. For $k \geqslant 2$, let A be the leading block formed by the first $k-1$ groups and $D = M_{kk}$. A is PD (Theorem 8.30), with the diagonal blocks $M_{11}, \ldots, M_{k-1,k-1}$. By (a) and the induction hypothesis,

$$\det M \leqslant \det A \cdot \det M_{kk} \leqslant \det M_{11} \cdots \det M_{k-1,k-1} \cdot \det M_{kk}$$

Chapter 17. Generalized linear models

Exponential families and generalized linear models

We use the following distributions of a real random variable y. For an integer $m \geqslant 1$ and $0 < \pi < 1$, y has the **binomial distribution** $\mathrm{Bin}(m, \pi)$ if $\mathsf{P}(y = k) = \binom{m}{k}\pi^k(1-\pi)^{m-k}$ for $k = 0, 1, \dots, m$, where $\binom{m}{k} = \frac{m!}{k!\,(m-k)!}$; for $m = 1$, y is **Bernoulli**, with $\mathsf{P}(y = 1) = \pi = 1 - \mathsf{P}(y = 0)$. For $\mu > 0$, y has the **Poisson distribution** $\mathrm{Poisson}(\mu)$ if $\mathsf{P}(y = k) = e^{-\mu}\mu^k/k!$ for every integer $k \geqslant 0$. For $\sigma^2 > 0$, $N(\mu, \sigma^2)$ has the density $(2\pi\sigma^2)^{-1/2}\exp\bigl(-(y-\mu)^2/(2\sigma^2)\bigr)$ (Theorem 10.8), mean μ and variance σ^2 (Definition 10.1). We use without proof that $\mathsf{E}\,y = m\pi$ and $\operatorname{var} y = m\pi(1-\pi)$ if $y \sim \mathrm{Bin}(m, \pi)$, and $\mathsf{E}\,y = \operatorname{var} y = \mu$ if $y \sim \mathrm{Poisson}(\mu)$ [17, §3.2], [9, Propositions 7.2, 8.2]. For $0 < \pi < 1$ the **odds** are $\pi/(1-\pi)$ and $\operatorname{logit}\pi = \log\bigl(\pi/(1-\pi)\bigr)$; the **logistic function** $\operatorname{expit} t = e^t/(1+e^t)$ $(t \in \mathbb{R})$ takes values in $(0, 1)$, and direct computation gives $1 - \operatorname{expit} t = 1/(1+e^t)$, $\operatorname{logit}(\operatorname{expit} t) = t$ and $\operatorname{expit}(\operatorname{logit}\pi) = \pi$. To the facts of calculus stated after Definition 16.17 we add: e^t is C^2 on $\mathbb{R}$, with derivative e^t.

Definition 17.1 ([9, Definitions 8.32, 8.36, 8.9, Proposition 8.35], [24, §4.4.2, (4.55)–(4.58)], [19, (4.42)])**.** A **generalized linear model** (GLM) with canonical link consists of a known $X \in \mathbb{R}^{n\times p}$ of any rank, with rows $x_1', \dots, x_n'$, known numbers $o_1, \dots, o_n$, the **offsets** (often 0), an unknown $\beta \in \mathbb{R}^p$, and independent responses $y_1, \dots, y_n$. Each y_i has a density, or probabilities, of the **one-parameter exponential family** form

$$f_i(y_i; \theta_i, \phi) = \exp\left\{\frac{y_i\theta_i - b_i(\theta_i)}{\phi} + c_i(y_i, \phi)\right\}, \qquad \theta_i = \eta_i = o_i + x_i'\beta$$

with the **natural parameter** θ_i equal to the **linear predictor** η_i, the **dispersion** $\phi > 0$, and the functions b_i, c_i of one of the three families below, the same family for all i. The number $\mu_i = b_i'(\theta_i)$ is the mean of y_i (shown after the definition), and the inverse of b_i', which maps μ_i to η_i, is the **canonical link**: $\mu \mapsto \mu$, $\mu \mapsto \operatorname{logit}(\mu/m_i)$ or $\mu \mapsto \log\mu$ in the three families below.

- **Normal linear model**: $y_i \sim N(\mu_i, \sigma^2)$, $o_i = 0$, $\theta_i = \mu_i = x_i'\beta$, $\phi = \sigma^2$, $b_i(\theta) = \theta^2/2$, $c_i(y, \phi) = -y^2/(2\phi) - \frac{1}{2}\log(2\pi\phi)$; this is the normal linear model of Definition 14.8, with p columns.
- **Logistic regression**: $y_i \sim \mathrm{Bin}(m_i, \pi_i)$ with known m_i, $o_i = 0$, $\theta_i = \operatorname{logit}\pi_i = x_i'\beta$, $\phi = 1$, $b_i(\theta) = m_i\log(1+e^\theta)$, $c_i(k, \phi) = \log\binom{m_i}{k}$. So $\pi_i = \operatorname{expit}(x_i'\beta)$ and $\mu_i = m_i\pi_i$; for $m_i = 1$ the responses are binary.
- **Poisson regression with exposure**: $y_i \sim \mathrm{Poisson}(\mu_i)$ with $\mu_i = t_i e^{x_i'\beta}$, that is, $\log\mu_i = \log t_i + x_i'\beta$, for known **exposures** $t_i > 0$ (offsets $o_i = \log t_i$), such as policy-years in insurance; $\theta_i = \log\mu_i$, $\phi = 1$, $b_i(\theta) = e^\theta$, $c_i(k, \phi) = -\log k!$.

The likelihood and the maximum likelihood estimates (MLE) are those of Definition 14.8: the likelihood is the density $f(y; \vartheta)$ of $y = (y_1, \dots, y_n)'$ at the observed y, as a function of the parameters ϑ (here β and ϕ). For a discrete y, $f(y; \vartheta)$ is instead the probability that y takes the observed value. By independence (the paragraphs before Definition 10.1 and before Theorem 10.8), it is the product $f_1(y_1; \theta_1, \phi)\cdots f_n(y_n; \theta_n, \phi)$.

The three families have this form [9, Proposition 8.35], [17, Examples 3.4.1, 3.4.4]. For $\mathrm{Bin}(m, \pi)$ with $\theta = \operatorname{logit}\pi$, $\pi = \operatorname{expit}\theta$ and $1 - \pi = 1/(1+e^\theta)$, so $\pi^k(1-\pi)^{m-k} = \bigl(\pi/(1-\pi)\bigr)^k(1-\pi)^m = e^{k\theta}(1+e^\theta)^{-m}$ and $\log\mathsf{P}(y = k) = k\theta - m\log(1+e^\theta) + \log\binom{m}{k}$. For $\mathrm{Poisson}(\mu)$ with $\theta = \log\mu$, $\log(e^{-\mu}\mu^k/k!) = k\theta - e^\theta - \log k!$. For $N(\mu, \sigma^2)$, $-(y-\mu)^2/(2\sigma^2) = (y\mu - \mu^2/2)/\sigma^2 - y^2/(2\sigma^2)$. The

functions b are C^2 on $\mathbb{R}$, with

$$\begin{aligned}
&\text{binomial:} && b'(\theta) = \frac{me^\theta}{1+e^\theta} = m\pi, && b''(\theta) = \frac{me^\theta}{(1+e^\theta)^2} = m\pi(1-\pi)\\
&\text{Poisson:} && b'(\theta) = e^\theta = \mu, && b''(\theta) = e^\theta = \mu\\
&\text{normal:} && b'(\theta) = \theta = \mu, && b''(\theta) = 1
\end{aligned}$$

So in each family $\mathsf{E}\, y = b'(\theta)$ and $\operatorname{var} y = \phi\, b''(\theta) > 0$, by the facts in the first paragraph.

Theorem 17.2 ([9, Proposition 8.37, Theorem 8.40], [24, §4.4.2])**.** In a GLM (Definition 17.1), let $\theta = o + X\beta$ with $o = (o_1, \dots, o_n)'$, and let

$$\mu = \begin{pmatrix} b_1'(\theta_1) \\ \vdots \\ b_n'(\theta_n) \end{pmatrix}, \qquad W = \operatorname{diag}\Big(b_1''(\theta_1), \dots, b_n''(\theta_n)\Big)$$

both evaluated at β. The log-likelihood, as a function of $\beta \in \mathbb{R}^p$ with ϕ fixed, is

$$\ell(\beta) = \frac{1}{\phi} \sum_{i=1}^n \Big(y_i\theta_i - b_i(\theta_i)\Big) + \sum_{i=1}^n c_i(y_i, \phi)$$

Here an **MLE** of β (for the given ϕ) is a maximizer of ℓ on $\mathbb{R}^p$; for the binomial and Poisson families $\phi = 1$, and for the normal family the maximizers do not depend on ϕ and, if $\mathrm{SSE} > 0$, they are the first components $\hat\beta$ of the MLEs $(\hat\beta, \mathrm{SSE}/n)$ of Theorem 14.9(d).

(a) ℓ is C^2 on $\mathbb{R}^p$, $\mathrm{D}\ell(\beta) = (y-\mu)'X/\phi$, so that the gradient, the **score**, is $X'(y-\mu)/\phi$, and $\mathrm{H}\ell(\beta) = -X'WX/\phi$. The matrix $X'WX$ is PSD for every β, and PD if $\operatorname{rank} X = p$.

(b) $\hat\beta$ is an MLE if and only if it solves the **likelihood equations**

$$X'(y - \hat\mu) = 0, \qquad \hat\mu_i = b_i'(o_i + x_i'\hat\beta)$$

which do not involve ϕ. If $\operatorname{rank} X = p$, there is at most one MLE.

(c) Let $\operatorname{rank} X = p$. The Newton–Raphson step for the stationary points of ℓ from β (Example 16.44) is

$$\beta^{\mathrm{new}} = \beta - \mathrm{H}\ell(\beta)^{-1}\Big(\mathrm{D}\ell(\beta)\Big)' = (X'WX)^{-1}X'Wz, \qquad z = X\beta + W^{-1}(y-\mu)$$

the weighted least-squares estimate (Proposition 13.21) for the **working response** z and the weight matrix W; it is the unique maximizer of the second-order Taylor polynomial of ℓ at β. Repeating the step is **iteratively reweighted least squares** (IRLS).

Proof. By Definition 17.1 the likelihood is $\prod_i f_i(y_i; \theta_i, \phi)$, and its logarithm is ℓ.

(a)

- Each b_i is C^2 on $\mathbb{R}$ (the paragraph after Definition 17.1), and $\theta_i = o_i + x_i'\beta$ is a polynomial in β; so ℓ is C^2 on $\mathbb{R}^p$ (rules and chain rule of calculus, after Definition 16.17).
- By Theorems 16.24(a) and 16.22(a),(c), $\mathrm{d}\theta_i = x_i'\,\mathrm{d}\beta$ and $\mathrm{d}\, b_i(\theta_i) = b_i'(\theta_i)\, x_i'\,\mathrm{d}\beta = \mu_i x_i'\,\mathrm{d}\beta$. By Theorem 1.4(c), the row $(y-\mu)'X$ is $\sum_i (y_i - \mu_i)x_i'$, so

$$\mathrm{d}\ell = \frac{1}{\phi}\sum_{i=1}^n (y_i - \mu_i)\, x_i'\,\mathrm{d}\beta = \frac{1}{\phi}(y-\mu)'X\,\mathrm{d}\beta$$

and $\mathrm{D}\ell(\beta) = (y-\mu)'X/\phi$ (Theorem 16.21(a)); the gradient is its transpose $X'(y-\mu)/\phi$.

- With $\mathrm{d}\beta$ fixed, $\mathrm{d}\mu_i = b_i''(\theta_i)\, x_i'\, \mathrm{d}\beta$, a number equal to its transpose $b_i''(\theta_i)(\mathrm{d}\beta)' x_i$. By Theorem 1.4(d), $X'WX = \sum_i x_i\, b_i''(\theta_i) x_i'$ (the columns of X' are the x_i, the rows of WX are the $b_i''(\theta_i) x_i'$). So

$$\mathrm{d}^2\ell = -\frac{1}{\phi}\sum_{i=1}^{n}(\mathrm{d}\mu_i)\, x_i'\, \mathrm{d}\beta = -\frac{1}{\phi}(\mathrm{d}\beta)'\Big(\sum_{i=1}^{n} b_i''(\theta_i) x_i x_i'\Big)\mathrm{d}\beta = -\frac{1}{\phi}(\mathrm{d}\beta)' X'WX\, \mathrm{d}\beta$$

  $X'WX$ is symmetric, so $\mathrm{H}\ell(\beta) = -X'WX/\phi$ (Theorem 16.32(a)).

- W is diagonal with positive diagonal entries (the paragraph after Definition 17.1), hence PD (Proposition 8.23(a)). By Theorem 8.24(a),(b) with $A = W$ and $P = X$, $X'WX$ is PSD, and PD if $\operatorname{rank} X = p$.

(b)

- $\mathbb{R}^p$ is open and convex. If $\hat\beta$ is an MLE, ℓ has a global, hence a local, maximum at $\hat\beta$, so $\mathrm{D}\ell(\hat\beta) = 0$ (Theorem 16.36(a)); by (a) and $\phi > 0$, this is $X'(y - \hat\mu) = 0$.
- Conversely, let $X'(y - \hat\mu) = 0$. The function $-\ell$ is C^2, $\mathrm{D}(-\ell)(\hat\beta) = 0$, and its Hessian $X'WX/\phi$ is PSD at every β by (a). By Theorem 16.36(d), $-\ell$ has a global minimum at $\hat\beta$: $\ell(\beta) \leqslant \ell(\hat\beta)$ for every β, and $\hat\beta$ is an MLE.
- If $\operatorname{rank} X = p$, the Hessian of $-\ell$ is PD at every β, and the global minimum is strict (Theorem 16.36(d)): $\ell(\beta) < \ell(\hat\beta)$ for $\beta \neq \hat\beta$, so no other β is an MLE.

(c)

- By (a), $X'WX$ is PD, hence invertible (Theorem 8.21(c)), and so is $\mathrm{H}\ell(\beta) = -X'WX/\phi$. The step of Example 16.44 for the function ℓ is $\beta - (-X'WX/\phi)^{-1}X'(y-\mu)/\phi = \beta + (X'WX)^{-1}X'(y-\mu)$, in which ϕ cancels.
- As $\beta = (X'WX)^{-1}X'WX\beta$, this is $(X'WX)^{-1}X'W\big(X\beta + W^{-1}(y-\mu)\big) = (X'WX)^{-1}X'Wz$, and by Proposition 13.21 (W PD, $\operatorname{rank} X = p$) it is the unique minimizer of $(z - Xb)'W(z - Xb)$.
- The Hessian $X'WX/\phi$ of $-\ell$ at β is PD, so by the paragraph after Example 16.44, applied to $-\ell$, β^{new} is the unique minimizer of the second-order Taylor polynomial q of $-\ell$ at β; and $-q$ is the second-order Taylor polynomial of ℓ. □

The theorem does not assert that an MLE exists. For instance, in a logistic regression whose design has the column $\mathbf{1}_n$ and whose responses are all 0, the likelihood equation $\mathbf{1}_n'(y - \hat\mu) = 0$ would require $\sum_i m_i \operatorname{expit}(x_i'\hat\beta) = 0$, which is impossible, as expit > 0. Nor does the theorem assert that the IRLS iterates converge. For the normal family, $b_i'' = 1$, so $W = I_n$ and $\mu = X\beta$: the likelihood equations $X'(y - X\hat\beta) = 0$ are the normal equations (Definition 13.1), as in Theorem 14.9(d); and $z = X\beta + (y - X\beta) = y$, so that for $\operatorname{rank} X = p$ one IRLS step from any β gives the least-squares estimate $(X'X)^{-1}X'y$, as in Example 16.44 [24, Example 4.36].

The rank-general case and weighted projections

Proposition 17.3. In a GLM (Definition 17.1), let $\operatorname{rank} X = r \geqslant 1$, and let $X = X_1 C$ be a full-rank factorization (Theorem 4.21): $X_1 \in \mathbb{R}^{n\times r}$ has full column rank and $C \in \mathbb{R}^{r\times p}$ full row rank. Let $\ell_1(\gamma)$ ($\gamma \in \mathbb{R}^r$) be the log-likelihood of the GLM with the same responses, family, offsets and ϕ and the design X_1.

(a) $\ell(\beta) = \ell_1(C\beta)$, and β is an MLE if and only if $C\beta$ is an MLE of the design X_1, of which there is at most one. If β^0 is an MLE, the MLEs are the vectors $\beta^0 + u$ with $u \in \mathcal{N}(X)$, and all of them give the same $X\hat{\beta}$, the same linear predictors $\hat{\eta} = o + X\hat{\beta}$ and the same fitted means $\hat{\mu}$.

(b) If $a \in \mathcal{C}(X')$, then $a'\hat{\beta}$ is the same number for all MLEs $\hat{\beta}$.

(c) At any β, with W and z as in Theorem 17.2, the system $X'WXb = X'Wz$ (unknown $b \in \mathbb{R}^p$) is consistent, and its solutions are the minimizers of $(z - Xb)'W(z - Xb)$; one of them is $GX'Wz$ for any g-inverse G of $X'WX$. All of them give the same $Xb = X_1\gamma^{\text{new}}$, where γ^{new} is the IRLS step of Theorem 17.2(c) for the design X_1 from $\gamma = C\beta$.

Proof. (a)

- $\theta = o + X\beta = o + X_1(C\beta)$, and ℓ depends on β only through θ (Theorem 17.2); so $\ell(\beta) = \ell_1(C\beta)$. As C has full row rank, $\mathcal{C}(C) = \mathbb{R}^r$ (Theorem 4.13(c)): every $\gamma \in \mathbb{R}^r$ is $C\beta'$ for some $\beta' \in \mathbb{R}^p$.
- If $C\beta$ maximizes ℓ_1, then $\ell(\beta') = \ell_1(C\beta') \leqslant \ell_1(C\beta) = \ell(\beta)$ for every β': β is an MLE. Conversely, if β is an MLE and $\gamma = C\beta' \in \mathbb{R}^r$, then $\ell_1(\gamma) = \ell(\beta') \leqslant \ell(\beta) = \ell_1(C\beta)$: $C\beta$ maximizes ℓ_1.
- X_1 has full column rank r, so ℓ_1 has at most one maximizer $\hat{\gamma}$ (Theorem 17.2(b)). If β^0 is an MLE, $\hat{\gamma} = C\beta^0$, and the MLEs are the solutions of $C\beta = C\beta^0$, that is, the vectors $\beta^0 + u$ with $u \in \mathcal{N}(C) = \mathcal{N}(X)$ (Theorem 4.21(b)). Each of them gives $X\beta = X_1C\beta = X_1\hat{\gamma}$, hence the same $\hat{\eta} = o + X\hat{\beta}$ and $\hat{\mu}_i = b_i'(\hat{\eta}_i)$.

(b) $a = X'c$ for some $c \in \mathbb{R}^n$, so $a'\hat{\beta} = c'X\hat{\beta} = c'X_1\hat{\gamma}$ for every MLE $\hat{\beta}$, by (a).

(c)

- By Proposition 13.21 (W PD, X of any rank), the minimizers of $Q(b) = (z - Xb)'W(z - Xb)$ are the solutions of $X'WXb = X'Wz$. W, μ and z depend on β only through $X\beta = X_1\gamma$, so they are also those of the design X_1 at $\gamma = C\beta$; and $Q(b) = Q_1(Cb)$ with $Q_1(h) = (z - X_1h)'W(z - X_1h)$ ($h \in \mathbb{R}^r$).
- By Proposition 13.21 for X_1 (rank r), Q_1 has exactly one minimizer, which is $\gamma^{\text{new}} = (X_1'WX_1)^{-1}X_1'Wz$, the step of Theorem 17.2(c) for X_1. As in (a), b minimizes Q if and only if Cb minimizes Q_1, that is, $Cb = \gamma^{\text{new}}$; such b exist ($\mathcal{C}(C) = \mathbb{R}^r$), so the system is consistent, and $Xb = X_1Cb = X_1\gamma^{\text{new}}$ for all of them.
- As the system is consistent, $X'Wz \in \mathcal{C}(X'WX)$, and $GX'Wz$ is a solution (Theorem 12.2(a)). □

Proposition 17.4 ([5, Theorems 14.12.11, 14.12.3, Corollary 14.12.4])**.** Let $X \in \mathbb{R}^{n\times p}$, let $W \in \mathbb{R}^{n\times n}$ be PD, let G be a g-inverse of $X'WX$, and put

$$P_{X,W} = XGX'W = X(X'WX)^{-}X'W$$

(a) Let $T = W^{1/2}$ (Theorem 8.46), and let $P = TXG(TX)'$, the orthogonal projection matrix onto $\mathcal{C}(TX)$ of Theorem 12.17 for $A = TX$. Then $P_{X,W} = T^{-1}PT$, and $P_{X,W}$ does not depend on the choice of G.

(b) $P_{X,W}X = X$, $P_{X,W}^2 = P_{X,W}$, and $WP_{X,W}$ is symmetric.

(c) $\mathcal{C}(P_{X,W}) = \mathcal{C}(X)$, $\operatorname{rank} P_{X,W} = \operatorname{rank} X$, and $\operatorname{rank}(I_n - P_{X,W}) = n - \operatorname{rank} X$.

(d) For every $z \in \mathbb{R}^n$, $P_{X,W}z = Xb$ for every minimizer b of $(z - Xb)'W(z - Xb)$. In particular, in a GLM the IRLS step of Proposition 17.3(c) gives $Xb = P_{X,W}z$.

Proof. • T is PD, hence symmetric and invertible, and $T^2 = W$ (Theorem 8.46(a),(c)); so $(TX)'(TX) = X'T^2X = X'WX$, and G is a g-inverse of $(TX)'(TX)$.

• By Theorem 12.17 for $A = TX$: $P = TXGX'T$ does not depend on G, $P' = P = P^2$, $P(TX) = TX$, and $I_n - P$ has rank $n - \operatorname{rank}(TX)$, where $\operatorname{rank}(TX) = \operatorname{rank} X$ (Theorem 4.15(d)).

(a) $T^{-1}PT = T^{-1}TXGX'TT = XGX'W = P_{X,W}$. As P does not depend on G, neither does $P_{X,W}$.

(b) $P_{X,W}X = T^{-1}P(TX) = T^{-1}TX = X$; $P_{X,W}^2 = T^{-1}P^2T = T^{-1}PT = P_{X,W}$; and $WP_{X,W} = T^2T^{-1}PT = TPT$, with $(TPT)' = T'P'T' = TPT$.

(c) $\mathcal{C}(P_{X,W}) = \mathcal{C}\big(X(GX'W)\big) \subseteq \mathcal{C}(X)$ and $\mathcal{C}(X) = \mathcal{C}(P_{X,W}X) \subseteq \mathcal{C}(P_{X,W})$ (Theorem 4.15(a)); so the two are equal and have the same dimension. $I_n - P_{X,W} = T^{-1}(I_n - P)T$ has the rank $n - \operatorname{rank} X$ of $I_n - P$ (Theorem 4.15(d)).

(d) By Proposition 13.21, the minimizers b are the solutions of $X'WXb = X'Wz$, that is, of $(TX)'(TX)b = (TX)'(Tz)$, the normal equations of the design TX and the response Tz. By Theorem 13.2(c), $G(TX)'Tz = GX'Wz$ is one of them, and all of them give the same TXb, hence the same Xb (T is invertible), which is $XGX'Wz = P_{X,W}z$. In a GLM, the solutions of Proposition 17.3(c) are these minimizers. □

For $W = I_n$, $P_{X,W} = X(X'X)^{-}X'$ is the hat matrix H of Theorem 13.4. For other W it is idempotent but in general not symmetric. By (d), $P_{X,W}z$ is the point Xb of $\mathcal{C}(X)$ nearest to z when the length of a vector v is measured by $\|Tv\| = (v'Wv)^{1/2}$; so each IRLS step replaces the working response z by this projection $P_{X,W}z$ of it.

Grouped data and examples

In logistic regression the odds are $\pi_i/(1-\pi_i) = e^{x_i'\beta}$: increasing x_{ij} by 1, the other entries of x_i fixed, multiplies the odds by e^{β_j}, the **odds ratio**. In Poisson regression with exposure, the expected count per unit of exposure is $\mu_i/t_i = e^{x_i'\beta}$; increasing x_{ij} by 1 multiplies it by e^{β_j}, the **rate ratio** [9, Theorem 7.11, Proposition 8.10], [19, §4.3, §4.6.2]. When column j of X is the indicator of a group, e^{β_j} compares that group with the observations that have $x_{ij} = 0$ and the same other entries: in the baseline coding $[\mathbf{1}_n\ d_2\ \cdots\ d_g]$ of Theorem 13.10, the observations of group 1; in $[\mathbf{1}_n\ d_1\ \cdots\ d_g]$ there are no such observations, as every row has exactly one indicator entry 1.

Proposition 17.5 ([9, Corollaries 8.12, 7.16, 8.14])**.** In a GLM (Definition 17.1), let the n observations fall into g nonempty groups, and let $d_l \in \mathbb{R}^n$ be the **indicator** of group l, with entry 1 in the rows of group l and 0 elsewhere (as in Theorem 13.10); $\sum_{i\in l}$ is the sum over the observations i of group l.

(a) If $\mathbf{1}_n \in \mathcal{C}(X)$, every MLE satisfies $\sum_{i=1}^n (y_i - \hat\mu_i) = 0$.

(b) If $d_l \in \mathcal{C}(X)$, every MLE satisfies $\sum_{i\in l} \hat\mu_i = \sum_{i\in l} y_i$.

(c) Let $\mathcal{C}(X) = \mathcal{C}(D)$ with $D = [d_1 \ \cdots \ d_g]$, for instance $X = [\mathbf{1}_n \ d_1 \ \cdots \ d_g]$, $[\mathbf{1}_n \ d_2 \ \cdots \ d_g]$ or D (Theorem 13.10).

- In logistic regression, let $s_l = \sum_{i\in l} y_i$ and $n_l = \sum_{i\in l} m_i$. An MLE exists if and only if $0 < s_l < n_l$ for every l; the MLEs are then the β with $x_i'\beta = \operatorname{logit}(s_l/n_l)$ for every i of every group l, and $\hat\pi_i = s_l/n_l$.
- In Poisson regression with exposures, let $Y_l = \sum_{i\in l} y_i$ and $\tau_l = \sum_{i\in l} t_i$. An MLE exists if and only if $Y_l > 0$ for every l; the MLEs are then the β with $x_i'\beta = \log(Y_l/\tau_l)$ for every i of every group l, and $\hat\mu_i = t_i Y_l/\tau_l$.

Proof. (a) If $\mathbf{1}_n = Xa$, then by Theorem 17.2(b), $\sum_i (y_i - \hat\mu_i) = \mathbf{1}_n'(y - \hat\mu) = a'X'(y - \hat\mu) = 0$.

(b) In the same way, $d_l = Xa_l$ gives $\sum_{i\in l}(y_i - \hat\mu_i) = d_l'(y - \hat\mu) = a_l'X'(y - \hat\mu) = 0$.

(c)

- Every column of X lies in $\mathcal{C}(D)$, so $X = DF$ for some $F \in \mathbb{R}^{g\times p}$ (Theorem 1.4(a),(b)); row i of D is e_l' for i in group l, so $x_i'\beta = (F\beta)_l$ is the same number c_l for all i of group l, and $X\beta = Dc$ with $c = F\beta$.
- Logistic. $\mu_i = m_i \operatorname{expit}(c_l)$ for i in group l. If β is an MLE, (b) gives $n_l \operatorname{expit}(c_l) = s_l$; as $0 < \operatorname{expit} < 1$, this has a solution only if $0 < s_l < n_l$, and then $c_l = \operatorname{logit}(s_l/n_l)$. Conversely, let $0 < s_l < n_l$ for every l, let $c_l = \operatorname{logit}(s_l/n_l)$, and let β satisfy $X\beta = Dc$ (there is such a β, as $Dc \in \mathcal{C}(D) = \mathcal{C}(X)$). Then $\hat\pi_i = s_l/n_l$, the entries of $D'(y - \hat\mu)$ are $s_l - n_l\, s_l/n_l = 0$, and $X'(y - \hat\mu) = F'D'(y - \hat\mu) = 0$: β is an MLE (Theorem 17.2(b)).
- Poisson. $\mu_i = t_i e^{c_l}$ for i in group l, and (b) reads $\tau_l e^{c_l} = Y_l$, which has a solution if and only if $Y_l > 0$, namely $c_l = \log(Y_l/\tau_l)$; the converse is as in the logistic case. □

Example 17.6. Four batches of 8 insects are exposed to an insecticide. Batch i has the dose $a_i \in \{0, 1\}$ (low, high) and the strain $b_i \in \{0, 1\}$, with $(a_i, b_i) = (0, 0), (1, 0), (0, 1), (1, 1)$, and $y = (2, 4, 4, 6)'$ of the insects die. Fit the logistic regression $y_i \sim \operatorname{Bin}(8, \pi_i)$, $\operatorname{logit}\pi_i = \beta_0 + \beta_1 a_i + \beta_2 b_i$: carry out one IRLS step from $\beta = 0$, and find the MLE.

Solution. With X and $X'X$ below, $\det(X'X) = 4$, and $(X'X)^{-1} = \frac{1}{4}\operatorname{adj}(X'X)$ (Theorem 5.24):

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad X'X = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{4}\begin{pmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{pmatrix}$$

so $\operatorname{rank} X = 3$ (Theorem 13.2(b)). By the paragraph after Definition 17.1, $\mu_i = 8\pi_i$ and $W = \operatorname{diag}\big(8\pi_i(1 - \pi_i)\big)$, and $\phi = 1$.

- At $\beta = 0$: $\pi_i = \frac{1}{2}$, $\mu = 4\mathbf{1}_4$, $W = 2I_4$, $y - \mu = (-2, 0, 0, 2)'$ and $z = X\beta + W^{-1}(y - \mu) = (-1, 0, 0, 1)'$. The score $X'(y - \mu) = (0, 2, 2)'$ is not 0, $X'Wz = X'(y - \mu)$, and

$$X'WX = 2X'X = \begin{pmatrix} 8 & 4 & 4 \\ 4 & 4 & 2 \\ 4 & 2 & 4 \end{pmatrix}, \qquad \beta^{(1)} = (X'WX)^{-1}X'Wz = \frac{1}{8}\begin{pmatrix} 3 & -2 & -2 \\ -2 & 4 & 0 \\ -2 & 0 & 4 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

by Theorem 17.2(c).

- The likelihood equations $X'(y-\hat\mu)=0$ (Theorem 17.2(b)) hold if $\hat\mu=y$, that is, $\hat\pi=\frac18 y=(\frac14,\frac12,\frac12,\frac34)'$, with $\operatorname{logit}\hat\pi=(-\log 3,0,0,\log 3)'$. The equations $\beta_0=-\log 3$, $\beta_0+\beta_1=0$, $\beta_0+\beta_2=0$, $\beta_0+\beta_1+\beta_2=\log 3$ are solved by $\hat\beta=(-\log 3,\log 3,\log 3)'$. As $\operatorname{rank}X=3$, it is the only MLE. For each strain, the high dose multiplies the odds of death by $e^{\hat\beta_1}=3$, and for each dose, the strain $b=1$ multiplies them by $e^{\hat\beta_2}=3$.
- At $\hat\beta$, $\hat W=\operatorname{diag}\big(8\hat\pi_i(1-\hat\pi_i)\big)=\operatorname{diag}(\frac32,2,2,\frac32)$, and $\mathrm{H}\ell(\hat\beta)=-X'\hat WX$ with

$$X'\hat WX=\begin{pmatrix}7&\frac72&\frac72\\ \frac72&2&\frac32\\ \frac72&\frac32&\frac72\end{pmatrix}$$

  which is PD (Theorem 17.2(a)).

Example 17.7. Three groups of 36 patients receive the treatments 1, 2, 3, and 12, 24, 18 of them recover. With one row for each group, $y_l\sim\operatorname{Bin}(36,\pi_l)$ $(l=1,2,3)$, so $n=3$ and $d_l=e_l$. Fit the one-way logistic model $\operatorname{logit}\pi_l=\beta_0+\alpha_l$, that is, $\beta=(\beta_0,\alpha_1,\alpha_2,\alpha_3)'$ and

$$X=[\mathbf{1}_3\ d_1\ d_2\ d_3]=\begin{pmatrix}1&1&0&0\\1&0&1&0\\1&0&0&1\end{pmatrix}$$

(Theorem 13.10): find the MLEs, the fitted probabilities and the odds ratios of the treatments.

Solution.

- $\mathcal{C}(X)=\mathcal{C}(D)$ with $D=[d_1\ d_2\ d_3]=I_3$ (Theorem 13.10(c)), and $0<s_l<36$ for each group. By Proposition 17.5(c), the MLEs exist, $\hat\pi=(\frac13,\frac23,\frac12)'$, and they are the β with

$$X\beta=\begin{pmatrix}\operatorname{logit}\frac13\\ \operatorname{logit}\frac23\\ \operatorname{logit}\frac12\end{pmatrix}=\begin{pmatrix}-\log 2\\ \log 2\\ 0\end{pmatrix}$$

- $\operatorname{rank}X=3<4$ and $\mathcal{N}(X)=\operatorname{span}\{(-1,1,1,1)'\}$ (Theorem 13.10(b)). One MLE is $\beta^0=(0,-\log 2,\log 2,0)'$, so by Proposition 17.3(a) the MLEs are $\beta^0+t(-1,1,1,1)'=(-t,\ t-\log 2,\ t+\log 2,\ t)'$ $(t\in\mathbb{R})$, all with the same $\hat\pi$. Their first entry $-t$ takes every real value: the data do not determine β_0.
- $\alpha_2-\alpha_1=a'\beta$ with $a=(0,-1,1,0)'=X'(-1,1,0)'\in\mathcal{C}(X')$, and likewise $\alpha_3-\alpha_1$, with $a=X'(-1,0,1)'$. By Proposition 17.3(b) they have the same value for all MLEs: $\hat\alpha_2-\hat\alpha_1=2\log 2$ and $\hat\alpha_3-\hat\alpha_1=\log 2$. So the estimated odds ratios of treatments 2 and 3 to treatment 1 are 4 and 2: the estimated odds of recovery are $\frac{24}{12}=2$, $\frac{12}{24}=\frac12$ and 1 for treatments 2, 1, 3.

We state without proof, as it needs limit theorems that are not in this book: in a GLM with $\operatorname{rank}X=p$ and ϕ known ($\phi=1$ for the binomial and Poisson families), under conditions on X (for instance, bounded rows x_i and $X'WX/n$ converging to a PD matrix), for large n an MLE exists with probability close to 1, and $\hat\beta$ is approximately $N_p\big(\beta,\phi(X'\hat WX)^{-1}\big)$ [9, Theorem 8.41, Corollary 7.21], [24, Theorem 4.18], where $\hat W$ is W at $\hat\beta$. Binomial rows are covered, with $\sum_i m_i$ in place of n: a row with m_i trials and response y_i contributes $(y_i-m_i\pi_i)x_i$ to the score and $m_i\pi_i(1-\pi_i)x_ix_i'$ to $X'WX$ (the sums in the proof of Theorem 17.2(a)), as do m_i Bernoulli rows with the same x_i and responses of sum y_i. Accordingly, the **standard error** of $c'\hat\beta$ is taken to be $\operatorname{SE}(c'\hat\beta)=\sqrt{\phi\,c'(X'\hat WX)^{-1}c}$, with $\phi(X'\hat WX)^{-1}$ in place of the $\sigma^2(X'X)^{-1}$ of Theorem 14.9(a); for the normal family, where $W=I_n$ and $\phi=\sigma^2$ is known, the normal distribution of $\hat\beta$ is exact. For $H_0\colon K\beta=k_0$ with $K\in\mathbb{R}^{q\times p}$ of rank q and $k_0\in\mathbb{R}^q$, the **Wald statistic** $(K\hat\beta-k_0)'\big(\phi K(X'\hat WX)^{-1}K'\big)^{-1}(K\hat\beta-k_0)$ and the **likelihood-ratio statistic** $2\big(\ell(\hat\beta)-\ell(\tilde\beta)\big)$, where $\tilde\beta$ maximizes ℓ subject to $K\beta=k_0$, are both approximately χ^2_q under H_0, and H_0 is rejected at approximate level α when the statistic exceeds $\chi^2_{q,1-\alpha}$ (Definition 11.7(c)).

Exercises

Exercise 17.1. Use the data of Example 17.7 (three groups of 36 patients, 12, 24, 18 recoveries, one row for each group).

(a) In the baseline coding $X_2 = [\mathbf{1}_3\ d_2\ d_3]$ (Example 13.11), $\operatorname{logit}\pi_l = \beta_0 + \beta_2 d_{l2} + \beta_3 d_{l3}$ (d_{l2}, d_{l3} the entries of d_2, d_3), find the MLE $\hat\beta = (\hat\beta_0, \hat\beta_2, \hat\beta_3)'$ and the odds ratios $e^{\hat\beta_2}$, $e^{\hat\beta_3}$, and compare them with Example 17.7.

(b) At the MLE, find $\hat W$, $X_2'\hat W X_2$ and its inverse, and $\mathrm{SE}(\hat\beta_2)$, $\mathrm{SE}(\hat\beta_3)$.

(c) In the coding $X_1 = [d_1\ d_2\ d_3] = I_3$ without intercept (Theorem 13.10), carry out one IRLS step from $\gamma = 0$.

(d) In the coding $X = [\mathbf{1}_3\ d_1\ d_2\ d_3]$ of Example 17.7, at $\beta = 0$, find $X'WX$, check that $G_0 = \frac{1}{9}\operatorname{diag}(0,1,1,1)$ is a g-inverse of it, compute $b = G_0X'Wz$, and compare Xb with the step of (c).

Solution. (a) $\mathcal{C}(X_2) = \mathcal{C}(I_3)$ (Theorem 13.10(c)), so by Proposition 17.5(c) the MLE satisfies $X_2\hat\beta = (-\log 2, \log 2, 0)'$, as in Example 17.7: $\hat\beta_0 = -\log 2$, $\hat\beta_0 + \hat\beta_2 = \log 2$, $\hat\beta_0 + \hat\beta_3 = 0$. So $\hat\beta = (-\log 2,\ 2\log 2,\ \log 2)'$, the only MLE ($\operatorname{rank} X_2 = 3$, Theorem 17.2(b)), and $e^{\hat\beta_2} = 4$, $e^{\hat\beta_3} = 2$. These are the odds ratios of Example 17.7: $\hat\beta_2$, $\hat\beta_3$ equal $\hat\alpha_2 - \hat\alpha_1$, $\hat\alpha_3 - \hat\alpha_1$, as both codings give the same fitted probabilities $\hat\pi = (\frac13, \frac23, \frac12)'$ (Proposition 17.5(c)).

(b) $\hat W = \operatorname{diag}\big(36\hat\pi_l(1-\hat\pi_l)\big) = \operatorname{diag}(36\cdot\frac13\cdot\frac23,\ 36\cdot\frac23\cdot\frac13,\ 36\cdot\frac12\cdot\frac12) = \operatorname{diag}(8,8,9)$, and X_2 is invertible, with the inverse below ($X_2X_2^{-1} = I_3$ by multiplication):

$$X_2 = \begin{pmatrix}1&0&0\\1&1&0\\1&0&1\end{pmatrix}, \qquad X_2'\hat W X_2 = \begin{pmatrix}25&8&9\\8&8&0\\9&0&9\end{pmatrix}, \qquad X_2^{-1} = \begin{pmatrix}1&0&0\\-1&1&0\\-1&0&1\end{pmatrix}$$

By Theorem 2.13(b),(c), $(X_2'\hat W X_2)^{-1} = X_2^{-1}\hat W^{-1}(X_2^{-1})'$, with

$$X_2^{-1}\hat W^{-1} = \begin{pmatrix}\frac18&0&0\\-\frac18&\frac18&0\\-\frac18&0&\frac19\end{pmatrix}, \qquad (X_2'\hat W X_2)^{-1} = \begin{pmatrix}\frac18&-\frac18&-\frac18\\-\frac18&\frac14&\frac18\\-\frac18&\frac18&\frac{17}{72}\end{pmatrix}$$

With $\phi = 1$ and the paragraph after Example 17.7, $\mathrm{SE}(\hat\beta_2)^2 = \frac14 = \frac18 + \frac18$ and $\mathrm{SE}(\hat\beta_3)^2 = \frac{17}{72} = \frac18 + \frac19$: so $\mathrm{SE}(\hat\beta_2) = \frac12$ and $\mathrm{SE}(\hat\beta_3) = \frac{\sqrt{34}}{12}$.

(c) At $\gamma = 0$: $\pi_l = \frac12$, $\mu = 18\mathbf{1}_3$, $W = 36\cdot\frac12\cdot\frac12 I_3 = 9I_3$, $y - \mu = (-6, 6, 0)'$ and $z = \gamma + W^{-1}(y-\mu) = (-\frac23, \frac23, 0)'$. By Theorem 17.2(c), $\gamma^{\text{new}} = (X_1'WX_1)^{-1}X_1'Wz = (9I_3)^{-1}\,9z = z = (-\frac23, \frac23, 0)'$ (the MLE is $\hat\gamma = (-\log 2, \log 2, 0)'$).

(d) At $\beta = 0$, $X\beta = 0$, so π, μ, $W = 9I_3$ and z are those of (c), and $X'Wz = X'(y-\mu) = (0, -6, 6, 0)'$.

$$X'WX = 9X'X = \begin{pmatrix}27&9&9&9\\9&9&0&0\\9&0&9&0\\9&0&0&9\end{pmatrix}, \qquad X'WX\,G_0 = \begin{pmatrix}0&1&1&1\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}$$

and $X'WX\,G_0\,X'WX = X'WX$: the first row of the product is the sum of rows 2, 3, 4 of $X'WX$, which is $(27, 9, 9, 9)$, and the other rows are rows 2, 3, 4 themselves. So G_0 is a g-inverse of $X'WX$ (Definition 12.1). Then

$$b = G_0X'Wz = \frac19\begin{pmatrix}0\\-6\\6\\0\end{pmatrix} = \begin{pmatrix}0\\-\frac23\\\frac23\\0\end{pmatrix}, \qquad Xb = \begin{pmatrix}-\frac23\\\frac23\\0\end{pmatrix} = \gamma^{\text{new}}$$

as Proposition 17.3(c) asserts for the full-rank factorization $X = X_1S$, $S = [\mathbf{1}_3\ I_3]$ (Theorem 13.10(a)), although b itself depends on the g-inverse.

Exercise 17.2. A motor portfolio has three rating classes A, B, C. Policy i is observed for t_i years (its exposure) and has y_i claims. Class A has total exposure 1000 policy-years and 24 claims, class B 500 policy-years and 48 claims, class C 750 policy-years and 36 claims. Assume the Poisson regression $\log \mu_i = \log t_i + \beta_0 + \beta_1 d_{Bi} + \beta_2 d_{Ci}$, where d_B, d_C are the indicators of classes B and C.

(a) Show that, apart from a term free of β, the log-likelihood depends on the data only through the class totals; find the MLE and the rate ratios of classes B and C to class A.

(b) Find $X'\hat{W}X$ at the MLE, its inverse, and $\mathrm{SE}(\hat{\beta}_1)$, $\mathrm{SE}(\hat{\beta}_2)$.

Solution. (a) Let Y_l, τ_l be the total count and exposure of class l, and $\eta_A = \beta_0$, $\eta_B = \beta_0 + \beta_1$, $\eta_C = \beta_0 + \beta_2$, so that $x_i'\beta = \eta_l$ for i in class l. By Theorem 17.2 with $b_i(\theta) = e^\theta$, $c_i(k, \phi) = -\log k!$ and $\phi = 1$,

$$\ell(\beta) = \sum_i \Big(y_i(\log t_i + x_i'\beta) - t_i e^{x_i'\beta} - \log y_i! \Big) = \sum_{l=A,B,C} \Big(Y_l \eta_l - \tau_l e^{\eta_l} \Big) + \sum_i (y_i \log t_i - \log y_i!)$$

$\mathcal{C}(X) = \mathcal{C}([d_A \ d_B \ d_C])$ (Theorem 13.10(c)) and every $Y_l > 0$, so by Proposition 17.5(c), $e^{\hat{\eta}_l} = Y_l/\tau_l$: $\frac{24}{1000} = \frac{3}{125}$, $\frac{48}{500} = \frac{12}{125}$, $\frac{36}{750} = \frac{6}{125}$. Hence $\hat{\beta}_0 = \log \frac{3}{125}$, $\hat{\beta}_1 = \log\Big(\frac{12}{125}\Big/\frac{3}{125}\Big) = \log 4$ and $\hat{\beta}_2 = \log\Big(\frac{6}{125}\Big/\frac{3}{125}\Big) = \log 2$, the only MLE ($X = [\mathbf{1}_n \ d_B \ d_C]$ has rank 3, Theorem 13.10(c)). So the rate ratios of classes B and C to class A are $e^{\hat{\beta}_1} = 4$ and $e^{\hat{\beta}_2} = 2$.

(b) Here $\hat{W} = \mathrm{diag}(\hat{\mu}_1, \dots, \hat{\mu}_n)$ ($b_i''(\theta) = e^\theta$). By Theorem 1.4(d), $X'\hat{W}X = \sum_i \hat{\mu}_i x_i x_i'$; x_i is $x_A = (1,0,0)'$, $x_B = (1,1,0)'$ or $x_C = (1,0,1)'$ according to the class, and $\sum_{i\in l} \hat{\mu}_i = Y_l$ (Proposition 17.5(b)). The matrix with rows x_A', x_B', x_C' is X_2 of Exercise 17.1(b), with the inverse computed there; so, by Theorem 1.4(d) again, $X'\hat{W}X = \sum_l Y_l x_l x_l' = X_2' \,\mathrm{diag}(24, 48, 36) X_2$:

$$X_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad X'\hat{W}X = \begin{pmatrix} 108 & 48 & 36 \\ 48 & 48 & 0 \\ 36 & 0 & 36 \end{pmatrix}, \qquad X_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

By Theorem 2.13(b),(c), $(X'\hat{W}X)^{-1} = X_2^{-1} \,\mathrm{diag}(\frac{1}{24}, \frac{1}{48}, \frac{1}{36})(X_2^{-1})'$, with

$$X_2^{-1} \,\mathrm{diag}\Big(\tfrac{1}{24}, \tfrac{1}{48}, \tfrac{1}{36}\Big) = \begin{pmatrix} \frac{1}{24} & 0 & 0 \\ -\frac{1}{24} & \frac{1}{48} & 0 \\ -\frac{1}{24} & 0 & \frac{1}{36} \end{pmatrix}, \qquad (X'\hat{W}X)^{-1} = \begin{pmatrix} \frac{1}{24} & -\frac{1}{24} & -\frac{1}{24} \\ -\frac{1}{24} & \frac{1}{16} & \frac{1}{24} \\ -\frac{1}{24} & \frac{1}{24} & \frac{5}{72} \end{pmatrix}$$

So $\mathrm{SE}(\hat{\beta}_1)^2 = \frac{1}{16} = \frac{1}{24} + \frac{1}{48}$ and $\mathrm{SE}(\hat{\beta}_2)^2 = \frac{5}{72} = \frac{1}{24} + \frac{1}{36}$ ($\phi = 1$; the paragraph after Example 17.7): $\mathrm{SE}(\hat{\beta}_1) = \frac{1}{4}$ and $\mathrm{SE}(\hat{\beta}_2) = \frac{\sqrt{10}}{12}$.

Exercise 17.3. For X and $\hat{W}$ of Example 17.6, compute $P_{X,\hat{W}} = X(X'\hat{W}X)^{-1}X'\hat{W}$. Check that $P_{X,\hat{W}}^2 = P_{X,\hat{W}}$, that $\hat{W}P_{X,\hat{W}}$ is symmetric while $P_{X,\hat{W}}$ is not, that $\mathrm{tr}\, P_{X,\hat{W}} = \mathrm{rank}\, X$, and that $I_4 - P_{X,\hat{W}}$ has rank 1 (Proposition 17.4).

Solution. By Example 17.6, $X'\hat{W}X = \frac{1}{2}B$ with B below. Expanding along the first row, $\det B = 14(49-9) - 7(49-21) + 7(21-49) = 560 - 196 - 196 = 168$, and by Theorem 5.24

$$B = \begin{pmatrix} 14 & 7 & 7 \\ 7 & 7 & 3 \\ 7 & 3 & 7 \end{pmatrix}, \qquad \mathrm{adj}\, B = \begin{pmatrix} 40 & -28 & -28 \\ -28 & 49 & 7 \\ -28 & 7 & 49 \end{pmatrix}$$

so $(X'\hat{W}X)^{-1} = 2B^{-1} = \frac{2}{168}\,\mathrm{adj}\, B = \frac{1}{84}\,\mathrm{adj}\, B$. Then

$$X(X'\hat{W}X)^{-1} = \frac{1}{84}\begin{pmatrix} 40 & -28 & -28 \\ 12 & 21 & -21 \\ 12 & -21 & 21 \\ -16 & 28 & 28 \end{pmatrix}, \qquad X'\hat{W} = \begin{pmatrix} \frac{3}{2} & 2 & 2 & \frac{3}{2} \\ 0 & 2 & 0 & \frac{3}{2} \\ 0 & 0 & 2 & \frac{3}{2} \end{pmatrix}$$

and $P_{X,\hat{W}} = \frac{1}{14}M$ with

$$M = \begin{pmatrix} 10 & 4 & 4 & -4 \\ 3 & 11 & -3 & 3 \\ 3 & -3 & 11 & 3 \\ -4 & 4 & 4 & 10 \end{pmatrix}, \qquad M^2 = \begin{pmatrix} 140 & 56 & 56 & -56 \\ 42 & 154 & -42 & 42 \\ 42 & -42 & 154 & 42 \\ -56 & 56 & 56 & 140 \end{pmatrix} = 14M$$

so $P_{X,\hat{W}}^2 = \frac{1}{196}M^2 = \frac{1}{14}M = P_{X,\hat{W}}$. Further,

$$\hat{W}P_{X,\hat{W}} = \frac{1}{28}\begin{pmatrix} 30 & 12 & 12 & -12 \\ 12 & 44 & -12 & 12 \\ 12 & -12 & 44 & 12 \\ -12 & 12 & 12 & 30 \end{pmatrix}$$

is symmetric, while $P_{X,\hat{W}}$ is not: its $(1,2)$ entry is $\frac{2}{7}$ and its $(2,1)$ entry $\frac{3}{14}$. $\operatorname{tr} P_{X,\hat{W}} = \frac{10+11+11+10}{14} = 3 = \operatorname{rank} X$ (Example 17.6), as it must be: $P_{X,\hat{W}}$ is idempotent of rank $\operatorname{rank} X$ (Proposition 17.4(b),(c)), and an idempotent matrix has trace equal to its rank (Theorem 6.29(c)). Finally,

$$I_4 - P_{X,\hat{W}} = \frac{1}{14}\begin{pmatrix} 4 & -4 & -4 & 4 \\ -3 & 3 & 3 & -3 \\ -3 & 3 & 3 & -3 \\ 4 & -4 & -4 & 4 \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 4 \\ -3 \\ -3 \\ 4 \end{pmatrix}(1,\ -1,\ -1,\ 1)$$

is a nonzero matrix whose columns are multiples of one vector, so its rank is $1 = 4 - \operatorname{rank} X$.

Problems

Problem 17.1 ([9, Exercise 8.3])**.** In a Poisson regression (Definition 17.1) whose design X has the column $\mathbf{1}_n$ (an intercept), let $\hat{\beta}$ be an MLE and $e = y - \hat{\mu}$ the vector of residuals. Show that

(a) $e_1 + \cdots + e_n = 0$;

(b) the residuals and each column u of X have sample covariance 0 (Definition 9.13), so that their sample correlation is 0 whenever neither e nor u is constant (Definition 9.17).

Solution. (a) This is Proposition 17.5(a), as $\mathbf{1}_n \in \mathcal{C}(X)$.

(b) By Theorem 17.2(b), $X'e = 0$, so $u'e = \sum_i u_i e_i = 0$ for each column u of X. By (a), $\bar{e} = 0$, so

$$\sum_{i=1}^{n}(u_i - \bar{u})(e_i - \bar{e}) = \sum_{i=1}^{n} u_i e_i - \bar{u}\sum_{i=1}^{n} e_i = 0$$

and the sample covariance $\frac{1}{n-1}\sum_i (u_i - \bar{u})(e_i - \bar{e})$ is 0. The sample correlation $\operatorname{cor}(u, e) = S_{ue}/\sqrt{S_{uu}S_{ee}}$ has the numerator $S_{ue} = 0$, so it is 0 when u and e are not constant (for the column $\mathbf{1}_n$ itself it is not defined).

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